Pierre Deligne

Collected Papers

English edition

A source-aligned Codex edition

Editorial note

This volume is intended for continuous mathematical reading. Each work is presented as a self-contained English text, retaining the source's mathematical notation, numbered structure, diagrams, and bibliography as closely as the edition permits.
The French and English editions are independent books rather than an interleaved bilingual layout. Texts are prepared by source-aligned transcription and translation against identified printed witnesses. Detailed source identities, correction history, and archival evidence accompany the public record instead of interrupting the papers themselves.
Coverage note. Included in numerical order: D001–D012; D014–D016; D018; D021; D022; D023; D025–D030; D034–D036; D038–D040; D043. Gaps: D013; D017; D019–D020; D024; D031–D033; D037; D041–D042. A later complete work remains eligible despite an earlier gap.

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CONGRUENCES FOR THE NUMBER OF SUBGROUPS OF ORDER p^k IN A FINITE GROUP

P. DELIGNE

The method used by Serre [1, p. 147] to prove the Sylow theorem gives results more general than that theorem. We spell them out here.

We use the following notation: p denotes a prime number; for every integer n, $v(n)$ is the exponent of the largest power of p dividing n; finally, G is a finite group of order g.

For completeness, recall the following lemma (Lazard [2, p. 71]).

Lemma 1.

a) *If $n = \sum c_i p^i$, with $0 \le c_i < p$ the base-p expansion of n, then*

$$v(n!) = \frac{1}{p-1}\left(n - \sum c_i\right).$$

b)

$$v\binom{r}{s} \ge v(r) - v(s).$$

For a), one has[1]

$$v(p^k!) = \sum_{r=1}^{k} \left\lfloor \frac{p^k}{p^r} \right\rfloor = \sum_{r=1}^{k} p^{k-r} = \sum_{i=0}^{k-1} p^i = \frac{p^k - 1}{p-1},$$

which proves the assertion in this case. It is clear that, if $0 \le c < p$, then $v((cp^k)!) = c\, v(p^k!)$, and that, if $m < p^{v(n)}$, then

$$v((n+m)!) = v(n!) + v(m!).$$

These additivity properties are the same as those of the right-hand side, whence the formula.

Part b) follows easily by returning to the definitions and taking account of the algorithm which computes the digits c_i of a sum from the digits of its summands. Equality holds when r is a power of p.

Proposition 1.

$$\binom{p^k n}{p^k m} \equiv \binom{n}{m} \pmod{p^{v(n)+2}}$$

if $p \ne 2$, or if n is odd. The congruence is always true modulo $p^{v(n)+1}$.

Proof. Let X and Y be two indeterminates. One has

$$(X+Y)^{p^k} = X^{p^k} + Y^{p^k} + pQ,$$

where Q is a polynomial of degree $< p^k$ in each variable. The binomial formula then gives

$$(X+Y)^{p^k n} = (X^{p^k} + Y^{p^k})^n + np(X^{p^k} + Y^{p^k})^{n-1} Q \tag{1}$$

[1] The source's intermediate sum is negative; we replace it by the exact Legendre calculation (source defect R1).

modulo the largest power of p which divides all the numbers $p^i\binom{n}{i}$, $i \geq 2$. These have valuation at least

$$i + v(n) - v(i) \geq v(n) + 2$$

if $p \neq 2$. If $p = 2$, one always has $i > v(i)$ (Cantor), which gives the exponent $v(n) + 1$.

The binomial coefficient $\binom{p^k n}{p^k m}$ is the coefficient of $X^{p^k m} Y^{p^k(n-m)}$.[2] The second term on the right-hand side of (1) contains no term of this type, since none of its exponents is divisible by p^k. Thus the coefficient of this monomial on the right-hand side of (1) is $\binom{n}{m}$. This proves the assertion when $p \neq 2$. If $p = 2$, the preceding already proves the congruence modulo $2^{v(n)+1}$. In addition, $Q \equiv X^{2^{k-1}} Y^{2^{k-1}} \pmod 2$, so $Q^2 \equiv X^{2^k} Y^{2^k} \pmod 4$, and the next term in the binomial expansion gives

$$\binom{2^k n}{2^k m} \equiv \binom{n}{m} + 2n(n-1)\binom{n-2}{m-1} \pmod{2^{v(n)+2}},$$

which gives the asserted improvement when n is odd. □

Theorem.

Let S be a Sylow p-subgroup of G, let $s = v(g)$, and let d_k be the number of subgroups of G of order p^{s-k}, for $0 \leq k \leq s$. Then:

(i) $d_k \equiv 1 \pmod p$*;*

(ii) *if S is cyclic or abelian of type $(p, \dots, p)$, then d_k is congruent modulo p^{k+1} to the number of subgroups of S of index p^k;*

(iii) *if S is not cyclic, then*

$$d_1 \equiv 1 + p \pmod{p^2}.$$

Let G act by left translations on the set of its subsets with p^k elements. If such a subset P of G has stabilizer H, then P is a union of right cosets of H, and hence H is a group of order p^l with $l \leq k$; one has $l = k$ if and only if P is a translate, on the right or on the left, of a subgroup of order p^k. The number of elements in the orbit of P is divisible by p^{s-l}. Therefore $\binom{g}{p^k}$ is congruent modulo p^{s-k+1} to the number of subsets of G which are translates of a subgroup of order p^k. By Proposition 1, it is also congruent to gp^{-k}. There are $d_{s-k}\, g\, p^{-k}$ such subsets. Thus

$$gp^{-k} \equiv d_{s-k}\, g\, p^{-k} \pmod{p^{s-k+1}},$$

and the first assertion follows.

We now consider case (ii), with S abelian of type $(p, \dots, p)$. Let $G_{i,j}$ be the number of subgroups of order p^i in a subgroup of order p^j; this is the number of points of a Grassmannian. For $i > j$, $G_{i,j} = 0$; for $i \leq j$,

$$G_{i,j} = \frac{(p^j - 1) \cdots (p^j - p^{i-1})}{(p^i - 1) \cdots (p^i - p^{i-1})} = \widetilde{G}_i \widetilde{G}_{j-i} \widetilde{G}_j^{-1},$$

where

$$\widetilde{G}_i = [(1 - p^i) \cdots (1 - p)]^{-1} \quad (i \geq 0), \qquad \widetilde{G}_i = 0 \quad (i < 0).$$

Here $\widetilde{G}_i = (-1)^i G_i$, where the source defines $G_i = [(p^i - 1) \cdots (p - 1)]^{-1}$. This sign normalization leaves $G_{i,j}$ unchanged.[3]

[2] The printed source instead has $X^{p^k m} Y^{p^k n}$. For $m > 0$ that monomial has total degree $p^k(m+n) > p^k n$, so its coefficient in $(X+Y)^{p^k n}$ is zero. The displayed $Y^{p^k(n-m)}$ is the source repair recorded as R2 in the editorial register.

[3] The source later claims $G_k^{-1} \equiv G_{k+1}^{-1} \pmod{p^{k+1}}$, which is false for its printed definition. The displayed normalization gives $\widetilde{G}_{k+1}^{-1} = \widetilde{G}_k^{-1}(1 - p^{k+1})$ and repairs the argument exactly (source defect A1).

We know that p^i divides the number u_i of subsets of G with p^s elements whose stabilizer has order p^{s-i}. Put $g = p^s h$. Counting in two ways the pairs formed by a subgroup of order p^{s-i} and a subset with p^s elements fixed by it gives, for $0 \le i, j \le s$,

$$\begin{cases} d_i \binom{p^i h}{p^i} = \sum_j G_{s-i,s-j} u_j & (2) \\ p^i \mid u_i & (3) \\ d_s = 1 & (4) \end{cases}$$

Now work in the localization of $\mathbf{Z}$ at p, or in $\mathbf{Z}_p$. Let e_i be the quotient of the left-hand side of (2) by $\binom{p^s h}{p^s}\widetilde{G}_i$. By Proposition 1, and since $\widetilde{G}_i$ is prime to p for $i \ge 0$, there are numbers v_i, $0 \le i \le s$, such that

$$\begin{cases} d_i \equiv \widetilde{G}_i e_i \pmod{p^{i+1}} & (5) \\ e_i = \sum_j \widetilde{G}_{i-j} v_j & (2') \\ p^i \mid v_i & (3') \\ e_s = \widetilde{G}_s^{-1} & (4') \end{cases}$$

Analogous identities hold in the group S. To show that they determine d_i modulo p^{i+1}, it is enough to check that the identities, for $0 \le i, j \le s$,

$$\begin{cases} e_i = \sum_j \widetilde{G}_{i-j} v_j & (2'') \\ p^i \mid v_i & (3'') \\ e_s = 0 & (4'') \end{cases}$$

imply $p^{i+1} \mid e_i$.

From $(2'')$ one obtains

$$e_i - e_{i+1} = \sum_j (\widetilde{G}_{i-j} - \widetilde{G}_{i+1-j}) v_j.$$

Since $\widetilde{G}_k^{-1} \equiv \widetilde{G}_{k+1}^{-1} \pmod{p^{k+1}}$, one also has

$$p^{i+1-j} \mid \widetilde{G}_{i-j} - \widetilde{G}_{i+1-j}, \qquad p^{i+1} \mid e_i - e_{i+1}.$$

The conclusion follows from $(4'')$.

The same technique, in a simpler form, applies when S is cyclic.

We turn to case (iii). The subgroups of S of index p are all normal. Their intersection S' is the Frattini subgroup of S. For every subgroup T of S, if $TS' = S$, then $T = S$. In particular, if S is not cyclic, then S/S' is not cyclic either, and the subgroups of S of index p are identified with the hyperplanes of a vector space over $\mathbf{Z}/p$ of dimension > 1; their number is congruent to $1 + p \pmod{p^2}$.

The preceding equations leave the following system.[4]

$$\begin{cases} d_0 h = u_0, \\ d_1 \binom{ph}{p} = (1+p)u_0 + u_1 \pmod{p^2}, \\ \binom{p^s h}{p^s} = u_0 + u_1 \pmod{p^2}. \end{cases}$$

[4] In its third line the source prints the lower argument p; the next source line itself uses p^s. We restore p^s here (source defect R3). The final subscript after u is damaged in the scan; the counting identities force the intended right-hand side $u_0 + u_1$.

By Proposition 1, $\binom{p^s h}{p^s} \equiv \binom{ph}{p} \equiv h \pmod{p^2}$. Hence

$$d_1 h \equiv (1+p) d_0 h + (h - d_0 h) \pmod{p^2}, \qquad \text{and}$$
$$d_1 \equiv 1 + p d_0 \pmod{p^2}.$$

The assertion follows because $d_0 \equiv 1 \pmod{p}$.

References

[1] Serre, J.-P., *Corps locaux. Act. sc. et ind.* Hermann, 1962.
[2] Lazard, M., Groupes analytiques p-adiques. *Publ. Math. I.H.E.S.*, no. 26, 1965.

Appendix to *Residues and Duality* (Lect. Notes in Math. 20, Springer-Verlag 1966, pp. 404–421)

APPENDIX: COHOMOLOGY WITH PROPER SUPPORT AND CONSTRUCTION OF THE FUNCTOR $f^!$.

by P. DELIGNE[1]

Verdier has shown that, in the topological case, the formalism of Poincare duality reduces to local problems upstairs (see [1]). To transpose his construction to the schematic setting, one needs a theory of cohomology "with proper support" for coherent sheaves.

Unless explicitly stated otherwise, all preschemes considered are noetherian and all presheaves are quasi-coherent.

No. 1. Formalities of pro-objects.

Proposition 1. *Let C be a U-category*[2] *in which finite inductive limits exist. Let h be a functor $C^\circ \to (ens)$. The following conditions are equivalent:*

(i) *h is an inductive limit, over a small*[3] *filtered ordered set, of representable functors.*

(ii) *h is an inductive limit, over a small filtered category, of representable functors.*

(iii) *h transforms finite projective limits into finite inductive limits, and there is a small set of objects such that every element of an $h(X)$ factors through one of them.*

The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are trivial, and (iii) $\Rightarrow$ (ii) is standard: h is the limit of the h_F over the category of pairs (F, α), where $\alpha \in h(F)$ and F lies in the subcategory of C stable under finite inductive limits generated by the given small set. It remains to prove that (ii) $\Rightarrow$ (i). If $\mathcal{L}$ and $\mathcal{M}$ are two filtered categories, a functor $F : \mathcal{L} \to \mathcal{M}$ is called cofinal if $\varinjlim_{\mathcal{L}} h_{F(L)}$ is the final functor. For every $G : \mathcal{M} \to \mathcal{N}$, one then has $\varinjlim_{\mathcal{M}} G = \varinjlim_{\mathcal{L}} GF$. Thus it remains to prove:

Lemma 1. *For every small filtered category $\mathcal{L}$, there exists a cofinal functor from a small filtered ordered set to $\mathcal{L}$.*

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[1] This is an expanded version of a letter from P. Deligne to R. Hartshorne (letter of March 3, 1966). The footnotes were added by the copyist.

[2] U denotes a fixed universe throughout.

[3] "small" means "$\in U$".

The first projection $\mathcal{L} \times \mathbb{N} \to \mathcal{L}$ is a cofinal functor, where $\mathbb{N}$ has its natural order. This permits one to reduce to the case where $\mathrm{Ob}\,\mathcal{L}$, preordered by the relation $\mathrm{Hom}(X, Y) \neq \varnothing$, has no greatest element. Order by inclusion the set E of finite subcategories of $\mathcal{L}$ having a single final object, where finite means having a finite set of arrows. Define a functor from E to $\mathcal{L}$ by sending each of these categories to its final object. Under the hypotheses made, E is filtered and this functor is cofinal.

The functors satisfying the equivalent conditions of Proposition 1 are the Ind-objects of C. They form a full subcategory $\mathrm{Ind}\,C$ of $\underline{\mathrm{Hom}}(C^\circ, (\mathrm{Ens}))$, stable under inductive limits. The assignment $C \mapsto \mathrm{Ind}\,C$ is a functor (cat) $\to$ (cat). If filtered inductive limits exist in C, then for every $h \in \mathrm{Ind}\,C$ the functor $X \mapsto \mathrm{Hom}(h, h_X)$ is corepresentable; hence there is a functor, denoted $\varinjlim$, from $\mathrm{Ind}\,C$ to C. In particular, one always has a functor $\mathrm{Ind}\,\mathrm{Ind}\,C \to \mathrm{Ind}\,C$, and every functor $C \to \mathrm{Ind}\,D$ extends to a functor $\mathrm{Ind}\,C \to \mathrm{Ind}\,D$. In $\mathrm{Ind}\,C$, filtered inductive limits are exact and will be denoted "$\varinjlim$"; in particular, identifying C with a full subcategory of $\mathrm{Ind}\,C$, if $(X_i)_{i\in I}$ is a filtered inductive system in C, then

$$\varinjlim_{i\in I} X_i = \varinjlim_{i\in I} \text{“}\varinjlim\text{”} X_i.$$

A functor on $\mathrm{Ind}\,C$ is representable if and only if it transforms finite projective limits into finite inductive limits, filtered inductive limits into filtered inductive limits, and its restriction to C satisfies the smallness condition of Proposition 1(iii). Indeed, Proposition 1 then shows that its restriction to C is an Ind-object, to which the functor is everywhere equal because of the condition on limits.

The preceding statements, except Proposition 1(iii) and the preceding assertion, remain true if one uses everywhere limits of sequences, or countable limits, which is the same thing.

By duality one defines the category $\mathrm{pro}\,C$ of pro-objects, namely the full subcategory of $\underline{\mathrm{Hom}}(C, (\mathrm{Ens}))^\circ$.

Proposition 2. *Let X be a quasi-compact, quasi-separated prescheme, not necessarily noetherian. The category of quasi-coherent sheaves on X is equivalent to the category of Ind-objects of the category of quasi-coherent sheaves of finite presentation on X.*

The map is "$\varinjlim$" $\mathcal{F}_i \mapsto \varinjlim \mathcal{F}_i$. If $\mathcal{F}$ is of finite presentation, then for every filtered inductive system $(\mathcal{G}_i)_{i\in I}$,

$$\mathrm{Hom}_X(\mathcal{F}, \varinjlim \mathcal{G}_i) = \varinjlim \mathrm{Hom}_X(\mathcal{F}, \mathcal{G}_i).$$

By a finite gluing argument, which commutes with $\varinjlim$, one reduces to the affine case. Thus the preceding functor is fully faithful. Its image is stable under filtered inductive limits and finite sums. It remains only to prove that for every $x \in X$, every $\mathcal{F}$ on X, and every $s \in \mathcal{F}_x$, there exists a sheaf $\mathcal{G}$ of finite presentation and a map from $\mathcal{G}$ to $\mathcal{F}$ whose image contains s.

The sheaf $\mathcal{F}$ will be a limit of images of sheaves of finite presentation, and each of these will be a quotient of a sheaf of finite presentation by a limit of sub-sheaves of finite type.

On a quasi-compact neighbourhood U of x, let $f : \mathcal{C} \to \mathcal{F}$ be a map with $\mathcal{C}$ of finite presentation and with s in its image. It is enough to extend $\mathcal{C}$ and f to all of X; proceeding step by step, it is enough to know how to extend them to $U \cup V$ when V is affine. This amounts to carrying out an extension from $U \cap V$ to V.

Lemma 2. *Let $\mathcal{C}$ be a sheaf of finite presentation on a quasi-compact open U of an affine scheme X, let $\mathcal{F}$ be a sheaf on X, and let f be a map from $\mathcal{C}$ to $\mathcal{F}|U$. It is possible to extend $\mathcal{C}$ and f to $\overline{\mathcal{C}}$ and $\overline{f}$, defined on X, with $\overline{\mathcal{C}}$ of finite presentation.*

Let ι be the inclusion of U in X, and let $\mathcal{C}_1$ be the fiber product of $\iota_*\mathcal{C}$ and $\mathcal{F}$ over $\iota_*\iota^*\mathcal{F}$. Then $\mathcal{C}_1$ extends $\mathcal{C}$ and maps to $\mathcal{F}$. If one writes $\mathcal{C}_1$ as a limit of its sub-sheaves of finite type, then, since U is quasi-compact and $\mathcal{C}$ is of finite presentation, one of these already extends $\mathcal{C}$. Writing the latter as a quotient of $\mathcal{O}^n$ by a limit of sub-sheaves of finite type of $\mathcal{O}^n$, one sees in the same way that $\mathcal{C}_1$ may be replaced by a sheaf of finite presentation.

Corollary 1. *A contravariant additive functor $F : (q\ coh) \to Ab$ is representable if and only if it is left exact and transforms filtered inductive limits into filtered inductive limits.*

Corollary 2. *Every sheaf of finite presentation defined on a quasi-compact open U of X extends to a sheaf of finite presentation on X.*

Let $\mathcal{A}$ be an abelian category. We shall have to work essentially in $\operatorname{pro} D^b\mathcal{A}$, the full subcategory of $\operatorname{pro} D(\mathcal{A})$ consisting of the "$\varprojlim$" K_i for which the cohomology of K_i is uniformly bounded as i varies. The functor $\mathbb{H}^p$ extends to a functor $\operatorname{pro} D(\mathcal{A}) \to \operatorname{pro} \mathcal{A}$.

Proposition 3. *The functors $\mathbb{H}^p$ form a conservative[4] family of functors on $\operatorname{pro} D^b\mathcal{A}$.*

[4]That is, if u is an arrow such that all $\mathbb{H}^p(u)$ are invertible, then u is invertible.

Let $f : K \to L$ be an arrow of pro $D^b\mathcal{A}$ such that the maps $\mathbb{H}^p(K) \to \mathbb{H}^p(L)$ are isomorphisms in pro $\mathcal{A}$. We must check that, for every M in $D^b(\mathcal{A})$,

$$\mathrm{Hom}(K, M) \longleftarrow \mathrm{Hom}(L, M)$$

is an isomorphism. Write $K = \text{“}\varprojlim\text{”}K_\alpha$. There is a convergent spectral sequence

$$\mathrm{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \mathrm{Ext}^{p+q}(K_\alpha, M),$$

whence, on passing to the limit, another convergent spectral sequence

$$\varinjlim \mathrm{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \varinjlim \mathrm{Ext}^{p+q}(K_\alpha, M),$$

since q is uniformly bounded. Using the fact that every functor, here Ext, passes to pro-objects, and using the functor $\varinjlim : \mathrm{Ind\ Ab} \to \mathrm{Ab}$, we may rewrite this as

$$\varinjlim \mathrm{Ext}^p(\mathbb{H}^q K, M) \Longrightarrow \varinjlim \mathrm{Ext}^{p+q}(K, M) = \mathrm{Hom}(K[-p-q], M).$$

This spectral sequence remains valid for $M \in D(\mathcal{A})$. The proposition follows immediately from the existence of these spectral sequences.

No. 2. Fundamental lemmas.

Recall that all preschemes considered are noetherian.

Proposition 4 (Duality theorem for an open immersion)**.** *Let U be an open subset of X, let j be the inclusion, let $\mathcal{J}$ be an ideal defining the closed complement, let $\mathcal{F}$ be a coherent sheaf on U, and let $\mathcal{G}$ be quasi-coherent on X. Let $\overline{\mathcal{F}}$ be a coherent extension of $\mathcal{F}$. Then*

$$\mathrm{Hom}_U(\mathcal{F}, j^*\mathcal{G}) = \mathrm{Hom}_X(\text{“}\varprojlim\text{”}\mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}). \tag{5}$$

The map $\varinjlim \mathrm{Hom}_X(\mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}) \to \mathrm{Hom}_U(\mathcal{F}, \mathcal{G})$ is injective: if the image under f of $\mathcal{J}^n\overline{\mathcal{F}}$ in $\mathcal{G}$ has its support in the complement of U, then it is annihilated by a power of $\mathcal{J}$, say $\mathcal{J}^k$, and the image of $\mathcal{J}^{n+k}\overline{\mathcal{F}}$ is zero.

Now assume that X is affine, and let $f \in \mathrm{Hom}_U(\mathcal{F}, \mathcal{G})$. For k sufficiently large, every section s of $\mathcal{J}^k\overline{\mathcal{F}}$ over X has an image over U which extends to a section of $\mathcal{G}$ over X. Replacing $\overline{\mathcal{F}}$ by $\mathcal{J}^k\overline{\mathcal{F}}$, we may

[5]This formula is of course well known; cf., for example, P. Gabriel's thesis.

suppose that we have a diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R & \longrightarrow & \mathcal{O}^n & \longrightarrow & \overline{\mathcal{F}} & \longrightarrow & 0 \\ & & \vdots & & \| & & \vdots & & \\ 0 & \longrightarrow & R_1 & \longrightarrow & \mathcal{O}^n & \longrightarrow & \mathcal{G} & & \end{array}$$

in which the dotted arrows are defined on U.

For ℓ sufficiently large, by Krull's theorem and by the fact that $R/(R \cap R_1)$ is concentrated outside U, one has $R \cap \mathcal{J}^\ell \cdot \mathcal{O}^n \subset R_1$, which permits f to be extended to a map $\overline{f} : \mathcal{J}^\ell \overline{\mathcal{F}} \to \mathcal{G}$.

In the general case, let U_i be a finite affine covering of X, and let U_{ijk} be finite affine coverings of $U_{ij} = U_i \cap U_j$. Since $\varinjlim$ commutes with finite products, one obtains

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Hom}_U(\mathcal{F}, j^*\mathcal{G}) & \longrightarrow & \prod_i \mathrm{Hom}_{U_i \cap U}(\mathcal{F}, j^*\mathcal{G}) & \rightrightarrows & \prod_{ijk} \mathrm{Hom}_{U_{ijk} \cap U}(\mathcal{F}, j^*\mathcal{G}) \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & \varinjlim \mathrm{Hom}_X(\mathcal{J}^n\overline{\mathcal{F}}, j^*\mathcal{G}) & \longrightarrow & \prod_i \varinjlim \mathrm{Hom}_{U_i}(\mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}) & \rightrightarrows & \prod_{ijk} \varinjlim \mathrm{Hom}_{U_{ijk}}(\mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}) \end{array}$$

which completes the proof.

Proposition 4 gives an explicit formula for the functor (coh on U) $\to$ pro (coh on X) left adjoint to j^*. We shall denote this functor by $j_!$ (extension by zero)[6]; it follows from Krull's theorem that it is exact. This also follows from the fact that, since X is noetherian, an injective object of the category of quasi-coherent sheaves on X, restricted to U, remains injective as a quasi-coherent sheaf on U.

Proposition 5 (Independence of the compactification)**.**

$$\begin{array}{ccc} U & \hookrightarrow & X \\ \| & & \downarrow f \\ U & \hookrightarrow & Y \end{array}$$

Let $f : X \to Y$ be a proper morphism inducing an isomorphism between the open subset U of Y and $f^{-1}(U)$. Let $\mathcal{J}$ be a sheaf of ideals defining $Y \setminus U$, and put $\mathcal{I} = f^*\mathcal{J}$. If $\mathcal{F}$ is coherent on X, then

(i) *for $k > 0$, the projective system $\mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$ is essentially zero, that is, it defines the zero pro-object;*

(ii) *for $k = 0$, for all sufficiently large n, $f_* \mathcal{I}^{n+1} \mathcal{F} = \mathcal{J} \cdot f_* \mathcal{I}^n \mathcal{F}$.*

[6]This terminology and notation plainly conflict with the generally accepted ones, for example in Godement's book and in SGA 1962. The reader may prefer to read $\mathfrak{I}_!$ instead of $j_!$.

One has

$$\bigoplus_{n=0}^{\infty} \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \quad \text{is a finite type } \bigoplus_{n=0}^{\infty} \mathcal{J}^n\text{-module (EGA III).}$$

Thus, for n sufficiently large, the map $\mathcal{J} \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \to \mathbb{R}^k f_* \mathcal{I}^{n+1} \mathcal{F}$ is surjective; (ii) follows. If $k > 0$, $\mathbb{R}^k f_*$ is zero on U, so there exists N such that $\mathcal{J}^N \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$ is zero for every n. The diagram

$$\begin{array}{ccccc} \mathcal{J}^p \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} & \longrightarrow & \mathbb{R}^k f_* \mathcal{I}^{n+p} \mathcal{F} & \longrightarrow & 0 \quad (\text{if } n \geq n_0) \\ \downarrow & & \downarrow & & \\ \mathcal{J}^p \cdot \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} & \longrightarrow & \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} & & \end{array}$$

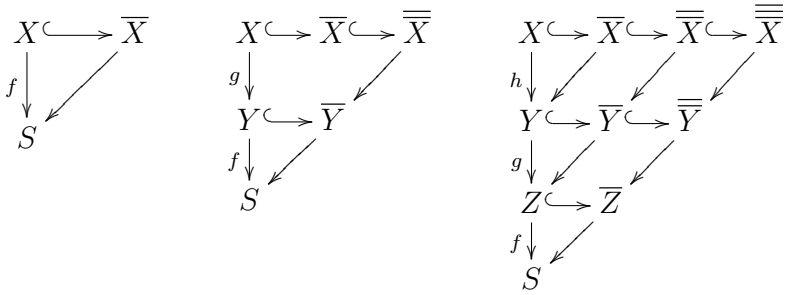
proves that for N sufficiently large and $k > 0$ the map

$$\mathbb{R}^k f_* \mathcal{I}^{n+N} \mathcal{F} \to \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$$

is zero, which is (i).

No. 3. Definition of $\mathbb{R}f_!$.

A morphism f, respectively a composable pair of morphisms, respectively a triple, will be called compactifiable if one can respectively find commutative diagrams



in which the horizontal arrows are open immersions and the oblique arrows are proper. Every compactifiable morphism is separated and of finite type, and Nagata asserts in [2] that his proof gives the converse for integral noetherian schemes, but the hypotheses he imposes on the base are not clear.[7]

We propose, for every compactifiable f, to define

$$\mathbb{R}f_! : \operatorname{pro} D^b_{\mathrm{coh}}(X) \to \operatorname{pro} D^b_{\mathrm{coh}}(S).$$

Recall that the category $D^b_{\mathrm{coh}}(S)$, the bounded derived category of coherent sheaves on S, is a full subcategory of the derived category of

[7]Mumford is said to have checked, via Nagata's proof, that every separated morphism of finite type of noetherian schemes is compactifiable.

all sheaves, not necessarily quasi-coherent, on S; its essential image consists of complexes with bounded coherent cohomology.

If f is proper, one has $\mathbb{R}f_* : D^b_{\mathrm{coh}}(X) \to D^b_{\mathrm{coh}}(S)$ of finite dimension, and it induces $\mathbb{R}f_! : \mathrm{pro}\, D^b_{\mathrm{coh}}(X) \to \mathrm{pro}\, D^b_{\mathrm{coh}}(S)$. If f is an open immersion, $f_! : (\text{coh on } X) \to \mathrm{pro}\,(\text{coh on } S)$ induces

$$D^b(\text{coh on } X) \to D^b\,\mathrm{pro}\,(\text{coh on } S),$$

which maps to $\mathrm{pro}\, D^b_{\mathrm{coh}}(S)$, and this map extends to

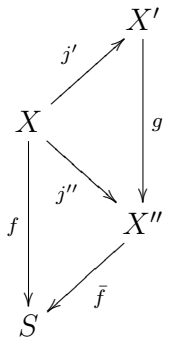
$$\mathbb{R}f_! : \mathrm{pro}\, D^b_{\mathrm{coh}}(X) \to \mathrm{pro}\, D^b_{\mathrm{coh}}(S).$$

If a coherent complex K on X is extended to $\overline{K}$ on S, its image is "$\varprojlim$" $\mathcal{J}^n\overline{K}$, where $\mathcal{J}$ defines $S \setminus X$.

For a composite $f = gh$, with g proper and h an open immersion, set $\mathbb{R}f_! = \mathbb{R}g_!\,\mathbb{R}h_!$. One must verify:

Independence of the compactification.

Consider a commutative diagram



where j' and j'' are open immersions, and $\bar{f}$ and g are proper maps. We have $j'X = g^{-1}j''X \cap j'X$. We must prove that

$$\mathbb{R}\bar{f}_*\,\mathbb{R}j''_! = \mathbb{R}(\bar{f}g)_*\,\mathbb{R}j'_!,$$

or equivalently, since $\mathbb{R}(\bar{f}g)_* = \mathbb{R}\bar{f}_*\,\mathbb{R}g_*$, that

$$\mathbb{R}j''_! = \mathbb{R}g_*\,\mathbb{R}j'_!.$$

One reduces to the case where X is dense in X', so as to be in the situation of Proposition 5. Let K be a bounded coherent complex on X, extended to $\overline{K}$ on X'. Then $g_*\overline{K}$ extends K on X'', and one has a map

$$g_*\mathcal{J}'^n\overline{K} \to \mathbb{R}g_*\mathcal{J}'^n\overline{K}, \tag{1}$$

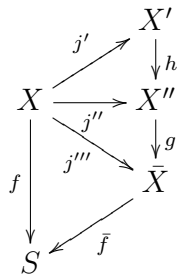
where $\mathcal{J}'$ denotes a sheaf of ideals defining $X' \setminus X$. Proposition 5(ii) shows that $j''_!K =$ "$\varprojlim$" $g_*\mathcal{J}'^n\overline{K}$. Passing to the "limit", one obtains

$$\text{“}\varprojlim\text{”}\,\mathbb{R}^p g_*\mathcal{J}'^n\overline{K}^q \Longrightarrow \text{“}\varprojlim\text{”}\,\mathbb{R}^{p+q}g_*\mathcal{J}'^n\overline{K}.$$

Proposition 5(i) shows that this spectral sequence degenerates, and Proposition 3 implies that the arrow deduced from (1) by passing to the "limit" is an isomorphism. This gives the desired formula.

It remains to verify the following.

(a) For every diagram of the form



one has a compatibility

$$\begin{aligned}\mathbb{R}\bar{f}_*\,\mathbb{R}j'''_! &= \mathbb{R}\bar{f}_*\,\mathbb{R}g_*\,\mathbb{R}j''_!\\ &= \mathbb{R}\bar{f}_*\,\mathbb{R}g_*\,\mathbb{R}h_*\,\mathbb{R}j'_!\\ &= \mathbb{R}\bar{f}_*\,\mathbb{R}(gh)_*\,\mathbb{R}j'_!\\ &= \mathbb{R}f_*\,\mathbb{R}j'''_!.\end{aligned}$$

Since two compactifications can always be dominated by a third, for example $X' \times_S X''$, this is enough to prove independence of the choices.

(b) For every compactifiable pair, an identification

$$\mathbb{R}(fg)_! = \mathbb{R}f_!\,\mathbb{R}g_!.$$

(c) For every compactifiable triple, a compatibility

$$\begin{array}{ccc}\mathbb{R}(fgh)_! & = & \mathbb{R}(fg)_!\,\mathbb{R}h_!\\ \| & & \|\\ \mathbb{R}f_!\,\mathbb{R}(gh)_! & = & \mathbb{R}f_!\,\mathbb{R}g_!\,\mathbb{R}h_!\end{array}$$

One would then have proved:

Theorem 1. *For $f : X \to Y$ compactifiable, $K \in pro\, D^b_{coh}(X)$ and $L \in pro\, D^b_{coh}(Y)$, put*

$$\mathrm{Hom}_f(K, L) = \mathrm{Hom}_Y(\mathbb{R}f_!K, L).$$

Modulo compactifiability questions, this makes the categories

$$pro\, D^b_{coh}(X)$$

into a cofibrant category over noetherian preschemes, with the arrows compactifiable.

No. 4. Definition of $\mathbb{R}f^!$.

It follows from a general theorem of Verdier [1] that the functor $\mathbb{R}f_* : D(X) \to D(S)$ has a right adjoint $D^+(S) \to D^+(X)$. It deserves a name, say $\mathbb{R}f^!$, only when f is proper, and its definition is not directly suited for computation.

First, one must make explicit a procedure for computing $\mathbb{R}f_*$ at the level of complexes, of the form

$$\mathbb{R}f_*K = f_*C^*(K),$$

where C^* is a finite acyclic resolution depending functorially and exactly on the object to which it is applied, and compatible with filtered inductive limits. Then, for every injective I on S, the functor $\mathrm{Hom}(f_*C^p\mathcal{F}, I)$ is exact in the quasi-coherent sheaf $\mathcal{F}$ on X and transforms $\varinjlim$ into $\varprojlim$; hence it is representable by $f_p^!I$, an injective quasi-coherent sheaf. The $f_p^!I$ form a complex, which defines $\mathbb{R}f^! : D^+S \to D^+X$, right adjoint to $\mathbb{R}f_* : D(X) \to D(S)$.

Here is how to define such a resolution.

1) If S is separated: take a finite covering of X by opens affine over S, for instance affine opens, and put $\mathbb{R}f_*K = f_*\check{C}(\mathcal{U}, K)$, the alternating Cech complex.

2) Otherwise: the canonical flasque resolutions have two defects here.

(a) $f_*C^*\mathcal{F}$ is no longer quasi-coherent. This causes no problem as long as, as here, quasi-coherent injectives are injective as sheaves.

(b) C^* does not commute with filtered inductive limits. This defect is easy to correct: write a quasi-coherent sheaf as $\mathcal{F} = \varinjlim \mathcal{F}_i$, with the $\mathcal{F}_i$ of finite presentation, and put $C'^*\mathcal{F} = \varinjlim C^*\mathcal{F}_i$. This does not depend on the choices.

For an open immersion, take $\mathbb{R}f^!K = f^*K$.

For a composite morphism $f = gh$, with g proper and h an open immersion, take $\mathbb{R}f^! = \mathbb{R}g^!\,\mathbb{R}h^!$. Combining Proposition 4 with the preceding construction gives:

Theorem 2 (Duality)**.** *Let $f : X \to S$ be compactifiable, $f = gh$. The functors*

$$\mathbb{R}f_! : \mathit{pro}\, D^b_{coh}(X) \to \mathit{pro}\, D^b_{coh}(S) \quad \textit{and} \quad \mathbb{R}f^! : D^+(S) \to D^+(X)$$

are adjoint to one another.

One should verify the independence of the compactification and a formalism of compatibilities and identifications analogous to the case of $\mathbb{R}f_!$. Particular cases of the formula sheet follow from the adjunction formula, by uniqueness of a genuine adjoint.

In the general case one may use the following proposition.

Proposition 6. *Let $f : X \to S$ be compactifiable and let $L \in D^+_{\mathrm{qc}}(S)$. The fiber at $x \in X$ of $\mathcal{H}^p\mathbb{R}f^!L$ is given by*

$$(\mathcal{H}^p\mathbb{R}f^!L)_x = \varinjlim_{x\in U} \mathrm{Hom}^p_S(\mathbb{R}f_!(U, \mathcal{O}_U), L).$$

One may assume that X is proper and that L is given as a complex of injective quasi-coherent sheaves. Let U be an affine open of X, and let $\mathcal{J}$ be an ideal defining $X \setminus U$. Then

$$\begin{aligned}(\mathcal{H}^p\mathbb{R}f^!L)(U) &= \mathbb{H}^p(\mathbb{R}f^!L(U)) = \mathbb{H}^p\,\mathrm{Hom}_U(\mathcal{O}, \mathbb{R}f^!L)\\ &= \mathbb{H}^p \varinjlim \mathrm{Hom}_X(\mathcal{J}^n, \mathbb{R}f^!L) = \varinjlim \mathrm{Hom}^p_S(\mathbb{R}f_*\mathcal{J}^n, L)\\ &= \mathrm{Hom}^p_S(\mathbb{R}f_!(U, \mathcal{O}_U), L),\end{aligned}$$

and the proposition follows.

We shall assume for $\mathbb{R}f^!$ a variance formalism analogous to that of Theorem 1 for $\mathbb{R}f_!$.

If one wants to make the definition of $\mathbb{R}f^!$ more explicit in the proper case, one can observe that if a quasi-coherent sheaf $\mathcal{F}$ represents a functor F, then for every open U and every $\mathcal{J}$ defining $X \setminus U$, one has

$$\mathcal{F}(U) = \varinjlim \mathrm{Hom}(\mathcal{J}^n, \mathcal{F}) = \varinjlim F(\mathcal{J}^n).$$

No. 5. Computation of $\mathbb{R}f^!$.

For a finite morphism, $\mathbb{R}f^!$ is what one wants; to see this, it is enough to take the identity as resolving functor in the computation of $\mathbb{R}f_*$. For projective space, the uniqueness of an adjoint leaves no choice; one uses the fact that the adjunction formula is available for $\mathbb{R}f_! = \mathbb{R}f_* : D(X) \to D(S)$. We shall return to the arrow. For an open immersion, $\mathbb{R}f^! = \mathbb{R}f^*$.

Thus for every quasi-projective open U of X over S, one can compute the restriction of $\mathbb{R}f^!$ to U. In particular, if one can check locally that $\mathbb{R}^q f^!K = 0$ for $q \neq p$, then one knows $\mathbb{R}f^!K$, since an object of the derived category with only one non-zero cohomology sheaf is determined by that sheaf. Recall an important case where this condition is satisfied, with $K = \mathcal{O}$.

Proposition 7. *Let $f : X \to Y$ be compactifiable. Suppose that locally on X and on Y one can find a diagram*

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle g}\searrow \quad \swarrow{\scriptstyle h} & \\ & S & \end{array}$$

where g and h are locally complete intersections of relative dimensions d' and d'', and let $d = d' - d''$.

Then $\mathbb{R}f^!\mathcal{O}$ is reduced to a single cohomology sheaf, situated in degree $-d$, and that sheaf is invertible.

Indeed, $\mathbb{R}g^!\mathcal{O} = \mathbb{R}f^!\,\mathbb{R}h^!\mathcal{O}$. We know, by local calculations in projective space, that $\mathbb{R}g^!\mathcal{O}$ and $\mathbb{R}h^!\mathcal{O}$ are of the indicated type, situated in degrees $-d'$ and $-d''$. Since everything is local, one may suppose that $\mathbb{R}^{-d''}h^!\mathcal{O}$ is isomorphic to $\mathcal{O}$, and the assertion follows.

To pass from this to more general complexes, one would first have to prove formulas of the form

$$\mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L) = \mathbb{R}f^!(K) \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L.$$

Only the definition of the arrow is problematic; its being an isomorphism is a local question. Of course boundedness restrictions are needed. I have checked none of the following in detail.

First method for defining the arrow.

Assume that K and L lie in $D^b_{\mathrm{coh}}(S)$, not merely in the quasi-coherent setting, and also that

(a) $\mathbb{R}f^!K$ is bounded, it being automatically coherent, which is a local problem;

(b) $K \overset{\mathbb{L}}{\otimes} L$ is bounded;

(c) $\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L$ is bounded.

To define

$$\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L \to \mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L),$$

it is enough by duality to define

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L) \to K \overset{\mathbb{L}}{\otimes} L.$$

By Kunneth,

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L) = \mathbb{R}f_!\,\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} L,$$

and the required arrow is induced by the adjunction arrow $\mathbb{R}f_!\,\mathbb{R}f^!K \to K$.

Second method.

Assume that f is compactifiable, flat, and locally a relative complete intersection. Then $\mathbb{R}f^!\mathcal{O} = \omega[d]$, where ω is invertible. Let $\mathbb{R}f_!$ denote a procedure for computing $\mathbb{R}f_!$ at the level of complexes, namely such a procedure for a compactification $\bar{f} : \bar{X} \to S$ of f, assumed to be of

the type described in No. 4 and such that $C^p\mathcal{F}$ is zero for $p > d$, which can be arranged by truncation. Then there is an arrow

$$\mathbb{R}^d\bar{f}_*\mathcal{F}[-d] \to \mathbb{R}\bar{f}_*\mathcal{F} \quad \text{for every } \mathcal{F} \text{ on } \bar{X},$$

since the last cohomology sheaf maps to its complex. From this one obtains morphisms of complexes

$$\mathrm{Hom}_S(\mathbb{R}\bar{f}_*\mathcal{F}, I) \to \mathrm{Hom}_S(\mathbb{R}^d\bar{f}_*\mathcal{F}[-d], I) \quad \text{for every complex } I \text{ on } S.$$

Let I be injective. The functor $\mathbb{R}^d\bar{f}_*\mathcal{F}[-d]$ is right exact in $\mathcal{F}$, so the functor in $\mathcal{F}$ on the right of the arrow is representable. Define I_1 by

$$\mathrm{Hom}_S(\mathbb{R}^d\bar{f}_*\mathcal{F}, I) = \mathrm{Hom}_{\bar{X}}(\mathcal{F}, I_1) \quad \text{and} \quad \mathbb{R}f_1^!I = I_1[d].$$

The preceding construction defines

$$\mathbb{R}f^!I \to \mathbb{R}f_1^!I,$$

and, for a complex of injectives, $\mathbb{R}f_1^!$ is defined term by term.

It remains to evaluate $\mathbb{R}f_1^!I$ on affine opens U of X. It is given by $\varinjlim \mathrm{Hom}(\mathbb{R}^d f_!\mathcal{J}^n, I)$. Since this problem is local upstairs, it is easy to show by projective methods that the dual of the ind-object

$$\text{“}\varprojlim\text{”}\,\mathbb{R}^d f_!\mathcal{J}^n = \mathbb{R}^d f_!(U, \mathcal{O})$$

is $f_*(U, \omega)$, represented as the limit of its submodules of sections having a pole of order at least n outside U, with $n \to \infty$.

This gives

$$\mathbb{R}f_1^!I = \omega \otimes f^*I[d],$$

and the general formula

$$\mathbb{R}f^!K = f^*K \otimes \omega[d].$$

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LEFSCHETZ THEOREM AND DEGENERATION CRITERIA FOR SPECTRAL SEQUENCES

by P. DELIGNE[1]

1. General theorems

Let $\mathcal{A}$ be an abelian category and let $D^b(\mathcal{A})$ be the full subcategory of its derived category formed by the bounded complexes. Recall that if T is a cohomological functor (see [8] or [9], p. 10) from $D^b(\mathcal{A})$ to an abelian category $\mathcal{B}$, Verdier [8] has defined a spectral sequence, functorial in $X \in D^b(\mathcal{A})$:

$$E_2^{pq} = T(H^q(X)[p]) \Longrightarrow T(X[p+q]). \tag{1.1}$$

Proposition (1.2). Let $X \in D^b(\mathcal{A})$. The following conditions are equivalent:

(i) for every T as above, the spectral sequence (1.1) degenerates;

(ii) the object X is isomorphic, in $D^b(\mathcal{A})$, to $\sum_i H^i(X)[-i]$.

(i) ⇒ (ii). Denote by T_i the cohomological functor in K defined by

$$T_i(K) = \mathrm{Hom}_{D^b(\mathcal{A})}(H^i(X), K).$$

For each of the T_i ($i \in \mathbb{Z}$), the spectral sequence (1.1) is written

$$E_2^{pq} = \mathrm{Ext}^p(H^i(X), H^q(K)) \Longrightarrow \mathrm{Hom}_{D^b(\mathcal{A})}(H^i(X), K[p+q]), \tag{1.3}$$

and the homomorphism from $\mathrm{Hom}(H^i(X), K[i])$ to

$$E_2^{0i} = \mathrm{Hom}(H^i(X), H^i(K))$$

which follows from it is none other than the one coming from the functoriality of H^0.

If the spectral sequences (1.3) degenerate, the maps

$$\mathrm{Hom}(H^i(X)[-i], K) \longrightarrow \mathrm{Hom}(H^i(X), H^i(K))$$

are surjective. By hypothesis, this is so for $K = X$; hence there exist morphisms a_i from $H^i(X)[-i]$ to X, inducing the identity on the i-th cohomology object. Moreover, since $H^i(X) = 0$ except for finitely many values of i, the a_i vanish except for finitely many i, which allows one to define

$$a = \sum_i a_i : \sum_i H^i(X)[-i] \longrightarrow X.$$

By construction, a induces the identity on cohomology, hence is an isomorphism.

(ii) ⇒ (i). The spectral sequence (1.1) is an additive functor in X, and degenerates when X has only one nonzero cohomology object; hence it also degenerates if X is a sum of complexes having only one nonzero cohomology object.

Remark (1.4). It follows from the preceding proof that it is enough to verify condition (i) for cohomological functors of the form $\mathrm{Hom}_{D^b(\mathcal{A})}(A, *)$, with $A \in \mathrm{Ob}(\mathcal{A})$. By duality, it would also be enough to consider only the functors $\mathrm{Hom}_{D^b(\mathcal{A})}(*, A)$, since condition (ii) is self-dual.

[1] Aspirant at the F.N.R.S.

Theorem (1.5). Let $X \in D^b(\mathcal{A})$, let n be an integer, and let u be an endomorphism of degree 2 of X,

$$u : X \longrightarrow X[2],$$

whose iterates induce isomorphisms

$$u^i : H^{n-i}(X) \xrightarrow{\sim} H^{n+i}(X) \qquad (i \geq 0).$$

Under these conditions, X is isomorphic to $\sum_i H^i(X)[-i]$ in the category $D^b(\mathcal{A})$.

We verify condition (i) of (1.2). Let T be a cohomological functor from $D^b(\mathcal{A})$ to $\mathcal{B}$, and put $T^p(K) = T(K[p])$.

The primitive part of $H^{n-i}(X)$, denoted ${}_0H^{n-i}(X)$, is the kernel of the homomorphism

$$u^{i+1} : H^{n-i}(X) \longrightarrow H^{n+i+2}(X).$$

One checks that, for $i \geq 0$, the maps

$$\begin{aligned} \bigoplus_{k\geq 0} u^k &: \bigoplus_{k\geq 0} {}_0H^{n-i-2k}(X) \xrightarrow{\sim} H^{n-i}(X), \\ \bigoplus_{k\geq 0} u^{k+i} &: \bigoplus_{k\geq 0} {}_0H^{n-i-2k}(X) \xrightarrow{\sim} H^{n+i}(X) \end{aligned} \tag{1.6}$$

are isomorphisms.

Since the spectral sequence (1.1) is functorial in X and compatible with translations of degrees, the endomorphism u of X defines an endomorphism of degree 2 of the spectral sequence,

$$u : E_r^{pq} \longrightarrow E_r^{p,q+2},$$

commuting with the differentials d_r ($r \geq 2$).

When $r = 2$, the endomorphism u of the spectral sequence

$$u : T^p(H^q(X)) \longrightarrow T^p(H^{q+2}(X))$$

is obtained from the morphism $u : H^q(X) \to H^{q+2}(X)$ by functoriality of T^p.

Let $i \geq 0$. Denote by ${}_0E_r^{p,n-i}$ the kernel of the homomorphism

$$u^{i+1} : E_r^{p,n-i} \longrightarrow E_r^{p,n+i+2}$$

(the primitive part of the spectral sequence). For $r = 2$, the decompositions (1.6) induce, by functoriality of each T^p, decompositions

$$\begin{aligned} \bigoplus_{k\geq 0} u^k &: \bigoplus_{k\geq 0} {}_0E_2^{p,n-i-2k} \xrightarrow{\sim} E_2^{p,n-i}, \\ \bigoplus_{k\geq 0} u^{k+i} &: \bigoplus_{k\geq 0} {}_0E_2^{p,n-i-2k} \xrightarrow{\sim} E_2^{p,n+i}. \end{aligned} \tag{1.7}$$

We prove by induction on r that the differentials d_r ($r \geq 2$) are zero. The induction hypothesis implies that $E_2 = E_r$, so that the decomposition (1.7) applies to the term E_r of the spectral sequence. Since u commutes with the d_r, it will be enough to prove that d_r vanishes on the primitive part of E_r.

The following diagram is commutative:

$$\begin{array}{ccc} {}_0E_r^{p,n-i} & \xrightarrow{d_r} & E_r^{p+r,n-i-r+1} \\ \downarrow{\scriptstyle u^{i+1}} & & \downarrow{\scriptstyle u^{i+1}} \\ E_r^{p,n+i+2} & \xrightarrow{d_r} & E_r^{p+r,n+i-r+3}. \end{array}$$

The left vertical arrow is zero by definition of primitivity. The right vertical arrow is a monomorphism, because the arrow

$$u^{i+r-1} : E_r^{p+r,n-(i+r-1)} \longrightarrow E_r^{p+r,n+(i+r-1)}$$

is an isomorphism[2] and $r - 1 \geq 1$. It follows that d_r is zero.

[2]The printed source gives $E_r^{p,n+(i+r-1)}$ as the target; the index $p + r$, required by the bidegree of d_r, is restored here.

Remark (1.8). One should take care that in general one cannot choose an isomorphism between X and $\sum_i H^i(X)[-i]$ which transforms u into the sum of the maps $H^i(u) : H^i(X) \to H^{i+2}(X)$. One easily obtains a counterexample by taking $H^i(X) = 0$ for $i \neq n$.

One can define an isomorphism, "more beautiful than the others", between X and $\sum_i H^i(X)[-i]$, but it will not be used here.

Remark (1.9). Let $X \in D^b(\mathcal{A})$, let n and s be integers, and let u be an endomorphism of degree s (respectively $2s$) of X. Suppose that $H^i(X) = 0$ when i is not of the form ks $(0 \leq k \leq n)$ (respectively $0 \leq k \leq 2n$), and that u^{n-2i} (respectively u^i) induces an isomorphism between $H^{is}(X)$ and $H^{(n-i)s}(X)$ (respectively between $H^{(n-i)s}(X)$ and $H^{(n+i)s}(X)$). Under these hypotheses, the proof of (1.5) still shows that X is isomorphic to $\sum_i H^i(X)[-i]$.

Remark (1.10). Let $(X_i)_{i\in\mathbb{Z}}$ be a family of objects of $D^b(\mathcal{A})$ and let u be a family of degree-2 homomorphisms $u_i : X_i \to X_{i+1}[2]$. If, for all i and j, u_i induces an isomorphism between $H^{n-j}(X_i)$ and $H^{n+j}(X_{i+j})$, it is still true that each X_i is isomorphic to the sum of its cohomology objects. To see this, it is enough to apply the preceding arguments formally to

$$u : \sum_i X_i \longrightarrow \sum_i X_i,$$

even if this sum does not exist in $D^b(\mathcal{A})$.

The same remark applies to the variant (1.9).

Theorem (1.11). Let $X \in D^b(\mathcal{A})$ and suppose that X admits degree-0 endomorphisms $\pi_i : X \to X$ such that $H^j(\pi_i) = \delta_{ij}$. Then X is isomorphic to $\sum_i H^i(X)[-i]$ in the category $D^b(\mathcal{A})$.

We again apply criterion (1.2) (i). The diagram

$$\begin{array}{ccc} E_r^{p,q} & \xrightarrow{d_r} & E_r^{p+r,q-r+1} \\ \downarrow{\scriptstyle \pi_q} & & \downarrow{\scriptstyle \pi_q} \\ E_r^{p,q} & \xrightarrow{d_r} & E_r^{p+r,q-r+1} \end{array}$$

is commutative; the first column is the identity and the second is zero, so that $d_r = 0$.

Corollary (1.12). Let $X \in D^b(\mathcal{A})$ be the sum of a finite family of objects $X_k \in D^b(\mathcal{A})$. If

$$X \simeq \sum_i H^i(X)[-i],$$

then, for every k,

$$X_k \simeq \sum_i H^i(X_k)[-i].$$

Let j_k be the canonical injection of X_k into X, let pr_k be the canonical projection of X onto X_k, and let π_i be an endomorphism of X such that $H^j(\pi_i) = \delta_{ij}$. For every k, the $\mathrm{pr}_k \, \pi_i j_k$ satisfy the hypothesis of (1.11).

Remark (1.13). One checks that, in order that the isomorphism between X and $\sum_i H^i(X)[-i]$ can be chosen so that the π_i are identified with the pr_i, it is necessary and sufficient that the π_i be pairwise orthogonal idempotents. There is then a unique isomorphism with this property and inducing the identity on cohomology.

2. Applications

Let $f : X \to Y$ be a continuous map between topological spaces, or even a morphism of topoi, let A be a sheaf of rings on X, and let $\mathcal{F}$ be a sheaf of A-modules. If $u \in H^2(X, A)$, then u defines a degree-2 endomorphism in $D^b(X)$, again denoted

$$u : \mathcal{F} \longrightarrow \mathcal{F}[2].$$

By functoriality, u defines a degree-2 endomorphism

$$Rf_*(u) : Rf_*\mathcal{F} \longrightarrow Rf_*\mathcal{F}[2].$$

We shall say that $(X, f, u, \mathcal{F}, n)$ satisfies the Lefschetz condition, or that $\mathcal{F}$ satisfies the Lefschetz condition relative to u, if, for every $i \geq 0$, the maps obtained by iterating i times the arrow $Rf_*(u)$,

$$Rf_*(u)^i : R^{n-i}f_*\mathcal{F} \longrightarrow R^{n+i}f_*\mathcal{F} \qquad (i \geq 0),$$

are isomorphisms. Theorem (1.5) gives here:

Proposition (2.1). If $\mathcal{F}$ satisfies the Lefschetz condition relative to u, then, in $D^b(Y)$,

$$Rf_*\mathcal{F} \simeq \sum_i R^i f_*\mathcal{F}[-i]$$

and in particular the Leray spectral sequence

$$H^p(Y, R^q f_*\mathcal{F}) \Longrightarrow H^{p+q}(X, \mathcal{F})$$

degenerates.

If one had wished, one could have supposed that $\mathcal{F}$ was an (A, f^*B)-bimodule, where B is a sheaf of rings on Y, and stated (2.1) in $D^b(Y, B)$.

(2.2) The case of schemes, endowed with the étale topology, does not enter into the framework of (2.1). In what follows we shall suppose that the theory of $\mathbb{Z}_\ell$-sheaves and $\mathbb{Q}_\ell$-sheaves (not necessarily constructible) has been made, and that the derived category of the category of these "sheaves" is reasonable. As far as the degeneration statements for spectral sequences are concerned, the reader who has no confidence in Jouanolou's future work may verify, by returning to the proof of (1.5), that one can dispense with it. Let ℓ be a prime number. Throughout what follows, assume that ℓ is invertible on all schemes under consideration.

Recall that on every scheme X the $\mathbb{Z}_\ell$-sheaf $\mathbb{Z}_\ell(1)$ is defined by the formula

$$\mathbb{Z}_\ell(1) = \varprojlim_n \mu_{\ell^n}.$$

This sheaf is an invertible $\mathbb{Z}_\ell$-module, which permits one to define, for every $\mathbb{Z}_\ell$-sheaf or $\mathbb{Q}_\ell$-sheaf $\mathcal{F}$, the "sheaf" $\mathcal{F}(i)$ by the formula

$$\mathcal{F}(i) = \mathcal{F} \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(1)^{\otimes i} \qquad (i \in \mathbb{Z}).$$

For every morphism $f : X \to Y$ of schemes and every pair of $\mathbb{Z}_\ell$- or $\mathbb{Q}_\ell$-sheaves $\mathcal{F}$ and $\mathcal{G}$ on X and Y respectively, one has

$$Rf_*(\mathcal{F})(i) = Rf_*(\mathcal{F}(i)) \qquad \text{and} \qquad f^*(\mathcal{G}(i)) = f^*(\mathcal{G})(i). \tag{2.3}$$

If $\mathcal{L}$ is an invertible sheaf on X, its first ℓ-adic Chern class lies in $H^2(X, \mathbb{Z}_\ell(1))$.

If $f : X \to Y$ is a morphism of schemes, if $\mathcal{F}$ is a $\mathbb{Z}_\ell$-sheaf, respectively a $\mathbb{Q}_\ell$-sheaf, on X, and if $u \in H^2(X, \mathbb{Z}_\ell(1))$, respectively $u \in H^2(X, \mathbb{Q}_\ell(1))$, then $\mathcal{F}$ will be said to satisfy the Lefschetz condition relative to u if, for every $i \geq 0$, the maps obtained by iterating i times the arrow $Rf_*(u)$,

$$Rf_*(u)^i : R^{n-i}f_*\mathcal{F} \longrightarrow R^{n+i}f_*\mathcal{F}(i) \qquad (i \geq 0),$$

are isomorphisms. By (2.3), this implies that all the arrows

$$Rf_*(u)^i : R^{n-i}f_*\mathcal{F}(k) \longrightarrow R^{n+i}f_*\mathcal{F}(k+i)$$

are isomorphisms. Applying (1.10), one obtains:

Proposition (2.4). With the preceding notation, if $\mathcal{F}$ satisfies the Lefschetz condition relative to u, then, in $D^b(Y, \mathbb{Z}_\ell)$, respectively $D^b(Y, \mathbb{Q}_\ell)$, one has

$$Rf_*\mathcal{F} \simeq \sum_i R^i f_*\mathcal{F}[-i] \tag{2.5}$$

and in particular the Leray spectral sequence

$$H^p(Y, R^q f_*\mathcal{F}) \Longrightarrow H^{p+q}(X, \mathcal{F})$$

degenerates.

If g is a morphism of schemes from Y to S, (2.4) also implies the degeneration of the Leray spectral sequence

$$R^p g_* R^q f_*\mathcal{F} \Longrightarrow R^{p+q}(gf)_*\mathcal{F}.$$

(2.6.1) It remains to review a few interesting cases in which the Lefschetz condition is satisfied; the justifying details are given in (2.7). Let $f : X \to Y$, u, $\mathcal{F}$, and n be given. In the topological case, it is known from Godement [2], II, (4.7.1), that if f is proper and Y locally paracompact, the formation of higher direct images commutes with passage to fibers, so that the Lefschetz condition is checked fiber by fiber. In the case of schemes, the same reduction applies, the reference to Godement being replaced by SGA 4, XII, (5.1). Still in the case of schemes, if f is proper and smooth and $\mathcal{F}$ is twisted constant, that is, smooth in a newer terminology, then the $R^i f_*\mathcal{F}$ are still twisted constant (cf. SGA 4, XVI, (2.2)); hence, if Y is connected, it is even enough to verify the Lefschetz condition on one geometric fiber.

In all the examples given, f will be a proper smooth continuous map, respectively a proper smooth morphism of schemes, of relative topological dimension $2n$, respectively algebraic dimension n. A continuous map is called smooth if locally on the source it is isomorphic to the projection from $\mathrm{R}^n \times Y$ to Y.

(2.6.2) If f is a proper smooth morphism of Kähler varieties, the constant sheaf $\mathbb{C}$ satisfies the Lefschetz condition relative to the fundamental 2-class of X.

(2.6.3) If X is a complex relative scheme (see [3]), projective and smooth over the locally paracompact topological space Y, and endowed with a relatively ample invertible sheaf $\mathcal{O}(1)$, the constant sheaf $\mathbb{C}$ satisfies the Lefschetz condition relative to the first Chern class of $\mathcal{O}(1)$.

(2.6.4) Suppose that f is a projective smooth morphism of schemes, with Y connected, and that a relatively ample invertible sheaf $\mathcal{O}(1)$ on X is given. If one can lift to characteristic 0 a geometric fiber of f, and its polarization, then the constant $\mathbb{Q}_\ell$-adic sheaf $\mathbb{Q}_\ell$ satisfies the Lefschetz condition relative to the ℓ-adic first Chern class of $\mathcal{O}(1)$.

It is known that one can lift to characteristic 0 in the preceding sense in the following cases:

- X is a Severi–Brauer variety over Y;
- X is a nonsingular complete intersection of hypersurfaces in a Severi–Brauer variety over Y, for the polarization induced by the canonical polarization of the Severi–Brauer variety;
- X is a polarized abelian scheme over Y (Mumford, to appear).

It is conjectured that (2.6.4) remains true when no lifting to characteristic 0 is known; this is proved only for surfaces.

(2.6.5) If f is a projective smooth morphism of relative dimension 2 and if X is endowed with a relatively ample invertible sheaf $\mathcal{O}(1)$, then the constant sheaf $\mathbb{Q}_\ell$ satisfies the Lefschetz condition relative to the first Chern class of $\mathcal{O}(1)$.

(2.7) In cases (2.6.2) and (2.6.3), the given cohomology 2-class on X induces on the fibers of f the 2-class defined by the Kähler structure of the fibers. Thus, by (2.6.1), to prove (2.6.2) and (2.6.3) it suffices to prove (2.6.2) in the special case where Y is a point. The assertion then follows from Hodge theory and the Hodge–Lepage decomposition (Weil [10], IV, no. 6, corollary to theorem 5). The proof given in loc. cit. still applies if $\mathcal{F}$ is

a locally constant sheaf of finite-dimensional $\mathbb{C}$-vector spaces, endowed with a locally constant positive definite Hermitian structure.

By (2.6.1), to verify (2.6.4) it suffices to treat the case where Y is the spectrum of a field of characteristic 0, and the Lefschetz principle allows one to suppose $Y = \mathrm{Spec}(\mathbb{C})$.

It is then known (cf. SGA 4, XI, (4.4)) that, for every twisted constant $\mathbb{Q}_\ell$-adic sheaf $\mathcal{F}$ on X, if $\mathcal{F}^{an}$ denotes the locally constant sheaf of $\mathbb{Q}_\ell$-vector spaces on the topological space X^{an} deduced from $\mathcal{F}$, then the algebraic and transcendental cohomology $\mathbb{Q}_\ell$-vector spaces of $\mathcal{F}$ and $\mathcal{F}^{an}$ are isomorphic:

$$H^i(X, \mathcal{F}) \xrightarrow{\sim} H^i(X^{an}, \mathcal{F}^{an}). \tag{2.7.2}$$

Choosing an embedding of $\mathbb{Q}_\ell$ into $\mathbb{C}$, one obtains isomorphisms

$$H^i(X, \mathcal{F}) \otimes_{\mathbb{Q}_\ell} \mathbb{C} \xrightarrow{\sim} H^i(X^{an}, \mathcal{F}^{an}) \otimes_{\mathbb{Q}_\ell} \mathbb{C}. \tag{2.7.3}$$

Over $\mathbb{C}$, the exponential map defines the so-called canonical isomorphism between $\mathbb{Z}/\ell^n$ and μ_{ℓ^n}, hence between $\mathbb{Z}_\ell$ and $\mathbb{Z}_\ell(1)$. The image, under the composite isomorphism

$$H^2(X, \mathbb{Z}_\ell(1)) \longrightarrow H^2(X, \mathbb{Z}_\ell) \longrightarrow H^2(X^{an}, \mathbb{Z}) \otimes \mathbb{Z}_\ell,$$

of the ℓ-adic first Chern class of $\mathcal{O}(1)$ coincides with the transcendental first Chern class of $\mathcal{O}(1)$. The isomorphism (2.7.3) therefore reduces the case of (2.6.4) to which we had come to (2.6.3).

For the proof of (2.6.5), see Kleiman [6], pp. 28 ff. The essential point is the following theorem (Weil [11], p. 127).

(2.8) Let S be a smooth projective surface over a field k, and let D be a smooth curve on S such that $\mathcal{O}(D)$ is ample. Then the composite map

$$\mathrm{Pic}^0(S) \longrightarrow \mathrm{Pic}^0(D) = \mathrm{Alb}(D) \longrightarrow \mathrm{Alb}(S)$$

is an isogeny.[3]

Remark (2.9). The degeneration theorem for the spectral sequence (2.2) deduced from (2.6.4) was conjectured by Grothendieck, from considerations of "weights", when Y is projective and smooth over an algebraically closed field.

Remark (2.10). Serre pointed out to me that the degeneration theorems for spectral sequences deduced from (2.6.2) and (2.6.3) can also be proved by an easy extension of the method used by Blanchard ([1], II, 1), at least if Poincaré duality is available on the base. Conversely, the method followed here completes Theorems II (1.1) and II (1.2) of loc. cit.; the proof of the implication (ii) $\Rightarrow$ (i) is essentially Blanchard's, who had to suppose the higher direct-image sheaves constant.

Theorem (2.11). Let $f : X \to Y$ be a proper smooth continuous map (2.6.1) of topological spaces of relative dimension $2n$, with Y locally paracompact, let k be a commutative field of characteristic zero, and let $u \in H^0(Y, R^2 f_* k)$. Suppose that:

a) the sheaf $R^1 f_* k$ is constant, i.e. simple;

b) the iterated cup products

$$u^i \wedge : R^{n-i} f_* k \longrightarrow R^{n+i} f_* k$$

are isomorphisms.

Then the following conditions are equivalent:

(i) the class u is induced by an element of $H^2(X, k)$;

(ii) the transgression

$$d_2 : H^0(Y, R^2 f_* k) \longrightarrow H^2(Y, f_* k)$$

is zero on u;

[3] The printed source gives $\mathrm{Pic}^0(X)$ and $\mathrm{Alb}(X)$ here, although the surface introduced in the statement is S; S is restored.

(iii) in $\mathrm{D}^b(Y,k)$, one has

$$Rf_*k \simeq \bigoplus_i R^i f_*k[-i].$$

The implication (i) $\Rightarrow$ (iii) is contained in (2.1). Assertion (iii) implies the degeneration of the whole Leray spectral sequence

$$H^p(Y, R^q f_*k) \Longrightarrow H^{p+q}(X,k), \tag{2.12}$$

and a fortiori (ii). It remains to prove that (ii) implies (i).

Let $x \in Y$ and $X_x = f^{-1}(x)$. The induced class $u_x^n \in H^{2n}(X_x,k)$ is nonzero on every connected component of X_x, which is therefore orientable.[4] Let Tr_u denote the composite map

$$H^{2n}(X_x,k) \xrightarrow{(u_x^n\wedge)^{-1}} H^0(X_x,k) = H_0(X_x,k) \xrightarrow{\mathrm{Tr}} k.$$

Poincaré duality implies that the bilinear form

$$H^{n-i}(X_x,k) \times H^{n+i}(X_x,k) \longrightarrow k, \qquad (x,y) \longmapsto \mathrm{Tr}_u(xy)$$

is a perfect duality. This construction globalizes and gives

$$\mathrm{Tr}_u : R^{2n} f_*k \longrightarrow k.$$

First prove that in (2.12), $d_2u \in H^2(Y, R^1f_*k)$ is zero. If $x \in H^0(Y, R^1f_*k)$, then for degree reasons $xu^n = 0$, hence

$$d_2(xu^n) = d_2x\, u^n - nxu^{n-1}d_2u = 0.$$

By b), the map

$$u^{n-1}\wedge : H^0(Y, R^1f_*k) \longrightarrow H^0(Y, R^{2n-1}f_*k)$$

is an isomorphism. Thus, if (ii) holds, for every $y \in H^0(Y, R^{2n-1}f_*k)$ one has

$$y\, d_2u = 0,$$

and in particular

$$\mathrm{Tr}_u(y\, d_2u) = 0. \tag{2.13}$$

Let V be a k-vector space such that R^1f_*k is isomorphic to the constant sheaf V. Then $d_2u \in H^2(Y,V)$ and, by (2.13), for every linear form w on V, the image of d_2u under w in $H^2(Y,k)$ is zero. Thus $d_2u = 0$.

If $d_2u = 0$, the proof of (1.5) shows that all differentials d_2 of (2.12) are zero, so that $E_2 = E_3$. For degree reasons, $u^{n+1} = 0$, hence

$$d_3u^{n+1} = (n+1)u^n d_3u = 0.$$

Since $d_3u \in H^3(Y, R^0f_*k)$, b) implies that $d_3u = 0$. For degree reasons $d_ru = 0$ for $r > 3$, and u therefore comes from an element of $H^2(X,k)$.

(2.14) In the scheme-theoretic setting, the preceding result remains valid, but it is usable only in the "geometric" case, more precisely when $\mathbb{Q}_\ell$ is isomorphic to $\mathbb{Q}_\ell(1)$ on Y.

(2.15) Corollary (1.12) allows some of the preceding statements to be extended to certain nonconstant sheaves.

Proposition (2.16). Let g and f be two composable continuous maps, respectively two morphisms of schemes,

$$X \xrightarrow{g} Y \xrightarrow{f} Z,$$

and let $\mathcal{F}$ be a sheaf, respectively a $\mathbb{Z}_\ell$-sheaf or $\mathbb{Q}_\ell$-sheaf, on X. Suppose that

a) $Rg_*\mathcal{F} \in \mathrm{D}^b(Y)$ and $Rg_*\mathcal{F} \simeq \bigoplus_i R^i g_*\mathcal{F}[-i]$;

b) $R(fg)_*\mathcal{F} \in \mathrm{D}^b(Z)$ and $R(fg)_*\mathcal{F} \simeq \bigoplus_i R^i(fg)_*\mathcal{F}[-i]$.

[4] After introducing x, the printed source switches to X_s and u_s^n; the subscript is regularized to x here.

Then, for every k, one has

$$Rf_*(R^k g_*\mathcal{F}) \simeq \bigoplus_i R^i f_*(R^k g_*\mathcal{F})[-i].$$

By b), the complex

$$R(fg)_*\mathcal{F} = Rf_*Rg_*\mathcal{F} \simeq \bigoplus_k Rf_*(R^k g_*\mathcal{F}[-k])$$

is a sum of its cohomology objects; by (1.12), the complexes $Rf_*R^k g_*\mathcal{F}$, which up to a shift are direct factors thanks to a), have the same property.

Proposition (2.17). Let $\mathcal{E}$ be a locally free vector bundle on a scheme S, let $\mathcal{F}$ be a twisted constant ℓ-adic sheaf on $\mathbb{P}(\mathcal{E})$, and let f be the projection from $\mathbb{P}(\mathcal{E})$ to S. In the derived category, one has

$$Rf_*\mathcal{F} \simeq \bigoplus_i R^i f_*\mathcal{F}[-i].$$

Projective space is simply connected and has torsion-free cohomology (SGA 1, XI, 1.1 and SGA 4, XVI, 2.2). Hence

$$\mathcal{F} = f^*f_*\mathcal{F}, \qquad Rf_*\mathcal{F} = Rf_*\mathbb{Z}_\ell \overset{\mathrm{L}}{\otimes} f_*\mathcal{F},$$

which reduces the proof to the case $\mathcal{F} = \mathbb{Z}_\ell$. Let u be the Chern class of the invertible sheaf $\mathcal{O}(1)$ on $\mathbb{P}(\mathcal{E})$. Then $\mathbb{Z}_\ell$ is known to satisfy the Lefschetz condition relative to u, and (2.2) concludes the proof.

(2.18) From (2.17), with the help of (2.16), one obtains the same result for flag bundles of all kinds defined by $\mathcal{E}$. In the topological setting, the preceding applies directly to vector bundles over $\mathbb{C}$, and extends to vector bundles over $\mathbb{H}$ by (1.9). For real vector bundles, one must restrict to the case where $\mathcal{F}$ comes from a sheaf of 2-torsion on the base.

Remark (2.19). Let X be an abelian scheme of relative dimension g over a base Y. For every integer n, multiplication by n, denoted $[n]$, defines an endomorphism

$$[n]^* : Rf_*\mathbb{Q}_\ell \longrightarrow Rf_*\mathbb{Q}_\ell.$$

Taking rational linear combinations, with denominators bounded in terms of g, of these endomorphisms, one constructs endomorphisms

$$\pi_i : Rf_*\mathbb{Q}_\ell \longrightarrow Rf_*\mathbb{Q}_\ell$$

such that $H^j(\pi_i) = \delta_{ij}$. Applying (1.11), one concludes, without any projectivity hypothesis, that

$$Rf_*\mathbb{Q}_\ell \simeq \bigoplus_i R^i f_*\mathbb{Q}_\ell[-i].$$

The idea used here is due to Lieberman.

3. The infinitesimal form of the semicontinuity theorem

I recall in this paragraph some results which one manages to find in EGA III, ğ 7. The formulation (3.4) was pointed out to me by L. Illusie.

(3.0) Throughout this paragraph, A denotes a fixed noetherian local ring, $\mathfrak{m}$ its maximal ideal, and k its residue field. Let $D^-_{\mathrm{coh}}(A)$ denote the subcategory of the derived category of A-modules formed by bounded-above complexes of finite-type A-modules.

Proposition (3.1). Let M be a finite-type module over A.

(i) For every homomorphism, not necessarily local, from A to an Artinian ring B, and every finite-type B-module N, one has

$$\lg_B(M \otimes_A N) \leq \lg_B(N) \cdot \dim_k(M \otimes_A k). \tag{3.1.1}$$

(ii) The following conditions on M are equivalent:

a) M is a free module;

b) for all B and N as in (i), one has

$$\lg_B(M \otimes_A N) = \lg_B(N) \cdot \dim_k(M \otimes_A k); \tag{3.1.2}$$

c) for every $n \in \mathbb{N}$, condition (3.1.2) is satisfied for $B = N = A/\mathfrak{m}^n$;

d) for every $\mathfrak{p} \in \mathrm{Ass}(A)$, $M_\mathfrak{p}$ is flat over $A_\mathfrak{p}$, and condition (3.1.2) is satisfied for $B = N = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$.

If A has no embedded components, these conditions are also equivalent to:

e) for every maximal point $\mathfrak{p}$ of $\mathrm{Spec}(A)$ one has

$$\lg_{A_\mathfrak{p}}(M_\mathfrak{p}) = \lg(A_\mathfrak{p}) \cdot \dim_k(M \otimes_A k). \tag{3.1.2'}$$

The proof, an application of Nakayama's lemma, is left to the reader.

(3.2) For every complex K and every integer n, let $\tau_{\geq n}(K)$ denote the complex defined by

$$\tau_{\geq n}(K)^i = \begin{cases} 0, & i < n, \\ \mathrm{coker}(d^{n-1}), & i = n, \\ K^i, & i > n, \end{cases} \tag{3.2.1}$$

so that

$$H^i(\tau_{\geq n}(K)) = \begin{cases} 0, & i < n, \\ H^i(K), & i \geq n. \end{cases} \tag{3.2.2}$$

Let K be an object of $D^-_{\mathrm{coh}}(A)$ (cf. (3.0)). Replacing K, if necessary, by an isomorphic complex in the derived category, one may suppose the components of K to be free of finite type.

For all B and N as in (3.1) (i), one has at the level of complexes

$$\tau_{\geq n}(K \otimes^L_A N) \simeq \tau_{\geq n}(K) \otimes_A N.$$

Taking the Euler–Poincaré characteristics of the two sides gives

$$\sum_{i \geq 0} (-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) = \lg_B(\mathrm{coker}(d^{n-1}) \otimes_A N) + \sum_{i>0} (-1)^i \lg_B(K^{n+i} \otimes_A N).$$

Subtract from this identity the analogous identity for $B = N = k$, multiplied by $\lg_B(N)$. By (3.1) (ii), $a \Leftrightarrow b$, and since the K^i are free, one obtains

$$\begin{aligned} &\sum_{i \geq 0} (-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) - \lg_B(N) \sum_{i \geq 0} (-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big) \\ &\qquad = \lg_B(\mathrm{coker}(d^{n-1}) \otimes_A N) - \lg_B(N) \dim_k(\mathrm{coker}(d^{n-1}) \otimes_A k), \end{aligned} \tag{3.2.3}$$

which allows one to apply (3.1) to the first member of (3.2.3).

Theorem (3.3) (exchange and semicontinuity theorems). Let $K \in \operatorname{Ob} D^-_{\mathrm{coh}}(A)$ (cf. (3.0)) and $n \in \mathbb{Z}$.

(i) For every homomorphism from A to an Artinian ring B and every finite-type B-module N, one has

$$\sum_{i\geq 0}(-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) \leq \lg_B(N) \sum_{i\geq 0}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.3.1}$$

(ii) The following conditions on K and n are equivalent:

a) The cohomological functor in the A-module N, $H^*(K \otimes^L_A N)$, satisfies the exchange property in degree $* = n$, i.e. the following equivalent conditions on K and n are satisfied:

a.1) the functor $H^n(K \otimes^L_A N)$ is left exact;
a.2) there is a module Q such that this functor is isomorphic to $\operatorname{Hom}_A(Q, N)$;
a.3) the functor $H^{n-1}(K \otimes^L_A N)$ is right exact;
a.4) there is a module P such that this functor is isomorphic to $P \otimes_A N$;
a.5) the map $H^{n-1}(K) \to H^{n-1}(K \otimes^L_A k)$ is surjective;
a.6) there exists a bounded-above complex K', quasi-isomorphic to K and with flat components, respectively free components of finite type, such that $d'^{\,n-1} = 0$.

b) For all B and N as in (i), one has

$$\sum_{i\geq 0}(-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) = \lg_B(N) \sum_{i\geq 0}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.3.2}$$

If A has no embedded components, these conditions are also equivalent to:

c) for every maximal point $\mathfrak{p}$ of $\operatorname{Spec}(A)$ one has

$$\sum_{i\geq 0}(-1)^i \lg_{A_\mathfrak{p}} H^{n+i}(K_\mathfrak{p}) = \lg(A_\mathfrak{p}) \sum_{i\geq 0}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.3.2'}$$

Remark (3.3.3). One could have lengthened the list by adding the conditions analogous to (3.1) (ii) c) and d).

Assertion (i) follows from (3.2.3) and (3.1) (i). By EGA III, (7.4.2), and its proof, conditions a.1), a.3), and a.6) are equivalent and, if K is represented by a bounded-above complex of finite-type free modules, they also mean that $\operatorname{coker}(d^{n-1})$ is free; taking (3.1) (ii) $a \Leftrightarrow b \Leftrightarrow e$ and (3.2.3) into account proves the equivalence of a.1), a.3), a.6), b), and c). Trivially, a.6) $\Rightarrow$ a.2) $\Rightarrow$ a.1) and a.6) $\Rightarrow$ a.4) $\Rightarrow$ a.3). The equivalence of condition a.5) with the others is contained in EGA III, (7.5.2).

Corollary (3.4). Let A, K, and n be as in (3.3), and let $k \in \mathbb{N}$.

(i) For all B and N as in (3.3) (i), one has

$$\sum_{0\leq i\leq 2k}(-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) \leq \lg_B(N) \sum_{0\leq i\leq 2k}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.4.1}$$

(ii) The following conditions on K, n, and k are equivalent:

a) The cohomological functor in N, $H^*(K \otimes^L_A N)$, satisfies the exchange property in degrees $* = n$ and $* = n + 2k + 1$.

a' The complex K is quasi-isomorphic to a bounded-above complex K' with flat components, respectively free components of finite type, such that $d'^{\,n-1} = d'^{\,n+2k} = 0$.

b) For all B and N as in (i), equality holds in (3.4.1).

If A has no embedded components, these conditions are also equivalent to:

c) for every maximal point $\mathfrak{p}$ of $\operatorname{Spec}(A)$, equality holds in (3.4.1) for $B = N = A_\mathfrak{p}$.

(3.4.2) For $k = 0$, these are properties only of the single hypertor $H^n(K \otimes^L_A N)$, considered as a functor in the A-module N. They can also be expressed by saying that $H^n(K)$ is free and that, for all N, the canonical map from $H^n(K) \otimes_A N$ to $H^n(K \otimes^L_A N)$ is an isomorphism.

(3.4.3) On the other hand, if K is a perfect complex, one deduces from (3.4), for $n \ll 0$, a result analogous to (3.3), with the summation over $i \geq 0$ in (3.3.1, 2, 2') replaced by a summation over $i \leq 0$.

Assertion (i) and the equivalence of a), b), and c) follow from (3.3) and from the identity

$$\sum_{0\leq i\leq 2k}(-1)^i x_i = \sum_{0\leq i}(-1)^i x_i + \sum_{0\leq i}(-1)^i x_{i+2k+1}.$$

To prove a'), one successively applies (3.3) (ii) $a \Leftrightarrow a.6)$ to (K, n) and to $(\tau_{\geq n}(K), n+2k+1)$.

(3.5) When A is Artinian, (3.4) gives, for $k = 0$, the inequality

$$\lg_A H^n(K) \leq \lg(A)\dim_k\big(H^n(K \otimes^L_A k)\big), \tag{3.5.1}$$

and, if equality holds in (3.5.1), then the exchange condition is satisfied in degrees n and $n+1$, and $H^n(K)$ is a free A-module.

4. The Gysin morphism in Hodge cohomology

This paragraph is written from notes of A. Grothendieck.

(4.0) Let $f : X \to Y$ be a proper morphism between schemes X and Y which are smooth and purely of relative dimensions n and m over a base S. We suppose that S is noetherian and has a dualizing complex, in order to refer to [4] for the definition of $\mathrm{R}f^!$; this restriction is probably unnecessary. Finally, put $d = n - m$.

(4.1) Every relative differential form on Y defines, by inverse image, a relative differential form on X, hence a morphism

$$f^*\Omega^p_{Y/S} = Lf^*\Omega^p_{Y/S} \longrightarrow \Omega^p_{X/S}. \tag{4.1.1}$$

By the duality theory for smooth morphisms and the transitivity of $\mathrm{R}f^!$, one has canonically

$$\mathrm{R}f^!\Omega^m_{Y/S} = \Omega^n_{X/S}[d]. \tag{4.1.2}$$

Apply to $\Omega^p_{Y/S}$ and $\Omega^m_{Y/S}$ the induction formula

$$\mathrm{R}f^!\,\mathrm{RHom}(K, L) = \mathrm{RHom}(Lf^*K, \mathrm{R}f^!L),$$

which is obtained readily from the definition of $\mathrm{R}f^!$ by biduality (Hartshorne [4], chap. VII, ğ 3). Since

$$\mathrm{RHom}(\Omega^p_{Y/S}, \Omega^m_{Y/S}) \simeq \Omega^{m-p}_{Y/S},$$

one obtains

$$\mathrm{R}f^!\Omega^{m-p}_{Y/S} = \mathrm{RHom}(Lf^*\Omega^p_{Y/S}, \Omega^n_{X/S})[d].$$

The arrow (4.1.1) therefore defines, by transposition, an arrow[5]

$$\Omega^{n-p}_{X/S} = \mathrm{RHom}(\Omega^p_{X/S}, \Omega^n_{X/S}) \longrightarrow \mathrm{R}f^!\Omega^{m-p}_{Y/S}[d]. \tag{4.1.3}$$

By duality theory, giving an arrow (4.1.3) amounts to giving an arrow

$$\mathrm{R}f_*\Omega^{n-p}_{X/S} \longrightarrow \Omega^{m-p}_{Y/S}[-d]. \tag{4.1.4}$$

Definition (4.2). Let X, Y, f, S, n, m, and d be as in (4.0). The arrow (4.1.4)

$$\mathrm{Tr}_f : \mathrm{R}f_*\Omega^p_{X/S} \longrightarrow \Omega^{p-d}_{Y/S}[-d]$$

is called the Gysin morphism, as are the induced maps

$$\mathrm{Tr}_f : H^q(X, \Omega^p_{X/S}) \longrightarrow H^{q-d}(Y, \Omega^{p-d}_{Y/S}).$$

[5] In the first member of (4.1.3), the printed source switches once from p to q; the index p used in (4.1.1), (4.1.4), and (4.2) is retained here.

Proposition (4.3). Let $f : X \to Y$ be a proper birational morphism of smooth schemes over a field k. The maps

$$f^* : H^q(Y, \Omega^p_Y) \longrightarrow H^q(X, \Omega^p_X)$$

are injective.

One reduces to the case where X and Y are purely of dimension n. The composite morphism

$$\mathrm{Tr}_f \circ f^* : \Omega^p_Y \longrightarrow \mathrm{R}f_* f^* \Omega^p_Y \longrightarrow \mathrm{R}f_* \Omega^p_X \longrightarrow \Omega^p_Y$$

is the identity, because it coincides with the identity on a dense open subset. Applying the functor $H^q(Y, \ )$, one finds that the composite morphism

$$\mathrm{Tr}_f \circ f^* : H^q(Y, \Omega^p_Y) \longrightarrow H^q(X, \Omega^p_X) \longrightarrow H^q(Y, \Omega^p_Y)$$

is the identity, which proves (4.3).

5. Hodge and De Rham cohomology

(5.1) Let $f : X \to S$ be a smooth morphism of schemes. By definition, the relative De Rham cohomology sheaves of X over S are given by

$$H^*_{\mathrm{DR}}(X/S) = \mathrm{R}^* f_*(\Omega^\bullet_{X/S}),$$

where $\Omega^\bullet_{X/S}$, the relative De Rham complex, is an S-linear differential complex. There is a spectral sequence

$$E_1^{pq} = \mathrm{R}^q f_* \Omega^p_{X/S} \Longrightarrow H^{p+q}_{\mathrm{DR}}(X/S). \tag{5.1.1}$$

(5.2) When $S = \mathrm{Spec}(\mathbb{C})$ and X is proper and smooth, GAGA ([7]) shows that the cohomology of each coherent sheaf Ω^p_X, and hence the hypercohomology of $\Omega^\bullet_X$, are the same in the algebraic sense (Zariski topology) and in the transcendental sense (usual topology of X^{an}); in particular, since $\Omega^\bullet_{X^{\mathrm{an}}}$ is a resolution of the constant sheaf $\mathbb{C}$, one has

$$H^n_{\mathrm{DR}}(X) = H^n(X^{\mathrm{an}}, \mathbb{C}),$$

and complex conjugation on the constant sheaf $\mathbb{C}$ induces a complex conjugation, of transcendental nature, on $H^n_{\mathrm{DR}}(X)$. The spectral sequence (5.1.1) is here

$$E_1^{pq} = H^q(X, \Omega^p_X) \Longrightarrow H^{p+q}_{\mathrm{DR}}(X); \tag{5.2.1}$$

let $F^p(H^n(X))$ denote the decreasing filtration which it defines on its abutment. By Weil [10], chap. IV, no. 4, th. 2, the spectral sequence (5.2.1) can be computed as the spectral sequence of the double complex $H^0(X^{\mathrm{an}}, \Omega^{p,q})$, filtered by the first degree, where $\Omega^{p,q}$ denotes the sheaf of C^∞ differential forms bihomogeneous of bidegree (p, q) (Weil [10], chap. II, no. 1).

Suppose now that X is projective, so that X^{an} is a Kähler variety. The harmonic forms then form a double subcomplex (Weil [10], chap. II, no. 6, cor. 1 to th. 2) of the preceding double complex, on which d' and d'' vanish. The inclusion of this subcomplex induces an isomorphism on the E_1 terms of the spectral sequences defined by these double complexes, filtered by the first degree, as one sees by considering the operator H (harmonic component), a retraction of the double complex $H^0(X^{\mathrm{an}}, \Omega^{p,q})$ onto the subcomplex of harmonic forms and satisfying

$$1 - H = d'(2\delta' G) + (2\delta' G)d'$$

(Weil [10], chap. IV, no. 1: (I) and lemma 3; no. 3: cor. 2 and chap. II, no. 5, th. 2 (IX)).

If $H^{p,q}(X)$ denotes the image in $H^{p+q}(X^{\mathrm{an}}, \mathbb{C})$ of the space of harmonic forms of type (p, q), one therefore has

$$H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X), \tag{5.2.2}$$

$$F^p(H^n(X)) = \bigoplus_{i \geq p} H^{i,n-i}(X), \tag{5.2.3}$$

$$\overline{H^{p,q}(X)} = H^{q,p}(X) \quad \text{(complex conjugation)}, \tag{5.2.4}$$

and the spectral sequence (5.2.1) degenerates.

Returning to the general case where X is only assumed proper and smooth over $\mathbb{C}$, put

$$H^{p,q}(X) = F^p(H^{p+q}(X)) \cap \overline{F^q(H^{p+q}(X))}, \tag{5.2.5}$$

so that

$$\overline{H^{p,q}(X)} = H^{q,p}(X). \tag{5.2.6}$$

By (5.2.3), this notation is compatible with the one used when X is projective. Put also

$$h^{p,q} = \dim H^q(X, \Omega_X^p). \tag{5.2.7}$$

Proposition (5.3). Let X be a proper smooth scheme over $\mathbb{C}$:

(i) the spectral sequence (5.2.1) degenerates;

(ii) one has

$$F^p(H^n(X)) = \bigoplus_{i \geq p} H^{i,n-i}(X), \tag{5.3.1}$$

and in particular

$$H^n(X) = \bigoplus_{p+q=n} H^{p,q}(X) \quad \text{(direct sum)}, \tag{5.3.2}$$

$$h^{p,q} = h^{q,p} = \dim H^{p,q}(X). \tag{5.3.3}$$

One reduces to the case where X is connected. Let N be its dimension. By Chow's lemma and resolution of singularities (Hironaka [5]), there exists a projective birational morphism $g : X' \to X$, with X' projective and smooth over $\mathbb{C}$. The spectral sequence

$$E_1^{pq} = H^q(X, \Omega_X^p) \Longrightarrow H_{\mathrm{DR}}^{p+q}(X) \tag{5.3.4}$$

maps, by inverse image, into the analogous spectral sequence

$$E_1'^{pq} = H^q(X', \Omega_{X'}^p) \Longrightarrow H_{\mathrm{DR}}^{p+q}(X'), \tag{5.3.5}$$

which degenerates by (5.2). By (4.3), the morphism $g^* : E_1 \to E_1'$ is injective. One concludes, by induction on $r \geq 1$, that $d_r = 0$, that $E_r = E_{r+1}$, and that g^* injects E_{r+1} into E'_{r+1}:

$$\begin{array}{ccc} E_r^{pq} & \xrightarrow{d_r} & E_r^{p+r,q-r+1} \\ {\scriptstyle g^*}\downarrow & & {\scriptstyle g^*}\downarrow \\ E_r'^{pq} & \xrightarrow{0} & E_r'^{p+r,q-r+1}. \end{array}$$

Since the spectral sequences (5.3.4) and (5.3.5) degenerate, and since g^* is injective on their initial term, it is still injective on their abutment. From formulas (5.2.3), (5.2.4), applied to X', it follows that

$$F^p(H^n(X')) \cap \overline{F^{n-p+1}(H^n(X'))} = \{0\};$$

hence one obtains the analogous formula

$$F^p(H^n(X)) \cap \overline{F^{n-p+1}(H^n(X))} = \{0\}, \tag{5.3.6}$$

because g^* sends $F^p(H^n(X))$ to $F^p(H^n(X'))$ and commutes with complex conjugation. In particular,

$$\sum_{i \geq p} h^{i,n-i} + \sum_{i \geq n-p+1} h^{i,n-i} \leq \dim H^n(X) \leq \sum_i h^{i,n-i},$$

which may be written

$$\sum_{i \geq p} h^{i,n-i} \leq \sum_{i \leq n-p} h^{i,n-i}. \tag{5.3.7}$$

By Serre duality, $h^{i,j} = h^{N-i,N-j}$, inequality (5.3.7), for $N-n$, implies the opposite inequality, so that

$$\dim F^p(H^n(X)) + \dim \overline{F^{n-p+1}(H^n(X))} = \dim H^n(X).$$

By (5.3.6), therefore,

$$H^n(X) = F^p(H^n(X)) \oplus \overline{F^{n-p+1}(H^n(X))}. \tag{5.3.8}$$

The decomposition (5.3.8) induces on $F^{p-1}(H^n(X))$, which contains one of the factors, a decomposition

$$F^{p-1}(H^n(X)) = F^p(H^n(X)) \oplus H^{p-1,n-p+1}(X),$$

and formula (5.3.1) follows by descending induction on p; (5.3.2) and (5.3.3) follow at once from (5.3.1) and (5.2.6), completing the proof of (5.3).

Corollary (5.4). Under the hypotheses of (5.3), one has:

(i) A complex cohomology class a on X lies in $H^{p,q}(X)$ if and only if it can be represented by a closed form α of type (p,q).

(ii) Every cohomology class can be represented by a form α satisfying $d'\alpha = d''\alpha = 0$.

(iii) If the form α satisfies $d'\alpha = d''\alpha = 0$, the following conditions are equivalent:

a) $(\exists\beta)\ \alpha = d\beta$;

b) $(\exists\beta)\ \alpha = d'\beta$;

c) $(\exists\beta)\ \alpha = d''\beta$.[6]

By definition, $a \in H^{p,q}(X)$ if and only if in the class of a there exist closed forms α_1 and α_2 whose components of type (p',q') are zero for $p' < p$ (respectively $q' < q$). Then there exists β such that $\alpha_1 - \alpha_2 = d\beta$. Let β_1 (respectively β_2) be the sum of the components of β of type (p',q') with $p' \geq p$ (respectively $q' \geq q$); one has $\beta = \beta_1 + \beta_2$ and

$$\alpha = \alpha_1 - d\beta_1 = \alpha_2 + d\beta_2$$

is a closed form of type (p,q) in the class of a. This proves (i). It suffices to prove (ii) for a bihomogeneous, hence represented by a closed bihomogeneous form α by (i). Such a form automatically satisfies $d'\alpha = d''\alpha = 0$.

If α satisfies $d'\alpha = d''\alpha = 0$, its bihomogeneous components are closed, and each of the conditions a), b), c) is equivalent to the conjunction of the analogous conditions for the various components of α, trivially for b) and c), and by (5.3) (ii) and (5.4) (i) for a). To prove (iii), suppose therefore that α is closed of type (p,q). The cohomology class a defined by α lies in $H^{p,q}(X)$, hence in $F^p(H^{p+q}(X))$, and is zero if and only if it already lies in $F^{p+1}(H^{p+q}(X))$, i.e. if its image in $H^q(X,\Omega_X^p)$ is zero. This proves a) $\Leftrightarrow$ c), and the equivalence a) $\Leftrightarrow$ b) follows by conjugation.

Theorem (5.5). Let S be a scheme of characteristic 0 and let $f : X \to S$ be a proper smooth morphism. Then:

(i) the sheaves $\mathrm{R}^q f_*\Omega^p_{X/S}$ are locally free of finite type and their formation is compatible with arbitrary base change;

(ii) the spectral sequence (5.1.1)

$$E_1^{pq} = \mathrm{R}^q f_*\Omega^p_{X/S} \Longrightarrow H^{p+q}_{\mathrm{DR}}(X/S)$$

degenerates;

(iii) at every point of S, the sheaves $\mathrm{R}^q f_*\Omega^p_{X/S}$ and $\mathrm{R}^p f_*\Omega^q_{X/S}$ have the same rank (Hodge symmetry).

The second assertion of (i) follows from the first and from EGA III, (7.8.5), applied to each of the $\Omega^p_{X/S}$; it implies that it suffices to verify (iii) for S the spectrum of a field.

Standard arguments allow one to reduce successively to the case where S is affine, since the question is local on S; noetherian, by passage to an inductive limit of rings; the spectrum of a noetherian local ring, since the question is local; the spectrum of a complete noetherian local ring, by faithful flatness of completion; and finally the spectrum of an Artinian local ring. To make this last reduction, write the complete noetherian local ring R as the projective limit of its Artinian quotients $R/\mathfrak{m}^n$. By the comparison theorem EGA III, (4.1.5), the E_1 term

[6] (Added in proofs.) One can show that these conditions are also equivalent to the existence of a form β such that $\alpha = d'd''\beta$.

of the spectral sequence (5.1.1) is the projective limit of the E_1 terms of the analogous spectral sequences over the $R/\mathfrak{m}^n$, so that if these degenerate, then (5.1.1) also degenerates. If assertion (i) is true over the $R/\mathfrak{m}^n$, these E_1 terms form a projective system of free modules, obtained from one another by reduction modulo $\mathfrak{m}^n$, so that their projective limit is a free module.

Suppose therefore that $S = \mathrm{Spec}(A)$ with A an Artinian local ring with residue field k of characteristic 0. This ring admits a field of representatives (EGA, 0_{IV}, (19.8.6), (i), for $W = k$, $u =$ identity) and can therefore be regarded as a local k-algebra of finite dimension. The Lefschetz principle permits one to suppose $k = \mathbb{C}$, in which case (iii) follows from (i) and (5.3.3).

To complete the proof of (5.5), it remains to prove the first assertion of (i), and (ii), under the additional hypothesis:

$$S = \mathrm{Spec}(A) \quad \text{for a finite-dimensional local } \mathbb{C}\text{-algebra } A \text{ with maximal ideal } \mathfrak{m}. \tag{5.5.1}$$

The spectral sequence (5.1.1) becomes

$$E_1^{pq} = H^q(X, \Omega^p_{X/S}) \Longrightarrow \mathrm{R}^{p+q}\Gamma(\Omega^\bullet_{X/S}). \tag{5.5.2}$$

Lemma (5.5.3). Under the hypotheses (5.5) and (5.5.1), the complex $\Omega^{\mathrm{an},\bullet}_{X/S}$ is a resolution of the constant sheaf A on the topological space X^{an}.

Since the complex $\Omega^{\mathrm{an},\bullet}_{X/S}$ is A-linear, its $\mathfrak{m}$-adic filtration is a filtration by subcomplexes. For this filtration,

$$\mathrm{Gr}\,\Omega^{\mathrm{an},\bullet}_{X/S} = \mathrm{Gr}\,A \otimes_{\mathbb{C}} \mathrm{Gr}^0\,\Omega^{\mathrm{an},\bullet}_{X/S}.$$

The complex $\mathrm{Gr}^0\,\Omega^{\mathrm{an},\bullet}_{X/S}$ is a resolution of the constant sheaf $\mathbb{C}$, since it is just the De Rham complex of X_{red}. Thus the complex $\mathrm{Gr}\,\Omega^{\mathrm{an},\bullet}_{X/S}$ is a resolution of $\mathrm{Gr}\,A$, and the assertion follows.

The space X is compact, so by GAGA (cf. (5.2)) the algebraic and transcendental hypercohomologies of $\Omega^\bullet_{X/S}$ coincide, and by (5.5.3) they also coincide with the cohomology of X^{an} with values in A. Hence

$$\mathrm{lg}_A\,\mathrm{R}^n\Gamma(\Omega^\bullet_{X/S}) = \mathrm{lg}(A)\,\dim_{\mathbb{C}}\mathrm{R}^n\Gamma(\Omega^\bullet_{X_{\mathrm{red}}}). \tag{5.5.4}$$

By (3.5.1) and EGA III, (6.10.5), one knows that

$$\mathrm{lg}_A\,H^q(X, \Omega^p_{X/S}) \leq \mathrm{lg}(A)\,\dim_{\mathbb{C}} H^q(X_{\mathrm{red}}, \Omega^p_{X_{\mathrm{red}}}), \tag{5.5.5}$$

with equality possible only if $H^q(X, \Omega^p_{X/S})$ is A-free.

From (5.5.2) one obtains

$$\sum_{p+q=n} \mathrm{lg}_A\,H^q(X, \Omega^p_{X/S}) \geq \mathrm{lg}_A\,\mathrm{R}^n\Gamma(\Omega^\bullet_{X/S}), \tag{5.5.6}$$

equality being possible for all n only if the spectral sequence (5.5.2) degenerates. Chaining these inequalities, one finds

$$\mathrm{lg}(A) \sum_{p+q=n} \dim_{\mathbb{C}} H^q(X_{\mathrm{red}}, \Omega^p_{X_{\mathrm{red}}}) \geq \mathrm{lg}(A)\mathrm{R}^n\Gamma(\Omega^\bullet_{X_{\mathrm{red}}}), \tag{5.5.7}$$

equality being possible only if the conclusions of (5.4) are satisfied. This equality follows from (5.3) (i) applied to X_{red}.

6. Application to coherent cohomology

Theorem (6.1). Let S be a scheme of characteristic 0, and let $f : X \to S$ be a projective smooth morphism of relative dimension n. For every p, for example $p = 0$, one has in $D^b(S, \mathcal{O}_S)$

$$\mathrm{R}f_*\Omega^p_{X/S} \simeq \bigoplus_q \mathrm{R}^q f_*\Omega^p_{X/S}[-q].$$

Let u denote the first Chern class of a relatively ample invertible sheaf on X, with values in $H^1(X, \Omega^1_{X/S})$. By cup product, the class u defines homomorphisms in $D^b(S)$,

$$u\wedge : \mathrm{R}f_*\Omega^p_{X/S} \longrightarrow \mathrm{R}f_*\Omega^{p+1}_{X/S}[1],$$

and a degree-two endomorphism

$$u\wedge : \bigoplus_p \mathrm{R}f_*\Omega^p_{X/S}[-p] \longrightarrow \left(\bigoplus_p \mathrm{R}f_*\Omega^p_{X/S}[-p]\right)[2]. \tag{6.1.1}$$

Lemma (6.2). The endomorphism (6.1.1) satisfies the hypotheses of (1.5).

By (5.4) (i) and the Lefschetz principle, it suffices to prove (6.2) when $S = \mathrm{Spec}(\mathbb{C})$. The assertion then follows from the Lefschetz theorem (see (2.6.2) or (2.6.3)) and from the Hodge decomposition.

By (1.5), the complex $\bigoplus_p \mathrm{R}f_*\Omega^p_{X/S}[-p]$ is isomorphic, in the derived category, to the sum of its cohomology objects, and the same is true for each of the $\mathrm{R}f_*\Omega^p_{X/S}$, which, up to a shift, are direct factors of it (1.12).

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Ordinary abelian varieties over a finite field

Pierre Deligne (Bures-sur-Yvette)

We give a down-to-earth description of the category of ordinary abelian varieties over a finite field $\mathbb{F}_q$. The result obtained was inspired by Ihara [2], ch. V (see also [3]).

1. Let p be a prime number, let $\mathbb{F}_p$ be the field $\mathbb{Z}/(p)$, let $\overline{\mathbb{F}_p}$ be an algebraic closure of $\mathbb{F}_p$, and, for each power q of p, let $\mathbb{F}_q$ be the subfield with q elements of $\overline{\mathbb{F}_p}$. For every algebraic extension k of $\mathbb{F}_p$, let $W_0(k)$ denote the henselian discrete valuation ring, essentially of finite type over $\mathbb{Z}$, absolutely unramified, with residue field k; let $W(k)$ be the ring of Witt vectors over k, i.e. the completion of $W_0(k)$, let $W = W(\overline{\mathbb{F}_p})$, and let φ be an embedding of W in the field $\mathbb{C}$ of complex numbers. We denote by $\mathbb{Z}(1)$ the subgroup $2\pi i\mathbb{Z}$ of $\mathbb{C}$. The exponential map defines an isomorphism between $\mathbb{Z}(1) \otimes \mathbb{Z}_\ell$ and

$$\mathbb{Z}_\ell(1)(\mathbb{C}) = \varprojlim_n \mu_{\ell^n}(\mathbb{C}).$$

We denote by A^* the abelian variety dual to an abelian variety A. For every field k, let $\bar{k}$ denote an algebraic closure of k.

2. Let A be an abelian variety of dimension g, defined over a field k of characteristic p. Recall that A is called ordinary if the following equivalent conditions are satisfied:

(I) A has p^g points of order dividing p with values in $\bar{k}$.

(II) The "Hasse–Witt matrix"
$$F^* : H^1(A^{(p)}, \mathcal{O}_{A^{(p)}}) \longrightarrow H^1(A, \mathcal{O}_A)$$
is invertible.

(III) The neutral component of the group scheme A_p, the kernel of multiplication by p, is of multiplicative type, hence geometrically isomorphic to a power of μ_p.

If $k = \mathbb{F}_q$, if F is the Frobenius endomorphism of A, and if $P_{cA}(F;x)$ is its characteristic polynomial, these conditions are further equivalent to:

(IV) At least half of the roots of $P_{cA}(F;x)$ in $\overline{\mathbb{Q}}_p$ are p-adic units. In other words, if $n = \dim A$, the reduction modulo p of the polynomial $P_{cA}(F;x)$ is not divisible by x^{n+1}.

3. Let A be an ordinary abelian variety over $\overline{\mathbb{F}_p}$. Let $\widetilde{A}$ denote the canonical Serre–Tate lift [4] of A over W. Recall that $\widetilde{A}$ depends functorially on A and is characterized by the fact that the p-divisible group $T_p(\widetilde{A})$ over W attached to $\widetilde{A}$ [5] is the product of the p-divisible groups, uniquely determined by 2. (III), lifting respectively the neutral component and the largest étale quotient of $T_p(A)$. The canonical lift $\widetilde{A}$ is also the unique lift of A such that every endomorphism of A lifts to $\widetilde{A}$. Let $T(A)$ denote the integral homology of the complex abelian variety $A_{\mathbb{C}}$ deduced from $\widetilde{A}$ and φ by extension of scalars from W to $\mathbb{C}$:

$$T(A) = H_1(\widetilde{A} \otimes_\varphi \mathbb{C}).$$

It is known that $\widetilde{A}$ descends uniquely to $W_0(\overline{\mathbb{F}_p})$, so that $A_{\mathbb{C}}$ depends only on A and on the restriction of φ to $W_0(\overline{\mathbb{F}_p})$. The free $\mathbb{Z}$-module $T(A)$ has rank $2\dim(A)$; it is functorial in A. Moreover, if ℓ is a prime number distinct from p, one has functorially

$$T(A) \otimes \mathbb{Z}_\ell = T_\ell(A). \tag{3.1}$$

The canonical lift of the abelian variety A^* dual to A is the dual of $\widetilde{A}$, so that $(A_{\mathbb{C}})^* = A^*_{\mathbb{C}}$ and $T(A)$ and $T(A^*)$ are in perfect duality with values in $\mathbb{Z}(1)$:

$$T(A) \otimes T(A^*) \longrightarrow \mathbb{Z}(1). \tag{3.2}$$

It is indispensable to use $\mathbb{Z}(1)$ rather than $\mathbb{Z}$ in order to obtain a theory invariant under complex conjugation. The pairings (3.2) are compatible, via (3.1), with the pairings

$$T_\ell(A) \otimes T_\ell(A^*) \longrightarrow \mathbb{Z}_\ell(1);$$

a morphism $\xi : A \to A^*$ defines a polarization of A if and only if $\xi_{\mathbb{C}} : A_{\mathbb{C}} \to A^*_{\mathbb{C}}$ defines a polarization of $A_{\mathbb{C}}$. Put

$$T'_p(A) = \mathrm{Hom}(\mathbb{Q}_p/\mathbb{Z}_p, A(\overline{\mathbb{F}_p}))$$

and

$$T''_p(A) = \mathrm{Hom}_{\mathbb{Z}_p}(T'_p(A^*), \mathbb{Z}(1) \otimes \mathbb{Z}_p).$$

These $\mathbb{Z}_p$-modules are covariant functors of A.

By definition of the canonical lift, the p-divisible group $T_p(\widetilde{A})$ is the sum of the constant pro-étale group $T'_p(A)$ and of the Cartier dual of $T'_p(A^*)$. For every morphism $u : A \to B$, the induced morphism

$$u : T_p(\widetilde{A}) \to T_p(\widetilde{B})$$

is identified with the sum of $u|T'_p(A) : T'_p(A) \to T'_p(B)$ and the Cartier transpose of ${}^t u|T'_p(B^*) : T'_p(B^*) \to T'_p(A^*)$. Over $\mathbb{C}$, one has canonically $\mathbb{Z}(1)/(p^n) \sim \mu_{p^n}$, whence an isomorphism of functors:

$$T_{(p)}(A) = T(A) \otimes \mathbb{Z}_p = T'_p(A) \oplus T''_p(A). \tag{3.2}$$

4. Recall that, if $\varphi : X \to Y$ is an isogeny between complex abelian varieties, the exact homotopy sequence reduces to a short exact sequence

$$0 \longrightarrow H_1(X) \longrightarrow H_1(Y) \longrightarrow \mathrm{Ker}(\varphi) \longrightarrow 0.$$

The abelian varieties which are quotients of X by a finite subgroup, and these finite subgroups of X, correspond bijectively to the sublattices of $H_1(X) \otimes \mathbb{Q}$ containing $H_1(X)$.

Let A be an ordinary abelian variety over $\overline{\mathbb{F}_p}$. If n is an integer prime to p, the finite group subschemes of order n of A, of $\widetilde{A}$, and of $A_{\mathbb{C}}$ correspond bijectively, and still correspond to the overlattices R of $T(A)$ such that $[R : T(A)] = n$.

Put

$$V'_p = T'_p(A) \otimes \mathbb{Q}_p, \qquad V''_p(A) = T''_p(A) \otimes \mathbb{Q}_p.$$

The finite group subschemes of order p^k of A are products of an étale subgroup and an infinitesimal subgroup. The étale subgroups of order p^k of A correspond to those subgroups of order p^k of $A_{\mathbb{C}}$ for which the corresponding overlattice R of $T(A)$ is contained in $T_{(p)}(A) + V'_p(A)$. By duality, the infinitesimal subgroups of A correspond to the overlattices R of $T(A)$, p-isogenous to $T(A)$, i.e. such that $[R : T(A)]$ is a power of p, and contained in $T_{(p)}(A) + V''_p(A)$.

In all, the finite subgroups of A, or the abelian varieties quotient of A, correspond bijectively to the overlattices R of $T(A)$ such that

$$R \otimes \mathbb{Z}_p = (R \otimes \mathbb{Z}_p \cap V'_p) + (R \otimes \mathbb{Z}_p \cap V''_p). \tag{4.1}$$

5. In particular, $A^{(p)}$, the quotient of A by the largest infinitesimal subgroup of A killed by p for A ordinary, is defined by the overlattice $T(A)^{(p)}$ of $T(A)$, p-isogenous to $T(A)$, such that

$$T(A)^{(p)} \otimes \mathbb{Z}_p = T'_p(A) + \frac{1}{p} T''_p(A).$$

6. Let A be an abelian variety over $\mathbb{F}_q$ and let $F : x \mapsto x^q$ be its Frobenius endomorphism. Recall that A is uniquely determined by the pair $(\overline{A}, F)$ deduced from (A, F) by extension of scalars from $\mathbb{F}_q$ to $\overline{\mathbb{F}_q}$; the endomorphism F of $\overline{A}$ factors through the relative Frobenius

$$\mathrm{Fr}^{(q)} : \overline{A} \longrightarrow \overline{A}^{(q)}$$

and an isomorphism

$$F' : \overline{A}^{(q)} \longrightarrow \overline{A}.$$

If A is ordinary, $T(A)$ will denote the $\mathbb{Z}$-module $T(\overline{A})$ endowed with the endomorphism F induced by the Frobenius endomorphism of A. By 5, by what precedes, and by (3.2), the lattices $T(A)$ and $F(T(A))$ are p-isogenous and one has

$$F(T'_p(A)) = T'_p(A), \tag{6.1}$$

$$F(T''_p(A)) = qT''_p(A). \tag{6.2}$$

7. Theorem. The functor

$$A \longmapsto (T(A), F)$$

is an equivalence of categories between the category of ordinary abelian varieties over $\mathbb{F}_q$ and the category of finite-type free $\mathbb{Z}$-modules T endowed with an endomorphism F satisfying the following conditions:

(a) F is semisimple, and its eigenvalues have complex absolute value $q^{1/2}$.

(b) At least half of the roots in $\overline{\mathbb{Q}}_p$ of the characteristic polynomial of F are p-adic units; in other words, if T has rank d, the reduction modulo p of the polynomial $P_{cT}(F;x)$ is not divisible by $x^{\lfloor d/2 \rfloor + 1}$.

(c) There exists an endomorphism V of T such that $FV = q$.

If condition (a) is satisfied, conditions (b) and (c) are equivalent to:

(d) The module $T \otimes \mathbb{Z}_p$ admits a decomposition, stable under F, into two $\mathbb{Z}_p$-submodules T'_p and T''_p of the same dimension, such that $F|T'_p$ is invertible and $F|T''_p$ is divisible by q.

Proof. (A) We prove that (a)+(b)+(c) imply (d). If α is a complex eigenvalue of F, then $\bar{\alpha}$ is another, with the same multiplicity, and $\alpha\bar{\alpha} = q$. Apart from those equal to $\pm q^{1/2}$, the eigenvalues of F in $\mathbb{C}$, hence in $\overline{\mathbb{Q}}_p$, fall into pairs of roots α and q/α. The roots α and q/α cannot both be p-adic units; it follows from (b) that $\pm q^{1/2}$ is not an eigenvalue of F, that half of the eigenvalues of F in $\overline{\mathbb{Q}}_p$ are p-adic units, say $\alpha_1, \ldots, \alpha_{d/2}$, and that the other half consists of $\beta_1 = q/\alpha_1, \ldots, \beta_{d/2} = q/\alpha_{d/2}$. Let

$$T_{(p)} = T \otimes \mathbb{Z}_p, \qquad V_p = T \otimes \mathbb{Q}_p,$$

let V'_p be the subspace of V_p which is the kernel of $\prod_i (F - \alpha_i)$, and let V''_p be the kernel of the endomorphism $\varphi = \prod_i (F - \beta_i)$. One has

$$V_p = V'_p \oplus V''_p.$$

Let T'_p be the projection of $T_{(p)}$ onto V'_p and let $T''_p = T_{(p)} \cap V''_p$. Since φ kills V''_p and preserves T, it sends T'_p into $T_{(p)} \cap V'_p \subset T'_p$. On the other hand,

$$\det(\varphi | V'_p) = \prod_{i,j} (\alpha_i - \beta_j)$$

is a p-adic unit, hence $\varphi(T'_p) = T'_p$ and $T_{(p)} \cap V'_p = T'_p$, so that

$$T_{(p)} = T'_p \oplus T''_p.$$

(B) *Full faithfulness.* Let A and B be two abelian varieties over $\mathbb{F}_q$, and let ψ be the map

$$\psi : \mathrm{Hom}(A, B) \longrightarrow \mathrm{Hom}_F(T(A), T(B)).$$

By Tate's theorem [7] and (3.1), the map

$$\psi_\ell : \mathrm{Hom}(A, B) \otimes \mathbb{Z}_\ell \longrightarrow \mathrm{Hom}_F(T(A), T(B)) \otimes \mathbb{Z}_\ell$$

is an isomorphism for $(\ell, p) = 1$, so that $\psi \otimes \mathbb{Q}$ is an isomorphism. It is known that $\mathrm{Hom}(A, B)$ is torsion-free, hence ψ is injective. Let $u : A \to B$ be a morphism such that $T(u)$ is divisible by n. The induced morphism

$$u_{\mathbb{C}} : \overline{A}_{\mathbb{C}} \longrightarrow \overline{B}_{\mathbb{C}}$$

is then divisible by n, hence so is $\widetilde{u} : \widetilde{\overline{A}} \to \widetilde{\overline{B}}$ at the generic point of W. The kernel of multiplication by n is flat over W; thus $\widetilde{u}$ vanishes on this kernel, $\widetilde{u}$ and u are divisible by n, and ψ is bijective.

(C) *Necessity.* That $(T(A), F)$ satisfies (a) follows from Weil; condition (d), which implies (b) and (c), follows from 6.

(D) *Isogenies.* Let (T_0, F) satisfy (a) (d) and let T be a lattice in $T_0 \otimes \mathbb{Q}$, stable under F, still satisfying (d). Suppose that (T_0, F) is the image of an abelian variety A over $\mathbb{F}_q$ and let us prove that (T, F) comes from an isogenous abelian variety. Replacing T by $\frac{1}{k}T$, which is isomorphic to it, one may suppose that $T \supset T_0$. Condition (d) implies that T satisfies (4.1), and T defines a subgroup H of $\overline{A}$, which is defined over $\mathbb{F}_q$, and

$$(T, F) = T(A/H).$$

(E) *Surjectivity.* The functor T induces a functor $T_{\mathbb{Q}}$ from the category of ordinary abelian varieties up to isogeny over $\mathbb{F}_q$ to the category of finite-rank $\mathbb{Q}$-vector spaces endowed with an automorphism F satisfying (a) (b). By (D), it is enough to prove that this functor $T_{\mathbb{Q}}$ is essentially surjective. It is even enough to show that every simple object (V, F) of the image category is attained. By Honda [1] (see also [6]), there exists an abelian variety A over $\mathbb{F}_q$ such that the characteristic polynomial of the Frobenius F_A of A is a power of that of F. The third characterization in 2.[1] of ordinary abelian varieties shows that A is ordinary. Moreover, $(T(A) \otimes \mathbb{Q}, F)$ is a sum of copies of (V, F), hence, by (B), the abelian variety up to isogeny $A \otimes \mathbb{Q}$ is a sum of copies of an abelian variety B satisfying

$$T(B) \otimes \mathbb{Q} = (V, F).$$

8. Let (T, F) be a pair satisfying the hypotheses of the theorem, let $2g$ be the rank of T, let A be the corresponding abelian variety over $\mathbb{F}_q$, and let $A_{\mathbb{C}}$ be the complex abelian variety deduced from it (3.). One has

$$T = H_1(A_{\mathbb{C}}),$$

so that $T \otimes \mathbb{R}$ is identified with the Lie algebra of $A_{\mathbb{C}}$ and, as such, is endowed with a complex structure. Here is, according to J.-P. Serre, how to reconstruct this complex structure in terms of T, F, and the restriction of φ to $W_0(\overline{\mathbb{F}_p})$.

Proposition. The complex structure defined above on $T \otimes \mathbb{R}$ is characterized by the following properties:

(I) The endomorphism F is $\mathbb{C}$-linear.

(II) If v is the valuation of the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ in $\mathbb{C}$ extending the valuation of $W_0(\overline{\mathbb{F}_p})$, the valuations of the g eigenvalues of this endomorphism are strictly positive.

Condition (I) is evident, and condition (II) follows from the fact that the action of F on the Lie algebra of A is congruent to zero modulo p. The uniqueness of the complex structure satisfying (I) and (II) follows easily from condition (b) satisfied by (T, F).

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[1] The printed source refers here to 1.; the characterizations of ordinarity are stated in 2., and that reference is restored here.

Pierre Deligne
I.H.E.S.
91, Bures-sur-Yvette, France

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THE IRREDUCIBILITY OF THE SPACE OF CURVES OF GIVEN GENUS

by P. DELIGNE and D. MUMFORD[1]

Fix an algebraically closed field k. Let M_g be the moduli space of curves of genus g over k. The main result of this note is that M_g is irreducible for every k. Of course, whether or not M_g is irreducible depends only on the characteristic of k. When the characteristic is 0, we can assume that $k = \mathbb{C}$, and then the result is classical. A simple proof appears in Enriques–Chisini [E, vol. 3, chap. 3], based on analyzing the totality of coverings of $\mathbf{P}^1$ of degree n, with a fixed number d of ordinary branch points. This method has been extended to char. p by William Fulton [F], using specializations from char. 0 to char. p provided that $p > 2g + 1$. Unfortunately, attempts to extend this method to all p seem to get stuck on difficult questions of wild ramification. Nowadays, the Teichmüller theory gives a thoroughly analytic but very profound insight into this irreducibility when $k = \mathbb{C}$. Our approach however is closest to Severi's incomplete proof ([Se], Anhang F; the error is on pp. 344–345 and seems to be quite basic) and follows a suggestion of Grothendieck for using the result in char. 0 to deduce the result in char. p. The basis of both Severi's and Grothendieck's ideas is to construct families of curves X, some singular, with $p_a(X) = g$, over non-singular parameter spaces, which in some sense contain enough singular curves to link together any two components that M_g might have.

The essential thing that makes this method work now is a recent "stable reduction theorem" for abelian varieties. This result was first proved independently in char. 0 by Grothendieck, using methods of etale cohomology (private correspondence with J. Tate), and by Mumford, applying the easy half of Theorem (2.5), to go from curves to abelian varieties (cf. [M$_2$]). Grothendieck has recently strengthened his method so that it applies in all characteristics (SGA 7, 1968). Mumford has also given a proof using theta functions in char. $\neq 2$. The result is this:

Stable Reduction Theorem. — Let R be a discrete valuation ring with quotient field K. Let A be an abelian variety over K. Then there exists a finite algebraic extension L of K such that, if $R_L =$ integral closure of R in L, and if $\mathcal{A}_L$ is the Néron model of $A \times_K L$ over R_L, then the closed fibre $\mathcal{A}_{L,s}$ of $\mathcal{A}_L$ has no unipotent radical.

We shall give two related proofs of our main result. One of these is quite elementary, and follows by quite standard techniques once the Stable Reduction Theorem for abelian varieties is applied, in § 2, to deduce an analogous stable reduction theorem for curves. The other proof is more powerful, and is based on the use of a larger category than the category of schemes, and on proving for the objects of this category many of the standard theorems for schemes, especially the Enriques–Zariski connectedness theorem (EGA 3, (4.3)). Unfortunately, this larger category is not quite a category — it is a simple type of 2-category; in fact, if X, Y are objects, then $\mathrm{Hom}(X, Y)$ is itself a category, but one in which all morphisms are isomorphisms. The objects of this 2-category we call algebraic stacks[2]. The moduli space M_g is just the "underlying coarse variety" of a more fundamental object, the moduli stack $\mathcal{M}_g$, studied in [M$_3$]. Full details on the basic properties and theorems for algebraic stacks will be given elsewhere. In this paper, we will only give definitions and state without proof the general theorems which we apply. Using the method of algebraic stacks, we can prove not only the irreducibility of M_g itself, but of all higher level moduli spaces of curves too (cf. § 5 below).

§ 1. Stable curves and their moduli.

The key definition of the whole paper is this:

Definition (1.1). — Let S be any scheme. Let $g \geq 2$. A stable curve of genus g over S is a proper flat morphism $\pi : C \to S$ whose geometric fibres are reduced, connected, 1-dimensional schemes C_s such that:

(i) *C_s has only ordinary double points;*

(ii) *if E is a non-singular rational component of C_s, then E meets the other components of C_s in more than 2 points;*

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[2]A slightly less general category of objects, called algebraic spaces, has been introduced and studied very deeply by M. Artin [A$_1$] and D. Knutson [K]. The idea of enlarging the category of varieties for the study of moduli spaces is due originally, we believe, to A. Weil.

(iii) $\dim H^1(\mathcal{O}_{C_s}) = g$.

We will study in this section three aspects of the theory of stable curves: their pluri-canonical linear systems, their deformations, and their automorphisms.

Suppose $\pi : C \to S$ is a stable curve. Since π is flat and its geometric fibres are local complete intersections, the morphism π is locally a complete intersection (i.e., locally, C is isomorphic as S-scheme to $V(f_1, \ldots, f_{n-1}) \subset \mathbf{A}^n \times U$, where $U \subset S$ is open, and $f_1, \ldots, f_{n-1} \in \Gamma(\mathcal{O}_{\mathbf{A}^n \times U})$ are a regular sequence). Therefore, by the theory of duality of coherent sheaves [H], there is a canonical invertible sheaf $\omega_{C/S}$ on C — the unique non-zero cohomology group of the complex of sheaves $f^!(\mathcal{O}_S)$. We need to know the following facts about $\omega_{C/S}$:

a) for all morphisms $f : T \to S$, $\omega_{C \times_S T/T}$ is canonically isomorphic to $f^*(\omega_{C/S})$;

b) if $S = \mathrm{Spec}(k)$, k algebraically closed, let $f : C' \to C$ be the normalization of C, $x_1, \ldots, x_n, y_1, \ldots, y_n$ the points of C' such that the $z_i = f(x_i) = f(y_i)$, $1 \leq i \leq n$, are the double points of C. Then $\omega_{C/S}$ is the sheaf of 1-forms η on C' regular except for simple poles at the x's and y's and with $\mathrm{Res}_{x_i}(\eta) + \mathrm{Res}_{y_i}(\eta) = 0$;

c) if $S = \mathrm{Spec}(k)$, and $\mathcal{F}$ is a coherent sheaf on C, then

$$\mathrm{Hom}(H^1(C, \mathcal{F}), k) \cong \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{F}, \omega_{C/S}).$$

Theorem (1.2). — *If $g \geq 2$ and C is a stable curve of genus g over an algebraically closed field k, then $H^1(C, \omega_{C/k}^{\otimes n}) = (0)$ if $n \geq 2$, and $\omega_{C/k}^{\otimes n}$ is very ample if $n \geq 3$.*

Proof. — Since C is stable, of genus $g \geq 2$, every irreducible component E of C either 1) has (arithmetic) genus ≥ 2 itself, 2) has genus 1, but meets other components of C in at least one point, or 3) is non-singular, rational and meets other components of C in at least three points. But by b) above, $\omega_{C/k} \otimes \mathcal{O}_E$ is isomorphic to $\omega_{E/k}(\sum Q_i)$, where $\{Q_i\}$ are the points where E meets the rest of C. Since the degree of $\omega_{E/k}$ is $2g_E - 2$, it follows that in any of the cases 1, 2 or 3, $\omega_{C/k} \otimes \mathcal{O}_E$ has positive degree. This shows immediately that $\omega_{C/k}$ is ample on each component E of C, hence is ample.

Next, by c) above, $H^1(\omega_{C/k}^{\otimes n})$ is dual to $H^0(\omega_{C/k}^{\otimes(1-n)})$. Since $\omega_{C/k} \otimes \mathcal{O}_E$ has positive degree, $\omega_{C/k}^{\otimes(1-n)} \otimes \mathcal{O}_E$ has no sections for any E, any $n \geq 2$; therefore $H^0(\omega_{C/k}^{\otimes(1-n)}) = (0)$ if $n \geq 2$, and so $H^1(\omega_{C/k}^{\otimes n}) = (0)$ if $n \geq 2$.

To prove that an invertible sheaf $\mathcal{L}$ on any scheme C, proper over k, is very ample, it suffices to show

a) for all closed points $x \neq y$

$$H^0(C, \mathcal{L}) \to H^0(C, \mathcal{L} \otimes k(x)) \oplus H^0(C, \mathcal{L} \otimes k(y))$$

is surjective,

b) for all closed points x,

$$H^0(C, \mathcal{L}) \to H^0(C, \mathcal{L} \otimes \mathcal{O}_x/\mathfrak{m}_x^2)$$

is surjective.

Using the exact sequence of cohomology, these both follow if $H^1(C, \mathfrak{m}_x \cdot \mathfrak{m}_y \cdot \mathcal{L}) = (0)$ for all closed points $x, y \in C$. In our case, $\mathcal{L} = \omega_C^{\otimes n}$, $n \geq 3$, so if we use duality, we must show:

$$(*) \qquad \mathrm{Hom}(\mathfrak{m}_x \cdot \mathfrak{m}_y, \omega_C^{\otimes -n}) = (0), \qquad \text{if } n \geq 2.$$

Source note. The printed formula has $\omega_X^{\otimes -n}$; the surrounding argument is entirely about C, so the required $\omega_C^{\otimes -n}$ is retained here. If x is a non-singular point, $\mathfrak{m}_x$ is an invertible sheaf. If x is a double point, let $\pi : C' \to C$ be the result of blowing up x, and let $x_1, x_2 \in C'$ be the two points in $\pi^{-1}(x)$. Then it is easy to check that for any invertible sheaf $\mathcal{L}$ on C:

$$\mathrm{Hom}(\mathfrak{m}_x, \mathcal{L}) \cong H^0(C', \pi^*\mathcal{L}),$$

$$\mathrm{Hom}(\mathfrak{m}_x^2, \mathcal{L}) \cong H^0(C', \pi^*\mathcal{L}(x_1 + x_2)).$$

Therefore, we have 3 cases of (*) to check:

Case 1. — x, y non-singular points of C, then

$$H^0(\omega_C^{-n}(x + y)) = (0), \qquad \text{if } n \geq 2.$$

Case 2. — x double point of C, $\pi : C' \to C$ blowing up x, $\{x_1, x_2\} = \pi^{-1}(x)$, and y a non-singular point of C. Then

$$H^0(\pi^*(\omega_C)^{-n}(y)) = (0), \qquad H^0(\pi^*(\omega_C)^{-1}(x_1 + x_2)) = (0), \qquad \text{if } n \geq 2.$$

Case 3. — x, y double points of C, $\pi : C' \to C$ blowing up x and y. Then

$$H^0(\pi^*(\omega_C)^{-n}) = (0) \qquad \text{if } n \geq 2.$$

Now since the degree of ω_C^{-n} $(n \geq 2)$ on all components E of C is less than or equal to -2, all of this is clear, except in those cases where $n = 2$, the degree of ω_C on some E is 1, and in which two poles are allowed on E. This occurs if:

(i) case 1, $p_a(E) = 1$, E meets $C - E$ at only one point, $x, y \in E$.

(ii) case 1, $p_a(E) = 0$, E meets $C - E$ at only three points, $x, y \in E$.

(iii) case 2, E a rational curve with one double point meeting $C - E$ at one point, $x =$ double point of E.

But in all these cases, C has components besides E and a section in the H^0 in question must definitely vanish on all these other components. So at the points where E meets $C - E$, the section has extra zeroes. Since the sheaf in question has degree 0 on E, the section is zero on E too. Q.E.D.

Corollary. — *Let $\pi : C \to S$ be any stable curve of genus $g \geq 2$. Then $\omega_{C/S}^{\otimes n}$ is relatively very ample if $n \geq 3$ and $\pi_*(\omega_{C/S}^{\otimes n})$ is a locally free sheaf on S of rank $(2n-1)(g-1)$.*

Proof. — In fact, since for all $s \in S$, $H^1(\omega_{C/S}^{\otimes n} \otimes \mathcal{O}_{C_s}) = (0)$, it follows from [EGA, chap. 3, § 7], that $\pi_*(\omega_{C/S}^{\otimes n})$ is locally free and that $\pi_*(\omega_{C/S}^{\otimes n}) \otimes k(s) \cong H^0(\omega_{C/S}^{\otimes n} \otimes \mathcal{O}_{C_s})$. Therefore the corollary follows. Q.E.D.

Taking $n = 3$, it follows that every stable curve C/S can be realized as a family of curves in $\mathbf{P}^{5g-6}$ with Hilbert polynomial:

$$P_g(n) = (6n - 1)(g - 1).$$

Following standard arguments ([M$_1$], p. 99), it is easy to prove that there is a subscheme

$$H_g \subset \mathrm{Hilb}^{P_g}_{\mathbf{P}^{5g-6}}$$

of "all" tri-canonically embedded stable curves. (Hilb is the Hilbert scheme over $\mathbb{Z}$.) To be precise, there is an isomorphism of functors:

$$\mathrm{Hom}(S, H_g) \cong \left\{ \begin{array}{l} \text{set of stable curves } \pi : C \to S, \text{ plus isomorphisms:} \\ \mathbf{P}(\pi_*(\omega_{C/S}^{\otimes 3})) \cong \mathbf{P}^{5g-6} \times S\,, \quad \text{(modulo isomorphism).} \end{array} \right.$$

We will denote by $Z_g \subset H_g \times \mathbf{P}^{5g-6}$ the universal tri-canonically embedded stable curve. The functor of stable curves itself is the sheafification of the quotient of functors:

$$H_g / \mathrm{PGL}(5g - 6).$$

We now consider the deformation theory of stable curves. Let k be any ground field. The deformation theory of X's smooth over k can be found in [SGA, 60–61]; for singular S's, the theory has been worked out in [Sc]. We shall indicate here the results of this theory for a scheme X which is

(i) one-dimensional;

(ii) generically smooth over k;

(iii) locally a complete intersection.

The advantages of this case are two-fold: first, the "cotangent complex" of Grothendieck, Lichtenbaum and Schlessinger reduces, in view of (ii) and (iii), to the single coherent sheaf $\Omega_{X/k}$, the Kähler differentials. Secondly, we have:

Lemma (1.3). — $\mathrm{Ext}^2(\Omega_{X/k}, \mathcal{O}_X) = (0)$.

Proof. — Use the spectral sequence:

$$H^p(X, \mathcal{E}xt^q(\Omega_{X/k}, \mathcal{O}_X)) \Rightarrow \mathrm{Ext}^{p+q}(\Omega_{X/k}, \mathcal{O}_X).$$

Then (i) $H^2(X, \mathcal{E}xt^0) = (0)$ since $\dim X = 1$. (ii) Since $\Omega_{X/k}$ is locally free except at a finite number of points, $\mathcal{E}xt^1(\Omega_{X/k}, \mathcal{O}_X)$ has 0-dimensional support, hence $H^1(X, \mathcal{E}xt^1) = (0)$. (iii) Locally, if we embed $X \subset \mathbf{A}^n$, then $\Omega_{X/k}$ has a free resolution of length 2:

$$0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_{\mathbf{A}^n} \otimes \mathcal{O}_X \to \Omega_{X/k} \to 0$$

where $\mathcal{I}$ = sheaf of ideals defining X. Therefore $\mathcal{E}xt^2 = (0)$. Q.E.D.

In Schlessinger's theory, the significance of Lemma (1.3) is that all obstructions vanish, i.e., deformations of X over base schemes $\mathrm{Spec}(A/\mathcal{J})$ (A = local Artin ring with residue field k) can always be embedded in deformations over $\mathrm{Spec}(A)$. Moreover, the theory says that there is a canonical one-one correspondence between $\mathrm{Ext}^1(\Omega_{X/k}, \mathcal{O}_X)$ and the first order deformations of X, i.e., proper, flat morphisms p, and isomorphisms α as follows:

$$\begin{array}{ccccc} X_1 & \supset & X_1 \times_{\mathrm{Spec}\, k[\varepsilon]} \mathrm{Spec}\, k & \xrightarrow{\sim_\alpha} & X \\ \downarrow p & & \downarrow & & \downarrow \\ \mathrm{Spec}\, k[\varepsilon]/\varepsilon^2 & \supset & \mathrm{Spec}\, k & = & \mathrm{Spec}\, k. \end{array}$$

Since the obstructions vanish, there is a versal formal deformation $\mathfrak{X}$ of X over the base scheme

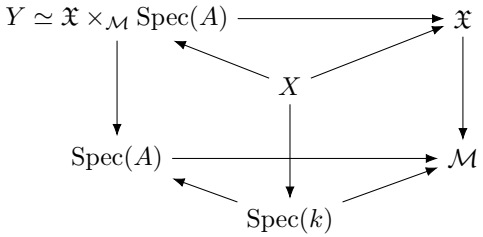
$$\mathfrak{M} = \mathrm{Spec}\, o_k[[t_1, \ldots, t_N]],$$

where $o_k = k$ if the char. is 0, or o_k is the complete regular local ring, with maximal ideal $p \cdot o_k$ and residue field k, if $\mathrm{char}(k) = p$, unique (by Cohen's structure theorem), and

$$N = \dim_k \mathrm{Ext}^1(\Omega_X, \mathcal{O}_C).$$

This means that $\mathfrak{X}$ is a formal scheme, proper and flat over $\mathcal{M}$, with fibre X over $\mathrm{Spec}(k)$ and the two properties:

a) Every deformation of X is induced from $\mathfrak{X}$, i.e., if A is a local Artin o_k-algebra with residue field k, and $p : Y \to \mathrm{Spec}(A)$ is proper and flat with fibre X over $\mathrm{Spec}(k)$, then there is a commutative diagram:



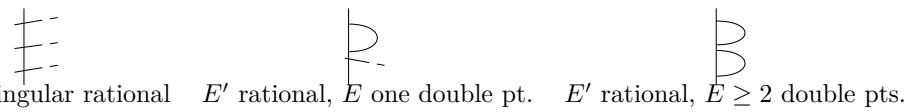
b) If $A = k[\varepsilon]/(\varepsilon^2)$, the above morphism f is uniquely determined by the diagram.

This implies that the tangent space to $\mathcal{M} \times_{o_k} k$ at its closed point is canonically isomorphic to $\mathrm{Ext}^1(\Omega_{X/k}, \mathcal{O}_X)$.

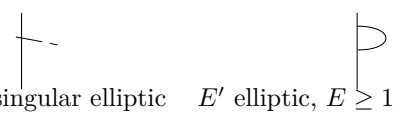
In case $\mathrm{Ext}^0(\Omega_{X/k}, \mathcal{O}_X) = (0)$, $\mathfrak{X}/\mathcal{M}$ is, in fact, *universal*: i.e., in property a), f is always unique, which means that the functor represented by $\mathcal{M}$ in the category of artin, local o_k-algebras is isomorphic to the functor of deformations Y/A of X. This fortunately holds for stable curves:

Lemma (1.4). — If X is a stable curve, $\mathrm{Ext}^0(\Omega_{X/k}, \mathcal{O}_X) = (0)$.

Proof. — We may assume that k is algebraically closed. Now a homomorphism from Ω_X to $\mathcal{O}_X$ is given by an everywhere regular vector field D on X. Such a vector field is given, in turn, by a regular vector field D' on the normalization X' of X which vanishes at all points of X' lying over the double points of X. In particular, D' and hence D vanishes identically on all components E of X whose normalization E' has genus ≥ 2. There remain the following possibilities for E:



E non-singular rational E' rational, E one double pt. E' rational, $E \geq 2$ double pts.



E non-singular elliptic E' elliptic, $E \geq 1$ double points

In all cases where E' is rational, note that D' has to have at least 3 zeroes; and where E' is elliptic, D' has to have at least one zero. So D' vanishes on all components E'. This proves that $\mathrm{Ext}^0(\Omega_{X/k}, \mathcal{O}_X) = (0)$. Q.E.D.

Schlessinger's theory also allows us to trace what happens to the singularities of X in this deformation $\mathcal{M}$. For each closed point $x \in X$, he studies deformations of the complete local ring $\widehat{\mathcal{O}}_{x,X}$ alone, i.e., flat A-algebras $\mathcal{O}$ plus isomorphisms:

$$\mathcal{O} \otimes_A k \simeq \widehat{\mathcal{O}}_{x,X}.$$

This is a functor of A exactly as before. Since $\mathrm{Ext}^2_{\widehat{\mathcal{O}}_x}(\widehat{\Omega}_{x/k}, \widehat{\mathcal{O}}_x) = (0)$, there are no obstructions to extending deformations. First order deformations with $A = k[\varepsilon]/(\varepsilon^2)$ are classified by $\mathrm{Ext}^1_{\widehat{\mathcal{O}}_x}(\widehat{\Omega}_{x/k}, \widehat{\mathcal{O}}_x)$. And there is a versal formal deformation $\widehat{\mathcal{O}}$, which is a complete local ring and a flat $o_k[[t_1, \ldots, t_N]]$-algebra, $N = \dim_k \mathrm{Ext}^1_{\widehat{\mathcal{O}}_x}(\widehat{\Omega}_{x/k}, \widehat{\mathcal{O}}_x)$, such that

$$\widehat{\mathcal{O}}/(p, t_1, \ldots, t_N) = \widehat{\mathcal{O}}_{x,X}.$$

Whenever X is smooth over k at x, $N = 0$, and the theory is uninteresting. The first non-trivial example is a k-rational ordinary double point with k-rational tangent lines:

$$\widehat{\mathcal{O}}_{x,X} \simeq k[[u, v]]/(u \cdot v).$$

Then $N = 1$, and

$$\widehat{\mathcal{O}} \simeq o_k[[u, v, t_1]]/(uv - t_1).$$

In other words, if $\mathcal{O}$ is any deformation of $\widehat{\mathcal{O}}_{x,X}$ and u, v are lifted suitably into $\mathcal{O}$, then $u \cdot v = w \in A$ and $\mathcal{O}$ is induced from $\widehat{\mathcal{O}}$ via the homomorphism $o_k[[t_1]] \to A$ taking t_1 to w. This is easy to prove.

Finally, Schlessinger's theory connects global deformations to local ones. Let $\mathfrak{X}/\mathcal{M}_{gl}$ with $\mathcal{M}_{gl} = \mathrm{Spec}(A)$ be the versal global deformation, let $x_1, \ldots, x_k \in X$ be the points where X is not smooth over k, and let $\widehat{\mathcal{O}}_i$ as A_i-algebra be the versal deformation of the local ring $\widehat{\mathcal{O}}_{x_i,X}$. Let $\mathcal{M}_{lo} = \mathrm{Spec}(A_1 \widehat{\otimes}_{o_k} \cdots \widehat{\otimes}_{o_k} A_k)$. Then we may consider the local rings

$$\widehat{\mathcal{O}}_{x_i,\mathfrak{X}}$$

of $\mathfrak{X}$ at x_i. There are o_k-homomorphisms $\varphi_i : A_i \to A$ such that $\widehat{\mathcal{O}}_{x_i,\mathfrak{X}} \simeq \widehat{\mathcal{O}}_i \otimes_{A_i} A$. Dualizing, we obtain a morphism

$$\Phi = \prod_i \mathrm{Spec}(\varphi_i) : \mathcal{M}_{gl} \to \mathcal{M}_{lo}$$

which describes exactly how the various singularities of X behave in the versal deformation $\mathfrak{X}$. The final fact that we need is:

Proposition (1.5). — *$\Phi : \mathcal{M}_{gl} \to \mathcal{M}_{lo}$ is formally smooth, i.e., there are isomorphisms*

$$\mathcal{M}_{gl} \simeq \mathrm{Spec}\, o_k[[t_1, \ldots, t_{N+M}]], \qquad \mathcal{M}_{lo} \simeq \mathrm{Spec}\, o_k[[t_1, \ldots, t_N]]$$

such that $\Phi^*(t_i) = t_i$, $1 \leq i \leq N$.

Proof. — In view of the functorial significance of $\mathcal{M}_{gl}$ and $\mathcal{M}_{lo}$, this follows if we prove that the natural map:

$$(*) \qquad \mathrm{Ext}^1_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X) \longrightarrow \prod_{i=1}^k \mathrm{Ext}^1_{\widehat{\mathcal{O}}_{x_i}}(\widehat{\Omega}_{x_i}, \widehat{\mathcal{O}}_{x_i})$$

is surjective; $(*)$ is the induced map $d\Phi$ on the tangent spaces to $\mathcal{M}_{gl}$ and $\mathcal{M}_{lo}$. Since Ω_X is an invertible $\mathcal{O}_X$-Module outside the x_i's, it follows that:

$$\begin{aligned} \prod_{i=1}^k \mathrm{Ext}^1_{\widehat{\mathcal{O}}_{x_i}}(\widehat{\Omega}_{x_i}, \widehat{\mathcal{O}}_{x_i}) &\simeq \prod_{i=1}^k \mathrm{Ext}^1_{\mathcal{O}_{x_i}}(\Omega_{x_i}, \mathcal{O}_{x_i}) \\ &\simeq H^0\big(X, \mathcal{E}xt^1_{\mathcal{O}_X}(\Omega_X, \mathcal{O}_X)\big). \end{aligned}$$

Therefore, $(*)$ is surjective by the spectral sequence used in Lemma (1.3) and the fact that $H^2(X, \mathcal{E}xt^0(\Omega_X, \mathcal{O}_X)) = (0)$. Q.E.D.

In particular, suppose C is a stable curve over an algebraically closed ground field k, and let $x_1, \ldots, x_k$ be the double points of C. Let $N = \dim \mathrm{Ext}^1(\Omega_X, \mathcal{O}_X)$. Then C has a universal formal deformation $\mathcal{C}/\mathcal{M}$ where

$\mathcal{M} = \operatorname{Spec} o_k[[t_1, \ldots, t_N]]$. Note that since in this case, the invertible sheaf $\omega_{\mathcal{C}/\mathcal{M}}$ is relatively ample, $\mathcal{C}$ is not only a formal scheme over $\mathcal{M}$, but also the formal completion of a unique scheme proper and flat over $\mathcal{M}$, which we will also denote by $\mathcal{C}$. $\mathcal{C}$ is clearly a stable curve over $\mathcal{M}$. Now each double point x_i has one modulus (cf. our example above) so the versal deformation space of the rings $\mathcal{O}_{x_i,C}$ is $o_k[[t'_1, \ldots, t'_k]]$. By Proposition (1.5), we may identify t_i with t'_i, and we conclude that for suitable u_i, v_i:

$$\widehat{\mathcal{O}}_{x_i,\mathcal{C}} \simeq o_k[[u_i, v_i, t_1, \ldots, t_N]]/(u_i v_i - t_i).$$

In particular, $t_i = 0$ is the locus in $\mathcal{M}$ where "x_i remains a double point".

The relation between the formal moduli space $\mathcal{M}$ of C and the local structure of H_g at a point x with $\kappa(x) = k$ corresponding to some tri-canonical model of C is exactly the same as in the case of non-singular curves ([M$_1$], chap. 5, § 2). Let $\widehat{\mathcal{O}}_x$ be the completion of the local ring $\mathcal{O}_{x,H_g}$, and let $T = \operatorname{Spec}(\widehat{\mathcal{O}}_x)$. Let x denote the closed point of T too. The universal family of stable curves $Z_g \subset H_g \times \mathbf{P}^{5g-6}$ induces a family $Z' \subset T \times \mathbf{P}^{5g-6}$, whose fibre Z'_x over x is C. Then there is a unique morphism $f : T \to \mathcal{M}$ such that $Z' \simeq \mathcal{C} \times_{\mathcal{M}} T$, with this isomorphism restricting to the identity on the fibres over x, both of which are C. I claim that via f, T is formally smooth over $\mathcal{M}$, i.e., $\widehat{\mathcal{O}}_x \simeq o_k[[t_1, \ldots, t_N, t_{N+1}, \ldots, t_m]]$. In fact, by choosing an isomorphism $\mathbf{P}(\pi_*(\omega^{\otimes 3}_{\mathcal{C}/\mathcal{M}})) \simeq \mathbf{P}^{5g-6} \times \mathcal{M}$, we obtain a tri-canonical embedding $\mathcal{C} \subset \mathbf{P}^{5g-6} \times \mathcal{M}$ of $\mathcal{C}$, hence a morphism $s : \mathcal{M} \to H_g$ such that $\mathcal{C}$, with this embedding, is the pull-back of Z_g. Then s factors through T and $s : \mathcal{M} \to T$ is a section of f. On the other hand, consider the action of $\mathrm{PGL}(5g-6)$ on H_g. Let S_x be the stabilizer of the k-valued point x. Then S_x is finite and reduced. Because if it were not, S_x would have a non-trivial tangent space at the origin, i.e., there would be a $k[\varepsilon]/(\varepsilon^2)$-valued point of $\mathrm{PGL}(5g-6)$ centered at the identity, which maps the embedded stable curve $C \subset \mathbf{P}^{5g-6}$ corresponding to x into itself. But this action is given by an everywhere regular derivation on C, and we have

seen that all such vanish. This means that this $k[\varepsilon]/(\varepsilon^2)$-valued automorphism is the identity at all points of C and, since C is connected and spans $\mathbf{P}^{5g-6}$, the automorphism is the identity everywhere. Thus S_x is finite and reduced. It follows that the action of $\mathrm{PGL}(5g-6)$ on T is formally free, and hence that T is formally a principal fibre bundle over $\mathcal{M}$ with group $\mathrm{PGL}(5g-6)$. Therefore T is formally smooth over $\mathcal{M}$ as required.

Putting this together with what we know about $\mathcal{M}$, we conclude the following:

Let k be any algebraically closed field,
let $H'_g = H_g \times \operatorname{Spec}(o_k)$, $Z'_g = Z_g \times \operatorname{Spec}(o_k)$,
let $x \in H'_g$ be a closed point,
let $C \subset Z'_g$ be the stable curve over x,
let $x_1, \ldots, x_k \in C$ be its double points.

Then

Theorem (1.6). — There are isomorphisms

$$\widehat{\mathcal{O}}_{x,H'_g} \simeq o_k[[t_1, \ldots, t_N]],$$

$$\widehat{\mathcal{O}}_{x_i,Z'_g} \simeq o_k[[u_i, v_i, t_1, \ldots, t_N]]/(u_i v_i - t_i).$$

Corollary (1.7). — H_g is smooth over $\mathbb{Z}$. In particular, for all algebraically closed fields k, $H_g \times \operatorname{Spec}(k)$ is a disjoint union of a finite number of non-singular algebraic varieties over k.

Let

$$H_g^0 = \{x \in H_g \mid \text{the corresponding stable curve } (Z_g)_x \text{ is non-singular}\},$$
$$S = \{x \in Z_g \mid \text{the projection } \pi : Z_g \to H_g \text{ is not smooth at } x\}.$$

Definition (1.8). — Let $p : X \to Y$ be a smooth morphism of finite type, with Y a noetherian scheme, and let $D \subset X$ be a relative Cartier divisor. Then D has normal crossings relative to Y if for all $x \in D$, the local equation $d = 0$ of D decomposes in the strict completion[3] $\widetilde{\mathcal{O}}_{x,X}$ of $\mathcal{O}_{x,X}$ as $d = d_1 \cdots d_k$, where $d_1, \ldots, d_k$ are linearly independent in

$$\widetilde{\mathfrak{m}}_{x,X}/\big(\widetilde{\mathfrak{m}}^2_{x,X} + \mathfrak{m}_{y,Y}\widetilde{\mathcal{O}}_{x,X}\big),$$

[3]The complete local ring, formally etale over $\mathcal{O}_{x,X}$ with residue field the separable closure of $\mathcal{O}_{x,X}/\mathfrak{m}_{x,X}$.

with $y = p(x)$.

Corollary (1.9). — $H_g^0 = H_g - S^*$, *where* S^* *is a divisor with normal crossings relative to* $\mathbb{Z}$. Z_g *and* S *are smooth over* $\mathbb{Z}$, *and the projection* $p : S \to S^*$ *is finite and an isomorphism at all points where* S^* *is smooth over* $\mathbb{Z}$, *i.e.,* S *is the normalization of* S^*.

Proof. — In the notation of Theorem (1.6), S^* is defined in $\widehat{\mathcal{O}}_{x,H_g'}$ by the local equation $t_1 \cdots t_k = 0$. And

$$\widehat{\mathcal{O}}_{x_i,Z_g'} \simeq o_k[[u_i, v_i, t_1, \ldots, t_{i-1}, t_{i+1}, \ldots, t_N]],$$

$$\widehat{\mathcal{O}}_{x_i,S} \simeq \widehat{\mathcal{O}}_{x_i,Z_g'}/(u_i, v_i) \simeq o_k[[t_1, \ldots, t_{i-1}, t_{i+1}, \ldots, t_N]].$$

Q.E.D.

Next we take up the isomorphisms and automorphisms of stable curves. Suppose $p : X \to S$, $q : Y \to S$ are two stable curves:

Definition (1.10). — $\underline{\mathrm{Isom}}_S(X, Y)$ *is the functor on* (Sch/S) *associating to each* S*-scheme* S' *the set of* S'*-isomorphisms between* $X \times_S S'$ *and* $Y \times_S S'$. *If* $X = Y$, *we denote* $\underline{\mathrm{Isom}}_S(X, X)$ *by* $\underline{\mathrm{Aut}}_S(X)$.

Since both X and Y have the canonical polarizations $\omega_{X/S}$, $\omega_{Y/S}$ respectively, any isomorphism $f : X \to Y$ must satisfy $f^*(\omega_{Y/S}) \simeq \omega_{X/S}$. Therefore, by Grothendieck's results on the representability of the Hilbert scheme and related functors [Gr$_1$], we conclude that $\underline{\mathrm{Isom}}_S(X, Y)$ is represented by a scheme $\mathrm{Isom}_S(X, Y)$, quasi-projective over S. Concerning this scheme, we have:

Theorem (1.11). — $\mathrm{Isom}_S(X, Y)$ *is finite and unramified over* S.

Proof. — To check that $\mathrm{Isom}_S(X, Y)$ is unramified, we may take S to be the spectrum of an algebraically closed field k, in which case $\mathrm{Isom}_S(X, Y)$ is either empty or isomorphic to $\mathrm{Aut}_k(X)$. A point of $\mathrm{Aut}_k(X)$ with values in $k[\varepsilon]/(\varepsilon^2)$ with image the identity may be identified with a vector field on X. By Lemma (1.4), stable curves have no non-zero vector fields. This proves that $\mathrm{Isom}_S(X, Y)$ is unramified over S, and since it is also of finite type over S, it is quasi-finite over S. It remains to check that $\mathrm{Isom}_S(X, Y)$ is proper over S.

Locally over S, X and Y are the pull-backs of the universal tri-canonically embedded stable curve by some morphisms from S to H_g, so that it suffices to prove the properness of $\mathrm{Isom}_S(X, Y)$ in the "universal" case where $S = H_g \times H_g$, X and Y being the two inverse images of the universal curve on H_g. In that case, the open subset of $\mathrm{Isom}_S(X, Y)$ corresponding to smooth curves is dense, so that the Theorem follows from the valuative criterion of properness which holds by:

Lemma (1.12). — *Let* X *and* Y *be two stable curves over a discrete valuation ring* R, *with algebraically closed residue field. Denote by* η *and* s *the generic and closed points of* $\mathrm{Spec}(R)$, *and assume that the generic fibres* X_η *and* Y_η *of* X *and* Y *are smooth. Then any isomorphism* φ_η *between* X_η *and* Y_η *extends to an isomorphism* φ *between* X *and* Y.

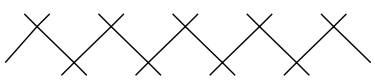
(A posteriori, it follows from Theorem (1.11) that the lemma holds for any valuation ring R and without assuming X_η or Y_η smooth.)

Proof. — Another way to put the lemma is that if we start with a smooth curve X_η of genus $g \geq 2$ over the quotient field K of R, there is, up to canonical isomorphism, at most one stable curve X over R with X_η as its generic fibre. We shall deduce this from the analogous uniqueness assertion for minimal models ([L] and [S]): given a smooth curve X_η of genus $g \geq 1$ over K, there is, up to canonical isomorphism, at most one regular 2-dimensional scheme $\widetilde{X}$, proper and flat over R, with X_η as its generic fibre, without exceptional curves of the first kind in $\widetilde{X}_s$.

Let z denote a generator of the maximal ideal of R and consider the affine plane curve C_n over R given by:

$$xy = z^n.$$

Let $\widetilde{C}_n$ denote the scheme obtained by: 1) blowing up the maximal ideal at the unique singularity of C_n; 2) blowing up the maximal ideal at the unique singularity of this scheme..., and so on $\left[\frac{n}{2}\right]$ times. It is easy to check that $\widetilde{C}_n$ is a regular scheme whose special fibre is the same as that of C_n except that the singular point is replaced by a sequence of $n-1$ projective lines as follows:



Now suppose x is a singular point of the stable curve X over R. At x, X is formally isomorphic as scheme over R to one of the schemes C_n, so we may blow up X the same way we blew up C_n. If we do this for all singular points of X, we get a regular scheme $\widetilde{X}$ with generic fibre X_η. In addition, any non-singular rational component of $\widetilde{X}_s$ is linked to the other irreducible components by at least two points, hence it is not exceptional of first kind. Therefore $\widetilde{X}$ is the minimal model of X_η. Note finally that C_n is a normal scheme, hence so is X; therefore X is the unique normal scheme obtained from $\widetilde{X}$ by contracting all non-singular rational components of $\widetilde{X}_s$ linked to the other irreducible components by exactly two points. This proves that X is essentially unique. Q.E.D.

Another important fact about the automorphisms of stable curves is:

Theorem (1.13). — Let k be an algebraically closed field and X a stable curve over k. Let $\mathrm{Pic}^0(X)$ *denote the group of invertible sheaves on X of degree 0 on each component. Then the map (of ordinary groups):*

$$\mathrm{Aut}_k(X) \longrightarrow \mathrm{Aut}_k(\mathrm{Pic}^0(X))$$

is injective.

Proof. — Let φ be an automorphism of X inducing the identity on $\mathrm{Pic}^0 X$.

Lemma (1.14). — If X is smooth, then φ is the identity.

Proof. — If not, by the Lefschetz-Weil fixed point formula, the number n of fixed points of φ, counted with their multiplicities, is

$$n = 1 - \mathrm{Tr}(\varphi, T_l(\mathrm{Pic}^0(X))) + 1 = 2 - 2g < 0,$$

which is absurd. Q.E.D.

Lemma (1.15). — If X is irreducible, then φ is the identity.

Proof. — Let φ' be the action of φ on the normalization X' of X. Each singular point of X, together with an ordering of its two inverse images in X', defines a distinct morphism from $\mathbb{G}_m$ to $\mathrm{Pic}^0(X)$, so that the inverse image S of the singular locus of X is pointwise fixed by φ'. One has either

a) $\mathrm{genus}(X') \geq 2$: then conclude by Lemma (1.14);

b) $\mathrm{genus}(X') = 1$, $|S| \geq 2$, then φ' is a translation on X' leaving a point fixed, and so φ' is the identity;

c) X' is the projective line, $|S| \geq 4$ and φ' is a projectivity leaving more than three points fixed, so is the identity. Q.E.D.

Let Γ be the following (unoriented) graph:

(i) The set of vertices of Γ is the set Γ^0 of irreducible components of X,

(ii) the set of edges of Γ is the set Γ^1 of the singular points of X which lie on two distinct irreducible components,

(iii) an edge $x \in \Gamma^1$ has for extremities the irreducible components on which x lies.

Lemma (1.16). — If φ induces the identity on Γ, then φ is the identity.

Proof. — If X_1 is an irreducible component of X, then $\varphi(X_1) = X_1$ and φ leaves fixed the points of intersection of X_1 with the other components. In addition, $\mathrm{Pic}^0(X)$ maps onto $\mathrm{Pic}^0(X_1)$ so that φ acts trivially on $\mathrm{Pic}^0(X_1)$. Either:

a) $\mathrm{genus}(X_1) \geq 2$ and $\varphi|X_1$ is the identity by Lemma (1.15);

b) $\mathrm{genus}(X_1) = 1$, φ acts by a translation and leaves a point fixed, so is the identity on X_1;

c) X_1 is the projective line and φ leaves fixed at least three points, so is the identity on X_1. Q.E.D.

Lemma (1.17). — (i) Any edge in Γ has distinct extremities.

(ii) Any vertex which is the extremity of 0, 1 or 2 edges is fixed by φ.

(iii) φ acts trivially on $H^1(\Gamma, \mathbb{Z})$.

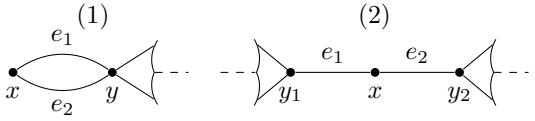
Proof. — It is easy to check that the subgroup of $\mathrm{Pic}^0(X)$ corresponding to invertible sheaves whose restriction to each irreducible component of X is trivial is canonically isomorphic to

$$H^1(\Gamma, \mathbb{Z}) \otimes \mathbb{G}_m.$$

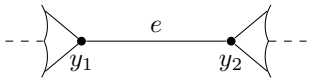
This implies (iii), and (i) is trivial.

The morphism from $\mathrm{Pic}^0(X)$ to the product $\prod_i \mathrm{Pic}^0(X_i)$, extended over the irreducible components of X, is surjective, so that if $\mathrm{Pic}^0(X_i) \neq \{e\}$, then $\varphi(X_i) = X_i$. This is the case, unless X_i is a projective line, linked to the other components in at least three points. Q.E.D.

We prove now that if an automorphism φ of any finite graph Γ has the properties stated in Lemma (1.17), it is the identity. Make induction on the sum of the number of vertices and edges of Γ. If Γ has an isolated point x, then $\varphi(x) = x$ so let $\Gamma^* = \Gamma - \{x\}$. Then $\varphi =$ identity on Γ^* by induction, so $\varphi =$ identity on Γ too. If Γ has an extremity x, then $\varphi(x) = x$, and again let Γ^* be Γ minus x and the edge abutting at x. Then Γ^* has all the properties Γ has, so $\varphi =$ identity on Γ^*, hence $\varphi =$ identity on Γ. If Γ has a vertex x on which only 2 edges abut, we have one of the two cases:



In the first case, $\varphi(y) = y$, and let Γ^* be Γ minus x, e_1 and e_2. Then $\varphi =$ identity on Γ^* and $\varphi(e_i) = e_i$ too, since if φ reverses the e_i's, this contradicts (iii). In the second case, let Γ^* be Γ minus x, and with e_1 and e_2 identified:



Then $\varphi =$ identity on Γ^*, so $\varphi =$ identity on Γ. Next, say Γ has an edge e with extremities x and y such that $\varphi(x) = x$, $\varphi(y) = y$, $\varphi(e) = e$. Let Γ^* be Γ minus e. Then $\varphi =$ identity on Γ^*, so $\varphi =$ identity on Γ. If none of these reductions are possible, we must be in a situation where a) every vertex is the abutment of at least three edges and b) no edge is left fixed. It is easily seen that the first Betti number b_1 of any of the connected components of Γ is at least 2. Let

$$n_0 = \text{number of fixed vertices,}$$
$$n_1 = \text{number of edges reversed by } \varphi.$$

Then, unless $\Gamma = \varnothing$, the Lefschetz fixed point formula reads:

$$n_0 + n_1 = b_0 - b_1 < 0,$$

which is impossible. Q.E.D.

§2. Degenerations of curves and their jacobians.

We consider the situation:

$K =$ discretely-valued field;
$R =$ integers in K, $\quad k = R/\mathfrak{M} =$ residue field (assumed algebraically closed);
$S = \mathrm{Spec}(R)$, $\quad \eta$ and s its generic and closed points respectively;
$C =$ a curve, smooth, geometrically irreducible and proper over K, of genus $g \geq 2$;
$J =$ the jacobian variety of C;
$\mathscr{J} =$ the Néron model of J over R (cf. [N]);
$\mathscr{J}^0 \subset \mathscr{J}$ the open subgroup scheme with $\mathscr{J}^0_s =$ identity component of $\mathscr{J}_s$;
$\mathscr{C} =$ the minimal model of C over R.

A word about the existence and uniqueness of $\mathscr{C}$ is needed. We recall that $\mathscr{C}$ is to be a regular scheme, flat and proper over R, with generic fibre $\mathscr{C}_\eta = C$ such that for any other regular scheme $\mathscr{C}'$, flat over R, with generic fibre $\mathscr{C}'_\eta = C$, the birational map $\mathscr{C}' \to \mathscr{C}$ is a morphism. Šafarevič in [Š] and Lichtenbaum [L] have proven that such a $\mathscr{C}$ (which is obviously unique) exists, provided that there is some regular $\mathscr{C}'$, proper and flat over R, with

generic fibre C. And, in fact, that $\mathscr{C}$ is projective over R. To construct such a $\mathscr{C}$, proceed as follows: first let $\mathscr{C}''$ be any scheme, projective and flat over R with generic fibre C. Let $\widehat{R}$ be the completion of R, and let

$$\widehat{\mathscr{C}''} = \mathscr{C}'' \times_{\operatorname{Spec} R} \operatorname{Spec} \widehat{R}.$$

Then $\widehat{\mathscr{C}''}$ is an excellent surface, so by [Ab] and by unpublished results of Hironaka, there is a sheaf of ideals $\widehat{\mathcal{J}}$ with support in the singular locus of $\widehat{\mathscr{C}''}$ such that blowing up $\widehat{\mathcal{J}}$ leads to a regular surface $\widehat{\mathscr{C}'}$. But then

$$\widehat{\mathcal{J}} \supset \mathfrak{M}^n \cdot \mathcal{O}_{\widehat{\mathscr{C}''}}$$

for some n, so $\widehat{\mathcal{J}}$ is induced by a unique sheaf of ideals $\mathcal{J} \subset \mathcal{O}_{\mathscr{C}''}$. Let $\mathscr{C}'$ be obtained by blowing up $\mathcal{J}$. Then

$$\widehat{\mathscr{C}'} \simeq \mathscr{C}' \times_{\operatorname{Spec} R} \operatorname{Spec} \widehat{R}$$

so $\mathscr{C}'$ is regular. $\mathscr{C}'$ is also projective over R, hence a projective $\mathscr{C}$ exists.

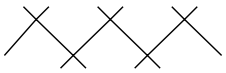
Definition (2.1). — J has stable reduction if $\mathscr{J}_s$ has no unipotent radical.

Definition (2.2). — C has stable reduction in sense 1 if $\mathscr{C}_s$ is reduced and has only ordinary double points. C has stable reduction in sense 2 if there is a stable curve $\mathscr{C}'$ over R with generic fibre $\mathscr{C}'_\eta = C$.

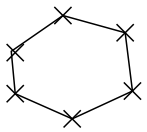
Note that if a stable $\mathscr{C}'$ exists, then by Theorem (1.11) it is unique.

Proposition (2.3). — *The two senses of stable reduction for C are equivalent.*

Proof. — Say a stable $\mathscr{C}'$ exists. Blowing up the singularities of $\mathscr{C}'$ as in Lemma (1.12), we obtain the minimal model $\mathscr{C}$ of C and it is seen that $\mathscr{C}_s$ is reduced with only ordinary double points. Conversely, suppose the minimal model $\mathscr{C}$ has this property. Let $E_1, \ldots, E_n$ be the non-singular rational components of $\mathscr{C}_s$ which meet the other components in only two points. Then the E_i's divide into several chains of the type:



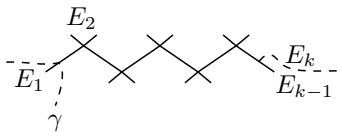
unless the entire fibre consisted of E_i's and has the type:



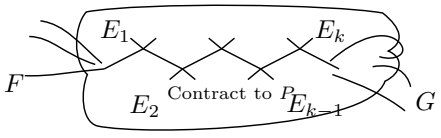
$n \geq 2$
$\mathscr{C}_s$ = loop of E_i's
$(E_i^2)_{\text{on } \mathscr{C}} = -2$.

But in this case $\operatorname{genus}(C) = \operatorname{genus}(\mathscr{C}_s) = 1$, which contradicts our assumption. Now, according to Theorem (27.1) of Lipman [Li] (generalizing a result of Artin [A$_2$], which works for surfaces of finite type over a field) any set of k non-singular rational curves connected in a chain as above on a regular surface X, with self-intersection -2 on X, can be blown down to a rational double point P of type A_k on a normal surface X_0. Reversing the process and blowing up a rational double point of type A_k, it is easy to see that non-singular branches γ on X, crossing transversally only the first or the last rational curve in the chain:

Source note. The printed prose says "with self-intersection 2 on X"; the displayed loop immediately above and the contraction argument require self-intersection -2, which is retained here.



are still non-singular branches when mapped to X_0; and if γ_1, γ_2 intersect E_1 or E_k in distinct points, then they cross transversally on X_0. Therefore, suppose we blow down all the chains of E_i's on $\mathscr{C}$. Let $\mathscr{C}'$ be the normal surface so obtained. Then if one of these chains fits into $\mathscr{C}_s$ like this:



then on $\mathscr{C}'_s$, the images of F and G still have only ordinary double points, each has one non-singular branch through the singular point P, and these branches cross transversally. Therefore $\mathscr{C}'$ is a stable curve over R. Q.E.D.

We are now ready to prove the key result on which our proof of irreducibility depends:

Theorem (2.4). — J has stable reduction if and only if C has stable reduction.

Proof. — The connection between $\mathscr{C}$ and $\mathscr{J}$ is based on the following result of Raynaud [R]:

Theorem (2.5). — If $\mathscr{C}$ and $\mathscr{J}$ are as above, and the greatest common denominator d of the multiplicities of the components of $\mathscr{C}_s$ is 1, then $\mathscr{J}^0$ represents the functor $\underline{\mathrm{Pic}}^0(\mathscr{C}/S)$.

(This result is not stated as such in [R]. It comes out like this, in the terminology of his paper:

a) Condition (N) is verified and $p_*(\mathcal{O}_{\mathscr{C}}) = \mathcal{O}_S$, so

b) $\mathscr{C}$ is cohomologically flat over S in dim. 0 by Theorem 4;

c) therefore $\underline{\mathrm{Pic}}^0$ is representable and separated over S by Theorem 3;

d) since $E = \overline{\{0\}}$ and $Q = P/E$, we find $P^0 = Q^0$, and since $\mathscr{C}$ is 1-dimensional over S, P^0 and Q^0 are smooth over S; therefore $R^0 = Q^0$ and $\widetilde{R} = R$.

e) Then by Theorem 5, Pic^0 is the identity component of the Néron model of $\mathrm{Pic}^0(\mathscr{C}_\eta) = J$.)

Source note. The printed text says "in the terminology of is paper"; the evident omitted letter is restored as "his paper" above.

Now assume that C has stable reduction. Then $\mathscr{C}_s$ is reduced so $d = 1$. By Theorem (2.5), $\mathscr{J}^0 = \mathrm{Pic}^0(\mathscr{C}/S)$. Therefore $\mathscr{J}^0_s = \mathrm{Pic}^0(\mathscr{C}_s/k)$. Since $\mathscr{C}_s$ is reduced with only ordinary double points, its generalized jacobian $\mathrm{Pic}^0(\mathscr{C}_s/k)$ is an extension of an abelian variety by a torus, i.e., has no unipotent radical. Therefore J has stable reduction.

The converse is more difficult. Assume J has stable reduction. We first prove that C has stable reduction under the additional hypothesis that C has a K-rational point.[4] In this case, $\mathscr{C}$ has an R-rational point, and since $\mathscr{C}$ is regular, sections of $\mathscr{C}$ over R pass through components of $\mathscr{C}_s$ of multiplicity one. Therefore $d = 1$, and

Theorem (2.5) applies. In particular, $\mathrm{Pic}^0(\mathscr{C}_s/k) = \mathscr{J}^0_s$, so $\mathrm{Pic}^0(\mathscr{C}_s/k)$ has no unipotent radical. We apply:

Lemma (2.6). — Let D be a complete 1-dimensional scheme over k such that $H^0(\mathcal{O}_D) \simeq k$, and such that the generalized jacobian of D has no unipotent radical. Then

(i) $\chi(\mathcal{O}_D) = \chi(\mathcal{O}_{D_{\mathrm{red}}})$;

(ii) *the singularities of D_{red} are all transversal crossings of a set of non-singular branches (i.e., analytically isomorphic to the union of the coordinate axes in $\mathbf{A}^n$).*

Proof. — Let $\mathcal{J} \subset \mathcal{O}_D$ be the ideal of nilpotent elements. Filtering $\mathcal{J}$ by a chain of ideals $\mathcal{J}_k$ such that $\mathcal{J}\mathcal{J}_k \subset \mathcal{J}_{k+1}$, and using the exact sequence

$$0 \longrightarrow \mathcal{J}_k/\mathcal{J}_{k+1} \xrightarrow{a\mapsto 1+a} (\mathcal{O}_D/\mathcal{J}_{k+1})^* \longrightarrow (\mathcal{O}_D/\mathcal{J}_k)^* \longrightarrow 0,$$

it is easy to deduce that $\mathrm{Pic}^0(D/k)$ is an extension of $\mathrm{Pic}^0(D_{\mathrm{red}}/k)$ by a unipotent group. Therefore, since by assumption $\mathrm{Pic}^0(D/k)$ has no unipotent subgroups,

$$\mathrm{Pic}^0(D/k) \simeq \mathrm{Pic}^0(D_{\mathrm{red}}/k).$$

Since $H^1(\mathcal{O}_D)$, resp. $H^1(\mathcal{O}_{D_{\mathrm{red}}})$, is naturally isomorphic to the Zariski tangent space to $\mathrm{Pic}^0(D/k)$, resp. $\mathrm{Pic}^0(D_{\mathrm{red}}/k)$, it follows that

$$H^1(\mathcal{O}_D) \simeq H^1(\mathcal{O}_{D_{\mathrm{red}}}),$$

hence (i) is proven. Let $\pi : C \to D_{\mathrm{red}}$ be the normalization of D_{red} and let D^* be the local ringed space which, as topological space is D, and whose structure sheaf is given by

$$\Gamma(U, \mathcal{O}_{D^*}) = \{f \in \Gamma(U, \pi_*(\mathcal{O}_C)) \mid x_1, x_2 \in \pi^{-1}(U),\ f(x_1) = f(x_2) \text{ if } \pi(x_1) = \pi(x_2)\}.$$

It is easy to check that D^* is a 1-dimensional scheme whose singularities are all transversal crossings of a set of non-singular branches and that π factors:

$$C \xrightarrow{\pi'} D^* \xrightarrow{\pi''} D_{\mathrm{red}}.$$

[4] This is in fact the only case which will be needed in our application to questions of irreducibility.

I claim that $\pi'' : D^* \to D_{\mathrm{red}}$ is an isomorphism. Filter $\mathcal{O}_{D^*}/\mathcal{O}_{D_{\mathrm{red}}}$ so as to obtain a chain of coherent $\mathcal{O}_{D_{\mathrm{red}}}$-algebras

$$\mathcal{O}_{D^*} = \mathcal{O}^{(0)} \supset \mathcal{O}^{(1)} \supset \cdots \supset \mathcal{O}^{(k)} = \mathcal{O}_{D_{\mathrm{red}}}$$

such that $\ell(\mathcal{O}^{(n)}/\mathcal{O}^{(n+1)}) = 1$. Equivalently, this factors π'':

$$D^* = D_0 \longrightarrow D_1 \longrightarrow \cdots \longrightarrow D_k = D_{\mathrm{red}},$$

where $\mathcal{O}^{(n)} \simeq \mathcal{O}_{D_n}$, and all arrows are homeomorphisms. If $\{x_n\} = \mathrm{Supp}(\mathcal{O}^{(n)}/\mathcal{O}^{(n+1)})$, then we get an exact sequence:

$$0 \longrightarrow \mathcal{O}^*_{D_{n+1}} \longrightarrow \mathcal{O}^*_{D_n} \longrightarrow k_{x_n} \longrightarrow 0$$

(k_y denotes the residue field at y, as sheaf on D), and hence:

$$0 \longrightarrow k \longrightarrow H^1(\mathcal{O}^*_{D_{n+1}}) \longrightarrow H^1(\mathcal{O}^*_{D_n}) \longrightarrow 0.$$

It follows easily that $\mathrm{Pic}^0(D_{n+1}/k)$ is an extension of $\mathrm{Pic}^0(D_n/k)$ by G_a. But $\mathrm{Pic}^0(D_{\mathrm{red}}/k)$ has no unipotent subgroups, and this can only happen if $D_{\mathrm{red}} = D^*$. Q.E.D. for lemma.

We apply the lemma to $\mathscr{C}_s$. According to Lichtenbaum [L] and Šafarevič [Š], there is a divisor K on $\mathscr{C}$ such that for all positive divisors D lying over the closed point of $\mathrm{Spec}(R)$, we have:

$$\chi(\mathcal{O}_D) = -\frac{(D \cdot (D + K))}{2}.$$

Source note. The printed display has the extraneous lower-case factor kK in place of K. The standard adjunction formula, and every immediately following use of it, require $D + K$ as delivered above.

Let $E_1, \ldots, E_n$ be the components of $\mathscr{C}_s$, $d_1, \ldots, d_n$ their multiplicities. Then conclusion (i) of the lemma implies that:

$$\Big(\sum_i d_i E_i \cdot \big(\sum_i d_i E_i + K\big)\Big) = \Big(\sum_i E_i \cdot \big(\sum_i E_i + K\big)\Big).$$

But $((\sum_i d_i E_i) \cdot E_k) = 0$, all k, since $\sum_i d_i E_i$ is the divisor of a function $\pi \in R$ if (π) = maximal ideal of R. Therefore:

$$(*) \qquad \Big(\big(\sum_i (d_i - 1)E_i\big) \cdot K\Big) = \Big(\sum_i E_i \cdot \sum_i E_i\Big).$$

Note that at least one d_i equals 1 since $\mathscr{C}$ has a section over $\mathrm{Spec}(R)$ and every section must pass through a component of $\mathscr{C}_s$ of multiplicity 1. Moreover the intersection matrix $(E_i \cdot E_j)$ is negative semidefinite, with one-dimensional degenerate subspace generated by $\sum_i d_i E_i$, hence if some $d_i > 1$, it follows that $(\sum_i E_i \cdot \sum_i E_i) < 0$. Therefore, by $(*)$, $(E_{i_0} \cdot K) < 0$ for some i_0. Then we have:

Source note. The printed text says "negative indefinite" while simultaneously specifying a one-dimensional degenerate subspace. The mathematically consistent reading "negative semidefinite" is delivered above.

a) $(E_{i_0} \cdot K) < 0$;

b) $(E_{i_0} \cdot E_{i_0}) < 0$;

c) $(E_{i_0} \cdot (E_{i_0} + K)) = -2\chi(\mathcal{O}_{E_{i_0}}) \geq -2$;

hence in fact $(E_{i_0} \cdot E_{i_0}) = (E_{i_0} \cdot K) = -1$, so E_{i_0} is an exceptional curve of the first kind. This contradicts our assumption that on $\mathscr{C}$ all possible curves have been blown down. Therefore $d_i = 1$, all i.

This proves that $\mathscr{C}_s$ is reduced. By conclusion (ii) of the lemma, plus the fact that the dimension of its Zariski tangent-space is everywhere one or two (since $\mathscr{C}_s$ lies on a regular surface $\mathscr{C}$), we deduce that $\mathscr{C}_s$ has only double points. This proves that C has stable reduction in the case that $C(K) \neq \varnothing$.

In the general case, C will acquire a rational point in a finite extension K' of K. Let S' be the spectrum of the localisation at some maximal ideal of the integral closure of R in K'; S' is the spectrum of a valuation ring and is faithfully flat over S. We put $S'' = S' \times_S S'$ and denote by C' and C'' the inverse images of C on $S'_\eta = \mathrm{Spec}(K')$ and S''_η respectively.

$$\begin{array}{ccccc} C'' & \rightrightarrows & C' & \longrightarrow & C \\ \downarrow & & \downarrow & & \downarrow \\ S'' & \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} & S' & \xrightarrow{\;p\;} & S \end{array}$$

Let $\overline{C}'$ be the stable curve on S' having C' as generic fibre. The restriction, C', of $\overline{C}'$ to S'_η carries a descent datum with respect to p, i.e. an isomorphism, φ_η, between the restrictions of $\mathrm{pr}_1^*(\overline{C}')$ and $\mathrm{pr}_2^*(\overline{C}')$ to S''_η. It remains only to extend the isomorphism φ_η to an isomorphism, φ, between the $\mathrm{pr}_i^*(\overline{C}')$. Then φ will be a descent datum for $\overline{C}'$ with respect to p. As $\overline{C}'$ is canonically polarized (1.2), this descent datum will be effective, and so define a stable curve, $\overline{C}$, over S with generic fibre C.

Because $\mathscr{J}_s^0$ has no unipotent radical, the inverse image, $p^*\mathscr{J}^0$, of $\mathscr{J}^0$ on S' is the identity component of the Néron Model of the jacobian of C', so that, defining $q = p \circ \mathrm{pr}_1 = p \circ \mathrm{pr}_2$, one has

$$\mathrm{Pic}^0(\overline{C}'/S') = p^*\mathscr{J}^0,$$

$$\mathrm{Pic}^0(\mathrm{pr}_1^*\overline{C}') = \mathrm{Pic}^0(\mathrm{pr}_2^*\overline{C}') = q^*\mathscr{J}^0.$$

We denote by T the closed subscheme of $\mathrm{Isom}(S'', \mathrm{pr}_1^*(\overline{C}'), \mathrm{pr}_2^*(\overline{C}'))$ corresponding to those isomorphisms which induce (via the preceding identifications) the identity on the inverse image of $\mathscr{J}^0$.

By (1.11), T is finite and unramified over S'', and by (1.13) T is radicial over S''. We conclude that the morphism from T to S'' identifies T with a closed subscheme X of S''. As X contains S''_η, which is schematically dense in S'', we have that X is S'' and T "is" the desired section, φ, of $\mathrm{Isom}(S'', \mathrm{pr}_1^*(\overline{C}'), \mathrm{pr}_2^*(\overline{C}'))$ over S'' which extends φ_η. Q.E.D.

Combining Proposition (2.3), Theorem (2.4), and the stable reduction theorem for abelian varieties quoted in the introduction, we obtain the most important consequence:

Corollary (2.7). — Let R be a discrete valuation ring with quotient field K. Let C be a smooth geometrically irreducible curve over K of genus $g \geq 2$. Then there exists a finite algebraic extension L of K and a stable curve $\mathscr{C}_L$ over R_L, the integral closure of R in L, with generic fibre $\mathscr{C}_{L,\eta} = C \times_K L$.

§3. Elementary derivation of the theorem.

Let k be an algebraically closed field of char. $p \neq 0$. We use the notation of § 1, except that we will now denote by H_g the product $H_g \times \mathrm{Spec}(k)$ of the previous H_g with $\mathrm{Spec}(k)$: it is a disjoint union of non-singular varieties $H_{g,1}, \ldots, H_{g,n}$ over k and is the subscheme of $\mathrm{Hilb}^{P_g}_{\mathbf{P}^{5g-6}/k}$ of tri-canonical stable curves. Similarly, H_g^0 is the open dense subset of H_g of tri-canonical non-singular curves. By the results of [M$_1$], we know that a coarse geometric quotient

$$M_g^0 = H_g^0/\mathrm{PGL}(5g-6)$$

exists, that it is a disjoint union of normal varieties over k and is the coarse moduli space for non-singular curves of genus g. Let $S^* = H_g - H_g^0$. Then everything decomposes into the same set of components.

Let

$$\begin{array}{rcl} H_{g,i} & = & \text{components of } H_g, \\ \text{then } H_{g,i}^0 & = & H_{g,i} \cap H_g^0 = \text{components of } H_g^0, \\ \text{and } M_{g,i}^0 & = & H_{g,i}^0/\mathrm{PGL}(5g-6) = \text{components of } M_g^0, \\ \text{and } S^* & = & \text{disjoint union of } S_1^*, \ldots, S_n^*, \quad S_i^* = H_{g,i} \cap S^*. \end{array}$$

We want to prove that M_g^0, or equivalently H_g^0, or equivalently H_g is irreducible. We shall use: (i) the fact that these statements are true in char. 0; (ii) the inductive assumption that these statements are true for smaller genus.

Step I. — No component of M_g^0 is complete (i.e., proper over k).

Proof. — Here we use the char. 0 result. By [M$_1$], there is a scheme X, quasi-projective over $\mathrm{Spec}(W(k))$, $W(k)$ the Witt vectors, whose closed fibre is M_g^0, and whose generic fibre X_η is the char. 0 coarse moduli space over the quotient field of $W(k)$. In particular, X_η is known to be connected. Since X is quasi-projective over $W(k)$, we can embed X as an open dense subset of a scheme $\overline{X}$ projective over $W(k)$. $\overline{X}_\eta$ is still connected, hence by the connectedness theorem of Enriques-Zariski [EGA 3], the closed fibre $\overline{X}_0$ of $\overline{X}$ is connected. But if Y were a complete variety which is a component of M_g^0, then: a) Y is an open subset of M_g^0, which is an open subset of $\overline{X}_0$, and: b) since Y is proper/k, Y would be a closed subset of $\overline{X}_0$ too. Therefore, $Y = \overline{X}_0$, hence M_g^0 is itself irreducible and complete. On the other hand, if A_g is the coarse moduli space of principally polarized g-dimensional abelian varieties, then the map associating to each curve its jacobian defines a morphism:

Source note. The printed text has the typographical form "principally pclarized"; "principally polarized" is restored above.

$$\theta : M_g^0 \to A_g.$$

If M_g^0 were complete, the image of θ would be closed. But it is well known that the closure of the image of θ contains all products of lower dimensional jacobians too, so it is not closed. Q.E.D.

Step II. — No component of H_g consists entirely of non-singular curves, i.e., $S_i^* \neq \varnothing$ for all i.

Proof. — Here we combine Step I with the result of § 2. Take any i. Let $T = \operatorname{Spec} k[[t]]$. Since $M_{g,i}^0$ is not complete, there is a morphism φ of the generic point T_η of T into $M_{g,i}^0$ which does not extend to a morphism of T into $M_{g,i}^0$. Now we replace T by its normalization T' in a finite algebraic extension and let $\varphi' : T'_\eta \to M_{g,i}^0$ be the induced morphism, which still does not extend to a morphism from T' to $M_{g,i}^0$. By the results of § 2, if T' is chosen suitably there exists a stable curve $\pi : C' \to T'$ over T' whose generic fibre C'_η is a non-singular curve corresponding to the morphism $\varphi' : T'_\eta \to M_g^0$ via the functorial properties of the moduli space. Since T' is the spectrum of a local ring, we can choose an isomorphism

$$\mathbf{P}\big(\pi_*(\omega_{C'/T'}^{\otimes 3})\big) \simeq \mathbf{P}^{5g-6} \times T'$$

and get a tri-canonical embedding $C' \subset \mathbf{P}^{5g-6} \times T'$. C', with this embedding, is then induced from the universal tri-canonically embedded stable curve by a morphism $\psi : T' \to H_g$. Since the generic fibre of C' is C'_η, we get a commutative diagram:

$$\begin{array}{ccc} T' & \xrightarrow{\psi} & H_g \\ \cup & & \cup \\ T'_\eta & \xrightarrow{\operatorname{res}(\psi)} & H_g^0 \\ {\scriptstyle\varphi'}\searrow & & \downarrow \\ & M_{g,i}^0 & \subset M_g^0. \end{array}$$

If $\operatorname{Im}(\psi) \subset H_g^0$, then φ' would extend to a morphism from T' to M_g^0, hence from T' to $M_{g,i}^0$, and this is a contradiction. Moreover, $\psi(T_\eta)$ must be a point of $H_{g,i}^0$, since its image in M_g^0 is in $M_{g,i}^0$. Therefore the image x of the closed point of T' by ψ is in the closure of $H_{g,i}^0$, i.e., in $H_{g,i}$, but not in $H_{g,i}^0$ itself. Q.E.D.

Step III. — S^* is connected.

Proof. — This will follow using only the induction assumption of irreducibility for lower genera. Let $Z \subset H_g \times \mathbf{P}^{5g-6}$ be the universal tri-canonically embedded stable curve. Let $S \subset Z$ be the set of points where Z is not smooth over H_g. As we proved in § 1, S is non-singular and is the normalization of S^*. In particular, this shows that if $x \in S^*$, then the corresponding curve Z_x has exactly one double point if and only if x is a non-singular point of S^*. Stable curves C of genus g with exactly one double point belong to one of the following types:

$$\text{type } 0: \quad \text{} \quad \begin{array}{l} C \text{ irreducible} \\ \text{normalization } C' \text{ of } C \text{ has genus } g-1; \end{array}$$

$$\text{type } k: \quad \text{} \quad \begin{array}{l} C \text{ has two non-singular components } C_1, C_2 \\ \text{genus}(C_1) = k \\ \text{genus}(C_2) = g-k \end{array}$$

where $1 \leq k \leq \left[\frac{g}{2}\right]$. If $0 \leq k \leq \left[\frac{g}{2}\right]$, let

$$S^*(k) = \{x \in S^* \mid Z_x \text{ has one double point and is of type } k\}.$$

Then the open dense subset of S^* of non-singular points is the disjoint union of open subsets $S^*(0), \ldots, S^*(\left[\frac{g}{2}\right])$. We first check:

$(*)$ Each set $S^*(k)$ is irreducible.

Proof of $(*)$. — Take the case $k = 0$: the cases $1 \leq k \leq \left[\frac{g}{2}\right]$ are similar. Let

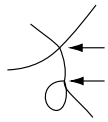
$$T = \{\{x_1, x_2\} \mid x_1 \neq x_2 \text{ and } \pi(x_1) = \pi(x_2) \in H_{g-1}^0\} \subset Z_{g-1} \times_{H_{g-1}} Z_{g-1}.$$

T is smooth with irreducible fibres over H_{g-1}^0, hence T is irreducible. Consider the correspondence relating $S^*(0)$ and T:

$$W = \left\{ \begin{array}{l} \text{set of pairs } x \in S^*(0),\ \{x_1, x_2\} \in T \text{ such that if } y = \pi(x_1) = \pi(x_2), \\ \text{then there exists a birational morphism } f : (Z_{g-1})_y \to (Z_g)_x \\ \text{such that } f(x_1) = f(x_2). \end{array} \right.$$

It is easy to check that W is Zariski-closed. Moreover, for any $\{x_1, x_2\} \in T$, $W \cap (S^*(0) \times \{x_1, x_2\})$ is an orbit in $S^*(0)$ under $\mathrm{PGL}(5g-6)$: so these are non-empty irreducible subsets all of the same dimension. It follows that W itself is irreducible. But the projection from W to $S^*(0)$ is surjective, so $S^*(0)$ is irreducible. Q.E.D. for $(*)$.

Now for any k, $1 \leq k \leq \left[\frac{g}{2}\right]$, choose any stable curve $C(k)$ of the type:



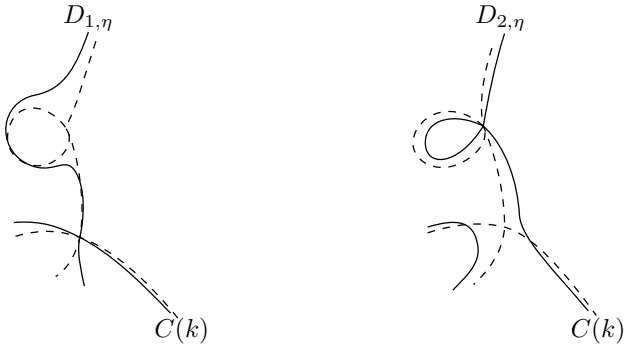
one non-singular component $C(k)'$ of genus k,
one component $C(k)''$ with one double point, normalization of genus $g-k-1$.

Let $P(k)$ be a point of H_g such that $Z_{P(k)} \simeq C(k)$. Step III will be completed if we prove:

$(**)$ $P(k)$ is in the closure of $S^*(0)$ and of $S^*(k)$, hence $S^*(0)$, $S^*(k)$ both lie in the same topological component of S^*.

Proof of $(**)$. — Let $T = \mathrm{Spec}\, k[[t]]$. Using the fact that S^* has two branches through $P(k)$, one for each of the double points of $C(k)$, we see that there exist two morphisms

$$f_1, f_2 : T \to S^*,$$
$$f_1(T_s) = f_2(T_s) = P(k), \qquad T_s = \text{closed point of } T,$$



such that if $\pi_1 : D_1 \to T$, $\pi_2 : D_2 \to T$ are the two stable curves over T induced by f_1 and f_2, then: a) the closed fibres $D_{1,s}$, $D_{2,s}$ are $C(k)$; b) there are sections $s_1 : T \to D_1$, $s_2 : T \to D_2$ whose images are non-smooth points of π_1, π_2 and such that $s_1(T_s)$ and $s_2(T_s)$ are the two double points Q_1 and Q_2 of $C(k)$ respectively; and c) the generic fibres $D_{1,\eta}$, $D_{2,\eta}$ have only one double point (cf. figure). I claim:

A) $D_{1,\eta}$ is of type (k);

B) $D_{2,\eta}$ is of type (0).

To prove A), let D_1' be the result of blowing up the subscheme $s_1(T)$ of D_1. Then D_1' is still flat and proper over T and its special fibre is the special fibre of D_1 with Q_1 blown up (use the fact that formally at Q_1, D_1 is isomorphic to $k[[t,x,y]]/(xy)$ with the section given by $x = y = 0$). Therefore the special fibre of D_1' is the disjoint union of $C(k)'$, $C(k)''$, so the general fibre of D_1' is the disjoint union of two irreducible curves which specialize to $C(k)'$, $C(k)''$ respectively. Since $D_{1,\eta}$ has only one double point, $D'_{1,\eta}$ is non-singular, so $D'_{1,\eta}$ is the disjoint union of two non-singular irreducible curves which must then have the same genera as $C(k)'$ and $C(k)''$, i.e., k, $g-k$. Thus $D_{1,\eta}$ has type (k).

To prove B), it suffices to check that $D_{2,\eta}$ is geometrically irreducible. If not, $D_{2,\eta}$ would have two components meeting at the single point $s_2(T_\eta)$. Since D_2 is smooth over T at each generic point of its special fibre, distinct geometric components of $D_{2,\eta}$ have to have specializations which are distinct components of $(D_2)_s$. Then $(D_2)_s$ would have two components meeting at the point $Q_2 = s_2(T_s)$. This is false, so B) is proven.

Now because of A), $f_1(T_\eta) \in S^*(k)$, hence $P(k) = f_1(T_s)$ is in the closure of $S^*(k)$. And because of B), $f_2(T_\eta) \in S^*(0)$, hence $P(k) = f_2(T_s)$ is in the closure of $S^*(0)$ too. Q.E.D. for $(**)$.

This completes the proof of Step III since we now see that all irreducible components of S^* are part of the same topological component. Finally, from Steps II and III, we see that

a) S^*, being connected, is part of a single component of H_g, while

b) each component of H_g contains part of S^*.

Thus H_g is irreducible, as was to be proven.

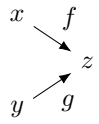
§4. Some results on algebraic stacks.

The proofs of the results stated in this section will be given elsewhere.

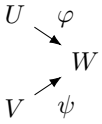
Let C be a category and let $p : \mathscr{S} \to C$ be a category over C. For each $U \in \mathrm{Ob}\, C$, we denote by $\mathscr{S}_U$ the fibre $p^{-1}(U)$. The category $\mathscr{S}$ is *fibered in groupoids* over C if the following two conditions are verified:

a) For all $\varphi : U \to V$ in C and $y \in \mathrm{Ob}\, \mathscr{S}_V$ there is a map $f : x \to y$ in $\mathscr{S}$ with $p(f) = \varphi$.

b) Given a diagram



in $\mathscr{S}$, let



be its image in C. Then for all $\chi : U \to V$ such that $\varphi = \psi\chi$, there is a unique $h : x \to y$ such that $f = g \cdot h$ and $p(h) = \chi$.

Source note. The printed first diagram labels its lower-left object Y; the surrounding notation requires the y retained here.

Condition b) implies that the $f : x \to y$ whose existence is asserted in a) is unique up to canonical isomorphism.

Assume that for each $\varphi : U \to V$ in C and each $Y \in \mathrm{Ob}\, \mathscr{S}_V$, such an $f : x \to y$ has been chosen. This x will be written as $\varphi^* y$. Then, φ^* "is" a functor from $\mathscr{S}_V$ to $\mathscr{S}_U$ and if $\varphi\psi$ is a composite morphism in C, the functors $(\varphi\psi)^*$ and $\psi^*\varphi^*$ are canonically isomorphic.

We propose the terminology "stack" for the French word "champ" of non-abelian cohomology (Giraud [G]).

Definition (4.1). — Let C be a category with a Grothendieck topology. We assume products and fibre products exist in C. A stack in groupoids over C is a category over C, $p : \mathscr{S} \to C$ such that:

(i) *$\mathscr{S}$ is fibered in groupoids over C.*

(ii) *For any $U \in \mathrm{Ob}\, C$ and any objects x, y in $\mathscr{S}_U$ the functor from C/U to (sets) which to any $\varphi : V \to U$ associates $\mathrm{Hom}_{\mathscr{S}_V}(\varphi^* x, \varphi^* y)$ is a sheaf.*

(iii) *If $\varphi_i : V_i \to U$ is a covering family in C, any descent datum relative to the φ_i, for objects in $\mathscr{S}$, is effective.*

For each $x \in \mathrm{Ob}\, \mathscr{S}_U$, there are given isomorphisms between the inverse images of $x_i = \varphi_i^* x$ and $x_j = \varphi_j^* x$ over $V_{ij} = V_i \times_U V_j$, and the pull-backs of these isomorphisms on $V_{ijk} = V_i \times_U V_j \times_U V_k$ satisfy a "cocycle" condition. In (iii) it is required that reciprocally, any such "descent datum" be defined by some $x \in \mathscr{S}_U$.

In what follows, for the sake of brevity, we will use "stack" to mean "stack in groupoids".

If $U \in \mathrm{Ob}\, C$ and if $\mathscr{S}$ is a stack over C, the fibre $\mathscr{S}_U$ will be called the category of sections of $\mathscr{S}$ over U.

Let C be as in (4.1). The stacks over C are the objects of a 2-category [B] (stacks/C): 1-morphisms are functors from one stack to another, compatible with the projection into C; and 2-morphisms are morphisms of functors. In this 2-category every 2-morphism is an isomorphism. Products and 2-fibre products exist in this 2-category.

To each $X \in \mathrm{Ob}\, C$ is associated the "representable" stack over C whose category of sections over U is the discrete category whose objects are the morphisms from U to X. This stack will be denoted simply X. For any stack $\mathscr{S}$, the category

$$\mathrm{Hom}(X, \mathscr{S})$$

is canonically equivalent to the category of sections of $\mathscr{S}$ over X. Because of this, $\mathscr{S}$ is sometimes said to "classify" its sections over variable $X \in \mathrm{Ob}\, C$. In the category $\mathrm{Hom}(\mathscr{S}, X)$, all morphisms are identities, i.e., $\mathrm{Hom}(\mathscr{S}, X)$ is just a set.

Let us denote also by C the 2-category having the same objects and morphisms as C, and in which the identities are the only 2-morphisms. The above construction then identifies C with a full sub-2-category of (stacks/C).

For each $S \in \mathrm{Ob}\, C$, the category C/S satisfies the assumptions of (4.1). Any stack $\mathscr{S}_0$ over C/S (an "S-stack") gives rise to a stack $\mathscr{S}$ over C; a section (φ, η) of $\mathscr{S}$ over $U \in \mathrm{Ob}\, C$ consists of

(i) a morphism $\varphi : U \to S$;

(ii) a section η of $\mathscr{S}_0$ over (U, φ).

Definition (4.2). — A 1-morphism of stacks over C, $F : \mathscr{S}_1 \to \mathscr{S}_2$, will be called representable if for any X in C and any 1-morphism $x : X \to \mathscr{S}_2$, the fibre product $X \times_{\mathscr{S}_2} \mathscr{S}_1$ is a representable stack.

In down to earth terms, this means the following:

(i) for any $f : Y \to X$ in C, the category whose objects are pairs

$$\{\text{a section, } y, \text{ of } \mathscr{S}_1 \text{ over } Y;\ \text{an isomorphism } F(y) \xrightarrow{\sim} f^*(x)\}$$

is equivalent to a category $S(f)$ in which all morphisms are identities;

(ii) the functor $f \mapsto \operatorname{Ob} S(f)$ is representable by some $g : Z \to X$. Such a Z represents the fibre product $X \times_{\mathscr{S}_2} \mathscr{S}_1$.

Let P be a property of morphisms in C, stable by change of base and of a local nature on the target.

Definition (4.3). — A representable morphism $F : \mathscr{S}_1 \to \mathscr{S}_2$ of stacks over C has property P if for any 1-morphism $x : X \to \mathscr{S}_2$ the morphism in C deduced by base change: $F' : X \times_{\mathscr{S}_2} \mathscr{S}_1 \to X$ has that property.

Proposition (4.4). — Let $\mathscr{S}$ be a stack. The diagonal map

$$\mathscr{S} \to \mathscr{S} \times \mathscr{S}$$

is representable if and only if for all $X, Y \in \operatorname{Ob} C$ and 1-morphisms $x : X \to \mathscr{S}$, $y : Y \to \mathscr{S}$, the fibre product $X \times_{\mathscr{S}} Y$ is representable.

If $X \in \operatorname{Ob} C$ and x, y are sections of $\mathscr{S}$ over X, we denote by $\underline{\mathrm{Isom}}(X, x, y)$ the sheaf on C/X which to every Z over X associates the set of isomorphisms between the inverse images of x and y over Z. Then the object representing $\mathscr{S} \times_{(\mathscr{S} \times \mathscr{S})} X$ (the product taken with the map $(x, y) : X \to \mathscr{S} \times \mathscr{S}$) is just $\mathrm{Isom}(X, x, y)$. If $X \to \mathscr{S}$, $Y \to \mathscr{S}$ are 1-morphisms, then the object representing $X \times_{\mathscr{S}} Y$ is just $\mathrm{Isom}(X \times Y, p_1^* x, p_2^* y)$.

Henceforth, C will be the category of schemes with the etale topology (SGAD, IV, (6.3)).

Definition (4.5). — A stack $\mathscr{S}$ is quasi-separated if the diagonal morphism from $\mathscr{S}$ to $\mathscr{S} \times \mathscr{S}$ is representable, quasi-compact and separated.

Source note. The printed definition reads "the diagonal morphism from to $\mathscr{S}$ to $\mathscr{S} \times \mathscr{S}$"; the duplicated "to" is omitted here.

Definition (4.6). — A stack $\mathscr{S}$ is an algebraic stack[5] *if*

(i) *$\mathscr{S} \to \mathscr{S} \times \mathscr{S}$ is representable;*

(ii) *there exists a 1-morphism $x : X \to \mathscr{S}$ such that for all $y : Y \to \mathscr{S}$, the projection morphism $X \times_{\mathscr{S}} Y \to Y$ is surjective and etale (i.e., x is etale and surjective).*

The 2-category of algebraic stacks contains the representable stacks and is stable under products and fibre products. If $\mathscr{S}$ is a quasi-separated algebraic stack, the diagonal map is unramified and quasi-affine.

Definition (4.7). — An algebraic stack $\mathscr{S}$ is separated if $\mathscr{S} \to \mathscr{S} \times \mathscr{S}$ is proper (or, equivalently, finite). A 1-morphism $f : \mathscr{S}_1 \to \mathscr{S}_2$ is separated (resp. quasi-separated) if for any morphism $x : X \to \mathscr{S}_2$ from a separated scheme X to $\mathscr{S}_2$, the fibre product $\mathscr{S}_1 \times_{\mathscr{S}_2} X$ is separated (resp. quasi-separated).

Example (4.8). — Let X be a scheme over S. Let G be a group scheme over S, etale, separated and of finite type over S, which operates on X. We will denote by $[X/G]$ the S-stack whose category of sections over an S-scheme T is the category of principal homogeneous spaces (p.h.s.) E over T, with structural group G (i.e., a p.h.s. under G_T), provided with a G-morphism $\varphi : E \to X$. The principal homogeneous space $G \times X$ over X (G acting only on the first factor) plus the G-morphism $G \times X \to X$ (given by the action of G on X) is a section of $[X/G]$ over X. The corresponding morphism $q : X \to [X/G]$ is etale and surjective, so that $[X/G]$ is an algebraic stack. In addition, X is a principal homogeneous space over $[X/G]$; the stack $[X/G]$ is representable if and only if X is a principal homogeneous space over a scheme Y, in which case

$$[X/G] \sim Y.$$

[5] This definition is the "right" one only for quasi-separated stacks. It will however be sufficient for our purposes.

If $X = S$, then $[X/G] = [S/G]$ might be called the "classifying stack" of G over S.

Example (4.9). — Suppose a stack $\mathscr{S}$ has the property that in each category $\mathscr{S}_X$ the only morphisms are the identity morphisms. Then $\mathscr{S}_X$ is just a set $\mathscr{F}(X) = \mathrm{Ob}\,\mathscr{S}_X$, and this set, under pull-back, is a contravariant functor $\mathscr{F}$ in X. Conditions (ii) and (iii) of (4.1) assert that the functor $\mathscr{F}$ is a sheaf on C. Artin and Knutson [K] have defined an *algebraic space* to be a sheaf $\mathscr{F}$ such that:

(i) for any morphisms $X \to \mathscr{F}$, $Y \to \mathscr{F}$ of representable functors to $\mathscr{F}$, the fibre product $X \times_{\mathscr{F}} Y$ is representable;

(ii) there exists a morphism $X \to \mathscr{F}$, represented by surjective, etale morphisms of schemes.

This is exactly what we have called an algebraic stack in this case.

Definition (4.10). — *Let $\mathscr{S}$ be an algebraic stack. The etale site $\mathscr{S}_{et}$ of $\mathscr{S}$ is the category with objects the etale morphisms*

$$x : X \to \mathscr{S}$$

and where a morphism from (X, x) to (Y, y) is a morphism of schemes $f : X \to Y$ plus a 2-morphism between the 1-morphism $x : X \to \mathscr{S}$ and $y \cdot f : X \to \mathscr{S}$. A collection of morphisms $f_i : (X_i, x_i) \to (X, x)$ is a covering family if the underlying family of morphisms of schemes is surjective.

The site $\mathscr{S}_{et}$ is in a natural way ringed. When we speak of sheaves on $\mathscr{S}$ we mean sheaves on $\mathscr{S}_{et}$.

We now explain how many concepts from the theory of schemes may be applied to algebraic stacks.

Let $\mathcal{P}$ be a property of morphisms of schemes, stable by etale change of base, and of a local nature (for the etale topology) on the target.

For instance: being an open immersion with dense image, being dominant, birational...

A representable morphism of algebraic stacks $f : T_1 \to T_2$ is said to have property $\mathcal{P}$ if for one (and hence for every) surjective etale morphism $x : X \to T_2$, the morphism of schemes deduced by base change

$$f' : X \times_{T_2} T_1 \longrightarrow X$$

has that property.

Let $\mathcal{P}$ be a property of morphisms of schemes, which, at source and target, is of a local nature for the etale topology. This means that, for any family of commutative squares

$$\begin{array}{ccc} X_i & \xrightarrow{g_i} & X \\ {\scriptstyle f_i}\downarrow & & \downarrow{\scriptstyle f} \\ Y_i & \xrightarrow[h_i]{} & Y \end{array}$$

where the g_i (resp. h_i) are etale and cover X (resp. Y):

$$\mathcal{P}(f) \Longleftrightarrow \forall i\, \mathcal{P}(f_i).$$

For instance: f flat, smooth, etale, unramified, normal, locally of finite type, locally of finite presentation.

If $f : T_1 \to T_2$ is a morphism of algebraic stacks, we say f has property $\mathcal{P}$ if for one, then necessarily for every, commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{x} & T_1 \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[y]{} & T_2 \end{array}$$

where X and Y are schemes and x, y are etale and surjective, f' has property $\mathcal{P}$.

Similarly, if $\mathcal{P}$ is a property of schemes, of a local nature for the etale topology, an algebraic stack T will be said to have property $\mathcal{P}$ if for one (and hence for every) surjective etale morphism $x : X \to T$, X has property $\mathcal{P}$. This applies to, for instance, the properties of being regular, normal, locally noetherian, of characteristic p, reduced, Cohen–Macaulay...

An algebraic stack T will be called *quasi-compact* if there exists a surjective etale morphism $x : X \to T$ with X quasi-compact. A morphism $f : T_1 \to T_2$ of algebraic stacks will be called *quasi-compact* if for any quasi-compact

scheme X over T_2, the fibre product $T_1 \times_{T_2} X$ is quasi-compact. It is enough to test the condition for a surjective family $f_i : X_i \to T_2$. We define a morphism $f : T_1 \to T_2$ to be of *finite type*, if it is quasi-compact, and locally of finite type; of *finite presentation*, if it is quasi-compact, quasi-separated, and locally of finite presentation. An algebraic stack is *noetherian*, if it is quasi-compact, quasi-separated, and locally noetherian.

The key point in what follows will be the definition of a "proper morphism" and the analogue of Chow's lemma.

A morphism $f : T_1 \to T_2$ is said to have a property $\mathcal{P}$, locally on T_2, if there exists a surjective etale morphism $x : X \to T_2$ such that the morphism f' deduced from f by the change of base by x has property $\mathcal{P}$.

Definition (4.11). — A morphism $f : T_1 \to T_2$ is proper if it is separated, of finite type and if, locally over T_2, there exist commutative diagrams

$$\begin{array}{ccc} T_3 & \xrightarrow{g} & T_1 \\ & {\scriptstyle h}\searrow \quad \swarrow{\scriptstyle f} & \\ & T_2 & \end{array}$$

with g surjective and h representable and proper.

Source note. The printed definition reads "there exists commutative diagrams"; the verb is corrected here without changing the plural construction.

The following form of Chow's lemma will be sufficient for our purposes.

Theorem (4.12). — Let S be a noetherian scheme and f be a morphism from an etale site T to S. We assume f to be separated and of finite type. Then, there exists a commutative diagram

$$\begin{array}{ccccc} T & \xleftarrow{g} & T' & \xrightarrow{j} & T'' \\ {\scriptstyle f}\downarrow & & {\scriptstyle f'}\swarrow \quad \swarrow{\scriptstyle f''} & & \\ S & \xleftarrow{\qquad\qquad} & & & \end{array}$$

in which T' and T'' are schemes and such that

(i) *g is proper, surjective and generically finite;*

(ii) *j is an open immersion;*

(iii) *f'' is projective.*

Using (4.12), it is easy to extend the cohomological theory of coherent sheaves to the present situation. In fact if $f : T_1 \to T_2$ is a proper morphism of noetherian algebraic stacks and if $\mathscr{F}$ is a coherent sheaf on T_1, then the $R^i f_*(\mathscr{F})$ are coherent sheaves on T_2.

However, the $R^i f_*(\mathscr{F})$ don't need to be zero for i large enough. Let S be a scheme and G a finite group. We denote by p the projection $p : [S/G] \to S$. Quasi-coherent (resp. coherent) sheaves of modules on $[S/G]$ may (and will) be identified with quasi-coherent (resp. coherent) sheaves of modules on S, on which G acts. One has

$$R^i p_*(\mathscr{F}) \simeq H^i(G, \mathscr{F}).$$

In general, the (quasi-coherent sheaf) cohomology of algebraic stacks appears as a mixture of finite group cohomology and of scheme cohomology.

The *disjoint sum* T of a family $(T_i)_{i \in I}$ of stacks is the stack a section of which over a scheme X consists of

(i) a decomposition $X = \coprod_i X_i$ of X;

(ii) a section of T_i over X_i for each i.

The *void stack* $\varnothing$ is the one represented by the void scheme.

A stack is *connected* if it is non-void and is not the disjoint sum of two non-void stacks.

Proposition (4.13). — A locally noetherian algebraic stack is in one and only one way the disjoint sum of a family of connected algebraic stacks (called its connected components).

We denote by $\pi_0(T)$ the set of connected components of a locally noetherian algebraic stack T. If $x : X \to T$ is surjective and etale, $\pi_0(T)$ is the cokernel of the two maps

$$\pi_0(X \times_T X) \rightrightarrows \pi_0(X) \longrightarrow \pi_0(T).$$

Proposition (4.14). — Let T be an algebraic stack of finite type over a field k. Then, T is connected if and only if there exists a connected scheme X, of finite type over k, and a surjective morphism from X to T.

An open subset U of an algebraic stack T is a full subcategory $U \subset T$ which is an algebraic stack, which contains together with any $t \in \mathrm{Ob}(T)$ all isomorphic t' and such that the inclusion $j : U \to T$ is representable by open immersions. The open subsets of T correspond bijectively to the open subsets of its etale site.

Source note. The printed sentence has "The open subsets of T corresponds"; subject–verb agreement is corrected here.

For each open subset U of T, there exists one and only one full subcategory $T - U$ of T, which is an algebraic stack, which contains together with any $t \in \mathrm{Ob}(T)$ all isomorphic t' and such that

(i) $T - U$ is reduced;

(ii) the inclusion map $i : T - U \to T$ is representable by closed immersions;

(iii) for any etale surjective morphism $x : X \to T$, the inverse image of $T - U$ on X is the complement of the inverse image of U.

An algebraic stack F in T satisfying (i) and (ii) is a *closed subset* of T, and the functor $U \mapsto T - U$ is an isomorphism of the set of open and the set of closed subsets of T. If F satisfies only (ii), F_{red} satisfies (i) and (ii) so that F defines a closed subset of T.

An algebraic stack T is *irreducible* if it is not the union of two closed subsets, non-void and distinct from T.

Proposition (4.15). — A noetherian algebraic stack T is in one and only one way the union of irreducible closed subsets, none of which contains any other. They are called the irreducible components of T. If U is an open dense subset of T, the irreducible components of U are the non-void intersections of U with the irreducible components of T.

Each irreducible component of T is contained in a connected component of T. Conversely:

Proposition (4.16). — The connected components of a normal noetherian algebraic stack are irreducible.

Theorem (4.17). — Let f be a morphism of finite type from an algebraic stack T to a noetherian scheme S. For $s \in S$, let $n(s)$ be the number of connected components of the geometric fibre of T at s. Then

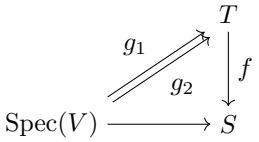
(i) *$n(s)$ is a constructible function of s;*

(ii) *if f is proper and flat, then $n(s)$ is lower-semi-continuous;*

(iii) *if f is proper flat, and has geometrically normal fibres, then $n(s)$ is constant.*

Let $f : T \to S$ be a morphism of finite type from an algebraic stack T to a noetherian scheme S. Assume that the diagonal map $T \to T \times_S T$ is separated and quasi-compact.

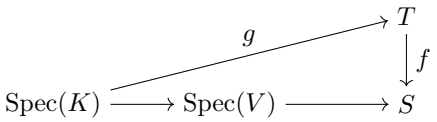
Theorem (4.18) (Valuative criterion for separation). — The morphism f is separated if and only if, for any complete discrete valuation ring with algebraically closed residue field and any commutative diagram



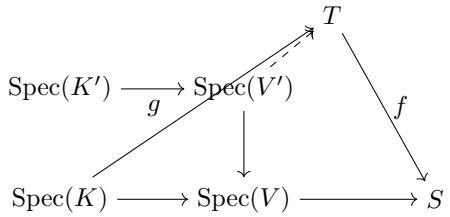
any isomorphism between the restrictions of g_1 and g_2 to the generic point of $\mathrm{Spec}(V)$ can be extended to an isomorphism between g_1 and g_2.

This criterion is nothing other but the valuative criterion of properness (EGA, II, 7.3.8) applied to the (representable) diagonal morphism.

Theorem (4.19) (Valuative criterion for properness). — If f is separated, then f is proper if and only if, for any discrete valuation ring V with field of fractions K and any commutative diagram



there exists a finite extension K' of K such that g extends to $\mathrm{Spec}(V')$*, where V' is the integral closure of V in K'*



To prove a given f is proper, it suffices to verify the above criterion under the additional hypothesis that V is complete and has an algebraically closed residue field. Further, given a dense open subset U of T, it is enough to test only g's which factor through U.

Proposition (4.20). — Let $\mathscr{S}$ be an algebraic stack. The functor which, to any algebraic stack over $\mathscr{S}$, $f : \mathscr{T} \to \mathscr{S}$, associates the $\mathcal{O}_{\mathscr{S}}$-sheaf of algebras $f_*\mathcal{O}_{\mathscr{T}}$ induces an equivalence of categories between:

(i) *the category of algebraic stacks representable and affine over $\mathscr{S}$;*

(ii) *the dual of the category of quasi-coherent $\mathcal{O}_{\mathscr{S}}$-algebras.*

Let $\mathscr{A}$ be a quasi-coherent sheaf of $\mathcal{O}_{\mathscr{S}}$-algebras on an algebraic stack $\mathscr{S}$. For each etale morphism $x : U \to \mathscr{S}$, with U affine, let $\mathscr{A}'(U)$ be the integral closure of $\Gamma(U, \mathcal{O}_{\mathscr{S}})$ in $\mathscr{A}(U)$. By (EGA, II, 6.3.4), the $\mathscr{A}'(U)$ for variable U are the sections over U of a quasi-coherent sheaf $\mathscr{A}'$ on $\mathscr{S}$, which will be called the integral closure of $\mathcal{O}_{\mathscr{S}}$ in $\mathscr{A}$.

Let $f : \mathscr{T} \to \mathscr{S}$ be representable and affine. The algebraic stack which is associated by (4.20) to the integral closure of $\mathcal{O}_{\mathscr{S}}$ in $f_*\mathcal{O}_{\mathscr{T}}$ will be called the normalization of $\mathscr{S}$ with respect to $\mathscr{T}$. Its formation is compatible with any etale change of basis.

Theorem (4.21). — Let $\mathscr{S}$ be a quasi-separated stack over a noetherian scheme S. Assume that

(i) *the diagonal map $\mathscr{S} \to \mathscr{S} \times_S \mathscr{S}$ is representable and unramified;*

(ii) *there exists a scheme X of finite type over S and a smooth and surjective S-morphism from X to $\mathscr{S}$.*

Then, $\mathscr{S}$ is an algebraic stack of finite type over S.

M. Artin has developed powerful methods to relate pro-representability of a stack to the existence of etale surjective maps $x : X \to \mathscr{S}$.

§ 5. Second proof of the irreducibility theorem.

Let $\mathcal{M}_g$ ($g \geq 2$) be the stack whose category of sections over a scheme S is the category of stable curves of genus g over S, the morphisms being the isomorphisms of schemes over S. By (1.11), the diagonal morphism

$$\Delta : \mathcal{M}_g \longrightarrow \mathcal{M}_g \times \mathcal{M}_g$$

is representable, finite and unramified.

We saw in § 1 that the stack classifying the tricanonically embedded stable curves of genus g is represented by a scheme H_g, smooth and of finite type over $\mathrm{Spec}(\mathbb{Z})$. The "forgetful" morphism

$$H_g \longrightarrow \mathcal{M}_g$$

is representable, smooth and surjective. Indeed, if $p : C \to S$ is a stable curve over S, defining a morphism

$$\gamma : S \longrightarrow \mathcal{M}_g,$$

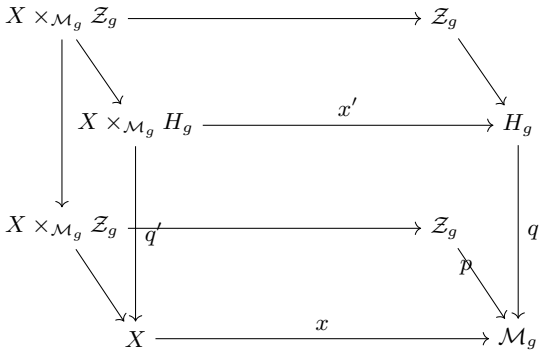
then the fibre product $H_g \times_{\mathcal{M}_g} S$ is the scheme, smooth over S, of isomorphisms between the standard projective space of dimension $5g - 6$ over S and $\mathbf{P}(p_*(\omega_{C/S}^{\otimes 3}))$. We deduce from this and (4.21) that:

Proposition (5.1). — $\mathcal{M}_g$ is a separated algebraic stack of finite type over $\mathrm{Spec}(\mathbb{Z})$.

Let us denote by $\mathcal{M}_g^0$ the open subset of $\mathcal{M}_g$ which "consists of" smooth curves, and by $\mathcal{Z}_g$, the "universal curve" over $\mathcal{M}_g$, the algebraic stack classifying pointed stable curves.

Theorem (5.2). — The algebraic stacks $\mathcal{M}_g$ and $\mathcal{Z}_g$ are proper and smooth over $\mathrm{Spec}(\mathbb{Z})$ *and the complement of $\mathcal{M}_g^0$ in $\mathcal{M}_g$ is a divisor with normal crossings relative to* $\mathrm{Spec}(\mathbb{Z})$.

Proof. — Let $x : X \to \mathcal{M}_g$ be etale and consider the following commutative diagram of algebraic stacks of finite type over Spec($\mathbb{Z}$):



In this diagram, the four horizontal arrows are etale and the four vertical arrows are smooth and surjective. As H_g and $\mathcal{Z}_g$ (p. 78) are smooth over Spec($\mathbb{Z}$), so are X and $X \times_{\mathcal{M}_g} \mathcal{Z}_g$. In addition,

$$q'^{-1}x^{-1}(\mathcal{M}_g^0) = x'^{-1}(H_g^0)$$

is the complement of a divisor with normal crossings relative to Spec($\mathbb{Z}$), so that the inclusion of $x^{-1}(\mathcal{M}_g^0)$ in X has the same property. This being true for any x, $\mathcal{M}_g$ and $\mathcal{Z}_g$ are smooth over Spec($\mathbb{Z}$) and $\mathcal{M}_g^0$ is the complement of a divisor with normal crossings relative to Spec($\mathbb{Z}$). In particular $\mathcal{M}_g^0$ is dense in $\mathcal{M}_g$.

We may now use the valuative criterion for properness in its modified form (4.19) to deduce the properness of $\mathcal{M}_g$ from the stable reduction theorem. The properness of $\mathcal{Z}_g$ then results from that of p.

(5.3) Let $p : X \to S$ be a stable smooth curve of genus $g \geq 2$ over S. If $k \in \mathbb{N}$ is invertible in $\mathcal{O}_S$, the sheaf $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ on S_{et} is locally free over $\mathbb{Z}/k\mathbb{Z}$, of rank $2g$, and the cup-product is a non degenerate alternating form

$$R^1p_*(\mathbb{Z}/k\mathbb{Z}) \otimes R^1p_*(\mathbb{Z}/k\mathbb{Z}) \longrightarrow R^2p_*(\mathbb{Z}/k\mathbb{Z}) \simeq \mu_k^{\otimes -1}.$$

Locally on S_{et}, μ_k is isomorphic to $\mathbb{Z}/k\mathbb{Z}$, and thus, locally, $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ is provided with a non degenerate symplectic form with values in $\mathbb{Z}/k\mathbb{Z}$, which is determined up to a unit in $\mathbb{Z}/k\mathbb{Z}$, or, as we shall say, $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ is provided with an homogeneous symplectic structure. The constant sheaf $(\mathbb{Z}/k\mathbb{Z})^{2g}$ will be provided with the homogeneous symplectic structure induced by the standard symplectic structure of $(\mathbb{Z}/k\mathbb{Z})^{2g}$.

Definition (5.4). — A Jacobi structure of level k on X is an isomorphism (respecting their homogeneous symplectic structures) between $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ and $(\mathbb{Z}/k\mathbb{Z})^{2g}$.

(5.5) Given a section s of p, and a set of prime numbers $\mathbf{P}$ including all residue characteristics of S, the specialisation theorem for the fundamental group enables one to construct a pro-object $\pi_1(X/S, s)^{(\mathbf{P})}$ of the category (l.c. $\mathrm{gr}^{(\mathbf{P})}$) of locally constant sheaves of finite groups of order prime to $\mathbf{P}$, with the properties

(i) $\mathrm{Hom}_S(\pi_1(X/S, s)^{(\mathbf{P})}, G) = R^1p_*(X \bmod s, p^*G) = R^1p_*(\mathrm{Ker}(p^*G \to s_*G))$ functorially in $G \in$ (l.c. $\mathrm{gr}^{(\mathbf{P})}$);

(ii) the formation of $\pi_1(X/S, s)^{(\mathbf{P})}$ is compatible with any change of base.

If G and H are two sheaves of groups on S_{et}, we define the sheaf of exterior morphisms from H to G, $\mathrm{Hom}^{\mathrm{ext}}(H, G)$, as the quotient of $\mathrm{Hom}(H, G)$ by the action of H induced by its action on itself by inner automorphism.

The sheaf

$$\mathrm{Hom}_S^{\mathrm{ext}}(\pi_1(X/S, s)^{(\mathbf{P})}, G) \simeq R^1p_*(p^*G) \qquad (\text{for } G \in (\text{l.c. } \mathrm{gr}^{(\mathbf{P})}))$$

is "independent" of the choice of s.

We shall denote it as

$$\mathrm{Hom}_S^{\mathrm{ext}}(\pi_1(X/S)^{(\mathbf{P})}, G).$$

As $p : X \to S$ admits sections locally for the etale topology, this sheaf makes sense without assuming p to have a global section. It is independent of the choice of $\mathbf{P}$, so long as $\mathbf{P}$ is prime to the order of G and includes all residue characteristics of S.

Definition (5.6). — Let G be a finite group of order n prime to $\mathbf{P}$. A Teichmüller structure of level G on X is a surjective exterior homomorphism from $\pi_1(X/S)$ to G.

The finite generation of $\pi_1(X/S)$ (SGA, 60/61, exp. 10) implies:

Lemma (5.7). — The sheaf on (Sch/S) *of the Teichmüller structures of level* G *on* X *is represented by an etale covering of* S.

We denote by ${}_G\mathcal{M}_g^0$ the stack classifying the stable smooth curves of genus g and characteristic prime to n, with a Teichmüller structure of level G.

For any algebraic stack $\mathscr{M}$, we denote by $\mathscr{M}[1/n]$ its open subset $\mathscr{M} \times \mathrm{Spec}(\mathbb{Z}[1/n])$.

Lemma (5.7) may now be rephrased:

Proposition (5.8). — The "forgetful" morphism

$$ {}_G\mathcal{M}_g^0 \longrightarrow \mathcal{M}_g^0[1/n] $$

is representable, finite and etale.

The stack ${}_G\mathcal{M}_g^0$ is thus an algebraic stack. Let ${}_G\mathcal{M}_g$ be the normalisation of $\mathcal{M}_g[1/n]$ with respect to ${}_G\mathcal{M}_g^0$. The stack ${}_G\mathcal{M}_g$, being representable and finite over $\mathcal{M}_g[1/n]$, is proper over $\mathrm{Spec}(\mathbb{Z}[1/n])$.

Theorem (5.9). — The geometric fibres of the projection of ${}_G\mathcal{M}_g$ *onto* $\mathrm{Spec}(\mathbb{Z}[1/n])$ *are normal, and, fibre by fibre,* ${}_G\mathcal{M}_g^0$ *is dense in* ${}_G\mathcal{M}_g$.

Proof. — We will use Abhyankar–Artin's lemma, in its "absolute" form:

Lemma (5.10). — Let D *be a divisor with normal crossings on an excellent regular scheme* X, Y *an etale covering of* $X-D$ *and* $\overline{Y}$ *the normalisation of* X *with respect to* Y. *Assume that the generic points of the irreducible components of* D *are all of characteristic* 0. *Then every geometric point of* X *has an etale neighbourhood* $x : X' \to X$ *such that, on* X'*:*

(i) D *becomes a union of regular divisors* $(D_i)_{i\in I}$, D_i *of equation* $t_i = 0$.

(ii) *There exists an integer* k *prime to residue characteristics of* X' *such that* $\overline{Y}$ *becomes isomorphic to a disjoint union of quotients (by subgroups of* μ_k^I*) of the covering of* X' *obtained by extracting the* k^{th}*-roots of the* t_i*'s.*

If $x : X \to \mathcal{M}_g[1/n]$ is an etale morphism, then $X^0 = x^{-1}(\mathcal{M}_g^0)$ is the complement in X of a divisor with normal crossing relative to $\mathrm{Spec}(\mathbb{Z})$, $X_1^0 = X \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g^0)$ is an etale covering of X^0 and $X_1 = X \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g)$ is the normalisation of X with respect to this covering of X^0.

Source note. The printed formula has $X_1 = S \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g)$; since the morphism under discussion is $x : X \to \mathcal{M}_g[1/n]$ and X_1 is its base change, the required factor X is retained above.

By the explicit local description (5.10), for any prime number l prime to n, $X_1 \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ is the normalisation of $X \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ with respect to $X_1^0 \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$. As this is true for any modular family, we get (5.9).

Corollary (5.11). — The geometric fibres of the projection

$$ {}_G\mathcal{M}_g \longrightarrow \mathrm{Spec}(\mathbb{Z}[1/n]) $$

all have the same number of connected components.

Proof. — By (4.17), all geometric fibres $\mathcal{M}_g \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ of $\mathcal{M}_g$ over $\mathrm{Spec}(\mathbb{Z}[1/n])$ have the same number of connected components. These connected components are irreducible (4.16). Furthermore ${}_G\mathcal{M}_g^0 \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ is dense in ${}_G\mathcal{M}_g \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ and thus has the same number of connected (or irreducible) components as ${}_G\mathcal{M}_g \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$, and this number is independent of l.

(5.12) Let us denote by Π the fundamental group of an oriented closed differentiable surface S_0 of genus g. The group Π may be defined by generators and relations as follows:

(i) generators: x_i for $1 \le i \le 2g$;

(ii) relation: $(x_1, x_{g+1}) \cdots (x_i, x_{g+i}) \cdots (x_g, x_{2g}) = e$, where $(a, b) = aba^{-1}b^{-1}$.

Let (e_i) be the standard basis of $\mathbb{Z}^{2g}$. The morphism φ from Π to $\mathbb{Z}^{2g}$ with $\varphi(x_i) = e_i$ identifies $\Pi/(\Pi, \Pi) = H_1(\Pi, \mathbb{Z})$ with $\mathbb{Z}^{2g}$.

Source note. The printed formula has $\varphi(X_i) = e_i$ with an upper-case X_i; the generators immediately above are x_i, which is retained here.

The surface S_0 is a $K(\Pi, 1)$ and thus

$$ H^2(\Pi, \mathbb{Z}) = H^2(S_0, \mathbb{Z}) = \mathbb{Z}, $$

and the cup product defines a symplectic structure on $\Pi/(\Pi,\Pi)$. This structure is identified by φ with the standard symplectic structure of $\mathbb{Z}^{2g}$.

We denote by $\mathrm{Aut}^0(\Pi)$ the subgroup of $\mathrm{Aut}(\Pi)$ which acts trivially on $H^2(\Pi,\mathbb{Z})$. Dehn has proved that each exterior automorphism of Π is induced by a diffeomorphism of S_0 onto itself and that the map induced by φ:

$$\mathrm{Aut}^0(\Pi) \longrightarrow \mathrm{Sp}_{2g}(\mathbb{Z}),$$

is surjective (see [Ma]).

Theorem (5.13). — The number of connected components of any geometric fibre of the projection of ${}_G\mathcal{M}_g^0$ onto $\mathrm{Spec}(\mathbb{Z}[1/n])$ *is equal to the number of orbits of* $\mathrm{Aut}^0(\Pi)$ *in the set of exterior epimorphisms from* Π *to* G.

Proof. — By (5.11), it suffices to prove that ${}_G\mathcal{M}_g^0 \times \mathrm{Spec}(\mathbb{C})$ has the said number of connected components. As follows from (4.14),

$$\pi_0({}_G\mathcal{M}_g^0 \times \mathrm{Spec}(\mathbb{C})) = \pi_0({}_GM_g)$$

where ${}_GM_g$ is the coarse moduli scheme classifying stable smooth curves of genus g over $\mathbb{C}$ with a Teichmüller structure of level G.

Source note. The printed proof begins "By (5.13)", a self-reference to the theorem being proved; Corollary (5.11) is the result that reduces the claim to one geometric fibre, and is cited above. The printed phrase "As results from (4.14)" is corrected to "As follows from (4.14)".

Recall that a Teichmüller curve of genus g is a stable smooth curve C of genus g over $\mathbb{C}$ together with an exterior isomorphism φ of the transcendental fundamental group $\pi_1(C)$ with Π, which induces a symplectic isomorphism[6] between

$$H_1(C,\mathbb{Z}) \simeq \pi_1(C)/(\pi_1(C),\pi_1(C))$$

and $\Pi/(\Pi,\Pi)$. By Teichmüller's theory [W], the analytic space T_g classifying Teichmüller curves of a given genus $g \geq 2$ is homeomorphic to a ball, and hence connected.

If ψ is a surjective homomorphism from Π to G, the map

$$(C,\varphi) \longmapsto (C,\psi\varphi)$$

defines a morphism $t_\psi : T_g \to {}_GM_g$. Two such maps t_ψ and $t_{\psi'}$ have the same image if and only if $\psi = \psi'\sigma$ for $\sigma \in \mathrm{Aut}^0(\Pi)$, and

$${}_GM_g = \coprod_{\psi \bmod \mathrm{Aut}^0(\Pi)} t_\psi(T_g)$$

which implies (5.13).

(5.14) Let us denote by ${}_n\mathcal{M}_g^0$ the algebraic stack classifying stable smooth curves with a Jacobi structure of level n. This algebraic stack "is" a true scheme for $n \geq 3$ (by [S]).

If φ is a Jacobi structure of level n on a stable smooth curve $p : X \to S$, we define the "multiplicator" $\mu(\varphi)$ of φ by the commutative diagram

$$\begin{array}{ccc} \Lambda^2(\mathbb{Z}/n\mathbb{Z})^{2g} & \longrightarrow & \mathbb{Z}/n\mathbb{Z} \\ \downarrow{\scriptstyle \Lambda^2\varphi} & & \downarrow{\scriptstyle \mu(\varphi)} \\ \Lambda^2 R^1p_*(\mathbb{Z}/n\mathbb{Z}) & \xrightarrow{\Lambda} & \mu_n^{\otimes -1} \simeq R^2p_*(\mathbb{Z}/n\mathbb{Z}) \end{array}$$

The scheme of isomorphisms between $\mathbb{Z}/n\mathbb{Z}$ and $\mu_n^{\otimes -1}$ is $\mathrm{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$; thus $\varphi \mapsto \mu(\varphi)$ defines a morphism μ from S to $\mathrm{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$. This being true for any X and S, μ is induced by

$$\mu : {}_n\mathcal{M}_g^0 \longrightarrow \mathrm{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n]).$$

Theorem (5.15). — The geometric fibres of the morphism μ are connected.

Proof. — By definition, ${}_n\mathcal{M}_g^0$ is open and closed in ${}_G\mathcal{M}_g^0$ for $G = (\mathbb{Z}/n\mathbb{Z})^{2g}$. The group $\mathrm{GL}_{2g}(\mathbb{Z}/n\mathbb{Z})$ acts on G, and thus on ${}_G\mathcal{M}_g^0$. One has:

[6] An arbitrary isomorphism φ induces an isomorphism between $H_1(C,\mathbb{Z})$ and $\Pi/(\Pi,\Pi)$ which is always symplectic up to sign.

(i) the open subset ${}_n\mathcal{M}_g^0$ of ${}_G\mathcal{M}_g^0$ is stable under the subgroup $H = \mathrm{CSp}_{2g}(\mathbb{Z}/n\mathbb{Z})$ of symplectic similitudes;

(ii) ${}_G\mathcal{M}_g^0 = \coprod_{\sigma \in \mathrm{GL}_{2g}(\mathbb{Z}/n\mathbb{Z})/H} \sigma({}_n\mathcal{M}_g^0)$.

Source note. The printed coproduct is indexed by G/H, although $G = (\mathbb{Z}/n\mathbb{Z})^{2g}$ and H has just been defined as a subgroup of $\mathrm{GL}_{2g}(\mathbb{Z}/n\mathbb{Z})$. The required coset space of the acting group is written explicitly above.

It now follows from (5.11) that all geometric fibres of μ have the same number of connected components.

Source note. The printed phrase "It now results from (5.11)" is corrected to "It now follows from (5.11)".

Consider the geometric fibre of μ at the standard complex place of $\mathrm{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$. This fibre is the algebraic stack classifying complex stable smooth curves C provided with a symplectic isomorphism

$$H^1(C, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} (\mathbb{Z}/n\mathbb{Z})^{2g}.$$

Reasoning as in (5.13), we are reduced to proving

Lemma (5.16). — The homomorphism

$$\mathrm{Aut}^0(\Pi) \longrightarrow \mathrm{Sp}_{2g}(\mathbb{Z}/n\mathbb{Z})$$

is surjective.

Proof. — This follows from Dehn's theory (5.12) and from the fact that the groups Sp_{2g}, being split semi-simple simply connected groups, are generated by their unipotent elements, so that the reduction map

$$\mathrm{Sp}_{2g}(\mathbb{Z}) \longrightarrow \mathrm{Sp}_{2g}(\mathbb{Z}/n\mathbb{Z})$$

is surjective.

Source note. The printed proof says "This results from Dehn's theory" and names the groups Sp_n. The English construction is corrected to "This follows", and the group is restored to Sp_{2g}, as required by both the lemma and the displayed reduction map.

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MODULAR FORMS AND ℓ-ADIC REPRESENTATIONS

Séminaire Bourbaki, 21st year, 1968/69, no. 355
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Pierre Deligne

Editorial note. The controlling source for this English edition is an English retype dated 28 June 2004. The published French Bourbaki original, pp. 139–172, controls the French edition and adjudicates damaged English-retype passages. Evident non-mathematical typographical slips are regularized in the running text; substantive or non-obvious restorations are disclosed in place and registered in the source-alignment ledger delivered with this edition.

1 Introduction

Let

$$D(q) = q\prod_{n=1}^{\infty}(1-q^n)^{24} = \sum_{n=1}^{\infty}\tau(n)q^n \qquad (|q|<1)$$

and

$$\Delta(z) = D(e^{2\pi iz}) \qquad (\operatorname{Im} z > 0).$$

It is known that the function Δ is, up to a constant factor, the unique cusp form of weight 12 for the group $\mathrm{SL}_2(\mathbb{Z})$. Put, for p prime,

$$H_p(X) = 1 - \tau(p)X + p^{11}X^2.$$

Following Hecke's theory, the Dirichlet series

$$L_\tau(s) = \sum \tau(n)n^{-s} = \prod_p \frac{1}{H_p(p^{-s})}$$

extends to an entire function of s, and the function

$$(2\pi)^{-s}\Gamma(s)L_\tau(s)$$

is invariant under $s \mapsto 12 - s$. Ramanujan's conjecture asserts that the roots of the polynomial H_p have absolute value $p^{-11/2}$; equivalently, $|\tau(p)| < 2p^{11/2}$.

These properties, proven or conjectural, are similar to conjectural properties of zeta functions of algebraic varieties over $\mathbb{Q}$. Suggested by this, as a first approximation, one is tempted to interpret L_τ as the zeta function of such a variety.

For each prime number ℓ, let K_ℓ be the maximal extension of $\mathbb{Q}$ unramified away from ℓ and, for $p \neq \ell$, let F_p be the inverse in $\mathrm{Gal}(K_\ell/\mathbb{Q})$ of the Frobenius element φ_p relative to p. This object is well defined up to conjugation.

Translating the preceding in terms of ℓ-adic cohomology, Serre conjectured the existence, for each ℓ, of a representation of $\mathrm{Gal}(K_\ell/\mathbb{Q})$ in a $\mathbb{Q}_\ell$-vector space W_ℓ of rank 2 such that, for each $p \neq \ell$,

$$H_p(X) = \det(1 - F_pX; W_\ell).$$

Moreover, the representation W_ℓ should lie in the range of application of the Weil conjectures, and Ramanujan's conjecture should be a particular case of them.

This program has been carried out by Kuga–Shimura [4] in the analogous case of modular forms relative to certain subgroups of $\mathrm{SL}_2(\mathbb{R})$ with compact quotient. Reduced to the present case, the fundamental idea of Sato–Kuga–Shimura is the following: if E is the universal elliptic curve over the moduli scheme S of elliptic curves (forgetting that it does not exist) and if E^k is the fiber product of E with itself over S, repeated k times, then $L_\tau(s)$ is essentially the zeta function of E^k for $k = 10 = 12 - 2$.

We show below how to resolve the difficulties created by the points at infinity, and how to construct the representations W_ℓ having the properties indicated above. For more historical details and for applications, see Serre [6].

Source note. The retyped source has "the difficulties creeted by the points." The evident spelling is regularized, and "at infinity" is restored from the subject of the construction that follows.

Notation.

- We denote by $\mathbb{A}$ the ring of adeles of $\mathbb{Q}$, by $\mathbb{A}_f$ the ring of finite adeles, the restricted product over all prime numbers of the fields $\mathbb{Q}_p$, and, for S a finite set of prime numbers, put

$$\mathbb{A}_f^S = \prod_{p\in S} \mathbb{Q}_p \times \prod_{p\notin S} \mathbb{Z}_p \subset \mathbb{A}_f.$$

  Source note. In the second product the retyped source prints $\mathbb{Q}_p$. The restricted-product definition and the immediately following identity $\widehat{\mathbb{Z}} = \mathbb{A}_f^{\varnothing}$ require $\mathbb{Z}_p$, as delivered here. For $S = \varnothing$, one writes $\widehat{\mathbb{Z}} = \mathbb{A}_f^{\varnothing}$.

- If X is a topological space, or the étale site of a scheme, and G is a set, G also denotes the constant sheaf on X defined by G.
- We denote by $\mathbb{G}_a$ and $\mathbb{G}_m$ the additive and multiplicative groups.
- An elliptic curve is an abelian variety of dimension one, and in particular is equipped with an origin.
- If L is an invertible sheaf and $n \in \mathbb{Z}$, L^n denotes the nth tensor power $L^{\otimes n}$.
- We denote by $\overline{\mathbb{Q}}$ the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$.
- The symbol $\square$ marks the end of a proof, or its absence.

2 The Shimura Isomorphism

(2.1) An elliptic curve over a complex analytic space S is a proper and flat morphism of analytic spaces $f : E \to S$, equipped with a section e, whose fibers are elliptic curves. An elliptic curve over S admits one and only one S-group law

$$\mu : E \times_S E \to E$$

whose unit section is e. Associated with an elliptic curve are the following.

(a) The invertible sheaf

$$\omega_E = e^*\Omega^1_{E/S}.$$

The relative Lie algebra $\mathrm{Lie}_S(E)$ is the invertible sheaf ω^{-1} dual to ω. One has

$$f_*\Omega^1_{E/S} \xrightarrow{\sim} \omega.$$

(b) The local system of free rank-2 $\mathbb{Z}$-modules $R^1f_*\mathbb{Z}$. One puts

$$T_{\mathbb{Z}}(E) = (R^1f_*\mathbb{Z})^\vee, \qquad T_{\mathbb{Q}}(E) = T_{\mathbb{Z}}(E) \otimes \mathbb{Q},$$

the local system of homology of E over S.

The exponential mapping defines an exact sequence of sheaves of sections

$$0 \longrightarrow T_{\mathbb{Z}}(E) \xrightarrow{\alpha} \omega^{-1} \longrightarrow E \longrightarrow 0,$$

so that the elliptic curve E can be reconstructed from the map α.

The local system $\bigwedge^2 R^1f_*\mathbb{Z} \simeq R^2f_*\mathbb{Z}$ is canonically isomorphic to $\mathbb{Z}$. An isomorphism between $\mathbb{Z}^2$ and $R^1f_*\mathbb{Z}$ is said to be admissible if it induces 1 on second exterior powers. Denote by $\mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C})$ the set of $\mathbb{R}$-vector-space isomorphisms between $\mathbb{R}^2$ and $\mathbb{C}$ which respect the natural orientations of $\mathbb{R}^2$ and $\mathbb{C}$, defined by $e_1 \wedge e_2 > 0$ and $1 \wedge i > 0$. Such a homomorphism is determined by its restriction to $\mathbb{Z}^2$, and one puts

$$\mathrm{Hom}^+(\mathbb{Z}^2, \mathbb{C}) = \mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C}).$$

Source note. The French original prints that an admissible isomorphism "induces -1 (sic)" and that the elements of Hom^+ "do not respect (sic)" the orientations. The plus sign, the displayed orientations, the Poincaré half-plane in 2.2(ii), and the ensuing construction require respectively 1 and "respect," as delivered here. This space is equipped with the complex structure obtained from its inclusion in the complex vector space $\mathrm{Hom}(\mathbb{Z}^2, \mathbb{C})$. Over this space one arranges a "universal" exact sequence

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{\alpha} \mathbb{G}_a \longrightarrow E_0 \longrightarrow 0.$$

Proposition (2.2)**.**

(i) The functor which associates with each analytic space S the set of isomorphism classes of elliptic curves E over S, equipped with isomorphisms

$$\omega_E \simeq \mathbb{G}_a, \qquad R^1 f_* \mathbb{Z} \simeq \mathbb{Z}^2$$

(the last being admissible), is represented by the analytic space $\mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C})$*, equipped with a universal elliptic curve E_0.*

(ii) The functor which associates with each analytic space S the set of isomorphism classes of elliptic curves over S, equipped with an admissible isomorphism $R^1 f_* \mathbb{Z} \simeq \mathbb{Z}^2$, is represented by the analytic space

$$X = \mathbb{C}^\times \backslash \mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C}),$$

the Poincaré half-plane.

(iii) The space $\mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C})$ *is a principal homogeneous space of the group $\mathbb{G}_m$ over X.*

One can again regard X as the set of complex structures on $\mathbb{R}^2$. This space is, by (ii), equipped with a universal elliptic curve E_X, whose local system of real cohomology is canonically isomorphic to $\mathbb{R}^2$. Let ω be the invertible sheaf associated with E_X.

The coherent analytic sheaf

$$R^1 f_* \mathbb{R} \otimes_{\mathbb{R}} \mathcal{O}_X$$

is the relative de Rham cohomology sheaf of E_X over X, and as such it is inserted into an exact sequence, the Hodge filtration,

$$0 \longrightarrow \omega \longrightarrow R^1 f_* \mathbb{R} \otimes_{\mathbb{R}} \mathcal{O}_X \xrightarrow{q} \omega^{-1} \longrightarrow 0$$

since, by Serre duality, $\omega^{-1} \simeq R^1 f_* \mathcal{O}_{E_X}$.

The functorial description in 2.2(ii) evidently gives a right action of the group $\mathrm{SL}_2(\mathbb{Z})$ on (X, E_X): to $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and to the elliptic curve E equipped with $\alpha : \mathbb{Z}^2 \xrightarrow{\sim} R^1 f_* \mathbb{Z}$, one associates $(E, \alpha \circ \gamma)$. If one regards X, equipped with

$$q : \mathbb{R}^2 \otimes_{\mathbb{R}} \mathcal{O}_X \simeq R^1 f_* \mathbb{R} \otimes_{\mathbb{R}} \mathcal{O}_X \longrightarrow \omega^{-1},$$

as classifying the complex structures on $\mathbb{R}^2$, one has the same evident action of the group $\mathrm{GL}_2^+(\mathbb{R})$ on $(X, \mathbb{R}^2, \omega, q)$.

(2.3) Choose a basis (x_1, x_2) of $\mathbb{R}^2$ such that $x_1 \wedge x_2 > 0$. A point

$$(f : \mathbb{R}^2 \to \mathbb{C}) \mod \mathbb{C}^\times$$

of X is located by

$$z = \frac{f(x_1)}{f(x_2)} \qquad (\mathrm{Im}\, z > 0),$$

and the map q is identified with

$$q : \mathbb{R}^2 \to \mathbb{G}_a, \qquad a x_1 + b x_2 \longmapsto a z + b.$$

This gives an evident trivialization of ω^{-1} over X, which is not equivariant. Relative to this trivialization, a section $f(z)$ of ω^k over X is transformed by an element

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \in \mathrm{GL}^+(\mathbb{R}^2)$$

(with respect to the basis (x_1, x_2)) by

$$f \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} (z) = (cz + d)^{-k} f\left(\frac{az + b}{cz + d}\right).$$

Source note. In the first matrix on the printed line, both the French original and the English retype have $\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right)^{-1}$. The fractional-linear formula on that same line and the determinant matrix immediately below both require $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)^{-1}$, as delivered here. From the identity

$$dz = (cz + d)^2\, d\left(\frac{az + b}{cz + d}\right) \begin{vmatrix} a & b \\ c & d \end{vmatrix}^{-1},$$

one deduces that dz is a section of $\omega^{-2} \Omega^1_X$ invariant under $\mathrm{SL}_2(\mathbb{R})$. This section is everywhere nonzero and defines an equivariant isomorphism of $\mathrm{SL}_2(\mathbb{R})$-sheaves between ω^2 and Ω^1_X.

(2.4) Let Γ be a discrete subgroup of $\mathrm{SL}_2(\mathbb{R})$ which has no elements of finite order and has quotient of finite volume. One then knows that the quotient space X/Γ is identified with a smooth projective curve $\overline{X/\Gamma}$ minus a finite number of points. The group Γ acts on X without fixed points. The equivariant local system $\mathbb{R}^2$ on X, as well as the exact sequence

$$0 \to \omega \to \mathbb{R}^2 \otimes_{\mathbb{R}} \mathcal{O} \to \omega^{-1} \to 0,$$

therefore define a local system U over X/Γ and an exact sequence

$$0 \to \omega \to U \otimes_{\mathbb{R}} \mathcal{O} \to \omega^{-1} \to 0. \tag{2.5}$$

In the particular case where $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$, these structures are obtained from an elliptic curve E over X/Γ whose inverse image is the equivariant elliptic curve E_X over X.

(2.6) The points at infinity of X/Γ are described as follows (see [9]).

(a) They correspond to conjugacy classes in Γ of nontrivial subgroups of Γ which are maximal among the subgroups of Γ consisting of unipotent elements.

(b) Let $\Gamma_0 \subset \Gamma$ be one such subgroup and choose a basis (x_1, x_2) of $\mathbb{R}^2$ such that, in this basis, Γ_0 is represented as the set of matrices

$$\left\{ \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix} \;\middle|\; n \in \mathbb{Z} \right\}.$$

Let z be the coordinate (2.3) on X defined by (x_1, x_2). There exists N such that the subset

$$X_N = \{z \mid \operatorname{Im} z > N\}$$

of X is disjoint from all of its conjugates γX_N for $\gamma \in \Gamma/\Gamma_0$, so that $X_N/\Gamma_0 \hookrightarrow X/\Gamma$. The function $q = e^{2\pi i z}$ establishes an isomorphism between X_N/Γ_0 and the punctured disk

$$0 < |q| < e^{-2\pi N}.$$

If P_{Γ_0} is the point of $\overline{X/\Gamma} - X/\Gamma$ associated with Γ_0, this isomorphism extends to an isomorphism of a neighborhood of P_{Γ_0} with the disk $0 \le |q| < e^{-2\pi N}$.

Source note. The retyped source prints the two disk inequalities without absolute-value bars around the complex coordinate q. The disk description requires $|q|$, which is restored in both inequalities above.

By virtue of (2.3), the sections of ω over X_N which are invariant under Γ_0 are identified with periodic holomorphic functions of period one on X_N. One again denotes by ω the invertible sheaf over $\overline{X/\Gamma}$ which extends ω and such that, in a neighborhood of a point P_{Γ_0}, the section of ω over X_N/Γ_0 defined by the function 1 extends to an invertible section on X_N/Γ_0.

(2.7) Over X/Γ one has the two invertible sheaves Ω^1 and ω^2, and an isomorphism φ (2.3) between their restrictions to X/Γ. From the formula

$$dq = d(e^{2\pi i z}) = 2\pi i e^{2\pi i z}\, dz = 2\pi i q\, dz$$

it follows that the map

$$\varphi : \Omega^1 \to \omega^2$$

extends to $\overline{X/\Gamma}$, and has a simple zero at each of the points at infinity.

Definition (2.8)**.** *The space of cusp forms of weight $k+2$, relative to the group Γ, is the space of global sections*

$$H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k).$$

By virtue of (2.7), this space is again identified with the space of global sections of ω^{k+2} vanishing at infinity, that is, at each "point."

(2.9) Denote by U^k the kth symmetric power of the local system U over X/Γ. The map (2.5) induces a map

$$\iota^k : \omega^k \to U^k \otimes_{\mathbb{R}} \mathcal{O},$$

hence a map, again denoted by ι^k,

$$\iota^k : \Omega^1 \otimes \omega^k \to \Omega^1(U^k),$$

where the target is the sheaf of holomorphic differential forms over X/Γ with coefficients in the local system U^k.

The de Rham resolution of $U^k \otimes_{\mathbb{R}} \mathbb{C}$,

$$0 \to U^k \otimes_{\mathbb{R}} \mathbb{C} \to U^k \otimes_{\mathbb{R}} \mathcal{O} \xrightarrow{d} U^k \otimes_{\mathbb{R}} \Omega^1 \to 0,$$

induces a map

$$\delta : H^0(X/\Gamma, \Omega^1(U^k)) \to H^1(X/\Gamma, U^k \otimes \mathbb{C}).$$

Moreover, the cohomology space $H^1(X/\Gamma, U^k \otimes \mathbb{C})$ is equipped with a natural complex conjugation, such that δ defines a conjugate-linear mapping $\overline{\delta}$ from the complex conjugate of $H^0(X/\Gamma, \Omega^1(U^k))$ into $H^1(X/\Gamma, U^k \otimes \mathbb{C})$. One thus obtains a map

$$\mathrm{sh}_0 = \delta \circ H^0(\iota^k) \oplus \overline{\delta} \circ H^0(\overline{\iota^k})$$

$$\mathrm{sh}_0 : H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k)} \longrightarrow H^1(X/\Gamma, U^k \otimes \mathbb{C}).$$

For an arbitrary sheaf F on a space Y, one denotes by $\widetilde{H}^i(Y, F)$ the image of cohomology with compact supports $H^i_c(Y, F)$ in cohomology without supports $H^i(Y, F)$.

Theorem 4.2.6 of [9] is essentially equivalent to the following theorem; in loc. cit. k is supposed even, but the same proof works in general.

Theorem (2.10, Shimura [7])**.** *There exists an isomorphism* sh *making the following diagram commutative:*

$$\begin{array}{ccc} H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k)} & \xrightarrow{\mathrm{sh}} & \widetilde{H}^1(X/\Gamma, U^k \otimes \mathbb{C}) \\ \bigcap & & \bigcap \\ H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k)} & \xrightarrow{\mathrm{sh}_0} & H^1(X/\Gamma, U^k \otimes \mathbb{C}). \end{array}$$

One calls sh *the Shimura isomorphism.*

(2.11) In the particular case where Γ is a subgroup of finite index of $\mathrm{SL}_2(\mathbb{Z})$, the elliptic curve E over X/Γ furnishes an elliptic curve scheme over the algebraic curve $\overline{X/\Gamma}$; equivalently, its modular invariant is meromorphic at infinity. It therefore admits a Néron model $\overline{E}$ over $\overline{X/\Gamma}$. One can show that the fibers of $\overline{E}$ at the points at infinity are of multiplicative type, and that over all of $\overline{X/\Gamma}$ one has

$$\omega = e^* \Omega^1_{\overline{E}/\overline{X/\Gamma}}.$$

In this particular case, one has $U = R^1 f_* \mathbb{Z} \otimes \mathbb{R}$, so that the target of the Shimura isomorphism can be rewritten as

$$\widetilde{H}^1(X/\Gamma, U^k \otimes \mathbb{C}) \simeq \widetilde{H}^1(X/\Gamma, \mathrm{Sym}^k(R^1 f_* \mathbb{Z})) \otimes_{\mathbb{Z}} \mathbb{C}.$$

Source note. The printed text twice has the typographical form "fournishes"; both occurrences are delivered as "furnishes."

3 Hecke Operators and the Fundamental ℓ-adic Representation

(3.1) Recall (cf. [3]) that the category of "locally constant" constructible $\mathbb{Z}_\ell$-sheaves, abbreviated lcc, over a scheme S is the category of projective systems $(F_n)_{n \in \mathbb{N}}$ on the étale site $S_{\text{ét}}$ of S satisfying:

(i) F_n is a locally constant sheaf of $\mathbb{Z}/(\ell^n)$-modules of finite type;

(ii) if $n \leq m$, then

$$F_m \otimes_{\mathbb{Z}/(\ell^m)} \mathbb{Z}/(\ell^n) \xrightarrow{\sim} F_n.$$

The lcc $\mathbb{Z}_\ell$-sheaves form a stack of abelian categories over S; the stack of lcc $\mathbb{Q}_\ell$-sheaves is the quotient of this stack by the thick substack of lcc $\mathbb{Z}_\ell$-sheaves killed by a power of ℓ. One denotes by $\otimes\mathbb{Q}_\ell$ the canonical functor from the category of lcc $\mathbb{Z}_\ell$-sheaves into that of lcc $\mathbb{Q}_\ell$-sheaves.

If S is connected and equipped with a geometric point s, the category of lcc $\mathbb{Z}_\ell$-sheaves, respectively $\mathbb{Q}_\ell$-sheaves, over S is equivalent, by the functor "fiber over s," to the category of continuous representations of the fundamental group $\pi_1(S,s)$ on $\mathbb{Z}_\ell$-modules of finite type, respectively on finite-dimensional $\mathbb{Q}_\ell$-vector spaces.

If T is a finite set of prime numbers, an lcc $\mathbb{A}_T^f$-sheaf consists of the giving, for each prime number ℓ, of an lcc $\mathbb{Z}_\ell$-sheaf if $\ell \notin T$, and an lcc $\mathbb{Q}_\ell$-sheaf if $\ell \in T$. For $T = \varnothing$, one speaks of lcc $\widehat{\mathbb{Z}}$-sheaves rather than of lcc $\mathbb{A}_\varnothing^f$-sheaves.

For general T, the category of lcc $\mathbb{A}_T^f$-sheaves is the inductive limit of the categories of lcc $\mathbb{A}_{T'}^f$-sheaves for $T' \subset T$ finite. One puts

$$\mathbb{Z}_\ell = \varprojlim_n \mathbb{Z}/(\ell^n), \qquad \mathbb{Q}_\ell = \mathbb{Z}_\ell \otimes \mathbb{Q}_\ell, \qquad \widehat{\mathbb{Z}} = (\mathbb{Z}_\ell)_\ell, \qquad \mathbb{A}_T^f = \widehat{\mathbb{Z}} \otimes \mathbb{A}_T^f.$$

Source note. The English retype's obvious spelling slips on this page—including "Shiumra," "particluar," "schme," "constent," and "numbesr"—are regularized without changing their mathematical content.

The stack of elliptic curves up to isogeny over S is the stack obtained from the stack of elliptic curves over S by "formally inverting isogenies." One denotes by $\otimes\mathbb{Q}$ the functor associating to an elliptic curve its underlying elliptic curve up to isogeny. For S quasi-compact, one has

$$\mathrm{Hom}(E,F) \otimes \mathbb{Q} \xrightarrow{\sim} \mathrm{Hom}(E \otimes \mathbb{Q}, F \otimes \mathbb{Q}),$$

and, for S normal, every elliptic curve up to isogeny over S underlies an elliptic curve over S.

(3.2) Let $f : E \to S$ be an elliptic curve over a scheme S. One denotes by $T_\ell(E)$ the projective system of kernels E_{ℓ^n} of multiplication by ℓ^n on E, the transition maps from E_{ℓ^n} to E_{ℓ^m}, for $n \geq m$, being multiplication by ℓ^{n-m}. Proceeding in the same way for $\mathbb{G}_m$, one puts $T_\ell(\mathbb{G}_m) = \mathbb{Z}_\ell(1)$. If ℓ is invertible on S, then $T_\ell(E)$ and $\mathbb{Z}_\ell(1)$ are $\mathbb{Z}_\ell$-sheaves on S. One defines $T_\infty(E)$ to be the relative Lie algebra of E over S, that is, the invertible dual of the invertible sheaf ω of (2.1(a)).

Suppose S is of characteristic 0. One then defines the $\widehat{\mathbb{Z}}$-sheaf $T_f(E)$ over S to be the system of the $T_\ell(E)$, and puts

$$V_f(E) = T_f(E) \otimes \mathbb{A}_f.$$

If $u : E \to F$ is an isogeny, then u induces an isomorphism of $V_f(E)$ onto $V_f(F)$ and of $T_\infty(E)$ onto $T_\infty(F)$; the functors V_f and T_∞ therefore factor through the category of elliptic curves up to isogeny over S.

Proposition (3.3)**.** *Let S be a scheme of characteristic 0; let $E_1(S)$ be the category of elliptic curves over S; and let $E_2(S)$ be the category of triples composed of an elliptic curve up to isogeny E over S, a $\widehat{\mathbb{Z}}$-sheaf T which is a twisted form of $\widehat{\mathbb{Z}}^2$, and an isomorphism*

$$\beta : V_f(E) \xrightarrow{\sim} T \otimes \mathbb{A}_f.$$

Then the functor

$$I : E \longmapsto \big(E \otimes \mathbb{Q},\ T_f(E),\ V_f(E) \simeq T_f(E) \otimes \mathbb{A}_f\big)$$

from $E_1(S)$ to $E_2(S)$ is an equivalence of categories.

The question is local on S, which one may suppose quasi-compact. If $f : E \to F$ is a morphism of S-elliptic curves, and if f is an isogeny, then one has an exact sequence

$$0 \to T_f(E) \to T_f(F) \to \ker(f) \to 0. \tag{3.4}$$

The morphism f is divisible by n if and only if it kills the kernel E_n of multiplication by n, because the "multiplication by n" map $E/E_n \to E$ is an isomorphism. By (3.4), this holds if and only if $T_f(f)$ is divisible by n, and one deduces from this that $\mathrm{Hom}_S(E,F)$ is the subgroup of $\mathrm{Hom}_S(E \otimes \mathbb{Q}, F \otimes \mathbb{Q})$ consisting of the morphisms f such that $V_f(f)$ sends $T_f(E)$ into $T_f(F)$. The functor I is thus fully faithful.

Source note. The canonical French original says "pleinement fidèle," whereas the inherited English retype says "faithfully flat." Since I is a functor between categories and the preceding argument identifies its Hom sets, the

French reading "fully faithful" is retained. The English retype's obvious case slips $u : e \to F$ and $E_1(s)$ are normalized to $u : E \to F$ and $E_1(S)$.

Let $X \in \mathrm{Ob}(E_2(S))$. Locally on S, X is defined by an elliptic curve up to isogeny $E \otimes \mathbb{Q}$, and by a "lattice" T in $V_f(E)$ which, for almost all ℓ, coincides with $T_\ell(E)$. For $q \in \mathbb{Q}$, $(E \otimes \mathbb{Q}, T)$ is isomorphic to $(E \otimes \mathbb{Q}, qT)$, which allows us to suppose that $T_f(E) \subset T$. The quotient $K = T/T_f(E)$ is then canonically isomorphic to a finite subgroup of E, and X is the image of E/K under I (cf. 3.4). □

Corollary (3.5). *The functor $F_1(S)$, respectively $F_1'(S)$, which associates with each scheme S of characteristic* 0 *the set of isomorphism classes of elliptic curves over S equipped with an isomorphism*

$$\alpha : T_f(E) \xrightarrow{\sim} \widehat{\mathbb{Z}}^2$$

respectively also with an isomorphism

$$\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathbb{G}_a,$$

is isomorphic to the functor $F_2(S)$, respectively $F_2'(S)$, which associates with S the set of isomorphism classes of elliptic curves up to isogeny over S, equipped with an isomorphism

$$\beta : V_f(F) \xrightarrow{\sim} \mathbb{A}_f^2$$

respectively also with an isomorphism

$$\beta_\infty : T_\infty(F) \xrightarrow{\sim} \mathbb{G}_a.$$

Proposition (3.6). *The functor F_1 (respectively F_1') is represented by a scheme M_∞ (respectively M_∞') over $\mathbb{Q}$.*

Let $n \geq 3$ be an integer. The functor which associates with each scheme S the set of isomorphism classes of elliptic curves, equipped with an isomorphism

$$\alpha_n : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$$

respectively also with $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$, is represented by an affine curve M_n (respectively by an affine surface M_n') over $\mathrm{Spec}(\mathbb{Z}[1/n])$. For $n \mid m$, the morphism from M_m to M_n defined by

$$(E, \alpha_m : E_m \xrightarrow{\sim} (\mathbb{Z}/m)^2) \longmapsto (E, (n/m)\alpha_m : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2)$$

is finite and étale over $\mathrm{Spec}(\mathbb{Z}[1/m])$, and one has

$$M_\infty = \varprojlim M_n.$$

One proceeds in the same way to represent F_1'. □

Source note. The canonical French original gives $\mathcal{O}_S$ and the factor $(n/m)\alpha_m$. The inherited readings $\mathcal{O}_X$ and $(m/n)\alpha_m$, incompatible with the functor's base and with restriction of level for $n \mid m$, are corrected in both editions. The English retype's "schem" and duplicated phrase "One goes one proceeds" are also regularized to "scheme" and "One proceeds."

(3.7) The scheme M_∞ (respectively M_∞') is equipped with a universal elliptic curve $f_\infty : E_\infty \to M_\infty$ and an isomorphism

$$\alpha : T_f(E_\infty) \xrightarrow{\sim} \widehat{\mathbb{Z}}^2,$$

respectively also with an isomorphism

$$\alpha_\infty : T_\infty(E_\infty) \xrightarrow{\sim} \mathbb{G}_a.$$

Following (3.5), the scheme M_∞ (respectively M_∞') represents the functor F_2 (respectively F_2'), which gives an evident left action of the adelic group $\mathrm{GL}_2(\mathbb{A}_f)$ on

$$(M_\infty, E_\infty \otimes \mathbb{Q}, \alpha \otimes \mathbb{A}_f)$$

respectively on

$$(M_\infty', E_\infty \otimes \mathbb{Q}, \alpha \otimes \mathbb{A}_f, \alpha_\infty).$$

On the functor this action is given, for $g \in \mathrm{GL}_2(\mathbb{A}_f)$, by

$$g : (F, \beta : V_f(F) \xrightarrow{\sim} \mathbb{A}_f^2, \beta_\infty) \longmapsto (F, g \circ \beta : V_f(F) \xrightarrow{\sim} \mathbb{A}_f^2, \beta_\infty).$$

Source note. Both printed witnesses abbreviate the induced trivialization as $\alpha \otimes \mathbb{A}_f^2$ and write $V_f(E)$ inside a tuple whose elliptic curve is F. Since $\alpha : T_f(E_\infty) \to \widehat{\mathbb{Z}}^2$, scalar extension is $\alpha \otimes \mathbb{A}_f$, and the action's domain is $V_f(F)$; these type-required readings are delivered above. The French original also uses the same universal curve E_∞ in both tuples; the inherited prime on E'_∞ is therefore removed in both editions. Shafarevich first noted this fact.

Let Y be a scheme over $\mathbb{C}$, equal to the projective limit of schemes Y_i of finite type over $\mathbb{C}$, with finite transition maps. The locally compact ringed space Y^{an}, equal to the projective limit of the Y_i^{an}, depends only on Y and not on its representation as a projective limit. If Y is a scheme over $\mathbb{Q}$, equal to the projective limit of schemes Y_i of finite type over $\mathbb{Q}$ with finite transition maps, one puts $Y^{\mathrm{an}} = (Y \otimes \mathbb{C})^{\mathrm{an}}$. This applies to M_∞ and M'_∞.

Proposition (3.8)**.** *One has canonically*

$$M_\infty'^{\mathrm{an}} \simeq \mathrm{Hom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2)/\,\mathrm{GL}_2(\mathbb{Q})$$

and

$$M_\infty^{\mathrm{an}} \simeq \mathbb{C}^\times \backslash\, \mathrm{Hom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2)/\,\mathrm{GL}_2(\mathbb{Q}),$$

and, less canonically,

$$M_\infty'^{\mathrm{an}} \simeq \mathrm{GL}_2(\mathbb{A})/\,\mathrm{GL}_2(\mathbb{Q}), \qquad M_\infty^{\mathrm{an}} \simeq K_\infty \backslash\, \mathrm{GL}_2(\mathbb{A})/\,\mathrm{GL}_2(\mathbb{Q}),$$

where K_∞ is the maximal compact subgroup at infinity, enlarged by the real homotheties. These isomorphisms are compatible with the action of $\mathrm{GL}_2(\mathbb{A}_f)$.

Source note. The IAS English retype contains unresolved "[a/the, bad copy]" and "[bad copy]" placeholders here. The canonical French original reads exactly that K_∞ is the maximal compact subgroup at infinity, enlarged by the real homotheties; that source-established wording is delivered above.

The notion of elliptic curve up to isogeny, and its variants, extends to the complex analytic case. On the other hand, an isogeny $\varphi : E \to F$ induces an isomorphism φ^* between the local systems of rational cohomology of E and F, which allows us to define this last object for an elliptic curve up to isogeny. Let S be a complex analytic space. With the aid of (2.1), one sees that to give an elliptic curve up to isogeny over S is equivalent to giving an invertible sheaf T_∞, a local system $T_\mathbb{Q}$ of $\mathbb{Q}$-vector spaces, and a morphism $\alpha : T_\mathbb{Q} \to T_\infty$ inducing an isomorphism $T_\mathbb{Q} \otimes \mathbb{R} \to T_\infty$ point by point.

Let n be an integer and let K_n be the kernel of the natural mapping

$$\prod_\ell \mathrm{GL}_2(\mathbb{Z}_\ell) \to \mathrm{GL}_2(\mathbb{Z}/n).$$

Let G_1 be the functor which associates with S the set of isomorphism classes of elliptic curves $f : E \to S$ over S, equipped with an isomorphism

$$\varphi : \mathbb{Q}^2 \xrightarrow{\sim} T_\mathbb{Q}(E),$$

an isomorphism $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$, and an isomorphism $\alpha_n : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$. As in (3.3), one sees that G_1 is isomorphic to the functor G_2 which associates with S the set of isomorphism classes of elliptic curves up to isogeny E over S, equipped with $\varphi : \mathbb{Q}^2 \xrightarrow{\sim} T_\mathbb{Q}(E)$, $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$, and an isomorphism $V_f(E) \xrightarrow{\sim} \mathbb{A}_f^2$ given locally over S up to composition with an element of K_n.

Such an object is determined by the composite map φ' (given locally modulo K_n) obtained from

$$\varphi' : \mathbb{Q}^2 \xrightarrow{\varphi} T_\mathbb{Q}(E) \longrightarrow T_\infty(E) \times V_f(E) \xrightarrow{\sim} \mathcal{O}_S \times \mathbb{A}_f^2.$$

One has

$$E = \mathcal{O}_S/\varphi'\big(\mathbb{Q}^2 \cap \varphi'^{-1}(T_\infty(E) \times T_f(E))\big) = \widehat{\mathbb{Z}}^2 \backslash (\mathcal{O}_S \times \mathbb{A}_f^2)/\varphi'(\mathbb{Q}^2),$$

so that, cf. 2.2, G_1 and G_2 are represented by

$$K_n \backslash\, \mathrm{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2).$$

Suppose still that $n \geq 3$, so that $\mathrm{GL}_2(\mathbb{Q})$ acts freely on the preceding space. The analytic space M_n^{an}, respectively $M_n'^{\mathrm{an}}$, represents the functor analogous in analytic geometry to the functor represented by M_n, respectively M'_n, because this functor, say X, is representable and the arrow $X \to M_n^{\mathrm{an}}$, respectively $X \to M_n'^{\mathrm{an}}$, induces a

bijection on sets of points with values in any finite-rank $\mathbb{C}$-algebra. Consequently one deduces from the preceding that

$$M_n'^{\mathrm{an}} \simeq K_n \backslash \operatorname{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2) / \operatorname{GL}_2(\mathbb{Q}).$$

Proceeding in the same way for M_n, one obtains the first assertion of (3.8) by passage to the limit on n.

A point x of

$$\operatorname{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2) / \operatorname{GL}_2(\mathbb{Q})$$

is identified with a "lattice" L_x of $\mathbb{C} \times \mathbb{A}_f^2$, and the corresponding curve is

$$E_x \simeq \widehat{\mathbb{Z}}^2 \backslash (\mathbb{C} \times \mathbb{A}_f^2) / L_x,$$

equipped with

$$V_f(E_x) \simeq L_x \otimes \mathbb{A}_f \xrightarrow{\sim} \mathbb{A}_f^2.$$

The last assertion of (3.8) follows easily. □

Denote by $f_n : E \to M_n$ the universal elliptic curve over M_n. The integer k being fixed, one makes the following definition.

Definition (3.9). *One denotes by W (or by ${}^kW$ if confusion is possible) the $\mathbb{Q}$-vector space*

$$W = \varinjlim_n \widetilde{H}^1(M_n^{\mathrm{an}}, \operatorname{Sym}^k(R^1 f_{n*}\mathbb{Q})) = \varinjlim_n {}_n W.$$

Source note. The retype's page-level slips "beause," "arros," and "Once deduces" are regularized. In the description of the curve corresponding to x, the printed $V_f(E)$ is delivered as $V_f(E_x)$, the only object just defined there. The English retype then says that W "doesn't depend on" the universal curve, reversing the canonical French "ne dépend que de." The source-coherent reading "depends only on" is delivered below in both editions. The level index n, visible in the French f_{n*}, is also restored in the first formula defining W.

This vector space depends only on the universal elliptic curve, up to isogeny, $f_\infty : E \to M_\infty$, so that, by transport of structure, it is equipped with a left action of the adelic group $\operatorname{GL}_2(\mathbb{A}_f)$.

If ℓ is a prime number, the vector space $W_\ell = W \otimes \mathbb{Q}_\ell$ admits a purely algebraic definition, in terms of the ℓ-adic cohomology of the scheme over the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ obtained from M_n by extension of scalars:

$$W_\ell = \varinjlim_n \widetilde{H}^1(M_n \otimes \overline{\mathbb{Q}}, \operatorname{Sym}^k(R^1 f_{n*}\mathbb{Q}_\ell)) = \varinjlim_n {}_n W_\ell. \tag{3.10}$$

Thus the Galois group of $\overline{\mathbb{Q}}$ over $\mathbb{Q}$ acts, by transport of structure, on W_ℓ and on the finite-level spaces.

Finally, the space M_n^{an} is the disjoint union of quotients of the Poincaré half-plane by congruence subgroups of $\mathrm{SL}_2(\mathbb{Z})$, so that, denoting by ω the invertible sheaf on M_n defined by E, Shimura's theory (2.10) gives

$$W_\infty = W \otimes \mathbb{C} = \varinjlim_n \left(H^0(\overline{M}_n^{\mathrm{an}}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{M}_n^{\mathrm{an}}, \Omega^1 \otimes \omega^k)} \right). \tag{3.11}$$

Source note. The inherited transcription dropped the bars on the compactified curves $\overline{M}_n^{\mathrm{an}}$ in both summands of (3.11); they are restored above. The retype's opening "if is equipped" is regularized to "it is equipped." This decomposition of $W \otimes \mathbb{C}$ into two complex-conjugate subspaces, one of which is the space of all cusp forms of weight $k+2$ relative to a general congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$, is analogous to a Hodge decomposition, of type

$$(0, k+1) + (k+1, 0).$$

The action of the adelic group commutes with the action of the Galois group and respects the preceding decomposition.

While the ℓ-adic local system $R^1 f_* \mathbb{Q}_\ell$ is trivial on M_∞, I do not know whether there is a relation between W_ℓ and

$$\varinjlim_n \big(\widetilde{H}^1(M_n \otimes \overline{\mathbb{Q}}, \mathbb{Q}_\ell) \big) \otimes \operatorname{Sym}^k(\mathbb{Q}_\ell^2).$$

(3.12) Let $n \geq 3$ be an integer and let K_n be as in (3.8). Then one has

$$W^{K_n} = {}_nW.$$

This follows by passage to the limit and from the fact that, in rational cohomology, the cohomology of the quotient of a space by a finite group is obtained by taking the invariants of that group in the cohomology.

Put, for p prime,

$$W^{(p)} = W^{\mathrm{GL}_2(\mathbb{Z}_p)}.$$

By passage to the limit, one obtains

$$W^{(p)} = \varinjlim_{(n,p)=1} {}_nW.$$

On this cohomology space there again act:

(i) the subgroup $\prod_{\ell \neq p} \mathrm{GL}_2(\mathbb{Q}_\ell)$ of $\mathrm{GL}_2(\mathbb{A}_f)$, because this subgroup centralizes $\mathrm{GL}_2(\mathbb{Z}_p)$;

(ii) the Hecke algebra $\mathcal{H}(\mathrm{GL}_2(\mathbb{Q}_p), \mathrm{GL}_2(\mathbb{Z}_p))$, which is the algebra of integral measures on the discrete space $\mathrm{GL}_2(\mathbb{Q}_p)/\mathrm{GL}_2(\mathbb{Z}_p)$ invariant on the left by $\mathrm{GL}_2(\mathbb{Z}_p)$. This subalgebra of the group algebra of $\mathrm{GL}_2(\mathbb{Q}_p)$ acts on W and respects $W^{(p)}$. It acts already on each ${}_nW$ for n prime to p.

The Hecke algebra admits as a basis the measures associated with characteristic functions of the double cosets of $\mathrm{GL}_2(\mathbb{Z}_p)$ in $\mathrm{GL}_2(\mathbb{Q}_p)$, and one knows that

$$\mathcal{H}(\mathrm{GL}_2(\mathbb{Q}_p), \mathrm{GL}_2(\mathbb{Z}_p)) = \mathbb{Z}[T_p, R_p, R_p^{-1}],$$

where T_p and R_p are the double cosets of

$$\begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} p^{-1} & 0 \\ 0 & p^{-1} \end{pmatrix}.$$

(3.13) Let p be a prime number, let $n \geq 3$ be an integer prime to p, and let $F_{n,p}$ be the functor which associates to each scheme S the set of isomorphism classes of commutative diagrams of S-schemes

$$\begin{array}{ccccc} & & & & \mathbb{Z}/(n)^2 & & \\ & & \alpha \nearrow & & & \nwarrow \alpha' & \\ E_n & \longrightarrow & & & F_n & & \\ \cap & & & & \cap & & \\ E & \xrightarrow{\varphi} & & & F & & \end{array} \tag{3.14}$$

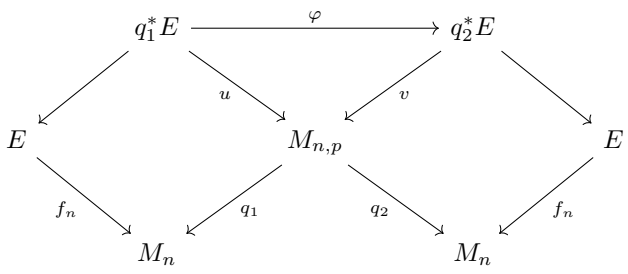
where φ is a p-isogeny between elliptic curves and α is an isomorphism. One denotes by q_1 and $q_2 : F_{n,p} \to M_n$ the morphisms of functors associating to a diagram (3.14) the subdiagrams (E, E_n, α) and (F, F_n, α').

Proposition (3.15)**.** *The functor $F_{n,p}$ is represented by a scheme $M_{n,p}$, and the morphisms $q_1, q_2 : M_{n,p} \to M_n$ are finite.*

The automorphism σ of $F_{n,p}$ sending $\varphi : E \to F$ to ${}^t\varphi : F \to E$ exchanges q_1 and q_2; it suffices therefore to consider q_1. This morphism identifies $F_{n,p}$ with the functor of subgroups of order p of the universal elliptic curve E over M_n, so that, by the theory of Hilbert schemes, $F_{n,p}$ is representable and $M_{n,p}$ is proper over M_n. If s is a geometric point of M_n, $q_1^{-1}(s)$ is the set of subgroups of order p of E_s, and has $p+1$ elements if $\mathrm{char}(k(s)) \neq p$, and only one, the kernel of Frobenius, if $\mathrm{char}(k(s)) = p$. □

One can show that $M_{n,p}$ is regular, and that q_1 and q_2 are finite flat; we do not use this delicate result, but content ourselves here to note that over $\mathrm{Spec}(\mathbb{Z}[1/p])$, each q_i makes $M_{n,p}$ an étale covering of degree $p+1$ of M_n.

The morphisms q_i can be inserted into a commutative diagram



(3.16)

where (φ, u, v) is part of the universal diagram (3.14).

Source note. In the inherited TeX, diagram (3.16) attached u and v to the lower object $M_{n,p}$ and collapsed the two diagonal pullback maps. The source has $u : q_1^* E \to M_{n,p}$ and $v : q_2^* E \to M_{n,p}$, with separate unlabeled diagonal maps from each pullback to E; the native diagram above restores that topology.

One denotes by I_p the morphism from M_n to M_n corresponding to the morphism of functors $(E, \alpha) \mapsto (E, \alpha/p)$:

$$\begin{array}{ccc} E & \longrightarrow & E \\ \downarrow & & \downarrow \\ M_n & \xrightarrow{I_p} & M_n \end{array} \qquad I_p^*(E, \alpha) = (E, \alpha/p). \tag{3.17}$$

I_p^* is an automorphism of $\widetilde{H}^i(M_n^{\mathrm{an}}, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}))$. It is tiresome, but routine, to prove the following.

Proposition (3.18)**.**

(i) The endomorphism T_p of ${}_nW$ is expressed, with the aid of (3.16), as the composite map

$$\widetilde{H}^1(M_n^{\mathrm{an}}, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q})) \xrightarrow{q_2^*} \widetilde{H}^1(M_{n,p}^{\mathrm{an}}, \mathrm{Sym}^k(R^1 v_*\mathbb{Q}))$$
$$\xrightarrow{\varphi^*} \widetilde{H}^1(M_{n,p}^{\mathrm{an}}, \mathrm{Sym}^k(R^1 u_*\mathbb{Q})) \xrightarrow{q_{1*}} \widetilde{H}^1(M_n^{\mathrm{an}}, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q})).$$

where q_{1*} is the trace morphism for the covering q_1.

(ii) Similarly, $R_p = p^k I_p^*$.

□

The distrustful reader could forget the adelic preliminaries and define T_p by (i). When $n = 1$ or 2, one puts ${}_nW = W^{K_n}$, so that

$${}_1W = {}_nW^{\mathrm{GL}_2(\mathbb{Z}/(n))}.$$

If S_{k+2} denotes the space of cusp forms, for the group $\mathrm{SL}_2(\mathbb{Z})$, of weight $k+2$, the Shimura isomorphism (3.11) induces an isomorphism

$${}_1^kW_\infty = {}_1^kW \otimes \mathbb{C} = S_{k+2} \oplus \overline{S}_{k+2}.$$

It is tiresome, but routine, to prove the following.

Proposition (3.19)**.** *The endomorphism T_p of ${}_1^kW_\infty$ is identified, via the Shimura isomorphism, with the direct sum of the Hecke operator on S_{k+2}, including the factor p^{k-1}, and its conjugate.*

□

(3.20) One has canonically

$$\bigwedge^2 R^1 f_{n*}\mathbb{Z}_\ell \simeq R^2 f_{n*}\mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(-1),$$

so that $\mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)$ is equipped with a bilinear form, symmetric for k even and alternating for k odd, with values in $\mathbb{Z}_\ell(-k)$. The form induced by tensoring with $\mathbb{Q}_\ell$ is nondegenerate.

If F is an lcc $\mathbb{Q}_\ell$-sheaf over a smooth scheme X of pure dimension n over an algebraically closed field k, then Poincaré duality gives

$$H^i(X, F)^\vee \simeq H_c^{2n-i}(X, \mathrm{Hom}(F, \mathbb{Q}_\ell(n))),$$
$$H_c^i(X, F)^\vee \simeq H^{2n-i}(X, \mathrm{Hom}(F, \mathbb{Q}_\ell(n))),$$
$$\widetilde{H}^i(X, F)^\vee \simeq \widetilde{H}^{2n-i}(X, \mathrm{Hom}(F, \mathbb{Q}_\ell(n))).$$

Taking $X = \overline{M}_n$ and $F = \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q}_\ell)$ in these considerations, one defines a nondegenerate bilinear form ${}_n(\ ,\ )$ on ${}_n^kW_\ell$ with values in $\mathbb{Q}_\ell(-k-1)$. This form is symmetric for k odd and alternating for k even. This is the ℓ-adic analogue of the Petersson inner product. If $n \mid m$, and if the covering $\psi : M_m \to M_n$ is of degree d, then one has

$${}_m(\psi^* x, \psi^* y) = d \cdot {}_n(x, y).$$

Source note. The printed text spells the name "Peterson" here and again in the proof of (5.6), and "Pertersson" at the end. The standard name "Petersson" is restored throughout. The same page's evident spelling slip "isomprhism" is regularized to "isomorphism."

4 The Congruence Formula

In this section we fix integers $k \geq 0$ and $n \geq 3$, and prime numbers p and ℓ. We suppose that p is prime to ℓ and n. We denote by $f_n : E \to M_n$ the universal elliptic curve over M_n, equipped with $\alpha : E_n \xrightarrow{\sim} \mathbb{Z}/(n)^2$.

Whatever the scheme Y, one denotes by a the unique morphism of Y to $\mathrm{Spec}(\mathbb{Z})$ or, if necessary, to a subscheme of $\mathrm{Spec}(\mathbb{Z})$. If Y is separated and of finite type over $\mathrm{Spec}(\mathbb{Z})$, and if F is a $\mathbb{Z}_\ell$- or $\mathbb{Q}_\ell$-sheaf on Y, one denotes by $R^i a_*(Y, F)$, respectively $R^i a_!(Y, F)$, respectively $R^i\widetilde{a}(Y, F)$, the $\mathbb{Z}_\ell$- or $\mathbb{Q}_\ell$-sheaf over $\mathrm{Spec}(\mathbb{Z})$ that is the ith higher direct image of F under a, respectively the ith direct image with proper supports, respectively $\mathrm{Im}(R^i a_!(Y, F) \to R^i a_*(Y, F))$. For $m \in \mathbb{N}$, one puts

$$Y[1/m] = Y \times \mathrm{Spec}(\mathbb{Z}[1/m]).$$

Theorem (4.1, Igusa [1])**.** *The scheme M_n can be compactified to a scheme of curves M_n^* that is projective and smooth over* $\mathrm{Spec}(\mathbb{Z}[1/n])$*, such that $M_n^* \setminus M_n$ is an étale covering of* $\mathrm{Spec}(\mathbb{Z}[1/n])$*.*

The schemes M_n are formally smooth, thus smooth over $\mathrm{Spec}(\mathbb{Z})$. The modular invariant j of the universal curve over M_n is a morphism of M_n to the affine line $\mathbb{A}^1$ over $\mathrm{Spec}(\mathbb{Z}[1/n])$. The morphism j is finite, and is an étale covering away from the sections 0 and 1728 of $\mathbb{A}^1$; in fact:

(a) Two elliptic curves over an algebraically closed field with the same j-invariant are isomorphic (e.g. [8] 6.3), thus the geometric fibers of j are finite. The schemes M_n and $\mathbb{A}^1$ being smooth of the same dimension over $\mathrm{Spec}(\mathbb{Z})$, j is quasi-finite and flat.

(b) If E is an elliptic curve over the field of fractions K of a discrete valuation ring R, of j-invariant in R, and whose points of order n are rational over K, then E has good reduction. The valuative criterion for properness shows thus that j is proper.

(c) If E and F are two elliptic curves over a scheme S, of the same j-invariant, and if j and $j - 1728$ are invertible, then the scheme $\mathrm{Isom}(S; E, F)$ of isomorphisms between E and F is étale over S ([8] 6.3). In the diagram

$$\begin{array}{ccc} \mathrm{Isom}(M_n \times_{\mathbb{A}^1} M_n; \mathrm{pr}_1^* E, \mathrm{pr}_2^* E) & \sim & M_n \times \mathrm{GL}_2(\mathbb{Z}/(n)) \\ \downarrow u & & \downarrow v \\ M_n \times_{\mathbb{A}^1} M_n & \xrightarrow{\mathrm{pr}_1} & M_n, \end{array}$$

where $j \neq 0, 1728$, u and v are étale surjective, thus pr_1 is étale and, by flat descent, j is étale.

The section at infinity of the projective line $\mathbb{P}^1 \supset \mathbb{A}^1$ over $\mathrm{Spec}(\mathbb{Z}[1/n])$ is a regular divisor whose generic point is of characteristic 0 in a regular scheme. It follows from a theorem of Abhyankar (see [5]) that along this divisor $j = \infty$, M_n is tamely ramified over $\mathbb{P}^1$, and that the normalization M_n^* of $\mathbb{P}^1$ in M_n satisfies (4.1). □

Source note. The English retype's "The schemes M_n is" is regularized to "are." Both printed witnesses spell Abhyankar's name "Abyankhar," and the English retype renders the standard French term *modérément ramifié* as the unresolved "moderately (tamely?) ramified." The standard name and English term "tamely ramified" are delivered above.

It follows from the same theorem that the lcc $\mathbb{Z}_\ell$-sheaves $R^i f_{n*}\mathbb{Z}_\ell$ on $M_n[1/\ell]$ are tamely ramified at infinity. Hence, from (4.1) and the specialization theorems for ℓ-adic cohomology (see [5]), it follows that $R^i a_*(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$, $R^i a_!(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$, and therefore $R^i\widetilde{a}(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$ are lcc $\mathbb{Z}_\ell$-sheaves over $\mathrm{Spec}(\mathbb{Z}[1/n, 1/\ell])$, whose formation is compatible with all base changes.

Corollary (4.2)**.** *The Galois module ${}_nW_\ell$ is the fiber of the lcc $\mathbb{Q}_\ell$-sheaf*

$$R^1\widetilde{a}(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)) \otimes \mathbb{Q}_\ell$$

over $\mathrm{Spec}(\mathbb{Z}[1/n, 1/\ell])$ *at the geometric point $\overline{\mathbb{Q}}$. It is unramified away from n and ℓ.*

□

Let the two commutative diagrams over $M_n \otimes \mathbb{F}_p$

$$\begin{array}{ccc} & \mathbb{Z}/(n)^2 & \\ {\scriptstyle\alpha}\nearrow & & \nwarrow{\scriptstyle\alpha^{(p)}} \\ E_n & \longrightarrow & E_n^{(p)} \\ \cap & & \cap \\ E & \xrightarrow{F} & E^{(p)} \end{array} \qquad \text{and} \qquad \begin{array}{ccc} & \mathbb{Z}/(n)^2 & \\ {\scriptstyle p\alpha^{(p)}}\nearrow & & \nwarrow{\scriptstyle\alpha} \\ E_n^{(p)} & \longrightarrow & E_n, \\ \cap & & \cap \\ E^{(p)} & \xrightarrow{V} & E \end{array}$$

be written more briefly

$$F : (E, \alpha) \to (E^{(p)}, \alpha^{(p)}) \qquad \text{and} \qquad V : (E^{(p)}, p\alpha^{(p)}) \to (E, \alpha),$$

where F is the Frobenius morphism and V, its transpose, is the Verschiebung. These diagrams define morphisms Φ_1 and Φ_2 of $M_n \otimes \mathbb{F}_p$ to $M_{n,p}$. These morphisms are finite, considered as sections of q_1 and q_2, and define a morphism

$$\Phi = \Phi_1 \coprod \Phi_2 : M_n \otimes \mathbb{F}_p \coprod M_n \otimes \mathbb{F}_p \to M_{n,p} \otimes \mathbb{F}_p.$$

Source note. The inherited arrays incorrectly repeated the labels F and V on the induced arrows between the n-torsion subgroup schemes. In the source those upper horizontal arrows are unlabeled; only the maps between the elliptic curves carry F and V. The native diagrams above restore that distinction. The printed English witness also has four bare inclusion signs $\cap$, the central conjunction "and," F and V above the lower arrows, and a terminal comma after the right-hand E_n; those details are retained here. Let Φ^h be the restriction of Φ to the opens M_n^h and $M_{n,p}^h$ of $M_n \otimes \mathbb{F}_p$ and $M_{n,p} \otimes \mathbb{F}_p$ corresponding to curves of nonzero Hasse invariant h.

Proposition (4.3)**.** *Φ^h is an isomorphism.*

Let $\varphi : E_1 \to E_2$ be a p-isogeny between elliptic curves of invertible Hasse invariant over a scheme S of characteristic p. At each geometric point of S, either the kernel $\ker(\varphi)$ of φ is étale over S, or its Cartier dual, isomorphic to $\ker({}^t\varphi)$, is étale over S. The property "$\ker(\varphi)$ is étale" is an open property, so that locally on S, either $\ker(\varphi)$ is purely infinitesimal, or $\ker({}^t\varphi)$ is. The only infinitesimal subgroup of order p of E_1 or E_2 being the kernel of Frobenius, in the first case φ is isomorphic to $F : E_1 \to E_1^{(p)}$, and in the second case ${}^t\varphi$ is isomorphic to $F : E_2 \to E_2^{(p)}$ and φ to $V : E_2^{(p)} \to E_2$. □

Proposition (4.4)**.**

(i) The scheme $M_{n,p}$ is smooth over $\mathrm{Spec}(\mathbb{Z})$ *away from points of characteristic p where $h = 0$.*

(ii) The morphisms q_1 and q_2 induce finite flat morphisms q_1' and q_2' from the normalization $M'_{n,p}$ of $M_{n,p}$ to M_n.

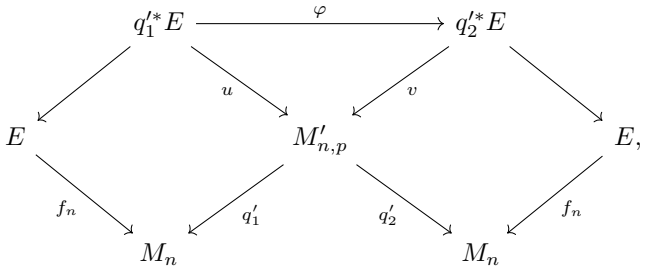
(iii) The morphism Φ can be factored through a surjective morphism

$$\Phi' : M_n \otimes \mathbb{F}_p \coprod M_n \otimes \mathbb{F}_p \to M'_{n,p} \otimes \mathbb{F}_p.$$

The automorphism σ of (3.15) exchanges φ and ${}^t\varphi$, so that it suffices to prove (i) at the points of characteristic p of $M_{n,p}$ where the kernel of φ is infinitesimal: there is no obstruction to infinitesimally lifting an elliptic curve and the infinitesimal part of the kernel of multiplication by p.

Where $p = h = 0$, the fiber of the finite morphism (3.15) $q_i : M_{n,p} \to M_n$ is reduced to a point, so that the open of smoothness of $M_{n,p}$ is dense in $M_{n,p}$ and $M'_{n,p}$ is everywhere of dimension 2. The scheme M_n being regular, following EGA 0_{IV} 16.5.1 and 17.3.5(ii), the morphism $q_i : M'_{n,p} \to M_n$ is flat. Finally, (iii) results from the fact that Φ is finite and $M_n \otimes \mathbb{F}_p$ is a normal curve. □

The Hecke endomorphism T_p of ${}_nW_\ell$, such as is made explicit in (3.18), is the tensorization with $\mathbb{Q}_\ell$ of the fiber at the geometric point $\overline{\mathbb{Q}}$ of $\mathrm{Spec}(\mathbb{Z}[1/n, 1/\ell])$ of the endomorphism, again denoted by T_p, of $R^1\widetilde{a}(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$ defined by the correspondence



(4.5)

with

$$T_p = q'_{1*}\, \varphi^*\, q_2'^* \qquad \text{(cf. 3.18)}.$$

Source note. As in the inherited version of (3.16), the inherited array for (4.5) attached u and v to the lower moduli object and collapsed two distinct pullback-to-curve maps. The native diagram restores the source's full nine-arrow topology and retains the printed English witness's terminal comma after the right-hand E. The endomorphisms R_p and I_p can be interpreted in the same way.

Lemma (4.6). *Let X, Y, Z_1, and Z_2 be four schemes that are separated and of finite type over a noetherian scheme S, let F be a $\mathbb{Z}_\ell$-sheaf on X, let G be a $\mathbb{Z}_\ell$-sheaf on Y, and denote by a each of the structural maps from X, Y, Z_1 or Z_2 to S. Suppose we have the following commutative diagram of S-schemes and morphisms of sheaves:*

$$y_1^*G \xrightarrow{z_1} x_1^*F \qquad\qquad y_2^*G \xrightarrow{z_2} x_2^*F$$

$$\begin{array}{ccccc} & & Z_1 & \xrightarrow{\quad f \quad} & Z_2 & & \\ & \overset{x_1}{\swarrow} & & \overset{x_2}{\searrow}\ \overset{y_1}{\swarrow} & & \overset{y_2}{\searrow} & \\ X & & & & & & Y \end{array}$$

Suppose that $f^*z_2 = z_1$, that y_1 and y_2 are proper, that x_1 and x_2 are finite and flat, and that, for every geometric point s of Z_2, the multiplicity of s in its fiber $x_2^{-1}(x_2(s))$ is equal to the sum of the multiplicities in their fibers, for x_1, of the geometric points of Z_1 in the inverse image of s under f. Then the diagram

$$\begin{array}{ccccccc} R^i\widetilde{a}(Y,G) & \xrightarrow{y_1^*} & R^i\widetilde{a}(Z_1,G) & \xrightarrow{z_1} & R^i\widetilde{a}(Z_1,F) & \xrightarrow{x_{1*}} & R^i\widetilde{a}(X,F) \\ \| & & \uparrow{\scriptstyle f^*} & & \uparrow{\scriptstyle f^*} & & \| \\ R^i\widetilde{a}(Y,G) & \xrightarrow[y_2^*]{} & R^i\widetilde{a}(Z_2,G) & \xrightarrow[z_2]{} & R^i\widetilde{a}(Z_2,F) & \xrightarrow[x_{2*}]{} & R^i\widetilde{a}(X,F) \end{array}$$

commutes.

Source note. The inherited arrays lost the shared $Z_1 \to Z_2$ object pair in the structural diagram and misaligned the vertical maps in the two cohomological diagrams. The diagrams are rebuilt natively with the source's shared objects and its distinct diagonal, vertical, equality, pullback, and trace arrows. An earlier repair duplicated $Z_1 \to Z_2$ into two disconnected copies; direct replay of the French original at printed p. 164 exposed that remaining topology error, which is now repaired in both editions.

This lemma results from the analogous lemmas for $R^ia_!$ and R^ia_*. The commutativity of the first squares is trivial. The last is rewritten

$$\begin{array}{ccccc} R^i\widetilde{a}(Z_1, x_1^*F) & \xleftarrow{\sim} & R^i\widetilde{a}(X, x_{1*}x_1^*F) & \xrightarrow{\mathrm{Tr}} & R^i\widetilde{a}(X,F) \\ \uparrow & & \uparrow & & \| \\ R^i\widetilde{a}(Z_2, x_2^*F) & \xleftarrow{\sim} & R^i\widetilde{a}(X, x_{2*}x_2^*F) & \xrightarrow[\mathrm{Tr}]{} & R^i\widetilde{a}(X,F) \end{array}$$

and one recalls the definition of the trace in verifying that the square

$$\begin{array}{ccc} x_{1*}x_1^*F & \xrightarrow{\mathrm{Tr}} & F \\ \uparrow & & \| \\ x_{2*}x_2^*F & \xrightarrow{\mathrm{Tr}} & F \end{array}$$

is commutative. □

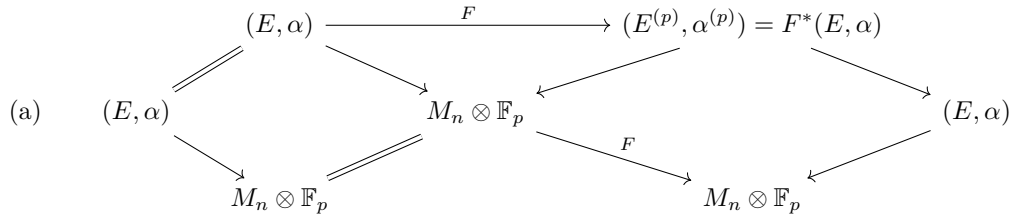
(4.7) One denotes by $T_p/\mathbb{F}_p$ the endomorphism induced by T_p on the restriction to $\mathrm{Spec}(\mathbb{F}_p)$ of the lcc $\mathbb{Z}_\ell$-sheaf $R^1\widetilde{a}(M_n, \mathrm{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell)$. One has

$$R^1\widetilde{a}(M_n, \mathrm{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell)|\,\mathrm{Spec}(\mathbb{F}_p) \xrightarrow{\sim} R^1\widetilde{a}(M_n \otimes \mathbb{F}_p, \mathrm{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell);$$

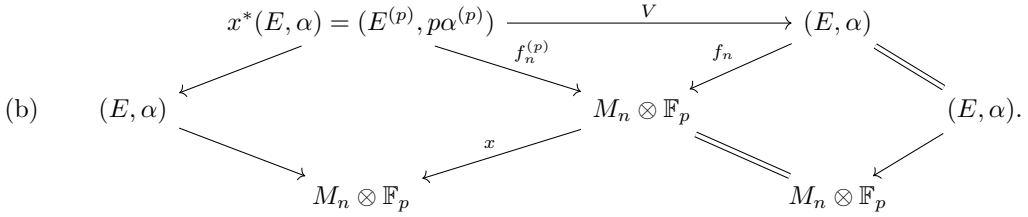
the formation of the trace morphism for a finite flat morphism is compatible with change of base, so that $T_p/\mathbb{F}_p$ can be constructed, over the model (3.18), starting from the fiber over $\mathbb{F}_p$ of the "correspondence" (4.5). Lemma (4.6), applied to the commutative diagram

$$\begin{array}{ccccc} & & M_n \otimes \mathbb{F}_p \coprod M_n \otimes \mathbb{F}_p & \xrightarrow{\Phi'} & M'_{n,p} \otimes \mathbb{F}_p & & \\ & \swarrow & & \overset{q'_1}{\swarrow}\ \overset{q'_2}{\searrow} & & \searrow & \\ M_n \otimes \mathbb{F}_p & & & & & & M_n \otimes \mathbb{F}_p, \end{array}$$

furnishes then a decomposition of $T_p/\mathbb{F}_p$ into the sum of endomorphisms defined by the two following correspondences:



where F is the absolute Frobenius. One recognizes in this correspondence the geometric Frobenius



Source note. The later English retype and the inherited arrays duplicated the shared upper row of the base-change diagram into two disconnected triangles; the native diagram restores the two unique upper objects and their four arrows from the canonical French original. The inherited arrays for correspondences (a) and (b) retained only four of the eight descending source relations and rendered two equality relations as arrows. The native diagrams restore every diagonal, both equality relations in each correspondence, and the labels $f_n^{(p)}$, f_n, x, and F at their printed attachment sites. The canonical French original places the terminal period for (b) in the cell to the right of the lower-right object; the later English retype places it after the middle-right object, and each independent edition preserves its printed punctuation. Both printed witnesses likewise place the final period of the composite x ladder in a separate terminal cell to the right of the lower-right object; both editions preserve it there. The map x is the composite map $I_p^{-1} \circ F$:

$$\begin{array}{ccccc} (E,\alpha) & \longleftarrow & (E,p\alpha) & \longleftarrow & (E^{(p)},p\alpha^{(p)}) \\ \downarrow & & \downarrow & & \downarrow \\ M_n\otimes\mathbb{F}_p & \xleftarrow{I_p^{-1}} & M_n\otimes\mathbb{F}_p & \xleftarrow{F} & M_n\otimes\mathbb{F}_p \end{array} \quad .$$

The corresponding endomorphism is thus the composite of

$$V : R^1\widetilde{a}(M_n\otimes\mathbb{F}_p, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)) \xrightarrow{V^*} R^1\widetilde{a}(M_n\otimes\mathbb{F}_p, \mathrm{Sym}^k(R^1 f_{n*}^{(p)}\mathbb{Z}_\ell)) \xrightarrow{\mathrm{Tr}_F} R^1\widetilde{a}(M_n\otimes\mathbb{F}_p, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$$

and of

$$I_p^* = \mathrm{Tr}_{I_p^{-1}} : \quad \text{endomorphism of } R^1\widetilde{a}(M_n\otimes\mathbb{F}_p, \mathrm{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell).$$

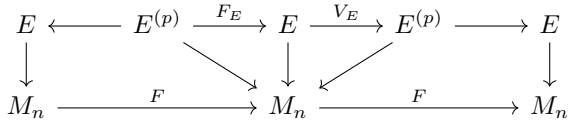
Proposition (4.8)**.** *One has $T_p/\mathbb{F}_p = F + I_p^* V$, and:*

(i) F can be identified with the inverse of the "arithmetic" Frobenius element φ_p of the Galois group $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ *acting on $\widetilde{H}^1(M_n\otimes\overline{\mathbb{F}}_p, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$;*

(ii) F and V are transposes of one another, relative to the scalar product (3.20);

(iii) $FV = VF = p^{k+1}$.

For the relation (i) between the geometric and arithmetic Frobenius, one is referred to the exposé of C. Houzel (SGA 5.XV). The composite VF is the composite of the homomorphisms obtained from the following maps:



Source note. The inherited array collapsed the printed five-object $E \leftarrow E^{(p)} \to E \to E^{(p)} \to E$ diagram to three objects and thereby omitted both intermediate Frobenius twists and their diagonal structure maps. The complete native diagram above restores all five objects and all eleven source arrows. Thus

$$VF = \mathrm{Tr}_F \circ F_E^* \circ V_E^* \circ F^*.$$

The map $F_E^* V_E^* = (F_E V_E)^* = (p \cdot 1_E)^*$ acts by multiplication by p^k on $\mathrm{Sym}^k R^1 f_{n*} \mathbb{Z}_\ell$, so that

$$VF = p^k \,\mathrm{Tr}_F \circ F^* = p^k \cdot p = p^{k+1},$$

because $F : M_n \to M_n$ is of degree p.

By transport of structure, φ_p respects the scalar product (3.20) with values in $\mathbb{Q}_\ell(-k-1)$, a group on which φ_p acts by multiplication by p^{-k-1}. One has thus

$$(Fx, y) = p^{k+1}(\varphi_p F x, \varphi_p y) = (x, p^{k+1} F^{-1} y) = (x, Vy).$$

□

The following theorem, synonymous with (4.8), goes back to Eichler.

Theorem (4.9, The Congruence Formula)**.** *Let $K_{n,\ell}$ be the largest subextension of $\overline{\mathbb{Q}}$ that is unramified away from n and ℓ, let φ_p be a Frobenius element relative to p in $\mathrm{Gal}(K_{n,\ell}/\mathbb{Q})$, let F be the endomorphism φ_p^{-1} of ${}_nW_\ell$, and let V be the transpose of F relative to the scalar product (3.20). Then*

$$T_p = F + I_p^* V, \qquad FV = p^{k+1}, \qquad 1 - T_p X + p R_p X^2 = (1 - FX)(1 - I_p^* VX).$$

□

5 Weil implies Ramanujan

If p is a prime number and X a scheme over $\mathbb{F}_p$, then one denotes by $\overline{\mathbb{F}}_p$ an algebraic closure of $\mathbb{F}_p$, by $F : X \to X$ the "geometric" Frobenius endomorphism, and one puts $\overline{X} = X \otimes \overline{\mathbb{F}}_p$; ℓ always denotes a prime number different from p.

By the "Weil conjectures" one means to claim the following: let X be a projective smooth scheme over $\mathbb{F}_p$ and let ℓ be a prime number different from p. Then the eigenvalues of the endomorphism F^* of $H^i(\overline{X}, \mathbb{Q}_\ell)$ are algebraic integers all of whose complex conjugates have absolute value $p^{i/2}$.

With the hypotheses and notations of (4.9), one has, recalling that $(p, n) = 1$:

Theorem (5.1)**.** *If the Weil conjectures are true, then the eigenvalues of the endomorphism F of ${}_n^k W_\ell$ are algebraic integers, all of whose complex conjugates have absolute value $p^{(k+1)/2}$.*

Source note. The printed statement says "the absolute values of the endomorphism" and prints $p^{k+1/2}$. The theorem concerns the eigenvalues of F, and the weight is $k+1$; the mathematically required reading $p^{(k+1)/2}$ is delivered above.

Assume the Weil conjectures.

Lemma (5.2, modulo Weil)**.** *Let X be a smooth scheme over $\mathbb{F}_p$, which can be represented as an open in a smooth projective scheme X^*. Then the eigenvalues of the endomorphism F^* of $\widetilde{H}^i(X, \mathbb{Q}_\ell)$ are algebraic integers of absolute value $p^{i/2}$.*

Source note. The printed phrase "the absolute values of [the] endomorphism" is corrected to "the eigenvalues of the endomorphism"; the displayed absolute value $p^{i/2}$ is unchanged.

The natural map of $H_c^i(X, \mathbb{Q}_\ell)$ to $H^i(X, \mathbb{Q}_\ell)$ can be factored through $H^i(X^*, \mathbb{Q}_\ell)$:

$$H_c^i(X, \mathbb{Q}_\ell) \to H^i(X^*, \mathbb{Q}_\ell) \to H^i(X, \mathbb{Q}_\ell),$$

so that, as a $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$-module, $\widetilde{H}^i(X, \mathbb{Q}_\ell)$ is the quotient of a subobject of $H^i(X^*, \mathbb{Q}_\ell)$. □

Lemma (5.3, modulo Weil)**.** *Let S be a smooth scheme over $\mathbb{F}_p$ and $f : A \to S$ an abelian scheme over S. Suppose that A can be represented as an open in a projective smooth scheme A^* over $\mathbb{F}_p$. Then the geometric Frobenius F^* of $\widetilde{H}^i(S, R^j f_* \mathbb{Q}_\ell)$ has eigenvalues that are algebraic integers of absolute value $p^{(i+j)/2}$.*

Source note. The printed exponent is $i + j/2$. Since the cohomological weight is $i + j$, the required exponent is $(i + j)/2$, as delivered above and used in the proof via (5.2).

Let m be an integer > 1, and consider the Leray spectral sequences

$$E : E_2^{ij} = H^i(\overline{S}, R^j f_* \mathbb{Q}_\ell) \Longrightarrow H^{i+j}(\overline{A}, \mathbb{Q}_\ell), \qquad {}_cE : {}_cE_2^{ij} = H_c^i(\overline{S}, R^j f_* \mathbb{Q}_\ell) \Longrightarrow H_c^{i+j}(\overline{A}, \mathbb{Q}_\ell).$$

The endomorphism of multiplication by m, $\psi_m = m \cdot 1_A$, defines endomorphisms of E and ${}_cE$ that fit into a commutative diagram

$$\begin{array}{ccc} {}_cE & \longrightarrow & E \\ \psi_m^* \downarrow & & \psi_m^* \downarrow \\ {}_cE & \longrightarrow & E \end{array} .$$

ψ_m^* acts on $R^j f_* \mathbb{Q}_\ell$ by multiplication by m^j, so that ψ_m^* acts by multiplication by m^j on the terms ${}_cE_r^{ij}$ and E_r^{ij} of ${}_cE$ and E. The maps d_r for $r \geq 2$ commute with ψ_m^*, and send E_r^{ij}, respectively ${}_cE_r^{ij}$, to $E_r^{i'j'}$, respectively ${}_cE_r^{i'j'}$, with $j \neq j'$. These maps are zero, and E_2^{ij}, respectively ${}_cE_2^{ij}$, can be identified with a subspace of $H^{i+j}(\overline{A}, \mathbb{Q}_\ell)$, respectively $H_c^{i+j}(\overline{A}, \mathbb{Q}_\ell)$, where $\psi_m^* = m^j$. Consequently, $\widetilde{H}^i(S, R^j f_* \mathbb{Q}_\ell)$ is identified, as a Galois module, with a subspace of $\widetilde{H}^{i+j}(A, \mathbb{Q}_\ell)$ where $\psi_m^* = m^j$, and one applies (5.2). The trick used here is due to Lieberman. □

Source note. The printed proof says that the differentials are nonzero. Since they commute with ψ_m^* but go between the distinct eigenspaces m^j and $m^{j'}$ with $j \neq j'$, they are zero; this is precisely the degeneration used in the following identification. Both printed witnesses place the final period of the commutative diagram in a separate cell to the right of the lower-right object; this edition preserves that placement.

Let $f_n : E \to M_n \otimes \mathbb{F}_p$ be the universal elliptic curve over $M_n \otimes \mathbb{F}_p$, and let $f_{n,k} : E_k \to M_n \otimes \mathbb{F}_p$ be its k-fold fiber product with itself. The Künneth formula shows that the $\mathbb{Q}_\ell$-sheaf $R^k f_{n,k*} \mathbb{Q}_\ell$ admits as a direct factor the k-fold tensor power of $R^1 f_{n*} \mathbb{Q}_\ell$; the latter in turn contains as a direct factor the $\mathbb{Q}_\ell$-sheaf $\mathrm{Sym}^k(R^1 f_{n*} \mathbb{Q}_\ell)$. Theorem 5.1 results thus from (5.3) and the following lemma.

Lemma (5.4)**.** *The scheme E_k is an open in a smooth projective scheme E_k^* over $\mathbb{F}_p$.*

Let E^* be the Néron minimal model of E over $M_n^* \otimes \mathbb{F}_p$ (4.1). The scheme E^* is smooth and projective over $\mathbb{F}_p$. Since $n \geq 3$ and the points of order n of E form a trivial covering of $M_n \otimes \mathbb{F}_p$, the Néron model is "semistable," case a or b_m in Néron's classification. In particular, the projection $f_n : E^* \to M_n^*$ has only a finite number of nonsmooth points, and at these points f_n is nondegenerate, presenting an ordinary quadratic singularity.

Let E_k^{**} be the k-fold fiber product of E^* over M_n^*. To prove (5.4), it suffices to resolve the singularities of E_k^{**} without touching the open E_k. We first prove:

Lemma (5.5)**.** *Let V be the subvariety of affine space over a field k, with coordinates $(X_i)_{0 \leq i \leq r}$, $(Y_i)_{0 \leq i \leq r}$, and $(T_i)_{1 \leq i \leq s}$, with equation*

$$X_0 Y_0 = X_1 Y_1 = \cdots = X_r Y_r.$$

Let $\mathfrak{m}$ be the ideal of $\mathcal{O}_V$ generated by the monomials obtained from the monomial $\prod_{i=0}^r X_i^i$ by a permutation of the coordinates which respects the set of pairs $\{X_i, Y_i\}$, $0 \leq i \leq r$. Then $\mathfrak{m} = \mathcal{O}_V$ away from the singular locus of V, and the variety $\widetilde{V}$ obtained from V by blowing up $\mathfrak{m}$ is smooth over k.

The singular locus is the place where the four coordinates X_i, Y_i, X_j, Y_j for $i \neq j$ vanish. The open affine of $\widetilde{V}$ defined by the element $\prod_{i=0}^r X_i^i$ of the ideal $\mathfrak{m}$ of blowing up is the spectrum of the regular ring

$$k[Y_0/X_1, X_0/X_1, X_1/X_2, \ldots, X_{r-1}/X_r, X_r, T_1, \ldots, T_s]$$

(for the proof, note that $X_i/X_{i+1} = Y_{i+1}/Y_i$), and 5.5 results. □

One still shows that, locally for the étale topology, the singularities of E_k^{**} are isomorphic to those of V for $r = k - 1$, and that this allows the definition on E_k^{**} of an ideal $\mathfrak{m}$ analogous to the ideal $\mathfrak{m}$ of (5.5). Blowing up this ideal, one obtains E_k^*. □

An approximation to the following theorem has been proven by Ihara [2].

Theorem (5.6)**.** *The Weil conjectures imply the Ramanujan conjecture.*

We note first that (5.1) is true for $n = 1$, because ${}_1^k W_\ell$ is the Galois submodule of ${}_m^k W_\ell$ invariant under $\mathrm{GL}_2(\mathbb{Z}/(m))$. On ${}_1^k W_\ell$, I_p^* induces the identity, and (4.8) reduces to

$$1 - T_p X + p^{k+1} X^2 = (1 - FX)(1 - VX).$$

The endomorphisms F and V are transposes of each other, so that

$$\det(1 - FX; {}^k_1W_\ell) = \det(1 - VX; {}^k_1W_\ell).$$

The action of T_p on ${}^k_1W_\ell$ is induced by its action on ${}^k_1W$, and is compatible with the decomposition of ${}^k_1W \otimes \mathbb{C}$ into the sum of the space S_{k+2} of cusp forms relative to $\mathrm{SL}_2(\mathbb{Z})$ of weight $k+2$, and of the complex conjugate space. From the Hermitian property for T_p, for the Petersson inner product, and from (3.19), one deduces then that

$$\det(1 - T_pX + p^{k+1}X^2; {}^k_1W_\ell) = \det(1 - T_pX + p^{k+1}X^2; S_{k+2})^2,$$

and

$$\det(1 - T_pX + p^{k+1}X^2; S_{k+2})^2 = \det(1 - FX; {}^k_1W_\ell)^2,$$

so that

$$\det(1 - T_pX + p^{k+1}X^2; S_{k+2}) = \det(1 - FX; {}^k_1W_\ell). \tag{5.7}$$

Resume the notations of section 1 and make $k = 10$. By virtue of Hecke's theory and of (3.19), (5.7) can be rewritten

$$H_p(X) = \det(1 - FX; {}^{10}_1W_\ell),$$

and one applies (5.1). □

One verifies in the same way that the Weil conjectures imply Petersson's generalization of the Ramanujan conjecture.

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SGA: *Séminaire de géométrie algébrique du Bois-Marie*, mimeographed notes by I.H.E.S., to appear from North-Holland.

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GRIFFITHS'S WORK

by Pierre DELIGNE

This talk contains part of Griffiths's fundamental results on families of Hodge structures. It contains none of his results on algebraic cycles.

1. Hodge structures.

Let $H_{\mathbb{R}}$ be a real vector space of finite dimension.

Definition 1.1.- *A real Hodge structure on $H_{\mathbb{R}}$ is a bigrading of*

$$H_{\mathbb{C}} = H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$$

$$H_{\mathbb{C}} = \bigoplus_{p,q} H^{pq},$$

such that H^{pq} and H^{qp} are complex conjugates.

The Hodge numbers of a real Hodge structure H are the integers

$$h^{pq} = \dim_{\mathbb{C}}(H^{pq}).$$

One says that H has weight n if $h^{pq} = 0$ for $p + q \neq n$. A variant of 1.1 is the following.

Definition 1.2.- *A real Hodge structure of weight n on $H_{\mathbb{R}}$ is a finite decreasing filtration F of $H_{\mathbb{C}}$ (the Hodge filtration) such that, for $p + q = n + 1$, one has*

$$F^p(H_{\mathbb{C}}) \oplus \overline{F}^q(H_{\mathbb{C}}) \xrightarrow{\sim} H_{\mathbb{C}}. \tag{1.2.1}$$

One passes from 1.1 to 1.2 by putting

$$F^p(H_{\mathbb{C}}) = \sum_{p' \geq p} H^{p'q'} \tag{1.2.2}$$

and one passes from 1.2 to 1.1 by putting, for $p + q = n$,

$$H^{pq} = F^p(H_{\mathbb{C}}) \cap \overline{F}^q(H_{\mathbb{C}}). \tag{1.2.3}$$

1.3. Let $\mathbb{S}$ denote the real algebraic group obtained from the multiplicative group $\mathbb{G}_m$ by Weil restriction of scalars from $\mathbb{C}$ to $\mathbb{R}$. We have $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$; there is a natural morphism w from $\mathbb{G}_m$ to $\mathbb{S}$ which, on real points, is the inclusion of $\mathbb{R}^*$ into $\mathbb{C}^*$, and there is a natural morphism t from $\mathbb{S}$ to $\mathbb{G}_m$ which, on real points, is the norm. The composite tw is the squaring map. A new variant of 1.1 is the following.

Definition 1.4.- *A real Hodge structure on $H_{\mathbb{R}}$ is an action (defined over $\mathbb{R}$) of the algebraic group $\mathbb{S}$ on $H_{\mathbb{R}}$.*

If z denotes the defining isomorphism $z : \mathbb{S}(\mathbb{R}) \xrightarrow{\sim} \mathbb{C}^*$, one passes from 1.1 to 1.4 by defining H^{pq} as the subspace of $H_{\mathbb{C}}$ on which $\mathbb{S}(\mathbb{R})$ acts by multiplication by $z^p \overline{z}^q$.

If $H_{\mathbb{R}}$ is endowed with a real Hodge structure, the Weil automorphism C is defined as multiplication by i^{p-q} on H^{pq}. This automorphism is real and is simply the action of the element i of $\mathbb{S}(\mathbb{R}) \subset \mathbb{C}^*$.

Definition 1.5.- *A polarization of a real Hodge structure H of weight n is a bilinear form Ψ on $H_{\mathbb{R}}$, invariant under the compact subgroup $\mathrm{Ker}(t)$ of $\mathbb{S}$, and such that the form $\Psi(x, Cy)$ is symmetric and positive definite.*

From the identity $C^2 = (-1)^n$, one obtains

$$\Psi(x,y) = (-1)^n \Psi(x, C^2 y) = (-1)^n \Psi(Cy, Cx) = (-1)^n \Psi(y,x)$$

so that Ψ is alternating or symmetric according to the parity of n. The variance property under $\mathbb{S}$ and the positivity property are written as follows:

(1.5.1) the orthogonal complement of $F^p(H_{\mathbb{C}})$ with respect to Ψ is $F^{n-p}(H_{\mathbb{C}})$;

(1.5.2) for nonzero x in H^{pq}, one has $i^{p-q}\Psi(x, \overline{x}) > 0$

(Riemann bilinear relations).

Definition 1.6.- *(i) A Hodge structure H of weight n consists of a finitely generated free $\mathbb{Z}$-module $H_{\mathbb{Z}}$ (the "integral lattice") and a real Hodge structure of weight n on $H_{\mathbb{R}} = H_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{R}$.*

(ii) A polarization of a Hodge structure H of weight n is a polarization Ψ of the underlying real Hodge structure that takes integral values on the integral lattice $H_{\mathbb{Z}}$.

2. Hodge theory.

2.1. Let $X \subset \mathbb{P}^r(\mathbb{C})$ be a nonsingular projective complex algebraic variety, pure of dimension d, and let $\eta \in H^2(X, \mathbb{Z})$ be the cohomology class of a hyperplane section of X, i.e., the first Chern class of $\mathcal{O}(1)$.

The holomorphic de Rham complex Ω_X^* is a resolution of the constant sheaf $\mathbb{C}$ (the holomorphic Poincaré lemma), whence a spectral sequence

(2.1.1) $$E_1^{pq} = H^q(X, \Omega_X^p) \Rightarrow H^{p+q}(X, \mathbb{C}).$$

2.2. Hodge theory asserts that

(A) the spectral sequence (2.1.1) degenerates ($E_1 = E_\infty$).

(B) the filtration of $H^n(X, \mathbb{C})$ to which it abuts is a Hodge structure of weight n (1.2) on $H^n(X, \mathbb{R})$.

By (A), one has

$$H^{pq} \simeq H^q(\Omega_X^p).$$

(C) For $n \leq d$, let $P_{\mathbb{Q}}^n$ denote the kernel of the iterated cup-product

$$\eta^{d-n+1} : H^n(X, \mathbb{Q}) \longrightarrow H^{2d-n+2}(X, \mathbb{Q})$$

(the primitive part of cohomology), and let $P_{\mathbb{Z}}^n$ be the intersection of $P_{\mathbb{Q}}^n$ with the image of $H^n(X, \mathbb{Z})$ in $H^n(X, \mathbb{Q})$. Since the class η is of type $(1,1)$, $P_{\mathbb{R}}^n = P_{\mathbb{Z}}^n \otimes \mathbb{R}$ is a Hodge substructure of $H^n(X, \mathbb{R})$. Up to a sign depending only on d and n, the form

$$\Psi(x,y) = \int \eta^{d-n} \wedge x \wedge y$$

is a polarization of the Hodge structure P^n.

(D) The natural map

$$\bigoplus_{n-2k \leq d} \eta^k \wedge : \bigoplus P_{\mathbb{Q}}^{n-2k} \longrightarrow H^n(X, \mathbb{Q})$$

is an isomorphism (the Hodge-Lepage decomposition).

3. Families of Hodge structures.

3.1. Let $f : X \to S$ be a projective and smooth morphism of analytic spaces, of pure relative dimension d. For simplicity, we suppose that S is nonsingular and that f is provided with a factorization through $\mathbb{P}_S^r = S \times \mathbb{P}^r(\mathbb{C})$. The fibers $X_s = f^{-1}(s)$ therefore form an analytic family, parametrized by S, of nonsingular subvarieties of $\mathbb{P}^r(\mathbb{C})$.

3.2. From the C^∞ point of view, f is a locally trivial fibration. The cohomology algebra of the fibers of f therefore forms a local system on S, denoted

$$R^*_{\mathbb{Z}}(f) = \sum R^n f_* \mathbb{Z}.$$

For every abelian sheaf F on S (for example $\mathbb{Q}$, $\mathbb{C}$, $\mathcal{O}$), put

$$R^n_F(f) = F \otimes R^n_{\mathbb{Z}}(f). \tag{3.2.1}$$

Since f is proper, one also has

$$R^n_F(f) \xrightarrow{\sim} R^n f_*(f^* F).$$

The cohomology classes of the hyperplane sections of the X_s define a locally constant section

$$\eta \in H^0(S, R^2_{\mathbb{Z}}(f)). \tag{3.2.2}$$

For $n \leq d$, the $P^n_{\mathbb{Z}}(X_s)$ therefore form a local subsystem $P^n_{\mathbb{Z}}(f)$ of $R^n_{\mathbb{Q}}(f)$. Putting $P^n_F(f) = F \otimes P^n_{\mathbb{Z}}(f)$, one has by definition

$$P^n_{\mathbb{Q}}(f) = \mathrm{Ker}\big(\eta^{d-n+1} \wedge : R^n_{\mathbb{Q}}(f) \to R^{2d-n+2}_{\mathbb{Q}}(f)\big), \tag{3.2.3}$$

$$P^n_{\mathbb{Z}}(f) = P^n_{\mathbb{Q}}(f) \cap \mathrm{Im}\big(R^n_{\mathbb{Z}}(f) \to R^n_{\mathbb{Q}}(f)\big). \tag{3.2.4}$$

The polarization form of 2.2 (C)

$$\Psi(x, y) = \pm \int_{X_s} \eta^{d-n} \wedge x \wedge y \tag{3.2.5}$$

is locally constant on S, i.e., it defines

$$\Psi : P^n_{\mathbb{Z}}(f) \otimes P^n_{\mathbb{Z}}(f) \longrightarrow \mathbb{Z}. \tag{3.2.6}$$

3.3. For F a coherent analytic sheaf on S, $f^{\cdot} F$ will denote its inverse image as a sheaf, and

$$f^* F = \mathcal{O}_X \otimes_{f^{\cdot}\mathcal{O}_S} f^{\cdot} F$$

its inverse image as a sheaf of modules. For $s \in S$, $F_{(s)}$ denotes the $\mathcal{O}_{S,s}$-module of germs of sections of F at s; the fiber of F at s is the vector space

$$F_s = F_{(s)} \otimes_{\mathcal{O}_{S,s}} \mathbb{C}.$$

Recall finally that the relative de Rham complex $\Omega^*_{X/S}$ is the quotient of Ω^*_X with components $\Omega^1_{X/S} = \Omega^1_X / f^* \Omega^1_S$ and $\Omega^p_{X/S} = \bigwedge^p \Omega^1_{X/S}$. If $T^1_{X/S}$, the dual of $\Omega^1_{X/S}$, is the relative tangent bundle, then one has "contraction" pairings

$$\lrcorner : T^1_{X/S} \otimes \Omega^p_{X/S} \longrightarrow \Omega^{p-1}_{X/S}. \tag{3.3.1}$$

3.4. By the relative holomorphic Poincaré lemma, $\Omega^*_{X/S}$ is a resolution of $f^{\cdot}\mathcal{O}_S$. Hence one has a spectral sequence

$$E_1^{pq} = R^q f_* \Omega^p_{X/S} \Rightarrow R^n f_*(f^{\cdot}\mathcal{O}_S) = R^n_{\mathcal{O}}(f). \tag{3.4.1}$$

It follows from 2.2 (A) that the E_1^{pq} are locally free, that the spectral sequence (3.4.1) degenerates ($E_1 = E_\infty$) and that its formation commutes with arbitrary base change. The filtration of $R^n_{\mathcal{O}}(f)$ to which it abuts (the Hodge filtration) therefore induces on $R^n_{\mathcal{O}}(f)_s \simeq H^n(X_s, \mathbb{C})$ the Hodge filtration of 2.2: the latter varies holomorphically with $s \in S$.

Definition 3.5.- *The Gauss-Manin connection on the holomorphic vector bundle* $R^n_{\mathcal{O}}(f)$ *(identified here with its sheaf of holomorphic sections) is the holomorphic and integrable connection*

$$\nabla : R^n_{\mathcal{O}}(f) \longrightarrow \Omega^1_S \otimes R^n_{\mathcal{O}}(f)$$

whose horizontal local sections ($\nabla v = 0$) *are the local sections of* $R^n_{\mathbb{C}}(f)$.

Theorem of transversality 3.6.- *The Hodge filtration F on $R^n_{\mathcal{O}}(f)$ satisfies*

$$\nabla F^p(R^n_{\mathcal{O}}(f)) \subset \Omega^1_S \otimes F^{p-1}(R^n_{\mathcal{O}}(f)).$$

Proof (following Katz-Oda [10] and a personal communication from Katz). The question is local on S. Let $\mathcal{U} = (U_i)_{0\leq i\leq N}$ be a finite covering of X by Stein opens. For $Q \subset [0, N]$, let $U_Q = \bigcap_{i\in Q} U_i$, let j_Q be the inclusion of U_Q in X, and put

$$f_*(U_Q, \Omega^p_{X/S}) = (fj_Q)_*(\Omega^p_{X/S}|U_Q).$$

We denote by $f_*(\mathcal{U}, \Omega^*_{X/S})$ the double complex of sheaves on S with components

$$f_*(\mathcal{U}, \Omega^*_{X/S})^{pq} = \bigoplus_{\#Q=q+1} f_*(U_Q, \Omega^p_{X/S}) \tag{3.6.1}$$

whose first differential is induced by the exterior differential and whose second differential is the "Čech" differential. The spectral sequence 3.4.1 is the spectral sequence of this double complex obtained from the filtration by the first degree.

Let v be a (holomorphic) vector field on S, and suppose that on each U_i a vector field v_i lifting v is given. We denote by $\theta(v_i)$ the endomorphism of $f_*(\mathcal{U}, \Omega^*_{X/S})$ satisfying, for $Q = (i_1, \ldots, i_q)$ with $i_1 < \cdots < i_q$,

$$\theta(v_i) f_*(U_Q, \Omega^p_{X/S}) \subset f_*(U_Q, \Omega^p_{X/S}) \oplus \bigoplus_{i_0<i_1} f_*(U_{\{i_0\}\cup Q}, \Omega^{p-1}_{X/S}),$$

and having as coordinates the Lie derivative

$$\mathcal{L}_{v_{i_1}} : f_*(U_Q, \Omega^p_{X/S}) \longrightarrow f_*(U_Q, \Omega^p_{X/S})$$

and the contractions

$$(-1)^p(v_{i_1} - v_{i_0})\lrcorner : f_*(U_Q, \Omega^p_{X/S}) \longrightarrow f_*(U_{\{i_0\}\cup Q}, \Omega^{p-1}_{X/S}).$$

A simple calculation shows that $\theta(v_i)$ commutes with d.

Lemma 3.7.- *On the cohomology sheaves $R^n_{\mathcal{O}}(f)$ of $f_*(\mathcal{U}, \Omega^*_{X/S})$, the endomorphism $\theta(v_i)$ induces ∇_v.*

Let $\mathcal{C}^\infty$ be the sheaf of C^∞ functions on S. If $f_*(\mathcal{U}, \Omega^*_{\infty X/S})$ is the C^∞ analogue of (3.6.1), constructed in terms of the complex of relative C^∞ differential forms with complex values on X, then

$$R^n_{\mathcal{C}^\infty}(f) = \mathcal{H}^n(f_*(\mathcal{U}, \Omega^*_{\infty X/S})).$$

Since the sheaves $\Omega^p_{\infty X/S}$ are soft, this formula remains valid for every covering $\mathcal{U}'$.

If $(v'_i)_{0\leq i\leq N}$ are C^∞ liftings of v on the U_i, the construction 3.6 still defines endomorphisms $\theta(v'_i)$ of $R^n_{\mathcal{C}^\infty}(f)$. If (v'_i) and (v''_i) are two systems of liftings, a simple calculation shows that, at the level of complexes,

$$\theta(v'_i) - \theta(v''_i) = dH - Hd, \tag{3.7.1}$$

where the only nonzero coordinates of H are the contractions

$$v'_{i_1} - v''_{i_1}\lrcorner : f_*(U_Q, \Omega^p_{\infty X/S}) \longrightarrow f_*(U_Q, \Omega^{p-1}_{\infty X/S}),$$

$(Q = \{i_1, \ldots, i_q\},\ i_1 < \cdots < i_q)$.

Thus the endomorphism $\theta(v'_i)$ of $R^n_{\mathcal{C}^\infty}(f)$ is independent of the choice of the v'_i. Via the injection of $R^n_{\mathcal{O}}(f)$ into $R^n_{\mathcal{C}^\infty}(f)$, it is compatible with the endomorphism $\theta(v_i)$ of 3.7. It therefore remains only to verify that $\theta(v'_i) = \nabla_v$ on $R^n_{\mathcal{C}^\infty}(f)$. One verifies that $\theta(v'_i)$, which is independent of the v'_i, is also independent of $\mathcal{U}$. Taking $\mathcal{U} = \{X\}$ and a lift v' of v, one then has

$$f_*(\mathcal{U}, \Omega^*_{\infty X/S}) = f_*(\Omega^*_{\infty X/S}),$$

and $\theta(v')$ is the Lie derivative along v', visibly equal to ∇_v.

3.8. We finish the proof of 3.6. By construction, one has

$$\theta(v_i)\left(\bigoplus_{p'\geq p} f_*(\mathcal{U},\Omega^*_{X/S})^{p',q'}\right)\subset \bigoplus_{p'\geq p-1} f_*(\mathcal{U},\Omega^*_{X/S})^{p',q'}. \tag{3.8.1}$$

Since the spectral sequence (3.4.1) is obtained from the double complex $f_*(\mathcal{U},\Omega^*_{X/S})$, it follows from 3.7 and (3.8.1) that

$$\theta(v_i)F^p(R^n_{\mathcal{O}}(f)) = \nabla_v F^p(R^n_{\mathcal{O}}(f))\subset F^{p-1}(R^n_{\mathcal{O}}(f)). \tag{3.8.2}$$

Since this is true locally on S and for every v, it implies 3.6.

3.9. By the Leibniz formula

$$\nabla(fh) = df\cdot h + f\cdot\nabla h$$

the map induced by ∇

$$\mathrm{def}(\nabla):\mathrm{Gr}^p_F(R^n_{\mathcal{O}}(f))\longrightarrow \Omega^1_S\otimes \mathrm{Gr}^{p-1}_F(R^n_{\mathcal{O}}(f))$$

is $\mathcal{O}$-linear. By 3.4.1 it is identified with

$$\mathrm{def}(\nabla): R^q f_*\Omega^p_{X/S}\longrightarrow \Omega^1_S\otimes R^{q+1}f_*\Omega^{p-1}_{X/S}. \tag{3.9.1}$$

It follows from formula 3.7 for ∇_v that

Proposition 3.10.-
The homomorphism (3.9.1) is the cup product, via the pairing (3.3.1), with the Kodaira–Spencer class

$$c\in\Omega^1_S\otimes R^1f_*(T^1_{X/S})$$

which expresses how X_s deforms with s.

Definition 3.11.- *(i) A family H of real Hodge structures of weight n on S consists of*

a) a local system of real vector spaces $H_{\mathbb{R}}$ on S;

b) a finite holomorphic filtration F of the locally free analytic sheaf

$$H_{\mathcal{O}} = H_{\mathbb{R}}\otimes_{\mathbb{R}}\mathcal{O},$$

these data satisfying the following conditions:

(H.1) The natural connection ∇ on $H_{\mathcal{O}}$ is such that

$$\nabla F^p(H_{\mathcal{O}})\subset\Omega^1_S\otimes F^{p-1}(H_{\mathcal{O}});$$

(H.2) At every point $s\in S$, F defines on $(H_{\mathbb{R}})_s$ a Hodge structure of weight n.

(ii) A polarization of H is a locally constant bilinear form Ψ on $H_{\mathbb{R}}$ which at every point $s\in S$ induces a polarization of $(H_{\mathbb{R}})_s$.

3.12. A family of Hodge structures H of weight n on S is a local system $H_{\mathbb{Z}}$ of finitely generated free $\mathbb{Z}$-modules on S, together with a family of real Hodge structures on $H_{\mathbb{R}} = \mathbb{R}\otimes H_{\mathbb{Z}}$. A polarization of H is a polarization Ψ of $H_{\mathbb{R}}$ taking integral values on $H_{\mathbb{Z}}$.

With these definitions, one can reformulate 2.2 (C) and 3.6 by saying that, under the hypotheses of 3.1, $P^n_{\mathbb{Z}}(f)$ is a polarized family of Hodge structures on S.

4. The regularity theorem.

4.1. When one starts with a projective and smooth morphism $f : X \to S$ of schemes of finite type over $\mathbb{C}$, such that $f^{\mathrm{an}} : X^{\mathrm{an}} \to S^{\mathrm{an}}$ satisfies the hypotheses of 3.1, some of the objects constructed in no. 3 are obtained from purely algebraic objects by applying the "passage to the analytic category" functor. They are:

a) the sheaves of modules $R^n_{\mathcal{O}}(f)$, their Hodge filtration, the spectral sequence 3.4.1, and the Gauss–Manin connection;

b) the algebra structure (cup-product) on $R^*_{\mathcal{O}}(f)$, and the "primitive part" $P^n_{\mathcal{O}}(f)$;

c) the product of the polarization form Ψ by a suitable power of $2\pi i$.

By contrast, the "integral lattice" $R^*_{\mathbb{Z}}(f)$, and the real structure thereby obtained on the complex cohomology of the fibers of f, are transcendental in nature.

4.2. Indeed, let S be a smooth scheme of finite type over $\mathbb{C}$, and let X be a closed subscheme of $\mathbb{P}^r(\mathbb{C}) \times S$, such that the projection $f : X \to S$ is a smooth morphism of pure relative dimension d.

By the relative variant of GAGA, for every coherent algebraic sheaf F on X one has

$$(R^n f_* F)^{\mathrm{an}} \xrightarrow{\sim} R^n f^{\mathrm{an}}_*(F^{\mathrm{an}}).$$

More generally, if K^* is a complex of coherent algebraic sheaves whose

differentials d_i are (algebraic) differential operators which are $f^{\cdot}\mathcal{O}_S$-linear, then the direct images in hypercohomology satisfy

$$\big(R^n f_*(K^*)\big)^{\mathrm{an}} \xrightarrow{\sim} R^n f^{\mathrm{an}}_*(K^{*\mathrm{an}})$$

(this case reduces to the preceding one via one of the hypercohomology spectral sequences).

Take K to be the relative algebraic de Rham complex $\Omega^*_{X/S}$. One obtains (in the notation of 3.2)

$$\big(R^n f_* \Omega^*_{X/S}\big)^{\mathrm{an}} \xrightarrow{\sim} R^n f^{\mathrm{an}}_*\big((\Omega^*_{X/S})^{\mathrm{an}}\big) \xleftarrow{\sim} R^n f^{\mathrm{an}}_*\big(f^{\cdot\mathrm{an}}\mathcal{O}_S^{\mathrm{an}}\big) = R^n_{\mathcal{O}}(f).$$

Moreover, the spectral sequence (3.4.1) comes from a purely algebraic spectral sequence

$$E_1^{pq} = R^q f_* \Omega^p_{X/S} \Longrightarrow R^{p+q} f_* \Omega^*_{X/S}. \tag{4.2.1}$$

According to 3.4, the E_1^{pq} are locally free (since the $E_1^{pq\,\mathrm{an}}$ are), the spectral sequence (4.2.1) degenerates, and its formation is compatible with every change of base. The Hodge filtration to which it abuts is therefore, ipso facto, algebraic.

The cup-product is defined in terms of the exterior product in $\Omega^*_{X/S}$. The de Rham cohomology class of a hyperplane section can be defined algebraically (up to multiplication by a power of $2\pi i$), from the logarithmic differential

$$\mathrm{d}f/f : \mathcal{O}_X^* \longrightarrow \Omega^1_{X/S},$$

the primitive part $P^n_{\mathcal{O}}(f)$ and Ψ are also algebraic in nature.

Finally, the construction 3.6–3.7 of the Gauss–Manin connection carries over easily to the algebraic case (Katz–Oda [10]).

4.3. Let D be a Riemann surface isomorphic to the unit disk, let 0 be a point of D, put $D^* = D - \{0\}$, let V be a holomorphic vector bundle on D^*, let ∇ be a holomorphic connection on V, and let V_o be the local system of horizontal sections of V. One has

$$V \xrightarrow{\sim} V_o \otimes_{\mathbb{C}} \mathcal{O}.$$

The (commutative) fundamental group $\pi_1(D^*) \simeq \mathbb{Z}$ acts on V_o. We denote by T the monodromy transformation, i.e. the action on V_o or on V of the positive generator of $\pi_1(D^*)$.

There exists U such that

$$T = \exp(2\pi i U).$$

Let $z : D \to \mathbb{C}$ be a coordinate, with $z(0) = 0$, and let $\widetilde{D}^*$ be the (universal) covering of D^* on which $\log z$ is defined. For v a section of V_o on $\widetilde{D}^*$,

$$\exp(-\log z \cdot U)(v)$$

is the inverse image of a holomorphic section of V on D^*.

This construction defines an isomorphism

$$m_{U,z} : \mathcal{O} \otimes_{\mathbb{C}} H^0(\widetilde{D}^*, V_o) \xrightarrow{\sim} V,$$

and hence a natural extension V_U of V to D, depending only on U. A section v of V on D^* will be called ∇-moderate if, as a section of V_U, it is meromorphic at 0. This notion does not depend on the choice of U.

4.4. Let S be the complement, in a smooth projective curve $\overline{S}$, of a finite set T, and let V be an algebraic vector bundle on S. An (algebraic) connection ∇ on V is called regular if, for every $t \in T$, the meromorphic sections of V in a neighborhood of t are ∇-moderate.

Theorem 4.5.- *For $f : X \to S$ a projective and smooth morphism of schemes, the Gauss–Manin connection on $R^n f_* \Omega^*_{X/S}$ is regular.*

This theorem is proved in [2]. A completely different proof, proceeding by arithmetic methods, is due to Katz [9].

4.6. Let D^* be as in 4.3, let H be a family of real Hodge structures on D^*, let T be the monodromy transformation of $H_{\mathbb{C}}$, let U be an endomorphism of $H_{\mathbb{C}}$ such that $T = \exp(2\pi i U)$, and let $H_{\mathcal{O},U}$ be the corresponding locally free extension of $H_{\mathcal{O}}$ to D.

One says that H is regular at 0 if the Hodge filtration of $H_{\mathcal{O}}$ extends to $H_{\mathcal{O},U}$. This condition does not depend on U. Theorem 4.5 has the following corollary, whose conclusion is purely analytic.

Corollary 4.7.- *Under the hypotheses of 4.5, the families of Hodge structures $R^n(f)$ on S^{an} are regular in a neighborhood of every point $t \in T$.*

One may hope that, in fact, every polarized family of Hodge structures on D^* is automatically regular.

5. Moduli Spaces.

5.1. Let G be a real algebraic group, let $C \in G(\mathbb{R})$, and let V be a real representation of G. A C-polarization of V is a bilinear form Ψ on V, invariant under G, and such that $\Psi(x, Cy)$ is symmetric and positive definite.

Lemma 5.2.- *If G is connected (by which one means that $G(\mathbb{C})$ is connected), the following conditions are equivalent:*

(i) G admits a representation $\rho : G \to \mathrm{GL}(V)$, with finite kernel $\mathrm{Ker}(\rho)$, which is C-polarizable.

(ii) Every representation of G is C-polarizable.

(iii) G is reductive and $\mathrm{ad}\, C$ is a Cartan involution (of G).

The proof is left to the reader.

Remark 5.3.- There exist in $G(\mathbb{R})$ elements C satisfying the conditions of 5.2 if and only if G is reductive and admits a compact maximal torus. In this case:

a) Such an element C is contained in a single maximal compact subgroup, namely its centralizer.

b) For G adjoint, the correspondence $C \longmapsto$ (centralizer of C) is a bijection between the set of these elements and the set of maximal compact subgroups of $G(\mathbb{R})$.

5.4. Let G be a real algebraic group, endowed with homomorphisms

$$\mathbb{G}_m \xrightarrow{w} G \xrightarrow{t} \mathbb{G}_m \tag{5.4.1}$$

such that w is central and $tw(x) = x^2$. A representation $\rho : G \to \mathrm{GL}(V)$ will be called homogeneous of weight n if $\rho w(x) = x^n$.

By a morphism from $\mathbb{S}$ to G we shall always mean a homomorphism $h : \mathbb{S} \to G$ making the diagram

$$\begin{array}{ccccc} \mathbb{G}_m & \xrightarrow{w} & \mathbb{S} & \xrightarrow{t} & \mathbb{G}_m \\ \| & & \downarrow h & & \| \\ \mathbb{G}_m & \xrightarrow{w} & G & \xrightarrow{t} & \mathbb{G}_m \end{array}$$

commutative.

If $h : \mathbb{S} \to G$ is a morphism and $\rho : G \to \mathrm{GL}(V)$ a real representation of G, we shall denote by h_V the real Hodge structure (1.4) $\rho \circ h$ on V. For ρ homogeneous of weight n, a polarization of V is a polarization Ψ of (V, h_V) which satisfies

$$\Psi(gx, gy) = t(g)^n \Psi(x, y). \tag{5.4.2}$$

We shall say that h is positive if every representation of G is polarizable.

5.5. The group G, endowed with (5.4.1), is uniquely determined by $G^o = \mathrm{Ker}(t)$, endowed with the central element, of order dividing 2, $\varepsilon = w(-1)$. Indeed G is identified with the quotient of $\mathbb{G}_m \times G^o$ by the subgroup generated by $(-1, \varepsilon)$.

From this point of view, a morphism $h : \mathbb{S} \to G$ is identified with $h^o : \mathbb{S}^o \to G^o$ such that $h^o(-1) = \varepsilon$. A homogeneous representation of weight n of G is identified with a representation ρ of G^o such that $\rho(\varepsilon) = (-1)^n$, and a polarization of this representation is identified with an $h^o(i)$-polarization (5.1). According to 5.2, if G^o is connected, h is a positive morphism if and only if G is reductive and $\mathrm{ad}\, h^o(i)$ is a Cartan involution of G^o.

5.6. We shall henceforth suppose that G^o is reductive and connected.

Let h be a morphism from $\mathbb{S}$ to G. For $\rho : G \to \mathrm{GL}(V)$ a representation of G, denote by F_h the Hodge filtration of $V_{\mathbb{C}}$ deduced from h_V. For the adjoint representation $\mathrm{Lie}(G)$,

$$F_h^0(\mathrm{Lie}(G)_{\mathbb{C}}) = \bigoplus_{p \geq 0} \mathrm{Lie}(G_{\mathbb{C}})^{p,-p} \tag{5.6.1}$$

is the Lie algebra of a parabolic subgroup $P(h)$ of $G_{\mathbb{C}}$. For every representation $\rho : G \to \mathrm{GL}(V)$ of G, $P(h)$ respects the Hodge filtration F_h of $V_{\mathbb{C}}$; if $\mathrm{Ker}(\rho) \cap G^o$ is central, then $P(h)$ is exactly the subgroup of G which respects this filtration.

Finally, denote by $H(h)$ the centralizer of h in $G(\mathbb{R})$.

5.7. Let X be a conjugacy class of morphisms from $\mathbb{S}$ to G. We shall denote by $\check{X}$ the set of conjugates in $G(\mathbb{C})$ of $P(h)$, for $h \in X$.

Lemma 5.8.- *The $G(\mathbb{R})$-equivariant map $P : h \mapsto P(h)$ identifies X with an open subset of the flag space $\check{X}$.*

Let $h \in X$ and let $\rho : G \to \mathrm{GL}(V)$ be a homogeneous representation of G such that $\mathrm{Ker}(\rho) \cap G^o$ is finite. The set X is identified with the set of Hodge structures on V, conjugate under $G(\mathbb{R})$ to h_V, while $\check{X}$ is identified with the set of filtrations on $V_{\mathbb{C}}$, conjugate under $G(\mathbb{C})$ to F_h. Since a homogeneous Hodge structure is determined by its Hodge filtration, the map P is injective. It is identified with the map

$$G(\mathbb{R})/H(h) \longrightarrow G(\mathbb{C})/P(h)$$

whose differential at the origin is

$$\mathrm{Lie}(G)/\mathrm{Lie}(H) \longrightarrow \mathrm{Lie}(G_{\mathbb{C}})/\mathrm{Lie}(P(h)),$$

i.e.

$$\mathrm{Lie}(G)/(F^0 \cap \overline{F^0})(\mathrm{Lie}(G)) \longrightarrow F^{-\infty}/F^0(\mathrm{Lie}(G_{\mathbb{C}})).$$

This differential is bijective, whence 5.8.

We have seen in passing that

$$H(h) = G(\mathbb{R}) \cap P(h). \tag{5.8.1}$$

5.9. Let V be a real representation of G, let $V(\check{X},\mathbb{R})$ be the constant sheaf on $\check{X}$ with value V, put $V(\check{X},\mathbb{C}) = \mathbb{C}\otimes V(\check{X},\mathbb{R})$ and $V(\check{X},\mathcal{O}) = \mathcal{O}\otimes_{\mathbb{C}} V(\check{X},\mathbb{C})$, and let ∇ be the natural integrable connection of $V(\check{X},\mathcal{O})$. The $G(\mathbb{R})$-equivariant map $h\mapsto h_V$ from X to the Hodge structures on V extends uniquely to a $G(\mathbb{C})$-equivariant, and hence holomorphic, map from $\check{X}$ to the filtrations of $V_{\mathbb{C}}$. This map corresponds to a filtration F of $V(\check{X},\mathcal{O})$, the Hodge filtration. The system $(V(\check{X},\mathcal{O}),\nabla,F)$ is $G(\mathbb{C})$-equivariant. The real structure $V(\check{X},\mathbb{R})$ of $V(\check{X},\mathbb{C})$ is $G(\mathbb{R})$-equivariant. Finally, at $h\in X$, F induces on $V_{\mathbb{C}}$ the Hodge filtration F_h.

5.10. The action of $G(\mathbb{C})$ on $\check{X}$ defines a morphism

$$\mathrm{Lie}(G)(\check{X},\mathcal{O})\longrightarrow T_{\check{X}}, \tag{5.10.1}$$

where $T_{\check{X}}$ is the tangent bundle. At $h\in X$, the kernel of (5.10.1) is $\mathrm{Lie}(P(h)) = F_h^0(\mathrm{Lie}(G_{\mathbb{C}}))$ (5.6), so that (5.10.1) factors through a $G(\mathbb{C})$-equivariant isomorphism

$$F^{-\infty}/F^0\big(\mathrm{Lie}(G)(\check{X},\mathcal{O})\big)\xrightarrow{\sim} T_{\check{X}}. \tag{5.10.2}$$

Let $T^1_{\check{X}}$ denote the holomorphic subbundle of $T_{\check{X}}$ which is the image of $F^{-1}(\mathrm{Lie}(G)(\check{X},\mathcal{O}))$. It is easily verified that if v is a local section of $T^1_{\check{X}}$, then

$$\nabla_v\big(F^p(V(\check{X},\mathcal{O}))\big)\subset F^{p-1}(V(\check{X},\mathcal{O})),$$

whatever the representation $\rho: G\to \mathrm{GL}(V)$ may be. The converse is true when $\mathrm{Ker}(\rho)\cap G^o$ is central.

5.11. Suppose now that X is a conjugacy class of positive morphisms from $\mathbb{S}$ to G. For $h\in X$, put $C_h = h(i)$, and denote by $K(h)$ the centralizer of C_h in $G(\mathbb{R})$. The group $K^o(h) = K(h)\cap G^o(\mathbb{R})$ is compact. Let V be a representation of G and let Ψ be a G-invariant bilinear form on V. If Ψ is a polarization of (V,h_V) for one $h\in X$, then Ψ is a polarization of (V,h_V) for every $h\in X$. Indeed, putting $\Phi_h(x,y) = \Psi(x,C_hy)$, one has

$$\Phi_{gh} = g(\Phi_h). \tag{5.11.1}$$

We shall then say that Ψ is a polarization of V. By hypothesis, every representation of G is polarizable.

5.12. Let Ψ be a polarization of the adjoint representation of G on $\mathrm{Lie}(G)$. By 5.10.2, Ψ induces, at each point $h\in X$, a positive definite Hermitian form on

$$(T_{\check{X}})_h\simeq F_h^{-\infty}/F_h^0(\mathrm{Lie}(G)_{\mathbb{C}})\simeq\bigoplus_{p<0}\mathrm{Lie}(G)^{p,-p}. \tag{5.12.1}$$

The polarization Ψ therefore defines a Hermitian structure, invariant under $G(\mathbb{R})$, on X.

If Z is the center of G, one already has

$$F^{-\infty}/F^0\big(\mathrm{Lie}(G/Z)(\check{X},\mathcal{O})\big)\xrightarrow{\sim} T_{\check{X}}, \tag{5.12.2}$$

and a polarization Ψ of $\mathrm{Lie}(G/Z)$ already defines a structure of Hermitian variety on X. One may take for Ψ the negative of the Killing form of G/Z.

Put

$$\begin{cases}(T^v_{\check{X}})_h = \bigoplus_{p<0}(\mathrm{Lie}(G/Z))^{2p,-2p},\\ (T^h_{\check{X}})_h = \bigoplus_{p<0}(\mathrm{Lie}(G/Z))^{2p-1,-2p+1}.\end{cases} \tag{5.12.3}$$

Thus one obtains an orthogonal C^∞ decomposition of the tangent bundle into two complex subbundles. One has

$$(T^1_{\check{X}})_h\subset(T^h_{\check{X}})_h\qquad(h\in X). \tag{5.12.4}$$

5.13. The subgroup $K(h)_{\mathbb{C}} \cap P(h)$ of $K(h)_{\mathbb{C}}$ is parabolic, so that $K(h)_{\mathbb{C}} \subset K(h) \cdot P(h)$ and $K(h) \cdot h = K(h)_{\mathbb{C}} \cdot h$ is a compact complex analytic subvariety of X. These varieties are the fibers of the C^∞, $G(\mathbb{R})$-equivariant fibration

$$X = G/H(h) \longrightarrow G/K(h).$$

It is easily verified that T_X^v is the tangent bundle to this fibration.

5.14. Recall that on a holomorphic Hermitian bundle F there exists a unique Hermitian connection ∇ such that $\nabla'' = \partial''$. The curvature R of the bundle is, by definition, the curvature of this connection. It is a form of type $(1,1)$ with values in the Lie algebra of the unitary group of the bundle.

If F is the constant bundle defined by a vector space F_o, and if the Hermitian form is written

$$\Phi(x,y) = \Phi_o(hx,y), \tag{5.14.1}$$

then, for $x : S \to F_o$ a C^∞ section of F, one has

$$\nabla x = \partial x + h^{-1}\partial' h \cdot x. \tag{5.14.2}$$

It follows that, for u and v two tangent fields,

$$R(u,v) = \partial_u(h^{-1}\partial'_v h) - \partial_v(h^{-1}\partial'_u h) - h^{-1}\partial'_{[uv]}h + [h^{-1}\partial'_u h, h^{-1}\partial'_v h]. \tag{5.14.3}$$

5.15. Let F be the tangent bundle of a Hermitian complex analytic variety. For u a tangent vector to S, denote by $u_{1,0}$ and $u_{0,1} = \overline{u_{1,0}}$ its components of type $(1,0)$ and $(0,1)$ in the complexification of the tangent bundle. The holomorphic curvature of S in the direction u is the real number

$$R(u) = \Phi(R(u_{1,0}, u_{0,1})(u), u)/\Phi(u,u)^2. \tag{5.15.1}$$

If X is endowed with the Hermitian structure 5.12, one has the fundamental result ([8], §9):

Theorem 5.16.- *There exists* $\lambda < 0$ *such that* $R(u) \leq \lambda$ *in every direction belonging to* T_X^h.

Proof (a sketch). By homogeneity it is enough to prove 5.16 at a point $e \in X$. Put $H = H(e)$, $P = P(e)$, let N^+ be the unipotent radical of P, and let N^- be the conjugate complex subgroup whose Lie algebra is

$$\mathfrak{n}^- = \bigoplus_{p<0} \mathrm{Lie}(G)^{p,-p}.$$

In the calculations which follow, near e, we identify X with N^- by the maps

$$X = G/H \hookrightarrow G(\mathbb{C})/P \hookleftarrow N^-.$$

Left translations on N^- permit one to identify the tangent bundle with $\mathfrak{n}^-$. We shall denote by t^+, t^o, and t^- the orthogonal projections of $\mathrm{Lie}(G)_{\mathbb{C}}$ onto $\mathfrak{n}^-$, $\mathrm{Lie}(H_{\mathbb{C}})$, and $\mathfrak{n}^+ = \mathrm{Lie}(N^+)$. For $g = np$ ($g \in G(\mathbb{R})$, $n \in N^-$, $p \in P$), the differential, at e, of the action of g on X is identified with

$$t^- \operatorname{ad} p : \mathfrak{n}^- \longrightarrow \mathfrak{n}^-.$$

The Hermitian structure is therefore written

$$\Phi_n(x,y) = \Phi_e(t^- \operatorname{ad} p^{-1} x, t^- \operatorname{ad} p^{-1} y).$$

Denoting by Ψ the polarization form and by C the Cartan involution associated to e, one has

$$\begin{aligned}
\Phi_n(x,y) &= \Psi(t^- \operatorname{ad} p^{-1} x,\ t^- C\, \overline{\operatorname{ad} p^{-1} y}) \\
&= \Psi(t^- \operatorname{ad} p^{-1} x,\ \operatorname{ad} C(\overline{p})^{-1} \cdot C\overline{y}) \\
&= \Psi(\operatorname{ad} C(\overline{p}) \cdot t^- \cdot \operatorname{ad} p^{-1} \cdot x,\ C\overline{y}) \\
&= \Phi_e(\operatorname{ad} C(\overline{p}) \cdot t^- \cdot \operatorname{ad} p^{-1} \cdot x,\ y).
\end{aligned}$$

One verifies that, for $n = \exp(u)$ with $u \sim 0$, to third order one has

$$p = \exp\bigl(\overline{u} - \tfrac{1}{2}t^o[u,\overline{u}] - t^+[u,\overline{u}]\bigr)$$

and

$$h = \operatorname{ad} C(\overline{p})\, t^- \operatorname{ad} p^{-1} = \exp(H),$$

for

$$H = t^- \circ \operatorname{ad}(Cu - \overline{u} + [u,\overline{u}]) - \tfrac{1}{2}[t^- \operatorname{ad} Cu, t^- \operatorname{ad}\overline{u}].$$

Under the hypotheses of 5.14, if $h = \exp(H)$ and if $H = 0$ at $s_o \in S$, one deduces from 5.14.3 that, at s_o,

$$\begin{aligned} R(u,v) = (\partial_u\partial'_v - \partial_v\partial'_u)H - \tfrac{1}{2}[\partial''_u H, \partial'_v H] + \tfrac{1}{2}[\partial''_v H, \partial'_u H] \\ + (\partial'_u\partial'_v - \partial'_v\partial'_u - \partial'_{[uv]})H. \end{aligned}$$

Applying this formula, one sees that the curvature is given, at e, by

$$R(u,v)_e = S(u,\overline{v}) - S(v,\overline{u}),$$

with

$$\begin{aligned} S(u,\overline{v}) &= -t^- \circ \operatorname{ad}[u,\overline{v}] + \tfrac{1}{2}[t^- \operatorname{ad} Cu, t^- \operatorname{ad}\overline{v}] + \tfrac{1}{2}[-t^- \operatorname{ad}\overline{v}, t^- \operatorname{ad} Cu] \\ &= [t^- \operatorname{ad} Cu, t^- \operatorname{ad}\overline{v}] - t^- \circ \operatorname{ad}[u,\overline{v}]. \end{aligned}$$

For any $u, v \in \mathfrak{n}^-$, if u_{10} and u_{01} are the components of type (10) and (01) of u, as in 5.15, then, by the invariance of Ψ,

$$\begin{aligned} \Phi(R(u_{10},u_{01})(v),v) &= \Psi(S(u,\overline{u})(v), C\overline{v}) \\ &= \Psi\bigl([Cu, t^-[\overline{u},v]] - t^-[\overline{u}[Cu,v]] - t^-[[u\overline{u}],v], C\overline{v}\bigr) \\ &= -\Psi\bigl(t^-[\overline{u},v],[Cu,C\overline{v}]\bigr) + \Psi\bigl([Cu,v],[\overline{u},C\overline{v}]\bigr) - \Psi\bigl([u,\overline{u}],[v,C\overline{v}]\bigr). \end{aligned}$$

For $u = v \in T^h_X$, one has $Cu = -u$ and

$$\begin{aligned} \Phi(R(u_{10},u_{01})(u),u) &= \Psi\bigl(t^-[u,\overline{u}],[u,\overline{u}]\bigr) + \Psi\bigl([u,\overline{u}],[v,\overline{v}]\bigr) \\ &= -\Phi\bigl(t^-[u,\overline{u}], t^-[u,\overline{u}]\bigr) - \Phi\bigl([u,\overline{u}],[u,\overline{u}]\bigr) < 0, \end{aligned}$$

which proves 5.16.

A generalization of Schwarz's lemma allows one to deduce from 5.16 (see [1], III, no. 10):

Corollary 5.17.- *Let $f : D \to X$ be a holomorphic map from the unit disk into X. Endow D with the hyperbolic metric d for which it has curvature -1, and X with the metric 4.12. Then, if $f(D)$ is tangent to T^h_X, one has*

$$d(f(x),f(y)) \leq |\lambda|\, d(x,y).$$

6. The Monodromy Theorem, after A. Borel.

6.1. Let S be a connected complex analytic variety, endowed with a base point $s_o \in S$, and let $p : \widetilde{S} \to S$ be its universal covering. Let H be a polarized family of Hodge structures of weight n on S. Denote by H_o the polarized Hodge structure which is the fiber of H at s_o.

The polarization Ψ is a bilinear form on $H_{o\mathbb{Q}}$. Denote by G^o the algebraic group, defined over $\mathbb{Q}$, which is the neutral component of the group of automorphisms of $(H_{o\mathbb{Q}}, \Psi)$; by $\varepsilon \in G^o$ the homothety of ratio $(-1)^n$; by G the group deduced, by the procedure 5.5, from (G^o, ε); and by Γ the discrete group of automorphisms of $(H_{o\mathbb{Z}}, \Psi)$. One has

$$T : \pi_1(S, s_o) \longrightarrow \Gamma. \tag{6.1.1}$$

6.2. For each point $s \in \widetilde{S}$, one has a natural isomorphism

$$H_{o\mathbb{Z}} \simeq H_{p(s),\mathbb{Z}},$$

and the Hodge structure $H_{p(s)}$ is identified with a Hodge structure on $H_{o,\mathbb{Z}}$, defined by $h(s) : \mathbb{S} \to G_{\mathbb{R}}$. The morphisms $h(s)$ are positive and mutually conjugate, because they form a connected continuous family and G is reductive. If X is the corresponding moduli space, defined in § 5, one therefore has a map

$$h : \widetilde{S} \longrightarrow X.$$

The family H is the pullback along h of the system of Hodge structures on X defined in (5.9) by the natural weight-n representation of G on H_o.

The morphism h is equivariant, via (5.5.1), for the actions of $\pi_1(S, s_o)$ and Γ on $\widetilde{S}$ and X.

By 3.4 and 3.6, the map h is holomorphic and $\mathrm{Im}(dh) \subset T^1_X$ (cf. 5.10).

6.3. Suppose that S is the punctured unit disk

$$D^* = \{z \mid 0 < |z| < 1\}$$

and take for the universal covering of D^* the Poincaré half-plane Y, endowed with

$$\exp(2\pi i y) : Y \longrightarrow D^*.$$

If $T \in \Gamma$ is the monodromy transformation (4.3) of $H_{\mathbb{Z}}$, the map h satisfies

$$h(y+1) = Th(y). \tag{6.3.1}$$

Choose $h_o \in X$. If $h(y) = g_y \cdot h_o$, the distance in X, for a metric (5.12), satisfies

$$d(h(y+1), h(y)) = d(Tg_y h_o, g_y h_o) = d(g_y^{-1} T g_y h_o, h_o). \tag{6.3.2}$$

By 5.17, if Y is endowed with its natural hyperbolic distance, the map h decreases distances (suitably normalized). In Y, one has

$$d(y, y+1) \sim (\mathrm{Im}(y))^{-1}$$

and therefore

$$d(g_y^{-1} T g_y h_o, h_o) \longrightarrow 0 \qquad \text{for } \mathrm{Im}(y) \to \infty. \tag{6.3.3}$$

The same argument applies to T^2; moreover, $T^2 \in G(\mathbb{R})$, and (6.2.3) implies that a limit of conjugates of T^2 lies in the compact subgroup $H(h_o)$ of $G(\mathbb{R})$. The eigenvalues of T^2 (and of T) are therefore all of absolute value 1. These eigenvalues form a set of algebraic integers stable under conjugation. Now,

Lemma 6.4.- *If all the complex conjugates of an algebraic integer* α *have absolute value* 1, *then* α *is a root of unity.*

The eigenvalues of T are therefore roots of unity, which proves

Monodromy Theorem 6.5 (A. Borel).- *Let* H *be a polarized family of Hodge structures on the punctured disk* D^*. *There exist integers* N *and* M *such that the monodromy transformation* T *satisfies*

$$(T^N - I)^M = 0.$$

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SHIMURA'S WORK(*)

by Pierre DELIGNE

Let X/Γ be the quotient of a Hermitian symmetric domain X by a discrete arithmetic group Γ (assumed to be defined by congruence conditions; see 1.7). By Baily and Borel, X/Γ is a complex algebraic variety. In papers [10] through [14], Shimura shows, among other things, that many of these varieties can be defined over number fields which he explicitly identifies. This is the subject of the present report.

To obtain clean statements, one is forced to use adelic language. This leads one to consider, for a given "congruence condition", a scheme over $\mathbb{C}$, a disjoint union of finitely many varieties of the type X/Γ (see §§ 1 and 2). In many cases, this scheme can be defined over a number field E, independent of the congruence condition under consideration. Its geometric connected components, however, will be defined over class fields of E. This phenomenon, although essential, will be neglected in the rest of this introduction.

In a small number of cases, X/Γ can be interpreted as the set of isomorphism classes of complex abelian varieties, endowed with some additional algebraic structures (polarizations, endomorphisms, structures on the points of order n). The solution, over $\mathbb{Q}$, of a corresponding moduli problem then provides a scheme M over an explicit number field F, whose complex points are X/Γ. We shall call M a "model" of X/Γ. The fundamental case is that of polarized abelian varieties of the principal series, endowed with a level-n structure (1.6, 1.11, 4.16, 4.17 and 4.21).

In other cases, X/Γ can be interpreted as the set of isomorphism classes of complex abelian varieties A endowed with some structures as above, and with some integral cohomology classes of type (p,p) $a_i \in H^{2p}(A \times \cdots \times A, \mathbb{C})$ (cf. Mumford [7]). In this case, the idea will be to construct a model M of X/Γ as a subscheme of a model M' already constructed for a variety X'/Γ'. Roughly speaking, one first constructs in M' the points of M corresponding to abelian varieties of C.M. type (= maximal complex multiplication). One then defines M as the closure of the set of these points. This construction rests on the detailed knowledge of abelian varieties of C.M. type provided by [16].

In the cases considered above, one has much information on the fields of definition of the points of M corresponding to abelian varieties of C.M. type. This makes it possible to give partial solutions to Hilbert's twelfth problem (cf. the introduction of [15] and 3.15, 3.16).

Looking at what these cases have in common, one arrives at the notion of a "canonical model" (§ 3). One shows that there is at most one "canonical model" for X/Γ (5.5). Descent techniques then make it possible to construct some canonical models which seem far from being solutions of moduli problems ([12] [13] [14] and [5]). For a description of these "strange models", see §§ 5 and 6.

We shall denote by $\mathbb{Z}/n(1)$ both the group scheme μ_n of nth roots of unity and the group of nth roots of unity in a fixed algebraically closed field $\bar{k}$. For $\bar{k} = \mathbb{C}$, the exponential map identifies $\mathbb{Z}/n(1)$ with $\mathbb{Z}/n$. The $\mathbb{Z}/n(1)$ form a projective system: the transition maps are $\varphi_{n,m} : x \longmapsto x^{n/m}$. We put

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \simeq \prod_p \mathbb{Z}_p, \qquad \widehat{\mathbb{Z}}(1) = \varprojlim_n \mathbb{Z}/n\mathbb{Z}(1),$$

$$\mathbb{A}^f = \mathbb{Q} \otimes \widehat{\mathbb{Z}} \simeq \prod_p{}' \mathbb{Q}_p \quad \text{(restricted product)}, \qquad \mathbb{A}^f(1) = \mathbb{Q} \otimes \widehat{\mathbb{Z}}(1) = \mathbb{A}^f \otimes_{\widehat{\mathbb{Z}}} \widehat{\mathbb{Z}}(1),$$

and we denote by $\mathbb{A}$ the ring of adeles of $\mathbb{Q}$.

We shall use the following notation:

$\pi_0(X)$ (for X a topological space): the set of connected components of X. In what follows, $\pi_0(X)$, endowed with the quotient topology from X, will be discrete or compact totally disconnected.

G^o: the connected component of the identity of a topological group G (or of the base point of a pointed topological space).

(*) Text written in March 1971

$G(K)$, G_K, $G \otimes_F K$: for G a scheme over F (for instance an algebraic group over F) and K an F-algebra, $G(K)$ denotes the set of K-valued points of G, and G_K or $G \otimes_F K$ denotes the scheme over K obtained from G by extension of scalars.
E^*: for E an algebra of finite rank over a field F, E^* denotes the algebraic group over F of invertible elements of E. For E a number field and $F = \mathbb{Q}$, $E^*(\mathbb{A})/E^*(\mathbb{Q})$ is the idele class group of E.

We shall say that an algebraic group G over $\mathbb{Q}$ satisfies the Hasse principle if $H^1(\mathbb{Q}, G) \underset{\text{dfn}}{=} H^1(\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), G(\overline{\mathbb{Q}}))$ injects into the product, over all places, of the $H^1(\mathbb{Q}_v, G_{\mathbb{Q}_v})$. We shall use the following theorems.
(0.1) Hasse principle: a simply connected semisimple group, without a factor of type E_8, satisfies the Hasse principle [3].
(0.2) A simply connected semisimple group G over a non-Archimedean local field K satisfies $H^1(K, G) = 0$ (a proof independent of the classification is announced in Bruhat–Tits [1]).
(0.3) Strong approximation theorem: let G be a simply connected semisimple group over a global field K. If G is K-simple, and if v is a place of K such that $G(K_v)$ is noncompact, then $G(K_v)G(K)$ is dense in $G(\mathbb{A})$ (for a proof independent of the classification, see Platonov [9]).
(0.4) Real approximation theorem: let G be a connected (linear) algebraic group over $\mathbb{Q}$. Then $G(\mathbb{Q})$ is dense in $G(\mathbb{R})$ (to prove this, one reduces easily to the case of tori, which is a corollary of ([4] 5.1)).

ğ 1. The adelic language.

1.1. Let G be a reductive group over $\mathbb{Q}$. We denote by G' the derived group of the identity component of G, by C the center of this identity component, and we put $T = G/G'$. There are exact sequences of algebraic groups

$$0 \longrightarrow C \longrightarrow G \longrightarrow G/C \longrightarrow 0, \tag{1.1.1}$$

$$0 \longrightarrow G' \longrightarrow G \xrightarrow{\nu} T \longrightarrow 0, \tag{1.1.2}$$

and the canonical maps from C to T and from G' to G/C are isogenies with kernel $C \cap G'$.
1.2. Let $\mathbb{S}$ be the real algebraic group of invertible elements of the $\mathbb{R}$-algebra $\mathbb{C}$. It is also the real algebraic group obtained from the multiplicative group $\mathbb{G}_m$ over $\mathbb{C}$ by Weil restriction of scalars from $\mathbb{C}$ to $\mathbb{R}$. One has $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$, and $\mathbb{S}_{\mathbb{C}} \simeq \mathbb{G}_{m,\mathbb{C}} \times \mathbb{G}_{m,\mathbb{C}}$. The group $\mathrm{Hom}(\mathbb{S}_{\mathbb{C}}, \mathbb{G}_{m,\mathbb{C}})$ of characters of $\mathbb{S}_{\mathbb{C}}$ has as a basis the characters z and $\bar{z}$ such that the composite

$$\mathbb{C}^* = \mathbb{S}(\mathbb{R}) \longrightarrow \mathbb{S}(\mathbb{C}) \xrightarrow{z,\bar{z}} \mathbb{C}^*$$

is respectively the identity and complex conjugation.

Let V be a real vector space. For any representation $h : \mathbb{S} \to \mathrm{GL}(V)$, we denote by V^{pq} the subspace of $V_{\mathbb{C}}$ on which $\mathbb{S}$ acts by the character $z^p \bar{z}^q$. The V^{pq} form a bigrading of $V_{\mathbb{C}}$, and the construction $h \mapsto V^{pq}$ identifies the representations of $\mathbb{S}$ in $\mathrm{GL}(V)$ with Hodge bigradings of $V_{\mathbb{C}}$, i.e. with bigradings of $V_{\mathbb{C}}$ such that $V^{pq} = \overline{V^{qp}}$. We shall denote by $F(h)$ the Hodge filtration of $V_{\mathbb{C}}$:

$$F(h)^p = \bigoplus_{p' \geq p} V^{p'q'}.$$

1.3. We denote by $r : \mathbb{G}_{m,\mathbb{C}} \to \mathbb{S}_{\mathbb{C}}$ the $\mathbb{C}$-homomorphism such that $(z^p \bar{z}^q) \circ r = (x \longmapsto x^p)$, and by $w : \mathbb{G}_{m,\mathbb{R}} \to \mathbb{S}$ the $\mathbb{R}$-homomorphism such that $(z^p \bar{z}^q) \circ w = (x \longmapsto x^{p+q})$. For a representation $h : \mathbb{S} \to \mathrm{GL}(V)$ and $v^{pq} \in V^{pq}$, one has

$$hr(x) \cdot v^{pq} = x^p \cdot v^{pq} \qquad \text{and}$$
$$hw(x) \cdot v^{pq} = x^{p+q} \cdot v^{pq}.$$

The composite $w : \mathbb{R}^* = \mathbb{G}_m(\mathbb{R}) \xrightarrow{w} \mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ is the evident inclusion.
1.4. Let $h : \mathbb{S} \to G_{\mathbb{R}}$ be a homomorphism. For any representation $\rho : G \to \mathrm{GL}(V)$ of G on a $\mathbb{Q}$-vector space V, ρh defines a bigrading of $V_{\mathbb{C}}$ (1.2). If V is faithful, and if G is the subgroup of $\mathrm{GL}(V)$ fixing some tensors s_i, the construction $h \mapsto (V^{pq})$ identifies the homomorphisms h with Hodge bigradings of $V_{\mathbb{C}}$ such that the s_i are of type $(0, 0)$.
1.5. Let $h_o : \mathbb{S} \to G_{\mathbb{R}}$ be a homomorphism satisfying the following conditions (cf. [7] and [2]; (1.5.2) is motivated by the Griffiths transversality theorem).

(1.5.1) The image of $h_o w : \mathbb{G}_m \to G_{\mathbb{R}}$ is central. We shall call $h_o w$ the weight of h_o;

(1.5.2) the Hodge bigrading on the complexification $\mathrm{Lie}(G)_{\mathbb{C}}$ of the adjoint representation space (1.2) is of type $(-1,1)+(0,0)+(1,-1)$;

(1.5.3) the automorphism $\mathrm{ad}\, h_o(i)$ of G, an involution by (1.5.1), induces a Cartan involution of G'.

Let K_∞ be the centralizer of h_o in $G(\mathbb{R})$; it contains the center of $G(\mathbb{R})$, and $K_\infty \cap G'(\mathbb{R})^o$ is a maximal compact subgroup of $G'(\mathbb{R})^o$, equal to the centralizer of $h_o(i)$ in $G'(\mathbb{R})^o$. The space $K_\infty\backslash G(\mathbb{R})$ is identified with the set X of conjugates of h_o by elements of $G(\mathbb{R})$. One checks that it has a unique complex structure such that, for every representation $\rho : G \to \mathrm{GL}(V)$, the filtration $F(\rho h)$ of $V_{\mathbb{C}}$ depends holomorphically on $h \in X$. This structure is right invariant under $G(\mathbb{R})$, and the connected components of X are Hermitian symmetric domains. We shall denote by X^o the connected component of h_o.

1.6. Example I. Let $\mathrm{Gp}(V)$ be the group of symplectic similitudes of a real vector space V endowed with a non-degenerate alternating form ψ. There is then a unique conjugacy class X of homomorphisms $h : \mathbb{S} \to \mathrm{Gp}(V)$ satisfying (1.5.1) and having weight -1, i.e. weight $hw : \mathbb{G}_m \to \mathrm{Gp}(V) : x \longmapsto$ the homothety with ratio x^{-1}. The space X is identified with the union of two Siegel half-spaces.

Construction 1.5 identifies the elements $h \in X$ with the decompositions $V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1}$ of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$ such that $V^{pq} = \overline{V^{qp}}$, $V^{0,-1}$ is totally isotropic for ψ, and, if C denotes the endomorphism of V which induces multiplication by i^{p-q} on V^{pq}, the symmetric form $\psi(x, Cy) = \psi(x, h(i)y)$ on V is definite (> 0 or < 0).

1.7. Let Γ be an arithmetic subgroup of $G(\mathbb{Q})$. We are interested in the case where Γ is defined by congruence conditions. If $V_{\mathbb{Z}}$ is an integral lattice in a faithful representation V of G, this means that there exists n such that Γ contains, with finite index, the group $\Gamma_n \subset G(\mathbb{Q})$ of $\gamma \in G(\mathbb{Q})$ such that $\gamma \in G(\mathbb{R})^o$, $\gamma V_{\mathbb{Z}} = V_{\mathbb{Z}}$, and $\gamma \equiv \mathrm{Id} \pmod{n}$. By Baily and Borel, the quotient X^o/Γ carries a natural quasi-projective scheme structure over $\mathbb{C}$. Let K_n be the compact open subgroup of $G(\mathbb{A}^f)$ consisting of those $k = (k_p)$ such that $k_p \cdot V_{\mathbb{Z}} \otimes \mathbb{Z}_p = V_{\mathbb{Z}} \otimes \mathbb{Z}_p$ and $k_p \equiv 1 \pmod{n}$ for $p \mid n$. The group Γ_n is then the intersection, inside $G(\mathbb{A})$, of $G(\mathbb{Q})$ and $G(\mathbb{R})^o \times K_n$.

1.8. Let K be a compact open subgroup of $G(\mathbb{A}^f)$. The space $K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}) = K\backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q}) = X \times (K\backslash G(\mathbb{A}^f))/G(\mathbb{Q})$ is then a finite disjoint union of spaces of the type considered in 1.7. It is therefore a quasi-projective scheme over $\mathbb{C}$. We shall denote it by ${}_K M_{\mathbb{C}}(G, h_o)$ (or simply ${}_K M_{\mathbb{C}}(G)$ or ${}_K M_{\mathbb{C}}$).

As K becomes smaller, these schemes form a projective system of schemes with finite and surjective transition morphisms

$$ {}_K M_{\mathbb{C}}(G, h_o) \qquad (K \to 0). \tag{1.8.1}$$

We shall denote by $M_{\mathbb{C}}(G, h_o)$ (or simply $M_{\mathbb{C}}(G)$, or $M_{\mathbb{C}}$) the quasi-compact and separated scheme (not of finite type for $G \neq \{1\}$) that is the projective limit of (1.8.1)

$$ M_{\mathbb{C}}(G, h_o) = \varprojlim_K K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}). \tag{1.8.2}$$

It should be regarded as an avatar of the projective system (1.8.1). The group $G(\mathbb{A}^f)$ acts on $M_{\mathbb{C}}$, or, if one prefers, on the pro-object defined by the projective system (1.8.1), but of course not on the individual ${}_K M_{\mathbb{C}}$ (cf. 3.2). One has

$$K\backslash M_{\mathbb{C}} = {}_K M_{\mathbb{C}},$$

so that

$$ M_{\mathbb{C}} = \varprojlim_K K\backslash M_{\mathbb{C}} \qquad (K \text{ compact open in } G(\mathbb{A}^f)). \tag{1.8.3}$$

This amounts to saying that $G(\mathbb{A}^f)$ acts continuously on $M_{\mathbb{C}}$.

The following construction will be useful for interpreting the ${}_K M_{\mathbb{C}}$ as moduli schemes.

1.9. Let H be an algebraic group over $\mathbb{Q}$ and V a faithful representation of H. We define an H-structure $\bar{\iota}$ on a vector space W to be an isomorphism from V to W, given modulo $H(\mathbb{Q})$: $\bar{\iota} \in \mathrm{Isom}(V, W)/H(\mathbb{Q})$. If $H(\mathbb{Q})$ is the group of automorphisms of a structure s of type Σ on V, then an H-structure on W is identified with a structure t of the same type on W, isomorphic to s (with $\bar{\iota} = \mathrm{Isom}((V, s), (W, t))$). Let W be endowed with an H-structure $\bar{\iota}$. The algebraic group $\iota H \iota^{-1} \subset \mathrm{GL}(W)$ ($\iota \in \bar{\iota}$) depends only on $\bar{\iota}$. We denote it by $H^{\bar{\iota}}$. Let A be a $\mathbb{Q}$-algebra. The set $\mathrm{Isom}_{\bar{\iota}}(W \otimes A, V \otimes A)$ of permissible isomorphisms from $W \otimes A$ to $V \otimes A$ is the set of isomorphisms k such that, for $\iota \in \bar{\iota}$, one has $k\iota \in H(A)$.

1.10. Let (G, h_o) be as in 1.5, let V be a faithful linear representation of G, let K be a compact open subgroup of $G(\mathbb{A}^f)$, and consider objects H of the following type: H consists of

(a) a vector space $H_{\mathbb{Q}}$ over $\mathbb{Q}$, endowed with a G-structure $\bar{\iota}$ (1.9);

(b) a homomorphism $h : \mathbb{S} \to G_{\mathbb{R}}^{\bar{\iota}}$ such that, for $\iota \in \bar{\iota}$, $\iota^{-1} h \iota : \mathbb{S} \to G_{\mathbb{R}}$ is conjugate to h_o;

(c) a class $\bar{k} \in K \backslash \mathrm{Isom}_{\bar{\iota}}(H \otimes \mathbb{A}^f, V \otimes \mathbb{A}^f)$ (1.9).

1.11. Example I (continued from 1.6). Let V be a $\mathbb{Q}$-vector space endowed with a non-degenerate alternating form ψ, let G be the group of symplectic similitudes of V, let $V_{\mathbb{Z}}$ be an integral lattice on which ψ has integral values and discriminant 1, and let h_o be as in 1.6. Take $K = \prod_\ell \mathrm{Gp}(V_{\mathbb{Z}} \otimes \mathbb{Z}_\ell)$. A datum (a) is interpreted as a vector space H of the same dimension as V, endowed with an alternating form ψ given up to a rational factor. A datum (b) is interpreted as a Hodge bigrading of $H_{\mathbb{C}}$, of type 1.6. A datum (c) is interpreted as an integral lattice $H_{\mathbb{Z}}$ in H, on which a rational multiple of ψ has integral values and discriminant 1: $\bar{k}$ is the set of k such that, for every ℓ, $V_{\mathbb{Z}} \otimes \mathbb{Z}_\ell = k_\ell(H_{\mathbb{Z}} \otimes \mathbb{Z}_\ell)$. We may normalize ψ by imposing the conditions that it have integral values and discriminant 1 on $H_{\mathbb{Z}}$, and that $\psi(x, Cx) > 0$ (cf. 1.6).

Let n be an integer, and let K_n be the subgroup of K consisting of those k such that $k \equiv 1 \pmod{n}$. For K_n, a datum (c) is interpreted as $H_{\mathbb{Z}}$ as above, together with a symplectic similitude $H_{\mathbb{Z}}/nH_{\mathbb{Z}} \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}$.

1.12. Under the hypotheses of 1.10, the points of ${}_K M_{\mathbb{C}}(G, h_o)$ are in bijective correspondence with the isomorphism classes of objects H of type 1.10. Indeed, let $H = (H_{\mathbb{Q}}, \bar{\iota}, h, \bar{k})$. Given H and $\iota \in \bar{\iota}$, we associate the morphism $\iota^{-1} h \iota : \mathbb{S} \to G_{\mathbb{R}}$, an element of $X \simeq K_\infty \backslash G(\mathbb{R})$, and $\bar{k}\iota$ in $K \backslash G(\mathbb{A}^f)$, hence an element of $K \backslash X \times G(\mathbb{A}^f)$; to H alone we associate only an element of ${}_K M_{\mathbb{C}}(G, h_o) = K \backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q}) = K_\infty \times K \backslash G(\mathbb{A})/G(\mathbb{Q})$, which uniquely determines the isomorphism class of H.

In certain cases (see 4.11), to give an object H as above amounts to giving an abelian variety endowed with some additional structures. One then obtains an interpretation of ${}_K M_{\mathbb{C}}(G)$ (and of $M_{\mathbb{C}}(G)$) as the coarse moduli scheme for abelian varieties of this type.

1.13. Let G^i $(i = 1, 2)$ be two reductive groups, let $h^i : \mathbb{S} \to G^i_{\mathbb{R}}$ be homomorphisms satisfying the conditions of 1.5, and let X^i be the set of conjugates of h^i by an element of $G^i(\mathbb{R})$. Let $G = G^1 \times G^2$, let $h : \mathbb{S} \to G_{\mathbb{R}}$ be the morphism (h^1, h^2), and let $X = X^1 \times X^2$ be the set of conjugates of h. There is an evident isomorphism

$$M_{\mathbb{C}}(G^1, h^1) \times M_{\mathbb{C}}(G^2, h^2) = M_{\mathbb{C}}(G, h). \tag{1.13.1}$$

For K^i compact open in $G^i(\mathbb{A}^f)$ and $K = K^1 \times K^2$, one likewise has

$${}_{K^1} M_{\mathbb{C}}(G^1, h^1) \times {}_{K^2} M_{\mathbb{C}}(G^2, h^2) = {}_K M_{\mathbb{C}}(G, h). \tag{1.13.2}$$

1.14. Let G^1, G^2, h^1, h^2 be as in 1.13. We shall denote by $u : (G^1, h^1) \to (G^2, h^2)$ a homomorphism $u : G^1 \to G^2$ such that $uX^1 \subset X^2$. Such a homomorphism defines

$$u : M_{\mathbb{C}}(G^1, h^1) \longrightarrow M_{\mathbb{C}}(G^2, h^2). \tag{1.14.1}$$

For K^i compact open in $G^i(\mathbb{A}^f)$, with $u(K^1) \subset K^2$, it defines

$$u(K^1, K^2) : {}_{K^1} M_{\mathbb{C}}(G^1, h^1) \longrightarrow {}_{K^2} M_{\mathbb{C}}(G^2, h^2). \tag{1.14.2}$$

Proposition 1.15. With the notation of 1.14, suppose that G^1 is a subgroup of G^2. Then, for every compact open subgroup K^1 of $G^1(\mathbb{A}^f)$, there exists a compact open subgroup $K^2 \supset K^1$ of $G^2(\mathbb{A}^f)$ such that $u(K^1, K^2)$ identifies ${}_{K^1} M_{\mathbb{C}}(G^1, h^1)$ with a closed subscheme of ${}_{K^2} M_{\mathbb{C}}(G^2, h^2)$.

If the K^i are small enough that the map from $X^i \times (K^i \backslash G^i(\mathbb{A}^f))$ to ${}_{K^i} M_{\mathbb{C}}(G^i, h^i)$ is étale, then the map $u(K^1, K^2)$ is finite and unramified. It suffices to treat this case, and to show that $u(K^1, K^2)$ is injective for $K^2 \supset K^1$ sufficiently small. The graph of the equivalence relation $u(K^1, K^2)(x) = u(K^1, K^2)(y)$ is a closed subscheme of $({}_{K^1} M_{\mathbb{C}}(G^1, h^1))^2$, decreasing with K^2, hence stationary for $K^2 \supset K^1$ sufficiently small. It is enough to show that

$$u(K^1) : {}_{K^1} M_{\mathbb{C}}(G^1, h^1) \longrightarrow \varprojlim_{K^2 \supset K^1} {}_{K^2} M_{\mathbb{C}}(G^2, h^2) = \mathrm{dfn}\, {}_{K^1} M_{\mathbb{C}}(G^2, h^2)$$

is injective. One has

$${}_{K^1} M_{\mathbb{C}}(G^1, h^1) = K^1 \backslash M_{\mathbb{C}}(G^1, h^1),$$
$${}_{K^1} M_{\mathbb{C}}(G^2, h^2) = K^1 \backslash M_{\mathbb{C}}(G^2, h^2),$$

so it is enough to prove the

Variant 1.15.1. $u : M_{\mathbb{C}}(G^1, h^1) \longrightarrow M_{\mathbb{C}}(G^2, h^2)$ is injective.

This assertion follows easily from the following two lemmas.

Lemma 1.15.2. Let $U^i \subset C^i(\mathbb{Q})$ be the group of units of the centre of G^i, and put $G^i(\mathbb{Q})^\wedge = \varprojlim_n \big(G^i(\mathbb{Q})/(U^i)^n\big)$. Then

$$M_{\mathbb{C}}(G^i, h^i) = K^i_\infty \backslash G^i(\mathbb{A})/G^i(\mathbb{Q})^\wedge.$$

This is a corollary of Chevalley's theorem, according to which the $(U^i)^n$ are congruence subgroups of U^i.

Lemma 1.15.3. $u : G^1(\mathbb{A}^f)/G^1(\mathbb{Q})^\wedge \longrightarrow G^2(\mathbb{A}^f)/G^2(\mathbb{Q})^\wedge$ is injective.

By left translation, it is enough to show that $u^{-1}(\{e\}) = \{e\}$: consider the diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & G^1(\mathbb{Q})^\wedge & \longrightarrow & G^2(\mathbb{Q})^\wedge & \longrightarrow & (G^2/G^1)(\mathbb{Q})^\wedge \\
 & & \downarrow & & \downarrow & & \downarrow\ ! \\
0 & \longrightarrow & G^1(\mathbb{A}^f) & \longrightarrow & G^2(\mathbb{A}^f) & \longrightarrow & (G^2/G^1)(\mathbb{A}^f) \\
 & & \downarrow & & \downarrow & & \\
 & & G^1(\mathbb{A}^f)/G^1(\mathbb{Q})^\wedge & \longrightarrow & G^2(\mathbb{A}^f)/G^2(\mathbb{Q})^\wedge & &
\end{array}$$

where $(G^2/G^1)(\mathbb{Q})^\wedge = \varprojlim_n (G^2/G^1)(\mathbb{Q})/(U^2/U^1)^n$.

§ 2. Connected components.

2.1. Let G be a connected reductive group over $\mathbb{Q}$. We use the notation of 1.1 and assume that G' admits no non-trivial invariant subgroup H (defined over $\mathbb{Q}$) and such that $H(\mathbb{R})$ is compact. This hypothesis does not exclude the possibility that $G'_{\mathbb{R}}$ has compact factors. We denote by $\widetilde{G}'$ the simply connected covering of the derived group G'.

Proposition 2.2. (i) The group $G(\mathbb{A}^f)$ acts transitively on the set $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$ of connected components of $G(\mathbb{A})/G(\mathbb{Q})$.

(ii) The group $\widetilde{G}'(\mathbb{A})$ acts trivially on $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$.

(iii) The quotient of $G(\mathbb{A})$ by the image of $\widetilde{G}'(\mathbb{A})$ is a commutative group.

Assertion (i) follows from (0.4): $G(\mathbb{A}) = G(\mathbb{A}^f)G(\mathbb{R})^0G(\mathbb{Q})$. Since $\widetilde{G}'$ is simply connected, $\widetilde{G}'(\mathbb{R})$ is connected. By (0.3), $\widetilde{G}'(\mathbb{R}) \cdot \widetilde{G}'(\mathbb{Q})$ is dense in $\widetilde{G}'(\mathbb{A})$, so that $\widetilde{G}'(\mathbb{A})/\widetilde{G}'(\mathbb{Q})$ is connected. For $x \in G(\mathbb{A})$ one has $\widetilde{G}'(\mathbb{A}) \cdot x = x \cdot \widetilde{G}'(\mathbb{A})$; hence the image of $\widetilde{G}'(\mathbb{A}) \cdot x$ in $G(\mathbb{A})/G(\mathbb{Q})$ is an image of $\widetilde{G}'(\mathbb{A})/\widetilde{G}'(\mathbb{Q})$: it is connected, and (ii) follows.

Let Z be the centre of $\widetilde{G}'$. Assertion (iii) follows from the fact that the commutator map $xyx^{-1}y^{-1} : G \times G \longrightarrow G$ has a factorization

$$G \times G \longrightarrow G/C \times G/C \longleftarrow \widetilde{G}'/Z \times \widetilde{G}'/Z \longrightarrow \widetilde{G}' \longrightarrow G.$$

Definition 2.3. We denote by $\pi(G)$ the commutative quotient of $G(\mathbb{A})$ (or $G(\mathbb{A}^f)$) that acts faithfully on $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$.

In other words, $\pi(G)$ is $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$, endowed with its "evident" commutative group structure. We recall the following theorem.

Theorem 2.4. Under the hypotheses of 2.1, if G' is simply connected, then the canonical homomorphism

$$\pi(G) = \pi_0(G(\mathbb{A})/G(\mathbb{Q})) \longrightarrow \pi_0(T(\mathbb{A})/T(\mathbb{Q})) \tag{2.4.1}$$

is bijective.

To prove the surjectivity of (2.4.1), by 2.2(i) applied to T it is enough to show that $G(\mathbb{A}^f)$ maps onto $T(\mathbb{A}^f)$. Since the kernel G' of $\nu : G \to T$ is connected, one knows that, for almost all p, $\nu(G(\mathbb{Z}_p)) = T(\mathbb{Z}_p)$; this formula has a meaning for almost all p, and is a corollary of Lang's theorem applied to G' reduced modulo p. It is therefore enough to prove that, for every p, $\nu(G(\mathbb{Q}_p)) = T(\mathbb{Q}_p)$, which follows from (0.2) applied to G'.

Theorem 2.4 is equivalent to the following more concrete statement.

Variant 2.5. For every compact open subgroup K of $G(\mathbb{A}^f)$, the map

$$\nu : \pi_0(K\backslash G(\mathbb{A})/G(\mathbb{Q})) \longrightarrow \pi_0(\nu(K)\backslash T(\mathbb{A})/T(\mathbb{Q})) \tag{2.5.1}$$

is bijective.

Indeed, $\pi_0(G(\mathbb{A})/G(\mathbb{Q})) = \varprojlim_K \pi_0(K\backslash G(\mathbb{A})/G(\mathbb{Q}))$, and similarly for T. Also, $\pi_0(K\backslash G(\mathbb{A})/G(\mathbb{Q})) = G(\mathbb{R})^0\backslash K\backslash G(\mathbb{A})/G(\mathbb{Q})$. We show that if $x, y \in G(\mathbb{A})$ have the same image in the right-hand side of (2.5.1), then they have the same image in the left-hand side. Left translation by y^{-1}, which replaces K by $y^{-1}Ky$, allows us to suppose that $y = e$. Modifying x by an element of $G(\mathbb{R})^0 \times K$, we may suppose that $\nu(x) = \tau$ with $\tau \in T(\mathbb{Q})$. By the Hasse principle (0.1) for G', the equation $\nu(y) = \tau$ has a solution in $G(\mathbb{Q})$. Correcting x by y, we may suppose that $x \in G'(\mathbb{A})$. By 2.2(ii), the image of x in $G(\mathbb{A})/G(\mathbb{Q})$ is in the neutral component, and a fortiori x and y have the same image in the left-hand side of 2.5.1.

This proof seems to use, via (0.1), that G' has no factor of type E_8. In fact, the class in $H^1(\mathbb{Q}, G')$ which obstructs the equation $\nu(y) = \tau$ comes from the centre of G', and the E_8 factors, being adjoint, count for nothing.

Proposition 2.6. One has $K_\infty = C(\mathbb{R}) \cdot \big(K_\infty \cap G'(\mathbb{R})^0\big)$. The group $K_\infty \cap G'(\mathbb{R})^0$ is connected, and

$$\pi_0(K_\infty) \xrightarrow{\sim} \mathrm{Im}\big(\pi_0(C(\mathbb{R})) \longrightarrow \pi_0(G(\mathbb{R}))\big),$$

Let $h' : \mathbb{S} \to G_{\mathbb{R}} \to (G/C)_{\mathbb{R}}$. The centralizer of h' in $(G/C)(\mathbb{R})$ is connected, because it is a compact group whose complexification is connected (as the centralizer of a torus). In particular, the image of K_∞ in $G/C(\mathbb{R})$ lies in the neutral component, the image of $G'(\mathbb{R})^0$, and $K_\infty = C(\mathbb{R}) \cdot \big(K_\infty \cap G'(\mathbb{R})^0\big)$. The group $K_\infty \cap G'(\mathbb{R})^0$ is connected as a maximal compact subgroup of a connected group. This proves 2.6.

2.7. The image of $\pi_0(K_\infty)$ in $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$ depends only on the conjugacy class of h, and $\pi_0(M_{\mathbb{C}}(G,h)) = \varprojlim_K \pi_0(K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}))$ is identified with $\pi(G)/\pi_0(K_\infty)$.

In particular, if G' is simply connected, one has

$$\pi_0(M_{\mathbb{C}}(G,h)) \xrightarrow{\sim} \pi_0(T(\mathbb{A})/T(\mathbb{Q}))/\pi_0(K_\infty). \tag{2.7.1}$$

More concretely, for every compact open subgroup K of $G(\mathbb{A}^f)$, $\pi_0({}_KM_{\mathbb{C}}(G,h))$ is then a principal homogeneous space under $\nu(K_\infty \times K)\backslash T(\mathbb{A})/T(\mathbb{Q})$.

§ 3. Models.

Let G be a reductive group over $\mathbb{Q}$, let $h : \mathbb{S} \to G_{\mathbb{R}}$ satisfy the conditions of 1.5, and let E be a field endowed with a complex embedding $\rho : E \to \mathbb{C}$. Put $M_{\mathbb{C}}(G) = M_{\mathbb{C}}(G,h)$.

Definition 3.1. A model over E of $M_{\mathbb{C}}(G)$ consists of

(a) a scheme M over E, endowed with a continuous action of $G(\mathbb{A}^f)$;

(b) an isomorphism m from $M \otimes_{E,\rho} \mathbb{C}$ to $M_{\mathbb{C}}(G)$ compatible with the action of $G(\mathbb{A}^f)$.

Let F be a finite extension of E, endowed with a complex embedding extending that of E. If $M_E(G,h)$ is a model of $M_{\mathbb{C}}(G,h)$ over E, we denote by $M_F(G,h)$ the model $M_E(G,h) \otimes_E F$ of $M_{\mathbb{C}}(G,h)$ over F.

Remark 3.2. Giving a scheme M over E, endowed with a continuous action of $G(\mathbb{A}^f)$, is equivalent to giving

(a) for every compact open subgroup K of $G(\mathbb{A}^f)$, a scheme ${}_KM$ over E;

(b) for compact open subgroups K, L of $G(\mathbb{A}^f)$ and for $x \in G(\mathbb{A}^f)$ such that $xKx^{-1} \subset L$, a homomorphism $J_{L,K}(x) : {}_KM \longrightarrow {}_LM$, these homomorphisms satisfying

(α) $J_{M,L}(y)J_{L,K}(x) = J_{M,K}(yx)$,

(β) $J_{K,K}(x) = \mathrm{Id}$ if $x \in K$,

(γ) for K normal in L, J defines an action of L/K on ${}_KM$, and $J_{L,K}(e)$ defines $(L/K)\backslash {}_KM \xrightarrow{\sim} {}_LM$.

One has

$$M = \varprojlim_K {}_KM \qquad \text{and} \tag{3.2.1}$$

$${}_KM = K\backslash M. \tag{3.2.2}$$

3.3. Let $M(G)$ be a model over E of $M_{\mathbb{C}}(G)$. For K compact open in $G(\mathbb{A}^f)$, put ${}_KM(G) = K\backslash M(G)$. The ${}_KM(G)$ are quasi-projective schemes over E, not necessarily geometrically connected. The structural isomorphism m induces isomorphisms

$${}_KM(G) \otimes_{E,\rho} \mathbb{C} \xrightarrow{\sim} {}_KM_{\mathbb{C}}(G). \tag{3.3.1}$$

3.4. Suppose that G satisfies the hypotheses of 2.1, and let (M, m) be a model of $M_{\mathbb{C}}(G, h)$ over E. Let $\overline{E}$ be the algebraic closure of E in $\mathbb{C}$. Since M is defined over E, the Galois group $\mathrm{Gal}(\overline{E}/E)$ acts on the profinite set

$$\pi_0(M \otimes_E \overline{E}) = \varprojlim_K \pi_0({}_K M \otimes_E \overline{E}) \xrightarrow[m]{\sim} \pi_0(M_{\mathbb{C}}(G, h)).$$

The group $G(\mathbb{A}^f)$ acts on $\pi_0(M \otimes_E \overline{E})$ through its quotient $\pi(G)/\pi_0(K_\infty)$, and $\pi_0(M \otimes_E \overline{E})$ is a principal homogeneous space under the commutative group $\pi(G)/\pi_0(K_\infty)$ (2.6). The action of $\mathrm{Gal}(\overline{E}/E)$ commutes with that of $G(\mathbb{A}^f)$, hence with that of $\pi(G)/\pi_0(K_\infty)$. The action of an element σ of $\mathrm{Gal}(\overline{E}/E)$ can therefore only be the translation defined by an element $\lambda(\sigma)$ of $\pi(G)/\pi_0(K_\infty)$, and λ is a homomorphism

$$\lambda_M : \mathrm{Gal}(\overline{E}/E) \longrightarrow \pi(G)/\pi_0(K_\infty). \tag{3.4.1}$$

Assume that E is a number field. Class field theory identifies the largest abelian quotient of $\mathrm{Gal}(\overline{E}/E)$ with $\pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q}))$, and identifies (3.4.1) with

$$\lambda_M : \mathrm{Gal}(\overline{E}/E)^{\mathrm{ab}} = \pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q})) \longrightarrow \pi(G)/\pi_0(K_\infty). \tag{3.4.2}$$

In the special case where G' is simply connected, the homomorphism (3.4.2) is identified, by (2.7.1), with

$$\lambda_M : \pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q})) \longrightarrow \pi_0(T(\mathbb{A})/T(\mathbb{Q}))/\pi_0(K_\infty). \tag{3.4.3}$$

Definition 3.5. The homomorphisms (3.4.1), (3.4.2), and (3.4.3) are called the reciprocity law of the model M. If G' is simply connected, and if λ_M comes from a homomorphism of algebraic groups over $\mathbb{Q}$ from E^* to T, the latter will also be called the "reciprocity law" of M.

3.6. There is a rule, applicable to each of the models constructed by Shimura, for determining E and the reciprocity law λ_M from G and h. The field E, for arbitrary G, and λ_M, for simply connected G', are described as follows.

3.7. The composite homomorphism hr (1.3)

$$hr : \mathbb{G}_m \xrightarrow{r} \mathbb{S}_{\mathbb{C}} \xrightarrow{h} G_{\mathbb{C}}$$

is a homomorphism over $\mathbb{C}$ between algebraic groups defined over $\mathbb{Q}$. The subfield E of $\mathbb{C}$ is the field of definition $E(G, h)$ of the conjugacy class of this homomorphism.

For G semisimple adjoint, the field $E(G, h)$ can be described as follows. Let Δ be the Dynkin diagram of $G_{\mathbb{C}}$, and let $|\Delta|$ be the set of vertices of Δ. The Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on Δ.

Let H be a maximal torus of $G_{\mathbb{C}}$, endowed with a system of simple roots $(\alpha_i)_{i\in|\Delta|}$. The α_i identify H with $\mathbb{G}_{m,\mathbb{C}}^{|\Delta|}$. For every homomorphism $u : \mathbb{G}_{m,\mathbb{C}} \to G_{\mathbb{C}}$, there is then a unique homomorphism $u' : \mathbb{G}_{m,\mathbb{C}} \to H = \mathbb{G}_{m,\mathbb{C}}^{\Delta}$: $x \longmapsto (x^{n_i})_{i\in|\Delta|}$ with $n_i \geq 0$, which is conjugate to u. The construction $u \mapsto \eta(u) = (n_i)_{i\in|\Delta|}$ identifies the set of conjugacy classes of maps from $\mathbb{G}_{m,\mathbb{C}}$ to $G_{\mathbb{C}}$ with the set of functions from $|\Delta|$ to $\mathbb{N}$. It follows that $E(G, h)$ is defined by the subgroup of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which stabilizes $\eta(hr)$.

Proposition 3.8. Let $G/C = \prod_{i=1}^{g} G_i$ be the decomposition of the adjoint group G/C into $\mathbb{Q}$-simple factors, and let h_i be the composite $h_i : \mathbb{S} \xrightarrow{h} G_{\mathbb{R}} \longrightarrow (G_i)_{\mathbb{R}}$.

(i) The subfield $E(G, h)$ of $\mathbb{C}$ is the compositum of the subfields $E(T, \nu \circ h)$ and $E(G_i, h_i)$ $(1 \leq i \leq g)$ of $\mathbb{C}$.

(ii) If the opposition involution of the Dynkin diagram of G_i respects $\eta(h_i r)$, then $E(G_i, h_i)$ is totally real. Otherwise, $E(G_i, h_i)$ is a totally imaginary quadratic extension of a totally real number field.

Assertion (i) is easy to check, and it is enough to prove (ii) for G $\mathbb{Q}$-simple and adjoint. Let Δ be its Dynkin diagram. The hypotheses 1.5 imply that $G_{\mathbb{R}}$ has a compact maximal torus; this implies that complex conjugation acts on Δ by the opposition involution ι. This involution is in the centre of the automorphism group of Δ.

Let E' be the Galois extension of $\mathbb{Q}$ defined by the subgroup of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which acts trivially on Δ. Then the image σ of complex conjugation is central in $\mathrm{Gal}(E'/\mathbb{Q})$; hence E' is totally real or a totally imaginary quadratic extension of a totally real field. Finally, σ is the identity on $E(G, h) \subset E'$ if and only if ι respects $\eta(hr)$, and the assertion follows.

3.9. We use the notation of 3.6. Suppose that G' is simply connected, that E contains $E(G, h)$, and that the complex embedding of E induces that of $E(G, h)$. The composite morphism

$$r''(h) : \nu\, h\, r : \mathbb{G}_{m,\mathbb{C}} \xrightarrow{r} \mathbb{S}_{\mathbb{C}} \xrightarrow{h} G_{\mathbb{C}} \xrightarrow{\nu} T_{\mathbb{C}}$$

depends only on the conjugacy class of hr. It is therefore defined over E, i.e. it arises, by extension of scalars from E to $\mathbb{C}$, from a homomorphism of algebraic groups over E, again denoted $r''(h) : \mathbb{G}_m \longrightarrow T_E$. Apply Weil restriction of scalars, and let $r'(h) : E^* \to T$ be the composite

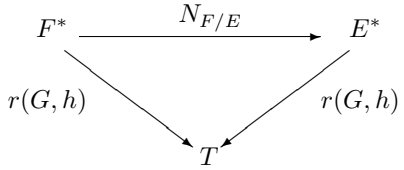
$$E^* = R_{E/\mathbb{Q}}(\mathbb{G}_{m,E}) \xrightarrow{R_{E/\mathbb{Q}}(r''(h))} R_{E/\mathbb{Q}}(T_E) \xrightarrow{N_{E/\mathbb{Q}}} T.$$

The reciprocity law of M is the inverse $r(G,h)$ of $r'(h)$:

$$r(G,h) = r'(h)^{-1} : E^* \longrightarrow T. \tag{3.9.1}$$

3.10. Example II. Let H be a torus and let $h : \mathbb{S} \to H_{\mathbb{R}}$. The conditions of 1.5 are automatically satisfied. The ${}_KM_{\mathbb{C}}(H,h)$ are finite sets. A finite reduced scheme over the field $E(H,h)$ (3.7) is identified with a Galois set. It follows that, up to unique isomorphism, there is a unique model of $M_{\mathbb{C}}(H,h)$ over $E(H,h)$ whose reciprocity law is given by 3.9.1. It is denoted $M(H,h)$.

3.11. Let F be a finite extension of E, endowed with a complex embedding extending that of E. The diagram



is commutative. If $M_E(G,h)$ is a model over E of $M_{\mathbb{C}}(G,h)$ whose reciprocity law is (3.9.1), it follows that $M_F(G,h)$ (3.1) again has (3.9.1) as its reciprocity law.

3.12. Let $u : (G^1,h^1) \to (G^2,h^2)$ be as in 1.14, and let $M^i_{E^i}(G^i,h^i)$ be a model of $M_{\mathbb{C}}(G^i,h^i)$ over E^i. Let E be the compositum of the subfields E^1 and E^2 of $\mathbb{C}$. With the notation of 3.1 one has

$$M^i_E(G^i,h^i) \otimes_E \mathbb{C} = M_{\mathbb{C}}(G^i,h^i) \qquad (i = 1,2).$$

It therefore makes sense to ask whether

$$u : M_{\mathbb{C}}(G^1,h^1) \longrightarrow M_{\mathbb{C}}(G^2,h^2)$$

is defined over E.

Definition 3.13. Let G be a connected reductive group over $\mathbb{Q}$ satisfying the condition of 2.1, $h : \mathbb{S} \to G_{\mathbb{R}}$ be a homomorphism satisfying the conditions of 1.5, and E be an extension of $E(G,h)$ (3.7) endowed with a complex embedding extending that of $E(G,h)$. A model $M_E(G,h)$ of $M_{\mathbb{C}}(G,h)$ over E is called weakly canonical if, for every torus $u : H \to G$ in G, endowed with $h' : \mathbb{S} \to H_{\mathbb{R}}$ such that uh' is the conjugate of h by an element of $G(\mathbb{R})$, the morphism (1.14)

$$M_{\mathbb{C}}(H,h') \longrightarrow M_{\mathbb{C}}(G,h)$$

is defined, in the sense of (3.10) and (3.12), over the compositum $E(H,h') \cdot E \subset \mathbb{C}$. A model $M_E(G,h)$ is called canonical if it is weakly canonical and if $E = E(G,h)$.

3.14. Translation. Let M be a canonical model of $M_{\mathbb{C}}(G,h)$, with reciprocity law λ_M. If G' is simply connected, then λ_M is given by 3.9.1 (5.6). For every compact open subgroup K of $G(\mathbb{A}^f)$, again denote by $\nu(K)$ the image of K in $\pi(G)/\pi_0(K_\infty)$, and let $E(K)$ be the extension of $E(G,h)$ defined by the idèle class group $\lambda_M^{-1}(\nu(K))$. One verifies that $E(K)$ is the field of definition of the neutral component ${}_KM^0_{\mathbb{C}}$ of ${}_KM_{\mathbb{C}}$, considered as a subscheme of ${}_KM_{\mathbb{C}} = {}_KM \otimes_{E,\rho} \mathbb{C}$.

By definition, ${}_KM^0_{\mathbb{C}}$ is the scalar extension to $\mathbb{C}$ of a subscheme ${}_KM'$ of ${}_KM \otimes_E E(K)$. The scheme ${}_KM'/E(K)$ is geometrically connected over $E(K)$, and one can also describe ${}_KM'$ as the union of the connected components (over E) of ${}_KM$ that, after extension of scalars from E to $\mathbb{C}$, contain the origin:

$$\begin{array}{ccccc}
{}_KM & \hookleftarrow & {}_KM' & \hookleftarrow & {}_KM^0_{\mathbb{C}} \\
\downarrow & & \downarrow & & \downarrow \\
 & & \operatorname{Spec} E(K) & \hookleftarrow & \operatorname{Spec} \mathbb{C} \\
 & & \downarrow & & \\
\operatorname{Spec} E & = & \operatorname{Spec} E & & .
\end{array}$$

Shimura usually states his results in terms of the system of varieties ${}_KM'/E(K)$ and the homomorphisms of type 2.2 between them. For a typical statement, see [14, p. 146]. For the translation, cf. [14, 2.7].

3.15. For $u : (H, h') \longrightarrow (G, h)$ as in 3.13, the image of $M_{\mathbb{C}}(H, h')$ in ${}_KM_{\mathbb{C}}(G, h)$ is a finite set. The hypothesis is that this subset of ${}_KM_{\mathbb{C}}(G, h)$ be defined over $E(H, h')$, that its points be defined over an abelian extension of $E(H, h')$, and that $\mathrm{Gal}(\overline{E(H, h')}/E(H, h'))$ act on them in the prescribed manner.

We shall call "special" the points of ${}_KM_{\mathbb{C}}(G, h)$ lying in the image of some $M_{\mathbb{C}}(H, h')$.

Suppose given a projective embedding $q : {}_KM^0_{\mathbb{C}} \longrightarrow \mathbb{P}^r$ defined over $E(K)$. It may be interpreted as $\widetilde{q} : X^0 \longrightarrow \mathbb{P}^r$ (cf. 1.7, 1.8). If $uh' \in X$ lies in X^0, then the coordinates of $\widetilde{q}(uh')$ generate over $E(K)E(H, h')$ an abelian extension of $E(H, h')$ that can be described explicitly as a class field.

3.16. The classical example of such a situation is provided by the moduli scheme of elliptic curves (corresponding to $G = \mathrm{GL}_2$, $K = \mathrm{GL}_2(\widehat{\mathbb{Z}})$, h as in 1.6), and by its points corresponding to elliptic curves with complex multiplication.

Let j be the modular invariant, viewed as a function on the Poincaré upper half-plane X^0. If $\tau \in X^0$ lies in an imaginary quadratic field K, let $L(\tau) = \mathbb{Z} \oplus \mathbb{Z}\tau \subset \mathbb{C}$, and let $\mathcal{O}(\tau) = \{\, k \in K \mid kL(\tau) \subset L(\tau) \,\}$. The field $K(j(\tau))$ is the abelian extension of K whose Galois group is the ideal class group of $\mathcal{O}(\tau)$. If σ in this Galois group corresponds to an invertible $\mathcal{O}(\tau)$-module L, and if $L(\tau') \simeq L(\tau) \otimes_{\mathcal{O}(\tau)} L$, then $j(\tau) = j(\tau')^\sigma$.

For more exotic explicit examples, see [15], pp. 45–46.

ğ 4. Abelian varieties.

4.1. Let k be an algebraically closed field of characteristic 0. Let A be an abelian variety over k. For $n \in \mathbb{N}^+$, denote by A_n the kernel of multiplication by n. The A_n form a projective system, with $\varphi_{nm,n} : A_{nm} \longrightarrow A_n$ given by $x \longmapsto x^m$, and we put

$$\widehat{T}(A) = \varprojlim A_n, \qquad \widehat{V}(A) = \mathbb{A}^f \otimes_{\widehat{\mathbb{Z}}} \widehat{T}(A).$$

The $\widehat{\mathbb{Z}}$-module $\widehat{T}(A)$ is the product of the Tate modules (= Weil representations) $T_\ell(A)$. If $k = \mathbb{C}$, then

$$\widehat{T}(A) = \widehat{\mathbb{Z}} \otimes H_1(A, \mathbb{Z}), \tag{4.1.1}$$

$$\widehat{V}(A) = \mathbb{A}^f \otimes H_1(A, \mathbb{Q}). \tag{4.1.2}$$

4.2. The category of abelian varieties up to isogeny over k is the category whose objects are the abelian varieties over k and in which $\mathrm{Hom}_{\mathrm{iso}}(A, B) = \mathrm{Hom}(A, B) \otimes \mathbb{Q}$. We denote by $A \otimes \mathbb{Q}$ the abelian variety up to isogeny "underlying" an abelian variety A. Every additive functor from the category of abelian varieties to a $\mathbb{Q}$-linear additive category factors through the functor $A \mapsto A \otimes \mathbb{Q}$. Thus the contravariant functor "dual variety" $A \mapsto A^*$ extends to abelian varieties up to isogeny. The functor $A \mapsto \widehat{V}(A)$ extends in the same way. Moreover, if A_0 is an abelian variety up to isogeny, it is equivalent to give B together with an isomorphism $B \otimes \mathbb{Q} = A_0$, or to give $\widehat{T}(B) \subset \widehat{V}(A_0)$.

4.3. Let A be an abelian variety over k. Let $\mathrm{NS}(A)$ denote the group of algebraic-equivalence classes of invertible sheaves on A. An effective polarization (resp. a polarization) of A is defined to be an element of $\mathrm{NS}(A)$, respectively of $\mathrm{NS}(A) \otimes \mathbb{Q}$, a positive multiple of which is defined by a projective embedding of A. A homogeneous polarization is an element of $(\mathrm{NS}(A) \otimes \mathbb{Q})/\mathbb{Q}^*$ which is the class of a polarization.

Recall that $\mathrm{NS}(A) \otimes \mathbb{Q}$ is identified, by a bijection $p \mapsto p'$, with the set of elements of $\mathrm{Hom}(A, A^*) \otimes \mathbb{Q}$ equal to their transpose. An isomorphism $u \in \mathrm{Hom}(A \otimes \mathbb{Q}, B \otimes \mathbb{Q})$ induces isomorphisms from $\mathrm{Hom}(A, A^*) \otimes \mathbb{Q}$ to $\mathrm{Hom}(B, B^*) \otimes \mathbb{Q}$, and from $\mathrm{NS}(A) \otimes \mathbb{Q}$ to $\mathrm{NS}(B) \otimes \mathbb{Q}$, and carries polarizations to polarizations. This makes it possible to speak of a polarization of an abelian variety up to isogeny.

4.4. A polarization p of A defines a positive involution on $\mathrm{End}(A) \otimes \mathbb{Q}$, $u \longmapsto p'^{-1}u^*p'$, which depends only on the homogeneous polarization $\mathbb{Q}^*.p$.

Let F be a product of totally real fields, and let $\rho : F \longrightarrow \mathrm{End}(A) \otimes \mathbb{Q}$.

Let $\mathrm{NS}_\rho(A) \subset \mathrm{NS}(A)$ be the set of p such that $p'.\rho(f) = \rho(f)^*p'$ for $f \in F$.

The vector space $\mathrm{NS}_\rho(A) \otimes \mathbb{Q}$ is endowed, for $f \in F$, with an action of F: $(f.p)' = p' \circ \rho(f) = \rho(f)^* \circ p'$. A weak polarization of A (relative to ρ, F) is an element of $(\mathrm{NS}_\rho(A) \otimes \mathbb{Q})/F^*$ which is the class of a polarization. The set of weak polarizations of A depends only on $A \otimes \mathbb{Q}$. The restriction to the centralizer of F of the involution on $\mathrm{End}(A) \otimes \mathbb{Q}$ defined by a polarization $p \in \mathrm{NS}_\rho(A)$ depends only on the weak polarization $F^*.p$.

4.5. If A_0^* is the dual of an abelian variety up to isogeny A_0, then $\widehat{V}(A_0)$ and $\widehat{V}(A_0^*)$ are in duality with values in $\mathbb{A}^f(1)$. If p is a polarization of A_0, the polarization form $\psi_p(x,y) = \langle x, p'(y)\rangle$ is a nondegenerate alternating form on the free $\mathbb{A}^f$-module $\widehat{V}(A_0)$, with values in $\mathbb{A}^f(1)$.

The facts recalled in 4.1 and 4.5 extend to abelian schemes over an arbitrary base.

4.6. Set $k = \mathbb{C}$. The additive functor $A \mapsto H_1(A, \mathbb{Q})$ extends to abelian varieties up to isogeny over $\mathbb{C}$ (4.2). For an abelian variety up to isogeny A_0, $H_1(A_0, \mathbb{Q})$ and $H_1(A_0^*, \mathbb{Q})$ are in duality. Via 4.1.2 and the isomorphism of $\mathbb{A}^f(1)$ with $\mathbb{A}^f$ defined by the exponential, this duality is compatible with the one considered in 4.5. The alternating form $\psi_p(x,y) = \langle x, p'(y)\rangle$ on $H_1(A_0, \mathbb{Q})$ defined by a polarization p of A_0 is again called the polarization form.

The Hodge structure on $H_1(A_0, \mathbb{Q})$ is the homomorphism $h : \underline{\mathbb{S}} \longrightarrow \mathrm{GL}(H_1(A_0, \mathbb{Q}))_{\mathbb{R}}$, such that $\underline{\mathbb{S}}$ acts on $H_1(A_0, \mathbb{Q}) \otimes \mathbb{C}$ through the characters z^{-1} and $\bar{z}^{-1}$, and such that $F(h)^\circ$ (1.5) is the kernel of the exponential map $H_1(A_0, \mathbb{Q}) \otimes \mathbb{C} \longrightarrow \mathrm{Lie}(A)$.

Theorem 4.7. (Riemann). The constructions

$$(A, p) \longmapsto (H_1(A, \mathbb{Q}), \psi_p, H_1(A, \mathbb{Z}), h),$$

$$(V, \psi, V_{\mathbb{Z}}, h) \longmapsto F(h)^\circ \backslash V \otimes \mathbb{C}/V_{\mathbb{Z}}$$

establish an equivalence between

(a) polarized abelian varieties over $\mathbb{C}$;

(b) systems consisting of a vector space V over $\mathbb{Q}$, a nondegenerate alternating form ψ on V, an integral lattice $V_{\mathbb{Z}}$ in V and a homomorphism h from $\underline{\mathbb{S}}$ into the group $\mathrm{Gp}(V)_{\mathbb{R}}$ of symplectic similitudes of $V \otimes \mathbb{R}$, of type 1.6 and such that

$$\psi(x, h(i)x) > 0 \qquad (x \in V \otimes \mathbb{R},\ x \neq 0).$$

Variant 4.8. It is equivalent to give an abelian variety up to isogeny A over $\mathbb{C}$, endowed with a homogeneous polarization, or to give a vector space over $\mathbb{Q}$, endowed with a nondegenerate alternating form ψ given up to scalar, together with $h : \underline{\mathbb{S}} \longrightarrow \mathrm{Gp}(V)$ of type 1.6.

4.9. Let L be a semisimple algebra with involution over $\mathbb{Q}$, and let V be a vector space over $\mathbb{Q}$, endowed with a faithful L-module structure and with a nondegenerate alternating form ψ such that

$$\psi(\ell x, y) = \psi(x, \ell^* y). \tag{4.9.1}$$

Let G denote the algebraic group over $\mathbb{Q}$ of L-linear symplectic similitudes of V. The group $G(\mathbb{Q})$ is the set of $g \in \mathrm{GL}_L(V)$ for which there exists $\mu(g) \in \mathbb{Q}^*$ satisfying

$$\psi(gx, gy) = \mu(g)\psi(x, y). \tag{4.9.2}$$

Suppose given a homomorphism $h_0 : \underline{\mathbb{S}} \longrightarrow G_{\mathbb{R}}$ such that, via h_0, $\underline{\mathbb{S}}$ acts on $V_{\mathbb{C}}$ through the characters z^{-1} and $\bar{z}^{-1}$ and such that the form $\psi(x, h_0(i)y)$ is symmetric and positive definite. The conditions (1.5.1) to (1.5.3) are then satisfied by (G, h_0), and the involution of L is positive.

4.10. Apply (1.11) to (G, h_0) and to the representation V of G. A G-structure $\bar{\iota}$ on a vector space H may be interpreted, by 1.8, as specifying on H an L-module structure and an alternating form ψ, given up to scalar, such that H, endowed with these structures, is isomorphic to V. Let $\bar{\iota}$ be a G-structure on H. A datum 1.9 (b), $h : \underline{\mathbb{S}} \longrightarrow G^1_{\mathbb{R}}$ on H, then defines, by 4.8 applied to (H, ψ, h), an abelian variety up to isogeny A, endowed with a homogeneous polarization $\bar{p}$, such that H is $H_1(A, \mathbb{Q})$, h is the Hodge structure on $H_1(A, \mathbb{Q})$, and ψ is the polarization form. Moreover, the L-module structure on H comes from $\rho : L \longrightarrow \mathrm{End}(A)$, and H, endowed with $\bar{\iota}$ and h, is determined by $(A, \bar{p}, \rho)$.

For a triple $(A, \bar{p}, \rho)$ to arise from such an H, it is necessary and sufficient that

(4.10.1) $H_1(A, \mathbb{Q})$, endowed with its L-module structure and with the polarization form (given up to scalar), is isomorphic to V;

(4.10.2) For an isomorphism $t : V \longrightarrow H_1(A, \mathbb{Q})$, $t^{-1}h_1 t$ is conjugate to h_0.

Let t be the map from L to $\mathbb{C}$ given by

$$t(\ell) = \mathrm{Tr}(\ell; V_{\mathbb{C}}/F^0(h_0)). \tag{4.10.3}$$

Scholium 4.11. Let K be a compact open subgroup of $G(\mathbb{A}^f)$. The points of ${}_KM_{\mathbb{C}}(G,h_0)$ correspond bijectively to isomorphism classes of abelian varieties up to isogeny A, endowed with

(a) $\rho : L \longrightarrow \mathrm{End}(A)$ such that
$$\mathrm{Tr}(\rho(\ell), \mathrm{Lie}(A)) = t(\ell) \qquad (\ell \in L),$$

(b) a homogeneous polarization $\bar{p}$ that induces the given involution of L,

(c) a class mod K, $\bar{k}$, of symplectic similitudes L-linear
$k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$,

and such that

(*) the conditions (4.10.1) and (4.10.2) are satisfied.

We apply 1.11 and 4.10, observing that a datum 1.9 (c) is identified with a datum (c), and that the conditions in (b) and (a) follow from (4.10.1) and (4.10.2), respectively.

4.12. Let K be a compact open subgroup of $G(\mathbb{A}^f)$, let $V_{\mathbb{Z}}$ be an integral lattice in V such that $V_{\widehat{\mathbb{Z}}} = V_{\mathbb{Z}} \otimes \widehat{\mathbb{Z}}$ is K-invariant, let $L_{\mathbb{Z}}$ be an order in L such that $L_{\mathbb{Z}} V_{\mathbb{Z}} \subset V_{\mathbb{Z}}$, let $\psi_{\mathbb{Z}}$ be a positive rational multiple of the alternating form of V, with integral values on $V_{\mathbb{Z}}$, and let $V'_{\mathbb{Z}}$ be the largest lattice in V such that $\psi_{\mathbb{Z}}(V_{\mathbb{Z}}, V'_{\mathbb{Z}}) \subset \widehat{\mathbb{Z}}$. Let n be an integer, and put $K_n = \{\, g \in G(\mathbb{A}^f) \mid (g-1)V_{\widehat{\mathbb{Z}}} \subset nV_{\widehat{\mathbb{Z}}} \,\}$. Assume that n is large enough that $K \supset K_n$.

If A is as in 4.11, then $k^{-1}(V_{\widehat{\mathbb{Z}}}) \subset \widehat{V}(A)$ does not depend on $k \in \bar{k}$, and defines an abelian variety B with $B \otimes \mathbb{Q} \simeq A$ and $\widehat{T}(B) = k^{-1}(V_{\widehat{\mathbb{Z}}}) \subset \widehat{V}(A)$. The action of L on A induces an action of $L_{\mathbb{Z}}$ on B. There exists a unique effective polarization p of B, with $p \in \bar{p}$, such that $k^{-1}(V'_{\widehat{\mathbb{Z}}})$ is the largest "lattice" $\widehat{T}'(B) \supset \widehat{T}(B)$ satisfying $\psi_p(\widehat{T}(B), \widehat{T}'(B)) \subset \widehat{\mathbb{Z}}(1)$. Finally, the set $\bar{k}$ of L-linear symplectic isomorphisms from $\widehat{T}(B)$ to $V_{\widehat{\mathbb{Z}}}$ is the inverse image of its image $\bar{k}_n$ in $\mathrm{Isom}(B_n, V_{\mathbb{Z}}/nV_{\mathbb{Z}})$.

This again permits the points of ${}_KM_{\mathbb{C}}(G,h_0)$ to be interpreted as corresponding to isomorphism classes of systems $(B, p, \rho, \bar{k}_n)$ consisting of

(a) a polarized abelian variety (B,p), with complex multiplication by $L_{\mathbb{Z}}$, satisfying 4.11 (a) and (b);

(b) a class mod K/K_n, $\bar{k}_n$, of isomorphisms $k_n : B_n \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}$, which can be lifted to L-linear symplectic isomorphisms $k : \widehat{T}(B) \longrightarrow V_{\widehat{\mathbb{Z}}}$, and which satisfy (4.10.1) and (4.10.2).

4.13. Let F denote the subalgebra of the center of L fixed by $*$. It is a product of totally real fields. There exists a unique alternating F-bilinear form ${}_F\psi$ on V such that $\mathrm{Tr}_{F/\mathbb{Q}}\, {}_F\psi = \psi$.

Let G_1 denote the group of L-linear symplectic similitudes of ${}_F\psi$: $G_1(\mathbb{Q})$ is the set of $g \in \mathrm{GL}_L(V)$ for which there exists $\mu(g) \in F^*$ satisfying

$$ {}_F\psi(gx, gy) = \mu(g)\, {}_F\psi(x,y), \quad \text{i.e.} \tag{4.13.1}$$
$$ \psi(gx, gy) = \psi(\mu(g)x, y). \tag{4.13.2}$$

As above, we verify the

Variant 4.14. Let K be a compact open subgroup of $G_1(\mathbb{A}^f)$. The points of ${}_KM_{\mathbb{C}}(G_1,h_0)$ correspond bijectively to isomorphism classes of abelian varieties up to isogeny A, endowed with

(a) $\rho : L \longrightarrow \mathrm{End}(A)$ as in 4.11,

(b) a weak polarization (relative to F) $\bar{p}$ which induces the given involution of L,

(c) a class modulo K of L-linear symplectic similitudes $k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$ (for the $F \otimes \mathbb{A}^f$-bilinear forms on the two sides),

and such that

(*) conditions analogous to (4.10.1) and (4.10.2) are satisfied.

4.15. One can make 4.11 and 4.14 more precise by showing that ${}_KM_{\mathbb{C}}(G,h)$ is the coarse moduli scheme of the evident functor on schemes of finite type over $\mathbb{C}$.

4.16. Example I (continuation of 1.6, 1.11). Consider the special case of 4.9 in which $L = \mathbb{Q}$ and $\dim(V) = 2g$. Then $G = \mathrm{Gp}(V)$ (the group of symplectic similitudes). For $h_0 : \mathbb{S} \longrightarrow G_{\mathbb{R}}$ of type 1.6, one has $E(\mathrm{Gp}, h_0) = \mathbb{Q}$.

Let $V_{\mathbb{Z}}$, n, and K_n be as in 1.11. For every scheme S, let $F(S)$ be the set of isomorphism classes of principally polarized abelian schemes A/S endowed with a symplectic similitude $k_n : A_n \xrightarrow{\sim} (V_{\mathbb{Z}}/nV_{\mathbb{Z}})_S$. For $n \geq 3$, the classified objects have no nontrivial automorphisms, and the functor F is represented by a $\mathbb{Q}$-scheme ${}_{K_n}M$ [6]. By 1.11, 4.11, and 4.12, there is an isomorphism

$$m_n : {}_{K_n}M(\mathbb{C}) \xrightarrow{\sim} {}_{K_n}M_{\mathbb{C}}(\mathrm{Gp}, h_0).$$

In the language of 3.1, one has

Proposition 4.17. The $\mathbb{Q}$-scheme $M(\mathrm{Gp}, h_0) = \varprojlim_{K_n} {}_{K_n}M$ is a model over $\mathbb{Q} = E(\mathrm{Gp}, h_0)$ of $M_{\mathbb{C}}(\mathrm{Gp}, h_0)$.

4.18. Consider the special case of 4.9 in which V is a cyclic L-module. The groups G and G_1 are then contained in the center of L^*.

Let $\bar{E}$ be an algebraic closure of $E(G, h)$, and let M^+ be the set of isomorphism classes $[A, \rho, k, \bar{p}]$ of objects $(A, \rho, k, \bar{p})$ consisting of

(a) an abelian variety up to isogeny $A/\bar{E}$, endowed with a homogeneous polarization $\bar{p}$ and with $\rho : L \to \mathrm{End}(A)$ such that, with the notation of 4.10.3, for $\ell \in L$ one has

$$\mathrm{Tr}(\rho(\ell), \mathrm{Lie}(A)) = t(\ell) \in E(G, h);$$

(b) an $L \otimes \mathbb{A}^f$-linear isomorphism $k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$.

The abelian variety A is therefore of CM type. We refer to [16] for the proof of the following fundamental theorem.

Theorem 4.19. (Shimura–Taniyama). The Galois group $\mathrm{Gal}(\bar{E}/E(G, h))$ acts on M^+ through its largest abelian quotient $\pi_0(E(G, h)^*(\mathbb{A})/E(G, h)^*(\mathbb{Q}))$. For $\underline{e} \in E(G, h)^*(\mathbb{A})$, with image $\varphi(\underline{e})$ in abelianized Galois and with component $\underline{e}^f$ in $E(G, h)^*(\mathbb{A}^f)$, one has

$$\varphi(\underline{e})\,([A, \rho, k, \bar{p}]) = [A, \rho, r(G, h)(e^f) \cdot k, \bar{p}]. \tag{4.19.1}$$

Let τ be an embedding of $\bar{E}$ into $\mathbb{C}$ extending that of $E(G, h)$, let K be a compact open subgroup of $G(\mathbb{A}^f)$, and let ${}_KM(G, h)(\bar{E})$ be the set of isomorphism classes of objects $(A, \rho, \bar{k}, \bar{p})$ where

(a) A, ρ, and $\bar{p}$ are as above;

(b) $\bar{k}$ is a class modulo K of $L \otimes \mathbb{A}^f$-linear symplectic similitudes $k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$;

(c) the vector space $H_1(A_{\mathbb{C}}, \mathbb{Q})$, where $A_{\mathbb{C}}$ is defined via τ, endowed with the action of L and with the polarization form given up to scalar, is isomorphic to V.

By 4.19, condition (c) is independent of the choice of τ, so that the finite set ${}_KM(G, h)(\bar{E})$ is endowed with an action of $\mathrm{Gal}(\bar{E}/E(G, h))$; as a Galois set, it is identified with the set of $\bar{E}$-points of a finite étale $E(G, h)$-scheme ${}_KM(G, h)$. By 4.14, ${}_KM(G, h)(\mathbb{C}) = G(\mathbb{R}) \times K \backslash G(\mathbb{A})/G(\mathbb{Q})$, and by 4.19 one has:

Proposition 4.20. The $E(G, h)$-scheme $M(G, h) = \varprojlim_K {}_KM(G, h)$ is a canonical model of $M_{\mathbb{C}}(G, h)$.

Theorem 4.21. The model $M(\mathrm{Gp}, h_0)$ constructed in 4.17 is a canonical model.

Let (V, ψ) be as in 4.16, and let L, $\rho : L \to \mathrm{End}(V)$, G, and $h : \mathbb{S} \to G_{\mathbb{R}}$ be as in 4.18. We have $u : (G, h) \longrightarrow (\mathrm{Gp}, h_0)$. Let $K \subset G(\mathbb{A}^f)$ be such that $u(K) \subset K_n$ (4.16). If $\bar{E}$ is an algebraic closure of $E(G, h)$, define

$$u : {}_KM(G, h)(\bar{E}) \longrightarrow {}_{K_n}M(\mathrm{Gp}, h_0)(\bar{E})$$

by "forgetting the complex multiplication", associating to $[A, \rho, \bar{k}, \bar{p}]$ the triple consisting of

(a) the abelian variety B, equipped with an isomorphism $B \otimes \mathbb{Q} \simeq A$ and such that

$$\widehat{T}(B) = k^{-1}(V_{\widehat{\mathbb{Z}}}) \subset \widehat{V}(A) \qquad (k \in \bar{k});$$

(b) the polarization $\bar{p}$ of B

(c) the isomorphism $k : B_n \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}$ $(k \in \bar{k})$.

The maps u define a morphism of models

$$u : M(G, h) \longrightarrow M(\mathrm{Gp}, h_0) \otimes E(G, h).$$

We complete the proof by showing ([8]) that for every $u : (H, h') \longrightarrow (\mathrm{Gp}, h_0)$ as in 3.13, there exists $v : (G, h) \longrightarrow (\mathrm{Gp}, h_0)$ as above (for a suitable L) such that $uh' = vh$.

Variant 4.22. Let F be a totally real number field, let $n = [F : \mathbb{Q}]$, and let $\mathrm{Gp}_{2g}(F)$ be the group of symplectic similitudes of a symplectic F-vector space of dimension $2g$. Let $h_0 : \mathbb{S} \longrightarrow \mathrm{Gp}_{2g}(F)_{\mathbb{R}} \simeq \mathrm{Gp}_{2g}(\mathbb{R})^n$ be as in 1.6. One can construct a canonical model of $M_{\mathbb{C}}(\mathrm{Gp}_{2g}(F), h_0)$ from moduli of weakly polarized abelian schemes.

5. Construction techniques.

Definition 3.13 is usable only because there are "many" special points.

Theorem 5.1. Let G and h be as in 3.13. For every finite extension F of $E(G, h)$, there exists $(H, h', u : H \hookrightarrow G)$ as in 3.13 such that the extension $E(H, h')$ of $E(G, h)$ is linearly disjoint from F.

Let $Y = \mathrm{Hom}(\mathbb{G}_m, G)/G$ be the scheme of conjugacy classes of morphisms from $\mathbb{G}_m$ to G. The Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on the (infinite discrete) set $Y(\mathbb{C})$ of conjugacy classes of morphisms from $\mathbb{G}_{m\mathbb{C}}$ to $G_{\mathbb{C}}$. Let Y_o be the finite subscheme of Y such that $Y_o(\mathbb{C})$ is the Galois orbit of the class of hr. It is the spectrum of $E(G, h)$.

Let $V' = \mathrm{Lie}(G)$, regarded as an affine space over $\mathbb{Q}$, and let V be the open subscheme of regular elements of the Lie algebra. For v in V, let T_v denote the centralizer of v (a maximal torus). Let W be the moduli scheme of maximal tori T of G, endowed with $s : \mathbb{G}_m \longrightarrow T$ and $v \in \mathrm{Lie}(T)$ such that

(a) v is regular in $\mathrm{Lie}(G)$;

(b) the class of $s : \mathbb{G}_m \to G$ lies in Y_o.

The morphism $f : W \longrightarrow V : (T, s, v) \longmapsto v$ is finite étale and surjective (note that $T = T_v$). Let p be the morphism from W to $Y_o : (T, s, v) \longmapsto$ (the class of s).

(5.1.1)

$$\begin{array}{ccc} W & \xrightarrow{\quad f \quad} & V \\ \Big\downarrow{\scriptstyle p} & & \Big\downarrow \\ \mathrm{Spec}(E(G,h)) & \longrightarrow & \mathrm{Spec}(\mathbb{Q}) \end{array}$$

Lemma 5.1.2. The scheme W is irreducible, and the geometric fibers of p are (geometrically) irreducible.

Since Y_o is irreducible (the spectrum of an extension of $\mathbb{Q}$), it suffices to prove the second assertion. This follows from the fact that

(a) since G is connected, the scheme over $\mathbb{C}$ of homomorphisms from $\mathbb{G}_{m\mathbb{C}}$ to $G_{\mathbb{C}}$ conjugate to a given homomorphism is irreducible;

(b) for a given $i : \mathbb{G}_{m\mathbb{C}} \to G_{\mathbb{C}}$, by the theorem on conjugacy of maximal tori applied to the (connected) centralizer of $i(\mathbb{G}_{m\mathbb{C}})$, the scheme of maximal tori of $G_{\mathbb{C}}$ containing $i(\mathbb{G}_{m\mathbb{C}})$ is irreducible.

Let $U \subset V(\mathbb{R})$ be the set of $v \in V(\mathbb{R})$ such that $(T_v/C)(\mathbb{R})$ is compact. For the usual topology, U is open.

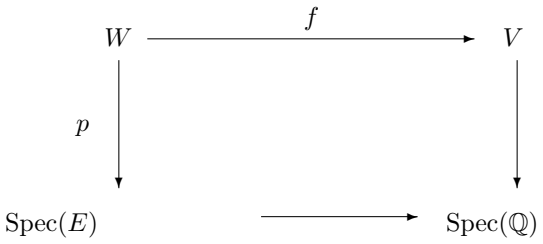
For $v \in U$ and $w \in W(\mathbb{C})$ such that $f(w) = v \in V(\mathbb{R}) \subset V(\mathbb{C})$, there exists a unique homomorphism $h(w) : \mathbb{S} \longrightarrow T_v$ such that $h(w) \circ r = s_w$: after extension of scalars to $\mathbb{C}$

$$h(w) = s_w \circ z \cdot \overline{s}_w \circ \overline{z}.$$

Let $v \in U$. By the theorem on conjugacy of maximal compact subgroups in $G/C(\mathbb{R})^\circ$, there exists $g \in G(\mathbb{R})$ (indeed one may take $g \in G'(\mathbb{R})^\circ$) such that $gT_vg^{-1} \subset K_\infty$. Since T_v is a maximal torus, gT_vg^{-1} contains the center of K_∞. It follows that there exists $w \in W(\mathbb{C})$ such that $f(w) = v$ and such that $h(w)$ is conjugate to h.

We conclude the proof of 5.1 by applying to diagram (5.1.1) and to U the following variant of Hilbert's irreducibility theorem, whose proof I owe to J.-P. Serre.

Lemma 5.13. Let E be a finite extension of $\mathbb{Q}$, and let there be a commutative diagram



in which

(i) V is a Zariski open subset of an affine space over $\mathbb{Q}$

(ii) the geometric fibers of p are irreducible, and W is reduced.

(iii) f is quasi-finite and dominant.

Then, for every open set $U \subset V(\mathbb{R})$ (in the usual topology) and every extension F of E, there exists $v \in V(\mathbb{Q}) \cap U$ such that $f^{-1}(v)$ is the spectrum of an extension of E linearly disjoint from F.

Let $W' = W \otimes_E F$. The hypotheses of 4.12 are again satisfied for W'/F and $f' : W' \longrightarrow W \longrightarrow V$, and one must construct $v \in V(\mathbb{Q}) \cap U$ such that $f'^{-1}(v) = f^{-1}(v) \otimes_E F$ is the spectrum of a field. This reduces us to the case $E = F$, which follows from Hilbert's irreducibility theorem (cf. Lang, *Diophantine Geometry*, Chap. VIII).

Proposition 5.2. For $x \in M_{\mathbb{C}}(G,h)$ and K compact open in $G(\mathbb{A}^f)$, the image in ${}_K M_{\mathbb{C}}(G,h)$ of $G(\mathbb{A}^f) \cdot x \subset M_{\mathbb{C}}(G,h)$ is dense.

This is a consequence of (0.4). The reader will verify:

Proposition 5.3. Let E be a field endowed with a complex embedding ρ, X a scheme over E, and Y a reduced closed subscheme of $X_{\mathbb{C}}$. Suppose there exist finite extensions E_i of E in $\mathbb{C}$, schemes $M_{i,\alpha}/E_i$ and E_i-morphisms $v_{i,\alpha} : M_{i,\alpha} \longrightarrow X \otimes_E E_i$, or $v_i : \coprod_\alpha M_{i,\alpha} \longrightarrow X \otimes_E E_i$, such that $E = \bigcap_i E_i$, that $v_i(\coprod_\alpha M_{i,\alpha} \otimes_{E_i} \mathbb{C}) \subset Y$ and that $v_i(\coprod_\alpha M_{i,\alpha} \otimes_{E_i} \mathbb{C})$ is dense in Y. Then Y is defined over E.

Corollary 5.4. Let $u : (G^1,h^1) \longrightarrow (G^2,h^2)$ be as in 1.14. Let $M_E(G^1,h^1)$ and $M_E(G^2,h^2)$ be weakly canonical models of the $M_{\mathbb{C}}(G^i,h^i)$ over a number field E ($E(G^i,h^i) \subset E \subset \mathbb{C}$). Then $v : M_{\mathbb{C}}(G^1,h^1) \longrightarrow M_{\mathbb{C}}(G^2,h^2)$ is defined over E (cf. 3.12)

Let $K^i \subset G^i(\mathbb{A}^f)$ be compact open subgroups such that $v(K^1) \subset K^2$. We prove that $v(K^1,K^2) : {}_{K^1}M_{\mathbb{C}}(G^1,h^1) \longrightarrow {}_{K^2}M_{\mathbb{C}}(G^2,h^2)$ is defined over E. Let $u : (H,h') \to (G^1,h^1)$ be as in 3.13 and set $F = E \cdot E(H,h')$. For $g \in G^1(\mathbb{A}^f)$, one has F-morphisms

$$a(g): \quad M_F(H,h') \longrightarrow M_F(G^1,h^1) \xrightarrow{g} M_F(G^1,h^1) \longrightarrow {}_{K^1}M_F(G^1,h^1),$$

$$b(g): \quad M_F(H,h') \longrightarrow M_F(G^2,h^2) \xrightarrow{v(g)} M_F(G^2,h^2) \longrightarrow {}_{K^2}M_F(G^2,h^2),$$

and, after extension of scalars to $\mathbb{C}$, $v(K^1,K^2) \circ a(g) = b(g)$.

By 5.1 and 5.2, one can apply 5.3 to $X = {}_{K^1}M_E(G^1,h^1) \times {}_{K^2}M_E(G^2,h^2)$, to the graph Y of v, to the $E_i = E \cdot E(H,h')$ for variable (H,h',u), and to $(a(g),b(g)) : M_{E_i}(H,h') \longrightarrow X_{E_i}$: it follows that the graph of v is defined over E.

In the special case $(G^1,h^1) = (G^2,h^2)$ and $u = \mathrm{Id}$, 5.4 says:

Corollary 5.5. Up to (unique) isomorphism, there exists at most one weakly canonical model of $M_{\mathbb{C}}(G,h)$ over E ($E(G,h) \subset E \subset \mathbb{C}$).

Corollary 5.6. If G' is simply connected, the reciprocity law of a weakly canonical model of $M_{\mathbb{C}}(G,h)$ over E ($E(G,h) \subset E \subset \mathbb{C}$) is given by 3.9.1.

The proof is the same as in the special case $v : (G,h) \to (T,vh)$ of 5.4.

Corollary 5.7. Let $v : (G^1,h^1) \hookrightarrow (G^2,h^2)$ be as in 1.15. If $M_{\mathbb{C}}(G^2,h^2)$ admits a weakly canonical model over E, with $E(G^2,h^2) \subset E(G^1,h^1) \subset E \subset \mathbb{C}$, then $M_{\mathbb{C}}(G^1,h^1)$ admits a weakly canonical model over E.

Let K^i be a compact open subgroup of $G^i(\mathbb{A}^f)$. Using 1.15, we are reduced to showing only that if $v(K^1) \subset K^2$ and ${}_{K^1}M_{\mathbb{C}}(G^1,h^1) \hookrightarrow {}_{K^2}M_{\mathbb{C}}(G^2,h^2)$, then the subscheme ${}_{K^1}M_{\mathbb{C}}(G^1,h^1)$ of ${}_{K^2}M_{\mathbb{C}}(G^2,h^2) = {}_{K^2}M_E(G^2,h^2) \otimes_E \mathbb{C}$ is defined over E.

Let $u : (H, h') \to (G^1, h^1)$ be as in 3.13, and for $g \in G^1(\mathbb{A}^f)$ let $a(g)$ be the composite homomorphism defined over $F = E \cdot E(H, h')$:

$$a(g) : M_F(H, h') \longrightarrow M_F(G^1, h^1) \xrightarrow{v(g)} M_F(G^2, h^2) \longrightarrow {}_{K^2}M_F(G^2, h^2).$$

By 5.1 and 5.2, one concludes by applying 5.3 to $X = {}_{K^2}M_E(G^2, h^2)$, to $Y = {}_{K^1}M_{\mathbb{C}}(G^1, h^1)$, to the $E_i = E \cdot E(H, h')$ for variable (H, h', u), and to $a(g) : M_{E_i}(H, h') \longrightarrow X_{E_i}$.

Example 5.8. For (G, h_0) as in 4.9, one constructs, by 5.7 and 4.21, a canonical model $M(G^\circ, h_0) \subset M(\mathrm{Gp}(V, \psi), h_0) \otimes_{\mathbb{Q}} E(G^\circ, h^\circ)$.

Variant 5.9. Let (G_1, h_0) be such that, with the notation of 4.22, there exists $u : (G_1, h_0) \hookrightarrow (\mathrm{Gp}_{2g}^{(F)}, h_0)$ (for example (G_1°, h_0) with (G_1, h_0) as in 4.13). An application of 5.7 yields a canonical model of $M(G_1, h_0)$.

Proposition 5.10. Let (G, h) be as in 3.13, let the $E_i : E(G, h) \subset E_i \subset \mathbb{C}$ be number fields, let E be the intersection of the E_i, and let $M_{E_i}(G, h)$ be a weakly canonical model of $M_{\mathbb{C}}(G, h)$ over E_i. Then there exists a unique model $M_E(G, h)$ of $M_{\mathbb{C}}(G, h)$ over E, such that for every i the model $M_E(G, h) \otimes_E E_i$ over E_i is isomorphic to $M_{E_i}(G, h)$.

In view of the uniqueness assertion, one may suppose the E_i finite in number. Let F be the extension of E in $\mathbb{C}$ generated by them, and let $\mathfrak{E}$ be the set of subextensions of F containing one of the E_i. Using 5.5, one obtains for every $F' \in \mathfrak{E}$ a model $M_{F'}(G, h)$, and for every inclusion $F' \subset F''$ $(F', F'' \in \mathfrak{E})$ an isomorphism of models $\alpha(F'', F') : M_{F'}(G, h) \otimes_{F'} F'' \longrightarrow M_{F''}(G, h)$.

Let $u : (H, h') \to (G, h)$ be as in 3.13, with $E(H, h')$ linearly disjoint from F over $E(G, h)$ (5.1). Put $E_1 = E(H, h') \cdot E$ and $F_1' = E_1 \cdot F' = E(H, h') \cdot F'$ $(F' \in \mathfrak{E})$.

Let $g \in G(\mathbb{A}^f)$ and let $K \subset G(\mathbb{A}^f)$ be a compact open subgroup. Since $M_{F'}(G, h)$ is weakly canonical, one has an F_1'-morphism

$$a'(g) : M_{F_1'}(H, h') \longrightarrow M_{F_1'}(G, h) \xrightarrow{g} M_{F_1'}(G, h) \longrightarrow {}_K M_{F_1'}(G, h).$$

Since $F_1' = E_1 \otimes_E F'$, one may "interpret" this as a morphism of F'-schemes

$$a(g) : M_{E_1}(H, h') \otimes_E F' \longrightarrow {}_K M_{F'}(G, h).$$

The $a(g)$ satisfy $\alpha(F'', F')\, a(g)_{F''} = a(g)$. One concludes by 5.2 and the following lemma, applied to $X^1 = \coprod_{g \in G(\mathbb{A}^f)} M_{E_1}(H, h')$. Its proof is left to the reader as an exercise.

Lemma 5.10.1. Let F/E be a finite extension of fields and let $\mathfrak{E}$ be a set of subfields of F, whose intersection is reduced to E, such that

$$F \in \mathfrak{E} \text{ and } F' \subset F'' \subset F \Longrightarrow F'' \in \mathfrak{E}.$$

(i) The functor that sends a scheme X/E to the system $\mathcal{X}$ of schemes $X_{F'}/F'$ $(F' \in \mathfrak{E})$ and the isomorphisms $(X_{F'}) \otimes_{F'} F'' \xrightarrow{\sim} X_{F''}$ $(F' \subset F'',\ F', F'' \in \mathfrak{E})$ is fully faithful.

(ii) For a system $\mathcal{X} = \{X(F')/F'\ (F' \in \mathfrak{E});\ \alpha(F'', F') : X(F') \otimes_{F'} F'' \xrightarrow{\sim} X(F'')\ (F' \subset F'',\ F', F'' \in \mathfrak{E})\}$ to come from a scheme X/E, it is enough that the $X(F')/F'$ be quasi-projective and that there exist a scheme X^1/E and a morphism $u : X^1_{\mathfrak{E}} \longrightarrow \mathcal{X}$ such that the $u(F') : X^1_{F'} \longrightarrow X(F')$ have dense image.

Remark 5.10.2. Under the hypotheses of 5.10, for $u : (H, h') \hookrightarrow (G, h)$ as in 3.13, the field of definition of $u : M_{\mathbb{C}}(H, h') \longrightarrow M_{\mathbb{C}}(G, h)$ is contained in the extension $\bigcap_i E(H, h') \cdot E_i$ of $E(H, h') \cdot E$.

Proposition 5.11. Let $h : \mathbb{S} \longrightarrow G_{\mathbb{R}}$ be as in 3.13 and let $\delta : \mathbb{S} \longrightarrow C^\circ_{\mathbb{R}}$ (C the center of G). If $E(G, h), E(C, \delta) \subset E \subset \mathbb{C}$ and if $M_{\mathbb{C}}(G, h)$ admits a weakly canonical model over E, then $M_{\mathbb{C}}(G, h.\delta)$ does as well.

Let K be a compact open subgroup of $G(\mathbb{A}^f)$ and let $L = K \cap C^\circ(\mathbb{A}^f)$. Over $\mathbb{C}$, the morphism

$$u : (G \times C^\circ, (h, \delta)) \longrightarrow (G, h\delta) : (g, c) \longmapsto gc$$

induces morphisms

$$u_K : {}_K M_{\mathbb{C}}(G, h) \times {}_L M_{\mathbb{C}}(C^\circ, \delta) \longrightarrow {}_K M_{\mathbb{C}}(G, h\delta).$$

Let d be the morphism $d : C^\circ \longrightarrow G \times C^\circ : c \longmapsto (c, c^{-1})$. The subgroup $d(C^\circ(\mathbb{A}^f))$ of $G \times C^\circ(\mathbb{A}^f)$ acts on $M_{\mathbb{C}}(G, h) \times M_{\mathbb{C}}(C^\circ, \delta)$ through the quotient $\pi_0(C^\circ(\mathbb{A}))$ of $C^\circ(\mathbb{A}^f)$, and u_K induces an isomorphism

$$\bar{u}_K : \big({}_K M_{\mathbb{C}}(G, h) \times {}_L M_{\mathbb{C}}(C^\circ, \delta)\big) / \big(L \backslash \pi_0(C^\circ(\mathbb{A}))\big) \xrightarrow{\sim} {}_K M_{\mathbb{C}}(G, h\delta).$$

Define ${}_KM_E(G,h\delta)$ to be the quotient

$$ {}_KM_E(G,h) \times {}_LM_E(C^\circ,\delta)/\big(L\backslash\pi_0(C^\circ(\mathbb{A}))\big) $$

and one verifies that this gives a weakly canonical model. By construction, one has

$$ M_E(G,h) \times_E M_E(C^\circ,\delta) \longrightarrow M_E(G,h\delta). \tag{5.11.1} $$

Conjecture 5.12. For every (G,h_0) as in 3.13, there exists a canonical model of $M_{\mathbb{C}}(G,h_0)$.

Suppose—this is a typical case—that G is $\mathbb{Q}$-simple and adjoint. Let $\widetilde{G}$ be the universal covering of G and let ω be a weight of $\widetilde{G}_{\mathbb{C}}$. If, with the notation of 3.7, the conjugacy class of h_0r is described by integers $(n_i)_{i\in|\Delta|}$, and if $\omega = \sum m_i\alpha_i$, put $\langle\omega, h_0r\rangle = \sum n_im_i$. So far, it has only been possible to attack 5.12 when $\widetilde{G}$ admits a nontrivial representation V such that all irreducible components V' of $V_{\mathbb{C}}$ satisfy:

$(*)$ as ω runs through the weights of V', $\langle\omega, h_0r\rangle$ takes at most two values.

Let σ be the opposition involution. Condition $(*)$ also means that the dominant weight ω of V' satisfies $\langle\omega, h_0r\rangle + \langle\sigma\omega, h_0r\rangle = 0$ or 1. There is no such representation V if G is exceptional (the noncompact factors of $G_{\mathbb{R}}$ are then of type $E_{6(-14)}$ or $E_{7(-25)}$), or of trialitarian type D_4, nor if G is of type D_n and $G_{\mathbb{R}}$ has noncompact factors of both types $D_n^{\mathbb{H}}$ and $D_n^{\mathbb{R},2}$.

§ 6. Strange models.

6.1. Let F be a totally real number field, and let

$$ {}_F\Sigma: \quad 0 \longrightarrow {}_FG' \longrightarrow {}_FG \longrightarrow \mathbb{G}_m \longrightarrow 0 \tag{6.1.1} $$

be a form over F of the exact sequence

$$ {}_F\Sigma_0: \quad 0 \longrightarrow \mathrm{Sp}_{2n} \longrightarrow \mathrm{Gp}_{2n} \xrightarrow{\mu} \mathbb{G}_m \longrightarrow 0, \tag{6.1.2} $$

where Gp denotes a group of symplectic similitudes.

Assume that, for every real place τ of F, $\Sigma \otimes_{F,\tau} \mathbb{R}$ is of one of the following types:

(a) ${}_F\Sigma \otimes_{F,\tau} \mathbb{R}$ is isomorphic to ${}_F\Sigma_0 \otimes_{F,\tau} \mathbb{R}$, or

(b) $G' \otimes_{F,\tau} \mathbb{R}$ is compact.

Let $\tau_1,\dots,\tau_g$ denote the real places of F, and suppose that the places of type (a) are $\tau_1,\dots,\tau_r$, with $r>0$.

6.2. The group ${}_FG$ may be described as follows:

(a) Choose a quaternion algebra B over F, indefinite at the places τ_i for $i \le r$, and definite for $i > r$. Let $x \mapsto \overline{x}$ denote its canonical involution.

(b) Choose a free B-module V, endowed with a nondegenerate F-bilinear form Φ, symmetric and such that

$$ \Phi(bx,y) = \Phi(x,\bar{b}\,y). $$

(c) Assume that, for $i > r$, the form deduced from Φ by extension of scalars through $\tau_i : F \to \mathbb{R}$ is positive definite.

(d) Take ${}_FG$ to be the group of B-linear similitudes of Φ; in other words, the elements $g \in G(\mathbb{Q})$ are the $g \in \mathrm{GL}_B(V)$ for which there exists $\mu(g) \in F^*$ satisfying

$$ \Phi(gx,gy) = \mu(g)\Phi(x,y). $$

6.3. Let

$$ \Sigma: \quad 0 \longrightarrow G' \longrightarrow G \longrightarrow F^* \longrightarrow 0 $$

denote the exact sequence of algebraic groups over $\mathbb{Q}$ deduced from ${}_F\Sigma$ by Weil restriction of scalars from F to $\mathbb{Q}$. We have

$$ G_{\mathbb{R}} = \prod_\tau {}_FG \otimes_{F,\tau} \mathbb{R} \qquad (\tau \text{ a real place}). \tag{6.3.1} $$

A morphism $h : \mathbb{S} \to G_{\mathbb{R}}$ is defined by its coordinates $h_i : \mathbb{S} \longrightarrow {}_F G \otimes_{F,\tau_i} \mathbb{R}$. By 1.6, there exists a unique conjugacy class of morphisms h such that, for $i \leq r$, h_i is of type 1.6, and for $i > r$, h_i is trivial. Let h_0 be in this class.

The field $E(G, h_0)$ is the totally real subfield of $\mathbb{C}$ generated by the elements $\sum_{i=1}^{r} \tau_i(f)$ for $f \in F$.

Theorem 6.4. With the preceding notation, $M_{\mathbb{C}}(G, h_0)$ admits a canonical model $M(G, h_0)$.

Let Z be a totally imaginary quadratic extension of F. Let $V' = V \otimes_F Z$, $L = B \otimes_F Z$, and let G' be the algebraic subgroup of $\mathrm{GL}_L(V')$ generated by G and Z^*:

$$0 \longrightarrow F^* \xrightarrow{(f, f^{-1})} G \times Z^* \xrightarrow{q} G' \longrightarrow 0. \tag{6.4.1}$$

Let $x \mapsto \widetilde{x}$ be the involution of L given by the tensor product of the involutions of B and Z. Up to a factor in F, there exists on Z a unique alternating F-bilinear form ψ_Z. If ${}_F\psi$ is the alternating F-bilinear form on V' which is the tensor product of Φ and ψ_Z, then

$${}_F\psi(\ell x, y) = {}_F\psi(x, \widetilde{\ell}\, y).$$

Put $\psi(x, y) = \mathrm{Tr}_{F/\mathbb{Q}}\big({}_F\psi(x, y)\big)$.

Choose, for every real place of F, a complex embedding of Z inducing it. Then $Z^*(\mathbb{R}) \simeq \prod_{i=1}^{g} \mathbb{C}^*$. Let h_Z be the homomorphism from $\mathbb{S}$ into $Z^*_{\mathbb{R}}$ with coordinates $h_i : \mathbb{S}(\mathbb{R}) = \mathbb{C}^* \longrightarrow \mathbb{C}^*$ $(1 \leq i \leq g)$ equal to $(z \mapsto 1)$ for $i \leq r$, and to $(z \mapsto z^{-1})$ for $i > r$.

For every $h : \mathbb{S} \to G_{\mathbb{R}}$ as in 6.3, put $h' = q \circ (h \times h_Z) : \mathbb{S} \longrightarrow G'_{\mathbb{R}}$.

Lemma 6.4.2. There exists $b \in L$ such that, for a suitable choice of ψ_Z, the form

$$\psi\big(x, b\, h'_0(i) y\big) \qquad \text{on} \quad V' \otimes_{\mathbb{Q}} \mathbb{R}$$

is symmetric and positive definite.

This is verified after extension of scalars from $\mathbb{Q}$ to $\mathbb{R}$.

This lemma and 5.9 make it possible to construct a canonical model of $M(G', h'_0)$ (since G' is contained in the group of symplectic similitudes of ${}_F\psi(x, by)$). Applying 5.11 and 5.7, one obtains a weakly canonical model of $M_{\mathbb{C}}(G, h_0)$ over the extension $E(Z^*, h_Z)$ of $E(G, h_0)$. The conclusion follows from 5.10, 5.10.2 and the following lemma, which can be deduced from 5.13.

Lemma 6.5. Let F be a totally real number field, endowed with a set S of real embeddings. Let F^* be the number field corresponding to the subgroup of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which stabilizes S. For Z a totally imaginary quadratic extension of F, endowed with a set T of complex embeddings, one above each element of S, let likewise $Z^* \supset F^*$ be the field of invariants of the stabilizer of T. Then, for every extension E of F^*, there exist Z and T such that Z^* is linearly disjoint from E.

Remark 6.6. The canonical map 1.13, 1.14

$$M_{\mathbb{C}}(G, h_0) \times M_{\mathbb{C}}(Z^*, h_Z) \longrightarrow M_{\mathbb{C}}(G', h'_0) \tag{6.6.1}$$

is interpreted as follows. Each conjugate of h defines on $V_{\mathbb{C}}$ a Hodge bigrading invariant under F:

$$V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1} \oplus V^{0,0} \qquad (\overline{V^{pq}} = V^{qp}).$$

One should beware that the grading (by weight) of $V_{\mathbb{C}}$ by the $V^n = \sum_{p+q=n} V^{pq}$ is not defined over $\mathbb{Q}$ if $r \neq g$.

Similarly, $Z_{\mathbb{C}}$ is bigraded:

$$Z_{\mathbb{C}} = Z^{-1,0} \oplus Z^{0,-1} \oplus Z^{0,0}.$$

The miracle is that the $\mathbb{Q}$-vector space $V' = V \otimes_F Z$, endowed with the tensor-product Hodge bigrading $V'_{\mathbb{C}} = V'^{-1,0}_{\mathbb{C}} \oplus V'^{0,-1}_{\mathbb{C}}$ (!), is the H_1 of an abelian variety.

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HODGE THEORY I

by PIERRE DELIGNE

We propose to give a heuristic dictionary between statements in ℓ-adic cohomology and statements in Hodge theory. This dictionary draws in particular on [3] and on Grothendieck's conjectural theory of motives [2]. So far it has mainly served to formulate conjectures in Hodge theory, and has sometimes suggested proofs of them.

Definition 1.1. A mixed Hodge structure H consists of

(a) a finitely generated $\mathbb{Z}$-module $H_{\mathbb{Z}}$ (the "integral lattice");

(b) a finite increasing filtration W on $H_{\mathbb{Q}} = H_{\mathbb{Z}} \otimes \mathbb{Q}$ (the "weight filtration");

(c) a finite decreasing filtration F on $H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$ (the "Hodge filtration").

These data are subject to the axiom:

There exists on $\mathrm{Gr}_W(H_{\mathbb{C}})$ a (unique) bigrading by subspaces $H^{p,q}$ such that

(i)
$$\mathrm{Gr}^n_W(H_{\mathbb{C}}) = \bigoplus_{p+q=n} H^{p,q};$$

(ii) the filtration F induces on $\mathrm{Gr}_W(H_{\mathbb{C}})$ the filtration
$$\mathrm{Gr}_W(F)^p = \bigoplus_{p' \geq p} H^{p',q'};$$

(iii) $\overline{H^{p,q}} = H^{q,p}$.

A morphism $f : H \to H'$ is a homomorphism $f_{\mathbb{Z}} : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ such that $f_{\mathbb{Q}} : H_{\mathbb{Q}} \to H'_{\mathbb{Q}}$ and $f_{\mathbb{C}} : H_{\mathbb{C}} \to H'_{\mathbb{C}}$ are respectively compatible with the filtrations W and F.

The Hodge numbers of H are the integers

$$h^{pq} = \dim H^{p,q} = h^{qp}. \tag{1.2}$$

One says that H is pure of weight n if $h^{pq} = 0$ for $p + q \neq n$ (i.e. if $\mathrm{Gr}^i_W(H) = 0$ for $i \neq n$). One also says that H is a Hodge structure of weight n.

The Tate Hodge structure $\mathbb{Z}(1)$ is the Hodge structure of weight -2, purely of type $(-1,-1)$, for which $\mathbb{Z}(1)_{\mathbb{C}} = \mathbb{C}$ and $\mathbb{Z}(1)_{\mathbb{Z}} = 2\pi i\mathbb{Z} = \mathrm{Ker}(\exp : \mathbb{C} \to \mathbb{C}^*) \subset \mathbb{C}$. Put $\mathbb{Z}(n) = \mathbb{Z}(1)^{\otimes n}$.

One can show that mixed Hodge structures form an abelian category. If $f : H \to H'$ is a morphism, then $f_{\mathbb{Q}}$ and $f_{\mathbb{C}}$ are strictly compatible with the filtrations W and F ([1], 2.3.5).

2. Let A be a normal integral domain of finite type over $\mathbb{Z}$, let K be its field of fractions, and let $\overline{K}$ be an algebraic closure of K. Let K_{nr} be the largest subextension of $\overline{K}$ unramified at every prime ideal of A. It is known—or one may take this as the definition—that

$$\pi_1(\mathrm{Spec}(A), \overline{K}) = \mathrm{Gal}(K_{nr}/K).$$

For every closed point x of $\mathrm{Spec}(A)$, defined by a maximal ideal m_x of A, the residue field $k_x = A/m_x$ is finite; the point x defines a conjugacy class of "Frobenius substitutions" $\varphi_x \in \pi_1(\mathrm{Spec}(A), \overline{K})$. Put $q_x = \#k_x$ and $F_x = \varphi_x^{-1}$.

Let K be a field of finite type over the prime field of characteristic p, let $\overline{K}$ be an algebraic closure of K, let ℓ be a prime number $\neq p$, and let H be a finitely generated $\mathbb{Z}_\ell$-module (or $\mathbb{Q}_\ell$-module) endowed with a continuous

action ρ of $\mathrm{Gal}(\overline{K}/K)$. In what follows we shall always suppose that there exists A as above, with ℓ invertible in A, such that ρ factors through

$$\pi_1(\mathrm{Spec}(A), \overline{K}) = \mathrm{Gal}(K_{nr}/K).$$

We shall say that H is pure of weight n if, for every closed point x of a nonempty open subset of $\mathrm{Spec}(A)$, the eigenvalues α of F_x acting on H are algebraic integers all of whose complex conjugates have absolute value

$$|\alpha| = q_x^{n/2}.$$

Principle 2.1. If the Galois module H "comes from algebraic geometry", there exists on $H_{\mathbb{Q}_\ell} = H \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ a (unique) increasing filtration W (the "weight filtration"), invariant under Galois, such that $\mathrm{Gr}_n^W(H)$ is pure of weight n.

One may think that $\mathrm{Gr}_n^W(H)$ is moreover semisimple.

When resolution of singularities is available, one can often give a conjectural definition of W, whose correctness follows from the Weil conjectures [5] (cf. 6).

Let μ be the subgroup of $\overline{K}^*$ formed by the roots of unity. The Tate module $\mathbb{Z}_\ell(1)$, defined by

$$\mathbb{Z}_\ell(1) = \mathrm{Hom}(\mathbb{Q}_\ell/\mathbb{Z}_\ell, \mu),$$

is pure of weight -2. Put $\mathbb{Z}_\ell(n) = \mathbb{Z}_\ell(1)^{\otimes n}$.

It is trivial that every morphism $f : H \to H'$ is strictly compatible with the weight filtration.

Principle 2.1 agrees with the fact that every extension of $\mathbb{G}_m$ ("weight -2") by an abelian variety ("weight $-1 > -2$") is trivial.

3. Translation. The Galois modules which appear in ℓ-adic cohomology have, over $\mathbb{C}$, mixed Hodge structures as analogues. In addition one has the dictionary

pure module of weight n	Hodge structure of weight n
weight filtration	weight filtration
Galois-equivariant homomorphism	morphism
Tate module $\mathbb{Z}_\ell(1)$	Tate Hodge structure $\mathbb{Z}(1)$.

4. Let X be a complex algebraic variety (= a scheme of finite type over $\mathbb{C}$, which we shall suppose separated). There exists a subfield K of $\mathbb{C}$, of finite type over $\mathbb{Q}$, such that X can be defined over K (i.e. arises by extension of scalars from K to $\mathbb{C}$ from a K-scheme X'). Let $\overline{K}$ be the algebraic closure of K in $\mathbb{C}$. The Galois group $\mathrm{Gal}(\overline{K}/K)$ then acts on the ℓ-adic cohomology groups $H^*(X, \mathbb{Z}_\ell)$; one has

$$H^*(X(\mathbb{C}), \mathbb{Z}) \otimes \mathbb{Z}_\ell = H^*(X, \mathbb{Z}_\ell) = H^*(X'_{\overline{K}}, \mathbb{Z}_\ell).$$

According to 3, one should expect the cohomology groups $H^n(X(\mathbb{C}), \mathbb{Z})$ to carry natural mixed Hodge structures. This can be proved (see [1], 3.2.5, for the case where X is smooth; the proof is algebraic, starting from classical Hodge theory [6]). For X projective and smooth, the Weil conjectures imply that $H^n(X, \mathbb{Z}_\ell)$ is pure of weight n, while classical Hodge theory endows $H^n(X, \mathbb{Z})$ with a Hodge structure of weight n. For every morphism $f : X \to Y$ and for K large enough, $f^* : H^*(Y, \mathbb{Z}_\ell) \to H^*(X, \mathbb{Z}_\ell)$ commutes with Galois (by transport of structure); likewise, $f^* : H^*(Y, \mathbb{Z}) \to H^*(X, \mathbb{Z})$ is a morphism of mixed Hodge structures. For X smooth, the cohomology class in $H^{2n}(X, \mathbb{Z}_\ell(n))$ of an algebraic cycle of codimension n, Z, defined over K, is Galois-invariant, i.e. defines

$$c(Z) \in \mathrm{Hom}_{\mathrm{Gal}}(\mathbb{Z}_\ell(-n), H^{2n}(X, \mathbb{Z}_\ell)).$$

Similarly, the cohomology class $c(Z) \in H^{2n}(X(\mathbb{C}), \mathbb{Z})$ is purely of type (n, n), i.e. corresponds to

$$c(Z) \in \mathrm{Hom}_{\mathrm{H.M.}}(\mathbb{Z}(-n), H^{2n}(X(\mathbb{C}), \mathbb{Z})).$$

5. If $f : H \to H'$ is a Galois-compatible morphism between $\mathbb{Q}_\ell$-vector spaces of different weights, then $f = 0$. Likewise, if $f : H \to H'$ is a morphism of pure mixed Hodge structures of different weights, then f is torsion. A more useful remark is the following.

Scholium 5.1. Let H and H' be Hodge structures of weights n and n', with $n > n'$. Let $f : H_{\mathbb{Q}} \to H'_{\mathbb{Q}}$ be a homomorphism such that $f : H_{\mathbb{C}} \to H'_{\mathbb{C}}$ respects F. Then $f = 0$.

6. Let X be a smooth projective variety over $\mathbb{C}$, let $D = \sum_{i=1}^{n} D_i$ be a normal-crossings divisor in X, a sum of smooth divisors, and let j be the inclusion into X of $U = X - D$. For $Q \subset [1, n]$, put

$$D_Q = \bigcap_{i \in Q} D_i.$$

In ℓ-adic cohomology, one has canonically

$$R^q j_* \mathbb{Z}_\ell = \bigoplus_{\#Q=q} \mathbb{Z}_\ell(-q)_{D_Q}, \tag{6.1}$$

and the Leray spectral sequence for j is

$$E_2^{pq} = \bigoplus_{\#Q=q} H^p(D_Q, \mathbb{Q}_\ell) \otimes \mathbb{Z}_\ell(-q) \Rightarrow H^{p+q}(U, \mathbb{Q}_\ell). \tag{6.2}$$

According to the Weil conjectures [5], $H^p(D_Q, \mathbb{Q}_\ell)$ is pure of weight p, so that E_2^{pq} is pure of weight $p + 2q$. As a quotient of a subobject of E_2^{pq}, E_r^{pq} is also pure of weight $p + 2q$. By 5, $d_r = 0$ for $r \geq 3$, since the weights $p + 2q$ and $p + 2q - r + 2$ of E_r^{pq} and $E_r^{p+r,q-r+1}$ are different. Hence $E_3^{pq} = E_\infty^{pq}$. Up to reindexing, the weight filtration of $H^*(U, \mathbb{Q}_\ell)$ is the abutment of (6.2):

$$\mathrm{Gr}_n^W(H^k(U, \mathbb{Q}_\ell)) = E_3^{2k-n,n-k}. \tag{6.3}$$

7. In integral cohomology, for the usual topology, the Leray spectral sequence for j is

$${}'E_2^{pq} = \bigoplus_{\#Q=q} H^p(D_Q, \mathbb{Z}) \Rightarrow H^{p+q}(U, \mathbb{Z}). \tag{7.1}$$

Since every D_Q is a nonsingular projective variety, ${}'E_2^{pq}$ is endowed with a Hodge structure of weight p. Put $E_2^{pq} = {}'E_2^{pq} \otimes \mathbb{Z}(-q)$ (a Hodge structure of weight $p + 2q$). As an abelian group, $E_2^{pq} = {}'E_2^{pq}$; it is useful to regard (7.1) as a spectral sequence with initial term E_2^{pq}. According to 3, one should expect

$$d_2 : E_2^{pq} \to E_2^{p+2,q-1}$$

to be a morphism of Hodge structures. This is proved by interpreting d_2 as a Gysin morphism. Thus E_3^{pq} is endowed with a Hodge structure of weight $p + 2q$. According to 3, one expects that, modulo torsion, the spectral sequence (6.4)[1] degenerates at E_3 ($E_3 = E_\infty$), and that the vanishing of the d_r ($r \geq 3$) is an application of 5.1. This program is carried out in [1] 3.2. There the weight filtration of $H^*(U, \mathbb{Q})$ is defined as the abutment of (7.1), up to the reindexing (6.3).

In fact, in order to endow cohomology groups H^* with a mixed Hodge structure, the key point up to now has always been to find a spectral sequence E abutting to H^* such that the ℓ-adic analogue of E_2^{pq} is conjecturally pure (of weight $p + 2q$); E_2^{pq} must then carry a natural Hodge structure (of weight $p + 2q$), and the filtration W is the abutment of E.

8. Let $\mathrm{Spec}(V)$ be the spectrum of a henselian discrete valuation ring (a henselian trait), with fraction field K and residue field k of finite type over the prime field of characteristic p. Let $\overline{K}$ be an algebraic closure of K and let H be a finite-dimensional vector space over $\mathbb{Q}_\ell$ ($\ell \neq p$), on which $\mathrm{Gal}(\overline{K}/K)$ acts continuously. According to Grothendieck, it is known ([4], appendix) that a finite-index subgroup of the inertia group I acts unipotently. Replacing V by a finite extension, one reduces to the case in which the action of all of I is unipotent (the semistable case); it then factors through the largest pro-ℓ quotient I_ℓ of I, canonically isomorphic to $\mathbb{Z}_\ell(1)$.

[1] The printed text refers to (6.4); the intended reference appears to be (7.1).

Principle 8.1. In the semistable case, if the Galois module H "comes from algebraic geometry", there exists a (unique) increasing filtration W of H (the "weight filtration") such that I acts trivially on $\mathrm{Gr}_n^W(H)$ and such that $\mathrm{Gr}_n^W(H)$, as a Galois module under $\mathrm{Gal}(\bar{k}/k) \simeq \mathrm{Gal}(\overline{K}/K)/I$, is pure of weight n.

Compare 2.1 and the appendix of [4].

When resolution of singularities is available, one can sometimes give a conjectural definition of W, whose validity follows from the Weil conjectures. With the help of resolution and Weil, it is often easy to show that in any case H has a filtration whose successive quotients are pure Galois modules under $\mathrm{Gal}(\bar{k}/k)$.

Suppose H is semistable. For $T \in I_\ell$, define $\log T$ as the finite sum

$$-\sum_{n>0} (\mathrm{Id} - T)^n/n.$$

The map $(T, x) \mapsto \log T(x)$ is identified with a homomorphism

$$M : \mathbb{Z}_\ell(1) \otimes H \longrightarrow H. \tag{8.2}$$

Since $\mathbb{Z}_\ell(1)$ has weight -2, one necessarily has (cf. 5)

$$M(\mathbb{Z}_\ell(1) \otimes W_n(H)) \subset W_{n-2}(H) \tag{8.3}$$

and M induces

$$\mathrm{Gr}(M) : \mathbb{Z}_\ell(1) \otimes \mathrm{Gr}_n^W(H) \longrightarrow \mathrm{Gr}_{n-2}^W(H). \tag{8.4}$$

8.5. If X is a nonsingular projective variety over an algebraically closed field k_0, define

$$L : \mathbb{Z}_\ell(-1) \otimes H^*(X, \mathbb{Z}_\ell) \longrightarrow H^*(X, \mathbb{Z}_\ell)$$

to be cup product with the cohomology class of a hyperplane section. One will note a formal analogy between L and M: just as M is defined by an action of $\mathbb{Z}_\ell(1)$, one may regard L as defined by an action of $\mathbb{Z}_\ell(-1)$; L raises degree by 2, and $\mathrm{Gr}\, M$ (8.4) lowers it by 2.

9. Let D be the unit disk, $D^* = D - \{0\}$, and let X

$$\begin{array}{ccc} X & \hookrightarrow & \mathbb{P}^r(\mathbb{C}) \times D \\ {\scriptstyle f}\searrow & & \swarrow{\scriptstyle \mathrm{pr}_2} \\ & D & \end{array}$$

be a family of projective varieties parametrized by D, with f proper and $f|D^*$ smooth. Keep the notation of 8, and recall that in the analogy between a henselian trait and a small neighborhood of 0 in the complex line, one has the following dictionary (note that the spectrum of the ring of germs of holomorphic functions at 0 is a henselian trait):

D	$\mathrm{Spec}(V)$
D^*	$\mathrm{Spec}(K)$
a universal covering $\widetilde{D}^*$ of D^*	$\mathrm{Spec}(\overline{K})$
fundamental group $\pi_1(D^*)$	inertia group I
(with $\pi_1(D^*) = \mathbb{Z} \simeq \mathbb{Z}(1)_{\mathbb{Z}}$)	(with $I_\ell = \mathbb{Z}_\ell(1)$)
X	projective scheme X over $\mathrm{Spec}(V)$
$X^* = f^{-1}(D^*)$	X_K
$\widetilde{X} = X \times_D \widetilde{D}^*$	$X_{\overline{K}}$
local system $R^i f_* \mathbb{Z}|D^*$	Galois module $H^i(X_{\overline{K}}, \mathbb{Z}_\ell)$
$H^i(\widetilde{X}, \mathbb{Z})$	$H^i(X_{\overline{K}}, \mathbb{Z}_\ell)$.

(9.1)

Note that $\widetilde{X}$ is homotopy equivalent to each of the fibers $X_t = f^{-1}(t)$ $(t \in D^*)$: $H^i(X_{\overline{K}}, \mathbb{Z}_\ell)$ again has $H^i(X_t, \mathbb{Z})$ as analogue, and the action of I corresponds to the monodromy transformation T.

Here again, it is known that a finite-index subgroup of $\pi_1(D^*)$ acts unipotently on $H^i(\widetilde{X}, \mathbb{Q}) = H^i(X_t, \mathbb{Q})$. Assume we are in the semistable case where all of $\pi_1(D^*)$ acts unipotently (this amounts to replacing D by a finite covering), and let T be the action of the canonical generator of $\pi_1(D^*)$.

By 3 and 8, one expects $H^i(\widetilde{X},\mathbb{Q}) \simeq H^i(X_t,\mathbb{Q})$ to be endowed with an increasing filtration W, $\mathrm{Gr}_n^W(H^i(\widetilde{X},\mathbb{Q}))$ to be endowed with a Hodge structure of weight n, $\log T(W_n) \subset W_{n-2}$, and $\log T$ to induce a morphism of Hodge structures

$$M_n : \mathbb{Z}(-1) \otimes \mathrm{Gr}_n^W(H^i) \longrightarrow \mathrm{Gr}_{n-2}^W(H^i).$$

One would moreover like (8.2), and not only (8.3) and (8.4), to have an analogue.

In fact, for each vector u in the tangent space to D at 0, one can define a mixed Hodge structure $\mathcal{H}_u$ on $H^i(\widetilde{X},\mathbb{Z})$. The filtration W and the Hodge structures on the $\mathrm{Gr}_n^W(H^i)$ are independent of u, and the dependence of $\mathcal{H}_u$ on u is expressed simply in terms of T. In analogy with (8.2), one finds that, for every u, $\log T$ induces a homomorphism of mixed Hodge structures

$$M : \mathbb{Z}(1) \otimes H^i(\widetilde{X},\mathbb{Z}) \longrightarrow H^i(\widetilde{X},\mathbb{Z}).$$

Finally, the analogy 8.5 is not misleading (but here the fact that $f|D^*$ is supposed proper and smooth is no doubt essential). One proves that

$$(\log T)^k : \mathrm{Gr}_{n+k}^W(H^n(\widetilde{X},\mathbb{Q})) \longrightarrow \mathrm{Gr}_{n-k}^W(H^n(\widetilde{X},\mathbb{Q}))$$

is an isomorphism for every k (cf. [6], IV 6, cor. to th. 5). This characterizes the filtration W. Up to now, an analogue of Hodge's positivity theorem (cf. [6], IV 7, cor. to th. 7) is available only in very special cases. One hopes that the mixed structures $\mathcal{H}_u$ determine, as $t \to 0$, the asymptotic behavior of the family of pure structures $H^i(X_t,\mathbb{Z})$ $(t \in D^*)$.

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I. H. E. S.
35, route de Chartres,
91-Bures-sur-Yvette
(France)

HODGE THEORY, II

by PIERRE DELIGNE[1]

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Introduction

According to Hodge, the cohomology space $H^n(X, \mathbb{C})$ of a compact Kähler variety X is endowed with a *Hodge structure* of weight n, that is, with a natural bigrading

$$H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{pq}$$

satisfying $\overline{H^{pq}} = H^{qp}$. We show here that the complex cohomology of a nonsingular algebraic variety, not necessarily compact, is endowed with a somewhat more general kind of structure, which presents $H^n(X, \mathbb{C})$ as a successive extension of Hodge structures of decreasing weights, lying between $2n$ and n, whose Hodge numbers $h^{pq} = \dim H^{pq}$ are zero for $p > n$ or $q > n$.

The reader will find in [14] an exposition of the yoga underlying this construction.

The proof, essentially algebraic, rests on the one hand on Hodge theory and on the other hand on Hironaka's resolution of singularities, which permits one, via a spectral sequence, to *express* the cohomology of a nonsingular quasi-projective algebraic variety in terms of the cohomology of nonsingular projective varieties.

Section 1 contains, besides a few elementary facts about filtrations collected for the reader's convenience, two key results:

a) Theorem (1.2.10), which will be used only through its corollary (2.3.5), giving the fundamental properties of mixed Hodge structures;

b) the *lemma on two filtrations* (1.3.16).

Section 2 recalls Hodge theory and introduces mixed Hodge structures. The heart of the paper is no. 3.2, which defines the mixed Hodge structure on $H^n(X, \mathbb{C})$ and establishes certain degeneracies of spectral sequences.

Section 4 gives various applications, all deduced from (4.1.1) and from the theory of the (K/k)-trace for Hodge structures which follows from it (4.1.2). The principal ones are (4.2.6) and (4.4.15).

[1] Work presented as a doctoral thesis at the University of Orsay.

1. Filtrations

1.1. Filtered objects

(1.1.1) Let $\mathcal{A}$ be an abelian category. We shall have to consider filtrations of type $\mathbb{Z}$, usually finite, on the objects of $\mathcal{A}$.

Definition (1.1.2). A decreasing, respectively increasing, filtration F of an object A of $\mathcal{A}$ is a family $(F^n(A))_{n\in\mathbb{Z}}$, respectively $(F_n(A))_{n\in\mathbb{Z}}$, of subobjects of A, satisfying

$$\forall n, m \qquad n \leq m \Rightarrow F^n(A) \supset F^m(A)$$

(resp. $\forall n, m \quad n \leq m \Rightarrow F_n(A) \subset F_m(A)$).

A filtered object is an object endowed with a filtration. When no confusion is possible, the same letter will often denote filtrations on different objects of $\mathcal{A}$.

If F is a decreasing, respectively increasing, filtration on A, put $F^{\infty}(A) = 0$ and $F^{-\infty}(A) = A$ (resp. $F_{-\infty}(A) = 0$ and $F_{\infty}(A) = A$).

The shifted filtrations of a decreasing filtration W are defined by

$$W[n]^p(A) = W^{n+p}(A).$$

(1.1.3) If F is a decreasing, respectively increasing, filtration of A, then the $F_n(A) = F^{-n}(A)$, respectively the $F^n(A) = F_{-n}(A)$, form an increasing, respectively decreasing, filtration of A. Thus in principle it is enough to consider decreasing filtrations; unless explicitly stated otherwise, the word *filtration* will henceforth mean *decreasing filtration*.

(1.1.4) A filtration F of A is called finite if there exist n and m such that $F^n(A) = A$ and $F^m(A) = 0$.

(1.1.5) A morphism from a filtered object (A, F) to a filtered object (B, F) is a morphism f from A to B satisfying

$$f(F^n(A)) \subset F^n(B) \qquad \text{for } n \in \mathbb{Z}.$$

The filtered objects (respectively, the objects with finite filtrations) of $\mathcal{A}$ form an additive category in which finite inductive and projective limits exist (and hence so do the kernels, cokernels, images, and coimages of a morphism).

A morphism $f : (A, F) \to (B, F)$ is called strict, or strictly compatible with the filtrations, if the canonical arrow from $\mathrm{Coim}(f)$ to $\mathrm{Im}(f)$ is an isomorphism of filtered objects (cf. (1.1.11)).

(1.1.6) Let $^\circ$ be the contravariant identity functor from $\mathcal{A}$ to the dual category $\mathcal{A}^\circ$. If (A, F) is a filtered object of $\mathcal{A}$, the objects $(A/F^{1-n}(A))^\circ$ are identified with subobjects of A°. The filtration on A° dual to F is defined by

$$F^n(A^\circ) = (A/F^{1-n}(A))^\circ.$$

The bidual of (A, F) is identified with (A, F). This construction identifies the dual of the category of filtered objects of $\mathcal{A}$ with the category of filtered objects of $\mathcal{A}^\circ$.

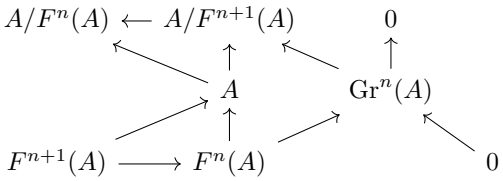
(1.1.7) If (A, F) is a filtered object of $\mathcal{A}$, its associated graded object is the object of $\mathcal{A}^{\mathbb{Z}}$ defined by

$$\mathrm{Gr}^n(A) = F^n(A)/F^{n+1}(A).$$

The convention (1.1.6) is justified by the simple formula

$$\mathrm{Gr}^n(A^\circ) = \mathrm{Gr}^{-n}(A)^\circ,$$

which is verified on the autodual diagram



(1.1.7.1)

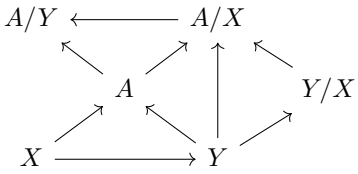
(1.1.8) Let (A, F) be a filtered object and let $j : X \hookrightarrow A$ be a subobject of A. The filtration induced on X by F is the unique filtration on X such that j is strictly compatible with the filtrations; one has

$$F^n(X) = j^{-1}(F^n(A)) = X \cap F^n(A).$$

Dually, the quotient filtration on A/X, the unique filtration such that $p : A \to A/X$ is strictly compatible with the filtrations, is given by

$$F^n(A/X) = p(F^n(A)) \simeq (X + F^n(A))/X \simeq F^n(A)/(X \cap F^n(A)).$$

Lemma (1.1.9). If X and Y are two subobjects of A, with $X \subset Y$, then on $Y/X \simeq \mathrm{Ker}(A/X \to A/Y)$ the quotient filtration from Y coincides with the filtration induced from A/X. In the diagram



the arrows are strict.

(1.1.10) The filtration of (1.1.9) on Y/X will be called the filtration induced by that of A. By (1.1.9), its definition is autodual.

In particular, if $\Sigma : A \xrightarrow{f} B \xrightarrow{g} C$ is a 0-sequence and if B is filtered, then

$$H(\Sigma) = \mathrm{Ker}(g)/\mathrm{Im}(f) = \mathrm{Ker}(\mathrm{Coker}(f) \to \mathrm{Coim}(g))$$

is endowed with a canonical induced filtration.

The reader will verify the following.

Proposition (1.1.11).

(i) Let $f : (A, F) \to (B, F)$ be a morphism of finitely filtered objects. For f to be strict, it is necessary and sufficient that the sequence

$$0 \to \mathrm{Gr}(\mathrm{Ker}(f)) \to \mathrm{Gr}(A) \to \mathrm{Gr}(B) \to \mathrm{Gr}(\mathrm{Coker}(f)) \to 0$$

be exact.

(ii) Let $\Sigma : (A, F) \to (B, F) \to (C, F)$ be a 0-sequence of strict morphisms. Then one has canonically

$$H(\mathrm{Gr}(\Sigma)) \simeq \mathrm{Gr}(H(\Sigma)).$$

In particular, if Σ is exact in $\mathcal{A}$, then $\mathrm{Gr}(\Sigma)$ is exact in $\mathcal{A}^{\mathbb{Z}}$.

In a category of modules, to say that a morphism $f : (A, F) \to (B, F)$ is strict means that every $b \in B$ of filtration $\geq n$ ($b \in F^n(B)$) which is in the image of A is already in the image of $F^n(A)$:

$$f(F^n(A)) = f(A) \cap F^n(B).$$

(1.1.12) If $\otimes : \mathcal{A}_1 \times \cdots \times \mathcal{A}_n \to \mathcal{B}$ is a multiadditive right exact functor, and if A_i is a finitely filtered object of $\mathcal{A}_i$ $(1 \leq i \leq n)$, one defines a filtration on $\bigotimes_{i=1}^n A_i$ by

$$F^k\left(\bigotimes_{i=1}^n A_i\right) = \sum_{\sum k_i = k} \operatorname{Im}\left(\bigotimes_{i=1}^n F^{k_i}(A_i) \to \bigotimes_{i=1}^n A_i\right)$$

(sum of subobjects).

Dually, if H is left exact, put

$$F^k(H(A_i)) = \bigcap_{\sum k_i = k} \operatorname{Ker}\left(H(A_i) \to H(A_i/F^{k_i}(A_i))\right).$$

For exact H, the two definitions are equivalent. These definitions are extended to functors contravariant in certain variables by (1.1.6). In particular, for the left exact functor Hom, one puts

$$F^k(\operatorname{Hom}(A,B)) = \{f : A \to B \mid \forall n,\ f(F^n(A)) \subset F^{n+k}(B)\}.$$

Thus

$$\operatorname{Hom}((A,F),(B,F)) = F^0(\operatorname{Hom}(A,B)).$$

Under the preceding hypotheses, one has evident morphisms

$$\bigotimes_{i=1}^n \operatorname{Gr}(A_i) \to \operatorname{Gr}\left(\bigotimes_{i=1}^n A_i\right)$$

and

$$\operatorname{Gr} H(A_i) \to H(\operatorname{Gr}(A_i)).$$

For exact H, these are mutually inverse isomorphisms.

These constructions are compatible with the composition of functors, in a sense which we leave to the reader not to make explicit.

1.2. Opposite filtrations

(1.2.1) Let A be an object of $\mathcal{A}$ endowed with two filtrations F and G. By definition, $\operatorname{Gr}_F^m(A)$ is a quotient of a subobject of A, and as such is endowed with a filtration induced by G (1.1.10). Passing to the associated graded object, one defines a bigraded object $(\operatorname{Gr}_G^n \operatorname{Gr}_F^m(A))_{n,m\in\mathbb{Z}}$. By a lemma of Zassenhaus, $\operatorname{Gr}_G^n \operatorname{Gr}_F^m(A)$ and $\operatorname{Gr}_F^m \operatorname{Gr}_G^n(A)$ are canonically isomorphic: if one defines the induced filtrations (1.1.10) as quotient filtrations of filtrations induced on a subobject, one has

$$\begin{aligned} \operatorname{Gr}_G^n \operatorname{Gr}_F^m(A) &\approx (F^m(A) \cap G^n(A))/((F^{m+1}(A) \cap G^n(A)) + (F^m(A) \cap G^{n+1}(A))) \\ &\qquad\qquad \| \\ \operatorname{Gr}_F^m \operatorname{Gr}_G^n(A) &\approx (G^n(A) \cap F^m(A))/((G^{n+1}(A) \cap F^m(A)) + (G^n(A) \cap F^{m+1}(A))). \end{aligned}$$

(1.2.2) Let H be a third filtration of A. It induces a filtration on $\operatorname{Gr}_F(A)$, and hence on $\operatorname{Gr}_G \operatorname{Gr}_F(A)$. It also induces a filtration on $\operatorname{Gr}_F \operatorname{Gr}_G(A)$. One should note that these filtrations do not in general correspond under the isomorphism (1.2.1). In the expression $\operatorname{Gr}_H \operatorname{Gr}_G \operatorname{Gr}_F(A)$, therefore, G and H play symmetric roles, but F and G do not.

Definition (1.2.3). Two finite filtrations F and $\overline{F}$ on A are called n-opposite if

$$\operatorname{Gr}_F^p \operatorname{Gr}_{\overline{F}}^q(A) = 0 \qquad \text{for } p+q \neq n.$$

(1.2.4) If $A^{p,q}$ is a bigraded object of $\mathcal{A}$ such that:

a) $A^{p,q} = 0$ except for finitely many pairs (p, q);

b) $A^{p,q} = 0$ for $p + q \neq n$,

then one defines two finite n-opposite filtrations of $A = \sum_{p,q} A^{p,q}$ by putting

$$F^p(A) = \sum_{p' \geq p} A^{p',q'}, \tag{1.2.4.1}$$

$$\overline{F}^q(A) = \sum_{q' \geq q} A^{p',q'}. \tag{1.2.4.2}$$

One has

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = A^{p,q}. \tag{1.2.4.3}$$

Conversely:

Proposition (1.2.5).

(i) Let F and $\overline{F}$ be two finite filtrations on A. For F and $\overline{F}$ to be n-opposite, it is necessary and sufficient that

$$\forall p, q \qquad p + q = n + 1 \Rightarrow F^p(A) \oplus \overline{F}^q(A) \xrightarrow{\sim} A.$$

(ii) If F and $\overline{F}$ are n-opposite, and if one puts

$$\begin{cases} A^{p,q} = 0 & \text{for } p + q \neq n, \\ A^{p,q} = F^p(A) \cap \overline{F}^q(A) & \text{for } p + q = n, \end{cases}$$

then A is the direct sum of the $A^{p,q}$, and F and $\overline{F}$ are deduced from the bigrading $A^{p,q}$ of A by the procedure (1.2.4).

Proof. (i) The condition $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q > n$ means that

$$F^p \cap \overline{F}^q = (F^{p+1} \cap \overline{F}^q) + (F^p \cap \overline{F}^{q+1}) \qquad (p + q > n).$$

By hypothesis, $F^p \cap \overline{F}^q$ is zero for $p+q$ sufficiently large; by descending induction one deduces that the condition $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q > n$ is equivalent to the condition $F^p(A) \cap \overline{F}^q(A) = 0$ for $p + q > n$. Dually ((1.1.6), (1.1.7), (1.1.10)) the condition $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q < n$ is equivalent to $A = F^p(A) + \overline{F}^q(A)$ for $(1 - p) + (1 - q) > -n$, that is, for $p + q \leq n + 1$, and (i) follows.

(ii) If F and $\overline{F}$ are n-opposite, we prove by descending induction on p that

$$\bigoplus_{p' \geq p} A^{p',q'} \xrightarrow{\sim} F^p(A). \tag{1.2.5.1}$$

a) For $F^p(A) = 0$, the assertion is evident.

b) The decomposition $A = F^{p+1}(A) \oplus \overline{F}^{n-p}(A)$ induces on $F^p(A) \supset F^{p+1}(A)$ a decomposition

$$F^p(A) = F^{p+1}(A) \oplus (F^p(A) \cap \overline{F}^{n-p}(A)),$$

and the induction concludes the proof. For p sufficiently small, $F^p(A) = A$. By (1.2.5.1), the $A^{p,q}$ therefore form a bigrading of A, and F satisfies (1.2.4.1). That $\overline{F}$ satisfies (1.2.4.2) follows by symmetry.

(1.2.6) The constructions (1.2.4) and (1.2.5) establish quasi-inverse equivalences of categories between objects of $\mathcal{A}$ endowed with two finite n-opposite filtrations and bigraded objects of $\mathcal{A}$ of the type considered in (1.2.4).

Definition (1.2.7). Three finite filtrations W, F, and $\overline{F}$ on an object A of $\mathcal{A}$ are called opposite if

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^n(A) = 0 \qquad \text{for } p+q+n \neq 0.$$

This condition is symmetric in F and $\overline{F}$. It means that F and $\overline{F}$ induce on $W^n(A)/W^{n+1}(A)$ two $(-n)$-opposite filtrations. Put

$$A^{p,q} = \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^{-p-q}(A),$$

whence decompositions (1.2.4), (1.2.5)

$$W^n(A)/W^{n+1}(A) = \bigoplus_{p+q=-n} A^{p,q} \tag{1.2.7.1}$$

which make $\mathrm{Gr}_W(A)$ into a bigraded object.

Lemma (1.2.8). Let W, F, and $\overline{F}$ be three finite opposite filtrations on A, and let σ be a sequence $(p_i, q_i)_{i\geq 0}$ of pairs of integers satisfying:

a) $p_i \leq p_j$ and $q_i \leq q_j$ for $i \geq j$;

b) For $i > 0$, $p_i + q_i = p_0 + q_0 - i + 1$.

Put $p = p_0$, $q = q_0$, $n = -p - q$, and

$$A_\sigma = \left(\sum_{0\leq i}(W^{n+i}(A) \cap F^{p_i}(A))\right) \cap \left(\sum_{0\leq i}(W^{n+i}(A) \cap \overline{F}^{q_i}(A))\right).$$

Then the projection from $W^n(A)$ to $\mathrm{Gr}_W^n(A)$ induces an isomorphism

$$A_\sigma \xrightarrow{\sim} A^{p,q} \subset \mathrm{Gr}_W^n(A).$$

We prove by induction on k the following assertion:

$(*_k)$ The projection from $W^n(A)/W^{n+k}(A)$ to $\mathrm{Gr}_W^n(A)$ induces an isomorphism from

$$(((\sum_{i<k}(W^{n+i}(A) \cap F^{p_i}(A))) + W^{n+k}(A)) \cap ((\sum_{i<k}(W^{n+i}(A) \cap \overline{F}^{q_i}(A))) + W^{n+k}(A)))/W^{n+k}(A)$$

to $A^{p,q} \subset \mathrm{Gr}_W^n(A)$.

For $k = 1$, this is exactly the definition of $A^{p,q}$. By (1.2.5)(i), one has

$$F^{p_k}(\mathrm{Gr}_W^{n+k}(A)) \oplus \overline{F}^{q_k}(\mathrm{Gr}_W^{n+k}(A)) \xrightarrow{\sim} \mathrm{Gr}_W^{n+k}(A). \tag{1.2.8.1}$$

Put

$$\begin{aligned}
B &= \sum_{i<k}(W^{n+i}(A) \cap F^{p_i}(A)) \\
C &= \sum_{i<k}(W^{n+i}(A) \cap \overline{F}^{q_i}(A)) \\
B' &= (W^{n+k}(A) \cap F^{p_k}(A)) + W^{n+k+1}(A) \\
C' &= (W^{n+k}(A) \cap \overline{F}^{q_k}(A)) + W^{n+k+1}(A) \\
D &= W^{n+k}(A) \\
E &= W^{n+k+1}(A).
\end{aligned}$$

Formula (1.2.8.1) becomes

$$B' + C' = D,$$

and

$$B' \cap C' = E.$$

Moreover, since $p_k \leq p_i$ $(i \leq k)$,

$$B \cap D \subset F^{p_k}(A) \cap W^{n+k}(A) \subset B'$$

and since $q_k \leq q_i$ $(i \leq k)$,

$$C \cap D \subset \overline{F}^{q_k}(A) \cap W^{n+k}(A) \subset C'.$$

The assertion $(*_{k+1})$ therefore follows from $(*_k)$ and from the following lemma.

Lemma (1.2.9). Let B, C, B', C', D, and E be subobjects of A. Assume that

$$B' + C' = D, \qquad B' \cap C' = E,$$
$$B \cap D \subset B', \qquad C \cap D \subset C'.$$

Then,

$$((B + B') \cap (C + C'))/E \xrightarrow{\sim} ((B + D) \cap (C + D))/D.$$

To prove surjectivity, one writes

$$\begin{aligned}((B + B') \cap (C + C')) + D &= (((B + B') \cap (C + C')) + B') + C' \\ &= ((B + B') \cap (C + C' + B')) + C' \\ &= (B + B' + C') \cap (C + C' + B') \\ &= (B + D) \cap (C + D).\end{aligned}$$

To prove injectivity, one writes

$$(B + B') \cap (C + C') \cap D = ((B + B') \cap D) \cap ((C + C') \cap D).$$

Since $B' \subset D$, one has

$$(B + B') \cap D = (B \cap D) + B' = B';$$

similarly,

$$(C + C') \cap D = C',$$

and

$$(B + B') \cap (C + C') \cap D = B' \cap C' = E.$$

Finally, the proof of (1.2.8) is completed by observing that (1.2.8) is equivalent to $(*_k)$ for large k.

Theorem (1.2.10). Let $\mathcal{A}$ be an abelian category, and provisionally denote by $\mathcal{A}'$ the category of objects of $\mathcal{A}$ endowed with three opposite filtrations W, F, and $\overline{F}$. The morphisms in $\mathcal{A}'$ are the morphisms in $\mathcal{A}$ compatible with the three filtrations.

(i) $\mathcal{A}'$ is an abelian category.

(ii) The kernel (respectively, the cokernel) of an arrow $f : A \to B$ in $\mathcal{A}'$ is the kernel (respectively, the cokernel) of f in $\mathcal{A}$, endowed with the filtrations induced from those of A (respectively, the quotient filtrations induced from those of B).

(iii) Every morphism $f : A \to B$ in $\mathcal{A}'$ is strictly compatible with the filtrations W, F, and $\overline{F}$; the morphism $\mathrm{Gr}_W(f)$ is compatible with the bigradings of $\mathrm{Gr}_W(A)$ and $\mathrm{Gr}_W(B)$; the morphisms $\mathrm{Gr}_F(f)$ and $\mathrm{Gr}_{\overline{F}}(f)$ are strictly compatible with the filtration induced by W.

(iv) The functors *forget the filtrations*, Gr_W, Gr_F, $\mathrm{Gr}_{\overline{F}}$, and

$$\begin{aligned}\mathrm{Gr}_W \mathrm{Gr}_F &\simeq \mathrm{Gr}_F \mathrm{Gr}_W \simeq \mathrm{Gr}_{\overline{F}} \mathrm{Gr}_F \mathrm{Gr}_W \\ &\simeq \mathrm{Gr}_{\overline{F}} \mathrm{Gr}_W \simeq \mathrm{Gr}_W \mathrm{Gr}_{\overline{F}}\end{aligned}$$

from $\mathcal{A}'$ to $\mathcal{A}$ are exact.

Let $\sigma_0(p, q)$ and $\sigma_1(p, q)$ denote the sequences

$$\sigma_0(p, q) = (p, q), (p, q), (p, q - 1), (p, q - 2), (p, q - 3), \ldots$$
$$\sigma_1(p, q) = (p, q), (p, q), (p - 1, q), (p - 2, q), (p - 3, q), \ldots$$

and, with the notation of (1.2.8), put

$$A_i^{p,q} = A_{\sigma_i(p,q)} \qquad (i = 0, 1).$$

If $f : A \to B$ is compatible with W, F, and $\overline{F}$, then

$$f(A_i^{p,q}) \subset B_i^{p,q} \qquad (i = 0, 1). \tag{1.2.10.1}$$

Assertion (iii) therefore follows from the following lemma.

Lemma (1.2.11). The $A_i^{p,q}$ form a bigrading of A. One has

$$W^n(A) = \sum_{n+p+q\leq 0} A_i^{p,q} \qquad (i = 0, 1), \tag{1.2.11.1}$$

$$F^p(A) = \sum_{p'\geq p} A_0^{p',q'}, \tag{1.2.11.2}$$

$$\overline{F}^q(A) = \sum_{q'\geq q} A_1^{p',q'}. \tag{1.2.11.3}$$

By symmetry, it is enough to prove the assertions for $i = 0$. Put $A_0 = \bigoplus A_0^{p,q}$, and define on A_0 filtrations W and F by the formulae of (1.2.11). The canonical map i from A_0 to A is compatible with the filtrations W and F. Moreover, by (1.2.8), $\mathrm{Gr}_W(i)$ is an isomorphism, and it induces isomorphisms of graded objects

$$\sum_{p+q=n} A_0^{p,q} \xrightarrow{\sim} \mathrm{Gr}_W^{-n}(A) = \sum_{p+q=n} A^{p,q}. \tag{1.2.11.4}$$

The morphism i is therefore an isomorphism, and the $A_0^{p,q}$ form a bigrading of A. Formula (1.2.11.1) then expresses that $\mathrm{Gr}_W(i)$ is an isomorphism. By (1.2.11.4), $\mathrm{Gr}_F \mathrm{Gr}_W(i)$ is an isomorphism, hence so are $\mathrm{Gr}_W \mathrm{Gr}_F(i)$ and $\mathrm{Gr}_F(i)$. Formula (1.2.11.2) follows.

(1.2.12) We prove (1.2.10). Let $f : A \to B$ be in $\mathcal{A}'$ and endow $K = \mathrm{Ker}(f)$ with the filtrations induced from those of A. By (1.2.11), $\mathrm{Gr}_W(K) \hookrightarrow \mathrm{Gr}_W(A)$; moreover, the filtration F (respectively, $\overline{F}$) of K induces on $\mathrm{Gr}_W(K)$ the inverse-image filtration of the filtration F on $\mathrm{Gr}_W(A)$. Finally, the subobject $\mathrm{Gr}_W(K)$ of $\mathrm{Gr}_W(A)$ is compatible with the bigrading of $\mathrm{Gr}_W(A)$:

$$\mathrm{Gr}_W(K) = \bigoplus_{p,q} (\mathrm{Gr}_W(K) \cap A^{p,q}).$$

It follows that

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^n(K) \hookrightarrow \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^n(A);$$

the filtrations W, F, and $\overline{F}$ of K are therefore opposite, and K is a kernel of f in $\mathcal{A}'$. Together with the dual result, this proves (ii).

If f is an arrow of $\mathcal{A}'$, the canonical morphism from $\mathrm{Coim}(f)$ to $\mathrm{Im}(f)$ is an isomorphism in $\mathcal{A}$; by (iii), it is also an isomorphism in $\mathcal{A}'$, which is therefore abelian.

The functor *forget the filtrations* is exact by (ii). The exactness of the other functors in (iv) follows at once from (ii), (iii), and (1.1.11)(i) or (ii).

(1.2.13) Let A be an object of $\mathcal{A}$ endowed with a finite increasing filtration W, and with two finite decreasing filtrations F and $\overline{F}$. The construction (1.1.3) associates to $W_\bullet$ a decreasing filtration $W^\bullet$. We shall say that the filtrations $W_\bullet$, F, and $\overline{F}$ are opposite if the filtrations $W^\bullet$, F, and $\overline{F}$ are opposite, that is, if for every n the filtrations induced by F and $\overline{F}$ on

$$\mathrm{Gr}_n^W(A) = W_n(A)/W_{n-1}(A)$$

are n-opposite.

Theorem (1.2.10) carries over trivially to this variant.

1.3. The lemma on two filtrations

(1.3.1) Let K be a differential complex of objects of $\mathcal{A}$, endowed with a filtration F. This filtration is called biregular if it induces on each component of K a finite filtration. Recall the definition of the terms $E_r^{pq}(K, F)$, or simply E_r^{pq}, of the spectral sequence defined by F. Put

$$Z_r^{pq} = \mathrm{Ker}\left(d : F^p(K^{p+q}) \to K^{p+q+1}/F^{p+r}(K^{p+q+1})\right),$$

and define B_r^{pq} dually by the formula

$$K^{p+q}/B_r^{pq} = \operatorname{coker}\left(d : F^{p-r+1}(K^{p+q-1}) \to K^{p+q}/F^{p+1}(K^{p+q})\right).$$

These formulae still make sense for $r = \infty$.

One should note that the use made here of the notation B_r^{pq} is different from that of Godement [TF].

By definition one has:

$$E_r^{pq} = \operatorname{Im}\left(Z_r^{pq} \to K^{p+q}/B_r^{pq}\right) \tag{1.3.1.1}$$

$$= Z_r^{pq}/(B_r^{pq} \cap Z_r^{pq}) \tag{1.3.1.2}$$

$$= \operatorname{Ker}\left(K^{p+q}/B_r^{pq} \to K^{p+q}/(Z_r^{pq} + B_r^{pq})\right). \tag{1.3.1.3}$$

One can also write

$$B_r^{p*} \cap Z_r^{p*} \underset{\text{dfn}}{=} (dF^{p-r+1} + F^{p+1}) \cap (d^{-1}F^{p+r} \cap F^p)$$

$$= (dF^{p-r+1} \cap F^p) + (F^{p+1} \cap d^{-1}F^{p+r}), \tag{1.3.1.4}$$

since $dF^{p-r+1} \subset d^{-1}F^{p+r}$ and $F^{p+1} \subset F^p$.

For $r < \infty$, the E_r form a complex graded by the degree $p - r(p+q)$, and E_{r+1} is expressed as the cohomology of this complex:

$$E_{r+1}^{p,q} = H\big(E_r^{p-r,q+r-1} \xrightarrow{d_r} E_r^{p,q} \xrightarrow{d_r} E_r^{p+r,q-r+1}\big). \tag{1.3.1.5}$$

For $r = 0$ one has

$$E_0^{**} = \operatorname{Gr}_F^*(K^*). \tag{1.3.1.6}$$

Proposition (1.3.2). Let K be a complex endowed with a biregular filtration F. The following conditions are equivalent:

(i) the spectral sequence defined by F degenerates ($E_1 = E_\infty$).

(ii) the morphisms $d : K^i \to K^{i+1}$ are strictly compatible with the filtrations.

We check this when $\mathcal{A}$ is a category of modules. For fixed p and q, the hypothesis that the arrows d_r with source $E_r^{p,q}$ are zero for $r \geq 1$ means that, if $x \in F^p(K^{p+q})$ satisfies $dx \in F^{p+1}(K^{p+q+1})$, then there exists y in K^{p+q} such that $dy = 0$ and such that x and y have the same image in $E_1^{p,q}$. Modifying y by a boundary and putting $z = x - y$, one obtains

$$\forall x \in F^p(K^{p+q})\ \big(dx \in F^{p+1}(K^{p+q+1}) \Rightarrow \exists z \in F^{p+1}(K^{p+q}),\ dz = dx\big)$$

or, in other words,

$$F^{p+1}(K^{p+q+1}) \cap dF^p(K^{p+q}) = dF^{p+1}(K^{p+q}). \tag{1}$$

If this condition holds for all p and q, then by induction on r one has

$$F^{p+r} \cap dF^p = dF^{p+r},$$

which, for $p + r$ large, becomes

$$F^p \cap dK = dF^p. \tag{2}$$

Assertion (2) trivially implies (i), and is equivalent to (ii). This proves (1.3.2).

(1.3.3) If (K, F) is a filtered complex, we shall denote by $\operatorname{Dec}(K)$ the complex K endowed with the shifted filtration

$$\operatorname{Dec}(F)^p K^n = Z_1^{p+n,-p}.$$

This filtration is compatible with the differentials:

$$dZ_1^{p+n,-p} \subset F^{p+n+1}(K^{n+1}) \cap \operatorname{Ker}(d) \subset Z_\infty^{p+n+1,-p} \subset Z_1^{p+n+1,-p}.$$

Since

$$Z_1^{p+1+n,-p-1} \subset F^{p+1+n}(K^n) \subset B_1^{p+n,-p} \subset Z_1^{p+n,-p}, \tag{1.3.3.1}$$

the evident map from $Z_1^{p+n,-p}/Z_1^{p+1+n,-p-1}$ to $Z_1^{p+n,-p}/B_1^{p+n,-p}$ is a morphism

$$u : E_0^{p,n-p}(\operatorname{Dec} K) \longrightarrow E_1^{p+n,-p}(K). \tag{1.3.3.2}$$

Proposition (1.3.4).

(i) The morphisms (1.3.3.2) form a morphism of graded complexes from $E_0(\mathrm{Dec}\, K)$ to $E_1(K)$.

(ii) This morphism induces an isomorphism on cohomology.

(iii) Step by step, via (1.3.1.5), it induces isomorphisms of graded complexes

$$E_r(\mathrm{Dec}(K)) \xrightarrow{\sim} E_{r+1}(K) \qquad (r \geq 1).$$

Proof. Let F' be the filtration of K defined by

$$F'^p(K^n) = \mathrm{Dec}(F)^{p-n}(K^n) = Z_1^{p,n-p}.$$

There are trivial isomorphisms, compatible with the d_r and with (1.3.1.5),

$$E_r^{p,n-p}(\mathrm{Dec}\, K) = E_{r+1}^{p+n,-p}(K, F'). \tag{1.3.4.1}$$

The map u is deduced from (1.3.4.1) and from the identity morphism

$$(K, F') \longrightarrow (K, F).$$

This proves (i), and it remains to verify that for $r \geq 2$,

$$E_r^{p,q}(K, F') \simeq E_r^{p,q}(K, F).$$

Indeed one has

$$Z_r^{p,q}(K, F') = Z_r^{p,q}(K, F) \qquad (r \geq 1),$$

and

$$Z_r^{p,q}(K, F') \cap B_r^{p,q}(K, F') = Z_r^{p,q}(K, F) \cap B_r^{p,q}(K, F) \qquad (r \geq 2),$$

and then applies (1.3.1.2).

(1.3.5) The construction (1.3.3) is not self-dual. The dual construction consists in defining

$$\mathrm{Dec}^*(F)^p K^n = B_1^{p+n-1,-p+1}.$$

One then has morphisms

$$E_0^{p,n-p}(\mathrm{Dec}\, K) \longrightarrow E_1^{p+n,p}(K) \longrightarrow E_0^{p,n-p}(\mathrm{Dec}^*\, K)$$

and, for $r \geq 1$, isomorphisms

$$E_r^{p,n-p}(\mathrm{Dec}\, K) \xrightarrow{\sim} E_{r+1}^{p+n,p}(K) \xrightarrow{\sim} E_r^{p,n-p}(\mathrm{Dec}^*\, K).$$

Recall that a morphism of complexes is called a quasi-isomorphism if it induces an isomorphism on cohomology.

Definition (1.3.6).

(i) A morphism $f : (K, F) \to (K', F')$ of filtered complexes with biregular filtration is a filtered quasi-isomorphism if $\mathrm{Gr}_F(f)$ is a quasi-isomorphism, i.e. if the $E_1^{p,q}(f)$ are isomorphisms.

(ii) A morphism $f : (K, F, W) \to (K', F', W')$ between biregular bifiltered complexes is a bifiltered quasi-isomorphism if $\mathrm{Gr}_F \mathrm{Gr}_W(f)$ is a quasi-isomorphism.

(1.3.7) Let K be a differential complex of objects of $\mathcal{A}$, endowed with two filtrations F and W. Let $E_r^{p,q}$ be the spectral sequence defined by W. The filtration F induces several filtrations on the $E_r^{p,q}$, which we shall compare.

(1.3.8) Formula (1.3.1.2) identifies $E_r^{p,q}$ with a quotient of a subobject of K^{p+q}. Thus the term $E_r^{p,q}$ receives a filtration F_d induced by F, called the first direct filtration.

(1.3.9) Dually, formula (1.3.1.3) identifies $E_r^{p,q}$ with a subobject of a quotient of K^{p+q}, and hence gives a new filtration F_{d*} induced by F, the second direct filtration.

Lemma (1.3.10). On E_0 and E_1 one has $F_d = F_{d*}$. For $r = 0$ or 1, one has $B_r^{p,q} \subset Z_r^{p,q}$, and one applies (1.1.9).

(1.3.11) Formula (1.3.1.5) identifies $E_{r+1}^{p,q}$ with a quotient of a subobject of $E_r^{p,q}$. The recurrent filtration F_r of the $E_r^{p,q}$ is defined by the following conditions:

(i) on $E_0^{p,q}$, $F_r = F_d = F_{d*}$.

(ii) on $E_{r+1}^{p,q}$, the recurrent filtration is the one induced by the recurrent filtration of $E_r^{p,q}$.

(1.3.12) The definitions (1.3.8) and (1.3.9) still make sense for $r = \infty$. If the filtration of K is biregular, the direct filtrations of $E_\infty^{p,q}$ coincide with those of $E_r^{p,q} = E_\infty^{p,q}$ for r large enough, and the recurrent filtration of $E_\infty^{p,q}$ is defined by requiring it to coincide with that of $E_r^{p,q}$ for r large enough.

The filtrations F and W each induce a filtration of $H^*(K)$, and $E_\infty^{**} = \mathrm{Gr}_W^*(H^*(K))$. The filtration F of $H^*(K)$ therefore induces a new filtration on $E_\infty^{p,q}$.

Proposition (1.3.13).

(i) For the first direct filtration, the morphisms d_r are compatible with filtrations. If $E_{r+1}^{p,q}$ is regarded as a quotient of a subobject of $E_r^{p,q}$, then the first direct filtration on $E_{r+1}^{p,q}$ is finer than the filtration F' induced by the first direct filtration on $E_r^{p,q}$; one has $F_d(E_{r+1}^{p,q}) \subset F'(E_{r+1}^{p,q})$.

(ii) Dually, the morphisms d_r are compatible with the second direct filtration, and the second direct filtration on $E_{r+1}^{p,q}$ is less fine than the filtration induced by that of $E_r^{p,q}$.

(iii) $F_d(E_r^{p,q}) \subset F_r(E_r^{p,q}) \subset F_{d*}(E_r^{p,q})$.

(iv) On $E_\infty^{p,q}$, the filtration induced by the filtration F of $H^*(K)$ (1.3.12) is finer than the first direct filtration and less fine than the second.

(i) is evident, (ii) is its dual, and (iii) follows by induction. The first assertion of (iv) is easy to check, and the second is dual to it.

(1.3.14) Denote by $\mathrm{Dec}(K)$, respectively by $\mathrm{Dec}^*(K)$, the complex K endowed with the filtrations $\mathrm{Dec}(W)$ and F, respectively $\mathrm{Dec}^*(W)$ and F.

It is clear from (1.3.4.1) that the isomorphism (1.3.4) transforms the first direct filtration of $E_r(\mathrm{Dec}\, K)$ into the first direct filtration of $E_{r+1}(K)$, for $r \geq 1$. The dual isomorphism (1.3.5) transforms the second direct filtration of $E_r(\mathrm{Dec}^*\, K)$ into the second direct filtration of $E_{r+1}(K)$.

Lemma (1.3.15). If the filtration F is biregular, and if, on the $\mathrm{Gr}_W^p(K)$, the morphisms d are strictly compatible with the filtration induced by F, then:

(i) the morphism (1.3.3.2) of graded complexes filtered by F

$$u : \mathrm{Gr}_{\mathrm{Dec}(W)}(K) \longrightarrow E_1(K, W)$$

is a filtered quasi-isomorphism;

(ii) dually, the morphism (1.3.5)

$$u : E_1(K, W) \longrightarrow \mathrm{Gr}_{\mathrm{Dec}^*(W)}(K)$$

is a filtered quasi-isomorphism.

By duality it is enough to prove (i). By (1.3.3) and (1.3.4), the complex $E_1(K, W)$, filtered by F, is a quotient of the filtered complex $\mathrm{Gr}_{\mathrm{Dec}(W)}(K)$. Let U be the filtered kernel complex; it is acyclic by (1.3.4) (ii). The long exact cohomology sequence associated with the exact sequence of complexes

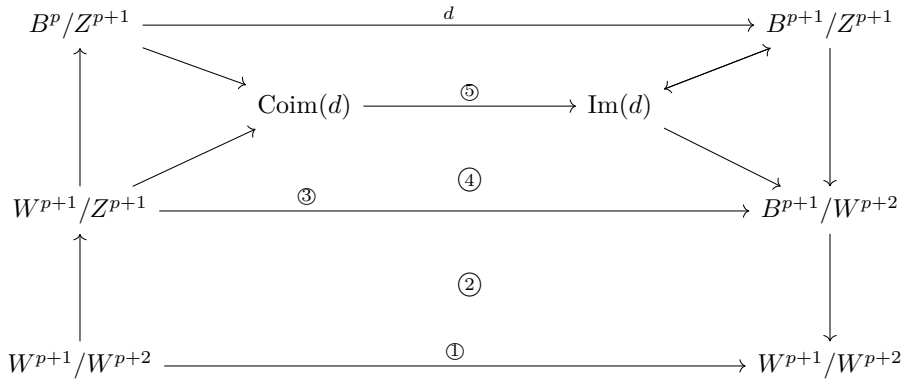
$$0 \to \mathrm{Gr}_F(U) \to \mathrm{Gr}_F(\mathrm{Gr}_{\mathrm{Dec}(W)}(K)) \to \mathrm{Gr}_F(E_1(K, W)) \to 0$$

shows that u is a filtered quasi-isomorphism if and only if $\mathrm{Gr}_F(U)$ is an acyclic complex. By (1.3.2), and because U is acyclic, this amounts to requiring that the differentials of U be strictly compatible with the filtration F. From (1.3.3.1) one obtains that U is the sum, over p, of the complexes

$$(U^p)^n = B_1^{p+n,-p}/Z_1^{p+1+n,-p-1},$$

endowed with the filtration induced by F.

Each differential d of each of the complexes U^p is included in a commutative diagram of filtered objects of the following type, where, for simplicity, the total or complementary degree has been omitted:



By hypothesis, the morphism (1) is strict. Since the square (2) is just the canonical decomposition of (1), the arrow (3) is a filtered isomorphism. The arrows of the trapezium (4) are isomorphisms; hence they are filtered isomorphisms, because (3) is one. That (5) is a filtered isomorphism means that d is strict. This proves (1.3.15).

Theorem (1.3.16). Let K be a complex endowed with two filtrations W and F, with F biregular. Let $r_0 \geq 0$ be an integer, and suppose that, for $0 \leq r < r_0$, the differentials of the graded complex $E_r(K, W)$ are strictly compatible with the filtration F_r. Then, for $r \leq r_0 + 1$, one has

$$F_d = F_r = F_{d*} \quad \text{on } E_r^{p,q}.$$

The theorem is proved by induction on r_0. For $r_0 = 0$ the hypothesis is empty, and one applies (1.3.10) and (1.3.13), (iii). For $r_0 \geq 1$, by the induction hypothesis, one has $F_d = F_r = F_{d*}$ on $E_r^{p,q}$ for $r \leq r_0$.

By (1.3.15), the morphism $u : E_0(\mathrm{Dec}\, K) \to E_1(K)$ is a filtered quasi-isomorphism. It therefore induces a filtered isomorphism from $H^*(\mathrm{Dec}\, K)$ to $H^*(E_1(K))$:

$$u : (E_1(\mathrm{Dec}\, K), F_r) \xrightarrow{\sim} (E_2(K), F_r).$$

Step by step, it follows that the canonical isomorphism from $E_s(\mathrm{Dec}\, K)$ to $E_{s+1}(K)$ ($s \geq 1$) is a filtered isomorphism for the recurrent filtration. On $E_1(\mathrm{Dec}(K))$, $F_r = F_d$ (1.3.10), and we already know (1.3.14) that u' is a filtered isomorphism

$$u' : (E_1(\mathrm{Dec}\, K), F_d) \xrightarrow{\sim} (E_2(K), F_d).$$

Thus on $E_2(K)$ one has $F_d = F_r$. Together with the dual result, this proves (1.3.16) for $r_0 = 1$.

Assume now that $r_0 \geq 2$. Then the arrows d_1 of $E_1(K)$ are strictly compatible with the filtrations; hence so are the arrows d_0 of $E_0(\mathrm{Dec}\, K)$ (indeed, u induces an isomorphism of spectral sequences, and one applies the criterion (1.3.2)).

For $0 < s < r_0 - 1$, the isomorphism $(E_s(\mathrm{Dec}\, K), F_r) \simeq (E_{s+1}(K), F_r)$ shows that the d_s are strictly compatible with the recurrent filtrations. By the induction hypothesis one therefore has $F_d = F_r$ on $E_s(\mathrm{Dec}\, K)$ for $s \leq r_0$. The isomorphism $(E_s(\mathrm{Dec}\, K), F_d) \simeq (E_{s+1}(K), F_d)$ (1.3.13) then shows that $F_d = F_r$ on $E_r(K)$ for $r \leq r_0 + 1$. Together with the dual result, this proves (1.3.16).

Corollary (1.3.17). Under the general hypotheses of (1.3.16), suppose that, for every r, the differentials d_r are strictly compatible with the recurrent filtrations of the E_r. Then, on E_∞, the filtrations F_d, F_r, and F_{d*} coincide, and coincide with the filtration induced by the filtration F of $H^*(K)$.

This follows at once from (1.3.16) and (1.3.13) (iv).

1.4. Hypercohomology of filtered complexes.

In this section we recall some standard constructions in hypercohomology. We do not use the language of derived categories, which would be the most natural here.

Throughout this section, by a "complex" we shall mean a "complex bounded below".

(1.4.1) Let T be a left exact functor from an abelian category $\mathcal{A}$ to an abelian category $\mathcal{B}$. Assume that every object of $\mathcal{A}$ injects into an injective object; the derived functors $R^iT : \mathcal{A} \to \mathcal{B}$ are therefore defined. An object A of $\mathcal{A}$ will be called acyclic for T if $R^iT(A) = 0$ for $i > 0$.

(1.4.2) Let (A, F) be a filtered object with a finite filtration, and let TF be the filtration of TA by its subobjects TF^pA (these are subobjects because T is left exact). If $\mathrm{Gr}_F(A)$ is T-acyclic, then the $F^p(A)$ are T-acyclic as successive extensions of T-acyclic objects. The image under T of the sequence

$$0 \to F^{p+1}(A) \to F^p(A) \to \mathrm{Gr}^p(A) \to 0$$

is therefore exact, and

$$\mathrm{Gr}_{TF}\, TA \xrightarrow{\sim} T\,\mathrm{Gr}_F\, A. \tag{1.4.2.1}$$

(1.4.3) Let A be an object with two finite filtrations F and W such that $\mathrm{Gr}_F\,\mathrm{Gr}_W\, A$ is T-acyclic. The objects $\mathrm{Gr}_F\, A$ and $\mathrm{Gr}_W\, A$ are then T-acyclic, as are the $F^q(A) \cap W^p(A)$. The sequences

$$0 \to T(F^q \cap W^{p+1}) \to T(F^q \cap W^p) \to T\big((F^q \cap W^p)/(F^q \cap W^{p+1})\big) \to 0$$

are therefore exact, and $T(F^q(\mathrm{Gr}^p_W(A)))$ is the image in $T(\mathrm{Gr}^p_W(A))$ of $T(F^p \cap W^q)$. The diagram

$$\begin{array}{ccccc} T(F^q \cap W^p) & \longrightarrow & T(F^q\,\mathrm{Gr}^p_W\, A) & \longrightarrow & T\,\mathrm{Gr}^p_W\, A \\ \downarrow & & & & \downarrow \\ TF^q \cap TW^p & & \longrightarrow & & \mathrm{Gr}^p_{TW}\, TA \end{array}$$

then shows that the isomorphism (1.4.2.1) relative to W carries the filtration $\mathrm{Gr}_{TW}(TF)$ to the filtration $T(\mathrm{Gr}_W(F))$.

(1.4.4) Let K be a complex of objects of $\mathcal{A}$. The *hypercohomology objects* $R^iT(K)$ are computed as follows:

a) Choose a quasi-isomorphism $i : K \to K'$ such that the components of K' are acyclic for T. For example, one may take for K' the simple complex associated with a Cartan–Eilenberg injective resolution of K.

b) Put
$$R^iT(K) = H^i(T(K')).$$

One checks that $R^iT(K)$ does not depend on the choice of K', depends functorially on K, and that a quasi-isomorphism $f : K_1 \to K_2$ induces *isomorphisms*

$$R^iT(f) : R^iT(K_1) \longrightarrow R^iT(K_2).$$

(1.4.5) Let F be a biregular filtration of K. A *filtered T-acyclic resolution* of K is a filtered quasi-isomorphism $i : K \to K'$ from K to a biregular filtered complex such that the $\mathrm{Gr}^p(K'^n)$ are acyclic for T. If K' is such a resolution, the K'^n are acyclic for T and the filtered complex (cf. (1.4.2)) $T(K')$ defines a spectral sequence

$$E_1^{p,q} = R^{p+q}T(\mathrm{Gr}^p\, K) \Rightarrow R^{p+q}T(K).$$

This is independent of the choice of K'. It is called the *hypercohomology spectral sequence of the filtered complex* K. It depends functorially on K, and a filtered quasi-isomorphism induces an isomorphism of spectral sequences.

The differentials d_1 of this spectral sequence are the connecting morphisms defined by the short exact sequences

$$0 \to \mathrm{Gr}^{p+1}\, K \to F^pK/F^{p+2}K \to \mathrm{Gr}^p\, K \to 0.$$

(1.4.6) Let K be a complex. Denote by $\tau_{\leq p}(K)$ the following subcomplex:

$$\tau_{\leq p}(K)^n = \begin{cases} K^n & \text{for } n < p \\ \mathrm{Ker}(d) & \text{for } n = p \\ 0 & \text{for } n > p. \end{cases}$$

The *filtration*, called *canonical*, of K by the $\tau_{\leq p}(K)$ is obtained by shifting the trivial filtration G for which $G^0(K) = K$ and $G^1(K) = 0$. For the canonical filtration one has

$$\begin{aligned} E_1^{p,q} &= 0 \quad \text{if } p+q \neq -p \\ H^{-p} &\qquad \text{if } p+q = -p. \end{aligned}$$

A quasi-isomorphism $f : K \to K'$ is automatically a filtered quasi-isomorphism for the canonical filtrations.

(1.4.7) The subcomplexes $\sigma_{\geq p}(K)$ of K:

$$\sigma_{\geq p}(K)^n = \begin{cases} 0 & \text{if } n < p \\ K^n & \text{if } n \geq p \end{cases}$$

define a biregular filtration, the *stupid filtration* of K.

The hypercohomology spectral sequences attached to the stupid or canonical filtrations of K are the two *hypercohomology spectral sequences* of K.

Example (1.4.8). — Let $f : X \to Y$ be a continuous map between topological spaces, and let $\mathcal{F}$ be an abelian sheaf on X. Let $\mathcal{F}^*$ be a resolution of $\mathcal{F}$ by f_*-acyclic sheaves. Then $R^i f_* \mathcal{F} \simeq \mathcal{H}^i(f_* \mathcal{F}^*)$. Take as T the functor $\Gamma(Y, \cdot)$. The hypercohomology spectral sequence of the complex $f_* \mathcal{F}^*$, endowed with its canonical filtration (1.4.6),

$$E_1^{p,q} = H^{2p+q}(Y, R^{-p} f_* \mathcal{F}) \Rightarrow H^{p+q}(X, \mathcal{F})$$

is, up to the renumbering $E_r^{p,q} \mapsto E_{r+1}^{2p+q,-p}$, the *Leray spectral sequence* for f and $\mathcal{F}$.

(1.4.9) Let (K, W, F) be a biregular bifiltered complex. To this complex one associates:

a) A spectral sequence

$$_W E_1^{p,n-p} = H^n(\mathrm{Gr}_W^p(K)) \Rightarrow H^n(K),$$

whose differentials $_W d_1$ are the connecting morphisms deduced from the short exact sequences

$$0 \to \mathrm{Gr}_W^{p+1}(K) \to W^p(K)/W^{p+2}(K) \to \mathrm{Gr}_W^p(K) \to 0;$$

b) An analogous spectral sequence for the filtration F;

c) Exact squares

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{Gr}_F^{p+1} \mathrm{Gr}_W^{q+1} K & \longrightarrow & F^p/F^{p+2}(\mathrm{Gr}_W^{q+1} K) & \longrightarrow & \mathrm{Gr}_F^p \mathrm{Gr}_W^{q+1} K & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{Gr}_F^{p+1}(W^q/W^{q+2}(K)) & \longrightarrow & F^p/F^{p+2}(W^q/W^{q+2}(K)) & \longrightarrow & \mathrm{Gr}_F^p(W^q/W^{q+2}(K)) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{Gr}_F^{p+1} \mathrm{Gr}_W^q K & \longrightarrow & F^p/F^{p+2} \mathrm{Gr}_W^q K & \longrightarrow & \mathrm{Gr}_F^p \mathrm{Gr}_W^q K & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

The exterior rows and columns of this square define connecting morphisms

$$ {}_{F,W}d_1 : H^n \operatorname{Gr}_F^p \operatorname{Gr}_W^q K \longrightarrow H^{n+1} \operatorname{Gr}_F^{p+1} \operatorname{Gr}_W^q K, $$

$$ {}_{W,F}d_1 : H^n \operatorname{Gr}_F^p \operatorname{Gr}_W^q K \longrightarrow H^{n+1} \operatorname{Gr}_F^p \operatorname{Gr}_W^{q+1} K. $$

These morphisms satisfy

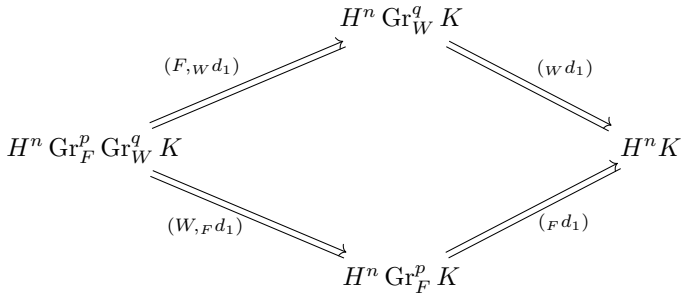
$$ {}_{FW}d_1 \circ {}_{WF}d_1 + {}_{WF}d_1 \circ {}_{FW}d_1 = 0 $$
$$ {}_{FW}d_1^2 = 0 \qquad {}_{WF}d_1^2 = 0. $$

The morphisms ${}_{FW}d_1$ are the d_1 morphisms of the spectral sequences $E^{(q)}$, with term $E_1^{(q)p,n-p}$ equal to

$$ E_1^{p,q,n-p-q} \underset{\text{dfn}}{=} H^n(\operatorname{Gr}_F^p \operatorname{Gr}_W^q K) \Rightarrow H^n(\operatorname{Gr}_W^q K) = {}_W E_1^{q,n-q}, \tag{1.4.9.1} $$

defined by the filtered complex $\operatorname{Gr}_W^q(K)$. This spectral sequence abuts to the filtration induced by F on $H^* \operatorname{Gr}_W^q K$. Similarly, the ${}_{WF}d_1$ are the d_1 of spectral sequences with the same initial terms

$$ E_1^{p,q,n-p-q} = H^n(\operatorname{Gr}_F^p \operatorname{Gr}_W^q K) \Rightarrow H^n(\operatorname{Gr}_F^p K). \tag{1.4.9.2} $$



These constructions are symmetric in F and W, *via* the isomorphism

$$ \operatorname{Gr}_F^p \operatorname{Gr}_W^q \simeq \operatorname{Gr}_W^q \operatorname{Gr}_F^p . $$

(1.4.10) One may also interpret the ${}_{W,F}d_1$ as the initial morphisms of a morphism of spectral sequences (1.4.9.2), abutting to ${}_W d_1$. Indeed, let C^q be the cone of the morphism

$$ W^q(K)/W^{q+2}(K) \longrightarrow \operatorname{Gr}_W^q(K). $$

In the diagram

$$ \Sigma : \operatorname{Gr}_W^{q+1}(K)[1] \xrightarrow{u} C^q \xleftarrow{i} \operatorname{Gr}_W^q(K), $$

u is a quasi-isomorphism, and one has

$$ {}_W d_1 = H(u)^{-1} \circ H(i). $$

In fact, u is even a filtered quasi-isomorphism (for F), and the preceding construction defines a morphism from the spectral sequence defined by $(\operatorname{Gr}_W^q(K), F)$ to the one defined by $(\operatorname{Gr}_W^{q+1}(K)[1], F)$, a morphism which abuts to ${}_W d_1$. The initial term of this morphism, deduced from $\operatorname{Gr}_F(\Sigma)$, is precisely ${}_{W,F}d_1$.

(1.4.11) These constructions pass unchanged to hypercohomology. Let K be a complex endowed with two biregular filtrations F and W.

A *bifiltered T-acyclic resolution* of K is a bifiltered quasi-isomorphism $i : K \to K'$ such that the $\operatorname{Gr}_F^p \operatorname{Gr}_W^n(K'^m)$ are T-acyclic. Such resolutions always exist. In the special case where $\mathcal{A}$ is the category of sheaves of A-modules on a topological space X, and where T is the functor Γ from $\mathcal{A}$ to A-modules, one example of a bifiltered T-acyclic resolution of K is the simple complex associated with the Godement double-resolution complex $\mathcal{C}^*(K)$ of K, filtered by the $\mathcal{C}^*(F^p(K))$ and the $\mathcal{C}^*(W^n(K))$. Since $\mathcal{C}^*$ is exact, one indeed has

$$ \operatorname{Gr}_F \operatorname{Gr}_W(\mathcal{C}^*(K)) \simeq \mathcal{C}^*(\operatorname{Gr}_F \operatorname{Gr}_W(K)). $$

No other case will be needed here.

If K' is a bifiltered T-acyclic resolution of K, the complex TK' is filtered by the TF^pK' and the TW^qK' (1.4.3). Moreover, $\mathrm{Gr}^n_W(K')$ is a filtered T-acyclic resolution (for F) of $\mathrm{Gr}^n_W(K)$, $\mathrm{Gr}^n_F(K')$ is a filtered T-acyclic resolution (for W) of $\mathrm{Gr}^n_F(K)$, $\mathrm{Gr}^n_F\,\mathrm{Gr}^m_W(K')$ is a T-acyclic resolution of $\mathrm{Gr}^n_F\,\mathrm{Gr}^m_W(K)$, and

$$T\,\mathrm{Gr}_F\,K' \approx \mathrm{Gr}_F\,TK' \qquad \text{(as a } W\text{-filtered complex)}$$
$$T\,\mathrm{Gr}_W\,K' \approx \mathrm{Gr}_W\,TK' \qquad \text{(as an } F\text{-filtered complex)}$$
$$T\,\mathrm{Gr}_F\,\mathrm{Gr}_W\,K' \approx \mathrm{Gr}_F\,\mathrm{Gr}_W\,TK'.$$

Lemma (1.4.12). — Under the hypotheses of (1.4.11):

(i) The initial terms of the hypercohomology spectral sequences

$$ {}_WE_1^{q,n-q} = R^nT(\mathrm{Gr}^q_W\,K) \Rightarrow R^nT(K) \tag{1}$$

$$ {}_FE_1^{p,n-p} = R^nT(\mathrm{Gr}^p_F\,K) \Rightarrow R^nT(K) \tag{2}$$

are the abutments of the hypercohomology spectral sequences of the filtered complexes $\mathrm{Gr}^q_W\,K$ and $\mathrm{Gr}^p_F\,K$, with E_1 term given by

$$E_1^{p,q,n-p-q} \underset{\text{dfn}}{=} R^nT(\mathrm{Gr}^p_F\,\mathrm{Gr}^q_W\,K) \Rightarrow {}_WE_1^{q,n-q} \qquad (q \text{ fixed}) \tag{3}$$

$$E_1^{p,q,n-p-q} \underset{\text{dfn}}{=} R^nT(\mathrm{Gr}^p_F\,\mathrm{Gr}^q_W\,K) \Rightarrow {}_FE_1^{p,n-p} \qquad (p \text{ fixed}). \tag{4}$$

(ii) The filtration of ${}_WE_1^{p,n-p}$, the abutment of the spectral sequence (3), is the filtration of ${}_WE_1^{p,n-p}(TK')$ induced by the filtration F of TK'.

(iii) The differentials of the complexes $\mathrm{Gr}^n_W(T(K))$ are strictly compatible with the filtration F if and only if the hypercohomology spectral sequences (3) degenerate at E_1.

(iv) The morphisms d_1 of the spectral sequence (4) are the initial terms of degree-one morphisms of spectral sequences (3) abutting to the morphisms d_1 of the spectral sequence (1).

Assertions (i) and (iv) follow from (1.4.9) and (1.4.10) applied to TK', *via* the isomorphisms (1.4.11). Assertion (ii) is then immediate from the definition of the recurrent filtration F (identical with the direct filtrations by (1.3.10) and (1.3.13) (iii)), and assertion (iii) follows from (1.3.2).

2. Hodge structures

2.1. Pure structures.

(2.1.1) In what follows, $\mathbb{C}$ will denote an algebraic closure of $\mathbb{R}$: we do not assume that a root i of the equation $x^2+1=0$ has been chosen. The theory will be invariant under complex conjugation (cf. (2.1.14)).

(2.1.2) Let S denote the real algebraic group $\mathbb{C}^*$, obtained by Weil restriction of scalars from $\mathbb{C}$ to $\mathbb{R}$ of the group $\mathbb{G}_m$:

$$S = \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m.$$
$$S(\mathbb{R}) = \mathbb{C}^*.$$

The group S is a torus, i.e. it is connected and of multiplicative type. It is therefore described by the finitely generated free abelian group

$$X(S) = \mathrm{Hom}(S_{\mathbb{C}}, \mathbb{G}_m) = \mathrm{Hom}(S, \mathbb{G}_m)(\mathbb{C})$$

of its complex characters, endowed with the action of $\mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \mathbb{Z}/(2)$.

The group $X(S)$ has generators z and $\bar{z}$, inducing respectively the identity and complex conjugation:

$$\mathbb{C}^* = S(\mathbb{R}) \longrightarrow S(\mathbb{C}) \longrightarrow \mathbb{G}_m(\mathbb{C}) = \mathbb{C}^*.$$

Complex conjugation exchanges z and $\bar{z}$.

(2.1.3) There is a canonical map

$$w : \mathbb{G}_m \longrightarrow S \tag{2.1.3.1}$$

which, on real points, induces the inclusion of $\mathbb{R}^*$ into $\mathbb{C}^*$. One has

$$zw = \bar{z}w = \mathrm{Id}\,. \tag{2.1.3.2}$$

There is also a map

$$N : S \longrightarrow \mathbb{G}_m \tag{2.1.3.3}$$

which, on real points, is identified with the norm $N_{\mathbb{C}/\mathbb{R}} : \mathbb{C}^* \to \mathbb{R}^*$. One has

$$N = z\bar{z} \tag{2.1.3.4}$$

and

$$N \circ w = (x \mapsto x^2). \tag{2.1.3.5}$$

Definition (2.1.4). — *A real Hodge structure is a finite-dimensional real vector space V endowed with an action of the real algebraic group S.*

(2.1.5) By the general theory of groups of multiplicative type, giving a real Hodge structure on V is equivalent to giving a bigrading V^{pq} of $V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$ such that $\overline{V^{pq}} = V^{qp}$. The action of S and the bigrading determine one another *via* the condition:

$$\text{On } V^{pq}, \quad S \text{ acts by multiplication by } z^p \bar{z}^q. \tag{2.1.5.1}$$

(2.1.6) Let $(\mathbb{C}, \times)$ be the multiplicative monoid, and put

$$\overline{S} = \prod_{\mathbb{C}/\mathbb{R}} (\mathbb{C}, \times).$$

If V is a real vector space, giving an action of $\overline{S}$ on V is equivalent to giving on V a bigrading such that $\overline{V^{pq}} = V^{qp}$ and $V^{pq} = 0$ for $p < 0$ or $q < 0$.

(2.1.7) Let V be a real Hodge structure, defined by a representation σ of S and by a bigrading V^{pq}. The grading of $V_{\mathbb{C}}$ by the $V^n_{\mathbb{C}} = \sum_{p+q=n} V^{pq}$ is then defined over $\mathbb{R}$. It is called the *weight grading*. On $V^n = V \cap V^n_{\mathbb{C}}$, the representation σw of $\mathbb{G}_m$ is multiplication by x^n.

We shall say that V is *of weight n* if $V^{pq} = 0$ for $p + q \neq n$, i.e. if σw is multiplication by x^n.

(2.1.8) Let V be a real Hodge structure. The *Hodge filtration* on $V_{\mathbb{C}}$ is defined by

$$F^p(V_{\mathbb{C}}) = \sum_{p' \geq p} V^{p'q'}.$$

By (1.2.6), one has

Proposition (2.1.9). — *Let n be an integer. The construction (2.1.8) establishes an equivalence of categories between:*

a) the category of real Hodge structures of weight n;

b) the category of pairs consisting of a finite-dimensional real vector space V and a filtration F on $V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$ which is n-opposed to its complex conjugate $\overline{F}$.

Definition (2.1.10). — *A Hodge structure H of weight n consists of*

a) a finitely generated $\mathbb{Z}$-module $H_{\mathbb{Z}}$ (the "integral lattice");

b) a real Hodge structure of weight n on $H_{\mathbb{R}} = \mathbb{R} \otimes_{\mathbb{Z}} H_{\mathbb{Z}}$.

(2.1.11) A morphism $f : H \to H'$ is a homomorphism $f : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ such that $f_{\mathbb{R}} : H_{\mathbb{R}} \to H'_{\mathbb{R}}$ is compatible with the action of S (i.e. such that $f_{\mathbb{C}}$ is compatible with the bigrading, or with the Hodge filtration).

Hodge structures of weight n form an abelian category. If H has weight n and H' has weight n', one defines a Hodge structure $H \otimes H'$ of weight $n + n'$ by the formulas:

a) $(H \otimes H')_{\mathbb{Z}} = H_{\mathbb{Z}} \otimes H'_{\mathbb{Z}}$;

b) the action of S on $(H \otimes H')_{\mathbb{R}} = H_{\mathbb{R}} \otimes H'_{\mathbb{R}}$ is the tensor product of the actions of S on $H_{\mathbb{R}}$ and $H'_{\mathbb{R}}$.

The bigrading (respectively, the Hodge filtration) of $(H \otimes H')_{\mathbb{C}} = H_{\mathbb{C}} \otimes H'_{\mathbb{C}}$ is the tensor product of the bigradings (respectively, of the Hodge filtrations (cf. (1.1.12))) of $H_{\mathbb{C}}$ and $H'_{\mathbb{C}}$.

Similarly one defines the Hodge structure $\mathrm{Hom}(H, H')$ (of weight $n' - n$), the Hodge structures $\bigwedge^p H$ (of weight pn), and the dual Hodge structure H^*.

The preceding internal *Hom*, and the group of homomorphisms, are related by the *Remark* **(2.1.11.1).** — $\mathrm{Hom}(H, H')$ is the subgroup of

$$\mathrm{Hom}(H, H')_{\mathbb{Z}} = \mathrm{Hom}_{\mathbb{Z}}(H_{\mathbb{Z}}, H'_{\mathbb{Z}})$$

formed by the elements of type $(0, 0)$.

The actions of S on $H_{\mathbb{R}}$, $H'_{\mathbb{R}}$, and $\mathrm{Hom}(H, H')_{\mathbb{R}} = \mathrm{Hom}(H_{\mathbb{R}}, H'_{\mathbb{R}})$ are indeed related by

$$s(f(x)) = s(f)(s(x)).$$

For f to be of type $(0, 0)$, i.e. invariant under S, therefore means that it commutes with the action of S.

(2.1.12) For A a noetherian subring of $\mathbb{R}$, an *A-Hodge structure of weight n* is defined as consisting of a finite-type A-module H_A and of a real Hodge structure of weight n on $H_{\mathbb{R}} = \mathbb{R} \otimes_A H_A$. These definitions are used mainly for $A = \mathbb{Q}$. An A-Hodge structure consists of a finite-type A-module H_A and of a real Hodge structure on $H_{\mathbb{R}} = H_A \otimes_A \mathbb{R}$, such that the weight grading is defined over the field of fractions of A.

Definition (2.1.13). — *The Tate Hodge structure $\mathbb{Z}(1)$ is the Hodge structure of weight -2, of rank 1, purely of bidegree $(-1, -1)$, with integral lattice $2\pi i\mathbb{Z} \subset \mathbb{C}$.*

The action of S is therefore multiplication by the inverse of the norm (2.1.3.3). For $n \in \mathbb{Z}$, define $\mathbb{Z}(n)$ to be the n-th tensor power of $\mathbb{Z}(1)$: $\mathbb{Z}(n)$ is the Hodge structure of weight $-2n$, rank 1, purely of bidegree $(-n, -n)$, with integral lattice $(2\pi i)^n\mathbb{Z} \subset \mathbb{C}$. The action of S is multiplication by $N(x)^{-n}$.

(2.1.14) The choice in $\mathbb{C}$ of a solution i of the equation $x^2 + 1 = 0$ determines on every complex variety X purely of dimension n an orientation $\mathrm{or}_i(X)$. When i is changed to $-i$, one has

$$\mathrm{or}_{-i}(X) = (-1)^n \, \mathrm{or}_i(X).$$

The choice of i also defines an element C of order 4 in $S(\mathbb{R})$: the image of i under the isomorphism $S(\mathbb{R}) \simeq \mathbb{C}^*$.

Finally it defines an isomorphism between $\mathbb{Z}$ and the integral lattice of $\mathbb{Z}(n)$: multiplication by $(2\pi i)^n$.

Whenever i, an orientation of X, C, or an identification $\mathbb{Z} \simeq \mathbb{Z}(n)_{\mathbb{Z}}$ occurs in a formula, it will in principle be understood that they are subordinated to the same choice of i, and that changing i to $-i$ would give an equivalent definition or formula.

Definition (2.1.15). — *A polarization of a Hodge structure H of weight n is a homomorphism*

$$(x, y) : H \otimes H \longrightarrow \mathbb{Z}(-n)$$

such that the real bilinear form $(2\pi i)^n(x, Cy)$ on $H_{\mathbb{R}}$ is symmetric and positive definite.

(2.1.16) The *real Tate Hodge structure* is the real Hodge structure $\mathbb{R}(1)$ underlying $\mathbb{Z}(1)$. Similarly one defines $\mathbb{R}(n)$ underlying $\mathbb{Z}(n)$. A polarization of a *real* Hodge structure H of weight n is a homomorphism

$$(x, y) : H \otimes H \longrightarrow \mathbb{R}(-n)$$

such that the real bilinear form $(2\pi i)^n(x, Cy)$ on $H_{\mathbb{R}}$ is symmetric and positive definite. A polarization is entirely determined by the positive-definite quadratic form $(2\pi i)^n(x, Cy)$ on $H_{\mathbb{R}}$, subject only to the condition that it be invariant under the compact subtorus of S, the kernel of N.

One has

$$(x, y) = (Cx, Cy) = (y, C^2x) = (-1)^n(y, x).$$

The form (x, y) is therefore symmetric or alternating according to the parity of n.

(2.1.17) The reader will generalize these definitions to A-Hodge structures of weight n (2.1.12).

2.2. Hodge theory.

(2.2.1) Let X be a compact Kähler variety, for example smooth projective. By the holomorphic Poincaré lemma, the De Rham complex Ω_X^* is a resolution of the constant sheaf $\mathbb{C}$. Thus one has an isomorphism

$$H^*(X, \mathbb{C}) \sim \mathbb{H}^*(X, \Omega_X^*),$$

and the bête filtration of Ω_X^* (1.4.5) defines the hypercohomology spectral sequence

(2.2.1.1) $$E_1^{pq} = H^q(X, \Omega_X^p) \Rightarrow H^{p+q}(X, \mathbb{C}),$$

whose abutment is the *Hodge filtration* of $H^*(X, \mathbb{C})$.

By Hodge theory [9], [12], one has:

(A) The spectral sequence (2.2.1.1) degenerates: $E_1 = E_\infty$.

(B) The Hodge filtration of $H^n(X, \mathbb{C})$ is n-opposed to the complex conjugate filtration.

(2.2.2) Let V be a local system of real vector spaces on X, i.e. a sheaf of $\mathbb{R}$-vector spaces locally isomorphic to a constant sheaf $\mathbb{R}^n$. Suppose that on V there exists a bilinear form

$$Q : V \otimes V \longrightarrow \mathbb{R}$$

which is locally constant and *definite*. For X connected, this is the case if V is defined by a representation of a finite quotient of the fundamental group of X.

The assertions (A) and (B) above remain valid exactly as stated, without any change in the proofs, for cohomology with coefficients in $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$: the spectral sequence

$$E_1^{pq} = H^q(X, \Omega_X^p(V)) \Rightarrow H^{p+q}(X, V_{\mathbb{C}})$$

deduced from the De Rham resolution of $V_{\mathbb{C}}$ by $\Omega_X^*(V_{\mathbb{C}})$ degenerates and abuts to a filtration on $H^n(X, V_{\mathbb{C}})$ which is n-opposed to the complex conjugate filtration.

The vector space $H^n(X, V)$ is therefore endowed with a canonical real Hodge structure of weight n.

(2.2.3) It is shown in [2] that the statements (2.2.1) remain valid for X a complete nonsingular algebraic variety, not necessarily Kähler. The proof in *loc. cit.*, based on a reduction to the projective case *via* Chow's lemma and resolution of singularities, extends to the setting (2.2.2).

(2.2.4) Let $\mathcal{L}$ be an invertible sheaf. Here are two ways to define the class $c_1(\mathcal{L})$.

(2.2.4.1) The sheaf $\mathcal{L}$ defines an element c in $H^1(X, \mathcal{O}^*)$. Its image under $df/f : \mathcal{O}^* \to \Omega_X^1$ lies in $H^1(X, \Omega_X^1)$. More precisely, df/f defines a morphism of complexes

$$d\log : \mathcal{O}^*[-1] \longrightarrow [\,0 \to \Omega_X^1 \to \Omega_X^2 \to \cdots] = \sigma_{\geq 1}(\Omega_X^*).$$

This complex maps to Ω_X^*, whence

$$d\log : \mathcal{O}^*[-1] \longrightarrow \Omega_X^*,$$

and the image of c under $d\log$ lies in $H^2(\Omega_X^*)$. This construction would still make sense for an algebraic variety over an arbitrary field k. For $k = \mathbb{C}$, one also has

$$H^2(X, \mathbb{C}) \sim H^2(X, \Omega_X^*)$$

and hence a class $$c_1'(\mathcal{L}) \in H^2(X, \mathbb{C}).$$

(2.2.4.2) The exponential exact sequence

$$0 \to \mathbb{Z}(1) \to \mathcal{O} \to \mathcal{O}^* \to 0$$

defines a homomorphism

$$\partial : H^1(X, \mathcal{O}^*) \longrightarrow H^2(X, \mathbb{Z}(1)),$$

hence a class $\partial c = c_1''(\mathcal{L}) \in H^2(X, \mathbb{Z}(1))$.
If i is chosen, this class is identified with

$$c_1'''(\mathcal{L}) \in H^2(X, \mathbb{Z}),$$

and for $\mathcal{L} = \mathcal{O}(D)$, $c_1'''(\mathcal{L})$ is just the integral cohomology class defined by D (the orientations being defined by i).

(2.2.5) We prove that:

(2.2.5.1) For α the natural injection of $\mathbb{Z}(1)$ into $\mathbb{C}$, one has

$$-\alpha c_1''(\mathcal{L}) = c_1'(\mathcal{L}).$$

(2.2.5.2) For β the natural injection of $\mathbb{Z}$ into $\mathbb{C}$, one has

$$-\beta c_1'''(\mathcal{L}) = \frac{1}{2\pi i} c_1'(\mathcal{L}).$$

This is verified by contemplating the following diagram of complexes, in which $\simeq$ denotes a quasi-isomorphism, and whose upper rectangle is anticommutative up to homotopy

$$\begin{array}{ccccc}
\mathbb{C} & \xrightarrow{\simeq} & \Omega_X^* & \longleftarrow & \sigma_{\geq 1}\Omega^* \\
\| & & & & \| \\
\mathbb{C} & \longleftarrow & [\mathbb{C} \to \mathcal{O}] & \xrightarrow[d]{\simeq} & \sigma_{\geq 1}\Omega^* \\
\uparrow & & \uparrow & & \uparrow{\scriptstyle d\log} \\
\mathbb{Z}(1) & \longleftarrow & [\mathbb{Z}(1) \to \mathcal{O}] & \xrightarrow[\exp]{\simeq} & \mathcal{O}^*[-1]
\end{array}$$

(2.2.6) Let X be a nonsingular projective variety purely of dimension n. A choice of i defines an orientation of X and an isomorphism $\mathbb{Z}(1) \simeq \mathbb{Z}$. The corresponding trace morphism:

$$H^{2n}(X, \mathbb{Z}(n)) \longrightarrow \mathbb{Z}$$

which follows from it does not depend on the choice of i.

According to Hodge, for $i \leq n$ the morphism

$$L^{n-i} = \wedge c_1''(\mathcal{O}(1))^{n-i} : H^i(X, \mathbb{Z}) \longrightarrow H^{2n-i}(X, \mathbb{Z}(n-i)),$$

is an isomorphism and, combined with Poincaré duality

$$H^i(X, \mathbb{Z}) \otimes H^{2n-i}(X, \mathbb{Z}(n-i)) \longrightarrow H^{2n}(X, \mathbb{Z}(n-i)) \longrightarrow \mathbb{Z}(-i),$$

it furnishes a polarization on the *primitive part* $\mathrm{Ker}(L^{n-i+1})$ of $H^i(X, \mathbb{Z})$. It follows that the rational Hodge structures $H^i(X, \mathbb{Q})$ are polarizable.

2.3. Mixed structures.

Definition (2.3.1). — *A mixed Hodge structure H consists of:*

a) a finitely generated $\mathbb{Z}$-module $H_\mathbb{Z}$ (the "integral lattice").

b) a finite increasing filtration W_n of $H_\mathbb{Q} = \mathbb{Q} \otimes_\mathbb{Z} H_\mathbb{Z}$, called the weight filtration.

c) a finite filtration F^p of $H_\mathbb{C} = \mathbb{C} \otimes_\mathbb{Z} H_\mathbb{Z}$, called the Hodge filtration.

It is required that on $H_\mathbb{C}$, the filtration $W_\mathbb{C}$ deduced from W by extension of scalars, the filtration F, and its complex conjugate $\overline{F}$ form a system $(W_\mathbb{C}, F, \overline{F})$ of three opposed filtrations ((1.2.7) and (1.2.13)).

(2.3.2) Let W again denote the filtration of $H_{\mathbb{Z}}$ which is the inverse image of the filtration W of $H_{\mathbb{Q}}$. The axiom of mixed Hodge structures means that, for every n, the filtration F induces on $\mathbb{C} \otimes_{\mathbb{Z}} \mathrm{Gr}_n^W(H_{\mathbb{Z}})$ a filtration n-opposed to its complex conjugate. By (2.1.9), $\mathrm{Gr}_n^W(H_{\mathbb{Z}})$ is endowed with a Hodge structure of weight n, with Hodge filtration induced by F.

Example (2.3.3). — If H is a Hodge structure of weight n, one defines a mixed Hodge structure with the same integral lattice and the same Hodge filtration by putting

$$W_i(H_{\mathbb{Q}}) = 0 \quad \text{for } i < n \qquad \text{and} \qquad W_i(H_{\mathbb{Q}}) = H_{\mathbb{Q}} \quad \text{for } i \geq n.$$

(2.3.4) A *morphism* $f : H \to H'$ of Hodge structures is a homomorphism $f : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ compatible with the filtrations W and F (and therefore compatible with $\overline{F}$). One immediately deduces from (1.2.10) the following theorem.

Theorem (2.3.5). —

(i) The category of mixed Hodge structures is abelian.

(ii) The kernel (respectively, cokernel) of a morphism $f : H \to H'$ has as integral lattice the kernel (respectively, cokernel) K of $f : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$, with $K \otimes \mathbb{Q}$ and $K \otimes \mathbb{C}$ endowed with the induced (respectively, quotient) filtrations from the filtrations W and F of $H_{\mathbb{Q}}$ and $H_{\mathbb{C}}$ (respectively, of $H'_{\mathbb{Q}}$ and $H'_{\mathbb{C}}$).

(iii) Every morphism $f : H \to H'$ is strictly compatible with the filtration W of $H_{\mathbb{Q}}$ and $H'_{\mathbb{Q}}$, and with the filtration F of $H_{\mathbb{C}}$ and $H'_{\mathbb{C}}$. It induces morphisms of $\mathbb{Q}$-Hodge structures:

$$\mathrm{Gr}_n^W(f) : \mathrm{Gr}_n^W(H_{\mathbb{Q}}) \longrightarrow \mathrm{Gr}_n^W(H'_{\mathbb{Q}})$$

and morphisms strictly compatible with the filtration induced by $W_{\mathbb{C}}$:

$$\mathrm{Gr}_F^p(f) : \mathrm{Gr}_F^p(H_{\mathbb{C}}) \longrightarrow \mathrm{Gr}_F^p(H'_{\mathbb{C}}).$$

(iv) The functor Gr_n^W is an exact functor from the category of mixed Hodge structures to the category of $\mathbb{Q}$-Hodge structures of weight n.

(v) The functor Gr_F^p is exact.

(2.3.6) Let H be a mixed Hodge structure. The $W_n(H_{\mathbb{Z}})$, endowed with the filtrations induced by W and F, then form mixed Hodge substructures $W_n(H)$ of H. The quotient $W_n(H)/W_{n-1}(H)$ is identified with $\mathrm{Gr}_n^W(H_{\mathbb{Z}})$ endowed with its Hodge structure (2.3.2), (2.3.3). This Hodge structure will be denoted by $\mathrm{Gr}_n^W(H)$.

(2.3.7) Put $H^{pq} = \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_{p+q}^W(H_{\mathbb{C}}) = \big(\mathrm{Gr}_{p+q}^W(H)\big)^{p,q}$. The *Hodge numbers* of H are the integers

$$h^{pq} = \dim_{\mathbb{C}} H^{pq}.$$

Thus the Hodge number h^{pq} of H is the Hodge number h^{pq} of the Hodge structure $\mathrm{Gr}_{p+q}^W(H)$.

(2.3.8) A *$\mathbb{Q}$-mixed Hodge structure* H is defined as consisting of a finite-dimensional vector space $H_{\mathbb{Q}}$ over $\mathbb{Q}$, a finite increasing filtration W of $H_{\mathbb{Q}}$, and a finite decreasing filtration F of $H_{\mathbb{C}}$, the filtrations $W_{\mathbb{C}}$, F, and $\overline{F}$ being opposed. Theorem (2.3.5) generalizes trivially to this variant.

3. Hodge theory of nonsingular algebraic varieties

3.1. Logarithmic poles and residues.

(3.1.1) We recall some classical properties of "logarithmic poles" of holomorphic differential forms. The reader will find proofs in [3], II, (3.1) to (3.7), for example.

(3.1.2) A divisor Y in a smooth complex analytic variety X will be said *to have normal crossings* if the inclusion of Y into X is locally isomorphic to the inclusion of a union of coordinate hyperplanes in $\mathbb{C}^n$; this does not imply that Y is a union of smooth divisors. Let Y be a normal-crossings divisor in X, and let j be the inclusion of $X^* = X - Y$ in X. Let $\Omega^1_X\langle Y\rangle$ denote the locally free $\mathcal{O}$-submodule of $j_*\Omega^1_{X^*}$ generated by Ω^1_X and by the $\frac{dz_i}{z_i}$, where z_i is a local equation of a local irreducible component of Y. The *sheaf* $\Omega^p_X\langle Y\rangle$ *of differential p-forms on* X *with logarithmic pole along* Y is, by definition, the locally free subsheaf $\bigwedge^p \Omega^1_X\langle Y\rangle$ of $j_*\Omega^p_{X^*}$.

Proposition (3.1.3). —

(i) A section α of $j_*\Omega^p_{X^*}$ belongs to $\Omega^p_X\langle Y\rangle$ if and only if α and $d\alpha$ have at worst simple poles along the divisor Y.

(ii) The $\Omega^p_X\langle Y\rangle$ form the smallest subcomplex of $j_*\Omega^*_{X^*}$, stable under exterior product, containing Ω^*_X, and containing the logarithmic differential df/f of every local section, meromorphic along Y, of $j_*\mathcal{O}^*_{X^*}$.

The complex $\Omega^*_X\langle Y\rangle$ is called the *logarithmic De Rham complex* of X along Y. By (3.1.3), (ii), this complex is contravariant in the pair (X, X^*).

(3.1.4) Locally on X, Y is a union of smooth divisors Y_i, and Y^n (resp. $\widetilde{Y}^n$) denotes the union (resp. the disjoint sum) of the intersections n at a time of the Y_i. The Y^n glue to a subspace Y^n of X, and the $\widetilde{Y}^n$ glue to the normalization of Y^n. One has $\widetilde{Y}^0 = Y^0 = X$ and puts $\widetilde{Y} = \widetilde{Y}^1$.

The two-element set of *orientations* of a finite set E with n elements is defined to be the set of generators of $\bigwedge^n \mathbb{Z}^E$. For $n \geq 2$, this set is the set of conjugacy classes, under the alternating group, of total orderings of E.

If to each point y of $\widetilde{Y}^n$ one associates the set of the n local components of Y which contain the image in X of a neighborhood of y in $\widetilde{Y}^n$, then one defines on $\widetilde{Y}^n$ a local system E_n of n-element sets. The local system of orientations of these sets is a torsor under $\mathbb{Z}/(2)$. This torsor defines, *via* the inclusion of $\mathbb{Z}/(2)$ into $\mathbb{C}^*$, a rank-one complex local system ε^n on $\widetilde{Y}^n$, endowed with an isomorphism

$$(\varepsilon^n)^{\otimes 2} \simeq \mathbb{C}.$$

One has

$$\varepsilon^n \simeq \bigwedge^n \mathbb{C}^{E_n}.$$

Locally on $\widetilde{Y}^n$, ε^n is endowed with two opposite isomorphisms $\pm\alpha : \varepsilon^n \xrightarrow{\sim} \mathbb{C}$. Put

$$\varepsilon^n_{\mathbb{Z}} = \alpha^{-1}\big((2\pi i)^{-n}\mathbb{Z}\big).$$

The local system ε^n, endowed with $\varepsilon^n_{\mathbb{Z}}$, should be regarded as a twisted form of $\mathbb{Z}(-n)$ on $\widetilde{Y}^n$.

Let ε^n_X (resp. $(\varepsilon^n_X)_{\mathbb{Z}}$) denote the direct image of ε^n (resp. $\varepsilon^n_{\mathbb{Z}}$) under the map from $\widetilde{Y}^n$ to X. One has

(3.1.4.1) $$\varepsilon^n_X \simeq \bigwedge^n \varepsilon^1_X \qquad (n \geq 0).$$

If Y is a union of distinct smooth divisors $(Y_i)_{i\in I}$, the choice of a total order on I trivializes the ε^n.

(3.1.5) Let $W_n(\Omega^p_X\langle Y\rangle)$ denote the submodule of $\Omega^p_X\langle Y\rangle$ formed by linear combinations of products

$$\alpha \wedge \frac{dt_{i(1)}}{t_{i(1)}} \wedge \cdots \wedge \frac{dt_{i(m)}}{t_{i(m)}} \qquad (m \leq n),$$

with α holomorphic and with the $t_{i(j)}$ local equations of distinct local components Y_j of Y. The *weight filtration* of $\Omega^*_X\langle Y\rangle$ is the *increasing* filtration by the subcomplexes $W_n(\Omega^*_X\langle Y\rangle)$. One has

(3.1.5.1) $$W_n(\Omega^p_X\langle Y\rangle) \wedge W_m(\Omega^q_X\langle Y\rangle) \subset W_{n+m}(\Omega^{p+q}_X\langle Y\rangle).$$

Letting i_n denote the map from $\widetilde{Y}^n$ to X, one checks that the correspondence

$$\alpha \wedge \frac{dt_{i(1)}}{t_{i(1)}} \wedge \cdots \wedge \frac{dt_{i(n)}}{t_{i(n)}} \longmapsto (\alpha \mid Y_{i(1)} \cap \cdots \cap Y_{i(n)}) \otimes (\text{orientation } i(1)\ldots i(n))$$

defines isomorphisms of complexes

(3.1.5.2) $$\mathrm{Res} : \mathrm{Gr}_n^W(\Omega_X^*\langle Y\rangle) \simeq i_{n*}\Omega^*_{\widetilde{Y}_n}(\varepsilon^n)[-n]$$

(the *"Poincaré residue"*).

(3.1.6) The following interpretation, in terms of (3.1.5.2), of the Leray spectral sequence for the inclusion of X^* in X was pointed out to me by N. Katz. It will make it possible to verify a point which I had initially considered evident ((3.2.5), (ii), first part).

(3.1.7) Every point of X admits a fundamental system of Stein open neighborhoods whose trace on X^* is Stein. For $\mathcal{F}$ a coherent analytic sheaf on X^*, one therefore has $R^i j_*\mathcal{F} = 0$ for $i > 0$. The De Rham complex $\Omega^*_{X^*}$ is therefore a resolution of the constant sheaf $\mathbb{C}$ by sheaves acyclic for the functor j_*. Hence

(3.1.7.1) $$H^*(X^*, \mathbb{C}) \xrightarrow{\sim} \mathbb{H}^*(X^*, \Omega^*_{X^*}) \xleftarrow{\sim} \mathbb{H}^*(X, j_*\Omega^*_{X^*})$$

and the Leray spectral sequence for the morphism j is identified with the hypercohomology spectral sequence for $j_*\Omega^*_{X^*}$, corresponding to the filtration τ by the subcomplexes $\tau_{\leq -n}(j_*\Omega^*_{X^*})$ (1.4.6).

Proposition (3.1.8). — *The morphisms of filtered complexes*

$$(\Omega_X^*\langle Y\rangle, W) \xleftarrow{\alpha} (\Omega_X^*\langle Y\rangle, \tau) \xrightarrow{\beta} (j_*\Omega^*_{X^*}, \tau)$$

are filtered quasi-isomorphisms. They define an isomorphism between the Leray spectral sequence for j in complex cohomology and the hypercohomology spectral sequence of the filtered complex $(\Omega_X^*\langle Y\rangle, W)$ on X.

By (1.4.5) and (3.1.7), it suffices to prove the first assertion. In [3], II, (6.9), or in [1], one finds the proof that β is a quasi-isomorphism, hence a filtered quasi-isomorphism. One can also compute directly the cohomology sheaves of the two sides: those of $\Omega_X^*\langle Y\rangle$ are determined by (3.1.5.2), whereas those of $j_*\Omega^*_{X^*}$ are the $R^i j_*\mathbb{C}$, which can be computed topologically.

For $n \geq p$ one has

$$W_n(\Omega_X^p\langle Y\rangle) = \Omega_X^p\langle Y\rangle,$$

so that α is a morphism from $\Omega_X^*\langle Y\rangle$, endowed with τ, to $\Omega_X^*\langle Y\rangle$, endowed with the decreasing filtration associated with W (1.1.3). By (3.1.5.2), one has

(3.1.8.1) $$\mathcal{H}^i(\mathrm{Gr}_n^W(\Omega_X^*\langle Y\rangle)) = \begin{cases} 0 & \text{for } i \neq n \\ \varepsilon_X^n & \text{for } i = n, \end{cases}$$

from the first line of this formula one deduces that α is a filtered quasi-isomorphism. This proves (3.1.8).

By (3.1.7), the isomorphism (3.1.8.1) defines an isomorphism

(3.1.8.2) $$R^n j_*\mathbb{C} \simeq \mathcal{H}^n(j_*\Omega^*_{X^*}) \simeq \mathcal{H}^n(\Omega_X^*\langle Y\rangle) \simeq \varepsilon_X^n.$$

The isomorphisms (3.1.4.1) correspond, *via* (3.1.8.2), to the cup product.

Proposition (3.1.9). — *The canonical morphism from $R^n j_*\mathbb{Z}$ to $R^n j_*\mathbb{C}$ identifies, via (3.1.8.2), the sheaf $R^n j_*\mathbb{Z}$ with $(\varepsilon_X^n)_{\mathbb{Z}}$ (3.1.4).*

The question is local on X. One can therefore suppose that X is an open polydisc D^m, with $D = \{z \in \mathbb{C} \mid |z| < 1\}$, and that $Y = \bigcup_{k=1}^{\ell} Y_k$, $Y_k = \mathrm{pr}_k^{-1}(0)$. The fiber at 0 of $R^n j_*\mathbb{Z}$ is then the integral cohomology of $X^* = D^{*\ell} \times D^{(m-\ell)}$, with $D^* = \{z \in \mathbb{C} \mid 0 < |z| < 1\}$. The space X^* has the homotopy type of a torus; its cohomology is therefore torsion-free, and the cup product defines isomorphisms

$$\bigwedge^n (R^1 j_*\mathbb{Z})_0 \xrightarrow{\sim} (R^n j_*\mathbb{Z})_0.$$

It is therefore enough to prove (3.1.9) for $n = 1$.

The integral homology $H_1(X^*)$ is generated by the loops γ_k winding around the various Y_k. One has

$$\oint_{\gamma_k} \frac{dz_k}{z_k} = \pm 2\pi i;$$

the integral cohomology is therefore generated by the $\frac{1}{2\pi i}\frac{dz_k}{z_k}$, and this proves (3.1.9).

(3.1.10) Let $\mathcal{F}$ be a coherent analytic sheaf on X^*, given as the restriction to X^* of a coherent analytic sheaf $\mathcal{F}'$ on X. The meromorphic direct image $j_*^m\mathcal{F}$ of $\mathcal{F}$ is the inductive limit

$$j_*^m\mathcal{F} = \varinjlim_n \mathcal{F}'(nY).$$

Locally on X, Y is the sum of a finite family $(Y_i)_{i\in I}$ of smooth divisors, and the *filtration by order of pole* P on $j_*^m\mathcal{O}_{X^*}$ is defined by the formula

(3.1.10.1)
$$P^p(j_*^m\mathcal{O}_{X^*}) = \sum_{\mathbf{n}\in A_p} \mathcal{O}_X\Big(\sum_i (n_i+1)Y_i\Big)$$

where

$$A_p = \{(n_i)_{i\in I} \mid \sum_i n_i \leq -p \ \text{ and } \ \forall i,\ n_i \geq 0\}.$$

This construction globalizes and provides on $j_*^m\mathcal{O}_{X^*}$ an exhaustive filtration such that $P^p = 0$ for $p > 0$.

The *filtration by order of pole* of the complex $j_*^m\Omega_{X^*}^* = j_*^m\mathcal{O}_{X^*}\otimes\Omega_X^*$ is the filtration

(3.1.10.2)
$$P^p(j_*^m\Omega_{X^*}^k) = P^{p-k}(j_*^m\mathcal{O}_{X^*})\otimes\Omega_X^k.$$

The filtration P induces on the subcomplex $\Omega_X^*\langle Y\rangle$ of $j_*^m\Omega_{X^*}^*$ the bête filtration by the $\sigma_{\geq p}(\Omega_X^*\langle Y\rangle)$, a filtration also called the *Hodge filtration* F.

Proposition (3.1.11). — *The inclusion morphism*

$$(\Omega_X^*\langle Y\rangle, F) \longrightarrow (j_*^m\Omega_{X^*}^*, P)$$

is a filtered quasi-isomorphism.

This statement was suggested to me by [4]. A proof appears in [3], II, (3.13).

3.2. Mixed Hodge theory.

Recall that from now on, by a scheme we mean a scheme of finite type over $\mathbb{C}$*, and by a sheaf on* S *a sheaf on* S^{an}*.*

(3.2.1) Let X be a smooth separated scheme. By Nagata [11], X is a Zariski open subset of a complete scheme $\overline{X}$. By Hironaka [8], one can take $\overline{X}$ smooth, and such that $Y = \overline{X} - X$ is a normal-crossings divisor.

The reader who would like to avoid the reference to Nagata may suppose X quasi-

projective. The smooth completion $\overline{X}$ can then be chosen projective and such that Y is a union of smooth divisors. When one restricts oneself to such compactifications, Hodge theory is needed only in its standard form (2.2.1).

(3.2.2) By (3.1.7) and (3.1.8), one has

$$H^*(X,\mathbb{C}) \simeq \mathbb{H}^*(\overline{X}, \Omega_{\overline{X}}^*\langle Y\rangle).$$

The *Hodge filtration* F on the complex $\Omega_{\overline{X}}^*\langle Y\rangle$ is defined to be the filtration $F^p = \sigma_{\geq p}$ by bête truncations (1.4.5). Thus $\Omega_{\overline{X}}^*\langle Y\rangle$ carries two filtrations: F and W (3.1.5).

(3.2.3) We shall use the existence of bifiltered resolutions $i : \Omega_{\overline{X}}^*\langle Y\rangle \longrightarrow K^*$ such that the $\mathrm{Gr}_F^p\,\mathrm{Gr}_n^W(K^j)$ are sheaves acyclic for the functor Γ:

$$H^i(\overline{X}, \mathrm{Gr}_F^p\,\mathrm{Gr}_n^W(K^j)) = 0 \qquad \text{for } i > 0.$$

Here are two ways to construct them:

a) One may take for K^* the canonical Godement resolution $\mathcal{C}^*(\Omega^*_{\overline{X}}\langle Y\rangle)$, filtered by the $\mathcal{C}^*(W_n(\Omega^*_{\overline{X}}\langle Y\rangle))$ and the $\mathcal{C}^*(F^p(\Omega^*_{\overline{X}}\langle Y\rangle))$. This is a bifiltered resolution because $\mathcal{C}^*$ is an exact functor.

b) One may take for K^* the d''-resolution of $\Omega^*_{\overline{X}}\langle Y\rangle$. Let $\Omega^{pq}_{\overline{X}}$ be the sheaf of C^∞ forms of type (p,q); then K^* is the simple complex associated with the double complex of the $\Omega^p_{\overline{X}}\langle Y\rangle \otimes_{\mathcal{O}} \Omega^{0,q}_{\overline{X}}$ (a subcomplex of the $j_*\Omega^{**}_X$). This complex is filtered by the $F^p(\Omega^*_{\overline{X}}\langle Y\rangle)\otimes\Omega^{0,*}_{\overline{X}}$ and by the $W_n(\Omega^*_{\overline{X}}\langle Y\rangle)\otimes\Omega^{0,*}_{\overline{X}}$; to prove that this is a bifiltered resolution, one uses the fact that the sheaf $\mathcal{O}_\infty$ of complex-valued C^∞ functions on $\overline{X}$ is flat over $\mathcal{O}$ (a corollary of Malgrange's C^∞ preparation theorem). The sheaves $\mathrm{Gr}_F\,\mathrm{Gr}_W(K^*)$ are fine, because they are sheaves of modules over the soft sheaf $\mathcal{O}_\infty$.

(3.2.4) With the notation of (3.2.3), the complex cohomology of X appears as the cohomology of the bifiltered complex $\Gamma(\overline{X},K^*)$. Thus there are two spectral sequences abutting to $H^*(X,\mathbb{C})$. With the notation of (3.1.4), they are

$$ {}_W E_1^{pq} = H^{p+q}(\overline{X},\varepsilon^{-p}_{\overline{X}}[p]) = H^{2p+q}(\widetilde{Y}^p,\varepsilon^{-p}) \Rightarrow H^n(X,\mathbb{C}) \tag{3.2.4.1}$$

$$ {}_F E_1^{pq} = H^q(\overline{X},\Omega^p_{\overline{X}}\langle Y\rangle) \Rightarrow H^n(X,\mathbb{C}). \tag{3.2.4.2}$$

The first of these, up to the renumbering ${}_W E_1^{pq} \mapsto E_2^{2p+q,-p}$, is just the Leray spectral sequence of the inclusion j.

Theorem (3.2.5). —

(i) On the terms ${}_W E_r^{pq}$ of the spectral sequence (3.2.4.1), the first direct filtration, the second direct filtration, and the recurrent filtration defined by F coincide.

(ii) The filtration on $H^n(X,\mathbb{C})$ which is the abutment of the spectral sequence ${}_W E$ is deduced from a filtration W of $H^n(X,\mathbb{Q})$. Neither it, nor the filtration F which is the abutment of the spectral sequence ${}_F E$, depends on the chosen compactification $\overline{X}$ of X or on the choice of K^*.

(iii) The filtrations $W[n]$ (1.1.2) and F define on $H^n(X,\mathbb{Z})$ a mixed Hodge structure, functorial in X.

By (3.1.7), the spectral sequence ${}_W E$ is the Leray spectral sequence for j_* (up to renumbering). It is therefore obtained by tensoring with $\mathbb{C}$ a spectral sequence of $\mathbb{Q}$-vector spaces, and the first assertion of (ii) is true. Complex conjugation acts on the ${}_W E_r$; it can be computed via (3.1.10).

Lemma (3.2.6). — *The hypercohomology spectral sequences of the filtered complexes $\mathrm{Gr}^W_n(\Omega^*_{\overline{X}}\langle Y\rangle)$, endowed with the filtration induced by the Hodge filtration, degenerate at E_1.*

Define Y^n and $\widetilde{Y}^n$ as in (3.1.4), and let $i_n : \widetilde{Y}^n \to \overline{X}$. By (3.1.5.2), one has

$$\mathrm{Gr}^W_n(\Omega^*_{\overline{X}}\langle Y\rangle) \simeq i_{n*}\Omega^*_{\widetilde{Y}^n}(\varepsilon^n)[-n].$$

Moreover, the Hodge filtration induces the bête filtration (1.4.5), so that the spectral sequence (3.2.6) is obtained, by translations in the degrees, from the classical spectral sequence

$$E_1^{pq} = H^q(\widetilde{Y}^n,\Omega^p_{\widetilde{Y}^n}(\varepsilon^n)) \Rightarrow H^{p+q}(\widetilde{Y}^n,\varepsilon^n).$$

If $\overline{X}$ is projective and Y is a union of smooth divisors, then $\widetilde{Y}^n$ is projective, ε^n is a trivial local system, and classical Hodge theory (2.2.1) gives the degeneration (3.2.6). For the general case, one must refer to [2] (see (2.2.2), (2.2.3)).

Hodge theory further gives that the Hodge filtration on $H^k(\widetilde{Y}^n,\varepsilon^n)$ is k-opposed to its complex conjugate (complex conjugation being defined in terms of $\varepsilon^n_{\mathbb{Z}}$ (3.1.4)). Taking account here of the translations in the degrees, one has:

Lemma (3.2.7). — *The filtration on*

$$ {}_W E_1^{-n,k+n} = H^k(\overline{X},\mathrm{Gr}^W_n(\Omega^*_{\overline{X}}\langle Y\rangle)) \approx H^{k-n}(\widetilde{Y}^n,\varepsilon^n),$$

which is the abutment of the spectral sequence (3.2.6), is $(k+n)$-opposed to its complex conjugate.

Lemma (3.2.8). — *The differentials d_1 of the spectral sequence ${}_WE$ are strictly compatible with the filtration F.*

On the terms E_1, there is only one filtration induced by F to consider ((1.3.10) and (1.3.13), (iii)), and d_1 is compatible with this filtration ((1.3.13), (i)). This filtration is the abutment of the spectral sequence (3.2.6) ((1.4.8), (iii)). By (3.2.7), the arrow d_1

$$d_1 : H^k(\overline{X}, \mathrm{Gr}_n^W(\Omega^*_{\overline{X}}\langle Y\rangle)) \longrightarrow H^{k+1}(\overline{X}, \mathrm{Gr}_{n-1}^W(\Omega^*_{\overline{X}}\langle Y\rangle))$$

that is,

(3.2.8.1)
$$d_1 : H^{k-n}(\widetilde{Y}^n, \varepsilon^n) \longrightarrow H^{k-n+2}(\widetilde{Y}^{n-1}, \varepsilon^{n-1})$$

is compatible with filtrations that are $(k+n)$-opposed to their complex conjugates. Since d_1 commutes with complex conjugation, it respects the bigrading, and hence the filtration defined by F and $\overline{F}$; this proves (3.2.8). Moreover, the cohomology of the complex E_1 will still be bigraded.

Lemma (3.2.9). — *On ${}_WE_2^{pq}$, the recurrent filtration F is q-opposed to its complex conjugate.*

We prove by induction on r that:

Lemma (3.2.10). — *For $r \geq 0$, the differentials d_r of the spectral sequence ${}_WE$ are strictly compatible with the recurrent filtration F. For $r \geq 2$, they are zero.*

For $r = 0$ (resp. $r = 1$), one applies (3.2.6) and (1.4.8), (iii) (resp. (3.2.8)). For $r \geq 2$, it suffices to prove that $d_r = 0$. By induction and by (1.3.16), one may suppose that on the terms ${}_WE_s$ ($s \geq r+1$) one has $F_d = F_r = F_{d*}$, and that ${}_WE_r = {}_WE_2$. By (1.3.13), (i), d_r is therefore compatible with the filtration F_r.

On ${}_WE_r^{pq} = {}_WE_2^{pq}$, the filtration F_r is q-opposed to its complex conjugate. The morphism

$$d_r : {}_WE_r^{pq} \longrightarrow {}_WE_r^{p+r,q-r+1}$$

therefore satisfies, for $r - 1 > 0$,

$$\begin{aligned} d_r({}_WE_r^{pq}) &= d_r\left(\sum_{a+b=q} \left(F^a({}_WE_r^{pq}) \cap \overline{F}^b({}_WE_r^{pq})\right)\right) \\ &\subset \sum_{a+b=q} \left(F^a({}_WE_r^{p+r,q-r+1}) \cap \overline{F}^b({}_WE_r^{p+r,q-r+1})\right) = 0. \end{aligned}$$

This proves (3.2.10), which by (1.3.16) implies (3.2.5) (i).

By (1.3.17), the filtration of ${}_WE_\infty^{pq}$ induced by the filtration F of $H^{p+q}(X, \mathbb{C})$ is q-opposed to its complex conjugate. Since $q = -p + (p+q)$, this proves the first part of (3.2.5) (iii).

(3.2.11) We prove (ii) and (iii), which will complete the proof.

A. *Independence of the choice of K^*.* The filtrations F and W of $H^*(X, \mathbb{C})$ are the abutments of the hypercohomology spectral sequences of $\Omega^*_{\overline{X}}\langle Y\rangle$ for the filtrations F and W. These entire spectral sequences do not depend on the choice of K^*.

B. *Functoriality.* Let $f : X_1 \to X_2$ be a morphism of schemes. Suppose given a morphism of smooth compactifications

(3.2.11.1)
$$\begin{array}{ccc} X_1 & \xrightarrow{f} & X_2 \\ {\scriptstyle j_1}\downarrow & & \downarrow{\scriptstyle j_2} \\ \overline{X}_1 & \xrightarrow[\overline{f}]{} & \overline{X}_2, \end{array}$$

where $Y_i = \overline{X}_i - X_i$ are normal-crossings divisors. The canonical morphism (see (3.1.3)) from $\overline{f}^*\Omega^*_{\overline{X}_2}\langle Y_2\rangle$ to $\Omega^*_{\overline{X}_1}\langle Y_1\rangle$ is then a morphism of bifiltered complexes; on hypercohomology it induces a morphism compatible with F and W, and therefore $f^* : H^n(X_2, \mathbb{Z}) \longrightarrow H^n(X_1, \mathbb{Z})$ is a morphism of mixed Hodge structures, for the structures defined by the compactifications $\overline{X}_i$.

C. *Independence of the compactification.* With the notation of B, if f is an isomorphism, then f^* is a bijective morphism of mixed Hodge structures, hence an isomorphism (2.3.5).

If $\overline{X}_1$ and $\overline{X}_2$ are two smooth compactifications of X, with $Y_i = \overline{X}_i - X_i$ a normal-crossings divisor, there exists a third smooth compactification $\overline{X}$, with $Y = \overline{X} - X$ a normal-crossings divisor, fitting into a commutative diagram

$$\begin{array}{ccccc} & & X & & \\ & \swarrow & \downarrow & \searrow & \\ \overline{X}_1 & \longleftarrow & \overline{X} & \longrightarrow & \overline{X}_2 \end{array}$$

Namely, one takes for $\overline{X}$ a resolution of the singularities of the closure of the diagonal image of X in $\overline{X}_1 \times \overline{X}_2$. The identity map from $H^n(X, \mathbb{Z})$, endowed with the mixed Hodge structure defined by $\overline{X}_1$, to $H^n(X, \mathbb{Z})$, endowed with that defined by $\overline{X}_2$, is therefore a composition of two isomorphisms.

To finish the proof of (ii) and (iii), one observes that every morphism f fits into a diagram (3.2.11.1): choose compactifications $\overline{X}_1'$ and $\overline{X}_2$ of X_1 and X_2, and then take for $\overline{X}_1$ a resolution of the singularities of the closure of the image of X_1 in $\overline{X}_1' \times \overline{X}_2$.

Definition (3.2.12). — *The mixed Hodge structure on the cohomology of a smooth separated algebraic variety is the mixed Hodge structure (3.2.5), (iii).*

Corollary (3.2.13). — *With the preceding notation:*

(i) The spectral sequence (3.2.4.1) degenerates at E_2, i.e. the Leray spectral sequence for the inclusion $j : X^* \hookrightarrow \overline{X}$ degenerates at E_3 ($E_3 = E_\infty$).

(ii) The spectral sequence (3.2.4.2)

$$_F E_1^{pq} = H^q(\overline{X}, \Omega^p_{\overline{X}}\langle Y\rangle) \Longrightarrow H^{p+q}(X, \mathbb{C})$$

degenerates at E_1.

(iii) The spectral sequence defined by the sheaf $\Omega^p_{\overline{X}}\langle Y\rangle$, endowed with the filtration W:

$$\begin{aligned} E_1^{-n,k+n} &= H^k(\overline{X}, \mathrm{Gr}^W_n(\Omega^p_{\overline{X}}\langle Y\rangle)) \\ &\approx H^k(\widetilde{Y}^n, \Omega^{p-n}_{\widetilde{Y}^n}(\varepsilon^n)) \Longrightarrow H^k(\overline{X}, \Omega^p_{\overline{X}}\langle Y\rangle) \end{aligned}$$

degenerates at E_2.

Assertion (i) was proved in (3.2.10).

Consider the four spectral sequences of (1.4.12), relative to the bifiltered complex $\Omega^*_{\overline{X}}\langle Y\rangle$. By (i), one has

$$\sum_n \dim H^n(X, \mathbb{C}) = \sum_{p,q} \dim {}_W E_2^{p,q}.$$

The terms ${}_W E_1^{n,k-n}$ are, for fixed n, the abutment of a spectral sequence *degenerate at* E_1 (3.2.6), with initial terms $E_1^{p,n,k-p-n}$ (notation of (1.4.12)), the initial terms of spectral sequence (iii). Since ${}_W d_1$ is strictly compatible with the filtration F that is the abutment of this spectral sequence (3.2.7), and is (1.4.12) the abutment of a morphism between these spectral sequences, beginning with the differentials of (iii), one has

$$\mathrm{Gr}^p_F({}_W E_2^{n,k-n}) \approx H^*(E_1^{p,n-1,k-n-p} \to E_1^{p,n,k-n-p} \to E_1^{p,n+1,k-n-p})$$

and $\mathrm{Gr}^*_F({}_W E_2^{**})$ is the sum of the terms E_2^{***} of the spectral sequences (iii). Therefore

$$\textbf{(3.2.13.1)} \qquad \sum_n \dim H^n(X, \mathbb{C}) = \sum \dim E_2^{***}.$$

On the other hand,

$$\sum_n \dim H^n(X, \mathbb{C}) \leq \sum \dim {}_F E_1^{**},$$

with equality if and only if the spectral sequence (ii) degenerates at E_1, and

$$\sum \dim {}_F E_1^{**} \leq \sum \dim E_2^{***},$$

with equality if and only if the spectral sequences (iii) degenerate at E_2. Comparing with (3.2.13.1), one obtains (3.2.13).

Corollary (3.2.14). — *Let ω be a meromorphic differential p-form on $\overline{X}$, holomorphic on X and having at worst logarithmic poles along Y. Then the restriction $\omega|X$ of ω to X is closed, and if the cohomology class in $H^p(X, \mathbb{C})$ defined by ω is zero, then $\omega = 0$.*

This is the special case ${}_F E_1^{p0} = {}_F E_\infty^{p0}$ of (3.2.13), (ii).

Corollary (3.2.15). —

(i) If X is a complete smooth algebraic variety, the mixed Hodge structure on $H^n(X, \mathbb{Z})$ is the classical Hodge structure of weight n.

(ii) The Hodge numbers h^{pq} of the mixed Hodge structure of $H^n(X, \mathbb{Z})$ (X smooth algebraic) can be nonzero only for $p \leq n$, $q \leq n$, and $p + q \geq n$.

Assertion (i) is clear. To prove (ii), one observes that, for Y a union of smooth divisors, the rational Hodge structure $\mathrm{Gr}_k^W(H^n(X, \mathbb{Q}))$ is a quotient of a subobject of a Hodge structure of weight $n + k$, namely

$$H^{n-k}(\widetilde{Y}^n, \mathbb{Q}) \otimes \mathbb{Q}(-k).$$

(3.2.16) Let X be a smooth separated scheme. It is known that X admits smooth compactifications $\overline{X}$, and that the subgroup of $H^n(X, \mathbb{Z})$ which is the image of $H^n(\overline{X}, \mathbb{Z})$ is independent of the choice of $\overline{X}$ (cf. [6], (9.1) to (9.4)).

Corollary (3.2.17). — *Under the hypotheses (3.2.16), the image of $H^n(\overline{X}, \mathbb{Q})$ in $H^n(X, \mathbb{Q})$ is $W_n(H^n(X, \mathbb{Q}))$* (W denoting the weight filtration (3.2.12)).

One may suppose that $\overline{X} - X$ is a normal-crossings divisor. The assertion then follows from the fact that $W[-n]$ is the abutment of the Leray spectral sequence for the inclusion $j : X \hookrightarrow \overline{X}$.

Corollary (3.2.18). — *Let f be a morphism from a proper smooth scheme Y to a smooth scheme X admitting a smooth compactification $\overline{X}$:*

$$Y \xrightarrow{f} X \xhookrightarrow{j} \overline{X}.$$

Then the groups $H^n(X, \mathbb{Q})$ and $H^n(\overline{X}, \mathbb{Q})$ have the same image in $H^n(Y, \mathbb{Q})$.

Since f^* and $(jf)^*$ are strictly compatible with the weight filtration ((3.2.5), (iii)), it is enough to prove that $\mathrm{Gr}^W(f^*)$ and $\mathrm{Gr}^W(f^* j^*)$ have the same image in $\mathrm{Gr}^W(H^n(Y, \mathbb{Q}))$. By (3.2.17) and (3.2.15), $\mathrm{Gr}_n^W(j^*)$ is an isomorphism, while $\mathrm{Gr}_m^W(f^*) = 0$ for $m \neq n$ because $\mathrm{Gr}_m^W(H^n(Y, \mathbb{Q})) = 0$ for $m \neq n$.

***Remark* (3.2.19).** — By (3.1.11) and (1.4.5), under the hypotheses of (3.2.5), the Hodge filtration on $H^n(X, \mathbb{C})$ is the abutment of the hypercohomology spectral sequence of the complex $j_*^m \Omega_{X^*}^*$, endowed with the filtration by order of pole (3.1.10): this spectral sequence coincides with (3.2.4.2).

4. Applications and complements

4.1. The fixed-part theorem.

***Theorem* (4.1.1).** — *Let S be a smooth separated scheme, and let $f : X \to S$ be a proper smooth morphism.*

(i) In rational cohomology, the Leray spectral sequence

$$E_2^{pq} = H^p(S, R^q f_* \mathbb{Q}) \Longrightarrow H^{p+q}(X, \mathbb{Q})$$

degenerates ($E_2 = E_\infty$).

(ii) If $\overline{X}$ is a nonsingular compactification of X, the canonical morphism

$$H^n(\overline{X},\mathbb{Q}) \longrightarrow H^0(S,R^nf_*\mathbb{Q})$$

is surjective.

For f projective and smooth, assertion (i) is proved in [2]. The proof of [2] is rewritten more readably in [5]; it does not use the smoothness of S.

Fix f, and prove that (i) $\Rightarrow$ (ii). We reduce to the case where S is connected and nonempty. If $s \in S$ is a point of S, the local system $R^nf_*\mathbb{Q}$ is entirely described by its fiber $(R^nf_*\mathbb{Q})_s$ at s, and by the action of the fundamental group $\pi = \pi_1(S,s)$ on this fiber. One has

$$H^0(S,R^nf_*\mathbb{Q}) \xrightarrow{\sim} [(R^nf_*\mathbb{Q})_s]^\pi.$$

If $X_s = f^{-1}(s)$, the composite arrow

$$H^0(S,R^nf_*\mathbb{Q}) \longrightarrow (R^nf_*\mathbb{Q})_s \simeq H^n(X_s,\mathbb{Q})$$

is therefore injective. Consider the arrows

$$H^n(\overline{X},\mathbb{Q}) \xrightarrow{a} H^n(X,\mathbb{Q}) \xrightarrow{b} H^0(S,R^nf_*\mathbb{Q}) \xrightarrow{c} H^n(X_s,\mathbb{Q}).$$

The arrows cb and cba have the same image by (3.2.18). The arrow b is an edge homomorphism of the Leray spectral sequence, hence is surjective by hypothesis. Since c is injective, b and ba have the same image and ba is surjective. This proves (ii) for f projective. We now deduce the general case. Again reduce to the case where S is connected and nonempty.

By Chow's lemma and resolution of singularities, there exists a scheme X' which is quasi-projective and smooth, and a projective birational morphism

$$p : X' \longrightarrow X.$$

There then exists (by Bertini, or Sard) a nonempty Zariski open subset S_1 of S such that $X'_1 = (fp)^{-1}(S_1)$ is smooth over S_1. Finally, let $X_1 = f^{-1}(S_1)$, let $\overline{X}$ be a smooth compactification of X, and let $\overline{X}'_1$ be a smooth compactification of X'_1 which dominates $\overline{X}$.

$$\begin{array}{ccccc}
\overline{X} & \longleftarrow & & \overline{p} & \overline{X}'_1 \\
\uparrow & & & & \uparrow \\
X & \hookleftarrow & X_1 & \xleftarrow{p} & X'_1 \\
{\scriptstyle f}\downarrow & & {\scriptstyle f_1}\downarrow & & {\scriptstyle f'_1}\downarrow \\
S & \underset{i}{\hookleftarrow} & S_1 & = & S_1
\end{array}$$

If $s \in S_1$, the morphism $i_* : \pi_1(S_1,s) \to \pi_1(S,s)$ is surjective because the topological codimension of $S - S_1$ is ≥ 2. Therefore

$$H^0(S,R^nf_*\mathbb{Q}) \xrightarrow{\sim} H^0(S_1,R^nf_{1*}\mathbb{Q}),$$

and it suffices to prove that

$$H^n(\overline{X},\mathbb{Q}) \longrightarrow H^0(S_1,R^nf_{1*}\mathbb{Q})$$

is surjective.

The vertical arrows of the diagram

$$\begin{array}{ccccc}
H^n(\overline{X},\mathbb{Q}) & \longrightarrow & H^n(X_1,\mathbb{Q}) & \longrightarrow & H^0(S_1,R^nf_{1*}\mathbb{Q}) \\
\downarrow{\scriptstyle \overline{p}^*} & & \downarrow{\scriptstyle p^*} & & \downarrow{\scriptstyle p^*} \\
H^n(\overline{X}'_1,\mathbb{Q}) & \longrightarrow & H^n(X'_1,\mathbb{Q}) & \longrightarrow & H^0(S_1,R^nf'_{1*}\mathbb{Q})
\end{array}$$

admit as left inverses the *Gysin morphisms* $\overline{p}_!$, $p_!$, and $p_!$, defined by Poincaré duality as transposes of the arrows analogous to the preceding ones in cohomology with proper support. Moreover, the diagram

$$\begin{array}{ccc} H^n(\overline{X},\mathbb{Q}) & \xrightarrow{\quad u \quad} & H^0(S_1, R^n f_{1*}\mathbb{Q}) \\ \uparrow{\scriptstyle \overline{p}_!} & & \uparrow{\scriptstyle p_!} \\ H^n(\overline{X}'_1,\mathbb{Q}) & \xrightarrow[\quad v \quad]{} & H^0(S_1, R^n f'_{1*}\mathbb{Q}) \end{array}$$

is commutative. Thus u is a direct factor of v. The latter being surjective (since f'_1 is projective), the same is true of u.

We now prove (i) in the general case. We reduce to the case where f is purely of relative dimension n. Let $f \times f$ be the projection of $X \times_S X$ onto S.

Let δ be the image of the cohomology class of the diagonal of $X \times_S X$ in $H^0(S, R^n(f \times f)_*\mathbb{Q})$. By Künneth,

$$R^n(f \times f)_*\mathbb{Q} = \sum_{p+q=n} (R^p f_*\mathbb{Q} \otimes R^q f_*\mathbb{Q}).$$

Let δ'_{pq} $(p+q=n)$ denote the components of δ in this decomposition, and let δ_{pq} be classes in $H^n(X \times_S X)$ whose images are the δ'_{pq}.

The δ_{pq} define in the derived category $D^+(S)$ homomorphisms

$$\delta_p : Rf_*\mathbb{Q} \longrightarrow Rf_*\mathbb{Q}$$

such that $\mathcal{H}^q(\delta_p)$ is zero for $p \neq q$, and is the identity for $p = q$. By [2], one therefore has in $D^+(S)$

(4.1.1.1) $$Rf_*\mathbb{Q} \approx \sum_p R^p f_*\mathbb{Q}[-p],$$

and the Leray spectral sequence degenerates.

***Corollary* (4.1.2).** — *Let $f : X \to S$ be a proper smooth morphism whose target is a reduced connected separated scheme S. Let $(R^n f_*\mathbb{Q})^0$ be the largest constant local subsystem of $R^n f_*\mathbb{Q}$, with fiber $H^0(S, R^n f_*\mathbb{Q})$. Then, for each $s \in S$, $(R^n f_*\mathbb{Q})^0_s$ is a Hodge substructure of $(R^n f_*\mathbb{Q})_s \simeq H^n(X_s,\mathbb{Q})$, and the Hodge structure induced on $H^0(S, R^n f_*\mathbb{Q})$ is independent of s.*

Let $s \in S$ and $X_s = f^{-1}(s)$. If S is smooth, and if $\overline{X}$ is a smooth compactification of X, then by (4.1.1) the subspace $(R^n f_*\mathbb{Q})^0_s$ of $(R^n f_*\mathbb{Q})_s$ is the image of $H^n(\overline{X},\mathbb{Q})$. Since the restriction map

$$H^n(\overline{X},\mathbb{Q}) \longrightarrow H^n(X_s,\mathbb{Q})$$

is a morphism of Hodge structures, its image is a Hodge substructure, and the Hodge structure induced on $H^0(S, R^n f_*\mathbb{Q})$, a quotient of that of $H^n(\overline{X},\mathbb{Q})$, is independent of s.

In the general case, (4.1.2) also means that if a is a global section of $R^n f_*\mathbb{C}$, then its components $a^{p,q}$ of type (p,q), *a priori* only continuous sections of the complex vector bundle defined by $R^n f_*\mathbb{C}$, are in fact locally constant, i.e. are sections of $R^n f_*\mathbb{C}$. This is so because $a^{p,q}$ is continuous and locally constant on the smooth locus (which is dense) of S, by what precedes.

(4.1.3) In the case where S is compact, a generalization of (4.1.2) is proved analytically in Griffiths [5].

As corollaries of (4.1.2), let us cite:

(4.1.3.1) (Griffiths [5]) Let $f : X \to S$ be a proper smooth morphism as in (4.1.2). If a global section a of $R^n f_*\mathbb{C}$ is of Hodge type (p,q) at one point, then a is of type (p,q) everywhere. In particular, if $n = 2$ and if, at one point s, a is the cohomology class of a divisor of X_s, then a is at every point the cohomology class of a divisor and, for S smooth, is even defined by a divisor D on X.

(4.1.3.2) (Grothendieck [7]) Let S be a reduced connected scheme of finite type over $\mathbb{C}$, and let $f_1 : X_1 \to S$ and $f_2 : X_2 \to S$ be two abelian schemes over S. If a morphism

$$u : R^1 f_{2*}\mathbb{Z} \longrightarrow R^1 f_{1*}\mathbb{Z}$$

comes at one point s of S from a morphism of abelian varieties

$$\widetilde{u}_s : (X_1)_s \longrightarrow (X_2)_s,$$

then u comes from one (and only one) morphism of abelian schemes

$$\widetilde{u} : X_1 \longrightarrow X_2.$$

(4.1.3.3) (Cf. Katz [10]) Let S be a smooth connected scheme, let s be a point of S, let $f : X \to S$ be a proper smooth morphism, and let P be a direct factor of the local system $R^i f_*\mathbb{Q}$ which is pointwise a Hodge substructure. The following conditions are equivalent:

a) The Hodge structure of P is locally constant.

b) The representation P_s of $\pi_1(S, s)$ factors through a finite quotient of $\pi_1(S, s)$.

c) There exists a nonempty finite étale covering $u : S' \to S$ such that u^*P is a constant family of Hodge structures.

4.2. The semisimplicity theorem.

(4.2.1) Let S be a topological space. A *continuous family of Hodge structures* on S consists of:

a) A local system $H_\mathbb{Z}$ of finitely generated $\mathbb{Z}$-modules on S.

b) For every point $s \in S$, a Hodge structure on the fiber $(H_\mathbb{Z})_s$, this structure varying continuously with s.

A continuous family H of Hodge structures on S is said to be *of weight* n if the fibers H_s ($s \in S$) have weight n.

Similarly, a *continuous family of* $\mathbb{Q}$*-Hodge structures* is defined as a local system of $\mathbb{Q}$-vector spaces endowed at each point with a continuously varying $\mathbb{Q}$-Hodge structure.

A *polarization* of a continuous family H of $\mathbb{Q}$-Hodge structures of weight n is a morphism of local systems from $H_\mathbb{Q} \otimes H_\mathbb{Q}$ into the constant local system $\mathbb{Q}(-n)_\mathbb{Q}$, which at every point $s \in S$ defines a polarization of H_s.

(4.2.2) Suppose S is connected, and let $\mathcal{C}$ be a strictly full subcategory of the category of continuous families of $\mathbb{Q}$-Hodge structures on S. We shall consider the following conditions:

(4.2.2.1) $\mathcal{C}$ is stable under direct factors, direct sums, and tensor products; the constant Tate families $\mathbb{Q}(n)$ ($n \in \mathbb{Z}$) belong to $\mathcal{C}$.

(4.2.2.2) Every homogeneous Hodge structure (= of some weight n) in $\mathcal{C}$ is polarizable.

(4.2.2.3) For every $H \in \mathrm{Ob}\,\mathcal{C}$, there exists a local system $H'_\mathbb{Z}$ of free $\mathbb{Z}$-modules on S such that $H'_\mathbb{Z} \otimes \mathbb{Q} \approx H_\mathbb{Q}$.

(4.2.2.4) For every H in $\mathcal{C}$, the largest constant local subsystem H' of H is a constant family of Hodge substructures of H.

***Lemma* (4.2.3).** — *If $\mathcal{C}$ satisfies conditions (4.2.2.1) and (4.2.2.2), then:*

(i) $\mathcal{C}$ is a semisimple abelian subcategory of the abelian category of continuous families of $\mathbb{Q}$-Hodge structures on S.

(ii) If $H \in \mathrm{Ob}\,\mathcal{C}$, then its dual H^* and the $\bigwedge^p H$ belong to $\mathcal{C}$; if $H_1, H_2 \in \mathrm{Ob}\,\mathcal{C}$, then $\mathrm{Hom}(H_1, H_2) \in \mathrm{Ob}\,\mathcal{C}$.

If $H \in \mathrm{Ob}\,\mathcal{C}$ and H_1 is a subobject of H, in the category of continuous families of $\mathbb{Q}$-Hodge structures, we prove that H_1 is a direct factor of H in that category. We may suppose H homogeneous. If ψ is a polarization form for H, the orthogonal complement of H_1 for ψ is indeed a subobject of H complementary to H_1. This proves (i).

If $H \in \mathrm{Ob}\,\mathcal{C}$ has weight n, a polarization of H defines an isomorphism between H^* and $H \otimes \mathbb{Q}(n)$. By (4.2.2.1), one therefore has $H^* \in \mathrm{Ob}\,\mathcal{C}$. For arbitrary H in $\mathcal{C}$, decomposing H into its homogeneous components, one has

$$H^* = \bigoplus_n (H^n)^*,$$

whence again $H^* \in \mathrm{Ob}\,\mathcal{C}$. Finally, $\bigwedge^p H$ is a direct factor in $\bigotimes^p H$, and $\mathrm{Hom}(H_1, H_2) \simeq H_1^* \otimes H_2$.

Let $f : X \to S$ be a proper smooth morphism of reduced schemes. The sheaf $R^i f_* \mathbb{Q}$ is then a local system, and for $s \in S$,

$$(R^i f_* \mathbb{Q})_s \simeq H^i(X_s, \mathbb{Q})$$

is endowed with a $\mathbb{Q}$-Hodge structure. This varies continuously with s.

***Definition* (4.2.4).** — *Let S be a smooth connected scheme. A continuous family H of $\mathbb{Q}$-Hodge structures on S^{an} will be called algebraic if there exist a nonempty Zariski open subset U of S, an integer k, and a projective smooth morphism $f : X \to U$ such that $H|U$ is a direct factor of $Rf_*\mathbb{Q} \otimes \mathbb{Q}(k)$.*

***Proposition* (4.2.5).** —

(i) The category of algebraic continuous families of $\mathbb{Q}$-Hodge structures on S satisfies the conditions of (4.2.2).

(ii) If a continuous family H of Hodge structures on S is such that its restriction to a dense Zariski open subset U of S is algebraic, then H is algebraic.

(iii) If $f : X \to S$ is proper and smooth, then $Rf_*\mathbb{Q}$ is algebraic.

Assertion (ii) is evident. We prove (i). Let $\mathcal{C}_0$ be the set of continuous families of Hodge structures on S that are of the form $Rf_*\mathbb{Q}$ (f projective and smooth). Then:

a) By the Künneth formula,

$$R(f \times g)_*\mathbb{Q} \simeq Rf_*\mathbb{Q} \otimes Rg_*\mathbb{Q},$$

$\mathcal{C}_0$ is stable under tensor product.

b) $\mathcal{C}_0$ is stable under direct sum:

$$R(f \amalg g)_*\mathbb{Q} \simeq Rf_*\mathbb{Q} \oplus Rg_*\mathbb{Q}.$$

c) The homogeneous components of every $H \in \mathcal{C}_0$ are polarizable (cf. (2.2.6)).

d) The objects $H \in \mathcal{C}_0$ satisfy (4.2.2.3), since

$$Rf_*\mathbb{Q} \simeq Rf_*\mathbb{Z} \otimes \mathbb{Q}.$$

e) The objects $H \in \mathcal{C}_0$ satisfy (4.2.2.4), by (4.1.2).

Let $\mathcal{C}_1$ be the set of direct factors of objects of $\mathcal{C}_0$. Then $\mathcal{C}_1$ is stable under tensor product, direct sum, and direct factor, and satisfies (4.2.2.2) to (4.2.2.4). Moreover $\mathbb{Q}(-1)$ belongs to $\mathcal{C}_1$, since it is of the form $R^2 f_*\mathbb{Q}$ for the projective bundle $f : \mathbb{P}^1_S \to S$. The category $\mathcal{C}_2$ of the $H \otimes \mathbb{Q}(k)$ ($k \in \mathbb{N}$) therefore satisfies (4.2.2.1) to (4.2.2.4).

To deduce (i), it suffices to note that if H is a continuous family of Hodge structures, and U is a dense Zariski open subset of S, then a subobject, a polarization, or an integral lattice of $H|U$ extends uniquely to H.

We prove (iii). If $f : X \to S$ is projective and smooth, then there exist a dense Zariski open subset U of S and a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\ p\ } & X|U \\ & {\scriptstyle f'}\searrow \quad \swarrow{\scriptstyle f|U} & \\ & U & \end{array}$$

with f' projective and smooth and p birational and surjective (apply Chow's lemma and resolution of singularities to the generic fiber of f). The restriction map

$$p^* : (Rf_*\mathbb{Q})|U \longrightarrow Rf'_*\mathbb{Q}$$

is then a split injection of continuous families of Hodge structures, with left inverse the Gysin morphism p_*; hence $Rf_*\mathbb{Q}$ is algebraic.

Theorem (4.2.6)[(1)]. — *Let S be a connected, locally connected and locally simply connected topological space, endowed with a base point s. Let $\mathcal{C}$ be a category of continuous families of Hodge structures on S satisfying the conditions (4.2.2). Then, if $H \in \mathrm{Ob}\,\mathcal{C}$, the representation of $\pi_1(S,s)$ on the fiber $(H_{\mathbb{Q}})_s$ is semisimple.*

Let $H \in \mathrm{Ob}\,\mathcal{C}$. The local system $H_{\mathbb{Q}}$ defines a complex local system $H_{\mathbb{C}} = H_{\mathbb{Q}} \otimes \mathbb{C}$. For every complex local system V on S, denote by V^c the complex vector bundle which it defines, identified with the sheaf of its continuous sections.

By definition, the group S (2.1.2) acts on $H^c_{\mathbb{C}}$. A subbundle of $H^c_{\mathbb{C}}$ will be called *horizontal*, or *locally constant*, if it is defined by a local subsystem of $H_{\mathbb{C}}$.

Lemma (4.2.7). — *Let V be a rank-one local subsystem of $H_{\mathbb{C}}$. Suppose that a tensor power $V^{\otimes n}$ $(n \geq 1)$ of V is a trivial local system, i.e. that $\pi_1(S,s)$ acts on V_s through a finite group (necessarily cyclic). Then, for $t \in S$, tV^c is again locally constant.*

For tV^c to be locally constant, it suffices that

$$(tV^c)^{\otimes n} = tV^{c\otimes n} \subset \bigotimes^n H_{\mathbb{C}}$$

be locally constant. But $V^{\otimes n}$ is generated by a global horizontal section v, and by hypothesis tv is again horizontal (4.2.2.4).

We proceed by induction on $\dim(H_{\mathbb{Q}})_s$. We may suppose H homogeneous and nonzero. Let d be the minimal dimension of the nonzero complex local subsystems of $H_{\mathbb{C}}$. The sum W of all local subsystems of $H_{\mathbb{C}}$ of dimension d (which are automatically simple) is "defined over $\mathbb{Q}$", i.e. is of the form $W_{\mathbb{Q}} \otimes \mathbb{C}$ for $W_{\mathbb{Q}}$ a local subsystem of $H_{\mathbb{Q}}$. By construction, W_s is a semisimple complex $\pi_1(S,s)$-module, hence $(W_{\mathbb{Q}})_s$ is a semisimple $\pi_1(S,s)$-module over $\mathbb{Q}$.

Let $H_{\mathbb{Z}}$ be a local system of free $\mathbb{Z}$-modules such that $H_{\mathbb{Z}} \otimes \mathbb{Q} \simeq H_{\mathbb{Q}}$, and let $W_{\mathbb{Z}} = H_{\mathbb{Z}} \cap W_{\mathbb{Q}}$. If W has dimension e, the local system $(\bigwedge^e W)^{\otimes 2}$ is trivial. Indeed:

$$\bigwedge^e W \simeq \bigwedge^e W_{\mathbb{Z}} \otimes \mathbb{C},$$

and $\pi_1(S,s)$ can act on $(\bigwedge^e W_{\mathbb{Z}})_s$ only by ± 1.

Let V be a complex local subsystem of dimension d of $H_{\mathbb{C}}$, and let V' be a complement of V in W (semisimplicity of W). One has

$$\bigwedge^e W \simeq \bigwedge^d V \otimes \bigwedge^{e-d} V'.$$

Apply (4.2.7) to the local subsystem $\bigwedge^d V \otimes \bigwedge^{e-d} V'$ of $\bigwedge^d H_{\mathbb{C}} \otimes \bigwedge^{e-d} H_{\mathbb{C}}$. We find that, for $t \in S$,

$$t(\bigwedge^d V \otimes \bigwedge^{e-d} V')^c = \bigwedge^d tV^c \otimes \bigwedge^{e-d} tV'^c \subset \bigwedge^d H_{\mathbb{C}} \otimes \bigwedge^{e-d} H_{\mathbb{C}}$$

is locally constant. Hence $\bigwedge^d tV^c \subset \bigwedge^d H_{\mathbb{C}}$ and $tV^c \subset H_{\mathbb{C}}$ are locally constant.

By definition of W, one has $tV^c \subset W^c$: W^c is stable under S, and $W_{\mathbb{Q}}$ is a Hodge substructure of H. If ψ is a polarization of H, H is therefore the direct sum of W and of its orthogonal complement, both in $\mathcal{C}$, and the induction concludes.

Corollary (4.2.8). — *Under the hypotheses of (4.2.6):*

(i) *The algebra A of endomorphisms of the local system $H_{\mathbb{Q}}$ is semisimple. It admits one (and only one) $\mathbb{Q}$-Hodge structure such that, at each point s, $A \otimes H_s \to H_s$ is a morphism of Hodge structures.*

---

(1) *(Added in proof.)* Let S be a smooth connected scheme. A *family of Hodge structures on S* is a continuous family H of Hodge structures on S satisfying the following conditions:

a) The Hodge filtration of $(H_{\mathbb{C}})_s$ varies holomorphically with s, i.e. corresponds to a filtration F of $H_{\mathcal{O}} = H_{\mathbb{Z}} \otimes \mathcal{O}$.

b) The covariant derivative ∇ satisfies $\nabla F^p \subset \Omega^1_S \otimes F^{p-1}$.

Let $\mathcal{C}$ be the category of those continuous families of $\mathbb{Q}$-Hodge structures on S which underlie a family of Hodge structures, and whose homogeneous direct factors are polarizable. It is clear that $\mathcal{C}$ satisfies (4.2.2.1) to (4.2.2.3). From results of W. Schmid (August 1970, unpublished) and P. A. Griffiths [5] one can deduce that $\mathcal{C}$ satisfies (4.2.2.4). This theorem, of which (4.1.2) is a corollary, makes it possible to apply (4.2.6) and its corollaries to objects of $\mathcal{C}$.

(ii) *The center of A is of type $(0,0)$ and the composition law $\cdot : A \otimes A \to A$ is a morphism of Hodge structures.*

(iii) *Let W be a complex local subsystem of dimension d of $H_{\mathbb{C}}$:*

a) For $t \in S$, $tW^c \subset H_{\mathbb{C}}$ is locally constant and defines a local subsystem tW isomorphic to W.

b) A power $(\bigwedge^d W)^{\otimes n}$ $(n \geq 1)$ of the local system $\bigwedge^d W$ is a trivial local system.

The algebra A is the commutant of $\pi_1(S,s)$ in $(H_{\mathbb{Q}})_s$. Its semisimplicity therefore follows from (4.2.6) and Bourb., *Alg.*, chap. 8, § 5, no. 2, prop. 4. Moreover, A is the fiber of the largest constant local subsystem of $\mathrm{Hom}(H,H)$. Since $\mathrm{Hom}(H,H) \in \mathcal{C}$ ((4.2.3), (ii)), the existence of the Hodge structure in (i) follows from hypothesis (4.2.2.4). It is clear that the composition law is a morphism of Hodge structures. It follows that the center Z of A is a Hodge substructure: $Z_{\mathbb{C}}$ is bigraded, as the intersection of the kernels of the bihomogeneous morphisms $[x,\ ]$ for x bihomogeneous. Finally Z, and hence $Z_{\mathbb{C}}$, is semisimple, and for $(p,q) \neq (0,0)$, Z^{pq} is zero because it is nilpotent.

A complex local subsystem W of $H_{\mathbb{C}}$ is defined by a projector e of $A_{\mathbb{C}}$. For $t \in S$, tW^c is defined by the projector $t.e$, and is therefore locally constant. Since the group S is connected, to verify that tW is isomorphic to W, it suffices to verify that for t in a sufficiently small neighborhood of 1, the $\mathbb{C}[\pi_1(S,s)]$-module tW_s is isomorphic to W_s. Let H_i be the isotypic components of the $\mathbb{C}[\pi_1(S,s)]$-module $(H_{\mathbb{C}})_s$. One has

$$W_s = \bigoplus_i (W_s \cap H_i)$$

and

$$tW_s = \bigoplus_i (tW_s \cap H_i).$$

Moreover, for t sufficiently close to 1, tW_s is close to W_s in the Grassmannian, and hence

(4.2.8.1) $$\dim(tW_s \cap H_i) \leq \dim(W_s \cap H_i).$$

In fact, since $\dim(tW_s) = \dim(W_s)$, equality holds in (4.2.8.1). The respective isotypic components of W_s and tW_s therefore have the same length, and W_s is isomorphic to tW_s.

We prove (iii), *b*). It suffices to treat the case where H is homogeneous and W_s is a simple $\mathbb{C}[\pi_1(S,s)]$-module. Let H_1 be the isotypic component of $(H_{\mathbb{C}})_s$ which contains W_s. By *a*), H_1 is stable under S. If W_s has dimension d and H_1 has length k, one has

$$\bigwedge^{kd} H_1 \simeq (\bigwedge^{d} W_s)^{\otimes k}$$

as a $\mathbb{C}[\pi_1(S,s)]$-module. If χ is the character of $\pi_1(S,s)$ defined by $\bigwedge^d W$, the character χ_1 defined by $\bigwedge^{kd} H_1$ is therefore χ^k. Put $H_2 = H_1 + \overline{H_1}$. Since H_1 is stable under S, H_2 is defined by a real Hodge substructure of $(H_{\mathbb{R}})_s$. Every polarization form on H therefore induces on H_2 a nondegenerate bilinear form invariant under $\pi_1(S,s)$.

If $\dim(H_2) = e$, the representation $(\bigwedge^e H_2)^{\otimes 2}$ of $\pi_1(S,s)$ is therefore trivial. Thus:

a) For H_1 real: χ^{2k} is trivial;

b) For H_1 nonreal ($H_1 \neq \overline{H_1}$): $\chi^{2k} \cdot \bar{\chi}^{2k}$ is trivial.

In both cases one has $|\chi| = 1$.

The representation of $\pi_1(S,s)$ on $(H_{\mathbb{C}})_s$ is obtained by extension of scalars from a representation defined over $\mathbb{Q}$ (namely $(H_{\mathbb{Q}})_s$). The conjugate representations of W therefore occur in $(H_{\mathbb{C}})_s$; there are at most $N = \dim(H)$ of them, and for every automorphism σ of $\mathbb{C}$ one has

$$|\sigma(\chi)| = 1.$$

For $\gamma \in \pi_1(S,s)$, $\chi(\gamma)$ is an algebraic integer with at most N complex conjugates, all of absolute value 1; $\chi(\gamma)$ is therefore a k-th root of unity, with $k \leq N$. Hence $\chi^{N!} = 1$, which proves b).

Corollary (4.2.9). — *Let S be a smooth connected separated scheme, endowed with a base point $s \in S$. Let $f : X \to S$ be a morphism of schemes such that $R^n f_* \mathbb{Q}$ is a local system on S, let G be the Zariski closure of the image of $\pi_1(S,s)$ in $\mathrm{Aut}_{\mathbb{C}}((R^n f_* \mathbb{C})_s)$, and let G^0 be the identity component of G. Then:*

a) *If f is proper and smooth, G^0 is semisimple.*

b) *In general, G^0 has no quotient of multiplicative type (i.e., the radical of G^0 is unipotent).*

We prove *a*). After replacing S by a finite étale covering, we reduce to the case $G = G^0$. By ((4.2.5), (iii)), the local system $R^n f_*\mathbb{Q}$ underlies an algebraic family H of Hodge structures, so (4.2.8) applies to it ((4.2.5), (i)). The group G is reductive, since it has a faithful semisimple representation, namely $(R^n f_*\mathbb{C})_s$. If G (assumed connected) were not semisimple, it would admit a nontrivial complex representation ρ of rank one; since H_s is faithful, ρ would be a direct summand of $((H_{\mathbb{C}} + H_{\mathbb{C}}^*)^{\otimes m})_s$ for some m. This contradicts (4.2.8), (iii), *b*), because $(H + H^*)^{\otimes m}$ is algebraic and no tensor power of ρ is trivial.

Let π be a group, and consider the following property of a representation ρ of π on a finite-dimensional complex vector space V:

> $(*)$ The identity component G^0 of the Zariski closure G of $\rho(\pi)$ in $\mathrm{Aut}(V)$ has no quotient of multiplicative type.

Lemma (4.2.10). — *Let $0 \to V' \to V \to V'' \to 0$ be an exact sequence of representations of the preceding kind. Then V satisfies $(*)$ if (and only if) V' and V'' satisfy $(*)$.*

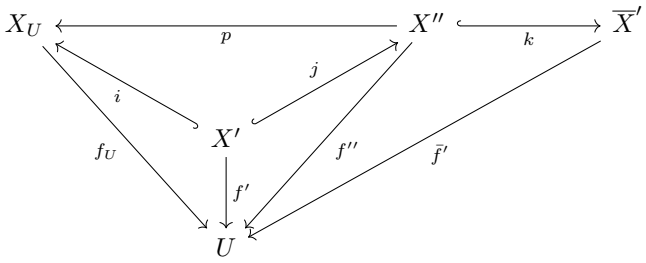
Let G, G', and G'' be the groups defined by V, V', and V''. Since, like π, G leaves V' invariant, there is a morphism

$$u = (u', u'') : G \longrightarrow G' \times G'',$$

with unipotent kernel, and such that u' and u'' are surjective. Let N be the identity component of the kernel of u''. Since u' is surjective, $u'(N)$ is normal in G', and therefore has no quotient of multiplicative type. If χ is a character of G^0, then χ factors through $u(G^0)$, is trivial on N, a power of χ factors through G''^0, and χ is trivial.

We prove (4.2.9), *b)*, by dévissage and induction on the relative dimension of f. We may suppose X reduced.

First suppose that X is quasi-projective. Then there exist a nonempty open subset U of S, a dense open subset X' of $X_U = f^{-1}(U)$ smooth over U, and a compactification $\overline{X}'$ of X' over U, proper and smooth over U. One can choose $\overline{X}'$ so that an open X'' of $\overline{X}'$ contains X' and is proper over X_U



We have long exact sequences of sheaves

$$\begin{gathered}
\to R^i f_{U*}(i_!\mathbb{Q}) \to R^i f_{U*}\mathbb{Q} \to R^i f_{U*}(\mathbb{Q}/i_!\mathbb{Q}) \to \\
\to R^i f''_*(j_!\mathbb{Q}) \to R^i f''_*\mathbb{Q} \to Rf''_*(\mathbb{Q}/j_!\mathbb{Q}) \to \\
\to R^i f''_!\mathbb{Q} \to R^i \bar{f}'_*\mathbb{Q} \to R^i \bar{f}'_*(\mathbb{Q}/k_!\mathbb{Q}) \to
\end{gathered}$$

For U sufficiently small, these sheaves are local systems and the induction hypothesis applies to $R^* f_{U*}(\mathbb{Q}/i_!\mathbb{Q})$, to $R^* f''_*(\mathbb{Q}/j_!\mathbb{Q})$, and to $R^* \bar{f}'_*(\mathbb{Q}/k_!\mathbb{Q})$. By *a)*, condition $(*)$ is satisfied by $R^* \bar{f}'_*\mathbb{Q}$. By (4.2.10), it is therefore satisfied by $R^* f''_!\mathbb{Q}$, and by the dual local system $R^* f''_*\mathbb{Q}$ (Poincaré duality). By (4.2.10), condition $(*)$ is satisfied by $R^* f''_*(j_!\mathbb{Q})$, which is isomorphic to $R^* f_{U*}(i_!\mathbb{Q})$ because p is proper, and also by $R^* f_{U*}\mathbb{Q}$. Since the morphism $\pi_1(U) \to \pi_1(S)$ is surjective, $R^i f_*\mathbb{Q}$ satisfies (4.2.9), *b)*.

To pass from the quasi-projective case to the general case, it suffices to use the Leray spectral sequence for a covering (U_i) of X by quasi-projective open subsets:

$$R^* f_*(\bigcap_i U_i, \mathbb{Q}) \Rightarrow R^* f_*\mathbb{Q},$$

and (4.2.10).

4.3. Complement to [2].

In [2], (5.4) (footnote), I assert without proof that

Proposition (4.3.1). — *Let X be a proper smooth scheme over $\mathbb{C}$. If a C^∞ form on X satisfies $d'\alpha = d''\alpha = 0$ and is cohomologous to zero, then there exists β such that $\alpha = d'd''\beta$.*

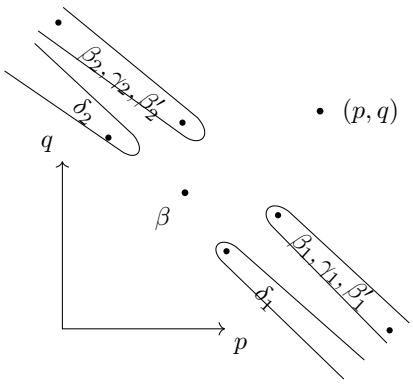
Here is the proof. As in *loc. cit.*, we reduce to the case where α is bihomogeneous, of bidegree (p, q). Since the spectral sequence of the double complex $H^0(X, \mathcal{A}^{pq})$ degenerates, the exterior differential d is strictly compatible with the filtration by first degree p (1.3.2). If $\alpha \sim 0$, there is therefore a form β_1 such that $d\beta_1 = \alpha$ and such that the components $\beta_1^{p'q'}$ of β_1 are zero for $p' < p$.

By complex conjugation, there similarly exists β_2 such that $d\beta_2 = \alpha$ and $\beta_2^{p'q'} = 0$ for $q' < q$. One has $d(\beta_1 - \beta_2) = 0$.

Let γ be a form satisfying $d'\gamma = d''\gamma = 0$ in the cohomology class of $\beta_1 - \beta_2$ (*loc. cit.*). Let γ_1 (resp. γ_2) be the sum of the components of γ of type (p', q') with $p' \geq p$ (resp. $q' \geq q$). Setting $\beta_1' = \beta_1 - \gamma_1$ and $\beta_2' = \beta_2 + \gamma_2$, one still has $d\beta_1' = d\beta_2' = \alpha$; moreover, $\beta_1' - \beta_2' = \beta_1 - \beta_2 - \gamma$ is cohomologous to zero: there exists δ such that

$$d\delta = \beta_1' - \beta_2'.$$

Let δ_1 be the sum of the components of type (p', q') of δ with $p' < p - 1$, let δ_2 be the sum of the components for which $q' > q - 1$, and let β be the component of type $(p - 1, q - 1)$.



One has $\beta_2' = d\delta_2 + d''\beta$ and $\alpha = d\beta_2' = dd\delta_2 + dd''\beta = d'd''\beta$.

4.4. Homomorphisms of abelian schemes.

We propose to show how, in some cases, one can remove from (4.1.3.2) the hypothesis of algebraicity at one point.

(4.4.1) For $f : X \to S$ a proper smooth morphism, we shall denote by $R_i f_* \mathbb{Z}$ the local system on S^{an} of the homology of the fibers of f. At each point $s \in S$, $(R_i f_* \mathbb{Z})_s \simeq H_i(X_s, \mathbb{Z})$ is endowed with a Hodge structure of weight $-i$, dual to the Hodge structure of $H^i(X_s, \mathbb{Z})$. This structure varies continuously with $s \in S$.

(4.4.2) If $f : X \to S$ is an abelian scheme, the exponential defines an exact sequence of sheaves on S^{an}

$$0 \to R_1 f_* \mathbb{Z} \to \mathrm{Lie}(X) \to X \to 0,$$

and the Hodge filtration is the short exact sequence

$$0 \to \mathrm{Ker}(\alpha) \to R_1 f_* \mathbb{Z} \otimes \mathcal{O} \xrightarrow{\alpha} \mathrm{Lie}(X) \to 0.$$

It is known that

Reminder (4.4.3). — *Let S be a smooth scheme of finite type over $\mathbb{C}$. The construction (4.4.2) establishes an equivalence of categories between:*

a) the category of abelian schemes over S;

b) the category of polarizable continuous families of Hodge structures H of type $(-1,0)+(0,-1)$ on S, such that $H_{\mathbb{Z}}$ is torsion-free and the Hodge filtration varies holomorphically on S.

Essential surjectivity of the functor a)↦b) follows from Borel [13]; full faithfulness follows from the fact that, for X_1 and X_2 two abelian schemes over S, the S-scheme $\mathrm{Hom}_S(X_1, X_2)$ is unramified over S, and therefore has the same algebraic and holomorphic sections.

(4.4.4) Let Z be a commutative field, H a central simple algebra over Z, and $*$ a Z-linear anti-automorphism of H. After extension of scalars, one has $H \simeq \mathrm{End}(V)$ for a vector space V over Z, and $*$ is then transposition with respect to a bilinear form Φ on V, unique up to scalar. Suppose that $*$ is involutive; the forms $\Phi(X,Y)$ and $\Phi(Y,X)$ are then proportional: $\Phi(X,Y) = \lambda\Phi(Y,X)$, with $\lambda^2 = 1$. *Put* $\varepsilon_* = \lambda$. If $[H:Z] = d^2$, then

$$\mathrm{Tr}(* : H \to H) = \varepsilon_* \cdot d.$$

Every Z-linear automorphism C of H is inner: $Cx = cxc^{-1}$. If C is involutive and commutes with $*$, the elements c^2 and cc^* are central, so that $c^* = \lambda c$ with λ central. Since $c^{**} = c$, one has $\lambda^2 = 1$. *Put* $\varepsilon_C = \lambda$.

One checks easily that the involution $C \circ *$ satisfies

$$\varepsilon_{C\circ *} = \varepsilon_C \cdot \varepsilon_*.$$

Reminder (4.4.5). — *Let X be a simple abelian variety over a field K and let H be the division algebra $\mathrm{End}_K(X) \otimes \mathbb{Q}$. Every polarization of X defines a positive involution $*$ of H:*

$$\mathrm{Tr}(hh^*) > 0 \qquad \textit{if } h \neq 0.$$

If Z_0 is the subfield invariant under $*$ in the center Z of H, then Z_0 is totally real and one is in one of the following cases:

a) $Z = Z_0$; for every real place ρ of Z, $H \otimes_\rho \mathbb{R}$ is a matrix algebra over $\mathbb{R}$, and $\varepsilon_* = 1$.

b) $Z = Z_0$; for every real place ρ of Z, $H \otimes_\rho \mathbb{R}$ is a matrix algebra over $\mathbb{H}$, and $\varepsilon_* = -1$.

c) Z is a totally imaginary quadratic extension of Z_0.

(4.4.6) Let $f_i : X_i \to S$ $(i = 1, 2)$ be two abelian schemes over a smooth scheme S. The free abelian group

$$\mathrm{Hom}(R_1f_{1*}\mathbb{Z}, R_1f_{2*}\mathbb{Z}) \simeq \Gamma\big(S, \mathrm{Hom}(R_1f_{1*}\mathbb{Z}, R_1f_{2*}\mathbb{Z})\big)$$

is then endowed with a Hodge structure of weight 0 (4.2.8). Moreover, by (2.1.11.1) and (4.4.3), the group $\mathrm{Hom}_S(X_1, X_2)$ is the component of type $(0,0)$ of this group.

(4.4.7) Let S be a nonempty smooth connected scheme with generic point η, and let $f : X \to S$ be an abelian scheme over S. The center Z of $H = \mathrm{End}(R_1f_*\mathbb{Z})$ is then of type $(0,0)$ (4.2.8), and is therefore contained in $\mathrm{End}_S(X) \simeq \mathrm{End}_{k(\eta)}(X_\eta)$.

A polarization of X defines an involution $*$ of H. Its restriction to $\mathrm{End}_S(X)$ is positive, so that Z is a product of totally real fields or totally imaginary quadratic extensions of totally real fields.

Let C denote the actions of $i \in S(\mathbb{R})$ on $H_{\mathbb{R}} = H \otimes \mathbb{R}$ and on the $(R_1f_*\mathbb{R})_s$, for $s \in S$. On $H_{\mathbb{R}}$, $C^2 = 1$, and C commutes with $*$; on $(R_1f_*\mathbb{R})_s$, $C^2 = -1$. If ψ is the alternating form on $R_1f_*\mathbb{Z}$, with integral values, deduced from the polarization of X, then

$$\psi(hx, Cy) = \psi(x, h^*Cy) = \psi(x, C \cdot C(h^*) \cdot y).$$

It follows that

$$\mathrm{Tr}(h \cdot C(h^*)) > 0 \qquad \text{for } h \neq 0. \tag{4.4.7.1}$$

(4.4.8) Now suppose that X_η is simple, so that Z is a field. Let $\rho : Z \to \mathbb{C}$ be a complex embedding. The algebra

$$H_\rho = H \otimes_{Z,\rho} \mathbb{C}$$

is then a matrix algebra and a bigraded direct factor in $H_\mathbb{C} = H \otimes_\mathbb{Q} \mathbb{C}$. Choose an isomorphism

$$H_\rho \simeq \mathrm{End}(V_\rho). \tag{4.4.8.1}$$

Since the infinitesimal generators of S act on H_ρ by derivations, necessarily inner, it follows that V_ρ admits a bigrading, unique up to translation, such that (4.4.8.1) is a bigraded isomorphism. Let $(R_1 f_* \mathbb{C})_\rho$ be the bigraded local system

$$(R_1 f_* \mathbb{C}) \otimes_{Z \otimes \mathbb{C}, \rho} \mathbb{C},$$

a direct factor of $R_1 f_* \mathbb{C}$. Denote by T_ρ the local system

$$T_\rho = \mathrm{Hom}_{H_\rho}(V_\rho, (R_1 f_* \mathbb{C})_\rho).$$

One has

$$V_\rho \otimes_\mathbb{C} T_\rho \xrightarrow{\sim} (R_1 f_* \mathbb{C})_\rho \tag{4.4.8.2}$$

(a bigraded isomorphism). Since $(R_1 f_* \mathbb{C})_\rho$ has type $(-1,0)+(0,-1)$, one of the following conditions is satisfied:

a) V_ρ is bihomogeneous;

b) T_ρ is bihomogeneous.

It is clear that condition a) (respectively b)) is simultaneously satisfied for ρ and for $\bar{\rho}$. If condition a) is satisfied at every place, then H is of type $(0,0)$. If condition b) is satisfied at every place, then the Hodge structure on $R_1 f_* \mathbb{Q}$ is locally constant.

(4.4.9) Now suppose that Z is totally real. For every real embedding φ of Z, put $\varepsilon_{H,\varphi} = +1$ if $H \otimes_{Z,\varphi} \mathbb{R}$ is a matrix algebra and $\varepsilon_{H,\varphi} = -1$ otherwise. If a complex embedding ρ induces a real embedding φ, put $\varepsilon_{H,\rho} = \varepsilon_{H,\varphi}$.

Let $\rho : Z \to \mathbb{C}$ factor through $\varphi : Z \to \mathbb{R}$. Complex conjugation on H_ρ is induced by an antilinear automorphism σ_1 of V_ρ, unique up to a real scalar factor, with scalar square. Since σ_1 commutes with σ_1^2, σ_1^2 is real and, normalizing σ_1, one may suppose that $\sigma_1^2 = \pm 1$. Then

$$\sigma_1^2 = \varepsilon_{H,\rho}. \tag{4.4.9.1}$$

The automorphism of complex conjugation on

$$(R_1 f_* \mathbb{C})_\rho = ((R_1 f_* \mathbb{Q}) \otimes_{Z,\varphi} \mathbb{R}) \otimes_\mathbb{R} \mathbb{C}$$

is "compatible" with (4.4.8.2) and may be written

$$\sigma = \sigma_1 \otimes \sigma_2.$$

Since $\sigma^2 = 1$, by (4.4.9.1) one has

$$\sigma_1^2 = \sigma_2^2 = \varepsilon_{H,\rho}. \tag{4.4.9.2}$$

Let Φ be a polarization form on $R_1 f_* \mathbb{Z}$. This form is necessarily of the type

$$\Phi = \mathrm{Tr}_{Z/\mathbb{Q}}(\psi),$$

where ψ is a Z-bilinear alternating form on $R_1 f_* \mathbb{Q}$.

Let ψ_ρ be the induced form on $V_\rho \otimes_\mathbb{C} T_\rho$. The alternating form ψ_ρ is invariant under $\pi_1(S,s)$. Since T_ρ is irreducible, it can be written

$$\psi_\rho = A_\rho \otimes B_\rho, \tag{4.4.9.3}$$

with A_ρ on V_ρ, and B_ρ on T_ρ, invariant under $\pi_1(S,s)$.

Since T_ρ is irreducible, B_ρ is symmetric or alternating. Since ψ_ρ is alternating, A_ρ is then alternating or symmetric. Since the T_ρ are conjugate to one another, which case occurs is independent of ρ. Put

$$A_\rho(X,Y) = \varepsilon_V A_\rho(Y,X), \tag{4.4.9.4}$$

$$B_\rho(X,Y) = \varepsilon_T B_\rho(Y,X). \tag{4.4.9.5}$$

Thus

$$\varepsilon_V \varepsilon_T = -1 \tag{4.4.9.6}$$

and it is clear from (4.4.4) that if $*$ is the involution of H deduced from ψ,

$$\varepsilon_* = \varepsilon_V. \tag{4.4.9.7}$$

In case (4.4.8), a) (respectively b)), normalize the bigradings so that V_ρ (respectively T_ρ) has bidegree $(0,0)$. Then for nonzero x, y in V_ρ and T_ρ:

$$\begin{aligned} &\text{case a)} \quad A_\rho(x, \sigma_1 x)\, B_\rho(y, C\sigma_2 y) > 0; \\ &\text{case b)} \quad A_\rho(x, C\sigma_1 x)\, B_\rho(y, \sigma_2 y) > 0. \end{aligned}$$

Normalizing A_ρ and B_ρ (only their tensor product being given), we may assume that

$$\begin{aligned} &\text{case a)} \quad A_\rho(x, \sigma_1 x) > 0 \quad \text{and} \quad B_\rho(y, C\sigma_2 y) > 0; \\ &\text{case b)} \quad A_\rho(x, C\sigma_1 x) > 0 \quad \text{and} \quad B_\rho(y, \sigma_2 y) > 0. \end{aligned}$$

Moreover, A_ρ and B_ρ, like Φ, are invariant under the compact subtorus $\mathrm{Ker}(N)$ of S, and in particular under C. In case a) one has

$$0 < A_\rho(\sigma_1 x, \sigma_1\sigma_1 x) = \varepsilon_V \cdot \varepsilon_{H,\rho} \cdot A_\rho(x, \sigma_1 x),$$

whence $\varepsilon_V \cdot \varepsilon_{H,\rho} = 1$. Similarly, in case b), one finds $\varepsilon_T \cdot \varepsilon_{H,\rho} = 1$.

Lemma (4.4.10). If $\varepsilon_T = \varepsilon_{H,\rho}$, we are in case b); if $\varepsilon_T = -\varepsilon_{H,\rho}$, we are in case a).

Proposition (4.4.11). Let S be a nonempty smooth connected scheme with generic point η, let $f : X \to S$ be an abelian scheme over S, let $H = \mathrm{End}(R_1 f_* \mathbb{Q})$, and let Z be the center of H. Suppose that:

(a) X_η is simple over $k(\eta)$.

(b) X does not become constant over any finite covering of S.

(c) One of the following conditions is satisfied (cf. (4.4.5)):

- (c_1) Z is imaginary quadratic;
- (c_2) Z is totally real, and for every real embedding $\varphi : Z \to \mathbb{R}$, $H \otimes_{Z,\varphi} \mathbb{R}$ is a matrix algebra over $\mathbb{R}$;
- (c_3) Z is totally real, and for every real embedding $\varphi : Z \to \mathbb{R}$, $H \otimes_{Z,\varphi} \mathbb{R}$ is a matrix algebra over $\mathbb{H}$.

Then

$$\mathrm{End}(X) \xrightarrow{\sim} \mathrm{End}(R_1 f_* \mathbb{Z}).$$

We show that we are in case (4.4.8), a), or (4.4.8), b) at all places of Z simultaneously. This is clear from (4.4.8) under hypothesis (c_1), and follows from (4.4.10) under hypotheses (c_2) or (c_3), because not only ε_T but also $\varepsilon_{H,\rho}$ is then independent of ρ.

Condition (4.4.8), b) cannot hold at every place of Z, for the Hodge structure would then be locally constant, which by (4.1.3.3) and (4.4.3) contradicts b). Hence condition a) holds at every place of Z, so that H is of type $(0,0)$ (4.4.8). In view of (4.4.3), this completes the proof.

Proposition (4.4.12). Let $f : X \to S$ be an abelian scheme over a smooth scheme S. The following conditions are equivalent:

(i) For every abelian scheme $g : Y \to S$, one has

$$\mathrm{Hom}_S(X,Y) \xrightarrow{\sim} \mathrm{Hom}_S(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z}).$$

(ii) Condition (i) holds for $X = Y$, and the center Z of $\mathrm{End}_S(X) \otimes \mathbb{Q}$ has no complex place $\rho : Z \to \mathbb{C}$ such that the direct summand $R_1 f_* \mathbb{Q} \otimes_{Z,\rho} \mathbb{C}$ of $R_1 f_* \mathbb{C}$ is purely of Hodge type $(-1, 0)$.

One reduces to the case where S is nonempty and connected, with generic point η, and where X_η is a simple abelian variety over $k(\eta)$. If (i) or (ii) is satisfied, then $D = \mathrm{End}(X_\eta) \otimes \mathbb{Q}$ is a division algebra, and, for $s \in S$, $(R_1 f_* \mathbb{Q})_s$ is an irreducible representation of $\pi_1(S, s)$.

To prove that (ii) implies (i), one may suppose that Y_η is simple and that $(R_1 g_* \mathbb{Q})_s$ is isotypic of type $(R_1 f_* \mathbb{Q})_s$. One then has an isomorphism of continuous families of $\mathbb{Q}$-Hodge structures (with D of type $(0,0)$)

$$R_1 g_* \mathbb{Q} \approx R_1 f_* \mathbb{Q} \otimes_D \mathrm{Hom}(R_1 f_* \mathbb{Q}, R_1 g_* \mathbb{Q}).$$

One has $D \otimes \mathbb{C} \simeq M_n(Z \otimes \mathbb{C})$. If $e \in D \otimes \mathbb{C}$ is transformed, under such an isomorphism, into the idempotent e_{11}, one still has an isomorphism of bigraded vector spaces

$$(R_1 g_* \mathbb{C})_s \approx (R_1 f_* \mathbb{C})_s \otimes_{Z \otimes \mathbb{C}} e\big(\mathrm{Hom}(R_1 f_* \mathbb{C}, R_1 g_* \mathbb{C})\big).$$

If condition (ii) is satisfied, and if $e\big(\mathrm{Hom}(R_1 f_* \mathbb{C}, R_1 g_* \mathbb{C})\big)$ is not purely of type $(0,0)$, then $(R_1 g_* \mathbb{C})_s$ could not be purely of type $(-1,0)+(0,-1)$, which is absurd. It follows that $\mathrm{Hom}(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z})$ is purely of type $(0,0)$, and (i) follows from (4.4.3) and (2.1.11.1).

Let Z be a totally imaginary quadratic extension of a totally real number field Z^0. If $Z \otimes \mathbb{R} \simeq \mathbb{C} \oplus \cdots \oplus \mathbb{C}$, define an action of S on $Z \otimes \mathbb{R}$ by scalar multiplication, sending $s \in S(\mathbb{R}) \simeq \mathbb{C}^*$ to $(s^2 \cdot N(s)^{-1}, 1, \ldots, 1)$. The pair consisting of Z and this action of S on $Z \otimes \mathbb{R}$ is a polarizable Hodge structure Z_ρ, of type $(-1,1)+(0,0)+(1,-1)$, depending on Z and on the chosen place $\rho : Z \to \mathbb{C}$.

Let X be an abelian scheme over S, with X_η simple. If the center Z of $\mathrm{End}(R_1 f_* \mathbb{Q})$ is not totally real, it is a totally imaginary quadratic extension of a totally real number field. If the second hypothesis of (ii) is violated, the direct summand $R_1 f_* \mathbb{Q} \otimes_Z Z_\rho$ of the continuous family of polarizable Hodge structures $R_1 f_* \mathbb{Q} \otimes_{\mathbb{Q}} Z_\rho$ is purely of type $(-1,0)+(0,-1)$.

By (4.4.3), the choice of an integral lattice in $R_1 f_* \mathbb{Q} \otimes_Z Z_\rho$ defines an abelian scheme $g : Y \to S$, and $\mathrm{Hom}(R_1 f_* \mathbb{Q}, R_1 g_* \mathbb{Q})$ is not purely of type $(0,0)$, which violates (i).

***Corollary* (4.4.13).** — *Let S be a smooth connected scheme with generic point η, and let $f : X \to S$ be an abelian scheme over S of relative dimension $g \leq 3$. Suppose that on no finite étale covering of S does X admit a constant abelian subscheme. Then, for every abelian scheme $g : Y \to S$, one has*

$$\mathrm{Hom}(X, Y) \xrightarrow{\sim} \mathrm{Hom}(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z})$$

and

$$\mathrm{Hom}(Y, X) \xrightarrow{\sim} \mathrm{Hom}(R_1 g_* \mathbb{Z}, R_1 f_* \mathbb{Z}).$$

By duality, it suffices to prove the first assertion. We may also reduce to the case where X_η is simple. We verify that X satisfies one of the conditions (c) of (4.4.11). Put $H = M_c(D)$, where D is a division algebra of rank b^2 over its center Z, which is known to be either totally real or totally imaginary. Putting $a = [Z : \mathbb{Q}]$, one has

$$ab^2 c \mid 2g \leq 6. \tag{4.4.13.1}$$

1) If Z is totally imaginary but not an imaginary quadratic field, one must have $a = 2g = 4$. For $s \in S$, the commutant of the action of Z on $(R_1 f_* \mathbb{Q})_s$ is then commutative, so the action of $\pi_1(S, s)$ is abelian and therefore factors through a finite group ((4.2.8), (iii), b) or (4.2.9)). This is absurd ((4.1.3.3) and (4.4.3)).

   With the notation of (4.4.8), one could also note that the T_ρ are then of rank one, so condition (4.4.8), b) holds at every place and the Hodge structure is locally constant.

2) If Z is totally real and conditions (c_1) and (c_2) fail, then $a \geq 2$ and $b \geq 2$, which is absurd.

Let us verify that X satisfies the conditions of (4.4.12). If Z is imaginary quadratic and there exists $\rho : Z \to \mathbb{C}$ such that $R_1 f_* \mathbb{Q} \otimes_{Z,\rho} \mathbb{C}$ is purely of Hodge type $(-1,0)$, then the Hodge structure on $R_1 f_* \mathbb{Q}$ is locally constant, which is absurd. This proves (4.4.13).

(4.4.14) Let $f : X \to S$ be a polarized abelian scheme of dimension g over a smooth connected base S. If $s \in S$, the fundamental group $\pi_1(S, s)$ acts on the fiber $H^1(X_s, \mathbb{Z})$ of $R^1 f_* \mathbb{Z}$ at s. The polarization of X defines an alternating form on $H^1(X_s, \mathbb{Z})$, with values in $\mathbb{Z}(-1)$, nondegenerate over $\mathbb{Q}$, and the action of $\pi_1(S, s)$ respects this form. Consider the hypothesis:

(4.4.14.1) The morphism defined by X from S into one of the irreducible components of the corresponding moduli scheme of polarized abelian varieties is dominant.

The following result was suggested to me by J.-P. Serre.

***Corollary* (4.4.15).** — *Let S be a smooth connected scheme and let $f : X \to S$ be a polarized abelian scheme over S satisfying (4.4.14.1). Then, for every abelian scheme $g : Y \to S$, one has*

$$\mathrm{Hom}(X, Y) \xrightarrow{\sim} \mathrm{Hom}(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z}).$$

Let $s \in S$. By (4.4.11) and (4.4.12), it suffices to show that the representation of $\pi_1(S, s)$ on $(R_1 f_* \mathbb{Q})_s$ is absolutely irreducible, so that $H = \mathbb{Q}$. This follows from the following lemma.

***Lemma* (4.4.16).** — *If X satisfies (4.4.14.1), then the image of $\pi_1(S, s)$ in the group of symplectic automorphisms of $H^1(X_s, \mathbb{Z})$ has finite index.*

Let n be an integer. After replacing S by a surjective étale covering corresponding to a subgroup of finite index in $\pi_1(S, s)$, we may assume that the local system ${}_nX = \mathrm{Ker}(n \cdot \mathrm{id} : X \to X)$ is trivial on S. Replacing X by an isogenous abelian scheme, we may further assume that X is principally polarized. Suppose $n \geq 3$.

The abelian scheme X then defines a dominant morphism x from S to the "Jacobi" moduli scheme M_n of principally polarized abelian schemes equipped with a symplectic isomorphism between ${}_nX$ and $(\mathbb{Z}/n)^{2g}$; moreover, X is the pullback via x of the universal abelian scheme over M_n. It is known that $\pi_1(M_n, x(s))$ maps isomorphically onto the subgroup of $\mathrm{Sp}(H^1(X_s, \mathbb{Z}))$ which is the kernel of reduction modulo n.

It therefore remains to establish the following:

***Lemma* (4.4.17).** — *Let $p : S \to M$ be a dominant morphism of smooth connected schemes. The subgroup $p(\pi_1(S, s))$ of $\pi_1(M, p(s))$ then has finite index.*

Let t be a closed point of the generic fiber of p, and let T' be its closure in S. There exists a dense Zariski open subset U of M such that $T = p^{-1}(U) \cap T'$ is an étale cover of U.

There is no loss of generality in taking $s \in T$. In the diagram

$$\begin{array}{ccc} \pi_1(T, s) & \longrightarrow & \pi_1(S, s) \\ \downarrow{\scriptstyle a} & & \downarrow{\scriptstyle c} \\ \pi_1(U, p(s)) & \xrightarrow[b]{} & \pi_1(M, p(s)) \end{array}$$

the arrow a has image of finite index, because T is a finite étale cover, and the arrow b is surjective, because $M - U$ has real codimension ≥ 2. Thus the arrow c has image of finite index.

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A CONGRUENCE BETWEEN MULTINOMIAL COEFFICIENTS

by P. DELIGNE
Institut des Hautes Études Scientifiques, Bures sur Yvette

Let p be a prime. For every nonzero integer m, let $v(m)$ denote the exponent of the highest power of p dividing m. Set also $v(0) = \infty$.

For a k-tuple $\mathfrak{n} = (n_1, \dots, n_k)$ of nonnegative integers, set

$$|\mathfrak{n}| = \sum_1^k n_i, \qquad a\mathfrak{n} = (an_1, \dots, an_k), \qquad X^{\mathfrak{n}} = \prod_1^k X_i^{n_i}, \qquad \mathfrak{n}! = \prod_1^k n_i!$$

and denote by $C(\mathfrak{n})$ the coefficient of the monomial $X^{\mathfrak{n}}$ in

$$\left(\sum_1^k X_i\right)^{|\mathfrak{n}|}$$

(multinomial coefficients).

Thus

$$\left(\sum_1^k X_i\right)^n = \sum_{|\mathfrak{n}|=n} C(\mathfrak{n}) X^{\mathfrak{n}} \tag{1}$$

$$C(\mathfrak{n}) = \frac{|\mathfrak{n}|!}{\mathfrak{n}!} \tag{2}$$

For $k = 2$, these are the binomial coefficients.

$$C(n_1, n_2) = \binom{n_1 + n_2}{n_1} \tag{3}$$

Proposition.— *Let $\mathfrak{n}$ be a k-tuple of nonnegative integers, and let $n = |\mathfrak{n}|$. If p is a prime ≥ 5, then*

$$C(p^l \mathfrak{n}) \equiv C(\mathfrak{n}) \pmod{p^{v(n)+3}} \tag{4}$$

Example.— For $p = 5$, $k = 2$, $l = 1$, and $\mathfrak{n} = (1, 1)$, one has

$$\binom{10}{5} = 252 = \binom{2}{1} + 2 \cdot 5^3$$

Proof.— It is known, or follows from (2), that

$$\left(\sum_1^k X_i\right)^{p^l} = \left(\sum_1^k X_i^{p^l}\right) + pQ \tag{5}$$

where Q is a polynomial with *integer* coefficients, of degree $< p^l$ in each variable. The binomial formula then gives

$$\left(\sum_1^k X_i\right)^{p^l \cdot n} = \left(\sum_1^k X_i^{p^l}\right)^n + \sum_1^n p^i \binom{n}{i} R_i \tag{6}$$

with

$$R_i = \left(\sum_1^k X_i^{p^l}\right)^{n-i} \cdot Q^i \tag{7}$$

For two polynomials A and B with integer coefficients, write $A \doteq B \pmod r$ if the coefficients of $X^{p^l \mathfrak{n}}$ in A and B are congruent modulo r. With this notation, (4) becomes

$$\left(\sum_1^k X_i\right)^{p^l \cdot n} \doteq \left(\sum_1^k X_i^{p^l}\right)^n \pmod{p^{v(n)+3}} \tag{8}$$

Lemma 1.

$$v\left(\binom{n}{i}\right) \geq v(n) - v(i) \tag{9}$$

Let the additive group $\mathbb{Z}/(n)$ act on itself by translation[1]; if an i-element subset P of $\mathbb{Z}/(n)$ has as stabilizer[2] a subgroup H of $\mathbb{Z}/(n)$ ($H \cdot P = P$), then P is a union of cosets of H, and $\#H \mid i$. The cardinality $r_p = n/\#H$ of the orbit of P under $\mathbb{Z}/(n)$, in $\mathcal{P}(\mathbb{Z}/(n))$, therefore satisfies

$$p^{v(n)-v(i)} \mid r_p,$$

hence, since the set of i-element subsets of $\mathbb{Z}/(n)$ is a disjoint union of orbits,

$$p^{v(n)-v(i)} \mid \binom{n}{i}$$

Lemma 2. — *For $p \neq 2, 3$ and $i \geq 3$, we have*

$$p^i \binom{n}{i} \equiv 0 \pmod{p^{v(n)+3}} \tag{10}$$

We have

$$v\left(p^i \binom{n}{i}\right) = i + v\binom{n}{i} \geq v(n) + (i - v(i)) \qquad \text{(Lemma 1)}$$

For $3 \leq i < p$, we have $v(i) = 0$ and $i - v(i) \geq 3$. For $i \geq p$, we have $i - v(i) \geq i - \log i / \log p$. This last function of i is increasing for $i \geq p$, and equals $p - 1 \geq 3$ when $i = p$.

Lemma 3. — *The coefficient of $X^{p^l \mathfrak{n}}$ in R_1 is zero.*

None of the monomials $X^{\mathbf{r}}$ occurring in Q has an exponent $\mathbf{r}$ divisible by p^l. The polynomial

$$R_1 = \left(\sum_1^k X_i^{p^l}\right)^{n-1} \cdot Q$$

inherits this property.

It follows from Lemmas 2 and 3 that

$$\left(\sum_1^k X_i\right)^{p^l n} \doteq \left(\sum_1^k X_i^{p^l}\right)^n + \binom{n}{2} p^2 R^2, \pmod{p^{v(n)+3}} \tag{11}$$

It already follows from this congruence, by Lemma 1, that

$$C(p^l \mathfrak{n}) \equiv C(\mathfrak{n}) \pmod{p^{v(n)+2}} \tag{12}$$

The only monomials $X^{\mathbf{r}}$ with exponent divisible by p^l which occur in $(pQ)^2$ are the monomials

$$X_i^{p^l} \cdot X_j^{p^l} \qquad (i < j)$$

[1] To each $x \in \mathbb{Z}/(n)$ there is thus associated the permutation of $\mathbb{Z}/(n)$ that sends y to $x + y$.

[2] If a group G acts on a set E, the stabilizer of a subset P of E is the set of $g \in G$ for which P is stable.

The coefficient of each is

$$\sum_{i\neq 0,p^l} \binom{p^l}{i}\binom{p^l}{p^l-i} = \sum \binom{p^l}{i}^2 - 2 = \binom{2p^l}{p^l} - 2 \tag{13}$$

and

$$\binom{2p^l}{p^l} - 2 \equiv \binom{2p}{p} - 2 \pmod{p^3}$$

by (12).

Thus

$$\left(\sum_1^k X_i\right)^{p^l\cdot n} \doteq \left(\sum_1^k X_i^{p^l}\right)^n + \binom{n}{2}\left[\binom{2p}{2} - 2\right]\cdot R_2 \pmod{p^{v(n)+3}}$$

with R a polynomial with integer coefficients.

Since

$$v\left(\binom{n}{2}\right) \geq v(n) \qquad \text{(Lemma 1)},$$

the proposition follows from

Lemma 4. — *For a prime $p \geq 5$, one has:*

$$\binom{2p}{p} \equiv 2 \pmod{p^3}.$$

One has

$$p^{-1}\binom{p}{i} \equiv (-1)^{i-1}\cdot i^{-1} \pmod{p}$$

for $1 \leq i \leq p-1$, and hence

$$p^{-2}\left[\binom{2p}{p} - 2\right] = \sum_1^{p-1} p^{-2}\binom{p}{i}^2 \equiv \sum_1^{p-1}(1/i)^2 \pmod{p} \tag{14}$$

Define $z \in \mathbb{Z}/(p)$ by

$$z = \sum_{i\in\mathbb{Z}/(p)^*} 1/i^2$$

For $u \in \mathbb{Z}/(p)^*$, one has

$$z = \sum_{i\in\mathbb{Z}/(p)^*} 1/(u^{-1}i)^2 = u^2 z$$

and

$$(u^2-1)z = 0.$$

Since $p \neq 2, 3$, there exists u such that $u^2 \neq 1$, and $z = 0$. Formula (14) therefore simplifies to

$$p^{-2}\left[\binom{2p}{p} - 2\right] \equiv 0 \pmod{p},$$

which completes the proof.

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The Weil Conjecture for K3 Surfaces

Pierre Deligne* (Bures-sur-Yvette)

1. Statement of the Theorem

Let $\mathbb{F}_q$ be a field with q elements, let $\overline{\mathbb{F}}_q$ be an algebraic closure of $\mathbb{F}_q$, let $\varphi \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ be the Frobenius substitution $x \mapsto x^q$, and let $F = \varphi^{-1}$ be the "geometric Frobenius."

Let X be a scheme (separated and of finite type) over $\mathbb{F}_q$, and let $\overline{X}$ be the scheme over $\overline{\mathbb{F}}_q$ obtained from it by extension of scalars. For every closed point x of X, let $\deg(x) = [k(x) : \mathbb{F}_q]$ be the degree of the residue-field extension over $\mathbb{F}_q$. The zeta function $Z(X,t) \in \mathbb{Z}[[t]]$ is defined by

$$Z(X,t) = \prod_{x \in X} (1 - t^{\deg(x)})^{-1} \qquad (x \text{ a closed point of } X),$$

$$\log Z(X,t) = \sum_{n>0} \#X(\mathbb{F}_{q^n}) \frac{t^n}{n}.$$

For every prime number ℓ prime to q, the ℓ-adic cohomology with proper supports $H^i_c(\overline{X}, \mathbb{Q}_\ell)$ of $\overline{X}$ is a finite-dimensional vector space over $\mathbb{Q}_\ell$, on which $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ acts by transport of structure. By Grothendieck, one has

$$Z(X,t) = \prod_i \det(1 - Ft, H^i_c(\overline{X}, \mathbb{Q}_\ell))^{(-1)^{i+1}}.$$

Take X to be a nonsingular projective surface. In this case it is known that the factors

$$P_i(t) = \det(1 - Ft, H^i(\overline{X}, \mathbb{Q}_\ell))$$

of $Z(X,t)$ are polynomials with integer coefficients independent of ℓ. The Weil conjecture asserts here that the roots of $P_i(t)$ have complex absolute value $q^{-i/2}$. This conjecture is invariant under finite extension of the finite base field. For X as above, and $i \neq 2$, it follows from Weil [9]. If, for simplicity, X is assumed geometrically connected, one indeed has

$$P_0(t) = 1 - t, \qquad P_4(t) = 1 - q^2 t,$$
$$P_1(t) = \det(1 - \varphi t; T_\ell(\mathrm{Pic}^0(X)_{\mathrm{red}})), \qquad P_3(t) = P_1(qt).$$

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Let k be a field of characteristic 0, and let X be a nonsingular projective geometrically connected surface over k. We say that X is a K3 surface if X is regular and has trivial canonical divisor, that is, if $H^1(X, \mathcal{O}) = 0$ and Ω^2_X is isomorphic to $\mathcal{O}_X$. For the remarkable properties of these surfaces, we refer to [8]. We shall mainly use the following facts.

1.1. Lemma. *(i) One has $h^{2,0} = h^{0,2} = 1$.*

(i) One has $H^2(X, T^1_X) = 0$ (infinitesimal deformations are unobstructed), and the map induced by interior product

$$H^1(X, T^1_X) \otimes H^0(X, \Omega^2_X) \xrightarrow{\sim} H^1(X, \Omega^1_X) \tag{1.1.1}$$

is bijective, while $H^0(X, T^1_X) = 0$ (X has no infinitesimal automorphisms).

One has

$$h^{2,0} =_{\text{dfn}} \dim H^0(X, \Omega^2_X) = \dim H^0(X, \mathcal{O}_X) = 1,$$

and (i) follows by Hodge symmetry. Since $H^1(X, \mathcal{O}_X) = 0$, Hodge symmetry and Serre duality give

$$h^{0,1} = h^{1,0} = h^{2,1} = h^{1,2} = 0.$$

Since $\Omega^2_X = \mathcal{O}_X$, the "interior product" isomorphism

$$T^1_X \otimes \Omega^2_X \xrightarrow{\sim} \Omega^1_X$$

induces isomorphisms $T^1_X \simeq \Omega^1_X$. The vanishing of $H^2(X, T^1_X)$ therefore follows from that of $h^{1,2}$, and it is clear that (1.1.1) is bijective. The same argument gives $H^0(X, T^1_X) = 0$.

1.2. Lemma. *Let Y_0 be a nonsingular projective surface over a finite field $\mathbb{F}_q$. The following conditions are equivalent.*

(i) *There exists a complete discrete valuation ring V of unequal characteristic with finite residue field k, a proper smooth morphism $f_V : Y \to \operatorname{Spec}(V)$ whose generic fiber is a K3 surface, and $i : \mathbb{F}_q \to k$ such that $Y_0 \otimes_{\mathbb{F}_q} k \simeq Y \otimes_V k$.*

(ii) *There exists an integral scheme S whose fraction field has characteristic 0, a point $s \in S$, a proper smooth morphism $f : Y \to S$ whose generic fiber is a K3 surface, and $i : \mathbb{F}_q \to k(s)$ such that $Y_0 \otimes_{\mathbb{F}_q} k(s) \simeq Y_s$.*

The proof uses a standard passage to the limit. If the equivalent conditions of 1.2 are satisfied, we say that Y_0 lifts to a K3 surface.

1.3. Theorem. *A nonsingular projective surface over a finite field $\mathbb{F}_q$ which lifts to a K3 surface satisfies the Weil conjecture.*

This theorem was obtained independently by Pjateckii-Shapiro and Šafarevitch.

It applies, for example, to surfaces of degree 4 in $\mathbb{P}^3$. The Betti numbers of a K3 surface are $(1, 0, 22, 0, 1)$; the theorem therefore implies

that

$$\left|\#X(\mathbb{F}_q) - \#\mathbb{P}^2(\mathbb{F}_q)\right| \leq 21q.$$

Grothendieck's conjectural theory of motives (in particular his theory of the "motivic Galois group") was very useful to me in constructing the proof. In this language (which will not be used below), the idea is, roughly speaking, to construct, by reduction from $\mathbb{C}$ to $\mathbb{F}_q$, an abelian variety A and an injective "morphism" of motives

$$H^2(X) \hookrightarrow H^1(A) \otimes H^1(A),$$

in order to deduce the Weil conjecture for X from Weil's theorem for A. However, we do not define this "morphism" by an algebraic cycle; it is therefore not a morphism in the sense of [5]. In Hodge theory, its mode of definition has the following analogue (which will not be used below).

1.4. Lemma. *Let H be a variation of Hodge structures of weight n on a scheme S. If the local system $H_{\mathbb{Z}}$ has no global sections other than the section s and its multiples, then at every point of S the section s is of type $(n/2, n/2)$.*

1.5. Notations. 1.5.1. Scalar extensions will be denoted by subscripts: if M_A is a module over a ring (commutative with identity) A (or a scheme over A, for example an algebraic group over A), then M_B denotes the module over the A-algebra B (or the scheme over B) obtained from it by extension of scalars.

1.5.2. If a group G acts on a set X, we write $g * x$ for the transform of x by g.

1.5.3. When an equality is the definition of one of its sides, we write it $=_{\text{dfn}}$.

2. Polarized Hodge Structures

To work with Hodge structures, we shall use the formalism set out in [4] and briefly recalled below.

2.1. Let $\underline{S}$ be the real algebraic group of invertible elements of the $\mathbb{R}$-algebra $\mathbb{C}$, let $w : \mathbb{G}_{m_{\mathbb{R}}} \to \underline{S}$ be the inclusion of $\mathbb{R}^*$ into $\mathbb{C}^*$, and let $t : \underline{S} \to \mathbb{G}_{m_{\mathbb{R}}}$ be the inverse of the norm; then $tw(x) = x^{-2}$. Let V be a finite-dimensional real vector space. It amounts to the same thing to give:

a) $\rho : \underline{S} \to \mathrm{GL}(V)$ of weight n: $\rho w(x) * v = x^n \cdot v$;

b) a Hodge bigrading of weight n of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$:

$$V_{\mathbb{C}} = \bigoplus_{p+q=n} V^{p,q} \qquad \text{with} \qquad \overline{V^{p,q}} = V^{q,p};$$

c) a Hodge filtration of weight n of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$: this is a decreasing filtration such that

$$F^p(V_{\mathbb{C}}) \oplus \overline{F^{n-p+1}(V_{\mathbb{C}})} \xrightarrow{\sim} V_{\mathbb{C}}.$$

One has

$$F^p(V_{\mathbb{C}}) = \bigoplus_{p' \geq p} V^{p',q'};$$

for $z \in \underline{S}(\mathbb{R}) = \mathbb{C}^*$ and $v^{p,q} \in V^{p,q}$, one has

$$z * v^{p,q} = z^p \bar{z}^q v^{p,q}.$$

Put $C = \rho(i)$; this is the *Weil operator* C. One has $Cv^{p,q} = i^{p-q} v^{p,q}$. For $\mathscr{E}$ a subset of $\mathbb{Z} \times \mathbb{Z}$, a Hodge bigrading is said to be of *type* $\mathscr{E}$ if the *Hodge numbers* $h^{pq} = \dim V^{pq}$ vanish for $(p,q) \notin \mathscr{E}$.

2.2. Here a *Hodge structure of weight* n H will mean a free finite-type $\mathbb{Z}$-module $H_{\mathbb{Z}}$ together with a Hodge bigrading of weight n of $H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$. The *Tate Hodge structure* $\mathbb{Z}(n)$ is the Hodge structure of type $\{(-n,-n)\}$ given by

$$\mathbb{Z}(n)_{\mathbb{C}} = \mathbb{C} \qquad \text{and} \qquad \mathbb{Z}(n)_{\mathbb{Z}} = (2\pi i)^n \mathbb{Z} \subset \mathbb{C}.$$

A polarization of a Hodge structure H of weight n is a morphism of Hodge structures

$$\psi : H \otimes H \longrightarrow \mathbb{Z}(-n)$$

such that, on $H_{\mathbb{R}} = H_{\mathbb{Z}} \otimes \mathbb{R}$, the real bilinear form

$$(2\pi i)^n \psi(x, Cy)$$

is symmetric and positive definite. The form ψ is then $(-1)^n$-symmetric.

A *rational Hodge structure of weight* n consists of a $\mathbb{Q}$-vector space $H_{\mathbb{Q}}$ and a Hodge bigrading of weight n of $H_{\mathbb{C}} = H_{\mathbb{Q}} \otimes \mathbb{C}$. We denote by $\mathbb{Q}(n)$ the rational Hodge structure underlying $\mathbb{Z}(n)$. The definition of polarizations is left to the reader.

The following statement reformulates a theorem of Riemann.

2.3. Scholium. *The functor $A \mapsto H^1(A, \mathbb{Z})$ identifies polarized abelian varieties with polarized Hodge structures of type $\{(0,1),(1,0)\}$.*

From the remainder of this subsection, we shall use only 2.11(iii′) ⇒ (ii). To prove this implication, one may restrict to connected groups, which makes 2.5 unnecessary or obvious.

2.4. Let G be a real linear algebraic group. We say that G is *compact* if $G(\mathbb{R})$ is compact and every connected component of $G(\mathbb{C})$ has a real point. The obvious functor is then an equivalence from the category of algebraic linear representations of G to the category of linear representations of the compact group $G(\mathbb{R})$.

2.5. Lemma. *Every real algebraic subgroup H of a compact real algebraic group G is compact.*

It is known that the map

$$(g, \ell) \longmapsto g \cdot \exp(i\ell) : G(\mathbb{R}) \times \mathrm{Lie}(G) \longrightarrow G(\mathbb{C})$$

is bijective. It is enough to prove that, for every real subgroup H of G and every $h = g \cdot \exp(i\ell) \in H(\mathbb{C})$, one has $\ell \in \mathrm{Lie}(H)$. Since $\bar{h} \in H(\mathbb{C})$, one has

$$\exp(2i\ell) = \bar{h}^{-1}h \in H(\mathbb{C}).$$

Regard $G(\mathbb{C})$ as the set of real points of a real algebraic group. Let $\overline{J} \subset H(\mathbb{C})$ be, within this group, the Zariski closure of $J = \{\exp(ni\ell) \mid n \in 2\mathbb{Z}\}$. The elements $g \in J$, and hence the elements $g \in \overline{J}$, satisfy $g^{-1} = \bar{g}$, whence $\mathrm{Lie}(\overline{J}) \subset i\,\mathrm{Lie}(G)$. The Lie algebra $\mathrm{Lie}(\overline{J})$ is abelian, and $\exp(\mathrm{Lie}(\overline{J}))$ is a group, necessarily of finite index in $\overline{J}$. There are therefore n and $\ell' \in \mathrm{Lie}(\overline{J})$ such that

$$\exp(ni\ell) = \exp(\ell').$$

Thus $i\ell \in \mathrm{Lie}(\overline{J}) \subset \mathrm{Lie}(H)$ and $\ell \in \mathrm{Lie}(H)$, completing the proof.

2.6. Lemma. *Let G be a real algebraic group. The following conditions are equivalent:*

(i) *G is compact;*

(ii) *for every real (respectively complex) linear representation of G, there exists a positive-definite G-invariant symmetric bilinear (respectively hermitian sesquilinear) form;*

(iii) *for a faithful representation of G, there exists a form as in (ii).*

For G connected, these conditions are further equivalent to

(iii') *for a representation of G with finite kernel, there exists a form as in (ii).*

For G compact, G-invariance is equivalent to $G(\mathbb{R})$-invariance, and (ii) is classical. If (iii) holds, G is a subgroup of an orthogonal or unitary group, and 2.5 applies. Condition (iii') implies that $G(\mathbb{R})$ is compact (as a finite covering of a compact group); if G is connected, G is therefore compact.

2.7. Let G be a real linear algebraic group. For every involutive automorphism σ of G, let $G^{(\sigma)}$ be the real form of $G_{\mathbb{C}}$ defined by the antilinear involution $g \mapsto \sigma(\bar{g})$. We say that σ is a *Cartan involution* if $G^{(\sigma)}$ is compact.

Let C be an element of $G(\mathbb{R})$ whose square is central. A real (respectively complex) representation of G will be called C-polarizable if there exists on V a G-invariant bilinear (respectively sesquilinear) form ψ such that $\psi(x, Cy)$ is symmetric (respectively hermitian) and positive definite. We then say that ψ is a C-polarization of V. A special case of the following lemma is stated without proof in [4].

2.8. Lemma. *a) Whether ψ is a C-polarization of V depends only on the $G(\mathbb{R})$-conjugacy class of C.*

b) The following conditions are equivalent:

(i) $\mathrm{ad}\, C$ *is a Cartan involution of* G*;*

(ii) *every real (respectively complex) representation of* G *is* C*-polarizable;*

(iii) G *has a faithful real (respectively complex)* C*-polarizable representation.*

If G *is connected, these conditions are also equivalent to*

(iii') G *has a real (respectively complex)* C*-polarizable representation with finite kernel.*

To prove a), note that $\psi(x, gCg^{-1}y) = \psi(g^{-1}x, Cg^{-1}y)$. A sesquilinear form ψ on a complex representation of G is G-invariant if and only if

$$\psi(gx, \bar{g}y) = \psi(x, y) \qquad \text{for } g \in G(\mathbb{C}).$$

This condition is equivalent to

$$\psi(gx, CC^{-1}\bar{g}Cy) = \psi(x, Cy) \qquad \text{for } g \in G(\mathbb{C})$$

(replace y by Cy), i.e. to the $G^{(\mathrm{ad}\, C)}$-invariance of $\psi(x, Cy)$. The "respectively" version of 2.8 therefore follows from 2.6. The non-"respectively" version follows from it:

a) the complexification (respectively realification) of a real (respectively complex) C-polarizable representation is C-polarizable;

b) a polarization of V induces a polarization of every subrepresentation of V;

c) for V real (respectively complex), one has $V \hookrightarrow V \otimes_{\mathbb{R}} \mathbb{C}$.

2.9. Let G be a reductive group over $\mathbb{Q}$, and let t and w be morphisms defined over $\mathbb{Q}$

$$\mathbb{G}_m \xrightarrow{w} G \xrightarrow{t} \mathbb{G}_m,$$

with $tw(x) = x^{-2}$ and w central. Suppose also given a commutative diagram

$$\begin{array}{ccccc} \mathbb{G}_{m,\mathbb{R}} & \xrightarrow{w} & \underline{S} & \xrightarrow{t} & \mathbb{G}_{m,\mathbb{R}} \\ \| & & \downarrow h & & \| \\ \mathbb{G}_{m,\mathbb{R}} & \xrightarrow{w} & G_{\mathbb{R}} & \xrightarrow{t} & \mathbb{G}_{m,\mathbb{R}}. \end{array}$$

A representation defined over $\mathbb{Q}$, $V_{\mathbb{Q}}$, of G is called *homogeneous of weight* n if $w(x) * v = x^n \cdot v$ $(v \in V)$. Every representation is a sum of homogeneous representations. Let $V_{\mathbb{Q}}$ be a representation of weight n. Then h endows $V_{\mathbb{Q}}$ with a rational Hodge structure of weight n.

We shall regard $\mathbb{Q}(n)_{\mathbb{Q}}$ as a representation of G by the rule

$$g * x = t(g)^n \cdot x.$$

This convention is reasonable, since $\underline{S}$ acts on $\mathbb{Q}(n)_{\mathbb{R}}$ by the same formula. A *polarization* of a weight-n representation $V_{\mathbb{Q}}$ of G is a G-invariant form

$$\psi : V_{\mathbb{Q}} \otimes V_{\mathbb{Q}} \longrightarrow \mathbb{Q}(n)$$

such that $\psi(x, h(i)y)$ is a symmetric positive-definite form on $V_{\mathbb{R}}$. Being G-invariant, a polarization ψ is also $\underline{S}$-invariant and hence a polarization of the rational Hodge structure $V_{\mathbb{Q}}$.

2.10. Lemma. *A representation $V_{\mathbb{Q}}$ is polarizable if and only if the representation $V_{\mathbb{R}}$ is $h(i)$-polarizable.*

Let $P_{\mathbb{Q}}$ (resp. $P_{\mathbb{R}}$) be the space of G-invariant $(-1)^n$-symmetric bilinear forms on $V_{\mathbb{Q}}$ (resp. $V_{\mathbb{R}}$). One has $P_{\mathbb{R}} = \mathbb{R} \otimes P_{\mathbb{Q}}$. The $h(i)$-polarizations form an open subset of $P_{\mathbb{R}}$, and the polarizations of $V_{\mathbb{Q}}$ are the intersection of this open subset with $P_{\mathbb{Q}}$. The lemma follows.

2.11. Proposition. *a) Whether ψ is a polarization of V depends only on the $G(\mathbb{R})$-conjugacy class of h.*

b) The following conditions are equivalent:

(i) $\operatorname{ad} h(i)$ *is a Cartan involution of* $\operatorname{Ker}(t)_{\mathbb{R}}$*;*

(ii) *every homogeneous representation of G is polarizable;*

(iii) *G has a faithful family of polarizable homogeneous representations.*

If G is connected, these conditions are equivalent to

(iii') *G has a polarizable representation ρ such that* $\operatorname{Ker}(\rho) \cap \operatorname{Ker}(t)$ *is finite.*

Assertion a) follows from 2.8.a). If G is connected, then $\operatorname{Ker}(t)$ is connected: otherwise t would be of the form t_0^n, with $n > 1$. Since $twx = x^{-2}$, one would have $n = 2$ and $w(-1) \notin \operatorname{Ker}(t)^0$. This is absurd, since

$$w(-1) \in h\big(\operatorname{Ker}(t : \underline{S} \longrightarrow \mathbb{G}_{m,\mathbb{R}})\big),$$

which is connected. With this established, b) follows from 2.10 and 2.8.b).

3. Review of the Spinor Group

For the material recalled in this section, see [3].

3.1. Let V be a free module over a ring A, equipped with a quadratic form Q. The module V is identified with a submodule of the Clifford algebra $C(V)$; in $C(V)$ one has

$$v \cdot v = Q(v).$$

We denote by $C^+(V)$ the even part of $C(V)$.

If V is a free module over A, equipped with a symmetric bilinear form ψ, we denote by $C(V)$ (or by $C(V,\psi)$) the Clifford algebra of V equipped with the quadratic form $\psi(x,x)$.

3.2. Suppose that A is a field of characteristic 0, and that Q is nondegenerate. We denote by $\mathrm{CSpin}(V)$ the *Clifford group*. It is the algebraic group of invertible elements g of $C^+(V)$ such that $gVg^{-1} = V$. We define a morphism from $\mathrm{CSpin}(V)$ to $\mathrm{SO}(V)$ by

$$g \longmapsto (v \longmapsto gvg^{-1}).$$

The kernel of this morphism consists only of the scalars: there is an exact sequence of algebraic groups

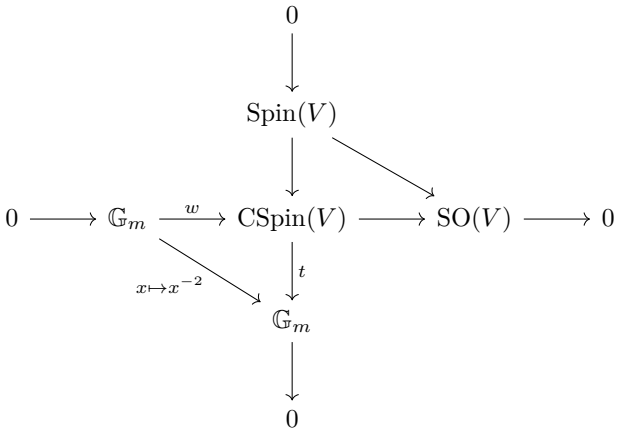
$$0 \longrightarrow \mathbb{G}_m \xrightarrow{w} \mathrm{CSpin}(V) \longrightarrow \mathrm{SO}(V) \longrightarrow 0.$$

The spin group $\mathrm{Spin}(V)$ is the algebraic subgroup of $\mathrm{CSpin}(V)$ that is the kernel of the spinor norm

$$N : \mathrm{CSpin}(V) \longrightarrow \mathbb{G}_m.$$

Let t denote the inverse of N; there is a commutative diagram

(3.2.1)



By definition, one also has

$$\alpha : \mathrm{CSpin}(V) \hookrightarrow C^+(V)^*. \tag{3.2.2}$$

3.3. We denote by $(C^+(V))_s$ the representation of $\mathrm{CSpin}(V)$ on $C^+(V)$ given by

$$g * x = \alpha(g) \cdot x. \tag{3.3.1}$$

We denote by $(C^+(V))_{\mathrm{ad}}$ the representation of $\mathrm{CSpin}(V)$ on $C^+(V)$ obtained from the action of $\mathrm{SO}(V)$ on $C^+(V)$ by transport of structure. One has

$$g *_{\mathrm{ad}} x = \alpha(g) \cdot x \cdot \alpha(g)^{-1}. \tag{3.3.2}$$

The action of $\mathrm{CSpin}(V)$ on $(C^+(V))_s$ is compatible with the right $C^+(V)$-module structure of $(C^+(V))_s$. There is an isomorphism of representations

$$\mathrm{End}_{C^+(V)}((C^+(V))_s) = (C^+(V))_{\mathrm{ad}}. \tag{3.3.3}$$

There is an isomorphism of representations

$$(C^+(V))_{\mathrm{ad}} = \bigoplus \bigwedge^{2i} V. \tag{3.3.4}$$

3.4. Suppose that A is algebraically closed. We distinguish two cases.

1) $\dim(V)$ *odd*: $C^+(V)$ is here a matrix algebra. Let W be a simple $C^+(V)$-module; W is a *spin representation* of $\mathrm{CSpin}(V)$:

$$g * W = \alpha(g) \cdot W.$$

There are isomorphisms of representations

$$(C^+(V))_s = \text{a sum of copies of } W, \tag{3.4.1}$$

$$(C^+(V))_{\mathrm{ad}} = \mathrm{End}_A(W). \tag{3.4.2}$$

2) $\dim(V)$ *even*: $C^+(V)$ is here a product of two matrix algebras. Let W_1 and W_2 be two nonisomorphic simple $C^+(V)$-modules, and let $W = W_1 + W_2$; W_1 and W_2 are the *half-spin representations.*

There are isomorphisms of representations

$$(C^+(V))_s = \text{a sum of copies of } W, \tag{3.4.3}$$

$$(C^+(V))_{\mathrm{ad}} = \mathrm{End}_A(W_1) \times \mathrm{End}_A(W_2). \tag{3.4.4}$$

In both cases, $C^+(V)$ is the bicommutant of the representation W.

3.5. Proposition. *Let* Γ *be a Zariski-dense subgroup of* $\mathrm{Spin}(V)$ *and let* a *be an automorphism of the* A*-algebra* $C^+(V)$ *that commutes with the action of* Γ *given by* $\gamma \mapsto \alpha(\gamma) \cdot x \cdot \alpha(\gamma)^{-1}$. *Then* a *is the identity.*

It is enough to treat the case in which A is an algebraically closed field and Γ is the whole spin group (indeed, commuting with a is a Zariski-closed condition).

With the notation of 3.4, let us identify $C^+(V)$ with a subalgebra of $\mathrm{End}_A(W)$.

There is an automorphism $\bar{a}$ of W such that

$$a(x) = \bar{a} x \bar{a}^{-1} \qquad (x \in C^+(V)).$$

For $\gamma \in \mathrm{Spin}(V)$, by hypothesis one has

$$\alpha(\gamma)\bar{a}\alpha(\gamma)^{-1}\bar{a}^{-1} \cdot x \cdot \bar{a}\alpha(\gamma)\bar{a}^{-1}\alpha(\gamma)^{-1} = x \qquad (x \in C^+(V));$$

the commutator $(\alpha(\gamma), \bar{a}) = \alpha(\gamma)\bar{a}\alpha(\gamma)^{-1}\bar{a}^{-1}$ therefore lies in the center of $C^+(V)$.

1) $\dim(V)$ *odd*: here $C^+(V) = \mathrm{End}_A(W)$, and $\det_W((\alpha(\gamma), \bar{a})) = 1$, so that $\alpha(\gamma)\bar{a}\alpha(\gamma)^{-1}\bar{a}^{-1}$ is a root of unity ν. Since $\mathrm{Spin}(V)$ is connected, and since for $\gamma = e$ one has $\nu = 1$, one even has

$$(\alpha(\gamma), \bar{a}) = 1, \qquad \text{i.e.} \qquad \alpha(\gamma)\bar{a} = \bar{a}\alpha(\gamma).$$

2) $\dim(V)$ *even*: here $C^+(V) = \mathrm{End}_A(W_1) \times \mathrm{End}_A(W_2)$, and $\bar{a}$ either permutes the W_i or preserves them. In both cases, from

$$\det_{W_i}(\alpha(\gamma)) = 1 \qquad (i = 1, 2)$$

one deduces that

$$\det_{W_i}((\alpha(\gamma), \bar{a})) = 1 \qquad (i = 1, 2)$$

and one concludes as above that

$$\alpha(\gamma)\bar{a} = \bar{a}\alpha(\gamma).$$

In both cases, $\bar{a}$ lies in the commutant of the representation, and therefore commutes with the bicommutant $C^+(V)$: a is the identity.

4. Construction of abelian varieties

In this paragraph and part of the next, we present results of [7].

4.1. Let $V_{\mathbb{Z}}$ be a free $\mathbb{Z}$-module of rank $n+2$ endowed with a bilinear form of nonzero discriminant

$$B : V_{\mathbb{Z}} \otimes V_{\mathbb{Z}} \longrightarrow \mathbb{Z}.$$

Assume that B has index $(n+, 2-)$. We are interested in weight-0 morphisms $h : \underline{S} \to \mathrm{SO}(V_{\mathbb{R}})$ such that

1) B is a polarization of the representation $V_{\mathbb{Q}}$ of $\mathrm{SO}(V_{\mathbb{Q}})$;

2) the Hodge numbers of $V_{\mathbb{C}}$ are

$$h^{-1,1} = h^{1,-1} = 1 \qquad \text{and} \qquad h^{0,0} = n.$$

4.2. A morphism h as above lifts uniquely to $\tilde{h} : \underline{S} \to \mathrm{CSpin}(V_{\mathbb{R}})$, making the diagram

$$\begin{array}{ccccc}
\mathbb{G}_{m,\mathbb{R}} & \xrightarrow{w} & \underline{S} & \xrightarrow{t} & \mathbb{G}_{m,\mathbb{R}} \\
\| & & \downarrow{\scriptstyle \tilde{h}} & & \| \\
\mathbb{G}_{m,\mathbb{R}} & \longrightarrow & \mathrm{CSpin}(V_{\mathbb{R}}) & \xrightarrow{t} & \mathbb{G}_{m,\mathbb{R}}.
\end{array}$$

commutative.

4.3. Lemma. *With respect to $\tilde{h}$, every representation of* $\mathrm{CSpin}(V_{\mathbb{Q}})$ *is polarizable (2.9).*

We apply criterion 2.11. b)(iii') to the representation $V_{\mathbb{Q}}$ of $\mathrm{CSpin}(V_{\mathbb{Q}})$.

4.4. Lemma. *The Hodge structure on $C^+(V_{\mathbb{Q}})_{\mathrm{ad}}$ defined by $\tilde{h}$ is of type $\{(-1,1),(0,0),(1,-1)\}$.*

We have

$$C^+(V_{\mathbb{Q}})_{\mathrm{ad}} \simeq \bigoplus \bigwedge^{2i} V_{\mathbb{Q}},$$

and the conclusion follows from $h^{-1,1}(V_{\mathbb{Q}}) = 1$.

4.5. Proposition. *The Hodge structure on $C^+(V_{\mathbb{Q}})_s$ defined by $\tilde{h}$ is of type $\{(0,1),(1,0)\}$. Thus the Hodge structure $C^+(V_{\mathbb{Q}})_s$, endowed with the lattice $C^+(V_{\mathbb{Z}})$, defines an abelian variety.*

First suppose that n is odd. After extension of scalars to $\mathbb{C}$, $C^+(V_{\mathbb{Q}})$ becomes a matrix algebra

$$C^+(V_{\mathbb{C}}) \simeq \mathrm{End}(W).$$

On W, $\mathrm{CSpin}(V_{\mathbb{Q}})_{\mathbb{C}}$ acts by $g \cdot w = \alpha(g) \cdot w$: this is the spin representation. The representation $C^+(V_{\mathbb{C}})_s$ is a sum of $\dim(W)$ copies of W, so it is enough to prove that the weight-one representation W is purely of type $\{(0,1),(1,0)\}$. Otherwise, the representation $\mathrm{End}(W) \simeq C^+(V_{\mathbb{C}})_{\mathrm{ad}}$ could not be purely of type $\{(-1,1),(0,0),(1,-1)\}$.

For n even, similarly,

$$C^+(V_{\mathbb{C}}) \simeq \mathrm{End}(W') \oplus \mathrm{End}(W''),$$

where W' and W'' are the half-spin representations. As above, one proves that W' and W'' are of type $\{(0,1),(1,0)\}$, and hence so is $C^+(V_{\mathbb{C}})_s$.

The second assertion follows from 2.3 and 3.3.

4.6. The construction given in this paragraph conceals the following fact (which we shall not use), visible from the Dynkin diagram and the real reason why everything works.

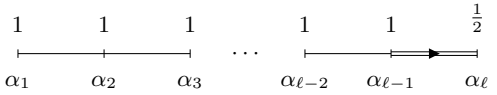
Let $V_{\mathbb{Z}}$ be a polarized Hodge structure of weight 0 and type $\{(1,-1),(0,0),(-1,1)\}$, with $h^{1,-1} = 0$. Let

$$r : \mathbb{G}_m \longrightarrow \mathrm{SO}(V_{\mathbb{C}})$$

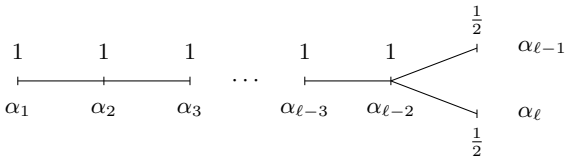
be the homomorphism such that $r(x) \cdot v^{pq} = x^p v^{pq}$ for v^{pq} of type (p,q). Let T be a maximal torus of $\mathrm{SO}(V_{\mathbb{C}})$ equipped with a system of simple roots Φ, let $\widetilde{T}$ be the inverse image of T in $\mathrm{Spin}(V_{\mathbb{C}})$, and let $(w_\alpha)_{\alpha\in\Phi}$ be the fundamental weights of $\widetilde{T}$. The homomorphism r has a unique conjugate

$$r_0 : \mathbb{G}_m \longrightarrow T$$

such that the rational numbers $\langle w_\alpha, r_0\rangle \in \frac{1}{2}\mathbb{Z}$ are all positive. One checks that r_0 is the homomorphism $H_{\alpha_1} : \mathbb{G}_m \to T$ corresponding to the root denoted below by α_1. The numbers $\langle w_\alpha, r_0\rangle$ are as follows:



for $\mathrm{SO}(2, 2\ell - 1)$,



for $\mathrm{SO}(2, 2\ell - 2)$.

Let w be a dominant weight of a spin or half-spin representation, and let σ be the opposition involution. The point is that

$$\langle w, r_0\rangle + \langle \sigma w, r_0\rangle = 1.$$

5. Construction of Families of Abelian Varieties

We shall use the following theorem of A. Borel (to appear).

5.1. Theorem (A. Borel). *Let X/Γ be the quotient of a Hermitian symmetric domain by a torsion-free arithmetic group. It is known [2] that X/Γ is naturally a quasi-projective algebraic variety. Let S be a reduced scheme over $\mathbb{C}$ and let $f : S^{\mathrm{an}} \to X/\Gamma$ be a morphism of analytic spaces. Then f is algebraic.*

Let S be smooth over $\mathbb{C}$. For the notion of a variation of Hodge structures $\underline{H}$ on S, and of a polarization of $\underline{H}$, we refer to Griffiths [6]. We write $\underline{H}_{\mathbb{Z}}$ for the local system of free $\mathbb{Z}$-modules on S defined by $\underline{H}$, and $\underline{H}_s$ for the fiber Hodge structure of $\underline{H}$ at $s \in S$.

For X the Siegel upper half-space, Theorem 5.1 has the following corollary, which generalizes 2.3.

5.2. Scholium. *The functor $(a : A \to S) \mapsto R^1 a_* \mathbb{Z}$ identifies polarized abelian schemes A over S with polarized variations of Hodge structures of type $\{(0,1),(1,0)\}$ on S.*

5.3. In the category of variations of Hodge structures on S, all the usual tensor operations make sense. Thus

a) if $\underline{H}$ and $\underline{H}'$ are variations of Hodge structures, there are variations of Hodge structures

$$\underline{H} \otimes \underline{H}', \qquad \underline{\mathrm{Hom}}(\underline{H}, \underline{H}'), \qquad \bigwedge^n \underline{H}, \qquad \underline{H}^\vee, \qquad \underline{\mathrm{End}}(\underline{H}), \ldots.$$

b) if $\underline{H}$ is a variation of Hodge structures of weight 0, and ψ is a polarization of $\underline{H}$ (or any symmetric morphism of Hodge structures $\underline{H} \otimes \underline{H} \to \mathbb{Z}(0)$), then $C(\underline{H}, \psi)$ and $C^+(\underline{H}, \psi)$ are defined. At every point $t \in S$, with the notation of 3.4, one has

$$C^+(\underline{H}, \psi)_t = C^+(\underline{H}_t, \psi)_{\mathrm{ad}}.$$

5.4. Let $(V_{\mathbb{Z}}, B)$ be as in 4.1, and let X be the set of $h : \underline{S} \to \mathrm{SO}(V_{\mathbb{R}})$ of the type considered in 4.1. Giving such an h is equivalent to giving a two-dimensional subspace $V_{\mathbb{R}}^-$ of $V_{\mathbb{R}}$, on which B is negative definite, together with an orientation of $V_{\mathbb{R}}^-$: for $z = \tau \cdot e^{i\theta} \in \underline{S}(\mathbb{R}) = \mathbb{C}^*$, $h(z)$ induces the identity on the orthogonal complement of $V_{\mathbb{R}}^-$ and the rotation through angle 2θ on $V_{\mathbb{R}}^-$. The group $\mathrm{O}(V_{\mathbb{R}})$ acts on X by transport of structure: $(g * h)(z) = g \cdot h(z) \cdot g^{-1}$, and the description above makes it geometrically evident that X is a homogeneous space under $\mathrm{SO}(V_{\mathbb{R}})$. The stabilizer of $h \in X$ is a subgroup $\mathrm{SO}(2) \times \mathrm{SO}(n)$, the identity component of a *maximal* compact subgroup.

5.5. For $h \in X$, let F_h be the corresponding Hodge filtration of $V_{\mathbb{C}}$. The function $h \mapsto F_h^1$ identifies X with an open subset of the space of isotropic lines in $V_{\mathbb{C}}$. This construction endows X with a complex structure for which F_h is a holomorphic function of h. It is known that, for this structure, the (two) connected components of X are Hermitian symmetric domains.

5.6. Let $W_{\mathbb{Q}}$ be a weight-n representation of the algebraic group $\mathrm{CSpin}(V_{\mathbb{Q}})$: $w(x) * y = x^n \cdot y$. Let $\underline{W}_{\mathbb{Q}}$ be the constant local system on X with fiber $W_{\mathbb{Q}}$. The group $\mathrm{CSpin}(V_{\mathbb{Q}})$ acts on $(X, \underline{W}_{\mathbb{Q}})$ by

$$\gamma * (w \text{ at } h \in X) = (\gamma w \text{ at } \gamma h \gamma^{-1} \in X).$$

Each $h \in X$ defines $\tilde{h} : \underline{S} \to \mathrm{CSpin}(V_{\mathbb{R}})$, hence a $\mathbb{Q}$-Hodge structure on the fiber of $\underline{W}_{\mathbb{Q}}$ at $h \in X$. Thus one obtains a variation of $\mathbb{Q}$-Hodge structures of weight n on X, again denoted $\underline{W}_{\mathbb{Q}}$:

a) the Hodge filtration is a holomorphic function of h;

b) the transversality axiom is satisfied (we shall use it only in a trivial case).

This construction is compatible with tensor products. In particular, a polarization $\psi : W_{\mathbb{Q}} \otimes W_{\mathbb{Q}} \to \mathbb{Q}(-n)$ of the representation $W_{\mathbb{Q}}$ defines a polarization of $\underline{W}_{\mathbb{Q}}$.

Let $W_{\mathbb{Z}}$ be an integral lattice in $W_{\mathbb{Q}}$ and let Γ be a subgroup of $\mathrm{CSpin}(V_{\mathbb{Q}})$, discrete in $\mathrm{CSpin}(V_{\mathbb{R}})$, acting freely on X and satisfying $\Gamma W_{\mathbb{Z}} = W_{\mathbb{Z}}$. Then the constant local system $\underline{W}_{\mathbb{Z}}$ on X, with fiber $W_{\mathbb{Z}}$, underlies a Γ-equivariant variation of Hodge structure $\underline{W}$ on X as above. Passing to the quotient, it defines a variation of Hodge structures $\underline{W}$ on X/Γ.

If $W_{\mathbb{Q}}$ is a representation of $\mathrm{SO}(V_{\mathbb{Q}})$, the same construction applies to discrete subgroups of $\mathrm{SO}(V_{\mathbb{Q}})$.

5.7. Proposition. *Let $(\underline{H}, \psi)$ be a polarized variation of Hodge structures of type $\{(-1,1),(0,0),(1,-1)\}$, with $h^{1,-1} = 1$, on a smooth connected scheme S. There exist*

a) *a finite surjective étale cover $u : S_1 \to S$;*

b) *an abelian scheme A over S_1;*

c) *a $\mathbb{Z}$-algebra C, and $\mu : C \hookrightarrow \mathrm{End}_{S_1}(A)$;*

d) *an isomorphism of variations of Hodge structures*

$$u : C^+(\underline{H}, \psi) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_* \mathbb{Z})$$

which induces an isomorphism of local systems of algebras

$$u_{\mathbb{Z}} : C^+(\underline{H}_{\mathbb{Z}}, \psi) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_* \mathbb{Z}).$$

Let $(V_{\mathbb{Z}}, B)$ be a $\mathbb{Z}$-module equipped with a symmetric bilinear form, isomorphic to $(H_{s\mathbb{Z}}, \psi)$, and retain the notation of 5.4. Let n be an integer,

$$\Gamma = \{\gamma \in \mathrm{SO}(V_{\mathbb{Z}}) \mid \gamma \equiv 1 \pmod{n}\}$$

and

$$\widetilde{\Gamma} = \{\gamma \in \mathrm{CSpin}(V_{\mathbb{Q}}) \mid \alpha(\gamma) \equiv 1 \pmod{n} \text{ in } C^+(V_{\mathbb{Z}})\}.$$

Assume n is large enough that Γ and $\widetilde{\Gamma}$ are torsion-free.

There is a finite étale cover $v : S_1' \to S$ such that the local system $v^* \underline{H}_{\mathbb{Z}/n\mathbb{Z}}$ is trivial. Choose an isomorphism σ_n between this local system and the constant local system $\underline{V}_{\mathbb{Z}/n\mathbb{Z}}$. Let $s \in S_1'$ and let $\sigma : V_{\mathbb{Z}} \xrightarrow{\sim} H_{s\mathbb{Z}}$ be an isometry such that $\sigma \equiv \sigma_n \pmod{n}$ (we choose σ_n so that such σ exist). The Hodge structure h_s on H_s defines $\sigma^{-1}(h_s) \in X$. The image of $\sigma^{-1}(h_s)$ in X/Γ depends only on s. We thus define

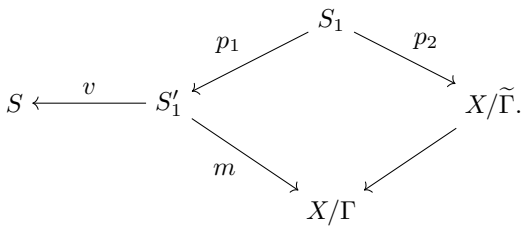
$$m : S_1' \longrightarrow X/\Gamma$$

and an isomorphism compatible with the polarizations

$$\underline{H} \simeq m^* \underline{V}.$$

Let

$$S_1 = S_1' \times_{X/\Gamma} X/\widetilde{\Gamma} :$$



On S_1, one has

a) an isomorphism $(vp_1)^*H \simeq (mp_1)^*\underline{V}$;

b) a variation of Hodge structure, the pullback by p_2 of the one on $X/\widetilde{\Gamma}$ defined by the representation $(C^+(V_{\mathbb{Q}}))_s$ and the lattice $C^+(V_{\mathbb{Z}})$.

By 5.2, 5.6 and 4.5, this variation of Hodge structures corresponds to an abelian scheme $a : A \to S_1$.

c) Let C be the algebra $C^+(V_{\mathbb{Z}})$. The abelian scheme constructed in b) has complex multiplication by C, and the variation of Hodge structures $\underline{\mathrm{End}}_C(R^1a_*\mathbb{Z})$ is

$$p_2^*((\underline{C}^+(V_{\mathbb{Z}}))_{\mathrm{ad}}).$$

Combining this with a), one obtains an isomorphism of local systems of algebras and of variations of Hodge structures

$$\underline{C}^+(vp_1^*H) \simeq \underline{\mathrm{End}}_C(R^1a_*\mathbb{Z}).$$

This proves 5.7.

6. $K3$ Surfaces

6.1. A polarized $K3$ surface over a field K of characteristic 0 is a $K3$ surface over K, say Y, endowed with an element $\eta \in \mathrm{NS}(Y) = \mathrm{Pic}(Y)$ which is the class of an ample invertible sheaf. For $K = \mathbb{C}$, we shall identify $\mathrm{NS}(Y)$ with a subset of $H^2(Y, \mathbb{Z})$.

6.2. A *family of $K3$ surfaces* parametrized by a scheme S of characteristic 0 is a proper flat morphism $f : Y \to S$ whose fibers are $K3$ surfaces. A *polarization* of $f : Y \to S$ is a section η of $\mathrm{Pic}_S(Y)$ which, for every $s \in S$, is a polarization of $Y_s = f^{-1}(s)$.

For every prime number ℓ, we identify η with a section of $R^2f_*\mathbb{Z}_\ell$; we denote by $P^2f_*\mathbb{Z}_\ell$ the orthogonal complement of η (for the cup product); it is a subsheaf of $R^2f_*\mathbb{Z}_\ell$ in $\mathbb{Z}_\ell$-modules. The cup product

$$R^2f_*\mathbb{Z}_\ell \otimes R^2f_*\mathbb{Z}_\ell \longrightarrow R^4f_*\mathbb{Z}_\ell$$

and the trace morphism

$$R^4f_*\mathbb{Z}_\ell \xrightarrow{\sim} \underline{\mathbb{Z}}_\ell(-2)$$

induce

$$\psi_\ell : P^2f_*\mathbb{Z}_\ell(1) \otimes P^2f_*\mathbb{Z}_\ell(1) \longrightarrow \underline{\mathbb{Z}}_\ell. \tag{6.2.1}$$

6.3. Take S to be a scheme of finite type over $\mathbb{C}$. Then $R^2f_*\mathbb{Z}$ is a variation of Hodge structures on S; η is everywhere of type $(1,1)$ and its orthogonal complement $P^2f_*\mathbb{Z}$ is again a variation of Hodge structures. As above, the cup product defines

$$\psi : P^2f_*\mathbb{Z}(1) \otimes P^2f_*\mathbb{Z}(1) \longrightarrow \underline{\mathbb{Z}}. \tag{6.3.1}$$

This time, the negative of ψ is a polarization of the variation of Hodge structures $P^2 f_* \mathbb{Z}(1)$. Moreover, ψ_ℓ is obtained from ψ via the isomorphism

$$(P^2 f_* \mathbb{Z}(1)_{\mathbb{Z}}) \otimes \mathbb{Z}_\ell \simeq P^2 f_* \mathbb{Z}_\ell(1).$$

For every $s \in S$, the fiber at s, $(P^2 f_* \mathbb{Z}(1))_s$, is the primitive part of the cohomology $H^2(Y_s, \mathbb{Z})$. The Hodge structure on $(P^2 f_* \mathbb{Z}(1))_s$ is of type $\{(-1,1),(0,0),(1,-1)\}$, with $h^{-1,1} = 1$ ((1.1)(i)).

The action of $\pi_1(S,s)$ on $H^2(Y_s, \mathbb{Z})$ induces

$$\pi : \pi_1(S,s) \longrightarrow \mathrm{O}(P^2(Y_s, \mathbb{Z})(1)_{\mathbb{Z}}, \psi). \tag{6.3.2}$$

6.4. Proposition. *For every complex polarized $K3$ surface Y_0, there exists a family of polarized $K3$ surfaces $f : Y \to S$ as above, with $Y_s \simeq Y_0$ for some $s \in S$, such that the image of π in (6.3.2) has finite index.*

Let $f_T^\wedge : Y_T^\wedge \to T^\wedge$ be the versal formal deformation of Y_0. Since $H^0(Y_0, T^1) = 0$, $T^\wedge$ is even a formal moduli scheme for Y_0, but this is immaterial. Since $H^2(Y_0, T^1) = 0$ ((1.1)(i)), $T^\wedge$ is the spectrum of a formal power-series ring over $\mathbb{C}$. From $f_T^\wedge$ one obtains a (non-polarized) variation of Hodge structures on $T^\wedge$. By (1.1.1), this is the universal deformation of {the Hodge structure $H^2(Y_0, \mathbb{Z})$, endowed with the cup product}.

Let $\bar{\eta}$ be the image of η under the composite map

$$H^2(Y_0, \mathbb{Z}) \longrightarrow H^2_{\mathrm{DR}}(Y_T^\wedge / T^\wedge) \longrightarrow R^2 f_{T*}^\wedge \mathcal{O}.$$

Let $S^\wedge$ be the formal subscheme of $T^\wedge$ with equation $\bar{\eta} = 0$, and let $f^\wedge : Y^\wedge \to S^\wedge$ be induced by $f_T^\wedge$. Since $h^{0,2} = 1$, $S^\wedge \subset T^\wedge$ is defined by one equation; this equation is part of a regular system of parameters, so $S^\wedge$ is a power-series ring. On $S^\wedge$, η comes from an ample invertible sheaf $\mathcal{O}(1)$ on $Y^\wedge$; hence $f^\wedge$ is algebraizable. Moreover, on $S^\wedge$, the polarized variation of Hodge structure $(P^2(Y^\wedge/S^\wedge, \mathbb{Z})(1), \psi)$ is a universal deformation of $(P^2(Y_0, \mathbb{Z})(1), \psi)$.

Put $S^\wedge = \mathrm{Spec}(A^\wedge)$, and express $A^\wedge$ as an inductive limit of rings of finite type over $\mathbb{C}$. Since $Y^\wedge$ is algebraizable, there is a Cartesian diagram

$$\begin{array}{ccc} Y^\wedge & \longrightarrow & Y \\ \downarrow{\scriptstyle f^\wedge} & & \downarrow{\scriptstyle f} \\ S^\wedge & \longrightarrow & S \end{array}$$

with S of finite type over $\mathbb{C}$ and Y a polarized family of $K3$ surfaces over S (we do not assume that $S^\wedge$ is a completion of S). We may take S normal and connected; let $s \in S$ be the image of the closed point of $S^\wedge$.

In fact, by Artin's approximation theorem [1], one can find S such that $S^\wedge$ is a completion of S. Then S is smooth in a neighborhood of s. The simplifications resulting from such a choice are mostly psychological.

Let us show that $f : Y \to S$ satisfies 6.2. After replacing S by a finite étale cover, one defines, as in 5.7, a holomorphic map

$$m : S \longrightarrow X/\Gamma$$

from S to a moduli space of polarized Hodge structures such that the inverse image of the universal family of Hodge structures on X/Γ is $(P^2(Y/S, \mathbb{Z})(1), \psi)$.

By Borel (5.1), m is a morphism of schemes. The morphism m is dominant, since the composite $m^\wedge : S^\wedge \to X/\Gamma$ is dominant. It follows (see, for example, P. Deligne, Théorie de Hodge II. Publ. Math. IHES 40, Lemma 4.4.17) that $m(\pi_1(S, s))$ has finite index in $\pi_1(X/\Gamma, m(s)) = \Gamma$, which itself has finite index in the orthogonal group.

6.5. Proposition. *Let Y_0 be a polarized $K3$ surface over a field K of characteristic 0. There exist:*

a) *a scheme S of finite type over $\mathbb{Q}$, and a family $f : Y \to S$ of polarized $K3$ surfaces, parametrized by S;*

b) *a finite extension K' of K and $v : \mathrm{Spec}(K') \to S$ such that v^*Y is isomorphic to $Y_0 \otimes_K K'$;*

c) *an abelian scheme $a : A \to S$ over S;*

d) *a $\mathbb{Z}$-algebra C and $\mu : C \to \mathrm{End}_S(A)$;*

e) *an isomorphism of sheaves of $\mathbb{Z}_\ell$-algebras on S*

$$u : C^+(P^2 f_* \mathbb{Z}_\ell(1), \psi_\ell) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_* \mathbb{Z}_\ell).$$

We reduce to the case where K is of finite type over $\mathbb{Q}$. Choose a complex embedding $\sigma : K \to \mathbb{C}$ and apply 6.4 to $Y_0 \otimes_K \mathbb{C}$ to obtain a family of polarized $K3$ surfaces $f_1 : Y_1 \to S_1$. Apply 5.7 to f_1; one obtains a family $f_2 : Y_2 \to S_2$ still satisfying the conclusion of 6.4, and, over S_2, an abelian scheme $a_2 : A_2 \to S_2$ with complex multiplication $\mu_2 : C \to \mathrm{End}_{S_2}(A_2)$ and an isomorphism

$$u_2 : C^+(P^2 f_{2*} \mathbb{Z}(1), \psi) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_{2*} \mathbb{Z})$$

of local systems of algebras. The system of complex schemes and morphisms

$$\overline{E}_2 = (S_2, f_2, Y_2, a_2, A_2, \mu_2)$$

is defined by finitely many equations. Hence there exists an extension K_3 of K of finite type, with $K \subset K_3 \subset \mathbb{C}$, and, over K_3, a system

$$\overline{E}_3 = (S_3; f_3, Y_3, a_3, A_3, \mu_3)$$

which, after extension of scalars from K_3 to $\mathbb{C}$, gives $\overline{E}_2$.

6.5.1. Lemma. *There exists a unique isomorphism of sheaves of $\mathbb{Z}_\ell$-algebras*

$$u_3 : C^+(P^2 f_{3*}\mathbb{Z}_\ell(1), \psi_\ell) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_{3*}\mathbb{Z}_\ell).$$

It is enough to verify that such an isomorphism exists and is unique after extending scalars to an algebraic closure of K_3, for example $\mathbb{C}$. Existence holds by construction, and uniqueness follows from 3.5. Indeed, for $s \in S_2$, the image of $\pi_1(S_2, s)$ in $\mathrm{O}(P^2(Y_{2s}, \mathbb{Z}), \psi)$ has finite index and therefore contains a Zariski-dense subgroup of SO.

Let T be an integral scheme of finite type over $\mathbb{Q}$, with function field K_3. For a suitable T, $\overline{E}_3$ is the generic fiber of a system $\overline{E} = (S, f, Y, a, A, \mu)$ over T, and u_3 extends to

$$u : C^+(P^2 f_*\mathbb{Z}_\ell(1), \psi_\ell) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_*\mathbb{Z}_\ell).$$

The existence of K' as in b) is easily verified, and this completes the proof.

6.6. We prove Theorem 1.3. Let $f_V : Y_V \to \mathrm{Spec}(V)$ be as in 1.2(i). Let K be the fraction field of V, and apply 6.5 to the surface $Y_K = Y \otimes_V K$ over K, endowed with any polarization. After replacing V by its normalization in a finite extension of K, one may suppose that there exists $\overline{E} = (S, f, Y, a, A, \mu, u)$ as in 6.5, such that Y_K is the inverse image of Y under $v : \mathrm{Spec}(K) \to S$. Taking the inverse image gives an abelian variety with complex multiplication (A_K, μ) over K.

Let $\overline{K}$ be an algebraic closure of K, and let $\overline{k}$ be the corresponding algebraic closure of k (the residue field of the normalization of V in $\overline{K}$). From u one obtains an isomorphism of $\mathrm{Gal}(\overline{K}/K)$-modules

$$C^+(P^2(Y_{\overline{K}}, \mathbb{Z}_\ell)(1), \psi_\ell) \xrightarrow{\sim} \mathrm{End}_C(H^1(A_{\overline{K}}, \mathbb{Z}_\ell)). \tag{6.6.1}$$

Since Y_V is smooth over V, the inertia group I acts trivially on

$$H^2(Y_{\overline{K}}, \mathbb{Z}_\ell) \simeq H^2(Y_{\overline{k}}, \mathbb{Z}_\ell).$$

The polarization defines an invariant η in $H^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$, whose orthogonal complement is $P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$, and I acts trivially on

$$C^+(P^2(Y_{\overline{K}}, \mathbb{Z}_\ell)(1), \psi_\ell) \simeq C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell).$$

Since I acts trivially on $\mathrm{End}_C(H^1(A_{\overline{K}}, \mathbb{Z}_\ell))$ and commutes with the complex multiplication, it acts through the center of $C \otimes \mathbb{Q}_\ell$. It is also known that a subgroup of finite index in I acts unipotently on $H^1(A_{\overline{K}}, \mathbb{Z}_\ell)$; it follows that I acts through one of its finite quotients. After a further finite extension of K, one may therefore suppose that A_K has good reduction over V.

For $A_{\overline{k}}$, the special fiber of its reduction, (6.6.1) reduces to an isomorphism of $\mathrm{Gal}(\overline{k}/k)$-modules

$$C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell) \xrightarrow{\sim} \mathrm{End}_C(H^1(A_{\overline{k}}, \mathbb{Z}_\ell)). \tag{6.6.2}$$

By Weil's theory for A_k, the eigenvalues of Frobenius acting on $C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell)$ are algebraic numbers all of whose complex conjugates have absolute value 1. We have

$$\bigwedge^2(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)) \subset C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell),$$

and, since $\dim P^2(Y_{\overline{k}}, \mathbb{Z}_\ell) > 2$ (as is the case here), it follows that the eigenvalues of Frobenius acting on $P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$ likewise have complex absolute value 1. The same assertion holds for $H^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$, and the theorem follows.

7. Remarks

7.1. Let H be a rational Hodge structure. We shall call the *Mumford–Tate group* of H the smallest algebraic subgroup G of $\mathrm{GL}(H_{\mathbb{Q}})$, defined over $\mathbb{Q}$, such that $G(\mathbb{R})$ contains the image of $h : \underline{S} \to \mathrm{GL}(H_{\mathbb{R}})$. For polarizable H, G is reductive. This definition differs from Mumford's: if H has weight $n \neq 0$, our G contains the scalar homotheties. By 2.9, every homogeneous representation of G carries a natural Hodge structure.

One of the key points in the proof of 1.3 was that the Hodge structure $H^2(X, \mathbb{Q})$, for X a $K3$ surface, "is expressible" in terms of abelian varieties, in the following sense.

7.2. Definition. A rational Hodge structure H is expressible in terms of abelian varieties if it belongs to the smallest category of rational Hodge structures stable under direct factors, direct sums, and tensor products, and containing the $H^1(A, \mathbb{Q})$ for A an abelian variety, as well as the $\mathbb{Q}(n)$.

In this paper, we show that this property of $K3$ surfaces is exceptional.

Consider the following property of a rational Hodge structure with Mumford–Tate group G.

($*$) The Hodge structure on the adjoint representation $\mathrm{Lie}(G)$ of G is purely of type $\{(-1,1),(0,0),(1,-1)\}$.

7.3. Proposition. *If a rational Hodge structure H is expressible in terms of abelian varieties, then condition ($*$) holds.*

Indeed,

(a) the Hodge structures $H^1(A, \mathbb{Q})$ and $\mathbb{Q}(n)$ satisfy ($*$);

(b) condition ($*$) is stable under direct factors, direct sums, and tensor products.

7.4. Analogous arguments show that if H is expressible in terms of abelian varieties, then the reductive group $G_{\mathbb{C}}$ has no "factors" of exceptional type.

7.5. Proposition. *Let $\underline{H}$ be a polarized variation of Hodge structures on an irreducible scheme S. For $s \in S$ outside a meagre subset of S, there exists a finite-index subgroup of $\pi_1(S,s)$ whose image in $\mathrm{GL}(H_{s\mathbb{Z}})$ is contained in the Mumford–Tate group of H_s.*

At every point of S, the Mumford–Tate group G_s of H_s may be described as follows. It is the algebraic subgroup of $\mathrm{GL}(H_{s\mathbb{Q}})$ that fixes all tensors $t \in H_{s\mathbb{Q}}^{\otimes n} \otimes H_{s\mathbb{Q}}^{*\otimes n}$ $(n > 0)$ that are rational of type $(0,0)$. From this description it follows readily that, outside a meagre subset M of S, G_s is locally constant: for $s \notin M$, the largest subspace of type $(0,0)$, $V(n)_s \subset H_{s\mathbb{Q}}^{\otimes n} \otimes H_{s\mathbb{Q}}^{*\otimes n}$, is locally constant and defines a local system $V(n)$ on S. This local system underlies a polarized variation of Hodge structures of type $(0,0)$; hence $V(n)_s$ carries a positive-definite quadratic form invariant under $\pi_1(S,s)$, and $\pi_1(S,s)$ acts on $V(n)_s$ through a finite group. One concludes by noting that there is a finite sequence of integers n_i such that G_s is the algebraic subgroup of $\mathrm{GL}(H_{s\mathbb{Q}})$ acting trivially on the $V(n_i)_s$.

7.6. Consider a Lefschetz pencil of hypersurfaces of degree d and odd dimension $2r-1$.

$$\begin{array}{ccc} X & \hookrightarrow & \mathbb{P}^{2r}(\mathbb{C}) \times \mathbb{P}^1(\mathbb{C}) \\ {\scriptstyle f}\downarrow & \swarrow_{pr_2} & \\ \mathbb{P}^1(\mathbb{C}). & & \end{array}$$

Let $U \subset \mathbb{P}^1(\mathbb{C})$ be the complement of the finite set of $t \in \mathbb{P}^1(\mathbb{C})$ for which the hypersurface $X_t = f^{-1}(t)$ is singular, and let $s \in U$. The cup product is an alternating form ψ on $H^{2r-1}(X_s, \mathbb{Z})$, preserved by monodromy, whence

$$m : \pi_1(U,s) \longrightarrow \mathrm{Sp}(H^{2r-1}(X_s,\mathbb{Z}), \psi).$$

By a theorem of Kajdan and Margulis, the image of m is Zariski-dense in the symplectic group. By 7.5, for s outside a meagre (in fact, countable) subset of $\mathbb{P}^1(\mathbb{C})$, the Mumford–Tate group of the rational Hodge structure $H^{2r-1}(X_s,\mathbb{Q})$ is therefore the entire group of symplectic similitudes. If this Hodge structure is not of Hodge level at most one (i.e. if some h^{pq} with $|q-p| \neq 1$ is nonzero), it follows from 7.3 that it is not expressible in terms of abelian varieties.

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Pierre Deligne
Institut des Hautes Études Scientifiques
35, Route de Chartres
F-91 Bures-sur-Yvette
France

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Buildings of Generalized Braid Groups

Pierre Deligne (Bures-sur-Yvette)

Introduction

The main result of this work is the following.

Theorem. Let V be a finite-dimensional real vector space, let $\mathcal{M}$ be a finite set of homogeneous hyperplanes of V, let V_C be the complexification of V, and let

$$Y = V_C - \bigcup_{M \in \mathcal{M}} M_C$$

Assume that the connected components of $V - \bigcup_{M \in \mathcal{M}} M$ are open simplicial cones. Then Y is a $K(\pi, 1)$.

Let V be as above, and let $W \subset \mathrm{GL}(V)$ be a finite group generated by reflections. Suppose that no nonzero vector of V is fixed by W :

$$V^W = 0$$

Let Φ be any Euclidean structure on V invariant under W, and let $\mathcal{M}$ be the set of hyperplanes M such that the orthogonal reflection with respect to M belongs to W. It is then known that $(V, \mathcal{M})$ satisfies the hypothesis of the theorem, and that W acts freely on the corresponding space Y_W. The quotient $X_W = Y_W/W$ is therefore also a $K(\pi, 1)$.

This result had been conjectured by Brieskorn. It is new only for W of type H_3, H_4, E_6, E_7, or E_8 (Brieskorn [2]). We also give a new proof that the fundamental group of X_W is the generalized braid group $\widetilde{W}$ corresponding to W (Brieskorn [3]).

Reduced to the particular case considered above, the plan of the proof is as follows.

a) We construct a building $I(\widetilde{W})$ on which $\widetilde{W}$ acts, and a set $\mathcal{S}$ of spheres in $I(\widetilde{W})$, isomorphic to the unit sphere of V. The group $\widetilde{W}$ acts strictly simply transitively on $\mathcal{S}$.

b) We show that, up to homotopy, $I(\widetilde{W})$ is the wedge of the spheres $S \in \mathcal{S}$.

c) We relate X_W and $I(\widetilde{W})$.

The description of $\mathcal{S}$ and the possibility of c) had emerged during a conversation with Brieskorn and Tits in the spring of 1970. The ideas required to establish b) were supplied to me by Garside [4], whom I follow very closely in many places.

In Section 4, we determine the center of $\widetilde{W}$ and solve the word problem and the conjugacy problem in $\widetilde{W}$. These results were obtained independently by E. Brieskorn and K. Saito, who, like us, use Garside's methods [4].

The geometric language used and the techniques of proof are essentially due to Tits; I was able to familiarize myself with his ideas by attending one of his seminars and through conversations.

I am pleased to express my gratitude to him here.

0. Notation

(0.1) $L^+(D)$: the free monoid with identity generated by a set D.

(0.2) L(D): the free group generated by a set D.

(0.3) A^- : the closure of a subset A of a topological space.

(0.4) For x, y in a monoid with identity and $n \geq 0$, set

$$\mathrm{prod}(n; x, y) = xyxy \ldots \quad (n \text{ factors});$$

one has $\mathrm{prod}(2n; x, y) = (xy)^n$, $\mathrm{prod}(2n+1; x, y) = (xy)^n x$.

1. Galleries

(1.1) Let $\mathcal{M}$ be a finite set of hyperplanes (the walls) in a finite-dimensional real vector space V. We shall use the terminology (walls, chambers, faces, facets) of [1] V § 1 (the facets form a partition of V). The support P of a facet F is the intersection of the walls containing F; F is open in P. For every chamber A and every wall M, let $D_M(A)$ denote the set of chambers on the same side of M as A. If A and B are two chambers, let $D(A, B)$ denote the intersection of the $D_M(A)$ for which A and B are on the same side of M. Departing from the terminology of [1] IV 1 ex 15, we shall say that two chambers A and B are adjacent if A $\neq$ B and A and B have a face in common. We shall use the following elementary facts throughout.

(1.2) Lemma. (i) Let M be a wall of a chamber B. There exists a unique chamber B' adjacent to B and having M as a wall. M is the only wall separating B from B'.
(ii) Let B_1, B_2, B_3 be three chambers, and let $\mathcal{M}(B_i, B_j)$ be the set of walls separating B_i from B_j. Then

$$\mathcal{M}(B_1, B_3) = (\mathcal{M}(B_1, B_2) - \mathcal{M}(B_2, B_3)) \cup (\mathcal{M}(B_2, B_3) - \mathcal{M}(B_1, B_2))$$

A gallery of length n ($n \geq 0$), with source A and target B, is a sequence of chambers $(C_0, \ldots, C_n)$ with A = C_0, C_{i+1} adjacent to C_i ($0 \leq i \leq n$), and $C_n = B$. The composite of two galleries $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_m)$
is defined if $C_n = C'_0$, and is then
$GG' = (C_0, \ldots, C_n, C'_1, \ldots, C'_m)$.
If G is a gallery with source A, A.G denotes its target.
The opposite gallery G* of a gallery $G = (C_0, \ldots, C_n)$ is the gallery $(C_n, \ldots, C_0)$. One has $(GG')^* = G'^*$ G*.
The antipodal gallery $-G$ of a gallery $G = (C_0, \ldots, C_n)$ is the sequence of antipodal chambers $-G = (-C_0, \ldots, -C_n)$.
One has $-(GG') = (-G)(-G')$, and $-(G^*) = (-G)^*$.
The distance d(A, B) from a chamber A to a chamber B is the least length of a gallery from A to B. A gallery from A to B is minimal if it has length d(A, B).

(1.3) Proposition. The distance d(A, B) is the number of walls separating A from B. For a gallery G from A to B to be minimal, it is necessary and sufficient that it cross once the walls separating A from B and zero times the others.

It is clear that every gallery from A to B crosses every wall separating A from B, and it is enough to prove the existence of a gallery G from A to B whose length equals the number k of walls separating A from B. We proceed by induction on k. The intersection of the $D_M(A)$, for M a wall of A, is reduced to A. Thus either A = B, in which case we take $G = (A)$, or there is a wall M of A separating A from B. Let A' be the chamber adjacent to A having M as a wall. M is the only wall separating A from A', so a wall N separates A' from B if and only if $M \neq N$ and N separates A from B. By the induction hypothesis, there is a gallery G' of length $k - 1$ from A' to B, and we take $G = (AA')G'$.

(1.4) Corollary. Let A, B, C be three chambers. The following conditions are equivalent.
(i) d(A, C) = d(A, B) + d(B, C), i.e. there exists a minimal gallery from A to C passing through B.
(ii) For a wall to separate A from C, it is necessary and sufficient that it separate A from B or B from C. In other words, $\mathcal{M}(A, B) \subset \mathcal{M}(A, C)$ (cf. 1.2(ii)).
(iii) $B \in D(A, C)$.

(1.5) Proposition. Let P be an intersection of walls, let V_P = V/P, let $pr_P : V \longrightarrow V_P$ be the projection, and let $\mathcal{M}_P$ be the set of hyperplanes N of V_P such that $pr_P^{-1}(N)$ is a wall. Denote by π'_P the unique map from the facets of $(V, \mathcal{M})$ to those of $(V_P, \mathcal{M}_P)$ such that $pr_P(F) \subset \pi'_P(F)$.
(i) π'_P respects the incidence relation $F_1 \subset F_2^-$: if $F_1 \subset F_2^-$, then
codim(F_1 in F_2^-) $\geq$ codim($\pi'_P[F_1]$ in $\pi'_P[F_2]$).
π'_P takes chambers to chambers; if A and B are adjacent, either $\pi'_P(A) = \pi'_P(B)$, or $\pi'_P(A)$ and $\pi'_P(B)$ are adjacent.
(ii) Let F be a facet with support P (1.1). The restriction of π'_P to the set of facets E of $(V, \mathcal{M})$ such that $F \subset E^-$ is bijective. Like its inverse π_F, it preserves codimension, the incidence $E_1 \subset E_2^-$, and hence adjacency of chambers. We shall also write π_F for the bijection pr_P^{-1} between intersections of walls in V_P and intersections of walls in V containing P.
(iii) If $C = \pi_F(C')$, then, for every chamber X of $(V, \mathcal{M})$,
$\pi_F^{-1}D(C, X) = D(C', \pi'_P(X))$ and $\pi_F^{-1}\mathcal{M}(C, X) = \mathcal{M}_P(C', \pi'_P(X))$.

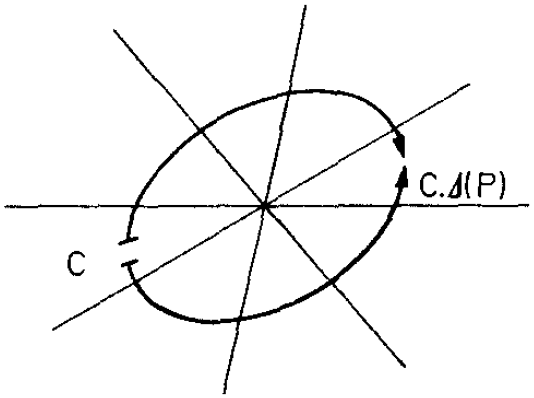
For $X = \pi_F(X')$, one has
$\pi_F(D(C', X')) = D(C, X)$ and $\pi_F(\mathcal{M}_P(C', X')) = \mathcal{M}(C, X)$;
π_F then induces a bijection between minimal galleries from C' to X' and from C to X.
The verification is left to the reader ((iii) follows from 1.4(ii) $\Longleftrightarrow$ (iii)).

(1.6) Notation. (i) For F a facet with support P, V_P, $\mathcal{M}_P$, pr_P, π'_P are also denoted V_F, $\mathcal{M}_F$, pr_F, π'_F.
(ii) Let P be an intersection of walls and C a chamber. Suppose that there is a facet of C (i.e. in C^-) with support P. This facet is then unique. Denote it by F(P), and set

$$C.\Delta(P) = \pi_{F(P)}(-\pi^{-1}_{F(P)}(C))$$

(1.7) Lemma. (i) A wall separates C from $C.\Delta(P)$ if and only if it contains P ((1.5)(iii)).
(ii) If M is a wall of C, then $C.\Delta(M)$ is the unique chamber adjacent to C and having M as a wall (cf. 1.5(iii), 1.2(i)).
(iii) Let $M \neq N$ be two walls of C whose intersection P contains a facet of C open in P. There are exactly two minimal galleries from C to $C.\Delta(P)$. One begins with $(C, C.\Delta(M))$; the other begins with $(C, C.\Delta(N))$.
Assertion (iii) is verified by reduction to rank two (cf. 1.5(iii)), where the drawing is
(1.7.1)

(1.7.1)



From now on we assume the following condition.

(1.8) Hypothesis. The chambers are open simplicial cones.
In other words, each chamber is the set of points with coordinates > 0 in a suitable basis of V.

(1.9.1) Remark. Let P be an intersection of walls.
(i) $(V_P, \mathcal{M}_P)$ still satisfies (1.8);
(ii) the set $\mathcal{M}^P$ of traces on P of the walls not containing P still satisfies (1.8).
If $\mathcal{M}$ is defined by a "Weyl group" W (see the introduction), $\mathcal{M}^P$ is in general no longer of this type.

(1.9.2) Remark. Hypothesis (1.8) ensures that if P is an intersection of walls of a chamber C, the condition of (1.6) is satisfied. The chamber $C.\Delta(P)$ is therefore defined.

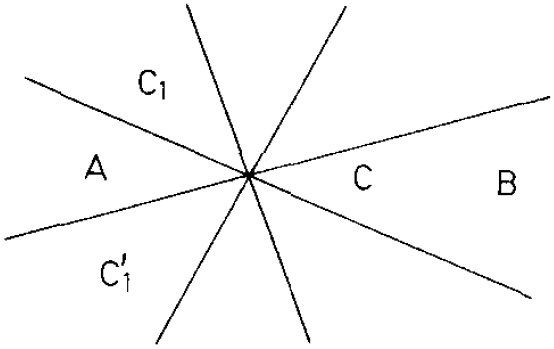
(1.10) Definition. Two galleries G and G' with the same endpoints are equivalent (notation: $G \sim G'$) if there exists a sequence of galleries $G = G_0, \ldots, G_n = G'$ $(n \geq 0)$ such that G_{j+1} is obtained from G_j in the following way $(0 \leq j < n)$:
a) there are decompositions $G_j = E_1 F E_2$, $G_{j+1} = E_1 F' E_2$;
b) there is a chamber C and two walls M, N of C such that F and F' are the two minimal galleries from C to $C.\Delta(M \cap N)$.

Composition of galleries, the operations $G \longrightarrow G^*$ and $G \longrightarrow -G$, the source and target maps, and the "length" function are compatible with this equivalence relation and therefore pass to the quotient. We shall generally denote classes of galleries by lowercase letters. We say that a class of galleries e begins (respectively ends) with a class of galleries f if there exists g such that $e = fg$ (respectively $e = gf$). We verify:

(1.11) Proposition. Equivalent galleries cross each wall the same number of times.

(1.12) Proposition. Two minimal galleries with the same endpoints are equivalent.

Let $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_n)$, with endpoints A and B. We proceed by induction on the length of the galleries under consideration. The case $n = 0$ is trivial, and the induction hypothesis allows us to suppose that $C_1 \neq C'_1$. Let M and M' be the walls separating $\mathrm{A} = C_0 = C'_0$ from C_1 and C'_1, and let $C = A.\Delta(M \cap M')$. The walls M and M' separate A from B. By reduction to rank two (1.5)(iii), it follows that every wall separating A from C separates A from B



(dessin dans $V_{M\cap M'}$; on omet d'écrire $\pi'_{M\cap M'}(\ )$).

(drawing in $V_{M\cap M'}$; we omit writing $\pi'_{M\cap M'}(\ )$).

Let F (respectively F') be the minimal gallery from A to C that passes through C_1 (respectively C'_1), and let E be a minimal gallery from C to B. The galleries FE and $F'E$ are minimal and equivalent. The minimal galleries G and FE (respectively G' and FE') begin with (A, C_1) (respectively with (A, C'_1)). The induction hypothesis implies that they are equivalent, and (1.12) follows.

(1.13) Notation. (i) Let $u(A, B)$ denote the equivalence class of minimal galleries from A to B.

(ii) For I a set of walls of a chamber C, with intersection P, set $\Delta(P) = u(C, C.\Delta(P))$ (cf. (1.9.2)). For I the set of all walls, set $\Delta = u(C, -C)$.

The interest of the notation lies in its ambiguity with respect to C.

(1.14) Proposition. Let A be a chamber and $\mathcal{G}$ a set of classes of galleries of bounded length with source A. Suppose that $\mathcal{G}$ satisfies the conjunction (i) of

(i_a) $(A) \in \mathcal{G}$.

(i_b) If $gh \in \mathcal{G}$, then $g \in \mathcal{G}$.

(i_c) Let g have target B, and let M, N be two walls of B. If $g\Delta(M)$ and $g\Delta(N)$ are in $\mathcal{G}$, then $g\Delta(M \cap N) \in \mathcal{G}$.

Then

(ii) There exists a unique class of galleries x with source A such that

$\mathcal{G} = \{\mathrm{g} \mid x \text{ begins with g}\}$.

Uniqueness is clear: if x begins with y and $x \neq y$, then y has strictly smaller length than x. Let x have maximal length in $\mathcal{G}$. To ensure that x has the required property, it is enough to prove the following assertion.

(*) Let g and M be such that (a) x begins with g and (b) x does not begin with $g\Delta(M)$, while $g\Delta(M) \in \mathcal{G}$. Then there exist g' and M' still satisfying (a) and (b), with g' strictly longer than g.

Since $g\Delta(M) \in \mathcal{G}$, maximality of x implies that $g \neq x$, and hence that x begins with $g\Delta(N)$ for a suitable N, necessarily distinct from M. By (i_c), one then has $g\Delta(M \cap N) \in \mathcal{G}$.

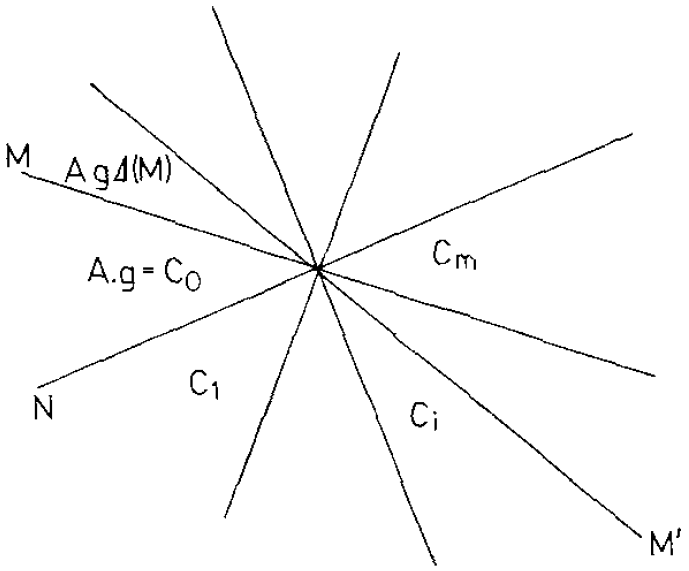
Since $g\Delta(M \cap N)$ begins with $g\Delta(M)$, x does not begin with $g\Delta(M \cap N)$.

Let $(C_0, \ldots, C_m)$ be the minimal gallery from A.g to $A.g\Delta(M \cap N)$ that begins with (A.g, $A.g\Delta(N)$). Let i ($0 < \mathrm{i} < m$) be the largest i such that x begins with $g' = g(C_0, \ldots, C_i)$. Then g' and the wall M' between C_i and C_{i+1} satisfy (a)(b).

(projection in $V_{M\cap N}$).

(1.15) Proposition. Let A be a chamber and $\mathcal{B}$ a set of chambers. For $\mathcal{B}$ to satisfy the conjunction (i) consisting of

(i_a) $A \in \mathcal{B}$.



(projection dans $V_{M \cap N}$).

(i_b) If $B \in \mathcal{B}$, then $D(A, B) \subset \mathcal{B}$.

(i_c) Let M and N be two walls of a chamber B. If A and B are on the same side of M and of N, and $B.\Delta(M)$ and $B.\Delta(N)$ belong to $\mathcal{B}$, then $B.\Delta(M \cap N) \in \mathcal{B}$.

it is necessary and sufficient that

(ii) There exists a unique chamber C such that $\mathcal{B} = D(A, C)$.

It is easy to verify that (ii) $\Rightarrow$ (i). To prove that (i) $\Rightarrow$ (ii), apply (1.14) to the set $\mathcal{G}$ of the $u(A, B)$ for $B \in \mathcal{B}$ (note that, by (1.3) and (1.11), every gallery G of class g such that some $u(A, B)$ begins with g is automatically minimal), and apply (1.4)(i) $\Longleftrightarrow$ (iii) to the gallery class $x = u(A, X)$ thereby obtained.

Applying criterion (1.15), we find:

(1.16) Corollary. Let A and B be two adjacent chambers separated by the wall M, and let C be on the same side of M as B. There exists C' such that

$$D(A, C) \cap D_M(B) = D(B, C')$$

(1.17) Corollary. (i) Let A, C_1, C_2 be three chambers. There exists C such that

(1.17.1) $D(A, C) = D(A, C_1) \cap D(A, C_2)$.

(ii) Let A and C_1 be two chambers, and let M be a wall of A. There exists C' such that

(1.17.2) $D(A, C') = D(A, C_1) \cap D_M(A)$.

In fact, (ii) is the special case of (i) with $C_2 = -(A.\Delta(M))$.

(1.18) Lemma. Let M, M', M'' be three distinct walls of a chamber A, $A_1 = A.\Delta(M)$, $B = A.\Delta(M \cap M' \cap M'')$, $B' = A.\Delta(M \cap M')$, and $B'' = A.\Delta(M \cap M'')$. For every chamber C, if $B' \in D(A_1, C)$ and $B'' \in D(A_1, C)$, then $B \in D(A_1, C)$.

The facet F of A open in $M \cap M' \cap M''$ is a facet of all the chambers A, A_1, B, B', and B''. Applying (1.5)(iii), we may therefore reduce to the case dim $V = 3$. Applying (1.17)(ii), we may further suppose that A_1 and C are on the same side of M. Make these assumptions, and suppose that B', $B'' \in D(A_1, C)$. Since M' (respectively M'') separates A_1 from B' (respectively B''), A_1 and C are separated by M' and M'' (and are on the same side of M); hence A and C are separated by M, M', and M'', and $C = -A = B$.

The following key result is inspired by [4].

(1.19) Proposition. (i) Let A, B, C be three chambers, F a gallery from A to B, and G_1, G_2 two galleries from B to C. If $FG_1 \sim FG_2$, then $G_1 \sim G_2$.

(ii) Similarly, if $G_1E \sim G_2E$, then $G_1 \sim G_2$.
(iii) Let g be an equivalence class of galleries with source A. There exists a chamber C such that g begins with $u(A, B)$ if and only if $B \in D(A, C)$.

Statement (ii) follows from (i) by passing to opposite galleries.

Let $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_n)$ be two galleries of length n with the same endpoints. If $n \geq 1$, set $G_1 = (C_1, \ldots, C_n)$ and $G'_1 = (C'_1, \ldots, C'_n)$. If moreover $C_1 \neq C'_1$, let M and M' be the walls separating $C_0 = C'_0$ from C_1 and C'_1, and set $\mathrm{A} = C_0.\Delta(M \cap M')$. Consider the following relation $R_n(G, G')$ between galleries G and G' as above.

$R_n(G, G') \Longleftrightarrow$ one of the following holds:

$\alpha)$ $n = 0$;

$\beta)$ $n \neq 0$, $C_1 = C'_1$, and $G_1 \sim G'_1$;

$\gamma)$ $n \neq 0$, $C_1 \neq C'_1$, and there exists a gallery F from A to $C_n = C'_n$ such that $G_1 \sim u(C_1, \mathrm{A}).\mathrm{F}$ and $G'_1 \sim u(C'_1, \mathrm{A}).\mathrm{F}$.

We shall prove by induction on n the following assertion:

(A_n) R_n is an equivalence relation.

Assume (A_i) for $\mathrm{i} < n$, and prove (1.19.1) through (1.19.3) below.

(1.19.1) For galleries of length $\mathrm{i} < n$, $R_i(G, G') \Longleftrightarrow G \sim G'$.

It is trivial that $R_i(G, G') \Rightarrow G \sim G'$; and, with the notation of (1.10), if $G \sim G'$, then $R_i(G_j, G_{j+1})$.

(1.19.2) Assertion (i) is true for FG_1 of length $< n$.

This follows from (1.19.1) and the second clause in the definition of the R_i.

(1.19.3) Assertion (iii) is true for g of length $< n$.

This follows from (1.15) applied to the set $\mathcal{B}$ of chambers B such that g begins with $u(A, B)$: hypothesis (i_c) of (1.15) is verified using (1.19.2), (1.18), and the third clause in the definition of the R_i.

It remains to prove that if G, G', and G'' are three galleries of length n from A to C, and $R_n(G, G')$ and $R_n(G, G'')$, then $R_n(G', G'')$. If $n = 0$, or $C_1 = C'_1$, or $C_1 = C''_1$, this is trivial.

For $n > 0$, denote by M, M', M'' the walls separating $\mathrm{A} = C_0 = C'_0 = C''_0$ from C_1, C'_1, and C''_1. Distinguish two cases.

Case 1. $C_1 \neq C'_1 = C''_1$. Let $B = A.\Delta(M \cap M')$. By hypothesis,

$(C_1, \ldots, C_n) \sim u(C_1, B)F' \sim u(C_1, B)F''$,

$(C'_1, \ldots, C'_n) \sim u(C'_1, B)F'$,

$(C''_1, \ldots, C''_n) \sim u(C''_1, B)F''$.

It follows from (1.19.2) that $F' \sim F''$, and hence that $R_n(G', G'')$.

Case 2. $C_1 \neq C'_1 \neq C''_1 \neq C_1$. Let $B_1 = A.\Delta(M' \cap M'')$, $B' = A.\Delta(M \cap M')$, $B'' = A.\Delta(M \cap M'')$, and $B = A.\Delta(M \cap M' \cap M'')$. We have

$$G_1 = (C_1, \ldots, C_n) \sim u(C_1, B')F' \sim u(C_1, B'')F''$$

$$G'_1 = (C'_1, \ldots, C'_n) \sim u(C'_1, B')F'$$

$$G''_1 = (C''_1, \ldots, C''_n) \sim u(C''_1, B'')F''$$

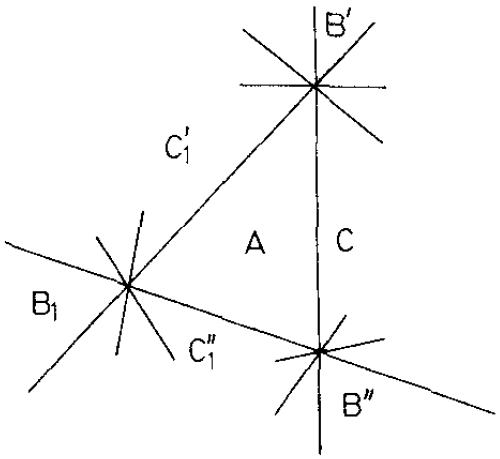
so that the class g_1 of G_1 begins with $u(C_1, B')$ and $u(C_1, B'')$. It then follows from (1.19.3) and Lemma (1.18) that g_1 begins with $u(C_1, B) : G_1 \sim u(C_1, \mathrm{B}).\mathrm{F}$. By (1.19.2), $F' \sim u(B', \mathrm{B}).\mathrm{F}$ and $F'' \sim u(B'', \mathrm{B}).\mathrm{F}$; hence $G_1 \sim u(C_1, \mathrm{B}).\mathrm{F}$, $G'_1 \sim u(C'_1, \mathrm{B}).\mathrm{F}$, and $G''_1 \sim u(C''_1, \mathrm{B}).\mathrm{F}$. Therefore,

$G'_1 \sim u(C'_1, B_1)u(B_1, \mathrm{B}).\mathrm{F}$

$G''_1 \sim u(C''_1, B_1)u(B_1, \mathrm{B}).\mathrm{F}$,

so that $R_n(G', G'')$.

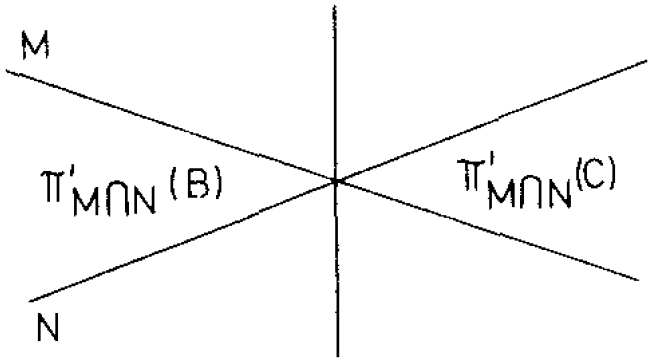
(drawing on the unit sphere of $V_{M \cap M' \cap M''}$).



(dessin sur la sphère unité de $V_{M\cap M'\cap M''}$).

(1.20) Corollary. Conditions (i) and (ii) of (1.14) are equivalent.

Assume (ii) and prove (i_c). Under the hypotheses of (i_c), if $x = \mathrm{gh}$ (h with source B), then (1.19)(i) implies that h begins with $\Delta(M)$ and $\Delta(N)$. Let C be the chamber guaranteed by (1.19)(iii) for h. Since M and N separate B from C, $\pi'_{M\cap N}(C)$ can only be $\pi'_{M\cap N}(B).\Delta$, and (i_c) follows from (1.5)(iii).



(1.21) Corollary. For every chamber A, let n(A, i) be the number of classes of galleries with source A and length i. Set

$$f_A = \sum{}_0^{\infty} n(A, i)t^i \in \mathbb{Z}[[t]]$$

For every chamber A, and for P running through the intersections of walls of A (including $\{0\}$ and V), one has

$$\sum{}_P (-1)^{\mathrm{codim}(P)} t^{d(A, A.\Delta(P))} f_{A.\Delta(P)} = 1$$

This expresses that

a) the number of classes of galleries of length i that begin with $\Delta(P)$ is $n(A.\Delta(P), \mathrm{i} - \mathrm{d}(\mathrm{A}, A.\Delta(P)))$ ((1.19)(i));

b) classes of galleries that begin with $\Delta(P)$ and $\Delta(Q)$ begin with $\Delta(P \cap Q)$ (by (1.19)(iii)).

(1.22) Algorithm. Let $G = (A_0, \ldots, A_n)$ be a gallery of length $n \geq 1$, let $G_1 = (A_1, \ldots, A_n)$, let M be the wall separating A_0 from A_1, let C be the chamber whose existence is guaranteed by (1.19)(iii) (for G), and let C_1 be the analogous chamber for G_1. One can compute C recursively using the following formula (cf. (1.16)):

(1.22.1) $D(A_1, C) = D_M(A_1) \cap D(A_1, C_1)$.

Since $A_1 \in D(A, C)$, M indeed separates A from C and $C \in D_M(A_1)$. By (1.19)(i), one also has $D(A_1, C) \subset D(A_1, C_1)$, giving the inclusion $\subset$. Finally, if $B \in D_M(A_1) \cap D(A_1, C_1)$, then G_1 begins with $u(A_1, B)$ and G begins with $u(A, A_1).u(A_1, B) = u(A, B)$, which proves (1.22.1).

(1.23) Corollary. Let G and H be two composable galleries. Let A be the source of G, B that of H, and let C be guaranteed by (1.19)(iii): one has $H \sim u(B, C).H'$. Then, for every chamber D, for the class of GH to begin with $u(A, D)$, it is necessary (and sufficient) that the class of Gu(B, C) begin with $u(A, D)$.

(1.24) Proposition. Let A be a chamber, P an intersection of walls of A, $F = \mathrm{F(P)}$ (1.6), and g an equivalence class of galleries with source A. There exists a class of galleries g' of $(V_P, \mathcal{M}_P)$, with source $\pi_F^{-1}(A)$, such that, for every gallery H of $(V_P, \mathcal{M}_P)$, g begins with $\pi_F(H)$ if and only if g' begins with H.

Let $\mathcal{G}$ be the set of classes of galleries h with source $\pi_F^{-1}(A)$ in $(V_P, \mathcal{M}_P)$ such that g begins with $\pi_F(h)$. Apply (1.14) to $(V_P, \mathcal{M}_P)$ and to $\mathcal{G}$. By (1.19), condition (i) is satisfied (cf. the proof of (1.20)), and (1.24) follows from (1.14)(ii).

(1.25) One may regard the set of galleries as the set of arrows of a category $\mathrm{Gal}_0(V, \mathcal{M})$ whose objects are the chambers. Similarly, equivalence classes of galleries are the arrows of a quotient category $\mathrm{Gal}_+(V, \mathcal{M})$. Let A, B, and C be three chambers, E a gallery from A to B, and F a gallery from B to C. Although the general categorical conventions are different, we shall continue to write EF for the composite of E and F. The operation * (respectively $G \longrightarrow -G$) is an anti-equivalence (respectively an equivalence) of $\mathrm{Gal}_0(V, \mathcal{M})$ or $\mathrm{Gal}_+(V, \mathcal{M})$ with itself, inducing the identity (respectively $C \longrightarrow -C$) on the set of objects.

When there is no risk of confusion, we shall write simply Gal_0 and Gal_+ for $\mathrm{Gal}_0(V, \mathcal{M})$ and $\mathrm{Gal}_+(V, \mathcal{M})$ (or for a category $\mathrm{Gal}_0(V_P, \mathcal{M}_P)$ or $\mathrm{Gal}_+(V_P, \mathcal{M}_P)$). By abuse of language, we shall often call the arrows of Gal_+ galleries.

(1.26) Lemma. (i) In Gal_+, for every gallery g, one has

$$g\Delta = \Delta(-g)$$

(ii) For any g and h with source A in Gal_+, there exists n such that $g\Delta^n$ begins with h. If h is composed of k galleries $u(A_i, A_{i+1})$, one may take $n = k$.

Proceeding by induction, it is enough to prove (i) for g of length one. For $g = (B, C)$, one has

$$g\Delta = u(B,C)u(C,-C) = u(B,C)u(C,-B)u(-B,-C) = u(B,-B)u(-B,-C) = \Delta(-g)$$

For every chamber B, $\Delta = u(A, -A)$ begins with $u(A, B)$. Hence (ii) follows from (i) by induction on k.

The following proposition follows at once from (1.26), from its transform by *, and from (1.19)(i), (ii).

(1.27) Proposition. (i) In Gal_+, the set of all arrows admits a left and right calculus of fractions.

(ii) Let $\mathrm{Gal}(V, \mathcal{M})$, or simply Gal, be the category (a groupoid) obtained from Gal_+ by making all arrows invertible: call positive the arrows of Gal lying in the image of Gal_+. The canonical functor from Gal_+ to Gal is faithful, and every arrow g of Gal can be written in the forms $g = g_1\Delta^{-n} = \Delta^{-n}g_2$ with g_1 and g_2 positive.

If a category C has only one object A and the monoid $\mathrm{Hom}(A, \mathrm{A})$ is cancellative, saying that all arrows of C admit a left and right calculus of fractions means that $\mathrm{Hom}(A, \mathrm{A})$ satisfies the Öre condition on the left and on the right. The "theory of the calculus of fractions" used in (1.27) and below is an immediate generalization of the theory of Öre for embedding such a monoid in a group.

(1.28) For every wall M, the number of times that g in Gal_+ crosses M is well defined (1.11). This function of g extends additively to $g \in \mathrm{Gal}$. More generally, for every intersection of walls P, one could use the universal property of Gal to define functors

$$\pi'_P : \mathrm{Gal}(V, \mathcal{M}) \longrightarrow \mathrm{Gal}(V_P, \mathcal{M}_P)$$

(1.29) Let A be a chamber, P an intersection of walls of A, and $F = \mathrm{F(P)}$ (1.6). The function π_F induces a functor

(1.29.1) $\pi_F : \mathrm{Gal}_0(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}_0(V, \mathcal{M})$.

The image of this functor is stable under equivalence, and it induces a faithful functor

(1.29.2) $\pi_F : \mathrm{Gal}_+(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}_+(V, \mathcal{M})$.

From the theory of the calculus of fractions and (1.19)(i) follows the

(1.30) Proposition. Under the preceding hypotheses, the functor deduced from (1.29.2)

$$\pi_F : \mathrm{Gal}(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}(V, \mathcal{M})$$

is faithful.

(1.31) Proposition. Let C be a chamber, I and J two sets of walls of C, $K = \mathrm{I} \cap J$, P, Q, and R the intersections of the walls in I, J, and K, and F(P), F(Q), and F(R) the corresponding facets in C (1.6). The facet F(R) is the smallest facet containing F(P) and F(Q). In the groupoid Gal(V, $\mathcal{M}$), one has

$$\pi_{F(P)}(\mathrm{Gal}(V_P, \mathcal{M}_P)) \cap \pi_{F(Q)}(\mathrm{Gal}(V_Q, \mathcal{M}_Q)) = \pi_{F(R)}(\mathrm{Gal}(V_R, \mathcal{M}_R))$$

A chamber having F(P) and F(Q) as facets also has F(R) as a facet. This proves (1.31) for the objects.

Let $g \in \mathrm{Hom}_{\mathrm{Gal}}(A, B)$, and suppose that

$$g = \pi_{F(P)}(g_1') = \pi_{F(Q)}(g_2')$$

By (1.27)(ii), one has
$g_1' = \Delta(P)^{-n} g_1''$ and $g_2' = \Delta(Q)^{-n} g_2''$
with g_i'' positive, for $n \geq 0$ sufficiently large. Let $g_1 = \pi_{F(P)} g_1''$ and $g_2 = \pi_{F(Q)} g_2''$.
By (1.26)(ii) (transformed by *), $\Delta^n \Delta(P)^{-n}$ is positive; in $\mathrm{Gal}_+(V, \mathcal{M})$ one has

(1.31.1) $(\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2$.

(1.31.2) Lemma. If $h \in \mathrm{Hom}_{\mathrm{Gal}_+}(A, B)$ begins both with $(\Delta^n \Delta(P)^{-n})$ and with $(\Delta^n \Delta(Q)^{-n})$, then h begins with $(\Delta^n \Delta(R)^{-n})$.

Let us deduce (1.31) from (1.31.2). By (1.31.2), one has
$(\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2 = (\Delta^n \Delta(R)^{-n}) g_3$
with g_3 positive. Thus $g = \Delta(R)^{-n} g_3$. The positive gallery $g_3 = \Delta(R)^n\, g$ belongs to the image of $\pi_{F(P)}$ and of $\pi_{F(Q)}$. By (1.28), it therefore crosses zero times every wall not containing P or Q, i.e. not containing R. It consequently belongs to the image of $\pi_{F(R)}$.

Let us prove (1.31.2). Set A(P) = $A.\Delta.\Delta(P)$, and similarly for Q and R. The gallery $\Delta\Delta(P)^{-1}$ with source A is $u(A, \mathrm{A(P)})$; the one with source A(P) is $u(A(P), \mathrm{A})$. It follows from (1.26)(i) that $\Delta\Delta(P) = \Delta(P)\Delta$. Hence

(1.31.3) $\Delta^n \Delta(P)^{-n} = u(A, \mathrm{A(P)}).u(\mathrm{A(P)}, \mathrm{A})u(\mathrm{A}, \mathrm{A(P)}) \ldots$ (n factors).

(1.31.4) Lemma. $A(R) \in D(A(P), \mathrm{A(Q)})$.

With the notation of (1.2), $\mathcal{M}(A\Delta, \mathrm{A(P)})$ is the set of walls containing P, and $\mathcal{M}(A\Delta, \mathrm{A(Q)})$ the set of those containing Q. By (1.2), $\mathcal{M}(A(P), \mathrm{A(Q)})$ is the set of those containing P or Q, but not R, namely $\mathcal{M}(A(P), \mathrm{A(R)}) \cup \mathcal{M}(A(Q), \mathrm{A(R)})$; apply (1.4).

By (1.19)(iii), a gallery class that begins with $\Delta\Delta(P)^{-1}$ and $\Delta\Delta(Q)^{-1}$ therefore also begins with $\Delta\Delta(R)^{-1}$, and (1.31.2) follows by induction from

(1.31.5) Lemma. Let P and R be intersections of walls of a chamber A, with $P \subset R$. Let h be in Gal_+ with source A. If h begins both with $\Delta^n \Delta(P)^{-n}$ ($n > 0$) and with $\Delta\Delta(R)^{-1}$, then h begins with $(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-1}$.

We may suppose that $n \geq 2$. Set $h = (\Delta\Delta(P)^{-1})h'$. With the preceding notation, h' then begins both with $\Delta\Delta(P)^{-1} = u(A(P), \mathrm{A})$ and with $u(A(P), \mathrm{A(R)})$. Apply (1.19)(iii).

The chamber C of loc. cit. is then separated from A(P) by every wall M that separates A(P) from A (i.e. $M \not\supset P$) or A(P) from A(R) (i.e. $M \supset P$ and $M \not\supset R$). Thus C is separated from $A(P)\Delta$ only by walls $M \supset R$, and $C \in D(A(P).\Delta,\ A(P).\Delta\Delta(R))$. Hence h' begins both with $(\Delta\Delta(R)^{-1})$ and with $(\Delta\Delta(P)^{-1})^{n-1}$. Proceeding by induction on n, one deduces that h' begins with $(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-2}$, so that h begins with $(\Delta\Delta(P)^{-1})(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-2}$; the conclusion follows by noting that

$$\Delta\Delta(P)^{-1}\Delta\Delta(R)^{-1} = \Delta\Delta(R)^{-1}\Delta\Delta(P)^{-1}$$

(1.32) Proposition. For F a facet of a chamber C, spanning an intersection of walls P, one has

$$\pi_F^{-1}(\mathrm{Gal}_+(V,\mathcal{M})) = \mathrm{Gal}_+(V_P,\mathcal{M}_P) \subset \mathrm{Gal}(V_P,\mathcal{M}_P)$$

Suppose that $\pi_F(\Delta^{-n}g) = h$ lies in $\mathrm{Gal}_+(V, \mathcal{M})$. Then $\pi_F(g) = \Delta(P)^n\ h$ and, by (1.28), every gallery H representing h crosses only walls passing through F. Thus $h = \pi_F(h_1)$, with h_1 in $\mathrm{Gal}_+(V_P, \mathcal{M}_P)$, and the conclusion follows from (1.30).

2. Buildings

(2.1) Let $(V, \mathcal{M})$ be as in § 1 (satisfying (1.8)), and let $r = \dim V$. Let S be the sphere of radius one in V, for any Euclidean structure on V (more intrinsically, one could set $S = V - \{0\}/\mathbb{R}_+^*$). The hyperplanes $M \in \mathcal{M}$ cut out a triangulation of S, and we shall also denote by S the corresponding simplicial space (simplicial scheme) and its geometric realization. We transfer to S the terminology used for V (chambers, facets, adjacency, gallery, ...).

(2.2) Choose a "fundamental chamber" A_0 in S. We denote by I_+ the space constructed below; it depends on V, $\mathcal{M}$, and A_0, and is equipped with $q : I_+ \longrightarrow S$.

a) Let $\mathcal{L}_+$ be the set of equivalence classes of galleries g with source A_0,

$$\mathcal{L}_+ = \coprod_B \mathrm{Hom}_{\mathrm{Gal}_+}(A_0, B)$$

b) Let Z be the disjoint sum, indexed by $g \in \mathcal{L}_+$, of the closed simplex of S that is the closure of the open simplex of dimension $r-1$ which is the target of g :
$Z_+ = \coprod_{g\in\mathcal{L}_+}$ (target of g)$^-$.
Let q' be the evident map from Z_+ onto S.

c) I_+ (equipped with q) is a quotient of Z_+ (equipped with q'). If G is a gallery from A_0 to B and C is a chamber adjacent to B, one glues (target of g)$^-$ and (target of $g(BC)$)$^-$ along the common closed face of their images in S.

(2.3) The space I_+ is decomposed into facets, images of facets of Z_+, each mapping bijectively onto a facet of S. Its chambers (respectively
faces and vertices) are its facets of dimension $r-1$ (respectively $r-2$ and 0).

The vertices of a facet F are the vertices contained in the closure $\overline{F}$ of F. The chambers of I_+ are indexed by $\mathcal{L}_+$; denote by $\widetilde{A}_0.g$ the one with index g and by $\widetilde{A}_0$ the one with index (A_0).

Let $g \in \mathrm{Hom}_{\mathrm{Gal}_+}(A_0, \mathrm{A})$, let $\widetilde{A} = \widetilde{A}_0.g$, and let $H = (C_0, \ldots, C_n)$ be a gallery from A to B, of class h.
Set
$\widetilde{A}.h = \widetilde{A}_0.gh$
The chambers $\widetilde{A}.(C_0, \ldots, C_i)$ $(0 \leq \mathrm{i} \leq n)$ form a gallery in I_+. Galleries obtained in this way will be called positive.

(2.4) Lemma. Let F be a facet of S, let A and B be two chambers of S having F as a facet, let $g \in \mathrm{Hom}_{\mathrm{Gal}_+}(A_0, \mathrm{A})$, and let $h = \mathrm{Hom}_{\mathrm{Gal}_+}(A_0, B)$. The following conditions are equivalent:

(a) in I_+, the facets $q^{-1}(F)$ of $\widetilde{A}_0.g$ and $\widetilde{A}_0.h$ coincide;

(b) $g^{-1}h \in \mathrm{Hom}_{\mathrm{Gal}}(A, B)$ lies in $\pi_F(\mathrm{Gal})$;

(c) there exist g_1 and h_1 in $\pi_F(\mathrm{Gal}_+)$ such that $gg_1 = hh_1$.

Condition (b) is an equivalence relation among galleries from A_0 to a chamber having F as a facet. It follows that (a) $\Rightarrow$ (b). That (b) $\Rightarrow$ (c) follows from (1.26)(i). Finally, (c) $\Rightarrow$ (a) is immediate from the definitions.

From (2.4)(a) $\Longleftrightarrow$ (b) and (1.31), one immediately obtains the following result.

(2.5) Proposition. Each facet of I_+ is uniquely determined by the set of its vertices. Thus I_+ may be described as the geometric realization of the following simplicial scheme.

a) For x a vertex of S, let $\mathcal{G}(x)$ be the set of g in Gal_+ with source A_0 and target a chamber having x as a vertex. Let $\mathcal{G}(x)/\pi_x$ be the quotient of $\mathcal{G}(x)$ by the equivalence relation (2.4)(b) (for $F = x$). Then the set of vertices is

$$\coprod_x \mathcal{G}(x)/\pi_x$$

b) For a set E of vertices to span a simplex, it is necessary and sufficient that there exist a gallery g with source A_0 such that, for $y \in E$, g lies in $\mathcal{G}(qy)$ and y is the image of g in $\mathcal{G}(qy)/\pi_{qy}$.

(2.6) Proposition. Let F be a facet of I_+. There exists a gallery class g such that, for F to be a facet of a chamber $B = \widetilde{A}_0.h$, it is necessary and sufficient that

$$h = g\pi_F(h')$$

for a suitable h' in Gal_+.

Take g to be a gallery of minimum length such that F is a facet of $\mathrm{A} = \widetilde{A}_0.g$. Let $B = \widetilde{A}_0.h$ be a chamber having F as a facet. By (2.4)(a) $\Longleftrightarrow$ (c), there exist galleries h_1 and h_2 in $(V_{qF}, \mathcal{M}_{qF})$ such that

$$g\pi_F(h_1) = h\pi_F(h_2)$$

Apply the transform by * of (1.24). In view of the minimality of g, one finds that h_1 ends with h_2; $h_1 = h'h_2$. By (1.19)(ii), one has

$$g\pi_F(h') = h,$$

which proves (2.6).

(2.7) Notation. Let A be a chamber of I_+. Denote by $S(A)$ the union of the closed chambers $(\mathrm{A}.(\mathrm{qA},\ B))^-$, as B ranges over the chambers of S.

One verifies that $q|S(A)$ is an isomorphism between $S(A)$ and S.

(2.8) Definition. $\widehat{I}_+$ is the space obtained from I_+ by "filling in" the spheres $S(A)$.

More precisely, let B^r be a ball whose boundary is S. When working simplicially, take B^r to be the cone on the simplicial space S. Define $\widehat{I}_+$ from I_+ by attaching a family of copies $b(A)$ of B^r, indexed by the set of chambers of I_+. The attaching maps are

$$\partial b(A) \xrightarrow{\sim} S \xleftarrow{q} S(A).$$

The attaching maps are simplicial, so $\widehat{I}_+$ again appears as the geometric realization of a simplicial scheme (whose vertices are those of I_+ together with the "centers of the balls"). The space $\widehat{I}_+$ is equipped with an evident (simplicial) map

$$q : \widehat{I}_+ \longrightarrow B^r$$

(2.9) Proposition. The space $\widehat{I}_+$ is contractible.

The space I_+ is therefore the wedge of the spheres $S(A)$.

Let $(I_+)_n$ be the union of the closed chambers $(\widetilde{A}_0.g)^-$ for g of length $\leq\ n$, and let $(\widehat{I}_+)_n$ be the union of $(I_+)_n$ and the set of balls $b(A)$ for which $\partial b(A) \subset (I_+)_n$.

One has $(\widehat{I}_+)_0 = \widetilde{A}_0$ (contractible), and

$$\widehat{I}_+ = \varinjlim (\widehat{I}_+)_n$$

The proposition follows from

(2.9.1) Lemma. $(\widehat{I}_+)_n$ is a deformation retract of $(\widehat{I}_+)_{n+1}$.

We must construct a continuous family $(\varphi_t)_{0\leq t\leq 1}$ of continuous maps from $(\widehat{I}_+)_{n+1}$ to itself, with $\varphi_0 = \mathrm{Id}$, $\varphi_t|(\widehat{I}_+)_n = \mathrm{Id}$, and $\varphi_1((\widehat{I}_+)_{n+1}) = (\widehat{I}_+)_n$.

Let $\mathrm{A} = \widetilde{A}_0.g$ be a chamber of $(I_+)_{n+1}$ that is not in $(I_+)_n$, and let F be a facet of A.

(2.9.2) Lemma. Suppose that there exists a chamber $B = \widetilde{A}_0.h$ in $(I_+)_{n+1}$, distinct from A, having F as a facet. Then there exists a chamber $B' = \widetilde{A}_0.h'$ in $(I_+)_n$ and a wall M of qB', containing qF, such that $\mathrm{A} = B'.\Delta(M)$.

Let g_0 be the gallery considered in (2.6). One has

$g = g_0\pi_F(g_1)$ and $h = g_0\pi_F(h_1)$.

The hypotheses imply that the length $\lg(g_1) \neq 0$ (otherwise, since A $\neq B$, one would have $\lg(h) > \lg(g) =$ n+1). For a suitable wall M of qA, one then has $g_1 = h'\Delta(M)$, and set $B' = \widetilde{A}_0.h'$.

For the closed chamber A^-, $A^- \cap (I_+)_n$ is therefore the union of a nonempty set of closed faces; and, in $(I_+)_{n+1}$, the points of $A^- - (A^- \cap (I_+)_n)$ belong only to the single closed chamber A^-.

We distinguish two cases.

Case 1. $A^- \cap (I_+)_n \neq \partial A^-$.

In this case, there is no sphere $S(B)$ with

$A \subset S(B) \subset (I_+)_{n+1}$.

We take $\varphi_t|A^-$ to be a collapse of A^- onto $A^- \cap (I_+)_n$.

Case 2. $A^- \cap (I_+)_n = \partial A^-$.

Set A $= \widetilde{A}_0.g$. For every wall M of qA, it then follows from (2.7.2) that g ends with $\Delta(M)$.

By ((1.19)(iii)), g therefore ends with Δ : one has A $= B.\Delta$, with B uniquely determined by A, by (1.19)(ii). In passing from $(\widehat{I}_+)_{n+1}$ to $(\widehat{I}_+)_n$, A and the interior of the ball $b(B)$ disappear. We take $\varphi_t|b(B)$ to be a collapse of $b(B)$ onto $S(B) - A$.

By what was seen in case 1, every ball that disappears is of the preceding type. This completes the construction of φ_t and proves (2.7).

(2.10) The space I, equipped with q : I $\longrightarrow S$, is defined like I_+, replacing Gal_+ by Gal.

a) Set $\mathcal{L} = \coprod_B \mathrm{Hom}_{\mathrm{Gal}}(A_0, B)$.

b) Set $Z = \coprod_{g\in\mathcal{L}}$ (target of $g)^-$.

c) Let q' be the evident map from Z onto S. The space I, equipped with q, is a quotient of Z equipped with q'.

For $g \in \mathcal{L}$ with target B and C a chamber adjacent to B, one glues (target of $g)^-$ and (target of $g(BC))^-$ along the common closed face of their images in S. In other words, if two

chambers B and C have a common facet F, if $g \in \mathrm{Hom}_{\mathrm{Gal}}(A_0, B)$ and $h \in \mathrm{Hom}_{\mathrm{Gal}}(A_0, C)$, then in I the closed facets $q'^{-1}(F)^-$ of (target of $g)^-$ and of (target of $h)^-$ are identified if and only if

$hg^{-1} \in \pi_F \mathrm{Hom}_{\mathrm{Gal}}(\pi_F^{-1}(B), \pi_F^{-1}(C))$.

(2.11) As for I_+, one defines the chambers, faces, and open or closed facets of I. One defines $\widetilde{A}_0$ and $\widetilde{A}.h$ (for $\widetilde{A}$ a chamber and h in Gal with source $q\widetilde{A}$) as in (2.3). The analogue of (2.4)(a) $\Longleftrightarrow$ (b) is here trivially true. As in (2.5), one then deduces from (1.31) that each facet of I is uniquely determined by the set of its vertices, and I appears as the geometric realization of a simplicial scheme admitting a description of type (2.5). For every chamber A of I, define a sphere $S(A)$ as in (2.7); q induces an isomorphism from $S(A)$ onto S. As in (2.8), define a building $\widehat{I}$, equipped with q : $\widehat{I} \longrightarrow B^r$, by "filling in" all the spheres $S(A)$.

(2.12) Besides q and its decomposition into chambers, I has the following additional structure:

— A "composition" $\widetilde{B}.g$ is defined which, to a chamber $\widetilde{B}$ of I with image B in S and to $g \in \mathrm{Hom}_{\mathrm{Gal}}(B, C)$, associates a chamber of I with image C in S. The construction $g \longmapsto \widetilde{B}.g$ defines an isomorphism from the space analogous to I obtained by taking B as fundamental chamber to the space I (for $A_0 = B$, one obtains automorphisms of I). In particular,

$\widetilde{B}.(gh) = (\widetilde{B}.g).h$.

(2.13) For every chamber A of I lying over A_0, define a map

$i_A : I_+ \longrightarrow$ I (respectively $\widehat{I}_+ \longrightarrow \widehat{I}$),

compatible with the projection onto S (respectively B^r), by sending the closed chamber $(A.g)^-$ of I_+ (respectively the ball $b(A.g)$ of $\widehat{I}_+$) to the chamber (respectively the ball) of the same name in I (respectively $\widehat{I}$).

(2.14) Proposition. (i) i_A identifies I_+ with a closed subspace of I and $\widehat{I}_+$ with a closed subspace of $\widehat{I}$.

(ii) One has

I $= \varinjlim_n i_{\widetilde{A}_0.\Delta^{-2n}}(I_+)$ and $\widehat{I} = \varinjlim_n i_{\widetilde{A}_0.\Delta^{-2n}}(\widehat{I}_+)$.

The assertions (i) or (ii) for $\widehat{I}_+$ and $\widehat{I}$ follow from the assertions (i) or (ii) for I_+ and I.

We prove (i) for I_+ and I. Let B and C be two chambers of S having a common facet F. Let $g \in \mathrm{Hom}_{\mathrm{Gal}_+}(A_0, B)$ and $h \in \mathrm{Hom}_{\mathrm{Gal}_+}(A, C)$. Suppose that the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are equal in I.

One then has

$g^{-1}h = \pi_F(e)$ for $e \in \mathrm{Hom}_{\mathrm{Gal}}(\pi_F^{-1}B,\, \pi_F^{-1}C)$.

By (1.26), one may write $e = e_1 e_2^{-1}$ with e_i in Gal_+. By (1.29), one then has $g\pi_F(e_1) = h\pi_F(e_2)$ in Gal_+, and the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are therefore already equal in I_+.

To prove (ii), it suffices to observe that every gallery g with source A_0 in Gal can be written $g = \Delta^{-2n}g'$ with g' in Gal_+.

From (2.14) one obtains $\pi_i(\widehat{I}) = \varinjlim \pi_i(\widehat{I}_+)$. Since I is a CW complex, (2.9) gives the following result.

(2.15) Theorem. The space $\widehat{I}$ is contractible.

The space I is therefore the wedge of the spheres $S(A)$, as A ranges over the chambers of I.

3. Coverings

(3.1) Let V_C be the complexification of V, and M_C the complexification of M for $M \in \mathcal{M}$, and let

$$Y = V_C - \bigcup_{M \in \mathcal{M}} M_C$$

In this section we construct, by gluing, a space $\widetilde{Y}$ above Y. The gluing data will be described using I and $\widehat{I}$. The facets of S and the facets of V other than $\{0\}$ are in canonical bijection, and we shall constantly pass from one to the other. In general, we use the same notation for corresponding facets; when necessary, the correspondence will be denoted $F \longmapsto [F]$.

(3.2) Lemma. Let A, B, and C be three chambers of I, with $C \subset S(A) \cap S(B)$. Then q induces an isomorphism from $S(A) \cap S(B)$ onto the intersection of S with the closed half-spaces $D'_M(qC)$ containing qC and bounded by a wall M separating qA from qB.

The intersection of the $D'_M(qC)$ is a union of closed chambers. If C' is one of them, qA and qB lie on the same side of every wall separating qC from C'. It follows that the lifts to $S(A)$ and $S(B)$ of a minimal gallery from qC to C' coincide, and $(C')^- \subset q(S(A) \cap S(B))$.

Conversely, if a facet F is not in the intersection of the $D'_M(qC)$, there is a wall M such that $F \not\subset M$ which separates qA from qB and F from qC. Suppose, impossibly, that $F \subset q(S(A) \cap S(B))$. Let D be a chamber of S having F as a facet, and let $(C_0,\, C_1\, \ldots\, C_{n+1})$ be a gallery from qC to D. Let M_i be the wall separating C_i from C_{i+1}, and let α_i (respectively β_i) $= \pm 1$ according as C_{i+1} and A (respectively B) are or are not on the same side of M_i. By hypothesis, F is a facet of $C\Delta(M_0)^{\alpha_0} \ldots \Delta(M_n)^{\alpha_n}$ and of $C.\Delta(M_0)^{\beta_0} \ldots \Delta(M_n)^{\beta_n}$.

By (2.4) and (1.24), one then has (since $M \not\supset F$)

$$\sum\nolimits_{M_i = M} \alpha_i = \sum\nolimits_{M_i = M} \beta_i,$$

which is absurd: one side is one, the other -1.

(3.3) Lemma. Let $\mathcal{R}$ and $\mathcal{I}$ be the "real part" and "imaginary part" maps from V_C to V. Let $v \in V_C$, and let F be the facet of V such that $\mathcal{R}v \in F$. For v to belong to Y, it is necessary and sufficient that $pr_F(\mathcal{I}v)$ lie in a chamber of $(V_F,\, \mathcal{M}_F)$.

This is clear.

What follows has as its intuitive basis the fact that to every "walk" in I (made by following a gallery of the form $\Delta(M_1)^{\varepsilon_1} \ldots \Delta(M_k)^{\varepsilon_k}$, $\varepsilon_i = \pm 1$) there corresponds a path ch (a path class) in Y. The image under $\mathcal{R}$ of ch in V passes from chamber to chamber, successively crossing the walls $M_1, \ldots, M_k$ at points m_i. For $\varepsilon_i = 1$ (respectively -1), the path passes through one of the two components R of $\mathcal{R}^{-1}(m_i) \cap Y$ (in bijection under $\mathcal{I}$ with $V - M$) : the one for which $\mathcal{I}R$ contains (respectively does not contain) the preceding chamber.

(3.4) Notations. (i) Let C be a chamber of $(V,\, \mathcal{M})$. Denote by Y(C) the open subset of Y consisting of the $x \in V_C$ satisfying the following condition:

— if F is the facet of V such that $\mathcal{R}x \in F$, then $pr_F\, \mathcal{I}x \in \pi'_F(C)$.

(ii) Let A and B be two chambers of I. If $S(A) \cap S(B)$ contains a chamber C, temporarily denote by $Y'(A,\, B)$ the intersection of the open half-spaces $D'_M(qC)^\circ$, for M as in 3.2. Set

Y(A, B) $= Y(qA) \cap Y(qB) \cap \mathcal{R}^{-1}Y'(A,\, B) \subset V_C$.

If $S(A) \cap S(B)$ contains no chamber, set Y(A, B) $= \varnothing$.

(iii) For a facet F of V, its star $\mathrm{Et}(F)$ is the union of the open facets G of $(V,\, \mathcal{M})$ such that $F \subset G^-$.

(iv) For a facet F of V and a chamber C of $(V_F,\, \mathcal{M}_F)$, set

V(F, C) $= \{v \in V_C \mid \mathcal{R}v \in \mathrm{Et}(F)$ and $pr_F\, \mathcal{I}v \in C\}$.

(3.5) Lemma. Let G be a facet of V. The V(F, C), for F a facet such that $G \subset F^-$ and C a chamber of V_F, form an open covering of $Y \cap \mathcal{R}^{-1}(\mathrm{Et}\ G)$. In particular (take $G = \{0\}$), the V(F, C) cover Y.

If $x \in Y \cap \mathcal{R}^{-1}(\mathrm{Et}\ G)$ and $\mathcal{R}x \in F$, then x lies in one of the V(F, C) (3.3).

(3.6) Let Y' be the disjoint sum, indexed by the balls $b(A)$ of $\widehat{I}$,

$Y' = \coprod_{b(A)} \mathrm{Y(qA)}$.

Denote by $Y'(b(A))$ the component indexed by $b(A)$. There is an evident projection $p' : Y' \longrightarrow Y$. Let R be the following relation on Y' :

For $x \in Y'(b(A))$ and $y \in Y'(b(B))$, R(x, y) means that $p'(x) = p'(y)$ and that $p'(x)$, $p'(y) \in Y(A, B)$.

It is an equivalence relation because (cf. (3.2)) Y(A, B) $\cap\, Y(B, C) \subset Y(A, C)$. One has

(3.6.1) The R-saturation of an open subset of Y' is again open.

Set $\widetilde{Y} = Y'/R$. There is an evident projection

$$p : \widetilde{Y} \longrightarrow Y$$

For every ball $b(A)$ of $\widehat{I}$, denote by $\widetilde{Y}(b(A))$ the image of $Y'(b(A))$ in $\widetilde{Y}$.

The following theorem implies the one with which the introduction begins.

(3.7) Theorem. $\widetilde{Y}$ is a contractible covering of Y.

A. $\widetilde{Y}$ is a covering of Y.

We shall show, for V(F, C) as in (3.4)(iv), that $p^{-1}V(F, C)$ is a sum of copies of V(F, C).

a) Let $b(A)$ be a ball of $\widehat{I}$. For B a chamber such that $F \subset B^-$, let the following be the chamber of I:

$$C'_B = A.u(A, B)u(\pi_F(C), B)^{-1}$$

By construction, $B \subset q(S(A) \cap S(C'_B))$, and in particular, if $F \neq \{0\}$,

(3.7.1) $[F] \subset q(S(A) \cap S(C'_B))$.

We prove that

(3.7.2) $\widetilde{Y}(b(A)) \cap p^{-1}V(F, C) \subset \bigcup_{F \subset B^-} \widetilde{Y}(b(C'_B))$.

Let x be in the left-hand side. Let G be the facet of V to which $\mathcal{R}px$ belongs. One has $F \subset G^-$. Choose B to be a chamber having G as a facet. By hypothesis, one has

$pr_G\ \mathcal{I}px \in \pi'_G(A)$ and $pr_F\ \mathcal{I}px \in C$.

Thus $\pi'_G(A) = \pi'_G(\pi_F\ C)$. It then follows from (3.2) that every chamber having G as a facet lies in $q(S(A) \cap S(C'_B))$, that px lies in Y(A, C'_B), and hence that $x \in \widetilde{Y}(b(C'_B))$.

b) Set $Y'(F, C) = \bigcup_{q(C')=C}(Y'(b(C')) \cap p'^{-1}(V(F, C)))$. By a), set-theoretically,

$p^{-1}V(F, C) = Y'(F, \mathrm{C})/\mathrm{R}$.

This is even an isomorphism of topological spaces, by (3.6.1), since $Y'(F, C)$ is open in Y'. By (3.2), the equivalence relation

induced by R on $\bigcup_{q(C')=C} Y'(b(C'))$ is trivial. We conclude by observing that, for every C' above C,

V(F, C) $\subset Y(b(C'))$.

B. **An open covering of $\widetilde{Y}$.**

(3.7.3) Notations. (i) For a facet F of I,

$\widetilde{Y}(F) = \bigcup_{F \subset S(C)}(\widetilde{Y}(b(C)) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(qF))$.

(ii) For $o(A)$, the center of a ball $b(A)$ of $\widehat{I}$,

$\widetilde{Y}(o(A)) = \widetilde{Y}(b(A)) \cap p^{-1}V(\{0\}, \mathrm{A})$.

(3.7.4) Lemma. (i) The $\widetilde{Y}(s)$, for s a vertex of $\widehat{I}$, form a covering of $\widetilde{Y}$ whose nerve is $\widehat{I}$ (i.e., a family of $\widetilde{Y}(s)$ has nonempty intersection if and only if the corresponding s are the vertices of a facet of $\widehat{I}$).

(ii) If a facet F of I has vertices $s_0, \ldots, s_p$, then

$\widetilde{Y}(F) = \widetilde{Y}(s_0) \cap \cdots \cap \widetilde{Y}(s_p)$.

(iii) If $F \subset S(A)$, then $\widetilde{Y}(o(A)) \cap \widetilde{Y}(F)$ is contractible; $\widetilde{Y}(o(A))$ is contractible.

(3.7.4.1) Let F_i $(i = 1, 2)$ be two facets of I and C_i $(i = 1, 2)$ two chambers of I such that $F_i \subset S(C_i)$ and $(\widetilde{Y}(b(C_1)) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(qF_1)) \cap (\widetilde{Y}(b(C_2)) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(qF_2)) \neq \varnothing$.

Then F_1 and F_2 span a facet F of I (unique by (2.5) – (2.11)), $F \subset S(C_i)$, and the preceding intersection coincides with

$\widetilde{Y}(b(C_1)) \cap \widetilde{Y}(b(C_2)) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(F)$.

By hypothesis,

$Y(C_1, C_2) \cap \mathcal{R}^{-1}\,\mathrm{Et}(qF_1) \cap \mathcal{R}^{-1}\,\mathrm{Et}(qF_2) \neq \varnothing$;

F_1 and F_2 lie in $S(C_1) \cap S(C_2)$ and span a facet F there. Moreover, $\mathrm{Et}(qF_1) \cap \mathrm{Et}(qF_2) = \mathrm{Et}(qF)$, whence the stated formula.

From (3.7.4.1), one deduces that $\widetilde{Y}(F_1) \cap \widetilde{Y}(F_2)$ is empty if F_1 and F_2 do not span a common facet F, and otherwise equals

$\bigcap_{F \subset S(C)} \widetilde{Y}(bC) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(F) = \widetilde{Y}(F)$

(take $C_1 = C_2$). This proves (ii) and part of (i).

(3.7.4.2) If $o(A) \neq o(B)$, then $\widetilde{Y}(o(A)) \cap \widetilde{Y}(o(B)) = \varnothing$.

If $qA = qB$, one has $\widetilde{Y}(b(A)) \cap \widetilde{Y}(b(B)) = \varnothing$. If $qA \neq qB$, one has

$p\widetilde{Y}(o(A)) \cap p\widetilde{Y}(o(B)) = \varnothing$.

(3.7.4.3) If $F \not\subset S(A)$, then $\widetilde{Y}(o(A)) \cap \widetilde{Y}(F) = \varnothing$.

If $F \subset S(B)$, then $F \not\subset S(A) \cap S(B)$, and

$\mathrm{Et}(qF) \cap Y(A, B) = \mathrm{Et}(qF) \cap p(\widetilde{Y}(b(A)) \cap \widetilde{Y}(b(B))) = \varnothing$.

It is clear that:

(3.7.4.4) If $F \subset S(A)$, then p induces an isomorphism from $\widetilde{Y}(o(A)) \cap \widetilde{Y}(F)$ onto $Y(qA) \cap \mathcal{R}^{-1}\,\mathrm{Et}(qF)$.

Lemma (3.7.4) then follows from:

(3.7.5) Lemma. For every facet F of V and every chamber A,

$Y(A) \cap \mathcal{R}^{-1}\,\mathrm{Et}(F)$

is contractible.

Let v_0 be a vector such that $\mathcal{R}v_0 \in F$ and $\mathcal{I}v_0 \in A$. Then, for $0 \leq t \leq 1$, $\varphi_t : v \longmapsto v_0 + t(v - v_0)$ is a map of Y(A) and of $\mathcal{R}^{-1}\,\mathrm{Et}(F)$ into themselves. One has $\varphi_0(Y(A) \cap \mathcal{R}^{-1}\,\mathrm{Et}(F)) = \{v_0\}$ and $\varphi_1 = \mathrm{Id}$.

C. **Study of $\widetilde{Y}(F)$ (F a facet of I).**

By (3.5), (3.7.1), and (3.7.2), one has

(3.7.6) $\widetilde{Y}(F) = \bigcup_{F \subset B^-} \widetilde{Y}(b(B)) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(F)$.

Choose a "fundamental chamber" $A_{F,0}$ in $(V_F, \mathcal{M}_F)$, and denote the corresponding building by I_F. Also choose $\widetilde{A}^I_{F,0}$ in I above $\pi_F(A_{F,0})$. For every chamber $B = \widetilde{A}_0.g_F$ of I_F, denote by $\pi_F(B)$ the chamber $\widetilde{A}^I_{F,0}.\pi_F(g_F)$ of I. By (2.10), (3.7.6) may be rewritten

$$\widetilde{Y}(F) = \bigcup_{B \text{ in } I_F} \widetilde{Y}(b(\pi_F(B))) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(F).$$

Let

$$Y_F = (V_F)_C - \bigcup_{M \in \mathcal{M}_F} M_C,$$

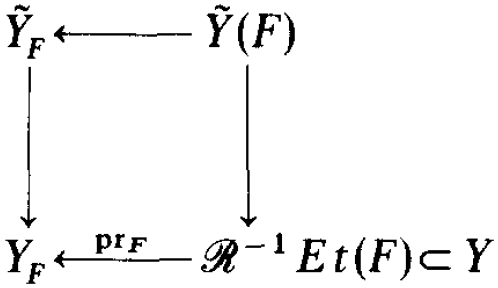
and let $\widetilde{Y}_F$ be the covering of Y_F defined using I_F.

For any B and C in I_F, it follows from (1.30) that the chambers of S in $Y(\pi_F(B), \pi_F(C)) \cap \mathrm{Et}(F)$ are exactly the $\pi_F(D)$, for D in $Y_F(B, C)$.

The gluings defining $\widetilde{Y}(F)$ and $\widetilde{Y}_F$ are therefore compatible, and there is a unique Cartesian diagram

(3.7.7)

(3.7.7)



which sends $\widetilde{Y}(b(\pi_F(B))) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(F)$ into $\widetilde{Y}_F(b(B))$ for B a chamber of I_F. The first vertical arrow therefore also the second are coverings.

It is easily verified that the map pr_F of (3.7.7) is a homotopy equivalence, whence

(3.7.8) Lemma. *$\widetilde{Y}_F$ and $\widetilde{Y}(F)$ have the same homotopy type.*

D. **End of the proof.**

We prove (3.7) by induction on $r = dim\ V$. Let $\mathcal{U}$ be the open covering of $\widetilde{Y}$ by the $\widetilde{Y}(s)$, for s a vertex of $\widehat{I}$. By (3.7.4)(i), $\widehat{I}$ is the nerve of $\mathcal{U}$. By (3.7.4), (3.7.8), and the induction hypothesis, the nonempty intersections of open sets belonging to $\mathcal{U}$ are contractible. This implies [6] that $\widetilde{Y}$ and $\widehat{I} = \mathrm{Nerf}(\mathcal{U})$ have the same homotopy type. The conclusion follows from (2.15).

4. Braid groups

(4.1) Let V be a finite-dimensional real vector space and $W \subset \mathrm{GL}(V)$ a finite group generated by reflections. Assume that $V^W = \{0\}$. Let Φ be any Euclidean structure on V invariant under W, and let $\mathcal{M}$ be the set of hyperplanes M such that the orthogonal reflection with respect to $\mathcal{M}$ belongs to W. One knows that

a) $(V, \mathcal{M})$ satisfies 1.6 ([1] V 3.9 prop. 7);

b) W permutes the chambers of $(V, \mathcal{M})$ in a strictly simply transitive manner ([1] V 3.2 Th. 1);

c) if $x \in V$ belongs to the closed chamber $\overline{A}$, its stabilizer is generated by the reflections in the walls of A containing x ([1] V 3.3 prop. 1);

d) (follows from c)) the group W acts freely on

$$Y_W = V_C - \bigcup_{M\in\mathcal{M}} M_C.$$

By b), if C_0 and C_1 are two chambers, there is a unique $w \in W$ such that $wC_0 = C_1$; these w induce a transitive system of bijections between the walls of the different chambers. Passing to the quotient, one obtains a set D with dim(V) elements and, for each chamber C, a bijection φ_C from D onto the set of walls of C. One has

(4.1.1) $\varphi_{wC}(i) = w\varphi_C(i)$ $(w \in W)$.

For the definition of the Coxeter matrix $(m_{ij})_{i,j\in D}$ and that of the Coxeter graph, we refer to [1] V 3.4.

(4.2) Choose a chamber A_0 of $(V, \mathcal{M})$. For $i \in D$, let w_i be the reflection in the wall $\varphi_{A_0}(i)$. [1] V 3.2 Th. 1 says that W is generated by the w_i, these generators being subject only to the relations

$(w_iw_j)^{m_{ij}} = e$.

(4.3) Let $X_W = Y_W/W$: by (4.1)d), Y_W is a covering of X_W, with group W. Let $y_0 \in A_0$ have image $x_0 \in X_W$. For each $i \in D$, let ℓ_i' be the homotopy class of paths in Y_W, from y_0 to $w_i(y_0)$, which, for $y \in A_0$, contains the broken line with successive vertices y_0, $y_0 + iy_0$, $w_i(y_0) + iy_0$, $w_i(y_0)$. The image ℓ_i of ℓ_i' in X_W is a loop based at x_0.

(4.4) Theorem. (i) *The fundamental group $\pi_1(X_W, x_0)$ is generated by the ℓ_i; these are subject only to the relations*

(4.4.1) $\mathrm{prod}(m_{ij};\, \ell_i,\, \ell_j) = \mathrm{prod}(m_{ji};\, \ell_j,\, \ell_i)$.

(ii) The universal covering of X_W is contractible: X_W is a $K(\pi;\, 1)$.

Assertion (i) is proved in Brieskorn [3]. Another proof will be sketched in (4.18) below. As explained in the introduction, (ii) follows from (3.7) and (4.1).

(4.5) In § 1, we have drawn heavily on Garside's study [4] of Artin's braid group; from the results obtained we can now derive information about all groups of the type $\pi_1(X_W,\, x_0)$ (Garside had already considered some of them). The translation rests on (4.6), (4.7) below.

(4.6) Let $G = (C_0,\, \ldots,\, C_n)$ be a gallery and M_i the wall separating C_{i-1} from C_i $(1 \leq i \leq n)$. One has $\varphi_{C_{i-1}}^{-1}(M_i) = \varphi_{C_i}^{-1}(M_i)$. Let $\ell(G)$ be the element $\varphi_{C_1}^{-1}(M_1) \ldots \varphi_{C_n}^{-1}(M_n)$ of $L^+(D)$ (0.1).

The map ℓ induces a bijection from the set of galleries G with a given source to $L^+(D)$. One has

(4.6.1) $\ell(w(G)) = \ell(G)$ (for $w \in W$);

(4.6.2) $\ell(G_0 G_1) = \ell(G_0)\ell(G_1)$ (for composable G_0 and G_1).

Let $w : L(D) \longrightarrow W$ be the homomorphism extending the map $i \longmapsto w_i$ from D to W.

(4.7) Proposition. For every gallery G with source A_0, one has

$A_0.G = w(\ell(G))(A_0)$.

This is an avatar of the identity

$[(w_1 \ldots w_{n-1}) w_n (w_1 \ldots w_{n-1})^{-1}]$

$\cdot [(w_1 \ldots w_{n-2}) w_{n-1} (w_1 \ldots w_{n-2})^{-1}] \ldots [w_1] = w_1 \ldots w_n$.

(4.8) From this proposition one deduces that, for $w \in W$, the integer $d(A_0,\, w(A_0))$ is the least length of words in the w_i that equal w; more precisely, ℓ induces a bijection from the set of minimal galleries from A_0 to wA_0 to the set of words $m \in L(D)$, of minimal length, such that $w = w(m)$.

(4.9) Definition. (i) The braid monoid G^+ of type D is the monoid with unit generated by D, the generators $i \in D$ being subject only to the relations

(4.9.1) $\mathrm{prod}(m_{ij};\, \mathrm{i},\, j) = \mathrm{prod}(m_{ij};\, \mathrm{j},\, \mathrm{i})$.

(ii) The braid group G (or generalized braid group) of type D is the group generated by D, the generators being subject to (4.9.1).

(4.10) Notice that W also admits the presentation

(4.10.1) $\mathrm{prod}(m_{ij};\, w_i,\, w_j) = \mathrm{prod}(m_{ij};\, w_j,\, w_i)$,

(4.10.2) $w_i^2 = e$.

so that $i \longmapsto w_i$ extends to a homomorphism from G to W.

(4.11) One checks from the definitions that two galleries G_0 and G_1 with the same source are equivalent if and only if $\ell(G_0)$ and $\ell(G_1)$ have the same image in G^+. Taking account of (4.9), (1.12) therefore admits the following translation ([1] IV 1.6 prop. 5).

(4.12) Reminder. Let $w \in W$. The image in G^+ of the words $m \in L(D)$, of minimal length among those such that $w = w(m)$, depends only on w.

This image will be denoted $r(w)$; r is a section of $w : G^+ \longrightarrow W$. By (4.7), one has $r(w) = \ell(u(A_0,\, wA_0))$.

Similarly, [1] IV 1.4 lemma 2 corresponds to (1.11).

(4.13) The "quotient" category of $\mathrm{Gal}_+(V, \mathcal{M})$ by W is defined as follows:
a) Its set of objects is reduced to one element: it is
$\mathrm{Ob}(\mathrm{Gal}_+(V, \mathcal{M}))/W$.
b) The monoid of its arrows is $\mathrm{Fl}(\mathrm{Gal}_+(V, \mathcal{M}))/W$, and passage to the quotient
$\mathrm{Gal}_+(V, \mathcal{M}) \longrightarrow \mathrm{Gal}_+(V, \mathcal{M})/W$
is a functor.

This category, having only one object, is identified with the monoid of its arrows. By (4.6.1), (4.6.2), and (4.11), ℓ identifies this monoid with G^+.

Similarly one defines $\mathrm{Gal}(V, \mathcal{M})/W$. This category has only one object and, by its universal property, ℓ extends to an isomorphism from the group of its arrows to G :

$$\begin{array}{ccccc} \mathrm{Gal}_+(V, \mathcal{M}) & \longrightarrow & \mathrm{Gal}_+(V, \mathcal{M})/W & \xrightarrow[\ell]{\sim} & G^+ \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Gal}(V, \mathcal{M}) & \longrightarrow & \mathrm{Gal}(V, \mathcal{M})/W & \xrightarrow[\ell]{\sim} & G \end{array}$$

(4.14) Theorem. (i) In G^+, left and right translations are injective.
(ii) G^+ satisfies the Öre condition on the left and on the right. It follows from (i) that G^+ embeds in G.
(iii) For J a subset of D, let G_J^+ (resp. G_J) be the submonoid (resp. subgroup) of G^+ (resp. G) generated by the $i \in J$. Then the evident map identifies G_J^+ (resp. G_J) with the braid monoid (resp. braid group) of type J.
(iv) One has $G_J^+ = G_J \cap G^+$.
(v) If J and K are two subsets of D, one has $G_J \cap G_K = G_{J \cap K}$.
(vi) Let n(i) be the number of elements of G^+ of length i, and put $f = \sum n(i)t^i$. For $J \subset D$, let m(J) be the number of walls passing through the intersection of the walls of an (arbitrary) chamber A, indexed by J (= *the* number of reflections in the Weyl group W_J). One has

$$f = (\sum {}_{J \subset D}(-1)^{|J|} t^{m(J)})^{-1}$$

These assertions translate respectively: (i): (1.1)(i) and (ii); (ii): (1.26)(ii) and its transform by passage to opposite galleries (cf. also (4.16) below); (iii): (1.30); (iv): (1.32); (v) : (1.31); (vi): (1.21) (the f_A of loc. cit. are all equal to f).

Let x, $y \in G^+$. We say that x begins with y if there exists z in G^+ with $x = yz$. Proposition (1.19)(iii) translates as follows

(4.15) Lemma. Let $x \in G^+$. There exists a chamber C such that, for every $w \in W$, x begins with $r(w)$ if and only if $wA_0 \in D(A_0, C)$.

(4.16) Let J be a subset of D and P the intersection of the walls $\varphi_{A_0}(i)$ $(i \in J)$. Let $\Delta(J)$ be the element $\ell(u(A_0, A_0.\Delta(P)))$ of G^+; put $\Delta = \Delta(D)$. Let $i \longmapsto \bar{\imath} = \varphi_{-C}^{-1}\varphi_C(i)$ be the opposition involution of D. We also denote by $w \longmapsto \overline{w}$ the automorphisms of $L^+(D)$, G^+, and G that send i to $\bar{\imath}$ (one has $m_{ij} = m_{\overline{ij}}$). Translating (1.26) with the aid of (4.4), one finds that for g in G^+ or G,

(4.16.1) $g\Delta = \Delta\overline{g}$.

In particular, Δ^2 is central.

For every $i \in D$, Δ "begins" with i: one has $\Delta = ix$, in G^+. Hence

(4.17) Proposition. G is obtained from G^+ by making the central element Δ^2 invertible.

(4.18) The description (2.11), (2.5) – (2.7) of the building I attached to $(V, \mathcal{M})$ can be translated as follows.
a) The set of vertices of I is

$$I^0 = \coprod_{i \in D} G/G_{D-i}$$

b) For a set of vertices to span a simplex, it is necessary and sufficient that it be contained in the set of classes gG_{D-i}, for some $g \in G$.

c) The fundamental sphere $S_0 = S(A_0)$ is the union of the fundamental simplex, spanned by the lateral classes of e in the G/G_{D-i}, and of its transforms by the $r(w)$ $(w \in W)$.

The preceding description makes evident a left action of G on I. It respects the set of spheres $S(A)$ and the correspondence $A \longmapsto S(A)$.

By transport of structure, G acts on the covering $\widetilde{Y}_W$ of Y_W (3.6), and one has an equivariant morphism $(G - \widetilde{Y}_W) \longrightarrow (W - Y_W)$.

It follows that G is the fundamental group of $X_W = Y_W/W$, and (4.4)(i).

(4.19) For completeness, let us show that the word problem, the conjugacy problem, and the question of computing the center of G can be solved as in [4]. If the Coxeter graph D of W is a disjoint union of graphs D_α, the braid group of type D is the product of the braid groups of types D_α. The essential case is therefore that of a connected graph D.

(4.20) The word problem: there is a decision procedure for determining whether two elements m and n of L(D) have the same image in G.

a) Whether m and n in $L^+(D)$ have the same image in G^+ is decidable, because the imposed relations do not change the length of the words, and there are only finitely many words of a given length. One may also use (1.22).

b) For any m_i $(i = 1, 2)$ in the free group L(D) generated by D, one can compute $m_i' \in L^+(D)$ and $k \geq 0$ such that m_i and $m_i'\Delta^{-k}$ have the same image in G (cf. (4.17)); m_1 and m_2 then have the same image in G if and only if m_1' and m_2' have the same image in G^+.

(4.21) Theorem. If the Coxeter graph D is connected (nonempty) and the opposition involution is trivial (resp. nontrivial), the center of G is infinite monogenic, generated by Δ (resp. Δ^2).

Consider the following property of $x \in G^+$.

(*) For every $i \in D$, there exists $i' \in D$ such that $ix = xi'$.

It is enough to prove that, if x satisfies (*) and $x \neq e$, then $x = \Delta y$ with $y \in G^+$. By induction on the length of x, it will follow that if x satisfies (*), x is of the form Δ^k. In particular, the center of G^+ is contained in the set of the Δ^k $(k \geq 0)$; by (4.14)(i) and (4.17), the center of G is contained in the set of the Δ^k $(k \in \mathbb{Z})$, and the assertion follows from (4.16.1).

Let $x \in G^+$ satisfy (*). Let C be as in (4.15), and let w be the image in W of $\ell u(A_0, C)$. We may write $x = r(w)y$, with $y \in G^+$. Let $i \in D$. By (*), $ix = r(w)yi'$ begins with $r(w)$; by (1.23), it follows that ir(w) begins with $r(w) : ir(w) = r(w)i''$. In particular, for every i, there exists i'' with $iw = wi''$; this means that every wall of A_0 is also a wall of wA_0. Among the $2^{|D|}$ open simplicial cones bounded by the walls of A_0, only A_0 and $-A_0$ are chambers: they are the only ones whose angles between the faces are all $\leq \pi/2$ (the connectedness of D is used here). If $x \neq e$, then $w \neq e$ and hence $wA_0 = -A_0$. Since then $\Delta = r(w)$, this completes the proof.

(4.22) The derived group. Let D' be the graph having D as its set of vertices, two vertices being joined by an edge if and only if m_{ij} is odd. Let D_0 be the set of connected components of D'. Define an epimorphism

Lg: $G \longrightarrow \mathbb{Z}^{D_0}$

by sending i to the basis vector indexed by the connected component of i in D'. One verifies at once that Lg identifies $\mathbb{Z}^{D_0}$ with the largest abelian quotient of G.

(4.23) The conjugacy problem. The problem is to give a decision procedure for determining whether two elements x and y of G (given explicitly as images of words in L(D)) are conjugate. Since Δ^2 is central, $\Delta^{-2k}a$ and $\Delta^{-2k}b$ are conjugate if and only if a and b are. By

(4.17), it therefore suffices to treat the case where x, $y \in G^+$. Similarly, if x and y are conjugate, there exists a in G^+ such that $xa = ay$. If x and y are conjugate, $\mathrm{Lg}(x) = \mathrm{Lg}(y)$. There are only finitely many $x \in G^+$ of given "length" $\mathrm{Lg}(x)$, and W is finite. The existence of a decision procedure then follows from (4.20) and the following lemma.

(4.24) Lemma. Let x, y be in G^+. If x and y are conjugate, there exists a sequence $x_0 = x$, x_1, ..., $x_n = y$ of elements of G^+ and a sequence of elements w_i of W such that

$x_{i+1} r(w_i) = r(w_i) x_i$.

Let $a \in G^+$ be such that $xa = ay$. Put $a = r(w_0) \dots r(w_{n-1})$, where w_i is the element of W of maximum length such that $r(w_i) \dots r(w_{n-1})$ can be written in the form $r(w_i) b_i$ with $b_i \in G^+$ (4.15). Let $a_i = r(w_0) \dots r(w_i)$. We shall prove that $x_i = a_i^{-1} x a_i$ lies in G^+, so that the x_i and w_i answer the problem.

It follows from (1.23) that, for every $w \in W$, if xa begins with $r(w)$, then $xr(w_0)$ also begins with $r(w)$. In particular, since $xa = ay = r(w_0) b_0 y$ begins with $r(w_0)$, $xr(w_0) = r(w_0) x_1$ with $x_1 \in G^+$. The proof is completed by induction on n.

Remark. Let σ be an automorphism of the Coxeter graph D, and again denote by σ the corresponding automorphism of G. One has $\Delta^\sigma = \Delta$. The preceding arguments still apply to the question of deciding, for x, $y \in G$, whether there exists a in G such that $xa = a^\sigma y$. We reduce to taking x, y, and a in G^+. In that case and, with the preceding notation, if $xa = a^\sigma y$, the $(a_i^{-1})^\sigma x a_i$ lie in G^+. Taking for σ the involution $i \longmapsto \bar{\imath}$, one obtains a criterion analogous to (4.24) for the conjugacy of Δx and Δy (x, $y \in G^+$).

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Pierre Deligne
Institut des Hautes Études Scientifiques
35 Route de Chartres
F-91440 Bures-sur-Yvette
France

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UNIRATIONAL NON-RATIONAL VARIETIES

[after M. ARTIN and D. MUMFORD]
by Pierre DELIGNE

Editorial note. This independent English edition follows the frozen French text and the IAS authority. Mathematical notation, cross-references, and typographical peculiarities of the printed source are retained rather than silently corrected. The seven source objects are reproduced from authority-grounded image derivatives. Running headers and printed page numbers are excluded from the body.

This exposé contains a description of the variety constructed by Artin and Mumford, and the proof that it is unirational and non-rational. It also contains the statements of the theorems proved by Clemens and Griffiths [4], and by Manin and Iskovskih [6], for constructing other examples. I attended talks by Artin and Mumford on the same subject and drew heavily on them.

In everything that follows, the expression "algebraic variety" will mean "a separated scheme of finite type over $\mathbb{C}$, reduced and irreducible". We shall write $k(X)$ for the field of rational functions on an algebraic variety X.

1. Rationality and unirationality

Let K be an extension of finite type of $\mathbb{C}$. There then exists an algebraic variety X such that K is isomorphic to $k(X)$. X is called a *model* of K. By [5], K always admits a projective and smooth model.

Let K and L be two extensions of finite type of $\mathbb{C}$, with models X and Y. The homomorphisms $a^* : K \hookrightarrow L$ correspond bijectively to the dominant rational maps $a_* : Y \dashrightarrow X$, by the formula $a^*(f) = f \circ a_*$. By [5], if X and Y are projective and smooth, every rational map $a_* : Y \dashrightarrow X$ admits a decomposition

$$a = b\, c_n^{-1} \cdots c_1^{-1}$$

(1.1)

$$\begin{array}{ccccccccc} Y_{n+1} & \xrightarrow{c_n} & \cdots & \to & Y_2 & \xrightarrow{c_1} & Y_1 & = & Y \\ \downarrow{\scriptstyle b} & & & & & & & & \\ X & & & & & & & & \end{array}$$

where b and the c_i are morphisms (defined everywhere), and where Y_{i+1} is obtained from Y_i by blowing up a smooth subvariety Z_i (of codimension ≥ 2).

If $a^* : K \to L$ is an isomorphism, the preceding result can be applied to $((a^*)^{-1}, Y_{n+1}, X)$. By iteration, one obtains a commutative diagram of morphisms of schemes

(1.2)

$$\begin{array}{ccccccccc} \cdots & Y^2 & \xrightarrow{y^2} & Y^1 & = & Y^1 & \xrightarrow{y^1} & Y & \\ & \downarrow{\scriptstyle a^2} & & \uparrow{\scriptstyle b^1} & & \downarrow{\scriptstyle a^1} & & \downarrow{\scriptstyle a} & \\ \cdots & X^1 & = & X^1 & \xrightarrow{x^1} & X & = & X & . \end{array}$$

In this diagram, the horizontal arrows x^i, y^i are composites of blow-ups with nonsingular centers, as in (1.1); the a^i and b^i are birational.

An invariant of a projective and smooth variety X is called *birational* if it depends only on $k(X)$. The birational character of an invariant is often verified with the aid of (1.2) (for a typical example, see 2.1). Invariants usable for a field K are generally defined as the value of a birational invariant of any projective and smooth model of K.

Definition 1.3.- Let X be an algebraic variety of dimension n. We say that X is unirational if the following conditions are satisfied:

(i) $k(X)$ is a subfield of a purely transcendental extension $\mathbb{C}(T_1, \ldots, T_r)$ of $\mathbb{C}$;

(ii) a finite extension of $k(X)$ is isomorphic to $\mathbb{C}(T_1, \ldots, T_n)$;

(iii) (for X projective and smooth), there exists a diagram (1.1) with $Y = \mathbb{P}^r(\mathbb{C})$;

(iv) (for X projective and smooth), there exists a diagram (1.1) with $Y = \mathbb{P}^n(\mathbb{C})$.

One may suppose X projective and smooth, and one has

$$\text{(ii)} \Rightarrow \text{(i)} \Leftrightarrow \text{(iii)} \Rightarrow \text{(iv)} \Leftrightarrow \text{(ii)}.$$

Definition 1.4.- We say that X is rational if, alternatively,

(i) $k(X)$ is a purely transcendental extension of $\mathbb{C}$;

(ii) (for X projective and smooth), there exists a diagram (1.2) with $Y = \mathbb{P}^n(\mathbb{C})$.

The most evident birational invariants do not permit one to distinguish unirational varieties from rational varieties.

Proposition 1.5.- Let X be a projective and smooth unirational algebraic variety of dimension $n > 0$.

(i) The plurigenera

$$P_k = \dim H^0\big(X, (\Omega^n_X)^{\otimes k}\big) \qquad (k > 0)$$

vanish. More generally, $H^0\big(X, (\Omega^1_X)^{\otimes k}\big) = 0$ for $k > 0$ (see [9], Remark, page 02).

(ii) X is simply connected ([10]).

For $n = 1$ or 2, property (i) above already characterizes rational varieties. A unirational variety of dimension ≤ 2 is therefore rational, and a subextension of transcendence degree ≤ 2 of $\mathbb{C}(T_1, \ldots, T_k)$ is automatically purely transcendental. More precisely, one has the following:

a) For $n = 1$, if $H^0(X, \Omega^1_X) = 0$, then $X \simeq \mathbb{P}^1(\mathbb{C})$. By the Riemann–Roch theorem, for every $x \in X$, $\mathcal{O}(x)$ indeed defines an embedding of X into $\mathbb{P}^1(\mathbb{C})$.

b) For $n = 2$, the Castelnuovo–Enriques criterion asserts that X is rational if $P_a \simeq P_2 = 0$, i.e. if

$$H^0(X, \Omega^1_X) = H^0\big(X, (\Omega^2_X)^{\otimes 2}\big) = 0$$

(see [9]).

The symbols $P_a \simeq P_2$ and Ω^1_X in this criterion reproduce the printed source.

For $n > 2$, it has been very difficult to define or calculate birational invariants capable of distinguishing between rational and unirational varieties.

2. Some birational invariants

A) Artin and Mumford

Proposition 2.1.- The torsion subgroup $H^3(X, \mathbb{Z})_{\text{tors}}$ of $H^3(X, \mathbb{Z})$ is a birational invariant of the projective and smooth variety X.

If a variety X_2 is obtained from a smooth variety X_1 by blowing up a smooth subvariety $Z \subset X_1$ of codimension $r \geq 2$, then (SGA 5 VII)

$$H^n(X_2, \mathbb{Z}) \simeq H^n(X_1, \mathbb{Z}) \oplus \bigoplus_{1}^{r-1} H^{n-2i}(Z, \mathbb{Z}).$$

The lower summation index 1, without an explicit $i =$, is retained from the printed formula. In particular,

$$H^3(X_2, \mathbb{Z}) \simeq H^3(X_1, \mathbb{Z}) \oplus H^1(Z, \mathbb{Z}).$$

Since $H^1(Z, \mathbb{Z})$ is torsion-free, one therefore has

$$H^3(X_1, \mathbb{Z})_{\text{tors}} \simeq H^3(X_2, \mathbb{Z})_{\text{tors}}.$$

Suppose that $k(X)$ is isomorphic to $k(Y)$, and let there be a diagram (1.2) From it one obtains a diagram

$$\begin{array}{ccccccc}
H^3(X,\mathbb{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(X_1,\mathbb{Z})_{\text{tors}} & = & H^3(X_1,\mathbb{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(X_2,\mathbb{Z})_{\text{tors}} \\
 & & \uparrow a^{1*} & & \downarrow b^{1*} & & \uparrow a^{2*} \\
H^3(Y,\mathbb{Z})_{\text{tors}} & = & H^3(Y,\mathbb{Z})_{\text{tors}} & \xrightarrow{\sim} & H^3(Y_1,\mathbb{Z})_{\text{tors}} & = & H^3(Y_1,\mathbb{Z})_{\text{tors}} \,.
\end{array}$$

Since $a_2^* b_1^*$ is bijective, b_1^* is injective. Since $b_1^* a_1^*$ is bijective and b_1^* injective, a_1^* is bijective, and the assertion follows.

The example of Artin and Mumford is based on the

Construction 2.2.- We shall construct a unirational variety X, projective and smooth, of dimension 3, such that

$$H^3(X,\mathbb{Z})_{\text{tors}} \neq 0 \qquad \big(\text{we shall have } H^3(X,\mathbb{Z}) \simeq \mathbb{Z}/(2)\big).$$

Multiplying this variety by $\mathbb{P}^r(\mathbb{C})$, one obtains an analogous example in every dimension ≥ 3.

B) Clemens and Griffiths

Let X be a projective and smooth variety of dimension 3, such that $H^0(X,\Omega_X^3) = 0$. The Hodge decomposition of $H^3(X,\mathbb{C})$ then reduces to

$$H^3(X,\mathbb{C}) = H^{2,1} \oplus H^{1,2}$$

and, according to Weil [11], the complex torus

$$J(X) = H^3(X,\mathbb{Z}) \backslash H^3(X,\mathbb{C}) / H^{2,1}$$

is an abelian variety: the *intermediate Jacobian* of X. Let

$$\psi : H^3(X,\mathbb{Z}) \otimes H^3(X,\mathbb{Z}) \longrightarrow H^6(X,\mathbb{Z}) \simeq \mathbb{Z}$$

be the cup product. By the Poincaré duality theorem, the alternating form on $H^3(X,\mathbb{Z})/\text{torsion}$ deduced from ψ has discriminant one. If $H^1(X,\mathbb{Z}) = 0$, it follows from Weil [11] that this form defines a principal polarization on $J(X)$.

Proposition 2.3 ([4] 3.26).- If X is rational, the polarized abelian variety $J(X)$ is a product of Jacobians.

In the proof sketch that follows, the sign $\simeq$ denotes an isomorphism of *polarized* abelian varieties. One verifies successively:

a) $J(\mathbb{P}^3(\mathbb{C})) = 0$.

b) If Y'' is obtained from the nonsingular projective variety Y' by blowing up a smooth curve Z with Jacobian $J(Z)$ (respectively, a point), one has ([4] 3.11)

$$J(Y'') \simeq J(Y') \times J(Z) \qquad \big(\text{respectively, } J(Y'') \simeq J(Y')\big).$$

c) If a morphism $b : Y' \to X$ is birational, $J(X)$ is a direct factor in $J(Y')$: there exists a principally polarized abelian variety of the principal series B such that

$$J(Y') \simeq J(X) \times B.$$

d) Let $(A_i)_{1 \leq i \leq n}$ be nonzero principally polarized abelian varieties. Suppose that no A_i admits a decomposition $A_i \simeq A_i' \times A_i''$ (for example, that A_i is a Jacobian). Then every decomposition of $A = \prod A_i$ is of the form

$$A \simeq \left(\prod_{i \in I} A_i\right) \times \left(\prod_{i \notin I} A_i\right) \quad \text{for } I \subset [1,n] \quad ([4]\ 3.23).$$

If X is rational, there exists a diagram 2.1 with $Y = \mathbb{P}^3(\mathbb{C})$ and b birational. By a) and b), $J(Y_{n+1})$ is a product of Jacobians. By c) and d), $J(X)$ is one as well. The reference "diagram 2.1" reproduces the printed source.

Take X to be a cubic hypersurface in $\mathbb{P}^4(\mathbb{C})$. One has

$$H^1(X,\mathbb{Z}) = H^3(X,\mathbb{Z})_{\text{tors}} = H^0(X,\Omega^3_X) = 0, \qquad \dim J(X) = \dim H^1(X,\Omega^2_X) = 5,$$

and X is unirational by [7]. Let S be the surface parametrizing the lines contained in X, and let $\alpha : S \to \text{Alb}(S)$ be the map of S into its Albanese variety. One of the essential results of [4] is the following

THEOREM 2.4 ([4] 11.19 AND 13.4).- There exists an isomorphism

$$J(X) \xrightarrow{\sim} \text{Alb}(S),$$

which carries the divisor Θ of $J(X)$ to the image of $S \times S$ under the map

$$(x,y) \xrightarrow{\delta} \alpha(x) - \alpha(y).$$

Clemens and Griffiths then manage to use results of [1] to prove that $(\text{Alb}(S), \delta(S \times S))$ cannot be the polarized Jacobian of a curve. Another proof of the irrationality of X is suggested in the appendix to [4]. It is a matter of proving

THEOREM 2.5.- The singular locus of the divisor Θ of $J(X)$ has dimension 0, hence codimension > 4. It is even reduced to the image under δ of the diagonal of $S \times S$.

It is known that this is never the case for the divisor Θ of a Jacobian.

C) Manin and Iskovskih

The group of birational automorphisms of a variety X (equal to the group of $\mathbb{C}$-automorphisms of $k(X)$) is a birational invariant of X. It is this invariant that Manin and Iskovski use to show that a smooth quartic hypersurface in $\mathbb{P}^4(\mathbb{C})$ is never rational:

THEOREM 2.6 ([6]).- Let X and Y be two smooth quartic hypersurfaces in $\mathbb{P}^4(\mathbb{C})$. Every birational isomorphism $a : X \longrightarrow Y$ is automatically biregular.

Since certain quartic hypersurfaces are unirational, [7] this theorem provides a new example of unirational non-rational varieties.

3. The example of Artin and Mumford

Consider in $\mathbb{P}^2(\mathbb{C})$ a configuration consisting of

a) two smooth cubic curves C_1 and C_2, with equations $f_1 = 0$ and $f_2 = 0$, meeting transversely;

b) a smooth conic Q, with equation $q = 0$, which meets each C_i in three distinct points of tangency.

We shall construct

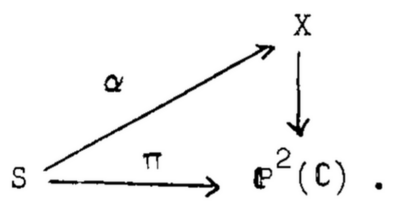
a) a vector bundle V of rank 3 on $\mathbb{P}^2(\mathbb{C})$;

b) a quadratic form $\Phi : V \to \mathcal{L}$ ($\mathcal{L}$ an invertible sheaf) of discriminant $f_1 f_2$ (if Δ, a section of $\mathcal{L}^{\otimes 3} \otimes (\bigwedge^3 V)^{\otimes(-2)}$, is the discriminant, the subscheme of $\mathbb{P}^2(\mathbb{C})$ with equation $\Delta = 0$ is $C_1 \cup C_2$).

Let $X \subset \mathbb{P}(V^*)$ be the conic bundle over $\mathbb{P}^2(\mathbb{C})$, degenerating along $C_1 \cup C_2$, with homogeneous equation $\Phi = 0$. The following conditions will be satisfied.

c) Φ has rank ≥ 2 at every point: for $s \in C_1 \cup C_2$, the fibre $X_s = f^{-1}(s)$ is the union of two concurrent lines; the only singular points of X are the singular points of the fibres X_s for $s \in C_1 \cap C_2$.

d) Let S be the double cover of $\mathbb{P}^2(\mathbb{C})$, ramified along Q, and let $X_S = X \times_{\mathbb{P}^2(\mathbb{C})} S$. Then X_S/S has a section



e) For $s \in (C_1 \cup C_2) - Q$, the two points $\alpha(t)$ for which $\pi(t) = s$ lie in the two irreducible components of X_s.

Locally (for the usual topology) on $C_1 \cup C_2$, the restriction of the bundle X to $C_1 \cup C_2$ is obtained by gluing two projective-line bundles along a section. However, when one traverses a loop in $C_1 \cup C_2$, these two bundles may be exchanged: there exists an unramified double cover $(C_1 \cup C_2)^*$ of $C_1 \cup C_2$ whose two local sections correspond, locally, to the two projective-line bundles of which $X|_{C_1 \cup C_2}$ is the union. Let $\pi_i : C_i^* \to C_i$ be the double cover of C_i induced by $(C_1 \cup C_2)^*$. We shall use from e) only the following corollary

LEMMA 3.1.- C_i^* is irreducible.

By e), C_i^* is the normalization of the double cover of C_i induced by S. The lemma follows from the fact that the section q of $\mathcal{O}(2)|_{C_i}$ is not the square of a section of $\mathcal{O}(1)|_{C_i}$; if q were a square, the three intersection points of Q with C_i would indeed be collinear.

To the double cover C_i^* of C_i correspond a character of order 2 of the fundamental group of C_i and a local system A_i, locally isomorphic to $\mathbb{Z}$, defined by the exact sequence of sheaves on C_i

$$0 \longrightarrow A_i \longrightarrow \pi_{i*}\mathbb{Z} \xrightarrow{\mathrm{Tr}} \mathbb{Z} \longrightarrow 0,$$

where Tr is the "sum of the values on the two sheets of the covering". One has

$$H^0(C_i, A_i) = 0, \qquad H^1(C_i, A_i) = \mathbb{Z}/2, \qquad H^2(C_i, A_i) = \mathbb{Z}/2. \tag{3.2}$$

We shall use d) only to prove the following fact

LEMMA 3.3.- X is unirational.

The surface S is a quadric in $\mathbb{P}^3(\mathbb{C})$, hence is rational. Since X_S/S has a section, there is a birational isomorphism $X_S \sim S \times \mathbb{P}^1(\mathbb{C})$, and X is the image of the rational variety X_S.

The nine singular points of X are ordinary quadratic points. Each can be resolved as follows (cf. [3])

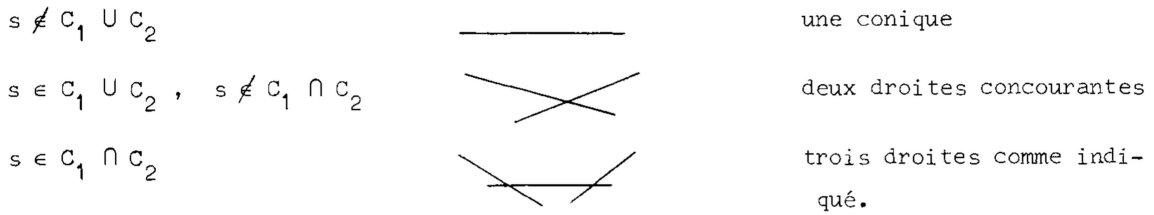
a) Blow it up; this resolves X and replaces the singular point by a quadric P, isomorphic to $\mathbb{P}^1(\mathbb{C}) \times \mathbb{P}^1(\mathbb{C})$.

b) Choose a projection $P \xrightarrow{\beta} \mathbb{P}^1(\mathbb{C})$, and contract in the blow-up $\widetilde{X}_1$ of X the fibres of this projection; this is possible because β is smooth, because its

fibres are isomorphic to $\mathbb{P}^1(\mathbb{C})$, and because the normal bundle of P in $\widetilde{X}_1$ induces on each fibre a bundle $\mathcal{O}(-1)$ ([8]).

Let $\widetilde{X}$ be the resulting variety. There is a map from $\widetilde{X}$ into X, and $\widetilde{X}$ is obtained from X by replacing each singular point by a projective line.

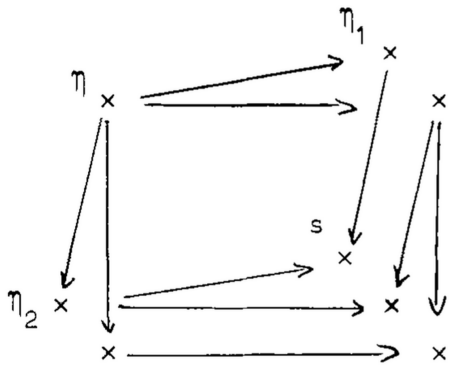
The fibres of the projection $p : \widetilde{X} \to \mathbb{P}^2(\mathbb{C})$ are of the following three types



English key: $s \notin C_1 \cup C_2$, a conic; $s \in C_1 \cup C_2$ and $s \notin C_1 \cap C_2$, two concurrent lines; $s \in C_1 \cap C_2$, three lines as shown.

When one passes from the generic point η of $\mathbb{P}^2(\mathbb{C})$ to the generic point η_i of C_i $(i = 1, 2)$, and to $s \in C_1 \cap C_2$, the irreducible components specialize as indicated in the following diagram

(3.4)



(the crosses represent irreducible components, all the multiplicities being one). The tedious verification of this local result is omitted. From it one obtains

LEMMA 3.5.- One has

$$R^0p_*\mathbb{Z} = \mathbb{Z}, \qquad R^1p_*\mathbb{Z} = 0,$$

and an exact sequence of sheaves

$$0 \longrightarrow A_1 \oplus A_2 \longrightarrow R^2p_*\mathbb{Z} \xrightarrow{\mathrm{Tr}} \mathbb{Z} \longrightarrow 0.$$

We can now use the Leray spectral sequence of p to compute $H^*(\widetilde{X}, \mathbb{Z})$. The essential fact is the following (which implies the irrationality of X).

LEMMA 3.6.- One has

$$H^3(\widetilde{X}, \mathbb{Z}) = \mathbb{Z}/(2).$$

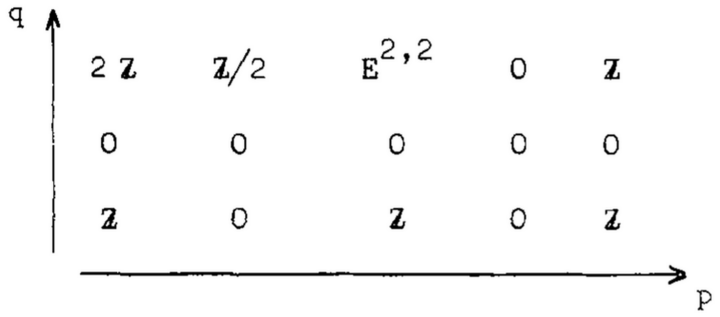
Let us compute the E_2-terms of the Leray spectral sequence of p. One has

$$R^0p_*\mathbb{Z} = \mathbb{Z} \quad \text{hence} \quad E_2^{p,0} = \mathbb{Z}, 0, \mathbb{Z}, 0, \mathbb{Z} \quad \text{for } p = 0 \text{ to } 4,$$
$$R^1p_*\mathbb{Z} = 0 \quad \text{hence} \quad E_2^{p,1} = 0.$$

To compute $E_2^{p,2} = H^p(R^2p_*\mathbb{Z})$, one uses the long exact sequence defined by the short exact sequence of sheaves 3.5. One finds

$$0 \longrightarrow E^{0,2} \xrightarrow{a} \mathbb{Z} \xrightarrow{j} \mathbb{Z}/2 \oplus \mathbb{Z}/2 \longrightarrow E^{1,2} \longrightarrow 0,$$
$$0 \longrightarrow \mathbb{Z}/2 \oplus \mathbb{Z}/2 \longrightarrow E^{2,2} \longrightarrow \mathbb{Z} \longrightarrow 0,$$
$$E^{3,2} = 0 \qquad \text{and} \qquad E^{4,2} = \mathbb{Z}.$$

One easily verifies that $a(E^{0,2}) = 2\mathbb{Z}$ (this fact is, moreover, not necessary for the proof that $H^3(\widetilde{X}, \mathbb{Z})_{\mathrm{tors}} \neq 0$). The E_2^{pq} are therefore as follows



The differentials d_r can only be zero, so that $E_2^{pq} = E_\infty^{pq}$, and the lemma follows.

It is apparent that it was essential to have two curves $C_i \subset \mathbb{P}^2(\mathbb{C})$ along which X degenerates, for a group $\mathbb{Z}/2$ is killed in passing from

$$H^1\big(\mathrm{Ker}(R^2p_*\mathbb{Z} \to \mathbb{Z})\big) \quad \text{to} \quad H^1R^2p_*\mathbb{Z} = E^{1,2}.$$

It remains to carry out the promised constructions. Artin and Mumford know how to prove *a priori* the existence of a conic bundle of the required type. I shall confine myself to giving equations that define one.

1) The image of $f_1 f_2$ in $H^0(Q, \mathcal{O}(6))$ is the square of $g_1 \in H^0(Q, \mathcal{O}(3))$, because Q is rational and the zeros of $f_1 f_2$ on Q are double. This g_1 lifts to $g \in H^0(\mathbb{P}^2(\mathbb{C}), \mathcal{O}(3))$ because $H^1(\mathbb{P}^2(\mathbb{C}), \mathcal{O}(1)) = 0$. Since $f_1 f_2 - g^2$ vanishes on Q, it is a multiple of q:

$$f_1 f_2 = g^2 + qd.$$

2) Take

$$V = \mathcal{O}(-1) \oplus \mathcal{O}(-2) \oplus \mathcal{O},$$

and the form

$$\Phi : V \longrightarrow \mathcal{O} : (x, y, z) \longmapsto qx^2 + 2gxy - dy^2 - z^2.$$

One has $\Delta = g^2 + qd = f_1 f_2$. At a point where Φ had rank ≤ 1, one would have $\Delta = q = d = 0$, whence $g = 0$, and $\Delta = g^2 + qd$ would have a double zero. This is impossible because $C_1 \cap C_2 \cap Q = \varnothing$. Finally, S is identified with the subscheme of X having equation $y = 0$, and d) and e) follow.

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THE CONSTANTS OF THE FUNCTIONAL EQUATIONS OF L-FUNCTIONS

By P. Deligne

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Introduction

The essential results of this article are the following.

(A) We give a simple proof of Langlands' theorem [9], already proved up to sign by Dwork [6], which makes it possible to write the constant in the functional equations of Artin L-functions, or of Weil L-functions [19], as a product of local constants.

The theorem to be proved is local, and Langlands' proof had the elegance of being purely local. Essentially, the problem is to prove a multiplicative identity between somewhat generalized Gauss sums whenever one has a $\mathbb{Z}$-linear relation between characters of local Galois groups induced by characters of one-dimensional representations of subgroups. Langlands constructed a generating set of such relations, and reduced the identities to be proved to identities between Gauss sums, which he proved using Stickelberger's theorem.

The proof presented here is very close to the proof [5] of the Hasse–Davenport identity (cf. also Weil [18]). It uses a global argument (4.12), and the fact that the local constants to be defined are essentially trivial at infinity, in the unramified case, and in certain very ramified cases (4.14).

(B) Let K be a function field and

$$\rho : \mathrm{Gal}(\overline{K}/K) \longrightarrow \mathrm{GL}(n, \mathbb{Q}_\ell)$$

an ℓ-adic representation of the Galois group of K, unramified almost everywhere. Grothendieck showed that the Euler product over the places of K, with p^{-s} replaced by an indeterminate t,

$$Z(\rho, t) = \prod_v \det\bigl(1 - F_v t^{\deg(v)}, (\mathbb{Q}_\ell^n)^{\rho(I_v)}\bigr)^{-1} \in \mathbb{Q}_\ell[[t]],$$

is a rational function, and that a functional equation relates $Z(\rho, t)$ to

$$Z^\vee(\rho, t) : \qquad Z^\vee(\rho, p^{-1}t^{-1}) = \varepsilon\, Z(\rho, t),$$

where the constant ε is a monomial at^b. This theory is summarized in §10. Result (A) suggests a formula expressing ε as a product of local constants. We prove that this formula is true when ρ belongs to an infinite compatible system of ℓ-adic representations (9.3 and 9.9). According to Weil [16] and Jacquet–Langlands [8], this formula, applied to the compatible system of ℓ-adic representations defined by an elliptic curve over K, makes it possible to attach to every elliptic curve over K a modular form for $\mathrm{GL}(2, K)$.

It would be very interesting to know whether the formula obtained remains valid for an arbitrary ℓ-adic representation.

Here is the content of the various sections, and their interdependence.

The beginning of §1 gathers a few results on induced representations of finite groups which will be useful to us.

The end of §1, from 1.11 to the end, is not used in the rest of the article. For a solvable group G, it describes a set R of $\mathbb{Z}$-linear relations between the characters of G induced by one-dimensional characters of subgroups, and proves that every such relation is a linear combination of elements of R. The result and its proof are implicitly contained in [9].

Sections 2 and 3 serve mainly to fix notation. In §2 we recall the structure of the Galois group of a local field [10], define the Weil groups, a variant of Galois groups, and give a few statements from class field theory. The reader should note that throughout this article the isomorphism of local class field theory sends a uniformizer to the inverse of the Frobenius substitution; the reason is given in 3.6. Section 3 recalls Tate's thesis and Artin [1].

Section 4 contains the proof of result (A) above. The formula sheet in §5 gives a list of fundamental properties of the local constants constructed in §4, and a few identities between Gauss sums that follow from them.

Sections 6, 7, and 8, the latter depending only on §2, are preliminaries to the proof of (B), given in §9. In the statements of §§3 and 4, we were very careful to use only rational formulas, involving neither passage to a limit nor square roots, even of a positive integer. We harvest the benefit of this in §§6 and 7. In §6, for K a non-archimedean local field and V a modular representation of $W(\overline{K}/K)$ over a field A of characteristic $\ell \neq p$, we define, by transport of structure for $\ell = 0$ and by reduction modulo ℓ for $\ell \neq 0$, a local constant

$$\varepsilon(V, \psi, dx) \in A^*.$$

In §7, for K global of characteristic p and V a modular representation of $W(\overline{K}/K)$, we prove, by reduction modulo ℓ, a functional equation for the corresponding L-function $L \in \Lambda(t)$.

In §8 we give the isomorphism classes of ℓ-adic representations of the Weil group of a local field a purely algebraic description, independent of the topology of $\mathbb{Q}_\ell$. This is used to define compatibility of ℓ-adic and ℓ'-adic representations. The notion introduced is finer than Serre's [12].

Section 9 gives the proof of (B) (9.3 and 9.9). The idea is that, to prove an identity, it is enough to prove it modulo ℓ for infinitely many ℓ. We also give, for function fields, a partial answer to a question of Serre by showing (9.8) that two ℓ- and ℓ'-adic representations giving the same characteristic polynomial of Frobenius, assumed to lie in $\mathbb{Q}[t]$, at almost all places satisfy compatibility at the residual places.

Section 10 summarizes Grothendieck's theory of L-functions (cf. [7] and SGA 5), which is used crucially in §9. In 10.11 I try to explain the point of his method. The odd results 10.13 follow immediately from what precedes. I do not know their meaning.

§1. Virtual representations of a finite group

1.1. Let K be a commutative field. The most important case for us will be that in which K is algebraically closed of characteristic 0, for example $K = \mathbb{C}$. Let G be a group. We shall write $R_K(G)$ for the Grothendieck group of the category of finite-dimensional $K[G]$-modules over K. It is identified with the free $\mathbb{Z}$-module with basis the set of isomorphism classes of irreducible K-linear representations of G. It is a commutative ring with unit, for tensor product over K, and even a λ-ring in the sense of Grothendieck. Its elements are the virtual representations of G over K.

To define a homomorphism f from $R_K(G)$ to a commutative group X, it is enough:

a) to define $f(V)$ for V a representation of G over K;

b) to check that for every exact sequence of representations

$$0 \longrightarrow V' \longrightarrow V \longrightarrow V'' \longrightarrow 0,$$

one has $f(V) = f(V') + f(V'')$.

Applying this rule, one defines the dimension of a virtual representation

$$\dim : R_K(G) \longrightarrow \mathbb{Z},$$

and its determinant

$$\det : R_K(G) \longrightarrow \mathrm{Hom}(G, K^*).$$

If H is a subgroup of finite index in G, define induction

$$\mathrm{Ind}_H^G : R_K(H) \longrightarrow R_K(G)$$

as corresponding to extension of scalars

$$V \longmapsto K[G] \otimes_{K[H]} V.$$

This is an exact functor, and therefore passes to virtual representations. For $f : G \to H$, define restriction

$$\mathrm{Res}_G^H : R_K(H) \longrightarrow R_K(G)$$

as restriction of scalars from $K[H]$ to $K[G]$. For f surjective one sometimes speaks rather of inflation.

For every group G, write G^{ab} for the largest abelian quotient of G. A K-quasi-character χ of G is a homomorphism

$$\chi : G \longrightarrow K^*;$$

we identify it with the K-quasi-character of G^{ab} through which it factors, and denote by $[\chi]$ the corresponding one-dimensional representation. For G finite, we shall use the words character and quasi-character interchangeably. The word character will never mean the character of a representation of dimension > 1.

The following proposition is due to P. Gallagher [20].

Proposition 1.2. Let G be a group, H a subgroup of finite index, and

$$t : G^{\mathrm{ab}} \longrightarrow H^{\mathrm{ab}}$$

the transfer. For ρ a virtual representation of dimension 0 of H, and $x \in G^{\mathrm{ab}}$, one has

$$\det(\mathrm{Ind}_H^G(\rho))(x) = \det(\rho)(t(x)).$$

We shall prove, in only slightly greater generality, that if ε is the determinant of the permutation representation of G on G/H,

$$\varepsilon : G \longrightarrow \mathfrak{S}_{G/H} \longrightarrow \{\pm 1\},$$

then, for ρ of arbitrary dimension,

$$\det(\mathrm{Ind}_H^G(\rho))(x) = \varepsilon^{\dim \rho} \det(\rho)(t(x)).$$

This is fairly evident in topological terms. Let BG be the classifying space of G, and let B_H be that of H, regarded as a covering of B_G. Representations of G are identified with local systems of finite-dimensional K-vector spaces on B_G, and Ind_H^G with the functor f_*. The transfer $H^1(B_G, X) \to H^1(B_H, X)$, for $X = K^*$, corresponds to the norm $f_!$. If $\rho \in H^1(B_G, K^*)$ is the class of a rank-one local system of K-vector spaces, then $f_!(\rho)$ is the class of $N(L)$, the local system on B_G which, locally on B_G, is the tensor product of the inverse images of L by the $[G : H]$ disjoint sections.

Put

$$\det V = \bigwedge^{\dim V} V$$

for a local system V, and let ε be the local system $\det f_* K$. Proposition 1.2 follows from the existence of a canonical isomorphism, for V a local system on B_H,

$$\det f_* V \simeq \varepsilon^{\otimes \dim V} \otimes N(\det V).$$

The construction of this isomorphism is local on B_G, and reduces to the following construction.

Construction 1.3. For $(V_i)_{i \in I}$ a finite family of K-vector spaces of dimension d, and $\varepsilon = \det(K^I)$, one has canonically

$$\det\Big(\bigoplus_{i \in I} V_i\Big) \simeq \varepsilon^{\otimes d} \otimes \bigotimes_{i \in I} \det(V_i).$$

For $i_1, \ldots, i_N$ the sequence of elements of I, the map is

$$\begin{aligned}(e_{i_1,1} \wedge \cdots \wedge e_{i_1,d}) \wedge \cdots \wedge (e_{i_N,1} \wedge \cdots \wedge e_{i_N,d}) \\ \longmapsto (i_1 \wedge \cdots \wedge i_N)^{\otimes d} \otimes (e_{i_1,1} \wedge \cdots \wedge e_{i_1,d}) \otimes \cdots \otimes (e_{i_N,1} \wedge \cdots \wedge e_{i_N,d}).\end{aligned}$$

Parenthesis 1.4. Instead of a classifying space BG, one could more intrinsically have used, in the preceding argument, Grothendieck's classifying topos B_G (SGA 4, IV, 2.4, p. 315, for $E = (\mathrm{Ens})$).

Assume G finite. We shall say that K is large enough if it contains all roots of unity whose orders divide the l.c.m. of the orders of elements of G. A group H will be called p-elementary if it is the product of a p-group and a cyclic group; elementary if it is p-elementary for some prime p. We shall need the following variant of Brauer's theorem on induced characters, obtained by omitting everywhere "of dimension zero."

Proposition 1.5. If K is large enough, every virtual representation of dimension 0 of G is a sum of virtual representations

$$\mathrm{Ind}_H^G(\chi),$$

where χ is virtual of dimension 0, H is elementary, and χ is a linear combination of one-dimensional representations of H.

By Brauer's theorem applied to the trivial representation 1_G of G, there is a decomposition

$$1_G = \sum_H \mathrm{Ind}_H^G(z_H), \qquad H \text{ elementary.}$$

For y a virtual representation of dimension 0 of G, one then has

$$y = \sum_H y \cdot \mathrm{Ind}_H^G(z_H) = \sum_H \mathrm{Ind}_H^G(\mathrm{Res}_H^G(y) \cdot z_H),$$

with $\dim(\mathrm{Res}_H^G(y) \cdot z_H) = 0$. This reduces us to the case where G is elementary.

By linearity, one may suppose that y is of the form

$$\rho - \dim(\rho)\, 1_G,$$

with ρ irreducible, hence, since G is nilpotent, induced by a character χ of a subgroup H of G containing the center Z. Then

$$\rho - \dim(\rho)\, 1_G = \mathrm{Ind}_H^G(\chi - 1_H) + \big(\mathrm{Ind}_H^G(1_H) - [G:H]1_G\big).$$

The first term is of the desired type, and the second is the inflation of a virtual representation of dimension 0 of G/Z. One concludes by induction on the order of G, noting that $|G/Z| < |G|$ if $G \neq \{e\}$.

Scholium 1.6. A homomorphism

$$f : R_K(G) \longrightarrow X$$

is known once its value on 1_G and on the

$$\mathrm{Ind}_H^G(\chi - 1_H),$$

for H elementary and χ a character of H, are known.

1.7. Let A be a discrete valuation ring of unequal characteristic p, with fraction field K and residue field $k = A/\mathfrak{m}$. Recall the definition of the decomposition homomorphism ([11], III, 2.2)

$$d : R_K(G) \longrightarrow R_k(G) :$$

for V a representation of G over K, and E a G-invariant lattice,

$$d([V]) = [E \otimes_A k].$$

Proposition 1.8. Assume that K is sufficiently large. The kernel of the decomposition homomorphism d is then generated by the elements

$$\mathrm{Ind}_H^G(\chi') - \mathrm{Ind}_H^G(\chi'')$$

where H is an elementary subgroup and $\chi', \chi'' \in \mathrm{Hom}(H, K^*)$ are congruent modulo $\mathfrak{m}$.

Since the ring homomorphism d commutes with induction, the argument of 1.5 reduces us to the case where G is elementary, hence the product of a p-group P and a group H of order prime to p. Since K and k are sufficiently large, one has

$$R(G) = R(P) \otimes R(H),$$

and similarly over k. For a group of order prime to p, d is bijective. Applying Brauer's theorem to H, it is enough to prove 1.8 for a p-group. This case is trivial, since every character $\chi \in \mathrm{Hom}(P, K^*)$ of a p-group is congruent to 1 modulo $\mathfrak{m}$.

1.9. For the rest of this subsection, assume that $K = \mathbb{C}$, consider only finite groups, and abbreviate $R_{\mathbb{C}}(G)$ to $R(G)$. For G commutative, write G^* for the group of characters of G.

Let $R^+(G)$ be the free $\mathbb{Z}$-module with basis the set of G-conjugacy classes of pairs (H, χ), where H is a subgroup of G and χ a character of H. Define a multiplication in $R^+(G)$ by

$$(A, \alpha)\,(B, \beta) = \sum_{(x,y) \in G\backslash(G/A \times G/B)} \big(A^x \cap B^y,\ (\alpha^x|_{A^x \cap B^y})(\beta^y|_{A^x \cap B^y})\big), \tag{1.9.1}$$

where (x, y) runs through representatives of the orbits of G in $G/A \times G/B$, $A^x = xAx^{-1}$, and α^x is the character $a \mapsto \alpha(x^{-1}ax)$ of A^x. With this multiplication, $R^+(G)$ is a commutative ring with unit $(G, 1)$, and one defines a ring homomorphism

$$b : R^+(G) \longrightarrow R(G), \qquad (H, \chi) \longmapsto \mathrm{Ind}_H^G(\chi). \tag{1.9.2}$$

This homomorphism is surjective by Brauer's theorem. For χ a character of G, multiplication by (G,χ) is denoted $\cdot\chi$ and is called twisting by χ:

$$(H,\varphi)\cdot\chi = (H,\varphi\cdot\chi|_H). \tag{1.9.3}$$

For $G'\subset G$, define induction

$$\mathrm{Ind}_{G'}^{G} : R^+(G') \longrightarrow R^+(G), \qquad (H,\chi)\longmapsto (H,\chi). \tag{1.9.4}$$

For a morphism $u : G\to G'$, define restriction by

$$\mathrm{Res}_u(H,\chi) = \sum_{s\in u(G)\backslash G'/H} \big(u^{-1}(sHs^{-1}),\ \chi^s\circ u\big), \tag{1.9.5}$$

where s runs over a set of representatives of the double classes. There are commutative diagrams

$$\begin{array}{ccc} R^+(G') & \xrightarrow{\mathrm{Ind}} & R^+(G) \\ {\scriptstyle b}\downarrow & & \downarrow{\scriptstyle b} \\ R(G') & \xrightarrow[\mathrm{Ind}]{} & R(G) \end{array} \qquad \begin{array}{ccc} R^+(G') & \xrightarrow{\mathrm{Res}} & R^+(G) \\ {\scriptstyle b}\downarrow & & \downarrow{\scriptstyle b} \\ R(G') & \xrightarrow[\mathrm{Res}]{} & R(G), \end{array}$$

for the second, see [11], p. 75. For $R\in R^+(G)$ one has

$$R\cdot(H,\chi) = \mathrm{Ind}_H^G\big(\mathrm{Res}_H^G(R)\cdot\chi\big). \tag{1.9.6}$$

When u is surjective, one speaks of inflation.

Notation 1.9.8. For every additive function of representations F, write the same symbol for the function it defines on $R(G)$, and for $F\circ b$. Thus

$$\dim(v) = \dim(b(v)), \quad \ldots$$

Variants 1.10. (i) Let $R_0^+(G)$ be the subgroup of $R^+(G)$ generated by the $(H,\chi)-(H,1)$. As (H,χ) runs through the canonical basis of $R^+(G)$, except for the $(H,1)$, these elements form a basis of $R_0^+(G)$. One checks that $R_0^+(G)$ is an ideal of $R^+(G)$. Let $R^0(G)\subset R(G)$ be the set of virtual representations of dimension 0 of G. By 1.5, the natural map, induced by (1.9.2), from $R_0^+(G)$ to $R^0(G)$ is surjective.

(ii) For a topological group G, finite or not, $R^+(G)$ and $R_0^+(G)$ denote the analogous groups obtained by restricting to closed subgroups H of finite index and continuous quasi-characters. The functorialities of 1.9 generalize mutatis mutandis.

The rest of this subsection will not be used below. We give results drawn from Langlands [9].

Lemma 1.11. Suppose that $G = H\cdot C$, where C is commutative and normal, and let χ be a character of H. For every $\mu\in C^*$, let $G_\mu\subset G$ be the stabilizer of μ under conjugation. If

$$\chi|_{H\cap C} = \mu|_{H\cap C},$$

write $\{\chi,\mu\}$ for the character of $G_\mu = (H\cap G_\mu)\cdot C$ which extends $\chi|_{H\cap G_\mu}$ and μ. Then

$$\mathrm{Ind}_H^G(\chi) = \sum_{\substack{\mu\in C^*/G \\ \chi|_{H\cap C}=\mu|_{H\cap C}}} \mathrm{Ind}_{G_\mu}^G\{\chi,\mu\}.$$

The verification is left to the reader.

1.12. An element R of $\mathrm{Ker}(R^+(G)\to R(G))$ will be said to be of type I, II, or III if there exists a quotient A of a subgroup B of G such that R is obtained from an element S of one of the following respective types in $\mathrm{Ker}(R^+(A)\to R(A))$, by inflation from A to B, twisting by a character of B, and induction from B to G. Let ℓ be a prime number.

I. $A \simeq \mathbb{Z}/\ell\mathbb{Z}$, and

$$S = (e, 1) - \sum_{\chi \in A^*} (A, \chi).$$

II. A is a central extension of $(\mathbb{Z}/\ell\mathbb{Z})^2$ by an abelian group Z. Let H_i, $i = 1, 2$, be the inverse images in A of the first and second factors $\mathbb{Z}/\ell\mathbb{Z}$, and let χ_i be a character of H_i. Assume that $\chi_1|_Z = \chi_2|_Z$, and that this character of Z is nontrivial on a commutator.

$$\begin{array}{ccc} Z & \hookrightarrow & H_1 \\ \downarrow & & \downarrow \\ H_2 & \hookrightarrow & A \end{array}$$

The relation S is

$$S = (H_1, \chi_1) - (H_2, \chi_2).$$

III. A is a semidirect product $H \cdot C$, χ is a character of H, and the normal subgroup C is minimal among the nontrivial commutative normal subgroups of A. For every character μ of C, A_μ is the stabilizer of μ, and $\{\chi, \mu\}$ is the character of A_μ extending χ and μ. The relation S expresses

$$\mathrm{Ind}_H^A(\chi) = \sum_{\mu \in C^*/A} \mathrm{Ind}_{A_\mu}^A(\{\chi, \mu\}).$$

where μ runs over a set of representatives for the orbits of A in C^*.

These relations are special cases of 1.11: for I, $H = \{e\}$ and $C = A$; for II, $H = H_1$ and $C = H_2$; for III, H and C are as indicated.

Every relation obtained by induction, inflation, or twisting from a relation of type I, respectively II or III, is again of the same type.

Theorem 1.13 (Langlands [9], §18). If G is nilpotent, the kernel of

$$b : R^+(G) \longrightarrow R(G)$$

is generated, as a $\mathbb{Z}$-module, by the relations of type I and II.

Lemma 1.13.1. If G is commutative, $\mathrm{Ker}(b)$ is generated by the relations of type I.

One has to show that, for every subgroup H of G and every $\chi \in H^*$, the relation

$$(H, \chi) - \sum_{\psi \in (G/H)^*} (G, \tilde{\chi}\psi)$$

is a consequence of relations of type I. These are the following: for $H' \subset H'' \subset G$, with $[H'' : H']$ prime, and χ a character of H', always induced by a character of G,

$$\mathrm{Ind}_{H'}^G(\chi) - \sum_{\chi''|_{H'} = \chi} \mathrm{Ind}_{H''}^G(\chi'') = 0.$$

The general relation follows by taking a Jordan–Hölder series of G/H.

Lemma 1.13.2. Let C be a commutative normal subgroup of G, let T be a set of representatives of C^*/G, let $\mu \in T$, and let G_μ be the stabilizer of μ. Let (H_i) be a family of subgroups of G containing C, and let χ_i be a character of H_i, with $\chi_i|_C \in T$. If the relation

$$\sum_i n_i \ \mathrm{Ind}_{H_i}^G(\chi_i) = 0$$

holds in $R(G)$, then

$$\sum_{\chi_i|_C = \mu} n_i \ \mathrm{Ind}_{H_i}^{G_\mu}(\chi_i) = 0$$

holds in $R(G_\mu)$. The verification is left to the reader.

Let Z be the center of G, and prove 1.13 by induction on the order of G/Z. By 1.13.1, we may suppose G noncommutative. Let C be a commutative normal subgroup containing Z, with $[C : Z] = \ell$ prime, and with central image in G/Z.

Consider a relation

$$\sum_i n_i \operatorname{Ind}_{H_i}^G(\chi_i) = 0. \tag{1}$$

It is a consequence of relations induced by

$$\sum_{\chi_i|_{H_i\cap Z}=\chi} \operatorname{Ind}_{H_i}^G(\chi_i) = \operatorname{Ind}_{H_iZ}^G(\chi)$$

and of a relation (2) analogous to (1), with $H_i \supset Z$. The displayed relations are linear combinations of relations of type I; this follows from 1.13.1 applied to $H_iZ/\operatorname{Ker}(\chi_i)$.

The relation (2) is a consequence of relations induced from 1.11 for HC, H, C, χ, with $H \supset Z$,

$$\operatorname{Ind}_H^{HC}(\chi) = \sum_{\substack{\mu\in C^*/H \\ \chi|_{H\cap C}=\mu|_{H\cap C}}} \operatorname{Ind}_{G_\mu}^{HC}\{\chi, \mu\}, \tag{b}$$

and of a relation (3) analogous to (1), this time with $H_i \supset C$. Apply 1.13.2 to (3), to express it as a sum of relations induced by relations

$$\sum_i n_i \operatorname{Ind}_{H_i}^{G_\mu}(\chi_i) = 0.$$

The image of C in $G/\operatorname{Ker}(\mu)$ is central, and the induction hypothesis therefore applies to G_μ; the resulting relations are of the desired type.

It remains to prove that the relations (b) are of the desired type. If $HC \neq G$, this follows from the induction hypothesis applied to HC. If $HC = G$, then H is normal, because it contains Z and C is central modulo Z. Let A be the intersection of the kernels of the conjugates of χ. The relation (b) is the inflation of an analogous relation for G/A. If the induction hypothesis applies to G/A, we are done. Otherwise we are in the same situation as before, with $A = \{e\}$, and we conclude by the following lemma.

Lemma 1.13.3. With the preceding notation, if $A = \{e\}$, the relation (b) is of type II.

The group H is commutative, since characters separate its elements. The group Z is cyclic, since one character separates its elements. For $c \in C$ and $h \in H$, put

$$u(c, h) = chc^{-1}h^{-1}.$$

This is an element of Z, since C is central modulo Z. It depends biadditively on c and h, and defines

$$u : C/Z \otimes H/Z \longrightarrow Z.$$

Let c generate C/Z. Since Z is the center and $G \neq Z$, the map $u(c, \ ) : H/Z \to Z$ is injective, with nontrivial cyclic image killed by ℓ. Hence $|C/Z| = |H/Z| = \ell$, and G is a central extension of $C/Z \times H/Z$ by Z. The character χ is distinct from at least one of its conjugates, otherwise H would be central; it is therefore nontrivial on a commutator, and we are in situation II.

Theorem 1.14 (Langlands [9]). If G is solvable, the kernel of

$$b : R^+(G) \longrightarrow R(G)$$

is generated by the relations of type I, II, and III.

Lemma 1.14.1. If G is solvable, the $\mathbb{Z}$-submodule $N(G)$ of $R^+(G)$ generated by the relations of type I, II, and III is an ideal.

We prove 1.14 and 1.14.1 by simultaneous induction on the order of G.

(A) Theorem 1.14 for the proper subgroups of G implies 1.14.1 for G. Let $R \in R^+(G)$ be of type I, II, or III, let H be a subgroup, and let χ be a character of H. We prove that

$$R \cdot (H, \chi) \in N(G).$$

If $H = G$, this follows from the stability of $N(G)$ under twisting by a character of G. If $H \neq G$, then by (1.9.6)

$$R \cdot (H, \chi) = \mathrm{Ind}_H^G(\mathrm{Res}_H^G(R) \cdot \chi).$$

By the hypothesis, $\mathrm{Res}_H^G(R) \in N(H)$, and twisting followed by induction gives an element of $N(G)$.

(B) Lemma 1.14.1 for G, together with Theorem 1.14 for groups of smaller order, implies 1.14 for G. Let Z be the center of G. By 1.13, we may and shall suppose that G is not nilpotent. There are two cases.

(B1) $Z \neq \{e\}$. By Brauer's theorem in G/Z, there exist nilpotent subgroups $H_i \supset Z$ of G, and characters χ_i of H_i/Z, such that

$$\sum_i n_i \ \mathrm{Ind}_{H_i/Z}^{G/Z}(\chi_i) = 1_{G/Z}.$$

Since this relation lifts to a relation S in $R^+(G)$, and since the induction hypothesis applies to G/Z, one has $S - (G, 1) \in N(G)$. For $R \in \mathrm{Ker}(b)$, one has $R \cdot S \in N(G)$ by 1.14.1, while (1.9.6) reduces the terms to restrictions to the H_i, where the induction hypothesis applies. This proves the assertion.

(B2) $Z = \{e\}$. Let C be a nontrivial minimal commutative normal subgroup and, for $\mu \in C^*$, let G_μ be its stabilizer. Proceed as in the proof of 1.13. One finds that every relation $R \in \mathrm{Ker}(b)$ is a linear combination of:

(a) relations induced from 1.11 for HC, H, C, χ, with $H \subset G$, χ a character of H, and $H \not\supset C$;

(b) relations induced from identities in G_μ.

Since $Z = \{e\}$, one has $|G_\mu / \mathrm{Ker}(\mu)| < |G|$, and the induction hypothesis applies to the latter relations. Consider a relation of type (a). If $HC \neq G$, apply the induction hypothesis. If $HC = G$, then $H \cap C$ is normal, since it is normal in both H and C. By minimality of C, $H \cap C = \{e\}$, and the relation is of type III.

Remark 1.15. It is probably possible to extract from [9] a description of a system of generators of

$$\mathrm{Ker}(R_0^+(G) \longrightarrow R^0(G))$$

for suitable groups G. I have not had the courage to try.

2. Weil groups

2.1. Roots of unity. Let K be a separably closed field of characteristic exponent p. For every integer n, write $\mu_n(1)(K)$, or simply $\mu_n(1)$, for the group of n-th roots of unity in K. For $n \mid m$, there is a transition map

$$\mu_m(1) \longrightarrow \mu_n(1), \qquad x \longmapsto x^{m/n};$$

put

$$\widehat{\mathbb{Z}}(1) = \varprojlim_n \mu_n(1).$$

For a prime number ℓ, similarly put

$$\mathbb{Z}_\ell(1) = \varprojlim_r \mu_{\ell^r}(1).$$

The evident maps $\widehat{\mathbb{Z}}(1) \to \mathbb{Z}_\ell(1)$ induce an isomorphism

$$\widehat{\mathbb{Z}}(1) \xrightarrow{\sim} \prod_{\ell \neq p} \mathbb{Z}_\ell(1).$$

For every $\widehat{\mathbb{Z}}$-module V, put $V(1) = V \otimes_{\widehat{\mathbb{Z}}} \widehat{\mathbb{Z}}(1)$. For a $\mathbb{Z}_\ell$-module V, one has

$$V(1) \simeq V \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(1).$$

For $K = \mathbb{C}$, the group $\mu_n(1)$ is canonically isomorphic to the group of roots of unity in $\mathbb{C}$. For $x \in \mu_n(1)$ and $\xi \in \mathbb{Z}/n\mathbb{Z}$, the image of $\xi \otimes x$ in $\mathbb{Q}/\mathbb{Z}(1)$ is ξx, giving the usual identifications.

For $\ell \neq p$, $\mathbb{Z}_\ell(1)$ is a free rank-one $\mathbb{Z}_\ell$-module. For $i \in \mathbb{Z}$, write $\mathbb{Z}_\ell(i)$ for its i-th tensor power, and for $i < 0$ for the dual of $\mathbb{Z}_\ell(-i)$. For every $\mathbb{Z}_\ell$-module V, put again $V(i) = V \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(i)$.

2.2. Notation. Let K be a local field. We write $\| \ \|$ for the normalized absolute value of K. If dx is a Haar measure, then

$$\int f(ax)\, dx = \|a\|^{-1} \int f(x)\, dx,$$

i.e. $d(ax) = \|a\|\, dx$.

For $s \in \mathbb{C}$, write ω_s for the quasi-character

$$x \longmapsto \|x\|^s$$

from K^* to $\mathbb{C}^*$.

If K is nonarchimedean, i.e. $K \neq \mathbb{R}, \mathbb{C}$, write $\mathcal{O}$ for the valuation ring of the valuation v of K, π for a uniformizer, k for the residue field $\mathcal{O}/(\pi)$, p for the characteristic of k, and put

$$d = [k : \mathbb{F}_p], \qquad q = p^d = \#k.$$

Thus

$$\|x\| = q^{-v(x)}.$$

2.2.2. Suppose K is nonarchimedean. Let $\overline{K}$ be a separable closure of K, write $\overline{\mathcal{O}}$ for the integral closure of $\mathcal{O}$ in $\overline{K}$, let $\overline{k}$ be the residue field of $\overline{\mathcal{O}}$, an algebraic closure of k, and let v be the valuation of $\overline{K}$, with values in $\mathbb{Q}$, extending that of K.

There is a three-step filtration of the Galois group

$$\mathrm{Gal}(\overline{K}/K) \supset I \supset P$$

whose successive quotients are

$$\begin{array}{rcl} \mathrm{Gal}(\overline{K}/K)/I & \xrightarrow{\sim} & \mathrm{Gal}(\overline{k}/k) \xrightarrow{\sim} \widehat{\mathbb{Z}}, \\ I/P & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1)(\overline{k}), \\ P & : & \text{a pro-}p\text{-group.} \end{array}$$

It is obtained as follows; for proofs we refer to [10].

a) By transport of structure, $\mathrm{Gal}(\overline{K}/K)$ acts on $\overline{\mathcal{O}}$ and on $\overline{k}$. The inertia group I is the kernel of the reduction map

$$v' : \mathrm{Gal}(\overline{K}/K) \longrightarrow \mathrm{Gal}(\overline{k}/k).$$

The quotient $\mathrm{Gal}(\overline{k}/k)$ is canonically isomorphic to $\widehat{\mathbb{Z}}$, with topological generator the Frobenius substitution

$$\varphi : x \longmapsto x^q.$$

b) The wild inertia group P is the unique pro-p-Sylow subgroup of I. The equivariant morphism

$$t : I \longrightarrow \widehat{\mathbb{Z}}(1)(\overline{k}),$$

with kernel P, is characterized by the congruence, modulo the maximal ideal of $\overline{\mathcal{O}}$,

$$\frac{\sigma(x)}{x} = t(\sigma)\, v(x) \qquad (\sigma \in I,\ x \in \overline{K}).$$

For a prime $\ell \neq p$, write t_ℓ for the composite

$$I \longrightarrow \widehat{\mathbb{Z}}(1)(\overline{k}) \longrightarrow \mathbb{Z}_\ell(1)(\overline{k}).$$

2.2.3. Let k be a finite field with q elements, and let $\overline{k}$ be its algebraic closure. Let $W(\overline{k}/k)$ be the subgroup of $\mathrm{Gal}(\overline{k}/k) \simeq \widehat{\mathbb{Z}}$ generated by

$$\varphi : x \longmapsto x^q.$$

There is a canonical isomorphism

$$W(\overline{k}/k) \simeq \mathbb{Z}$$

with generator φ. The geometric Frobenius $F \in W(\overline{k}/k)$ is, by definition, φ^{-1}.

2.2.4. Let $K, \overline{K}$ be as in 2.2.2. The Weil group $W(\overline{K}/K)$, or absolute Weil group of K, is the subgroup of $\mathrm{Gal}(\overline{K}/K)$ consisting of the σ such that $v'(\sigma)$ is an integral power of Frobenius φ. It is endowed with the topology inducing the natural topology on I, and for which I is open. The group $\mathrm{Gal}(\overline{K}/K)$ is the completion of $W(\overline{K}/K)$ for the topology of open subgroups of finite index; if L is a finite extension of K in $\overline{K}$, the corresponding subgroup of $W(\overline{K}/K)$ is identified with $W(\overline{K}/L)$. We shall sometimes call geometric Frobenius elements those elements of $\mathrm{Gal}(\overline{K}/K)$ or $W(\overline{K}/K)$ whose image is F.

2.2.5. Suppose K is archimedean, with algebraic closure $\overline{K}$. Define $W(\overline{K}/K)$ as the following extension of $\mathrm{Gal}(\overline{K}/K)$ by $\overline{K}^*$:

1) $K \simeq \mathbb{R}$: $W(\overline{K}/K)$ is generated by $\overline{K}^*$ and an element F, with

$$F^2 = -1, \qquad FzF^{-1} = \overline{z} \quad (z \in \overline{K}^*).$$

2) $K \simeq \mathbb{C}$: $W(\overline{K}/K) = \overline{K}^*$.

2.3. Local class field theory provides an isomorphism between $W(\overline{K}/K)^{\mathrm{ab}}$ and K^*.

Assume K nonarchimedean. For a reason explained in 3.6, we normalize this isomorphism, i.e. choose it or its inverse, so that geometric Frobenius elements correspond to uniformizers:

$$\begin{array}{ccccccc} P & \hookrightarrow & I & \hookrightarrow & W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k) \simeq \mathbb{Z} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow{\scriptstyle -1} \\ 1+(\pi) & \hookrightarrow & \mathcal{O}^* & \hookrightarrow & K^* & \xrightarrow{\ v\ } & \mathbb{Z}. \end{array}$$

A representation of $W(\overline{K}/K)$ is unramified, respectively tamely ramified, if it is trivial on I, respectively on P. Similarly, a quasi-character χ of K^* is unramified, respectively tamely ramified, if it is trivial on $\mathcal{O}^*$, respectively on $1+(\pi)$. For ℓ prime to p, the action of $\mathrm{Gal}(\overline{K}/K)$ on $\mathbb{Z}_\ell(1)$ is unramified, described by the quasi-character

$$\omega_1 = \|x\|$$

of K^*.

For K complex, the isomorphism of local class field theory is the evident one on 2.2.5; for K real, it is the one which makes 2.3.1 and 2.3.2 below true.

Let $L \subset \overline{K}$ be a finite extension of K, so that $W(\overline{K}/L) \subset W(\overline{K}/K)$. The following diagrams, where N is the norm and t the transfer, are commutative ([10], XIII, §4):

$$\begin{array}{ccccc} W(\overline{K}/L) & \longrightarrow & W(\overline{K}/L)^{\mathrm{ab}} & \xrightarrow{\sim} & L^* \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle N_{L/K}} \\ W(\overline{K}/K) & \longrightarrow & W(\overline{K}/K)^{\mathrm{ab}} & \xrightarrow{\sim} & K^*, \end{array} \tag{2.3.1}$$

$$\begin{array}{ccc} W(\overline{K}/K)^{\mathrm{ab}} & \xrightarrow{\sim} & K^* \\ \downarrow{\scriptstyle t} & & \downarrow \\ W(\overline{K}/L)^{\mathrm{ab}} & \xrightarrow{\sim} & L^*. \end{array} \tag{2.3.2}$$

2.4. Let K be a function field in one variable over $\mathbb{F}_p$, and let k be its field of constants, the algebraic closure of $\mathbb{F}_p$ in K. Let $\overline{K}$ be a separable closure of K, let $\overline{k}$ be the algebraic closure of k in $\overline{K}$, and let $\mathrm{Gal}^0(\overline{K}/K)$ be the kernel of the restriction map from $\mathrm{Gal}(\overline{K}/K)$ to $\mathrm{Gal}(\overline{k}/k)$. The absolute global Weil group $W(\overline{K}/K)$ is, by analogy with 2.2.4, defined by the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Gal}^0(\overline{K}/K) & \longrightarrow & W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k) = \mathbb{Z} & \longrightarrow & 0 \\ & & \| & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{Gal}^0(\overline{K}/K) & \longrightarrow & \mathrm{Gal}(\overline{K}/K) & \longrightarrow & \mathrm{Gal}(\overline{k}/k) = \widehat{\mathbb{Z}} & \longrightarrow & 0. \end{array}$$

Let v be a place of K, and suppose a place of $\overline{K}$ above v has been chosen. There is then a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I_v & \longrightarrow & W(\overline{K}_v/K_v) & \longrightarrow & W(\overline{k}_v/k_v) = \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow{\scriptstyle [k_v:k]} & & \\ 0 & \longrightarrow & \mathrm{Gal}^0(\overline{K}/K) & \longrightarrow & W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k) = \mathbb{Z} & \longrightarrow & 0. \end{array}$$

Global class field theory provides an isomorphism

$$W(\overline{K}/K)^{\mathrm{ab}} \xrightarrow{\sim} \mathbb{A}_K^*/K^*$$

where $\mathbb{A}_K^*/K^*$ is the idèle-class group, making the diagrams

$$\begin{array}{ccc} W(\overline{K}_v/K_v)^{\mathrm{ab}} & \longrightarrow & W(\overline{K}/K)^{\mathrm{ab}} \\ \downarrow & & \downarrow \\ K_v^* & \longrightarrow & \mathbb{A}_K^*/K^* \end{array}$$

commutative.

2.5. For the definition of global Weil groups in the case of number fields, we refer to Weil [19] or to [2], XIV. We shall use them only incidentally.

3. Reminders on L-functions

3.1. Let K be a local field and use again the notation of 2.2.1. We denote by dx a Haar measure on K, by d^*x a Haar measure on K^*, and by ψ a nontrivial additive character of K.

3.2. For a quasi-character, i.e. a continuous homomorphism,

$$\chi : K^* \longrightarrow \mathbb{C}^*,$$

define $L(\chi) \in \mathbb{C} \cup \{\infty\}$ as follows.

$$K \simeq \mathbb{R}. \tag{3.2.1}$$

Put

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2),$$

and, for x the embedding of K in $\mathbb{C}$ and $N = 0$ or 1,

$$L(x^{-N}\omega_s) = \Gamma_{\mathbb{R}}(s).$$

$$K \simeq \mathbb{C}. \tag{3.2.2}$$

Put

$$\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s),$$

and, for z an embedding of K in $\mathbb{C}$ and $N \geq 0$,

$$L(z^{-N}\omega_s) = \Gamma_{\mathbb{C}}(s).$$

$$K \text{ nonarchimedean.} \tag{3.2.3}$$

Put

$$L(\chi) = 1 \quad (\chi \text{ ramified}), \qquad L(\chi) = \frac{1}{1-\chi(\pi)} \quad (\chi \text{ unramified}).$$

3.3. Given ψ and dx, put, for f a C^∞ function of compact support on K,

$$\widehat{f}(y) = \int_K f(x)\psi(xy)\,dx.$$

Define $\varepsilon(\chi, \psi, dx) \in \mathbb{C}^*$ by Tate's local functional equation, independent of f,

$$\frac{\displaystyle\int_{K^*} \widehat{f}(x)\,\omega_1\chi^{-1}(x)\,d^*x}{L(\omega_1\chi^{-1})} = \varepsilon(\chi,\psi,dx)\,\frac{\displaystyle\int_{K^*} f(x)\chi(x)\,d^*x}{L(\chi)}. \tag{3.3.1}$$

The same Haar measure d^*x is used on both sides. Both sides are defined by analytic continuation in χ in the families $\chi\omega_s$, $s \in \mathbb{C}$. In these families, ε is of the form Ae^{Bs}.

One has

$$\varepsilon(\chi, \psi, a\,dx) = a\,\varepsilon(\chi,\psi,dx), \tag{3.3.2}$$

$$\varepsilon(\chi, \psi(ax), dx) = \chi(a)\|a\|^{-1}\varepsilon(\chi,\psi,dx). \tag{3.3.3}$$

3.4. For K nonarchimedean, put:

$$n(\psi) = \text{the largest integer } n \text{ such that } \psi|_{\pi^{-n}\mathcal{O}} = 1;$$

$$a(\chi) = \begin{cases} 0, & \chi \text{ unramified,} \\ \text{the least } m \text{ such that } \chi|_{1+(\pi)^m} = 1, & \chi \text{ ramified;} \end{cases}$$

$$sw(\chi) = \begin{cases} 0, & \chi \text{ unramified or tamely ramified,} \\ a(\chi) - 1, & \chi \text{ ramified;} \end{cases}$$

and let γ be an element of K^* of valuation

$$v(\gamma) = n(\psi) + sw(\chi) + 1.$$

The local constant ε is given by (3.3.2), (3.3.3), and the following formulas.

$$K \simeq \mathbb{R}. \tag{3.4.1}$$

Let x be the embedding of K in $\mathbb{C}$, and $N = 0$ or 1. For $\psi = \exp(2\pi i x)$ and dx the Lebesgue measure,

$$\varepsilon(x^{-N}\omega_s, \psi, dx) = i^N.$$

$$K \simeq \mathbb{C}. \tag{3.4.2}$$

Let z be an embedding of K in $\mathbb{C}$, and $N \geq 0$. For

$$\psi = \exp(2\pi i\ \mathrm{Tr}_{\mathbb{C}/\mathbb{R}}(z))$$

and

$$dx = |dz \wedge d\bar{z}| = 2\,da\,db \quad (z = a + bi),$$

one has

$$\varepsilon(z^{-N}\omega_s, \psi, dx) = i^N.$$

3.4.3. K nonarchimedean. If χ is unramified,

$$\int_{\mathcal{O}} dx = 1, \qquad n(\psi) = 0,$$

then

$$\varepsilon(\chi, \psi, dx) = 1. \tag{3.4.3.1}$$

For χ ramified, one has

$$\varepsilon(\chi, \psi, dx) = \int_{K^*} \chi^{-1}(x)\psi(x)\, dx = \sum_n \int_{v(x)=n} \chi^{-1}(x)\psi(x)\, dx = \int_{\gamma^{-1}\mathcal{O}^*} \chi^{-1}(x)\psi(x)\, dx. \tag{3.4.3.2}$$

We shall see that it is natural to unify these two formulas as follows. If $\varepsilon_0(\chi, \psi, dx)$ is defined by

$$\varepsilon(\chi, \psi, dx) = \varepsilon_0(\chi, \psi, dx) \quad (\chi \text{ ramified}),$$

and

$$\varepsilon(\chi, \psi, dx) = \varepsilon_0(\chi, \psi, dx)\, \chi(\pi)^{-1} \quad (\chi \text{ unramified}), \tag{3.4.3.3}$$

then

$$\varepsilon_0(\chi, \psi, dx) = \int_{\gamma^{-1}\mathcal{O}^*} \chi^{-1}(x)\psi(x)\, dx. \tag{3.4.3.4}$$

From (3.4.3.3) and (3.4.3.4), one deduces that, for ω unramified,

$$\varepsilon_0(\chi\omega, \psi, dx) = \omega(\pi)^{n(\psi)+sw(\chi)+1}\varepsilon_0(\chi, \psi, dx), \tag{3.4.3.5}$$

and

$$\varepsilon(\chi\omega, \psi, dx) = \omega(\pi)^{n(\psi)+a(\chi)}\varepsilon(\chi, \psi, dx).$$

3.5. Suppose K nonarchimedean, and resume the notation of 2.2.1. Let V be a finite-dimensional vector space over a field E of characteristic 0, or over $\mathbb{C}$. The representations

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V)$$

will always be assumed continuous: an open subgroup of I acts trivially. For V the space of a representation and V^I the subspace of inertia invariants, put

$$Z(V; t) = \det(1 - Ft^d, V^I)^{-1}, \tag{3.5.1}$$

$$L(V) = Z(V; 1) \qquad (E \in E^* \cup \{\infty\}). \tag{3.5.2}$$

Let

$$\omega^t : W(\overline{K}/K) \longrightarrow E((t))^*$$

be the unramified character such that $\omega^t(F) = t^d$. In an evident sense, $\omega_s = \omega^{p^{-s}}$. One has

$$Z(V, t) = L(V \otimes \omega^t). \tag{3.5.3}$$

3.6. It is so that (3.2.3) and (3.5.2) are compatible that we normalized the local class field theory isomorphism so that geometric Frobenius elements correspond to uniformizers. Both (3.2.3) and (3.5.2) are very natural: 3.2.3 is forced from the analytic point of view (cf. 3.3.1), and 3.5.2 follows the tradition of geometers; it is geometric Frobenius which tends to act on the cohomology of algebraic varieties with algebraic integers as eigenvalues.

3.7. The irreducible continuous complex representations of $W(\overline{K}/K)$ are, for K complex, the quasi-characters. For K real, besides the quasi-characters of K^*, they are the representations induced by quasi-characters χ of $\overline{K}^*$ such that $\chi \neq \chi \circ \sigma$, where σ is complex conjugation of $\overline{K}$.

For any complex representation V, expressed in the Grothendieck group of representations of $W(\overline{K}/K)$ as a sum of simple representations, define $L(V)$ as follows:

$$K \simeq \mathbb{R}, \quad V = \sum_i [\chi_i] + \sum_j \mathrm{Ind}_{\overline{K}^*}^{W}(\chi_j), \tag{3.7.1}$$

$$L(V) = \prod_i L_K(\chi_i) \prod_j L_{\overline{K}}(\chi_j).$$

$$K \simeq \mathbb{C}, \quad V = \sum_i [\chi_i], \qquad L(V) = \prod_i L(\chi_i). \tag{3.7.2}$$

Proposition 3.8. Let K be a local field and $\overline{K}$ an algebraic closure of K.

(i) For every exact sequence

$$0 \longrightarrow V' \longrightarrow V \longrightarrow V'' \longrightarrow 0$$

of complex representations of $W(\overline{K}/K)$, one has

$$L(V) = L(V')L(V''). \tag{3.8.1}$$

(ii) Let L be a finite separable extension of K in $\overline{K}$, and let V_L be a complex representation of $W(\overline{K}/L)$. Let V_K be the representation induced to $W(\overline{K}/K)$. Then

$$L(V_K) = L(V_L). \tag{3.8.2}$$

Assertion (i) is trivial. In the archimedean case, assertion (ii) follows from the duplication formula

$$\Gamma_{\mathbb{C}}(s) = \Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1),$$

corresponding to

$$\mathrm{Ind}([\omega_s]) = [\omega_s] + [\chi^{-1}\omega_{s+1}].$$

We prove (ii) for K nonarchimedean. Let k and ℓ be the residue fields of K and L. Then

$$V_K^{I_K} = \left(\mathrm{Ind}_{W(\overline{K}/L)}^{W(\overline{K}/K)} V_L\right)^{I_K} \simeq \mathrm{Ind}_{W(\overline{\ell}/\ell)}^{W(\overline{k}/k)}(V_L^{I_L})$$

as $W(\overline{k}/k)$-representations, both sides being identified with

$$\mathrm{Hom}_{W(\overline{K}/L)}(W(\overline{k}/k), V_L),$$

the covariant functions from $W(\overline{k}/k)$ to V_L, $W(\overline{K}/L)$ acting on $W(\overline{k}/k)$ by translation; cf. the commutative diagram

$$\begin{array}{ccc} W(\overline{K}/L) & \longrightarrow & W(\overline{\ell}/\ell) \\ \downarrow & & \downarrow \\ W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k). \end{array}$$

This reduces (3.8.2) to the following well-known lemma.

Lemma 3.9. Let F be an endomorphism of V, let n be an integer, and let F_n be the endomorphism of

$$V_n = V \otimes \mathbb{C}^n$$

such that, for a basis $e_0, \ldots, e_{n-1}$ of $\mathbb{C}^n$,

$$F_n(v \otimes e_i) = v \otimes e_{i+1} \quad (0 \leq i < n-1), \qquad F_n(v \otimes e_{n-1}) = F(v) \otimes e_0.$$

Then

$$\det(1 - F_n t) = \det(1 - F t^n).$$

By the principle of continuation of algebraic identities, one may suppose F diagonal in a suitable basis of V. One is immediately reduced to the case $\dim(V) = 1$, where F is multiplication by a number a^n, $a \neq 0$. In the basis $f_i = a^{-i}(v \otimes e_i)$, one has $F_n(f_i) = a f_{i+1}$, for $i \in \mathbb{Z}/n\mathbb{Z}$. The eigenvectors of F_n are the

$$\sum_i \chi(i) f_i$$

where χ runs through the characters of $\mathbb{Z}/n\mathbb{Z}$, and the determinant identity follows.

3.10. Let K be a global field, an $\mathfrak{A}$-field in the sense of [15]. For every place v of K, write K_v for the local field which is the completion of K at v. The objects defined above for local fields will be denoted, for K_v, with a subscript v. Let $\mathbb{A}$ be the adele ring of K, let $\| \ \|$ be the absolute value satisfying

$$d(ax) = \|a\| \, dx$$

for a Haar measure dx on $\mathbb{A}$, and let ω_s be the quasi-character $\|x\|^s$ of $\mathbb{A}^*$. If x_v is the component of $x \in \mathbb{A}$ in K_v, then

$$\|x\| = \prod_v \|x_v\|_v.$$

Let dx denote the unique Haar measure on $\mathbb{A}$ such that

$$\int_{\mathbb{A}/K} dx = 1$$

(the Tamagawa measure), let d^*x be a Haar measure on $\mathbb{A}^*$, and let ψ be a character trivial on $\mathbb{A}/K$, with local components ψ_v. For f a C^∞ function with compact support on $\mathbb{A}$, put

$$\widehat{f}(y) = \int_{\mathbb{A}} f(x)\psi(xy) \, dx.$$

3.11. Tate's global functional equation [14] says, for χ a quasi-character of $\mathbb{A}^*/K^*$,

$$\int_{\mathbb{A}^*} \widehat{f}(x)\chi'(x)\, d^*x = \int_{\mathbb{A}^*} f(x)\chi(x)\, d^*x, \qquad \chi' = \omega_1\chi^{-1}, \tag{3.11.1}$$

the integrals being defined by analytic continuation in the families $\chi\omega_s$. Put

$$L(\chi) = \prod_v L(\chi_v) \tag{3.11.2}$$

by analytic continuation, and, for $dx = \otimes_v dx_v$,

$$\varepsilon(\chi) = \prod_v \varepsilon(\chi_v, \psi_v, dx_v). \tag{3.11.3}$$

The product (3.11.3) is independent of the choice of ψ and of the decomposition of dx, as follows from (3.3.2) and (3.3.3). If, for almost all v,

$$\int_{\mathcal{O}_v} dx_v = 1,$$

then almost all its factors are equal to 1. Formulas (3.3.1) and (3.11.1) give the global functional equation of Hecke L-functions:

$$\varepsilon(\chi)L(\chi') = L(\chi). \tag{3.11.4}$$

3.12. The results stated below are proved in [1] and [19]. Let K be a global field, $\overline{K}$ an algebraic closure of K, and V a complex representation of $\mathrm{Gal}(\overline{K}/K)$, always assumed continuous: the action of $\mathrm{Gal}(\overline{K}/K)$ factors through a finite Galois group $\mathrm{Gal}(K'/K)$. For every place v of K, choose a place of $\overline{K}$ above it, and let V_v be the representation obtained from V by restriction to the subgroup

$$W(\overline{K}_v/K_v)$$

of the decomposition group $\mathrm{Gal}(\overline{K}_v/K_v)$.

(A) For χ a quasi-character of $\mathbb{A}^*/K^*$, with local components χ_v, the infinite product

$$L(V,\chi) = \prod_v L(V_v \otimes \chi_v)$$

can be defined by analytic continuation in χ in the families $\chi\omega_s$. It converges for $\Re(s)$ large and extends to a meromorphic function of s.

(B) Let V^* denote the dual of V. One has

$$L(V,\chi) = \varepsilon(V,\chi)\, L(V^*,\chi'), \qquad \chi' = \omega_1\chi^{-1},$$

with $\varepsilon(V,\chi) \in \mathbb{C}^*$, of the form Ae^{Bs} as χ varies in the families $\chi\omega_s$.

(C) Let L be a finite extension of K in $\overline{K}$, let $N_{L/K}$ be the norm, let V_L be a representation of $\mathrm{Gal}(\overline{K}/L)$, and put

$$V_K = \mathrm{Ind}_{\mathrm{Gal}(\overline{K}/L)}^{\mathrm{Gal}(\overline{K}/K)}(V_L).$$

For χ a quasi-character of $\mathbb{A}_K^*/K^*$, one has

$$L(V_K,\chi) = L(V_L, \chi\circ N_{L/K}),$$

and similarly

$$\varepsilon(V_K,\chi) = \varepsilon(V_L, \chi\circ N_{L/K}).$$

(D) One also has

$$\varepsilon(V'\oplus V'',\chi) = \varepsilon(V',\chi)\varepsilon(V'',\chi),$$

which makes it possible to define $\varepsilon(V,\chi)$ when V is only a virtual representation.

The assertions (A) and (B) are proved by reduction to the case $\dim V = 1$, via Brauer's theorem. Assertion (C) follows from (B) and from the compatibility of norm and transfer recalled above.

4. Existence of local constants

Theorem 4.1. There exists a unique function ε, satisfying conditions (1) to (4) below, which associates a number

$$\varepsilon(V, \psi, dx) \in \mathbb{C}^*$$

to each isomorphism class of sextuples $(K, \overline{K}, \psi, dx, V, \rho)$, consisting of a local field K, an algebraic closure $\overline{K}$ of K, a nontrivial additive character $\psi : K \to \mathbb{C}^*$, a Haar measure dx on K, a finite-dimensional complex vector space V, and a representation

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V).$$

(1) For every exact sequence of representations

$$0 \to V' \to V \to V'' \to 0,$$

one has

$$\varepsilon(V, \psi, dx) = \varepsilon(V', \psi, dx)\varepsilon(V'', \psi, dx).$$

This condition shows that $\varepsilon(V, \psi, dx)$ depends only on the class of V in the Grothendieck group of representations of $W(\overline{K}/K)$, and allows $\varepsilon(V, \psi, dx)$ to be interpreted when V is only a virtual representation.

(2)

$$\varepsilon(V, \psi, a\,dx) = a^{\dim V}\varepsilon(V, \psi, dx).$$

In particular, for virtual V of dimension 0, $\varepsilon(V, \psi, dx)$ is independent of dx; it is denoted $\varepsilon(V, \psi)$.

(3) If L is a finite separable extension of K in $\overline{K}$, and V_K is the virtual representation of $W(\overline{K}/K)$ induced by a virtual representation of dimension zero V_L of $W(\overline{K}/L)$, then

$$\varepsilon(V_K, \psi) = \varepsilon(V_L, \psi \circ \mathrm{Tr}_{L/K}).$$

(4) If $\dim V = 1$, and ρ is defined by a quasi-character χ of K^*, then

$$\varepsilon(V, \psi, dx) = \varepsilon(\chi, \psi, dx).$$

4.2. Let

$$v = \sum_{i,j} n_{ij}(H_i, \chi_{ij}) \in R_0^+(W(\overline{K}/K))$$

(1.10). Assume that the H_i are pairwise nonconjugate. Let K_i be the extension of K defined by H_i, and again denote by χ_{ij} the quasi-character of K_i^* defined by χ_{ij}. For any choice of Haar measures dx_i on the K_i, and for ψ an additive character of K, put

$$\varepsilon(v, \psi) = \prod_{i,j} \varepsilon(\chi_{ij}, \psi \circ \mathrm{Tr}_{K_i/K}, dx_i)^{n_{ij}}. \tag{4.2.1}$$

It follows from (3.3.2) that $\varepsilon(v, \psi)$ is independent of the choice of the dx_i.

Lemma 4.3. With the notation 1.9.8, for $v \in R_0^+(W(\overline{K}/K))$,

$$\varepsilon(v, \psi(ax)) = \det(v)(a)\,\varepsilon(v, \psi).$$

It is enough to check this for $v = (H, \chi) - (H, 1)$, where H corresponds to an extension L of K. Let v_0 be the virtual representation $[\chi] - [1]$ of $W(\overline{K}/L)$. Formula (3.3.3) gives the assertion over L. It is known ([10], XII, §4) that the inclusion of K^* into L^* is identified with the transfer

$$W(\overline{K}/K)^{\mathrm{ab}} \longrightarrow W(\overline{K}/L)^{\mathrm{ab}}.$$

Since $b(v) = \mathrm{Ind}(v_0)$, the lemma follows from 1.2.

Lemma 4.4. The theorem is true in the archimedean case.

For $K \simeq \mathbb{C}$, define ε by (1) and (4), and (2) is automatic. For $K \simeq \mathbb{R}$, one checks from 2.9 that

$$b : R^+(W(\overline{K}/K)) \longrightarrow R(W(\overline{K}/K))$$

is surjective, with kernel generated by the $\chi_s - \chi_t$ where

$$\chi_s = (K^*, \omega_s) - [\omega_s] - [\chi^{-1}\omega_{s+1}].$$

For a virtual representation V, and $v \in R_0^+(W(\overline{K}/K))$ such that

$$b(v) = V - \dim(V)[1],$$

put

$$\varepsilon(V, \psi, dx) = \varepsilon(v, \psi)\, \varepsilon(1, \psi, dx)^{\dim V}. \tag{4.4.1}$$

We check that the right-hand side is independent of the choice of v, i.e. that, for v satisfying $b(v) = 0$, one has $\varepsilon(v, \psi) = 1$. By 4.3, it is enough to check this for $\psi(x) = \exp(2\pi i x)$. For this choice of ψ, the assertion follows from the preceding determination of $\mathrm{Ker}(b)$, and from the fact that the right-hand sides of (3.4.1), (3.4.2) are independent of s. It is clear that ε defined by (4.4.1) satisfies (1) to (4).

We now restrict to the nonarchimedean case.

4.5. Let K be a nonarchimedean local field, $\overline{K}$ a separable closure of K, and $K_{nr} \subset \overline{K}$ the maximal unramified extension of K. Let J be an open normal subgroup of the inertia group

$$I = \mathrm{Gal}(\overline{K}/K_{nr})$$

and let L be the corresponding extension of K_{nr}. For $\sigma \in I/J$, put

$$t_{I/J}(\sigma) = \inf\big(v_L(\sigma(x) - x) \mid x \text{ integral in } L\big),$$

and define functions $a_{I/J}$ and $sw_{I/J}$ by

$$a_{I/J}(\sigma) = -t_{I/J}(\sigma) \quad (\sigma \neq e), \tag{4.5.1}$$

$$sw_{I/J}(\sigma) = 1 - t_{I/J}(\sigma) \quad (\sigma \neq e),$$

$$\sum_\sigma a_{I/J}(\sigma) = \sum_\sigma sw_{I/J}(\sigma) = 0.$$

By Artin (see [10], VI, §2, th. 1), $a_{I/J}$ is the character of a representation $A_{I/J}$ of I/J, the Artin representation. By loc. cit., prop. 2, $sw_{I/J}$ is likewise the character of a representation $Sw_{I/J}$, the Swan representation. These representations are defined only up to isomorphism. $A_{I/J}$ is the sum of $Sw_{I/J}$ and the augmentation representation of I/J. For $I \supset J \supset K$, one has (loc. cit., prop. 3)

$$(A_{I/K})^{J/K} \xrightarrow{\sim} A_{I/J}, \qquad (Sw_{I/K})^{J/K} \xrightarrow{\sim} Sw_{I/J}. \tag{4.5.2}$$

If d is the valuation of the discriminant $\delta_{L/K_{nr}}$, i.e. the valuation of the different $\mathfrak{D}_{L/K}$ ([10], III, §3, prop. 6), and r denotes a regular representation, then (loc. cit., prop. 4)

$$A_{I/K}|_{J/K} = d\, r_{J/K} + A_{J/K}. \tag{4.5.3}$$

Let V be a representation of $W(\overline{K}/K)$, trivial on $J \subset I$, with character ψ. Define the Artin and Swan conductors of V by

$$a(V) = \dim \mathrm{Hom}_{I/J}(A_{I/J}, V) = sw(V) + \dim V - \dim V^I, \tag{4.5.4}$$

$$sw(V) = \dim \mathrm{Hom}_{I/J}(Sw_{I/J}, V) = \frac{1}{|I/J|} \sum_\sigma sw_{I/J}(\sigma)\psi(\sigma).$$

By 4.5.2, these conductors are independent of the choice of J. For V of dimension 1, defined by a quasi-character χ, one has (loc. cit., prop. 5)

$$a(V) = \text{the conductor } a(\chi) \text{ of } \chi. \tag{4.5.5}$$

The preceding notation is therefore compatible with that introduced in 3.4.

With the notation 1.9.8, one has:

Lemma 4.5. For $v \in R_0^+(W(\overline{K}/K))$ and χ an unramified quasi-character of K^*,

$$\varepsilon(v \cdot \chi, \psi) = \chi(\pi)^{a(v)} \varepsilon(v, \psi).$$

It suffices to verify this for $v = (H, \lambda) - (H, \mu)$, where H defines an extension L of K, and λ, μ are two quasi-characters of L^*. Let d be the valuation of the different of L/K, let f be the residue-degree, let $I \subset W(\overline{K}/K)$ and $J \subset W(\overline{K}/L)$ be the inertia groups, and choose $K \subset J$ small enough. We compute $a(v)$. The adjunction formula for induced representations, (4.5.3), and (4.5.5) give, for every quasi-character ν of H, i.e. of L^*,

$$a((H,\nu)) = \dim \mathrm{Hom}_{I/K}(A_{I/K}, \mathrm{Ind}_H^W(\nu)) = f \dim \mathrm{Hom}_{J/K}(A_{J/K} + d\, r_{J/K}, [\nu]|_J) = f\, a(\nu) + fd. \qquad (4.5.1')$$

Thus

$$a(v) = f(a(\lambda) - a(\mu)).$$

For ν a quasi-character of L^*, χ_1 an unramified quasi-character of L^*, and $n = n(\psi \circ \mathrm{Tr}_{L/K})$ as in 3.4, one has by (3.4.3.1) or (3.4.3.2)

$$\frac{\varepsilon(\nu\chi_1, \psi \circ \mathrm{Tr}, dx)}{\varepsilon(\nu, \psi \circ \mathrm{Tr}, dx)} = \chi_1(\pi_L)^{a(\nu)+n}.$$

Hence

$$\varepsilon(v \cdot \chi, \psi)\varepsilon(v, \psi)^{-1} = \chi \circ N_{L/K}\left(\pi_L^{a(\lambda)-a(\mu)}\right) = \chi(\pi)^{f(a(\lambda)-a(\mu))},$$

which is the announced formula.

Terminology 4.6. Let K_1 be a Galois extension, finite or not, of K, with Galois group G. Let α be a quasi-character of K^*, and ψ a nontrivial additive character of K. We shall say that there exists an α, ψ-theory of constants for K_1/K if there exists $\varepsilon_{\alpha,\psi}$ of the following type:

(0) For H an open subgroup of G, defining a finite extension L of K, for V a representation of H, and dx a Haar measure on L,

$$\varepsilon_{\alpha,\psi}(V, dx) \in \mathbb{C}^*$$

is defined.

(1) For every exact sequence of representations

$$0 \to V' \to V \to V'' \to 0,$$

one has

$$\varepsilon_{\alpha,\psi}(V, dx) = \varepsilon_{\alpha,\psi}(V', dx)\varepsilon_{\alpha,\psi}(V'', dx).$$

Thus $\varepsilon_{\alpha,\psi}$ still makes sense for virtual representations.

(2) One has

$$\varepsilon_{\alpha,\psi}(V, a\,dx) = a^{\dim V}\varepsilon_{\alpha,\psi}(V, dx).$$

In particular, for virtual V of dimension 0, $\varepsilon_{\alpha,\psi}(V, dx)$ is independent of dx; it is denoted $\varepsilon_{\alpha,\psi}(V)$.

(3) For $H_1 \subset H_2 \subset G$, defining $L_1 \supset L_2$, and V a virtual representation of dimension 0 of H_1, one has

$$\varepsilon_{\alpha,\psi}(\mathrm{Ind}_{H_1}^{H_2} V) = \varepsilon_{\alpha,\psi}(V).$$

(4) For V of dimension 1, defined by a character χ of L^*,

$$\varepsilon_{\alpha,\psi}(V, dx) = \varepsilon(\chi\,\alpha \circ N_{L/K},\, \psi \circ \mathrm{Tr}_{L/K},\, dx).$$

Lemma 4.7. Let K_1/K and α be as above, and let ψ be a nontrivial additive character of K. There exists at most one α,ψ-theory of constants for K_1/K. For one to exist, it is necessary and sufficient that, for every

$$v \in \mathrm{Ker}(R_0^+(G) \longrightarrow R(G)),$$

one have

$$\varepsilon(v\cdot\alpha,\psi) = 1.$$

Let $H \subset G$, L, and V be as above. By 1.5, there exists $v \in R_0^+(H)$ whose image in $R(H)$ is $V - \dim(V)[1]$. Necessarily

$$\varepsilon_{\alpha,\psi}(V,dx) = \varepsilon(v\cdot(\alpha\circ N_{L/K}),\, \psi\circ \mathrm{Tr}_{L/K})\,\varepsilon(\alpha\circ N_{L/K},\, \psi\circ \mathrm{Tr}_{L/K},\, dx)^{\dim V}, \tag{4.7.1}$$

which gives uniqueness. The hypothesis ensures that $\varepsilon_{\alpha,\psi}$, defined by this formula, is independent of the choice of v; one easily checks axioms (0) to (4).

Lemma 4.8.

(i) Let β be an unramified quasi-character of K^*, respectively let $\psi' = \psi(ax)$ be a nontrivial additive character. For there to exist an α,ψ-theory of constants for K_1/K, it is necessary and sufficient that there exist an $\alpha\beta,\psi$-theory, respectively an α,ψ'-theory. For H, L, V as above,

$$\psi_L = \psi\circ \mathrm{Tr}_{L/K}, \qquad \beta_L = \beta\circ N_{L/K}, \qquad \alpha_L = \alpha\circ N_{L/K},$$

one then has

$$\varepsilon_{\alpha\beta,\psi}(V,dx) = \beta_L\Big(\pi^{\,a(V\alpha_L)+\dim(V)n(\psi_L)}\Big)\,\varepsilon_{\alpha,\psi}(V,dx), \tag{4.8.1}$$

$$\varepsilon_{\alpha,\psi'}(V,dx) = \|a\|_L^{-\dim(V)}\det(V\alpha_L)(a)\,\varepsilon_{\alpha,\psi}(V,dx). \tag{4.8.2}$$

The first assertion follows from criterion 4.7 and from 4.5, respectively from 4.3. It is enough to prove (4.8.1) and (4.8.2) for $K = L$. Moreover, one may suppose either that $V \in R^0(\mathrm{Gal}(K_1/K))$, or that $V = [1]$. In the first case one applies 4.5 and 4.3. In the second, one uses (4.7.1) and the definition 3.4.3.

We shall say that there exists an α-theory of constants for K_1/K if, for every nontrivial additive character ψ, or for one such ψ, there exists an α,ψ-theory of constants for K_1/K. For $K_1 \supset L \supset K$, there is then an $\alpha\circ N_{L/K}$-theory of constants for K_1/L.

Reduction 4.9. Theorem 4.1 is implied by the following assertion:

($*$) for every nonarchimedean local field K and every finite Galois extension K_1/K,

there exists an unramified quasi-character α of K^* and an α-theory of constants for K_1/K.

By 4.8, assertion ($*$) implies that, for $\overline{K}$ a separable closure of K, and for any unramified quasi-character α of K^*, there exists an α-theory of constants for $\overline{K}/K$. The discussion below will allow us to pass from the Galois group to the Weil group.

4.10. Every representation V of $W(\overline{K}/K)$ is trivial on a suitable finite-index subgroup J of inertia I. Let F be a geometric Frobenius. Since I/J is finite, a suitable power F^n of F acts trivially on I/J by conjugation, and hence is central in $W(\overline{K}/K)/J$. We shall say that V has a *type* if a power F^m of F^n has only one eigenvalue a. The type of V is then the element of the inductive limit

$$\varinjlim_{n|m}\{X_n,\varphi_{n,m}\}, \qquad X_n = \mathbb{C}^*, \quad \varphi_{n,m}(x) = x^{m/n},$$

defined by $a \in X_m = \mathbb{C}^*$. Every irreducible representation has a type.

The representations of $W(\overline{K}/K)$ of a given type τ form an abelian category R_τ, and the category of all representations of $W(\overline{K}/K)$ is the sum of these subcategories. If $R_\tau(W(\overline{K}/K))$ is the Grothendieck group of R_τ, one therefore has

$$R(W(\overline{K}/K)) = \bigoplus_\tau R_\tau(W(\overline{K}/K)). \tag{4.10.1}$$

The semisimple representations of type 1 are exactly the continuous representations of $\mathrm{Gal}(\overline{K}/K)$. If X_τ is a quasi-character of type τ, and there are quasi-characters, even unramified ones, of all types, then twisting by X_τ defines an isomorphism

$$\otimes X_\tau : R(\mathrm{Gal}(\overline{K}/K)) \xrightarrow{\sim} R_\tau(W(\overline{K}/K)). \tag{4.10.2}$$

Let L be a finite separable extension of K in $\overline{K}$, and let f be the residue-degree. If a representation V of $W(\overline{K}/L)$ has type σ, the representation induced to $W(\overline{K}/K)$ has type σ^f. We say that V is of K-type τ if $\sigma^f = \tau$. The set of representations of $W(\overline{K}/L)$ of a given K-type is therefore stable under induction.

Variant 4.10.3. The preceding arguments apply to every group which is an extension of $\mathbb{Z}$ by a finite group, and to projective limits of such extensions. The case of global Weil groups will be useful later.

4.11. Assume $(*)$. For every type τ, let X_τ be an unramified character of type τ. Since there exists an X_τ-theory of constants, one can define ε on $R_\tau(W(\overline{K}/K))$ by

$$\varepsilon(V, \psi, dx) = \varepsilon_{X_\tau,\psi}(V \otimes X_\tau^{-1}, dx).$$

It follows from the uniqueness 4.7 that this ε is independent of the choice of X_τ. One then defines ε on $R(W(\overline{K}/K))$ by additivity (4.10.1). By 4.7, if a theory 4.1 exists, it is given by this ε.

We have arranged things so that ε satisfies (1), (2), (4), and (3) when V_L has a type. It remains to verify (3) for V_L of the form

$$[\lambda \circ N_{L/K}] - [\mu \circ N_{L/K}],$$

where λ and μ are two unramified quasi-characters of K^*, of given types. By (3.4.3.1), (3.3.2), and (3.3.3), one has

$$\varepsilon\big([\lambda \circ N_{L/K}] - [\mu \circ N_{L/K}], \psi \circ \mathrm{Tr}_{L/K}\big) = (\lambda\mu^{-1}) \circ N_{L/K}\left(\pi_L^{n(\psi\circ \mathrm{Tr}_{L/K})}\right).$$

On the other hand, by 4.8,

$$\varepsilon\big(\mathrm{Ind}([\lambda \circ N_{L/K}] - [\mu \circ N_{L/K}]), \psi\big) = (\lambda\mu^{-1})\left(\pi_K^{a(\mathrm{Ind}(\mu\circ N_{L/K}))+[L:K]\,n(\psi)}\right).$$

Let d be the valuation of the different of L/K, e the ramification index, and f the residue-degree. Formula (4.5.1) gives

$$a(\mathrm{Ind}(\mu \circ N_{L/K})) = fd.$$

The formula to be proved is therefore reduced to

$$f\, n(\psi \circ \mathrm{Tr}_{L/K}) = fd + [L:K]\, n(\psi),$$

that is,

$$n(\psi \circ \mathrm{Tr}_{L/K}) = d + e\, n(\psi),$$

which is essentially the definition of the different ([10], III, §3). The key point in the proof of $(*)$ is the following lemma.

Lemma 4.12. Let K'/K be a Galois extension of global fields. For every place w of K, choose a place of K' above it and denote it again by w. Let α be a quasi-character of the idèle-class group of K, and let v be a place of K which does not decompose in K'. If, for every finite place $w \neq v$ of K, there exists an α_w-theory of constants for K'_w/K_w, then there exists an α_v-theory of constants for K'_v/K_v.

Let ψ be a nontrivial character of $\mathbb{A}_K/K$, and let ψ_v be its v-component. We shall construct an α_v, ψ_v-theory of constants for K'_v/K_v.

One has

$$\mathrm{Gal}(K'/K) = \mathrm{Gal}(K'_v/K_v).$$

A subgroup H of $\mathrm{Gal}(K'_v/K_v)$ therefore defines, besides an extension L_v of K_v, an extension L of K with completion L_v. Let dx be the Tamagawa measure on $\mathbb{A}_L$, so that $\mathbb{A}_L/L$ has volume 1, and write

$$dx = \otimes_w dx_w,$$

with dx_v chosen in advance. Every representation V_v of H defines a representation V of $\mathrm{Gal}(K'/L)$, and, by restriction, representations V_w of the decomposition groups $\mathrm{Gal}(K'_w/K_w)$. A global constant $\varepsilon(V,\alpha)$ was defined in 3.12.

Let w be a place of L, and again write w for its image in K. For w infinite, one has by 4.4 local constants

$$\varepsilon\big(V_w, (\alpha\circ N_{L/K})_w, (\psi\circ \mathrm{Tr}_{L/K})_w, dx_w\big).$$

For w finite and different from v, one also has, by hypothesis, constants

$$\varepsilon_{\alpha_w,\psi_w}(V_w, dx_w).$$

Denoting both kinds by $\varepsilon_\alpha(V_w,\psi_w,dx_w)$, put

$$\varepsilon_{\alpha_v,\psi_v}(V_v,dx_v) = \frac{\varepsilon(V,\alpha)}{\prod\limits_{w\neq v}\varepsilon_\alpha(V_w,\psi_w,dx_w)}.$$

One checks that $\varepsilon_{\alpha_v,\psi_v}$ does not depend on the choice of the decomposition $dx=\otimes dx_w$, and satisfies axioms (0) to (4). The key point is 3.12(C).

Lemma 4.13. Let K'/K be a Galois extension of local fields. There exists a Galois extension of global fields ${}^1K'/{}^1K$, and a place v of ${}^1K$ which does not decompose in ${}^1K'$, such that K is identified with the completion of ${}^1K$ at v, and K' with a subextension of ${}^1K'_v/{}^1K_v$.

This is trivial: first take ${}^1K'/{}^1K$ with ${}^1K'_v\simeq K'\supset K={}^1K_v$, and then replace ${}^1K$ by the extension defined by a decomposition group at v.

Lemma 4.14. Let K be a global field, let S be a finite set of finite places of K, and let $m: S\to\mathbb{N}$. There exists a character α of the idèle-class group such that, for $w\in S$, the conductor of α_w is $\geq m(w)$, and such that, for every finite place outside S, α_w is unramified.

If K is a number field, one can indeed freely specify α on

$$\prod_{w\ \text{finite}}\mathcal{O}_w^*,$$

because this compact subgroup of $\mathbb{A}^*$ meets K^* only in $\{1\}$. If K is a function field, with field of constants k, one has

$$\prod_w \mathcal{O}_w^*\cap K^* = k^*,$$

and one observes that there exist characters of $\mathcal{O}_w^*$, of arbitrarily large conductor, which are trivial on k^*.

By 4.12, 4.13, and 4.14, assertion 4.9 $(*)$, and hence 4.1, follow from the following two lemmas, where K'/K is a finite Galois extension of nonarchimedean local fields, with Galois group G, and α is a quasi-character of K^*.

Lemma 4.15. If α and K'/K are unramified, there exists an α-theory of constants for K'/K.

Choose for ψ a character of K such that $n(\psi)=0$. It is then enough to take

$$\varepsilon_{\alpha,\psi}(V,dx) = \left(\int_L dx\right)^{\dim(V)}.$$

Lemma 4.16. For K'/K fixed, there exists m such that, if the conductor of α is $\geq m$, there exists an α-theory of constants for K'/K.

Let α have conductor m, and put

$$n=\left[\frac{m+1}{2}\right].$$

Then the function $\alpha(1+a)$ is additive in a for $v(a)\geq m-n$. If ψ is a nontrivial additive character of K, there is therefore a y, of valuation $-(m+n(\psi))$, well-defined modulo $\pi^{-(n+n(\psi))}$, such that

$$\alpha(1+a)=\psi(ya)\qquad \text{for } v(a)\geq n.$$

For χ a quasi-character of K^*, of conductor $\leq m-n$, one then has

$$\varepsilon(\chi\alpha,\psi,dx)=\sum_{u\in\mathcal{O}^*/(1+(\pi^n))}\int_{\pi^{-m-n(\psi)}u}\chi^{-1}(x)\alpha^{-1}(x)\psi(x)\,dx. \tag{4.16.1}$$

For $\pi^{-m-n(\psi)}u$ not contained in $y(1+(\pi^{m-n}))$, the corresponding partial integral is zero: it is the integral of a nontrivial additive character. Since χ is constant with value $\chi(y)$ on $y(1+(\pi^{m-n}))$, one has

$$\varepsilon(\chi\alpha,\psi,dx)=\chi^{-1}(y)\,\varepsilon(\alpha,\psi,dx).$$

We shall show that, for m sufficiently large, one obtains an α,ψ-theory of constants for K'/K by putting, for $H\subset G$ defining an extension L of K, and V a representation of H,

$$\varepsilon_{\alpha,\psi}(V,dx)=\det(V)^{-1}(y)\,\varepsilon(\alpha\circ N_{L/K},\psi\circ\mathrm{Tr}_{L/K},dx)^{\dim V}.$$

Axioms (0), (1), and (2) are trivial. Axiom (3) follows from the fact that the inclusion of K^* in L^* is identified with the transfer

$$W(\overline{K}/K)^{\mathrm{ab}}\longrightarrow W(\overline{K}/L)^{\mathrm{ab}}$$

([10], XIII, §4), and from 1.2. It remains to prove axiom (4).

Let $H\subset G$ correspond to L/K, and let e be the ramification index. Recall that, for $u\in\mathcal{O}_L$ and for n the least integer such that $ne\geq 2v(u)$,

$$N_{L/K}(1+u)=1+\mathrm{Tr}_{L/K}(u)\pmod{\pi^n}. \tag{4.16.2}$$

This is checked by writing the norm of $1+u$ as the product of its conjugates. We shall write $O(1)$ for any number bounded in terms of K'/K alone. By 4.16.2, if α has conductor m, then $\alpha\circ N_{L/K}$ has conductor $em-O(1)$. Moreover, for

$$\frac{v(a)}{e}\geq\left[\frac{m+1}{2}\right]+O(1),$$

one has

$$\alpha\circ N_{L/K}(1+a)=\alpha(1+\mathrm{Tr}(a))=\psi(y\ \mathrm{Tr}\,a)=\psi\circ\mathrm{Tr}_{L/K}(ya).$$

For every character χ of H, of conductor $O(1)$, one can repeat the reasoning used above, obtaining the desired formula

$$\varepsilon(\chi\,\alpha\circ N_{L/K},\psi\circ\mathrm{Tr}_{L/K},dx)=\chi^{-1}(y)\,\varepsilon(\alpha\circ N_{L/K},\psi\circ\mathrm{Tr}_{L/K},dx).$$

5. Formula list

5. Formula list.

(5.1) Besides the local constant $\varepsilon(V,\psi,dx)$ defined in 4.1, we shall also consider, in the nonarchimedean case, the constant ε_0 defined by

$$\varepsilon(V,\psi,dx)=\varepsilon_0(V,\psi,dx)\ \det(-F,V^I)^{-1}.$$

The constants ε and ε_0 have the following properties.

(5.2) ε and ε_0 are additive in V, and hence still make sense for virtual V.

(5.3) ε and ε_0 are independent of dx for virtual V of dimension 0. They are denoted $\varepsilon(V,\psi)$ and $\varepsilon_0(V,\psi)$. More precisely,

$$\varepsilon(V,\psi,a\,dx)=a^{\dim V}\varepsilon(V,\psi,dx),$$

and similarly for ε_0.

(5.4) One has

$$\varepsilon(V,\psi(ax),dx)=\det(V)(a)\,\|a\|^{-\dim V}\varepsilon(V,\psi,dx),$$

and similarly for ε_0.

(5.5) For K nonarchimedean and χ an unramified quasi-character of K^*, one has

$$\varepsilon(V\chi,\psi,dx)=\chi\Big(\pi^{a(V)+\dim(V)n(\psi)}\Big)\,\varepsilon(V,\psi,dx), \tag{5.5.1}$$

and

$$\varepsilon_0(V\chi,\psi,dx)=\chi\Big(\pi^{\dim(V)+sw(V)+\dim(V)n(\psi)}\Big)\,\varepsilon_0(V,\psi,dx).$$

In particular, if

$$\varepsilon(V,\psi,dx,s)=\varepsilon(V\omega_s,\psi,dx),$$

and similarly for ε_0, then

$$\varepsilon(V,\psi,dx,s)=\varepsilon(V,\psi,dx)\,q^{-(a(V)+\dim(V)n(\psi))s}, \tag{5.5.2}$$

and

$$\varepsilon_0(V,\psi,dx,s)=\varepsilon_0(V,\psi,dx)\,q^{-(sw(V)+\dim(V)(n(\psi)+1))s}.$$

For W an unramified representation, one similarly has

$$\varepsilon(V\otimes W,\psi,dx)=\det W\Big(\pi^{a(V)+\dim(V)n(\psi)}\Big)\,\varepsilon(V,\psi,dx)^{\dim W}, \tag{5.5.3}$$

$$\varepsilon_0(V\otimes W,\psi,dx)=\det W\Big(\pi^{sw(V)+\dim(V)(n(\psi)+1)}\Big)\,\varepsilon_0(V,\psi,dx)^{\dim W}.$$

(5.6) For L/K a finite separable extension, V_L a virtual representation of dimension 0 of $W(\overline{K}/L)$, and V_K the representation induced to $W(\overline{K}/K)$, one has

$$\varepsilon(V_K,\psi)=\varepsilon(V_L,\psi\circ\mathrm{Tr}_{L/K}), \tag{5.6.1}$$

and similarly for ε_0. In other words, there exists

$$\lambda(L/K,\psi,dx_L,dx_K)\in\mathbb{C}^*$$

such that, for V_L of arbitrary dimension,

$$\varepsilon(V_K,\psi,dx_K)=\lambda(L/K,\psi,dx_L,dx_K)^{\dim(V_L)}\varepsilon(V_L,\psi\circ\mathrm{Tr}_{L/K},dx_L). \tag{5.6.2}$$

The same formula holds for ε_0.

In Langlands [9], it is (5.6.2) which provides the framework of the theory. Moreover, he systematically takes dx to be the Haar measure self-dual for ψ, and omits it from the notation. The constants are also normalized to have complex absolute value 1. All of this forces on him another formula (5.4).

5.7. Let V^* be the dual of V, and let dx' be the Haar measure dual to dx, relative to ψ. One has

$$\varepsilon(V,\psi,dx)\,\varepsilon(V^*\omega_1,\psi(-x),dx')=1. \tag{5.7.1}$$

In particular, if K is nonarchimedean and V is unitary, so that $V^*=\overline{V}$, one has

$$|\varepsilon(V,\psi,dx)|^2=\left(\frac{dx'}{dx}\right)^{-\dim(V)}q^{a(V)+\dim(V)n(\psi)}. \tag{5.7.2}$$

5.8. ε^{-1} appears in Tate's local functional equation; putting $\chi'=\omega_1\chi^{-1}$,

$$\frac{\int\widehat{f}(x)\chi'(x)d^*x}{L(\chi')}=\varepsilon(\chi,\psi,dx)\,\frac{\int f(x)\chi(x)d^*x}{L(\chi)}\qquad\text{(Tate)}. \tag{5.8.1}$$

For K nonarchimedean, and $\gamma\in K^*$ of valuation $1+sw(\chi)+n(\psi)$, (3.4.3.4) gives

$$\varepsilon_0(\chi,\psi,dx)=\int_{\gamma^{-1}\mathcal{O}^*}\chi^{-1}(x)\psi(x)\,dx. \tag{5.8.2}$$

5.9. For K nonarchimedean, if χ is unramified, one has

$$\varepsilon(\chi, \psi, dx) = \chi(\pi^{n(\psi)})\, q^{n(\psi)} \int_{\mathcal{O}} dx.$$

In particular, if $n(\psi) = 0$ and $\int_{\mathcal{O}} dx = 1$, then

$$\varepsilon(\chi, \psi, dx) = 1.$$

5.10. Take K nonarchimedean, $a(\chi) \leq 1$, $n(\psi) = -1$, and

$$\int_{(\pi)} dx = 1.$$

Let χ_k and ψ_k be the characters of k^* and k defined by χ and ψ. One has

$$\varepsilon_0(\chi, \psi, dx) = -\tau(\chi_k, \psi_k), \tag{5.10.1}$$

where τ is the Gauss sum

$$\tau(\chi_k, \psi_k) = -\sum_{x \in k^*} \chi_k^{-1}(x)\psi_k(x), \tag{5.10.2}$$

equal to 1 for trivial χ_k. In particular,

$$\varepsilon_0([\chi] - [1], \psi) = \tau(\chi_k, \psi_k). \tag{5.10.3}$$

5.11. Let K be a global field. For V a representation of the global Weil group of K, ψ a nontrivial additive character of $\mathbb{A}_K/K$, and

$$dx = \bigotimes_v dx_v$$

the Tamagawa measure on $\mathbb{A}_K$, put

$$L(V) = \prod_v L(V_v), \tag{5.11.1}$$

$$\varepsilon(V) = \prod_v \varepsilon(V_v, \psi_v, dx_v). \tag{5.11.2}$$

The global functional equation is written

$$L(V) = \varepsilon(V)\, L(V^* \otimes \omega_1). \tag{5.11.3}$$

Proofs. Formulas 5.2, 5.3, 5.6, and 5.8 are 4.1, respectively (1), (2), (3), (4), together with 3.3. Formulas 5.4 and 5.5 follow from 4.8. It is enough to prove (5.7.1) for V defined by a character; the formula then follows from (5.8.1) and from the Fourier inversion formula. Formula (5.7.2) follows from (5.7.1), (5.3), and (5.4). Formula 5.9 follows from (3.4.3.1), and 5.10 from (5.8.2). Finally, 5.11 follows from the preceding facts and from (3.11.4), by known arguments [1], [19].

The restriction of 4.1 to the case of extensions and tamely ramified characters is equivalent to the following identities on Gauss sums (5.10.2). Here k denotes a finite field with q elements, and ψ a nontrivial additive character of k.

5.12. Hasse–Davenport. Let k' be an extension of k, of degree n, and let χ be a character of k^*. One has

$$\tau(\chi \circ N_{k'/k}, \psi \circ \mathrm{Tr}_{k'/k}) = \tau(\chi, \psi)^n. \tag{5.12.1}$$

5.13. Let χ be a character of k^*, and let n be an integer prime to q. Let X be a set of representatives of the isomorphism classes of pairs (k', χ'), consisting of an extension k' of k and a character χ' of k'^*, such that

$$\chi'(x^n) = \chi \circ N_{k'/k}(x), \tag{a}$$

and

$$\chi' \text{ is not of the form } \chi'' \circ N_{k'/k''} \tag{b}$$

for k'' intermediate between k and k', i.e.

$$\chi'^{q^{f'}} \neq \chi' \qquad (0 < f' < [k' : k]).$$

Put

$$\tau(\chi, \psi^n) = \lambda_n(\chi, \psi) \prod_{(k', \chi') \in X} \tau(\chi', \psi \circ \mathrm{Tr}_{k'/k}).$$

Then

$$\lambda_n(\chi, \psi) \text{ is independent of } \chi. \tag{5.13.1}$$

This identity splits into the following two identities.

5.14 (Langlands [9], 7.8). Let ℓ be a prime number prime to q, let f be the order of q in $(\mathbb{Z}/\ell)^*$, let k' be an extension of k of degree f, let T be a set of representatives of the $\mathrm{Gal}(k'/k)$-orbits in the nontrivial characters of order ℓ of k'^*, and let χ be a character of k^*. One has

$$\tau(\chi^\ell, \psi^\ell) \prod_{\mu \in T} \tau(\mu, \psi \circ \mathrm{Tr}_{k'/k}) = \tau(\chi, \psi) \prod_{\mu \in T} \tau(\chi \circ N_{k'/k} \cdot \mu, \psi \circ \mathrm{Tr}_{k'/k}).$$

5.15 (Langlands [9], 7.9). Let ℓ be a prime number dividing $q - 1$, let T be the set of characters of order ℓ of k^*, let χ be a character of k^* which is not an ℓ-th power, let k' be an extension of degree ℓ of k, and let χ' be a character of k'^* such that

$$\chi'(x^\ell) = \chi \circ N_{k'/k}(x).$$

One has

$$\tau(\chi, \psi^\ell) \prod_{\mu \in T} \tau(\mu, \psi) = \tau(\chi', \psi \circ \mathrm{Tr}_{k'/k}).$$

5.16. Proofs. If K'/K is a tamely ramified extension of nonarchimedean local fields, and if ψ is an additive character of K such that

$$n(\psi) = -1,$$

then still

$$n(\psi \circ \mathrm{Tr}_{K'/K}) = -1,$$

so that 5.10 applies both to K and to K'.

Let V be a tamely ramified representation of $W(\overline{K}/K)$. One can always write it as a sum of representations induced by tame characters χ_i of unramified extensions K_i/K:

$$V = \dim(V)[1] + \sum_i \mathrm{Ind}([\chi_i] - [1]) + \sum_i \big(\mathrm{Ind}([1]) - [K_i : K][1]\big).$$

Let k_i be the residue field of K_i, and let $\widetilde{\chi}_i$ be the character of k_i^* defined by χ_i. For dx a Haar measure on K such that

$$\int_{(\pi)} dx = 1,$$

one finds

$$\varepsilon_0(V, \psi, dx) = (-1)^{\dim V} \prod_i \tau(\widetilde{\chi}_i, \psi_k \circ \mathrm{Tr}_{k_i/k}). \tag{5.16.1}$$

Formula (5.12) is obtained by taking an unramified extension K'/K, and by writing that, for χ a character, here tamely ramified, of K^*, one has

$$\mathrm{Ind}(\chi \circ N_{K'/K} - [1]) = \sum_{\mu} ([\chi\mu] - [\mu]),$$

where μ runs through the unramified characters of order dividing $[K' : K]$. This formula (5.12) ensures that the right-hand side of (5.16.1) depends only on V.

The identities 5.13 are obtained by taking a totally ramified extension K'/K of degree n, and by combining 5.6.1 and (5.16.1) for V induced by a tame character χ of K'. Formulas (5.14) and (5.15) are the special case of (5.13.1) where n is a prime number ℓ. In 5.14, respectively 5.15, one supposes that the character χ of K'^* is, respectively is not, an ℓ-th power.

6. Reduction modulo ℓ of local constants

Let K be a nonarchimedean local field, let $\overline{K}$ be a separable closure of K, and resume the notation of 2.2. Let A be a commutative ring in which p is invertible. The exposition would be simplified, without loss of generality, by assuming A local, or even a field. For $x \in K$, $\|x\|$ has a meaning in A by hypothesis.

6.1. A Haar measure μ on K, with values in a $\mathbb{Z}[1/p]$-module M, is a finitely additive function on compact open subsets of K, with values in M, invariant under translation. The additive function defined by

$$\mu_0(a + x\mathcal{O}) = \|x\|$$

is the unique Haar measure with values in $\mathbb{Z}[1/p]$ such that

$$\mu_0(\mathcal{O}) = 1,$$

and every Haar measure with values in M is of the form

$$m\mu_0 \qquad (m = \mu(\mathcal{O}) \in M).$$

One can integrate a locally constant compactly supported function, with values in A, against a Haar measure with values in A.

6.2. Let

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}_A(V)$$

be a representation of the Weil group on a free, or projective, A-module V, and let J be an open normal subgroup of I on which ρ is trivial. For every prime number $\ell \neq p$, the Swan representation $Sw_{I/J}$ can be realized as a projective $\mathbb{Z}_\ell[I/J]$-module. If A is a $\mathbb{Z}_\ell$-algebra, then

$$\mathrm{Hom}_{I/J}(Sw_{I/J}, V)$$

is a projective A-module. If A is also local, it is free; its rank is the Swan conductor of V. More generally, one checks that the Swan conductor of V can always be defined as a locally constant integer-valued function on $\mathrm{Spec}(A)$. This conductor is invariant under extension of the scalar ring A.

6.3. If ζ is a p^n-th root of unity in A, the order of ζ is a locally constant function on $\mathrm{Spec}(A)$. Indeed, any two of the cyclotomic polynomials $\Phi_{p^i}(X)$ generate the unit ideal in $A[X]$, since $p \in A^*$. Thus A is a product of rings A_i, $1 \leq i \leq n$, the i-th coordinate of ζ satisfying

$$\Phi_{p^i}(\zeta) = 0;$$

on $\mathrm{Spec}(A_i)$, ζ has order p^i.

If ψ is a continuous additive character of K with values in A^*, then it takes its values in the p^n-th roots of unity. This makes it possible to define $n(\psi)$ as a locally constant function on $\mathrm{Spec}(A)$: at a prime ideal λ, $n(\psi)$ is $+\infty$ if ψ is trivial modulo λ, and otherwise the largest integer n such that

$$\psi|_{\pi^{-n}\mathcal{O}}$$

is trivial modulo λ, or in the localization A_λ, which is the same thing. By abuse of language, one says that ψ is nontrivial if $n(\psi)$ is nowhere $+\infty$.

6.4. For ψ a nontrivial additive character of K with values in A^*, for $\chi \in \mathrm{Hom}(K^*, A^*)$, and for dx a Haar measure with values in A, one can now define

$$\varepsilon_0(\chi, \psi, dx) \in A^*$$

by formula (3.4.3.4). More precisely, if $\mathrm{Spec}(A)$ is not connected, one first writes

$$A = \prod_i A_i$$

with $n(\psi)$ and $Sw(\chi)$ constant on $\mathrm{Spec}(A_i)$. The i-th projection of ε_0 is then defined by (3.4.3.4).

Theorem 6.5. There exists a unique function ε_0 of the following type:

(a) For K a nonarchimedean local field, $\overline{K}$ a separable closure of K, A a field of characteristic $\neq p$, V a representation of $W(\overline{K}/K)$ in a finite-dimensional vector space over A, ψ a nontrivial additive character with values in A^*, and dx a nonzero Haar measure with values in A, one has

$$\varepsilon_0(V, \psi, dx) \in A^*.$$

(b) ε_0 is invariant under extension of the field A.

(c) Let A be a discrete valuation ring of residual characteristic $\neq p$, let F be its fraction field, and let $s = A/\mathfrak{m}$ be its residue field. For V a representation of $W(\overline{K}/K)$ in a finite-rank free A-module, and for ψ and dx with values in A, one has

$$\varepsilon_0(V_F, \psi, dx) \in A^*,$$

and its reduction modulo $\mathfrak{m}$ is

$$\varepsilon_0(V_s, \overline{\psi}, \overline{dx}).$$

(d) Conditions (1) to (3) of 4.1 are satisfied.

(e) For $\dim V = 1$, ε_0 is given by 6.4.

By Galois descent, it is enough to treat the case where A is separably closed. If A has characteristic 0, the assertion, being purely algebraic, follows from the case $A = \mathbb{C}$ treated in 4.1. Thus suppose A has characteristic $\ell \neq 0$.

Let G be a finite quotient of $\mathrm{Gal}(\overline{K}/K)$, and let A_0 be a discrete valuation ring of unequal characteristic, with residue field $A_0/\mathfrak{m} = A$. Assume that ψ has been lifted to an additive character with values in A_0^*, that dx has been lifted to a Haar measure with values in A_0, and that the fraction field E of A_0 is large enough, in the sense of 1.4, relative to G.

For every character χ of a subgroup of G with values in E^*, in fact in A_0^*, the corresponding constant ε_0 is clearly in A_0. By 5.7.1, it is even in A_0^*. It follows that, for every representation V_E of G over E,

$$\varepsilon_0(V_E, \psi, dx) \in A_0^*.$$

The decomposition homomorphism

$$d : R_E(G) \longrightarrow R_A(G)$$

is known to be surjective: virtual representations lift. For $V \in R_A(G)$, choose $\widetilde{V} \in R_E(G)$ with $d(\widetilde{V}) = V$, and put

$$\varepsilon_0(V, \overline{\psi}, \overline{dx}) = \text{the reduction modulo } \mathfrak{m} \text{ of } \varepsilon_0(\widetilde{V}, \psi, dx).$$

By 1.8, to justify this definition it is enough to check that:

$$(*) \quad \text{for } H \subset G \text{ defining an extension } L \text{ of } K,$$

$$\psi_L = \psi \circ \mathrm{Tr}_{L/K}, \quad \chi', \chi'' \text{ characters of } H \text{ with values in } A_0^*,$$

if χ' and χ'' are congruent modulo $\mathfrak{m}$, then

$$\begin{aligned} \varepsilon_0\Big(\mathrm{Ind}_H^G([\chi'] - [\chi'']), \psi\Big) &= \varepsilon_0\big([\chi'] - [\chi''], \psi_L\big) \\ &\equiv 1 \pmod{\mathfrak{m}}. \end{aligned}$$

The Swan conductors of χ' and χ'' are both equal to the Swan conductor of χ' modulo $\mathfrak{m}$. For

$$y = \pi^{1+Sw(\chi')+n(\psi_L)},$$

the formulas of 3.4.3 give

$$\varepsilon_0(\chi', \psi_L, dx_L) = \int_{y^{-1}\mathcal{O}_L^*} \chi'^{-1}(x)\psi_L(x)\, dx_L,$$

and similarly for χ''. The numerator and denominator are units, and are clearly congruent modulo $\mathfrak{m}$. Hence

$$\frac{\varepsilon_0(\chi', \psi_L, dx_L)}{\varepsilon_0(\chi'', \psi_L, dx_L)} \equiv 1 \pmod{\mathfrak{m}}.$$

For V a representation of $\mathrm{Gal}(\overline{K}/K)$, one therefore obtains a function ε_0 with the announced properties. Its formal properties are the same as those made explicit in the formula list 5.1 in the case $A = \mathbb{C}$. To pass from $\mathrm{Gal}(\overline{K}/K)$ to $W(\overline{K}/K)$, one may therefore repeat the argument used in 4.9, with the help of the second formula (5.5.1).

Remark 6.6. Throughout this section it was essential to work with ε_0 rather than with ε. First, this permits the use of the Swan conductor, which is invariant under reduction modulo ℓ, rather than the Artin conductor; second, it avoids having to distinguish, in the definition of $\varepsilon_0(\chi, \psi, dx)$, between the unramified and tamely ramified cases. The meaning of ε_0 will appear more clearly in the next section.

7. Modular L-functions

7.1. Let X be a nonsingular projective curve over $\mathbb{F}_p$. Assume X irreducible, let K be its field of rational functions, and identify the closed points of X with the places of K. We do not assume that X is geometrically irreducible, i.e. that $\mathbb{F}_p$ is algebraically closed in K.

7.2. Let A be a noetherian ring. It is known what a constructible sheaf of A-modules on $X_{\text{ét}}$ is (SGA 4, IX, 2.3). For every $x \in X$, and for $k(\overline{x})$ a separably closed extension of $k(x)$, such a sheaf $\mathcal{G}$ has a fiber

$$\mathcal{G}_{\overline{x}}$$

at x, which is a finite-type A-module. It depends only on the separable closure of $k(x)$ in $k(\overline{x})$, and $\mathrm{Gal}(\overline{k(x)}/k(x))$ acts on $\mathcal{G}_{\overline{x}}$ by transport of structure.

Moreover, for v a place of K, i.e. a closed point of X, and $\overline{K}_v$ an algebraic closure of K_v, there is a specialization map

$$\mathrm{sp}_v : \mathcal{G}_{\overline{k(v)}} \longrightarrow \mathcal{G}_{\overline{K}_v}.$$

This map is equivariant; its image is therefore contained in $\mathcal{G}_{\overline{K}_v}^{I_v}$. We say that $\mathcal{G}$ is unramified at v if sp_v is an isomorphism. A constructible sheaf is unramified almost everywhere.

Once a separable closure $\overline{K}$ of K, and a place of $\overline{K}$ above each place of K, have been chosen, a constructible sheaf of A-modules can conversely be defined as follows:

(a) one gives the fibers $\mathcal{G}_{\overline{K}}$ and $\mathcal{G}_{\overline{k(v)}}$, respectively A-$\mathrm{Gal}(\overline{K}/K)$-modules or A-$\mathrm{Gal}(\overline{k(v)}/k(v))$-modules, of finite type over A;

(b) one gives the specialization maps, equivariant for the action of the decomposition groups,

$$\mathcal{G}_{\overline{k(v)}} \longrightarrow \mathcal{G}_{\overline{K}}.$$

For (a) and (b) to define a sheaf, it is enough that the specialization maps be isomorphisms almost everywhere.

7.3. It will be convenient to modify the preceding definition by replacing Galois groups by Weil groups. Let Y be a scheme over $\mathbb{F}_p$. Let $\overline{\mathbb{F}}_p$ be an algebraic closure of $\mathbb{F}_p$, and let

$$\overline{Y} = Y \otimes_{\mathbb{F}_p} \overline{\mathbb{F}}_p.$$

A Weil sheaf, respectively a constructible Weil sheaf of A-modules, $\mathcal{G}$ on Y, consists of a sheaf, respectively a constructible sheaf of A-modules, $\overline{\mathcal{G}}$ on $\overline{Y}$, together with an action of Frobenius on $\overline{\mathcal{G}}$, lying above its action on $\overline{Y}$.

Unneeded remark. Weil sheaves of sets on Y form a topos Y_{Weil}. It may be described as a 2-fiber product of topoi: identifying $\text{Spec}(\mathbb{F}_p)_{\text{ét}}$ with the classifying topos $B\widehat{\mathbb{Z}}$, one has

$$Y_{\text{Weil}} \simeq \overline{Y}_{\text{ét}} \times_{B\widehat{\mathbb{Z}}} B\mathbb{Z}.$$

We shall simply call the Weil sheaves on Y sheaves; sheaves on $Y_{\text{ét}}$ will be called étale sheaves.

7.4. A constructible A-module sheaf in the sense of 7.3 can be described in Galois terms as in 7.2, replacing everywhere

$$\text{Gal}(\overline{K}/K), \qquad \text{Gal}(\overline{K}_v/K_v), \qquad \text{Gal}(\overline{k_v}/k_v)$$

by

$$W(\overline{K}/K), \qquad W(\overline{K}_v/K_v), \qquad W(\overline{k_v}/k_v) = \mathbb{Z}.$$

7.5. Let K_v be a nonarchimedean local field. A Weil sheaf on $\text{Spec}(\mathcal{O}_v)$ is defined by paraphrasing 7.3. Let $\overline{K}_v$ be a separable closure of K_v. Giving a constructible Weil sheaf of A-modules $\mathcal{G}$ on $\text{Spec}(\mathcal{O}_v)$ is equivalent to giving:

a) the fiber $\mathcal{G}_{\overline{K}_v}$, an A-$W(\overline{K}_v/K_v)$-module, of finite type over A;

b) the fiber $\mathcal{G}_{\overline{k}_v}$, an A-$W(\overline{k}_v/k_v)$-module, of finite type over A;

c) the $W(\overline{K}_v/K_v)$-equivariant specialization map

$$\text{sp}_v : \mathcal{G}_{\overline{k}_v} \longrightarrow \mathcal{G}_{\overline{K}_v}.$$

Let v be a place of K, and put

$$X_v = \text{Spec}(\mathcal{O}_v).$$

By restriction, every constructible Weil sheaf of A-modules on X defines one on X_v.

A sheaf is flat if its fibers are flat A-modules.

7.6. Let K_v be a nonarchimedean local field. Put

$$\deg(v) = [k_v : \mathbb{F}_p],$$

and write ω_v^t for the unramified quasi-character of $W(\overline{K}_v/K_v)$, with values in $A(t)$, such that

$$\omega_v^t(F_v) = t^{\deg(v)}.$$

Let $\mathcal{G}$ be a flat constructible sheaf on $\text{Spec}(\mathcal{O}_v)$. Put

$$Z(\mathcal{G}_v, t) = \det(1 - F_v t^{\deg(v)}, \mathcal{G}_{\overline{k}_v})^{-1} = \det(1 - F_v, \mathcal{G}_{\overline{K}_v} \otimes \omega_v^t)^{-1}. \tag{7.6.1}$$

For p invertible in A, define a conductor

$$a(\mathcal{G}) = sw(\mathcal{G}_{\overline{K}_v}) + \dim(\mathcal{G}_{\overline{K}_v}) - \dim(\mathcal{G}_{\overline{k}_v}), \tag{7.6.2}$$

an integer if A is local, and in general a locally constant function on $\text{Spec}(A)$.

Assume that A is a field of characteristic $\neq p$. Let ψ be a nontrivial additive character of K_v with values in A^*, and let dx be a Haar measure with values in A, as in 6.5. Put

$$\varepsilon(\mathcal{G}_v, \psi, dx, t) = \varepsilon_0(\mathcal{G}_{\overline{K}_v} \otimes \omega_v^t, \psi, dx)\ \det(-F_v t^{\deg(v)}, \mathcal{G}_{\overline{k}_v})^{-1}. \tag{7.6.3}$$

This is a monomial. By (5.5.2), its degree in t is

$$a(\mathcal{G}) \deg(v).$$

Remark 7.7. Let M be a representation of $W(\overline{K}_v/K_v)$. In Galois terms, one defines sheaves $j_!M$ and j_*M on $\mathrm{Spec}(\mathcal{O}_v)$ by the specialization maps

$$0 \longrightarrow M, \qquad M^{I_v} \longrightarrow M.$$

The quantities ε and ε_0 of the preceding sections are identified with the quantity (7.6.3) relative respectively to j_*M and $j_!M$. Unlike j_*, the functor $j_!$ commutes with reduction modulo ℓ. This may explain why, in §6, we considered only ε_0.

7.8. Let $\mathcal{G}$ be a constructible sheaf of flat A-modules on X. For every place v of K, $\mathcal{G}$ induces a sheaf $\mathcal{G}_v$ on $\mathrm{Spec}(\mathcal{O}_v)$. Put

$$Z(\mathcal{G}, t) = \prod_v Z(\mathcal{G}_v, t) \in A[[t]]. \tag{7.8.1}$$

7.9. Assume that A is a field of characteristic $\neq p$. On the adele ring of K there is a unique Haar measure with values in $\mathbb{Z}[1/p]$, such that

$$\int_{\mathbb{A}/K} dx = 1.$$

One can write

$$dx = \bigotimes_v dx_v$$

where dx_v is a Haar measure on K_v, with values in $\mathbb{Z}[1/p]$, and, for almost all v,

$$\int_{\mathcal{O}_v} dx_v = 1.$$

The dx_v define Haar measures with values in A.

If there exists a nontrivial character ψ of $\mathbb{A}/K$, with values in A^*, put

$$\varepsilon(\mathcal{G}, t) = \prod_v \varepsilon(\mathcal{G}_v, \psi_v, dx_v, t). \tag{7.9.1}$$

Almost all terms in the product are equal to 1, and the product does not depend on the choice of ψ nor on the chosen decomposition of dx.

Let A' be obtained from A by adjoining a primitive p-th root of unity. There exists a ψ with values in A'^*. The corresponding global constant, being independent of ψ, is invariant under $\mathrm{Gal}(A'/A)$, hence lies in A. This makes it possible to define ε even if no character ψ exists.

7.10. Let i denote the inclusion of $\mathrm{Spec}(K)$ into X. If V is a representation of $W(\overline{K}/K)$, write i_*V for the sheaf on X with fiber over $\overline{K}$ equal to V, and for which the specialization maps are

$$\mathrm{sp}_v : V^{I_v} \hookrightarrow V.$$

Theorem 7.11. Let A be a field of characteristic $\ell \neq p$. Let $\mathcal{G}$ be a constructible sheaf of A-vector spaces, and let V be the fiber of $\mathcal{G}$ over $\overline{K}$.

(i) $Z(\mathcal{G}, t)$ is a rational function of t.

(ii) As $t \to \infty$, $Z(\mathcal{G}, t)$ is asymptotic to $\varepsilon(\mathcal{G}, t)$. In other words,

$$Z(\mathcal{G}, t)/\varepsilon(\mathcal{G}, t)$$

is a formal series in t^{-1}, with constant term 1.

(iii)

$$Z(i_*V, t) = \varepsilon(i_*V, t)\, Z(i_*V^*, pt^{-1}).$$

Let S be a finite set of places containing all the places where the representation V is ramified. Let j be the inclusion of $X - S$ into X, and let $j_!V$ be the sheaf on X, with generic fiber V, unramified outside S, and with zero fiber at $v \in S$. One checks immediately that (i) and (ii) are equivalent to

$$Z(j_!V, t) \text{ is a rational function of } t, \tag{i'}$$

$$\text{as } t \to \infty, \qquad Z(j_!V, t) \sim \varepsilon(j_!V, t). \tag{ii'}$$

Assume $\ell \neq 0$. For every representation W of $W(\overline{K}/K)$, unramified outside S, identified with a locally constant sheaf on $X - S$, put

$$R^0 j_* W = j_* W = i_* W,$$

and let $R^1 j_* W$ be the sheaf zero outside S, with fiber at $v \in S$ equal to $H^1(I_v, W)$. Put

$$Z(Rj_*W, t) = Z(R^0 j_* W, t)\, Z(R^1 j_* W, t)^{-1}. \tag{7.11.1}$$

Let P' be the kernel of

$$t_\ell : I_v \longrightarrow \mathbb{Z}_\ell(1).$$

For every representation W of I_v over A, one has canonically

$$H^i(I_v, W) = H^i(I_v/P', W^{P'}). \tag{7.11.2}$$

For every representation W of $\mathbb{Z}_\ell(1)$, one has canonically

$$H^0(\mathbb{Z}_\ell(1), W) = W^{\mathbb{Z}_\ell(1)} \quad \text{(invariants)}, \tag{7.11.3}$$

$$H^1(\mathbb{Z}_\ell(1), W) = W_{\mathbb{Z}_\ell(1)}(-1) \quad \text{(twisted coinvariants)},$$

$$H^i(\mathbb{Z}_\ell(1), W) = 0 \qquad (i \geq 2).$$

Since duality exchanges invariants and coinvariants, one has, cf. 2.3:

Lemma 7.11.4. For $\ell \neq 0$, $H^0(I_v, W)$ and $H^1(I_v, \omega_1 W^*)$ are canonically in duality.

We shall not use the fact that this duality can be interpreted as a cup product with values in

$$H^1(I_v, A(1)) = \mathrm{Hom}(I_v, A(1)) \simeq A,$$

generated by t_ℓ modulo ℓ.

For $\ell = 0$, in the course of this proof, put

$$H^1(I_v, W) := H^0(I_v, \omega_1 W^*)^*,$$

and define $Z(Rj_*W, t)$ by (7.11.1). This definition is justified by 7.11.5 and 7.11.6 below. For W a representation of $W(\overline{K}_v/K_v)$, put also

$$Z_v(Rj_*W, t) = \det(1 - Ft^{\deg(v)}, H^0(I_v, W))^{-1} \det(1 - Ft^{\deg(v)}, H^1(I_v, W)).$$

Let A be a discrete valuation ring of unequal characteristic $\ell \neq p$, and let W be a free A-module on which $W(\overline{K}_v/K_v)$ acts. Let η be the fraction field of A, and $s = A/\mathfrak{m}$ its residue field.

Lemma 7.11.5.

$$Z_v(Rj_*W_s, t) = Z_v(Rj_*W_\eta, t) \pmod{\mathfrak{m}}.$$

The proof below will lose its mystery in §10. Let

$$P' = \mathrm{Ker}(t_\ell : I \to \mathbb{Z}_\ell(1)),$$

let σ be a generator of $\mathbb{Z}_\ell(1)$, let F be a geometric Frobenius, and let $K^\bullet$ be the complex concentrated in degrees 0 and 1:

$$K^\bullet : \quad W^{P'} \xrightarrow{\sigma - 1} W^{P'}.$$

The endomorphism

$$\sum_{i=0}^{q-1} \sigma^i$$

of $W^{P'}$ is invertible: it is enough to see this after reduction modulo the maximal ideal of A. After reduction, since some power σ^{ℓ^n} is the identity, σ is unipotent, and q is the unique eigenvalue of $\sum \sigma^i$.

Let $\underline{F} = (F_0, F_1)$ be the endomorphism of $K^\bullet$ with components

$$F_0 = F, \qquad F_1 = F \circ \left(\sum_{i=0}^{q-1} \sigma^i \right)^{-1}.$$

The quantity

$$Z = \det(1 - F_0 t^{\deg(v)}, W^{P'})^{-1} \det(1 - F_1 t^{\deg(v)}, W^{P'})$$

is compatible with extension of scalars to η or to s. After extension of scalars to η, respectively to s, one has

$$Z = \det(1 - F t^{\deg(v)}, H^0(K^\bullet))^{-1} \det(1 - F t^{\deg(v)}, H^1(K^\bullet)).$$

One concludes by noting that

$$(H^i(K^\bullet_\eta), H^i(\underline{F})) \simeq (H^i(I_v, W_\eta), F),$$

and

$$(H^i(K^\bullet_s), H^i(\underline{F})) \simeq (H^i(I_v, W_s), F).$$

Variant 7.11.6. Suppose A is complete, and let W be an A-module on which $W(\overline{K}_v/K_v)$ acts continuously for the ℓ-adic topology, and no longer, as above, for the discrete topology. Put again

$$H^1(I_v, W_\eta) = (W_\eta^{P'})_{\mathbb{Z}_\ell(1)}(-1) \quad \text{(twisted coinvariants)},$$

and define $Z_v(Rj_*W, t)$ as before. The H^1 defined above can be interpreted as a continuous H^1:

$$H^1(I_v, W_\eta) = \varprojlim_n H^1(I_v, W/\ell^n) \otimes_A \eta. \tag{7.11.6.1}$$

The reduction formula 7.11.5 remains valid, with the same proof.

Lemma 7.11.7. Formula (iii) is equivalent to

$$Z(j_!V, t) = \varepsilon(j_!V, t)\, Z(Rj_*V, pt^{-1}). \tag{7.11.8}$$

Lemma 7.11.7 follows from the local identity

$$\det(1 - F_v t^{\deg(v)}, H^0(I_v, V))^{-1} = \det(-F_v t^{\deg(v)}, H^0(I_v, V))^{-1} \det(1 - F_v (pt^{-1})^{\deg(v)}, H^1(I_v, V^*))^{-1}.$$

If V is replaced by the representation $V \otimes \omega^t$ of $W(\overline{K}_v/K_v)$ over $A((t))$, this formula simplifies to

$$\det(1 - F_v^{-1}, H^0(I_v, V)) = \det(-F_v, H^0(I_v, V))\ \det(1 - F_v, H^1(I_v, \omega_1 \otimes V^*)),$$

which follows from local duality 7.11.4.

We prove 7.11. If $A = \mathbb{C}$, the infinite product $Z(j_*V, t)$ converges for $|t|$ sufficiently small, and

$$L(V, s) = Z(j_*V, q^{-s}).$$

Thus Z is a meromorphic function of t; the functional equation shows that it has moderate growth as $t \to \infty$, hence is a rational function of t. The functional equation of the L-function gives (iii).

The statements (i) and (iii) are purely algebraic. Since they are true for $A = \mathbb{C}$, and so are (i') and (7.11.8), they remain true for A of characteristic 0. Now suppose A has characteristic $\ell \neq 0$.

Let J be an open normal subgroup of $W(\overline{K}/K)$, containing I_v for $v \in S$. If a representation V of $W(\overline{K}/K)/J$ over A lifts to characteristic 0, the assertions (i') and (7.11.8) for V are obtained by reduction modulo ℓ, taking into account, for (7.11.8), 7.11.5. The quantities

$$Z(j_!V,t), \qquad Z(Rj_*V,t), \qquad \varepsilon(j_!V,t)$$

depend additively on V, an extension giving a product. To obtain (i) and (7.11.8) in general, it is therefore enough to know that, after extension of scalars, every representation of $W(\overline{K}/K)/J$ virtually lifts to characteristic 0. The method of 4.10, cf. 4.10.3, reduces this question to the case of finite groups, i.e. to Brauer's theorem.

Statement (ii') follows from (7.11.8) and from the fact that, for $t = 0$, $Z = 1$.

Proposition 7.12. If A is a field of characteristic p, and $\mathcal{G}$ a constructible sheaf of A-vector spaces, then $Z(\mathcal{G},t)$ is still a rational function of t.

In the proof of 7.11(i), we did not in fact use the hypothesis $\ell \neq p$.

8. ℓ-adic representations of nonarchimedean local Weil groups; Hodge structures and archimedean local Weil groups

8.1. Let K be a nonarchimedean local field. We use the notation of 2.1 and 2.2.

Let ℓ be a prime number $\neq p$, and let E_λ be a finite extension of $\mathbb{Q}_\ell$; more generally, one could take for E_λ a valued field extending $\mathbb{Q}_\ell$. A λ-adic representation V of $W(\overline{K}/K)$ is a representation continuous for the λ-adic topology

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V)$$

of $W(\overline{K}/K)$ on a finite-dimensional E_λ-vector space.

The following result is proved in [13].

Theorem 8.2 (Grothendieck). Let (V,ρ) be a λ-adic representation of $W(\overline{K}/K)$. There exists

$$N \in \mathrm{End}(V)(-1),$$

nilpotent, such that, for σ in a finite-index subgroup of I, one has

$$\rho(\sigma) = \exp(N\, t_\ell(\sigma)).$$

This $N \in \mathrm{End}(V)(-1)$ is unique, hence Galois-invariant; for $w \in W(\overline{K}/K)$, one has

$$\rho(w) N \rho(w)^{-1} = q^{v'(w)} N.$$

We shall use 8.2 to describe λ-adic representations of $W(\overline{K}/K)$ in terms independent of the topology of E_λ. This will be done twice: 8.3 is more intrinsic, and 8.4, which is enough for us, is more elementary.

8.3.1. Up to unique isomorphism, there exists a unique algebraic group G over $\mathbb{Q}_\ell$, isomorphic to $\mathbb{G}_a$, and endowed with an isomorphism

$$\mathbb{Q}_\ell(1) \xrightarrow{\sim} G(\mathbb{Q}_\ell).$$

It is denoted $\mathbb{G}_a(1)$.

8.3.2. Let $J \subset I$ be an open invariant subgroup of $W(\overline{K}/K)$, and put

$$J' = J \cap \mathrm{Ker}(t_\ell).$$

The quotient $W(\overline{K}/K)/J'$ is an extension of the discrete group $W(\overline{K}/K)/J$ by J/J'. Applying to this extension

$$t_\ell : J/J' \longrightarrow \mathbb{Z}_\ell(1) \longrightarrow \mathbb{G}_a(1),$$

one obtains an extension W_J^ℓ of $W(\overline{K}/K)/J$ by $\mathbb{G}_a(1)$, see Serre [12], II, 1.3:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & J/J' & \longrightarrow & W(\overline{K}/K)/J' & \longrightarrow & W(\overline{K}/K)/J & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle t_\ell} & & \downarrow{\scriptstyle \alpha} & & \| & & \\ 0 & \longrightarrow & \mathbb{G}_a(1) & \longrightarrow & W_J^\ell & \longrightarrow & W(\overline{K}/K)/J & \longrightarrow & 0. \end{array}$$

8.3.3. The W_J^ℓ form a projective system, and 8.2 means that every λ-adic representation of $W(\overline{K}/K)$ factors through an algebraic representation of some W_J^ℓ, i.e. through a representation of the group scheme

$$W^\ell(\overline{K}/K) = \varprojlim_J W_J^\ell.$$

8.3.4. Let I_J^ℓ be the inverse image of $I/J \subset W(\overline{K}/K)/J$, and put

$$I^\ell = \varprojlim_J I_J^\ell.$$

To $\bar{\sigma} \in I/J$, the image of $\sigma \in I$, associate the element $\alpha(\sigma)\, t(-t_\ell(\sigma))$ of W_J^ℓ. This section defines splittings

$$I_J^\ell = \mathbb{G}_a(1) \times I/J, \qquad I^\ell = \mathbb{G}_a(1) \times I.$$

There is a diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & W(\overline{K}/K) & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & I \times \mathbb{G}_a(1) & \longrightarrow & W^\ell(\overline{K}/K) & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & & & \\ 0 & \longrightarrow & \mathbb{G}_a(1) & \longrightarrow & \mathbb{G}_a(1) & & & & \end{array}$$

8.3.5. Let ${}'W^\ell$ be the semidirect product of $W(\overline{K}/K)$ by $\mathbb{G}_a(1)$, on which $W(\overline{K}/K)$ acts by

$$wxw^{-1} = q^{v'(w)}x.$$

There exist isomorphisms β between $W^\ell(\overline{K}/K)$ and ${}'W^\ell(\overline{K}/K)$ making the diagram

$$\begin{array}{ccccc} I \times \mathbb{G}_a(1) & \longrightarrow & W^\ell(\overline{K}/K) & \longrightarrow & W(\overline{K}/K) \\ \| & & \downarrow{\scriptstyle \beta} & & \| \\ I \times \mathbb{G}_a(1) & \longrightarrow & {}'W^\ell(\overline{K}/K) & \longrightarrow & W(\overline{K}/K) \end{array}$$

commutative, and one checks easily that any two are conjugate by an element of $\mathbb{G}_a(1)(\mathbb{Q}_\ell)$. The indeterminacy lies in the choice of a lift of a Frobenius $F \in W(\overline{K}/K)$, and one uses $q - 1 \neq 0$; see 8.4.3 below.

8.3.6. Let ${}'W$ be the group scheme over $\mathbb{Q}$, the semidirect product of $W(\overline{K}/K)$ by $\mathbb{G}_a$, on which $W(\overline{K}/K)$ acts by

$$wxw^{-1} = q^{v'(w)}x.$$

Every isomorphism

$$\mathbb{G}_a \xrightarrow{\sim} \mathbb{G}_a(1), \qquad \text{i.e. } \mathbb{Q}_\ell \xrightarrow{\sim} \mathbb{Q}_\ell(1),$$

defines an isomorphism

$${}'W_{\mathbb{Q}_\ell} \xrightarrow{\sim} {}'W^\ell,$$

hence a class of such automorphisms modulo $\mathbb{Q}_\ell^*$, acting on ${}'W^\ell$ through its action on $\mathbb{G}_a(1)$. One checks, see 8.4.3 below, that this automorphism group acts trivially on the set of isomorphism classes of representations of ${}'W^\ell$. In total one has obtained:

(8.3.7). The isomorphism classes of λ-adic representations of $W(\overline{K}/K)$ are in natural bijection with the isomorphism classes of representations over E_λ of the group scheme over $\mathbb{Q}$, ${}'W$.

Definition 8.4.1. Let E be a field of characteristic 0. A representation of $'W(\overline{K}/K)$ over E is a pair

$$\rho = (\rho', N')$$

consisting of:

(a) a finite-dimensional representation

$$\rho' : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V)$$

of $W(\overline{K}/K)$ over E;

(b) a nilpotent endomorphism N' of V, such that

$$\rho'(w) N' \rho'(w)^{-1} = q^{v'(w)} N'. \tag{8.4.1.1}$$

8.4.2. Choose a geometric Frobenius $F \in W(\overline{K}/K)$ and an isomorphism

$$\tau : \mathbb{Q}_\ell(1) \xrightarrow{\sim} \mathbb{Q}_\ell.$$

To a λ-adic representation (V, ρ) of $W(\overline{K}/K)$ one associates a representation (ρ', N') of $'W(\overline{K}/K)$ on E_λ as follows:

(a) For $\sigma \in I$, put, with N defined by 8.2,

$$\rho'(F^n \sigma) = \rho(F^n \sigma) \exp(-t_\ell(\sigma) N).$$

(b) $N' \in \mathrm{End}(V)$ corresponds to N, via τ.

Lemma 8.4.3. The isomorphism class of (ρ', N') depends only on ρ, not on the choices of F and τ.

Change F to $F\varpi$, obtaining (ρ'', N'). One has

$$\rho''(F^n \sigma) = \exp(n\, t_\ell(\varpi) N)\, \rho'(F^n \sigma) = A\, \rho'(F^n \sigma)\, A^{-1},$$

where

$$A = \exp((1 - q^{-1})^{-1} t_\ell(\varpi) N).$$

Changing τ to $a\tau$ changes (ρ', N') into (ρ', aN'). We shall show that, for every representation (ρ', N') of $'W(\overline{K}/K)$ over a field E, and every $a \in E^\times$,

$$(\rho', N') \simeq (\rho', aN').$$

Let $I' = \mathrm{Ker}(\rho')$, and choose n such that F^n is central in W/I', for example $\#\,\mathrm{Aut}(I/I')$. For α a conjugacy class in an algebraic closure of E, let V^α be the largest subspace of V, stable under F^n, such that the eigenvalues of $F^n | V^\alpha$ lie in α.

One checks:

(a) V^α is stable under $\rho'(W(\overline{K}/K))$, and $V = \bigoplus V^\alpha$;

(b) $N' V^\alpha \subset V^{q^{-n}\alpha}$.

If A is the linear transformation preserving the V^α, and such that

$$A|_{V^\alpha} = \lambda(\alpha),$$

with

$$\lambda(q^{-n}\alpha) = a\, \lambda(\alpha),$$

then

$$A(\rho', N') A^{-1} = (\rho', aN').$$

The conclusion 8.3.7 follows at once from 8.4.3.

8.5. Let (ρ', N') be a representation of $'W(\overline{K}/K)$ over E. Let u be the unipotent component of $\rho'(F)$:

$$\rho'(F) = \rho'(F)^{ss}\, u.$$

Since $N' \in \mathrm{End}(V)$ is an eigenvector for conjugation by $\rho'(F)$, u and N' commute. Since a power of F is central in $\mathrm{im}(\rho')$, a power of u, and hence u, centralizes $\mathrm{im}(\rho')$. If

$$w_1, w_2 \in W(\overline{K}/K) - I,$$

then $\rho'(w_1)$ and $\rho'(w_2)$ have a common power. It follows that u^n is the unipotent component of $\rho'(F^n\sigma)$, for $\sigma \in I$, and, for $n \neq 0$, also that of

$$\rho'(F^n\sigma)\exp(aN'),$$

which is conjugate to it by

$$\exp((1-q^{-n})^{-1}aN').$$

Put

$$\rho'^{ss}(F^n\sigma) = \rho'(F^n\sigma)u^{-n}. \tag{8.5.1}$$

Definition 8.6.

(i) (ρ'^{ss}, N') is the F-semisimplification of (ρ', N').

(ii) One says that (ρ', N') is F-semisimple if $\rho' = \rho'^{ss}$, i.e. if, for one, respectively for every,

$$w \in W(\overline{K}/K) - I,$$

and for one, respectively for every, $a \in E$,

$$\rho'(w)\exp(aN')$$

is semisimple.

For a λ-adic representation, its F-semisimplification is again defined by (8.5.1), and F-semisimplicity as in (ii).

8.7. The formalism of 8.3 or 8.4 makes it possible to compare λ-adic representations for different fields E_λ. To do this, introduce the following classical terminology.

Let F be an extension of E, and let ρ be a representation of $'W(\overline{K}/K)$ over F. We say that ρ is *defined over* E if, for every, or for one, algebraically closed extension $\overline{F}$ of F, $\rho_{\overline{F}}$ is isomorphic to its conjugates under $\mathrm{Aut}(\overline{F}/E)$.

Let F_1 and F_2 be two extensions of E, and let ρ_i be a representation of $'W(\overline{K}/K)$ over F_i. We say that ρ_1 and ρ_2 are compatible if they are defined over E, and if, for a common algebraically closed extension

$$F \supset F_1, F_2 \supset E,$$

the representations $(\rho_1)_F$ and $(\rho_2)_F$ are isomorphic.

Definition 8.8. Let E be a number field, let λ_1 and λ_2 be two finite places of E not dividing p, and let ρ_1, ρ_2 be λ_1- and λ_2-adic representations of $W(\overline{K}/K)$. We say that ρ_1 and ρ_2 are compatible if the corresponding representations of $'W(\overline{K}/K)$ are compatible in the sense of 8.7.

We leave to the reader the verification of the following pleasant exercise.

Proposition 8.9. Suppose that (V_1, ρ_1) and (V_2, ρ_2) are F-semisimple, and write N_i for the monodromy operator attached to (V_i, ρ_i). Let W be the unique increasing finite filtration of V_i such that

$$N_i W^k \subset W^{k-2}$$

and such that N_i^j induces an isomorphism

$$\mathrm{Gr}_W^j(V_i) \xrightarrow{N_i^j} \mathrm{Gr}_W^{-j}(V_i).$$

For ρ_1 and ρ_2 to be compatible, it is necessary and sufficient that the characters of the representations of $W(\overline{K}/K)$ on the $\mathrm{Gr}_W^j(V_1)$ and $\mathrm{Gr}_W^j(V_2)$ have values in E and be respectively equal.

Example 8.10. Take $E = \mathbb{Q}$. Let A be an abelian variety over K. The representations of

$$W(\overline{K}/K) \subset \mathrm{Gal}(\overline{K}/K)$$

on

$$V_\ell(A) = T_\ell(A) \otimes \mathbb{Q}_\ell$$

are F-semisimple and compatible.

Variant 8.11. Let G be a linear algebraic group over E, of characteristic 0. A representation of $'W$ in G, over E, is a pair (ρ', N') with

$$\rho' : W(\overline{K}/K) \longrightarrow G(E)$$

trivial on an open subgroup, and $N' \in \mathrm{Lie}(G)$, satisfying (8.4.1.1). For $E = E_\lambda$, the formulas (8.4.2) associate a representation of $'W$ in G to every representation, continuous for the λ-adic topology,

$$\rho : W \longrightarrow G(E_\lambda).$$

Its geometric isomorphism class, after extension of scalars to $\overline{E}_\lambda$, does not depend on the arbitrary choices made: the first part of the proof of 8.4.3 still applies; it remains to show that

$$(\rho', N') \simeq (\rho', aN').$$

For this, note that the algebraic subgroup H of G which centralizes $\mathrm{im}(\rho')$ and sends N' to a multiple contains an element $\rho'(F^m)$, hence an element g such that

$$\mathrm{ad}(g)(N') = q^m N'.$$

The character $\chi : H \to \mathbb{G}_m$ giving the action on N' can therefore only be surjective.

If G is defined over a number field E, one defines compatibility of λ-adic representations

$$W(\overline{K}/K) \longrightarrow G(E_\lambda)$$

as in 8.7 and 8.8. The formal discussion 8.5, 8.6 generalizes unchanged.

8.12. Local factors. Let (ρ', N') be a representation of $'W(\overline{K}/K)$ on an E-vector space V. By abuse of notation put

$$V^I = \mathrm{Ker}(N')^{\rho'(I)};$$

this is a representation of $W(\overline{k}/k)$. Define a conductor, a local L-factor, and local constants by

$$a(V) = sw(V, \rho') + \dim V - \dim V^I, \tag{8.12.1}$$

$$Z(V, t) = \det(1 - Ft^d, V^I)^{-1}, \tag{8.12.2}$$

$$\varepsilon_0(V, t) = \varepsilon_0(\rho', t), \tag{8.12.3}$$

$$\varepsilon(V, t) = \varepsilon_0(\rho', t)\ \det(-Ft, V^I)^{-1}. \tag{8.12.4}$$

If (ρ', N') is defined by a λ-adic representation ρ on V, then

$$V^I = V^{\rho(I)}, \qquad Z(V, t) = \det(1 - Ft, V^{\rho(I)})^{-1},$$

and

$$\varepsilon_0(V, t) = \varepsilon_0(\text{the semisimplification of } V, t).$$

The archimedean case. In the nonarchimedean case, what geometry supplies, namely ℓ-adic cohomology, are λ-adic representations, i.e. representations of $'W(\overline{K}/K)$, not representations of $W(\overline{K}/K)$.

In the archimedean case similarly, geometry supplies not representations of $W(\overline{K}/K)$, but Hodge structures. Here is the rule to use in passing from Hodge structures to representations.

(A) K complex. Geometry supplies vector spaces V over K, endowed with a Hodge bigrading

$$V = \bigoplus V^{pq}$$

(for example, if X is an algebraic variety over K, $H^n(X, K)$ has such a decomposition). One deduces a semisimple representation ρ of

$$W(\overline{K}/K) = K^*$$

on V by putting

$$\rho(z)v^{pq} = z^{-p}\overline{z}^{-q}v^{pq} \qquad (v^{pq} \in V^{pq}).$$

(B) K real. Geometry supplies vector spaces V over $\overline{K}$, endowed with a Hodge bigrading

$$V = \bigoplus V^{pq}$$

and an involution F_∞ such that

$$F_\infty(V^{pq}) = V^{qp}.$$

For example, if X is an algebraic variety over K, $\mathrm{Gal}(\overline{K}/K) = \mathbb{Z}/2\mathbb{Z}$ acts on $X(\overline{K})$ and $H^n(X(\overline{K}), \overline{K})$ has such a structure. One deduces a semisimple representation ρ of $W(\overline{K}/K)$ on V by putting

$$\rho(z)v^{pq} = z^{-p}\overline{z}^{-q}v^{pq} \qquad (z \in \overline{K}^* \subset W(\overline{K}/K),\ v^{pq} \in V^{pq}),$$

and

$$\rho(F)v^{pq} = i^{p+q}F_\infty(v^{pq}).$$

These formulas differ from those of [3], 1.12, moreover disfigured by a misprint on p. 21, 12, because at finite places in loc. cit. I used the inverse of the local class-field-theory isomorphism used here.

At finite places,

$$H^2(\mathbb{P}^1, \mathbb{Q}_\ell) \simeq \mathbb{Q}_\ell(1)$$

corresponds to the quasi-character ω_1. Similarly,

$$H_2(\mathbb{P}^1(\overline{K}), \overline{K}),$$

a Hodge structure of type $(-1, -1)$, with $F_\infty = -1$ if $K = \mathbb{R}$, corresponds to the quasi-character ω_1 of $W(\overline{K}/K)$.

9. Compatible systems of ℓ-adic representations

To simplify the exposition, in this section we consider only representations of Galois groups, rather than representations of Weil groups. As in 7.1, let X be a smooth projective curve over $\mathbb{F}_p$, irreducible over $\mathbb{F}_p$, with function field K.

9.1. Let E_λ be a local field, a finite extension of $\mathbb{Q}_\ell$, with $\ell \neq p$. Let W be a λ-adic representation of $\mathrm{Gal}(\overline{K}/K)$, i.e. a representation continuous for the topology of E_λ, almost everywhere unramified, of $\mathrm{Gal}(\overline{K}/K)$ in a finite-dimensional vector space over E_λ. Since the Galois group is compact, there exists in W an invariant lattice W^0, a lattice being a free $\mathcal{O}_\lambda$-submodule of W such that $E_\lambda W^0 = W$. The local factors

$$Z_v(W, t) = \det(1 - F_v t^{\deg(v)}, W_v^{I_v})^{-1}$$

are therefore in $\mathcal{O}_\lambda[[t]]$, as is their product $Z(W, t)$. Let $\mathrm{Gal}^0(\overline{K}/K)$ be the kernel of the map from $\mathrm{Gal}(\overline{K}/K)$ to $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$. By a theorem of Grothendieck, $Z(W, t)$ is a rational function; it is even a polynomial if

$$W^{\mathrm{Gal}^0(\overline{K}/K)} = W_{\mathrm{Gal}^0(\overline{K}/K)} = 0,$$

and it satisfies a functional equation

$$Z(W,t) = \varepsilon_{\mathrm{Gr}}(W,t)Z(W^*\omega_1, t^{-1}), \tag{9.1.1}$$

where $\varepsilon_{\mathrm{Gr}}$ is a monomial of degree

$$\sum_v \deg(v)\, a(i_*W_v)$$

(8.12.1). The reader will find in §10 a summary of Grothendieck's theory.

9.2. Let E be a number field, let $\mathcal{L}$ be an infinite set of finite places of E not dividing p, and let $\mathcal{O} \subset E$ be the ring of elements of E which are λ-integral for every $\lambda \in \mathcal{L}$. We again denote by λ the prime ideal of $\mathcal{O}$ defined by λ. Suppose given, for every $\lambda \in \mathcal{L}$, a λ-adic representation V_λ of $\mathrm{Gal}(\overline{K}/K)$. Assume that, for every place v of K, the λ-adic representations $V_{\lambda v}$ of $W(\overline{K}_v/K_v)$ are compatible in the strict sense (8.8). It would in fact suffice to know that their F-semisimplifications (8.6) are compatible; cf. also 9.8. The rational function

$$Z(V_\lambda, t) \in \mathcal{O}(t)$$

and the "constant", i.e. the monomial,

$$\varepsilon_\lambda(V,t) = \prod_v \varepsilon(V_{\lambda v}, \psi_v, dx_v, t) \in \mathcal{O}(t)$$

are then independent of λ; as usual, the ψ_v are the components of a nontrivial character of $\mathbb{A}/K$, and $\otimes dx_v$ is a Tamagawa measure. Denote this function and this constant by

$$Z(V,t), \qquad \varepsilon(V,t).$$

Theorem 9.3. Under the preceding hypotheses,

$$Z(V,t) = \varepsilon(V,t)Z(V^*, pt^{-1}). \tag{9.3.1}$$

Since $\mathcal{L}$ is infinite, the map

$$\mathcal{O} \longrightarrow \prod_{\lambda \in \mathcal{L}} \mathcal{O}/\lambda$$

is injective: it is enough to prove 9.3.1 after reduction modulo λ, for every $\lambda \in \mathcal{L}$. For this we shall put 9.3.1 in a form compatible with reduction modulo λ, and justified modulo λ by (7.11.8).

Let S be a finite set of places of K, containing all the places where V ramifies, and let j be the inclusion of $X - S$ in X. For every λ-adic representation W, unramified outside S, put

$$Z(j,W,t) = \prod_{v \notin S} \det(1 - F_v t^{\deg(v)}, W_v)^{-1},$$

$$Z(Rj_*W,t) = \prod_v \det(1 - F_v t^{\deg(v)}, W_v^{I_v})^{-1} \Big/ \prod_{v \in S} \det(1 - F_v t^{\deg(v)}, W_{v,I_v}(-1))^{-1},$$

$$\varepsilon(j,W,t) = \prod_{v \notin S} \varepsilon(W_v, \psi_v, dx_v, t) \prod_{v \in S} \varepsilon_0(W_v, \psi_v, dx_v, t).$$

An easy calculation, already made in 7.11.7, shows that (9.3.1) is equivalent to

$$Z(j,W,t) = \varepsilon(j,W,t)Z(Rj_*W,t). \tag{9.3.2}$$

One concludes by 7.11.6 and 7.11.8.

Stability 9.4. Let F be an extension of E, and let $\mathcal{L}_F$ be the inverse image of $\mathcal{L}$. A compatible system of λ-adic representations, $\lambda \in \mathcal{L}$, defines by extension of scalars a compatible system of λ-adic representations, $\lambda \in \mathcal{L}_F$. If χ is a character of the idèle-class group with values in the roots of unity of F, the $V_\lambda\chi$ still form a compatible system, to which 9.3 applies.

Corollary 9.5. Let χ be a character of the idèle-class group with values in the roots of unity of a finite extension of E (cf. 9.4). Let $S(\chi)$, respectively $S(V)$, be the set of places of K where χ, respectively V, ramifies. Suppose

$$S(V) \cap S(\chi) = \varnothing.$$

Then

$$\varepsilon\big((V - \dim V[1])([\chi] - [1]), t\big)$$

is, by definition,

$$\varepsilon(V\chi, t)\varepsilon(V, t)^{-1}\varepsilon(\chi, t)^{-\dim V}\varepsilon([1], t)^{\dim V}$$

and therefore

$$\prod_{v \in S(V)} \chi_v(\pi_v^{a(V_v)}) \cdot \prod_{v \in S(\chi)} \det V_v(\pi_v^{a(\chi_v)}).$$

This is a corollary of 5.5.3.

Example 9.6. For $E = \mathbb{Q}$ and $\mathcal{L}$ the set of primes other than p, if F is an elliptic curve over K, the $H^1(F, \mathbb{Q}_\ell)$ form a compatible system of ℓ-adic representations. If the modular invariant j is not constant, then, for every character χ as in 9.5, the corresponding rational functions $Z(F\chi, t)$ are polynomials and satisfy 9.5.

9.7. We refer to Weil [16] for the relation which 9.5 implies, when $\dim(V) = 2$, between $Z(V, t)$ and "modular forms" on $\mathrm{GL}(2, \mathbb{A})/\mathrm{GL}(2, K)$. The most interesting case is supplied by 9.6.

More precise results are obtained by invoking Jacquet–Langlands. Let, for example, E be an elliptic curve over K, with nonconstant modular invariant. By Langlands' dictionary, see [4], §3, for each place v of K, the representation of

$$\mathrm{Gal}(\overline{K}_v/K_v) \simeq W(\overline{K}_v/K_v)$$

on $H^1(E, \mathbb{Q}_\ell)$ defines an irreducible admissible representation π_v of $\mathrm{GL}_2(K_v)$. Since the representations $H^1(E, \mathbb{Q}_\ell)$, for varying ℓ, are compatible, π_v can be defined over $\mathbb{Q}$ and is independent of ℓ. We shall consider the corresponding complex representation. The representation π_v is infinite-dimensional, of weight 2, i.e. the center K_v^* of $\mathrm{GL}(2, K_v)$ acts by ω_2, and almost always has a $\mathrm{GL}(2, \mathcal{O}_v)$-invariant vector. By 9.3, 9.6, and [8], 11.5, the restricted tensor product

$$\pi(E) = \bigotimes_v \pi_v$$

appears as a direct factor in the admissible representation

$$L^0_{\omega_2}\big(\mathrm{GL}(2, \mathbb{A})/\mathrm{GL}(2, K)\big)$$

of $\mathrm{GL}(2, \mathbb{A})$ on the space of locally constant functions on $\mathrm{GL}(2, \mathbb{A})/\mathrm{GL}(2, K)$ which transform under the center by the quasi-character ω_2, have compact support modulo the center, and are cuspidal.

Theorem 9.8. Let λ, μ be places of E, and let V_λ, V_μ be λ- and μ-adic representations of $\mathrm{Gal}(\overline{K}/K)$. Let S be a finite set of places of K, and suppose that, for $v \notin S$, the semisimplifications of $V_{\lambda v}$ and $V_{\mu v}$ are compatible. Then, for every place v, the semisimplifications of $V_{\lambda v}$ and $V_{\mu v}$ are compatible.

If S is taken large enough to contain all places ramified for V_λ or V_μ, the hypothesis reduces to:

$(*)$ for $v \notin S$, the characteristic polynomials of F_v acting on $V_{\lambda v}$ and $V_{\mu v}$ belong to $E[t]$ and are equal.

Thus Theorem 9.8 is a partial answer, in the least interesting case of function fields, to question 1 of Serre [12], I, 12.

One should be careful about the following points.

(a) The conclusion concerns only the semisimplifications of the local representations, not their F-semisimplifications. In other words, with the notation of 8.4.2, only the semisimplification of ρ' is considered, and not N.

(b) For $\lambda = \mu$, $V_\lambda = V_\mu$, the theorem asserts that the semisimplifications of the local representations $V_{\lambda v}$ of $W(\overline{K}_v/K_v)$ are defined over E, i.e. have character with values in E.

We may, and do, suppose that S contains all places where I_v acts through an infinite group. For $v \notin S$, there is then no longer any distinction between the semisimplification of $V_{\lambda v}$ and its F-semisimplification.

Apply Grothendieck's theorem recalled in 9.1:

$$Z(V_\lambda, t) = \varepsilon_{\mathrm{Gr}}(V_\lambda, t) Z(V_\lambda^* \omega_1, t^{-1}).$$

This identity rewrites as

$$\prod_{v \in S} Z_v(V_\lambda, t) \prod_{v \in S} Z_v(V_\lambda^*, t^{-1})^{-1} = \varepsilon_{\mathrm{Gr}} \frac{\prod_{v \in S} Z_v(V_\lambda^* \omega_1, t^{-1})}{\prod_{v \in S} Z_v(V_\lambda \omega_1, t)}.$$

The left-hand side is in $E(T)$, and similarly for V_μ. Hence the right-hand side is in $E(T)$ and independent of λ. Decomposing the right-hand side as the product of a monomial and of a rational function equal to 1 at $t = 0$, one obtains from the local calculation of 7.11.7 that

$$\prod_{v \in S} \frac{\det(1 - F_v t^{\deg(v)}, V_{\lambda v}^{I_v})}{\det(1 - F_v t^{\deg(v)}, V_{\lambda v, I_v}(-1))}$$

is the same for V_λ and V_μ. The factors of this product are additive in $V_{\lambda v}$, an extension giving a product, by (7.11.6.1); hence they are the same for $V_{\lambda v}$ and for its semisimplification $V_{\lambda v}^{ss}$. For this latter representation, invariants and coinvariants of I_v are the same, and one obtains

$$\prod_{v \in S} \frac{\det(1 - F_v t^{\deg(v)}, V_{\lambda v}^{ss, I_v})}{\det(1 - F_v (pt)^{\deg(v)}, V_{\lambda v}^{ss, I_v})} \tag{9.8.1}$$

which is the same for V_λ and V_μ.

Let $v_0 \in S$, and twist V_λ and V_μ by a character χ of the idèle-class group which is unramified at v_0 and very ramified at the $v \in S$ other than v_0; χ has values in an extension of E, and for existence see 4.14. The hypotheses made on (V_λ, V_μ) are stable under twisting. For suitable χ, after twisting, the factors of 9.8.1 corresponding to $v \neq v_0$ become 1. The factor corresponding to v_0 undergoes a trivial modification, read from $\chi(F_v)$. Put

$$P_{\lambda v}(T) = \det(1 - F_v T, (V_{\lambda v}^{ss})^{I_v}).$$

Applying the invariance of (9.8.1) to $(V_\lambda \chi, V_\mu \chi)$, one finds that, for every $v \in S$, in $E(T)$,

$$\frac{P_{\lambda v}(T)}{P_{\lambda v}(qT)} = \frac{P_{\mu v}(T)}{P_{\mu v}(qT)}. \tag{9.8.2}$$

The transformation which sends a rational function R, equal to 1 at 0, to

$$R_q(T) = \frac{R(T)}{R(qT)}$$

is injective: for divisors, one has

$$\mathrm{div}(R) = \lim_N \mathrm{div} \left(\prod_0^N R_q(q^i T) \right)$$

(simple limit). From 9.8.2 one deduces that $P_{\lambda v}(T) \in E[T]$ and that

$$P_{\lambda v}(T) = P_{\mu v}(T). \tag{9.8.3}$$

The hypotheses imposed on V_λ and V_μ are preserved after replacing K by a finite extension K', replacing S by its inverse image. Since every finite local extension K'_v/K_v is induced by a global extension, because $\mathrm{Gal}(\overline{K}_v/K_v)$ is a closed subgroup of $\mathrm{Gal}(\overline{K}/K)$, one finds that 9.8.3 also holds for the representations of $W(\overline{K}_v/K'_v)$ defined by V_λ and V_μ. Taking for K'_v the various extensions for which the semisimplifications of these representations become unramified, one obtains, for every $F \in W(\overline{K}_v/K_v)$, $F \notin I_v$,

$$\det(1 - F_v t, V_\lambda) = \det(1 - F_v t, V_\mu).$$

Thus, outside I_v, the characters of the representations V_λ and V_μ coincide.

The same arguments, retaining from 9.8.3 only equality of degrees, show that for every open subgroup J of I_v and every character χ of J,

$$\dim\big((V_{\lambda v}^{ss}\otimes\chi)^J\big)=\dim\big((V_{\mu v}^{ss}\otimes\chi)^J\big).$$

By Brauer's theorem, this implies that

$$V_{\lambda v}^{ss}|_{I_v}\quad\text{and}\quad V_{\mu v}^{ss}|_{I_v}$$

have the same character. The characters of V_λ and V_μ therefore coincide on all of $W(\overline{K}_v/K_v)$, completing the proof.

Proposition 9.9. Let $(V_\lambda)_{\lambda\in\mathcal{L}}$ be an infinite system of λ-adic representations. Suppose that, for every, or almost every by 9.8, place v of K, the semisimplifications of the $V_{\lambda v}$ are compatible. Then, for each λ,

$$Z_\lambda(V,t)=\varepsilon_\lambda(V,t)Z_\lambda(V^*,pt^{-1}). \tag{9.9.1}$$

Put

$$Z_\lambda^{ss}(V,t)=\prod_v\det(1-F_vt^{\deg(v)},(V_{\lambda v}^{ss})^{I_v})^{-1},$$

$$\varepsilon_\lambda^{ss}(V,t)=\prod_v\varepsilon(V_{\lambda v}^{ss},\psi_v,dx_v,t).$$

A local calculation shows that (9.9.1) is equivalent to

$$Z_\lambda^{ss}(V,t)=\varepsilon_\lambda^{ss}(V,t)Z_\lambda^{ss}(V^*,pt^{-1}). \tag{9.9.2}$$

The quotient of the local factors at v of the two sides of (9.9.1), or of (9.9.2), is additive in $V_{\lambda v}$, and these quotients coincide in the semisimple case. The identity to be proved, (9.9.2), is independent of λ. One proves it modulo λ for each λ, and concludes as in 9.3.

10. Grothendieck's theory

10.1. Let X be a nonsingular projective curve over the finite field $\mathbb{F}_p$, let $\overline{\mathbb{F}}_p$ be an algebraic closure of $\mathbb{F}_p$, and let

$$\overline{X}=X\otimes_{\mathbb{F}_p}\overline{\mathbb{F}}_p$$

be obtained from X by extension of scalars from $\mathbb{F}_p$ to $\overline{\mathbb{F}}_p$. Let ℓ be a prime number prime to p, and let Λ be a noetherian ring killed by a power of ℓ.

Let G be a Weil sheaf, in the sense of 7.3 and 7.4, constructible in flat Λ-modules. We again denote by G the étale sheaf on $\overline{X}$ obtained from it.

10.2. The étale cohomology groups

$$H^i(\overline{X},G)$$

are finite-type Λ-modules. Since they need not be projective, they will be of no use to us. We shall have to use a finer object

$$R\Gamma(\overline{X},G)\in\mathrm{Ob}\,D_{\mathrm{parf}}(\Lambda)$$

which gives rise to them. Recall that $D_{\mathrm{parf}}(\Lambda)$ may be defined as the category of finite complexes of finite-type projective Λ-modules, the morphisms being morphisms of complexes taken up to homotopy. The object $R\Gamma(\overline{X},G)$ is a finite complex of finite-type projective Λ-modules, concentrated in degrees $0,1,2$ if desired, well-defined up to homotopy, and whose cohomology groups are the $H^i(\overline{X},G)$.

By transport of structure, the Frobenius substitution

$$\varphi\in\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$$

acts on $R\Gamma(\overline{X},G)$: it is an endomorphism of the complex $R\Gamma(\overline{X},G)$, well-determined up to homotopy, and inducing on cohomology the Frobenius automorphism.

10.3. For K a finite complex of projective Λ-modules and u an endomorphism of K, put

$$\det(1-ut,K)=\prod_i \det(1-ut,K^i)^{(-1)^i}\in\Lambda(t). \tag{10.3.1}$$

If u and v are homotopic endomorphisms, then

$$\det(1-ut,K)=\det(1-vt,K),$$

so that $\det(1-ut,K)$ is well-defined for $K\in\operatorname{Ob} D_{\mathrm{parf}}(\Lambda)$ and $u\in\operatorname{End}(K)$. Taking the negative of the coefficient of t in the formal power-series expansion of $\det(1-ut,K)$, one defines the trace

$$\operatorname{Tr}(u,K)=\sum_i (-1)^i \operatorname{Tr}(u,K^i). \tag{10.3.2}$$

If u is a homotopy auto-equivalence, i.e. induces an automorphism on cohomology, one can find (K',u'), isomorphic in $D_{\mathrm{parf}}(\Lambda)$ to (K,u), with u' an automorphism of the complex K'. One then puts

$$\det(u,K)=\prod_i \det(u,K^i)^{(-1)^i}, \tag{10.3.3}$$

and this quantity is independent of the arbitrary choices made. If u^{-1} is a homotopy inverse of u, then

$$\det(1-ut,K)=\det(-ut,K)\det(1-u^{-1}t^{-1},K). \tag{10.3.4}$$

If $D(K)$ is the dual of K, the complex with components

$$D(K)^i=\operatorname{Hom}(K^{-i},\Lambda),$$

and if $\check{u}$ is a contragredient of u, one also has

$$\det(1-ut,K)=\det(-ut,K)\det(1-\check{u}\,t^{-1},D(K)). \tag{10.3.5}$$

A sketch of the proof of the following theorem is given in 10.8.

Theorem 10.4 (Grothendieck). One has

$$Z(G,t)=\det(1-Ft,R\Gamma(\overline{X},G))^{-1}. \tag{10.4.1}$$

In applications, the hypotheses imposed on G are too restrictive. One should take as coefficients an element of $D_{\mathrm{parf}}(X,\Lambda)$, i.e. a finite complex $G^\bullet$ of Weil sheaves of Λ-modules on X, which locally for the étale topology on $\overline{X}$ is homotopic to a finite complex of constructible flat Λ-module sheaves. Using 10.3, one can define, for x a closed point of X,

$$Z_x(G^\bullet,t)=\det(1-F_x t^{\deg(x)},G^\bullet_x)^{-1}.$$

Put

$$Z(G^\bullet,t)=\prod_x Z_x(G^\bullet,t).$$

One also defines $R\Gamma(\overline{X},G^\bullet)$, whose cohomology groups are the hypercohomology groups of $\overline{X}$ with values in $G^\bullet$, and one has

$$Z(G^\bullet,t)=\det(1-Ft,R\Gamma(\overline{X},G^\bullet))^{-1}. \tag{10.4.2}$$

10.5. Every $G^\bullet\in D_{\mathrm{parf}}(X,\Lambda)$ has a dual with values in the dualizing complex $\Lambda(1)$, placed in degree -2:

$$D(G^\bullet)=R\operatorname{Hom}_\Lambda(G^\bullet,\Lambda(1)[2])\in D_{\mathrm{parf}}(X,\Lambda).$$

Its i-th cohomology sheaf is the $(2+i)$-th local hyper-Ext of $G^\bullet$ with $\Lambda(1)$. Poincaré duality takes the form

$$R\Gamma(\overline{X},D(G^\bullet))=D(R\Gamma(\overline{X},G^\bullet)). \tag{10.5.1}$$

Applying 10.3.5, one obtains the functional equation

$$Z(G,t)=\det(-Ft,R\Gamma(\overline{X},G))^{-1}\,Z(D(G),t^{-1}). \tag{10.5.2}$$

10.6. Let

$$j : U \hookrightarrow X$$

be an open subset of X, the complement of a finite set S of closed points, and let $G^\bullet \in D_{\mathrm{parf}}(U, \Lambda)$. Write

$$j_! : D_{\mathrm{parf}}(U, \Lambda) \longrightarrow D_{\mathrm{parf}}(X, \Lambda)$$

for the derived functor of the exact functor extension by zero, and

$$Rj_* : D_{\mathrm{parf}}(U, \Lambda) \longrightarrow D_{\mathrm{parf}}(X, \Lambda)$$

for the derived functor of the direct-image functor under j. The local duality theorem, proved in particular in dimension one, gives

$$D(j_! G^\bullet) = Rj_* D(G^\bullet). \tag{10.6.1}$$

For G a locally constant sheaf of projective Λ-modules on U, with dual $\check{G}$, one has

$$D(G) = \check{G}(1)[2],$$

and (10.5.2) takes the form

$$Z(j_! G, t) = \det(-Ft, R\Gamma(\overline{X}, j_! G))^{-1} Z(Rj_* \check{G}(1), t^{-1}). \tag{10.6.2}$$

10.7. We shall make this formula explicit. The Z-functions on both sides are products of local factors. For the left-hand side, the local factor at a closed point x of X is

$$Z_x(j_! G, t) = \begin{cases} \det(1 - F_x t^{\deg(x)}, G_x)^{-1}, & x \notin S, \\ 1, & x \in S. \end{cases}$$

For the right-hand side it is

$$Z_x(Rj_* \check{G}(1), t) = \begin{cases} \det(1 - F_x t^{\deg(x)}, \check{G}_x(1))^{-1}, & x \notin S, \\ \det(1 - F_x t^{\deg(x)}, R\Gamma(I_x, \check{G}_K(1)))^{-1}, & x \in S. \end{cases}$$

The last formula is read as follows: $\check{G}_K(1)$ defines a representation of $W(\overline{K}_x/K_x)$, to which one applies the derived functor of the functor of invariants under I. One thereby obtains a complex of projective Λ-modules on which Frobenius acts, and one performs construction 10.3. Explicitly:

(a) The functor of invariants under I is the composite of the functor of invariants under

$$P'_\ell = \mathrm{Ker}(t_\ell) : \quad I\text{-modules} \longrightarrow I/P'_\ell\text{-modules},$$

and the functor of invariants under $\mathbb{Z}_\ell(1)$. The first functor is exact and transforms a projective Λ-module into a projective Λ-module. For every I-module V, one has

$$R\Gamma(I, V) = R\Gamma(\mathbb{Z}_\ell(1), V^{P'_\ell}).$$

For every $\mathbb{Z}_\ell(1)$-module W, $R\Gamma(\mathbb{Z}_\ell(1), W)$ may be computed by taking a resolution of W by coinduced modules

$$\mathfrak{F}(\mathbb{Z}_\ell(1), M),$$

the module of locally constant functions from $\mathbb{Z}_\ell(1)$ to M, on which $\mathbb{Z}_\ell(1)$ acts by $f(x) \mapsto f(\tau x)$. For σ a generator of $\mathbb{Z}_\ell(1)$, one has an exact sequence

$$0 \longrightarrow W \longrightarrow \mathfrak{F}(\mathbb{Z}_\ell(1), W) \xrightarrow{\alpha_\sigma} \mathfrak{F}(\mathbb{Z}_\ell(1), W) \longrightarrow 0,$$

with maps $m \mapsto (i \mapsto im)$ and

$$f \longmapsto (i \mapsto f(\sigma i) - \sigma(f(i))).$$

Passing to invariants, one identifies $R\Gamma(\mathbb{Z}_\ell(1), W)$ with the complex

$$W \xrightarrow{1-\sigma} W. \tag{10.7.1}$$

When σ is replaced by σ^ν, $\nu \in \mathbb{Z}_\ell^*$, the unique commutative diagram

$$\begin{array}{ccccc} \mathfrak{F}(\mathbb{Z}_\ell(1), W) & \xrightarrow{\alpha_\sigma} & \mathfrak{F}(\mathbb{Z}_\ell(1), W) & \longrightarrow & 0 \\ \| & & \downarrow{\scriptstyle \beta} & & \\ \mathfrak{F}(\mathbb{Z}_\ell(1), W) & \xrightarrow[\alpha_{\sigma^\nu}]{} & \mathfrak{F}(\mathbb{Z}_\ell(1), W) & \longrightarrow & 0 \end{array}$$

gives, after passing to invariants,

$$\begin{array}{ccc} W & \xrightarrow{1-\sigma} & W \\ {\scriptstyle 1}\downarrow & & \downarrow{\scriptstyle \frac{1-\sigma^\nu}{1-\sigma}} \\ W & \xrightarrow[1-\sigma^\nu]{} & W, \end{array} \tag{10.7.2}$$

where, for n an integer converging in $\mathbb{Z}_\ell$ to ν,

$$\frac{1-\sigma^\nu}{1-\sigma}\, w = \lim_n \sum_{i=0}^{n-1} \sigma^i w. \tag{10.7.3}$$

It follows that $R\Gamma(\mathbb{Z}_\ell(1), W)$ may be represented by any of the complexes (10.7.1), these complexes, as σ varies, being identified by the isomorphisms (10.7.2).

Starting from a $W(\overline{K}/K)$-module V, Frobenius acts by transport of structure on

$$R\Gamma(I, V) = \left[V^{P'_\ell} \xrightarrow{1-\sigma} V^{P'_\ell}\right].$$

Whatever the geometric Frobenius F, and whatever the generator σ of $\mathbb{Z}_\ell(1)$, F defines

$$\begin{array}{ccc} V^{P'_\ell} & \xrightarrow{1-\sigma} & V^{P'_\ell} \\ {\scriptstyle F}\downarrow & & \downarrow{\scriptstyle F} \\ V^{P'_\ell} & \xrightarrow[1-\sigma^q]{} & V^{P'_\ell}, \end{array}$$

which, via (10.7.2), is identified with the automorphism of the complex

$$\left[V^{P'_\ell} \xrightarrow{1-\sigma} V^{P'_\ell}\right]$$

with components

$$\left(F,\ \frac{1-\sigma}{1-\sigma^q} \circ F\right),$$

where

$$\frac{1-\sigma}{1-\sigma^q} = \frac{1-(\sigma^q)^{1/q}}{1-\sigma^q}$$

is defined as in (10.7.3). Hence

$$\det(1 - Ft, R\Gamma(I, V)) = \det(1 - Ft, V^{P'_\ell}) \det\left(1 - \frac{1-\sigma}{1-\sigma^q} \circ Ft, V^{P'_\ell}\right)^{-1},$$

where F denotes any geometric Frobenius and σ any generator of $\mathbb{Z}_\ell(1)$.

Formula 10.6.2 is rewritten as

$$\begin{aligned} \prod_{x \notin S} \det(1 - F_x t^{\deg(x)}, G_x) = \det(-Ft, R\Gamma(j_! G))^{-1} \\ \cdot \prod_{x \notin S} \det(1 - F_x t^{-\deg(x)}, \check{G}_x(1))^{-1} \\ \cdot \prod_{x \in S} \Bigg[\det(1 - F_x t^{-\deg(x)}, \check{G}_x^{P'_\ell}(1))^{-1} \\ \cdot \det\left(1 - \frac{1-\sigma}{1-\sigma^q} F_x t^{-\deg(x)}, \check{G}_x^{P'_\ell}(1)\right) \Bigg]. \end{aligned} \tag{10.7.4}$$

10.8. Passing to the projective limit, the preceding results generalize to ℓ-adic, or λ-adic, sheaves, and one obtains the statements of 9.1. The reason, cf. 10.5.2, for the simple formula (9.1.1) is the local result that, for G locally constant on

$$U \xrightarrow{j} X,$$

one has

$$j_* D(G) = D(j_! G).$$

However, the formula obtained for the constant, the determinant of $-Ft$ acting on cohomology, is not very explicit, and there is no local theory for it.

Remark 10.9. Let X be a nonsingular projective curve over an algebraically closed field k, and let V be a locally constant sheaf of free Λ-modules on $X_{\text{ét}}$. For simplicity, assume X connected of genus $g \geq 1$ and Λ local. Here are two methods for attaching to V a free Λ-module of rank 1.

(a) Write $R\Gamma(X, V)$ as a finite complex K of free Λ-modules. Then

$$\det(R\Gamma(X, V)) = \bigotimes_i \det(K^i)^{(-1)^i},$$

where det denotes the maximal exterior power, depends, up to canonical isomorphism, only on $R\Gamma(X, V)$. We write $h(V)$ for its dual.

(b) Let

$$K = \sum_i n_i P_i$$

be a canonical divisor, $\mathcal{O}(K) \simeq \Omega^1_X$, and put

$$N_K(V) = \bigotimes_i \det(V_{P_i})^{\otimes n_i}.$$

The set of positive canonical divisors forms a projective space, and the $N_K(V)$ form a necessarily trivial local system on this projective space. Thus the $N_K(V)$, as K varies, are canonically isomorphic, and one puts

$$\langle \Omega^1, V \rangle = N_K(V).$$

If (X, V) is defined over $k_0 \subset k$, i.e. is obtained by extension of scalars from (X_0, V_0) over $\mathrm{Spec}(k_0)$, then $\mathrm{Gal}(k/k_0)$ acts on $h(V)$ and on $\langle \Omega^1, V \rangle$ by characters $\chi_h(V)$ and $\chi_{\Omega^1}(V)$, with values in Λ^*. These constructions generalize to the λ-adic case.

Proposition 10.9.1. Assume that one of the following conditions is satisfied:

$$\Lambda = \mathbb{Q}_\lambda \quad \text{and } V \text{ becomes trivial on a finite cover of } X; \tag{a}$$

$$V \text{ has rank 1.} \tag{b}$$

Then

$$\chi_h(V) = \chi_{\Omega^1}(V)\, \omega_{(1-g)\dim(V)}, \tag{10.9.1}$$

where ω_n is the character giving the Galois action on $\mathbb{Z}_\ell(n)$.

A standard argument using Chebotarev reduces us to the case where k_0 is a finite field and k is its algebraic closure. One has to show that χ_h and $\chi_{\Omega^1}\omega_{(1-g)\dim V}$ take the same value on Frobenius. In case (a), the Weil group $W(\overline{K}_0/K_0)$ of X_0 acts continuously, for the discrete topology, on $V_{\overline{K}_0}$, so that 9.3 can be applied.

Compare (9.3) with (10.7.4), for $S = \varnothing$. These are functional equations relating the same Z-functions. Expressing the fact that they have the same constant gives (10.9.1). It may be more convenient to make this calculation only for virtual V of dimension 0, and to verify (10.9.1) directly for $V = \mathbb{Z}_\ell$. In case (b), one proceeds in the same way using 10.12.1 below.

It would be very interesting to make (10.9.1) more precise by defining a canonical isomorphism between

$$h(V) \quad \text{and} \quad \langle \Omega^1, V \rangle \otimes \mathbb{Z}_\ell((1-g)\dim(V)),$$

and to generalize (10.9.1) to the ramified case.

Remark 10.10. Theorem 10.4 is probably still valid for Λ a noetherian ring of characteristic p, i.e. with $p\Lambda = 0$. The required trace formula should follow from the trace formula in coherent cohomology, the Woods Hole trace formula. The proof has not, however, been written down.

10.11. Proof of 10.4. In [7], 10.4 is proved only for ℓ-adic sheaves, because the method of proof, based on the Lefschetz trace formula on $\overline{X}$ for the powers of the Frobenius endomorphism, required division by an arbitrary integer to pass from Lefschetz to (10.4.1), through the logarithmic derivative of Z.

To obtain 10.4 as stated, one must use SGA 4, XVII, 5.5, the Lefschetz trace formula on the $\mathrm{Sym}^n(X)$, and expand both sides of 10.4.1 as series, the second expanding as

$$\det(1 - Ft, R\Gamma(\overline{X}, G))^{-1} = \sum_n \mathrm{Tr}(F, \mathrm{Sym}^n(R\Gamma(\overline{X}, G)))\, t^n,$$

where Sym^n is the derived functor of the nonadditive functor n-th symmetric power. Moreover, the method of proof of [7], by successive fibrations by curves, reduces the trace formula to be proved to the case of curves.

Trace formula 10.11.1. Let X be a projective smooth curve over $\mathbb{F}_q$, let U be an open subset of X, and let G be a locally constant Weil sheaf of projective Λ-modules. One has

$$\sum_{x \in U(\mathbb{F}_q)} \mathrm{Tr}(F_x, G_x) = \mathrm{Tr}(F, R\Gamma(\overline{X}, j_!G)).$$

In this formula, the Frobenii are relative to $\mathbb{F}_q$, and

$$\overline{X} = X \otimes_{\mathbb{F}_q} \overline{\mathbb{F}}_q.$$

One may suppose that $\mathbb{F}_q$ is the field of constants of the function field K of X; otherwise both sides are 0.

The method of proof presented by Grothendieck in the oral seminar SGA 5 consists in reducing to the Lefschetz formula for the number of fixed points of endomorphisms of complete curves, already proved in [17], by studying the cohomology of covers of $\overline{X}$ as representations of the group of the cover.

Let $\overline{K}$ be an algebraic closure of K, containing the function field of $\overline{X}$. The sheaf G defines a representation of $W(\overline{K}/K)$ on the projective Λ-module $G_{\overline{K}}$. The intersection J of $\mathrm{Gal}^0(\overline{K}/K)$ and the kernel of this representation defines a connected étale cover

$$f : \overline{U}' \longrightarrow \overline{U}$$

of $\overline{U}$ on which G becomes trivial. Put

$$H = W(\overline{K}/K)/J.$$

Then H is an extension of $\mathbb{Z}$ by the group H_0 of the cover $\overline{U}'/\overline{U}$, and $G_{\overline{K}}$ is a representation of H.

Let j' be the inclusion of $\overline{U}'$ in a nonsingular projective curve $\overline{X}'$. The ℓ-adic cohomology with proper support of $\overline{U}'$ is a representation of H. Standard arguments allow one to do better, and to define

$$R\Gamma_c(\overline{U}', \mathbb{Z}_\ell) = R\Gamma(\overline{X}', j'_!\mathbb{Z}_\ell) \in D_{\mathrm{parf}}(\mathbb{Z}_\ell[H_0]),$$

on which H acts semilinearly with respect to its action on $\mathbb{Z}_\ell[H_0]$, the action of $H_0 \subset H$ being the evident action. There is a canonical isomorphism

$$R\Gamma(\overline{X}, j_!G) = \left[R\Gamma(\overline{X}', j'_!\mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} G_{\overline{K}}\right]^{H_0}, \tag{10.11.2}$$

compatible with the Frobenius action on both sides.

Write $R\Gamma(\overline{X}', j'_!\mathbb{Z}_\ell)$ as a finite complex of projective $\mathbb{Z}_\ell[H_0]$-modules on which Frobenius acts. The components of this complex appear as $\mathbb{Z}_\ell[H^+]$-modules, for H^+ the submonoid of H, extension of $\mathbb{N} \subset \mathbb{Z}$ by H_0. We apply the following lemma to them.

Lemma 10.11.3. Let H^+ be a semidirect product of $\mathbb{N}$ by a finite group H_0, let M be a $\mathbb{Z}_\ell[H^+]$-module, projective of finite rank as a $\mathbb{Z}_\ell[H_0]$-module, let Λ be a $\mathbb{Z}_\ell$-algebra, and let V be a $\Lambda[H^+]$-module, projective of finite rank as a Λ-module. Let χ_M and χ_V be the characters of M and V.

(i) There exists a function χ_M^* on H, with values in $\mathbb{Z}_\ell$, such that, for $h \in H$, and $H_0(h)$ the centralizer of h in H_0, one has

$$\chi_M(h) = \mathrm{Tr}(h, M) = |H_0(h)|\, \chi_M^*(h).$$

(ii) For $F \in H/H_0$,

$$\mathrm{Tr}\big(F, (M \otimes_{\mathbb{Z}_\ell} V)^{H_0}\big) = \sum_{\substack{h \mapsto F \\ \mathrm{mod}\ H_0\text{-conjugacy}}} \chi_M^*(h)\chi_V(h).$$

Applying Lefschetz, one computes χ_M^* for $R\Gamma(\overline{X}', j_!'\mathbb{Z}_\ell)$, virtually the difference between $R\Gamma(\overline{X}', \mathbb{Z}_\ell)$ and an elementary quantity; 10.11.2 and 10.11.3 provide the desired formula.

10.12. The abelian case, and a fanciful application. Let

$$\chi : W(\overline{K}/K) \longrightarrow \Lambda^*$$

be a quasi-character with values in Λ^*, unramified outside a finite set S of places of K. Let j be the inclusion of $U = X - S$ in X, and let $j_!\lfloor\chi\rfloor$ be the rank-one sheaf on U defined by χ, extended by 0. Put

$$\varepsilon(j_!\lfloor\chi\rfloor, t) = \prod_{v \in S} \varepsilon_0(\chi_v\omega_v^t, \psi_v, dx_v) \prod_{v \notin S} \varepsilon(\chi_v\omega_v^t, \psi_v, dx_v).$$

Here ω^t is as in 7.6, ε_0 is defined in 6.4, and ε, for an unramified character, is defined by 3.4.3.3.

Proposition 10.12.1. One has

$$\det(-Ft, \mathrm{R}\Gamma(j_!\lfloor\chi\rfloor)) = \varepsilon(j_!\lfloor\chi\rfloor, t).$$

This proposition makes 10.7.4 more precise.

Let H be the quotient of $W(\overline{K}/K)$ by $\mathrm{Ker}(\chi)$. It is enough to treat the universal case where

$$\Lambda = \mathbb{Z}/\ell^n[H].$$

This case lifts to characteristic 0, and 10.12.1 is obtained by reduction.

10.13. Take for example, for Λ, the dual numbers over $\mathbb{Z}/\ell^n$:

$$\Lambda = \mathbb{Z}/\ell^n[\epsilon] \qquad (\epsilon^2 = 0),$$

let α be a continuous homomorphism from the idèle-class group $\mathbb{A}^*/K^*$ into the additive group $\mathbb{Z}/\ell^n$, and put

$$\chi = 1 + \alpha\epsilon.$$

The ramification of α is automatically tame. Let S be a finite set of places outside which α is unramified. Applying 10.4 to χ and to the trivial character, and taking the ratio, one finds that

$$L_S^+(\alpha, t) = \sum_{v \in S} \alpha_v(\pi_v) \frac{t^{\deg(v)}}{1 - t^{\deg(v)}} = \sum_n t^n \sum_{\deg(v)|n} \alpha_v(\pi_v) \tag{10.13.1}$$

lies in $\mathbb{Z}/\ell^n(t)$.

For every place v of K, choose a uniformizer π_v. Adjoin to $\mathbb{Z}/\ell^n$ a primitive p-th root of unity, and let ψ be a nontrivial additive character of $\mathbb{A}/K$, with values in the p-th roots of unity. Take the quotient of (10.7.4) for χ and of (10.7.4) for the trivial character, and use (10.12.1). After simplification, one obtains the following identity. In the left-hand side, the first two infinite sums are summed by (10.13.1). All sums run over all places of K:

$$\sum_v \alpha_v(\pi_v) \frac{t^{\deg(v)}}{1 - t^{\deg(v)}} + \sum_v \alpha_v(\pi_v) \frac{q_v t^{-\deg(v)}}{1 - q_v t^{-\deg(v)}} + \sum_v \alpha_v(-1) \frac{1}{1 - t^{\deg(v)}} \tag{10.13.2}$$

$$= -\sum_v n_v \alpha_v(\pi_v) - \sum_v \sum_{x \in k_v^*} \alpha_v(x)\psi_v(\pi_v^{1+n_v}x).$$

One checks elementarily that the validity of (10.13.2) does not depend on the choice of the uniformizers π_v, recalling that, for $x \in k_v^*$, one has $(q-1)\alpha(x) = 0$. In the left-hand side, the $\alpha_v(-1)$ are of order 2, hence vanish if $\ell \neq 2$.

11. Appendix. The calculation of ε modulo roots of unity

This appendix is a variation on a theme of Lakkis and Dwork. My attention was drawn to their results by J.-P. Serre.

We shall write $a \sim b$ for the following relation between invertible complex numbers:

$$a \sim b \quad : \quad a/b \text{ is the product of a root of unity and a power of } q^{1/2}.$$

Let K be a nonarchimedean local field, and resume the notation of 2.2.1, 2.2.2, and 3.1. The Haar measure dx is assumed to satisfy

$$\int_{\mathcal{O}} dx = q^{n/2},$$

with n an integer. Let V be a complex representation, of a finite quotient, of $\mathrm{Gal}(\overline{K}/K)$, and let V^P denote the invariants under wild inertia. This is a subrepresentation of V.

Theorem 11.1. One has

$$\varepsilon(V, \psi, dx) \sim \varepsilon(V^P, \psi, dx).$$

Lemma 11.2. Modulo the equivalence $\sim$, $\varepsilon(V, \psi, dx)$ is independent of ψ and dx. This follows immediately from 5.3 and 5.4. This lemma will allow us to abbreviate $\varepsilon(V, \psi, dx)$ to $\varepsilon(V)$.

Lemma 11.3. If the character of V is real, i.e. if V is isomorphic to V^*, then

$$\varepsilon(V) \sim 1.$$

One may suppose that $\int_{\mathcal{O}} dx = 1$ and that $n(\psi) = 0$. One then applies 5.7.1, 5.4, and 5.5.

Let L be a finite extension of K in $\overline{K}$, let W be a representation of $\mathrm{Gal}(\overline{K}/L)$, and let $\mathrm{Ind}(W)$ be the representation induced to $\mathrm{Gal}(\overline{K}/K)$.

Lemma 11.4. One has

$$\varepsilon_L(W) \sim \varepsilon_K(\mathrm{Ind}(W)).$$

Let W' be the trivial representation of the same dimension as W. For $\psi_L = \psi_K \circ \mathrm{Tr}_{L/K}$, (5.6.1) gives

$$\varepsilon(W - W', \psi_L) = \varepsilon(\mathrm{Ind}(W) - \mathrm{Ind}(W'), \psi_K),$$

and one concludes by applying 11.3 to W' and to $\mathrm{Ind}(W')$.

11.5. Let L and W be as above, let P_L be the wild inertia group of L, and regard W^{P_L} as a representation of

$$H = P \cdot \mathrm{Gal}(\overline{K}/L),$$

trivial on P. One has

$$\mathrm{Ind}(W)^P = \mathrm{Ind}_H^{\mathrm{Gal}(\overline{K}/K)}(W^{P_L}).$$

By 11.4, the theorem for W is therefore equivalent to the theorem for $\mathrm{Ind}(W)$. By Brauer, this reduces us to the case where V has dimension one, defined by a finite-order character χ of K^*. If χ is tame, $[\chi]^P = [\chi]$, and 11.1 is trivial. If χ is wildly ramified, $[\chi]^P = 0$, and 11.1 follows from the next lemma.

Lemma 11.6. For χ wildly ramified,

$$\varepsilon(\chi, \psi, dx) \sim 1.$$

Let m be the conductor of χ, put

$$n = \left[\frac{m+1}{2}\right],$$

and let $y \in K^*$ have valuation $-m - n(\psi)$, well-defined modulo $\pi^{n-m-n(\psi)}$, such that, cf. 4.16,

$$\chi(1+a) = \psi(ay) \qquad (v(a) \geq n).$$

Decompose χ as $\chi = \chi_1\chi_2$, where χ_1 has order prime to p and χ_2 has order a power of p. As in 4.16, one proves that

$$\varepsilon(\chi, \psi, dx) = \int \chi^{-1}(x)\psi(x)\,dx = \int_{y(1+(\pi^{m-n}))} \chi^{-1}(x)\psi(x)\,dx \tag{11.6.1}$$

$$= \chi_1(y) \int_{y(1+(\pi^{m-n}))} \chi_2^{-1}(x)\psi(x)\,dx.$$

If m is even, $m = 2n$, the integrand is constant and 11.6 is clear. For m odd, $m = 2n-1$, there are two ways to proceed.

(a) $\chi^{-1}(x)\psi(x)$ is a nondegenerate quadratic character on

$$y(1 + (\pi^{m-n}))/(1 + (\pi^n));$$

by Weil, the sum of its values is ~ 1.

(b) By (11.6.1), it is enough to prove that $\varepsilon(\chi_2, \psi, dx) \sim 1$. The square of this number, multiplied by a suitable power of q, belongs to a field of p^N-th roots of unity, and has valuation 1 at every place of this field, except possibly at the unique place dividing p. It is therefore a root of unity.

Corollary 11.7. Take ψ and dx such that $n(\psi) = -1$ and

$$\int_{(\pi)} dx = 1.$$

Then $\varepsilon(V, \psi, dx)$ is an algebraic integer.

Decomposing V as $V^P + W$, one sees that it is enough to treat the following two cases.

(a) V is tamely ramified. In this case, V is a sum of representations induced from one-dimensional representations of subgroups, and ε is, up to sign, a product of Gauss sums, cf. 5.10.

(b) $V^P = 0$. In this case, $\varepsilon \sim 1$, and it is enough to prove that the complex absolute value $|\varepsilon|$ is a positive power of q. This follows from 5.7.2, where $dx'/dx = q$.

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The Moduli Schemes of Elliptic Curves

P. Deligne and M. Rapoport

Standalone literal English edition aligned to all 174 authority pages

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Contents

THE MODULI SCHEMES OF ELLIPTIC CURVES

by P. Deligne(1) and M. Rapoport(2)

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(2) The second author considers the political activities of NATO to be incompatible with the peaceful activity of the mathematician and regrets, for this reason, that this congress was partially financed by that organization.

LES SCHÉMAS DE MODULES DE COURBES ELLIPTIQUES

par P. Deligne(1) et M. Rapoport(2)

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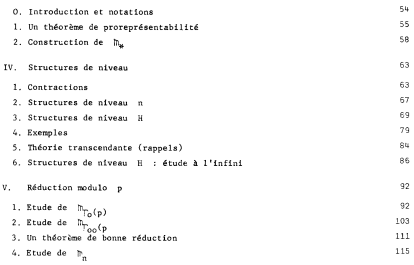

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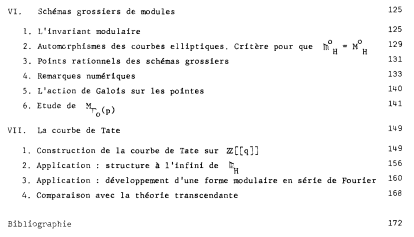

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Introduction

1. Let X be the Poincaré half-plane

$$X = \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\}.$$

The group $SL(2, \mathbb{R})$ acts on X by homographic transformations

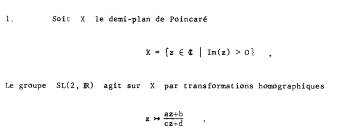
$$z \mapsto \frac{az+b}{cz+d}.$$

If Γ is a subgroup of $SL(2, \mathbb{Z})$ defined by congruence conditions, the Riemann surface X/Γ is the complement of a finite set of points (”at infinity”) in a compact Riemann surface. It is therefore an algebraic curve. Its points are in bijective correspondence with the isomorphism classes of elliptic curves equipped with a ”level structure” of a suitable kind. It is known from this interpretation that it admits as a field of definition a subfield of a cyclotomic field. In this article, we study the structure at infinity and the reduction modulo p of X/Γ.

Let us concentrate on the case in which Γ is the subgroup of SL(2, ℤ) formed by the matrices congruent modulo n to the identity matrix. Suppose that n ≥ 3, in which case Γ acts on X without a fixed point.

For z ∈ X, let E_z denote the quotient of ℂ by the lattice ℤ + ℤz. The addition law on ℂ descends to the quotient and equips E_z with the structure of an elliptic curve (= an abelian variety of dimension one). The elliptic curves E_z and E_z′ are isomorphic if and only if z and z′ are conjugate under SL(2, ℤ). The kernel E(z)_n of multiplication by n on E(z) is the image of (1/n)(ℤ + ℤz); we shall denote by α(z) the isomorphism

E(z)_n ≃ (ℤ/nℤ)^2 : (a+bz)/n ↦ (a,b).


1. Soit X le demi-plan de Poincaré

$$X = \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\} \quad .$$

Le groupe $SL(2, \mathbb{R})$ agit sur X par transformations homographiques

$$z \mapsto \frac{az+b}{cz+d} \quad .$$

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Authority page 5; collected-paper page 498; original folio 147.

For the pairs (E(z),α(z)) and (E(z′),α(z′)) to be isomorphic, it is necessary and sufficient that z and z′ be conjugate under Γ.

For every elliptic curve E over an algebraically closed field k of characteristic p not dividing n, the group E_n is equipped with a nondegenerate alternating form e_n with values in the nth roots of unity in k. A level-n structure on E is an isomorphism α:E_n ≃ (ℤ/n)^2. If α is a level-n structure, we shall denote by ζ(α) the primitive nth root of unity

ζ(α) = e_n(α^−1((1,0)), α^−1((0,1))).

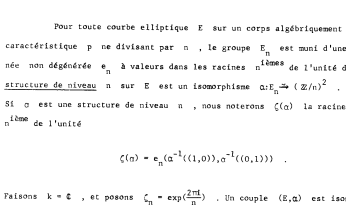
Put k = ℂ, and set ζ_n = exp(2πi/n). A pair (E,α) is isomorphic to a pair (E(z),α(z)) (z ∈ X) if and only if

ζ(α) = ζ_n

(or ζ_n^−1, according to the sign convention used in the definition of e_n).

2. The foregoing generalizes to families of elliptic curves parametrized by a scheme S. One finds that X/Γ represents the following functor F_n on schemes over ℂ: F_n(T) is the set of isomorphism classes of algebraic families, parametrized by T, of elliptic curves E equipped with a level-n structure such that ζ(α) = ζ_n.

The definition of F_n(T) remains meaningful when T is merely assumed to be a scheme over ℤ[ζ_n, 1/n]. Igusa proved in [11] that the resulting functor F_n is representable by a scheme M_n^∘[1/n] over ℤ[ζ_n,1/n]. Igusa's method is as follows.


Pour toute courbe elliptique E sur un corps algébriquement clos k de caractéristique p ne divisant par n, le groupe E_n est muni d'une forme alternée non dégénérée e_n à valeurs dans les racines $n^{\text{ièmes}}$ de l'unité de k. Une structure de niveau n sur E est un isomorphisme $\alpha : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$. Si α est une structure de niveau n, nous noterons ζ(α) la racine primitive $n^{\text{ième}}$ de l'unité

$$\zeta(\alpha) = e_n(\alpha^{-1}((1,0)), \alpha^{-1}((0,1))) \quad .$$

Faisons $k = \mathbb{C}$, et posons $\zeta_n = \exp(\frac{2\pi i}{n})$. Un couple (E,α) est isomorphe à un

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a) If M_n^∘[1/n] exists, then M_nm^∘[1/nm] exists. If E is an elliptic curve over an algebraically closed field, multiplication by m induces an isomorphism of E_nm/nE_nm with E_n; a level-nm structure α:E_nm ≃ (ℤ/nm)^2 thereby defines, by reduction modulo n, a level-n structure. This carries over to an arbitrary base scheme and defines a morphism of functors F_nm → F_n. This morphism is relatively representable. It makes M_nm^∘[1/nm] an unramified Galois covering, with group Ker(SL(2, ℤ/nm) → SL(2, ℤ/n)), of

M_n^∘[1/nm] ⊗_{ℤ[ζ_n]} ℤ[ζ_nm]

(⊗ denotes an extension of scalars).

b) If M_nm^∘[1/nm] exists, M_n^∘[1/nm] exists. One obtains M_n^∘[1/nm] by passing to the quotient.

c) M_3^∘[1/3] and M_4^∘[1/4] exist. Explicitly, one has

M_3^∘[1/3] = Spec(ℤ[ζ_3,1/3,μ,(μ^3−1)^−1]),

with universal elliptic curve the plane cubic with equation

X^3 + Y^3 + Z^3 = 3μ XYZ.

The zero section is (1,−1,0), and the basis of the group of points of order 3 is ((−1,0,1), (−1,ζ_3,0)).

At level 4, one has

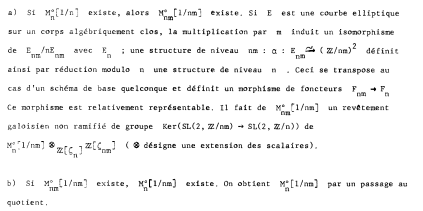
M_4^∘[1/4] = Spec(ℤ[i,1/2,σ,(σ(σ^4−1))^−1]),

where i = ζ_4, with universal elliptic curve the plane cubic whose nonhomogeneous equation is

y^2 = x(x−1)(x−λ), where

λ = (1/4)(σ + 1/σ)^2.

The zero section is the point at infinity, and the basis (r,s) of the group of points


a) Si $M_n^\circ[1/n]$ existe, alors $M_{nm}^\circ[1/nm]$ existe. Si E est une courbe elliptique sur un corps algébriquement clos, la multiplication par m induit un isomorphisme de E_{nm}/nE_{nm} avec E_n ; une structure de niveau nm : $\alpha : E_{nm} \xrightarrow{\sim} (\mathbb{Z}/nm)^2$ définit ainsi par réduction modulo n une structure de niveau n . Ceci se transpose au cas d'un schéma de base quelconque et définit un morphisme de foncteurs $F_{nm} \to F_n$ Ce morphisme est relativement représentable. Il fait de $M_{nm}^\circ[1/nm]$ un revêtement galoisien non ramifié de groupe $\mathrm{Ker}(SL(2,\mathbb{Z}/nm) \to SL(2,\mathbb{Z}/n))$ de $M_n^\circ[1/nm] \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{Z}[\zeta_{nm}]$ (⊗ désigne une extension des scalaires).

b) Si $M_{nm}^\circ[1/nm]$ existe, $M_n^\circ[1/nm]$ existe. On obtient $M_n^\circ[1/nm]$ par un passage au quotient.

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Authority page 7; collected-paper page 500; original folio 149.

of order 4 is given by

r = (1/2(σ + 1/σ), i(σ²+1)(σ−1)²/(4·σ²))

s = (−i/2(σ − 1/σ) + 1, −(σ²−1)(σ+1)²/(4·σ²)).

The formulas above for M_4^∘[1/4] are copied from Shioda [31]. Igusa used not M_4^∘[1/4], but the moduli scheme of elliptic curves equipped with a structure intermediate between a level-4 structure and a level-2 structure.

d) M_n^∘[1/n] is obtained by gluing M_n^∘[1/3n] and M_n^∘[1/4n].

3. The Riemann surface X/Γ is noncompact. Geometrically, this fact translates as follows: if E_η is an elliptic curve equipped with a level-n structure over $\mathbb{C}((T))$, it may happen that the minimal model of E_η over $\mathbb{C}[[T]]$ has bad reduction. In that case, the special fiber E'_o of the Néron model E' of E_η over $\mathbb{C}[[T]]$ is isomorphic to $\mathbb{C}^*\times\mathbb{Z}/kn$ for a suitable k. Let E_o be the subgroup of E'_o that is the union of the components of E'_o whose order in π_o(E'_o) divides n. This subgroup is isomorphic to $\mathbb{C}^*\times\mathbb{Z}/n$. It has the following advantage over E'_o: after extending scalars from $\mathbb{C}((T))$ to $\mathbb{C}((T^{\{1/\ell\}}))$, the number of connected components of E'_o is multiplied by ℓ, while E_o remains unchanged.

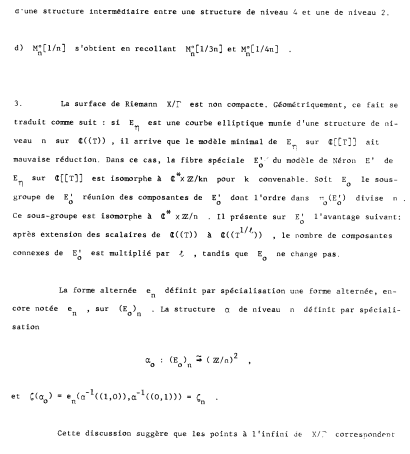
By specialization, the alternating form e_n defines an alternating form, again denoted e_n, on (E_o)_n. By specialization, the level-n structure α defines

$\alpha_o : (E_o)_n \simeq (\mathbb{Z}/n)^2$,

and

$\zeta(\alpha_o) = e_n(\alpha^{-1}((1,0)), \alpha^{-1}((0,1))) = \zeta_n$.

This discussion suggests that the points at infinity of X/Γ correspond


d'une structure intermédiaire entre une structure de niveau 4 et une de niveau 2.

d) $M_n^*[1/n]$ s'obtient en recollant $M_n^*[1/3n]$ et $M_n^*[1/4n]$.

3. La surface de Riemann X/Γ est non compacte. Géométriquement, ce fait se traduit comme suit : si E_η est une courbe elliptique munie d'une structure de niveau n sur $\mathbb{C}((T))$, il arrive que le modèle minimal de E_η sur $\mathbb{C}[[T]]$ ait mauvaise réduction. Dans ce cas, la fibre spéciale E'_o du modèle de Néron E' de E_η sur $\mathbb{C}[[T]]$ est isomorphe à $\mathbb{C}^*\times\mathbb{Z}/kn$ pour k convenable. Soit E_o le sous-groupe de E'_o réunion des composantes de E'_o dont l'ordre dans $\pi_o(E'_o)$ divise n. Ce sous-groupe est isomorphe à $\mathbb{C}^*\times\mathbb{Z}/n$. Il présente sur E'_o l'avantage suivant: après extension des scalaires de $\mathbb{C}((T))$ à $\mathbb{C}((T^{1/\ell}))$, le nombre de composantes connexes de E'_o est multiplié par ℓ , tandis que E_o ne change pas.

La forme alternée e_n définit par spécialisation une forme alternée, encore notée e_n , sur $(E_o)_n$. La structure α de niveau n définit par spécialisation

$$\alpha_o : (E_o)_n \xrightarrow{\sim} (\mathbb{Z}/n)^2 ,$$

et $\zeta(\alpha_o) = e_n(\alpha^{-1}((1,0)),\alpha^{-1}((0,1))) = \zeta_n$.

Cette discussion suggère que les points à l'infini de X/Γ correspondent

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Authority page 8; collected-paper page 501; original folio 150.

to the isomorphism classes of systems (E_o,e_n,α_n) as above. This is indeed the case.

In the text, we refine this modular interpretation of the set of points at infinity of X/Γ into a modular interpretation of the projective curve X/Γ, the compactification of X/Γ. Here is the principle.

For E_η over $\mathbb{C}((T))$ as above, the special fiber $\bar{E}'_o$ of the minimal model $\bar{E}'$ of E_η over $\mathbb{C}[[T]]$ is a cycle of kn projective lines:

[cycle of kn projective lines: see authority asset P0008-A01]

Let $\bar{E}$ be obtained from $\bar{E}'$ by contracting to a point the irreducible components of $\bar{E}'_o$ whose order does not divide n. The special fiber $\bar{E}_o$ of $\bar{E}$ is a cycle of n projective lines meeting transversely, and is a compactification of E_o (E_o identifies with the smooth locus $\bar{E}_o^{reg}$ of $\bar{E}_o$). By specialization, + and α define a group law on the smooth locus $\bar{E}_o^{reg}$ of $\bar{E}_o$, an action of $\bar{E}_o^{reg}$ on $\bar{E}_o$, and an isomorphism

$\alpha_o : (\bar{E}_o^{reg})_n \simeq (\mathbb{Z}/n)^2$.

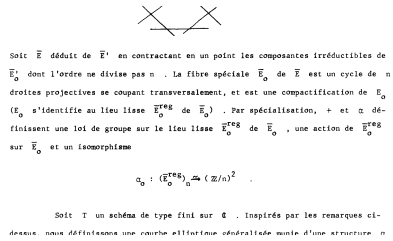
Let T be a scheme of finite type over $\mathbb{C}$. Inspired by the remarks above, we define a

generalized elliptic curve over T, equipped with a level-n structure α such that ζ(α) = ζ_n, as consisting of:

a) a proper flat morphism p : E → T;

b) a commutative group-scheme structure on the open subset E^reg of E where p is smooth; addition is assumed to extend to an action

+ : E^reg ×_T E → E;



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Authority page 9; collected-paper page 502; original folio 151.

c) an isomorphism of group schemes

(E^reg)_n ≃ (ℤ/n)²_T.

The geometric fibers of p are assumed either to be elliptic curves equipped with a level-n structure such that ζ(α) = ζ_n, or to be isomorphic to one of the systems (Ē_o,α_o) constructed above.

We prove that the curve X/Γ represents the following functor F̄_n on schemes of finite type over ℂ. F̄_n(T) is the set of isomorphism classes of generalized elliptic curves E over T, equipped with a level-n structure α such that ζ(α) = ζ_n.

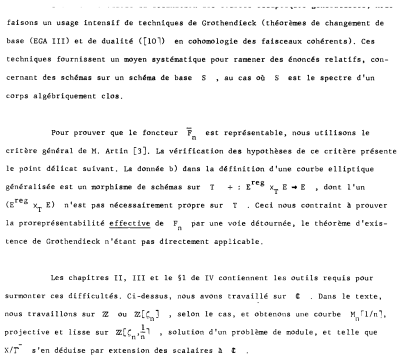
To make the definition of generalized elliptic curves viable, we make intensive use of Grothendieck techniques (base-change theorems (EGA III) and duality ([10]) in coherent-sheaf cohomology). These techniques provide a systematic means of reducing relative statements concerning schemes over a base scheme S to the case in which S is the spectrum of an algebraically closed field.

To prove that the functor F̄_n is representable, we use the general criterion of M. Artin [3]. Verifying the hypotheses of this criterion presents the following delicate point. Datum b) in the definition of a generalized elliptic curve is a morphism of schemes over T

+ : E^reg ×_T E → E,

whose source E^reg ×_T E is not necessarily proper over T. This forces us to prove the effective prorepresentability of F_n by an indirect route, since Grothendieck's existence theorem is not directly applicable.

Chapters II and III and §1 of IV contain the tools required to overcome these difficulties. Above, we worked over ℂ. In the text, we work over ℤ or ℤ[ζ_n], as appropriate, and obtain a curve M_n[1/n], projective and smooth over ℤ[ζ_n,1/n], solving a moduli problem, and such that X/Γ is obtained from it by extension of scalars to ℂ.



faisons un usage intensif de techniques de Grothendieck (théorèmes de changement de base (EGA III) et de dualité ([10]) en cohomologie des faisceaux cohérents). Ces techniques fournissent un moyen systématique pour ramener des énoncés relatifs, concernant des schémas sur un schéma de base S, au cas où S est le spectre d'un corps algébriquement clos.

Pour prouver que le foncteur $\overline{F}_n$ est représentable, nous utilisons le critère général de M. Artin [3]. La vérification des hypothèses de ce critère présente le point délicat suivant. La donnée b) dans la définition d'une courbe elliptique généralisée est un morphisme de schémas sur T $+ : E^{reg} \times_T E \to E$, dont l'un $(E^{reg} \times_T E)$ n'est pas nécessairement propre sur T. Ceci nous contraint à prouver la proreprésentabilité effective de F_n par une voie détournée, le théorème d'existence de Grothendieck n'étant pas directement applicable.

Les chapitres II, III et le §1 de IV contiennent les outils requis pour surmonter ces difficultés. Ci-dessus, nous avons travaillé sur $\mathbb{C}$. Dans le texte, nous travaillons sur $\mathbb{Z}$ ou $\mathbb{Z}[\zeta_n]$, selon le cas, et obtenons une courbe $M_n[1/n]$, projective et lisse sur $\mathbb{Z}[\zeta_n, \frac{1}{n}]$, solution d'un problème de module, et telle que X/Γ s'en déduise par extension des scalaires à $\mathbb{C}$.

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4. To speak of the ”reduction mod p of X/Γ” presupposes the choice of a model M_n of X/Γ over $\mathbb{Z}[\zeta_n]$. The definition of M_n given in the text is equivalent to the following: M_n is the normalization of the j-line (the projective line over $\mathbb{Z}[\zeta_n]$, to which M_n maps by the ”modular invariant of the universal elliptic curve” map) in the function field of M_n^∘[1/n]. Also let M_n^∘ be the affine curve whose coordinate ring is the normalization of $\mathbb{Z}[\zeta_n][j]$ in the coordinate ring of M_n^∘. It is shown in the text that

M_n[1/n] = M_n ⊗_{$\mathbb{Z}[\zeta_n]$} $\mathbb{Z}[\zeta_n,1/n]$ and

M_n^∘[1/n] = M_n^∘ ⊗_{$\mathbb{Z}[\zeta_n]$} $\mathbb{Z}[\zeta_n,1/n]$.

We prove the following results:

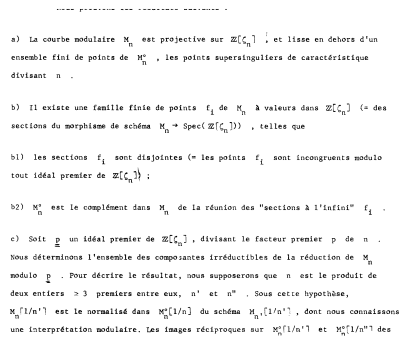
a) The modular curve M_n is projective over $\mathbb{Z}[\zeta_n]$, and smooth away from a finite set of points of M_n^∘, namely the supersingular points whose characteristic divides n.

b) There exists a finite family of points f_i of M_n with values in $\mathbb{Z}[\zeta_n]$ (= sections of the scheme morphism M_n → Spec($\mathbb{Z}[\zeta_n]$)), such that

b1) the sections f_i are disjoint (= the points f_i are incongruent modulo every prime ideal of $\mathbb{Z}[\zeta_n]$);

b2) M_n^∘ is the complement in M_n of the union of the ”sections at infinity” f_i.

c) Let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\zeta_n]$ dividing the prime factor p of n. We determine the set of irreducible components of the reduction of M_n modulo $\mathfrak{p}$. To describe the result, we suppose that n is the product of two relatively prime integers ≥ 3, n’ and n”. Under this hypothesis, M_n[1/n’] is the normalization in M_n^∘[1/n] of the scheme M_{n’}[1/n’], for which we know a modular interpretation. The pullbacks to M_{n’}^∘[1/n’] and M_{n”}^∘[1/n”] of the



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universal elliptic curves over M_{n'}^∘[1/n'] and M_{n''}^∘[1/n''] glue together and furnish a universal elliptic curve E over M_n^∘. The level structure

α : E_n ≃ (ℤ/n)²

over M_n^∘[1/n] extends to a morphism of group schemes over M_n^∘,

α^−1 : (ℤ/n)² → E_n.

The p-primary component of α^−1 is a morphism

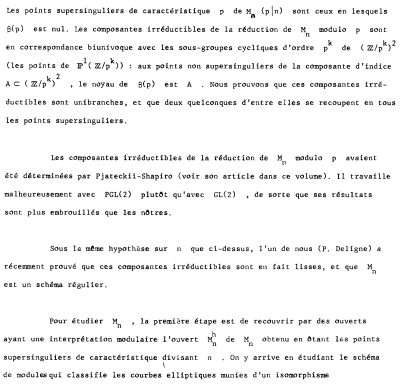
β(p) : (ℤ/p^k)² → E_{p^k}.

The supersingular points of characteristic p of M_𝔭 (p|n) are those at which β(p) is zero. The irreducible components of the reduction of M_n modulo 𝔭 are in bijective correspondence with the cyclic subgroups of order p^k of (ℤ/p^k)² (the points of ℙ¹(ℤ/p^k)): at the nonsupersingular points of the component indexed by A ⊂ (ℤ/p^k)², the kernel of β(p) is A. We prove that these irreducible components are unibranch, and that any two of them meet at every supersingular point.

The irreducible components of the reduction of M_n modulo p had been determined by Pjateckii-Shapiro (see his article in this volume). Unfortunately, he works with PGL(2) rather than GL(2), so that his results are more tangled than ours.

Under the same hypothesis on n as above, one of us (P. Deligne) recently proved that these irreducible components are in fact smooth, and that M_n is a regular scheme.

To study M_n, the first step is to cover by open subsets having a modular interpretation the open subset M_n^h of M_n obtained by removing the supersingular points whose characteristic divides n. This is achieved by studying the moduli scheme classifying elliptic curves equipped with an isomorphism



Les points supersinguliers de caractéristique p de $M_n(p|n)$ sont ceux en lesquels $\beta(p)$ est nul. Les composantes irréductibles de la réduction de M_n modulo p sont en correspondance biunivoque avec les sous-groupes cycliques d'ordre p^k de $(\mathbb{Z}/p^k)^2$ (les points de $\mathbb{P}^1(\mathbb{Z}/p^k)$) : aux points non supersinguliers de la composante d'indice $A \subset (\mathbb{Z}/p^k)^2$, le noyau de $\beta(p)$ est A. Nous prouvons que ces composantes irréductibles sont unibranches, et que deux quelconques d'entre elles se recoupent en tous les points supersinguliers.

Les composantes irréductibles de la réduction de M_n modulo p avaient été déterminées par Pjateckii-Shapiro (voir son article dans ce volume). Il travaille malheureusement avec $PGL(2)$ plutôt qu'avec $GL(2)$, de sorte que ses résultats sont plus embrouillés que les nôtres.

Sous la même hypothèse sur n que ci-dessus, l'un de nous (P. Deligne) a récemment prouvé que ces composantes irréductibles sont en fait lisses, et que M_n est un schéma régulier.

Pour étudier M_n, la première étape est de recouvrir par des ouverts ayant une interprétation modulaire l'ouvert M_n^h de M_n obtenu en ôtant les points supersinguliers de caractéristique divisant n. On y arrive en étudiant le schéma de modules qui classifie les courbes elliptiques munies d'un isomorphisme

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α : E_n ≃ μ_n × ℤ/n.

A modular interpretation makes it possible

a) to determine the set of points of M_n over an algebraically closed field;

b) to prove the smoothness of M_n^h over ℤ[ζ_n]. It remains, and this is the most delicate point, to study the supersingular points.

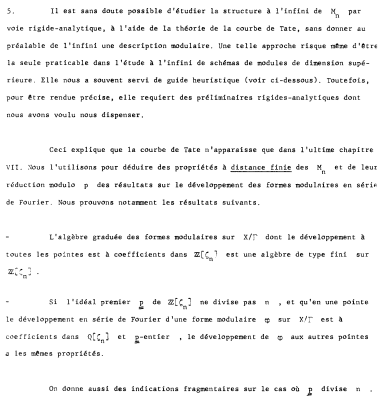
5. It is no doubt possible to study the structure of M_n at infinity by rigid- analytic methods, using the theory of the Tate curve, without first giving the points at infinity a modular description. Such an approach may even be the only practicable one for studying the boundary of higher-dimensional moduli schemes. It has often served us as a heuristic guide (see below). However, to make it precise, it requires rigid-analytic preliminaries that we wished to avoid.

This explains why the Tate curve appears only in the final chapter, Chapter VII. We use it to deduce properties at finite distance of the M_n, and of their reduction modulo p, from results on the Fourier-series expansions of modular forms. In particular, we prove the following results.

− The graded algebra of modular forms on X/Γ whose expansions at all cusps have coefficients in ℤ[ζ_n] is a finitely generated algebra over ℤ[ζ_n].

− If the prime ideal $\mathfrak{p}$ of ℤ[ζ_n] does not divide n, and if at one cusp the Fourier-series expansion of a modular form φ on X/Γ has coefficients in ℚ[ζ_n] and is $\mathfrak{p}$-integral, then the expansion of φ at the other cusps has the same properties.

Fragmentary indications are also given for the case where $\mathfrak{p}$ divides n.



5. Il est sans doute possible d'étudier la structure à l'infini de M_n par voie rigide-analytique, à l'aide de la théorie de la courbe de Tate, sans donner au préalable de l'infini une description modulaire. Une telle approche risque même d'être la seule praticable dans l'étude à l'infini de schémas de modules de dimension supérieure. Elle nous a souvent servi de guide heuristique (voir ci-dessous). Toutefois, pour être rendue précise, elle requiert des préliminaires rigides-analytiques dont nous avons voulu nous dispenser.

Ceci explique que la courbe de Tate n'apparaisse que dans l'ultime chapitre VII. Nous l'utilisons pour déduire des propriétés à distance finie des M_n et de leur réduction modulo p des résultats sur le développement des formes modulaires en série de Fourier. Nous prouvons notamment les résultats suivants.

- L'algèbre graduée des formes modulaires sur X/Γ dont le développement à toutes les pointes est à coefficients dans $\mathbb{Z}[\zeta_n]$ est une algèbre de type fini sur $\mathbb{Z}[\zeta_n]$.

- Si l'idéal premier $\underline{p}$ de $\mathbb{Z}[\zeta_n]$ ne divise pas n, et qu'en une pointe le développement en série de Fourier d'une forme modulaire φ sur X/Γ est à coefficients dans $\mathbb{Q}[\zeta_n]$ et $\underline{p}$-entier, le développement de φ aux autres pointes a les mêmes propriétés.

On donne aussi des indications fragmentaires sur le cas où $\underline{p}$ divise n.

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6. Our heuristic use of the Tate curve rests on the following principles.

a) With the aid of the Tate curve, one can guess or prove what happens in a neighborhood of infinity.

b) Outside the supersingular points, the situation is everywhere the same, and in particular the same as in a neighborhood of infinity.

To these principles is added the following one.

c) Geometrically, i.e. after extension of scalars (or of the residue field) to $\mathbb{F}_p$, the situation is the same at the various supersingular points of characteristic p.

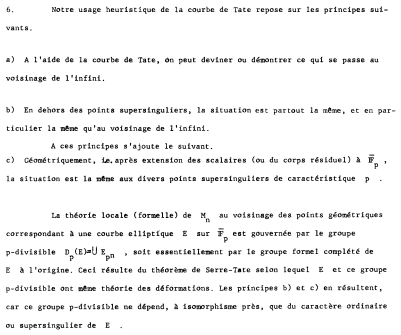
The local (formal) theory of M_n in a neighborhood of the geometric points corresponding to an elliptic curve E over $\mathbb{F}_p$ is governed by the p-divisible group

$$D_p(E) = \bigcup E_{p^n},$$

that is, essentially, by the completed formal group of E at the origin. This follows from the Serre–Tate theorem, according to which E and this p-divisible group have the same deformation theory. Principles b) and c) follow from it, since this p-divisible group depends, up to isomorphism, only on whether E is ordinary or supersingular.

7. Let H be an indefinite quaternion algebra, of discriminant D, over $\mathbb{Q}$, and let Δ be a congruence subgroup of the group of units of H. The group Δ is a discrete subgroup of $SL(2,\mathbb{R})$ with compact quotient. The complete curve X/Δ parametrizes abelian varieties A of dimension 2, equipped with a homomorphism from an order of H into End(A) ("false elliptic curves," in Serre's terminology). Our methods apply to the study of X/Δ and of its reduction modulo p, provided that p is prime to D.

The following remarks even make it possible to deduce, from theorems for M_n, theorems on the local structure of X/Δ.



6. Notre usage heuristique de la courbe de Tate repose sur les principes suivants.

a) A l'aide de la courbe de Tate, on peut deviner ou démontrer ce qui se passe au voisinage de l'infini.

b) En dehors des points supersinguliers, la situation est partout la même, et en particulier la même qu'au voisinage de l'infini.

A ces principes s'ajoute le suivant.

c) Géométriquement, i.e. après extension des scalaires (ou du corps résiduel) à $\bar{F}_p$, la situation est la même aux divers points supersinguliers de caractéristique p.

La théorie locale (formelle) de M_n au voisinage des points géométriques correspondant à une courbe elliptique E sur $\bar{F}_p$ est gouvernée par le groupe p-divisible $D_p(E)=\bigcup E_{p^n}$, soit essentiellement par le groupe formel complété de E à l'origine. Ceci résulte du théorème de Serre-Tate selon lequel E et ce groupe p-divisible ont même théorie des déformations. Les principes b) et c) en résultent, car ce groupe p-divisible ne dépend, à isomorphisme près, que du caractère ordinaire ou supersingulier de E.

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d) Let $\mathcal{O}$ be an order of H, with $\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Z}_p \simeq GL(2,\mathbb{Z}_p)$, and let e be a nontrivial idempotent of $\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Z}_p$. If A is an abelian scheme of dimension 2 over an algebraically closed field k of characteristic p, equipped with $\mu : \mathcal{O} \to \mathrm{End}(A)$, the completion $\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Z}_p$ acts on the p-divisible group $D_p(A) = \bigcup A_{p^n}$, and

$D_p A) = e \cdot D_p(A) \oplus (1-e) D_p(A)$.

The abelian scheme A, equipped with μ, and the p-divisible group $eD_p(A)$, have the same deformation theory.

e) the p-divisible group $eD_p(A)$ is isomorphic to $D_p(E)$ for an elliptic curve E that is either ordinary or supersingular.

The reduction modulo p of X/Δ was studied by Morita. He also obtains results in the much more difficult case where $p|D$.

8. In the literature on moduli schemes, the language of Weil's Foundations is sometimes used. According as one uses this language or Grothendieck's, emphasis is placed on one of the following two views of what a scheme X over a number field K is.

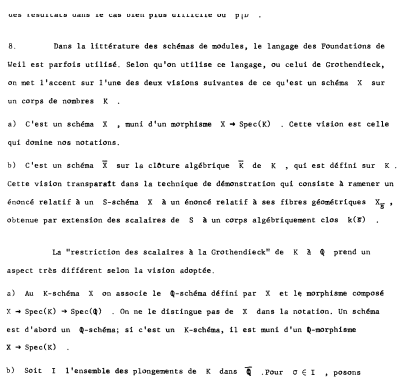
a) It is a scheme X, equipped with a morphism $X \to \mathrm{Spec}(K)$. This is the view that predominates in our notation.

b) It is a scheme $\bar{X}$ over the algebraic closure $\bar{K}$ of K, which is defined over K. This view shows through in the proof technique that consists in reducing a statement concerning an S-scheme X to a statement concerning its geometric fibers $X_{\bar{s}}$, obtained by extension of scalars from S to an algebraically closed field $k(\bar{s})$.

The "restriction of scalars à la Grothendieck" from K to $\mathbb{Q}$ takes on a very different appearance according to the view adopted.

a) To the K-scheme X one associates the $\mathbb{Q}$-scheme defined by X and the composite morphism $X \to \mathrm{Spec}(K) \to \mathrm{Spec}(\mathbb{Q})$. It is not distinguished from X in the notation. A scheme is first of all a $\mathbb{Q}$-scheme; if it is a K-scheme, it is equipped with a $\mathbb{Q}$-morphism $X \to \mathrm{Spec}(K)$.

b) Let I be the set of embeddings of K into $\bar{\mathbb{Q}}$. For $\sigma \in I$, set



des résultats dans le cas bien plus difficile où p|D .

8. Dans la littérature des schémas de modules, le langage des Foundations de Weil est parfois utilisé. Selon qu'on utilise ce langage, ou celui de Grothendieck, on met l'accent sur l'une des deux visions suivantes de ce qu'est un schéma X sur un corps de nombres K .

a) C'est un schéma X , muni d'un morphisme X → Spec(K) . Cette vision est celle qui domine nos notations.

b) C'est un schéma $\overline{X}$ sur la clôture algébrique $\overline{K}$ de K , qui est défini sur K . Cette vision transparaît dans la technique de démonstration qui consiste à ramener un énoncé relatif à un S-schéma X à un énoncé relatif à ses fibres géométriques $X_{\overline{s}}$, obtenue par extension des scalaires de S à un corps algébriquement clos k(s̄) .

La "restriction des scalaires à la Grothendieck" de K à ℚ prend un aspect très différent selon la vision adoptée.

a) Au K-schéma X on associe le ℚ-schéma défini par X et le morphisme composé X → Spec(K) → Spec(ℚ) . On ne le distingue pas de X dans la notation. Un schéma est d'abord un ℚ-schéma; si c'est un K-schéma, il est muni d'un ℚ-morphisme X → Spec(K) .

b) Soit I l'ensemble des plongements de K dans $\overline{\mathbb{Q}}$ .Pour σ ∈ I , posons

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$\bar{X}_\sigma = X \otimes_{K,\sigma} \bar{\mathbb{Q}}$. We have $X \otimes_{\mathbb{Q}} \bar{\mathbb{Q}} = \sum_{\sigma\in I} \bar{X}_\sigma$. The $\mathbb{Q}$-scheme deduced, as in a), from a K-scheme appears as the disjoint sum of the conjugates $\bar{X}_\sigma$ of $\bar{X}$, this sum being moreover defined over $\mathbb{Q}$. The $\mathbb{Q}$-morphism $X \to \mathrm{Spec}(K)$ is interpreted as the map from this sum to the index set $I = \mathrm{Spec}(K) \otimes_{\mathbb{Q}} \bar{\mathbb{Q}}$.

For $M_n^\circ[1/n]$, the situation is as follows: the scheme $M_n^\circ[1/n]$ represents the functor of isomorphism classes of elliptic curves E/T, equipped with an isomorphism

$\alpha : E_n \simeq (\mathbb{Z}/n)^2$.

Such a pair (E,α) defines (by the theory of the alternating form e_n) a primitive n-th root of unity $\zeta(\alpha)$ over T, hence a morphism of schemes

$(E,\alpha) \mapsto \zeta(\alpha) : M_n^\circ[1/n] \to \mathrm{Spec}(\mathbb{Z}[\zeta_n,1/n])$.

Viewed as a $\mathbb{Z}[\zeta_n,1/n]$-scheme, $M_n^\circ[1/n]$ classifies the $(E,\alpha)/T$, with T a $\mathbb{Z}[\zeta_n,1/n]$-scheme, such that $\zeta(\alpha) = \zeta_n$.

We have

$M_n^\circ[1/n] \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C} = X/\Gamma$;

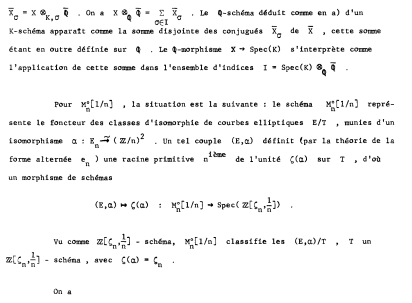
whereas, if one puts $X^\pm = \mathbb{C} - \mathbb{R}$,

$M_n^\circ[1/n] \otimes_{\mathbb{Z}} \mathbb{C} = X^\pm \times \mathrm{GL}(2,\mathbb{Z}/n) / \mathrm{GL}(2,\mathbb{Z})$

is the sum of $\varphi(n)$ copies of X/Γ.

9. An elliptic curve over an algebraically closed field k has one or more nontrivial automorphisms. This rules out representability of the functor that assigns to a scheme T the set of isomorphism classes of elliptic curves E over T: it is not even a sheaf. This can be remedied in two ways.

a) Systematically impose additional level structures that eliminate these automorphisms. This is what we have done in this introduction.



$\overline{X}_\sigma = X \otimes_{K,\sigma} \overline{\mathbb{Q}}$. On a $X \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} = \coprod_{\sigma \in \Gamma} \overline{X}_\sigma$. Le $\mathbb{Q}$-schéma déduit comme en a) d'un K-schéma apparaît comme la somme disjointe des conjugués $\overline{X}_\sigma$ de $\overline{X}$, cette somme étant en outre définie sur $\mathbb{Q}$. Le $\mathbb{Q}$-morphisme $X \to \mathrm{Spec}(K)$ s'interprète comme l'application de cette somme dans l'ensemble d'indices $\Gamma = \mathrm{Spec}(K) \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$.

Pour $M_n^*[1/n]$, la situation est la suivante : le schéma $M_n^*[1/n]$ représente le foncteur des classes d'isomorphie de courbes elliptiques E/T , munies d'un isomorphisme $\alpha : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$. Un tel couple (E,α) définit (par la théorie de la forme alternée e_n) une racine primitive $n^{\text{ième}}$ de l'unité $\zeta(\alpha)$ sur T , d'où un morphisme de schémas

$$(E,\alpha) \mapsto \zeta(\alpha) \quad : \quad M_n^*[1/n] \to \mathrm{Spec}\left(\mathbb{Z}\left[\zeta_n, \tfrac{1}{n}\right]\right) \quad .$$

Vu comme $\mathbb{Z}[\zeta_n, \frac{1}{n}]$ - schéma, $M_n^*[1/n]$ classifie les $(E,\alpha)/T$, T un $\mathbb{Z}[\zeta_n, \frac{1}{n}]$ - schéma , avec $\zeta(\alpha) = \zeta_n$.

On a

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b) Use the language of algebraic stacks, for which we refer to [8] and [17]. This is what we do in the text, for the following reasons.

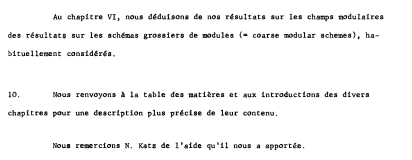
1. Imposing an additional level structure modifies the behavior at infinity of the objects studied (M_{ñm} ramifies over M_n along infinity).

2. The calculations and proofs become simpler when one does not have to concern oneself with troublesome additional structures.

In Chapter VI, from our results on modular stacks we deduce results on the coarse moduli schemes (= coarse modular schemes) usually considered.

10. We refer to the table of contents and to the introductions of the various chapters for a more precise description of their contents.

We thank N. Katz for the help he has given us.



Au chapitre VI, nous déduisons de nos résultats sur les champs modulaires des résultats sur les schémas grossiers de modules (= coarse modular schemes), habituellement considérés.

10. Nous renvoyons à la table des matières et aux introductions des divers chapitres pour une description plus précise de leur contenu.

Nous remercions N. Katz de l'aide qu'il nous a apportée.

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Notations

The work is divided into chapters; each chapter is divided into sections. Internal references within a single chapter do not mention the number of that chapter.

We shall freely use the definitions and notation of the theory of algebraic stacks as set out in [8] §4. However, contrary to the terminology of loc. cit., a morphism of categories fibered in groupoids

$\varphi : \mathfrak{M}_1 \to \mathfrak{M}_2$

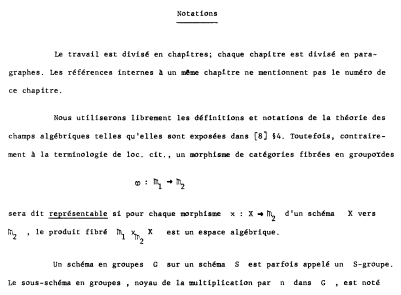
will be called representable if, for every morphism $x : X \to \mathfrak{M}_2$ from a scheme X to $\mathfrak{M}_2$, the fiber product $\mathfrak{M}_1 \times_{\{\mathfrak{M}_2\}} X$ is an algebraic space.

A group scheme G over a scheme S is sometimes called an S-group. The subgroup scheme that is the kernel of multiplication by n in G is denoted by G_n.

Let X be an S-scheme. If T is an S-scheme, we denote by X_T the fiber product X ×_S T. If S = Spec(A) and T = Spec(B), X_B, or X ⊗_A B, have the same meaning. For n ∈ ℕ, we set X[1/n] = X ⊗_ℤ ℤ[1/n].

We set ζ_n = exp(2πi/n).

For the principles governing our notation for modular stacks, we refer to §0 of Chapter III.



Notations

Le travail est divisé en chapitres; chaque chapitre est divisé en paragraphes. Les références internes à un même chapitre ne mentionnent pas le numéro de ce chapitre.

Nous utiliserons librement les définitions et notations de la théorie des champs algébriques telles qu'elles sont exposées dans [8] §4. Toutefois, contrairement à la terminologie de loc. cit., un morphisme de catégories fibrées en groupoïdes

$$\varpi : \mathcal{M}_1 \to \mathcal{M}_2$$

sera dit représentable si pour chaque morphisme $x : X \to \mathcal{M}_2$ d'un schéma X vers $\mathcal{M}_2$, le produit fibré $\mathcal{M}_1 \times_{\mathcal{M}_2} X$ est un espace algébrique.

Un schéma en groupes G sur un schéma S est parfois appelé un S-groupe. Le sous-schéma en groupes $\hat{n}$, noyau de la multiplication par n dans G, est noté

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I Preliminaries

This chapter gathers well-known results of which we shall have occasion to make use. The reader is invited to consult it only as needed.

1. Curve schemes.

1.0. In this section, a curve scheme over a scheme S means a proper and flat morphism of finite presentation, of relative dimension at most 1.

1.1. If C/S is a curve scheme over S, the Euler–Poincaré characteristic $\chi(\mathcal{O}_{C_s})$ of the fibers is locally constant (EGA III, 7.9.). The arithmetic genus g_s of C_s is defined by

$1 - g_s = \chi(C_s, \mathcal{O}_{C_s})$.

If the Euler–Poincaré characteristic of the fibers is constant with value $1 - g$, C|S is called a curve scheme of genus g, or simply a curve of genus g over S.

1.2. A morphism $p : X \to S$ is said to be Cohen–Macaulay if it is flat and of finite presentation and its (geometric) fibers are Cohen–Macaulay schemes.

A Cohen–Macaulay curve scheme over S is a curve scheme $p : C \to S$ that is Cohen–Macaulay and whose geometric fibers are purely of dimension one. Likewise, when we speak of smooth, reduced, or integral curves, ”purely of dimension one” will be understood.

For S = Spec(A), we shall speak interchangeably of curves over S and curves over A.

I Préliminaires

Ce chapitre rassemble des résultats bien connus dont nous aurons à faire usage. Le lecteur est invité à ne s'y reporter qu'en cas de besoin.

1. Schémas en courbes.

1.0. Dans ce paragraphe, on appelle schéma en courbes sur un schéma S un morphisme propre et plat de présentation finie de dimension relative au plus 1.

1.1. Si C/S est un schéma en courbes sur S, la caractéristique d'Euler-Poincaré $\chi(\mathcal{O}_{C_s})$ des fibres est localement constante (EGA III, 7.9.). On définit le genre arithmétique g de C_s par

$$1 - g_s = \chi(C_s, \mathcal{O}_{C_s}) .$$

Si la caractéristique d'Euler-Poincaré des fibres est constante de valeur $1 - g$, on appelle $C|S$ un schéma en courbes de genre g ou simplement une courbe de genre g sur S.

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2. Duality in the cohomology of coherent sheaves.

2.1. We propose to recall what Grothendieck duality theory provides for a morphism p : X → S when

a) p is Cohen–Macaulay, purely of relative dimension d; and

b) S is Noetherian.

In the applications we have in view, we could always reduce to the case in which p is moreover quasi-projective.

The relative dualizing complex Rp^!$\mathcal{O}$_S [10] has only one nonzero cohomology sheaf, that of degree −d; it is the sheaf of regular differentials

(2.1.1.) ω_{X/S} = $\mathcal{H}$^{−d}(Rp^!$\mathcal{O}$_S)

([10] V. 9.7.). This sheaf is flat over S. Its formation commutes with every base change S’ → S and with étale localization on X (see [10] or [34]).

2.2. If p is moreover proper, one defines a ”trace” morphism

(2.2.1) Tr : R^d p_*(ω_{X/S}) → $\mathcal{O}$_S

whose formation is compatible with every base change. The duality theorem asserts that, for every bounded complex of coherent sheaves K, the arrow deduced from Tr,

(2.2.2.) Rp_* R Hom_{$\mathcal{O}$_X}(K,ω_{X/S})[−d] ≃ R Hom_{$\mathcal{O}$_S}(Rp_*K,$\mathcal{O}$_S),

is an isomorphism.

If K is reduced to a locally free sheaf F placed in degree 0 and the sheaves R^i p_*(F) are locally free, (2.2.2.) becomes

(2.1.1.) $\omega_{X/S} = \underline{H}^{-d}(Rp^! \mathcal{O}_S)$

([10] V. 9.7.). Ce faisceau est plat sur S. Sa formation commute à tout changement de base $S' \to S$ et à la localisation étale sur X (voir [10] ou [34]).

2.2. Si p est de plus propre, on définit un morphisme "trace"

(2.2.1) $\mathrm{Tr} : R^d p_*(\omega_{X/S}) \to \mathcal{O}_S$

de formation compatible à tout changement de base. Le théorème de dualité affirme que, pour tout complexe borné de faisceaux cohérents K, la flèche déduite de Tr

(2.2.2.) $Rp_* R \underset{X}{\underline{\mathrm{Hom}}}_{\mathcal{O}}(K, \omega_{X/S})[-d] \to R \underset{\mathcal{O}_S}{\underline{\mathrm{Hom}}}(Rp_* K, \mathcal{O}_S)$

est un isomorphisme.

Si K est réduit à un faisceau localement libre F placé en degré 0 et que les faisceaux $R^i p_*(F)$ sont localement libres, (2.2.2.) devient

(2.2.3.) R^{d−j}p_*(F^∨ ⊗ ω_{X/S}) ≃ R^j p_*(F)^∨

(where ∨ denotes the dual of an $\mathcal{O}$_X-module, respectively of an $\mathcal{O}$_S-module).

ω_{X/S} is an invertible sheaf if and only if p is Gorenstein [10].

For S not necessarily Noetherian, one defines ω_{X/S} and Tr by passage to the limit; for K a bounded complex of locally free sheaves, (2.2.2.) remains valid and is proved by passage to the limit.

When S is the spectrum of a field k, ω_{X/S} will also be denoted ω_{X/k}, or simply ω_X.

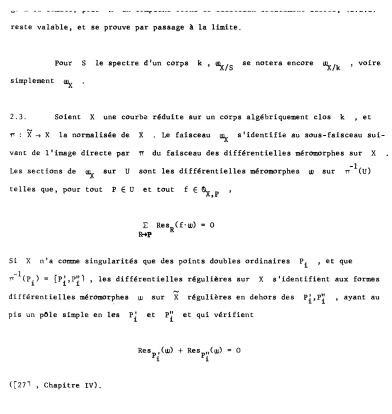
2.3. Let X be a reduced curve over an algebraically closed field k, and let π : $\tilde{X}$ → X be the normalization of X. The sheaf ω_X is identified with the following subsheaf of the direct image under π of the sheaf of meromorphic differentials on X. The sections of ω_X over U are the meromorphic differentials ω on π^{−1}(U) such that, for every P ∈ U and every f ∈ $\mathcal{O}$_{X,P},

∑_{R↦P} Res_R(f·ω) = 0.

If the only singularities of X are ordinary double points P_i, and π^{−1}(P_i) = {P'_i,P"_i}, the regular differentials on X are identified with the meromorphic differential forms ω on $\tilde{X}$ that are regular away from P'_i,P"_i, have at worst a simple pole at P'_i and P"_i, and satisfy

Res_{P'_i}(ω) + Res_{P"_i}(ω) = 0

([27], Chapter IV).


reste valable, et se prouve par passage à la limite.

Pour S le spectre d'un corps k, $\omega_{X/S}$ se notera encore $\omega_{X/k}$, voire simplement ω_X.

2.3. Soient X une courbe réduite sur un corps algébriquement clos k, et $\pi : \tilde{X} \to X$ la normalisée de X. Le faisceau ω_X s'identifie au sous-faisceau suivant de l'image directe par π du faisceau des différentielles méromorphes sur X. Les sections de ω_X sur U sont les différentielles méromorphes ω sur $\pi^{-1}(U)$ telles que, pour tout P ∈ U et tout $f \in \mathcal{O}_{X,P}$,

$$\sum_{R \mapsto P} \mathrm{Res}_R(f\cdot\omega) = 0$$

Si X n'a comme singularités que des points doubles ordinaires P_i, et que $\pi^{-1}(P_i) = \{P_i', P_i''\}$, les différentielles régulières sur X s'identifient aux formes différentielles méromorphes ω sur $\tilde{X}$ régulières en dehors des P_i', P_i'', ayant au pis un pôle simple en les P_i' et P_i'' et qui vérifient

$$\mathrm{Res}_{P_i'}(\omega) + \mathrm{Res}_{P_i''}(\omega) = 0$$

([27], Chapitre IV).

3. Review of the Picard functor.

3.1. Let p : X → S be a proper flat morphism of finite presentation. The relative Picard functor of X over S, denoted Pic_{X/S}, is the fppf sheaf associated with the presheaf

(3.1.1) S' ↦ Pic(X ×_S S') = the group of isomorphism classes of invertible sheaves on X ×_S S'.

The canonical morphism Pic(X)/Pic(S) → Pic_{X/S}(S) is injective if p_*$\mathcal{O}$_X = $\mathcal{O}$_S. It is bijective if, in addition, p has a section (TDTE V.2.4.).

3.2. One says that p is cohomologically flat in dimension 0 (EGA III, 7.), or simply cohomologically flat, if the formation of p_*$\mathcal{O}$_X commutes with every base change S' → S. A fundamental theorem of M. Artin asserts that, if p is proper, flat of finite presentation, and cohomologically flat, then Pic_{X/S} is representable by an algebraic space locally of finite presentation over S ([2] Thm. 7.3). If, in addition, p : X → S is a curve scheme, Pic_{X/S} is smooth over S.

If Pic_{X/S} is representable by an algebraic space, Pic^o_{X/S} denotes the subfunctor in groups of Pic_{X/S} that is the neutral component of Pic_{X/S}. If p : X → S is a curve scheme, it is representable by an algebraic space.

3.3. Let C be a proper curve over a field k. If $\mathcal{L}$ is an invertible sheaf on C, its degree deg_C($\mathcal{L}$) is defined by the Riemann–Roch formula

(3.3.1) χ($\mathcal{L}$) = deg_C($\mathcal{L}$) + χ($\mathcal{O}$_C).

Then (cf. [15])

(3.3.2) deg_C($\mathcal{L}$_1 ⊗ $\mathcal{L}$_2) = deg_C($\mathcal{L}$_1) + deg_C($\mathcal{L}$_1).

ment de base $S' \to S$. Un théorème fondamental de M. Artin affirme que, si p est propre, plat de présentation finie et cohomologiquement plat, alors $\mathrm{Pic}_{X/S}$ est représentable par un espace algébrique localement de présentation finie sur S ([2] Thm. 7.3). Si de plus, $p : X \to S$ est un schéma en courbes, $\mathrm{Pic}_{X/S}$ est lisse sur S.

Si $\mathrm{Pic}_{X/S}$ est représentable par un espace algébrique, on note $\mathrm{Pic}^{0}_{X/S}$ le sous-foncteur en groupes de $\mathrm{Pic}_{X/S}$ composante neutre de $\mathrm{Pic}_{X/S}$. Si $p : X \to S$ est un schéma en courbes, il est représentable par un espace algébrique.

3.3. Soit C un courbe propre sur un corps k. Si $\mathcal{L}$ est un faisceau inversible sur C, on définit son degré $\deg_C(\mathcal{L})$ par la formule (de Riemann-Roch) :

(3.3.1) $\chi(\mathcal{L}) = \deg_C(\mathcal{L}) + \chi(\mathcal{O}_C)$

On a alors (cf. [15])

(3.3.2) $\deg_C(\mathcal{L}_1 \otimes \mathcal{L}_2) = \deg_C(\mathcal{L}_1) + \mathrm{rk}_{\mathcal{O}_C}(\mathcal{L}_1)$

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(3.3.3) If the reduced irreducible components of C are the subschemes D_i, and if D_i has multiplicity n_i (= the length of the local ring of C at the generic point of D_i), then

deg_C($\mathcal{L}$) = ∑ n_i deg_{D_i}($\mathcal{L}$), where one has set

(3.3.4) deg_{D_i}($\mathcal{L}$) = deg_{D_i}($\mathcal{L}$ ⊗ $\mathcal{O}$_{D_i}).

3.4. Let p : C → S be a curve over S and let $\mathcal{L}$ be an invertible sheaf on C. The degree of $\mathcal{L}$ on the fibers C_s (s ∈ S) of C is locally constant and defines a morphism of abelian sheaves (for the fppf topology):

(3.4.1) deg : Pic_{C/S} → $\mathbb{Z}$.

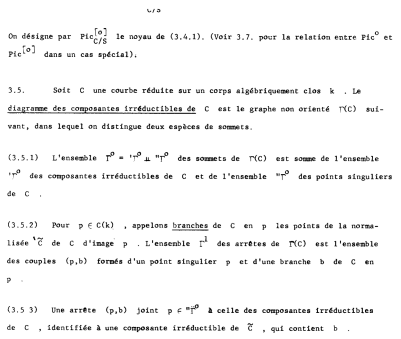
The kernel of (3.4.1) is denoted Pic^{[o]}_{C/S}. (See 3.7 for the relation between Pic^o and Pic^{[o]} in a special case.)

3.5. Let C be a reduced curve over an algebraically closed field k. The diagram of the irreducible components of C is the following unoriented graph Γ(C), in which two kinds of vertices are distinguished.

(3.5.1) The set Γ^o = 'Γ^o ⊔ "Γ^o of vertices of Γ(C) is the sum of the set 'Γ^o of irreducible components of C and the set "Γ^o of singular points of C.

(3.5.2) For p ∈ C(k), call the points of the normalization C̃ of C mapping to p the branches of C at p. The set Γ^1 of edges of Γ(C) is the set of pairs (p,b) formed by a singular point p and a branch b of C at p.

(3.5 3) An edge (p,b) joins p ∈ ”Γ^o to the irreducible component of C, identified with an irreducible component of C̃, that contains b.



On désigne par $\mathrm{Pic}^{[o]}_{C/S}$ le noyau de (3.4.1). (Voir 3.7. pour la relation entre Pic° et $\mathrm{Pic}^{[o]}$ dans un cas spécial).

3.5. Soit C une courbe réduite sur un corps algébriquement clos k. Le diagramme des composantes irréductibles de C est le graphe non orienté Γ(C) suivant, dans lequel on distingue deux espèces de sommets.

(3.5.1) L'ensemble $\Gamma^0 = {}'\Gamma^0 \amalg {}''\Gamma^0$ des sommets de Γ(C) est somme de l'ensemble ${}'\Gamma^0$ des composantes irréductibles de C et de l'ensemble ${}''\Gamma^0$ des points singuliers de C.

(3.5.2) Pour $p \in C(k)$, appelons branches de C en p les points de la normalisée ${}^{\nu}C$ de C d'image p. L'ensemble Γ^1 des arrêtes de Γ(C) est l'ensemble des couples (p,b) formés d'un point singulier p et d'une branche b de C en p.

(3.5 3) Une arrête (p,b) joint $p \in {}''\Gamma^0$ à celle des composantes irréductibles de C, identifiée à une composante irréductible de ${}^{\nu}C$, qui contient b.

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Authority page 23; collected-paper page 516; original folio 165.

Examples 3.5.4. Denote the points of ’Γ^o by ○ and those of ”Γ^o by ⊗. The following curves have the following diagrams:

curve diagram

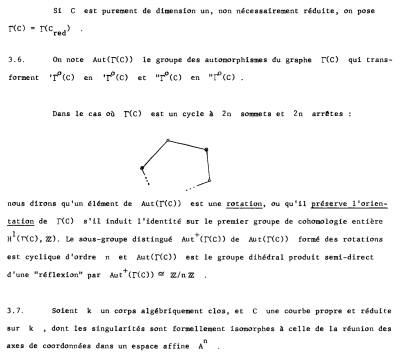
If C is purely of dimension one, but not necessarily reduced, set Γ(C) = Γ(C_red).

3.6. Let Aut(Γ(C)) denote the group of automorphisms of the graph Γ(C) that carry ’Γ^o(C) into ’Γ^o(C) and ”Γ^o(C) into ”Γ^o(C).

In the case where Γ(C) is a cycle with 2n vertices and 2n edges:

we shall say that an element of Aut(Γ(C)) is a rotation, or that it preserves the orientation of Γ(C), if it induces the identity on the first integral cohomology group H^1(Γ(C),ℤ). The distinguished subgroup Aut^+(Γ(C)) of Aut(Γ(C)) formed by the rotations is cyclic of order n, and Aut(Γ(C)) is the dihedral group that is the semidirect product of a ”reflection” by Aut^+(Γ(C)) ≃ ℤ/nℤ.

3.7. Let k be an algebraically closed field, and let C be a proper reduced curve over k whose singularities are formally isomorphic to that of the union of the coordinate axes in an affine space 𝔸^n.



Si C est purement de dimension un, non nécessairement réduite, on pose $\Gamma(C) = \Gamma(C_{red})$.

3.6. On note Aut(Γ(C)) le groupe des automorphismes du graphe Γ(C) qui transforment ${}'\Gamma^0(C)$ en ${}'\Gamma^0(C)$ et ${}''\Gamma^0(C)$ en ${}''\Gamma^0(C)$.

Dans le cas où Γ(C) est un cycle à 2n sommets et 2n arrêtes :

nous dirons qu'un élément de Aut(Γ(C)) est une rotation, ou qu'il préserve l'orientation de Γ(C) s'il induit l'identité sur le premier groupe de cohomologie entière $H^1(\Gamma(C), \mathbb{Z})$. Le sous-groupe distingué $\mathrm{Aut}^+(\Gamma(C))$ de Aut(Γ(C)) formé des rotations est cyclique d'ordre n et Aut(Γ(C)) est le groupe dihédral produit semi-direct d'une "réflexion" par $\mathrm{Aut}^+(\Gamma(C)) \cong \mathbb{Z}/n\mathbb{Z}$.

3.7. Soient k un corps algébriquement clos, et C une courbe propre et réduite sur k, dont les singularités sont formellement isomorphes à celle de la réunion des axes de coordonnées dans un espace affine A^n.

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The Picard scheme Pic_{C/k} decomposes as follows. Let (D_i)_{i∈I} be the irreducible components of C, let $\tilde{D}$_i be the normalization of D_i, and let $\tilde{C} = \amalg \tilde{D}$_i be the normalization of C.

a) The ”total degree” morphism

deg_I = (deg_{D_i})_{i∈I} : Pic_{C/k} → $\mathbb{Z}$^I

is surjective, with kernel Pic^o_{C/k}.

b) The morphism

Pic^o_{C/k} → Pic^o_{$\tilde{C}$/k} = ∏ Pic^o_{$\tilde{D}$_i/k}

is surjective, with image the largest quotient of Pic^o_{C/k} that is an abelian scheme. Its kernel is a torus, canonically isomorphic to

H^1(Γ(C),$\mathbb{Z}$) ⊗ $\mathbb{G}$_m.

If D is proper and purely of dimension one over k, there exists a unique curve C as above, equipped with a radicial morphism C → D; the map

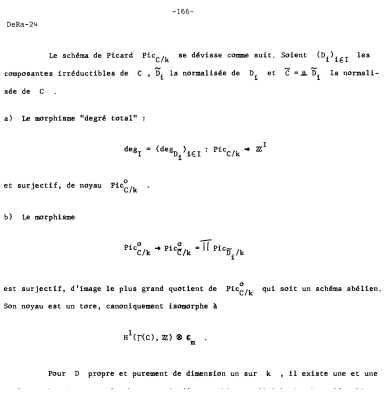
Pic_{D/k} → Pic_{C/k}

is surjective with connected unipotent kernel.

4. Non-smoothness subscheme.

Let f : X → S be a flat morphism of finite presentation, purely of relative dimension d.

It follows from the Jacobian criterion for smoothness that the subscheme X^sing of X defined by the Jacobian ideal, i.e. by the d-th determinantal ideal of Ω^1_{X/S}



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Le schéma de Picard $\mathrm{Pic}_{C/k}$ se dévisse comme suit. Soient $(D_i)_{i\in I}$ les composantes irréductibles de C, $\tilde{D}_i$ la normalisée de D_i et $\tilde{C} = \coprod \tilde{D}_i$ la normalisée de C.

a) Le morphisme "degré total" :

$$\deg_I = (\deg_{D_i})_{i\in I} : \mathrm{Pic}_{C/k} \to \mathbb{Z}^I$$

et surjectif, de noyau $\mathrm{Pic}^0_{C/k}$.

b) Le morphisme

$$\mathrm{Pic}^0_{C/k} \to \mathrm{Pic}^0_{\tilde{C}/k} = \prod \mathrm{Pic}^0_{\tilde{D}_i/k}$$

est surjectif, d'image le plus grand quotient de $\mathrm{Pic}^0_{C/k}$ qui soit un schéma abélien. Son noyau est un tore, canoniquement isomorphe à

$$H^1(\Gamma(C), \mathbb{Z}) \otimes \mathbb{G}_m .$$

Pour D propre et purement de dimension un sur k, il existe une et une

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(Bourbaki; Commutative Algebra, Chapter 7, §4, Exercise 10) has as its underlying topological space the set of points of X at which f is not smooth. We shall call it the non-smoothness subscheme of X [SGA 7, VI]. The locus on which f is smooth is denoted X^reg

5. Review of deformation theory.

5.1. Let Λ be a complete Noetherian local ring with residue field k. Two particularly useful cases are the following.

a) Λ = k.

b) k is perfect of characteristic p > 0 and Λ is the ring of Witt vectors W(k) (the unique complete discrete valuation ring with residue field k and maximal ideal (p)).

Let C_Λ denote the category of local Artinian Λ-algebras with residue field k, and let Ĉ_Λ denote the category of complete Noetherian local Λ-algebras with residue field k. A covariant functor F from C_Λ to (Sets) is said to be pro-representable if there exist R ∈ Ob Ĉ_Λ and an isomorphism of functors on C_Λ

(5.1.1) $\hat{\xi}$: Hom(R,A) ⟶̃ F(A).

The algebra R is then uniquely determined.

A covariant functor F from Ĉ_Λ to (Sets) induces by restriction a functor F|C_Λ : C_Λ → (Sets). If F|C_Λ is prorepresentable, it is said to be effectively prorepresentable if there exists ξ ∈ F(R) (with R as above) that defines (5.1.1).

5.2. Consider systems (M,B,α) of the following type

On désigne par C_Λ la catégorie des Λ - algèbres locales artiniennes de corps résiduel k et par $\hat{C}_\Lambda$ la catégorie des Λ-algèbres noethériennes locales complètes de corps résiduel k . Un foncteur covariant F de C_Λ vers (Ens) est dit pro-représentable s'il existe $R \in \mathrm{Ob}\, \hat{C}_\Lambda$ et un isomorphisme de foncteurs sur C_Λ .

(5.1.1) $\hat{\xi} : \mathrm{Hom}(R,A) \xrightarrow{\sim} F(A)$.

L'algèbre R est alors uniquement déterminée.

Un foncteur covariant F de $\hat{C}_\Lambda$ dans (Ens) induit par restriction un foncteur $F|C_\Lambda : C_\Lambda \to (\mathrm{Ens})$. Si $F|C_\Lambda$ est prorepréssentable, on dit qu'il est effectivement prorepréssentable s'il existe $\xi \in F(R)$ (R comme ci-dessus) qui définisse (5.1.1) .

5.2. Considérons les systèmes (M,B,α) du type suivant

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a) M is a complete Noetherian local Λ-algebra with residue field k;

b) B is a complete flat Noetherian local M-algebra, with residue field k, equipped with an isomorphism

α : B ⊗_M k ≃ k[[X,Y]]/(XY).

Such a system (Λ[[t]], A, α_{o/0}) is provided by the Λ[[t]]-algebra

A = Λ[[t,X,Y]]/(XY−t).

It is proved in [8] §1 that this last system is versal. For every (B,M,α), there exists a homomorphism Λ[[t]] → M such that (B,α) is deduced from (A,α_{o/0}) by extension of scalars:

B = A ⊗_{Λ[[t]]} M.

One easily deduces the following statement:

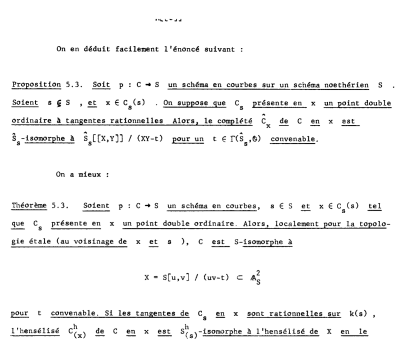
Proposition 5.3. Let p : C → S be a scheme of curves over a Noetherian scheme S. Let s ∈ S and x ∈ C_s(s). Suppose that C_s has at x an ordinary double point with rational tangent directions. Then the completion Ĉ_x of C at x is Ŝ_s-isomorphic to Ŝ_s[[X,Y]]/(XY−t) for a suitable t ∈ Γ(Ŝ_s,$\mathcal{O}$).

Better still:

Theorem 5.3. Let p : C → S be a scheme of curves, let s ∈ S, and let x ∈ C_s(s) be such that C_s has an ordinary double point at x. Then, locally for the étale topology (near x and s), C is S-isomorphic to

X = S[u,v]/(uv−t) ⊂ $\mathbb{A}$^2_S

for a suitable t. If the tangent directions of C_s at x are rational over k(s), the henselization C^h_{(x)} of C at x is S^h_{(s)}-isomorphic to the henselization of X at the



On en déduit facilement l'énoncé suivant :

Proposition 5.3. Soit p : C → S un schéma en courbes sur un schéma noethérien S. Soient s ∈ S, et x ∈ C_s(s). On suppose que C_s présente en x un point double ordinaire à tangentes rationnelles. Alors, le complété Ĉ_x de C en x est Ŝ_s-isomorphe à Ŝ_s[[X,Y]] / (XY-t) pour un t ∈ Γ(Ŝ_s,Ô) convenable.

On a mieux :

Théorème 5.3. Soient p : C → S un schéma en courbes, s ∈ S et x ∈ C_s(s) tel que C_s présente en x un point double ordinaire. Alors, localement pour la topologie étale (au voisinage de x et s), C est S-isomorphe à

X = S[u,v] / (uv-t) ⊂ $\mathbb{A}^2_S$

pour t convenable. Si les tangentes de C_s en x sont rationnelles sur k(s), l'hensélisé C^h_{(x)} de C en x est S^h_{(s)}-isomorphe à l'hensélisé de X en le

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point (0,0) lying over s, for a suitable t ∈ Γ(S^h_{(s)},$\mathcal{O}$).

This theorem follows at once from the special case in which S is assumed to be of finite type over $\mathbb{Z}$. It follows from 5.2 and R. Elkik's theory of algebraization of modules of isolated singularities.

6. Division points of elliptic curves.

Let E be an elliptic curve (an abelian variety of dimension one) over an algebraically closed field k.

For every integer n ≥ 1, the group scheme E_n, the kernel of multiplication by n, is known to be the spectrum of a Hopf algebra of dimension n^2 over k. If n is invertible in k, then E_n ≃ ($\mathbb{Z}$/n)^2.

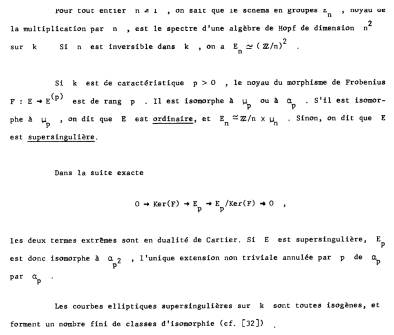
If k has characteristic p > 0, the kernel of the Frobenius morphism F : E → E^(p) has rank p. It is isomorphic to μ_p or to α_p. If it is isomorphic to μ_p, E is said to be ordinary, and E_n ≃ $\mathbb{Z}$/n × μ_n. Otherwise E is said to be supersingular.

In the exact sequence

0 → Ker(F) → E_p → E_p/Ker(F) → 0,

the two extreme terms are Cartier dual. If E is supersingular, E_p is therefore isomorphic to α_{p^2}, the unique nontrivial extension killed by p of α_p by α_p.

The supersingular elliptic curves over k are all isogenous and form a finite number of isomorphism classes (cf. [32]).



Pour tout entier $n \neq 1$, on sait que le schema en groupes μ_n, noyau de la multiplication par n, est le spectre d'une algèbre de Hopf de dimension n^2 sur k. Si n est inversible dans k, on a $E_n \simeq (\mathbb{Z}/n)^2$.

Si k est de caractéristique $p > 0$, le noyau du morphisme de Frobenius $F : E \to E^{(p)}$ est de rang p. Il est isomorphe à μ_p ou à α_p. S'il est isomorphe à μ_p, on dit que E est ordinaire, et $E_n \simeq \mathbb{Z}/n \times \mu_n$. Sinon, on dit que E est supersinguliere.

Dans la suite exacte

$$0 \to \mathrm{Ker}(F) \to E_p \to E_p/\mathrm{Ker}(F) \to 0 ,$$

les deux termes extrêmes sont en dualité de Cartier. Si E est supersinguliere, E_p est donc isomorphe à α_{p^2}, l'unique extension non triviale annulée par p de α_p par α_p.

Les courbes elliptiques supersinguliéres sur k sont toutes isogènes, et forment un nombre fini de classes d'isomorphie (cf. [32]).

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7. Review of commutative algebra.

We shall often use the following results from commutative algebra; for their proofs we refer to EGA.

(7.1) Let $f : X \to Y$ be a morphism of finite type between Noetherian schemes. If Y is Cohen–Macaulay and f is flat, then X is Cohen–Macaulay if and only if the fibers of f are. For example, if Y is Cohen–Macaulay and f is quasi-finite and flat, then X is Cohen–Macaulay.

If Y is regular, X is Cohen–Macaulay, and at the point $x \in X$ with image $y \in Y$ one has

(7.1 1) $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} k(y))$,

then f is flat at x. Condition (7.1.1) is satisfied if f is quasi-finite and dominant and X is irreducible.

EGA IV 6.3.5 and EGA IV 6.1.5.

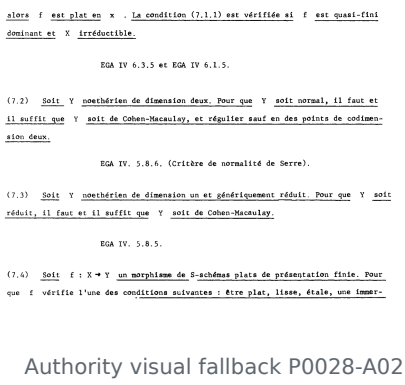
(7.2) Let Y be Noetherian of dimension two. For Y to be normal, it is necessary and sufficient that Y be Cohen–Macaulay and regular except at points of codimension two.

EGA IV. 5.8.6. (Serre's normality criterion).

(7.3) Let Y be Noetherian of dimension one and generically reduced. For Y to be reduced, it is necessary and sufficient that Y be Cohen–Macaulay.

EGA IV. 5.8.5.

(7.4) Let $f : X \to Y$ be a morphism of flat S-schemes of finite presentation. For f to satisfy one of the following conditions: to be flat, smooth, étale, an open immersion



alors f est plat en x. La condition (7.1.1) est vérifiée si f est quasi-fini dominant et X irréductible.

EGA IV 6.3.5 et EGA IV 6.1.5.

(7.2) Soit Y noethérien de dimension deux. Pour que Y soit normal, il faut et il suffit que Y soit de Cohen-Macaulay, et régulier sauf en des points de codimension deux.

EGA IV. 5.8.6. (Critère de normalité de Serre).

(7.3) Soit Y noethérien de dimension un et génériquement réduit. Pour que Y soit réduit, il faut et il suffit que Y soit de Cohen-Macaulay.

EGA IV. 5.8.5.

(7.4) Soit $f : X \to Y$ un morphisme de S-schémas plats de présentation finie. Pour que f vérifie l'une des conditions suivantes : être plat, lisse, étale, une imme[r]-

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an isomorphism, or a flat relative complete intersection, it is necessary and sufficient that for every geometric point $\bar{s}$ of S, $f_{\bar{s}} : X_{\bar{s}} \to Y_{\bar{s}}$ satisfy the stated condition.

For ”flat,” this is EGA IV 11.3.10. The properties ”smooth,” ”étale,” and ”flat relative complete intersection” are equivalent to ”flat” plus a property of the fibers. Finally, ”open immersion” is equivalent to ”étale and radicial” (EGA IV.17.9.1), where ”radicial” is a property of the fibers, and ”isomorphism” is equivalent to ”open immersion and surjective.”

8. Coarse moduli schemes

Let $\mathfrak{M}$ be an algebraic stack over a base scheme S.

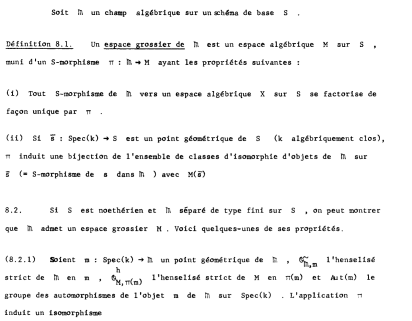
Definition 8.1. A coarse space of $\mathfrak{M}$ is an algebraic space M over S, equipped with an S-morphism $\pi : \mathfrak{M} \to M$ having the following properties:

(i) Every S-morphism from $\mathfrak{M}$ to an algebraic space X over S factors uniquely through π.

(ii) If $\bar{s} : \mathrm{Spec}(k) \to S$ is a geometric point of S (with k algebraically closed), π induces a bijection between the set of isomorphism classes of objects of $\mathfrak{M}$ over $\bar{s}$ (= an S-morphism from s to $\mathfrak{M}$) and $M(\bar{s})$.

8.2. If S is Noetherian and $\mathfrak{M}$ is separated and of finite type over S, one can show that $\mathfrak{M}$ admits a coarse space M. Here are some of its properties.

(8.2.1) Let $m : \mathrm{Spec}(k) \to \mathfrak{M}$ be a geometric point of $\mathfrak{M}$, let $\mathcal{O}_{\mathfrak{M},m}$ be the strict henselization of $\mathfrak{M}$ at m, let $\mathcal{O}^h_{M,\pi(m)}$ be the strict henselization of M at $\pi(m)$, and let Aut(m) be the group of automorphisms of the object m of $\mathfrak{M}$ over Spec(k). The map π induces an isomorphism


Soit $\mathfrak{M}$ un champ algébrique sur un schéma de base S.

Définition 8.1. Un espace grossier de $\mathfrak{M}$ est un espace algébrique M sur S, muni d'un S-morphisme $\pi : \mathfrak{M} \to M$ ayant les propriétés suivantes :

(i) Tout S-morphisme de $\mathfrak{M}$ vers un espace algébrique X sur S se factorise de façon unique par π.

(ii) Si $\bar{s} : \mathrm{Spec}(k) \to S$ est un point géométrique de S (k algébriquement clos), π induit une bijection de l'ensemble de classes d'isomorphie d'objets de $\mathfrak{M}$ sur $\bar{s}$ (= S-morphisme de s dans $\mathfrak{M}$) avec $M(\bar{s})$

8.2. Si S est noethérien et $\mathfrak{M}$ séparé de type fini sur S, on peut montrer que $\mathfrak{M}$ admet un espace grossier M. Voici quelques-unes de ses propriétés.

(8.2.1) Soient $m : \mathrm{Spec}(k) \to \mathfrak{M}$ un point géométrique de $\mathfrak{M}$, $\mathcal{O}^h_{\mathfrak{M},m}$ l'hensélisé strict de $\mathfrak{M}$ en m, $\mathcal{O}^h_{M,\pi(m)}$ l'hensélisé strict de M en $\pi(m)$ et Aut(m) le groupe des automorphismes de l'objet m de $\mathfrak{M}$ sur Spec(k). L'application π induit un isomorphisme

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$$\mathrm{Spec}(\mathcal{O}^h_{\mathfrak{M},m})/\mathrm{Aut}(m) \simeq \mathrm{Spec}(\mathcal{O}^h_{M,\pi(m)}).$$

Let $H \subset \mathrm{Aut}(m)$ be the subgroup of automorphisms of the object m of $\mathfrak{M}$ over k that extend to an automorphism of the object $\mathrm{Spec}(\mathcal{O}^h_{\mathfrak{M},m}) \to \mathfrak{M}$ of $\mathfrak{M}$ over $\mathrm{Spec}(\mathcal{O}^h_{\mathfrak{M},m})$. The group Aut(m)/H acts effectively on $\mathcal{O}^h_{\mathfrak{M},m}$.

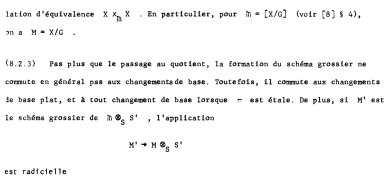
(8.2.2) If $u : X \to \mathfrak{M}$ is étale and surjective, M is the quotient of X by the equivalence relation $X \times_{\mathfrak{M}} X$. In particular, for $\mathfrak{M} = [X/G]$ (see [8] §4), one has $M = X/G$.

(8.2.3) No more than passage to the quotient does formation of the coarse scheme

commute in general with base changes. It does, however, commute with flat base changes, and with every base change when π is étale. Moreover, if M’ is the coarse scheme of $\mathfrak{M} \otimes_S S'$, the map

$M' \to M \otimes_S S'$

is radicial


lation d'équivalence $X \times_h X$. En particulier, pour $h = [X/G]$ (voir [8] § 4), on a $M = X/G$.

(8.2.3) Pas plus que le passage au quotient, la formation du schéma grossier ne commute en général pas aux changements de base. Toutefois, il commute aux changements de base plat, et à tout changement de base lorsque π est étale. De plus, si M' est le schéma grossier de $\mathfrak{M} \otimes_S S'$, l'application

$$M' \to M \otimes_S S'$$

est radicielle

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Authority page 31; collected-paper page 524; original folio 173.

II Generalized elliptic curves.

In this chapter, we define generalized elliptic curves over an arbitrary base scheme and prove the theorems that make the adopted definition workable.

1. Néron polygons.

1.1. Let $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$ be the disjoint union of n copies of $\mathbb{P}^1$, indexed by $\mathbb{Z}/n$ ($n \geq 1$). By gluing the i-th copy of $\mathbb{P}^1$ to the (i+1)-st, identifying the section 0 of the i-th copy with the section ∞ of the (i+1)-st, one obtains a genus-one curve C over Spec($\mathbb{Z}$), with normalization $\tilde{C}$.

For every scheme S, the scheme over S obtained from this one by extension of scalars is called the standard Néron polygon with n sides over S, or simply the standard n-gon over S. A Néron polygon (respectively, an n-gon) over an algebraically closed field k is a scheme over k isomorphic to one of the standard polygons (respectively, to the standard n-gon).

Examples.

n = 1

n = 3

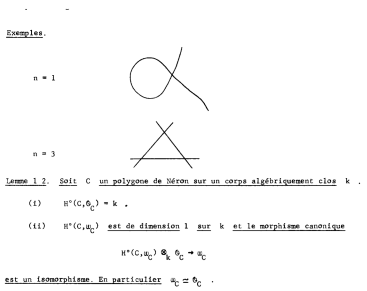
Lemma 1 2. Let C be a Néron polygon over an algebraically closed field k.

(i) $H^0(C,\mathcal{O}_C) = k$.

(ii) $H^0(C,\omega_C)$ has dimension 1 over k, and the canonical morphism

$H^0(C,\omega_C) \otimes_k \mathcal{O}_C \to \omega_C$

is an isomorphism. In particular, $\omega_C \simeq \mathcal{O}_C$.


Exemples.

n = 1

n = 3

Lemme 1.2. Soit C un polygone de Néron sur un corps algébriquement clos k .

(i) $H^0(C,\mathcal{O}_C) = k$.

(ii) $H^0(C,\omega_C)$ est de dimension 1 sur k et le morphisme canonique

$$H^0(C,\omega_C) \otimes_k \mathcal{O}_C \to \omega_C$$

est un isomorphisme. En particulier $\omega_C \simeq \mathcal{O}_C$.

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Proof: (i) follows from the fact that C is reduced and connected.

(ii) We may suppose that C is the standard n-gon.

Let η be the differential form on the normalization $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$ whose restriction to the i-th copy of $\mathbb{P}^1$ is dx/x. It is clear that η has at most simple poles, with residues +1 and −1 at the "0" and "∞," respectively, of the i-th copy of $\mathbb{P}^1$. The form η therefore defines a global section of ω_{C/S} (cf. I. 2.3.). It generates ω_C at every point of C.

Lemma 1.3. Let C be a reduced, connected genus-one curve over an algebraically closed field, whose only singularities are ordinary double points, and such that $\omega_C \simeq \mathcal{O}$. Then C is smooth or a Néron polygon.

Proof: Let $\pi : \tilde{C} \to C$ be the normalization of C, with irreducible components $(C_i)_{1\le i\le a}$; let $g_i \ge 0$ be the genus of C_i, and let b be the number of double points of C. The long exact cohomology sequence deduced from

$$0 \to \mathcal{O}_C \to \pi_*\mathcal{O}_{\tilde{C}} \to \text{cokernel} \to 0$$

and 1.2(i) gives

$a = \sum g_i + b$ and

$b \ge a-1 \ge 0$.

Let $b_i \ge 0$ be the number of points of C_i whose image is a singular point of C. One has

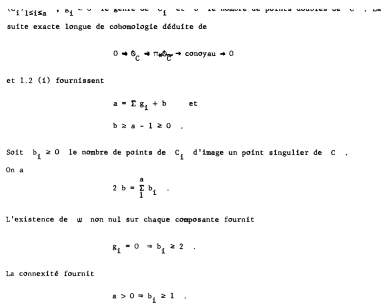
$2b = \sum_{i=1}^{a} b_i$.

The existence of a nonzero ω on each component gives

$g_i = 0 \Rightarrow b_i \ge 2$.

Connectedness gives

$a > 0 \Rightarrow b_i \ge 1$.



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Authority page 33; collected-paper page 526; original folio 175.

These inequalities have the following solutions:

a) a = 1, b = 0, g = 1 (the nonsingular case)

b) a = n, b = n, g_i = 0, b_i = 2 (n ≥ 1): the C_i are ≃ $\mathbb{P}^1$,

and on each C_i two points map to a double point of C. Since C is connected, it is a Néron polygon.

Definition 1.4. A stable genus-one curve C over a scheme S (or an algebraic space S, or an algebraic stack S) is a scheme of curves over S (cf. I.1.0.) every geometric fiber of which is either a proper smooth connected genus-one curve or a Néron polygon.

Proposition 1.5. : Let p : C → S be a proper flat morphism of finite presentation. The set of points s ∈ S such that C_s is a stable genus-one curve is an open subscheme of S.

Proof: The conditions that C_s be reduced, reduced and connected, reduced and purely one-dimensional, of genus one, and have only ordinary quadratic singularities are all open. That the condition ω ≃ $\mathcal{O}$ of 1.3 is open follows from the proof of 1.6 below.

Proposition 1.6. Let p : C → S be a stable genus-one curve.

(i) p_*$\mathcal{O}$_C = $\mathcal{O}$_S universally.

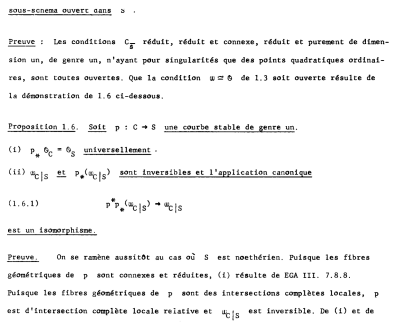
(ii) ω_{C/S} and p_*(ω_{C/S}) are invertible, and the canonical map

(1.6.1) p^*p_*(ω_{C/S}) → ω_{C/S}

is an isomorphism.

Proof. We immediately reduce to the case where S is Noetherian. Since the geometric fibers of p are connected and reduced, (i) follows from EGA III. 7.8.8.

Since the geometric fibers of p are local complete intersections, p is a relative local complete intersection and ω_{C/S} is invertible. From (i) and from



sous-schema ouvert dans S .

Preuve : Les conditions $C_{\bar{s}}$ réduit, réduit et connexe, réduit et purement de dimension un, de genre un, n'ayant pour singularités que des points quadratiques ordinaires, sont toutes ouvertes. Que la condition $\omega \simeq \mathcal{O}$ de 1.3 soit ouverte résulte de la démonstration de 1.6 ci-dessous.

Proposition 1.6. Soit $p : C \to S$ une courbe stable de genre un.

(i) $p_* \mathcal{O}_C = \mathcal{O}_S$ universellement .

(ii) $\omega_{C|S}$ et $p_*(\omega_{C|S})$ sont inversibles et l'application canonique

(1.6.1) $$p^* p_*(\omega_{C|S}) \to \omega_{C|S}$$

est un isomorphisme.

Preuve. On se ramène aussitôt au cas où S est noethérien. Puisque les fibres géométriques de p sont connexes et réduites, (i) résulte de EGA III. 7.8.8. Puisque les fibres géométriques de p sont des intersections complètes locales, p est d'intersection complète locale relative et $\omega_{C|S}$ est inversible. De (i) et de

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the fact that $R^2p_*\mathcal{O} = 0$, it follows that $R^1p_*\mathcal{O}$ is locally free and its formation is compatible with every base change S' → S (EGA III. 7.8.). Coherent-sheaf duality (I.2.) then shows that p_*(ω_{C/S}) is locally free and its formation is compatible with every base change. Since (ω_{C/S})_s = ω_{C_s/s}, it remains to prove (ii) when S is the spectrum of an algebraically closed field. In that case, 1.6.1 is well known when C is smooth, and follows from 1.2 for a Néron polygon.

The following Lemma 1.7 is a special case of I 3.7.

Lemma 1.7. Let C be a Néron polygon over an algebraically closed field k.

(i) Pic^o(C) = $\mathbb{G}_m \otimes H^1(\Gamma(C),\mathbb{Z})$ is isomorphic to $\mathbb{G}_m$.

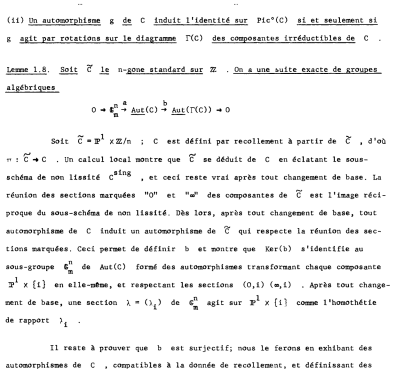
(ii) An automorphism g of C induces the identity on Pic^o(C) if and only if g acts by rotations on the component diagram Γ(C) of C.

Lemma 1.8. Let $\tilde{C}$ be the standard n-gon over $\mathbb{Z}$. There is an exact sequence of algebraic groups

$$0 \to \mathbb{G}_m^n \longrightarrow^a \mathrm{Aut}(C) \longrightarrow^b \mathrm{Aut}(\Gamma(C)) \to 0.$$

Let $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$; C is defined by gluing from $\tilde{C}$, whence $\pi : \tilde{C} \to C$. A local computation shows that $\tilde{C}$ is obtained from C by blowing up the non-smoothness subscheme C^sing, and this remains true after every base change. The union of the marked sections "0" and "∞" of the components of $\tilde{C}$ is the inverse image of the non-smoothness subscheme. Hence, after every base change, every automorphism of C induces an automorphism of $\tilde{C}$ that preserves the union of the marked sections. This defines b and shows that Ker(b) is identified with the subgroup $\mathbb{G}_m^n$ of Aut(C) consisting of the automorphisms that take every component $\mathbb{P}^1 \times \{i\}$ to itself and preserve the sections (0,i) and (∞,i). After every base change, a section λ = (λ_i) of $\mathbb{G}_m^n$ acts on $\mathbb{P}^1 \times \{i\}$ as scaling by the factor λ_i.

It remains to prove that b is surjective; we shall do so by exhibiting automorphisms of C compatible with the gluing data and defining



(ii) Un automorphisme g de C induit l'identité sur $\mathrm{Pic}^o(C)$ si et seulement si g agit par rotations sur le diagramme $\Gamma(C)$ des composantes irréductibles de C .

Lemme 1.8. Soit $\tilde{C}$ le n-gone standard sur $\mathbb{Z}$. On a une suite exacte de groupes algébriques

$$0 \to \mathfrak{G}_m^n \xrightarrow{a} \mathrm{Aut}(C) \xrightarrow{b} \mathrm{Aut}(\Gamma(C)) \to 0$$

Soit $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$; C est défini par recollement à partir de $\tilde{C}$, d'où $\pi : \tilde{C} \to C$. Un calcul local montre que $\tilde{C}$ se déduit de C en éclatant le sous-schéma de non lissité C^{sing} , et ceci reste vrai après tout changement de base. La réunion des sections marquées "0" et "∞" des composantes de $\tilde{C}$ est l'image réciproque du sous-schéma de non lissité. Dès lors, après tout changement de base, tout automorphisme de C induit un automorphisme de $\tilde{C}$ qui respecte la réunion des sections marquées. Ceci permet de définir b et montre que Ker(b) s'identifie au sous-groupe $\mathfrak{G}_m^n$ de Aut(C) formé des automorphismes transformant chaque composante $\mathbb{P}^1 \times \{i\}$ en elle-même, et respectant les sections $(0,i)$ (∞,i) . Après tout changement de base, une section $\lambda = (\lambda_i)$ de $\mathfrak{G}_m^n$ agit sur $\mathbb{P}^1 \times \{i\}$ comme l'homothétie de rapport λ_i .

Il reste à prouver que b est surjectif; nous le ferons en exhibant des automorphismes de C , compatibles à la donnée de recollement, et définissant des

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automorphisms of C whose images under b generate the dihedral group Aut(Γ(C)). They are

τ : (x,i) ↦ (x^{−1},−i)

σ_j : (x,i) ↦ (x,i+j) (j ∈ ℤ/n).

1.9. Let C be the standard n-gon over Spec(ℤ). Identify the normalization $\tilde{C}$ of C with $\mathbb{P}^1 \times \mathbb{Z}/n\mathbb{Z}$ and the inverse image of C^reg in $\tilde{C}$ with $\mathbb{G}_m \times \mathbb{Z}/n\mathbb{Z}$. An action of $\mathbb{G}_m \times \mathbb{Z}/n\mathbb{Z}$ on $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n\mathbb{Z}$ is obtained by setting

(a,i) + (b,j) = (a·b,i+j),

where a·b denotes the natural action of $\mathbb{G}_m$ on $\mathbb{P}^1$.

This action descends to the quotient and defines a morphism

(1.9.1) + : C^reg × C → C.

The restriction of + to C^reg × C^reg defines a commutative group-scheme structure on C^reg, and + is an action of the group C^reg on C.

Automorphisms of (C,+) are defined as follows:

a) the automorphism of $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$

τ : (x,i) ↦ (x^{−1},−i)

descends to the quotient and defines an automorphism τ of (C,+);

b) for every scheme S and every ζ ∈ μ_n(S), the automorphism u(ζ) of C_S

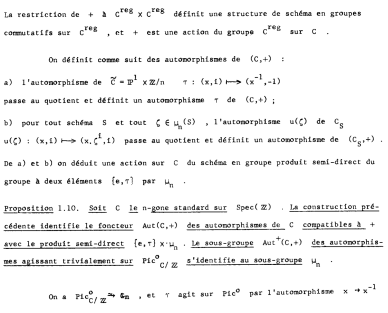
u(ζ) : (x,i) ↦ (x·ζ^i,i)

descends to the quotient and defines an automorphism of (C_S,+).

Parts a) and b) give an action on C by the semidirect-product group scheme of the two-element group {e,τ} by μ_n.

Proposition 1.10. Let C be the standard n-gon over Spec(ℤ). The preceding construction identifies the functor Aut(C,+) of automorphisms of C compatible with + with the semidirect product {e,τ} ×’ μ_n. The subgroup $\mathrm{Aut}^+(C,+)$ of automorphisms acting trivially on Pic^o_{C/ℤ} is identified with the subgroup μ_n.

One has Pic^o_{C/ℤ} ≃ $\mathbb{G}_m$, and τ acts on Pic^o by the automorphism x ↦ x^{−1}



La restriction de + à $C^{reg} \times C^{reg}$ définit une structure de schéma en groupes commutatifs sur C^{reg}, et + est une action du groupe C^{reg} sur C.

On définit comme suit des automorphismes de (C,+) :

a) l'automorphisme de $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$ $\tau : (x,i) \mapsto (x^{-1},-i)$ passe au quotient et définit un automorphisme τ de (C,+) ;

b) pour tout schéma S et tout $\zeta \in \mu_n(S)$, l'automorphisme $u(\zeta)$ de C_S $u(\zeta) : (x,i) \mapsto (x,\zeta^i,i)$ passe au quotient et définit un automorphisme de $(C_S,+)$.

De a) et b) on déduit une action sur C du schéma en groupe produit semi-direct du groupe à deux éléments $\{e,\tau\}$ par μ_n.

Proposition 1.10. Soit C le n-gone standard sur Spec($\mathbb{Z}$). La construction précédente identifie le foncteur Aut(C,+) des automorphismes de C compatibles à + avec le produit semi-direct $\{e,\tau\} \times \mu_n$. Le sous-groupe $Aut^0(C,+)$ des automorphismes agissant trivialement sur $Pic^0_{C/\mathbb{Z}}$ s'identifie au sous-groupe μ_n.

On a $Pic^0_{C/\mathbb{Z}} \xrightarrow{\sim} \mathbb{G}_m$, et τ agit sur Pic^0 par l'automorphisme $x \mapsto x^{-1}$

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of $\mathbb{G}_m$. Since $Aut(\mathbb{G}_m) \simeq \mathbb{Z}/2$, the diagram

$0 \to \mu_n \to \{e,\tau\} \times' \mu_n \to \{e,\tau\} \to 0 \downarrow \downarrow \downarrow 0 \to Aut^+(C,+) \to Aut(C,+) \to Aut(\mathbb{G}_m)$

shows that it is enough to verify that

$\mu_n \xrightarrow{\sim} Aut^+(C,+)$.

Let $g \in Aut^+(C,+)(S)$. From 1.8 and 1.7(ii), there exists $\lambda = (\lambda_i) \in \mathbb{G}_m^n$ such that g is $a(\lambda)$ (notation of 1.8). The condition that g commute with + then means that $\lambda_i \lambda_j = \lambda_{i+j}$: λ defines a homomorphism from $\mathbb{Z}/n$ to $\mathbb{G}_m$, and g is defined by a section of μ_n.

1.11. Let k be an algebraically closed field, and let (C,+) be obtained from the curve of 1.10 by extension of scalars from $\mathbb{Z}$ to k. One checks that, for $x \in C^{reg}(k)$, translation $y \mapsto x+y$ acts by rotations on the component diagram $\Gamma(C)$.

Definition 1.12. A generalized elliptic curve (over a base S) is a stable genus-one curve

$p : C \to S$,

equipped with a morphism

(1 12.1) $+ : C^{reg} \times_S C \to C$

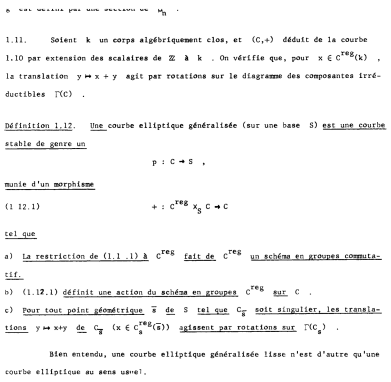
such that

a) The restriction of (1.1 .1) to C^{reg} makes C^{reg} a commutative group scheme.

b) (1.12.1) defines an action of the group scheme C^{reg} on C.

c) For every geometric point $\bar{s}$ of S such that $C_{\bar{s}}$ is singular, the translations $y \mapsto x+y$ of C_s ($x \in C_s^{reg}(\bar{s})$) act by rotations on $\Gamma(C_s)$.

Of course, a smooth generalized elliptic curve is just an elliptic curve in the usual sense.



1.11. Soient k un corps algébriquement clos, et (C,+) déduit de la courbe 1.10 par extension des scalaires de $\mathbb{Z}$ à k . On vérifie que, pour $x \in C^{reg}(k)$, la translation $y \mapsto x + y$ agit par rotations sur le diagramme des composantes irréductibles $\Gamma(C)$.

Définition 1.12. Une courbe elliptique généralisée (sur une base S) est une courbe stable de genre un

$$p : C \to S ,$$

munie d'un morphisme

$$(1\ 12.1) \qquad + : C^{reg} \times_S C \to C$$

tel que

a) La restriction de (1.1 .1) à C^{reg} fait de C^{reg} un schéma en groupes commutatif.

b) (1.12.1) définit une action du schéma en groupes C^{reg} sur C .

c) Pour tout point géométrique $\bar{s}$ de S tel que $C_{\bar{s}}$ soit singulier, les translations $y \mapsto x+y$ de $C_{\bar{s}}$ $(x \in C_{\bar{s}}^{reg}(\bar{s}))$ agissent par rotations sur $\Gamma(C_{\bar{s}})$.

Bien entendu, une courbe elliptique généralisée lisse n'est d'autre qu'une courbe elliptique au sens usuel.

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Proposition 1.13. Let (C,+) satisfy 1.12 a) and b). The set U of s ∈ S such that C_s/s satisfies c) is open and closed in S. Over U, C^reg acts trivially on Pic⁰_{C/S}.

Proof: Let x be a geometric point of C_s̄. Translation y ↦ x+y acts by translation if and only if it acts trivially on Pic⁰_{C_s}. Apply the following lemma (a variation on a theme of SGA 3) to Pic⁰_{C/S} ×_S C^reg over C^reg.

Lemma 1.14. Let p : A → S be a group scheme over S whose fibers are extensions of abelian varieties by tori. Let g and h be two endomorphisms of A. There exists an open and closed part of S such that, for every S-scheme T,

g_T = h_T if and only if T factors through V.

The lemma shows that the U of 1.13 is closed, since its complement is the image of an open (and closed) part C^reg−V of C^reg. It also proves that, if S = U, C^reg acts trivially on Pic⁰_{C/S}. We shall prove that U is open at the same time as 1.15 below.

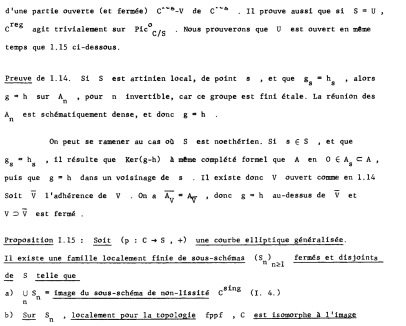
Proof of 1.14. If S is local Artinian, with point s, and g_s = h_s, then g = h on A_n, for n invertible, since this group is finite étale. The union of the A_n is schematically dense, and hence g = h.

We may reduce to the case where S is Noetherian. If s ∈ S and g_s = h_s, it follows that Ker(g−h) has the same formal completion as A at 0 ∈ A_s ⊂ A, and then that g = h in a neighborhood of s. There therefore exists an open V as in 1.14. Let V̄ be the closure of V. One has Ā_V = A_V̄, and hence g = h over V̄, and V ⊃ V̄ is closed.

Proposition 1.15: Let (p : C → S, +) be a generalized elliptic curve. There exists a locally finite family of closed and pairwise disjoint subschemes (S_n)_{n≥1} of S such that

a) ⋃_n S_n = image of the non-smoothness subscheme C^sing (I. 4.)

b) Over S_n, locally for the fppf topology, C is isomorphic to the inverse



d'une partie ouverte (et fermée) C^{reg}-V de C^{sing} . Il prouve aussi que si S = U , C^{reg} agit trivialement sur $Pic^{0}_{C/S}$. Nous prouverons que U est ouvert en même temps que 1.15 ci-dessous.

Preuve de 1.14. Si S est artinien local, de point s , et que $g_s = h_s$, alors g = h sur A_n , pour n invertible, car ce groupe est fini étale. La réunion des A_n est schématiquement dense, et donc g = h .

On peut se ramener au cas où S est noethérien. Si $s \in S$, et que $g_s = h_s$, il résulte que Ker(g-h) à même complété formel que A en $0 \in A_s \subset A$, puis que g = h dans un voisinage de s . Il existe donc V ouvert comme en 1.14 Soit $\overline{V}$ l'adhérence de V . On a $\overline{A_V} = A_{\overline{V}}$, donc g = h au-dessus de $\overline{V}$ et $V \supset \overline{V}$ est fermé .

Proposition 1.15 : Soit (p : C → S , +) une courbe elliptique généralisée. Il existe une famille localement finie de sous-schémas $(S_n)_{n\geq 1}$ fermés et disjoints de S telle que

a) $\bigcup_n S_n$ = image du sous-schéma de non-lissité C^{sing} (I. 4.)

b) Sur S_n , localement pour la topologie fppf , C est isomorphe à l'image

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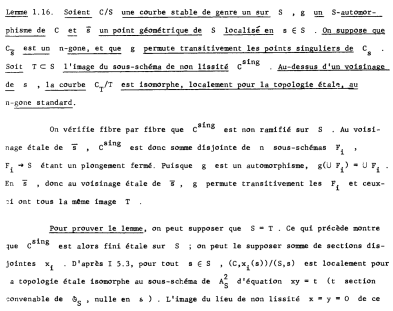
image of the standard generalized elliptic curve with n sides of 1.9.

Proof: The problem is local on S, and trivial in a neighborhood of the points of S where C_s is smooth. Let therefore s̄ be a geometric point of S, localized at s ∈ S, such that the curve C_s̄ is an n-gon. Let α be an isomorphism between C_s̄ and the standard polygon, such that α(e) is the point 1 of the component of index 0 ∈ ℤ/n, and index the components of C_s̄ by ℤ/n using α. Translation by x ∈ C_s̄^reg, in the component of index i, induces the rotation of Γ(C_s̄) taking the component of index 0 to that of index i: addition induces on π₀(C_s^reg) ≃ ℤ/n addition in ℤ/n, and, if x lies in the component of index one of C_s̄^reg, translation by x transitively permutes the singular points of C_s̄. In an étale neighborhood of s̄, C_reg/S admits sections x such that x(s̄) lies in the component of index one of C_s̄. The automorphism g = x + of C/S then satisfies the hypothesis of the following lemma.

Lemma 1.16. Let C/S be a stable genus-one curve over S, let g be an S-automorphism of C, and let s̄ be a geometric point of S localized at s ∈ S. Suppose that C_s̄ is an n-gon and that g transitively permutes the singular points of C_s. Let T ⊂ S be the image of the non-smoothness subscheme C^sing. Over a neighborhood of s, the curve C_T/T is isomorphic, locally for the étale topology, to the standard n-gon.

One verifies fiber by fiber that C^sing is unramified over S. In an étale neighborhood of s̄, C^sing is therefore a disjoint sum of n subschemes F_i, each F_i → S being a closed immersion. Since g is an automorphism, g(⋃ F_i) = ⋃ F_i. At s̄, and hence in an étale neighborhood of s̄, g transitively permutes the F_i, and all of these have the same image T.

To prove the lemma, we may suppose that S = T. What precedes shows that C^sing is then finite étale over S; we may suppose it is a sum of disjoint sections x_i. By I 5.3, for every s ∈ S, (C,x_i(s))/(S,s) is locally for the étale topology isomorphic to the subscheme of $\mathbb{A}^2$_S with equation xy = t, where t is a suitable section of $\mathcal{O}$_S, vanishing at s. The image of the non-smooth locus x = y = 0 of this



Lemme 1.16. Soient C/S une courbe stable de genre un sur S, g un S-automorphisme de C et $\bar{s}$ un point géométrique de S localisé en $s \in S$. On suppose que $C_{\bar{s}}$ est un n-gone, et que g permute transitivement les points singuliers de $C_{\bar{s}}$. Soit $T \subset S$ l'image du sous-schéma de non lissité C^{sing}. Au-dessus d'un voisinage de s, la courbe C_T/T est isomorphe, localement pour la topologie étale, au n-gone standard.

On vérifie fibre par fibre que C^{sing} est non ramifié sur S. Au voisinage étale de $\bar{s}$, C^{sing} est donc somme disjointe de n sous-schémas F_i, $F_i \to S$ étant un plongement fermé. Puisque g est un automorphisme, $g(\cup F_i) = \cup F_i$. En $\bar{s}$, donc au voisinage étale de $\bar{s}$, g permute transitivement les F_i et ceux-ci ont tous la même image T.

Pour prouver le lemme, on peut supposer que S = T. Ce qui précède montre que C^{sing} est alors fini étale sur S ; on peut le supposer somme de sections disjointes x_i. D'après I 5.3, pour tout $s \in S$, $(C,x_i(s))/(S,s)$ est localement pour la topologie étale isomorphe au sous-schéma de A^2_S d'équation xy = t (t section convenable de $\mathfrak{O}_S$, nulle en s). L'image du lieu de non lissité x = y = 0 de ce

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scheme is the subscheme t = 0 of S; the non-smooth locus can be étale only if t = 0 in a neighborhood of s. Locally for the étale topology, (C,x_i(s))/(S,s) is therefore isomorphic to the subscheme of $\mathbb{A}^2$_S with equation xy = 0.

Blow up C^sing in C to obtain $\tilde{C}$. A local calculation shows that $\tilde{C}$ is smooth over S, with fibers isomorphic to the normalizations of the fibers of C, and that the inverse image of C^sing in $\tilde{C}$ is étale over S.

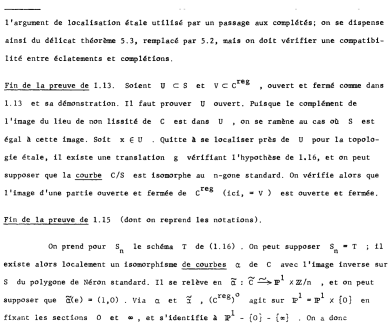
Let $\pi : \tilde{C} \to S'$ be the Stein factorization of $\tilde{C}$. The scheme S' is finite étale over S, and the geometric fibers of π are isomorphic to $\mathbb{P}^1$. Locally on S for the étale topology, $\tilde{C}$ is therefore a sum of n copies of $\mathbb{P}^1$. In an étale neighborhood of s, C is obtained from $\tilde{C}$ just as C_s̄ is obtained from $\tilde{C}$_s̄, and hence is a standard Néron polygon.

Variant: We easily reduce to supposing S Noetherian. We may then replace the étale-localization argument used above by passage to completions; this avoids the delicate theorem 5.3, replaced by 5.2, but one must verify compatibility between blowups and completions.

End of the proof of 1.13. Let U ⊂ S and V ⊂ C^reg be open and closed as in 1.13 and its proof. We must prove U open. Since the complement of the image of the non-smooth locus of C lies in U, we reduce to the case where S equals that image. Let x ∈ U. After localizing near U for the étale topology, there exists a translation g satisfying the hypothesis of 1.16, and we may suppose that the curve C/S is isomorphic to the standard n-gon. One then verifies that the image of an open and closed part of C^reg (here, = V) is open and closed.

End of the proof of 1.15 (whose notation we resume).

Take for S_n the scheme T of (1.16). We may suppose S_n = T; there then exists locally an isomorphism of curves α from C to the inverse image over S of the standard Néron polygon. It lifts to $\tilde{\alpha} : \tilde{C} \xrightarrow{\sim} \mathbb{P}^1 \times \mathbb{Z}/n$, and we may suppose that $\tilde{\alpha}(e) = (1,0)$. Via α and $\tilde{\alpha}$, $(C^{reg})^0$ acts on $\mathbb{P}^1 = \mathbb{P}^1 \times \{0\}$ while fixing the sections 0 and ∞, and is identified with $\mathbb{P}^1 - \{0\} - \{\infty\}$. Thus one has



l'argument de localisation étale utilisé par un passage aux complétés; on se dispense ainsi du délicat théorème 5.3, remplacé par 5.2, mais on doit vérifier une compatibilité entre éclatements et complétions.

Fin de la preuve de 1.13. Soient $U \subset S$ et $V \subset C^{reg}$, ouvert et fermé comme dans 1.13 et sa démonstration. Il faut prouver U ouvert. Puisque le complément de l'image du lieu de non lissité de C est dans U, on se ramène au cas où S est égal à cette image. Soit $x \in U$. Quitte à se localiser près de U pour la topologie étale, il existe une translation g vérifiant l'hypothèse de 1.16, et on peut supposer que la courbe C/S est isomorphe au n-gone standard. On vérifie alors que l'image d'une partie ouverte et fermée de C^{reg} (ici, $= V$) est ouverte et fermée.

Fin de la preuve de 1.15 (dont on reprend les notations).

On prend pour S_n le schéma T de (1.16). On peut supposer $S_n = T$; il existe alors localement un isomorphisme de courbes α de C avec l'image inverse sur S du polygone de Néron standard. Il se relève en $\tilde{\alpha} : \tilde{C} \xrightarrow{\sim} \mathbb{P}^1 \times \mathbb{Z}/n$, et on peut supposer que $\tilde{\alpha}(e) = (1,0)$. Via α et $\tilde{\alpha}$, $(C^{reg}/S)^0$ agit sur $\mathbb{P}^1 = \mathbb{P}^1 \times \{0\}$ en fixant les sections 0 et ∞, et s'identifie à $\mathbb{P}^1 - \{0\} - \{\infty\}$. On a donc

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$(C^{reg})^0 \simeq \mathbb{G}_m$ (compatibly with +). In a neighborhood of s, one may normalize α so that $\tilde{\alpha}(x^i) = (1,i)$ $(0 \le i < n)$. If $x^n = 1$, then α is an isomorphism of (C,+) with the curve (1.9). In the general case, $x^n = a \in \mathbb{G}_m(S)$; a admits an n-th root locally for the fppf topology of S (for the étale topology if n is invertible on S), and one replaces x by $x \cdot a^{-1/n}$.

The following proposition will play a key technical role in questions of representability.

Proposition 1.17. Let X/S be a generalized elliptic curve over S, with identity section $e \in X(S)$, and let $u : Y \to X$ be a finite étale cover of X, equipped with $e \in Y(S)$ over e. Suppose that the geometric fibers of Y/S are connected. There is then one and only one generalized-elliptic-curve structure on Y, with identity e, such that the diagram

$$\begin{array}{ccc} Y^{reg} \times_S Y & \longrightarrow & Y \\ \downarrow & & \downarrow u \\ X^{reg} \times_S Y & \xrightarrow{+} & X \end{array} \qquad (1.17.1)$$

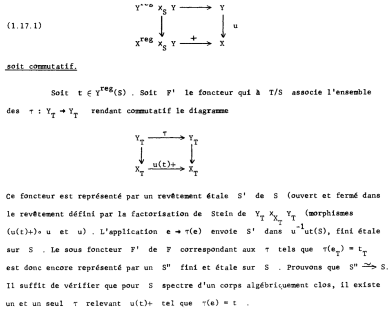
is commutative.

Let $t \in Y^{reg}(S)$. Let F' be the functor associating to T/S the set of $\tau : Y_T \to Y_T$ making the diagram

$$\begin{array}{ccc} Y_T & \xrightarrow{\tau} & Y_T \\ \downarrow & & \downarrow \\ X_T & \xrightarrow{u(t)+} & X_T \end{array}$$

commutative.

This functor is represented by an étale cover S' of S (open and closed in the cover defined by the Stein factorization of $Y_T \times_{\{X_T\}} Y_T$ (for the morphisms $(u(t)+)\circ u$ and u)). The map $e \mapsto \tau(e)$ sends S' into $u^{-1}ut(S)$, finite étale over S. The subfunctor F' of F corresponding to the τ such that $\tau(e_T) = t_T$ is therefore again represented by an S'' finite and étale over S. Let us prove that $S'' \xrightarrow{\sim} S$. It is enough to verify that, when S is the spectrum of an algebraically closed field, there exists one and only one τ lifting $u(t)+$ such that $\tau(e) = t$.



(1.17.1)

soit commutatif.

Soit $t \in Y^{reg}(S)$. Soit F' le foncteur qui à T/S associe l'ensemble des $\tau : Y_T \to Y_T$ rendant commutatif le diagramme

Ce foncteur est représenté par un revêtement étale S' de S (ouvert et fermé dans le revêtement défini par la factorisation de Stein de $Y_T \times_{X_T} Y_T$ (morphismes $(u(t)+)\circ u$ et u). L'application $e \to \tau(e)$ envoie S' dans $u^{-1}ut(S)$, fini étale sur S. Le sous foncteur F' de F correspondant aux τ tels que $\tau(e_T) = t_T$ est donc encore représenté par un S'' fini et étale sur S. Prouvons que $S'' \xrightarrow{\sim} S$. Il suffit de vérifier que pour S spectre d'un corps algébriquement clos, il existe un et un seul τ relevant $u(t)+$ tel que $\tau(e) = t$.

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This amounts to showing that the cover Y of X and its inverse image under u(t)+ : X → X are isomorphic. For smooth X, this follows from the fact that u(t) lies in the neutral component of X^reg. For X a Néron polygon, one observes that X has only one irreducible cover of a given degree.

The unique section of S” over S defines a morphism t+ : Y → Y. Since this construction is compatible with base changes, it defines

+ : Y^reg × Y → Y,

making (1.17.1) commutative, such that t + e = t, and characterized by these properties. It remains to prove that + makes Y a generalized elliptic curve, i.e. that various identities, such as x+y = y+x (x,y ∈ Y^reg(S) are satisfied. Since u(x+y) = u(x) + u(y) = u(y) + u(x) = u(y+x), x+y is a section of u^{−1}(u(y+x)(S)), étale over S, and it is enough to verify that x+y = y+x on the geometric fibers. For proper smooth fibers, a composition law with identity is automatically an abelian group law. For the others, the Néron polygon with n sides, with its standard + law, has as covers only Néron polygons with nk sides, with their standard + law.

The other identities to be verified are proved in the same way.

1.18. Let (C,+) be a generalized elliptic curve over S. One verifies fiber by fiber that the morphism

n : x ↦ nx : C^reg → C^reg

is flat. Its kernel C_n is therefore flat over S.

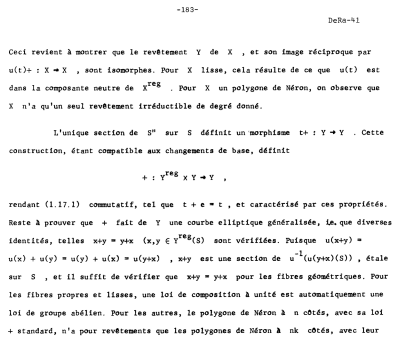
Suppose that S is the spectrum of an algebraically closed field k.

a) if C is smooth, one knows that C_n is finite of rank n^2

b) if C is an m-gon, one has C^reg ≃ $\mathbb{G}$_m × $\mathbb{Z}$/m.

Writing (n,m) for the gcd of n and m, one therefore has

C_n ≃ μ_n × $\mathbb{Z}$/((n,m))



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Ceci revient à montrer que le revêtement Y de X , et son image réciproque par u(t)+ : X → X , sont isomorphes. Pour X lisse, cela résulte de ce que u(t) est dans la composante neutre de X^{reg} . Pour X un polygone de Néron, on observe que X n'a qu'un seul revêtement irréductible de degré donné.

L'unique section de S" sur S définit un morphisme t+ : Y → Y . Cette construction, étant compatible aux changements de base, définit

+ : Y^{reg} × Y → Y ,

rendant (1.17.1) commutatif, tel que t + e = t , et caractérisé par ces propriétés. Reste à prouver que + fait de Y une courbe elliptique généralisée, i.e. que diverses identités, telles x+y = y+x (x,y ∈ Y^{reg}(S) sont vérifiées. Puisque u(x+y) = u(x) + u(y) = u(y) + u(x) = u(y+x) , x+y est une section de u^{-1}(u(y+x)(S)) , étale sur S , et il suffit de vérifier que x+y = y+x pour les fibres géométriques. Pour les fibres propres et lisses, une loi de composition à unité est automatiquement une loi de groupe abélien. Pour les autres, le polygone de Néron à n côtés, avec sa loi + standard, n'a pour revêtements que les polygones de Néron à nk côtés, avec leur

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In particular, C_n is finite of rank n.(n,m).

Recall the following lemma.

Lemma 1.19. Let f : X → S be a quasi-finite, flat, and separated morphism, with S Noetherian. If the rank of the fibers of f is constant, then f is finite.

Proof: Let us prove that f is proper. By the valuative criterion for properness, we may suppose that S is a trait. By the Main theorem of Zariski, X is an open subscheme of a scheme $\bar{X}$ finite over S, which may even be taken flat over S. Comparing the ranks of the special and generic fibers of X and $\bar{X}$, one finds that X = $\bar{X}$, whence 1.19.

Corollary 1.20. Let p : C → S be a generalized elliptic curve over S, and let n be an integer. Suppose that, for every geometric point $\bar{s}$ of S, C_$\bar{s}$ is smooth, or is an m-gon with n|m. Then C_n is finite locally free over S, of rank n^2.

We reduce to supposing S Noetherian, and apply 1.18 and 1.19.

2. Irreducible stable curves.

In this §, we prove that, for C/S a stable genus-one curve with irreducible geometric fibers, and for e ∈ C^reg(S), there exists on C one and only one generalized-elliptic-curve structure with identity e.

2.1 Certain preliminary results will hold for a proper flat morphism of finite presentation p : C → S satisfying the following condition.

(2.1.1) C is a Cohen-Macaulay genus-one curve over S, p_*$\mathcal{O}$_C = $\mathcal{O}$_S universally, and p^*p_*ω_{C/S} ⟶̃ ω_{C/S}.

A stable genus-one curve over S satisfies (2 1.1) (1.6). If a geometric fiber C_$\bar{s}$ of p satisfies (2.1.1), then C/S satisfies (2.1.1) in a neighborhood of s (for the condition: “Cohen-Macaulay,” see EGA IV 12.2.1.; for

ou un m-gone, avec n|m . Alors, C_n est fini localement libre sur S de rang n^2.

On se ramène à supposer S noethérien, et on applique 1.18, 1.19.

2. Courbes stables irréductibles.

Dans ce §, nous prouvons que, pour C/S une courbe stable de genre un, de fibres géométriques irréductibles, et pour e ∈ $C^{reg}(S)$, il existe sur C une et une seule structure de courbe elliptique généralisée d'unité e .

2.1 Certains résultats préliminaires vaudront pour un morphisme propre et plat de présentation finie p : C → S vérifiant la condition suivante.

(2.1.1) C est une courbe de Cohen-Macaulay de genre un sur S , $p_*\mathcal{O}_C = \mathcal{O}_S$ universellement et $p^*p_*\omega_{C/S} \xrightarrow{\sim} \omega_{C/S}$.

Une courbe stable de genre un sur S vérifie (2 1.1) (1.6). Si une fibre géométrique $C_{\bar{s}}$ de p vérifie (2.1.1), alors C/S vérifie (2.1.1) dans un voisinage de s (pour la condition : "Cohen-Macaulay", voir EGA IV 12.2.1.; pour

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p_* $\mathcal{O}$_C = $\mathcal{O}$_S, see EGA III. 7.8.8.; if p_* $\mathcal{O}$_C = $\mathcal{O}$_S universally, p_* ω_{C/S} is locally free with formation compatible with every base change, and we conclude as in 1.6). The special fibers of the Néron models of elliptic curves satisfy (2.1.1)(SGA 7 X 1.15).In particular, curves p : C → S of genus one with geometrically integral fibers (= elliptic curves, nodal plane cubics, or cuspidal plane cubics) satisfy (2 1.1).

For every section t of C/S, write m(t) for the ideal sheaf defining the subscheme t(S).

Proposition 2.2. Let S be a noetherian scheme, $\bar{s}$ a geometric point of S localized at s ∈ S, p : C → S satisfying (2.1.1), and $\mathcal{J}$ a coherent sheaf on C flat over S. Suppose that $\mathcal{J}$_{$\bar{s}$} on C_{$\bar{s}$} is isomorphic to a sheaf m(t_{$\bar{s}$}), for t_{$\bar{s}$} ∈ C_{$\bar{s}$}($\bar{s}$). Then, over

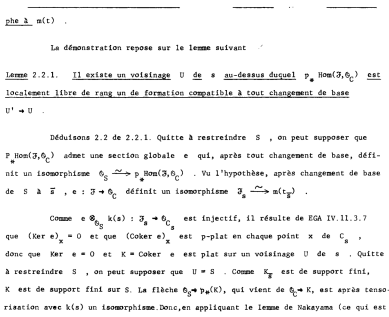
a neighborhood U of s in S, there exists a unique section t ∈ C(U) such that $\mathcal{J}$|_{p^{-1}(U)} is isomor- phic to m(t).

The proof rests on the following lemma.

Lemma 2.2.1. There is a neighborhood U of s over which p_* Hom($\mathcal{J}$,$\mathcal{O}$_C) is locally free of rank one, with formation compatible with every base change U' → U.

Let us deduce Proposition 2.2 from 2.2.1. After shrinking S, suppose that p_* Hom($\mathcal{J}$,$\mathcal{O}$_C) has a global section e which, after every base change, defines an isomorphism $\mathcal{O}$_S ≃ p_* Hom($\mathcal{J}$,$\mathcal{O}$_C). By the hypothesis, after base change from S to s̄, e : $\mathcal{J}$ → $\mathcal{O}$_C defines an isomorphism $\mathcal{J}$_{s̄} ≃ m(t_{s̄}).

Since e ⊗_{$\mathcal{O}$_S} k(s) : $\mathcal{J}$_s → $\mathcal{O}$_{C_s} is injective, EGA IV.11.3.7 gives (Ker e)_x = 0 and says that (Coker e)_x is p-flat at every x of C_s, hence Ker e = 0 and K = Coker e is flat near s. After shrinking S, suppose U = S. Since K_{s̄} has finite support, K has finite support over S. The map $\mathcal{O}$_S → p_*(K), induced by $\mathcal{O}$_C → K, becomes an isomorphism after tensoring with k(s). Thus, by Nakayama's lemma (which is



phe à m(t) .

La démonstration repose sur le lemme suivant .

Lemme 2.2.1. Il existe un voisinage U de s au-dessus duquel $P_*Hom(\mathcal{J},\mathcal{O}_C)$ est localement libre de rang un de formation compatible à tout changement de base U' → U .

Déduisons 2.2 de 2.2.1. Quitte à restreindre S , on peut supposer que $P_*Hom(\mathcal{J},\mathcal{O}_C)$ admet une section globale e qui, après tout changement de base, définit un isomorphisme $\mathcal{O}_S \xrightarrow{\sim} p_*Hom(\mathcal{J},\mathcal{O}_C)$. Vu l'hypothèse, après changement de base de S à $\bar{s}$, e : $\mathcal{J} \to \mathcal{O}_C$ définit un isomorphisme $\mathcal{J}_{\bar{s}} \xrightarrow{\sim} m(t_{\bar{s}})$.

Comme $e \otimes_{\mathcal{O}_S} k(s) : \mathcal{J}_s \to \mathcal{O}_{C_s}$ est injectif, il résulte de EGA IV.11.3.7 que $(\mathrm{Ker}\ e)_x = 0$ et que $(\mathrm{Coker}\ e)_x$ est p-plat en chaque point x de C_s , donc que Ker e = 0 et K = Coker e est plat sur un voisinage U de s . Quitte à restreindre S , on peut supposer que U = S . Comme $K_{\bar{s}}$ est de support fini, K est de support fini sur S. La flèche $\mathcal{O}_S \to p_*(K)$, qui vient de $\mathcal{O}_C \to K$, est après tensorisation avec k(s) un isomorphisme.Donc,en appliquant le lemme de Nakayama (ce qui est

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permissible, since the support of K is finite over S) this morphism is surjective and, because K is flat over S, it is also injective.

After shrinking S, we may therefore suppose that e is injective and that $\mathcal{O}$_S ≃ p_*(K). The subscheme of C defined by e($\mathcal{J}$) is consequently isomorphic to S and defines the unique section t such that $\mathcal{J}$ ≃ m(t).

Let us prove 2.2.1. The long exact cohomology sequence associated with

0 → m(t_{s̄}) → $\mathcal{O}$_{C_{s̄}} → $\mathcal{O}$(t_{s̄}) → 0

and (2 1.1) give

(2.2.2) $H^0(C_{\bar{s}},m(t_{\bar{s}})) = 0$

(2.2.3) $\dim_{k(\bar{s})} H^1(C_{\bar{s}},m(t_{\bar{s}})) = 1$.

It is therefore enough to prove the following lemma.

Lemma 2.3. With the notation of 2.2, suppose $H^0(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 0$ and $\dim_{k(\bar{s})} H^1(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 1$. There is then a neighborhood U of s in S over which p_* Hom($\mathcal{J}$,$\mathcal{O}$_C) is locally free of rank one, with formation compatible with every base change U' → U.

Proof. By hypothesis H^i(C_{s̄},$\mathcal{J}$_{s̄}) = 0 for i ≠ 1, while $H^1(C_{\bar{s}},\mathcal{J}_{\bar{s}})$ has rank one. It follows that near s, R^i p_*($\mathcal{J}$) vanishes for i ≠ 1, remains so after every base

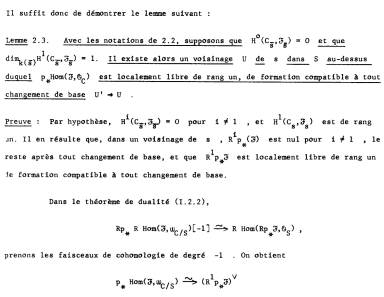
change, and $R^1p_*\mathcal{J}$ is locally free of rank one with formation compatible with every base change.

In the duality theorem (1.2.2),

Rp_* R Hom($\mathcal{J}$,ω_{C/S})[-1] ≃ R Hom(Rp_*$\mathcal{J}$,$\mathcal{O}$_S),

take the cohomology sheaves of degree −1. We obtain

p_* Hom($\mathcal{J}$,ω_{C/S}) ≃ ($R^1p_*\mathcal{J}$)^∨



Il suffit donc de démontrer le lemme suivant :

Lemme 2.3. Avec les notations de 2.2, supposons que $H^0(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 0$ et que $\dim_{k(\bar{s})} H^1(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 1$. Il existe alors un voisinage U de s dans S au-dessus duquel $p_*\mathrm{Hom}(\mathcal{J},\mathcal{O}_C)$ est localement libre de rang un, de formation compatible à tout changement de base U' → U .

Preuve : Par hypothèse, $H^i(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 0$ pour $i \neq 1$, et $H^1(C_{\bar{s}},\mathcal{J}_{\bar{s}})$ est de rang un. Il en résulte que, dans un voisinage de s, $R^ip_*(\mathcal{J})$ est nul pour $i \neq 1$, le reste après tout changement de base, et que $R^1p_*\mathcal{J}$ est localement libre de rang un de formation compatible à tout changement de base.

Dans le théorème de dualité (I.2.2),

$$Rp_* R\,\mathrm{Hom}(\mathcal{J},\omega_{C/S})[-1] \xrightarrow{\sim} R\,\mathrm{Hom}(Rp_*\mathcal{J},\mathcal{O}_S),$$

prenons les faisceaux de cohomologie de degré −1. On obtient

$$p_*\,\mathrm{Hom}(\mathcal{J},\omega_{C/S}) \xrightarrow{\sim} (R^1p_*\mathcal{J})^\vee$$

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(where ^∨ denotes the dual $\mathcal{O}$_S-module). In particular p_* Hom($\mathcal{J}$,ω_{C/S}) is locally free of rank one, compatible with every base change. It follows from (2.1.1) that p_*ω_{C/S} is locally free of rank one with formation compatible with every base change. Thus (2.1.1) also implies that, locally on S, ω_{C/S} is isomorphic to $\mathcal{O}$_C, and 2.3 follows.

Corollary 2.4. : Let p : C → S be a genus-one curve over S satisfying (2.1.1). Let $\mathcal{L}$ be an invertible sheaf on C defining an element of Pic⁰_{C/S}(S) (cf. I 3.2). For every section t of C over S, there is a unique section t’ such that, locally on S, m(t’) is isomorphic to m(t) ⊗ $\mathcal{L}$.

Proof. We reduce to the case where S is noetherian. By 2.2 we may suppose that S is the spectrum of an algebraically closed field. We treat a “universal” case.

Make the base change to T = C ×_S Pic⁰_{C/S} → S, take t to be the universal section and $\mathcal{L}$ the universal invertible sheaf. For every point of the form (x,1_{Pic}) (x ∈ C) of T the assertion is trivially true. Hence by 2.2 it holds on a neighborhood of C × 1_{Pic} ⊂ T. Since C is proper, there is an open neighborhood U of 1_{Pic} in Pic⁰_{C/S} over which the assertion holds above C × U ⊂ T. The assertion therefore holds on the product of C with the group generated by U, which is all of Pic⁰_{C/S}, and hence on T = C ×_S Pic⁰_{C/S}.

2.5. We write + for the morphism

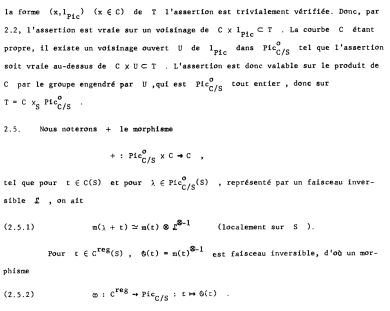
+ : Pic⁰_{C/S} × C → C,

such that, for t ∈ C(S) and λ ∈ Pic⁰_{C/S}(S), represented by an invertible sheaf $\mathcal{L}$,

(2.5.1) m(λ + t) ≃ m(t) ⊗ $\mathcal{L}$^{⊗−1} (locally on S).

For t ∈ C^reg(S), $\mathcal{O}$(t) = m(t)^{⊗−1} is invertible, giving a morphism

(2.5.2) φ : C^reg → Pic_{C/S} : t ↦ $\mathcal{O}$(t).



la forme $(x, 1_{Pic})$ $(x \in C)$ de T l'assertion est trivialement vérifiée. Donc, par 2.2, l'assertion est vraie sur un voisinage de $C \times 1_{Pic} \subset T$. La courbe C étant propre, il existe un voisinage ouvert U de 1_{Pic} dans $Pic^0_{C/S}$ tel que l'assertion soit vraie au-dessus de $C \times U \subset T$. L'assertion est donc valable sur le produit de C par le groupe engendré par U, qui est $Pic^0_{C/S}$ tout entier, donc sur $T = C \times_S Pic^0_{C/S}$.

2.5. Nous noterons $+$ le morphisme

$$+ : Pic^0_{C/S} \times C \to C,$$

tel que pour $t \in C(S)$ et pour $\lambda \in Pic^0_{C/S}(S)$, représenté par un faisceau inversible $\mathcal{L}$, on ait

(2.5.1) $\quad m(\lambda + t) \simeq m(t) \otimes \mathcal{L}^{\otimes -1}$ (localement sur S).

Pour $t \in C^{reg}(S)$, $\Phi(t) = m(t)^{\otimes -1}$ est faisceau inversible, d'où un morphisme

(2.5.2) $\quad \varphi : C^{reg} \to Pic_{C/S} : t \mapsto \Phi(t)$.

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We have

(2.5.3) φ(λ + t) = λ + φ(t).

Theorem 2.6. Let p : C → S be a genus-one curve satisfying (2.1.1).

(i) φ is an open immersion.

(ii) If p has geometrically integral fibers, the image of φ is a torsor under Pic⁰_{C/S}.

(iii) If p has geometrically reduced fibers, the action + of Pic⁰_{C/S} on C induces the trivial action of Pic⁰_{C/S} on Pic^[0]_{C/S}.

Proof. It is enough to prove (i) and (ii) when S is the spectrum of an algebraically closed field k (EGA IV. 17.8.2.). The images under φ of the various connected components of C^reg are then disjoint. Let D be one of them and e ∈ D(k). The maps λ ↦ λ + e and t ↦ φ(t) − φ(e) are inverse isomorphisms between D and Pic⁰. In particular φ is an isomorphism D ≃ φ(e) + Pic⁰, and (i), (ii) follow.

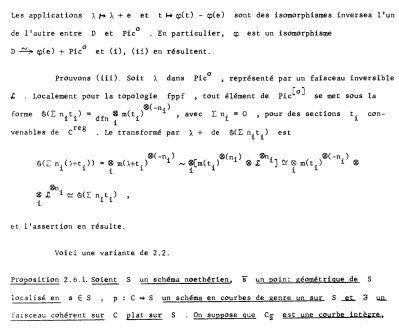
Let us prove (iii). Let λ ∈ Pic⁰, represented by an invertible sheaf $\mathcal{L}$. Locally for the fppf topology, every element of Pic^[0] has the form $\mathcal{O}(\Sigma\ n_i\ t_i) =_{dfn} \otimes_i m(t_i)^{\otimes(-n_i)}$, with Σ n_i = 0, for suitable sections t_i of C^reg. Transforming by λ + gives

$\mathcal{O}(\Sigma\ n_i(\lambda+t_i)) = \otimes_i\ m(\lambda+t_i)^{\otimes(-n_i)} \simeq \otimes_i\ [m(t_i)^{\otimes(n_i)} \otimes \mathcal{L}^{\otimes\ n_i}] \simeq \otimes_i\ m(t_i)^{\otimes(-n_i)} \otimes \otimes_i\ \mathcal{L}^{\otimes\ n_i} \simeq \mathcal{O}(\Sigma\ n_i\ t_i)$,

with Σ n_i = 0, so it remains equivalent to itself.

Here is a variant of 2.2.

Proposition 2.6.1. Let S be noetherian, s̄ a geometric point of S localized at s ∈ S, p : C → S a genus-one curve scheme over S, and $\mathcal{J}$ a coherent sheaf on C flat over S. Suppose that C_{s̄} is an integral curve,



Les applications $\lambda \mapsto \lambda + e$ et $t \mapsto \varphi(t) - \varphi(e)$ sont des isomorphismes inverses l'un de l'autre entre D et Pic^0. En particulier, φ est un isomorphisme $D \xrightarrow{\sim} \varphi(e) + \mathrm{Pic}^0$ et (i), (ii) en résultent.

Prouvons (iii). Soit λ dans Pic^0, représenté par un faisceau inversible $\mathcal{L}$. Localement pour la topologie fppf, tout élément de $\mathrm{Pic}^{[o]}$ se met sous la forme $\mathcal{O}(\sum n_i t_i) = \mathrm{dfn} \bigotimes_i m(t_i)^{\otimes(-n_i)}$, avec $\sum n_i = 0$, pour des sections t_i convenables de C^{reg}. Le transformé par $\lambda +$ de $\mathcal{O}(\sum n_i t_i)$ est

$$\mathcal{O}(\textstyle\sum n_i(\cdot + t_i)) = \bigotimes_i m(\cdot + t_i)^{\otimes(-n_i)} \sim \bigotimes_i [m(t_i)^{\otimes(n_i)} \otimes \mathcal{L}^{\otimes n_i}] \simeq \bigotimes_i m(t_i)^{\otimes(-n_i)} \otimes \bigotimes_i \mathcal{L}^{\otimes n_i} \simeq \mathcal{O}(\textstyle\sum n_i t_i),$$

et l'assertion en résulte.

Voici une variante de 2.2.

Proposition 2.6.1. Soient S un schéma noethérien, $\bar{s}$ un point géométrique de S localisé en $s \in S$, $p : C \to S$ un schéma en courbes de genre un sur S et $\mathcal{J}$ un faisceau cohérent sur C plat sur S. On suppose que $C_{\bar{s}}$ est une courbe intègre,

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that $\mathcal{J}_{\bar{s}}$ is torsion-free and generically of rank one, and that $\chi(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = -1$. Then over a neighborhood U of s there exists a unique section $t \in C(U)$ such that $\mathcal{J}|_{p^{-1}(U)}$ is isomorphic to m(t).

As observed in 2.1, C/S satisfies (2.1.1) near s; by 2.2 it is therefore enough to treat the case where S is the spectrum of an algebraically closed field, $S = \bar{s}$.

Under this hypothesis, let us prove $H^0(C,\mathcal{J}) = 0$. If a nonzero $f \in H^0(C,\mathcal{J})$ existed, the sequence

$$0 \to \mathcal{O}_C \to^{f} \mathcal{J} \to \mathcal{J}/\mathcal{O}_C\cdot f \to 0$$

would be exact, since $\mathcal{J}$ is torsion-free on integral C. We would obtain $\chi(C,\mathcal{J}) = \chi(C,\mathcal{O}_C) + \chi(C,\mathcal{J}/\mathcal{O}_C\cdot f) = 0 + \chi(C,\mathcal{J}/f\cdot\mathcal{J}) \geq 0$, contradicting the hypothesis.

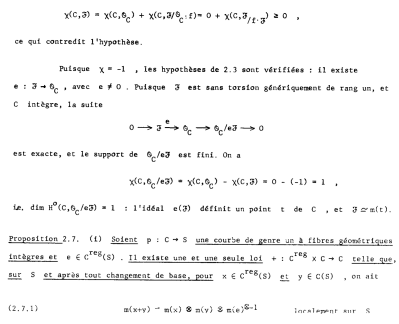
Since $\chi = -1$, the hypotheses of 2.3 hold: there is a nonzero $e : \mathcal{J} \to \mathcal{O}_C$. Since $\mathcal{J}$ is torsion-free of generic rank one and C is integral, the sequence

$$0 \to \mathcal{J} \to^{e} \mathcal{O}_C \to \mathcal{O}_C/e\mathcal{J} \to 0$$

is exact and $\mathcal{O}_C/e\mathcal{J}$ has finite support. We have $\chi(C,\mathcal{O}_C/e\mathcal{J}) = \chi(C,\mathcal{O}_C) - \chi(C,\mathcal{J}) = 0 - (-1) = 1$, i.e. $\dim H^0(C,\mathcal{O}_C/e\mathcal{J}) = 1$: the ideal $e(\mathcal{J})$ defines a point t of C, and $\mathcal{J} \simeq m(t)$.

Proposition 2.7. (i) Let $p : C \to S$ be a genus-one curve with geometrically integral fibers and $e \in C\hat{}reg(S)$. There is a unique law $+ : C\hat{}reg \times C \to C$ such that, over S and after every base change, for $x \in C\hat{}reg(S)$ and $y \in C(S)$,

(2.7.1) $m(x+y) \simeq m(x) \otimes m(y) \otimes m(e)^{\{\otimes -1\}}$ locally on S



$\chi(C,\mathcal{J}) = \chi(C,\mathcal{O}_C) + \chi(C,\mathcal{J}/\mathcal{O}_C\cdot f) = 0 + \chi(C,\mathcal{J}/f\cdot\mathcal{J}) \geq 0$,

ce qui contredit l'hypothèse.

Puisque $\chi = -1$, les hypothèses de 2.3 sont vérifiées : il existe $e : \mathcal{J} \to \mathcal{O}_C$, avec $e \neq 0$. Puisque $\mathcal{J}$ est sans torsion génériquement de rang un, et C intègre, la suite

$$0 \longrightarrow \mathcal{J} \xrightarrow{e} \mathcal{O}_C \longrightarrow \mathcal{O}_C/e\mathcal{J} \longrightarrow 0$$

est exacte, et le support de $\mathcal{O}_C/e\mathcal{J}$ est fini. On a

$$\chi(C,\mathcal{O}_C/e\mathcal{J}) = \chi(C,\mathcal{O}_C) - \chi(C,\mathcal{J}) = 0 - (-1) = 1,$$

i.e. $\dim H^0(C,\mathcal{O}_C/e\mathcal{J}) = 1$: l'idéal $e(\mathcal{J})$ définit un point t de C, et $\mathcal{J} \simeq m(t)$.

Proposition 2.7. (i) Soient $p : C \to S$ une courbe de genre un à fibres géométriques intègres et $e \in C^{reg}(S)$. Il existe une et une seule loi $+ : C^{reg} \times C \to C$ telle que, sur S et après tout changement de base, pour $x \in C^{reg}(S)$ et $y \in C(S)$, on ait

(2.7.1) $m(x+y) \simeq m(x) \otimes m(y) \otimes m(e)^{\otimes -1}$ localement sur S.

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(ii) This law makes C^reg a commutative group scheme with identity e acting on C; for this action, C^reg acts trivially on Pic⁰_{C/S}.

(iii) The symmetry x ↦ −x of C^reg extends to C and, for t ∈ C(S), locally on S,

(2.7.2) m(−t) ≃ Hom(m(t),$\mathcal{O}$(−2(e))).

(iv) If C/S is stable, + is the unique generalized elliptic-curve structure on C with identity e.

Proof. We may suppose S noetherian. By 2.1, condition (2.1.1) holds. Assertion (i) follows from 2.4. By 2.6, the morphism

φ_e : C^reg → Pic⁰_{C/S} : x ↦ φ(x) − φ(e)

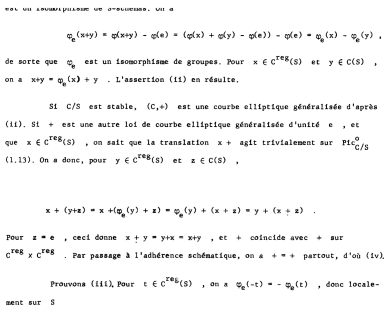
is an isomorphism of S-schemes. We have φ_e(x+y) = φ(x+y) − φ(e) = (φ(x) + φ(y) − φ(e)) − φ(e) = φ_e(x) − φ_e(y), so φ_e is a group isomorphism. For x ∈ C^reg(S) and y ∈ C(S), x+y = φ_e(x) + y, which proves (ii).

If C/S is stable, (C,+) is a generalized elliptic curve by (ii). If +′ is another generalized elliptic-curve law with identity e and x ∈ C^reg(S), translation by x +′ acts trivially on Pic⁰_{C/S} (1.13). Thus, for y ∈ C^reg(S) and z ∈ C(S),

x +′ (y+z) = x +′ (φ_e(y) + z) = φ_e(y) + (x +′ z) = y + (x +′ z).

Putting z = e gives x +′ y = y+x = x+y, and +′ agrees with + on C^reg × C^reg. By schematic closure, +′ = + everywhere, proving (iv).

Let us prove (iii). For t ∈ C^reg(S), φ_e(−t) = −φ_e(t), and therefore local- ly on S



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$\mathcal{O}$((-t)) ⊗ $\mathcal{O}$(-(e)) ≃ $\mathcal{O}$(-(t)) ⊗ $\mathcal{O}$((e)), i.e.

m(-t) ≃ m(t)^{⊗−1} ⊗ $\mathcal{O}$(-2(e)).

It remains to prove that, for arbitrary t, the second member of (2.7.2) is locally on S of the form m(x). The exact sequence

(2.7.3) 0 → m(t) → $\mathcal{O}$_C → t_*$\mathcal{O}$_S → 0

gives the exact sequence

(2.7.4) 0 → Hom(t_*$\mathcal{O}$_S,$\mathcal{O}$(-2(e))) → Hom($\mathcal{O}$_C,$\mathcal{O}$(-2(e))) →

→ Hom(m(t),$\mathcal{O}$(-2(e))) → Ext¹(t_*$\mathcal{O}$_S,$\mathcal{O}$(-2(e))) → 0.

Since ω_{C/S} is an invertible sheaf on C (cf. 1.6.1), $\mathcal{O}$(-2(e)) is locally on C isomorphic to ω_{C/S}. Hence, locally on C,

$\mathrm{Ext}^1(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) \simeq \mathrm{Ext}^1(t_*\mathcal{O}_S,\omega_{C/S})$.

We now apply the coherent-sheaf duality formula for the morphism $t : S \to C$:

(2.7.5) $R\,t_*\,R\,\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{O}_S,t^!\omega_{C/S}) \tilde{\to} R\,\mathrm{Hom}_{\mathcal{O}_C}(R\,t_*\mathcal{O}_S,\omega_{C/S})$.

Observe that

$t^!\omega_{C/S} = t^!p^!\mathcal{O}_S[1] \simeq (p\circ t)^!\mathcal{O}_S[1] = \mathcal{O}_S[1]$.

Moreover, $R^1t_*\mathcal{O}_S = 0$. Taking H^0 of the two sides of (2.7.5), we obtain

$\mathrm{Ext}^1(t_*\mathcal{O}_S,\omega_{C/S}) \simeq t_*\mathcal{O}_S$.

Furthermore, it is easy to see that $\mathrm{Hom}(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) = 0$. The resulting exact sequence

(2.7.6) $0 \to \mathcal{O}(-2(e)) \to \mathrm{Hom}(m(t),\mathcal{O}(-2(e))) \to t_*\mathcal{O}_S \to 0$

shows that $\mathrm{Hom}(m(t),\mathcal{O}(-2(e)))$ is flat over S, that its formation commutes with every base change $S' \to S$, and that its fibers are generically of rank 1 and torsion-free. To see that this sheaf is locally on S' of the form m(t'), it is enough to see that, for every geometric point $\bar{s}$ of S',

$\chi(C_{\bar{s}},\mathrm{Hom}(m(t),\mathcal{O}(-2(e)))) = -1$ (cf. 2.6).

But

$\deg_{C_{\bar{s}}}(\mathcal{O}(-2(e))) = -2$,

so

$\chi(C_{\bar{s}},\mathcal{O}(-2(e))) = -2$, and $\chi(C_{\bar{s}},t_*\mathcal{O}_{\bar{s}}) = 1$.

Thus indeed

$\mathcal{O}((-t))\mathcal{O}(-(e)) \simeq \mathcal{O}(-(t))\mathcal{O}((e))$, i.e.

$m(-t) \simeq m(t)^{\otimes -1} \otimes \mathcal{O}(-2(e))$.

Il reste à prouver que pour t quelconque, le second membre de (2.7.2) est, localement sur S , de la forme m(x) . La suite exacte

(2.7.3) $0 \longrightarrow m(t) \longrightarrow \mathcal{O}_C \longrightarrow t_*\mathcal{O}_S \longrightarrow 0$

donne la suite exacte

(2.7.4) $0 \longrightarrow \mathrm{Hom}(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) \longrightarrow \mathrm{Hom}(\mathcal{O}_C,\mathcal{O}(-2(e))) \longrightarrow$

$\longrightarrow \mathrm{Hom}(m(t),\ \mathcal{O}(-2(e))) \longrightarrow \mathrm{Ext}^1(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) \longrightarrow 0$.

Comme $\omega_{C/S}$ est un faisceau inversible sur C (cf. 1.6.1) , $\mathcal{O}(-2(e))$ est, localement sur C, isomorphe à $\omega_{C/S}$. Donc localement sur C $\mathrm{Ext}^1(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) \simeq \mathrm{Ext}^1(t_*\mathcal{O}_S,\omega_{C/S})$. On applique maintenant la formule de dualité des faisceaux cohérents pour le morphisme t : S → C :

(2.7.5) $Rt_*R\,\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{O}_S,t^!\omega_{C/S}) \xrightarrow{\sim} R\,\mathrm{Hom}_{\mathcal{O}_C}(Rt_*\mathcal{O}_S,\omega_{C/S})$.

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$\chi(C_{\bar{s}},\mathrm{Hom}(m(t),\mathcal{O}(-2(e)))) = -1$,

and consequently $\mathrm{Hom}(m(t),\mathcal{O}(-2(e))) = m(t')$ for a section t', which completes the proof.

Corollary 2.8. Let $p : C \to S$ be a generalized elliptic curve over S. The map

$x \mapsto -x : C^{\mathrm{reg}} \to C^{\mathrm{reg}}$

extends to an automorphism of C/S.

The required extension is unique because C^reg is schematically dense in C. The problem is therefore local for the fpqc topology. It is trivial over the open subset where C is smooth. Let s̄ be a geometric point of S, localized at s ∈ S, such that C_{s̄} is an n-gon. Near this point C_n is finite and flat and, locally for the fppf topology, admits a section t whose multiples meet the various components of C_{s̄}. Near s, t generates a subgroup H of C_n isomorphic to ℤ/n. The group H_{s̄} acts freely on C_{s̄}, and C_{s̄}/H is a 1-gon. On a neighborhood of s we then have:

a) H acts freely on C, and C/H is a generalized elliptic curve over S.

b) C/H has geometrically irreducible fibers (apply 1.15 to C/H).

c) C is a finite étale covering of C/H.

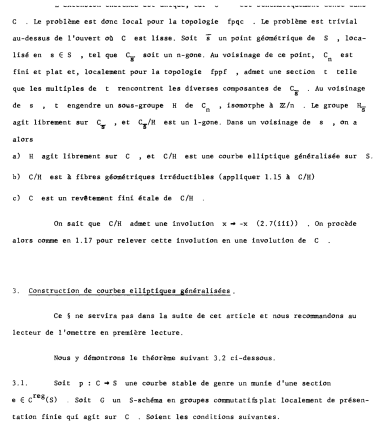
We know that C/H admits an involution x ↦ -x (2.7(iii)). We then proceed as in 1.17 to lift this involution to an involution of C.

3. Construction of generalized elliptic curves.

This section will not be used later in this article, and we recommend that the reader omit it on a first reading.

Here we prove Theorem 3.2 below.

3.1. Let p : C → S be a stable genus-one curve equipped with a section e ∈ C^reg(S). Let G be a commutative S-group scheme, flat and locally of finite presentation, acting on C. Consider the following conditions.



C . Le problème est donc local pour la topologie fpqc . Le problème est trivial au-dessus de l'ouvert où C est lisse. Soit $\bar{s}$ un point géométrique de S , localisé en s ∈ S , tel que $C_{\bar{s}}$ soit un n-gone. Au voisinage de ce point, C_n est fini et plat et, localement pour la topologie fppf , admet une section t telle que les multiples de t rencontrent les diverses composantes de $C_{\bar{s}}$. Au voisinage de s , t engendre un sous-groupe H de C_n , isomorphe à ℤ/n . Le groupe $H_{\bar{s}}$ agit librement sur $C_{\bar{s}}$, et $C_{\bar{s}}/H$ est un 1-gone. Dans un voisinage de s , on a alors

a) H agit librement sur C , et C/H est une courbe elliptique généralisée sur S.

b) C/H est à fibres géométriques irréductibles (appliquer 1.15 à C/H)

c) C est un revêtement fini étale de C/H .

On sait que C/H admet une involution x ↦ -x (2.7(iii)) . On procède alors comme en 1.17 pour relever cette involution en une involution de C .

3. Construction de courbes elliptiques généralisées.

Ce § ne servira pas dans la suite de cet article et nous recommandons au lecteur de l'omettre en première lecture.

Nous y démontrons le théorème suivant 3.2 ci-dessous.

3.1. Soit p : C → S une courbe stable de genre un munie d'une section e ∈ $C^{reg}(S)$. Soit G un S-schéma en groupes commutatifs plat localement de présentation finie qui agit sur C . Soient les conditions suivantes.

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(3.1.1) G acts trivially on Pic^o_{C/S}.

(3.1.2) For every geometric point s̄ of S, G(s̄) acts transitively on the set of irreducible components of C_{s̄}.

In the important special case where G is the constant group scheme ℤ, the action ζ of G is defined by the automorphism g = ζ(1) of C, and (3.1.1) and (3.1.2) become

(3.1.3) g acts trivially on Pic^o_{C/S}

(3.1.4) For every geometric point s̄ of S, g transitively permutes the irreducible components of S_{s̄}.

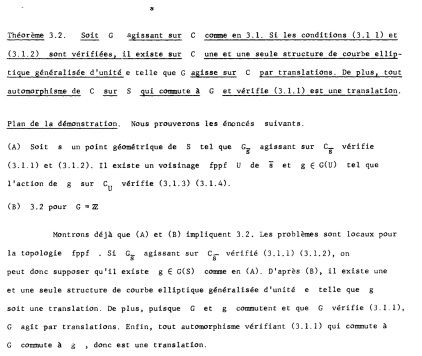
Theorem 3.2. Let G act on C as in 3.1. If conditions (3.1.1) and (3.1.2) hold, there is on C a unique generalized-elliptic-curve structure with identity e for which G acts on C by translations. Moreover, every automorphism of C over S that commutes with G and satisfies (3.1.1) is a translation.

Plan of the proof. We shall prove the following assertions.

(A) Let s be a geometric point of S such that G_{s̄} acting on C_{s̄} satisfies (3.1.1) and (3.1.2). There are an fppf neighborhood U of s̄ and an element g ∈ G(U) such that the action of g on C_U satisfies (3.1.3) and (3.1.4).

(B) Theorem 3.2 for G = $\mathbb{Z}$.

Let us first show that (A) and (B) imply 3.2. The questions are local for the fppf topology. If G_{s̄} acting on C_{s̄} satisfies (3.1.1) and (3.1.2), we may therefore suppose that there is a g ∈ G(S) as in (A). By (B), there is a unique generalized-elliptic-curve structure with identity e for which g is a translation. Moreover, since G and g commute and G satisfies (3.1.1), G acts by translations. Finally, every automorphism satisfying (3.1.1) that commutes with G also commutes with g, and hence is a translation.



Théorème 3.2. Soit G agissant sur C comme en 3.1. Si les conditions (3.1.1) et (3.1.2) sont vérifiées, il existe sur C une et une seule structure de courbe elliptique généralisée d'unité e telle que G agisse sur C par translations. De plus, tout automorphisme de C sur S qui commute à G est vérifie (3.1.1) est une translation.

Plan de la démonstration. Nous prouverons les énoncés suivants.

(A) Soit s un point géométrique de S tel que $G_{\bar{s}}$ agissant sur $C_{\bar{s}}$ vérifie (3.1.1) et (3.1.2). Il existe un voisinage fppf U de $\bar{s}$ et $g \in G(U)$ tel que l'action de g sur C_U vérifie (3.1.3) (3.1.4).

(B) 3.2 pour $G = \mathbb{Z}$

Montrons déjà que (A) et (B) impliquent 3.2. Les problèmes sont locaux pour la topologie fppf . Si $G_{\bar{s}}$ agissant sur $C_{\bar{s}}$ vérifié (3.1.1) (3.1.2), on peut donc supposer qu'il existe $g \in G(S)$ comme en (A). D'après (B), il existe une et une seule structure de courbe elliptique généralisée d'unité e telle que g soit une translation. De plus, puisque G et g commutent et que G vérifie (3.1.1), G agit par translations. Enfin, tout automorphisme vérifiant (3.1.1) qui commute à G commute à g , donc est une translation.

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Proof of (A). If C_{s̄} is smooth, take for U the open subset of S over which C is smooth, and take g = e. Otherwise C_{s̄} is an n-gon, and the image of G(s̄) in the group of automorphisms of the component diagram Γ(C_{s̄}) has order at least n. By (3.1.1), G(s̄) acts by rotations on Γ(C_{s̄}). Since the rotation group $\mathrm{Aut}^+(\Gamma(C_{\bar{s}}))$ is cyclic of order n, there is a g_{s̄} ∈ G(s) whose image generates this group, and g_{s̄} transitively permutes the irreducible components of C_{s̄}. Locally for the fppf topology there is a section g of G inducing g_{s̄}. We may apply 1.16 to g to verify (3.1.4) near s̄. That g satisfies 3.1.3 near s̄ follows from 1.13.

Proof of 3.2 for G = $\mathbb{Z}$.

Let ζ be the action of G on C and let g = ζ(1). Applying 1.16 to g, we reduce to the case in which the geometric fibers of C are all smooth or n-gons (with n fixed). In this case $g^n e$ lies fiber by fiber in the same connected component of C as e, and g^n is covered by the following lemma.

Lemma 3.3. Let h be an automorphism of C that commutes with g and satisfies (3.1.3). If, fiber by fiber, he lies in the same irreducible component of C as e, then h is of the form

$x \mapsto \lambda + x$

for a section λ of Pic^o_{C/S}.

Proof. We must have λ equal to the class of $\mathcal{O}$(he - e). Hypothesis 3.1.3 ensures that x ↦ λ + x commutes with g. Replacing h by h(x - λ), we reduce to the case he = e. Since h satisfies (3.1.3), for every section μ of Pic^o_{C/S} we have

$h(g^k(\mu+e)) = g^k(\mu+e)$.

Every section of C^reg is locally for the fppf topology of the form $g^k(\mu+e)$. Thus h acts trivially on C^reg, and hence on C.

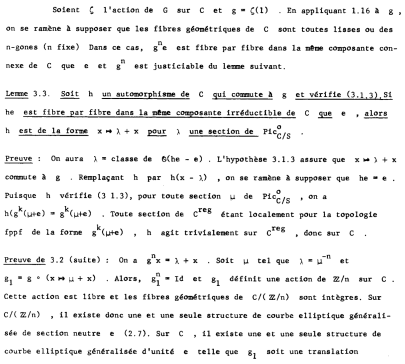
Continuation of the proof of 3.2. We have

$g^n x = \lambda + x$.

Choose μ such that $\lambda = \mu^{-n}$, and put

$g_1 = g \circ (x \mapsto \mu + x)$.

Then $g_1^n = \mathrm{Id}$ and g_1 defines an action of $\mathbb{Z}/n$ on C. This action is free, and the geometric fibers of $C/(\mathbb{Z}/n)$ are integral. On $C/(\mathbb{Z}/n)$ there is therefore a unique generalized-elliptic-curve structure with identity section e (2.7). On C there is a unique generalized-elliptic-curve structure with identity e for which g_1 is a translation



Soient ζ l'action de G sur C et g = ζ(1) . En appliquant 1.16 à g , on se ramène à supposer que les fibres géométriques de C sont toutes lisses ou des n-gones (n fixe) Dans ce cas, $g^n e$ est fibre par fibre dans la même composante connexe de C que e et g^n est justiciable du lemme suivant.

Lemme 3.3. Soit h un automorphisme de C qui commute à g et vérifie (3.1.3). Si he est fibre par fibre dans la même composante irréductible de C que e , alors h est de la forme x ↦ λ + x pour λ une section de $\mathrm{Pic}^o_{C/S}$.

Preuve : On aura λ = classe de δ(he - e) . L'hypothèse 3.1.3 assure que x ↦ λ + x commute à g . Remplaçant h par h(x - λ) , on se ramène à supposer que he = e . Puisque h vérifie (3 1.3), pour toute section μ de $\mathrm{Pic}^o_{C/S}$, on a $h(g^k(\mu+e)) = g^k(\mu+e)$. Toute section de C^{reg} étant localement pour la topologie fppf de la forme $g^k(\mu+e)$, h agit trivialement sur C^{reg} , donc sur C .

Preuve de 3.2 (suite) : On a $g^n x = \lambda + x$. Soit μ tel que $\lambda = \mu^{-n}$ et $g_1 = g \circ (x \mapsto \mu + x)$. Alors, $g_1^n = \mathrm{Id}$ et g_1 définit une action de $\mathbb{Z}/n$ sur C . Cette action est libre et les fibres géométriques de $C/(\mathbb{Z}/n)$ sont intègres. Sur $C/(\mathbb{Z}/n)$, il existe donc une et une seule structure de courbe elliptique généralisée de section neutre e (2.7). Sur C , il existe une et une seule structure de courbe elliptique généralisée d'unité e telle que g_1 soit une translation .

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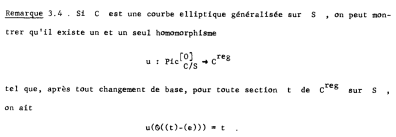
(cf. 1.17). By 3.3, for a section λ of Pic^o and for every generalized-elliptic-curve structure, (λ + e) + x equals λ + x. Thus the law obtained is also the unique one for which g is a translation. Finally, if h commutes with g and satisfies (3.1.3), locally there is a k such that hg^k is covered by 3.3, and h is a translation.

Remark 3.4. If C is a generalized elliptic curve over S, one can show that there is a unique homomorphism

u : Pic^[0]_{C/S} → C^reg

such that, after every base change, for every section t of C^reg over S,

$u(\mathcal{O}((t)-(e))) = t$.



Remarque 3.4 . Si C est une courbe elliptique généralisée sur S , on peut montrer qu'il existe un et un seul homomorphisme

$$u : \mathrm{Pic}^{[0]}_{C/S} \to C^{reg}$$

tel que, après tout changement de base, pour toute section t de C^{reg} sur S , on ait

$$u(\delta((t)-(e))) = t \ .$$

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III. Representability theorems.

0. Introduction and notation.

In this chapter and those that follow, we define and study various algebraic stacks classifying generalized elliptic curves equipped with various additional structures.

0.1. Write $\mathfrak{M}$ for the category fibered in groupoids over the category (Sch) of schemes defined as follows: $\mathfrak{M}$(S) is the category whose objects are generalized elliptic curves over S and whose morphisms are S-isomorphisms; the functor ”inverse image by u : S → T” is

E ↦ E ×_T S.

The category $\mathfrak{M}$ is a stack over (Sch) for the fpqc topology (2.1), but it is not an algebraic stack.

0.2. Let $\mathfrak{M}$_* be the following substack of $\mathfrak{M}$: $\mathfrak{M}$_* classifies generalized elliptic curves C/S such that, for every geometric point s̄ of S, the characteristic of k(s̄) does not divide the number of irreducible components of the geometric fiber C_{s̄}. The principal result 2.5 of this chapter is that $\mathfrak{M}$_* is an algebraic stack. The representability theorems of the following chapters will follow from this technical theorem. We shall deduce it from M. Artin’s general criteria. To apply them, the essential point will be to prove the effective prorepresentability of certain functors; this will be done in Section 1.

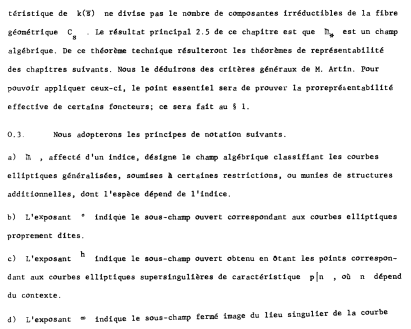
0.3. We adopt the following notational conventions.

a) $\mathfrak{M}$ with a subscript denotes the algebraic stack classifying generalized elliptic curves subject to certain restrictions or equipped with additional structures of the type indicated by the subscript.

b) The superscript o denotes the open substack corresponding to elliptic curves in the strict sense.

c) The superscript h denotes the open substack obtained by removing the points corresponding to supersingular elliptic curves in characteristic p dividing n, where n depends on the context.

d) The superscript ∞ denotes the closed substack that is the image of the singular locus of the curve



téristique de k(S) ne divise pas le nombre de composantes irréductibles de la fibre géométrique $C_{\bar{s}}$. Le résultat principal 2.5 de ce chapitre est que $\mathfrak{M}_g$ est un champ algébrique. De ce théorème technique résulteront les théorèmes de représentabilité des chapitres suivants. Nous le déduirons des critères généraux de M. Artin. Pour pouvoir appliquer ceux-ci, le point essentiel sera de prouver la prorepréSentabilité effective de certains foncteurs; ce sera fait au § 1.

0.3. Nous adopterons les principes de notation suivants.

a) $\mathfrak{M}$, affecté d'un indice, désigne le champ algébrique classifiant les courbes elliptiques généralisées, soumises à certaines restrictions, ou munies de structures additionnelles, dont l'espèce dépend de l'indice.

b) L'exposant * indique le sous-champ ouvert correspondant aux courbes elliptiques proprement dites.

c) L'exposant $^{\flat}$ indique le sous-champ ouvert obtenu en ôtant les points correspondant aux courbes elliptiques supersingulières de caractéristique $p|n$, où n dépend du contexte.

d) L'exposant $^{=}$ indique le sous-champ fermé image du lieu singulier de la courbe

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universal.

e) A notation $\mathfrak{m}[1/n]$ indicates either that a stack over $\mathbb{Z}[1/n]$ is being considered or, if $\mathfrak{m}$ has already been defined, denotes $\mathfrak{m} \otimes \mathbb{Z}[1/n]$.

f) The coarse moduli scheme defined by an algebraic stack (I.8) is denoted by the same symbol, with $\mathfrak{m}$ replaced by M.

We shall extend these notations to certain stacks over $\mathfrak{m}$, not necessarily defined by a "moduli problem" stricto sensu.

1. A prorepresentability theorem.

1.1 Let Λ be a complete local ring with residue field k and let C_o be a generalized elliptic curve over k. Assume that C_o is smooth or is the base change to k of the standard generalized elliptic curve with n sides (II.1.9). Let D be the deformation functor of C_o: for a complete local Noetherian Λ-algebra A with residue field k, D(A) is the set of isomorphism classes of generalized elliptic curves C/Spec(A), equipped with an isomorphism $C \otimes_A k \simeq C_o$. This section is devoted to the proof of the following theorem.

Theorem 1.2. Suppose that C_o is smooth or that n is prime to the characteristic exponent p of k. Then:

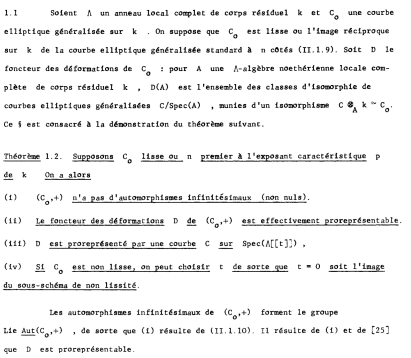
(i) (C_o,+) has no nonzero infinitesimal automorphisms.

(ii) The deformation functor D of (C_o,+) is effectively prorepresentable.

(iii) D is prorepresented by a curve C over $\mathrm{Spec}(\Lambda[[t]])$.

(iv) If C_o is not smooth, t may be chosen so that $t = 0$ is the image of the non-smoothness subscheme.

The infinitesimal automorphisms of (C_o,+) form the group Lie Aut(C_o,+), so (i) follows from (II.1.10). It follows from (i) and [25] that D is prorepresentable.



1.1 Soient Λ un anneau local complet de corps résiduel k et C_o une courbe elliptique généralisée sur k. On suppose que C_o est lisse ou l'image réciproque sur k de la courbe elliptique généralisée standard à n côtés (II.1.9). Soit D le foncteur des déformations de C_o : pour A une Λ-algèbre noethérienne locale complète de corps résiduel k, $D(A)$ est l'ensemble des classes d'isomorphie de courbes elliptiques généralisées $C/\mathrm{Spec}(A)$, munies d'un isomorphisme $C \otimes_A k \simeq C_o$. Ce § est consacré à la démonstration du théorème suivant.

Théorème 1.2. Supposons C_o lisse ou n premier à l'exposant caractéristique p de k. On a alors

(i) $(C_o,+)$ n'a pas d'automorphismes infinitésimaux (non nuls).

(ii) Le foncteur des déformations D de $(C_o,+)$ est effectivement prorepresentable.

(iii) D est prorepresenté par une courbe C sur $\mathrm{Spec}(\Lambda[[t]])$,

(iv) Si C_o est non lisse, on peut choisir t de sorte que $t = 0$ soit l'image du sous-schéma de non lissité.

Les automorphismes infinitésimaux de $(C_o,+)$ forment le groupe Lie $\mathrm{Aut}(C_o,+)$, de sorte que (i) résulte de (II.1.10). Il résulte de (i) et de [25] que D est proreprésentable.

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1.3. The case in which C_o is irreducible.

In this case the deformation functor of (C_o,+) is identified by II.2.7 with the functor D' of deformations of (C_o,e), the pair consisting of the curve C_o and the marked section e. This latter functor is effectively prorepresentable, since a formal scheme in complete curves is always projective and hence algebraizable. That D' satisfies (iii) follows from the two formulas

(1.3.1) $\dim \mathrm{Ext}^1(\Omega^1_{C_o}(e), \mathcal{O}_{C_o}) = 1$

(1.3.2) $\dim \mathrm{Ext}^2(\Omega^1_{C_o}(e), \mathcal{O}_{C_o}) = 0$.

Indeed, Ext^2 is the group in which the obstructions to deformations live. If it vanishes, D' is prorepresented by a formal power-series algebra over Λ. Ext^1 is identified with $D'(k[\varepsilon]/(\varepsilon^2))$ and determines the relative dimension.

Let us prove (1.3.1) and (1.3.2). Since ω_{C_o} is isomorphic to $\mathcal{O}_{C_o}$, duality (I.2.2) gives

$\dim \mathrm{Ext}^i(\Omega^1_{C_o}(e), \mathcal{O}_{C_o}) = \dim H^{1-i}(\Omega^1_{C_o}(e))$.

Formula (1.3.2) follows trivially. For (1.3.1), we may suppose that k is algebraically closed.

If C_o is smooth, $\Omega_{C_o}(e) \simeq \omega_{C_o}(e) \simeq \mathcal{O}_{C_o}(e)$ is an invertible sheaf of degree 1, whence $\dim H^0(C_o, \Omega_{C_o}(e)) = 1$ in this case.

Otherwise, the normalization $\pi : \tilde{C}_o \to C_o$ of C_o is isomorphic to $\mathbb{P}^1$, and a local calculation shows that the canonical morphism

$\Omega_{C_o}(e) \to \pi_* \Omega_{\tilde{C}_o}(e)$

is surjective with a kernel of length one, concentrated at the unique singular point of C_o. In this case the assertion follows from the long exact cohomology sequence: we use that

Ext s'identifie à $D'(k[\varepsilon]/(\varepsilon^2))$, et détermine la dimension relative.

Prouvons (1.3.1) et (1.3.2). Puisque ω_{C_o} est isomorphe à $\mathcal{O}_{C_o}$, on a par dualité (I.2.2)

$$\dim \mathrm{Ext}^1(\Omega^1_{C_o}(e), \mathcal{O}_{C_o}) = \dim H^{1-1}(\Omega^1_{C_o}(e)) .$$

La formule (1.3.2) en résulte trivialement. Pour (1.3.1), on peut supposer k algébriquement clos.

Si C_o est lisse, $\Omega_{C_o}(e) \simeq \omega_{C_o}(e) \simeq \mathcal{O}_{C_o}(e)$ est un faisceau inversible de degré 1, d'où $\dim H^0(C_o, \Omega_{C_o}(e)) = 1$ dans ce cas.

Sinon, le normalisé $\pi : \mathbb{P}^1_k \to C_o$ de C_o est isomorphe à $\mathbb{P}^1$ et un calcul local montre que le morphisme canonique

$$\Omega_{C_o}(e) \longrightarrow \pi_* \Omega_{\mathbb{P}^1}(e)$$

est surjectif à noyau de longueur un, concentré au seul point singulier de C_o. L'assertion dans ce cas résulte de la suite exacte de cohomologie : on utilise que

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$H^0(C, \pi_* \Omega_C(e)) \simeq H^0(\mathbb{P}^1, \Omega_{\mathbb{P}^1}(e)) = 0$ (since $\deg_{\mathbb{P}^1}(e) = -1$).

This proves (iii).

Let us prove (iv). Here C_o is the Néron 1-gon. The deformation theory of ordinary double points shows that there is a $u \in \Lambda[[t]]$ such that $u = 0$ is the image of the non-smooth locus of C. By (II.1.15), over $\mathrm{Spec}(\Lambda[[t]]/(u))$, C is isomorphic to the base change of the standard Néron 1-gon (1.1). The universal property of $\Lambda[[t]]$ then implies that $\mathrm{Spec}(\Lambda[[t]]/(u))$ is unramified over $\mathrm{Spec}(\Lambda)$, and hence that $\Lambda[[t]] \simeq \Lambda[[u]]$. This completes the proof for irreducible C_o.

1.4 The general case.

We may suppose that C_o is the standard Néron n-gon over k, obtained from $\mathbb{P}^1 \times \mathbb{Z}/n$ by gluing. Let H_o be the cyclic subgroup of order n in C_o, the image of $\{1\} \times \mathbb{Z}/n$.

Let D'' be the deformation functor of $(C_o,+,H_o)$: $D''(A)$ is the set of isomorphism classes of systems $(C,+,H)$, where $(C,+)$ is a generalized elliptic curve over A and H is a subgroup of C^{reg} isomorphic to $\mathbb{Z}/n$, equipped with an isomorphism of their special fiber with $(C_o,+,H_o)$.

Lemma 1.4.1. We have $D'' \simeq D$.

This follows because C_n^{reg} is finite étale over A (since n is prime to p).

1.4.2 For the rest of the proof we no longer suppose that n is prime to p (this hypothesis will be used only through 1.4.1). Let $(C,+,H)$ be as above (over a complete local A). The group H acts freely on C (because H_o acts freely on C_o), so C/H (which exists because C is projective) is flat over A. The law $+$ descends to the quotient, and $(C/H,+)$ is a deformation of the Néron 1-gon $(C_o/H_o,+)$. Writing D_1 for the deformation functor of $(C_o/H_o,+)$, we obtain

$\Lambda[[t]] \simeq \Lambda[[u]]$. Ceci achève la démonstration pour C_o irréductible.

1.4 Le cas général.

On peut supposer que C_o est le polygone de Néron standard à n côtés sur k, déduit de $\mathbb{P}^1_k \times \mathbb{Z}/n$ par recollement. Soit H_o le sous-groupe cyclique d'ordre n de C_o, image de $\{1\} \times \mathbb{Z}/n$.

Soit D'' le foncteur des déformations de $(C_o, +, H_o)$: $D''(A)$ est l'ensemble des classes d'isomorphie de systèmes $(C,+,H)$ ($(C,+)$ courbe elliptique généralisée sur A, H sous-groupe de C^{reg} isomorphe à $\mathbb{Z}/n$), munis d'un isomorphisme de leur fibre spéciale avec $(C_o,+,H_o)$.

Lemme 1.4.1. On a $D'' \xrightarrow{\sim} D$.

Cela résulte de ce que C_n^{reg} est fini étale sur A (car n est premier à p).

1.4.2 Dans la fin de la démonstration, on ne suppose plus que n est premier à p (cette hypothèse ne servira que via 1.4.1). Soit $(C,+,H)$ comme plus haut (sur A local complet). Le groupe H agit librement sur C (car H_o agit librement sur C_o), de sorte que C/H (qui existe car C est projectif) est plat sur A. La loi $+$ passe au quotient, et $(C/H,+)$ est une déformation du polygone de Néron à n côté $(C_o/H_o,+)$. Notant D_1 le foncteur des déformations de $(C_o/H_o,+)$, on obtient

in this way a morphism from D” to D_1.

Lemma 1.4.3. The morphism 1.4.2 from D” to D_1 is an isomorphism (here n is not assumed prime to p).

By EGA IV.18.1.2 when A is Artinian, together with the existence theorem in formal geometry, EGA III.5, when A is complete local, for every deformation (C,+) of (C_o/H_o,+) over A there exists one and only one deformation C of the curve C_o over A, equipped with $\pi_o : C_o \to C_o/H_o$. We then apply II.1.17.

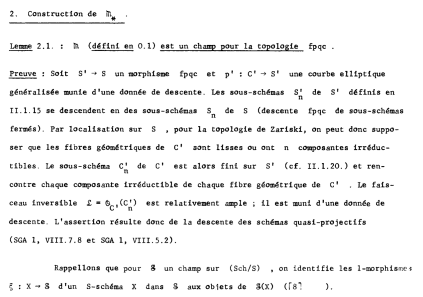
Let us complete the proof of 1.2. By 1.4.1 and 1.4.3, D is isomorphic to D_1, studied in 1.3, and therefore satisfies 1.2(ii) and (iii). Assertion (iv) follows from the analogous assertion for D_1, because the image of the non-smoothness subscheme is the same for C and C/H (notation of 1.4.2).

2. Construction of $\mathfrak{m}_*$.

Lemma 2.1. The stack $\mathfrak{m}$ (defined in 0.1) is a stack for the fpqc topology.

Proof. Let $S' \to S$ be an fpqc morphism and let $p' : C' \to S'$ be a generalized elliptic curve equipped with descent data. The subschemes S’_n of S’ defined in II.1.15 descend to subschemes S_n of S (fpqc descent of closed subschemes). By localizing on S for the Zariski topology, we may therefore suppose that the geometric fibers of C’ are smooth or have n irreducible components. The subscheme C’_n of C’ is then finite over S’ (cf. II.1.20) and meets every irreducible component of every geometric fiber of C’. The invertible sheaf $\mathcal{L} = \mathcal{O}_{C'}(C'_n)$ is relatively ample and is equipped with descent data. The assertion thus follows from descent for quasi-projective schemes (SGA 1, VIII.7.8 and SGA 1, VIII.5.2).

Recall that for a stack $\mathcal{G}$ over (Sch/S), the 1-morphisms $\xi : X \to \mathcal{G}$ from an S-scheme X to $\mathcal{G}$ are identified with the objects of $\mathcal{G}(X)$ ([8]).



2. Construction de $\mathfrak{M}_*$.

Lemme 2.1. : $\mathfrak{M}$ (défini en 0.1) est un champ pour la topologie fpqc .

Preuve : Soit $S' \to S$ un morphisme fpqc et $p' : C' \to S'$ une courbe elliptique généralisée munie d'une donnée de descente. Les sous-schémas S'_n de S' définis en II.1.15 se descendent en des sous-schémas S_n de S (descente fpqc de sous-schémas fermés). Par localisation sur S , pour la topologie de Zariski, on peut donc supposer que les fibres géométriques de C' sont lisses ou ont n composantes irréductibles. Le sous-schéma C'_n de C' est alors fini sur S' (cf. II.1.20.) et rencontre chaque composante irréductible de chaque fibre géométrique de C' . Le faisceau inversible $\mathcal{L} = \mathcal{O}_{C'}(C'_n)$ est relativement ample ; il est muni d'une donnée de descente. L'assertion résulte donc de la descente des schémas quasi-projectifs (SGA 1, VIII.7.8 et SGA 1, VIII.5.2).

Rappellons que pour $\mathcal{G}$ un champ sur (Sch/S) , on identifie les 1-morphismes $\xi : X \to \mathcal{G}$ d'un S-schéma X dans $\mathcal{G}$ aux objets de $\mathcal{G}(X)$ ([8]).

2.2. M. Artin proved a manageable criterion for showing that a functor is representable by an algebraic space ([2], [3]). The same proof gives the following criterion for verifying that a stack is algebraic.

Theorem 2.3 (M. Artin). Let S be a scheme of finite type over a field or over an excellent Dedekind ring. Let $\mathcal{G}$ be a category fibered in groupoids over (Sch/S).

$\mathcal{G}$ is an algebraic stack locally of finite type over S if (and only if):

(0) $\mathcal{G}$ is a stack for the étale topology.

(1) $\mathcal{G}$ is locally of finite presentation; that is, for every filtered projective system of affine S-schemes Spec(A_i), the canonical functor

lim→_i $\mathcal{G}$(Spec(A_i)) → $\mathcal{G}$(lim←_i Spec(A_i))

defines an equivalence of categories.

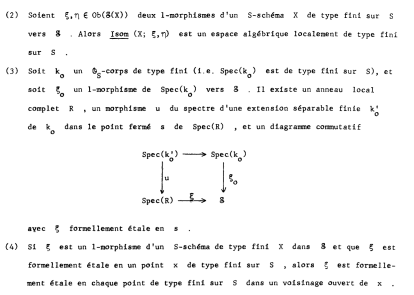
(2) Let ξ,η ∈ Ob($\mathcal{G}$(X)) be two 1-morphisms from an S-scheme X of finite type over S to $\mathcal{G}$. Then Isom(X; ξ,η) is an algebraic space locally of finite type over S.

(3) Let k_o be an $\mathcal{O}$_S-field of finite type (that is, Spec(k_o) is of finite type over S), and let ξ_o be a 1-morphism from Spec(k_o) to $\mathcal{G}$. There exist a complete local ring R, a morphism u from the spectrum of a finite separable extension k'_o of k_o into the closed point s of Spec(R), and a commutative diagram

Spec(k'_o) → Spec(k_o) | | u | | ξ_o ↓ ↓ Spec(R) —ξ—→ $\mathcal{G}$

with ξ formally étale at s.

(4) If ξ is a 1-morphism from an S-scheme X of finite type into $\mathcal{G}$ and ξ is formally étale at a point x of finite type over S, then ξ is formally étale at every point of finite type over S in an open neighborhood of x.


(2) Soient $\xi,\eta \in \mathrm{Ob}(\mathcal{G}(X))$ deux 1-morphismes d'un S-schéma X de type fini sur S vers $\mathcal{G}$. Alors $\underline{\mathrm{Isom}}(X;\xi,\eta)$ est un espace algébrique localement de type fini sur S.

(3) Soit k_o un $\mathcal{O}_S$-corps de type fini (i.e. $\mathrm{Spec}(k_o)$ est de type fini sur S), et soit ξ_o un 1-morphisme de $\mathrm{Spec}(k_o)$ vers $\mathcal{G}$. Il existe un anneau local complet R, un morphisme u du spectre d'une extension séparable finie k'_o de k_o dans le point fermé s de Spec(R), et un diagramme commutatif

$$\begin{array}{ccc} \mathrm{Spec}(k'_o) & \longrightarrow & \mathrm{Spec}(k_o) \\ \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle \xi_o} \\ \mathrm{Spec}(R) & \xrightarrow{\xi} & \mathcal{G} \end{array}$$

avec ξ formellement étale en s.

(4) Si ξ est un 1-morphisme d'un S-schéma de type fini X dans $\mathcal{G}$ et que ξ est formellement étale en un point x de type fini sur S, alors ξ est formellement étale en chaque point de type fini sur S dans un voisinage ouvert de x.

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Remarks 2.4.

(i) If the residue fields of finite type of S are perfect, then, to establish that $\mathcal{G}$ is an algebraic stack, one may replace (0)-(4) by (0), (1), (2), (3'), and (4), where (3') is the following condition.

(3') Let s ∈ S and let k_o be a finite extension of k(s). Suppose that u : Spec(k_o) → S is of finite type. By hypothesis k(s) is then perfect and k_o/k(s) is separable. There is therefore a unique complete local ring Λ(k_o) with residue field k_o, equipped with a formally étale morphism ū : Spec(Λ(k_o)) → S extending u.

For ξ_o ∈ Ob $\mathcal{G}$(k_o), write Def(ξ_o) for the following category over the dual of $\hat{C}$_{Λ(k_o)}: for A ∈ Ob $\hat{C}$_{Λ(k_o)}, an object of Def(ξ_o)(A) is an object ξ of $\mathcal{G}$(A), equipped with an isomorphism ξ_o $\tilde{\to}$ (the image of ξ in $\mathcal{G}$(k_o)). Condition (3') is the conjunction of (3'a) and (3'b):

(3'a) The objects of $\mathcal{G}$ have no infinitesimal automorphisms; more precisely, for k_o and ξ_o as above, and for ξ'_o the pullback of ξ_o to k_o[ε] (ε² = 0), one has Aut(ξ'_o) $\tilde{\to}$

Aut(ξ_o). This condition, together with (2), implies that the objects of Def(ξ_o)(A) have no nontrivial automorphisms.

(3'b) For k_o and ξ_o as above, the functor

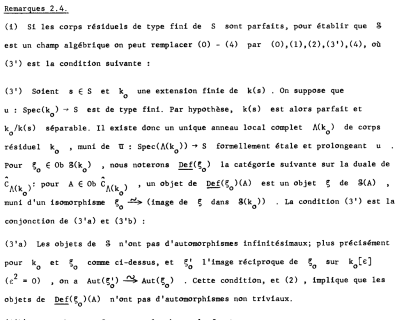
A → the set of isomorphism classes in Def(ξ_o)(A)

is effectively prorepresentable (by a complete local ring R and ξ ∈ $\mathcal{G}$(R) reducing to ξ_o).

It even suffices to verify (3'b) after extension of scalars to a finite extension of k_o.

(ii) If S is of finite type over a field or over an excellent Dedekind ring having infinitely many points, and if the complete local rings R that occur in condition (3'b) are all normal and of the same Krull dimension, condition (4) may be omitted.

(For a reader familiar with [2], (i) is easy to prove. For (ii), see [2], Thm. 3.9).



Remarques 2.4.

(1) Si les corps résiduels de type fini de S sont parfaits, pour établir que S est un champ algébrique on peut remplacer (0) - (4) par (0),(1),(2),(3'),(4), où (3') est la condition suivante :

(3') Soient $s \in S$ et k_o une extension finie de k(s). On suppose que $u : \mathrm{Spec}(k_o) \to S$ est de type fini. Par hypothèse, k(s) est alors parfait et $k_o/k(s)$ séparable. Il existe donc un unique anneau local complet $\Lambda(k_o)$ de corps résiduel k_o, muni de $\pi : \mathrm{Spec}(\Lambda(k_o)) \to S$ formellement étale et prolongeant u. Pour $\xi_o \in \mathrm{Ob}\, \mathcal{S}(k_o)$, nous noterons $\underline{\mathrm{Def}}(\xi_o)$ la catégorie suivante sur la duale de $\hat{C}_{\Lambda(k_o)}$: pour $A \in \mathrm{Ob}\, \hat{C}_{\Lambda(k_o)}$, un objet de $\underline{\mathrm{Def}}(\xi_o)(A)$ est un objet ξ de $\mathcal{S}(A)$, muni d'un isomorphisme $\xi_o \xrightarrow{\sim}$ (image de ξ dans $\mathcal{S}(k_o)$). La condition (3') est la conjonction de (3'a) et (3'b) :

(3'a) Les objets de $\mathcal{S}$ n'ont pas d'automorphismes infinitésimaux; plus précisément pour k_o et ξ_o comme ci-dessus, et ξ'_o l'image réciproque de ξ_o sur $k_o[\varepsilon]$ ($\varepsilon^2 = 0$), on a $\mathrm{Aut}(\xi'_o) \xrightarrow{\sim} \mathrm{Aut}(\xi_o)$. Cette condition, et (2), implique que les objets de $\underline{\mathrm{Def}}(\xi_o)(A)$ n'ont pas d'automorphismes non triviaux.

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The standard Néron n-gon defines a morphism

f_n : Spec(ℤ[1/n]) → 𝔪^*.

Theorem 2.5.

(i) 𝔪_* (defined in 0.2) is an algebraic stack smooth over Spec(ℤ).

(ii) The image 𝔪_*^∞ of the non-smoothness scheme of the universal generalized elliptic curve over 𝔪_* is the union of the images of f_n (n ≥ 1).

Proof. To prove that 𝔪_* is an algebraic stack, we use criterion 2.3 (with S = Spec(ℤ)). Let us review conditions (0) through (4).

(0) follows from 2.1.

(1) follows from standard arguments in EGA IV.8.8.

(2) Let X be an S-scheme of finite type and let ($p_1 : C_1 \to X,+$), ($p_2 : C_2 \to X,+$) be two generalized elliptic curves. Let us prove that Isom_X((C₁,+),(C₂,+)) is representable. This functor is an fpqc sheaf (2.1), so the problem is local. We may therefore suppose that the geometric fibers of C₁ are smooth or n-sided, for a suitable n, and that there is a φ : ℤ/n ↪ C₁ whose image meets every irreducible component of every geometric fiber of C₁. The morphism f ↦ f∘φ(1) maps Isom into the open subscheme of C_n^reg consisting of those g for which the g^k meet all components of the geometric fibers. The following assertion implies that this morphism is relatively representable and hence that Isom is representable.

(*) For C_i with smooth or n-sided fibers, equipped with H_i ⊂ C_i^reg, with H_i isomorphic to $\mathbb{Z}/n$ and meeting all components of all geometric fibers (i=1,2), Isom((C₁,H₁,+),(C₂,H₂,+)) is representable.

By II 1.17, the morphism

Isom((C₁,H₁,+),(C₂,H₂,+)) → Isom((C₁/H₁,e),(C₂/H₂,e))

is representable by an étale morphism, while the theory of the Hilbert functor implies that Isom((C₁/H₁,e),(C₂/H₂,e)) is representable.

(avec S = Spec(ℤ)) . Passons en revue les conditions (0) à (4) .

(0) résulte de 2.1.

(1) résulte par des arguments standard de EGA IV.8.8.

(2) Soient X un S-schéma de type fini et $(p_1 : C_1 \to X,+)$, $(p_2 : C_2 \to X,+)$ deux courbes elliptiques généralisées. Prouvons que $\mathrm{Isom}_X((C_1,+),(C_2,+))$ est représentable. Ce foncteur est un faisceau fpqc (2.1) de sorte que le problème est local. On peut donc supposer que les fibres géométriques de C_1 sont lisses ou à n côtés (n convenable) et qu'il existe $\varphi : \underline{\mathbb{Z}/n} \hookrightarrow C_1$ dont l'image rencontre chaque composante irréductible de chaque fibre géométrique de C_1 . Le morphisme $f \to f\circ\varphi(1)$ envoye $\underline{\mathrm{Isom}}$ dans le sous-schéma ouvert de C_n^{reg} formé des g tels que les g^k rencontrent toutes les composantes des fibres géométriques. Il résulte de l'assertion suivante que ce morphisme est relativement représentable, donc $\underline{\mathrm{Isom}}$ représentable.

(*) Pour C_i à fibres lisses ou à n côté, et muni de $H_i \subset C_i^{reg}$, H_i isomorphe à $\underline{\mathbb{Z}/n}$, rencontrant toutes les composantes de toutes les fibres géométriques (i=1,2), $\mathrm{Isom}((C_1,H_1,+),(C_2,H_2,+))$ est représentable.

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(3’) Let k_o be a Spec(ℤ)-field of finite type; it is therefore a finite field.

(3’)a) follows from 1.2(i).

(3’)b) follows from 1.2(ii).

(4) By 1.2(iii), the deformation functors are prorepresented by regular rings of dimension 2. By 2.4(ii), condition (4) is verified automatically.

The smoothness of $\mathfrak{m}_*$ follows from its formal smoothness in 1.2(iii). The second assertion of 2.5 follows from II 1.15 and 1.2(iv).

Remark 2.6. $\mathfrak{m}_*$ is horribly non-separated! It will follow from (IV.2.2) that the open substack $\mathfrak{m}_1$ of $\mathfrak{m}_*$ classifying generalized elliptic curves C/S with integral geometric fibers is proper and smooth over ℤ. The ”catch-all” $\mathfrak{m}_*$ will be useful to us for compactifying moduli of elliptic curves with level structure.

The open substack $\mathfrak{m}_(n)[1/n]$ of $\mathfrak{m}_*[1/n]$ classifying generalized elliptic curves C/S, with n invertible on S, whose geometric fibers are smooth or Néron n-gons is likewise proper and smooth over Spec(ℤ[1/n]). It is, however, less natural than $\mathfrak{m}_1[1/n]$.

Remarque 2.6. $\mathfrak{M}_*$ est horriblement non séparé! Il résultera de (IV.2.2) que le sous-champ ouvert $\mathfrak{M}_1$ de $\mathfrak{M}_*$ classifiant les courbes elliptiques généralisées C/S à fibres géométriques intègres est propre et lisse sur ℤ . Le "fourre-tout" $\mathfrak{M}_*$ nous sera utile pour compactifier des modules de courbes elliptiques avec niveau.

Le sous-champ ouvert $\mathfrak{M}_{(n)}[1/n]$ de $\mathfrak{M}_*[1/n]$ qui classifie les courbes elliptiques généralisées C/S (n inversible sur S) à fibres géométriques lisses ou des polygones de Néron à n côtés est lui aussi propre et lisse sur Spec(ℤ[1/n]). Il est toutefois moins naturel que $\mathfrak{M}_1[1/n]$.

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IV. Level structures.

In this chapter we define and study the moduli schemes of elliptic curves whose n-division points are equipped with additional structures. We study in detail the structure at infinity of these schemes. Their reduction modulo p, for p|n, will not be considered seriously until the following chapter.

1. Contractions.

1.1. Let p : C → S be a stable curve of genus one, and let H ⊂ C^reg be a finite locally free subscheme over S of the smooth locus of C. Fiber by fiber, one verifies that H is automatically a Cartier divisor on C. We suppose that the fibers of H are nonempty.

Proposition 1.2. There exists a unique S-scheme $\bar{C}$, equipped with u : C → $\bar{C}$, such that

(a) $\bar{C}$ is a stable curve of genus one over S;

(b) for every geometric point $\bar{s}$ of S, $\bar{C}_{\bar{s}}$ is obtained from $C_{\bar{s}}$ by contracting to a point each irreducible component of $C_{\bar{s}}$ that does not meet $H_{\bar{s}}$.

A. Existence. A limit argument reduces us to the case in which S is Noetherian. For every $\bar{s}$ and every n > 0,

$H^0(C_{\bar{s}}, \mathcal{O}(-nH_{\bar{s}})) = 0$.

By duality (I 2.2 and II 1.2), it follows that

$H^1(C_{\bar{s}}, \mathcal{O}(nH_{\bar{s}})) = 0$.

By EGA III 7.8, the $p_*\mathcal{O}(nH)$ for n > 0 are locally free and their formation is compatible with every base change. The same assertion holds for n = 0 (II.1.6). Let G be the homogeneous algebra

$G = \oplus_{n>0} p_* \mathcal{O}(nH)$.

When $C_{\bar{s}}$ is irreducible, $\mathcal{O}(H_{\bar{s}})$ is ample, and $G_{\bar{s}}$ is generated by

$\oplus_{n\leq A} H^0(\mathcal{O}(nH_{\bar{s}}))$,

where A is independent of s.

Suppose that $C_{\bar{s}}$ is a Néron polygon, and let $\bar{C}_{\bar{s}}$ be the Néron polygon obtained from it by contracting to a point every irreducible component that

1.1. Soit p : C → S une courbe stable de genre un et H ⊂ C^{reg} un sous-schéma fini localement libre sur S de l'ouvert de lissité de C. On vérifie fibre par fibre que H est automatiquement un diviseur de Cartier sur C. On suppose que les fibres de H sont non vides.

Proposition 1.2. Il existe un et un seul S-schéma $\bar{C}$, muni de u : C → $\bar{C}$, tel que

(a) $\bar{C}$ est une courbe stable de genre un sur S ;

(b) pour tout point géométrique $\bar{s}$ de S, $\bar{C}_{\bar{s}}$ se déduit de $C_{\bar{s}}$ en contractant en un point chacune des composantes irréductibles de $C_{\bar{s}}$ qui ne rencontrent pas $H_{\bar{s}}$.

A. Existence. Un argument de passage à la limite nous ramène à supposer S noethérien. Pour tout $\bar{s}$ et tout n > 0, on a $H^0(C_{\bar{s}}, \mathcal{O}(-nH_{\bar{s}})) = 0$. Par dualité

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does not meet H. One verifies that

$H^0(\bar{C}_{\bar{s}}, \mathcal{O}(n\bar{H}_{\bar{s}})) \xrightarrow{\sim} H^0(C_{\bar{s}}, \mathcal{O}(nH_{\bar{s}}))$.

On $\bar{C}_{\bar{s}}$, $\mathcal{O}(n\bar{H}_{\bar{s}})$ is ample, and $\bar{G}_{\bar{s}}$ is generated by

$\oplus_{n\leq B} H^0(\mathcal{O}(nH_{\bar{s}}))$,

where B is independent of s.

The homogeneous algebra G = ⊕_n p_*$\mathcal{O}$(nH) is therefore of finite type, and its homogeneous spectrum $\bar{C}$ is proper and flat over S, with formation compatible with every base change. The natural morphism u : C → $\bar{C}$ satisfies (a) and (b).

B. Uniqueness. Let

u C ————→ $\bar{C}$ ╲ p / p̄ ↘ ↙ S

satisfy (i) and (ii). Fiber by fiber, one verifies that u induces an isomorphism

$u^{-1}(\bar{C}^{reg}) \xrightarrow{\sim} C^{reg}$ (EGA IV 17.8.2).

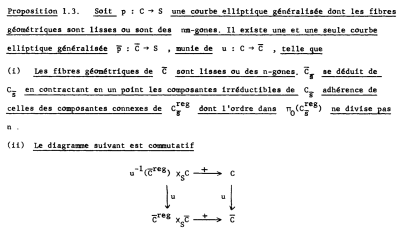
The arguments of A show that $\mathcal{O}$_$\bar{C}$(H) is ample and that $\bar{G}$ = ⊕ p̄_*$\mathcal{O}$_$\bar{C}$(nH) is flat and commutes with every base change. Fiber by fiber one verifies that $\bar{G} \xrightarrow{\sim} G$ and that $\bar{C}$ = Sph($\bar{G}$).

Proposition 1.3. Let p : C → S be a generalized elliptic curve whose geometric fibers are smooth or are nm-gons. There exists a unique generalized elliptic curve p̄ : $\bar{C}$ → S, equipped with u : C → $\bar{C}$, such that

(i) The geometric fibers of $\bar{C}$ are smooth or n-gons. $\bar{C}$_s̄ is obtained from C_s̄ by contracting to a point the irreducible components of C_s̄ that are the closures of those connected components of C_s̄^reg whose order in $\pi_0(C_{\bar{s}}^{reg})$ does not divide n.

(ii) The following diagram is commutative:

$u^{-1}(\bar{C}^{reg}) \times_S C$ —+—→ C │ u │ u ↓ ↓ $\bar{C}^{reg} \times_S \bar{C}$ —+—→ $\bar{C}$



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Authority page 65; collected-paper page 558; original folio 207.

Locally for the fppf topology on S, there exists an H ⊂ C as in 1.1 that meets exactly those irreducible components of the C_s̄ that we do not wish to contract; one may take H = C_n, which is finite and flat by (II.1.20). By 1.2, there exists a unique stable curve of genus one $\bar{C}$, equipped with u : C → $\bar{C}$, that satisfies (i). The subgroup $u^{-1}(\bar{C}^{reg})$ of C^reg acts on $\bar{C}$ by transport of structure; since $u^{-1}(\bar{C}^{reg}) \xrightarrow{\sim} C^{reg}$, this gives the desired generalized elliptic-curve structure on $\bar{C}$.

Example 1.4. Let C/S be a generalized elliptic curve and apply 1.3 with n = 1. Locally on S, the hypothesis of 1.3 is satisfied for a suitable m (II.1.15). Hence there exists a unique generalized elliptic curve c(C) over S, with irreducible geometric fibers, equipped with u : C → c(C), such that u induces an isomorphism

(1.4.1) $(C^{reg})^0 \xrightarrow{\sim} c(C)^{reg}$.

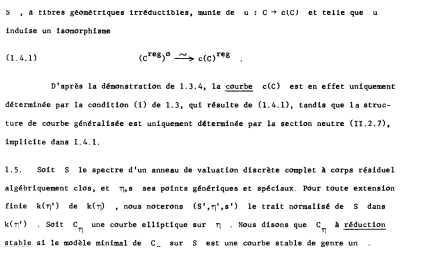
According to the proof of 1.3.4, the curve c(C) is in fact uniquely determined by condition (i) of 1.3, which follows from (1.4.1), while the generalized-curve structure is uniquely determined by the neutral section (II.2.7), implicit in 1.4.1.

1.5. Let S be the spectrum of a complete discrete valuation ring with algebraically closed residue field, and let η and s be its generic and special points. For every finite extension k(η') of k(η), write (S',η',s') for the normalization trait of S in k(η'). Let C_η be an elliptic curve over η. We say that C_η has stable reduction if the minimal model of C_η over S is a stable curve of genus one.

Proposition 1.6. (i) There exists a finite extension k(η') of k(η) such that the curve C_η' = C_η ⊗_η η' has stable reduction over η'.

(ii) If C_η has stable reduction, the minimal model C̃ of C_η over S has a unique generalized elliptic-curve structure extending that of C_η.

(iii) If, in addition, the points of order n of C_η are defined over k(η), then either the minimal model C̃ of C_η is smooth or C̃_s is an m-gon, with n|m, and there



S , à fibres géométriques irréductibles, munie de u : C → c(C) et telle que u induise un isomorphisme

(1.4.1) $(C^{reg})^0 \xrightarrow{\sim} c(C)^{reg}$.

D'après la démonstration de 1.3.4, la courbe c(C) est en effet uniquement déterminée par la condition (i) de 1.3, qui résulte de (1.4.1), tandis que la structure de courbe généralisée est uniquement déterminée par la section neutre (II.2.7), implicite dans 1.4.1.

1.5. Soit S le spectre d'un anneau de valuation discrète complet à corps résiduel algébriquement clos, et η,s ses points génériques et spéciaux. Pour toute extension finie k(η') de k(η) , nous noterons (S',η',s') le trait normalisé de S dans k(η') . Soit C_η une courbe elliptique sur η . Nous disons que C_η à réduction stable si le modèle minimal de C_ sur S est une courbe stable de genre un .

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exists a generalized elliptic curve C over S whose special fiber is smooth or an n-gon, together with an isomorphism α from its generic fiber to C_η.

(iv) The pair (C,α) is unique up to a unique isomorphism.

Proof. (i) This is a theorem of Néron and Kodaira ([4] or [20] or [14]).

(ii) The universal property of the Néron model ensures that e ∈ C_η(η) extends to a section of C̃^reg over S. If u ∈ C̃^reg(S), the translation u+ on C_η extends by transport of structure to an automorphism of C̃; similarly, x → −x extends to C̃.

Let U be the largest open subset of C̃ such that the group law + of C_η extends to + : U × C → C. It is enough to prove that U = C̃^reg. Once this is established, the identities expressing that + is a generalized elliptic-curve structure follow from the schematic density of C_η in C̃. For v ∈ C̃^reg(S), U is stable under translation by v: extend + to v+U by (v+u) + x = v+(u+x). Since U is also stable under étale localization on S, as follows by descent and the compatibility of minimal-model formation with étale localization on S, it is enough to show that U_s ≠ ∅.

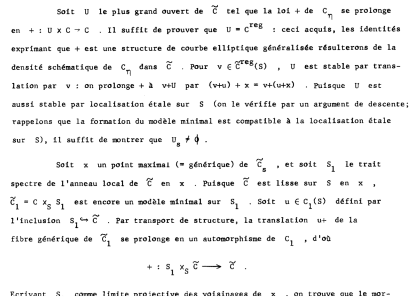
Let x be a maximal, hence generic, point of C̃_s, and let S_1 be the trait given by the spectrum of the local ring of C̃ at x. Since C̃ is smooth over S at x, $\tilde{C}_1 = \tilde{C} \times_S S_1$ is again a minimal model over S_1. Let u ∈ C_1(S) be defined by the inclusion $S_1 \hookrightarrow \tilde{C}$. By transport of structure, translation by u on the generic fiber of $\tilde{C}_1$ extends to an automorphism of C_1, whence

+ : $S_1 \times_S \tilde{C} \to \tilde{C}$.

Writing S_1 as the projective limit of the neighborhoods of x, one finds that the morphism + : C_η × C_η → C_η extends to + : U ×_S C̃ → C̃, with x ∈ U.

(iii) We know that C̃_n is quasi-finite and flat. By the hypothesis and the universal property of the Néron model, all geometric points of (C_η̄)_n come from sections of C̃_n

over S. This group is therefore finite over S, and (C̃_s)_n has rank n². Thus the special fiber C̃_s is smooth or an m-gon, with n|m (cf. II 1.13), and 1.3 is applied to obtain C by contraction.



Soit U le plus grand ouvert de $\bar{C}$ tel que la loi + de C_η se prolonge en $+ : U \times C \to C$. Il suffit de prouver que $U = C^{reg}$: ceci acquis, les identités exprimant que + est une structure de courbe elliptique généralisée résulterons de la densité schématique de C_η dans $\bar{C}$. Pour $v \in \bar{C}^{reg}(S)$, U est stable par translation par v : on prolonge $+ \grave{a}$ $v+U$ par $(v+u) + x = v+(u+x)$. Puisque U est aussi stable par localisation étale sur S (on le vérifie par un argument de descente; rappelons que la formation du modèle minimal est compatible à la localisation étale sur S), il suffit de montrer que $U_s \neq \emptyset$.

Soit x un point maximal (= générique) de $\bar{C}_s$, et soit S_1 le trait spectre de l'anneau local de $\bar{C}$ en x. Puisque $\bar{C}$ est lisse sur S en x, $\bar{C}_1 = C \times_S S_1$ est encore un modèle minimal sur S_1. Soit $u \in C_1(S)$ défini par l'inclusion $S_1 \hookrightarrow \bar{C}$. Par transport de structure, la translation $u+$ de la fibre générique de $\bar{C}_1$ se prolonge en un automorphisme de C_1, d'où

$$+ : S_1 \times_S \bar{C} \longrightarrow \bar{C} .$$

Ecrivant S_1 comme limite projective des voisinages de x, on trouve que le mor-

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(iv) Let C be a generalized elliptic curve over S, with generic fiber C_η, and suppose that C_s is smooth or an n-gon. At a non-smooth point u of C, the formal completion of C is isomorphic to the completion at the origin of the affine curve in A_S² with equation xy = t^k, for a suitable k. By homogeneity, k does not depend on the point chosen. For k > 1, C is singular at u. These singularities are resolved by repeatedly blowing up the singular points, a process that replaces u by k−1 projective lines:

u ⋀/⋀ gives ⋀ ···· ···· ···· ⋀

(cf. [8]). Let C̃ be the resolved curve. There is no exceptional curve of the first kind contained in C̃_s, so C̃ is the minimal model of C_η. The curve C is smooth if and only if the minimal model C̃ of C_η is smooth, in which case C = C̃. Otherwise C̃_s is a Néron nk-gon, and C is obtained from it by contraction in the manner used in the existence proof.

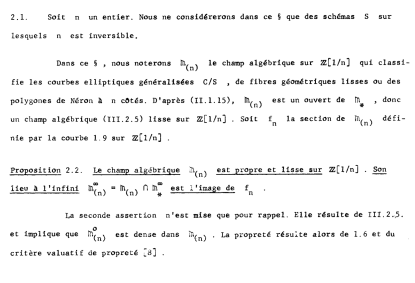
2. Level-n structures.

2.1. Let n be an integer. In this section we consider only schemes S on which n is invertible.

In this section, $\mathfrak{m}_1(n)$ denotes the algebraic stack over $\mathbb{Z}[1/n]$ classifying generalized elliptic curves C/S whose geometric fibers are smooth or Néron n-gons. By (II.1.15), $\mathfrak{m}_1(n)$ is an open substack of $\mathfrak{m}_1^*$ and hence an algebraic stack, smooth over $\mathbb{Z}[1/n]$ by (III.2.5). Let f_n be the section of $\mathfrak{m}_1(n)$ defined by curve 1.9 over $\mathbb{Z}[1/n]$.

Proposition 2.2. The algebraic stack $\mathfrak{m}_1(n)$ is proper and smooth over $\mathbb{Z}[1/n]$. Its locus at infinity $\mathfrak{m}_1(n)^\infty = \mathfrak{m}_1(n) \cap \mathfrak{m}_1^{*\infty}$ is the image of f_n.

The second assertion is included only as a reminder. It follows from III.2.5 and implies that $\mathfrak{m}_1(n)^0$ is dense in $\mathfrak{m}_1(n)$. Properness then follows from 1.6 and the valuative criterion for properness [8].



2.1. Soit n un entier. Nous ne considérerons dans ce § que des schémas S sur lesquels n est inversible.

Dans ce §, nous noterons $\mathfrak{M}_{(n)}$ le champ algébrique sur $\mathbb{Z}[1/n]$ qui classifie les courbes elliptiques généralisées C/S, de fibres géométriques lisses ou des polygones de Néron à n côtés. D'après (II.1.15), $\mathfrak{M}_{(n)}$ est un ouvert de $\mathfrak{M}_*$, donc un champ algébrique (III.2.5) lisse sur $\mathbb{Z}[1/n]$. Soit f_n la section de $\mathfrak{M}_{(n)}$ définie par la courbe I.9 sur $\mathbb{Z}[1/n]$.

Proposition 2.2. Le champ algébrique $\mathfrak{M}_{(n)}$ est propre et lisse sur $\mathbb{Z}[1/n]$. Son lieu à l'infini $\mathfrak{M}_{(n)}^\infty = \mathfrak{M}_{(n)} \cap \mathfrak{M}_*^\infty$ est l'image de f_n.

La seconde assertion n'est mise que pour rappel. Elle résulte de III.2.5. et implique que $\mathfrak{M}_{(n)}^0$ est dense dans $\mathfrak{M}_{(n)}$. La propreté résulte alors de I.6 et du critère valuatif de propreté [8].

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2.3. For C/S of the type under consideration, C_n^reg is finite étale of rank n^2 (II.1.20), hence locally isomorphic to $(\mathbb{Z}/n)^2$ in the étale topology. A level-n structure on C is an isomorphism

$\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$.

Definition 2.4. $\mathfrak{m}_n[1/n]$ is the following algebraic stack over $\mathbb{Z}[1/n]$: for a scheme S on which n is invertible, $\mathfrak{m}_n[1/n](S)$ is the category of generalized elliptic curves C/S whose geometric fibers are smooth or Néron n-gons, equipped with a level-n structure.

It is clear that $\mathfrak{m}_n[1/n]$ is finite étale and representable over $\mathfrak{m}_(n)$, and is in fact a principal homogeneous space under $GL(2,\mathbb{Z}/n)$. We write $\mathfrak{m}_n^\infty[1/n]$ for the inverse image of $\mathfrak{m}_(n)^\infty$, the image of the non-smoothness subscheme of the universal curve over $\mathfrak{m}_n[1/n]$. The following theorem follows immediately from 2.2.

Theorem 2.5. $\mathfrak{m}_n[1/n]$ is proper and smooth over $\mathbb{Z}[1/n]$, and $\mathfrak{m}_n^\infty[1/n]$ is finite étale over $\mathbb{Z}[1/n]$.

2.6. Recall that an algebraic stack $\mathcal{G}$ "is" an algebraic space when the objects it classifies have no automorphisms. More precisely, if for every algebraically closed field k the objects of $\mathcal{G}(k)$ have no nontrivial automorphisms, then

(a) for every scheme S, the objects of $\mathcal{G}(S)$ have no nontrivial automorphisms;

(b) the functor S ↦ (the set of isomorphism classes in $\mathcal{G}(S)$) is representable by an algebraic space.

Theorem 2.7. If $n \geq 3$, $\mathfrak{m}_n[1/n]$ is an algebraic space.

Let C be a generalized elliptic curve over an algebraically closed field k, either smooth or a Néron n-gon. One must prove that an automorphism σ of C that is trivial on C_n is the identity. When C is an elliptic curve, apply ([26], appendix to Exposé 17). When C is a Néron polygon, II.1.10 shows that Aut(C) acts faithfully on C_n.

2.6. Rappelons qu'un champ algébrique $\mathcal{G}$ "est" un espace algébrique si les objets qu'il classifie n'ont pas d'automorphismes. Plus précisément, si pour tout corps algébriquement clos, k, les objets de $\mathcal{G}(k)$ n'ont pas d'automorphisme non trivial, alors

a) pour tout schéma S, les objets de $\mathcal{G}(S)$ n'ont pas d'automorphisme non trivial;

b) le foncteur S ↦ (l'ensemble des classes d'isomorphie dans $\mathcal{G}(S)$) est représentable par un espace algébrique.

Théorème 2.7. Si $n \geq 3$, $\mathfrak{M}_n[1/n]$ est un espace algébrique.

Soient C une courbe elliptique généralisée sur un corps algébriquement clos k, lisse ou un polygone de Néron à n côtés. Il faut prouver qu'un automorphisme σ de C, trivial sur C_n, est l'identité. Pour C une courbe elliptique, on applique ([26], app. à Exp. 17). Pour C un polygone de Néron, on vérifie sur II.1.10 que Aut(C) agit fidèlement sur C_n.

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Variant 2.8. This argument, together with III.2.1, also immediately implies that the functor

F_n : S ↦ (the set of isomorphism classes of generalized elliptic curves over S equipped with a level-n structure)

is a sheaf for the fpqc topology. The arguments of III.2.5 then show that this functor satisfies M. Artin's representability criterion. To prove that F_n is representable by an algebraic space for n ≥ 3, it is therefore unnecessary first to generalize M. Artin's criterion to algebraic stacks.

Corollary 2.9. If n ≥ 3, $\mathfrak{m}$_n[1/n] is a projective smooth scheme over $\mathbb{Z}$[1/n].

Indeed, a regular algebraic space in curves over $\mathbb{Z}$ is always quasi-projective, and hence a scheme. Here is a more constructive proof: by transcendental methods (cf. §5), one verifies that after extension of scalars to $\mathbb{C}$, $\mathfrak{m}$_n^∞ meets every irreducible component of $\mathfrak{m}$_n. By specialization it follows that the divisor $\mathfrak{m}$_n^∞ meets every irreducible component of every geometric fiber of $\mathfrak{m}$_n; here one uses that $\mathfrak{m}$_n is proper and smooth. Consequently $\mathcal{O}$($\mathfrak{m}$_n^∞) is a relatively ample invertible sheaf.

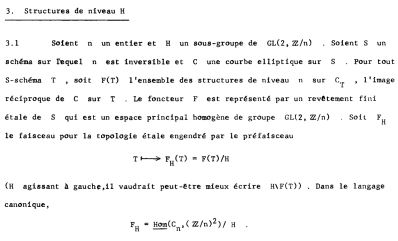
3. Level-H structures.

3.1. Let n be an integer and H a subgroup of GL(2,$\mathbb{Z}$/n). Let S be a scheme on which n is invertible and C an elliptic curve over S. For every S-scheme T, let F(T) be the set of level-n structures on C_T, the inverse image of C over T. The functor F is represented by a finite étale cover of S that is a principal homogeneous space under GL(2,$\mathbb{Z}$/n). Let F_H be the sheaf for the étale topology generated by the presheaf

T ↦ F_H(T) = F(T)/H.

(Since H acts on the left, it might be better to write H\F(T).) In canonical language,

F_H = Hom(C_n,($\mathbb{Z}$/n)²)/H.



3. Structures de niveau H

3.1 Soient n un entier et H un sous-groupe de GL(2, Z/n) . Soient S un schéma sur lequel n est inversible et C une courbe elliptique sur S . Pour tout S-schéma T , soit F(T) l'ensemble des structures de niveau n sur C_T , l'image réciproque de C sur T . Le foncteur F est représenté par un revêtement fini étale de S qui est un espace principal homogène de groupe GL(2, Z/n) . Soit F_H le faisceau pour la topologie étale engendré par le préfaisceau

$$T \longmapsto F_H(T) = F(T)/H$$

(H agissant à gauche,il vaudrait peut-être mieux écrire H\F(T)) . Dans le langage canonique,

$$F_H = \underline{\mathrm{Hom}}(C_n, (\mathbb{Z}/n)^2)/H .$$

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Authority page 70; collected-paper page 563; original folio 212.

This functor is represented by a finite étale cover of S.

A level-H structure on C is an element of F_H(S). An isomorphism α : C_n $\xrightarrow{\sim}$ ($\mathbb{Z}$/n)² defines a level-H structure, and two such isomorphisms define the same structure if and only if, locally on S, there is a g ∈ H such that gα = β. This g is uniquely determined and hence constant on each connected component of S. One should keep in mind that a level-H structure does not in general arise from an α : C_n $\xrightarrow{\sim}$ ($\mathbb{Z}$/n)². It arises from such an α only locally for the étale topology. For example, when H = GL(2,$\mathbb{Z}$/n), C has exactly one level-H structure; it comes from an α only if C_n is isomorphic to ($\mathbb{Z}$/n)² over S.

Definition 3.2. $\mathfrak{m}$_H^o[1/n] is the following algebraic stack: for a scheme S on which n is invertible, $\mathfrak{m}$_H^o[1/n](S) is the category of elliptic curves C/S equipped with a level-H structure, with isomorphisms as morphisms.

It is clear that $\mathfrak{m}$_H^o[1/n] is finite étale and locally representable over $\mathfrak{m}$_1^o[1/n].

Definition 3.3. The algebraic stack $\mathfrak{m}$_H is the normalization of $\mathfrak{m}$_1 in $\mathfrak{m}$_H^o[1/n].

Theorem 3.4. (i) $\mathfrak{m}$_H is proper over Spec($\mathbb{Z}$).

(ii) $\mathfrak{m}_H[1/n]$ is smooth over $\mathrm{Spec}(\mathbb{Z}[1/n])$, and $\mathfrak{m}_H^{o}[1/n]$ is the complement of a substack $\mathfrak{m}_H^{\infty}[1/n]$ that is finite étale over $\mathrm{Spec}(\mathbb{Z}[1/n])$.

Proof. (i) By definition, $\mathfrak{m}_H$ is finite and relatively representable over $\mathfrak{m}_1$. Its properness therefore follows from that of $\mathfrak{m}_1$.

(ii) The assertion follows from Abhyankar's lemma (SGA 1, XIII 5), which applies here because $\mathfrak{m}_1[1/n]$ is smooth over $\mathbb{Z}[1/n]$, because $\mathfrak{m}_1^{\infty}[1/n]$ is a divisor of $\mathfrak{m}_1[1/n]$, smooth over $\mathbb{Z}[1/n]$ and with generic point of characteristic zero, and because $\mathfrak{m}_H^{o}[1/n]$ is finite étale over $\mathfrak{m}_1[1/n] - \mathfrak{m}_1^{\infty}[1/n]$.

In 6.7 we shall give a "modular" proof of 3.4.

(ii) $\mathcal{M}_H[1/n]$ est lisse sur $\mathrm{Spec}\mathbb{Z}[1/n]$, et $\mathcal{M}_H^{o}[1/n]$ est le complément d'un sous-champ $\mathcal{M}_H^{\infty}[1/n]$ fini et étale sur $\mathrm{Spec}\mathbb{Z}[1/n]$).

Preuve (i) Par définition, $\mathcal{M}_H$ est fini et relativement représentable sur $\mathcal{M}_1$. Sa propreté résulte donc de celle de $\mathcal{M}_1$.

(ii) L'assertion résulte du lemme d'Abhyankar (SGA 1,XIII 5) applicable ici parce que $\mathcal{M}_1[1/n]$, est lisse sur $\mathbb{Z}[1/n]$, que $\mathcal{M}_1^{\infty}[1/n]$ est un diviseur de $\mathcal{M}_1[1/n]$, lisse sur $\mathbb{Z}[1/n]$, et dont le point générique est de caractéristique 0, et que $\mathcal{M}_H^{o}[1/n]$ est fini étale sur $\mathcal{M}_1[1/n] - \mathcal{M}_1^{\infty}[1/n]$.

En 6.7, nous donnerons de 3.4 une démonstration "modulaire".

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For H = {e}, we set $\mathfrak{m}_n = \mathfrak{m}_H$. This notation is justified by the following proposition.

Proposition 3.5. For H = {e}, $\mathfrak{m}_H[1/n]$ is the stack $\mathfrak{m}_n[1/n]$ of 2.4.

Construction 1.4 defines a morphism of stacks

$c : \mathfrak{m}_n[1/n] \to \mathfrak{m}_1[1/n] : (C,\alpha) \mapsto c(C)$.

Since $\mathfrak{m}_n[1/n]$ is normal, indeed smooth over $\mathbb{Z}[1/n]$, and proper over $\mathbb{Z}[1/n]$, and since plainly $\mathfrak{m}_n^{o}[1/n] \simeq \mathfrak{m}_H^{o}[1/n]$, it is enough to prove that c is finite and locally representable. Its fibers are plainly finite. To test local representability, it therefore suffices to verify the following (cf. 2.6).

Lemma 3.5.1. Let k be an algebraically closed field, let C be an object of $\mathfrak{m}_n[1/n](k)$, and let c(C) be its image in $\mathfrak{m}_1(k)$. The natural map $i : \mathrm{Aut}(C) \to \mathrm{Aut}(c(C))$ is injective.

For $n \geq 3$, Aut(C) is trivial (cf. 2.7). For n = 2, $\mathrm{Aut}(C) = \{\pm 1\}$. Finally, for n = 1, i is the identity.

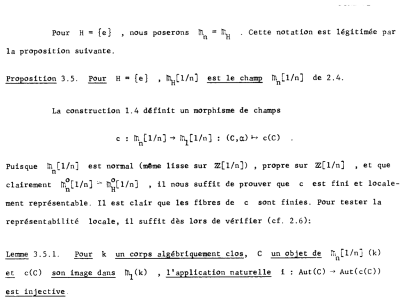
3.6. Let n and m be integers with n|m. Let $H \subset GL(2,\mathbb{Z}/n)$, and let H' be its inverse image in $GL(2,\mathbb{Z}/nm)$. For an elliptic curve E over S, a level-nm structure

$$\begin{array}{ccc} E_{nm} & \xrightarrow{\alpha'}\tilde{\to} & (\mathbb{Z}/nm)^2 \\ \big\downarrow x^m & & \big\downarrow \text{reduction} \\ E_n & \xrightarrow{\alpha}\tilde{\to} & (\mathbb{Z}/n)^2 \end{array}$$

defines a level-n structure α and makes the displayed square commute.

When nm is invertible on S, this construction defines an isomorphism from the sheaf of H'-structures on E to the sheaf of H-structures. Thus

$\mathfrak{m}_{H'}^{o}[1/nm] \simeq \mathfrak{m}_H^{o}[1/n][1/nm]$.



Pour $H = \{e\}$, nous poserons $\mathcal{M}_n = \mathcal{M}_H$. Cette notation est légitimée par la proposition suivante.

Proposition 3.5. Pour $H = \{e\}$, $\mathcal{M}_H[1/n]$ est le champ $\mathcal{M}_n[1/n]$ de 2.4.

La construction 1.4 définit un morphisme de champs

$$c : \mathcal{M}_n[1/n] \to \mathcal{M}_1[1/n] : (C,\alpha) \mapsto c(C) \quad .$$

Puisque $\mathcal{M}_n[1/n]$ est normal (même lisse sur $\mathbb{Z}[1/n]$) , propre sur $\mathbb{Z}[1/n]$, et que clairement $\mathcal{M}_n^0[1/n] \simeq \mathcal{M}_H^0[1/n]$, il nous suffit de prouver que c est fini et localement représentable. Il est clair que les fibres de c sont finies. Pour tester la représentabilité locale, il suffit dès lors de vérifier (cf. 2.6):

Lemme 3.5.1. Pour k un corps algébriquement clos, C un objet de $\mathcal{M}_n[1/n](k)$ et $c(C)$ son image dans $\mathcal{M}_1(k)$, l'application naturelle $i : \mathrm{Aut}(C) \to \mathrm{Aut}(c(C))$ est injective

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Since $\mathfrak{m}_H^o[1/n]$ is finite étale over $\mathfrak{m}_1^o[1/n]$, it is the normalization of $\mathfrak{m}_1^o[1/n]$ in $\mathfrak{m}_{H'}^o[1/nm]$. It follows that

$$\mathfrak{m}_{H'} \simeq \mathfrak{m}_H.$$

Thus the stack $\mathfrak{m}_H$ depends only on the open subgroup K of $GL(2,\hat{\mathbb{Z}})$ that is the inverse image of H. We shall sometimes denote it by $\mathfrak{m}_K$.

3.7. Let n and m be relatively prime integers, let $H \subset GL(2,\mathbb{Z}/n)$, and let $I \subset GL(2,\mathbb{Z}/m)$. Let K and L be the inverse images of H and I in $GL(2,\hat{\mathbb{Z}})$. Reduction modulo n and m induces an isomorphism

$$GL(2,\mathbb{Z}/nm) \xrightarrow{\sim} GL(2,\mathbb{Z}/n) \times GL(2,\mathbb{Z}/m).$$

Via this isomorphism, $K \cap L$ is the inverse image of $H \times I \subset GL(2,\mathbb{Z}/n) \times GL(2,\mathbb{Z}/m)$.

Construction 3.8. With the preceding notation,

$$\mathfrak{m}_K^o[1/n] \times_{\{\mathfrak{m}_1^o\}} \mathfrak{m}_L^o[1/m] \xrightarrow{\sim} \mathfrak{m}_{\{K\cap L\}}^o[1/nm].$$

Let C/S be an elliptic curve. The maps $x \to x^m$ and $x \to x^n$ define an isomorphism

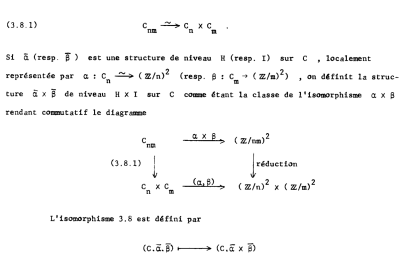
$$(3.8.1)\quad C_{nm} \xrightarrow{\sim} C_n \times C_m.$$

If $\bar{\alpha}$ (respectively $\bar{\beta}$) is a level-H (respectively level-I) structure on C, locally represented by $\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ (respectively $\beta : C_m \xrightarrow{\sim} (\mathbb{Z}/m)^2$), define the level-$H \times I$ structure $\bar{\alpha} \times \bar{\beta}$ on C to be the class of the isomorphism $\alpha \times \beta$ that makes the diagram commute:

$$\begin{array}{ccc} C_{nm} & \xrightarrow{\alpha\times\beta} & (\mathbb{Z}/nm)^2 \\ \big\downarrow (3.8.1) & & \big\downarrow \text{réduction} \\ C_n \times C_m & \xrightarrow{(\alpha,\beta)} & (\mathbb{Z}/n)^2 \times (\mathbb{Z}/m)^2 \end{array}$$

The isomorphism in 3.8 is defined by

$$(C,\bar{\alpha},\bar{\beta}) \mapsto (C,\bar{\alpha} \times \bar{\beta}).$$



(3.8.1) $\quad C_{nm} \xrightarrow{\sim} C_n \times C_m$.

Si $\bar{\alpha}$ (resp. $\bar{\beta}$) est une structure de niveau H (resp. I) sur C , localement représentée par $\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ (resp. $\beta : C_m \to (\mathbb{Z}/m)^2$) , on définit la structure $\bar{\alpha} \times \bar{\beta}$ de niveau $H \times I$ sur C comme étant la classe de l'isomorphisme $\alpha \times \beta$ rendant commutatif le diagramme

$$\begin{array}{ccc} C_{nm} & \xrightarrow{\alpha\times\beta} & (\mathbb{Z}/nm)^2 \\ \big\downarrow (3.8.1) & & \big\downarrow \text{réduction} \\ C_n \times C_m & \xrightarrow{(\alpha,\beta)} & (\mathbb{Z}/n)^2 \times (\mathbb{Z}/m)^2 \end{array}$$

L'isomorphisme 3.8 est défini par

$$(C,\bar{\alpha},\bar{\beta}) \longmapsto (C,\bar{\alpha} \times \bar{\beta})$$

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Proposition 3.9. With the notation of 3.7, the isomorphism of 3.8 extends to

(3.9.1) $\mathfrak{m}_K \times_{\mathfrak{m}_1} \mathfrak{m}_L \tilde{\to} \mathfrak{m}_{K\cap L}$.

Proof. It is enough to prove that $\mathfrak{m}_K \times_{\mathfrak{m}_1} \mathfrak{m}_L$ is normal. We prove this over $\mathbb{Z}[1/m]$; over $\mathbb{Z}[1/n]$ the argument is symmetric. The stack $\mathfrak{m}_L[1/m]$ is smooth over $\mathbb{Z}$ and hence flat over $\mathfrak{m}_1$ by (I.7.1). Thus $\mathfrak{m}_K \times_{\mathfrak{m}_1} \mathfrak{m}_L[1/m]$ is flat over $\mathfrak{m}_K$ and Cohen-Macaulay by (I.7.2 and I.7.1). Moreover $\mathfrak{m}_L^o[1/m]$ is étale over $\mathfrak{m}_1$, so $\mathfrak{m}_K \times_{\mathfrak{m}_1} \mathfrak{m}_L^o[1/m]$, being étale over $\mathfrak{m}_K$, is normal.

By Abhyankar's lemma, the ramification of $\mathfrak{m}_K[1/n]$ over $\mathfrak{m}_1[1/n]$ (respectively of $\mathfrak{m}_L[1/m]$ over $\mathfrak{m}_1[1/m]$) is tame. The ramification indices for $\mathfrak{m}_K$ and $\mathfrak{m}_L$ are therefore relatively prime; in fact they divide n and m. Hence $\mathfrak{m}_K[1/mn] \times_{\mathfrak{m}_1} \mathfrak{m}_L[1/mn]$ is smooth, and Serre's criterion (I.7.2) completes the proof.

Corollary 3.9.2. If n is the product of two relatively prime integers and $n \geq 3$, then $\mathfrak{m}_n$ is a scheme.

Write n=n'n". The stack $\mathfrak{m}_n[1/n']$ is a scheme because it is representable over $\mathfrak{m}_{n'}[1/n]$. Likewise $\mathfrak{m}_n[1/n"]$ is a scheme; therefore $\mathfrak{m}_n$ is a scheme.

Proposition 3.10. (i) The coarse space M_H defined by $\mathfrak{m}_H$ is proper and flat over $\mathrm{Spec}(\mathbb{Z})$.

(ii) It is the normalization of M_1, which is isomorphic to $\mathbb{P}^1$ by VI.1.1, in $\mathfrak{m}_H^o[1/n]$.

(iii) One has $M_H = M_n/H$.

(iv) The geometric fibers of $M_H[1/n] \to \mathrm{Spec}(\mathbb{Z}[1/n])$ are geometrically unibranch.

Proof. Properness of M_H follows from properness of $\mathfrak{m}_H$. The description in I.8.2 of the strict henselization of a local ring of M_H, together with normality of $\mathfrak{m}_H$, shows that M_H is normal. Similarly M_H is flat over $\mathbb{Z}$. The finite map $\mathfrak{m}_H \to \mathfrak{m}_1$ induces a finite map $M_H \to M_1$, and (ii) follows from (i) and normality.

Soit $n = n'.n"$. Le champ $\mathfrak{M}_H[1/n']$ est un schéma, car représentable sur $\mathfrak{M}_{n'}[1/n]$. De même, $\mathfrak{M}_H[1/n"]$, donc $\mathfrak{M}_n$, est un schéma.

Proposition 3.10. (i) L'espace grossier M_H défini par $\mathfrak{M}_H$ est propre et plat sur $\mathrm{Spec}(\mathbb{Z})$

(ii) C'est le normalisé dans $M_H^0[1/n]$ de M_1 (isomorphe à $\mathbb{P}^1$: voir VI 1.1).

(iii) On a $M_H = M_n/H$.

(iv) Les fibres géométriques de $M_H[1/n] \to \mathrm{Spec}(\mathbb{Z}[1/n])$ sont géométriquement unibranches.

Preuve La propreté de M_H résulte de celle de $\mathfrak{M}_H$. De la description de l'hensélisé strict d'un anneau local de M_H donnée en I.8.2, et de la normalité de $\mathfrak{M}_H$, il résulte que M_H est normal. De même, M_H est plat sur $\mathbb{Z}$. Puisque $\mathfrak{M}_H \to \mathfrak{M}_1$, donc $M_H \to M_1$ est fini, (ii) résulte de (i) et de la normalité.

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It remains to prove (iii) and (iv) when $n \geq 3$, by 3.6, in which case $M_n^o[1/n] = \mathfrak{m}_n^o[1/n]$. With the notation of [8], §4, one has

$\mathfrak{m}_H^o[1/n] = [M_n^o[1/n]/H]$,

and consequently $M_H^o[1/n] = M_n^o[1/n]/H$. Assertion (iii) now follows from (ii).

The map

$(M_n \otimes \bar{F}_p)/H \to (M_n/H) \otimes \bar{F}_p = M_H \otimes \bar{F}_p$

is radicial. If p does not divide n, then $M_n \otimes \bar{F}_p$ is smooth, so $(M_n \otimes \bar{F}_p)/H$ is smooth,

proving (iv).

3.11. The category of elliptic curves up to isogeny over a scheme S is obtained from the category of elliptic curves by inverting isogenies. Writing E⊗ℚ for the elliptic curve up to isogeny over S underlying E/S, one has

Hom_S(E⊗ℚ,F⊗ℚ)=Hom_S(E,F)⊗ℚ.

3.12. Let k be an algebraically closed field of characteristic zero. For every elliptic curve E over k, put

$\hat{T}(E)=\lim\leftarrow E_n$ (isomorphic to $\hat{\mathbb{Z}}^2$),

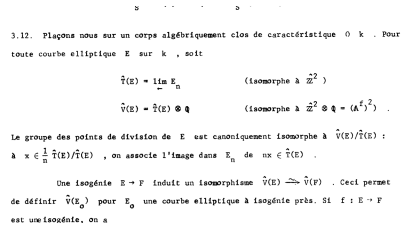
$\hat{V}(E)=\hat{T}(E)\otimes\mathbb{Q}$ (isomorphic to $\hat{\mathbb{Z}}^2\otimes\mathbb{Q}=(\mathbb{A}^f)^2$).

The group of division points of E is canonically isomorphic to $\hat{V}(E)/\hat{T}(E)$: to $x\in(1/n)\hat{T}(E)/\hat{T}(E)$ one associates the image in E_n of $nx\in\hat{T}(E)$.

An isogeny E→F induces an isomorphism $\hat{V}(E)\tilde{\to}\hat{V}(F)$. This permits one to define $\hat{V}(E_o)$ for an elliptic curve E_o up to isogeny. If f:E→F is an isogeny, then

$\mathrm{Ker}(f)=\mathrm{Ker}(\hat{V}(E)/\hat{T}(E)\to\hat{V}(F)/\hat{T}(F))=\mathrm{coker}(\hat{T}(E)\to\hat{T}(F))$.

The assignment $f\mapsto\hat{T}(F)\subset\hat{V}(E)$ identifies isogenies with source E with the lattices $\hat{T}'$ containing $\hat{T}(E)$. It follows that the functor



3.12. Plaçons nous sur un corps algébriquement clos de caractéristique 0 k . Pour toute courbe elliptique E sur k , soit

$\hat{T}(E) = \varprojlim E_n$ (isomorphe à $\hat{\mathbb{Z}}^2$)

$\hat{V}(E) = \hat{T}(E)\otimes\mathbb{Q}$ (isomorphe à $\hat{\mathbb{Z}}^2\otimes\mathbb{Q} = (\mathbb{A}^f)^2$).

Le groupe des points de division de E est canoniquement isomorphe à $\hat{V}(E)/\hat{T}(E)$: à $x \in \frac{1}{n}\hat{T}(E)/\hat{T}(E)$, on associe l'image dans E_n de $nx \in \hat{T}(E)$.

Une isogénie $E \to F$ induit un isomorphisme $\hat{V}(E) \xrightarrow{\sim} \hat{V}(F)$. Ceci permet de définir $\hat{V}(E_o)$ pour E_o une courbe elliptique à isogénie près. Si $f : E \to F$ est une isogénie, on a

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E ↦ (E⊗ℚ, $\hat{T}(E)\subset\hat{V}(E\otimes\mathbb{Q})$)

from elliptic curves to elliptic curves E_o up to isogeny equipped with a lattice $\hat{T}\subset\hat{V}(E)$ is an equivalence of categories.

3.13. Let E_o be an elliptic curve up to isogeny over k. A lattice $\hat{T}\subset\hat{V}(E_o)$ can be interpreted as a class, modulo $GL(2,\hat{\mathbb{Z}})$, of isomorphisms

$\beta:\hat{V}(E_o)\to(\mathbb{A}^f)^2$;

to β one associates $\beta^{-1}(\hat{\mathbb{Z}}^2)$.

Let $K\subset GL(2,\hat{\mathbb{Z}})$ be the inverse image of $H\subset GL(2,\mathbb{Z}/n)$. If $(E_o,\hat{T})$ corresponds to an elliptic curve E, then a level-H structure on E gives the set of isomorphisms $\beta:\hat{V}(E_o)\tilde{\to}(\mathbb{A}^f)^2$ such that $\beta^{-1}(\hat{\mathbb{Z}}^2)=\hat{T}$ and

$\beta_n:E_n=\hat{T}/n\hat{T}\tilde{\to}\hat{\mathbb{Z}}^2/n\hat{\mathbb{Z}}^2=(\mathbb{Z}/n\mathbb{Z})^2$

defines that level-H structure. The maps β form a left coset under K, and the set of elliptic curves with level-H structure $(E,\bar{\alpha})$ and $E\otimes\mathbb{Q}=E_o$ identifies with

$K\backslash\mathrm{Isom}(\hat{V}(E_o),(\mathbb{A}^f)^2)$.

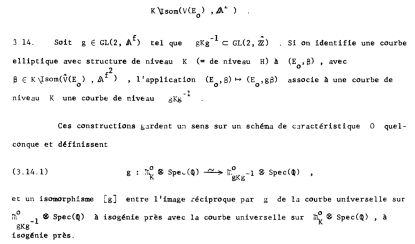
3.14. Let $g\in GL(2,\mathbb{A}^f)$ satisfy $gKg^{-1}\subset GL(2,\hat{\mathbb{Z}})$. Identifying an elliptic curve with level-K

structure, equivalently level-H structure, with (E_o,β), where β∈K\Isom(V̂(E_o),(𝔸^f)²), the assignment (E_o,β)↦(E_o,gβ) associates to a curve of level K a curve of level gKg⁻¹.

These constructions remain meaningful over any scheme of characteristic zero and define

(3.14.1) g:𝔪_K^o⊗Spec(ℚ)→̃𝔪_{gKg⁻¹}^o⊗Spec(ℚ),

together with an isomorphism [g] between the pullback by g of the universal curve over 𝔪_{gKg⁻¹}^o⊗Spec(ℚ), considered up to isogeny, and the universal curve over 𝔪_K^o⊗Spec(ℚ), considered up to isogeny.



K\Isom(V(E$_o$) , $\mathbb{A}^{\infty}$) .

3.14. Soit $g \in GL(2, \mathbb{A}^f)$ tel que $gKg^{-1} \subset GL(2, \hat{\mathbb{Z}})$. Si on identifie une courbe elliptique avec structure de niveau K (= de niveau H) à (E_o, β), avec $\beta \in K\backslash Isom(\hat{V}(E_o), \mathbb{A}^{f2})$, l'application $(E_o, \beta) \mapsto (E_o, g\beta)$ associe à une courbe de niveau K une courbe de niveau gKg^{-1}.

Ces constructions gardent un sens sur un schéma de caractéristique 0 quelconque et définissent

(3.14.1) $g : \mathfrak{M}^0_K \otimes Spec(\mathbb{Q}) \xrightarrow{\sim} \mathfrak{M}^0_{gKg^{-1}} \otimes Spec(\mathbb{Q})$,

et un isomorphisme [g] entre l'image réciproque par g de la courbe universelle sur $\mathfrak{M}^0_{gKg^{-1}} \otimes Spec(\mathbb{Q})$ à isogénie près avec la courbe universelle sur $\mathfrak{M}^0_K \otimes Spec(\mathbb{Q})$, à isogénie près.

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Passing to the inverse limit over K gives an action of GL(2,𝔸^f) on the scheme lim←_K 𝔪_K^o⊗Spec(ℚ) and on the universal elliptic curve up to isogeny.

3.15. We now make the composite map explicit:

(3.15.1) 𝔪_K^o⊗ℚ →̃ 𝔪_{gKg⁻¹}^o⊗ℚ ⟶c 𝔪_1^o⊗ℚ.

Choose n and m such that

$\hat{\mathbb{Z}}^2 \subset (1/m)g^{-1}(\hat{\mathbb{Z}}^2) \subset (1/n)\hat{\mathbb{Z}}^2$.

Let B be the subgroup of $(\mathbb{Z}/n\mathbb{Z})^2$ that is the image of $(1/m)g^{-1}(\hat{\mathbb{Z}}^2)/\hat{\mathbb{Z}}^2$ under multiplication by n on

$(1/n)\hat{\mathbb{Z}}^2/\hat{\mathbb{Z}}^2 \xrightarrow{\sim} (\mathbb{Z}/n\mathbb{Z})^2$.

To an elliptic curve with level-H structure (E,ᾱ), associate the elliptic curve $E/\alpha^{-1}(B)$, where $\alpha: E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ is any representative of ᾱ. The isogeny [g] is 1/m times the projection $E \to E/\alpha^{-1}(B)$.

Proposition 3.16. The isomorphism of 3.14.1 extends to an isomorphism

g:𝔪_K^o→̃𝔪_{gKg⁻¹}^o.

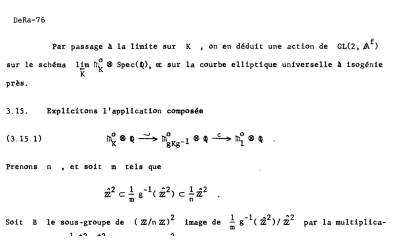
The isomorphism [g] extends to an isomorphism

[g]:g^*(the elliptic curve over 𝔪_{gKg⁻¹}^o)⊗ℚ ≃ (the elliptic curve over 𝔪_K^o)⊗ℚ.

Proof. We first show that (3.15.1) extends to

cg:𝔪_K^o→𝔪_1^o.

Let b be the order of B as in 3.15, and let I be the stack, relatively representable over 𝔪_1^o, that represents the functor of finite locally free subgroup schemes of rank b of the universal elliptic curve E. The theory of Hilbert schemes shows that the projection I→𝔪_1^o is proper and representable. By the next lemma it is finite.



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Par passage à la limite sur K, on en déduit une action de $GL(2, \mathbb{A}^f)$ sur le schéma $\varprojlim_K h^0_K \otimes \mathrm{Spec}(\mathbb{Q})$, m sur la courbe elliptique universelle à isogénie près.

3.15. Explicitons l'application composée

(3.15.1) $h^0_K \otimes \mathbb{Q} \xrightarrow{\sim} h^0_{gKg^{-1}} \otimes \mathbb{Q} \xrightarrow{c} h^0_1 \otimes \mathbb{Q}$.

Prenons n, et soit m tels que

$$\hat{\mathbb{Z}}^2 \subset \frac{1}{m} g^{-1}(\hat{\mathbb{Z}}^2) \subset \frac{1}{n}\hat{\mathbb{Z}}^2 .$$

Soit B le sous-groupe de $(\mathbb{Z}/n\mathbb{Z})^2$ image de $\frac{1}{m} g^{-1}(\hat{\mathbb{Z}}^2)/\hat{\mathbb{Z}}^2$ par la multiplica-

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Authority page 77; collected-paper page 570; original folio 219.

Lemma 3.17. Let E be an elliptic curve over an algebraically closed field k and let b be an integer. There are only finitely many subgroup schemes of E of rank b.

Proof of 3.17. Proceed by induction on b.

(a) There are only finitely many étale subgroup schemes of E of rank b: they identify with the subgroups of order b in E_b(k).

(b) If H⊂E is not étale, then k has characteristic p>0 and H contains Ker(F). Thus the non-étale subgroup schemes of E of rank b correspond to the subgroup schemes of rank b/p of E^(p)=E/Ker(F), and the induction hypothesis applies.

Proof of 3.16, continued. Construction 3.14 defines a section s over $\mathfrak{m}_K^o \otimes \mathrm{Spec}(\mathbb{Q})$ of $I \times_{\{\mathfrak{m}_1^o\}} \mathfrak{m}_K^o$.

By Zariski's Main Theorem and the normality of $\mathfrak{m}_K^o$, this section extends over $\mathfrak{m}_K^o$, defining a subgroup B of the pullback E, along c, of the universal elliptic curve over $\mathfrak{m}_1^o$. The required extension cg is defined by E/B, and the isomorphism [g] is defined as in 3.15.

The resulting morphism cg is quasi-finite and representable. It therefore identifies $\mathfrak{m}_K^o$ with an open substack of the normalization $\mathfrak{m}_{\{gKg^{-1}\}}^o$ of $\mathfrak{m}_1^o$ in

$\mathfrak{m}_K^o \otimes \mathrm{Spec}(\mathbb{Q}) \xrightarrow{g} \mathfrak{m}_{\{gKg^{-1}\}}^o \otimes \mathrm{Spec}(\mathbb{Q})$,

and hence gives

$g: \mathfrak{m}_K^o \hookrightarrow \mathfrak{m}_{\{gKg^{-1}\}}^o$.

This morphism is an isomorphism, with inverse g^{-1}.

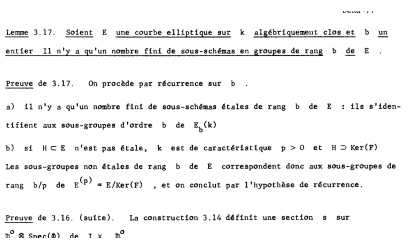
3.18. One can show that g extends to an isomorphism

$\mathfrak{m}_K \xrightarrow{\sim} \mathfrak{m}_{\{gKg^{-1}\}}$.

We shall only prove the analogous result for the coarse schemes.

Proposition 3.19. The morphism of coarse schemes induced by 3.16 extends to

$g: M_K \xrightarrow{\sim} M_{\{gKg^{-1}\}}$.



Lemme 3.17. Soient E une courbe elliptique sur k algébriquement clos et b un entier. Il n'y a qu'un nombre fini de sous-schémas en groupes de rang b de E.

Preuve de 3.17. On procède par récurrence sur b.

a) il n'y a qu'un nombre fini de sous-schémas étales de rang b de E : ils s'identifient aux sous-groupes d'ordre b de $E_b(k)$

b) si $H \subset E$ n'est pas étale, k est de caractéristique $p > 0$ et $H \supset \mathrm{Ker}(F)$. Les sous-groupes non étales de rang b de E correspondent donc aux sous-groupes de rang b/p de $E^{(p)} = E/\mathrm{Ker}(F)$, et on conclut par l'hypothèse de récurrence.

Preuve de 3.16. (suite). La construction 3.14 définit une section s sur $h^o \otimes \mathrm{Spec}(\mathbb{Q})$ de $I \times_{h^o}$

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Let $\Gamma \subset M_K \times M_{gKg^{-1}}$ be the closure of the graph G of $g: M_K^o \tilde{\to} M_{gKg^{-1}}^o$. Set-theoretically,

$$\Gamma = G \cup (\Gamma \cap (M_K^\infty \times_{\mathbb{Z}} M_{gKg^{-1}}^\infty)).$$

It follows that Γ is finite over both M_K and $M_{gKg^{-1}}$. By the Main Theorem and 3.10, the projections $\Gamma \to M_K$ and $\Gamma \to M_{gKg^{-1}}$ are isomorphisms. This proves 3.19.

3.20. Let C be an elliptic curve over S, with n invertible on S. The e_n pairing is well known to define an isomorphism

$$(3.20.1)\quad e_n : \Lambda^2 C_n \tilde{\to} \mu_n.$$

A level-n structure $\alpha : C_n \tilde{\to} (\mathbb{Z}/n)^2$ therefore defines an isomorphism

$$\det(\alpha)^{-1} : \mathbb{Z}/n \tilde{\to} \mu_n,$$

that is, a primitive n-th root of unity

$$d(\alpha) = \det(\alpha)^{-1}(1)$$

on S.

Let $\mathbb{Z}[\zeta_n]$ be the ring of integers in the field of n-th roots of unity. The construction $\alpha \mapsto d(\alpha)$ defines a morphism

$$d : \mathfrak{m}_n^o[1/n] \to \operatorname{Spec}(\mathbb{Z}[\zeta_n]).$$

This morphism extends to the normalization and defines

$$(3.20.2)\quad d : \mathfrak{m}_n \to \operatorname{Spec}(\mathbb{Z}[\zeta_n]).$$

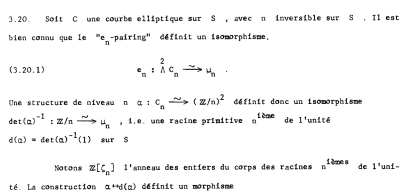
Similarly, a level-H structure α defines

$$d(\alpha) \in \operatorname{Isom}(\mathbb{Z}/n, \mu_n)/\det(H).$$

The group $(\mathbb{Z}/n)^*$ acts on $\mathbb{Z}[\zeta_n]$ by $\zeta_n \mapsto \zeta_n^i$. If $\mathbb{Z}[\zeta_n]^{\det H}$ is the subring of invariants under the action of $\det H \subset (\mathbb{Z}/n)^*$, the same construction gives

$$(3.20.3)\quad d : \mathfrak{m}_H \to \operatorname{Spec}(\mathbb{Z}[\zeta_n]^{\det H}),$$

which defines


3.20. Soit C une courbe elliptique sur S, avec n inversible sur S. Il est bien connu que le "e_n-pairing" définit un isomorphisme,

$$(3.20.1)\qquad e_n : \overset{2}{\Lambda} C_n \xrightarrow{\sim} \mu_n .$$

Une structure de niveau n $\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ définit donc un isomorphisme $\det(\alpha)^{-1} : \mathbb{Z}/n \xrightarrow{\sim} \mu_n$, i.e. une racine primitive $n^{\text{ième}}$ de l'unité $d(\alpha) = \det(\alpha)^{-1}(1)$ sur S

Notons $\mathbb{Z}[\zeta_n]$ l'anneau des entiers du corps des racines $n^{\text{ièmes}}$ de l'unité. La construction $\alpha \mapsto d(\alpha)$ définit un morphisme

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$$(3.20.4)\quad d : M_H \to \operatorname{Spec}(\mathbb{Z}[\zeta_n]^{\det H}).$$

3.21. Work over $\mathfrak{m}_{(m)}[1/m]$, with n dividing m. If C is the universal generalized elliptic curve, the e_n pairing extends to an isomorphism

$$\Lambda^2 C_n \simeq \mu_n.$$

In particular, for the standard Néron m-gon one obtains an isomorphism

(3.21.1) e_n:Λ²(𝔾_m×ℤ/m)_n≃μ_n.

Identify ℤ/n with the n-torsion subgroup (ℤ/m)_n by i↦(m/n)i. Then

(3.21.2) Λ²(𝔾_m×ℤ/m)_n=Λ²(μ_n×ℤ/n)→̃μ_n.

According to the sign conventions used to define e_n, the isomorphisms (3.21.1) and (3.21.2) are equal or opposite.

4. Examples.

4.1. Let Γ_o(n), Γ_{oo}(n), Γ'_{oo}(n), and Γ(n) be the subgroups of GL(2,$\hat{\mathbb{Z}}$) formed by the matrices with entries a,b,c,d satisfying, respectively,

Γ_o(n): c≡0 (mod n); Γ_{oo}(n): c≡0 and a≡1 (mod n); Γ'_{oo}(n): c≡0 and a≡d≡1 (mod n); Γ(n): the matrix is congruent to the identity (mod n).

Each of these is the inverse image of a subgroup H of GL(2,ℤ/n).

4.2. For Γ_o(n), H is the subgroup of GL(2,ℤ/n) that stabilizes ℤ/n×{0} in (ℤ/n)². If C is an elliptic curve over S and

Selon les conventions de signe utilisées pour définir e_n, (3.21.1) et (3.21.2) sont égaux ou opposés.

4. Exemples.

4.1. Nous noterons $\Gamma_o(n)$, $\Gamma_{oo}(n)$, $\Gamma'_{oo}(n)$ et $\Gamma(n)$ les sous-groupes de $GL(2,\hat{\mathbb{Z}})$ formé des matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ telles que :

$\Gamma_o(n)$: $c \equiv 0 \pmod n$;

$\Gamma_{oo}(n)$: $c \equiv 0 \pmod n$, $a \equiv 1 \pmod n$;

$\Gamma'_{oo}(n)$: $c \equiv 0 \pmod n$, $a \equiv d \equiv 1 \pmod n$;

$\Gamma(n)$: $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod n$.

Ces sous-groupes sont chacun image réciproque de sous-groupes H de

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Authority page 80; collected-paper page 573; original folio 222.

ᾱ is a level-H structure on C represented by an isomorphism α:C_n→̃(ℤ/n)², then the subgroup scheme α⁻¹(ℤ/n×{0}) of C_n depends only on ᾱ. When n is invertible on S, this gives an isomorphism between the sheaf of level-H structures on C and the sheaf of subgroup schemes of C locally isomorphic to ℤ/n. In other words:

Construction 4.3. The algebraic stack 𝔪^o_{Γ_o(n)}[1/n] classifies elliptic curves C/S, with n invertible on S, equipped with a subgroup scheme A locally isomorphic to ℤ/n.

4.4. For C/S and A as in 4.3, the subgroup C_n/A of the quotient curve C/A is locally isomorphic to ℤ/n. Hence there is a morphism

w:𝔪^o_{Γ_o(n)}[1/n]→𝔪^o_{Γ_o(n)}[1/n], (C,A)↦w(C,A)=(C/A,C_n/A).

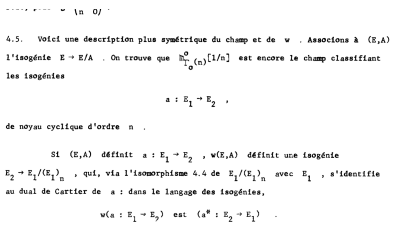
Multiplication by n factors through an isomorphism C/C_n→̃C. It yields an isomorphism

ww(C,A)=((C/A)/(C_n/A),(C/A)_n/(C_n/A))=(C/C_n,...)→̃(C,A),

and therefore w²≃Id: w is an involution. It is of the type described in 3.16 for the matrix with rows (0,1) and (n,0).

4.5. A more symmetric description is obtained by associating to (E,A) the isogeny E→E/A. The stack 𝔪^o_{Γ_o(n)}[1/n] also classifies isogenies a:E_1→E_2 with cyclic kernel of order n. If (E,A) defines a, then w(E,A) defines E_2→E_1/(E_1)_n; through the isomorphism of 4.4 between E_1/(E_1)_n and E_1, this is the Cartier dual of a. Thus

w(a:E_1→E_2)=(a^*:E_2→E_1).



4.5. Voici une description plus symétrique du champ et de w. Associons à (E,A) l'isogénie $E \to E/A$. On trouve que $\mathcal{M}^0_{\Gamma_0(n)}[1/n]$ est encore le champ classifiant les isogénies

$$a : E_1 \to E_2 ,$$

de noyau cyclique d'ordre n.

Si (E,A) définit $a : E_1 \to E_2$, $w(E,A)$ définit une isogénie $E_2 \to E_1/(E_1)_n$, qui, via l'isomorphisme 4.4 de $E_1/(E_1)_n$ avec E_1, s'identifie au dual de Cartier de a : dans le langage des isogénies,

$$w(a : E_1 \to E_2) \text{ est } (a^* : E_2 \to E_1) .$$

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Authority page 81; collected-paper page 574; original folio 223.

4.6. Let C/S and A be as in 4.3. For each prime divisor p of n, let A_p be the p-primary component of A, so that A=∏_{p|n}A_p, and let p^{n(p)} be the order of A_p, so n=∏_p p^{n(p)}. Put

w_p(E,A)=(E/A_p,(A+E_{p^{n(p)}})/A_p).

The w_p are pairwise commuting involutions of $\mathfrak{m}$^o_{Γ_o(n)}[1/n], and their composite is w.

4.7. The subgroup Γ_{oo}(n) of GL(2,$\hat{\mathbb{Z}}$) is the inverse image of the subgroup H of GL(2,$\mathbb{Z}$/n) that fixes (1,0) in $(\mathbb{Z}/n)^2$, or equivalently the homomorphism β:$\mathbb{Z}$/n→$(\mathbb{Z}/n)^2$, x↦(x,0). Reasoning as in 4.2 gives:

Construction 4.8. The algebraic stack $\mathfrak{m}$^o_{Γ_{oo}(n)}[1/n] classifies elliptic curves C/S, with n invertible on S, equipped with an embedding β:$\mathbb{Z}$/n↪C_n, equivalently a section β(1) having exact order n everywhere.

4.9. The group Γ'_{oo}(n) is the inverse image of the subgroup H that fixes $(1,0)\in(\mathbb{Z}/n)^2$ and the class of (0,1) in $(\mathbb{Z}/n)^2/(\mathbb{Z}/n\times\{0\})$. Thus $\mathfrak{m}$^o_{Γ'_{oo}(n)}[1/n] classifies elliptic curves C/S with n invertible, equipped with β:$\mathbb{Z}$/n↪C_n and an isomorphism γ:$\mathbb{Z}$/n→̃C_n/Im(β). The exterior product gives the chain of isomorphisms

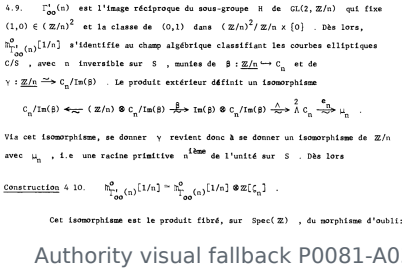
C_n/Im(β)←̃(ℤ/n)⊗C_n/Im(β)→̃^{β}Im(β)⊗C_n/Im(β)→̃^{Λ}Λ²C_n→̃^{e_n}μ_n.

Through it, choosing γ is equivalent to choosing an isomorphism $\mathbb{Z}$/n→̃μ_n, that is, a primitive n-th root of unity on S. Hence

Construction 4.10. $\mathfrak{m}$^o_{Γ'_{oo}(n)}[1/n]≃$\mathfrak{m}$^o_{Γ_{oo}(n)}[1/n]⊗$\mathbb{Z}$[ζ_n].

This is the fiber product over Spec($\mathbb{Z}$) of the forgetful morphism, defined by Γ'_{oo}(n)⊂Γ_{oo}(n), and the morphism d of (3.20.3).

Definition 4.11. The stack ´$\mathfrak{m}$_{Γ_o(n)}[1/n] classifies generalized elliptic curves C/S, with n invertible on S, equipped with a subgroup scheme A locally isomorphic to $\mathbb{Z}$/n and meeting every irreducible component of every geometric fiber of C.



4.9. $\Gamma'_{oo}(n)$ est l'image réciproque du sous-groupe H de $GL(2,\mathbb{Z}/n)$ qui fixe $(1,0) \in (\mathbb{Z}/n)^2$ et la classe de $(0,1)$ dans $(\mathbb{Z}/n)^2/\mathbb{Z}/n \times [0]$. Dès lors, $\mathcal{M}^0_{\Gamma'_{oo}(n)}[1/n]$ s'identifie au champ algébrique classifiant les courbes elliptiques C/S, avec n inversible sur S, munies de $\beta : \underline{\mathbb{Z}/n} \hookrightarrow C_n$ et de $\gamma : \underline{\mathbb{Z}/n} \xrightarrow{\sim} C_n/\mathrm{Im}(\beta)$. Le produit extérieur définit un isomorphisme

$$C_n/\mathrm{Im}(\beta) \xleftarrow{\sim} (\mathbb{Z}/n) \otimes C_n/\mathrm{Im}(\beta) \xrightarrow{\beta} \mathrm{Im}(\beta) \otimes C_n/\mathrm{Im}(\beta) \xrightarrow{\wedge} \overset{2}{\wedge} C_n \xrightarrow{e_\beta} \mu_n .$$

Via cet isomorphisme, se donner γ revient donc à se donner un isomorphisme de $\mathbb{Z}/n$ avec μ_n, i.e une racine primitive $n^{\text{ième}}$ de l'unité sur S. Dès lors

Construction 4.10. $\mathcal{M}^0_{\Gamma'_{oo}(n)}[1/n] \simeq \mathcal{M}^0_{\Gamma_{oo}(n)}[1/n] \otimes \mathbb{Z}[\zeta_n]$.

Cet isomorphisme est le produit fibré, sur $\mathrm{Spec}(\mathbb{Z})$, du morphisme d'oubli:

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Authority page 82; collected-paper page 575; original folio 224.

(We shall see in a moment that ′𝔪_{Γ_o(n)}[1/n]≃𝔪_{Γ_o(n)}[1/n].)

If an m-gon is classified by ′𝔪_{Γ_o(n)}, then m divides n. The map that forgets A therefore sends ′𝔪_{Γ_o(n)}[1/n] to 𝔪_*. It is étale, and hence the source is a smooth algebraic stack.

4.12. As in 3.5, define the morphism that forgets A and contracts as in (1.3):

c:′𝔪_{Γ_o(n)}[1/n]→𝔪_1[1/n].

Let k be algebraically closed and let C be an object of the source over k. The map Aut(C)→Aut(c(C)) is injective. This is trivial for smooth C. For an m-gon equipped with A⊂C, II.1.10 shows that x↦-x is the only nontrivial automorphism of (C,A). As in 3.4, ′𝔪_{Γ_o(n)}[1/n] is proper over ℤ[1/n]; the argument of 3.5 yields:

Construction 4.13. 𝔪_{Γ_o(n)}[1/n]≃′𝔪_{Γ_o(n)}[1/n].

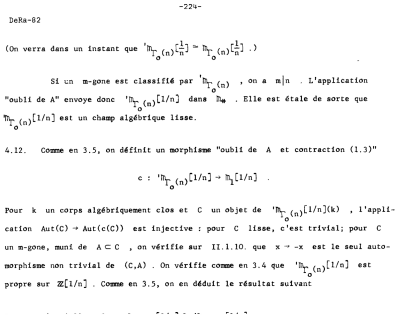
Analogous arguments give:

Construction 4.14. The algebraic stack 𝔪_{Γ_{oo}(n)}[1/n] classifies generalized elliptic curves C/S, with n invertible on S, equipped with an embedding β:ℤ/n↪C whose image meets every irreducible component of every geometric fiber.

Because ℤ[ζ_n][1/n] is étale over ℤ[1/n], normalization in 4.10 also gives:

Construction 4.15. 𝔪_{Γ'_{oo}(n)}[1/n]≃𝔪_{Γ_{oo}(n)}[1/n]⊗ℤ[ζ_n].

4.16. Let A be the subgroup of GL(2,ℤ/n) consisting of scalar matrices. Level-A structures will be called projective level-n structures. They play an important role in Ihara's work. More generally:



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(On verra dans un instant que $'\mathfrak{M}_{\Gamma_o(n)}[\frac{1}{n}] \simeq \mathfrak{M}_{\Gamma_o(n)}[\frac{1}{n}]$.)

Si un m-gone est classifié par $'\mathfrak{M}_{\Gamma_o(n)}$, on a $m|n$. L'application "oubli de A" envoye donc $'\mathfrak{M}_{\Gamma_o(n)}[1/n]$ dans $\mathfrak{M}_1$. Elle est étale de sorte que $\mathfrak{M}_{\Gamma_o(n)}[1/n]$ est un champ algébrique lisse.

4.12. Comme en 3.5, on définit un morphisme "oubli de A et contraction (1.3)"

$$c : '\mathfrak{M}_{\Gamma_o(n)}[1/n] \to \mathfrak{M}_1[1/n] \quad .$$

Pour k un corps algébriquement clos et C un objet de $'\mathfrak{M}_{\Gamma_o(n)}[1/n](k)$, l'application Aut(C) → Aut(c(C)) est injective : pour C lisse, c'est trivial; pour C un m-gone, muni de A ⊂ C , on vérifie sur II.1.10. que x → -x est le seul automorphisme non trivial de (C,A) . On vérifie comme en 3.4 que $'\mathfrak{M}_{\Gamma_o(n)}[1/n]$ est propre sur $\mathbb{Z}[1/n]$. Comme en 3.5, on en déduit le résultat suivant

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Authority page 83; collected-paper page 576; original folio 225.

Definition 4.17. Let C/S be a generalized elliptic curve over S, with n invertible on S, and suppose that the singular geometric fibers of C are n-gons. A projective level-n structure on C is a section of

Isom(C_n,(ℤ/n)²)/A.

Lemma 4.18. Let (C,α) be a generalized elliptic curve over an algebraically closed field, equipped with a projective level-n structure. If n≥2, the only nontrivial automorphism of C preserving α is x↦-x.

For an n-gon the lemma follows easily from II.1.10 and is true even for n=1. For smooth C it is a special case of the following lemma.

Lemma 4.19. Let n≥2 and let g be a finite-order automorphism of an abelian variety X over an algebraically closed field k. If g acts as a scalar on X_n, then g²=1.

By hypothesis there is r∈ℤ/n such that g-r vanishes on X_n. Hence g-r=n·h for some endomorphism h of X. If ζ is a root of the characteristic polynomial of g, then ζ is a root of unity and the algebraic integer ζ-r is divisible by n. Let R be the ring of integers of ℚ(ζ). The image of ζ in R/nR is therefore the image of r. Since R=ℤ[ζ], this is impossible unless ζ=±1. Since g is semisimple, the assertion follows.

Combining Lemma 4.18 for n-gons with the now familiar arguments gives:

Construction 4.20. For A as in 4.16, 𝔪_A[1/n] is the algebraic stack classifying generalized elliptic curves equipped with a projective level-n structure.

For (C,α) classified by 𝔪_A[1/n] and n≥2, Lemma 4.18 shows that Aut(C,α) is constant. Hence, by (1.8):

Proposition 4.21. For A as in 4.16 and n≥2, the map

si n = 1 . Si C est lisse, c'est un cas particulier du lemme suivant.

Lemme 4.19. Soient n ≥ 2 et g un automorphisme d'ordre fini d'une variété abélienne X sur un corps algébriquement clos k . Si g induit une homothétie sur X_n, alors $g^2 = 1$.

Par hypothèse, il existe r ∈ℤ/n tel que g - r soit nul sur X_n . Il existe donc un endomorphisme h de X tel que g-r = n.h . Si ζ est une racine du polynôme caractéristique de g , ζ est une racine de l'unité et l'entier algébrique ζ-r est divisible par n . Si R est l'anneau des entiers du corps ℚ(ζ) , l'image de ζ dans R/nR est donc égale à celle de r . Puisque R = ℤ[ζ] , c'est absurde si $\zeta \neq \pm 1$. Puisque g est semi-simple, 4.19 en résulte.

Du lemme 4.18 pour les n-gones, et d'arguments maintenant familiers, on déduit le résultat suivant.

Construction 4.20. Pour A comme en 4.16, h[1/n] est le champ algébrique classi-

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Authority page 84; collected-paper page 577; original folio 226.

𝔪_A[1/n]→M_A[1/n]

is étale.

Remark 4.22. When n=2, 𝔪_A is simply ′𝔪_2, so the map ′𝔪_2[1/2]→M_2[1/2] is étale.

5. Transcendental theory (review).

5.1. Let E be an elliptic curve over ℂ. The exponential map defines an exact sequence of analytic groups

0→H_1(E,ℤ)→^{ι}Lie(E)→^{exp}E(ℂ)→0.

If β:ℤ²→̃H_1(E,ℤ) is a basis of E, put

z(β)=ι(β(1,0))/ι(β(0,1))∈ℂ-ℝ.

For ι_1,ι_2∈Lie(E), with ι_2≠0, the quotient ι_1/ι_2 denotes the complex number λ satisfying ι_1=λι_2. If γ is the matrix with rows (a,c) and (b,d) in GL(2,ℤ), then

z(βγ)=(a z(β)+b)/(c z(β)+d).

5.2. The map x↦exp((1/n)ι(x)) induces an isomorphism

(5.2.1) H_1(E,ℤ)⊗ℤ/n→̃E_n.

Put X^±=ℂ-ℝ, and let X^+ be the upper half-plane. If α:E_n→̃(ℤ/n)² is a level-n structure, the class of

(z(β),α∘β), with z(β)∈X^± and α∘β∈GL(2,ℤ/n),

in X^±×GL(2,ℤ/n)/GL(2,ℤ) depends only on (E,α).

$\mathcal{H}_2[1/2] \to M_2[1/2]$ est donc étale.

5. Théorie transcendante (rappels).

5.1 Soit E une courbe elliptique sur $\mathbb{C}$. L'application exponentielle définit une suite exacte de groupes analytiques

$$0 \to H_1(E, \mathbb{Z}) \xrightarrow{\iota} \mathrm{Lie}(E) \xrightarrow{\exp} E(\mathbb{C}) \to 0 .$$

Si $\beta : \mathbb{Z}^2 \xrightarrow{\sim} H_1(E, \mathbb{Z})$ est une base de E, nous poserons

$$s(\beta) = s(\beta(1,0))/s(\beta(0,1)) \in \mathbb{C} - \mathbb{R}$$

(pour ι_1 et ι_2 deux éléments de $\mathrm{Lie}(E)$, avec $\iota_2 \neq 0$, on note ι_1/ι_2 le nombre complexe λ tel que $\iota_1 = \lambda\iota_2$).

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5.3. Let H and K be as in 3.6. Construction 5.2 defines an isomorphism

$M_H^o(\mathbb{C}) \tilde{\to} X^{\pm} \times (H\backslash GL(2,\mathbb{Z}/n)/GL(2,\mathbb{Z}))$.

There are also isomorphisms

$X \times (H\backslash GL(2,\mathbb{Z}/n))/SL(2,\mathbb{Z}) \tilde{\to} X^{\pm} \times (H\backslash GL(2,\mathbb{Z}/n))/GL(2,\mathbb{Z}) \tilde{\to} K\backslash X^{\pm} \times GL(2,\mathbb{A}^f)/GL(2,\mathbb{Q})$.

5.4. The reduction morphism $SL(2,\mathbb{Z}) \to SL(2,\mathbb{Z}/n)$ is surjective. Hence the connected components of $M_H^o(\mathbb{C})$ identify with the fibers of

(5.4.1) $X \times (H\backslash GL(2,\mathbb{Z}/n))/SL(2,\mathbb{Z}) \to^{\det} \det H\backslash(\mathbb{Z}/n)^*$.

Let $\mu_n^*(\mathbb{C})$ be the set of primitive n-th roots of unity in $\mathbb{C}$. After extension of scalars from $\mathbb{Z}$ to $\mathbb{C}$, (3.20.4) defines

(5.4.2) $M_H(\mathbb{C}) \to^{d} \det H\backslash\mu_n^*(\mathbb{C})$.

Through the bijection $x \mapsto \exp(2\pi i x/n) : \mathbb{Z}/n \tilde{\to} \mu_n(\mathbb{C})$, (5.4.1) identifies with the restriction of this map to $M_H^o(\mathbb{C})$. The geometric characteristic-zero fibers of (3.20.3) and (3.20.4) are therefore irreducible. The next corollary follows from Zariski's connectedness theorem.

Corollary 5.5. In every characteristic, the geometric fibers of $M_H \to \mathrm{Spec}(\mathbb{Z}[\zeta_n]^{\det H})$ in (3.20.4) are connected.

Corollary 5.6. In characteristic p prime to n, the geometric fibers of (3.20.4) are irreducible. They are connected and geometrically unibranch by 3.10(iv).

Soit $\mu_n^*(\mathbb{C})$ l'ensemble des racines primitives $n^{\text{èmes}}$ de l'unité dans $\mathbb{C}$. Par extension des scalaires de $\mathbb{Z}$ à $\mathbb{C}$, (3.20.4) définit une application.

(5.4.2) $M_H(\mathbb{C}) \xrightarrow{d} \det H\backslash\mu_n^*(\mathbb{C})$.

Via la bijection $x \mapsto \exp(\frac{2\pi i x}{n}) : \mathbb{Z}/n \xrightarrow{\sim} \mu_n(\mathbb{C})$, (5.4.1) s'identifie à la restriction de cette application à $M_H^0(\mathbb{C})$. Les fibres géométriques de caractéristiques 0 de (3.20.3) (3.20.4) sont donc irréductibles. Le corollaire suivant résulte dès lors du théorème de connexion de Zariski.

Corollaire 5.5. En toute caractéristique, les fibres géométriques de (3.20.4) : $M_H \to \mathrm{Spec}(\mathbb{Z}[\zeta_n]^{\det H})$ sont connexes.

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6. Level-H structures: study at infinity.

This section will not be used later in the article. We recommend omitting it on a first reading.

Let H be a subgroup of $GL(2,\mathbb{Z}/n)$. We seek a modular interpretation of $\mathfrak{m}_H[1/n]$ generalizing 3.5, 4.13, 4.14, 4.15, and 4.20. We shall first define modularly a stack $'\mathfrak{m}_H[1/n]$,

proper and smooth over $\mathbb{Z}[1/n]$, with $\mathfrak{m}_H^o[1/n]$ as a dense open substack. As in 3.5 we define a finite morphism $c:'\mathfrak{m}_H[1/n]\to\mathfrak{m}_1[1/n]$. We then prove the analogue of (3.5.1); the proof of 3.5 gives $'\mathfrak{m}_H[1/n]=\mathfrak{m}_H[1/n]$, the desired modular interpretation.

6.1. Let A be a group isomorphic to $(\mathbb{Z}/n)^2$, let $A'\subset A$ be a subgroup isomorphic to $\mathbb{Z}/n$, and let B contain A'. For a suitable divisor m of n, (A,A',B) is isomorphic to

$((\mathbb{Z}/n)^2,\mathbb{Z}/n\times\{0\},\mathbb{Z}/n\times m\mathbb{Z}/n\mathbb{Z})$.

Let μ also be isomorphic to $\mathbb{Z}/n$, let $\beta_0:\Lambda^2A\tilde{\to}\mu$, and denote by β_1 its restriction to Λ^2B.

The essential example is a generalized elliptic curve C over a complete trait S with points η and s. Let $\bar{\eta}$ be the spectrum of an algebraic closure of $k(\eta)$, and $\bar{s}$ the spectrum of the corresponding algebraic closure of k(s). If n is invertible on S, $C_{\bar{\eta}}$ is smooth, and $C_{\bar{s}}$ is an m-sided Néron polygon with m|n, take

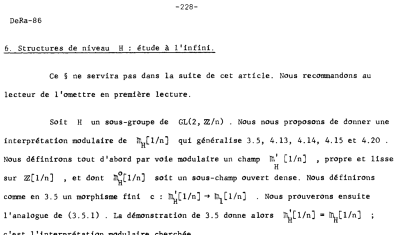
$\mu=\mu_n(\bar{\eta})\simeq\mu_n(\bar{s})$, $A=C_n(\bar{\eta})$, $A'=C_n^o(\bar{s})$, $B=(C_s^{reg})_n(\bar{s})$, and $\beta_0=e_n$.

The restriction β_1 of β_0 to Λ^2B depends only on the special fiber $C_{\bar{s}}$; see 3.21.

6.2. Let I be the set of isomorphisms $\alpha:A\to(\mathbb{Z}/n)^2$, and let $\bar{I}$ be the set of pairs $(\bar{\alpha},\beta)$, with $\bar{\alpha}:B\hookrightarrow(\mathbb{Z}/n)^2$ and $\beta:\mu\tilde{\to}\mathbb{Z}/n$, for which

$$\begin{array}{ccc}\Lambda^2\bar{\alpha}\ \Lambda^2B & \longrightarrow & \Lambda^2(\mathbb{Z}/n)^2\\ \beta_1\ \downarrow & & \downarrow\ s\\ \mu & \xrightarrow{\beta} & \mathbb{Z}/n\end{array}$$

commutes. This is diagram (6.2.1).



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6. Structures de niveau H : étude à l'infini.

Ce § ne servira pas dans la suite de cet article. Nous recommandons au lecteur de l'omettre en première lecture.

Soit H un sous-groupe de GL(2, ℤ/n) . Nous nous proposons de donner une interprétation modulaire de $\mathfrak{M}_H[1/n]$ qui généralise 3.5, 4.13, 4.14, 4.15 et 4.20 . Nous définirons tout d'abord par voie modulaire un champ $\mathfrak{M}'_H[1/n]$, propre et lisse sur $\mathbb{Z}[1/n]$, et dont $\mathfrak{M}^0_H[1/n]$ soit un sous-champ ouvert dense. Nous définirons comme en 3.5 un morphisme fini $c : \mathfrak{M}'_H[1/n] \to \mathfrak{M}_1[1/n]$. Nous prouverons ensuite l'analogue de (3.5.1) . La démonstration de 3.5 donne alors $\mathfrak{M}'_H[1/n] = \mathfrak{M}_H[1/n]$; c'est l'interprétation modulaire cherchée.

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Authority page 87; collected-paper page 580; original folio 229.

Define $c:I\to\bar{I}$ by $\alpha\mapsto(\alpha|_B,\Lambda^2\alpha\circ\beta_0^{-1})$. The group $GL(2,\mathbb{Z}/n)$ acts on I and $\bar{I}$ on the left by $g.\alpha=g\circ\alpha$ and $g.(\bar{\alpha},\beta)=(g\bar{\alpha},\det(g).\beta)$. Let U be the subgroup of Aut(A) acting trivially on A' and A/A'. It acts on the right by $\alpha.u=\alpha\circ u$ and $(\bar{\alpha},\beta)u=(\bar{\alpha}\circ(u|B),\beta)$. These actions are compatible with c.

Lemma 6.2.2. The map c is surjective.

The verification is left to the reader.

Subgroups U' of U correspond bijectively to subgroups B' of A containing A': U' maps to the invariant subgroup $A^{U'}$, while B' maps to the subgroup U(B') of U acting trivially on B'. Moreover $\bar{I}=I/U(B)$.

Let $\bar{a}\in H\backslash\bar{I}$ be the image of $\alpha\in I$. The set of $\alpha'\in I$ mapping to $\bar{a}$ is $H\alpha U(B)$. We call B adequate when U(B) is the subgroup of U stabilizing the image a of α in $H\backslash I$. Equivalently, in the square

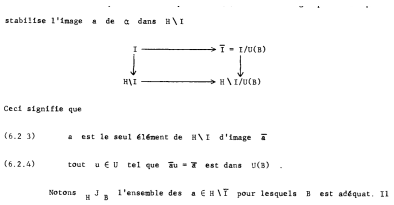
$$\begin{array}{ccc}I & \longrightarrow & \bar{I}=I/U(B)\\ \downarrow & & \downarrow\\ H\backslash I & \longrightarrow & H\backslash I/U(B),\end{array}$$

(6.2.3) a is the unique element of $H\backslash I$ mapping to $\bar{a}$, and (6.2.4) every $u\in U$ satisfying $\bar{a}u=\bar{a}$ lies in U(B).

Let {}_H J_B be the set of a∈H\Ī for which B is adequate. The preceding discussion gives:

Lemma 6.2.5. H\I→̃⨿_{A'⊂B'⊂A}{}_H J_{B'}, the disjoint sum over B'.

6.3. Suppose only a group B and a subgroup A' are given, with (A',B)


stabilise l'image a de α dans H\I

I → Ī = I/U(B)
↓ ↓
H\I → H\I/U(B)

Ceci signifie que

(6.2.3) a est le seul élément de H\I d'image ā

(6.2.4) tout u ∈ U tel que āu = ā est dans U(B) .

Notons $_H J_B$ l'ensemble des a ∈ H\Ī pour lesquels B est adéquat. II

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isomorphic to (ℤ/n×{0},ℤ/nℤ×mℤ/nℤ) for a suitable divisor m of n. Let μ be isomorphic to ℤ/n and let β_1:Λ²B↪μ. Again let Ī be the set of pairs (ᾱ,β), where ᾱ:B↪(ℤ/nℤ)^2 and β:μ→̃ℤ/n make diagram (6.2.1) commute.

The group GL(2,ℤ/n) acts on the left by g.(ᾱ,β)=(g∘ᾱ,det(g).β), and the group Ū of automorphisms of B inducing the identity on A' and B/A' acts on the right by (ᾱ,β)u=(ᾱ∘u,β). Let {}_H J_B be the set of a∈H\Ī such that:

(a) if ã∈Ī maps to ā and gã=ã, then g∈H;

(b) every u∈Ū satisfying āu=ā is the identity.

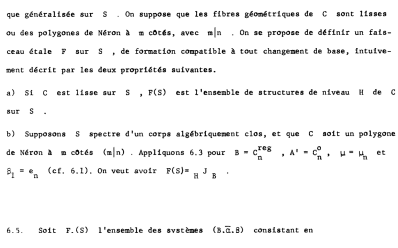
This notation agrees with that of 3.11 whenever the latter makes sense.

6.4. Let S be a scheme on which n is invertible, and let C be a generalized elliptic curve over S whose geometric fibers are smooth or m-sided Néron polygons with m|n. We define an étale sheaf F on S, compatible with arbitrary base change, by the following intended values.

(a) If C is smooth over S, F(S) is the set of level-H structures on C over S.

(b) If S is the spectrum of an algebraically closed field and C is an m-sided Néron polygon with m|n, apply 6.3 with B=C_n^{reg}, A'=C_n^o, μ=μ_n, and β_1=e_n. We require F(S)={}_H J_B.

6.5. Let F_1(S) be the set of systems (B,ᾱ,β) consisting of (a) a finite étale subgroup scheme B⊂C_n and (b) maps ᾱ:B↪(ℤ/n)^2 and β:μ_n→̃ℤ/n such that (c), locally on S, B contains a subgroup isomorphic to ℤ/n.


que généralisée sur S . On suppose que les fibres géométriques de C sont lisses ou des polygones de Néron à m côtés, avec m|n . On se propose de définir un faisceau étale F sur S , de formation compatible à tout changement de base, intuitivement décrit par les deux propriétés suivantes.

a) Si C est lisse sur S , F(S) est l'ensemble de structures de niveau H de C sur S .

b) Supposons S spectre d'un corps algébriquement clos, et que C soit un polygone de Néron à m côtés (m|n) . Appliquons 6.3 pour $B = C_n^{reg}$, $A' = C_n^o$, $\mu = \mu_n$ et $\beta_1 = e_n$ (cf. 6.1). On veut avoir $F(S) = {}_H J_\beta$.

6.5. Soit F.(S) l'ensemble des systèmes (B,ᾱ,β) consistant en

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(d) The diagram

$$\begin{array}{ccc} \bar{\alpha}\ \Lambda^2 B & \longrightarrow & \Lambda^2\mathbb{Z}/n \\ \big\downarrow e_n & & \big\downarrow s \\ \mu_n & \xrightarrow{\ \beta\ } & \mathbb{Z}/n \end{array}$$

commutes.

(e) For every geometric point $\bar{s}$ of S, if $C_{\bar{s}}$ is singular, $(\bar{\alpha},\beta)$ modulo H is of the type described in 6.4. In every case we require that if $g\in GL(2,\mathbb{Z}/n)$ satisfies $(g\bar{\alpha}_{\bar{s}},\det(g).\beta_{\bar{s}})=(\bar{\alpha}_{\bar{s}},$ then $g\in H$.

If C/S is smooth, a system $(B,\bar{\alpha},\beta)$ determines a unique level-H structure on C, locally represented by isomorphisms $\alpha: C_n \tilde{\to} (\mathbb{Z}/n)^2$ extending $\bar{\alpha}$ and having determinant β. Local existence follows from (6.2.1), and uniqueness modulo H from condition (e).

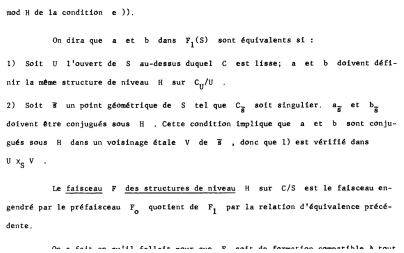
Elements $a,b\in F_1(S)$ are equivalent when:

(1) over the open $U\subset S$ where C is smooth, they define the same level-H structure on C_U/U;

(2) for every geometric point $\bar{s}$ with $C_{\bar{s}}$ singular, $a_{\bar{s}}$ and $b_{\bar{s}}$ are H-conjugate. This implies étale-locally near $\bar{s}$ that a and b are H-conjugate, so (1) holds over $U\times_S V$ for an étale neighborhood V of $\bar{s}$.

The sheaf F of level-H structures on C/S is generated by the presheaf F_0 obtained by quotienting F_1 by this equivalence relation. It is compatible with arbitrary base change, has the values prescribed in 6.4, and is represented by an étale S-scheme S'.

Definition 6.6. (i) A level-H structure on a generalized elliptic curve C/S as in 6.4 is an element of F(S).



mod H de la condition e)).

On dira que a et b dans $F_1(S)$ sont équivalents si :

1) Soit U l'ouvert de S au-dessus duquel C est lisse; a et b doivent définir la même structure de niveau H sur C_U/U .

2) Soit $\bar{s}$ un point géométrique de S tel que $C_{\bar{s}}$ soit singulier. $a_{\bar{s}}$ et $b_{\bar{s}}$ doivent être conjugués sous H . Cette condition implique que a et b sont conjugués sous H dans un voisinage étale V de $\bar{s}$, donc que 1) est vérifié dans $U \times_S V$.

Le faisceau F des structures de niveau H sur C/S est le faisceau engendré par le préfaisceau F_0 quotient de F_1 par la relation d'équivalence précédente.

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(ii) The algebraic stack $\acute{\mathfrak{m}}_H[1/n]$ over $\mathbb{Z}[1/n]$ classifies generalized elliptic curves C/S, with n invertible on S, whose geometric fibers are smooth or m-sided Néron polygons with m|n, equipped with a level-H structure.

Forgetting the level-H structure defines an étale morphism $\acute{\mathfrak{m}}_H[1/n]\to\mathfrak{m}_*$. Thus $\acute{\mathfrak{m}}_H[1/n]$ is smooth over $\mathbb{Z}[1/n]$, and $\acute{\mathfrak{m}}_H{}^\wedge\infty[1/n]$ is étale over $\mathbb{Z}[1/n]$. Contracting as in 1.3 for n=1 and forgetting the level structure defines $c:\acute{\mathfrak{m}}_H[1/n]\to\mathfrak{m}_1$.

Theorem 6.7. (i) The stack $\acute{\mathfrak{m}}_H[1/n]$ is proper and smooth over $\mathbb{Z}[1/n]$, and the image $\acute{\mathfrak{m}}_H{}^\wedge\infty[1/n]$ of the non-smoothness subscheme of the universal curve is finite étale over $\mathbb{Z}[1/n]$.

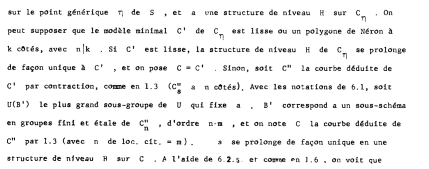
(ii) The morphism c is finite and representable. Consequently $\acute{\mathfrak{m}}_H[1/n]$ agrees with $\mathfrak{m}_H[1/n]$; see 3.5.

It remains to prove that $\acute{\mathfrak{m}}_H[1/n]$ is proper over $\mathbb{Z}[1/n]$ and that c is representable, that is, for every object C of $\acute{\mathfrak{m}}_H[1/n]$ over an algebraically closed field k, the map

(6.7) Aut(C)→Aut(c(C))

is injective.

Properness follows from the valuative criterion and 6.2.5. Let S be a complete trait with algebraically closed residue field, let C_η be an elliptic curve over the generic point η of S, and let a be a level-H structure on C_η. We may suppose that the minimal model C’ of C_η is smooth or a k-sided Néron polygon, with n|k. If C’ is smooth, the level-H structure extends uniquely to C’, and set C=C’. Otherwise let C” be obtained from C’ by contraction as in 1.3, so C”_s has n sides. With the notation of 6.1, let U(B’) be the largest subgroup of U fixing a. Then B’ corresponds to a finite étale subgroup scheme of C”_n of order n·m. Let C be the curve obtained from C” by 1.3, with the n of that reference equal to m. The structure a extends uniquely to a level-H structure on C. Using 6.2.5 and the argument of 1.6, one sees that


sur le point générique η de S, et a une structure de niveau H sur C_η. On peut supposer que le modèle minimal C' de C_η est lisse ou un polygone de Néron à k côtés, avec $n|k$. Si C' est lisse, la structure de niveau H de C_η se prolonge de façon unique à C', et on pose $C = C'$. Sinon, soit C'' la courbe déduite de C' par contraction, comme en 1.3 (C''_s a n côtés). Avec les notations de 6.1, soit $U(B')$ le plus grand sous-groupe de U qui fixe a. B' correspond à un sous-schéma en groupes fini et étale de C''_n, d'ordre $n\cdot m$, et on note C la courbe déduite de C'' par 1.3 (avec n de loc. cit. $= m$). a se prolonge de façon unique en une structure de niveau H sur C. A l'aide de 6.2.5, et comme en 1.6, on voit que

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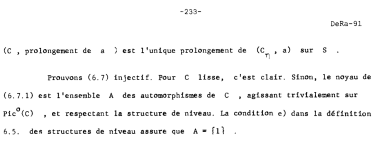
(C, an extension of a) is the unique extension of (C_η,a) over S.

Let us prove that (6.7) is injective. For C smooth, this is clear. Otherwise, the kernel of (6.7.1) is the set A of automorphisms of C that act trivially on $\mathrm{Pic}^0(C)$ and respect the level structure. Condition e) in Definition 6.5 of level structures ensures that A = {1}.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(C, prolongement de a) est l'unique prolongement de (C_η, a) sur S.

Prouvons (6.7) injectif. Pour C lisse, c'est clair. Sinon, le noyau de (6.7.1) est l'ensemble A des automorphismes de C, agissant trivialement sur $\mathrm{Pic}^0(C)$, et respectant la structure de niveau. La condition e) dans la définition 6.5. des structures de niveau assure que $A = \{1\}$.

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V. Reduction modulo p.

In this chapter we study the reduction modulo p of the stacks $\mathfrak{m}_K$, for K = Γ_0(p), Γ_∞(p), and Γ(p^k). Comparison of the results obtained for Γ_0(p) and Γ_∞(p) allows us to show that, in accordance with a conjecture of Shimura, certain abelian schemes studied by Shimura and Casselman [5] have good reduction over the ring of integers of suitable cyclotomic extensions of $\mathbb{Q}$.

1. Study of $\mathfrak{m}_{\Gamma_0(p)}$.

In this section we propose to give a modular interpretation of the stack $\mathfrak{m}_{\Gamma_0(p)}$ when p is prime. Some preliminary results will be used again in the following sections.

1.1. Let $\mathcal{G}$_n be the stack classifying generalized elliptic curves C/S with geometrically irreducible fibers, equipped with a subgroup scheme A locally isomorphic to μ_n (for the étale topology), that is, finite flat locally free of rank n with étale Cartier dual.

Proposition 1.2. The morphism "forget A" : $\mathcal{G}$_n → $\mathfrak{m}$_1 is étale; in particular, $\mathcal{G}$_n is an algebraic stack smooth over $\mathbb{Z}$.

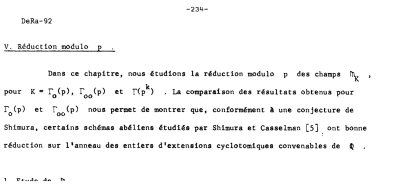
Let C/S be classified by $\mathfrak{m}$_1. It must be checked that the functor

F : (Sch/S) → (Sets),

which assigns to T/S the set of subgroups A of (C_T)_n locally isomorphic to μ_n, is represented by a scheme M étale over S. Representability follows from the theory of the Hilbert scheme; M is étale because it is formally étale (SGA 3 IX 3.6).

1.3. Let $\mathcal{B}$_n be the stack classifying generalized elliptic curves C/S whose geometric fibers are smooth or n-gons, equipped with a subgroup scheme B locally isomorphic to $\mathbb{Z}$/n (for the étale topology) that meets every irreducible component of every geometric fiber.

Let (C,B) be as above. The group scheme B acts freely on C



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DeRa-92

V. Réduction modulo p .

Dans ce chapitre, nous étudions la réduction modulo p des champs $\mathcal{M}_K$, pour $K = \Gamma_0(p)$, $\Gamma_{00}(p)$ et $\Gamma(p^N)$. La comparaison des résultats obtenus pour $\Gamma_0(p)$ et $\Gamma_{00}(p)$ nous permet de montrer que, conformément à une conjecture de Shimura, certains schémas abéliens étudiés par Shimura et Casselman [5] ont bonne réduction sur l'anneau des entiers d'extensions cyclotomiques convenables de $\mathbb{Q}$.

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and C/B has geometrically irreducible fibers. In the exact sequence

0 → B → C_n → A → 0,

the extreme terms are Cartier dual. The pair (C/B,C_n/B) is therefore classified by $\mathcal{G}$_n, and this defines

(1.3.1) w : $\mathcal{B}$_n → $\mathcal{G}$_n.

Proposition 1.4. The morphism 1.3.1 is an isomorphism. In particular, $\mathcal{B}$_n is an algebraic stack smooth over $\mathbb{Z}$.

Proof: Again denote by w the morphism from $\mathcal{G}$_n^{o} to $\mathcal{B}$_n^{o}

w : (C,A) ↦ (C/A,C_n/A).

A construction identical to that of IV.4.4 shows that this morphism is an inverse of the restriction w : $\mathcal{B}$_n^{o} → $\mathcal{G}$_n^{o} of w to $\mathcal{B}$_n^{o}.

Let (C,A) over S be classified by $\mathcal{G}$_n. It must be proved that there exist ($\tilde{C}$,$\tilde{A}$) over S, classified by $\mathcal{B}$_n, and an isomorphism β : w($\tilde{C}$,$\tilde{A}$) $\tilde{\to}$ (C,A), and that ($\tilde{C}$,$\tilde{A}$,β) is unique up to unique isomorphism. By abuse of notation, we again denote by β the composite morphism

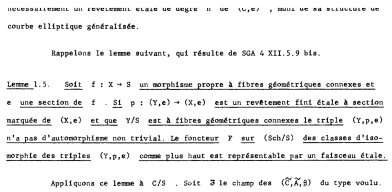
$\tilde{C}$ → $\tilde{C}$/$\tilde{A}$ $\tilde{\to}$^β C.

The curve with marked section ($\tilde{C}$,$\tilde{e}$) will necessarily be an étale covering of degree n of (C,e), endowed with its generalized elliptic-curve structure.

Recall the following lemma, which follows from SGA 4 XII.5.9 bis.

Lemma 1.5. Let f : X → S be a proper morphism with geometrically connected fibers, and let e be a section of f. If p : (Y,e) → (X,e) is a finite étale covering with marked section of (X,e), and Y/S has geometrically connected fibers, the triple (Y,p,e) has no nontrivial automorphism. The functor F on (Sch/S) of isomorphism classes of triples (Y,p,e) as above is represented by an étale sheaf.

Apply this lemma to C/S. Let $\mathcal{J}$ be the stack of triples (C̃,Ã,β) of the required type. The lemma shows that the objects of $\mathcal{J}$ have no nontrivial automorphism, and that the


nécessairement un revêtement étale de degré n de (C,e), muni de sa structure de courbe elliptique généralisée.

Rappelons le lemme suivant, qui résulte de SGA 4 XII.5.9 bis.

Lemme 1.5. Soit f : X → S un morphisme propre à fibres géométriques connexes et e une section de f. Si p : (Y,e) → (X,e) est un revêtement fini étale à section marquée de (X,e) et que Y/S est à fibres géométriques connexes le triple (Y,p,e) n'a pas d'automorphisme non trivial. Le foncteur F sur (Sch/S) des classes d'isomorphie des triples (Y,p,e) comme plus haut est représentable par un faisceau étale.

Appliquons ce lemme à C/S. Soit $\mathcal{J}$ le champ des $(\tilde{C},\tilde{A},\beta)$ du type voulu.

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functor [$\mathcal{J}$] of isomorphism classes of objects of $\mathcal{J}$ is the subfunctor of F formed by the triples (C̃,p;e) such that $p^{-1}(e)$ is locally isomorphic to $\mathbb{Z}/n$ and A is the image of C̃_n. These conditions are open (because A is of multiplicative type), so [$\mathcal{J}$] is represented by T étale over S.

Where C is smooth, one knows that T $\tilde{\to}$ S. Moreover, a 1-gon over an algebraically closed field k admits a unique étale covering by an n-gon. All geometric fibers of T/S are therefore reduced to one point, and T $\tilde{\to}$ S. This proves 1.4.

Let p be a prime number.

Theorem 1.6. $\mathfrak{m}_{\Gamma_0(p)}$ is the algebraic stack classifying generalized elliptic curves C/S equipped with a subgroup scheme A locally free of rank p that meets every irreducible component of every geometric fiber of C/S.

Proof: Recall [22] that a (commutative) group scheme locally free of rank p is killed by p. For A as above, one therefore has A ⊂ C_p.

Provisionally denote by $'\mathfrak{m}_{\Gamma_0(p)}$ the stack which assigns to each scheme S the category whose objects are the pairs (C,A) as in 1.6 and whose morphisms are the isomorphisms. In IV.4.3 we constructed an isomorphism between $\mathfrak{m}_{\Gamma_0(p)}{}^{o}[1/p]$ and $'\mathfrak{m}_{\Gamma_0(p)}{}^{o}[1/p]$. The isomorphism 1.6 will extend this isomorphism; reasoning as in IV.3.5, one finds that 1.6 follows from the conjunction of the following assertions:

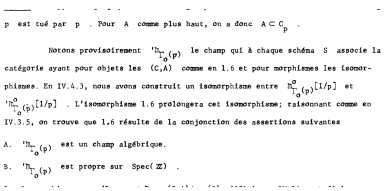
A. $'\mathfrak{m}_{\Gamma_0(p)}$ is an algebraic stack.

B. $'\mathfrak{m}_{\Gamma_0(p)}$ is proper over Spec($\mathbb{Z}$).

C. The morphism c : $'\mathfrak{m}_{\Gamma_0(p)} \to \mathfrak{m}_1$: (C,A) ↦ c(C), defined by IV.1.4, is finite and representable.

D. $'\mathfrak{m}_{\Gamma_0(p)}$ is normal.

E. $'\mathfrak{m}_{\Gamma_0(p)}{}^{o}[1/p]$ is dense in $'\mathfrak{m}_{\Gamma_0(p)}$.



p est tué par p . Pour A comme plus haut, on a donc A ⊂ C_p .

Notons provisoirement $'\mathcal{H}_{\Gamma_0(p)}$ le champ qui à chaque schéma S associe la catégorie ayant pour objets les (C,A) comme en 1.6 et pour morphismes les isomorphismes. En IV.4.3, nous avons construit un isomorphisme entre $\mathcal{H}^0_{\Gamma_0(p)}[1/p]$ et $'\mathcal{H}^0_{\Gamma_0(p)}[1/p]$. L'isomorphisme 1.6 prolongera cet isomorphisme; raisonnant comme en IV.3.5, on trouve que 1.6 résulte de la conjonction des assertions suivantes

A. $'\mathcal{H}_{\Gamma_0(p)}$ est un champ algébrique.

B. $'\mathcal{H}_{\Gamma_0(p)}$ est propre sur Spec(ℤ) .

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(1) Points at infinity of ´m_{Γ_0(p)}.

Let C be the one-sided Néron polygon over Spec(ℤ). One has C^{reg} ≃ G_m, C_p ≃ μ_p, and (C,C_p) is a section e_1 of ´m_{Γ_0(p)} over ℤ.

Let C be the p-gon over Spec(ℤ). One has C^{reg} ≃ G_m × ℤ/p, C_p ≃ μ_p × ℤ/p, and (C,e × ℤ/p) is a section e_2 of ´m_{Γ_0(p)} over ℤ.

Let k be an algebraically closed field. One checks that the only objects (C,A) of ´m_{Γ_0(p)}(k), with C singular, are e_1(k) and e_2(k).

(2) Proof of A (representability of ´m_{Γ_0(p)}).

Let $\mathcal{U}$_1 and $\mathcal{U}$_2 be the following two open substacks of ´m_{Γ_0(p)}: $\mathcal{U}$_1 classifies the curves (C,A) as in 1.6 for which the geometric fibers of C are irreducible, and $\mathcal{U}$_2 classifies the curves (C,A) for which the geometric fibers of C are smooth or p-gons.

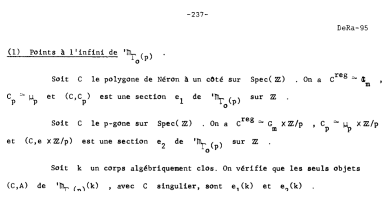
Lemma 1.7. The ”forgetful” morphism φ : $\mathcal{U}$_1 → m_1 is representable.

Proof. It must be seen that, for C/S a generalized elliptic curve with geometrically irreducible fibers over S, the following functor F : (Sch/S) → (Sets) is representable. To T/S, F assigns the set of finite locally free subgroup schemes of rank p of C_T^{reg}. The representability of F follows from the theory of Hilbert schemes.

By 1.7 and IV.2, $\mathcal{U}$_1 is an algebraic stack. We shall see below that $\mathcal{U}$_1 and $\mathcal{U}$_2 are isomorphic. The stack $\mathcal{U}$_2 is therefore algebraic, as is ´m_{Γ_0(p)}, the union of $\mathcal{U}$_1 and $\mathcal{U}$_2 by (1).

1.8. Let (C,A) be classified by $\mathcal{U}$_2. By II.1.20, C_p is locally free of rank p². One checks fiber by fiber that A acts freely on C, and this makes it possible to define

(1.8.1) w : $\mathcal{U}$_2 → $\mathcal{U}$_1 : (C,A) ↦ (C/A,C_p/A_p).



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DeRa-95

(1) Points à l'infini de $'\mathcal{H}_{\Gamma_0(p)}$.

Soit C le polygone de Néron à un côté sur Spec(ℤ) . On a $C^{reg} \simeq \mathbb{G}_m$, $C_p \simeq \mu_p$ et (C,C_p) est une section e_1 de $'\mathcal{H}_{\Gamma_0(p)}$ sur ℤ .

Soit C le p-gone sur Spec(ℤ) . On a $C^{reg} \simeq \mathbb{G}_m \times \mathbb{Z}/p$, $C_p \simeq \mu_p \times \mathbb{Z}/p$ et (C,e × ℤ/p) est une section e_2 de $'\mathcal{H}_{\Gamma_0(p)}$ sur ℤ .

Soit k un corps algébriquement clos. On vérifie que les seuls objets (C,A) de $'\mathcal{H}_{\Gamma_0(p)}(k)$, avec C singulier, sont $e_1(k)$ et $e_2(k)$.

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Authority page 96; collected-paper page 589; original folio 238.

Lemma 1.8.2. The morphism (1.8.1) is an isomorphism.

With the notation of 1.1 and 1.3, one has

$\mathcal{U}$_1 = $\mathcal{G}$_p ∪ m_{Γ_0(p)}° ↗ ↘ ´m_{Γ_0(p)}° = $\mathcal{U}$_1 ∩ $\mathcal{U}$_2 ´m_{Γ_0(p)} ↘ ↗ $\mathcal{U}$_2 = $\mathcal{B}$_p ∪ m_{Γ_0(p)}°.

Construction IV.4.4 again shows that w induces an involution of ′𝔪_{Γ_0(p)}°. In 1.4 we saw that w induces an isomorphism $\mathcal{G}$_p → $\mathcal{B}$_p, and 1.8.2 follows.

This completes the proof of (A). The proof has supplied us with the following construction.

Construction 1.9. By gluing w : $\mathcal{U}$_2 → $\mathcal{U}$_1 and w^{-1} : $\mathcal{U}$_1 → $\mathcal{U}$_2, one obtains an involution w : ′𝔪_{Γ_0(p)} → ′𝔪_{Γ_0(p)} that extends the involution (IV.4.4).

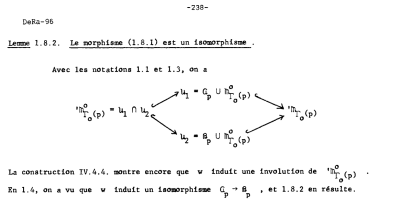
The union of $\mathcal{G}$_p and $\mathcal{B}$_p is the open substack of ′𝔪_{Γ_0(p)} corresponding to non-supersingular curves of characteristic p. Propositions 1.2 and 1.4 therefore give:

Proposition 1.10. ′𝔪_{Γ_0(p)} is smooth over Spec(ℤ) away from the supersingular points of characteristic p.

(3) Proof of (B) (properness of ′𝔪_{Γ_0(p)}).

By (1.10), ′𝔪_{Γ_0(p)}° is dense in ′𝔪_{Γ_0(p)}. The valuative criterion of properness therefore reduces us to proving the following lemma.

Lemma 1.11. Let (S,η,s) be a complete trait with algebraically closed residue field. Let (C,A) be an object of ′𝔪_{Γ_0(p)}°(η). There exists a finite extension k(η′) of k(η) such that, writing S′ for the normalization of S in k(η′), the image (C′,A′) of (C,A) in ′𝔪_{Γ_0(p)}°(η′) comes by base change from an object ($\bar{C}$,$\bar{A}$) of ′𝔪_{Γ_0(p)}(S′), unique up to unique isomorphism.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Lemme 1.8.2. Le morphisme (1.8.1) est un isomorphisme .

Avec les notations 1.1 et 1.3, on a

$'\mathfrak{h}^{o}_{\Gamma_0(p)} = \mathcal{U}_1 \cap \mathcal{U}_2$; $\mathcal{U}_1 = \mathcal{C}_p \cup \mathfrak{h}^{o}_{\Gamma_0(p)}$; $\mathcal{U}_2 = \mathcal{B}_p \cup \mathfrak{h}^{o}_{\Gamma_0(p)}$; $'\mathfrak{m}_{\Gamma_0(p)}$

La construction IV.4.4. montre encore que w induit une involution de $'\mathfrak{h}^{o}_{\Gamma_0(p)}$.
En 1.4, on a vu que w induit un isomorphisme $\mathcal{C}_p \to \mathcal{B}_p$, et 1.8.2 en résulte.

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Proof: After first making an extension of k(η), one may suppose that C has semistable reduction and that the special fiber of its minimal model $\tilde{C}$ is smooth or an m-gon, with p|m (IV.1.6). The group $\tilde{C}$_p is then finite over S (II.1.20), and the schematic closure of A is a finite locally free subgroup scheme of rank p of $\tilde{C}$. One obtains ($\bar{C}$,$\bar{A}$) by contracting the components of $\tilde{C}$_s that do not meet A (IV.1.3).

Uniqueness is proved as in (IV.1.6 (iv)): if ($\bar{C}$_1,$\bar{A}$_1) has generic fiber C, one obtains the minimal model $\tilde{C}$ of C by successively blowing up the singularities, as in (IV.1.6 (iv)). The inverse image of $\bar{A}$_1 is the schematic closure $\tilde{A}$ of A, and it is clear that $\bar{C}$ is obtained by contracting the irreducible components of $\tilde{C}$_s whose intersection with $\tilde{A}$ is empty; hence uniqueness.

(4) Proof of C (c finite and representable).

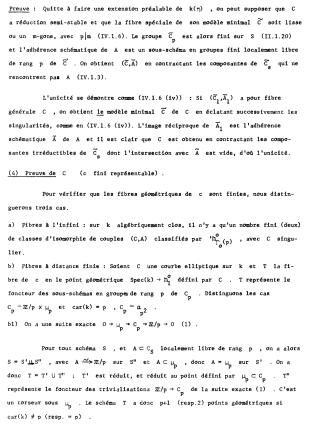
To verify that the geometric fibers of c are finite, we distinguish three cases.

a) Fibers at infinity: over an algebraically closed field k there are only finitely many (two) isomorphism classes of pairs (C,A), classified by ′𝔪_{Γ_0(p)}°, with C singular.

b) Fibers at finite distance: let C be an elliptic curve over k, and let T be the fiber of c over the geometric point Spec(k) → 𝔪_1° defined by C. The scheme T represents the functor of rank-p subgroup schemes of C_p. Distinguish the cases C_p ≃ ℤ/p × μ_p and, when char(k) = p, C_p ≃ α_{p^2}.

b1) There is an exact sequence 0 → μ_p → C_p → $\mathbb{Z}$/p → 0 (1).

For every scheme S and every A ⊂ C_S locally free of rank p, one then has S = S′ ∐ S″, with A $\tilde{\to}$ $\mathbb{Z}$/p over S″ and A ⊂ μ_p, hence A = μ_p over S′. Thus T = T′ ∪ T″; T′ is reduced and consists of the point defined by μ_p ⊂ C_p. The scheme T″ represents the functor of trivializations $\mathbb{Z}$/p → C_p of the exact sequence (1). It is a torsor under μ_p. Consequently T has p+1 (respectively 2) geometric points if char(k) ≠ p (respectively = p).



Preuve : Quitte à faire une extension préalable de k(η) , on peut supposer que C a réduction semi-stable et que la fibre spéciale de son modèle minimal $\overline{C}$ soit lisse ou un n-gone, avec p|n (IV.1.6). Le groupe $\overline{C}_p$ est alors fini sur S (II.1.20) et l'adhérence schématique de A est un sous-schéma en groupes fini localement libre de rang p de $\overline{C}$. On obtient $(\overline{C},\overline{A})$ en contractant les composantes de $\overline{C}_s$ qui ne rencontrent pas A (IV.1.3).

L'unicité se démontre comme (IV.1.6 (iv)) : Si $(\overline{C}_1,\overline{A}_1)$ a pour fibre générale C , on obtient le modèle minimal $\overline{C}'$ de C en éclatant successivement les singularités, comme en (IV.1.6 (iv)). L'image réciproque de $\overline{A}_1$ est l'adhérence schématique $\overline{A}'$ de A et il est clair que C est obtenu en contractant les composantes irréductibles de $\overline{C}'_s$ dont l'intersection avec $\overline{A}'$ est vide, d'où l'unicité.

(4) Preuve de C (c fini représentable) .

Pour vérifier que les fibres géométriques de c sont finies, nous distinguerons trois cas.

a) Fibres à l'infini : sur k algébriquement clos, il n'y a qu'un nombre fini (deux) de classes d'isomorphie de couples (C,A) classifiés par $\mathcal{M}^{ss}_{1,p}(p)$, avec C singulier.

b) Fibres à distance finie : Soient C une courbe elliptique sur k et T la fibre de c en le point géométrique Spec(k) → $\mathcal{M}^0_1$ défini par C . T représente le foncteur des sous-schémas en groupes de rang p de C_p . Distinguons les cas $C_p = \mathbb{Z}/p \times \mu_p$ et car(k) = p , $C_p = \alpha_{p^2}$.

b1) On a une suite exacte 0 → μ_p → C_p → $\mathbb{Z}/p$ → 0 (1) .

Pour tout schéma S , et A ⊂ C_S localement libre de rang p , on a alors S = S′⊔S″ , avec A $\overset{\sim}{\to}$ $\mathbb{Z}/p$ sur S″ et A ⊂ μ_p , donc A = μ_p sur S′ . On a donc T = T′ ∪ T″ ; T′ est réduit, et réduit au point défini par $\mu_p \subset C_p$. T″ représente le foncteur des trivialisations $\mathbb{Z}/p \to C_p$ de la suite exacte (1) . C'est un torseur sous μ_p . Le schéma T a donc p+1 (resp.2) points géométriques si car(k) ≠ p (resp. = p) .

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b2) C_p ≃ α_{p²}. Every subgroup of α_{p²} is infinitesimal. If it has order p, it equals the Frobenius kernel α_p, and T has only one geometric point.

Since ′𝔪_{Γ_0(p)} is proper, c, being proper with finite fibers, is finite. To prove that c is representable, one proceeds as in IV.3.5: one checks over ⟦a blank in the authority⟧ that, for a singular pair (C,A), Aut(C,A) = {±1}, so that

Aut(C,A) ↪ Aut(c(C)).

(5) The key point of the proof is the following lemma.

Lemma 1.12. The morphism c : ′𝔪_{Γ_0(p)}° → 𝔪_1° is finite and flat.

Proof: We shall prove 1.12 using the following criterion, which applies because 𝔪_1° is reduced.

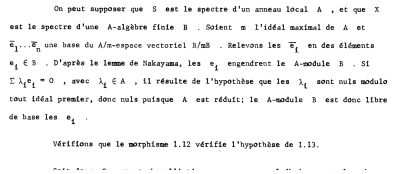
Lemma 1.13. Let f : X → S be a finite morphism of Noetherian schemes, with S reduced. If the rank of the geometric fibers of f is constant, then f is flat.

One may suppose that S is the spectrum of a local ring A and that X is the spectrum of a finite A-algebra B. Let 𝔪 be the maximal ideal of A, and let ē_1,...,ē_n be a basis of the A/𝔪-vector space B/𝔪B. Lift the ē_i to elements e_i ∈ B. By Nakayama's lemma, the e_i generate the A-module B. If ∑ λ_i e_i = 0, with λ_i ∈ A, the hypothesis implies that the λ_i vanish modulo every prime ideal, and hence vanish because A is reduced; thus the A-module B is free with basis the e_i.

Let us verify that the morphism 1.12 satisfies the hypothesis of 1.13.

Let C be an elliptic curve over an algebraically closed field k. The fiber T of c over the geometric point Spec(k) → 𝔪_1° defined by C represents the functor of rank-p subgroup schemes of C_p. We prove that it has rank p+1.

If C_p ≃ μ_p × ℤ/p, we saw in (4) that T is isomorphic to Spec(k) ⨿ μ_p,



On peut supposer que S est le spectre d'un anneau local A , et que X est le spectre d'une A-algèbre finie B . Soient m l'idéal maximal de A et $\bar{e}_1 \ldots \bar{e}_n$ une base du A/m-espace vectoriel B/mB . Relevons les $\bar{e}_i$ en des éléments $e_i \in B$. D'après le lemme de Nakayama, les e_i engendrent le A-module B . Si $\Sigma \lambda_i e_i = 0$, avec $\lambda_i \in A$, il résulte de l'hypothèse que les λ_i sont nuls modulo tout idéal premier, donc nuls puisque A est réduit; le A-module B est donc libre de base les e_i .

Vérifions que le morphisme 1.12 vérifie l'hypothèse de 1.13.

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and hence has rank p+1.

It remains to treat the case in which k has characteristic p and C_p ≃ α_{p²}. In this case T represents the functor of locally free rank-p subschemes A of 𝔾_a contained in α_{p²}.

Over an arbitrary base, a locally free rank-p subscheme A of 𝔾_a is defined by an equation

(*) X^p + ∑_{0}^{p-1} a_i X^i = 0.

For A to be stable under the homothety X ↦ nX (n ∈ (ℤ/p)*), it is necessary and sufficient that (*) be identical to the equation

X^p + ∑_{0}^{p-1} n^{p-i} a_i X^i.

In particular, if A is a subgroup scheme of 𝔾_a, its equation reduces to

X^p - aX = 0.

For A to be a subscheme of α_{p²}, it is necessary and sufficient that the left-hand side of this equation divide X^{p²}. Since

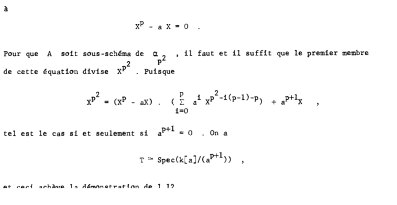
X^{p²} = (X^p - aX) · (∑_{i=0}^{p} a^i X^{p²-i(p-1)-p}) + a^{p+1}X,

this holds if and only if a^{p+1} = 0. Thus

T ≃ Spec(k[a]/(a^{p+1})),

which completes the proof of 1.12.

Since c is finite and flat and 𝔪_1° is regular, ′𝔪_{Γ_0(p)}° is Cohen–Macaulay (EGA IV.15 4.2), purely of relative dimension one over Spec(ℤ). At infinity, 𝔪_{Γ_0(p)} is even smooth (1.10); consequently the whole of ′𝔪_{Γ_0(p)} is Cohen–Macaulay and, again by 1.10, smooth over Spec(ℤ) outside a finite set (of codimension 2). Assertion (D) therefore follows from Serre's criterion for normality (EGA IV.5.8.6).



à

$$X^p - a X = 0 \ .$$

Pour que A soit sous-schéma de α_{p^2} , il faut et il suffit que le premier membre de cette équation divise X^{p^2} . Puisque

$$X^{p^2} = (X^p - aX) \ . \ \left(\sum_{i=0}^{p} a^i X^{p^2-1-(p-1)-p}\right) + a^{p+1}X \ ,$$

tel est le cas si et seulement si $a^{p+1} = 0$. On a

$$T \simeq \mathrm{Spec}(k[a]/(a^{p+1})) \ ,$$

et ceci achève la démonstration de 1.12

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1.14. Variants: Let n be an integer prime to p, let H ⊂ GL(2, ℤ/n), and let K be the inverse image of H in GL(2, $\hat{\mathbb{Z}}$). Theorem 1.6 generalizes to $\mathfrak{m}_{K\cap\Gamma_0(p)}[1/n]$:

(1.14.1) $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}[1/n]$ classifies generalized elliptic curves C/S whose geometric fibers are smooth, n-gons, or np-gons, equipped with an isomorphism α : $C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ and a locally free rank-p subgroup H of C such that $C_n \cdot H$ meets every geometric component of every geometric fiber.

(.14.2) For square-free n, $\mathfrak{m}_{\Gamma_0(n)}$ classifies generalized elliptic curves C/S equipped with a locally free rank-n subgroup H of C that meets every geometric component of every geometric fiber.

1.15. Let S be a scheme of characteristic p and E an elliptic curve over S. To E one associates the following two objects of $\mathfrak{m}_{\Gamma_0(p)}{}^{\circ}(S)$:

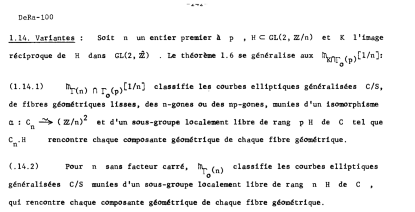
$\Phi_1(E) : (E, \mathrm{Ker}(F : E \to E^{(p)}))$

$\Phi_2(E) = w\Phi_1(E) : (E^{(p)}, \mathrm{Ker}(V : E^{(p)} \to E))$.

In these formulas, F denotes the Frobenius morphism and V its transpose, the Verschiebung: FV = VF = p. These constructions define a diagram

$$\begin{array}{ccccc} \mathfrak{m}_1{}^{\circ}\otimes\mathbb{F}_p & & \mathfrak{m}_1{}^{\circ}\otimes\mathbb{F}_p & & \\ & \searrow^{\Phi_1} \quad {}^{w\Phi_1}\swarrow & & & \\ & & \mathfrak{m}_{\Gamma_0(p)}{}^{\circ}\otimes\mathbb{F}_p & & \\ & & & \searrow^{c} \quad {}_{cw}\swarrow & \\ & & & & \mathfrak{m}_1{}^{\circ}\otimes\mathbb{F}_p \\ & & & & \mathfrak{m}_1{}^{\circ}\otimes\mathbb{F}_p \end{array}$$

The composites $c\Phi_1 : E \mapsto (E, \mathrm{Ker}\, F) \mapsto E$ and $cww\Phi_1$ are the identity. The composites $cw\Phi_1 : E \mapsto (E^{(p)}, \mathrm{Ker}\, V) \mapsto E^{(p)}$ are the Frobenius endomorphism of $\mathfrak{m}_1{}^{\circ} \otimes \mathbb{F}_p$.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1.14. Variantes : Soit n un entier premier à p , H ⊂ GL(2, ℤ/n) et K l'image réciproque de H dans GL(2, $\hat{\mathbb{Z}}$) . Le théorème 1.6 se généralise aux $\mathfrak{m}_{K\cap\Gamma_0(p)}[1/n]$:

(1.14.1) $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}[1/n]$ classifie les courbes elliptiques généralisées C/S, de fibres géométriques lisses, des n-gones ou des np-gones, munies d'un isomorphisme $\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ et d'un sous-groupe localement libre de rang p H de C tel que $C_n \cdot H$ rencontre chaque composante géométrique de chaque fibre géométrique.

(.14.2) Pour n sans facteur carré, $\mathfrak{m}_{\Gamma_0(n)}$ classifie les courbes elliptiques généralisées C/S munies d'un sous-groupe localement libre de rang n H de C , qui rencontre chaque composante géométrique de chaque fibre géométrique.

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Authority page 101; collected-paper page 594; original folio 243.

Theorem 1.16. (i) The maps Φ_1 and $\Phi_2 = w\Phi_1$ are closed immersions. Their images are the two irreducible components of $\mathfrak{m}_{\Gamma_0(p)}{}^{\circ} \otimes \mathbb{F}_p$.

(ii) These irreducible components meet transversely at the supersingular points. The stack $\mathfrak{m}_{\Gamma_0(p)} \otimes \mathbb{F}_p$, which is reduced, therefore has no singularities other than ordinary double points.

(iii) $\mathfrak{m}_{\Gamma_0(p)}$ is regular.

Use a superscript h to denote the stack obtained by removing the supersingular points of characteristic p. One checks that Φ_1 and $w\Phi_1$ define an isomorphism

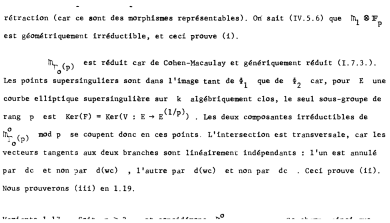
$\Phi_1 \amalg w\Phi_1 : \mathfrak{m}_1{}^{oh} \otimes \mathbb{F}_p \amalg \mathfrak{m}_1{}^{oh} \otimes \mathbb{F}_p \xrightarrow{\sim} \mathfrak{m}_{\Gamma_0(p)}{}^{oh} \otimes \mathbb{F}_p$.

The inverse isomorphism is defined by c and cw, restricted to the images of Φ_1 and $w\Phi_1$.

That Φ_1 (respectively $w\Phi_1$) is a closed immersion follows from the existence of a retraction (since these are representable morphisms). It is known (IV.5.6) that $\mathfrak{m}_1 \otimes \mathbb{F}_p$ is geometrically irreducible, and this proves (i).

The stack $\mathfrak{m}_{\Gamma_0(p)}$ is reduced because it is Cohen–Macaulay and generically reduced (I.7.3). The supersingular points lie in the images of both Φ_1 and Φ_2 because, for a supersingular elliptic curve E over an algebraically closed field k, the only subgroup of rank p is Ker(F) = Ker(V : E → E^{(1/p)}). Thus the two irreducible components of $\mathfrak{m}_{\Gamma_0(p)}{}^{\circ}$ modulo p meet at these points. The intersection is transverse because the tangent vectors to the two branches are linearly independent: one is killed by dc and not by d(wc), while the other is killed by d(wc) and not by dc. This proves (ii). We shall prove (iii) in 1.19.

Variant 1.17. Let $n \geq 3$ and consider $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ}$. This stack, as well as $\mathfrak{m}_{\Gamma(n)}{}^{\circ}$, is a scheme: the classified objects have no automorphisms (cf. IV.2.7). By 1.14 these schemes represent the functors of isomorphism classes of objects of the following type


rétraction (car ce sont des morphismes représentables). On sait (IV.5.6) que $\mathfrak{h}_1 \otimes \mathbb{F}_p$ est géométriquement irréductible, et cecí prouve (i).

$\mathfrak{h}_{\Gamma_0(p)}$ est réduit car de Cohen-Macaulay et génériquement réduit (I.7.3.). Les points supersinguliers sont dans l'image tant de Φ_1 que de Φ_2 car, pour E une courbe elliptique supersingulière sur k algébriquement clos, le seul sous-groupe de rang p est Ker(F) = Ker(V : E → $E^{(1/p)}$). Les deux composantes irréductibles de $\mathfrak{h}^{\circ}_{\Gamma_0(p)}$ mod p se coupent donc en ces points. L'intersection est transversale, car les vecteurs tangents aux deux branches sont linéairement indépendants : l'un est annulé par dc et non par d(wc), l'autre par d(wc) et non par dc. Cecí prouve (ii). Nous prouverons (iii) en 1.19.

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$\mathfrak{m}_{\Gamma(n)}{}^{\circ}$: (E,α_n) : α_n : E_n ⥲ $(\mathbb{Z}/n\mathbb{Z})^2$;

$\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ}$: (E,α_n,H) : α_n : E_n ⥲ $(\mathbb{Z}/n\mathbb{Z})^2$; H ⊂ E of rank p,

or, more symmetrically (cf. IV.4.5), set F = E/H.

(E →^h F, α_n) : h a p-isogeny and α_n : E_n ⥲_{h̄} F_n ⥲ $(\mathbb{Z}/n\mathbb{Z})^2$.

After reduction modulo p, there is a diagram analogous to (1.15.1). Its arrows (expressed as morphisms of functors) are

$\Phi_1 : \mathfrak{m}_{\Gamma(n)}{}^{\circ} \otimes \mathbb{F}_p \to \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} \otimes \mathbb{F}_p : (E,\alpha_n) \mapsto (E,\alpha_n, \mathrm{Ker}\, F)$

$w : \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} \xrightarrow{\sim} \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} : (E,\alpha_n,H) \mapsto (E/H,\alpha_n,E_p/H)$

$c : \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} \to \mathfrak{m}_{\Gamma(n)}{}^{\circ} : (E,\alpha_n,H) \mapsto (E,\alpha_n).$

Take care that the isomorphism $w^2(E,H) \simeq (E,H)$ defined in IV.4.4 induces an isomorphism $w^2(E,\alpha_n,H) \simeq (E,p\alpha_n,H)$. The composite $cw\Phi_1 : (E,\alpha_n) \mapsto (E^{(p)},\alpha_n)$ is Frobenius.

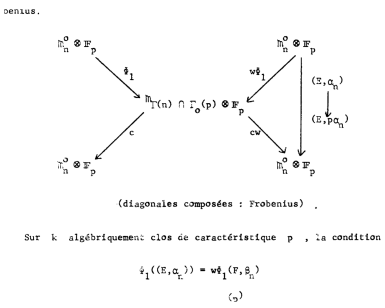
$\mathfrak{m}_n{}^{\circ} \otimes \mathbb{F}_p$ $\mathfrak{m}_n{}^{\circ} \otimes \mathbb{F}_p$ ╲ Φ_1 $w\Phi_1$ ╱ │ ↘ ↙ │ (E,α_n) $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} \otimes \mathbb{F}_p$ ↓ ↙ ↘ │ $(E,p\alpha_n)$ c ╱ ╲ cw $\mathfrak{m}_n{}^{\circ} \otimes \mathbb{F}_p$ $\mathfrak{m}_n{}^{\circ} \otimes \mathbb{F}_p$

(composite diagonals: Frobenius).

Over an algebraically closed field k of characteristic p, the condition

$\Phi_1((E,\alpha_r)) = w\Phi_1(F,\beta_n)$

means that F is supersingular, that $E = F^{(p)}$, and that the diagram



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$E_n = F_n^{(p)} \tilde{\leftarrow} F_n \mid \alpha_n \mid \beta_n \downarrow \downarrow (\mathbb{Z}/n\mathbb{Z})^2 = (\mathbb{Z}/n\mathbb{Z})^2$

is commutative, i.e. $(E,\alpha_n) = (F,\beta_n)^{(p)}$. Parts (i) and (ii) of 1.16 therefore admit the following variant.

Variant 1.18. The scheme $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}^{\circ} \otimes_{\mathbb{Z}} \mathbb{F}_p$ can be obtained by gluing two copies of $\mathfrak{m}_{\Gamma(n)}^{\circ} \otimes \mathbb{F}_p$ along the supersingular points: the supersingular point x of the second copy is identified with the point $x^{(p)}$ of the first copy.

1.19. Let us prove that $\mathfrak{m}_{\Gamma_0(p)}$ is regular. Away from the supersingular points, this follows from 1.10. Since at a supersingular point $\mathfrak{m}_{\Gamma_0(p)} \to \mathrm{Spec}(\mathbb{Z})$ has an ordinary quadratic singularity, (I.5.2) shows that there exists $k \in \mathbb{N}$ such that the completion of the strict local ring of $\mathfrak{m}_{\Gamma_0(p)}$ at such a point is isomorphic to

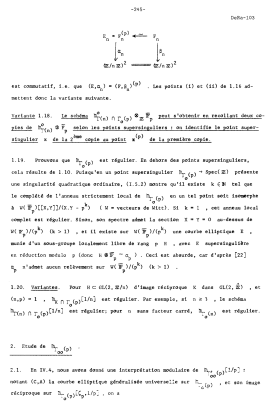
$W(\mathbb{F}_p)[[X,Y]]/(X.Y - p^k)$ (W = Witt vectors).

If $k = 1$, this complete local ring is regular. Otherwise, its spectrum has the section $X = Y = 0$ over $W(\mathbb{F}_p)/(p^k)$ $(k > 1)$, and over $W(\mathbb{F}_p)/(p^k)$ there exists an elliptic curve E equipped with a locally free subgroup H of rank p, with E supersingular after reduction modulo p (hence $H \otimes \mathbb{F}_p \simeq \alpha_p$). This is absurd, because by [22] α_p admits no lifting over $W(\mathbb{F}_p)/(p^k)$ when $k > 1$.

1.20. Variants. Let $H \subset GL(2, \mathbb{Z}/n)$, let K be its inverse image in $GL(2, \hat{\mathbb{Z}})$, and suppose $(n,p) = 1$. Then $\mathfrak{m}_{K\cap\Gamma_0(p)}[1/n]$ is regular. For example, if $n \geq 3$, the scheme $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}[1/n]$ is regular; for square-free n, $\mathfrak{m}_{\Gamma_0(n)}$ is regular.

2. Study of $\mathfrak{m}_{\Gamma'_\infty(p)}$.

2.1. In IV.4 we gave a modular interpretation of $\mathfrak{m}_{\Gamma'_\infty(p)}[1/p]$: denoting by (C,A) the universal generalized elliptic curve over $\mathfrak{m}_{\Gamma_0(p)}$, and also its inverse image over $\mathfrak{m}_{\Gamma_0(p)}[\zeta_p,1/p]$, one has



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$\mathfrak{m}_{\Gamma'_\infty(p)}[1/p] = \mathrm{Isom}_{\mathfrak{m}_{\Gamma_0(p)}[\zeta_p,1/p]}(\mathbb{Z}/p,A)$.

By definition, $\mathfrak{m}_{\Gamma'_\infty(p)}$ is the normalization of $\mathfrak{m}_{\Gamma_0(p)}[\zeta_p]$ in this stack $\mathrm{Isom}_{\mathfrak{m}_{\Gamma_0(p)}[\zeta$
One readily deduces the description below of the stack $\mathfrak{m}_{\Gamma'_\infty(p)}^h$ obtained from $\mathfrak{m}_{\Gamma'_\infty(p)}$ by removing the supersingular points of characteristic p.

2.2. Let $\mathcal{G}'_p \subset \mathfrak{m}_{\Gamma_0(p)}$ be the open substack classifying the pairs (C,A) for which A is locally isomorphic to μ_p (for the étale topology), and let $\mathcal{B}'_p \subset \mathfrak{m}_{\Gamma_0(p)}$ be the open substack classifying the pairs (C,A) for which A is locally isomorphic to $\mathbb{Z}/p$. Set

$\mathcal{U} = \mathrm{Isom}_{\mathcal{G}'_p}(\mu_p,A)$

$\mathcal{V} = \mathrm{Isom}_{\mathcal{B}'_p}(\mathbb{Z}/p,A)$;

the schemes $\mathcal{U}$ and $\mathcal{V}$ are finite étale over $\mathcal{G}'_p$ and $\mathcal{B}'_p$, respectively. Over $\mathbb{Z}[\zeta_p,1/p]$, $\mathbb{Z}/p$ and μ_p are canonically isomorphic, hence there is an isomorphism

$\mathcal{U}[\zeta_p,1/p] \simeq \mathcal{V}[\zeta_p,1/p]$.

Proposition 2.3. The stack $\mathfrak{m}_{\Gamma'_\infty(p)}^h$ is obtained by gluing $\mathcal{U}[\zeta_p]$ and $\mathcal{V}[\zeta_p]$ along the common open substack $\mathcal{U}[\zeta_p,1/p]$. In particular, $\mathfrak{m}_{\Gamma'_\infty(p)}^h$ is smooth over $\mathbb{Z}[\zeta_p]$.

To study $\mathfrak{m}_{\Gamma'_\infty(p)}$ near the supersingular points, we shall use the results of [22] recalled below.

2.4. The functor F that assigns to a scheme S the set of isomorphisms

$\chi : \mathbb{F}_p^* \tilde{\to} \mu_{p-1}$

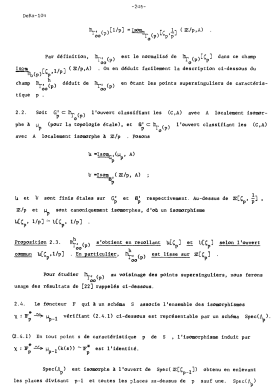
satisfying (2.4.1) below is represented by a scheme $\mathrm{Spec}(\Lambda_p)$.

(2.4.1) At every point s of S of characteristic p, the induced isomorphism

$\chi : \mathbb{F}_p^* \tilde{\to} \mu_{p-1}(k(s)) \simeq \mathbb{F}_p^*$

is the identity.

The scheme $\mathrm{Spec}(\Lambda_p)$ is isomorphic to the open subscheme of $\mathrm{Spec}(\mathbb{Z}[\zeta_{p-1}])$ obtained by removing the places dividing p-1 and all but one of the places above p. $\mathrm{Spec}(\Lambda_p)$



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is étale over Spec(ℤ), and 𝔽_p ↪̃ Λ_p ⊗ 𝔽_p.

Let S be equipped with χ satisfying (2.4.1) (so S is a Λ_p-scheme), and let A be a locally free group scheme of rank p over S. Let L be the submodule of the affine algebra 𝒪(A) of A (a sheaf of 𝒪_S-algebras) formed by those f such that

f(ix) = χ(i) f(x) (i ∈ 𝔽_p^*).

By [22], L is an invertible module over S, and there exists a unique section a of L^{⊗(1-p)} such that, for f as above,

f^p = ⟨a,f^{p-1}⟩.f in L ⊂ 𝒪(A).

Let A* be the Cartier dual of A, and let L* be defined by A*. The duality e between 𝒪(A) and 𝒪(A*) puts L and L* in duality. Let b ∈ (L*)^{⊗ 1-p} be the analogue for A* of the section a above. Then

a.b = w(χ)

for a certain number w(χ) ∈ Λ_p, with (w(χ)) = (p) in Λ_p.

The functor (on isomorphisms)

(A : a locally free group scheme of rank p) ↦ ((L,a,b) : L invertible, a ∈ L^{⊗(1-p)}, b ∈ L^{⊗(p-1)}, a.b = w(χ))

is an equivalence of categories.

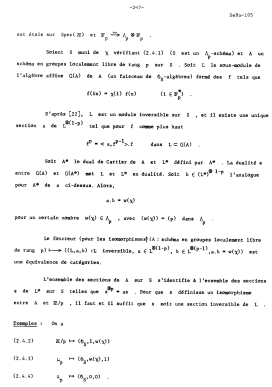
The set of sections of A over S is identified with the set of sections s of L* over S such that s^{⊗p} = as. For s to define an isomorphism between A and ℤ/p, it is necessary and sufficient that s be an invertible section of L*.

Examples: One has

(2.4.2) ℤ/p ↦ (𝒪_S,1,w(χ))

(2.4.3) μ_p ↦ (𝒪_S,w(χ),1)

(2.4.4) α_p ↦ (𝒪_S,0,0).



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2.5. Let an isomorphism ζ : $\mathbb{Z}/p \tilde{\to}$ μ_p be given over S (where S is a Λ_p[ζ_p,1/p]-scheme). The isomorphism ζ defines ζ* : $\mathcal{O}$(μ_p) → $\mathcal{O}$($\mathbb{Z}$/p) and, through (2.4.2) and (2.4.3), induces

ζ* : $\mathcal{O}$_S = L(μ_p) → L($\mathbb{Z}$/p) = $\mathcal{O}$_S,

namely multiplication by a number w(χ,ζ) ∈ Λ_p[ζ_p]. One has

w(χ,ζ)^{p-1} = w(χ).

An isomorphism s : $\mathbb{Z}/p \tilde{\to}$ A defines by duality

s* : A* $\tilde{\to}$ μ_p $\tilde{\leftarrow}$ $\mathbb{Z}$/p,

whose inverse is t. The sections s(1) and t(1) are identified with sections x and y of L* and L satisfying

x^{p-1} = a y^{p-1} = b

xy = w(χ,ζ).

2.6. Let ′$\mathfrak{m}$_{Γ′_∞(p)} ⊗ Λ_p be the following algebraic Λ_p[ζ_p]-stack over $\mathfrak{m}$_{Γ_0(p)} ⊗ Λ_p[ζ_p]: an object (C,A,x,y) of ′$\mathfrak{m}$_{Γ′_∞(p)} ⊗ Λ_p over a Λ_p[ζ_p]-scheme S consists of

a) an object (C,A) of $\mathfrak{m}$_{Γ_0(p)}. Let (L,a,b) be deduced from A.

b) sections x of L* and y of L such that x^{p-1} = a, y^{p-1} = b, and xy = w(χ,ζ).

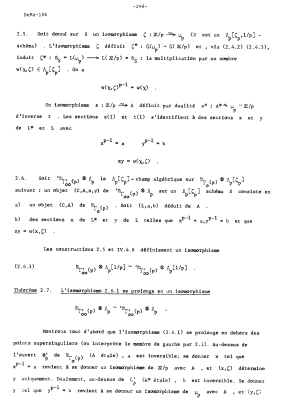
The constructions of 2.5 and IV.4.8 define an isomorphism

(2.6.1) $\mathfrak{m}$_{Γ′_∞(p)} ⊗ Λ_p[1/p] ≃ ′$\mathfrak{m}$_{Γ′_∞(p)} ⊗ Λ_p[1/p].

Theorem 2.7. The isomorphism 2.6.1 extends to an isomorphism

$\mathfrak{m}$_{Γ′_∞(p)} ⊗ Λ_p ≃ ′$\mathfrak{m}$_{Γ′_∞(p)} ⊗ Λ_p.

Let us first show that the isomorphism (2.6.1) extends away from the supersingular points (the left-hand side is interpreted using 2.3). Over the open substack $\mathcal{B}$′_p of $\mathfrak{m}$_{Γ_0(p)} (where A is étale), a is invertible; giving x such that x^{p-1} = a amounts to giving an isomorphism from $\mathbb{Z}$/p to A, and (x,ζ) uniquely determines y. Dually, over $\mathcal{G}$′_p (where A* is étale), b is invertible. Giving y such that y^{p-1} = b amounts to giving an isomorphism from μ_p to A, and (y,ζ)



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uniquely determines x. Where p is invertible, these two constructions are compatible through the isomorphism ζ : ℤ/p ⟶̃ μ_p.

The stack ´𝔪_{Γ´_∞(p)} ⊗ Λ_p is finite and representable over 𝔪_{Γ_0(p)} ⊗ Λ_p. It therefore suffices to prove that ´𝔪_{Γ´_∞(p)} ⊗ Λ_p is normal at the supersingular points. The following lemma shows that ´𝔪_{Γ´_∞(p)} ⊗ Λ_p is in fact regular at these points.

Lemma 2.8. Let s̄ be a supersingular geometric point of characteristic p of 𝔪_{Γ_0(p)}. Let Ŝ be the completion of the strict henselization of 𝔪_{Γ_0(p)} at s̄, equipped with its unique Λ_p-algebra structure. Let (C,A) over Ŝ be defined by Ŝ → 𝔪_{Γ_0(p)}, and let (L,a,b) be defined by A.

(i) Ŝ ≃ W(𝔽_p)[[u,v]]/(uv-w(χ)) with (L,a,b) ≃ (𝒪,u,v).

(ii) The normalization ´𝔪_{Γ´_∞(p)} ⊗_{𝔪_{Γ_0(p)}} Ŝ of Ŝ in Isom_{Ŝ[ζ_p,1/p]}(ℤ/p,A) is Ŝ-isomorphic to

W(𝔽_p)[[ζ_p,x,y]]/(xy-w(χ,ζ)) ≃ W(𝔽_p)[[x,y]]/(x^{p-1} y^{p-1} - w(χ)),

with x^{p-1} = u and y^{p-1} = v.

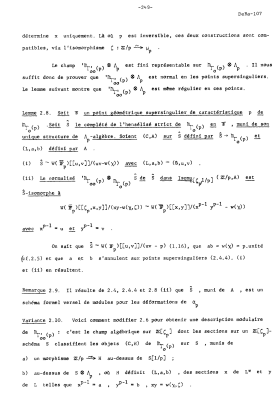
It is known that Ŝ ≃ W(𝔽_p)[[u,v]]/(uv - p) (1.16), that ab = w(χ) = p times a unit (cf. 2.5), and that a and b vanish at the supersingular points (2.4.4). Assertions (i) and (ii) follow.

Remark 2.9. It follows from 2.4, 2.4.4, and 2.8(ii) that Ŝ, equipped with A, is a versal formal moduli scheme for deformations of α_p.

Variant 2.10. Here is how to modify 2.6 to obtain a modular description of 𝔪_{Γ´_∞(p)}: it is the algebraic stack over ℤ[ζ_p] whose sections over a ℤ[ζ_p]-scheme S classify the objects (C,H) of 𝔪_{Γ_0(p)} over S, equipped with

a) a morphism ℤ/p ⟶̃ H over S[1/p];

b) over S ⊗ Λ_p, where H defines (L,a,b), sections x of L* and y of L such that x^{p-1} = a, y^{p-1} = b, and xy = w(χ,ζ).



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These two data are required to be compatible, through the dictionary of 2.4, over S ⊗ Λ_p[1/p].

2.11. With the notation of 2.2 and of 1.1 and 1.3, $\mathcal{U}$ ⊗ $\mathbb{F}$_p and $\mathcal{V}$ ⊗ $\mathbb{F}$_p are étale covers of $\mathcal{G}$_p ⊗ $\mathbb{F}$_p and $\mathcal{B}$_p ⊗ $\mathbb{F}$_p, respectively. The morphisms Φ_1 and wΦ_1 (cf. 1.15) identify $\mathcal{G}$_p ⊗ $\mathbb{F}$_p and $\mathcal{B}$_p ⊗ $\mathbb{F}$_p with open substacks of $\mathfrak{m}$_1 ⊗ $\mathbb{F}$_p; through this identification, the local structure theorem 2.8 shows that the covers $\mathcal{U}$ ⊗ $\mathbb{F}$_p and $\mathcal{V}$ ⊗ $\mathbb{F}$_p of $\mathfrak{m}$_1^h ⊗ $\mathbb{F}$_p are totally ramified at the supersingular points of $\mathfrak{m}$_1 ⊗ $\mathbb{F}$_p. In particular, $\mathcal{U}$ ⊗ $\mathbb{F}$_p and $\mathcal{V}$ ⊗ $\mathbb{F}$_p are geometrically irreducible, and $\mathfrak{m}$_{Γ′_∞(p)} ⊗ $\mathbb{F}$_p has two irreducible components, the inverse images of the irreducible components of $\mathfrak{m}$_{Γ_0(p)} ⊗ $\mathbb{F}$_p. The proof of 2.7 gives the following result.

Theorem 2.12. (i) The two irreducible components of $\mathfrak{m}$_{Γ′_∞(p)} ⊗ $\mathbb{F}$_p are smooth and meet transversely at the supersingular points.

(ii) The scheme $\mathfrak{m}$_{Γ′_∞(p)} is regular; the morphism $\mathfrak{m}$_{Γ′_∞(p)} → Spec($\mathbb{Z}$[ζ_p]) has only quadratic singularities.

(iii) The morphism $\mathfrak{m}$_{Γ′_∞(p)} → $\mathfrak{m}$_{Γ_0(p)} induces a bijection (an equivalence of categories) on the supersingular geometric points of characteristic p.

2.13. Let n be an integer ≥ 3 and prime to p. As in 1.17, one can make 2.12 more concrete by passing to the scheme $\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)}. Let $\mathcal{J}$_n^h be the étale cover Isom(μ_p,(C_p)^o) of $\mathfrak{m}$_{Γ(n)}^h ⊗ $\mathbb{F}$_p. It extends to a cover $\mathcal{J}$_n of $\mathfrak{m}$_{Γ(n)} ⊗ $\mathbb{F}$_p, totally ramified at the supersingular points, and

$\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)} ⊗_{$\mathbb{Z}$[ζ_p]} $\mathbb{F}$_p

is obtained by gluing two copies of $\mathcal{J}$_{n,p} ⊗_{$\mathbb{F}$_p} $\mathbb{F}$_p along the supersingular points, identifying x in the second copy with x^{(p)} in the first (as in 1.17). In particular, the morphism

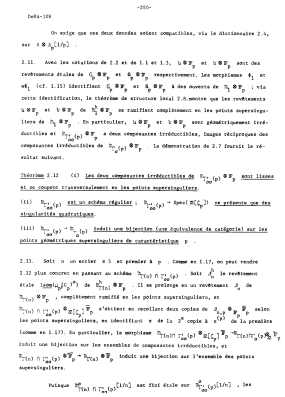
$\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)} ⊗_{$\mathbb{Z}$[ζ_p]} $\mathbb{F}$_p → $\mathfrak{m}$_{Γ(n)∩Γ_0(p)} ⊗_$\mathbb{Z}$ $\mathbb{F}$_p

induces a bijection on the sets of irreducible components, and

$\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)} ⊗ $\mathbb{F}$_p → $\mathfrak{m}$_{Γ(n)} ⊗ $\mathbb{F}$_p

induces a bijection on the set of supersingular points.

Since $\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)}°[1/n] is finite étale over $\mathfrak{m}$_{Γ′_∞(p)}°[1/n], the



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Authority page 109; collected-paper page 602; original folio 251.

the local properties are the same.

2.14. Let H be a subgroup of (ℤ/p)^*, and let d be its index. Write Γ_∞(H) and Γ′_∞(H) for the subgroups of GL(2, $\hat{\mathbb{Z}}$) that are the inverse images of the following subgroups of GL(2, ℤ/p):

Γ_∞(H) : (H ℤ/p) Γ′_∞(H) : (H ℤ/p) (0 (ℤ/p)^*) (0 H).

For H = {e} (respectively, H = (ℤ/p)^*), one has Γ_∞(H) = Γ_∞(p) and Γ′_∞(H) = Γ′_∞(p) (respectively, Γ_∞(H) = Γ′_∞(H) = Γ_0(p)).

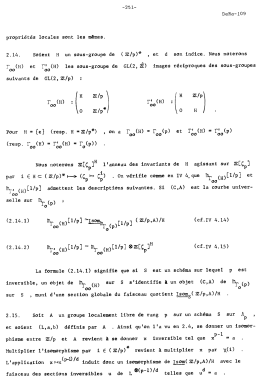
We denote by ℤ[ζ_p]^H the ring of invariants for H acting on ℤ[ζ_p] by i ∈ H ⊂ (ℤ/p)^* ↦ (ζ_p ↦ ζ_p^i). One verifies, as in IV.4, that 𝔪_{Γ_∞(H)}[1/p] and 𝔪_{Γ′_∞(H)}[1/p] admit the following descriptions. If (C,A) is the universal curve over 𝔪_{Γ_0(p)},

(2.14.1) 𝔪_{Γ_∞(H)}[1/p] ≃ Isom_{𝔪_{Γ_0(p)}[1/p]}(ℤ/p,A)/H (cf. IV.4.14)

(2.14.2) 𝔪_{Γ′_∞(H)}[1/p] ≃ 𝔪_{Γ_∞(H)}[1/p] ⊗ ℤ[ζ_p]^H (cf. IV.4.15)

Formula (2.14.1) means that if S is a scheme on which p is invertible, an object of 𝔪_{Γ_∞(H)} over S is identified with an object (C,A) of 𝔪_{Γ_0(p)} over S, equipped with a global section of the quotient sheaf Isom_S(ℤ/p,A)/H.

2.15. Let A be a locally free group scheme of rank p over a Λ_p-scheme S, and let (L,a,b) be defined by A. As seen in 2.4, giving an isomorphism between ℤ/p and A amounts to giving an invertible x such that x^{p-1} = a. Multiplying the isomorphism by i ∈ (ℤ/p)^* amounts to multiplying x by χ(i). The map x ↦ x^{(p-1)/d} therefore induces an isomorphism from Isom(ℤ/p,A)/H to the sheaf of invertible sections u of L^{⊗(1-p)/d} such that u^d = a.



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One verifies that

$w(\chi,\zeta)^{(p-1)/d} \in \Lambda_p[\zeta_p]^H$.

2.16. Let $'\mathfrak{m}_{\Gamma'_\infty(H)} \otimes \Lambda_p$ be the following algebraic stack over $\Lambda_p[\zeta_p]^H$: an object of $'\mathfrak{m}_{\Gamma'_\infty(H)} \otimes \Lambda_p$ over a $\Lambda_p[\zeta_p]^H$-scheme S is an object (C,A) of $\mathfrak{m}_{\Gamma_0(p)}$ over S, with A defining (L,a,b), together with sections u of $L^{\otimes(1-p)/d}$ and v of $L^{\otimes(p-1)/d}$ such that

$u^d = a$, $v^d = b$, $uv = w(\chi,\zeta)^{(p-1)/d}$.

The constructions of 2.14 and 1.15 define an isomorphism

(2.16.1) $'\mathfrak{m}_{\Gamma'_\infty(H)} \otimes \Lambda_p[1/p] \simeq \mathfrak{m}_{\Gamma'_\infty(H)} \otimes \Lambda_p[1/p]$.

2.17. As in 2.10, this definition can be modified to obtain a stack $'\mathfrak{m}_{\Gamma'_\infty(H)}$ over $\mathbb{Z}[\zeta_p]^H$: an object of $'\mathfrak{m}_{\Gamma'_\infty(H)}$ over a $\mathbb{Z}[\zeta_p]^H$-scheme S is an object (C,A) of $\mathfrak{m}_{\Gamma_0(H)}$ over S, equipped with

a) a global section u_1 of $\mathrm{Isom}_{S[1/p]}(\mathbb{Z}/p,A)/H$;

b) over $S \otimes \Lambda_p$, where A defines (L,a,b), sections u and v of $L^{\otimes(1-p)/d}$ and $L^{\otimes(p-1)/d}$, with $u^d = a$, $v^d = b$, and $uv = w(\chi,\zeta)^{(p-1)/d}$.

These data are required to be compatible, through (2.16.1), over $S \otimes \Lambda_p[1/p]$.

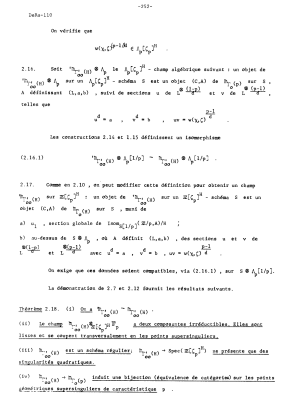
The proof of 2.7 and 2.12 gives the following results.

Theorem 2.18. (i) One has $\mathfrak{m}_{\Gamma'_\infty(H)} \simeq '\mathfrak{m}_{\Gamma'_\infty(H)}$.

(ii) The stack $\mathfrak{m}_{\Gamma'_\infty(H)} \otimes_{\mathbb{Z}[\zeta_p]^H} \mathbb{F}_p$ has two irreducible components. They are smooth and meet transversely at the supersingular points.

(iii) The stack $\mathfrak{m}_{\Gamma'_\infty(H)}$ is a regular scheme; the morphism $\mathfrak{m}_{\Gamma'_\infty(H)} \to \mathrm{Spec}(\mathbb{Z}[\zeta_p]^H)$ has only quadratic singularities.

(iv) The morphism $\mathfrak{m}_{\Gamma'_\infty(H)} \to \mathfrak{m}_{\Gamma_0(p)}$ induces a bijection (an equivalence of categories) on the supersingular geometric points of characteristic p.



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2.19. Let n be an integer prime to p, with n ≥ 3. Theorem 2.18 generalizes to the schemes $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)}[1/n]$.

(i) $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)}[1/n]$ is regular.

(ii) The morphism $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)}[1/n] \to \mathrm{Spec}(\mathbb{Z}[\zeta_p,1/p]^H)$ is smooth away from the supersingular points of characteristic p, where it has ordinary quadratic singularities.

(iii) The morphism $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes_{\mathbb{Z}[\zeta_p]^H} \mathbb{F}_p \to \mathfrak{m}_{\Gamma(n)} \otimes_{\mathbb{Z}} \mathbb{F}_p$ induces a bijection on supersingular points.

(iv) The morphism $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes_{\mathbb{Z}[\zeta_p]^H} \mathbb{F}_p \to \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)} \otimes_{\mathbb{Z}} \mathbb{F}_p$ induces a bijection between the sets of irreducible components.

(v) The scheme $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes_{\mathbb{Z}[\zeta_p]^H} \mathbb{F}_p$ is obtained by gluing two copies of $\mathcal{J}_n/H$ along the supersingular points (cf. 2.13).

3. A good-reduction theorem.

3.1. Let p be a prime, let $H \subset (\mathbb{Z}/p)^*$, let $\Gamma'_\infty(H)$ be the group of 2.14, let $\mathbb{Z}[\zeta_p]^H$ be as in 2.14, and let n be an integer prime to p. Assume n ≥ 3, so that $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes \mathbb{Q} = M_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes \mathbb{Q}$ is a scheme. It is a scheme over $\mathbb{Q}[\zeta_n,\zeta_p]^H$.

In this section, J(n,H) denotes the abelian variety over $\mathbb{Q}[\zeta_n,\zeta_p]^H$ given by the identity component of the Picard scheme of $M_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes \mathbb{Q}$:

$J(n,H) = \mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes \mathbb{Q} \,/\, \mathbb{Q}[\zeta_n,\zeta_p]^H)$.

For $H = (\mathbb{Z}/p)^*$, one has $\Gamma'_\infty(H) = \Gamma_0(p)$, and the map

$\mathfrak{m}_{\Gamma'_\infty(H)} \to \mathfrak{m}_{\Gamma_0(p)}$

induces a morphism

(3.1.1) $J(n,(\mathbb{Z}/p)^*) \otimes_{\mathbb{Q}[\zeta_n]} \mathbb{Q}[\zeta_n,\zeta_p]^H \to J(n,H)$

with finite kernel.

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Theorem 3.2. The abelian variety I(n,H) over ℚ[ζ_n,ζ_p]^H, defined as the cokernel of (3.1.1), has good reduction at the places dividing p.

Let v be a characteristic-p point of ℤ[ζ_n,ζ_p]^H, and let \overline{k(v)} be an algebraic closure of its residue field. Let ν be the number of supersingular points of M_{Γ(n)} ⊗_{ℤ[ζ]} \overline{k(v)}. The scheme M_{Γ(n)} ⊗_{ℤ[ζ]} \overline{k(v)} is known to be irreducible. It follows that the connected scheme M_{Γ(n)∩Γ(H)} ⊗_{ℤ[ζ_n,ζ_p]^H} \overline{k(v)} has two irreducible components meeting transversely at ν points (cf. 1.18), and that the Picard group variety over \overline{k(v)},

Pic^o(M_{Γ(n)∩Γ(H)} ⊗_{ℤ[ζ_n,ζ_p]^H} \overline{k(v)}/\overline{k(v)}),

is an extension of an abelian scheme by a torus of dimension ν - 1 (cf. I. 3.7). In particular:

Proposition 3.3. The abelian variety J(n,H) over ℚ[ζ_n,ζ_p]^H has semistable reduction at the places dividing p.

Suppose 0 → A → B → C → 0 is an exact sequence up to isogeny of abelian varieties over the fraction field of a discrete valuation ring. Let α_ab, α_m, and α_add denote the abelian, multiplicative, and unipotent ranks of the special fiber of the Néron model. Then

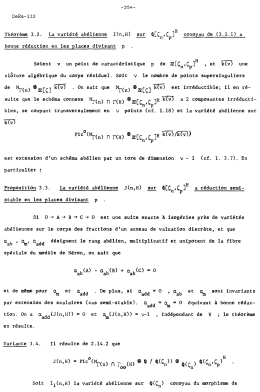
α_ab(A) - α_ab(B) + α_ab(C) = 0,

and likewise for α_m and α_add. Moreover, if α_add = 0, then α_ab and α_m are invariant under scalar extension (the semistable case). The condition α_add = α_m = 0 is equivalent to good reduction. Here α_add(J(n,H)) = 0 and α_m(J(n,H)) = ν-1, independently of H; the theorem follows.

Variant 3.4. It follows from 2.14.2 that

J(n,H) = Pic^o(M_{Γ(n)∩Γ_∞(H)} ⊗ ℚ / ℚ(ζ_n)) ⊗_{ℚ(ζ_n)} ℚ(ζ_n,ζ_p)^H.

Let I_1(n,H) be the abelian variety over ℚ(ζ_n) that is the cokernel of the morphism from



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a finite kernel defined by $\mathfrak{m}_{\Gamma_\infty(H)} \to \mathfrak{m}_{\Gamma_0(p)}$:

$$\mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma_0(p)} \otimes \mathbb{Q}/\mathbb{Q}(\zeta_n)) \to \mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma_\infty(H)} \otimes \mathbb{Q}/\mathbb{Q}(\zeta_n)) \to I_1(n,H) \to 0.$$

One has $I(n,H) = I_1(n,H) \otimes_{\mathbb{Q}(\zeta_n)} \mathbb{Q}(\zeta_n,\zeta_p)^H$, so 3.2 can still be formulated by saying that, after extension of scalars from $\mathbb{Q}(\zeta_n)$ to $\mathbb{Q}(\zeta_n,\zeta_p)^H$, $I_1(n,H)$ acquires good reduction at the places dividing p.

Variant 3.5. The abelian variety over $\mathbb{Q}(\zeta_p)^H$

$$RJ(n,H) =_{\mathrm{dfn}} \mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma'_\infty(H)} / \mathbb{Q}(\zeta_p)^H)$$

is obtained from J(n,H) by Weil restriction of scalars from $\mathbb{Q}(\zeta_n,\zeta_p)^H$ to $\mathbb{Q}(\zeta_p)^H$. The cokernel RI(n,H) of

$$\mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma_0(p)} / \mathbb{Q}) \otimes \mathbb{Q}(\zeta_p)^H \to \mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma'_\infty(p)} / \mathbb{Q}(\zeta_p)^H)$$

is obtained similarly from I(n,H) by Weil restriction of scalars. Since $\mathbb{Q}(\zeta_n)$ is unramified at p, RI(n,H) still has good reduction over $\mathbb{Q}(\zeta_p)^H$ at the places dividing p.

Variant 3.6. Let K be a subgroup of $GL(2,\hat{\mathbb{Z}})$ containing $\Gamma(n)$, and let $RI_1(K,H)$ be the abelian variety over $\mathbb{Q}$ that is the cokernel of

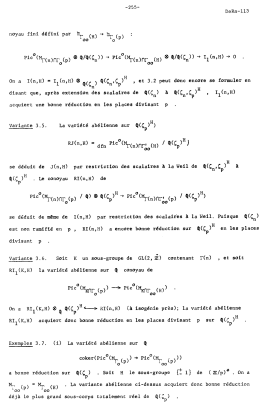
$$\mathrm{Pic}^o(M_{K\cap\Gamma_0(p)}) \to \mathrm{Pic}^o(M_{K\cap\Gamma_\infty(H)}).$$

One has $RI_1(K,H) \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_p)^H \leftrightarrow RI(n,H)$ (up to isogeny); hence the abelian variety $RI_1(K,H)$ acquires good reduction over $\mathbb{Q}(\zeta_p)^H$ at the places dividing p.

Examples 3.7. (i) The abelian variety over $\mathbb{Q}$

$$\mathrm{coker}(\mathrm{Pic}^o(M_{\Gamma_0(p)}) \to \mathrm{Pic}^o(M_{\Gamma_\infty(p)}))$$

has good reduction over $\mathbb{Q}(\zeta_p)$. Let H be the subgroup $\{\pm 1\}$ of $(\mathbb{Z}/p)^*$. One has $M_{\Gamma_\infty(p)} = M_{\Gamma_\infty(H)}$. The abelian variety in the preceding variant therefore acquires good reduction already over the maximal totally real subfield of $\mathbb{Q}(\zeta_p)$.



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(ii) Let p be a prime of the form 4n + 1, and let H ⊂ (ℤ/p)^* be the subgroup of squares. The abelian variety over ℚ

coker(Pic^o(M_{Γ_0(p)}) → Pic^o(M_{Γ_∞(H)}))

acquires good reduction everywhere over the real quadratic field ℚ(√p).

(iii) Let n be a square-free integer, let H be a subgroup of (ℤ/n)^*, and let Γ_∞(H) be the subgroup of GL(2,$\hat{\mathbb{Z}}$) that is the inverse image of

(H ℤ/n) (0 (ℤ/n)^*) ⊂ GL(2,ℤ/n).

If n = pm, then ℤ/n^* ≃ (ℤ/p)^* × (ℤ/m)^*. We set

H_p = (ℤ/p)^* ∩ H

and write H.(ℤ/p)^* for the subgroup of (ℤ/n)^* generated by H and (ℤ/p)^*. The abelian variety over ℚ

coker(Σ_{p|n} Pic^o(M_{Γ_∞(H.(ℤ/p)^*)}) → Pic^o(M_{Γ_∞(H)}))

acquires good reduction over ℚ(ζ_n)^H.

Proof. Let p be a prime divisor of n: n = mp with (m,p) = 1. Let us prove that the abelian variety A under consideration has good reduction at the places of ℚ(ζ_n)^H dividing p. Since ℚ(ζ_n)^{H_p} is unramified over ℚ(ζ_n)^H at the places dividing p, it is enough to prove that A has good reduction at the places of ℚ(ζ_n)^{H_p} dividing p. By 3.6,

B = coker(Pic^o(M_{Γ_0(p)∩Γ_∞(m)}) → Pic^o(M_{Γ_∞(H_p)}))

has good reduction at these places. The abelian variety

C = coker(Pic^o(M_{Γ_∞(H.(ℤ/p)^*)}) → Pic^o(M_{Γ_∞(H)}))

is isogenous to the one obtained from B by passing to the invariants under H (note that Γ_∞(H).(Γ_0(p) ∩ Γ_∞(m)) = Γ_∞(H.(ℤ/p)^*)). The abelian variety A is a quotient of

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C therefore also has good reduction.

(iv) Since M_{Γ_∞(H)} = M_{±Γ_∞(H)}, no generality is lost by assuming that -1 ∈ H. In particular, in (iii), ℚ(ζ_p)^H may be replaced by its totally real subfield.

Remark 3.8. The abelian varieties of 3.7(ii) are those studied by Casselman in [5].

4. Study of $\mathfrak{m}_n$

4.1. Let C be a generalized elliptic curve over a scheme S, and let n be an integer.

Assume that

a) the geometric fibers of C are smooth or m-gons, with n|m;

b) they are never supersingular of characteristic p|n.

Conditions a) and b) ensure that, locally for the étale topology, C_n is an extension of $\mathbb{Z}/n$ by μ_n. The e_n pairing defines an isomorphism of fppf sheaves

e_n : ∧^2 C_n ≃ μ_n.

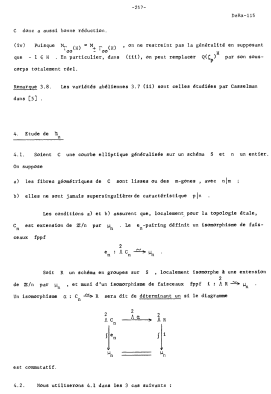
Let R be a group scheme over S, locally isomorphic to an extension of $\mathbb{Z}/n$ by μ_n, and equipped with an isomorphism of fppf sheaves i : ∧^2 R ≃ μ_n.

An isomorphism α : C_n ≃ R is said to have determinant one if the diagram

∧^2 C_n –∧^2 α–> ∧^2 R | e_n | i v v μ_n = μ_n

commutes.

4.2. We shall use 4.1 in the following three cases:



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a) S is a scheme over $\mathbb{Z}[\zeta_n, 1/n]$, and $R = (\mathbb{Z}/n)^2$.

In this case one has over S the map $\zeta : \mathbb{Z}/n \simeq \mu_n$, and i is the composite

$\wedge^2(\mathbb{Z}/n)^2 = \mathbb{Z}/n \simeq \mu_n$.

b) $R = \mu_n \times \mathbb{Z}/n$ and i is the evident isomorphism: if $a \in \mu_n$ and $b \in \mathbb{Z}/n$, then $i(a \wedge b) = b \cdot a$ (= a^b when the group law of μ_n is written multiplicatively).

c) S is a scheme over $\mathbb{Z}[\zeta_n]$ and $A \subset (\mathbb{Z}/n)^2$ is a cyclic subgroup of order n.

In this case, write R(A) for the S-group scheme obtained by the pushout

$A \to (\mathbb{Z}/n)^2 \to (\mathbb{Z}/n)^2/A \downarrow \zeta \downarrow \| A \otimes \mu_n \to R(A) \to (\mathbb{Z}/n)^2/A.$

One also has

$\wedge^2 R(A) \simeq (A \otimes \mu_n) \otimes (\mathbb{Z}/n)^2/A \simeq (A \otimes (\mathbb{Z}/n)^2/A) \otimes \mu_n \simeq \wedge^2(\mathbb{Z}/n)^2 \otimes \mu_n \simeq \mathbb{Z}/n \otimes \mu_n \simeq \mu_n$.

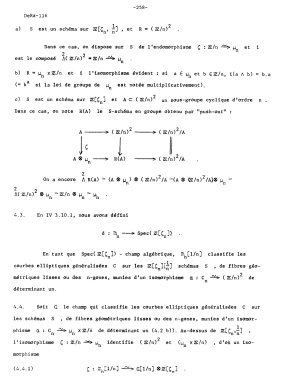
4.3. In IV.3.10.1 we defined

$d : \mathfrak{m}_n \to \mathrm{Spec}(\mathbb{Z}[\zeta_n])$.

As an algebraic stack over $\mathrm{Spec}(\mathbb{Z}[\zeta_n])$, $\mathfrak{m}_n[1/n]$ classifies generalized elliptic curves C over $\mathbb{Z}[\zeta_n][1/n]$-schemes S whose geometric fibers are smooth or n-gons, equipped with a determinant-one isomorphism $\alpha : C_n \simeq (\mathbb{Z}/n)^2$.

4.4. Let $\mathfrak{a}$ be the stack classifying generalized elliptic curves C over schemes S whose geometric fibers are smooth or n-gons, equipped with a determinant-one isomorphism $\alpha : C_n \simeq \mu_n \times \mathbb{Z}/n$ (4.2(b)). Over $\mathbb{Z}[\zeta_n, 1/n]$, the isomorphism $\zeta : \mathbb{Z}/n \simeq \mu_n$ identifies $(\mathbb{Z}/n)^2$ with $\mu_n \times \mathbb{Z}/n$, hence gives an isomorphism

(4.4.1) $\zeta : \mathfrak{m}_n[1/n] \simeq \mathfrak{a}[1/n] \otimes \mathbb{Z}[\zeta_n]$.



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We denote by $\mathcal{G}'$ the open substack of $\mathcal{G}$ classifying the (C,α) satisfying

(4.4.2) For every geometric point \overline{s} of S, if C_{\overline{s}} is an n-gon,

α((C^o_{\overline{s}})_n) = μ_n.

If (C,α) is an object of $\mathcal{G}'$ over S, then (C,α^{-1}(e × $\mathbb{Z}$/n)) is an object of the stack $\mathcal{B}$_n (1.3), hence a morphism

(4.4.3) γ : $\mathcal{G}'$ → $\mathcal{B}$_n.

Proposition 4.5. The morphism 4.3.3 is étale and separated. In particular, $\mathcal{G}'$ is a smooth algebraic stack over Spec($\mathbb{Z}$).

Let (C,β) be an object of $\mathcal{B}$_n over S. Then $\mathcal{G}'$ ×_{$\mathcal{B}$_n} S represents the following functor F : (Sch/S) → (Ens): F(T) is the set of pairs (A,u) consisting of a subgroup scheme A of C_n, locally isomorphic to μ_n, such that

A × B ⥲ C_n, together with u : B ⥲ $\mathbb{Z}$/n.

For such a pair (A,u), there is indeed a unique isomorphism

v : A ⥲ μ_n such that

α : C_n ↞ A × B –(v,u)⥲ μ_n × $\mathbb{Z}$/n$\mathbb{Z}$

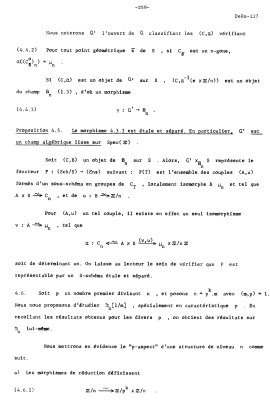
has determinant one. We leave it to the reader to verify that F is representable by an étale and separated S-scheme.

4.6. Let p be a prime divisor of n, and put n = p^k·m with (m,p) = 1. We propose to study $\mathfrak{m}$_n[1/m], especially in characteristic p. Gluing together the results obtained for the various p gives results on $\mathfrak{m}$_n itself.

We will exhibit the ”p-aspect” of a level-n structure as follows.

a) The reduction morphisms define

(4.6.1) $\mathbb{Z}$/n ⥲ $\mathbb{Z}$/p^k × $\mathbb{Z}$/m.



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b) The maps x ↦ x^m and x ↦ x^{p^k} define

(4.6.2) μ_n ⟶̃ μ_{p^k} × μ_m.

c) compatibility: if d | n, identify $\mathbb{Z}[\zeta_d]$ with the subring $\mathbb{Z}[\zeta_{n^{n/d}}]$ of $\mathbb{Z}[\zeta_n]$. With this convention, a $\mathbb{Z}[\zeta_n]$-scheme S is also a $\mathbb{Z}[\zeta_{p^k}]$-scheme and a $\mathbb{Z}[\zeta_m]$-scheme; the morphisms ζ : $\mathbb{Z}/p^k \to \mu_{p^k}$ and ζ : $\mathbb{Z}/m \to \mu_m$ are the components, via (4.6.1) and (4.6.2), of ζ : $\mathbb{Z}/n \to \mu_n$.

d) The maps x ↦ x^m and x ↦ x^{p^k} define

(4.6.3) C_n ⟶̃ C_{p^k} × C_m.

A morphism α : C_n ⟶̃ $(\mathbb{Z}/n)^2$ has determinant one if and only if its components, via (4.6.2) (4.6.4), namely α′ : C_{p^k} ⟶̃ $(\mathbb{Z}/p^k)^2$ and α″ : C_m ⟶̃ $(\mathbb{Z}/m)^2$, do.

e) We identify $\mathfrak{m}_n[1/n]$ with the stack classifying curves C over a $\mathbb{Z}[\zeta_n,1/n]$-scheme S as in 4.3, equipped with α′ : C_{p^k} ⟶̃ $(\mathbb{Z}/p^k)^2$ and α″ : C_m ⟶̃ $(\mathbb{Z}/m)^2$ of determinant one.

f) Likewise, $\mathcal{G}$ classifies the (C,α′,α″) such that α′ : C_{p^k} ⟶̃ μ_{p^k} × $\mathbb{Z}/p^k$, α″ : C_m ⟶̃ μ_m × $\mathbb{Z}/m$, and det(α′) = det(α″) = 1.

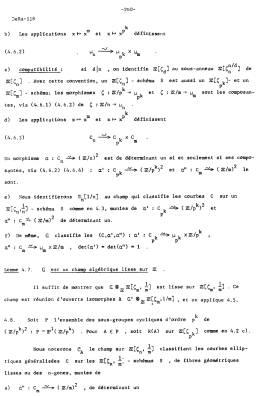
Lemma 4.7. $\mathcal{G}$ is a smooth algebraic stack over $\mathbb{Z}$.

It is enough to show that $\mathcal{G} \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_m,1/m]$ is smooth over $\mathbb{Z}[\zeta_m,1/m]$. This stack is a union of open substacks isomorphic to $\mathcal{G}' \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_m,1/m]$, and 4.5 applies.

4.8. Let P be the set of cyclic subgroups of order p^k in $(\mathbb{Z}/p^k)^2$: P ≃ $\mathbb{P}^1(\mathbb{Z}/p^k)$. For A ∈ P, let R(A) over $\mathbb{Z}[\zeta_{p^k}]$ be as in 4.2(c).

We denote by $\mathcal{C}_A$ the stack over $\mathbb{Z}[\zeta_n,1/m]$ classifying generalized elliptic curves C over $\mathbb{Z}[\zeta_n,1/m]$-schemes S whose geometric fibers are smooth or n-gons, equipped with

a) α″ : C_m ⟶̃ $(\mathbb{Z}/m)^2$, of determinant one



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b) α′ : C_{p^k} ⥲ R(A), with determinant one.

Since R(A) is isomorphic to μ_{p^k} × ℤ/p^k, the stack C_A is isomorphic to C ⊗_ℤ ℤ[ζ_n, 1/m], and hence is a smooth algebraic stack over ℤ[ζ_n, 1/m].

For p invertible, ζ : (ℤ/p^k)^2 ⥲ R(A) is an isomorphism, hence an isomorphism of stacks over ℤ[ζ_n, 1/n]

(4.8.1) $\mathfrak{m}$_n[1/n] ≃ C_A[1/p].

4.9. Let $\mathfrak{m}$_1^h be the complement in $\mathfrak{m}$_1 of the finite set of points corresponding to supersingular elliptic curves of characteristic dividing n. Define

c : C_A → $\mathfrak{m}$_1^h

by associating with (C,α′,α″) over S the curve c(C) (IV 1.4).

Lemma 4.10. The morphism c : C_A → $\mathfrak{m}$_1^h is representable and quasi-finite.

The verification is left to the reader.

4.11. Let C[1/m] be the stack obtained by gluing the stacks C_A (A ∈ P) along their common open C_A[1/p] (4.8.1). Let C be obtained by gluing the C[1/m] along their common open C[1/m,1/p] ≃ $\mathfrak{m}$_n[1/n]. The stack C is smooth over ℤ[ζ_n]; one has

(4.11.1) C[1/n] ≃ $\mathfrak{m}$_n[1/n]

and the morphisms 4.9 glue to a morphism

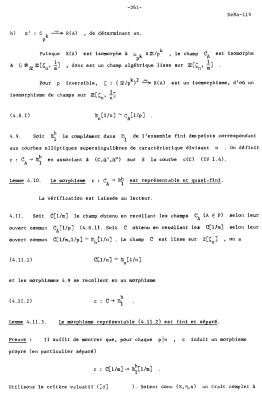
(4.11.2) c : C → $\mathfrak{m}$_1^h.

Lemma 4.11.3. The representable morphism (4.11.2) is finite and separated.

Proof. It suffices to show that, for every p|n, c induces a proper morphism (in particular, a separated one)

c : C[1/m] → $\mathfrak{m}$_1^h[1/m].

We use the valuative criterion ([8]). Thus let (S,η,s) be a complete trait with



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algebraically closed residue field, u : S → 𝔪_1^h[1/m], and v_η : η → C such that

c v_η = u_η.

We must show that, after replacing S by its normalization in a finite extension of k(η), u admits a lift v : S → C extending v_η, and that v is unique up to unique isomorphism. Since 𝔪_n^o[1/n] is dense in C, we may suppose that v_η sends η into 𝔪_n^o[1/n]. Since C[1/p] ≃ 𝔪_n[1/n] is finite over 𝔪_1[1/n] (cf. IV.2.5.), it is enough to consider the case in which s has characteristic p.

Let (C_η,α′_η,α″_η) be the elliptic curve with level-n structure over η that defines v_η. A preliminary finite extension of k(η) allows us to suppose that C_η extends to a generalized elliptic curve C over S, with special fiber either smooth or an n-gon. This curve is unique (IV 1.6). The group scheme C_m is finite étale over S, so α″_η extends uniquely to

α″ : C_m ⟶̃ (ℤ/m)^2.

By hypothesis, the special fiber C_s is not supersingular. The multiplicative subgroup (C_s)^o_{p^k} of (C_s)_{p^k} lifts uniquely to a subgroup of C_{p^k}, which becomes an extension

0 → A → C_{p^k} → B → 0

with A ≃ μ_n and B ≃ ℤ/n. Let A = α′(A_η) ∈ P. The isomorphism α′_η extends to

α′ : C_{p^k} ⟶̃ R(A),

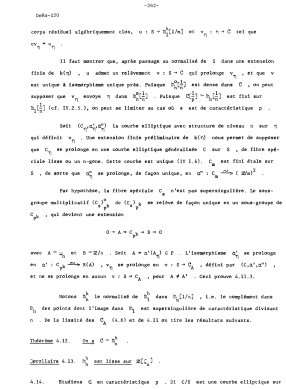
v_η extends to v : S → C_A, defined by (C,α′,α″), and does not extend to any v : S → C_{A′} for A ≠ A′. This proves 4.11.3.

Let 𝔪_n^h denote the normalization of 𝔪_1^h in 𝔪_n[1/n], i.e. the complement in 𝔪_n of the points whose image in 𝔪_1 is supersingular of characteristic dividing n. From the smoothness of the C_A (4.8) and from 4.11 we obtain the following results.

Theorem 4.12. One has C ≃ 𝔪_n^h.

Corollary 4.13. 𝔪_n^h is smooth over ℤ[ζ_n].

4.14. Let us study C in characteristic p. If C/S is an elliptic curve over



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S, with p nilpotent on S, whose geometric fibers are smooth and nonsupersingular or are n-gons, then (C_{p^k})^o is locally isomorphic to μ_{p^k} (for the étale topology), and giving an isomorphism α′ : C_{p^k} ⥲ μ_{p^k} × ℤ/p^k amounts to giving an isomorphism of extensions

(4.14.1) 0 → (C_{p^k})^o → C_{p^k} → C_{p^k}/(C_{p^k})^o → 0 ↓ α′_1 ↓ α′ ↓ α′_2 0 → μ_{p^k} → μ_{p^k} × ℤ/p^k → ℤ/p^k → 0.

This amounts to giving

a) a trivialization τ of the extension in the first row of 4.14.1;

b) isomorphisms α′_1 : C^o_{p^k} ⥲ μ_{p^k} and α′_2 : C_{p^k}/C^o_{p^k} ⥲ ℤ/p^k.

Giving τ amounts to giving the étale subgroup α′^{-1}(ℤ/p^k) of C_{p^k}, so that (C,α′) ↦ (C,τ) corresponds to the morphism $\mathcal{G} \to \mathcal{B}_{p^k}$ already considered. Moreover, if det(α′) = 1, α′_2 is uniquely determined by α′_1.

4.15. Let $\mathcal{J}_k^h \otimes \mathbb{Z}/p^N$ be the algebraic stack over ℤ/p^N classifying generalized elliptic curves with irreducible fibers C/S, with p^N = 0 on S, equipped with an isomorphism α′_1 : C^o_{p^k} ⥲ μ_{p^k}. The discussion in 4.14 shows that, after reduction modulo p^N, the stack $\mathcal{G} = \mathcal{G}(p^k)$ is a fiber product

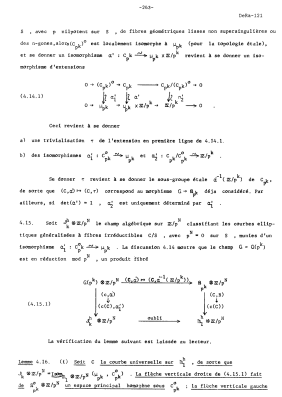
(4.15.1) $\mathcal{G}(p^k) \otimes \mathbb{Z}/p^N$ —(C,α′) ↦ (C,α′^{-1}(ℤ/p^k))⟶ $\mathcal{B}_{p^k} \otimes \mathbb{Z}/p^N$ │ (C,α′) │ (C,B) ↓ ↓ │ (c(C),α′_1) │ (c(C)) $\mathcal{J}_k^h \otimes \mathbb{Z}/p^N$ ——————— forgetful ———————→ $\mathfrak{m}_1^h \otimes \mathbb{Z}/p^N$.

Verification of the following lemma is left to the reader.

Lemma 4.16. (i) Let C be the universal curve over $\mathfrak{m}_1^h$, so that

$\mathcal{J}_k^h \otimes \mathbb{Z}/p^N = \mathrm{Isom}_{\mathfrak{m}_1^h \otimes \mathbb{Z}/p^N}(\mu_{p^k}, C^o_{p^k})$.

The right vertical arrow in (4.15.1) makes $\mathcal{B}_{p^k}^o \otimes \mathbb{Z}/p^N$ a principal homogeneous space under C^o_{p^k}; the left vertical arrow



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makes $\mathcal{G}$^o ⊗ $\mathbb{Z}$/p^N a principal homogeneous space under μ_{p^k}.

(ii) In particular, for N = 1, these arrows identify $\mathcal{J}$_k^h ⊗ $\mathbb{F}$_p with ($\mathcal{G}$ ⊗ $\mathbb{F}$_p)^{(p^k)} and $\mathfrak{m}$_1^h ⊗ $\mathbb{F}$_p with ($\mathcal{B}$_n ⊗ $\mathbb{F}$_p)^{(p^k)} (the arrow becoming an iterated Frobenius morphism) (a bijective radicial morphism between smooth schemes of dimension one is always an iterated Frobenius).

(iii) The horizontal arrows in (4.15.1) are étale, and make ($\mathcal{J}$_n^h ⊗ $\mathbb{Z}$/p^N) (respectively $\mathcal{G}$ ⊗ $\mathbb{Z}$/p^N) a principal homogeneous space under the group ($\mathbb{Z}$/p^k)^* over ($\mathfrak{m}$_1^h ⊗ $\mathbb{Z}$/p^N) (respectively $\mathcal{B}$_n ⊗ $\mathbb{Z}$/p^N), with the group ($\mathbb{Z}$/p^k)^* acting by (i,(C,α′_1)) ↦ (C,iα′_1).

Let $\mathcal{J}$_k ⊗ $\mathbb{Z}$/p be the normalization of $\mathfrak{m}$_1 ⊗ $\mathbb{Z}$/p in $\mathcal{J}$_k^h. We shall use the following theorem of Igusa [35] (see Katz [13] 4.3 (bis)).

Theorem 4.17 (Igusa). The covering $\mathcal{J}$_k ⊗ $\mathbb{Z}$/p → $\mathfrak{m}$_1 ⊗ $\mathbb{Z}$/p is totally ramified at the supersingular points.

4.18. For general n, $\mathcal{G}$^o = $\mathcal{G}$^o(n) is a fiber product

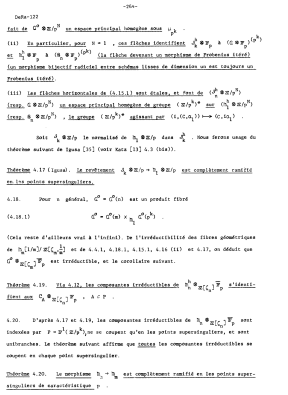
(4.18.1) $\mathcal{G}$^o = $\mathcal{G}$^o(m) ×_{$\mathfrak{m}$_1} $\mathcal{G}$^o(p^k).

(This moreover remains true at infinity.) From the irreducibility of the geometric fibers of $\mathfrak{m}$_m[1/m]/$\mathbb{Z}$[ζ_m,1/m] and from 4.4.1, 4.18.1, 4.15.1, 4.16(ii), and 4.17, it follows that $\mathcal{G}$^o ⊗_{$\mathbb{Z}$[ζ_m]} $\mathbb{F}$_p is irreducible, and the following corollary.

Theorem 4.19. Via 4.12, the irreducible components of $\mathfrak{m}$_n^h ⊗_{$\mathbb{Z}$[ζ_n]} $\mathbb{F}$_p identify with C_A ⊗_{$\mathbb{Z}$[ζ_n]} $\mathbb{F}$_p, A ∈ P.

4.20. By 4.17 and 4.19, the irreducible components of $\mathfrak{m}$_n ⊗_{$\mathbb{Z}$[ζ_n]} $\mathbb{F}$_p are indexed by P = $\mathbb{P}$^1($\mathbb{Z}$/p^k), meet only at the supersingular points, and are unibranch. The following theorem asserts that all the irreducible components meet at every supersingular point.

Theorem 4.20. The morphism $\mathfrak{m}$_n → $\mathfrak{m}$_m is totally ramified at the supersingular points of characteristic p.



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Proof. By 4.6, the validity of 4.20 depends only on p^k, not on m; we may therefore suppose that m ≥ 3, so that we have to deal only with schemes.

Let E_0 be a supersingular elliptic curve over $\bar{F}_p$, and let $E/\hat{S}$ be the formal moduli scheme of E_0 over $W(\bar{F}_p)$. We have $\hat{S} \simeq W(\bar{F}_p)[[t]]$; it is the completion of the strict henselization of $\mathfrak{m}_1$ at the given geometric point.

Let $\hat{T}$ be the normalization of $\hat{S}$ in the $\hat{S}[1/p]$-scheme Isom_$\hat{S}$(E_{p^k},$(\mathbb{Z}/p^k)^2$). Since completions and normalizations commute,

$\hat{T} = \mathfrak{m}_{p^k} \times_{\mathfrak{m}_1} \hat{S}$.

By 4.17, the irreducible components of $\hat{T} \otimes \bar{F}_p$ are naturally indexed by $P = \mathbb{P}^1(\mathbb{Z}/p^k)$. The connected components of $\hat{T}$ define a partition $\mathcal{P}$ of P. Assertion 4.20 means that $\hat{T}$ is always connected, i.e. that $\mathcal{P}$ is trivial.

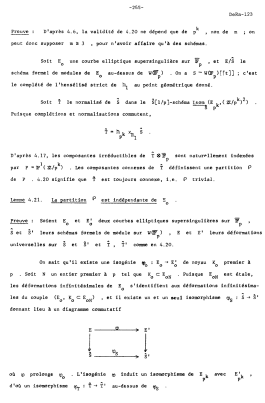
Lemma 4.21. The partition $\mathcal{P}$ is independent of E_0.

Proof. Let E_0 and E'_0 be two supersingular elliptic curves over $\bar{F}_p$, let $\hat{S}$ and $\hat{S}'$ be their formal moduli schemes over $W(\bar{F}_p)$, let E and E′ be their universal deformations over $\hat{S}$ and $\hat{S}'$, and let $\hat{T}$ and $\hat{T}'$ be as in 4.20.

It is known that there exists an isogeny $\varphi_0 : E_0 \to E'_0$ whose kernel K_0 is prime to p. Let N be an integer prime to p such that $K_0 \subset E_{oN}$. Since E_{oN} is étale, the infinitesimal deformations of E_0 identify with the infinitesimal deformations of the pair $(E_0, K_0 \subset E_{oN})$, and there exists a unique isomorphism $\varphi_S : \hat{S} \to \hat{S}'$ giving rise to a commutative diagram

$$\begin{array}{ccc} E & \xrightarrow{\varphi} & E' \\ \downarrow & & \downarrow \\ \hat{S} & \xrightarrow{\varphi_S} & \hat{S}' \end{array}$$

in which φ extends φ_0. The isogeny φ induces an isomorphism from E_{p^k} to E'_{p^k}, hence an isomorphism $\varphi_T : \hat{T} \to \hat{T}'$ above φ_S.



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The bijection between the set I(T̂ ⊗ F̄_p) of irreducible components of T̂ ⊗ F̄_p and P can be described as follows: on T̂ there is a morphism

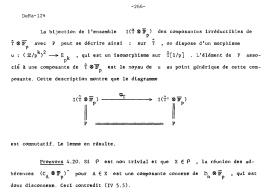
u : (ℤ/p^k)^2 → E_{p^k},

which is an isomorphism on T̂[1/p]. The element of P associated with a component of T̂ ⊗ F̄_p is the kernel of u at the generic point of that component. This description shows that the diagram

I(T̂ ⊗ F̄_p) —φ_T—→ I(T̂′ ⊗ F̄_p) ‖ ‖ P ═══════════ P

is commutative. The lemma follows.

Let us prove 4.20. If $\mathcal{P}$ is nontrivial and X ∈ $\mathcal{P}$, the union of the closures ($\mathcal{C}$_A ⊗ F̄_p)⁻ for A ∈ X is a connected component of $\mathfrak{m}$_n ⊗ F̄_p, which is therefore disconnected. This contradicts (IV 5.5).



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VI. Coarse moduli schemes.

In this chapter we study coarse moduli schemes, in particular the relation between $\mathfrak{m}$_H and M_H. Some classical results are proved by ”pure thought”. The result of § 5 is made more precise in the following chapter.

1. The modular invariant.

Theorem 1.1. The coarse space M_1 and $\mathfrak{m}_1$ is isomorphic to $\mathbb{P}^1_{\mathbb{Z}}$.

Characteristics 2 and 3 make the direct verification of 1.1 somewhat laborious. We shall deduce 1.1 from its corollary

(1.1.1) $M_{1\mathbb{C}} \simeq \mathbb{P}^1_{\mathbb{C}}$.

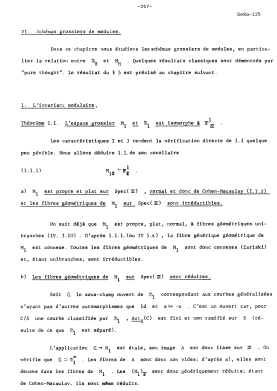
a) M_1 is proper and flat over Spec($\mathbb{Z}$), normal and hence Cohen–Macaulay (I.7.2), and the geometric fibers of M_1 over Spec($\mathbb{Z}$) are irreducible.

We already know that M_1 is proper, flat, normal, with geometrically unibranch fibers (IV. 3.10). By 1.1.1 (or IV 5.4), the geometric generic fiber of M_1 is connected. All geometric fibers of M_1 are therefore connected (Zariski) and, being unibranch, are irreducible.

b) The geometric fibers of M_1 over Spec($\mathbb{Z}$) are reduced.

Let G be the open substack of $\mathfrak{m}_1$ corresponding to generalized curves having no automorphisms other than Id and x ↦ -x. It is open because, for C/S a curve classified by $\mathfrak{m}_1$, Aut_S(C) is finite and unramified over S (a consequence of the fact that $\mathfrak{m}_1$ is separated).

The map G → M_1 is étale, so its image A is smooth over $\mathbb{Z}$. One verifies that G ⊃ $\mathfrak{m}_1^{\infty}$. The fibers of A are therefore nonempty; by a), they are dense in the fibers of M_1. The $(M_1)_{\bar{s}}$ are thus generically reduced; being Cohen–Macaulay, they are in fact reduced.



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VI Schémas grossiers de modules.

Dans ce chapitre nous étudions les schémas grossiers de modules, en particulier la relation entre D_N et M_1. Quelques résultats classiques sont démontrés par "pure thought". Le résultat du § 3 est précédé au chapitre suivant.

1. L'invariant modulaire.

Théorème 1.1. L'espace grossier M_1 et $\mathbb{P}^1_{\mathbb{Z}}$ sont isomorphe à $\mathbb{P}^1_{\mathbb{Z}}$.

Les caractéristiques 2 et 3 rendent la vérification directe de 1.1 quelque peu pénible. Nous allons déduire 1.1 de son corollaire

(1.1.1) $M_{1\mathbb{C}} \simeq \mathbb{P}^1_{\mathbb{C}}$

a) M_1 est propre et plat sur Spec($\mathbb{Z}$), normal et donc de Cohen-Macaulay (I.7.2), et les fibres géométriques de M_1 sur Spec($\mathbb{Z}$) sont irréductibles.

On sait déjà que M_1 est propre, plat, normal, à fibres géométriques unibranches (IV, 3.10). D'après 1.1.1 (ou IV 5.4), la fibre générique géométrique de M_1 est connexe. Toutes les fibres géométriques de M_1 sont donc connexes (Zariski) et, étant unibranches, sont irréductibles.

b) Les fibres géométriques de M_1 sur Spec($\mathbb{Z}$) sont réduites.

Soit G le sous-champ ouvert de D_1 correspondant aux courbes généralisées n'ayant pas d'autres automorphismes que Id et x↦-x . C'est un ouvert car, pour C/S une courbe classifiée par D_1, $\mathrm{Aut}_S(C)$ est fini et non ramifié sur S (résulte de ce que D_1 est séparé).

L'application G → M_1 est étale, son image A est donc lisse sur $\mathbb{Z}$. On vérifie que G ⊃ M_1^{∞}. Les fibres de A sont donc non vides; d'après a), elles sont denses dans les fibres de M_1. Les $(M_1)_{\bar{s}}$ sont donc génériquement réduits; étant de Cohen-Macaulay, ils sont même réduits.

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c) End of the proof.

By invariance of the arithmetic genus under specialization, and by (1.1.1), one finds that the geometric fibers of M_1 are integral and have arithmetic genus 0. They are therefore isomorphic to the projective line. In particular, M_1 is proper and smooth over $\mathbb{Z}$.

The Néron polygon with one side over Spec($\mathbb{Z}$) defines a section of M_1, with image M_1^{∞}. Fiber by geometric fiber one verifies that $H^1(M_1,\mathcal{O}(M_1^{\infty})) = 0$, and then that $H^0(M_1,\mathcal{O}(M_1^{\infty}))$ is locally free of rank 2 over $\mathbb{Z}$ and its formation is compatible with passage to geometric fibers. Since $\mathbb{Z}$ is principal, $H^0(M_1,\mathcal{O}(M_1^{\infty}))$ is free and

has a basis {1,j}. Fiber by geometric fiber one verifies that $\mathcal{O}(M_1^{\wedge}\infty)$ is very ample and that j defines an isomorphism

$M_1 \tilde{\to} \mathbb{P}^1_\mathbb{Z}$.

Proof of (1.1.1) Here are two classical proofs of (1.1.1).

(1) One verifies that $M_1^{\wedge}o(\mathbb{C})$ is the quotient of the Poincaré half-plane by $SL(2,\mathbb{Z})$ (IV 5.4). Drawing a fundamental domain, one finds that $M_1^{\wedge}o(\mathbb{C})$ is homeomorphic to $\mathbb{R}^2$. The space $M_1(\mathbb{C})$, obtained from it by adjoining one point, is therefore a sphere.

(2) Over $\mathbb{C}$, the moduli scheme classifying integral generalized elliptic curves equipped with an invariant differential is

$\mathrm{Spec}(\mathbb{C}[g_2,g_3])$ - (the point $g_2 = g_3 = 0$),

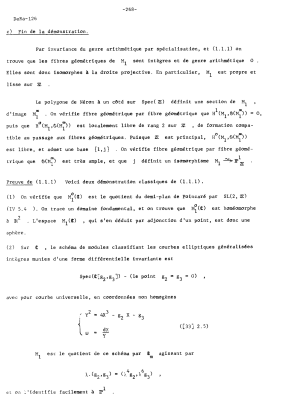
with universal curve, in nonhomogeneous coordinates,

$\lceil\ Y^2 = 4X^3 - g_2\ X - g_3\ \rfloor\ \omega = dX/Y$ ([33] 2.5)

M_1 is the quotient of this scheme by $\mathbb{G}_m$ acting by

$\lambda.(g_2,g_3) = (\lambda^4\ g_2,\lambda^6\ g_3)$,

and is easily identified with $\mathbb{P}^1$.



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The irreducibility of the fibers of M_1 over $\mathbb{Z}$ has the following corollary.

Corollary 1.2. Let k be an algebraically closed field of characteristic p. There are only finitely many isomorphism classes of supersingular elliptic curves over k.

The set of points of $(M_1)_k$ corresponding to generalized elliptic curves E for which the Frobenius kernel is of multiplicative type is open and contains the point at infinity. Its complement, which corresponds to supersingular curves, is therefore finite.

1.3. The function j introduced in 1.1 c) is not uniquely determined; the ambiguity is $j \mapsto \pm j + n$ ($n \in \mathbb{Z}$). A choice of j, recalled below, is classical.

Let C/S be a generalized elliptic curve with irreducible fibers over S. Tate's formulary [33] explains how to associate with C/S a section j of $\mathbb{P}^1$ over S, with $j = \infty$ wherever C is singular. By the universal property of M_1, this morphism $j : \mathfrak{m}_1 \to \mathbb{P}^1$ factors through $j : M_1 \to \mathbb{P}^1$. One readily checks that $j : M_1(\mathbb{C}) \to \mathbb{P}^1(\mathbb{C})$ is bijective [33].

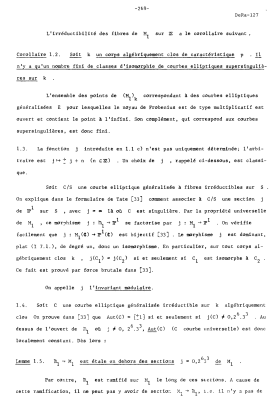
The morphism j is dominant, flat (I 7.1.), and of degree one, hence an isomorphism. In particular, over every algebraically closed field k, j(C_1) = j(C_2) if and only if C_1 is isomorphic to C_2. This is proved by brute force in [33].

The function j is called the modular invariant.

1.4. Let C be an irreducible generalized elliptic curve over an algebraically closed field k. It is proved in [33] that Aut(C) = {±1} if and only if j(C) ≠ 0,2^6.3^3. Over the open subset of $\mathfrak{m}$_1 where j ≠ 0, 2^6.3^3, Aut(C) (with C the universal curve) is therefore locally constant. Hence:

Lemma 1.5. $\mathfrak{m}$_1 → M_1 is étale away from the sections j = 0,2^6 3^3 of M_1.

By contrast, $\mathfrak{m}$_1 is ramified over M_1 along these sections. Because of this ramification, there can be no section M_1 → $\mathfrak{m}$_1, i.e. there is no



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Authority page 128; collected-paper page 621; original folio 270.

generalized elliptic curve over $\mathbb{P}$^1 with modular invariant j. Tate has nevertheless shown that over $\mathbb{P}$^1_$\mathbb{Z}$ minus the two sections 0 and 2^6 3^3, i.e. over

S = Spec $\mathbb{Z}$[1/j] (2^6 3^3 1/j - 1)^{-1},

there are two curves with invariant j. Put k = j-2^6 3^3, so that 1/j and 1/k are sections of $\mathcal{O}$_S and j/k is an invertible section.

Proposition 1.6 (Tate). The curves

(1.6.1) y^2 + xy = x^3 - (2^2 3^2/k) x - 1/k (1.6.2) y^2 + xy = x^3 + (2^2 3^2/j) x - (j-2^7 3^2)/j^2

have modular invariant j.

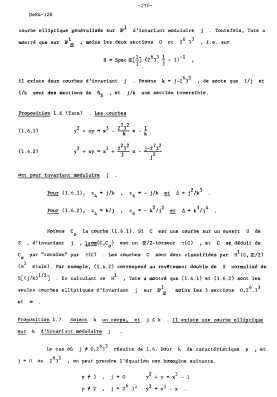
For (1.6.1), c_4 = j/k, c_6 = - j/k, and Δ = j^2/k^3. For (1.6.2), c_4 = k/j, c_6 = - k^2/j^2, and Δ = k^3/j^4.

Write C_o for the curve (1.6.1). If C is a curve over an open subset U of S with invariant j, Isom(C,C_o) is a $\mathbb{Z}$/2-torsor τ(C), and C is obtained from C_o by ”twisting” by τ(C). The curves C are therefore classified by H^1(U,$\mathbb{Z}$/2) (étale H^1). For example, (1.6.2) corresponds to the double cover of S with normalization S[(j/k)^{1/2}]. By calculating this H^1, Tate showed that (1.6.1) and (1.6.2) are the only elliptic curves with invariant j over $\mathbb{P}$^1_$\mathbb{Z}$ minus the three sections 0,2^6.3^3 and ∞.

Proposition 1.7. Let k be a field and let j ∈ k. There exists an elliptic curve over k with modular invariant j.

The case j ≠ 0,2^6 3^3 follows from 1.6. If k has characteristic p and j = 0 or 2^6 3^3, one may use the following affine equation.

p ≠ 3, j = 0 y^2 + y = x^3 - 1 p ≠ 2, j = 2^6.3^3 y^2 = x^3 - x.



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1.8. Let C be the universal elliptic curve over $\mathfrak{m}_1^o$, and let γ be the morphism

Aut(C) → Aut(Lie(C)) ≃ $\mathbb{G}_m$. We extract from [33] the following result about Aut(C).

Lemma 1.9. Over the substack of $\mathfrak{m}_1^o[1/6]$ defined by $c_4 = 0$ (resp. $c_6 = 0$) (i.e. $\sqrt[3]{j} = 0$, resp. $\sqrt[2]{(j-1728)} = 0$), γ is an isomorphism Aut(C) $\tilde{\to}$ μ_6 (resp. Aut(C) $\tilde{\to}$ μ_4).

2. Automorphisms of elliptic curves. Criterion for $\mathfrak{m}_H^o = M_H^o$.

2.1. For an algebraic stack to be an algebraic space, it is necessary and sufficient that its geometric points have no nontrivial automorphisms. Thus, if H ⊂ GL(2, $\mathbb{Z}/n$), in order that $\mathfrak{m}_H^o[1/n] = M_H^o[1/n]$ (resp. $\mathfrak{m}_H[1/n] = M_H[1/n]$), it is necessary and sufficient that an elliptic curve (resp. generalized elliptic curve) equipped with an H-level structure, over an algebraically closed field k of characteristic p ∤ n, have no nontrivial automorphism.

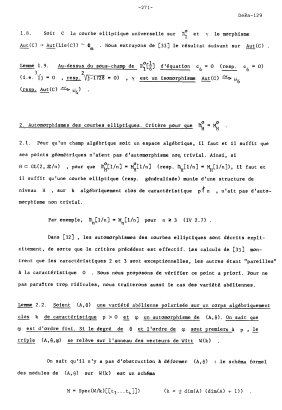
For example, $\mathfrak{m}_n[1/n] = M_n[1/n]$ for n ≥ 3 (IV 2.7).

In [32], the automorphisms of elliptic curves are described explicitly, so the preceding criterion is effective. The calculations of [33] show that characteristics 2 and 3 are exceptional, while the others are ”like” characteristic 0. We propose to verify this point a priori. So as not to seem too ridiculous, we shall also treat the case of abelian varieties.

Lemma 2.2. Let (A, θ) be a polarized abelian variety over an algebraically closed field k of characteristic p > 0, and let φ be an automorphism of (A, θ). It is known that φ has finite order. If the degree of θ and the order of φ are prime to p, the triple (A, θ, φ) lifts to the ring of Witt vectors W(k).

There is no obstruction to deforming (A, θ): the formal moduli scheme of (A, θ) over W(k) is a scheme

M = Spec(W/k)[[t_1...t_r]] (k = ½ dim(A) (dim(A) + 1)).



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Continuing the proof of Lemma 2.2, interpret φ as the action on (A, θ) of a cyclic group G whose order is prime to p. By transport of structure, G acts on the formal moduli scheme M: the automorphism φ of (A, θ) extends to an automorphism of M and of the universal deformation (A_M, θ_M) of (A, θ) over M. The formal moduli scheme of (A, θ, φ) is the G-fixed subscheme M^G of M. Since the order of G is prime to p, this subscheme is smooth over W(k), hence admits sections over W(k), and this proves 2.2.

Proposition 2.3. Let A be an abelian variety of dimension g over an algebraically closed field. Let φ be an automorphism of prime order q, and write

A^φ =_dfn Ker(φ - id_A).

There is an integer k such that

2.g = dim(A^φ) + k.(q-1).

Proof: Passing from A to A/A^φ, we reduce to the case in which A^φ is finite.

The characteristic polynomial of φ acting on A has integer coefficients, does not have 1 as a root, and has only qth roots of unity as roots. It is therefore a power of the cyclotomic polynomial, which has degree q-1. Comparing degrees proves the lemma.

Corollary 2.4. Let A be an abelian variety of dimension g over an algebraically closed field. Let φ be an automorphism of A of finite order. Every prime factor of the order of φ is smaller than 2g + 1.

Proof: This follows from 2.4.

Corollary 2.5. Let (A, θ) be a polarized abelian variety of dimension g over an algebraically closed field k of characteristic p, with polarization degree prime to p. Suppose that p > 2g + 1. If φ is an automorphism of (A, θ), then (A, θ, φ) lifts to characteristic 0 (over W(k)).

Proof: This follows from 2.2 and 2.4.

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2.6. An elliptic curve has a canonical principal polarization. By 2.6, if φ is an automorphism of an elliptic curve E over an algebraically closed field k of characteristic p > 3, (E,φ) lifts to characteristic 0 (over W(k)).

Theorem 2.7. For $\mathfrak{m}_H^o[1/6n]$ to be an algebraic space, it is necessary and sufficient that the inverse image Γ of H in $SL(2,\mathbb{Z})$ have no finite-order element other than 1.

Let $(E,\bar{\alpha})$ be an elliptic curve equipped with a level-H structure over an algebraically closed field k of characteristic p = 0 or p > 3, with $p \nmid n$. We must show that every automorphism φ of $(E,\bar{\alpha})$ is trivial. By 2.6, (E,φ), and hence $(E,\bar{\alpha},\varphi)$, lifts to characteristic 0, so it is enough to treat the case p = 0.

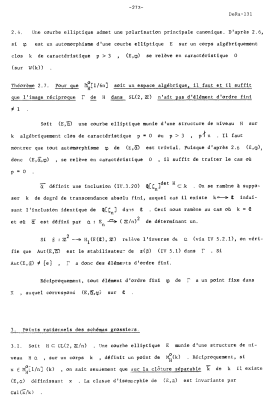
The structure $\bar{\alpha}$ defines an inclusion (IV.3.20) $\mathbb{Q}[\zeta_n]^{\det H} \subset k$. We may reduce to the case in which k has finite absolute transcendence degree; then there is an embedding $k \hookrightarrow \mathbb{C}$ inducing the identity inclusion of $\mathbb{Q}[\zeta_n]$ in $\mathbb{C}$. This reduces us to $k = \mathbb{C}$, with $\bar{\alpha}$ represented by a determinant-one isomorphism $\alpha : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$.

If $\beta : \mathbb{Z}^2 \to H_1(E(\mathbb{C}),\mathbb{Z})$ lifts the inverse of α (via IV 5.2.1), one verifies that $\mathrm{Aut}(E,\bar{\alpha})$ is the stabilizer of $z(\beta)$ (IV 5.1) in Γ. If $\mathrm{Aut}(E,\bar{\alpha}) \neq \{e\}$, Γ therefore has elements of finite order.

Conversely, every finite-order element φ of Γ has a fixed point in X, corresponding to $(E,\bar{\alpha},\varphi)$ over $\mathbb{C}$.

3. Rational points of coarse schemes.

3.1. Let $H \subset GL(2, \mathbb{Z}/n)$. An elliptic curve E equipped with a level-H structure α over a field k defines a point of $M_H^o(k)$. Conversely, if $x \in M_H^o[1/n](k)$, all one knows is that over the separable closure $\bar{k}$ of k there exists a pair (E,α) defining x. The isomorphism class of (E,α) is invariant under $\mathrm{Gal}(\bar{k}/k)$.



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If (E,α) comes from (E_0,α_0) over k, the set of k-isomorphism classes of curves (E_1,α_1) defining x is canonically H^1(Gal(k̄/k),Aut(E_0,α_0)(k̄)). The obstruction to the existence of an (E_0,α_0) lies in an H^2 (in Giraud's sense). In the special case where Aut(E,α) is commutative, Gal(k̄/k) acts on Aut(E,α), and the obstruction lies in H^2(Gal(k̄/k),Aut(E,α)

Proposition 3.2. Let k be a field of characteristic p and let H ⊂ GL(2, ℤ/n). Assume that p ∤ n. Every point x ∈ M_H^o(k) is defined by an elliptic curve E over k equipped with a level-H structure.

Proof (after J.P. Serre and J.S. Milne). Choose a separable closure k̄ of k and an object (E,α) over k̄ that defines the image of x in M_H^o(k̄). If H = GL(2, ℤ/n), which amounts to assuming n = 1, Proposition 3.2 reduces to 1.7. In the language of 3.1, a certain obstruction ε(E) is therefore zero. If Aut(E,α) is trivial, Proposition 3.2 is clear. If Aut(E) is reduced to ± 1 and Aut(E,α) is nontrivial, then Aut(E,α) $\xrightarrow{\sim}$ Aut(E); the obstructions ε(E,α) and ε(E) to defining (E,α) or E over k are therefore equal and zero. To complete the proof, it is enough to treat the following two cases.

Case 1: p ≠ 2,3. By 1.9, Aut(E) is then μ_2, μ_4, or μ_6. The automorphism group Aut(E,α) of (E,α) is a subgroup of Aut(E). Proposition 3.2 says that the obstruction class ε(E,α) ∈ H^2(Gal(k̄/k), Aut(E,α)) is zero. By 1.7 its image ε(E) in H^2(Gal(k̄/k), Aut(E)) is zero, and we apply the following lemma.

Lemma 3.3. Let n be prime to p.

(i) The subgroups of μ_n are the μ_m, for m|n.

(ii) The map H^2(Gal(k̄/k), μ_m) → Gal(k̄/k, μ_n) is injective.

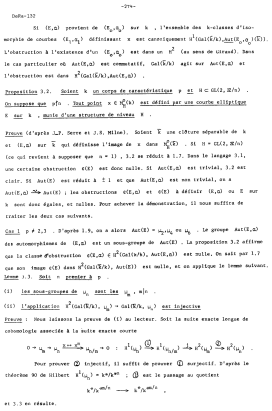
Proof. We leave the proof of (i) to the reader. Consider the long exact cohomology sequence associated with the short exact sequence

0 → μ_m → μ_n ⟶(x ↦ x^m)→ μ_{n/m} → 0 : H^1(μ_n) ⟶①→ H^1(μ_{n/m}) ⟶j→ H^2(μ_m) ⟶②→ H^2(μ_n).

To prove that ② is injective, it is enough to prove that ① is surjective. By Hilbert's Theorem 90, H^1(μ_n) = k^*/k^{*n}; ① is passage to the quotient

k^*/k^{*m/n} → k^*/k^{*m/n},

and 3.3 follows.



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Case 2: p ≠ 0 and Aut(E) is not reduced to ±1.

In this case, x is obtained by scalar extension from a point of M_H^o with values in a finite field (one has j = 0 or 2^6.3^3, and the field of definition of x is algebraic over the field generated by j(x)). We may therefore suppose that k is finite. Since Gal(k̄/k) = $\hat{\mathbb{Z}}$ and Aut(E,α) is finite, the Giraud H^2 in which the obstruction lies vanishes, and 3.2 follows. In other words, let x ∈ M_H^o($\mathbb{F}$_q); there is a finite extension $\mathbb{F}$_{q^n} of $\mathbb{F}$_q and a pair (E,α) over $\mathbb{F}$_{q^n} that defines x. The curve (E,α) is geometrically isomorphic to its conjugates under Gal($\mathbb{F}$_{q^n}/$\mathbb{F}$_q). After replacing $\mathbb{F}$_{q^n} by an extension, we may therefore suppose that there is an isomorphism

φ′ : (E,α)^(q) ⟶̃ (E,α).

This isomorphism defines descent data from $\mathbb{F}$_{q^n} to $\mathbb{F}$_q if and only if the nth power of φ : E —Fr→ E^(q) —φ′→ E is the Frobenius endomorphism F of E/$\mathbb{F}$_{q^n}. In general, φ^n = Fβ, where β has finite order m. Thus φ^{nm} = F^m, and after extending scalars to $\mathbb{F}$_{q^{nm}}, φ′ defines descent data from $\mathbb{F}$_{q^{nm}} to $\mathbb{F}$_q for (E,α). The descended curve solves the problem.

4. Numerical remarks

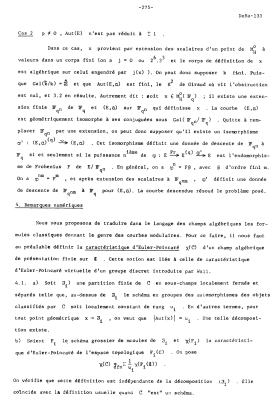
We propose to translate into the language of algebraic stacks the classical formulas giving the genera of modular curves. To do so, we must first define the Euler–Poincaré characteristic χ($\mathcal{C}$) of an algebraic stack of finite presentation over $\mathbb{C}$. This notion is related to Wall's virtual Euler–Poincaré characteristic of a discrete group.

4.1. a) Let ($\mathcal{Z}$_i) be a finite partition of $\mathcal{C}$ into separated locally closed substacks such that, over $\mathcal{Z}$_i, the group scheme of automorphisms of the objects classified by $\mathcal{C}$ is locally constant of rank u_i. In other words, for every geometric point x → $\mathcal{Z}$_i, we require |Aut(x)| = u_i. Such a decomposition exists.

b) Let F_i be the coarse moduli scheme of $\mathcal{Z}$_i, and let χ(F_i) be the Euler–Poincaré characteristic of the topological space F_i($\mathbb{C}$). Set

χ($\mathcal{C}$) =(dfn) Σ (1/u_i) χ(F_i($\mathbb{C}$)).

One checks that this definition is independent of the decomposition ($\mathcal{Z}$_i). It agrees with the usual definition when $\mathcal{C}$ "is" a scheme.



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The Euler–Poincaré characteristic has the following properties:

α) additivity: if $\mathcal{Z}$ is a closed substack of $\mathcal{C}$ with open complement $\mathcal{U}$,

$\chi(\mathcal{C}) = \chi(\mathcal{U}) + \chi(\mathcal{Z})$;

if $\mathcal{U}$ and $\mathcal{V}$ are two open substacks,

$\chi(\mathcal{U} \cup \mathcal{V}) + \chi(\mathcal{U} \cap \mathcal{V}) = \chi(\mathcal{U}) + \chi(\mathcal{V})$.

β) multiplicativity: if $f : \mathcal{C} \to \mathcal{D}$ is a morphism and the geometric fibers $\mathcal{C}$_s of f have constant Euler–Poincaré characteristic, then

$\chi(\mathcal{C}) = \chi(\mathcal{D}).\chi(\mathcal{C}_s)$.

γ) algebraic character: the definition extends readily and makes it possible to define $\chi(\mathcal{C})$ for an algebraic stack $\mathcal{C}$ of finite presentation over an algebraically closed field k of characteristic zero.

δ) Let $f : \mathcal{C} \to \mathcal{D}$ be a finite flat morphism (i.e. flat, proper, with finite fibers), and let s be a geometric point of $\mathcal{D}$. For a geometric point x of $\mathcal{C}$_s, let Ã_x be the local ring of $\mathcal{C}$_s at x. It is an Artinian ring. Set

(4.1.1) deg_s(f) = Σ_x (1/|Aut(x)|) dim_{k(s)}(A_x).

This number is locally constant in s. If it is constant, denote it by deg(f). If f is finite étale, then $\chi(\mathcal{C}_s) = \deg_s(f)$, and β) gives

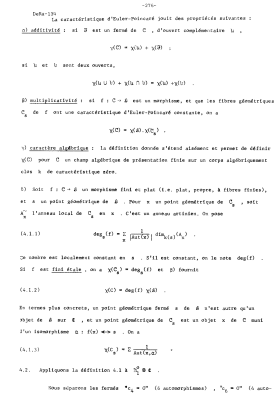
(4.1.2) $\chi(\mathcal{C}) = \deg(f)\, \chi(\mathcal{D})$.

More concretely, a closed geometric point s of $\mathcal{D}$ is simply an object of $\mathcal{D}$ over $\mathbb{C}$, and a geometric point of $\mathcal{C}$_s is an object x of $\mathcal{C}$ equipped with an isomorphism α : f(x) ↔ s. One has

(4.1.3) $\chi(\mathcal{C}_s) = \Sigma\, 1/\mathrm{Aut}(x,\alpha)$.

4.2. Apply Definition 4.1 to $\mathfrak{m}_1\text{^o} \otimes \mathbb{C}$.

We separate the closed substacks ”c_4 = 0” (6 automorphisms), ”c_6 = 0” (4 auto-



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Continuing the preceding sentence, the complementary open (2 automorphisms) has as coarse scheme the affine line with the points j = 0 and j = 1728 removed. We find

(4.2.1) $\chi(\mathfrak{m}_1^o \otimes \mathbb{C}) = 1/6 + 1/4 + -1/2 = -1/12,$

or, more pleasantly,

(4.2.2) $\chi(\mathfrak{m}_1^o \otimes \mathbb{C}) = \zeta(-1).$

For $H \subset GL(2, \mathbb{Z}/n)$, it follows from 4.1.5 that

(4.2.3) $\chi(\mathfrak{m}_H^o \otimes \mathbb{C}) = [GL(2, \mathbb{Z}/n) : H] \cdot -1/12.$

To pass from this to the classical formula for $\chi(M_H \otimes \mathbb{C})$, the cusps and the automorphisms must be taken into account. Consider $M_n \otimes \mathbb{C}$ for $n \geq 3$. Let $\pm U$ be the group of matrices $\pm(1\ u;\ 0\ 1)$, with $u \in \mathbb{Z}/n\mathbb{Z}$. There are $[GL(2, \mathbb{Z}/n) : \pm U]$ cusps (5.1), and the objects of $\mathfrak{m}_n \otimes \mathbb{C}$ have no nontrivial automorphisms (IV 2.7).

Therefore

$\chi(M_n(\mathbb{C})) = |GL(2, \mathbb{Z}/n)| \cdot -1/12 + [GL(2, \mathbb{Z}/n) : \pm U]$

$= -1/12\ n^4 \cdot \prod_{p|n} (1 - 1/p^2)(1 - 1/p) + n^3/2 \prod_{p|n} (1 - 1/p^2)(1 - 1/p).$

The curve $M_n(\mathbb{C})$ has $\varphi(n)$ connected components, all isomorphic. Their genus g is given by

$2g - 2 = \prod_{p|n} (1 - 1/p^2) \cdot (n^3/12 - n^2/2).$

4.3. Let C be a proper Cohen-Macaulay algebraic stack, pure of dimension one, over an algebraically closed field k (for example, $\mathfrak{m}_1 \otimes \mathbb{C}$). Let ω be an invertible sheaf on C. Its degree is defined as follows.

a) Choose a rational section f of ω.

b) Let x be a closed geometric point of C, and let $\tilde{O}_x$ be the henselian local ring of C at x. If f is regular at x, set $\deg_x(f) = \dim_k \tilde{O}_x/(f)$. If f^{-1} is regular there, set $\deg_x(f) = -\dim_k \tilde{O}_x/(f^{-1})$.

The definition continues on physical page 136.

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Continuing §4.3, set

deg(ω) = deg(C,ω) = Σ_x 1/|Aut(x)| deg_x(f).

When k is not algebraically closed, degree is defined after first extending scalars to $\bar{k}$. This degree is independent of the choice of f and agrees with the usual notion when C "is" a scheme.

It satisfies:

α) additivity in ω: deg(ω ⊗ ω') = deg(ω) + deg(ω').

β) coverings: Let f : C' → C be a finite flat morphism of constant degree. Then

deg(f*ω) = deg(f). deg(ω).

γ) specialization: Let S be a trait and C a proper flat Cohen-Macaulay algebraic stack, pure of relative dimension one over S. Let ω be an invertible sheaf on C. Then

deg(C_η,ω) = deg(C_s,ω).

4.4. Compute on $\mathfrak{m}_1$ the degree of ω, the dual of the Lie algebra of the universal elliptic curve. The discriminant Δ is a section of ω^(⊗12) vanishing simply at infinity. Since the point at infinity has two automorphisms, one finds

(4.4.1) deg(ω) = 1/24.

Starting from (4.4.1) and using (4.2.2) and 4.3(β), one obtains, for every H ⊂ GL(2, $\mathbb{Z}$/n),

(4.4.2) χ($\mathfrak{m}$_H^o ⊗ $\mathbb{C}$) = -2 deg($\mathfrak{m}$_H,ω).

4.5. Here is a direct proof of (4.4.2). At every point s of $\mathfrak{m}$_1^o over $\mathbb{C}$ corresponding to an elliptic curve E, deformation theory gives an isomorphism between the tangent space to $\mathfrak{m}$_1^o at s and H^1(E,T_E^1) ≃ H^1(E,O) ⊗ ω^(⊗-1) ≃ ω_s^(⊗-2). A study at infinity shows that the corresponding isomorphism Ω^1_($\mathfrak{m}$_1^o) ≃ ω^(⊗2) extends. The sentence continues on physical page 137.

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c) On pose

$$\deg(\omega) = \deg(C,\omega) = \sum_{x} \frac{1}{|\mathrm{Aut}(x)|} \deg_{k}(f) \ .$$

Pour k non algébriquement clos, on définit $\deg_k$ en étendant au préalable les scalaires à $\overline{k}$. Ce degré est indépendant du choix de C et coïncide avec la notion usuelle quand C "est" un schéma.

Il vérifie

α) additivité en ω : $\deg(\omega \otimes \omega') = \deg(\omega) + \deg(\omega')$.

β) changement : Soit $f : C' \to C$ un morphisme fini et plat de degré constant. On a

$$\deg(f^*\omega) = \deg(f).\ \deg(\omega) \ .$$

γ) spécialisation : Soient S un trait, $\mathcal{C}$ un champ algébrique propre et plat, de Cohen-Macaulay, purement de dimension relative un sur S . Soit ω un fuseau inversible sur $\mathcal{C}$. On a

$$\deg(\mathcal{C}_\eta,\omega) = \deg(\mathcal{C}_s,\omega) \ .$$

4.4. Calculons le degré sur $S_{\mathbb{Q}}$ de ω , le dual de l'algèbre de Lie de la courbe elliptique universelle. Le discriminant Δ est une section de $\omega^{\otimes 12}$, s'annulant simplement à l'infini. La point à l'infini admet deux automorphismes, on trouve

$$\deg(\omega) = \frac{1}{2.3} \ . \tag{4.4.1}$$

Partant de (4.4.1), et exploitant (4.2.2), 4.3. β) , on trouve que pour tout $N \in \mathrm{Cl}(\mathbb{Z}, \mathbb{Z}/4\mathbb{Z})$, on a

$$\chi(S_N^0 \otimes \mathbb{Q}) = -2\ \deg(B_{\mathbb{Q}},\omega) \ . \tag{4.4.2}$$

4.5. Tirt une démonstration directe de (4.4.2). En tout point x de S_N^0 , sur $\mathbb{C}$, correspondant à une courbe elliptique X , la théorie des déformations fournit un isomorphisme entre l'espace tangent à S_N^0 en x et $\mathrm{H}^1(x, \mathcal{T}_X) = \mathrm{H}^1(x, \Omega) \otimes \omega^{\otimes -1} = \omega_x^{\otimes -2}$. Une étude à l'infini montre que l'isomorphisme $\frac{dt}{t^2} = d\frac{1}{t}$ correspondant au pointage

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At infinity, the identification is

(4.5.1) $\Omega^1_{\mathfrak{m}_1}(\mathfrak{m}_1^\infty) \simeq \omega^{\otimes 2}$

In view of the multiplicativity properties of the two sides of (4.4.2), it is enough to consider $\mathfrak{m}_n$ for an integer $n \geq 3$. We then deal with genuine schemes, so that

$\chi(M_n \otimes \mathbb{C}) = -\deg(M_n, \Omega^1_{M_n})$.

Since the pullback of a differential form with logarithmic poles is again of this type, (4.5.1) gives us an isomorphism

(4.5.2) $\Omega^1_{\mathfrak{m}_H}(\mathfrak{m}_H^\infty) \simeq \omega^{\otimes 2}$;

if c is the number of points at infinity of $\mathfrak{m}_n$, we therefore have

$\chi(\mathfrak{m}_n^o \otimes \mathbb{C}) = \chi(\mathfrak{m}_n \otimes \mathbb{C}) - c = -\deg(\Omega^1) - c = -\deg(\Omega^1(\infty)) = -\deg(\omega^{\otimes 2}) = -2\deg(\omega)$.

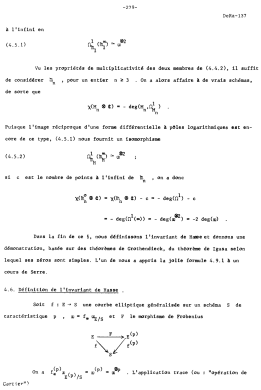
In the remainder of this section, we define the Hasse invariant and give a proof, based on theorems of Grothendieck, of Igusa's theorem that its zeros are simple. One of us learned the elegant formula 4.9.1 in a lecture by Serre.

4.6. Definition of the Hasse invariant.

Let $f : E \to S$ be a generalized elliptic curve over a scheme S of characteristic p, let $\omega = f_* \omega_{E/S}$, and let F be the Frobenius morphism

$E \xrightarrow{F} E^{(p)}$ $f\searrow$ $\swarrow f^{(p)}$ S

We have $f_*^{(p)} \omega_{E^{(p)}/S} = \omega^{(p)} = \omega^{\otimes p}$. The trace map (also called the "Cartier operation")



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Tr_F : F_* ω_{E/S} → ω_{E^(p)/S}

defines a morphism

(4.6.1) Tr_F : ω → ω^{⊗p},

i.e. a section

(4.6.2) h ∈ Γ(S,ω^{⊗(p-1)}).

This is the Hasse invariant. Here is a dual description: coherent-sheaf duality pairs ω with R^1 f_* $\mathcal{O}$, and likewise pairs ω^(p) with R^1 f_*^(p) $\mathcal{O}$. The transpose of (4.6.1) is the pullback

(4.6.3) F^* : R^1 f_*^(p) $\mathcal{O}$ → R^1 f_* $\mathcal{O}$.

If R^1 f_* $\mathcal{O}$ and R^1 f_*^(p) $\mathcal{O}$ are identified with ω^{⊗(-1)} and ω^{⊗(-p)}, then (4.6.3) becomes multiplication by h from (4.6.2).

On $\mathfrak{m}$_1 ⊗ $\mathbb{F}$_p, the Hasse invariant is known not to vanish at infinity, and its zeros correspond to supersingular elliptic curves (see Katz [13]).

The zeros of h on $\mathfrak{m}$_1 ⊗ $\mathbb{F}$_p are known to be simple. In 4.8 we give a direct proof based on Grothendieck's theory of deformations of abelian schemes; a special case of that theory is recalled below.

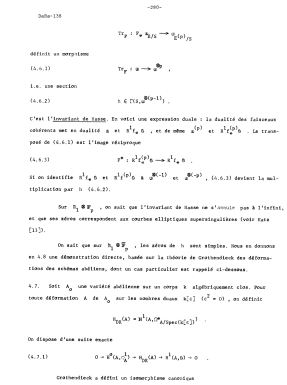
4.7. Let A_o be an abelian variety over an algebraically closed field k. For every deformation A of A_o over the dual numbers k[ε], where ε^2 = 0, define

H_DR(A) = H^1(A, Ω^*_{A/Spec(k[ε])}).

There is an exact sequence

(4.7.1) 0 → H^0(A, Ω^1_A) → H_DR(A) → H^1(A, $\mathcal{O}$) → 0.

Grothendieck defined a canonical isomorphism



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(4.7.2) H_DR(A) ≃ H_DR(A_o) ⊗_k k[ε],

and proved that

a) For every morphism f : A → B, the diagram

H_DR(B_o) ⊗ k[ε] —f_o ⊗ k[ε]→ H_DR(A_o) ⊗ k[ε] ‖ ‖ H_DR(B) —f*→ H_DR(A)

is commutative.

b) The map that assigns to each isomorphism class of deformations A the filtration (4.7.1) of H_DR(A_o) ⊗ k[ε], subject to lifting the filtration (4.7.1) of H_DR(A_o), is bijective.

4.8. Let E_o be a supersingular elliptic curve over k, let E be a nontrivial deformation of E_o over k[ε], and let us prove that h ≠ 0 (i.e. that h is a nonzero multiple of ε). Since E^(p) is the trivial deformation of E_o^(p), we have a commutative diagram

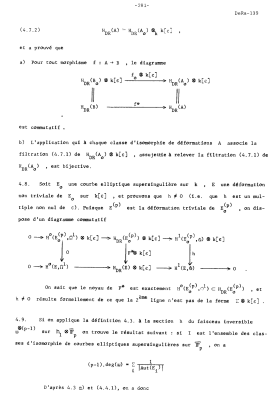
0 → H^0(E_o^(p),Ω^1) ⊗ k[ε] → H_DR(E_o^(p)) ⊗ k[ε] → H^1(E_o^(p),$\mathcal{O}$) ⊗ k[ε] ↓0 ↓F* ⊗ k[ε] ↓h 0 → H^0(E,Ω^1) → H_DR(E) ⊗ k[ε] → H^1(E,$\mathcal{O}$) → 0.

We know that the kernel of F* is exactly H^0(E_o^(p),Ω^1) ⊂ H_DR(E_o^(p)), and h ≠ 0 follows formally from the fact that the second row is not of the form Σ ⊗ k[ε].

4.9. If we apply Definition 4.3 to the section h of the invertible sheaf ω^{⊗(p-1)} on $\mathfrak{m}$_1 ⊗ $\mathbb{F}$_p, we obtain the following result: if I is the set of isomorphism classes of supersingular elliptic curves over $\mathbb{F}$_p, then

(p-1).deg(ω) = Σ_i 1/|Aut(E_i)|

By 4.3 α) and (4.4.1), we therefore have



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(4.9.1) Σ_{E supersingular} 1/|Aut(E)| = (p-1)/24.

For example, if p = 2 (respectively 3), only the elliptic curve with modular invariant 0 is supersingular. It has 24 (respectively 12) automorphisms. Likewise, if n ≥ 3 and (p,n) = 1, there are exactly (1-p)/2 χ(𝔪_n^o(ℂ)) supersingular points on 𝔪_n^o ⊗ 𝔽̄_p.

5. The Galois action on the cusps.

5.1. The set 𝔪_n^∞(ℂ) is the set of isomorphism classes, over ℂ, of singular generalized elliptic curves with n sides, equipped with a level-n structure.

Let (C,+) be the standard n-gon over ℂ, equipped with its generalized elliptic-curve structure II.1.9. Canonically,

C_n = μ_n × ℤ/n.

The automorphism group of (C,+) was computed in II 1.10. Its image in GL(μ_n × ℤ/n) is the group ± U of matrices

$$\pm \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \quad (u \in \mathrm{Hom}(\mathbb{Z}/n, \mu_n)).$$

The set of level-n structures on (C,+) is Isom(C_n,(ℤ/n)^2). Thus, canonically,

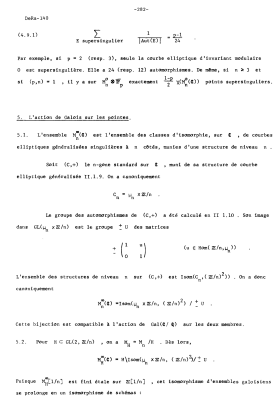
𝔪_n^∞(ℂ) = Isom(μ_n × ℤ/n, (ℤ/n)^2) / ± U.

This bijection is compatible with the action of Gal(ℂ/ℚ) on both sides.

5.2. For H ⊂ GL(2,ℤ/n), we have 𝔪_H = 𝔪_n/H. Consequently,

𝔪_H^∞(ℂ) = H\Isom(μ_n × ℤ/n, (ℤ/n)^2)/± U.

Since 𝔪_H^∞[1/n] is finite étale over ℤ[1/n], this isomorphism of Galois sets extends to an isomorphism of schemes:



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Construction 5.3. $\mathfrak{m}_H^\infty[1/n] = H\backslash\mathrm{Isom}_{\mathbb{Z}[1/n]}(\mu_n \times \mathbb{Z}/n, (\mathbb{Z}/n)^2) / \pm U$.

6. Study of $M_{\Gamma_o(n)}$.

6.1. In IV.4.5 we described $\mathfrak{m}^o_{\Gamma_o(n)}[1/n]$ as the stack over $\mathbb{Z}[1/n]$ classifying isogenies $E \to F$ with cyclic kernel of degree n. This exhibits two maps $\mathfrak{m}^o_{\Gamma_o(n)}[1/n] \to \mathfrak{m}_1^o[1/n]$, defined respectively by E and F and interchanged by an involution w (IV.4.4). By IV.3.19, the induced maps on the coarse schemes extend to

(6.1.1) c and $cw : M_{\Gamma_o(n)} \to M_1$.

6.2. The finite map $(c,cw) : M_{\Gamma_o(n)} \to M_1 \times M_1$ has as its image a divisor invariant under permutation of the coordinates. Since $M_{\Gamma_o(n)} \to M_1$ has degree equal to the number $P(n)$ of points of the projective line over $\mathbb{Z}/n$ and is generically injective, this divisor has bidegree $(P(n), P(n))$. The maps c and cw carry points at infinity to points at infinity. Since $M_1^o = \mathrm{Spec}(\mathbb{Z}[j])$ and $\mathbb{Z}[j,j']$ is factorial, the image divisor of

(6.2.1) $(c,cw) : M^o_{\Gamma_o(n)} \to M_1^o \times M_1^o$

is defined by an equation $\Phi_n \in \mathbb{Z}[j,j']$, the modular equation. We have

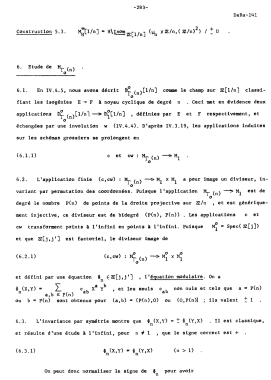
$\Phi_n(X,Y) = \Sigma_{a,b\le P(n)}\, c_{ab} X^a Y^b$,

and the only nonzero c_{ab} for which $a = P(n)$ or $b = P(n)$ occur at $(a,b) = (P(n),0)$ or $(0,P(n))$; they are ± 1.

6.3. Symmetry shows that $\Phi_n(X,Y) = \pm\Phi_n(Y,X)$. It is classical, and follows from a study at infinity, that for $n \neq 1$ the correct sign is $+$.

(6.3.1) $\Phi_n(X,Y) = \Phi_n(Y,X)$ $(n > 1)$.

We may therefore normalize the sign of Φ_n so as to have



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(6.3.2) Φ_n(X,Y) = X^{P(n)} + Y^{P(n)} + Σ_{a,b<P(n)} c_ab X^a Y^b c_ab = c_ba ∈ ℤ.

6.4. One should note that the map (6.2.1) is not necessarily injective. Over an algebraically closed field k in which n is invertible, there may indeed exist two cyclic-kernel n-isogenies

(6.4.1) φ_1 : E_1 → F_1 φ_2 : E_2 → F_2

that are not isomorphic, even though there are isomorphisms e : E_1 ≃→ E_2 and f : F_1 → F_2. The endomorphism g = e^{-1} φ_2^* f φ_1 of E_1 then has degree n^2, but its kernel is not the kernel of multiplication by n. Thus E_1 has complex multiplication.

Conversely, let g be an endomorphism of degree n^2 of an elliptic curve E, suppose Ker(g) ≠ E_n, and let A ⊂ Ker(g) be cyclic of order n. A pair (6.4.1) is obtained by taking

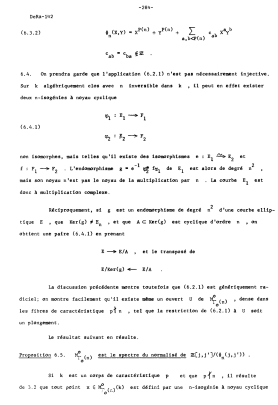
E → E/A, and the transpose of E/Ker(g) ← E/A.

The preceding discussion nevertheless shows that (6.2.1) is generically radicial; one easily shows that there is even an open subset U of M^o_{Γ_o(n)}, dense in the fibers of characteristic p ∤ n, such that the restriction of (6.2.1) to U is an immersion.

The following result follows.

Proposition 6.5. M^o_{Γ_o(n)} is the spectrum of the normalization of ℤ[j,j′]/(Φ_n(j,j′)).

If k is a field of characteristic p and p ∤ n, it follows from 3.2 that every point x ∈ M^o_{Γ_o(n)}(k) is defined by a cyclic-kernel n-isogeny



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E → F defined over k. It does not follow that if j,j’ belong to k and

Φ_n(j,j’) = 0,

there exists such an n-isogeny with j(E) = j and j(F) = j’.

If n is a prime p, the decomposition V.1.16 of $\mathfrak{m}_{\Gamma_o(p)}$ mod p translates as follows at the level of coarse schemes.

Theorem 6.6. (H. Weber-Lehrbuch der Algebra III, p.244 eq. (32)).

Φ_p(j,j’) ≡ (j-j’^p)(j^p-j’) mod p.

By studying infinity, one proves more generally that if (n,p) = 1, then

(6.6.1) $\Phi_{np}(j,j') \equiv \Phi_n(j^p,j')\,\Phi_n(j,j'^p)$ (p)

(6.6.2) $\Phi_{np^2}(j,j') \equiv \Phi_n(j^{p^2},j')\,\Phi_n(j,j')^{p-1}\,\Phi_n(j,j'^{p^2})$ (p)

(6.6.3) $\Phi_{np^k}(j,j') \equiv \prod_{\ell+m=k,\,K=\inf(\ell,m)} \Phi_n(j^{p^{\ell-K}},j'^{p^{m-K}})^{\varphi(p^K)}$ (p)

If p^2 | n, (6.6.3) gives

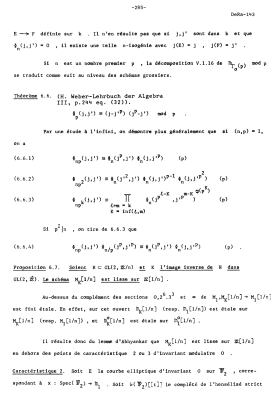
(6.6.4) $\Phi_{np}(j,j')\,\Phi_{n/p}(j^p,j'^p) \equiv \Phi_n(j^p,j')\,\Phi_n(j,j'^p)$ (p).

Proposition 6.7. Let H ⊂ GL(2,$\mathbb{Z}$/n), and let K be the inverse image of H in GL(2,$\hat{\mathbb{Z}}$). The scheme M_K[1/n] is smooth over $\mathbb{Z}$[1/n].

Over the complement of the sections 0, 2^6·3^3, and ∞ of M_1, the map M_K[1/n] → M_1[1/n] is finite étale. Indeed, over this open set $\mathfrak{m}$_K[1/n] (respectively $\mathfrak{m}$_1[1/n]) is étale over M_K[1/n] (respectively M_1[1/n]), and $\mathfrak{m}$_K^o[1/n] is étale over $\mathfrak{m}$_1^o[1/n].

It therefore follows from Abhyankar’s lemma that M_K[1/n] is smooth over $\mathbb{Z}$[1/n] away from the points of characteristic 2 or 3 with modular invariant 0.

Characteristic 2. Let E be the elliptic curve of invariant 0 over $\mathbb{F}$_2 corresponding to x : Spec($\mathbb{F}$_2) → $\mathfrak{m}$_1. Let W($\mathbb{F}$_2)[[t]] be the completion of the strict henselization



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of $\mathfrak{m}_1$ at x. The group A = Aut(E) acts on $W(\mathbb{F}_2)[[t]]$, and, in view of I.8, one must prove that for every subgroup B of A the algebra $W(\mathbb{F}_2)[[t]]^B$ is formally smooth over $W(\mathbb{F}_2)$. The group A embeds in GL(2,$\mathbb{F}_3$) (IV.2.7). In fact,

A = SL(2,$\mathbb{F}_3$).

It follows that if one of the algebras $W(\mathbb{F}_2)[[t]]^B$ were not formally smooth, then already one of the schemes M_H ($H \subset GL(2,\mathbb{Z}/3)$) would not be smooth over $\mathbb{Z}$.

The curve $M_3/\mathbb{Z}[\zeta_3]$ is known to have genus 0. Hence, for $H \subset GL(2,\mathbb{Z}/3)$, $M_H = M_3/H$ over $\mathbb{Z}[\zeta_3]^H$ has genus 0. The geometric fibers of $M_H[1/3]$ are also known to be generically smooth, irreducible, and reduced (being Cohen–Macaulay, with M_H normal). Since they have arithmetic genus 0, they are smooth.

Characteristic 3: The same argument applies, using the embedding $Aut(E) \subset GL(2,\mathbb{F}_4)$ and the fact that $M_4/\mathbb{Z}[\zeta_4]$ has genus 0.

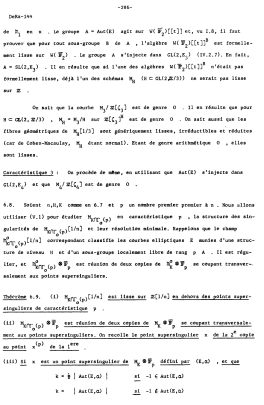
6.8. Let n,H,K be as in 6.7, and let p be a prime not dividing n. We shall use (V.1) to study $M_{K\cap\Gamma_0(p)}$ in characteristic p, the structure of the singularities of $M_{K\cap\Gamma_0(p)}[1/n]$, and their minimal resolution. Recall that the corresponding stack $\mathfrak{m}^o_{K\cap\Gamma_0(p)}[1/n]$ classifies elliptic curves E equipped with a level-H structure and a locally free subgroup A of rank p. It is regular, and $\mathfrak{m}^o_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$ is the union of two copies of $\mathfrak{m}_K^o \otimes \mathbb{F}_p$ meeting transversely at the supersingular points.

Theorem 6.9. (i) $M_{K\cap\Gamma_0(p)}[1/n]$ is smooth over $\mathbb{Z}[1/n]$ away from the supersingular points of characteristic p.

(ii) $M_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$ is the union of two copies of $M_K \otimes \mathbb{F}_p$ meeting transversely at the supersingular points. The supersingular point x of the second copy is glued to the point $x^{(p)}$ of the first.

(iii) If x is a supersingular point of $M_K \otimes \mathbb{F}_p$ defined by (E,α), and if

k = ½ |Aut(E,α)| when -1 ∈ Aut(E,α) k = |Aut(E,α)| when -1 ∉ Aut(E,α)



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Then, at the corresponding point of $M_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$, the scheme $M_{K\cap\Gamma_0(p)} \otimes W(\mathbb{F}_p)$ has a singularity of type A_{k-1}, formally isomorphic to

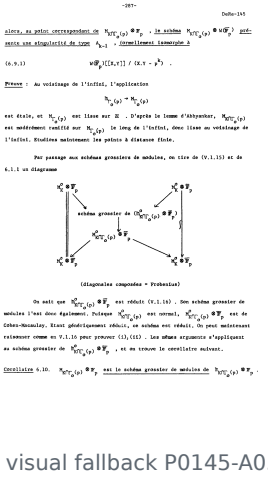
(6.9.1) $W(\mathbb{F}_p)[[X,Y]]/(X.Y-p^k)$.

Proof: In a neighborhood of infinity, the map $\mathcal{M}_{\Gamma_0(p)} \to M_{\Gamma_0(p)}$ is étale, and $M_{\Gamma_0(p)}$ is smooth over $\mathbb{Z}$. By Abhyankar's lemma, $M_{K\cap\Gamma_0(p)}$ is tamely ramified over $M_{\Gamma_0(p)}$ along infinity, and hence is smooth in a neighborhood of infinity. Let us now study the points at finite distance.

Passing to coarse moduli schemes, (V.1.15) and 6.1.1 yield the displayed diagram, with the central upper label "coarse moduli scheme of $\mathcal{M}^o_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$" and the caption "composed diagonals = Frobenius."

We know that $\mathcal{M}^o_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$ is reduced (V.1.16). Its coarse moduli scheme therefore is reduced as well. Since $M^o_{K\cap\Gamma_0(p)}$ is normal, $M^o_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$ is Cohen-Macaulay. Being generically reduced, this scheme is reduced. We may now argue as in V.1.16 to prove (i),(ii). The same arguments apply to the coarse moduli scheme of $\mathcal{M}^o_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$, and we obtain the following corollary.

Corollary 6.10. $M_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$ is the coarse moduli scheme of $\mathcal{M}_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$.



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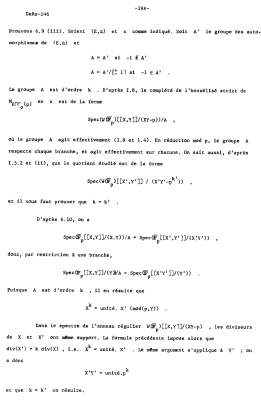
We prove 6.9 (iii). Let (E,α) and x be as indicated. Let A’ be the automorphism group of (E,α), and set

A=A’ if -1∉A’, A=A’/{±1} if -1∈A’.

The group A has order k. By I.8, the completion of the strict henselization of $M_{K\cap\Gamma_0(p)}$ at x is Spec(W($\mathbb{F}_p$)[[X,Y]]/(XY-p))/A, where A acts faithfully (I.8 and I.4). After reduction modulo p, A preserves each branch and acts faithfully on each one. We also know, by I.5.2 and (ii), that the quotient has the form Spec(W($\mathbb{F}_p$)[[X’,Y’]] /(X’Y’-p^{k’})), and we must prove k=k’.

By 6.10, Spec($\mathbb{F}_p$[[X,Y]]/(X.Y))/A = Spec($\mathbb{F}_p$[[X’,Y’]] /(X’.Y’)), and hence, upon restriction to one branch, Spec($\mathbb{F}_p$[[X,Y]]/(Y))/A = Spec($\mathbb{F}_p$[[X’Y’]] /(Y’)).

Since A has order k, it follows that X^k=unit·X’ (mod(p,Y)). In the spectrum of the regular ring W($\mathbb{F}_p$)[[X,Y]]/(XY-p), the divisors of X and X’ have the same support. The preceding formula therefore forces div(X’)=k div(X), i.e. X^k=unit·X’. The same argument applies to Y’; hence X’Y’=unit·p^k, and it follows that k=k’.



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Specialization invariance of the Euler-Poincaré characteristic with $\mathbb{Q}$-coefficients, together with 6.9(ii), gives the following topological Euler-Poincaré formula.

Corollary 6.11. Let s be the number of supersingular points on $M_K \otimes \mathbb{F}_p$. Then

(6.11.1) $\chi(M_{K\cap\Gamma_0(p)}) = 2\chi(M_K)-2s$.

If $M_{K\cap\Gamma_0(p)}$ is regarded as a geometrically irreducible curve over $R=\mathbb{Z}[\zeta_n]^H$, and if $s_0=s/[R:\mathbb{Z}]$ denotes the number of supersingular points of $M_K \otimes_R \mathbb{F}_p$, then 6.11 can be rewritten in terms of the genus as

(6.11.2) $g(M_{K\cap\Gamma_0(p)} \otimes_R \mathbb{C}) = 2g(M_K)+(s_0-1)$.

Corollary 6.12. $M_{K\cap\Gamma_0(p)} \otimes_R \mathbb{C}$ is an elliptic curve if and only if $M_K \otimes_R \mathbb{C}$ has genus 0 and $s_0=2$.

Reminder 6.13. Resolving the singularity (6.9.1) by blow-ups produces a chain of (k-1) projective lines of self-intersection -2. The locus p=0 in the resolution is reduced. This makes it possible to calculate the minimal resolution of $M_{K\cap\Gamma_0(p)}$.

The following lemma, taken from [20], will be proved in VII.2.7.

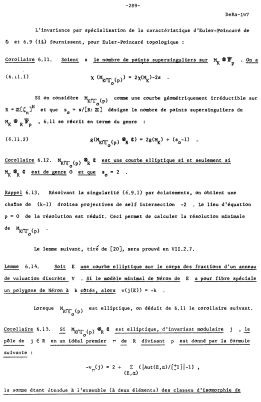
Lemma 6.14. Let E be an elliptic curve over the fraction field of a discrete valuation ring V. If the special fiber of the minimal Néron model of E is a Néron k-gon, then v(j(E))=-k.

When $M_{K\cap\Gamma_0(p)}$ is elliptic, Corollary 6.11 gives the following result.

Corollary 6.15. Suppose $M_{K\cap\Gamma_0(p)} \otimes_R \mathbb{C}$ is elliptic, with modular invariant j. The order of the pole of $j\in R$ at a prime ideal π of R dividing p is

$$-v_\pi(j) = 2 + \Sigma_{(E,\alpha)} (|\mathrm{Aut}(E,\alpha)/\{\pm 1\}|-1),$$

where the sum ranges over the two-element set of isomorphism classes of



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Deligne-Serre

l'invariance par spécialisation de la caractéristique d'Euler-Poincaré de G et 6.9 (iii) fournissent, pour Euler-Poincaré cyclotomique :

Corollaire 6.11. Soient ℓ le nombre de points supersinguliers sur $X_0 \otimes \mathbb{F}_p$. On a

$$(6.11.1) \qquad \chi(X_{H\cap\Gamma_0(p)}) = p\chi(X_0) - 2\ell \ .$$

Si on considère $X_{H\cap\Gamma_0(p)}$ comme une courbe géométriquement irréductible sur $R = \mathbb{Z}[\zeta_N, \frac{1}{N}]$ et que $e_p = \#(H\cap\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z}))$ désigne le nombre de points supersinguliers de $X_0 \otimes_R \mathbb{F}_p$, 6.11 se réécrit en terme du genre :

$$(6.11.2) \qquad g(X_{H\cap\Gamma_0(p)} \otimes_R k) = pg(X_0) + (e_p-1) \ .$$

Corollaire 6.12. $X_{H\cap\Gamma_0(p)} \otimes_R k$ est une courbe elliptique si et seulement si $X_0 \otimes_R k$ est de genre 0 et que $e_p = 2$.

Exemple 6.13. Résolvant la singularité (6,9,1) par éclatements, on obtient une chaîne de (k-1) droites projectives de self intersection -2 , le lieu d'équation $y = 0$ de la résolution est réduit. Ceci permet de calculer la résolution minimale de $X_{H\cap\Gamma_0(p)}$.

Le lemme suivant, tiré de [IV], sera prouvé en VII.2.7.

Lemme 6.14. Soit E une courbe elliptique sur le corps des fractions d'un anneau de valuation discrète V ; si le modèle minimal de Néron de E a pour fibre spéciale un polygone de Néron à k côtés, alors $v(j(E)) = -k$.

Lorsque $X_{H\cap\Gamma_0(p)}$ est elliptique, on déduit de 6.11 le corollaire suivant.

Corollaire 6.15. Si $X_{H\cap\Gamma_0(p)} \otimes_R \mathbb{C}$ est elliptique, d'invariant modulaire j , le pôle de $j \in R$ en un idéal premier π de R divisant p est donné par la formule suivante :

$$-v_\pi(j) = 2 + \sum_{(E,\alpha)} (|\mathrm{Aut}(E,\alpha)/\{\pm 1\}|-1) \ ,$$

la somme étant étendue à l'ensemble (à deux éléments) des classes d'isomorphie de

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Authority page 148; collected-paper page 641; original folio 290.

supersingular elliptic curves E, equipped with a level-H structure of determinant one, over an algebraic closure $\mathbb{F}_p$ of R/(π).

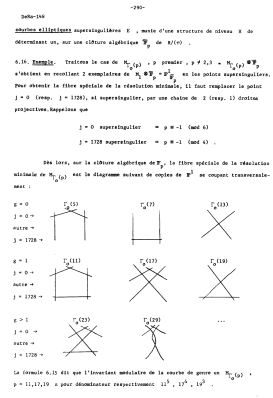
6.16. Example. Consider $M_{\Gamma_0(p)}$, where p is prime and p≠2,3. Its special fiber $M_{\Gamma_0(p)} \otimes \mathbb{F}_p$ is obtained by gluing two copies of $M_1 \otimes \mathbb{F}_p = \mathbb{P}^1_{\mathbb{F}_p}$ at the supersingular points.

To obtain the special fiber of the minimal resolution, a supersingular point j=0 must be replaced by a chain of two projective lines, while a supersingular point j=1728 must be replaced by a chain of one projective line. Recall that

j=0 is supersingular ⇔ p≡-1 (mod 6), j=1728 is supersingular ⇔ p≡-1 (mod 4).

Thus, over an algebraic closure of $\mathbb{F}_p$, the special fiber of the minimal resolution of $M_{\Gamma_0(p)}$ is represented by the displayed diagram of copies of $\mathbb{P}^1$ meeting transversely. Its left-hand labels are j=0, other, and j=1728.

Formula 6.15 says that the modular invariant of the genus-one curve $M_{\Gamma_0(p)}$, for p=11,17,19, has denominator 11^5,17^4,19^3, respectively.



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VII. The Tate curve.

In §1 of this chapter, we use a method due to Raynaud to construct the Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$ over $\mathbb{Z}[[q]]$. This enables us to give an explicit description of the formal completion of $\mathcal{M}_H$ along infinity (§2). In §§3 and 4, we deduce integrality theorems for the coefficients of the Fourier-series expansions of modular forms.

1. Construction of the Tate curve over $\mathbb{Z}[[q]]$.

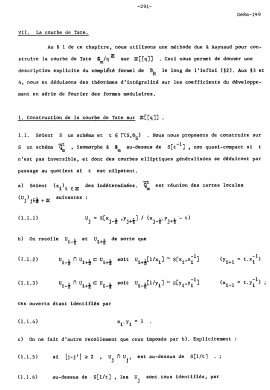
1.1. Let S be a scheme and $t \in \Gamma(S,\mathcal{O}_S)$. We propose to construct over S a scheme $\bar{\mathbb{G}}_m^t$, isomorphic to $\mathbb{G}_m$ over $S[t^{-1}]$, non-quasi-compact when t is not invertible, and from which generalized elliptic curves will be obtained by taking a quotient when t is nilpotent.

a) Let $(x_i)_{i\in\mathbb{Z}}$ be indeterminates. $\bar{\mathbb{G}}_m^t$ is the union of the local charts $(U_j)_{j\in 1/2+\mathbb{Z}}$:

(1.1.1) $U_j = S[x_{j-1/2}, y_{j+1/2}] / (x_{j-1/2} y_{j+1/2} - t)$.

b) Glue $U_{i-1/2}$ and $U_{i+1/2}$ so that their overlap is $U_{i+1/2}[1/x_i] \simeq S[x_i, x_i^{-1}]$, with $y_{i+1} = t\, x_i^{-1}$, and also $U_{i-1/2}[1/y_i] \simeq S[y_i, y_i^{-1}]$, with $x_{i-1} = t\, y_i^{-1}$; these opens are identified by $x_i\, y_i = 1$.

c) No gluings other than those imposed by b) are made. If $|j-j'| \geq 2$, $U_j \cap U_{j'}$ lies over $S[1/t]$; over $S[1/t]$, all the U_j are identified by



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x_i=t^{-i}x_0,

y_i=t^{-i}y_0.

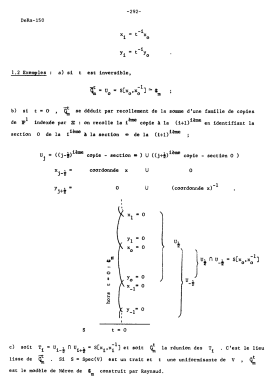
1.2 Examples: a) if t is invertible, Ḡ_m^t=U_0=S[x_0,x_0^{-1}]≃𝔾_m.

b) if t=0, Ḡ_m^t is obtained by gluing the disjoint union of a family of copies of $\mathbb{P}^1$ indexed by $\mathbb{Z}$: the ith copy is glued to the (i+1)st by identifying the zero section of the ith copy with the infinity section of the (i+1)st.

For each j, U_j is the union of the (j-1/2)th copy minus its infinity section and the (j+1/2)th copy minus its zero section, with x_{j-1/2}=(coordinate x)∪0 and y_{j+1/2}=0∪(coordina x)^{-1}.

The schematic shows an infinite vertical chain with successive junctions x_1=0, y_1=0, x_0=0, y_0=0, x_{-1}=0, y_{-1}=0. Overlapping braces label U_{1/2}, U_{-1/2}, and U_{1/2}∩U_{-1/2}=S[x_0,x_0^{-1}]. The chain meets the base S at the special fiber t=0, and the generic part is labeled “away from t=0: 𝔾_m.”

c) Let T_i=U_{i-1/2}∩U_{i+1/2}=S[x_i,x_i^{-1}], and let 𝔾_m^t be the union of the T_i. It is the smooth locus of Ḡ_m^t. If S=Spec(V) is a trait and t is a uniformizer of V, then 𝔾_m^t is the Néron model of 𝔾_m constructed by Raynaud.



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The maps

Ḡ_m^t(S) ← 𝔾_m^t(S) → 𝔾_m^t(K)=𝔾_m(K)=K^*

are bijective, where K is the fraction field of V. To t^i u∈K^*, with u∈V^*, corresponds the section x_i=u of T_i. This example motivates the definitions that follow.

d) Abuse of notation: for a invertible on S and k∈ℤ, we shall sometimes denote by at^k the section of T^k such that x_k(at^k)=a.

1.3. We define on Ḡ_m^t the structure of a commutative group scheme. Multiplication sends T_i×T_j to T_{i+j} and is given by x_{i+j}(a·b)=x_i(a)x_j(b). The intersection of distinct charts is the inverse image of S[1/t]; over S[1/t] the barred model equals 𝔾_m and this law is ordinary multiplication.

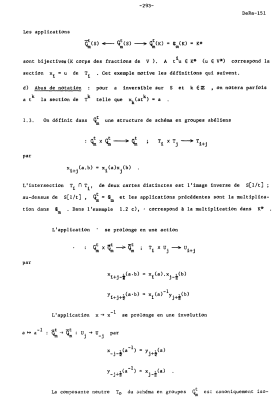
Multiplication extends to an action $\mathbb{G}_m^t \times \bar{G}_m^t \to \bar{G}_m^t$, carrying $T_i \times U_j$ to U_{i+j}, with

$x_{i+j-1/2}(a\cdot b)=x_i(a)x_{j-1/2}(b)$, $y_{i+j+1/2}(a\cdot b)=x_i(a)^{-1}y_{j+1/2}(b)$.

Inversion extends to $\bar{G}_m^t$ and carries U_j to U_{-j}, with

$x_{-j-1/2}(a^{-1})=y_{j+1/2}(a)$, $y_{-j+1/2}(a^{-1})=x_{j-1/2}(a)$.

The identity component T_0 of $\mathbb{G}_m^t$ is canonically iso-



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Les applications

$$\mathbb{G}_m^t(k) \longleftarrow \mathbb{G}_m^t(k) \longrightarrow \mathbb{G}_m^t(k) = \mathbb{G}_m(k) = k^*$$

avec bijectivité (k corps des fractions de V). A $t^k u \in k^*$ ($u \in V^*$) correspond la section $\pi_k \circ u$ de T_k. Cet exemple motive les définitions qui suivent.

d) Abus de notation : pour a inversible sur S et $k \in \mathbb{Z}$, on notera parfois $a.t^k$ la section de T^k telle que $x_k(a^k) = a$.

1.3. On définit dans $\mathbb{G}_m^t$ une structure de schéma en groupes abélien

$$\cdot : \mathbb{G}_m^t \times \mathbb{G}_m^t \longrightarrow \mathbb{G}_m^t \quad ; \quad T_i \times T_j \longrightarrow T_{i+j}$$

par

$$x_{i+j}(a.b) = x_i(a)x_j(b) .$$

L'intersection $T_i \cap T_{i'}$ de deux cartes distinctes est l'image inverse de $S[1/\ell]$; au-dessus de $S[1/\ell]$, $\mathbb{G}_m^t = \mathbb{G}_m$ et les applications précédentes sont la multiplication dans $\mathbb{G}_m$. Dans l'exemple 1.2 c), $\cdot$ correspond à la multiplication dans k^*.

L'application $\cdot$ se prolonge en une action

$$\cdot : \mathbb{G}_m^t \times \bar{\mathbb{G}}_m^t \longrightarrow \bar{\mathbb{G}}_m^t \quad ; \quad T_i \times U_j \longrightarrow U_{i+j}$$

par

$$x_{i+j-\frac{1}{2}}(a.b) = x_i(a).x_{j-\frac{1}{2}}(b)$$

$$y_{i+j+\frac{1}{2}}(a.b) = x_i(a)^{-1}y_{j+\frac{1}{2}}(b)$$

L'application $x \mapsto x^{-1}$ se prolonge en une involution

$\alpha \mapsto \alpha^{-1} : \bar{\mathbb{G}}_m^t \to \bar{\mathbb{G}}_m^t$; $U_j \to U_{-j}$ par

$$x_{-j-\frac{1}{2}}(a^{-1}) = y_{j+\frac{1}{2}}(a)$$

$$y_{-j+\frac{1}{2}}(a^{-1}) = x_{j-\frac{1}{2}}(a) .$$

Le composante neutre T_0 du schéma en groupes $\bar{\mathbb{G}}_m^t$ est canoniquement iso-

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-morphic to $\mathbb{G}_m$, via x_0.

1.4. If $u \in \Gamma(S,\mathcal{O}_S)^*$ and $m \geq 1$, define $[u]:\bar{G}_m^t \to \bar{G}_m^{ut^m}$. On the indicated charts its coordinates are

$x_i^{(ut^m)}\circ[u]=x_{mi+k}^{(t)}t^k u^{-i}$, $y_i^{(ut^m)}\circ[u]=y_{mi-k}^{(t)}t^k u^i$, $0 \leq k < m$.

For $t=0$ this contracts each T_i with i not congruent to 0 modulo m to a point. If $m=1$ it is an isomorphism; in particular, the Néron model is independent of the chosen uniformizer.

1.5. Suppose t is nilpotent. Let g be a section of $\mathbb{G}_m^t$ lying in T_k, $k \neq 0$. Let $\mathbb{Z}$ act on $\bar{G}_m^t$ by $(i,x) \mapsto g^i \cdot x$.

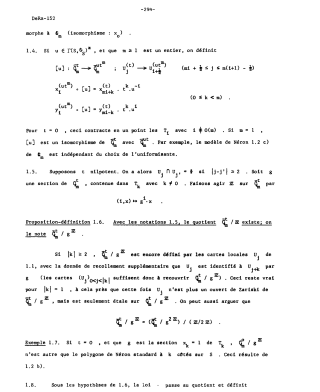
Proposition-definition 1.6. With the notation of 1.5, the quotient $\bar{G}_m^t/\mathbb{Z}$ exists and is denoted $\bar{G}_m^t/g^{\mathbb{Z}}$.

For $|k| \geq 2$ it is covered by the local charts U_j with the additional gluing that g identifies U_j with U_{j+k}. For $|k|=1$ the same description holds étale-locally rather than by Zariski opens. Equivalently,

$\bar{G}_m^t/g^{\mathbb{Z}}=(\bar{G}_m^t/g^{2\mathbb{Z}})/(\mathbb{Z}/2\mathbb{Z})$.

Example 1.7. For $t=0$ and g given by $x_k=1$ on T_k, the quotient is the standard Néron k-gon.

1.8. Under the hypotheses of 1.6, multiplication descends to the quotient and defines



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Authority page 153; collected-paper page 646; original folio 295.

a map

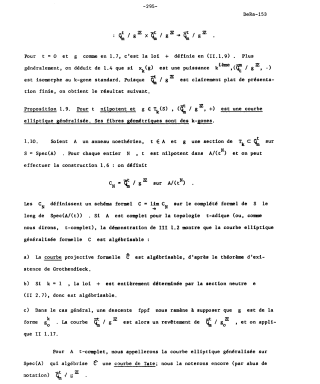
$\bar{G}_m\hat{}t/g\hat{}\mathbb{Z} \times \bar{G}_m\hat{}t/g\hat{}\mathbb{Z} \to \bar{G}_m\hat{}t/g\hat{}\mathbb{Z}$.

For t=0 and g as in 1.7, this is the addition law of (II.1.9). More generally, if x_k(g) is a kth power, the printed dot-marked quotient is isomorphic to the standard k-gon. Since the quotient is flat and of finite presentation, one obtains:

Proposition 1.9. For nilpotent t and g∈T_k(S), $(\bar{G}_m\hat{}t/g\hat{}\mathbb{Z},+)$ is a generalized elliptic curve whose geometric fibers are k-gons.

1.10. Let A be Noetherian, t∈A, and g a section of T_k over S=Spec(A). For every N define C_N=$\bar{G}_m\hat{}t/g\hat{}\mathbb{Z}$ over A/(t^N). The C_N define the direct-limit formal scheme C over the completion of S along Spec(A/(t)). If A is t-adically complete, the formal generalized elliptic curve is algebraizable: the formal projective curve $\hat{C}$ is algebraizable by Grothendieck existence; for k=1 the identity section determines the law; in general fppf descent reduces to g=g_0^k and II 1.17 applies to the resulting covering.

The algebraization over Spec(A) is called a Tate curve and is again denoted $\bar{G}_m\hat{}t/g\hat{}\mathbb{Z}$.



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Authority page 154; collected-paper page 647; original folio 296.

Proposition 1.11. The image of the nonsmoothness subscheme of the Tate curve $\bar{G}_m^t/g^{\mathbb{Z}}$ over Spec(A) is defined by t=0. It is the same image for the quotient, the formal scheme $\hat{C}$, and the barred model.

1.12. For nilpotent t, the identity components of $\bar{G}_m^t$ and $\bar{G}_m^t/g^{\mathbb{Z}}$ are canonically isomorphic to $\mathbb{G}_m$. Hence the formal completion of $\mathbb{G}_m$ along its identity is isomorphic to the completion of the quotient; their Lie algebras are identified with $\mathcal{O}$; and μ_n injects into the n-torsion.

For Noetherian t-complete A, formulas (1.12.1)–(1.12.3) give the corresponding formal-group isomorphism, Lie isomorphism, and inclusion $\mu_n \hookrightarrow (\bar{G}_m^t/g^{\mathbb{Z}})_n$. Tate curves therefore carry the canonical invariant differential dx/x.

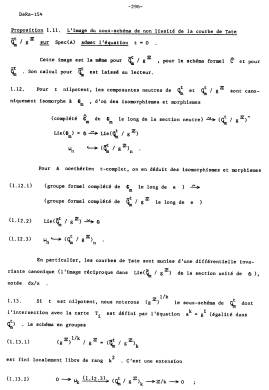
1.13. If t is nilpotent, let $(g^{\mathbb{Z}})^{1/k}$ be the subscheme whose intersection with T_i is defined by $a^k=g^i$. The group scheme

$$(g^{\mathbb{Z}})^{1/k}/g^{\mathbb{Z}}=(\bar{G}_m^t/g^{\mathbb{Z}})_k$$

is finite locally free of rank k^2 and is an extension

$$0\to\mu_k\to(\bar{G}_m^t/g^{\mathbb{Z}})_k\to\mathbb{Z}/k\to0,$$

whose explanation continues on page 155.



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Authority page 155; collected-paper page 648; original folio 297.

The projection to $\mathbb{Z}/k$ sends x to the residue class of the integers i for which x is the image of a section of T_i.

For Noetherian t-complete A, passage to the limit yields the exact sequence

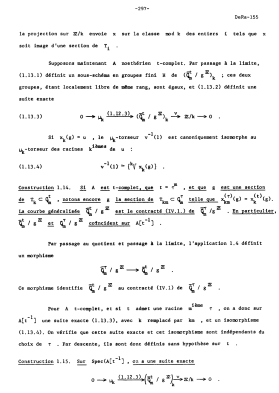
$$0\to\mu_k\to(\bar{G}_m^t/g^{\mathbb{Z}})_k\to{}^v\mathbb{Z}/k\to0. \quad (1.13.3)$$

If $x_k(g)=u$, the torsor $v^{-1}(1)$ is canonically the torsor of kth roots of u:

$$v^{-1}(1)\simeq\{\sqrt[k]{x_k(g)}\}. \quad (1.13.4)$$

Construction 1.14. If $t=\tau^m$ and g lies in T_k of $\bar{G}_m^t$, regard it as a section of T_{km} of $\bar{G}_m^\tau$ satisfying $x_{km}^{(\tau)}(g)=x_k^{(t)}(g)$. The t-curve is the contraction of the τ-curve, and the two coincide over $A[t^{-1}]$. The quotient map defines the contraction morphism. The exact sequence and root-torsor description are independent of τ and hence descend without a root hypothesis.

Construction 1.15. Over Spec(A[t^{-1}]) there is the same exact sequence with projection v.



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The torsor v^{-1}(1) is canonically the torsor of kth roots of g∈𝔾_m^t(A[t^{-1}])≃A[t^{-1}]^*.

Let ℤ[[q^{1/k}]] be the power-series ring in q^{1/k}. Denote by q the section of T_k in Ḡ_m^{q^{1/k}} for which x_k(g)=1.

Definition 1.16. The k-sided Tate curve over ℤ[[q^{1/k}]] is Ḡ_m^{q^{1/k}}/q^ℤ. For k=1 it is the Tate curve over ℤ[[q]]. It carries: an isomorphism Ĉ≃Ĝ_m; the invariant differential dx/x; injections μ_n↪C_n; and an isomorphism C_k≃μ_k×ℤ/k.

The last isomorphism is defined by μ_k and the section q^{1/k}. For one natural pairing convention, despite the printed e_k/e_n mismatch,

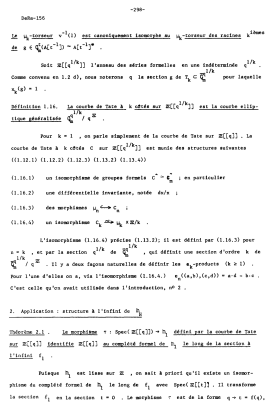
e_n((a,b),(c,d))=a·d-b·c.

2. Application: structure at infinity of $\mathcal{M}$_H

Theorem 2.1. The Tate-curve morphism τ:Spec(ℤ[[q]])→$\mathcal{M}$_1 identifies ℤ[[q]] with the formal completion of $\mathcal{M}$_1 along the section at infinity f_1. Smoothness gives a coordinate t with f_1 at t=0. The morphism has the form

q↦t=f(q),

and the proof continues on physical page 157.



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for a formal series f (that is, τ*(t) = f(q)). We know that the image, in $\mathcal{M}_1$, of the nonsmoothness subscheme of the universal curve is f_1 (IV 2.2). By 1.11, the image, in Spec($\mathbb{Z}$[[q]]), of the nonsmoothness subscheme of the Tate curve has equation q = 0. Since formation of this image commutes with base change, we have

(q) = (f(q)) in $\mathbb{Z}$[[q]],

and this implies that τ is an isomorphism.

Let H ⊂ GL(2, $\mathbb{Z}$/n).

Corollary 2.2. The fiber product

Spec($\mathbb{Z}$[[q]]) ×_{$\mathcal{M}_1$} $\mathcal{M}$_H

is the spectrum of the normalization of $\mathbb{Z}$[[q]] in the affine algebra of the finite étale scheme over $\mathbb{Z}$[[q]][1/n, 1/q]

Isom((𝔾_m^q/q^$\mathbb{Z}$)_n, ($\mathbb{Z}$/n)²)/H.

Proof. This follows from the definitions and from the fact that normalization and completion commute.

Since the group scheme (𝔾_m^q/q^$\mathbb{Z}$)_n over $\mathbb{Z}$[[q]][1/q] was computed in 1.15, 2.2 makes it possible to describe explicitly the structure of $\mathcal{M}$_H at infinity.

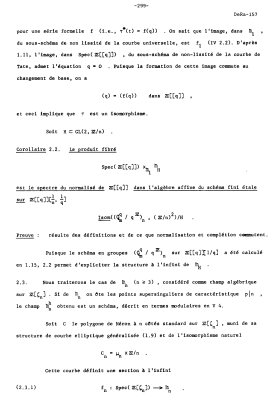
2.3. We shall treat the case of $\mathcal{M}$_n (n ≥ 3), considered as an algebraic stack over $\mathbb{Z}$[ζ_n]. If the supersingular points of characteristic p | n are removed from $\mathcal{M}$_n, the resulting stack $\mathcal{M}$_n^h is a scheme, described in modular terms in V 4.

Let C be the standard Néron n-gon over $\mathbb{Z}$[ζ_n], endowed with its structure of generalized elliptic curve (1.9) and with the natural isomorphism

C_n = μ_n × $\mathbb{Z}$/n.

This curve defines a section at infinity

(2.3.1) f_n : Spec($\mathbb{Z}$[ζ_n]) → $\mathcal{M}$_n.



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Over $\mathbb{Z}[\zeta_n][[q^{1/n}]]$, consider the generalized elliptic curve

$C' = \mathbb{G}_m^{q^{1/n}}/q^{\mathbb{Z}}$.

The map (1.16.3), $\mu_n \to C'_n$, and the section of order n "$q^{1/n}$" ($x_1 = 1$) define an isomorphism

$C'_n = \mu_n \times \mathbb{Z}/n$.

The corresponding morphism

(2.3.2) $\mathrm{Spec}(\mathbb{Z}[\zeta_n][[q^{1/n}]]) \to \mathcal{M}_n$

extends (2.3.1), and, by 1.14, the diagram

$$\begin{array}{ccc} \mathrm{Spec}(\mathbb{Z}[\zeta_n][[q^{1/n}]]) & \to & \mathcal{M}_n \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathbb{Z}[[q]]) & \to & \mathcal{M}_1 \end{array}$$

is commutative.

Corollary 2.4. The morphism (2.3.2) identifies $\mathbb{Z}[\zeta_n][[q^{1/n}]]$ with the formal completion of $\mathcal{M}_n$ along the section at infinity f_n.

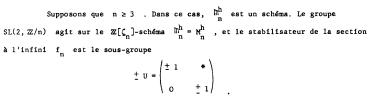
Proof. This can be deduced from 2.2. It is simpler to repeat the proof of 2.1, using the facts that

(a) $\mathcal{M}_n$ is smooth over $\mathbb{Z}[\zeta_n]$ in a neighborhood of infinity;

(b) the image on $\mathrm{Spec}(\mathbb{Z}[\zeta_n][[q^{1/n}]])$ of the nonsmooth locus of $\mathbb{G}_m^{q^{1/n}}$ is defined by the equation $q^{1/n} = 0$.

Suppose that $n \geq 3$. In this case, $\mathcal{M}_n^h$ is a scheme. The group $SL(2, \mathbb{Z}/n)$ acts on the $\mathbb{Z}[\zeta_n]$-scheme $\mathcal{M}_n^h = M_n^h$, and the stabilizer of the section at infinity f_n is the subgroup

$\pm U = \{ \pm \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \}$.



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Corollary 2.5. If n ≥ 3, the completion of $\mathcal{M}$_n^h along infinity is a sum of copies of Spec(ℤ[ζ_n][[q^{1/n}]]), indexed by SL(2, ℤ/n)/±U.

Corollary 2.6. Let C be a generalized elliptic curve over a complete local ring A. Suppose that C is not smooth; let (t) be the ideal of A that defines the nonsmoothness subscheme. Suppose that the special fiber C_s of C has k sides and that Gal(s̄/s) acts trivially on the diagram Γ(C_s̄) of the irreducible components of C_s̄. Then there exists u ∈ A* such that

C ≅ 𝔾_m^t/(ut^k)^ℤ.

Proof. Let X_v be the group scheme over ℤ[v, v^{-1}], an extension of ℤ/k by μ_k,

0 → μ_k → X_v →^{φ} ℤ/k → 0,

such that φ^{-1}(1) is the μ_k-torsor of kth roots of v. Let $\mathcal{G}$_v be the algebraic stack over ℤ[v, v^{-1}] classifying generalized elliptic curves E over a ℤ[v, v^{-1}]-scheme, smooth or k-sided, endowed with an isomorphism E_k ≅ X_v. After scalar extension to ℤ[v^{1/k}], X_v becomes isomorphic to μ_k × ℤ/k, and $\mathcal{G}$_v to $\mathcal{G}$ of V 4.4. The map

Spec(ℤ[v, v^{-1}][[τ]]) → $\mathcal{G}$_v

defined by the curve 𝔾_m^τ/(vτ^k)^ℤ therefore identifies Spec(ℤ[v, v^{-1}][[τ]]) with the formal completion of $\mathcal{G}$_v along a section at infinity.

Under the hypotheses of 2.6, there exists v ∈ A* such that C_k is isomorphic to X_v and C_s to 𝔾_m^τ/(vτ^k)^ℤ over k(s). The curve C is therefore the pullback of the curve 𝔾_m^τ/(vτ^k)^ℤ over ℤ[v, v^{-1}][[τ]], and 2.6 follows.

2.7. Let us prove VI 6.14. With the notation of loc. cit., we reduce to the case in which V is complete with separably closed residue field. Let C be the Néron model of the elliptic curve E over the fraction field, and apply 2.6 to C. We find

C ≅ 𝔾_m^t/(ut^k)^ℤ

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Corollaire 2.5. Si $n \geq 3$, le complété de $\mathcal{M}_n^h$ le long de l'infini est somme d'un ensemble de copies de $\mathrm{Spec}(\mathbb{Z}[\zeta_n][[q^{1/n}]])$, indexées par $\mathrm{SL}(2,\mathbb{Z}/n)/\pm\mathrm{U}$.

Corollaire 2.6. Soit C une courbe elliptique généralisée sur A local complet. On suppose que C n'est pas lisse; soit (t) l'idéal de A qui définit le sous-schéma de non lissité. On suppose que la fibre spéciale C_s de C à k côtés, et que $\mathrm{Gal}(\bar{s}/s)$ agit trivialement sur le diagramme $\Gamma(C_{\bar{s}})$ des composantes irréductibles de $C_{\bar{s}}$. Alors, il existe $u \in A^*$ tel que

$$C \simeq \mathbb{G}_m^t / (ut)^{k\mathbb{Z}} .$$

Preuve : Soit X_v le schéma en groupes sur $\mathbb{Z}[v,v^{-1}]$, extension de $\mathbb{Z}/k$ par μ_k,

$$0 \to \mu_k \to X_v \xrightarrow{\varphi} \mathbb{Z}/k \to 0 \quad ,$$

tel que $\varphi^{-1}(1)$ soit le μ_k-torseur des racines $k^{\text{ièmes}}$ de v. Soit $\mathcal{G}_1$ le champ algébrique sur $\mathbb{Z}[v,v^{-1}]$ qui classifie les courbes elliptiques généralisées E sur un $\mathbb{Z}[v,v^{-1}]$-schéma, lisses ou à k côtés, munies d'un isomorphisme $E_k \to X_v$. Après extension des scalaires à $\mathbb{Z}[v^{1/k}]$, X_v devient isomorphe à $\mu_k \times \mathbb{Z}/k$, et $\mathcal{G}_1$ à G de V 4.4. L'application

$$\mathrm{Spec}(\mathbb{Z}[v,v^{-1}][[\tau]]) \longrightarrow \mathcal{G}_1$$

définie par la courbe $\mathbb{G}_m^\tau/(v\tau^k)^{\mathbb{Z}}$ (identifie donc $\mathrm{Spec}\,\mathbb{Z}[v,v^{-1}][[\tau]]$ au complété formel de $\mathcal{G}_1$ le long d'une section à l'infini.

Sous les hypothèses de 2.6, il existe $v \in A^*$ tel que C_k soit isomorphe à X_v et C_s à $\mathbb{G}_m/(v\tau^k)^{\mathbb{Z}}$ sur $k(s)$. La courbe C est donc image réciproque de la courbe $\mathbb{G}_m^\tau/(v\tau^k)^{\mathbb{Z}}$ sur $\mathbb{Z}[v,v^{-1}][[\tau]]$, ce 2.6 en résulte.

2.7. Prouvons VI 6.14. Avec les notations de loc.cit., on se ramène à supposer V complet à corps résiduel séparablement clos. Soit C le modèle de Néron de la courbe elliptique E sur le corps des fractions, et appliquons 2.6 à C. On trouve

$$C \simeq \mathbb{G}_m^t/(ut)^{\mathbb{Z}}$$

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and, because C is regular, t is a uniformizer. By 2.1 and VI 1.1, we have

j(E)^{-1} = ±(ut^k) + ∑_{n>0} d_n(ut^k)^n,

and j(E) therefore has valuation −k.

3. Application: expansion of a modular form as a Fourier series.

We shall show in § 4 that, after scalar extension to $\mathbb{C}$, the notions introduced in this section coincide with the classical notions.

Definition 3.1. The algebra of integral modular forms is the graded algebra

$\oplus_k H^0(\mathcal{M}_1, \omega^{\otimes k})$.

In other words, an integral modular form of weight k is a rule, compatible with base change, which to C/S, an irreducible generalized elliptic curve over a scheme S, associates a section of $\omega^{\otimes k}$ over S.

Definition 3.2. The algebra of integral modular forms of level $H \subset GL(2, \mathbb{Z}/n)$ is the graded algebra

$\oplus_k H^0(\mathcal{M}_H, \omega^{\otimes k})$.

The following lemma makes it possible, in certain cases, to consider only the $\mathcal{M}_n$.

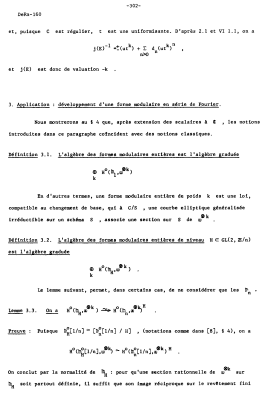
Lemma 3.3. We have

$H^0(\mathcal{M}_H, \omega^{\otimes k}) \simeq H^0(\mathcal{M}_n, \omega^{\otimes k})^H$.

Proof. Since $\mathcal{M}_H^\circ[1/n] \simeq [\mathcal{M}_n^\circ[1/n]/H]$ (notation as in [8], § 4), we have

$H^0(\mathcal{M}_H^\circ[1/n], \omega^{\otimes k}) \simeq H^0(\mathcal{M}_n^\circ[1/n], \omega^{\otimes k})^H$.

We conclude by the normality of $\mathcal{M}_H$: for a rational section of $\omega^{\otimes k}$ on $\mathcal{M}_H$ to be defined everywhere, it is enough that its pullback to the finite covering



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be so on $\mathcal{M}_n$.

Theorem 3.4. The algebra of integral modular forms of level H is a finitely generated $\mathbb{Z}$-algebra.

Proof. By 3.3, it is enough to treat the case of $\mathcal{M}_n$, for n large. In this case $\mathcal{M}_n$ = M_n is a projective scheme over $\mathbb{Z}$, and ω is an ample invertible sheaf on M_n. Indeed, the divisor of the section Δ of $\omega^{\otimes 12}$ is a multiple of M_n^∞ (on $\mathcal{M}_n$, $\mathrm{div}(\Delta) = \mathcal{M}_n^\infty$); it meets every irreducible component of every fiber of M_n over $\mathbb{Z}$, and is therefore ample. The theorem consequently follows from the general theorem EGA III, 2.3.4.1.

3.5. One may also consider modular forms with coefficients in algebras over $\mathbb{Z}$. Since we are not sure of the correct definition in the general case, we shall for the most part restrict ourselves to the case of the $\mathcal{M}$_n, viewed as $\mathbb{Z}[\zeta_n]$-stacks.

Definition 3.6. Let A be an algebra over $\mathbb{Z}[\zeta_n]$. A modular form of level n and weight k over the $\mathbb{Z}[\zeta_n]$-algebra A is an element of

$H^0(\mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} A, \omega^{\otimes k})$.

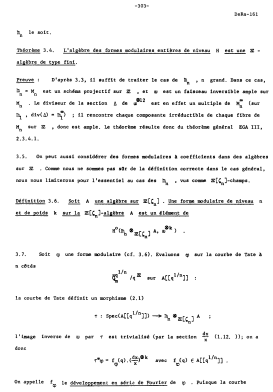
3.7. Let φ be a modular form (cf. 3.6). Evaluate φ on the n-sided Tate curve $\mathbb{G}_m^{q^{1/n}}/q^{\mathbb{Z}}$ over $A[[q^{1/n}]]$: the Tate curve defines a morphism (2.1)

$\tau : \mathrm{Spec}(A[[q^{1/n}]]) \to \mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} A$;

the pullback of ω by τ is trivialized (by the section dx/x (1.12)); hence

$\tau^*\varphi = f_\varphi(q)(dx/x)^{\otimes k}$, with $f_\varphi(q) \in A[[q^{1/n}]]$.

We call f_φ the Fourier-series expansion of φ. Since the curve



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$\mathcal{D}_N$ le sait.

Théorème 3.4. *L'algèbre des formes modulaires entières de niveau N est une $\mathbb{Z}$-algèbre de type fini.*

Preuve : D'après 3.3, il suffit de traiter le cas de $\mathcal{D}_N$, n grand. Dans ce cas, $\mathcal{D}_n = \mathcal{H}_n$ est un schéma projectif sur $\mathbb{Z}$, et ω est un faisceau inversible ample sur $\mathcal{H}_n$. Le diviseur de la section $\frac{dx}{x}$ de $\omega^{\otimes 2}$ est en effet un multiple de $\mathcal{H}_n^\infty$ (sur $\mathcal{D}_n$, $\mathrm{div}(\frac{dx}{x}) = \mathcal{H}_n^\infty$; il rencontre chaque composante irréductible de chaque fibre de $\mathcal{H}_n$ sur $\mathbb{Z}$, donc est ample. Le théorème résulte donc du théorème général EGA III, 2.3.4.1.

3.5. On peut aussi considérer des formes modulaires à coefficients dans des algèbres sur $\mathbb{Z}$. Comme nous ne sommes pas sûr de la définition correcte dans le cas général, nous nous limiterons pour l'essentiel au cas des $\mathcal{D}_N$, vus comme $\mathbb{Z}[\zeta_n]$-champs.

Définition 3.6. *Soit A une algèbre sur $\mathbb{Z}[\zeta_n]$. Une forme modulaire de niveau n et de poids k sur la $\mathbb{Z}[\zeta_n]$-algèbre A est un élément de*

$$H^0(\mathcal{D}_N \otimes_{\mathbb{Z}[\zeta_n]} A, \omega^{\otimes k}) .$$

3.7. Soit φ une forme modulaire (cf. 3.6). Evaluons φ sur la courbe de Tate A à cônde

$$\mathbb{G}_m^{1/n}/q^{\mathbb{Z}} \text{ sur } A[[t^{1/n}]] :$$

la courbe de Tate définit un morphisme (2.1)

$$\tau : \mathrm{Spec}(A[[t^{1/n}]]) \longrightarrow \mathcal{D}_N \otimes_{\mathbb{Z}[\zeta_n]} A :$$

L'image inverse de ω par τ est trivialisé (par la section $\frac{dx}{x}$ (1.12.)); on a donc

$$\tau^*\omega = f_\varphi(q).(\frac{dx}{x})^{\otimes k} \quad \text{avec} \quad f_\varphi(q) \in A[[t^{1/n}]] .$$

On appelle f_φ le *développement en série de Fourier* de φ. Puisque la courbe

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Tate n-gon is already defined over $\mathbb{Z}[[q^{1/n}]] \otimes_{\mathbb{Z}} A$ (the subring of $A[[q^{1/n}]]$ formed by the formal series whose coefficients generate a finitely generated $\mathbb{Z}$-module), we have

(3.7.1) $f_\varphi(q) \in \mathbb{Z}[[q^{1/n}]] \otimes_{\mathbb{Z}} A \subset A[[q^{1/n}]]$.

3.8. Let φ be a modular form (cf. 3.6). For $g \in SL(2, \mathbb{Z}/n)$, we denote by gφ the modular form which to a curve with level-n structure (C, α) over S associates the section

$(g\varphi)(C, \alpha) = \varphi(C, g^{-1}\alpha)$

of $\omega^{\otimes k}$ over S.

The $f_{g\varphi}$ are the expansions of φ at the cusps. If

$u = \pm\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$

and $f_\varphi = \sum c_m q^m$ $(m \in (1/n)\mathbb{Z})$, then

(3.8.1) $f_{u\varphi} = (-1)^k \sum c_m \zeta_n^{amn} q^m$.

Theorem 3.9. Let φ be a modular form of level n and weight k with coefficients in the $\mathbb{Z}[\zeta_n]$-algebra A.

(i) If the expansions at the cusps f_{gφ} vanish, then φ vanishes.

(ii) If B is a subalgebra of A and the f_{gφ} lie in B[[q^{1/n}]], then φ has coefficients in B (it is obtained by scalar extension from a unique form over B).

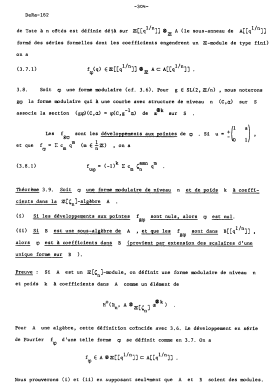
Proof. If A is a $\mathbb{Z}[\zeta_n]$-module, a modular form of level n and weight k with coefficients in A is defined to be an element of

$H^0(\mathcal{M}_n, A \otimes_{\mathbb{Z}[\zeta_n]} \omega^{\otimes k})$.

For A an algebra, this definition agrees with 3.6. The Fourier-series expansion f_φ of such a form φ is defined as in 3.7. We have

$f_\varphi \in A \otimes_{\mathbb{Z}} \mathbb{Z}[[q^{1/n}]] \subset A[[q^{1/n}]]$.

We shall prove (i) and (ii) under the sole assumption that A and B are modules.



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de Tate à n effets est définie déjà sur $\mathbb{Z}[\zeta_n^{1/n}]$; $\otimes_{\mathbb{Z}} A$ (le sous-anneau de $A[[q^{1/n}]]$ formé des séries formelles dont les coefficients engendrent un $\mathbb{Z}$-module de type fini) est à

(3.7.1) $\qquad f_\varphi(q) \in \mathbb{Z}[[q^{1/n}]] \otimes_{\mathbb{Z}} A \subset A[[q^{1/n}]]$.

3.8. Soit φ une forme modulaire (cf. 3.6). Pour $g \in SL(2, \mathbb{Z}/n)$, nous noterons $g\varphi$ la forme modulaire qui à une courbe avec structure de niveau n (C, ω) sur S associe la section $(g\varphi)(C, \omega) = \varphi(C, g^{-1}\omega)$ de $\omega^{\otimes k}$ sur S .

Les $f_{g\varphi}$ sont les développements aux pointes de φ . Si $u = \pm\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$, et que $f_\varphi = \sum a_m q^m$ $(a_m \in \frac{1}{n}\mathbb{Z})$, on a

(3.8.1) $\qquad f_{u\varphi} = (-1)^k \sum a_m \exp(2\pi i m/n)\, q^m$.

Théorème 3.9. Soit φ une forme modulaire de niveau n et de poids k à coefficients dans le $\mathbb{Z}[\zeta_n]$-algèbre A .

(i) Si les développements aux pointes $f_{g\varphi}$ sont nuls, alors φ est nul.

(ii) Si B est une sous-algèbre de A , et que les $f_{g\varphi}$ sont dans $B[[q^{1/n}]]$, alors φ est à coefficients dans B (provient par extension des scalaires d'une unique forme sur B).

Preuve : Si A est un $\mathbb{Z}[\zeta_n]$-module, on définit une forme modulaire de niveau n et poids k à coefficients dans A comme un élément de

$$H^0(\mathcal{M}_n, A \otimes_{\mathbb{Z}[\zeta_n]} \omega^{\otimes k}) .$$

Pour A une algèbre, cette définition coïncide avec 3.6. Le développement en série de Fourier f_φ d'une telle forme φ se définit comme en 3.7. On a

$$f_\varphi \in A \otimes \mathbb{Z}[[q^{1/n}]] \subset A[[q^{1/n}]] .$$

Nous prouverons (i) et (ii) en supposant seulement que A et B soient des modules.

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Let us prove (i). If A is a direct limit $\varinjlim A_i$, then

$H^0(\mathcal{M}_n, A \otimes \omega^k) = \varinjlim H^0(\mathcal{M}_n, A_i \otimes \omega^k)$.

It therefore suffices to treat the case in which A is a finitely generated $\mathbb{Z}[\zeta_n]$-module. If the f_{gφ} vanish, (2.4) shows that φ vanishes on the formal completion of $\mathcal{M}_n$ along infinity. The support of φ, disjoint from at least one point of every irreducible component of every geometric fibre of $\mathcal{M}_n$, is therefore finite over $\mathbb{Z}[\zeta_n]$. Since $\mathcal{M}_n$ is Cohen–Macaulay over $\mathbb{Z}[\zeta_n]$, and since $A \otimes \omega^k$ is locally the inverse image of a module over $\mathbb{Z}[\zeta_n]$, this is absurd if $\varphi \neq 0$.

Let us prove (ii). Let $\mathrm{Spec}(\hat{D})$ be the sum of the formal completions of $\mathcal{M}_n$ along the sections at infinity gf_n ($g \in SL(2, \mathbb{Z}/n)$). The expansions at the cusps define a map

$H^0(\mathcal{M}_n, A \otimes \omega^k) \to A \otimes_{\mathbb{Z}[\zeta_n]} \hat{D}$.

We prove (ii) by considering the following diagram, whose second row is exact because $\hat{D}$ is flat over $\mathbb{Z}[\zeta_n]$:

$0 \to H^0(\mathcal{M}_n, B \otimes \omega^k) \to H^0(\mathcal{M}_n, A \otimes \omega^k) \to H^0(\mathcal{M}_n, A/B \otimes \omega^k)$
$\downarrow \quad \downarrow \quad \downarrow$
$0 \to B \otimes \hat{D} \to A \otimes \hat{D} \to A/B \otimes \hat{D}$.

Theorem 3.10. (i) One has

$H^0(\mathcal{M}_n, \omega^k) \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{Q}(\zeta_n) \simeq H^0(\mathcal{M}_n \otimes \mathbb{Q}, \omega^k)$.

(ii) Let $\varphi \in H^0(\mathcal{M}_n \otimes \mathbb{Q}, \omega^k)$, let $f\varphi(q) = \sum a_i q^i$ ($i \in (1/n)\mathbb{Z}$, $a_i \in \mathbb{Q}(\zeta_n)$) be the Fourier-series expansion of φ at f_n, and let π be a place of $\mathbb{Q}(\zeta_n)$. Then

$$\nu\pi(f\varphi) = \deg(f_n)\ \inf(\nu\pi(a_i))$$

is the order of the zero (or the negative of the order of the pole) of φ at the generic point of the irreducible component of the fibre of $\mathcal{M}_n$ at π that meets the section at infinity f_n.

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From the point of view adopted here, this theorem is trivial; it follows from

Corollary 3.11. Let $\sum a_i q^i$ be the Fourier-series expansion of $\varphi \in H^0(\mathcal{M}_n \otimes \mathbb{Q}, \omega^k)$. The denominators of the a_i are bounded.

Corollary 3.12. Let π be a place of $\mathbb{Q}(\zeta_n)$ of characteristic p, let p^m be the largest power of p dividing n, and let g be an element of SL(2, $\mathbb{Z}$/n) whose image in SL(2, $\mathbb{Z}/p^m$) lies in the upper-triangular group. Then

$$\nu\pi(f\varphi) = \nu\pi(f_{g\varphi}).$$

Indeed, the cusps f_n and gf_n meet the same irreducible component in characteristic p.

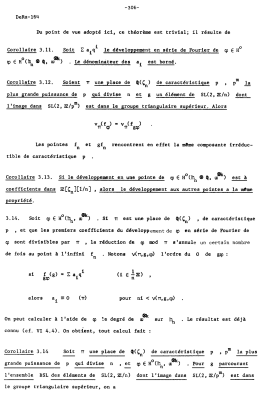
Corollary 3.13. If the expansion at one cusp of $\varphi \in H^0(\mathcal{M}_n \otimes \mathbb{Q}, \omega^k)$ has coefficients in $\mathbb{Z}[\zeta_n][1/n]$, then the expansions at the other cusps have the same property.

3.14. Let $\varphi \in H^0(\mathcal{M}_n, \omega^k)$. If π is a place of $\mathbb{Q}(\zeta_n)$ of characteristic p and the first coefficients of the Fourier-series expansion of φ are divisible by π, the reduction of φ modulo π vanishes to some order at the point at infinity f_n. Write $\nu(\pi,g,\varphi)$ for the order of the zero of $g\varphi$:

if $f_{g\varphi}(q) = \sum a_i q^i$ ($i \in (1/n)\mathbb{Z}$), then $a_i \equiv 0$ (π) for $ni < \nu(\pi,g,\varphi)$.

The degree of ω^k on $\mathcal{M}_n$ can be calculated using φ. The result is already known (cf. VI 4.4). Carrying out the calculation gives:

Corollary 3.14. Let π be a place of $\mathbb{Q}(\zeta_n)$ of characteristic p, let p^m be the largest power of p dividing n, and let $\varphi \in H^0(\mathcal{M}_n, \omega^k)$. As g runs through the set BSL of elements of SL(2, $\mathbb{Z}$/n) whose image in SL(2, $\mathbb{Z}/p^m$) lies in the upper-triangular group, one has



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$|SL(2, \mathbb{Z}/n)|^{-1} \sum_{g \in BSL} \nu(\pi, g, \varphi) \leq |\mathbb{P}^1(\mathbb{Z}/p^m)|^{-1}\, k/12.$

For n = 1, this result may be restated as follows.

Corollary 3.15. Let p be a prime number and let $\varphi \in H^0(\mathcal{M}_1, \omega^k)$ be a modular form with series expansion $f\varphi(q) = \sum a_n q^n$. Then

$\nu p(f\varphi) = \inf\{\nu p(a_i) \mid i \leq k/12\}$.

In other words, to verify that the coefficients of a modular form of weight k are divisible by a power of p, it suffices to verify this for those whose index is ≤ k/12.

3.16. One can also relate what happens on the various irreducible components in characteristic p. As an example, we shall treat the case of modular forms of level $\Gamma_0(p)$. The same arguments apply to the study, at p, of modular forms of level $\Gamma_0(p) \cap \Gamma(n)$, with n prime to p. Passing to such a level would spare us the use of fractional intersection numbers, introduced on the model of VI 4.

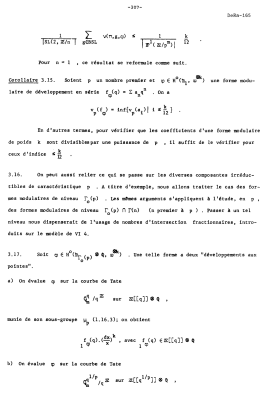
3.17. Let $\varphi \in H^0(\mathcal{M}_{\Gamma_0(p)} \otimes \mathbb{Q}, \omega^k)$. Such a form has two "expansions at the cusps."

a) Evaluate φ on the Tate curve $\overline{\mathbb{G}}_m\hat{}\,q/q^{\mathbb{Z}}$ over $\mathbb{Z}[[q]] \otimes \mathbb{Q}$, equipped with its subgroup μp (1.16.3); one obtains

${}_1f\varphi(q)\cdot(dx/x)^k$, with ${}_1f\varphi(q) \in \mathbb{Z}[[q]] \otimes \mathbb{Q}$.

b) Evaluate φ on the Tate curve

$\overline{\mathbb{G}}_m\hat{}\,\{q^{1/p}\}/q^{\mathbb{Z}}$ over $\mathbb{Z}[[q^{1/p}]] \otimes \mathbb{Q}$,



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equipped with its subgroup isomorphic to $\mathbb{Z}/p$ generated by q. One obtains

$_2f\varphi(q)\cdot(dx/x)^k$, with $_2f\varphi(q) \in \mathbb{Z}[[q^{1/p}]] \otimes \mathbb{Q}$.

For $\ell \neq p$, the ℓ-adic valuations of $_1f\varphi$ and $_2f\varphi$ (the lower bound of the ℓ-adic valuations of the coefficients) coincide. We shall write $\nu_1(\varphi)$ and $\nu_2(\varphi)$, or simply ν_1 and ν_2, for the p-adic valuations of $_1f\varphi$ and $_2f\varphi$.

3.18. We extend the involution w (V.1.3) to modular forms as follows. To an elliptic curve C/S equipped with a subgroup A ⊂ C[p], the form $w\varphi$ assigns the inverse image in ω^k of the section $\varphi(C/A, C[p]/A)$ of $(\omega_{(C/A)/S})^k$. One has

$ww\varphi = p^k\varphi$.

One checks that

$_1f_{w\varphi}(q) = p^k\ {}_2f\varphi(q^p)$,

$_2f_{w\varphi}(q) = {}_1f\varphi(q^{1/p})$.

Consequently

$(\nu_1(w\varphi), \nu_2(w\varphi)) = (k + \nu_2(\varphi), \nu_1(\varphi))$,

and $\nu_2 - \nu_1 + k/2$ changes sign when one passes from φ to $w\varphi$.

3.19. We propose to estimate $\nu_1 - \nu_2$. This number does not change when φ is multiplied by an integer; we may therefore suppose that

$\varphi \in H^0(\mathcal{M}_{\Gamma_0(p)}, \omega^k)$.

Assume that φ is not identically zero. If N_1 and N_2 are the two irreducible components of $\mathcal{M}_{\Gamma_0(p)} \otimes \mathbb{F}p$, corresponding respectively to curves (C,A) with A infinitesimal and A discrete, the divisor of φ may be written

$\mathrm{div}(\varphi) = D + \nu_1 N_1 + \nu_2 N_2$,

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where the divisor D meets N_1 and N_2 at only finitely many points. On the regular stack $\mathcal{M}_{\Gamma_0(p)}$, div(p) = $N_1 + N_2$, whence the vanishing of the intersection numbers

$(N_1 + N_2, N_1) = (N_1 + N_2, N_2) = 0$.

On $N_1 \simeq \mathcal{M}_1 \otimes \mathbb{F}p$, ω has degree 1/24; therefore

$(\mathrm{div}(\varphi), N_1) = k/24$.

Similarly, since on $\mathcal{M}_{\Gamma_0(p)}$ the degree of ω is $(1/24)[\mathcal{M}_{\Gamma_0(p)}:\mathcal{M}_1] = (p+1)/24$, one has

$\deg(D) = (D,N_1) + (D,N_2) = (p+1)k/24$.

By VI.4.9, $(N_2,N_1) = (p-1)/24$.

One has

$(D + \nu_1 N_1 + \nu_2 N_2, N_1) = (D,N_1) + \nu_1(N_1+N_2,N_1) + (\nu_2-\nu_1)(N_2,N_1)$

$= \tfrac{1}{2}((D,N_1)+(D,N_2)) + \tfrac{1}{2}((D,N_1)-(D,N_2)) + (\nu_2-\nu_1+k/2)(N_2,N_1) - (k/2)(N_2,N_1)$.

Substituting the expressions above gives

$\tfrac{1}{2}\cdot k\cdot(p+1)/24 + \tfrac{1}{2}((D,N_1)-(D,N_2)) + (\nu_2-\nu_1+k/2)(p-1)/24 - (k/2)\cdot(p-1)/24 = k/24$.

Thus

$(\nu_2-\nu_1+k/2)(p-1)/24 = \tfrac{1}{2}((D,N_2)-(D,N_1))$.

Proposition 3.20. One has

$\nu_2-\nu_1+k/2 = \tfrac{1}{2}\cdot 24/(p-1)\ ((D,N_2)-(D,N_1))$,

$(D,N_1) + (D,N_2) = k\cdot(p+1)/24$,

and hence

$|\nu_2-\nu_1+k/2| \le \tfrac{1}{2}\cdot k\cdot(p+1)/(p-1)$.

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In particular, if $_1f\varphi(q)$ is p-integral, then the p-adic valuation of $_1f_{w\varphi}(q)$ is $\geq -k/(p-1)$.

4. Comparison with transcendental theory.

4.1. We denote by D the open unit disc contained in $\mathbb{C}$, by q the usual coordinate on D (the inclusion $D \hookrightarrow \mathbb{C}$), and by D* the punctured disc $D - \{0\}$.

We denote by $D[q^{1/k}]$ the Riemann surface of $q^{1/k}$: it is a copy of the unit disc, whose usual coordinate is called $q^{1/k}$, equipped with the map $u : D[q^{1/k}] \to D$ such that

$q \circ u = (q^{1/k})^k$.

4.2. A construction identical to Construction 1.1 gives an analytic space $\overline{\mathbb{G}}_m\hat{}q$ over D; translation by the section q ($x_1 = 1$) defines an action of $\mathbb{Z}$ on $\overline{\mathbb{G}}_m\hat{}q$, and this action is free (because $|q| < 1$). Passing to the quotient gives an analytic space $\overline{\mathbb{G}}_m\hat{}q/q\mathbb{Z}$ over D, which is a family of generalized elliptic curves parametrized by D (the Tate curve).

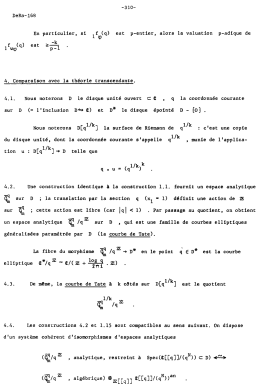
The fibre of the morphism $\overline{\mathbb{G}}_m\hat{}q/q\mathbb{Z} \to D^*$ at the point $q \in D^*$ is the elliptic curve $\mathbb{C}^*/q\mathbb{Z} \simeq \mathbb{C}/(\mathbb{Z} + (\log q)/(2\pi i)\cdot\mathbb{Z})$.

4.3. Similarly, the k-gon Tate curve over $D[q^{1/k}]$ is the quotient

$\overline{\mathbb{G}}_m\hat{}\{q^{1/k}\}/q\mathbb{Z}$.

4.4. Constructions 4.2 and 1.15 are compatible in the following sense. There is a coherent system of isomorphisms of analytic spaces

$(\overline{\mathbb{G}}_m\hat{}q/q\mathbb{Z}$, analytic, restricted to $\mathrm{Spec}(\mathbb{C}[[q]]/(q^N)) \subset D) \simeq ((\overline{\mathbb{G}}_m\hat{}q/q\mathbb{Z}$, algebraic$) \otimes_{\mathbb{Z}[[q]]} \mathbb{C}[[q]]/(q^N))\hat{}\{an\}$.



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Similarly, there are isomorphisms of analytic spaces over

$\mathbb{C}[[q^{1/k}]]/(q^N)$:

$(\overline{\mathbb{G}}_m\text{^}\{q^{1/k}\}/q^{\mathbb{Z}}$, analytic, restricted to $\mathrm{Spec}(\mathbb{C}[[q^{1/k}]]/(q^N)) \subset D[q^{1/k}]) \simeq ((\overline{\mathbb{G}}_m\text{^}\{q^{1/k}\}/q^{\mathbb{Z}}$, algebraic) $\otimes_{\{\mathbb{Z}[[q^{1/k}]]\}} \mathbb{C}[[q^{1/k}]]/(q^N))^{an}$;

these isomorphisms respect the structures 1.16 on the two sides.

4.5. Classically, a modular form of weight k and level n is a holomorphic function f on the Poincaré half-plane

$X = \{z \mid \mathrm{Im}(z) > 0\}$

satisfying the following conditions.

(A) If (a b; c d) $\in SL(2, \mathbb{Z})$ is congruent to (1 0; 0 1) mod n,

$f(z) = (cz+d)^{-k}\, f((az+b)/(cz+d))$.

This condition implies that, for $\gamma \in SL(2, \mathbb{Z}/n)$, the image of $\gamma_0 = (a\ b;\ c\ d) \in SL(2, \mathbb{Z})$, the function

$f|\gamma_0 : z \mapsto (cz+d)^{-k}\, f((az+b)/(cz+d))$

depends only on γ, not on the choice of γ_0. It is denoted $f|\gamma$.

(B) For every $\gamma \in SL(2, \mathbb{Z}/n)$,

$(f|\gamma)(z)$ remains bounded as $\mathrm{Im}(z) \to \infty$

(it is equivalent, and more honest, to require $f|\gamma$ to have polynomial growth in vertical strips as $\mathrm{Im}(z) \to \infty$).

Such a function is periodic with period n and therefore has a Fourier-series expansion

$f(z) = \sum a_m\, e\text{^}(2\pi i m z)$ $(m \in (1/n)\mathbb{Z},\ m \geq 0)$.

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Construction 4.6. Modular forms of weight k and level n are identified with the sections

$$H^0(\mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}, \omega^{\otimes k}) = H^0(\mathcal{M}_n, \omega^{\otimes k}) \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}.$$

Proof. Suppose first that $n \geq 3$. In that case, $\mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C} = M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ is a scheme. The analytic space underlying $M_n^\circ \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ is

$$X/\mathrm{Ker}(\mathrm{SL}(2, \mathbb{Z}) \to \mathrm{SL}(2, \mathbb{Z}/n)),$$

and the holomorphic sections of $\omega^{\otimes k}$ on this space are identified with the holomorphic functions on X satisfying 4.5 (A). The analytic space underlying $M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ is obtained from it by adding finitely many points. A holomorphic section of $\omega^{\otimes k}$ on $M_n^\circ \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ extends to $M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ if and only if it satisfies (B). By GAGA, these sections are the algebraic sections of $\omega^{\otimes k}$ on $M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$.

The case of arbitrary n is reduced to the preceding one by passing to the invariants under

$$\mathrm{Ker}(\mathrm{SL}(2, \mathbb{Z}/3n) \to \mathrm{SL}(2, \mathbb{Z}/n))$$

in the space of modular forms of weight 3n. One could also invoke GAGA for the algebraic stack $\mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$.

4.7. For $n \geq 3$, $(M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C})^{an}$ is also the analytic space classifying generalized elliptic curves C/S over analytic spaces S, equipped with a level-n structure α such that $\det(\alpha) = 1$. From this point of view, a modular form f is a rule that assigns to every such curve a section of $\omega^{\otimes k}$ over S. The Fourier-series expansion is then obtained by evaluating the form on the Tate curve (4.3) (and multiplying the result by a power of $2\pi i$ depending on the normalizations chosen). For $\varphi \in H^0(M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}, \omega^{\otimes k})$, compatibility 4.4 shows that this expansion coincides with the expansion defined purely algebraically in § 3.

4.8. From this point of view, the results of § 3 give arithmetic properties of the coefficients of the Fourier-series expansions at the various cusps of the functions on X satisfying 2.5 (A) and (B). For example:

- The graded algebra of modular forms of level n (2.5) is the tensor product with $\mathbb{C}$ over $\mathbb{Z}[\zeta_n]$ of the finite-type $\mathbb{Z}[\zeta_n]$-subalgebra consisting of those

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for which the Fourier-series expansions of the f|γ have coefficients in $\mathbb{Z}[\zeta_n]$.

- If the expansion of f has coefficients in $\mathbb{Z}[\zeta_n][1/n]$, that of f|γ has the same property.

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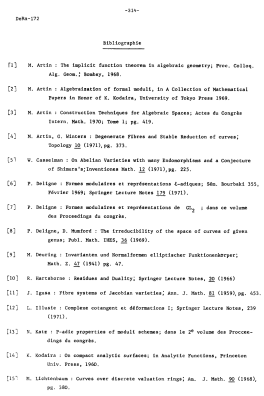

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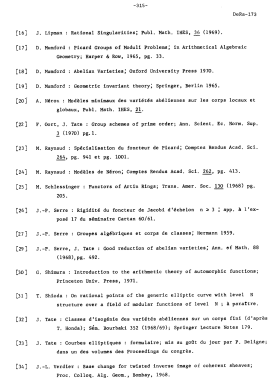

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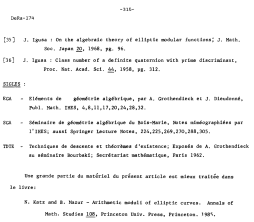

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MODULAR FORMS OF WEIGHT 1

BY PIERRE DELIGNE AND JEAN-PIERRE SERRE

To Henri Cartan,
on the occasion of his
70th birthday

Introduction

The Euler product decomposition and the functional equation of the Dirichlet series associated by Hecke with modular forms of weight 1 suggest that these forms correspond to Artin L-functions of degree 2 over $\mathbf{Q}$, in other words to representations of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ in $\mathrm{GL}_2(\mathbf{C})$. It is such a correspondence, conjectured by Langlands, that we establish here.

The first three sections are preliminary. Section 4 contains the statement of the main theorem, together with several supplements. The proof occupies §§ 5 through 9. Its principle is as follows: for every prime number ℓ, one first constructs a representation of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ in characteristic ℓ (cf. § 6); one then shows that the images of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ in these various representations are "small," which makes it possible to lift them to characteristic 0 and obtain the desired complex representation (§§ 7 and 8); the "smallness" in question itself follows from an average bound for the eigenvalues of the Hecke operators (Rankin, cf. § 5). Section 9 contains an estimate for the coefficients of modular forms of weight 1.

We point out that at one essential point (§ 6, Theorem 6.1) we have used results proved by one of us (P. Deligne), but for which no complete proof has yet been published; pending such a publication (as well as that of SGA 5, on which these results depend), we ask the reader to accept them.

§ 1. Review (analytic) of modular forms

1.1. Let N be an integer ≥ 1. With N we associate the subgroups

$$\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N).$$

of $\mathrm{SL}_2(\mathbf{Z})$ defined by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(N) \iff a \equiv d \equiv 1 \pmod{N} \quad \text{and} \quad b \equiv c \equiv 0 \pmod{N},$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N) \iff a \equiv d \equiv 1 \pmod{N} \quad \text{and} \quad c \equiv 0 \pmod{N},$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N) \iff c \equiv 0 \pmod{N}.$$

1.2. Let f be a function on the upper half-plane $H = \{z \mid \mathrm{Im}(z) > 0\}$. If k is an integer and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is an element of $\mathrm{SL}_2(\mathbf{R})$, set

$$(f \mid_k \gamma)(z) = (cz + d)^{-k} f(\gamma z), \qquad \text{where} \qquad \gamma z = \frac{az + b}{cz + d}.$$

Let Γ be a subgroup of $\mathrm{SL}_2(\mathbf{Z})$ containing $\Gamma(N)$. We say that f is modular of weight k on Γ if:
(1.2.1) $f \mid_k \gamma = f$ for every $\gamma \in \Gamma$;
(1.2.2) f is holomorphic on H;
(1.2.3) f is "holomorphic at the cusps," i.e., for every $\sigma \in \mathrm{SL}_2(\mathbf{Z})$, the function $f \mid_k \sigma$ has a power-series expansion in $e^{2\pi i z/N}$ with exponents ≥ 0.
If, in (1.2.3), "exponents ≥ 0" is replaced by "exponents > 0," one obtains the notion of a cusp form.
1.3. Let f be a modular form of weight k on $\Gamma(N)$. For f to be a modular form on $\Gamma_1(N)$, it is necessary and sufficient that $f(z + 1) = f(z)$, or equivalently that f have an expansion of the form

$$\sum_{n=0}^{\infty} a_n q^n, \qquad \text{where} \qquad q = e^{2\pi i z}.$$

In what follows, this expansion will be denoted by $f_\infty(q)$, or simply by f.
1.4. Let f be a modular form of weight k on $\Gamma_1(N)$. If $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is an element of $\Gamma_0(N)$, the form $f \mid_k \gamma$ depends only on the image of d in $(\mathbf{Z}/N\mathbf{Z})^*$; it is denoted by $f \mid R_d$. We have $f \mid R_{-1} = (-1)^k f$.
1.5. Let ε be a Dirichlet character modulo N, that is, a homomorphism

$$\varepsilon : (\mathbf{Z}/N\mathbf{Z})^* \longrightarrow \mathbf{C}^*.$$

We say that ε is even (respectively odd) if $\varepsilon(-1) = 1$ (respectively if $\varepsilon(-1) = -1$).

Let k be an integer of the same parity as ε [i.e., $\varepsilon(-1) = (-1)^k$]. A modular form of type (k, ε) on $\Gamma_0(N)$ is a modular form f of weight k on $\Gamma_1(N)$ such that

$$f \mid R_d = \varepsilon(d) f$$

for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$, i.e.,

$$f\left(\frac{az+b}{cz+d}\right) = \varepsilon(d)(cz+d)^k f(z) \quad \text{for every} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N).$$

(Note that if ε and k did not have the same parity, this formula would imply $f = 0$.)

Every modular form of weight k on $\Gamma_1(N)$ is a linear combination of forms of types (k, ε_i) on $\Gamma_0(N)$, where the ε_i are the various characters of $(\mathbf{Z}/N\mathbf{Z})^*$ having the same parity as k.

This follows (cf. [16], p. IV-13) by observing that the map

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto d \pmod N$$

induces an isomorphism from $\Gamma_0(N)/\Gamma_1(N)$ onto $(\mathbf{Z}/N\mathbf{Z})^*$.

1.6. HECKE OPERATORS. — Let $f = \sum a_n q^n$ be a modular form of type (k, ε) on $\Gamma_0(N)$, and let p be a prime number. Set

$$f \mid T_p = \sum a_{pn} q^n + \varepsilon(p) p^{k-1} \sum a_n q^{pn} \qquad \text{if } p \nmid N, \tag{1.6.1}$$

$$f \mid U_p = \sum a_{pn} q^n \qquad \text{if } p \mid N. \tag{1.6.2}$$

This gives another modular form of type (k, ε) on $\Gamma_0(N)$, which is a cusp form if f is.

1.7. PRIMITIVE FORMS. — We refer to [2] (when $\varepsilon = 1$) and to [6], [12], [13] (in the general case) for the definition of primitive cusp forms ("newforms") of type (k, ε) on $\Gamma_0(N)$.

If $f = \sum_{n=1}^{\infty} a_n q^n$ is such a form, then $a_1 = 1$, and f is an eigenform for the Hecke operators T_p and U_p, with corresponding eigenvalues a_p. It follows that the Dirichlet series

$$\Phi_f(s) = \sum_{n=1}^{\infty} a_n n^{-s} \tag{1.7.1}$$

has the Euler product

$$\Phi_f(s) = \prod_{p \mid N} \frac{1}{1 - a_p p^{-s}} \prod_{p \nmid N} \frac{1}{1 - a_p p^{-s} + \varepsilon(p) p^{k-1-2s}}. \tag{1.7.2}$$

1.8. If f is as above and $p \mid N$, the absolute value of a_p is given by the following rule ([12], [14]):

$$a_p = 0 \quad \text{if } p^2 \mid N \text{ and if } \varepsilon \text{ can be defined modulo } N/p;$$

$$|a_p| = p^{(k-1)/2} \quad \text{if } \varepsilon \text{ cannot be defined modulo } N/p;$$

$$|a_p| = p^{k/2-1} \quad \text{if } p^2 \nmid N \text{ and if } \varepsilon \text{ can be defined modulo } N/p.$$

(It will follow from Theorem 4.6 below that the last case does not occur when $k = 1$.)

1.9. Every form f of type (k, ε) on $\Gamma_0(N)$ can be written

$$f(z) = E(z) + \sum \lambda_i f_i(d_i z),$$

where E is an Eisenstein series and f_i is a primitive cusp form of type (k, ε) on $\Gamma_0(N_i)$, with N_i a divisor of N such that ε can be defined modulo N_i, and d_i a divisor of N/N_i. Moreover, this decomposition is unique in the evident sense.

§ 2. Review (geometric) of modular forms

2.1. Let k and N be integers ≥ 1, and let μ_N be the group scheme of Nth roots of unity. From the geometric point of view, a modular form of weight k on $\Gamma_1(N)$ is a rule that assigns, to every elliptic curve E equipped with an embedding $\alpha : \mu_N \longrightarrow E$, a section of the kth tensor power $\omega_E^{\otimes k}$ of ω_E, where ω_E is the dual of the Lie algebra of E.

More precisely:

(a) An elliptic curve over a scheme S is a proper smooth morphism $E \longrightarrow S$, equipped with a section $e : S \longrightarrow E$, whose geometric fibers are elliptic curves. When S is the spectrum of a commutative ring A, we also say that E is an elliptic curve over A. We set $\omega_E = e^*\Omega^1_{E/S}$; when $S = \mathrm{Spec}(A)$, ω_E is identified with an invertible A-module.

(b) Let R be a commutative ring in which N is invertible. A modular form of weight k on $\Gamma_1(N)$, meromorphic at infinity and defined over R, is a rule that assigns, to every elliptic curve E over an R-algebra A, equipped with an embedding $\alpha : \mu_N \longrightarrow E$, an element $f(E, \alpha)$ of $\omega_E^{\otimes k}$. This rule is required to be compatible with isomorphisms and with extension of scalars.

(c) We say that f is holomorphic at infinity if it extends to a rule $\tilde{f}$ defined for pairs (E, α) in which E is a generalized elliptic curve ([7], II. 1.12) and α is an embedding of μ_N into E whose image meets every irreducible component of every geometric fiber ([7], IV. 4.14). If it exists, the rule $\tilde{f}$ is unique.

Suppose that R is a field. We say that f is a cusp form if it is holomorphic at infinity and if $\tilde{f}(E, \alpha) = 0$ whenever E is a degenerate (i.e., nonsmooth) elliptic curve over an algebraically closed extension of R.

These notions could also be defined in terms of Laurent expansions in q (cf. [7], VII, § 3).

2.2. Let f be as above. If $d \in (\mathbf{Z}/N\mathbf{Z})^*$, the modular form $f \mid R_d$ is defined by

$$(f \mid R_d)(E, \alpha) = f(E, d\alpha). \tag{2.2.1}$$

If ε is a homomorphism from $(\mathbf{Z}/N\mathbf{Z})^*$ to R^*, we say that f is of type (k, ε) on $\Gamma_0(N)$ if $f \mid R_d = \varepsilon(d)f$ for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$.

2.3. Let p be a prime number not dividing N. The Hecke operator T_p is then defined on the spaces of modular forms. If f is such a form and (E, α) is defined over an algebraically closed field of characteristic $\neq p$, then

$$(f \mid T_p)(E, \alpha) = \frac{1}{p} \sum_{\varphi} \varphi^*(f(\varphi E, \varphi \circ \alpha)),$$

where φ runs through the classes of isogenies of degree p with source E (two isogenies being in the same class if their kernels are equal).

The operators T_p commute with one another and with the R_d.

2.4. Take $R = \mathbf{C}$. The datum of $\alpha : \mu_N \longrightarrow E$ is then equivalent to that of the point

$$\alpha(\exp(2\pi i/N)),$$

which has order N. To a modular form f as above, we associate a function (also denoted f) on the upper half-plane H by the rule

$$f(z) = f(E_z, 1/N)/(2\pi i\, du)^{\otimes k}, \tag{2.4.1}$$

where E_z denotes the elliptic curve $\mathbf{C}/(\mathbf{Z} \oplus z\mathbf{Z})$.

Put $f(z) = f_\infty(e^{2\pi i z})$. Formula (2.4.1) can be rewritten as

$$f_\infty(q) = f(\mathbf{C}^*/q^{\mathbf{Z}}, \mathrm{Id})/(dt/t)^{\otimes k} \qquad (0 < |q| < 1), \tag{2.4.2}$$

where Id is induced by the inclusion of μ_N in $\mathbf{C}^*$.

This construction identifies the spaces of modular forms in the sense of 2.1 and 2.2 with the spaces having the same names in § 1; the same holds for the operators T_p and R_d.

2.5. For the definition of the Tate curve $\mathbf{G}_m/q^{\mathbf{Z}}$ over the ring $\mathbf{Z}((q)) = \mathbf{Z}[[q]](q^{-1})$, we refer to [7], VII, § 1. This curve is equipped with an invariant differential dt/t and a natural embedding $\mathrm{Id} : \mu_N \longrightarrow \mathbf{G}_m/q^{\mathbf{Z}}$. If f is a modular form of weight k on $\Gamma_1(N)$, meromorphic at infinity and defined over a ring R, set

$$f_\infty(q) = f(\mathbf{G}_m/q^{\mathbf{Z}}, \mathrm{Id})/(dt/t)^{\otimes k} \in \mathbf{Z}((q)) \otimes R \subset R((q)).$$

(In this formula, $\mathbf{G}_m/q^{\mathbf{Z}}$ denotes the curve over $\mathbf{Z}((q)) \otimes R$ obtained from the Tate curve by extension of scalars.)

Put

$$f_\infty(q) = \sum a_n q^n \qquad \text{and} \qquad (f \mid R_d)_\infty(q) = \sum a_n(d) q^n, \quad d \in (\mathbf{Z}/N\mathbf{Z})^*;$$

if p is a prime number not dividing N, then

$$(f \mid T_p)_\infty(q) = \sum a_{pn} q^n + p^{k-1} \sum a_n(p) q^{np}. \tag{2.5.1}$$

In particular, if f is of type (k, ε) on $\Gamma_0(N)$, then

$$(f \mid T_p)_\infty(q) = \sum a_{pn} q^n + \varepsilon(p) p^{k-1} \sum a_n q^{pn}. \tag{2.5.2}$$

When $R = \mathbf{C}$, $f_\infty(q)$ is the series expansion in (2.4.2); this is proved in [7], VII, § 4 (at least for holomorphic f, the only case that concerns us). Formula (2.5.2) recovers (1.6.1).

2.6. If K is a field of characteristic 0, let S_K denote the vector space of cusp forms of weight k on $\Gamma_1(N)$ that are defined over K. We have

$$S_K = K \otimes_{\mathbf{Q}} S_{\mathbf{Q}}; \tag{2.6.1}$$

this follows by interpreting S_K as the space of sections of an invertible sheaf on the "algebraic stack" corresponding to $\Gamma_1(N)$ (cf. [7], VII, 3.2).

If K' is a subfield of K, a form $f \in S_K$ belongs to $S_{K'}$ if and only if the coefficients of the series $f_\infty(q)$ belong to K'. This follows by reducing to the case in which K is algebraically closed and observing that, for every K'-automorphism σ of K, the forms f and $\sigma(f)$ have the same series expansion and therefore coincide.

PROPOSITION 2.7. — Let L be the set of $f \in S_{\mathbf{C}}$ such that $(f \mid R_d)_\infty(q) \in \mathbf{Z}[[q]]$ for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$. Then:

(2.7.1) L is a free $\mathbf{Z}$-module of finite type, stable under the operators T_p and R_d.

(2.7.2) For every field K of characteristic 0, one has $S_K = K \otimes L$.

(2.7.3) The eigenvalues of the T_p on $S_{\mathbf{C}}$ are algebraic integers in a finite extension of $\mathbf{Q}$.

(2.7.4) If $f \in S_{\mathbf{C}} = \mathbf{C} \otimes L$ satisfies $f \mid T_p = a_p f$, then, for every automorphism σ of $\mathbf{C}$, the form $\sigma(f)$ satisfies $\sigma(f) \mid T_p = \sigma(a_p)\sigma(f)$. If f is of type (k, ε) on $\Gamma_0(N)$, then $\sigma(f)$ is of type $(k, \sigma(\varepsilon))$ on $\Gamma_0(N)$.

If $f \in S_{\mathbf{Q}}$, then $f \mid R_d \in S_{\mathbf{Q}}$ for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$, and the series $(f \mid R_d)_\infty(q)$ belong to $\mathbf{Z}[[q]] \otimes \mathbf{Q}$ and hence have bounded denominators. It follows that a nonzero multiple of f belongs to L, whence $\mathbf{Q} \otimes L = S_{\mathbf{Q}}$ and, by (2.6.1), $K \otimes L = S_K$ for every field K of characteristic 0. The fact that L is of finite type follows because the linear functionals "nth coefficients of $(f \mid R_d)_\infty(q)$" separate the elements of $S_{\mathbf{Q}}$.

The stability of L under the R_d (respectively the T_p) is evident (respectively follows from (2.5.1)). Assertions (2.7.3) and (2.7.4) follow.

REMARK 2.8. — The fact that the series $f_\infty(q)$, for $f \in S_{\mathbf{Q}}$, has bounded denominators was deduced here from the fact that the Tate curve is defined over $\mathbf{Z}((q)) \otimes \mathbf{Q}$. One could also have used Shimura's Theorem 3.5.2 [24], valid when $k \geq 2$, and reduced weight 1 to weight 13 by multiplication by Δ.

§ 3. Review of Galois representations

3.1. Let $\overline{\mathbf{Q}}$ be an algebraic closure of $\mathbf{Q}$, and let $G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$. We shall have to consider linear representations of G, that is, continuous homomorphisms

$$\rho : G \longrightarrow \mathrm{GL}_n(k),$$

where k is of one of the following types:
(a) the field $\mathbf{C}$ (with the discrete topology);
(b) a finite field (with the discrete topology);
(c) a finite extension of an ℓ-adic field $\mathbf{Q}_\ell$ (with its natural topology).
In the first two cases, the image of ρ is finite.

If p is a prime number, we say that ρ is unramified at p if it is trivial on the inertia group of a place of $\overline{\mathbf{Q}}$ extending p. We then denote by $F_{p,\rho}$ the image under ρ of the Frobenius substitution [1] at p; it is an element of $\mathrm{GL}_n(k)$ defined up to conjugacy. Set

$$P_{p,\rho}(T) = \det(1 - F_{p,\rho}T) = 1 - \mathrm{Tr}(F_{p,\rho})T + \cdots + (-1)^n \det(F_{p,\rho})T^n. \tag{3.1.1}$$

Knowledge of the polynomials $P_{p,\rho}(T)$ almost determines ρ. More precisely:
LEMMA 3.2. — Let X be a set of prime numbers of density 1, and let ρ and ρ' be two semisimple linear representations of G. Suppose that, for every $p \in X$, ρ and ρ' are unramified and that $P_{p,\rho}(T) = P_{p,\rho'}(T)$ (respectively that $\mathrm{Tr}(F_{p,\rho}) = \mathrm{Tr}(F_{p,\rho'})$ when k has characteristic 0). Then ρ and ρ' are isomorphic.

This follows from the Chebotarev density theorem, combined with the fact that a semisimple linear representation of a group is determined up to isomorphism by the corresponding characteristic polynomials (respectively by the traces if the field has characteristic 0) ([3], § 30.16).

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3.3. In what follows, we shall apply Lemma 3.2 to the particular case in which X is the set of prime numbers not dividing a given integer N; we shall then say that ρ and ρ' are unramified outside N.

3.4. When $k = \mathbf{C}$, the semisimplicity condition is automatically satisfied, since $\rho(G)$ and $\rho'(G)$ are finite.

§ 4. Results

(a) STATEMENT OF THE MAIN THEOREM

THEOREM 4.1. — Let N be an integer ≥ 1, let ε be a Dirichlet character modulo N such that $\varepsilon(-1) = -1$, and let f be a nonzero modular form of type $(1, \varepsilon)$ on $\Gamma_0(N)$. Suppose that f is an eigenform for the T_p, $p \nmid N$, with eigenvalues a_p.

[1] We adopt here Artin's conventions [1]. Our "Frobenius substitution" is therefore the element denoted φ in [5]; its inverse is the "geometric Frobenius."

There then exists a linear representation

$$\rho : G \longrightarrow \mathrm{GL}_2(\mathbf{C}), \qquad \text{where} \qquad G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}),$$

which is unramified outside N and such that

$$\mathrm{Tr}(F_{p,\rho}) = a_p \qquad \text{and} \qquad \det(F_{p,\rho}) = \varepsilon(p) \quad \text{for every } p \nmid N. \tag{4.1.1}$$

This representation is irreducible if and only if f is a cusp form.

The proof will be given in § 8.

COROLLARY 4.2. — The eigenvalues a_p are sums of two roots of unity; in particular, $|a_p| \leq 2$.

In other words, the "Ramanujan-Petersson conjecture" is true in weight 1; it is also known to be true in weights ≥ 2, cf. [5], 8.2.

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4.3. By Lemma 3.2, the representation ρ associated with f by 4.1 is unique up to isomorphism.

4.4. The formula $\det(F_{p,\rho}) = \varepsilon(p)$ shows that

$$\det(\rho) = \varepsilon,$$

where ε is identified with the character $G \longrightarrow \mathbf{C}^*$ corresponding to it under class field theory (it is simply the composite of ε with the homomorphism $G \longrightarrow (\mathbf{Z}/N\mathbf{Z})^*$ afforded by the action of G on the Nth roots of unity).

4.5. Let c denote the element of G corresponding to complex conjugation (for an embedding of $\overline{\mathbf{Q}}$ into $\mathbf{C}$). Since ε is odd, 4.4 shows that $\det(\rho(c)) = -1$; since c has order 2, this means that $\rho(c)$ is conjugate to the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

(b) ARTIN CONDUCTOR AND LOCAL FACTORS. — We retain the hypotheses and notation of Theorem 4.1.

THEOREM 4.6. — Suppose that f is a primitive cusp form, with coefficients a_n, $n \geq 1$. Let ρ be the corresponding representation of G. Then:

a. The Artin conductor of ρ is equal to N;

b. The Artin L-function $L(s, \rho)$ is equal to

$$\Phi_f(s) = \sum_{n=1}^{\infty} a_n n^{-s}.$$

(For the definition of the L-series and the conductor of a representation, see [1].)

COROLLARY 4.7. — The representation ρ is ramified at every prime divisor of N.

This follows from (a).

COROLLARY 4.8. — The function $L(s,\rho)$ is entire.

[In other words, the "Artin conjecture" holds for ρ, cf. (c) below.]

Indeed, Hecke theory shows that $\Phi_f(s)$ is entire.

Proof of 4.6. — It uses the functional equations satisfied by $\Phi_f(s)$ and $L(s,\rho)$ (compare [9], pp. 172-177).

(i) Put $\tilde{f} = \sum \overline{a_n} q^n$. Since f is primitive, there exists a constant $\lambda \neq 0$ such that $f(-1/Nz) = \lambda z \tilde{f}(z)$, cf. [12] and [13]. Mellin transformation then gives

$$\Psi_f(1-s) = \mu \widetilde{\Psi}_f(s) \qquad \text{with} \qquad \mu = i\lambda/N^{1/2},$$

where

$$\Psi_f(s) = N^{s/2}(2\pi)^{-s}\Gamma(s)\Phi_f(s) \qquad \text{and} \qquad \widetilde{\Psi}_f(s) = \Psi_{\tilde{f}}(s).$$

(ii) By 4.5, the "factor at infinity" of $L(s,\rho)$ is $(2\pi)^{-s}\Gamma(s)$. If M is the conductor of ρ, and if we set

$$\xi(s,\rho) = M^{s/2}(2\pi)^{-s}\Gamma(s)L(s,\rho),$$

then

$$\xi(1-s,\rho) = \nu \cdot \xi(s,\overline{\rho}) \qquad \text{with} \qquad \nu \in \mathbf{C}^*.$$

(iii) Put

$$F(s) = (N/M)^{s/2}\Psi_f(s)/\xi(s,\rho) \qquad \text{and} \qquad \widetilde{F}(s) = (N/M)^{s/2}\widetilde{\Psi}_f(s)/\xi(s,\overline{\rho}).$$

The preceding formulas show that

$$F(1-s) = \omega \cdot \widetilde{F}(s) \qquad \text{with} \qquad \omega = \mu/\nu.$$

But if p is a prime number not dividing N, the p-factors of $\Psi_f(s)$ and $\xi(s,\rho)$ agree by 4.1. Hence

$$F(s) = A^s \prod_{p|N} F_p(s),$$

where

$$A = (N/M)^{1/2} \qquad \text{and} \qquad F_p(s) = \frac{1 - a_p p^{-s}}{(1 - b_p p^{-s})(1 - c_p p^{-s})},$$

where $1 - a_p p^{-s}$ is the p-factor of $\Psi_f(s)$ and $(1 - b_p p^{-s})(1 - c_p p^{-s})$ is that of $\xi(s,\rho)$ (note that b_p and c_p may be zero). It remains only to show that A and the F_p are equal to 1. For this we use the following elementary lemma:

LEMMA 4.9. — Let $G(s) = A^s \prod_p G_p(s)$ and $H(s) = A^s \prod_p H_p(s)$ be two finite Euler products. Suppose that

$$G(1-s) = \omega \cdot H(s) \qquad \text{with} \qquad \omega \in \mathbf{C}^*; \tag{4.9.1}$$

(4.9.2) Each of G_p and H_p is a finite product of terms of the form $(1-\alpha_p^{(i)}p^{-s})^{\pm 1}$, with $|\alpha_p^{(i)}| < p^{1/2}$.
Then $A = 1$ and $G_p = H_p = 1$ for every p.
If H_p is not equal to 1, the function H has infinitely many zeros (or poles) of the form

$$(\log(\alpha_p^{(i)}) + 2\pi i n)/\log p, \qquad n \in \mathbf{Z},$$

and it is easy to see that these cannot all be zeros (or poles) of $G(1-s)$; indeed, the hypothesis $|\alpha_p^{(i)}| < p^{1/2}$ ensures that none of the $\alpha_p^{(i)}$ can equal a $p/\alpha_p^{(j)}$.

(iv) It remains to verify that a_p, b_p, c_p, and their conjugates satisfy (4.9.2), i.e., have absolute value $< p^{1/2}$. This is clear for b_p and c_p, which are either 0 or roots of unity. For a_p, one may invoke 1.8, which shows that $|a_p| \leq 1$; alternatively, one may use Rankin's inequality:

$$|a_n| = O(n^{1/2-1/5}), \qquad cf.\ [18];$$

applying it to $n = p^m$ and observing that $a_n = (a_p)^m$, one obtains

$$|a_p| \leq p^{1/2-1/5} < p^{1/2}.$$

This completes the proof of 4.6.

(c) CHARACTERIZATION OF THE REPRESENTATIONS ATTACHED TO FORMS OF WEIGHT 1. — Retain the notation of (a), and suppose that the form f under consideration is a cusp form. The corresponding representation

$$\rho : G \longrightarrow \mathrm{GL}_2(\mathbf{C})$$

then has the following properties:
(i) ρ is irreducible (4.1);
(ii) $\det\rho$ is an odd character (4.4);
(iii) For every continuous character $\chi : G \longrightarrow \mathbf{C}^*$, the Artin L-function $L(s, \rho\otimes\chi)$ is entire [this follows from 4.8 applied to the cusp form

$$f_\chi = \sum \chi(n) a_n q^n].$$

Conversely:

THEOREM 4.10. (WEIL-LANGLANDS). — Let $\rho : G \longrightarrow \mathrm{GL}_2(\mathbf{C})$ be a continuous representation of $G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ satisfying conditions (i), (ii), and (iii) above. Put

$$L(s,\rho) = \sum a_n n^{-s}, \qquad f = \sum a_n q^n, \qquad \varepsilon = \det\rho, \qquad N = \text{conduct. } \rho.$$

Then f is a primitive cusp form of type $(1, \varepsilon)$ on $\Gamma_0(N)$, and ρ is the representation attached to f.

According to Langlands (cf. [27], pp. 152 and 160), the constants in the functional equations of the series $\sum a_n\chi(n)n^{-s}$ satisfy the identity required to apply the Hecke-Weil characterization of modular forms ([12], [26]). It follows that f is modular of type $(1, \varepsilon)$ on $\Gamma_0(N)$; it is clear that f is an eigenform for the T_p and the U_p, and that the representation associated with it is isomorphic to ρ. By 4.1, f is a cusp form. Let f' be the unique primitive cusp form (on some $\Gamma_0(N')$, where N' is a suitable divisor of N) such that $f' \mid T_p = a_p f'$ for $p \nmid N$. By 4.6, the Dirichlet series associated with f' is $L(s, \rho) = \sum a_n n^{-s}$. It follows that $f' = f$, which proves that f is primitive.

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4.11. A generalization of Theorem 4.10 to all global fields may be found in [27], p. 163.

4.12. Condition (iii) (the Artin conjecture for the $\rho \otimes \chi$) may be replaced by the weaker condition:

(iii′) There exists an integer $M \geq 1$ such that, for every character χ whose conductor is relatively prime to M, the function $L(s, \rho \otimes \chi)$ is entire.

This follows from [26] (see also [12]).

4.13. If the Artin conjecture is true, the theorems above provide a bijection between "classes of irreducible degree-2 representations of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ with odd determinant" and "primitive cusp forms of weight 1."

4.14. Theorem 4.6 even gives a way to verify the Artin conjecture in particular cases. Given a representation ρ satisfying (i) and (ii), with conductor N and determinant ε, one can numerically determine the coefficients a_n of the series $L(s, \rho) = \sum a_n n^{-s}$ for n less than a given integer A, and one can seek to construct a primitive cusp form f of type $(1, \varepsilon)$ on $\Gamma_0(N)$ whose expansion begins with $\sum_{n \leq A} a_n q^n$. If A is sufficiently large,

$$\text{for example} \qquad A \geq (N/12) \prod_{p \mid N} (1 + p^{-1}),$$

such a form is unique if it exists (if it does not exist, the Artin conjecture is false). Once f has been obtained, a representation ρ_f corresponds to it; if one can prove that ρ_f is isomorphic to ρ, it follows that ρ satisfies (iii).

EXAMPLES. — If ρ is as above, the image of ρ in the group

$$\mathrm{PGL}_2(\mathbf{C}) = \mathrm{GL}_2(\mathbf{C})/\mathbf{C}^*$$

is either a dihedral group or one of the groups $\mathfrak{A}_4$, $\mathfrak{S}_4$, or $\mathfrak{A}_5$ ([22], prop. 16). In the dihedral case, ρ is induced by a degree-1 representation of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}(\sqrt{d}))$, where $\mathbf{Q}(\sqrt{d})$ is a quadratic extension of $\mathbf{Q}$. Condition (iii) is then satisfied, and ρ

corresponds indeed to a cusp form; the latter is a linear combination of theta series for binary forms of discriminant d, cf. [9], pp. 428-460. Non-dihedral examples have recently been constructed by Tate (for $N = 133, 229, 283, 331, \ldots$).

For one of these examples (the one with $N = 133$, corresponding to a group $\mathfrak{A}_4$), Tate, assisted by Atkin et al., was able to carry out the method outlined in 4.14 and prove the existence of a corresponding modular form—and hence also the Artin conjecture for the representation in question.

§ 5. Exploitation of a Result of Rankin

PROPOSITION 5.1. — Let f be a nonzero cusp form of type (k, ε) on $\Gamma_0(N)$. Suppose that f is an eigenform for the T_p, $p \nmid N$, with eigenvalues a_p. Then the series $\sum_{p \nmid N} |a_p|^2 p^{-s}$ converges for real $s > k$, and

$$\sum_{p \nmid N} |a_p|^2 p^{-s} \leq \log(1/(s-k)) + O(1) \qquad \text{as } s \longrightarrow k. \tag{5.1.1}$$

PROOF 5.2. — One immediately reduces to the case in which f is a primitive form $\sum_{n=1}^{\infty} a_n q^n$. For every $p \nmid N$, let $\varphi_p \in \mathrm{GL}_2(\mathbf{C})$ satisfy $\mathrm{Tr}(\varphi_p) = a_p$ and $\det(\varphi_p) = \varepsilon(p)p^{k-1}$. The Dirichlet series

$$\Phi_f(s) = \sum_{n=1}^{\infty} a_n n^{-s}$$

may then be written

$$\Phi_f(s) = \prod_{p|N} (1 - a_p p^{-s})^{-1} \prod_{p \nmid N} \det(1 - \varphi_p p^{-s})^{-1}, \qquad cf.\ (1.7.2).$$

Set

$$L(s) = \prod_{p \nmid N} \det(1 - \varphi_p \otimes \overline{\varphi}_p\, p^{-s})^{-1}.$$

This is an Euler product with four factors: if λ_p and μ_p denote the eigenvalues of φ_p, then

$$L(s) = \prod_{p \nmid N} [(1 - \lambda_p \overline{\lambda}_p p^{-s})(1 - \lambda_p \overline{\mu}_p p^{-s})(1 - \mu_p \overline{\lambda}_p p^{-s})(1 - \mu_p \overline{\mu}_p p^{-s})]^{-1}.$$

Using the formula $\lambda_p \overline{\lambda}_p \mu_p \overline{\mu}_p = |\varepsilon(p)p^{k-1}|^2 = p^{2k-2}$, one proves (cf., for example, [10], p. 33, or [12], [15]) that

$$L(s) = H(s)\zeta(2s - 2k + 2) \left(\sum_{n=1}^{\infty} |a_n|^2 n^{-s} \right),$$

where

$$H(s) = \prod_{p|N} (1 - p^{-2s+2k-2})(1 - |a_p|^2 p^{-s}).$$

According to [18] (cf. also [12], [13], and [15]), the series $\sum |a_n|^2 n^{-s}$ converges for $\mathrm{Re}(s) > k$, and its product with $\zeta(2s - 2k + 2)$ extends to a meromorphic function throughout the

complex plane, with its only pole at $s = k$. Since $|a_p| < p^{k/2}$ when $p \mid N$ (cf. 1.8), the function $H(s)$ is holomorphic and nonzero in $\mathrm{Re}(s) \geq k$. It follows that $L(s)$ is meromorphic throughout the complex plane and holomorphic for $\mathrm{Re}(s) \geq k$, with the sole exception of $s = k$, where it has a simple pole; moreover, $L(s) \neq 0$ for real $s > k$, because this is true of $H(s)$, $\zeta(2s - 2k + 2)$, and $\sum |a_n|^2 n^{-s}$.

Set

$$g_m(s) = \sum_{p \nmid N} |\operatorname{Tr}(\varphi_p^m)|^2 p^{-ms}/m \qquad \text{and} \qquad g(s) = \sum_{m=1}^{\infty} g_m(s).$$

The series $g(s)$ is a Dirichlet series with nonnegative coefficients. For sufficiently large s, a direct calculation shows that it equals $\log L(s)$. Since $L(s)$ is holomorphic and nonzero for real $s > k$, a classical lemma of Landau ([21], p. 112) implies that $g(s)$ converges for $\mathrm{Re}(s) > k$. Since $L(s)$ has a simple pole at $s = k$, one has

$$g(s) = \log(1/(s-k)) + O(1) \qquad \text{as } s \longrightarrow k.$$

But $g_1(s) = \sum |a_p|^2 p^{-s}$ is plainly $\leq g(s)$. Hence

$$\sum |a_p|^2 p^{-s} \leq \log(1/(s-k)) + O(1) \qquad \text{as } s \longrightarrow k.$$

REMARKS 5.3. — Proposition 5.1 can be strengthened in various ways. First, once the Petersson conjecture is available, an easy bound shows that the series

$$\sum_{p \nmid N} \sum_{m \geq 2} |\operatorname{Tr}(\varphi_p^m)|^2 p^{-ms}/m = g_2(s) + g_3(s) + \cdots$$

converges for $\mathrm{Re}(s) \geq k$, and this permits the inequality (5.1.1) to be replaced by the equality

$$\sum |a_p|^2 p^{-s} = \log(1/(s-k)) + O(1) \qquad \text{as } s \longrightarrow k. \tag{5.3.1}$$

On the other hand, a Hadamard-de la Vallée Poussin argument shows that $L(s) \neq 0$ for every s such that $\mathrm{Re}(s) \geq k$ (including the critical line $\mathrm{Re}(s) = k$), and applying the Wiener-Ikehara theorem to $L'(s)/L(s)$ gives

$$\sum_{p \leq x} |a_p|^2 p^{-(k-1)} \sim x \qquad \text{as } x \longrightarrow \infty, \tag{5.3.2}$$

cf. Rankin [19].

5.4. APPLICATION TO FORMS OF WEIGHT 1. — Let P be the set of prime numbers and let X be a subset of P. Set

$$\text{dens. sup } X = \limsup_{\substack{s \to 1 \\ s > 1}} \frac{\sum_{p \in X} p^{-s}}{\log(1/(s-1))}. \tag{5.4.1}$$

This is the upper density of X; it lies between 0 and 1.

PROPOSITION 5.5. — Retain the hypotheses of 5.1, and suppose in addition that the weight k of f is 1. Then, for every $\eta > 0$, there exists a set X_η of prime

numbers and a finite subset Y_η of $\mathbf{C}$ such that

$$\text{dens. sup } X_\eta \leq \eta \qquad \text{and} \qquad a_p \in Y_\eta \quad \text{for every } p \notin X_\eta.$$

By 2.7, the a_p are algebraic integers in a finite extension K of $\mathbf{Q}$. If c is a constant ≥ 0, let $Y(c)$ be the set of integers a of K such that $|\sigma(a)|^2 \leq c$ for every embedding σ of K into $\mathbf{C}$; this is a finite set. Let $X(c)$ be the set of p such that $a_p \notin Y(c)$; it is enough to prove that dens. sup $X(c) \leq \eta$ when c is sufficiently large.

We know from 2.7 that the $\sigma(a_p)$ are likewise eigenvalues of the T_p in weight 1. By (5.1.1), therefore,

$$\sum_\sigma \sum_p |\sigma(a_p)|^2 p^{-s} \leq r \log(1/(s-1)) + O(1) \qquad \text{as } s \longrightarrow 1,$$

where $r = [K : \mathbf{Q}]$. Since $\sum_\sigma |\sigma(a_p)|^2 \geq c$ if $p \in X(c)$, it follows that

$$c \sum_{p \in X(c)} p^{-s} \leq r \log(1/(s-1)) + O(1) \qquad \text{as } s \longrightarrow 1,$$

and hence

$$\text{dens. sup } X(c) \leq r/c.$$

It is therefore enough to take $c \geq r/\eta$.

REMARK 5.6. — By using (5.3.2) instead of (5.1.1) in the preceding proof, one could have replaced the "analytic" density (5.4.1) by the "natural" density (cf. [21], VI, no. 4.5). In any event, 5.5 is only of provisional interest: once Theorem 4.1 has been proved, we shall know that the set of the a_p is finite.

§ 6. ℓ-adic Representation and Reduction mod ℓ

(a) ℓ-ADIC REPRESENTATIONS. — We shall use the following result:

THEOREM 6.1. — Let f be a nonzero modular form of type (k, ε) on $\Gamma_0(N)$. Suppose that $k \geq 2$ and that f is an eigenform for the T_p, $p \nmid N$, with eigenvalues a_p. Let K be a finite extension of $\mathbf{Q}$ containing the a_p and the $\varepsilon(p)$, cf. (2.7.3). Let λ be a finite place of K of residual characteristic ℓ, and let K_λ be the completion of K at λ. There then exists a continuous semisimple linear representation

$$\rho_\lambda : G \longrightarrow \mathrm{GL}_2(K_\lambda), \qquad \text{where} \qquad G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}),$$

which is unramified outside $N\ell$ and satisfies

$$\mathrm{Tr}(F_{\rho_\lambda, p}) = a_p \qquad \text{and} \qquad \det(F_{\rho_\lambda, p}) = \varepsilon(p) p^{k-1} \quad \text{if } p \nmid N\ell. \tag{6.1.1}$$

By 3.2, condition (6.1.1) determines ρ_λ uniquely up to isomorphism.

REMARK 6.2. — If f is an Eisenstein series, the assertion above follows immediately from Hecke's results ([9], p. 690) by taking ρ_λ to be the direct sum of two degree-1 representations. When f is a cusp form, 6.1 is proved in a special case in [4]. The general case is not much more difficult. It is treated, by another method (inspired by Ihara) and in another language, by Langlands [11] (where, however, a "trace formula" that seems accessible but that no one has proved is admitted without proof). In a future work by one of us, 6.2 will be reproved by a method due to Piateckii-Shapiro [17].

COROLLARY 6.3. — Let $(f, N, k, \varepsilon, (a_p))$ and $(f', N', k', \varepsilon', (a'_p))$ be as in Theorem 6.1. If the set of prime numbers p such that $a_p = a'_p$ has density 1, then $k = k'$, $\varepsilon = \varepsilon'$, and $a_p = a'_p$ for every $p \nmid NN'$.

Indeed, the representations attached to f and f' (for the same choice of K and λ) are isomorphic by 3.2.

REMARKS

6.4. The image of G under ρ_λ is a compact subgroup of $\mathrm{GL}_2(K_\lambda)$, hence an ℓ-adic Lie group; it is not a finite group.

6.5. Once Theorem 4.1 has been proved, one sees easily [2] that 6.1 and 6.3 remain valid in weight 1; in that case, however, the image of the group G is finite.

(b) REDUCTION mod ℓ

6.6. Let $K \subset \mathbf{C}$ be a number field, let λ be a finite place of K, let O_λ be the corresponding valuation ring, $\mathfrak{m}_\lambda$ its maximal ideal, $k_\lambda = O_\lambda/\mathfrak{m}_\lambda$ its residue field, and ℓ the characteristic of k_λ. In what follows, we shall write "mod λ" for "mod $\mathfrak{m}_\lambda$."

Let f be a modular form of type (k, ε) on $\Gamma_0(N)$. We say that f is λ-integral (respectively, that $f \equiv 0 \pmod{\lambda}$) if the coefficients of the series $f_\infty(q)$ belong to O_λ (respectively, to $\mathfrak{m}_\lambda$). Suppose that f is λ-integral; we say that f is an eigenvector of T_p mod λ, with eigenvalue $a_p \in k_\lambda$, if

$$f \mid T_p - a_p f \equiv 0 \pmod{\lambda}. \tag{6.6.1}$$

THEOREM 6.7. — With the preceding notation, let f be a modular form of type (k, ε) on $\Gamma_0(N)$, with $k \geq 1$ and coefficients in K. Suppose that f is λ-integral, that $f \not\equiv 0 \pmod{\lambda}$, and that f is an eigenvector of the T_p mod λ, for $p \nmid N\ell$, with eigenvalues $a_p \in k_\lambda$. Let k_f be the subfield of k_λ generated by the a_p and the reductions mod λ of the $\varepsilon(p)$. There then exists a semisimple representation

$$\rho : G \longrightarrow \mathrm{GL}_2(k_f)$$

[2] This follows from the fact that the representation ρ of Theorem 4.1 is realizable over K: its image contains an element with distinct rational eigenvalues (the element $\rho(c)$ of 4.5), and this implies that its Schur index is 1 (cf. [20], IX a); it is therefore realizable over the field of values of its character.

which is unramified outside $N\ell$ and such that, for every $p \nmid N\ell$,

$$\mathrm{Tr}(F_{\rho,p}) = a_p \qquad \text{and} \qquad \det(F_{\rho,p}) \equiv \varepsilon(p)p^{k-1} \pmod{\lambda}. \tag{6.7.1}$$

Proof of Theorem 6.7

6.8. Let $(K', \lambda', f', k', \varepsilon', (a'_p))$ be as in Theorem 6.7, where K' contains K and λ' extends λ. If $a_p \equiv a'_p \pmod{\lambda'}$ and $\varepsilon(p)p^{k-1} \equiv \varepsilon'(p)p^{k'-1} \pmod{\lambda'}$ for every $p \nmid N\ell$, the theorem for f is equivalent to the theorem for f'. The second condition holds whenever $\varepsilon = \varepsilon'$ and $k \equiv k' \pmod{\ell - 1}$, and it implies the first provided that $f \equiv f' \pmod{\lambda'}$.

6.9. REDUCTION TO THE CASE $k \geq 2$. — For even $n > 2$, let E_n be the Eisenstein series of weight n on $\mathrm{SL}_2(\mathbf{Z})$, normalized so that its constant term is 1. If n is chosen divisible by $\ell - 1$, the series expansion of E_n is ℓ-integral, and $E_n \equiv 1 \pmod{\ell}$, cf. [25]. The product $f \cdot E_n$ is therefore congruent to f mod λ; its weight $k + n$ is congruent to k mod $(\ell - 1)$. By 6.8, the theorem for f is equivalent to the theorem for $f \cdot E_n$, which has weight > 2.

6.10. REDUCTION TO THE CASE IN WHICH f IS AN EIGENVECTOR OF THE T_p. — It is enough to verify that there exists an f' as in 6.8, with $(k', \varepsilon') = (k, \varepsilon)$, which is an eigenvector of the T_p. This follows from the next lemma, applied to the T_p acting on the O_λ-module M of modular forms of type (k, ε) on $\Gamma_0(N)$ with coefficients in O_λ:

LEMMA 6.11. — Let M be a free module of finite type over a discrete valuation ring O; let $\mathfrak{m}$ denote the maximal ideal of O, k its residue field, and K its field of fractions. Let $\mathcal{T}$ be a set of pairwise commuting endomorphisms of M. Let $f \in M/\mathfrak{m}M$ be a common nonzero eigenvector of the $T \in \mathcal{T}$, and let $a_T \in k$ be the corresponding eigenvalues. There then exists a discrete valuation ring O' containing O, with maximal ideal $\mathfrak{m}'$ such that $O \cap \mathfrak{m}' = \mathfrak{m}$, whose field of fractions K' is finite over K, and a nonzero element f' of

$$M' = O' \otimes_O M,$$

which is an eigenvector of the $T \in \mathcal{T}$, with eigenvalues a'_T such that $a'_T \equiv a_T \pmod{\mathfrak{m}'}$.

(Note that the assertion is not that the eigenvectors lift, but only that the eigenvalues do.)

Let $\mathcal{H}$ be the subalgebra of $\mathrm{End}(M)$ generated by $\mathcal{T}$. After making a finite extension of the scalars if necessary, we may suppose that $K \otimes \mathcal{H}$ is a product of Artinian rings with residue field K. Let $\chi : \mathcal{H} \longrightarrow k$ be the homomorphism such that $h \cdot f = \chi(h)f$ for every $h \in \mathcal{H}$. Since $\mathcal{H}$ is free over O, there exists a prime ideal $\mathfrak{p}$ of $\mathcal{H}$ contained in the maximal ideal $\ker(\chi)$ and such that $\mathfrak{p} \cap O = 0$; it is the kernel of a homomorphism $\chi' : \mathcal{H} \longrightarrow O$ whose reduction mod $\mathfrak{m}$ is χ. The ideal of $K \otimes \mathcal{H}$ generated by $\mathfrak{p}$ belongs to the support of the module $K \otimes M$; it follows that there exists a nonzero element f'' of $K \otimes M$ annihilated by this ideal, i.e. such that $hf'' = \chi'(h)f''$ for every $h \in \mathcal{H}$. We then take for f' a nonzero multiple of f'' belonging to M.

Variant. — Reduce to the case in which M is $\mathcal{T}$-indecomposable and the eigenvalues of the $T \in \mathcal{T}$ belong to K. Show that there then exists a basis $(e_1, \ldots, e_n)$ of M relative to which the elements T of $\mathcal{T}$ have upper triangular matrices (T_{ij}); use the indecomposability of M to prove that $T_{ii} \equiv a_T \pmod{\mathfrak{m}}$ for every T and every i. The element $f' = e_1$ then answers the question.

6.12. End of the proof of 6.7. — By 6.9 and 6.10, we may suppose that $k \geq 2$ and that f is an eigenvector of the T_p, $p \nmid N\ell$; since T_ℓ commutes with the T_p, we may also suppose that f is an eigenvector of T_ℓ if $\ell \nmid N$. Let

$$\rho_\lambda : G \longrightarrow \mathrm{GL}_2(K_\lambda)$$

be the representation associated with f by Theorem 6.1. After replacing ρ_λ by an isomorphic representation if necessary, we may suppose that $\rho_\lambda(G)$ is contained in $\mathrm{GL}_2(\widehat{O}_\lambda)$, where $\widehat{O}_\lambda$ is the ring of integers of K_λ (i.e. the completion of O_λ). Reducing ρ_λ mod λ gives a representation

$$\tilde{\rho}_\lambda : G \longrightarrow \mathrm{GL}_2(k_\lambda).$$

Let φ be the semisimplification of $\tilde{\rho}_\lambda$; it is a semisimple representation, unramified outside $N\ell$, and satisfies (6.7.1). The group $\varphi(G)$ is finite; by the Čebotarev density theorem, every element of $\varphi(G)$ is of the form $F_{\varphi,p}$, with $p \nmid N\ell$. From the definition of k_f, therefore:

(6.12.1) For every $s \in \varphi(G)$, the coefficients of the polynomial $\det(1 - sT)$ belong to k_f.

The existence of the desired representation $\rho : G \longrightarrow \mathrm{GL}_2(k_f)$ now follows from the next lemma:

LEMMA 6.13. — Let $\varphi : \Phi \longrightarrow \mathrm{GL}_n(k')$ be a semisimple representation of a group Φ over a finite field k'. Let k be a subfield of k' containing the coefficients of the polynomials $\det(1 - \varphi(s)T)$, $s \in \Phi$. Then φ is realizable over k, i.e. it is isomorphic to a representation $\rho : \Phi \longrightarrow \mathrm{GL}_n(k)$.

For φ to be realizable over k, it is enough to verify that φ is isomorphic to $\sigma(\varphi)$ for every k-automorphism σ of k': this follows from the fact that the Brauer group of a finite field is trivial, so there is no "Schur index" to consider. But φ and $\sigma(\varphi)$ have the same characteristic polynomials and are semisimple; they are therefore isomorphic by [3], th. 30.16.

§ 7. Bounding the Orders of Certain Subgroups of $\mathrm{GL}_2(\mathbf{F}_\ell)$

If ℓ is a prime number, let $\mathbf{F}_\ell$ denote the field $\mathbf{Z}/\ell\mathbf{Z}$ with ℓ elements.

7.1. Let η and M be two positive numbers. We shall consider the following property of a subgroup G of $\mathrm{GL}_2(\mathbf{F}_\ell)$:

$C(\eta, M)$. — There exists a subset H of G such that $|H| \geq (1 - \eta)|G|$, and the set of polynomials $\det(1 - hT)$, $h \in H$, has at most M elements.

(If X is a finite set, $|X|$ denotes its cardinality.)

We shall say that G is semisimple if the identity representation

$$G \longrightarrow \mathrm{GL}_2(\mathbf{F}_\ell)$$

is semisimple.

PROPOSITION 7.2. — Let $\eta < 1/2$ and $M \geq 0$. There exists a constant $A = A(\eta, M)$ such that, for every prime number ℓ and every semisimple subgroup G of $\mathrm{GL}_2(\mathbf{F}_\ell)$ satisfying $C(\eta, M)$, one has $|G| \leq A$.

Proof. — Let G be a semisimple subgroup of $\mathrm{GL}_2(\mathbf{F}_\ell)$. Recall (cf. [22], § 2, props. 15 and 16) that one of the following conditions holds:

(a) G contains $\mathrm{SL}_2(\mathbf{F}_\ell)$;

(b) G is contained in a Cartan subgroup T;

(c) G is contained in the normalizer of a Cartan subgroup T, but is not contained in T;

(d) the image of G in $\mathrm{PGL}_2(\mathbf{F}_\ell) = \mathrm{GL}_2(\mathbf{F}_\ell)/\mathbf{F}_\ell^*$ is isomorphic to $\mathfrak{A}_4$, $\mathfrak{S}_4$, or $\mathfrak{A}_5$.

We shall bound the order of G in each case.

Case (a). — Put $r = (G : \mathrm{SL}_2(\mathbf{F}_\ell))$. Then $|G| = r\ell(\ell^2 - 1)$. On the other hand, the number of elements of $\mathrm{GL}_2(\mathbf{F}_\ell)$ with a given characteristic polynomial is $\ell^2 + \ell$, ℓ^2, or $\ell^2 - \ell$ according as the polynomial in question has 2, 1, or 0 roots in $\mathbf{F}_\ell$. If G satisfies $C(\eta, M)$, then

$$(1 - \eta)r\ell(\ell^2 - 1) = (1 - \eta)|G| \leq |H| \leq M(\ell^2 + \ell),$$

and hence

$$(1 - \eta)r(\ell - 1) \leq M \qquad \text{and} \qquad \ell \leq 1 + \frac{M}{(1 - \eta)r} \leq 1 + \frac{M}{1 - \eta};$$

this gives a bound for ℓ, and therefore a fortiori a bound for $|G|$.

Case (b). — At most 2 elements of T have a given characteristic polynomial. The hypothesis $C(\eta, M)$ (with $\eta < 1$) therefore implies

$$(1 - \eta)|G| \leq 2M,$$

and hence the bound

$$|G| \leq \frac{2M}{1 - \eta}.$$

Case (c). — The group $G' = G \cap T$ has index 2 in G. If G satisfies $C(\eta, M)$, then G' satisfies $C(2\eta, M)$. Applying (b) to G', one obtains

$$|G| \leq \frac{4M}{1 - 2\eta}.$$

Case (d). — The image of G in $\mathrm{PGL}_2(\mathbf{F}_\ell)$ has order at most 60. The group

$$G \cap \mathrm{SL}_2(\mathbf{F}_\ell)$$

therefore has order at most 120, and G has at most 120 elements with a given determinant, and a fortiori with a given characteristic polynomial. If G satisfies $C(\eta, M)$, then

$$(1 - \eta)|G| \leq 120M, \qquad \text{whence} \qquad |G| \leq \frac{120M}{1 - \eta}.$$

§ 8. Proof of Theorem 4.1

We may suppose that the modular form f under consideration is either an Eisenstein series or a cusp form.

8.1. If f is an Eisenstein series, there exist characters χ_1 and χ_2 of $(\mathbf{Z}/N\mathbf{Z})^*$ such that $\chi_1\chi_2 = \varepsilon$ and $a_p = \chi_1(p) + \chi_2(p)$ for $p \nmid N$ (cf. [9], p. 690). We then take for ρ the reducible representation

$$\rho = \chi_1 \oplus \chi_2,$$

where the χ_i are identified with one-dimensional representations of G, cf. 4.4.

8.2. From now on, suppose that f is cuspidal. By 2.7, the a_p and the $\varepsilon(p)$ belong to the ring of integers O_K of a number field K, which we may suppose to be Galois over $\mathbf{Q}$. Let L be the set of prime numbers ℓ that split completely in K. For every $\ell \in L$, choose a place λ_ℓ of K above ℓ; the corresponding residue field is $\mathbf{F}_\ell$. By Theorem 6.7, there exists a continuous semisimple representation

$$\rho_\ell : \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \longrightarrow \mathrm{GL}_2(\mathbf{F}_\ell),$$

which is unramified outside $N\ell$ and satisfies

$$\det(1 - F_{\rho_\ell,p}T) \equiv 1 - a_pT + \varepsilon(p)T^2 \pmod{\lambda_\ell} \qquad \text{if } p \nmid N\ell.$$

Let G_ℓ be the subgroup of $\mathrm{GL}_2(\mathbf{F}_\ell)$ that is the image of ρ_ℓ.

LEMMA 8.3. — For every $\eta > 0$, there exists a constant M such that G_ℓ satisfies condition $C(\eta, M)$ of 7.1 for every $\ell \in L$.

By Proposition 5.5, there exists a subset X_η of the set P of prime numbers such that $\mathrm{dens.sup}\, X_\eta \leq \eta$ and the a_p, for $p \notin X_\eta$, form a finite set. Let $\mathcal{M}$ denote the finite set of polynomials $1 - a_pT + \varepsilon(p)T^2$, for $p \notin X_\eta$, and put $M = |\mathcal{M}|$.

The group G_ℓ satisfies $C(\eta, M)$ for every $\ell \in L$. Indeed, let H_ℓ be the subset of G_ℓ formed by the Frobenius elements $F_{\rho_\ell,p}$, $p \notin X_\eta$, and their conjugates. By the Čebotarev density theorem, $|H_\ell| \geq (1 - \eta)|G_\ell|$. On the other hand, if $h \in H_\ell$, the polynomial $\det(1 - hT)$ is the reduction modulo λ_ℓ of an element of $\mathcal{M}$, and therefore belongs to a set with at most M elements. Thus condition $C(\eta, M)$ is indeed satisfied.

LEMMA 8.4. — There exists a constant A such that $|G_\ell| \leq A$ for every $\ell \in L$.

This follows from the preceding lemma and Proposition 7.2.

8.5. Choose a constant A satisfying 8.4. After enlarging the field K if necessary (which decreases L), we may suppose that it contains all nth roots of unity for $n \leq A$. Let Y be the set of polynomials $(1 - \alpha T)(1 - \beta T)$, where α and β are roots of unity of order $\leq A$. If $p \nmid N$, then for every $\ell \in L$ with $\ell \neq p$ there exists $R(T) \in Y$ such that

$$1 - a_pT + \varepsilon(p)T^2 \equiv R(T) \pmod{\lambda_\ell}.$$

Since Y is finite, there is an R for which the preceding congruence holds for infinitely many ℓ, and hence

$$1 - a_p T + \varepsilon(p) T^2 = R(T),$$

that is, the polynomials $1 - a_p T + \varepsilon(p) T^2$ belong to Y.

8.6. Let L' be the set of $\ell \in L$ such that $\ell > A$ and such that, for $R, S \in Y$, $R \neq S$ implies $R \not\equiv S \pmod{\lambda_\ell}$; the set $L - L'$ is finite, and therefore L' is infinite. Let $\ell \in L'$. The order of the group G_ℓ is prime to ℓ. It follows, by a standard argument, that the identity representation $G_\ell \longrightarrow \mathrm{GL}_2(\mathbf{F}_\ell)$ is the reduction modulo λ_ℓ of a representation $G_\ell \longrightarrow \mathrm{GL}_2(O_{\lambda_\ell})$, where O_{λ_ℓ} is the valuation ring for λ_ℓ. Composing the latter with the canonical map $G \longrightarrow G_\ell$, we obtain a representation

$$\rho : G \longrightarrow \mathrm{GL}_2(O_{\lambda_\ell}).$$

By construction, ρ is unramified outside $N\ell$. If $p \nmid N\ell$, the eigenvalues of the Frobenius element $F_{\rho,p}$ are roots of unity of order $\leq A$ (since the image of ρ is isomorphic to G_ℓ, and hence has order $\leq A$); consequently $\det(1 - F_{\rho,p} T) \in Y$. On the other hand, since the reduction of ρ modulo λ_ℓ is ρ_ℓ, one has

$$\det(1 - F_{\rho,p} T) \equiv 1 - a_p T + \varepsilon(p) T^2 \pmod{\lambda_\ell}.$$

But the two polynomials $\det(1 - F_{\rho,p} T)$ and $1 - a_p T + \varepsilon(p) T^2$ belong to Y. Since they are congruent modulo λ_ℓ, they are equal, and

$$\det(1 - F_{\rho,p} T) = 1 - a_p T + \varepsilon(p) T^2 \qquad \text{for every } p \nmid N\ell.$$

Now replace ℓ by another prime number ℓ' in L'. We obtain a representation $\rho' : G \longrightarrow \mathrm{GL}_2(O_{\lambda_{\ell'}})$ having the same property as above, but for $p \nmid N\ell'$. In particular,

$$\det(1 - F_{\rho,p} T) = \det(1 - F_{\rho',p} T) \qquad \text{for } p \nmid N\ell\ell'.$$

By Lemma 3.2, this implies that ρ and ρ' are isomorphic as representations in $\mathrm{GL}_2(K)$, and a fortiori as complex representations. It follows that ρ is unramified outside N and that

$$\det(1 - F_{\rho,p} T) = 1 - a_p T + \varepsilon(p) T^2 \qquad \text{for every } p \nmid N.$$

8.7. It remains to show that ρ is irreducible. If it were not, it would be a sum of two one-dimensional representations; these would correspond to characters χ_1 and χ_2, unramified outside N, such that $\chi_1 \chi_2 = \varepsilon$ and

$$a_p = \chi_1(p) + \chi_2(p) \qquad \text{for } p \nmid N.$$

Then

$$\sum |a_p|^2 p^{-s} = 2 \sum p^{-s} + \sum \chi_1(p) \overline{\chi}_2(p) p^{-s} + \sum \chi_2(p) \overline{\chi}_1(p) p^{-s}.$$

As s tends to 1, $\sum p^{-s} = \log(1/(s-1)) + O(1)$. On the other hand, the character $\chi_1 \overline{\chi}_2$ is not 1 (otherwise one would have $\varepsilon = (\chi_1)^2$ and $\varepsilon(-1) = 1$); it follows (cf., for example, [21], VI.4.2) that

$$\sum \chi_1(p) \overline{\chi}_2(p) p^{-s} = O(1) \qquad \text{and} \qquad \sum \chi_2(p) \overline{\chi}_1(p) p^{-s} = O(1).$$

It follows that

$$\sum |a_p|^2 p^{-s} = 2\log(1/(s-1)) + O(1) \qquad \text{as } s \longrightarrow 1,$$

which contradicts Proposition 5.1 and completes the proof.

§ 9. Application to the Coefficients of Modular Forms of Weight 1

Let

$$f = \sum_{n=0}^{\infty} a_n e^{2\pi i n z/M}, \qquad M \geq 1,$$

be a modular form of weight 1 on a congruence subgroup of $\mathrm{SL}_2(\mathbf{Z})$.

(a) BOUND FOR $|a_n|$

THEOREM 9.1. — One has $|a_n| = O(d(n))$ as $n \longrightarrow \infty$.

(Recall that $d(n)$ denotes the number of divisors ≥ 1 of n.)

COROLLARY 9.2. — One has $|a_n| = O(n^{\delta})$ for every $\delta > 0$.

Indeed, $d(n)$ is known to have this property ([8], th. 315).

Proof of 9.1. — If n_0 is an integer ≥ 1, then $d(n_0 n)/d(n)$ lies between 1 and $d(n_0)$. It is therefore equivalent to prove estimate (9.1) for $f(z)$ or for $f(n_0 z)$, and this allows us to suppose that $M = 1$, i.e. that $f(z+1) = f(z)$. Using 1.5 and 1.9, we are reduced to the following two special cases:

(i) f is an Eisenstein series, in which case (9.1) follows from the formula for the a_n ([9], p. 475);

(ii) f is a primitive cusp form of type $(1, \varepsilon)$ on $\Gamma_0(N)$, for suitable N and ε. In this case one has the more precise result

$$|a_n| \leq d_N(n) \leq d(n), \tag{9.3}$$

where $d_N(n)$ is the number of positive divisors of n that are prime to N. Indeed, by the multiplicativity of a_n and $d_N(n)$, it is enough to verify (9.3) when n is a power p^m of a prime number p. We distinguish two cases:

(ii_1) $p \mid N$.

One has $a_n = (a_p)^m$, and Theorem 4.6 shows that a_p is either 0 or a root of unity. Hence

$$|a_n| \leq 1 = d_N(n).$$

(ii_2) $p \nmid N$.

If the polynomial $1 - a_p T + \varepsilon(p) T^2$ is written in the form $(1-\lambda T)(1-\mu T)$, then

$$a_n = \lambda^m + \lambda^{m-1}\mu + \cdots + \lambda\mu^{m-1} + \mu^m.$$

By Theorem 4.1, λ and μ are roots of unity. Thus

$$|a_n| \leq m + 1 = d_N(n).$$

REMARK 9.4. — If $f = \sum b_n e^{2\pi inz/M}$, $M \geq 1$, is a cusp form of weight $k \geq 2$ on a congruence subgroup of $\mathrm{SL}_2(\mathbf{Z})$, the same argument as above (using [5], 8.2) shows that

$$|b_n| = O(n^{(k-1)/2} d(n)) \qquad \text{as } n \longrightarrow \infty.$$

(b) MAXIMAL ORDER OF MAGNITUDE OF $|a_n|$. — It is known ([8], th. 317) that the "maximal" order of magnitude of $d(n)$ is $2^{\log n/\log\log n}$, in the sense that

$$\limsup \frac{\log d(n) \log\log n}{\log n} = \log 2.$$

The same result holds for the $|a_n|$:

PROPOSITION 9.5. — If $f \neq 0$, then

$$\limsup \frac{\log |a_n| \log\log n}{\log n} = \log 2.$$

LEMMA 9.6. — Let N be an integer ≥ 1. There exist sets X_N and Y_N of prime numbers, of densities > 0, such that:

(x) For every $p \in X_N$, one has $p \equiv 1 \pmod N$ and $g \mid T_p = 2g$ for every modular form g of weight 1 on $\Gamma_1(N)$;

(y) For every $p \in Y_N$, one has $p \equiv -1 \pmod N$ and $g \mid T_p = 0$ for every modular form g of weight 1 on $\Gamma_1(N)$.

Let $\rho_1, \ldots, \rho_h$ be the representations of G associated with the different systems of eigenvalues of the T_p acting on forms of type $(1, \varepsilon)$ on $\Gamma_0(N)$, as ε ranges over the odd characters of $(\mathbf{Z}/N\mathbf{Z})^*$. Let X_N be the set of $p \equiv 1 \pmod N$ such that

$$F_{\rho_i, p} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

for $i = 1, \ldots, h$, and let Y_N be the set of $p \equiv -1 \pmod N$ such that $F_{\rho_i, p}$ is conjugate to $\rho_i(c)$, cf. 4.5. By the Čebotarev density theorem, X_N and Y_N have densities > 0. If $p \in X_N$, then 2 is the only eigenvalue of T_p; since T_p is semisimple, one indeed has $g \mid T_p = 2g$ for every g. The same argument shows that $g \mid T_p = 0$ if $p \in Y_N$, since the trace of the matrix $\rho_i(c)$ is 0.

PROOF OF 9.5. — As in (a), we reduce to the case in which f is a modular form of weight 1 on $\Gamma_1(N)$. Let X_N be as in Lemma 9.6, and choose an integer m such that $a_m \neq 0$. If x is an integer ≥ 1, let $p_1, \ldots, p_{i(x)}$ be the distinct prime numbers $p \in X_N$ that are $\leq x$ and do not divide m. Put $n(x) = m p_1 p_2 \cdots p_{i(x)}$. Since the p_i belong to X_N, one has $f \mid T_{p_i} = 2f$ and $f \mid R_{p_i} = f$; by (2.5.1), this gives

$$a_{n(x)} = 2^{i(x)} a_m,$$

and hence

$$\log |a_{n(x)}| \sim i(x) \log 2 \qquad \text{as } x \longrightarrow \infty.$$

If c is the density of X_N, then $i(x) \sim cx/\log x$ and $\sum_{i \le i(x)} \log p_i \sim cx$. It follows that

$$\log |a_{n(x)}| \sim cx \log 2/\log x,$$

$$\log n(x) \sim cx,$$

$$\log \log n(x) \sim \log x,$$

and hence

$$\limsup \frac{\log |a_n| \log \log n}{\log n} \ge \log 2.$$

The opposite inequality follows from $|a_n| = O(d(n))$.

(c) NORMAL ORDER OF MAGNITUDE OF $|a_n|$. — The "normal" (i.e. most frequent) order of magnitude of $d(n)$ is $2^{\log \log n}$ (cf. [8], th. 432). That of $|a_n|$ is smaller:

PROPOSITION 9.7. — The set of n such that $a_n = 0$ has density 1.

(A subset S of $\mathbf{N}$ is said to have density c if the number of elements of S that are $\le x$ is $cx + o(x)$ as $x \longrightarrow \infty$.)

Here again, we may suppose that f is a modular form of weight 1 on $\Gamma_1(N)$. Let Y_N be as in Lemma 9.6. If $p \in Y_N$, then $f \mid T_p = 0$ and $f \mid R_p = -f$. By (2.5.1), it follows that if n is divisible by p but not by p^2, then $a_n = 0$. Now, if Y is a finite set of prime numbers, the set S_Y of integers n having the preceding property (for at least one $p \in Y$) has density

$$1 - \prod_{p \in Y} \left(1 - \frac{p-1}{p^2}\right).$$

Since Y_N has density > 0, the series $\sum_{p \in Y_N} 1/p$ diverges, and the product

$$\prod_{p \in Y_N} \left(1 - \frac{p-1}{p^2}\right)$$

has value 0. It follows that the union of the S_Y, $Y \subset Y_N$, has density 1, which proves the proposition.

REMARK 9.8. — For each x, let $M(x)$ denote the number of $n \le x$ such that $a_n \ne 0$. Proposition 9.7 says that

$$M(x) = o(x) \qquad \text{as } x \longrightarrow \infty.$$

Using Theorem 2 of [23], one can prove the following more precise result: there exists $\alpha > 0$ such that

$$M(x) = O(x/\log^{\alpha} x) \qquad \text{as } x \longrightarrow \infty.$$

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Pierre DELIGNE,
I.H.E.S.,
91440 Bures-sur-Yvette
and
Jean-Pierre SERRE,
Collège de France,
75231 Paris-Cedex 05

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Weights in the Cohomology of Algebraic Varieties

Pierre Deligne

1. Let X be a complex algebraic variety (i.e., a separated scheme of finite type over $\mathbb{C}$). We also write X for the usual topological space $X(\mathbb{C})$ underlying X. In this lecture, we describe a remarkable filtration on the rational cohomology groups of X, the *weight filtration*, and we give a collection of results concerning its properties. Its definition will be given in §12. For proofs, we refer to the works cited in the bibliography, where the theorems are often proved in more general settings; working over $\mathbb{C}$ allows us to use simultaneously Hodge theory, actions of Galois groups, and resolution of singularities.

The weight filtration is a finite increasing filtration. We denote it by W. It is also defined in relative situations (or in cohomology with proper support). It depends not only on the topological space X, but also on the way in which X is realized as an algebraic variety. It is compatible with the Künneth isomorphisms (i.e., under the isomorphism $H^*(X\times Y) = H^*(X)\otimes H^*(Y)$, one has

$$W_j\big(H^*(X\times Y)\big) = \sum_{j'+j''=j} W_{j'}\big(H^*(X)\big)\otimes W_{j''}\big(H^*(Y)\big)$$

and is functorial for algebraic morphisms. More precisely, if $f : X \to Y$ is algebraic, then $f^* : H^*(Y) \to H^*(X)$ is strictly compatible with the weight filtrations on $H^*(Y)$ and $H^*(X)$: if the class $x \in H^i(X)$ lies in the image of f^*, it has filtration $\leq i$ if and only if it is the image of a class of filtration $\leq i$. More generally, every natural map is strictly compatible with the weight filtrations.

The weight filtration is a discrete invariant; it is invariant under de-

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formation of the algebraic structure of X. More precisely, one has the following theorem.

2. THEOREM. *Let $f : X \to S$ be a morphism. For $t \in S$, let $X_t = f^{-1}(t)$. If the sheaf $R^i f_* \mathbb{Q}$ is locally constant (a local system), then there exists a weight filtration W on $R^i f_* \mathbb{Q}$ by local subsystems, such that the maps $r_t : (R^i f_* \mathbb{Q})_t \to H^i(X_t)$ are strictly compatible with the weight filtrations and, in particular, wherever r_t is an isomorphism, W induces the weight filtration on $H^i(X_t)$.*

3. Roughly speaking, the weight filtration expresses how the cohomology of X can be built in terms of the cohomology of nonsingular projective varieties. Here are some examples.

Example 3.1. If X is proper (= compact, for example projective) and smooth (= nonsingular), then $H^i(X) =_{\text{def}} H^i(X, \mathbb{Q})$ is pure of weight i: $\mathrm{Gr}_j^W(H^i(X)) = 0$ for $i \neq j$. In other words, $W_{i-1}(H^i(X)) = 0$ and $W_i(H^i(X)) = H^i(X)$.

Example 3.2. Let X be proper, smooth, connected, of dimension d, and let P be a point of X. Among the cohomology groups with support $H^i_{\{P\}}(X) = H^i(X \bmod (X - \{P\}))$, only the one in degree $2d$ is nonzero, and

$$H^{2d}_{\{P\}}(X) \simeq H^{2d}(X) = \mathbb{Q}. \tag{3.2.1}$$

By our principles, $H^{2d}_{\{P\}}(X)$ is therefore pure of weight $2d$. The inverse of the isomorphism (3.2.1) may be viewed as a Thom–Gysin isomorphism

$$\mathbb{Q} = H^0(P) \simeq H^{2d}_{\{P\}}(X);$$

one observes that it does not preserve weights. The general situation is as follows: for Y a smooth subvariety of pure codimension d in a smooth variety X, the Thom–Gysin isomorphism $H^i(Y) \simeq H^{i+2d}_Y(X)$ carries W^k to W^{k+2d}. Writing (n) for a shift of $2n$ in W ($W(n)_k = W_{k-2n}$), this becomes a filtered isomorphism

$$H^i(Y)(-d) \simeq H^{i+2d}_Y(X).$$

Example 3.3. Let X be proper and smooth, and let Y be a smooth (closed) subvariety of pure codimension d. There is an exact sequence

$$\cdots \to H^i(X) \to H^i(X - Y) \xrightarrow{\delta} H^{i+1}_Y(X) \to \cdots$$

By 3.1 and 3.2, $\mathrm{Gr}_j^W(H^i(X - Y)) = 0$ for $j \neq i, i+1$; W_i is the image of $H^i(X)$.

Example 3.4. Let X be proper and smooth except at an isolated singular point P. Suppose that the variety $\widetilde{X}$ obtained from X by blowing up P is smooth, and that the exceptional divisor D, the inverse image of P, is smooth: X is obtained from $\widetilde{X}$ (proper and smooth) by contracting D (proper and smooth) to a point. The space X has the homotopy type of [$\widetilde{X} \cup$ (a cone with base D)], whose cohomology is computed by Mayer–Vietoris; one obtains an exact sequence

$$\cdots \to H^{i-1}(D) \xrightarrow{\delta} H^i(X) \to H^i(\widetilde{X}) \oplus H^i(P) \to \cdots$$

which shows that $\mathrm{Gr}_j^W(H^i(X)) = 0$ for $j \neq i-1, i$; W_{i-1} is the image of δ. For

$i \neq 0$, it is also the kernel of $H^i(X) \to H^i(\widetilde{X})$, and $\mathrm{Gr}_i^W(H^i(X))$ is the image of $H^i(X)$ in $H^i(\widetilde{X})$.

Example 3.5. Flag varieties are proper and smooth. The weight filtration on their cohomology is therefore given by 3.1. If G is a connected reductive group, the same result holds for the cohomology of BG [1, III]. This makes it possible to calculate the weight filtration on the cohomology of G, which is related by transgression to that of BG. One finds that $W_iH^i(G) = 0$, and that $W_{i+1}H^i(G)$, zero for even i, is equal to the primitive part of the degree-i cohomology of G [1, III]. If $f : G \to H$ is an algebraic map between varieties of reductive groups, the inverse image of a primitive rational cohomology class of H is therefore again primitive. For other corollaries, see [1, III].

4. The weight filtration admits a grading in a very strong sense. There exist gradings $\underline{W}$ of the $H^i(X)$ that split W:

$$W_n(H^i(X)) = \bigoplus_{j \leq n} \underline{W}_j(H^i(X)) \tag{4.1}$$

and that are compatible with the cup product and with higher operations derived from the cup product (Massey products, ...). Since the latter are not defined everywhere, the meaning of "compatible with a grading" must be made precise. The simplest approach is to regard a grading as an action of the group G_m, i.e., an action of $\mathbb{Q}^*$ given by algebraic formulas: to a grading $\underline{W}$ one associates the action in which $\lambda \in \mathbb{Q}^*$ acts on $\underline{W}_j$ by multiplication by λ^j. "Compatibility" means that $\mathbb{Q}^*$ acts by automorphisms of $H^*(X)$, together with its cohomological grading, cup product, and the higher operations derived from the cup product.

5. For simplicity, suppose that X is connected, and let $\mathscr{M}$ be the minimal model of the rational homotopy type of X in the sense of Sullivan [7]. It is a differential graded algebra in degrees ≥ 0, connected ($\mathscr{M}^0 = \mathbb{Q}$), free (anti)commutative as a graded algebra, and generated by its indecomposable elements (i.e., $d\mathscr{M} \subset (\mathscr{M}^{>0})^2$). One has $H^*(\mathscr{M}) = H^*(X)$, and if X is simply connected, $\mathscr{M}^{>0}/(\mathscr{M}^{>0})^2 = (\pi_*(X) \otimes \mathbb{Q})^\vee$.

A more precise, and more convenient, statement than the one in §4 is that there exists a grading $\underline{W}$ satisfying (4.1), induced by a grading $\underline{W}$ of $\mathscr{M}$ (in degrees ≥ 0, the sum of gradings of the $\mathscr{M}^i$), such that d and the product are homogeneous of degree 0; in other words, $\mathbb{Q}^*$ acts by automorphisms of $\mathscr{M}$).

6. The mere existence of W and $\underline{W}$ imposes no restriction on the homotopy type of X: one may always declare that all of $\mathscr{M}$ has weight 0. Moreover, every finite polyhedron has the same homotopy type as an algebraic variety. Indeed, let S be a finite set furnished with a collection $\mathscr{S}$ of subsets (the simplices). Assume that every subset of an element of $\mathscr{S}$ also belongs to $\mathscr{S}$. Identify S with the set of basis vectors of $\mathbb{R}^S$, and for $\sigma \subset S$ let $|\sigma|$ be the simplex spanned by the $s \in \sigma$. We show that $|S| = \bigcup_{\sigma \in \mathscr{S}} |\sigma|$ has the homotopy type of an algebraic variety X. If $|\sigma|_{\mathbb{C}}$ denotes the complex affine space spanned by $\sigma \subset \mathbb{C}^S$, it is enough to take $X = \bigcup_{\sigma \in \mathscr{S}} |\sigma|_{\mathbb{C}}$, of which $|S|$ is a deformation retract. One can veri-

fy that in this example $H^*(X)$ is pure of weight-filtration degree 0.

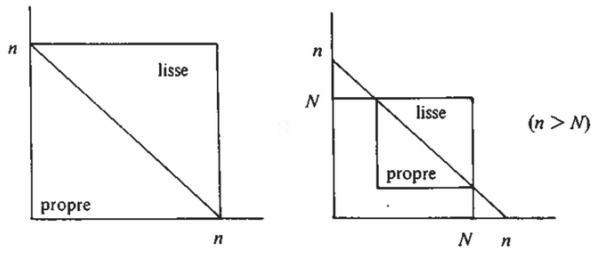
Nevertheless, as soon as the weights are nonzero, constraints appear on the homotopy type of X (cf. §10). The following rules help locate the weights.

7. For a cohomology group H of a given type, each rule will consist in describing a subset $\mathscr{E}$ of $\mathbb{Z}\times\mathbb{Z}$. The weights are controlled by $\mathscr{E}$ in the following sense: if $\mathrm{Gr}_n^W(H)\neq 0$, then there exists $(p,q)\in\mathscr{E}$ with $p+q=n$. This manner of speaking, which may seem artificial here, is not so: information other than the possible weights is controlled by the same region $\mathscr{E}$; cf. §15.

(7.1) $H^n(X)$ is controlled by the square $[0,n]\times[0,n]$.

(7.2) If X is proper, $H^n(X)$ is controlled by the part of this square below the second diagonal, namely $\{(p,q)\in[0,n]\times[0,n]\mid p+q\leq n\}$. If X is smooth, $H^n(X)$ is controlled by the part of this square above that diagonal, namely $\{(p,q)\in[0,n]\times[0,n]\mid p+q\geq n\}$.

(7.3) If $N=\dim X\leq n$, the square $[0,n]\times[0,n]$ may be replaced by the square $[n-N,N]\times[n-N,N]$.



8. Variants. **(8.1)** The compactly supported cohomology group $H_c^n(X)$ is controlled by the part of the square $[0,n]\times[0,n]$ below the second diagonal. If $N=\dim X\leq n$, this square may again be replaced by $[n-N,N]\times[n-N,N]$. In particular, $H_c^{2N}(X)$ is pure of weight $2N$.

(8.2) The "duality" between the proper and smooth cases follows from Poincaré duality: for X smooth and purely of dimension N,

$$H^n(X)=\mathrm{Hom}\big(H_c^{2N-n}(X),H_c^{2N}(X)\big)=\big(H_c^{2N-n}(X)^\vee\big)(-N).$$

This argument extends the results given for smooth X to the case in which X is a "rational homology manifold." For such an X, the image of $H_c^n(X)$ in $H^n(X)$ is pure of weight n.

(8.3) If X is only assumed normal, it remains true that $H^1(X)$ is controlled by $\{(0,1),(1,0),(1,1)\}$.

9. Let X be a connected variety, $x\in X$, and let $\Pi^{(n)}$ be the largest torsion-free nilpotent quotient of length n of $\pi_1(X,x)$. It is known that finite-index subgroups H of $\Pi^{(n)}$ that are sufficiently small have the following property.

(*) There exists a nilpotent Lie algebra of length n, $(\mathscr{H},[\ ,\ ])$, with $\mathscr{H}$ a free

$\mathbb{Z}$-module, such that the Campbell–Hausdorff formula

$$x \circ y = x + y + [x, y]/2 + [x, [x, y]]/12 + [y, [y, x]]/12 + \cdots$$

makes $\mathscr{H}$ into a group isomorphic to H.

Moreover, the Lie algebra $\mathscr{L}^{(n)} = \mathscr{H} \otimes \mathbb{Q}$ depends only on $\Pi^{(n)}$, not on H.

10. THEOREM (MORGAN). *If X is normal, $\mathscr{L}^{(n)}$ admits a grading $\underline{W}$ in degrees < 0, such that $\mathscr{L}^{(n)}$ is generated by elements of degrees -1 and -2, subject only to relations of degrees -2, -3, and -4, together with the vanishing of n-fold iterated commutators.*

It is more convenient to work with the inverse system $\mathscr{L}^{(\infty)}$ of the $\mathscr{L}^{(n)}$. Defining $H^i(\mathscr{L}^{(\infty)}) = \varinjlim H^i(\mathscr{L}^{(n)})$, one has

$$H^1(\mathscr{L}^{(\infty)}) = H^1(X), \qquad H^2(\mathscr{L}^{(\infty)}) \subset H^2(X).$$

Moreover, a grading $\underline{W}$ of $\mathscr{M}$ as in §5 defines compatible gradings of $\mathscr{L}^{(\infty)}$, its cohomology, and that of X. By (8.3) (respectively, (7.1)), $H^1(X)$ has weights 1 and 2 and $H^2(X)$ has weights ≤ 4. Consequently, $H^1(\mathscr{L}^{(\infty)})$ (the dual of the group $\mathscr{L}^{(\infty)}/[\mathscr{L}^{(\infty)}, \mathscr{L}^{(\infty)}]$ of generators of $\mathscr{L}^{(\infty)}$) has weights 1 and 2, and $H^2(\mathscr{L}^{(\infty)})$, the dual of the group of relations among generators, has weights ≤ 4. This proves the theorem.

11. For a proper smooth variety, the weight filtration reduces to very little (3.1). Its gradability in the sense of §4 implies the vanishing of all Massey products. Its gradability in the sense of §5 even implies that the rational homotopy type of X can be read off from its cohomology algebra—see [3].

12. Hodge theory. The cohomology group $H^i(X)$ of every complex algebraic variety is endowed with a mixed Hodge structure (W, F) (W is a filtration on $H^i(X)$, and F is a filtration on $H^i(X) \otimes \mathbb{C} = H^i(X, \mathbb{C})$, with W and F satisfying suitable axioms—see [1]). The weight filtration is, by definition, the filtration W.

13. l-adic theory. An algebraic variety X is defined by finitely many polynomial equations. The coefficients of these equations generate a finitely generated subring R of $\mathbb{C}$, with fraction field K, and X is obtained by extension of scalars to $\mathbb{C}$ from a scheme over K, or even over R. The following statements become true after replacing R by $R[1/f]$, with $f \in R$ sufficiently divisible.

(13.1) Let l be a prime number. The group $H^i(X) \otimes \mathbb{Q}_l$ is identified with the l-adic cohomology group $H^i(X, \mathbb{Q}_l)$. The latter is defined in a purely algebraic manner and therefore carries, by transport of structure, an action of $\mathrm{Aut}(\mathbb{C}/K)$.

(13.2) Let $\overline{K}$ be the algebraic closure of K in $\mathbb{C}$. The action factors through $\mathrm{Gal}(\overline{K}/K)$. It is unramified over $R[1/l]$, i.e., it factors through the Galois group of the largest extension $K_{\mathrm{nr}}^{(l)} \subset \overline{K}$ of K unramified over $R[1/l]$.

(13.3) Let $\mathfrak{m}$ be a maximal ideal of $R[1/l]$, and let $N(\mathfrak{m}) = \#R/\mathfrak{m}$ ($R/\mathfrak{m}$ is a finite field). To $\mathfrak{m}$ there corresponds a conjugacy class of Frobenius substitutions $\varphi_{\mathfrak{m}} \in \mathrm{Gal}(K_{\mathrm{nr}}/K)$, whose inverses are the geometric Frobenius substitutions $F_{\mathfrak{m}} = \varphi_{\mathfrak{m}}^{-1}$.

14. THEOREM. *For f sufficiently divisible, one has*

(i) *For every l and $\mathfrak{m}$, the eigenvalues α of $F_{\mathfrak{m}}$ (in a suitable finite extension of $\mathbb{Q}_l$) are algebraic integers. For each α, there is an integer $w(\alpha)$ such that all complex conjugates of α have absolute value $N(\mathfrak{m})^{w(\alpha)/2}$.*

(ii) *Choose a geometric Frobenius $F_{\mathfrak{m}}$, and let ${}_{\mathfrak{m}}W_j$ be the sum of the generalized eigenspaces corresponding to eigenvalues α of $F_{\mathfrak{m}}$ such that $w(\alpha) = j$. The filtration by the $\bigoplus_{j\le i} {}_{\mathfrak{m}}W_j$ is independent of $\mathfrak{m}$ and of the choice of $F_{\mathfrak{m}}$; it is rational, and its trace on $H^i(X)$ is independent of l; it is the weight filtration.*

Principle of proof. One expresses the cohomology of X, by means of a spectral sequence, in terms of the cohomology of proper smooth varieties, as in [1]. The spectral sequence abuts to a filtration W on $H^*(X)$, which is by definition the weight filtration of Hodge theory. This spectral sequence has an l-adic analogue; the l-adification W_l of W is therefore stable under a Galois group. The Weil conjecture implies that the eigenvalues of $F_{\mathfrak{m}}$ on $\mathrm{Gr}_j^{W_l}$ are algebraic integers whose complex absolute values are $N(\mathfrak{m})^{j/2}$, and the theorem follows.

15. In the cases considered in (7.1), (7.2), and (7.3), the region $\mathscr{E}$ described there controls not only the weights, but also the Hodge numbers and the divisibility of the eigenvalues of the $F_{\mathfrak{m}}$: in (7.1), (7.2), and (7.3), a region $\mathscr{E}$ has been assigned to a group H, and

(a) The nonzero Hodge numbers h^{pq} of the Hodge structures $\mathrm{Gr}_j^W(H)$ satisfy $(p,q) \in \mathscr{E}$.

(b) (Useful only in case (7.3).) If the $(p,q) \in \mathscr{E}$ with $p+q=j$ satisfy $p,q \ge k$, then (still for f sufficiently divisible) the eigenvalues of $F_{\mathfrak{m}}$ with complex absolute value $N(\mathfrak{m})^{j/2}$ are divisible by $N(\mathfrak{m})^k$.

One hopes that this is a special case of a general principle; cf. [4].

16. The following result is clear from the viewpoint of Hodge theory.

(16.1) For odd j, $\dim \mathrm{Gr}_j^W H^i(X)$ is even.

On the other hand, up to now I know how to prove §§ 2 and 5 only by l-adic methods. For §5, the method of §14 yields an l-adic grading; a device of Sullivan then gives the existence of rational gradings of the desired kind.

The functoriality properties of the weight filtration are evident from the l-adic viewpoint (because Galois commutes with everything that can be defined). It is nevertheless useful to prove them from the viewpoint of Hodge theory in order to obtain analogous properties for the filtration F. Morgan has obtained numerous results in this direction—enough to prove by Hodge theory a result slightly weaker than §10.

17. As §5 suggests, a weight filtration also exists on homotopy groups (tensored with $\mathbb{Q}$) under simple-connectivity hypotheses.

It also exists on groups of vanishing cycles (cf. [6]), and I hope that these filtrations will help us to understand the latter better.

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Institut des Hautes Etudes Scientifiques
91440 Bures-sur-Yvette, France

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Real Homotopy Theory of Kähler Manifolds

Pierre Deligne (Bures-sur-Yvette), Phillip Griffiths* (Cambridge, Mass.), John Morgan (New York) and Dennis Sullivan (Bures-sur-Yvette)

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This paper brings together the strengthened de Rham theory of the last author [12, 13] and [14] concerning the homotopy information of a manifold contained in the algebra of its differential forms, and the classical results [15] about the forms on a compact Kähler manifold to prove statements about the algebraic topology of compact Kähler manifolds, and consequently about smooth, complex, projective algebraic varieties.

The nature of the results is that the *homotopy type of a compact Kähler manifold, over the real numbers, is a formal consequence of the real cohomology ring.* For any manifold there is a differential graded algebra with rational structure constants whose isomorphism type is an invariant of the homotopy type of the space. The cohomology ring of this differential algebra is the rational cohomology ring of the space, but the algebra itself contains, for example, the higher order cohomology products. The isomorphism type of the differential algebra is equivalent to the "rational form of the nilpotent structure in the space". For a simply connected manifold this differential algebra is equivalent to the rational Postnikov tower of manifold. Our results here are that, for a compact Kähler manifold, from the real cohomology ring one can produce the real form of this differential algebra. These results complement the known structure of the cohomology ring of such a manifold. We prove no new results about it.

The initial motivation for these results was the relation in a Kähler manifold between harmonic forms and holomorphic forms. In particular the $(p, 0)$ harmonic forms for any Kähler metric are exactly the holomorphic forms. Consequently, products of $(p, 0)$ and $(q, 0)$ harmonic forms are harmonic[1], and thus there are no

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1 Chen in [2] was the first to exploit this idea to obtain higher order information about Kähler manifolds.

higher order products starting from $H^{*,0}$. The possibility that the full theorem would be true is related to the Weil conjectures. We were led to prove such a complete result by "using the Weil conjectures in characteristic p to guess results in characteristic 0 which can then be proved using Hodge theory". The relevant remark here is that the classical higher order cohomology products of Massey are multilinear operations, but the dimension of the product is always less than the sum of the dimensions of the various elements being multiplied. Thus, if we were working in a situation in characteristic p on which the Frobenius automorphism operated (for instance, the Čech cochains in étale cohomology theory), and if the eigenvalues of the Frobenius action are as predicted by the Weil conjectures[2], then the multilinearity of the product over the action would imply that the absolute value of the eigenvalues on the space of higher order products in a certain dimension would be different from the possible values in that dimension. The only way this can happen is for these spaces of higher order products to be zero.

In proving these results for compact Kähler manifolds we use the implications for the forms of the existence of a Kähler metric. If M is a complex manifold, form the differential algebras and algebra maps

$$\{H^*(M,\mathbb{R}), d=0\} \longleftarrow \{\mathrm{Ker}(d^c), d\} \longhookrightarrow \{\text{all } C^\infty \text{ forms}, d\}.$$

If M is a compact Kähler manifold, both maps above induce isomorphisms of cohomology. All our results about the homotopy type of compact Kähler manifolds can be deduced from this statement, which is itself a consequence of the dd^c lemma.

dd^c **Lemma.** *Let M be a compact Kähler manifold, and $d^c = J^{-1}dJ$, where J gives the complex structure in the cotangent bundle. Evidently, d^c is a real operator. If x is a form with*

1) $dx = 0 = d^c x$

and

2) $x = dy$ or $x = d^c y'$

then $x = dd^c z$ for some form z.

In Sections 1–4 we describe the machinery needed to deduce the consequences in homotopy theory of these statements about forms. Sections 1, 2, and 3 constitute a sketch of the theory in [12] and [14], (c.f. also [10]). Section 1 describes the construction of the minimal model (real Postnikov tower) of any differential algebra and proves a uniqueness result. To do this, some of the abstract homotopy theory of differential algebras is developed. For instance, we define the de Rham fundamental group of an algebra by using a minimal model for the algebra. This "group" is an inverse system of nilpotent Lie groups. We also define the higher homotopy groups of any simply connected differential algebra. These are vector spaces with a graded Lie algebra structure which should be thought of as the structure of "Whitehead products".

In Section 2 we assign to any simply connected[3], C^∞-manifold, M, the de Rham complex of C^∞-forms on M, $\mathcal{E}^*_M$, and its minimal model $\mathcal{M}_M$; and to any simply

2 Which is now [5].

3 Actually this form of theory works just as well for nilpotent spaces, see [13].

connected simplical complex, X, the differential algebra $\mathcal{E}^*_X$, of $\mathbb{Q}$-polynomial forms on X and its minimal model $\mathcal{M}_X$. The former algebras are over $\mathbb{R}$; the latter are over $\mathbb{Q}$. The minimal model $\mathcal{M}_M$ is defined to be the C^∞ de Rham homotopy type of M, and $\mathcal{M}_X$ is, similarly, the $\mathbb{Q}$-de Rham homotopy type of X. These constructions satisfy

a) the cohomology of the $\mathbb{Q}$-polynomial forms on X is the $\mathbb{Q}$-cohomology ring of X, and

b) the inclusions of the C^∞ forms on M and the ($\mathbb{Q}$-polynomial forms on a C^1-triangulation K of M)$\otimes\mathbb{R}$ into the piecewise C^∞ forms on K induce isomorphisms of cohomology.

The fact that the $\mathbb{Q}$-polynomial forms are a $\mathbb{Q}$-differential algebra is useful when we make the connection with homotopy theory in Section 3. There is an isomorphism between the category of $\mathbb{Q}$-minimal differential algebras and the category of towers of rational principal fibrations. The homotopy groups of algebras correspond to the homotopy groups of the fibration under this isomorphism. The main result is that, *for a simply connected space, the tower of rational principal fibrations built from the minimal model of the $\mathbb{Q}$-polynomial forms is the rational form of the Postnikov tower of the space. Thus the homotopy-theoretic information available in the differential algebra of $\mathbb{Q}$-polynomial forms on a simply connected simplical complex is exactly the rational homotopy type*. Consequently the differential algebra of C^∞ forms on a simply connected manifold determines the real form of the rational homotopy type of the manifold. There are analogous results concerning the nilpotent completion of the fundamental group for general simplical complexes and manifolds (see [14] Appendix N).

Section 4 discusses the question of when the minimal model of a differential algebra is determined by the cohomology ring of the algebra. We give examples of spaces where the higher order cohomology products are non-zero, and then show that a minimal model is a formal consequence of its cohomology ring if, and only if, all the higher order products vanish in a uniform way.

Section 5 is a discussion of the forms on a compact, complex manifold admitting a Kähler metric. From the identity (which is equivalent to the metric being Kähler)

$$\Delta_d = 2\Delta_\partial = 2\Delta_{\bar\partial}$$

between the various Laplacians, we prove the dd^c lemma. We then give a cohomological interpretation of this lemma, and show that it is valid on more general complex manifolds (e.g., Moišezon spaces).

In Section 6 we prove the main theorem that:

1) For any complex manifold for which the dd^c lemma is true (for example Kähler manifolds), the minimal model of the C^∞ differential forms is a formal consequence of the real cohomology ring.

2) Given a holomorphic map between two such manifolds, the effect of the map on the minimal models is determined by the map on the cohomology level.

The work of Sections 1, 2, and 3, allows us to translate this into the assertion that the real homotopy type of a compact Kähler manifold is a formal consequence of its real cohomology ring, and the real homotopy type of a holomorphic map

between compact Kähler manifolds is a formal consequence of the induced map on cohomology. In particular, if a compact Kähler manifold M is simply connected, then its real homotopy groups, $\pi_*(M) \otimes \mathbb{R}$, can be read off from the real cohomology ring. Likewise in this case, the effect of a holomorphic map on the real homotopy groups may be read off from the effect of the map on cohomology. In the non-simply connected case, the real nilpotent tower of the fundamental group can be read off from H^1 and the cup product $H^1 \otimes H^1 \to H^2$. The effect of a holomorphic map on the nilpotent tower of π_1 can similarly be read off from the map on H^1.

In Section 7 we give an application of this theory to a union of divisors with normal crossings. Using the fact that forms on a Kähler manifold can be naturally replaced by the cohomology, we give a different proof that a spectral sequence calculating the cohomology degenerates at E_2, i.e., $E_2 = E_\infty$. Here E_1 is built from the cohomologies of the various intersections, and d_1 is built from the various restriction maps.

The third author has studied the open complement of a divisor with normal crossings in a Kähler manifold where related but much more complicated results ensue [20].

Section 1. Homotopy Theory of Differential Algebras

This section, together with Sections 2 and 3, constitutes a sketch of the theory of the last author relating differential algebras and homotopy theory. For a more complete treatment of this theory see [6, 10, 12] and [13]. The discussion in this section is a completely algebraic one in which the fundamental group of any differential algebra and the higher homotopy groups (in fact the homotopy type) of an one-connected differential algebra are defined. This theory is valid for differential algebras over any field containing the rational numbers $\mathbb{Q}$, but the only fields arising in geometric applications are, in addition to $\mathbb{Q}$, the reals $\mathbb{R}$ and the complexes $\mathbb{C}$.

As the terminology suggests, these constructions are motivated by the theory of Postnikov towers and homotopy theory for spaces. The Hirsch lemma of Section 3 shows that the connection is stronger than just an analogy. In fact, the homotopy theory for differential algebras over $\mathbb{Q}$ and the theory of towers of principle fibrations with fibers $K(V, n)$, V a $\mathbb{Q}$-vector space, are equivalent.

A *differential graded algebra* (= differential algebra) is a graded algebra

$$\mathcal{A} = \bigoplus_{k \geq 0} \mathcal{A}^k$$

with a differential $d : \mathcal{A} \to \mathcal{A}$ of degree $+1$, such that

1) $\mathcal{A}$ is graded commutative, i.e.,

$$x \cdot y = (-1)^{k\ell} y \cdot x \qquad (x \in \mathcal{A}^k \text{ and } y \in \mathcal{A}^\ell)$$

2) d is a derivation, i.e.,

$$d(x \cdot y) = dx \cdot y + (-1)^k x \cdot dy \qquad (x \in \mathcal{A}^k),$$

and

3) $d^2 = 0$.

The cohomology $H^*(\mathcal{A})$ is an algebra. We shall always assume that it is finite dimensional in each degree. $\mathcal{A}$ is connected if $H^0(\mathcal{A})$ is the ground field, and $\mathcal{A}$ is one-connected if, in addition, $H^1(\mathcal{A}) = 0$. A map between two differential algebras $\mathcal{A}$ and $\mathcal{B}$ is an algebra homomorphism preserving the grading and d. Such a map, $f : \mathcal{A} \to \mathcal{B}$, induces an algebra map $f^* : H^*(\mathcal{A}) \to H^*(\mathcal{B})$.

Examples of Differential Algebras. (i) the de Rham complex $\mathcal{E}^*_M$ of C^∞ forms on a manifold M;

(ii) the cohomology ring of a CW complex X,

$$\{H^*(X;k); d = 0\} \qquad (k \text{ a field}).$$

The cochains on a simplicial complex form only a non-commutative differential algebra, even though they induce on cohomology a graded commutative algebra structure. If $\mathcal{A}^0$ is the ground field, then we define the augmentation ideal $A(\mathcal{A}) = \bigoplus_{k>0} \mathcal{A}^k$, and the graded space of indecomposables, $I(\mathcal{A}) = A(\mathcal{A})/A(\mathcal{A}) \cdot A(\mathcal{A})$. In such an algebra the derivation d is decomposable if for every $x \in \mathcal{A}$, $dx \in A \cdot A$. This is equivalent to saying that $d(x)$ is a sum of products of elements of positive degree. By the free algebra on a vector space V_n whose elements are of degree n, $\Lambda(V_n)$ or just (Λ_n), we mean the polynomial algebra generated by V_n if n is even, or the exterior algebra generated by V_n if n is odd. A tensor product (in the graded sense) of such algebras is again a free algebra. The algebraic analogue of a principal fibration is an *elementary extension*. An elementary extension of $(\mathcal{A}, d_{\mathcal{A}})$ is any algebra of the form $(\mathcal{B} = \mathcal{A} \otimes \Lambda(V_k), d_{\mathcal{B}})$ where $d_{\mathcal{B}}|_{\mathcal{A}} = d_{\mathcal{A}}$, $d_{\mathcal{B}}(V_k) \subset \mathcal{A}$, and V_k is a finite dimensional vector space.[4] We denote such extensions by $\mathcal{A} \otimes_d \Lambda(V_k)$. Note that if $\mathcal{A}$ is free as an algebra, then so is $\mathcal{B}$, and if d is decomposable in $\mathcal{A}$, then it is also decomposable in $\mathcal{B}$ if, and only if, $d(V_k) \subset A(\mathcal{A}) \cdot A(\mathcal{A})$. Our first main definition is that $\mathcal{M}$ is a *minimal differential algebra* if $\mathcal{M}$ may be written as an increasing union of sub-differential algebras

$$\text{ground field} \subset \mathcal{M}_1 \subset \mathcal{M}_2 \subset \cdots, \qquad \bigcup_{i\geq 0} \mathcal{M}_i = \mathcal{M}$$

with $\mathcal{M}_i \subset \mathcal{M}_{i+1}$ an elementary extension, and with d decomposable. Such a collection of sub-differential algebras is a *series for* $\mathcal{M}$. If $\mathcal{M}$ is minimal, then

1) d is decomposable, and

2) $\mathcal{M}$ is free as an algebra.

If we let $\{\mathcal{M}^{(i)}\}$ denote the subalgebra of $\mathcal{M}$ generated by elements of degree $\leq i$, then d being decomposable implies that these are sub-differential algebras. If $\mathcal{M}$ is any one-connected differential algebra satisfying 1) and 2), then $\mathcal{M}^1 = 0$, and the sequence $\mathcal{M}^{(2)} \subset \mathcal{M}^{(3)} \subset \cdots$ is a series for $\mathcal{M}$. Thus a one-connected algebra satisfying 1) and 2) has a canonical series. For any minimal differential algebra $\mathcal{M}$,

4 Always placed in a degree greater than zero.

$\mathcal{M}^{(1)}$ also has a canonical series

$$0 \subset \mathcal{M}_1^{(1)} \subset \mathcal{M}_2^{(1)} \subset \cdots, \qquad \bigcup_{i\geq 0} \mathcal{M}_i^{(1)} = \mathcal{M}^{(1)}$$

defined by: $\mathcal{M}_i^{(1)}$ = algebra generated by all $x \in \mathcal{M}^{(1)}$ such that $dx \in \mathcal{M}_{i-1}^{(1)}$.
Given a finitely generated free differential algebra $\mathcal{K}$ with decomposable d, we can form the dual space to the indecomposables

$$\mathcal{L}_{\mathcal{K}} = \bigoplus \mathcal{L}_i, \qquad \mathcal{L}_i = [I(\mathcal{K})^i]^*.$$

Since d is decomposable, it induces a map $d : I(\mathcal{K}) \to I(\mathcal{K}) \otimes I(\mathcal{K})$ whose dual is a map $\mathcal{L}_i \otimes \mathcal{L}_j \to \mathcal{L}_{i+j-1}$. This makes $\mathcal{L}$ into a graded Lie algebra. An elementary extension $\mathcal{K} \to \mathcal{K} \otimes_d \Lambda(V_n)$ with d decomposable dualizes to a central extension of Lie algebras

$$0 \to V_n^* \to \mathcal{L}_{\mathcal{K}\otimes_d \Lambda(V_n)} \to \mathcal{L}_{\mathcal{K}} \to 0$$

with $[v, x] = 0$ for $v \in V_n^*$ and $x \in \mathcal{L}_{\mathcal{K}\otimes_d \Lambda(V_n)}$. If $0 \subset \mathcal{M}_1 \subset \mathcal{M}_2 \subset \cdots$ is a series of a minimal algebra $\mathcal{M}$, then let $\mathcal{L}^i = [I(\mathcal{M}_i)]^*$. We have an inverse system of central extensions of Lie algebras

$$\cdots \to \mathcal{L}^3 \to \mathcal{L}^2 \to \mathcal{L}^1 \to 0.$$

By induction, each $\mathcal{L}^i$ is nilpotent. For a general differential algebra, $\mathcal{M}$, satisfying 1) and 2) above it is this extra condition of nilpotence on the dual Lie algebra, $\mathcal{L}$, which is equivalent to minimality. This nilpotence is automatic if $\mathcal{M}^1 = 0$, as is easily verified by a degree argument.
Example. Let $S = \Lambda_1(x, y, z)$, $dx = y \wedge z$, $dy = z \wedge x$, $dz = x \wedge y$. S is not minimal, and the dual Lie algebra to S is $\mathrm{so}(3)$ which is simple, not nilpotent.
In the one-connected case our inverse system of Lie algebras is eventually stable in each degree (at $\mathcal{L}^n$ we are adding generators in degree n), and thus coalesces into one Lie algebra $\mathcal{L} = \bigoplus_{n\geq 2} \mathcal{L}_n$. Each $\mathcal{L}_n$ is finitely generated, and we have $[\ ,\ ] : \mathcal{L}_n \otimes \mathcal{L}_k \to \mathcal{L}_{n+k-1}$. The vector space $\mathcal{L}_n$ is defined to be the n-th *de Rham homotopy group* of $\mathcal{M}$, and the graded Lie algebra bracket is called the Whitehead product. In the case of a non-simply connected minimal differential algebra, $\mathcal{M}$, we have associated to $\mathcal{M}^{(1)}$ a canonical series, and dually a canonical inverse system of nilpotent Lie algebras $\cdots \to \mathcal{L}_1^2 \to \mathcal{L}_1^1 \to 0$. $\mathcal{L}_1^n$ is nilpotent of order n. The Campbell-Hausdorff formula

$$X \cdot Y = X + Y + \tfrac{1}{2}[X, Y] + \cdots,$$

then defines nilpotent Lie group structures on each $\mathcal{L}_1^n$. This tower of nilpotent Lie groups is the *de Rham fundamental group* of $\mathcal{M}$.

The above discussion can be applied to any differential algebra by the following idea. If $\mathcal{A}$ is a differential algebra, then $\mathcal{M}_{\mathcal{A}} \xrightarrow{\rho} \mathcal{A}$ is a *k-stage minimal model for $\mathcal{M}$* if

1) $\mathcal{M}_{\mathcal{A}}$ is a minimal algebra generated in dimensions $\leq k$;
2) ρ induces an isomorphism on cohomology in dimensions $\leq k$ and an injection in dimension $k+1$.

We use the term minimal model for $k = \infty$.

To make applications the following theorem is essential. Recall we are assuming all Betti numbers are finite.

(1.1) Theorem. a) *Every one-connected differential algebra has a minimal model unique up to isomorphism.*

b) *Every connected differential algebra has a 1-stage minimal model unique up to isomorphism.*

Thus we define the *de Rham homotopy groups* of a one-connected differential algebra $\mathcal{A}$, denoted $\pi_*(\mathcal{A})$ (with Whitehead product), to be those of any minimal model for $\mathcal{A}$. Likewise, we define the *de Rham fundamental group* of a differential algebra to be the de Rham fundamental group of any minimal model for the algebra.

If $f : \mathcal{A} \to \mathcal{B}$ is an isomorphism on cohomology, then the de Rham fundamental groups of $\mathcal{A}$ and $\mathcal{B}$ are isomorphic. If $\mathcal{A}$ is one connected, then the de Rham homotopy groups of $\mathcal{A}$ and $\mathcal{B}$ are isomorphic. In the simply connected case we define $\mathcal{M}_{\mathcal{A}}$ to be the de Rham homotopy type of $\mathcal{A}$.

Examples. 1) Suppose $\mathcal{A} = \mathcal{E}^*_{S_n}$ is the de Rham complex on the n-sphere. Then

$$\mathcal{M}_{\mathcal{A}} \cong \begin{cases} \Lambda_n(y), & n \text{ odd}, \\ \Lambda_n(y) \otimes \Lambda_{2n-1}(z), \quad dz = y^2, & n \text{ even}. \end{cases}$$

In either case $\rho : \mathcal{M}_{\mathcal{A}} \to \mathcal{E}^*_{S_n}$ sends y to a generator of $H^n(S^n)$.

2) Suppose $\mathcal{A} = \mathcal{E}^*_{\mathbb{C}P_n}$. Then

$$\mathcal{M}_{\mathcal{A}} \cong \Lambda_2(x) \otimes_d \Lambda_{2n+1}(y), \qquad dy = x^{n+1}.$$

$\rho : \mathcal{M}_{\mathcal{A}} \to \mathcal{E}^*_{\mathbb{C}P_n}$ sends x to a closed form whose cohomology class generates $H^2(\mathbb{C}P^n)$. Thus in these cases the de Rham homotopy groups of $\mathcal{E}^*_{M^n}$ are equal the usual homotopy groups tensored with $\mathbb{R}$:

$$\pi_*(\mathcal{E}^*_{M^n}) \cong \pi_*(M^n) \otimes \mathbb{R}.$$

(cf. Serre [11]). It is a consequence of Sullivan's theory that a similar relation holds for all simply connected smooth manifolds.

For one-connected algebras the existence of the minimal model is straightforward. One inductively assumes the existence of $\mathcal{M}^{(k)} \xrightarrow{\rho} \mathcal{A}$ inducing an isomorphism on cohomology through dimension k and an injection on H^{k+1}. Then $\mathcal{M}^{(k)} \subset \mathcal{M}^{(k+1)}$ is an elementary extension on $V_{k+1} = C_{k+1} \oplus N_{k+1}$, where C_{k+1} is the subspace of closed elements included to make ρ surjective on H^{k+1}, and N_{k+1} is a space of nonclosed generators whose image under d, $d(N) \subset (\mathcal{M}^{(k)})^{k+2}$, makes exact a space of closed forms representing kernel ρ^* in degree $(k+2)$.

In the general case, the construction of $\mathcal{M}^{(1)}$ is similar: $\mathcal{M}^{(1)}_1 = \Lambda(H^1(\mathcal{A}))$ and $\mathcal{M}^{(1)}_i \subset \mathcal{M}^{(1)}_{i+1}$ is an elementary extension by $\Lambda_1(\text{Ker } H^2(\mathcal{M}^{(1)}_i) \to H^2(\mathcal{A}))$. It is necessary to keep killing the kernel in degree 2 repeatedly since at each step one is creating more kernel in that degree.

The uniqueness statements in these two cases require the development of the theory of homotopy for maps between differential algebras. Let (t, dt) denote the polynomials in the variable t of degree zero tensor the exterior algebra on its

differential, dt, in degree one. This is a contractible algebra which we think of as the real line. There are evaluations at $t = 0$ and at $t = 1$, which we think of as two points on the line. A map $H : \mathcal{A} \to \mathcal{B} \otimes (t, dt)$ is a homotopy from $f : \mathcal{A} \to \mathcal{B}$ to $g : \mathcal{A} \to \mathcal{B}$ if $H|_{t=0} = f$ and $H|_{t=1} = g$. (Note that a homotopy $H : X \times I \to Y$ induces a map

$$\mathcal{E}^*_Y \xrightarrow{H^*} \mathcal{E}^*_X \otimes \mathcal{E}^*_I.$$

Thus the contravariant nature of forms requires that (t, dt) be in the range rather than the domain.)

Given an elementary extension $\mathcal{N} = \mathcal{M} \otimes_d \Lambda(V_n)$, maps f and $g : \mathcal{N} \to \mathcal{A}$, and a homotopy $H : \mathcal{M} \to \mathcal{A} \otimes (t, dt)$ from $f|\mathcal{M}$ to $g|\mathcal{M}$, then H extends to a homotopy $\bar{H} : \mathcal{N} \to \mathcal{A} \otimes (t, dt)$ from f to g if and only if the homomorphism $V_n \xrightarrow{\sigma} H^{n+1}(\mathcal{A})$ given by

$$\left\{\sigma(x) = \text{cohomology class of } \left(f(x) - g(x) + (-1)^{\deg x} \int_{t=0}^{t=1} H(dx)\right)\right\}$$

is 0. If $\sigma = 0$, then an extension of H to $\bar{H}$ is determined by choosing linearly, for each x, $\alpha(x)$ such that $d(\alpha(x)) = \sigma(x)$. The formula for $\bar{H}$ is then

$$\bar{H}(x) = f(x) + (-1)^{\deg x} \int_0^t H(dx) - d(\alpha(x) \otimes t).$$

Thus we find a sequence of cohomological obstructions for building a homotopy whose algebraic structure is exactly analogous to the topological situation.

(1.2) Theorem (Lifting and lifting up to homotopy). *Given a diagram of solid arrows*

$$\begin{array}{ccc} & & \mathcal{A} \\ & & \downarrow{\scriptstyle \varphi} \\ \mathcal{M} & \to & \mathcal{B} \end{array}$$

with $\mathcal{M}$ minimal and φ^* an isomorphism on cohomology, then there is a $\tilde{f} : \mathcal{M} \to \mathcal{A}$ so that $\varphi\tilde{f}$ is homotopic to f. If φ is also onto, then we can choose $\tilde{f}$ so that $\varphi\tilde{f}$ is equal to f.

(1.3) Corollary. *Homotopy is an equivalence relation on the set of maps $\{f : \mathcal{M} \to \mathcal{A}\}$ for $\mathcal{M}$ minimal. We denote the set of equivalence classes by $[\mathcal{M}, \mathcal{A}]$.*

(1.4) Corollary. *If $\psi : \mathcal{A} \to \mathcal{B}$ is an isomorphism on cohomology, then ψ induces a bijection of sets $\psi : [\mathcal{M}, \mathcal{A}] \to [\mathcal{M}, \mathcal{B}]$.*

(1.5) Corollary. *If $f : \mathcal{A} \to \mathcal{B}$ is any map of differential algebras and $\rho_{\mathcal{A}} : \mathcal{M}_{\mathcal{A}} \to \mathcal{A}$ and $\rho_{\mathcal{B}} : \mathcal{M} \to \mathcal{B}$ are minimal models for $\mathcal{A}$ and $\mathcal{B}$, then f induces $\tilde{f} : \mathcal{M}_{\mathcal{A}} \to \mathcal{M}_{\mathcal{B}}$, unique up to homotopy, subject to the condition that*

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{f} & \mathcal{B} \\ \uparrow{\scriptstyle \rho_{\mathcal{A}}} & & \uparrow{\scriptstyle \rho_{\mathcal{B}}} \\ \mathcal{M}_{\mathcal{A}} & \xrightarrow{\tilde{f}} & \mathcal{M}_{\mathcal{B}} \end{array}$$

(1.6) Corollary. *Any two k-stage minimal models of an algebra are isomorphic by an isomorphism unique up to homotopy.*

Theorem (1.2) and its corollaries are sufficient for the proof of Theorem (1.1).

The first part of Theorem (1.2) is proved by induction using the cohomological form of the obstruction to extending a homotopy. The second part is proved by a direct algebraic argument. To prove Corollary (1.3) let (s,t,ds,dt) represent the plane and $(s,t,ds,dt)/\{s\cdot(t-1)=0, sdt=0\}$ represent the union of the lines $s=0$ and $t=1$. Both are contractible algebras.

$$\text{the lines } s=0, \quad t=1, \quad \text{and the diagonal } s=t$$

Two homotopies $H:\mathcal{M}\to\mathcal{A}\otimes(t,dt)$ and $H':\mathcal{M}\to\mathcal{A}\otimes(s,ds)$ with $H|_{t=1}=H'|_{s=0}$ define a map

$$H \mathbin{\widetilde{+}} H' : \mathcal{M}\to\mathcal{A}\otimes(s,t,ds,dt/\{s\cdot(t-1)=0, sdt=0\}).$$

Use Theorem (1.2) to produce a $\bar{H}:\mathcal{M}\to\mathcal{A}\otimes(s,t,ds,dt)$ which extends $H \mathbin{\widetilde{+}} H'$. Restricting $\bar{H}$ to the line $\{s=t\}$ gives the required homotopy from $H|_{t=0}$ to $H'|_{s=1}$.

The lifting up to homotopy theorem proves that the map $\psi:[\mathcal{M},\mathcal{A}]\to[\mathcal{M},\mathcal{B}]$ of Corollary (1.4) is onto if $\psi^*:H^*(\mathcal{A})\to H^*(\mathcal{B})$ is an isomorphism. That ψ is one-to-one is proved using a relative version of lifting up to homotopy.

Corollary (1.5) is an immediate consequence of (1.2).

To prove Corollary (1.6) we need to know that any map of minimal differential algebras inducing an isomorphism on cohomology induces an isomorphism on the subalgebras generated by elements in degree k. This is proved inductively over the canonical series.

Example. Let $f:S^{4n-1}\to S^{2n}$ be a C^∞ map. It induces $\tilde{f}:\Lambda_{2n}(y)\otimes\Lambda_{4n-1}(z)\to\Lambda_{4n-1}(x)$ unique up to homotopy. There are no homotopies here; thus $\tilde{f}$ is unique. It is completely determined by $\tilde{f}(z)=H(f)\cdot x$ for some real number $H(f)$. By explicit computation $H(f)=\int_{S^{4n-1}}\bar{u}\wedge f^*(\bar{y})$, where $\bar{y}=\rho(y_{2n})$ and $\bar{u}\in\mathcal{E}^{2n-1}(S^{4n-1})$ satisfies $d\bar{u}=f^*\bar{y}$. This is exactly Whitehead's formulation of the Hopf invariant of f as given in [17]. Thus, $H(f)$ is an integer. Also we see that the homotopy theoretic nature of f modulo torsion is determined by maps on forms, even though f^* is zero in cohomology.

As a general comment, to understand cohomology and maps on cohomology one need deal only with closed forms, but to detect the finer homotopy theoretic information one also needs to use non-closed forms. Differential geometric aspects of this philosophy have been given by Chern and Simons: "The manner in which a closed form which is zero in cohomology actually becomes exact contains geometric information".

Section 2. De Rham Homotopy Theory

There are two natural "geometric" differential algebras; one is the C^∞ de Rham forms on a smooth manifold, and the other is the $\mathbb{Q}$-polynomial forms on a simplical complex. Associated to each type of algebra is a de Rham theorem. Also, given a C^1 triangulation of a C^∞ manifold, the two algebras of forms are related through the piecewise C^∞ forms in that triangulation.

Let $\mathcal{E}^*_M$ be the C^∞ forms on M and $\mathcal{M}_M$ a minimal model for it. If M is simply connected, we call $\mathcal{M}_M$ the C^∞ de Rham homotopy type of M and $\pi_i(\mathcal{M}_M) = \pi_i(\mathcal{E}^*_M)$ the i-th C^∞ de Rham homotopy group of M. These vector spaces are given a graded Lie algebra structure called the de Rham-Whitehead product. These groups and their products are invariants of M. If $f : M \to N$ is an isomorphism on cohomology, then it induces an isomorphism $\tilde{f} : \mathcal{M}_N \cong \mathcal{M}_N$ and $f_\# : \pi_i(\mathcal{M}_M) \cong \pi_i(\mathcal{M}_N)$. More generally, given any smooth map $f : M \to N$, it induces $\tilde{f} : \mathcal{M}_N \to \mathcal{M}_M$ unique up to homotopy. The various homotopic $\tilde{f}$'s induce the same map $f_\# : \pi_i(\mathcal{M}_M) \to \pi_i(\mathcal{M}_N)$. There are analogous statements about the C^∞ de Rham fundamental group of any manifold.

In the next section we shall state how these homotopy invariants of manifolds are related to the more classical ones, thus justifying the terminology.

In particular we shall find that

1) $\pi_1(\mathcal{M}_M)$ is the real form of the nilpotent completion of $\pi_1(M)$.
2) If $\pi_1(M) = \{e\}$, then $\pi_n(\mathcal{M}_M) \cong \pi_n(M) \otimes \mathbb{R}$ by a natural isomorphism preserving the Whitehead product structures.
3) If $\pi_1(M) = \{e\}$, then $\mathcal{M}_M$ contains all the algebraic topology of M over real numbers. It is the real homotopy type of M.

Explanation. 1) If π is any group, then

$$\begin{array}{ccccccc} \cdots & \subset & [\pi,[\pi,\pi]] & \subset & [\pi,\pi] & \subset & \pi \\ & & \| & & \| & & \| \\ \cdots & \subset & \Gamma_2 & \subset & \Gamma_1 & \subset & \pi \end{array}$$

is the lower central series. The quotients

$$\cdots \to \pi/\Gamma_3 \to \pi/\Gamma_2 \to \pi/\Gamma_1 = \text{abelianization of } \pi$$

is a tower of nilpotent groups in the sense that $(n+1)$-fold commutators are 0 in π/Γ_n. This tower is the *nilpotent completion* of π. It is possible to "tensor" such nilpotent groups with $\mathbb{R}$ (or $\mathbb{Q}$, or $\mathbb{C}$), and produce real (rational or complex) nilpotent Lie groups, [1, 13]. These are the real (rational or complex) form of the nilpotent completion of π_1.

To give meaning to statement 3), much less prove it, we must relate $\mathcal{M}_M$ to some Postnikov tower related to M. As we will see in the next section it is possible to homotopy-theoretically tensor a Postnikov tower with $\mathbb{Q}$. However, there is no such direct process for tensoring with $\mathbb{R}$.[5] We must therefore relate $\mathcal{M}_M$ to a

5 Theorems (3.3) and (3.4) below provide a meaningful algebraic method for "tensoring" a space with $\mathbb{R}$. Finding a geometric process for tensoring a space with $\mathbb{R}$ presents interesting problems.

rational differential algebra which then may be related to a rational Postnikov tower. This leads us to the other class of differential algebras: the $\mathbb{Q}$ polynomial forms on a simplicial complex. The following discussion is motivated by Whitney's work [18], especially Chapter IV, Sections 24-29 and Chapter VII, Sections 10-12.

Let σ_n be an n-simplex embedded in $\mathbb{R}^{n+1}$ using barycentric coordinates. A $\mathbb{Q}$-polynomial form on σ_n is the restriction to the simplex of a differential form

$$\psi = \psi_{i_1 \dots i_\ell}(t)\, dt_{i_1} \wedge \cdots \wedge dt_{i_\ell}$$

in $\mathbb{R}^{n+1}$ where the $\psi_{i_1 \dots i_\ell}(t)$ are polynomials with $\mathbb{Q}$-coefficients. On a simplicial complex X, a $\mathbb{Q}$-polynomial form ψ is given by a $\mathbb{Q}$-polynomial form ψ_σ in each simplex $\sigma \subset X$ such that the compatibility condition, ω_σ restricted to τ equals ω_τ whenever σ contains τ as a face, is satisfied. The $\mathbb{Q}$-polynomial forms give a differential algebra $\mathcal{E}_X^*$ defined over $\mathbb{Q}$. Moreover, Stokes' theorem

$$\int_{\partial C} \psi = \int_C d\psi$$

is valid for simplical chains, and thus integration gives a chain map $\mathcal{E}_X^* \xrightarrow{i} C^*(X, \mathbb{Q})$. We obtains rational cochains because integration of polynomials is a rational operation.

(2.1) Theorem (p. l. de Rham Theorem, see [18, 14, 6]).

a) *The induced mapping $H^*(\mathcal{E}_X^*) \xrightarrow{i^*} H^*(X, \mathbb{Q})$ is an isomorphism of $\mathbb{Q}$-algebras.*

b) *If $K \xrightarrow{\psi} M$ is a C^1 triangulation of a C^∞ manifold, let $\tilde{\mathcal{E}}_M^*$ be the complex of piecewise C^∞ forms on M with respect to the triangulation K (i.e. forms whose restrictions to each simplex of K is C^∞). Then the inclusions*

$$\mathcal{E}_M^* \nearrow \quad \tilde{\mathcal{E}}_M^* \quad \mathcal{E}_K^* \nwarrow \otimes_{\mathbb{Q}} \mathbb{R}$$

induce isomorphisms on cohomology rings.

These two isomorphism theorems are proved similarly to the way Whitney proves the de Rham theorem and the isomorphism theorem for flat cochains, see [18]. For the fact that the isomorphism in a) is a ring map see [19].

To each simplicial complex X, we assign $\mathcal{E}_X^*$, the differential algebra of $\mathbb{Q}$-polynomial forms. From this we obtain $\mathcal{M}_{\mathcal{E}_X^*}^{(1)}$ and dually a natural tower of nilpotent Lie groups $\cdots \to \mathcal{L}_1^2(X) \to \mathcal{L}_1^1(X) \to 0$. This tower is the $\mathbb{Q}$-de Rham fundamental group of X. If X is simply connected, then $\mathcal{M}_{\mathcal{E}_X^*} = \mathcal{M}_X$ is an invariant of X. It is the $\mathbb{Q}$-de Rham homotopy type of X and its homotopy groups $\pi_*(\mathcal{M}_X)$, with their Whitehead products, are the $\mathbb{Q}$-de Rham homotopy groups of X. A simplicial map $F : X \to Y$ induces $\tilde{f} : \mathcal{M}_Y^{(1)} \to \mathcal{M}_X^{(1)}$ unique up to homotopy. All the homotopic $\tilde{f}$ induce the same map $\pi_1(\mathcal{M}_X) \xrightarrow{f_*} \pi_1(\mathcal{M}_Y)$, which is the map induced by f on the $\mathbb{Q}$-de Rham fundamental group. If X and Y are simply connected, then the various $f : \mathcal{M}_Y \to \mathcal{M}_X$ induce the same map $f_\# : \pi_*(\mathcal{M}_X) \to \pi_*(\mathcal{M}_Y)$.

If $K \xrightarrow{\psi} M$ is a C^1 triangulation of a C^∞ manifold, applying theorem 2.1, b) gives:

1) ($\mathbb{Q}$-de Rham fundamental group of K)$\otimes\mathbb{R} \cong C^\infty$ de Rham fundamental group of M: $\pi_1(\mathcal{M}_K) \otimes_{\mathbb{Q}} \mathbb{R} \cong \pi_1(\mathcal{M}_M)$,
2) If M is simply connected, then ($\mathbb{Q}$-de Rham homotopy type of K)$\otimes\mathbb{R} \cong C^\infty$ de Rham homotopy type of M: $\mathcal{M}_K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathcal{M}_M$, also $\pi_*(\mathcal{M}_K) \otimes_{\mathbb{Q}} \mathbb{R} \cong \pi_*(\mathcal{M}_M)$ by an isomorphism preserving Whitehead products.

In the next section we will relate all this to the classical homotopy theory for X and for M.

To do this it is necessary to be able to subdivide simplicial complexes (e.g., for the simplicial approximation theorem). Since we wish to preserve the $\mathbb{Q}$-polynomial forms, it is necessary to restrict to *rational barycentric subdivisions*, i.e., those where all new vertices have rational coordinates in the old vertices. With this restriction the $\mathbb{Q}$-polynomial forms on X induce $\mathbb{Q}$-polynomial forms on the subdivision. The category of simplicial complexes with $\mathbb{Q}$-subdivision and $\mathbb{Q}$-piecewise linear maps is equivalent to the usual homotopy category.

Section 3. Relation between De Rham Homotopy Theory and Classical Homotopy Theory

To relate the C^∞ de Rham homotopy type of a manifold M to the classical homotopy theory of M, we relate the $\mathbb{Q}$-de Rham homotopy type of some C^1 triangulation, K, to the classical homotopy type, and then use the relationship $\mathcal{M}_K \otimes \mathbb{R} \cong \mathcal{M}_M$. For another connection between differential (co-) algebras and homotopy theory compare [10], which motivated the questions answered in this section.

To each space X we can associate a tower of fibrations P_X, the Postnikov tower to X,

$$\begin{array}{ccc} & & \vdots \\ & & X_2 \\ & \nearrow_{f_2} & \downarrow_{p_2} \\ & & X_1 \\ & \nearrow_{f_1} & \downarrow_{p_1} \\ & \overset{f_0}{} & \\ X & \longrightarrow & X_0 \end{array}$$

such that

1) the fiber of $X_n \to X_{n-1}$ has only one non-zero homotopy group, that being in dimension n:
2) f_n is an isomorphism on homotopy groups in dimension $\leq n$, and
3) X_0 is a point.

Homotopy questions about X can be formulated in terms of this tower. To study such a tower with ordinary forms (i.e., forms taking values in a constant bundle), we need the fibrations ρ to be principal. They are principal if, and only if, $\pi_1(X_{n-1}) \cong \pi_1(X)$ acts trivially on π_n of the fiber (which is $\pi_n(X)$). Under weaker assumptions on X, namely that $\pi_1(X)$ is nilpotent and acts in a nilpotent manner in $\pi_n(X)$, the Postnikov tower admits a (finite) refinement into principal fibrations which can be treated algebraically using ordinary forms.

More generally every space maps to its universal tower of principal fibrations P_X (unique up to a pro-equivalence) which is convenient for our algebraic manipulations, but does not contain all the homotopy theoretic information in the space unless we are in the "nilpotent case" above.

Towers of principle fibrations can be inductively "tensored with $\mathbb{Q}$", see [13] for example, to produce a system of rational principal fibrations, i.e., ones in which the homotopy group of each fiber is a rational vector space. If X is simply connected (or even nilpotent) then $P_X \otimes \mathbb{Q}$ is the *rational Postnikov tower* of X, or the *rational homotopy type* of X. It contains the rational form of all the algebraic homotopy information in the space. We will see one way to construct this tower, $P_X \otimes \mathbb{Q}$, using $\mathbb{Q}$-polynomial forms on X below. To treat twisted fibrations directly one would need to use twisted forms (forms with coefficients in a flat vector bundle). The connection between forms and principal fibrations is achieved by the Hirsch lemma and its converse. Given a fibration $F \to E \xrightarrow{\pi} B$, an element $\alpha \in H^k(F;\mathbb{Q})$ is transgressive if it is possible to take a $\mathbb{Q}$-polynomial form representing α and extend it to a form $\tilde{\alpha}$ on E such that $d\tilde{\alpha} = \pi^*(\beta)$ for some closed form β in B. If $H^*(F)$ is a free algebra, we say that it is transgressive if it has an algebra basis consisting of transgressive elements.

(3.1) Lemma (Hirsch Lemma). Given a fibration $F \to E \xrightarrow{\pi} B$ with $H^*(F)$ free and transgressive and a differential algebra $\mathcal{A} \to \mathcal{E}_B^*$ giving an isomorphism on cohomology, then there is a map of differential algebras $\mathcal{A} \otimes_d H^*(F) \to \mathcal{E}_E^*$ also inducing an isomorphism on cohomology.

Proof. Pick an algebra basis $\{x_1, \dots, x_n\}$ for $H^*(F)$. Let $\tilde{\alpha}_i \in \mathcal{E}_E^*$ be a form which when restricted to F gives a closed form representing x_i. Let β_i be such that $d\tilde{\alpha}_i = \pi^*(\beta_i)$. Since $\rho : \mathcal{A} \to \mathcal{E}_B^*$ is an isomorphism on cohomology, we may pick $\tilde{\alpha}_i$ such that $\beta_i = \rho(a_i)$ for some closed form $a_i \in \mathcal{A}$.

Define $d : H^*(F) \to \mathcal{A}^{*+1}$ by $d(x_i) = a_i$, and $\tilde{\rho} : \mathcal{A} \otimes H^*(F) \to \mathcal{E}_E^*$ by $\tilde{\rho}|_{\mathcal{A}} = \pi^* \circ \rho$ and $\tilde{\rho}(x_i) = \tilde{\alpha}_i$. $\tilde{\rho}$ is a map of differential algebras. If one uses the $\mathbb{Q}$-polynomial forms on the base $\mathcal{E}_B^*$ for $\mathcal{A}$ with $\rho = id$, then an easy induction argument over the cells of the base using the five lemma shows that $\tilde{\rho}$ is an isomorphism on cohomology. An easy algebraic induction shows that $\rho \otimes id : \mathcal{A} \otimes_d H^*(F) \to \mathcal{E}_B^* \otimes H^*(F)$ is an isomorphism on cohomology if ρ is. Thus the Serre spectral sequence is, in effect, replaced in our $\mathbb{Q}$-theory by the algebraically simpler Hirsch lemma. The latter also contains more information than the rational Serre spectral sequence.

Examples of Such Fibrations. 1) *The frame bundle of a smooth manifold:* By the Chern-Weil theory, the fiber has free cohomology which is transgressive.

2) *Principal fibrations:* Using the Hirsch lemma, one proves inductively that $H^*(K(\pi, n); \mathbb{Q}) \cong \Lambda(\mathrm{Hom}(\pi, \mathbb{Q}))$.

Thus the cohomology of the fiber is free and generated in dimension n. It is transgressive, since this is the case in the universal principal fibration

$$K(\pi, n) \to P \to K(\pi, n+1).$$

We also need the converse of the Hirsch lemma. It is at this point that it is necessary to work with $\mathbb{Q}$-algebras and $\mathbb{Q}$-polynomial-forms rather than $\mathbb{R}$-algebras and classical forms.

(3.2) Lemma. Given a $\mathbb{Q}$-differential algebra, $\mathcal{A}$ and a space B with $\rho : \mathcal{A} \to \mathcal{E}_B^*$ inducing an isomorphism on $\mathbb{Q}$-cohomology, and an elementary extension $\mathcal{A} \otimes_d \Lambda_n$ of $\mathcal{A}$, then there is a principal fibration $F \to E \xrightarrow{\pi} B$, and $\tilde{\rho} : \mathcal{A} \otimes_d \Lambda_n \to \mathcal{E}_E^*$ inducing an isomorphism on $\mathbb{Q}$-cohomology. Furthermore, this fibration is unique if we require $\pi_*(F)$ to be a $\mathbb{Q}$-vector space.

Proof. Set $\Lambda_n = \Lambda(V_n^*)$. Then

$$d : V_n^* \to \{\text{closed forms in } \mathcal{A}^{n+1}\} \to H^{n+1}(\mathcal{A})$$

defines $k \in H^{n+1}(\mathcal{A}, V) \underset{\rho}{\overset{\cong}{\to}} H^{n+1}(B; V)$.

Form the principal fibration over B with fiber $K(V, n)$ and characteristic class $k \in H^{n+1}(B; V)$, i.e.,

$$\begin{array}{ccc} K(V,n) & \longrightarrow & K(V,n) \\ \downarrow & & \downarrow \\ E & \longrightarrow & P \\ \downarrow & & \downarrow \\ B & \xrightarrow{k} & K(V,n+1). \end{array}$$

To define $\rho : \mathcal{A} \otimes_d \Lambda_n \to \mathcal{E}_E^*$ inducing an isomorphism on $\mathbb{Q}$-cohomology, we need only know that

1) $H^*(K(V,n); \mathbb{Q}) \cong \Lambda(V^*) = \Lambda_n$ (This is true since we are working over $\mathbb{Q}$. The analogous statement over $\mathbb{R}$ is false.), and
2) the transgression $\tau : V^* \to H^{n+1}(B)$ equals k.

Thus given an algebra which is presented as a sequence of elementary extensions, $0 \subset \mathcal{A}_1 \subset \mathcal{A}_2 \subset \cdots \bigcup \mathcal{A}_i = \mathcal{A}$ (e.g., a minimal differential algebra with a series), we can form a tower of rational fibrations $X_1 \leftarrow X_2 \leftarrow \cdots$ using (3.2). Conversely given a tower of principal fibrations, we use (3.1) to build a sequence of elementary extensions of differential algebras. A main result, see [12] and [13], is that

1) the composition:

$$\left\{ \begin{array}{c} \text{tower of} \\ \text{principal} \\ \text{fibrations} \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{sequence of} \\ \text{elementary} \\ \text{extensions of} \\ \mathbb{Q}\text{-differential} \\ \text{algebras} \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{tower of} \\ \text{rational} \\ \text{principal} \\ \text{fibrations} \end{array} \right\}$$

tensors a tower with $\mathbb{Q}$, and that

2) the composition

$$\left\{\begin{array}{c}\text{sequence of}\\ \text{elementary}\\ \text{extensions of}\\ \mathbb{Q}\text{-differential}\\ \text{algebras}\end{array}\right\}\longrightarrow\left\{\begin{array}{c}\text{tower of}\\ \text{rational}\\ \text{principal}\\ \text{fibrations}\end{array}\right\}\longrightarrow\left\{\begin{array}{c}\text{sequence of}\\ \text{elementary}\\ \text{extensions of}\\ \mathbb{Q}\text{-differential}\\ \text{algebras}\end{array}\right\}$$

is the identity.

3) If a sequence of elementary extensions of differential algebras $0 \subset \mathcal{A}_1 \subset \mathcal{A}_2 \subset \cdots$ determines principal fibrations $X_1 \leftarrow X_2 \leftarrow \cdots$, then each $\mathcal{A}_i$ maps into $\mathcal{E}^*_{X_i}$ inducing an isomorphism on cohomology. Thus, if each $\mathcal{A}_i$ is minimal (i.e., d is decomposable), then $\mathcal{A}_i$ is the minimal model for $\mathcal{E}^*_{X_i}$.

(3.3) Theorem. a) If X is simply connected (or nilpotent), then the rational Postnikov tower of X and the rational de Rham homotopy type of X determine each other, $\mathcal{M}_X \longleftrightarrow P_X \otimes \mathbb{Q}$. (This justifies calling $\mathcal{M}_X$ the rational homotopy type of X.) In particular if X is simply connected $\pi_*(\mathcal{M}_X) \cong \pi_*(X) \otimes \mathbb{Q}$ by an isomorphism preserving Whitehead products.

b) $\pi_1(\mathcal{M}_X) \cong$ rational form of the nilpotent completion of $\pi_1(X)$, for any X.

(3.4) Corollary. a) If M^n is a simply connected C^∞ manifold, then the C^∞ de Rham homotopy type is the real form of an algebra with rational structure constants. Furthermore, this rational algebra can be taken to be that of the rational Postnikov tower of M. (This justifies calling $\mathcal{M}_M$ the real homotopy type of M.) In particular $\pi_*(\mathcal{M}_M) \cong \pi_*(M) \otimes \mathbb{R}$ by an isomorphism preserving Whitehead products, if M is simply connected.

b) The C^∞ de Rham fundamental group of M is the real form of the nilpotent completion of $\pi_1(M)$, for any M.

c) If we use complex valued forms on M, then a) and b) have complex analogues. Thus $\pi_1(\mathcal{E}^*_M \otimes_{\mathbb{R}} \mathbb{C})$ is the nilpotent completion of $\pi_1(M)$, and $\mathcal{M}_{\mathcal{E}^*_M \otimes_{\mathbb{R}} \mathbb{C}}$ is the complex homotopy type of M and contains $\pi_*(M) \otimes \mathbb{C}$, if M is simply connected.

There are also versions of these theorems for maps.

If $f : X \to Y$ is a simplical map, then

1) if X and Y are simply connected, the induced map on the $\mathbb{Q}$-de Rham homotopy type determines the map induced by f between the rational Postnikov towers of X and Y, and $f_\# : \pi_*(\mathcal{M}_X) \to \pi_*(\mathcal{M}_Y)$ equals $f_\# : \pi_*(X) \otimes \mathbb{Q} \to \pi_*(Y) \otimes \mathbb{Q}$,

2) if X and Y are not simply connected, then the map $f_\#$ on the $\mathbb{Q}$-de Rham fundamental group equals the map $f_\#$ on the $\mathbb{Q}$ form of the nilpotent completion of π_1.

If $f : M \to N$ is a C^∞ map between C^∞ manifolds, then

1) if M and N are simply connected, the induced map on C^∞ de Rham homotopy groups equals the induced map on classical homotopy groups with $\mathbb{R}$,

2) if M and N are non simply connected, the map induced by f on the C^∞ de Rham fundamental group equals the map induced by f on the real form of the nilpotent completion of $\pi_1(M)$.

Section 4. Formality of Differential Algebras

As we have seen, the abstract homotopy type of a differential algebra $\mathcal{A}$, $\mathcal{M}_{\mathcal{A}}$, contains more information in general than the cohomology ring. There are, for example, higher order products (Massey products) containing further invariants of the homotopy theory of the differential algebra (or space) under consideration. If $a \in H^a$, $b \in H^b$, and $c \in H^c$ satisfy $a \cdot b = 0 = b \cdot c$, then $a \cdot b \cdot c$ is zero for two reasons. Consequently, a difference element $\langle a, b, c\rangle$, in $H^{a+b+c-1}/\,\text{ideal}(a, c)$ can be formed. Let α, β, and γ be closed forms representing a, b, and c. Then $\alpha \wedge \beta = d\eta$ and $\beta \wedge \gamma = d\rho$, and so $\eta \wedge \gamma + (-1)^{a+1}\alpha \wedge \rho$ is a closed $(a + b + c - 1)$ form whose cohomology class is well defined modulo the ideal (a, c). This is the difference element. If such a quotient group is non zero, then there will always be homotopy theoretically distinct spaces (or differential algebras with the same cohomology rings.)

These higher order products are invariants of the homotopy type (i.e., minimal model) of a differential algebra. In one case of interest to us they all vanish, namely if the differential algebra is homotopy equivalent to (i.e., has the same minimal model as) a differential algebra whose d is identically 0. This leads us to the following definition. Let $\mathcal{M}$ be a minimal differential algebra and $H^*(\mathcal{M})$ the cohomology of $\mathcal{M}$, viewed as a differential algebra with $d = 0$.

Definition. (i) $\mathcal{M}$ is formal if there is a map of differential algebras $\psi : \mathcal{M} \to H^*(\mathcal{M})$ inducing the identity on cohomology.

(ii) The homotopy type of a differential algebra $\mathcal{A}$ is a *formal consequence of its cohomology* if its minimal model is formal.

(iii) The real (or complex) homotopy type of a manifold M is a formal consequence of the cohomology M if the de Rham homotopy type of the real (or complex) forms $\mathcal{E}^*_M$ is a formal consequence of its cohomology.

(iv) A $\mathbb{Q}$-polyhedron X is a formal consequence of its cohomology over $\mathbb{Q}$ if the differential algebra of $\mathbb{Q}$-polynomial forms $\mathcal{E}^*_X$ is a formal consequence of its cohomology.

Note. Formal differential co-algebras were used by Quillen in [10], to show, for example, that any graded, commutative algebra of finite type which is zero in degree one is the cohomology ring of some space.

Definition. Let $\mathcal{A}$, $\mathcal{B}$ be differential algebras with minimal models $\mathcal{M}_{\mathcal{A}}$, $\mathcal{M}_{\mathcal{B}}$ respectively. Suppose that $f : \mathcal{A} \to \mathcal{B}$ is a map of differential algebras inducing $f^* : H^*(\mathcal{A}) \to H^*(\mathcal{B})$ and $\hat{f} : \mathcal{M}_{\mathcal{A}} \to \mathcal{M}_{\mathcal{B}}$, defined up to homotopy by Corollary 5 of Section 1. Assume finally that $\mathcal{A}$ and $\mathcal{B}$ are formal consequences of their cohomology by maps $\mathcal{M}_{\mathcal{A}} \xrightarrow{\psi_{\mathcal{A}}} H^*(\mathcal{A})$ and $\mathcal{M}_{\mathcal{B}} \xrightarrow{\psi_{\mathcal{B}}} H^*(\mathcal{B})$. Then we say that the induced map $\hat{f}$ on homotopy types is a formal consequences of the induced map f^* on cohomology if the diagram

$$\begin{array}{ccc} \mathcal{M}_{\mathcal{A}} & \xrightarrow{\hat{f}} & \mathcal{M}_{\mathcal{B}} \\ {\scriptstyle \psi_{\mathcal{A}}}\downarrow & & \downarrow{\scriptstyle \psi_{\mathcal{B}}} \\ H^*(\mathcal{A}) & \xrightarrow{f^*} & H^*(\mathcal{B}) \end{array}$$

is homotopy commutative.

The simplest example of a manifold which is not formal is the following 3-manifold. Let N^3 denote the space of upper triangular matrices

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

with a, b, and c real numbers (N^3 is homeomorphic to $\mathbb{R}^3$). Let $\Gamma \subset N^3$ be the subgroup of integral matrices, and $M^3 = N^3/\Gamma$.

Projecting N^3 onto the a and c coordinates induces a fibration

$$S^1 \to M^3 \to T^2$$

The minimal model of M^3 is (by the Hirsch lemma)

$$\mathcal{M}_1^{(1)} = \Lambda_1(x,y); \qquad \mathcal{M}_2^{(1)} = \mathcal{M} = \Lambda_1(x,y) \otimes_d \Lambda_1(z) \qquad \text{and} \qquad dz = x \wedge y.$$

Thus $x \wedge z$, for example, is closed but not exact. In any algebra with $d = 0$ such a second order product is in the ideal of x. In this case $x \cdot H^1(M) = 0$. Thus if $\mathcal{M}$ were formal, $x \wedge z$ would be cohomologous to 0. Hence, $\mathcal{M}$ is not formal.

The simplest 1-connected example where a minimal model fails to be formal is

$$\mathcal{M} = \Lambda_2(x,y) \otimes \Lambda_3(\psi, \varphi),$$

$$dx = dy = 0, \qquad d\psi = x \wedge x \qquad \text{and} \qquad d\varphi = x \wedge y.$$

Again we have a higher order product which is closed but not exact. In dimension 5, $x\varphi + y\psi$ is closed but not exact. Since $(x \cdot H^3(\mathcal{M}) + y \cdot H^3(\mathcal{M}))$ is 0, the triple product $\langle x, x, y\rangle$ is non-zero in $\mathcal{M}$. Consequently, there can be no map of $\mathcal{M} \to \{H^*(\mathcal{M}), d = 0\}$ inducing an isomorphism in cohomology. These examples lead us to the following equivalent condition to formality.

Any minimal differential algebra is isomorphic as an algebra to

$$P[V_2 \oplus V_4 \oplus \cdots] \otimes \Lambda[V_1 \oplus V_3 \oplus V_5 \oplus \cdots]$$

where each vector space V_i contains elements of degree i only.

In V_i there is the subspace C_i of closed elements.

(4.1) Theorem. $\mathcal{M}$ is formal if, and only if, there is in each V_i a complement N_i to C_i, $V_i = C_i \oplus N_i$, such that any closed form, a, in the ideal, $I(\oplus N_i)$, is exact. Choosing such N_i is equivalent to choosing $\psi : \mathcal{M} \to H^*(\mathcal{M})$ giving the identity on cohomology.

Proof. Suppose first that we have complements N_i to C_i in V_i. They give projections $\pi : V_i \to C_i$. Define $\psi : V_i \to H^*(\mathcal{M})$ by $\psi(x) = \rho(\pi_i(x))$ where $\rho : C_i \to H^i(\mathcal{M})$ is the natural inclusion. We extend ψ to an algebra map $\psi : \mathcal{M} \to H^*(\mathcal{M})$ by multiplicativity. To show that $d\psi = \psi d$, we must prove $\psi d = 0$. Let $\alpha \in \mathcal{M}$. Then $d\alpha = \beta + \gamma$ where $\beta \in \Lambda(\oplus C_i)$ and $\gamma \in I(\oplus N_i)$. Since β is closed, so is γ. By hypothesis γ is then exact. Thus $\beta = d\alpha - \gamma$ is also exact. Thus, $\rho(\beta) \in H^*(\mathcal{M})$ is 0. This shows that $\psi(d\alpha) = \psi(\beta) = \rho(\beta) = 0$, and that $\psi : \mathcal{M} \to H^*(\mathcal{M})$ is a map of differential algebras inducing the identity on cohomology.

Conversely, given $\psi : \mathcal{M} \to H^*(\mathcal{M})$, we let $N_i = \ker \psi \cap V_i$. Because $C_i \to H^*(\mathcal{M})$ is an injection, $V_i = C_i \oplus N_i$. If $\alpha \in I(\oplus N_i)$ is closed, then $[\alpha] = \psi(\alpha) = 0$, and α is exact in $\mathcal{M}$ since ψ^* is an isomorphism on cohomology. Q.E.D.

The existence of the splitting N_i with the property that any closed $\alpha \in I(\oplus N_i)$ is exact is a way of saying that *one may make uniform choices so that the forms representing all Massey products and higher order Massey products are exact.* This is stronger than requiring each individual Massey product or higher order Massey product to vanish. The latter means that, given one such product, choices may be made to make the form representing it exact, and there may be no way to do this uniformly.

Section 5. The De Rham Complex of a Compact Kähler Manifold

In this section we shall review the basic facts regarding Hodge theory on a compact Kähler manifold which will be used later.

To begin with, an almost complex manifold is a smooth manifold M having an endomorphism $J : T(M) \to T(M)$ of the real tangent bundle which satisfies $J^2 = -1$. The complex tangent bundle $T_{\mathbb{C}}(M) = T(M) \otimes_{\mathbb{R}} \mathbb{C}$ splits into conjugate subbundles under the action of J

$$T_{\mathbb{C}}(M) = T'(M) \oplus T''(M),$$

$$J|T'(M) = \sqrt{-1},$$

$$T''(M) = \text{complex conjugate of } T'(M).$$

On the exterior powers of the cotangent bundle there is an induced decomposition

$$\Lambda^r T_{\mathbb{C}}(M)^* = \bigoplus_{p+q=r} \{\Lambda^p T'(M)^* \otimes \overline{\Lambda^q T'(M)^*}\} = \bigoplus_{p+q=r} \Lambda^{p,q} T_{\mathbb{C}}(M)$$

Using this, a differential form on M may be uniquely written as a sum of its (p, q) components $\psi = \sum_{p,q} \psi_{p,q}$. This produces a splitting of the complex valued forms on a complex manifold, $\mathcal{E}^*_M \otimes \mathbb{C}$, into $\bigoplus_{p,q\geq 0} \mathcal{E}^{p,q}_M$.

In particular, for a function f, $df = \partial f + \bar{\partial} f$ where $\partial f = (df)_{1,0}$. The Cauchy-Riemann operator $\bar{\partial}$ extends to a differentiation on the de Rham complex, and the almost-complex structure is integrable in case $\bar{\partial}^2 = 0$. By the theorem of Newlander-Nirenberg [9] this is the case if, and only if, there are local maps $z : U \to \mathbb{C}^n$ ($2n = \dim_{\mathbb{R}} M$) from open sets $U \subset M$ such that $dz_1, \ldots, dz_n$ span $T'(M)$. Since any two such maps are related by a biholomorphic change of variables, the integrability condition is equivalent to M being a complex manifold.

On a complex manifold M one defines $d^c = \sqrt{-1}(\bar{\partial} - \partial)$ with the properties that

$$d^c \text{ is real, in fact } d^c = J^{-1} dJ,$$

$$(d^c)^2 = 0,$$

and

$$dd^c = 2\sqrt{-1}\, \partial\bar{\partial} = -d^c d.$$

Thus $\mathcal{E}^*_M$ is a real differential algebra using either of the commuting operators d and d^c, and the resulting cohomologies will be denoted by $H_d(M)$ and $H_{d^c}(M)$.

The geometric nature of d^c may perhaps be best seen using polar coordinates $z = re^{i\theta}$ on $\mathbb{C}$ where

$$d = \frac{\partial}{\partial r} \otimes dr + \frac{\partial}{\partial \theta} \otimes d\theta,$$

$$d^c = r\frac{\partial}{\partial r} \otimes d\theta + \frac{1}{r}\frac{\partial}{\partial \theta} \otimes dr,$$

$$dd^c = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) \otimes dx \wedge dy.$$

On any complex manifold there are positive definite Hermitian metrics

$$(\xi, \eta) = S(\xi, \eta) + \sqrt{-1}A(\xi, \eta) = (J\xi, J\eta) \qquad \text{for } \xi, \eta \in T(M).$$

S and A determine one another. Because of the Hermitian symmetry, S is symmetric and A is alternating. Thus S is a Riemannian metric and A is a $(1,1)$ form on M. The Hermitian metric is defined to be Kähler if $dA = 0$, and a Kähler manifold is a complex manifold admitting a Kähler metric. The existence of a Kähler metric is a statement of close interaction between the complex structure and potential theory, and it has many deep consequences for the topology of the manifold [15].

There are several equivalent statements to the definition of a Kähler metric. The one given above, which is by now the usual one, has the advantage of making it obvious that a complex submanifold of a Kähler manifold is again a Kähler manifold. Thus, any smooth, complex projective algebraic variety is Kähler. The one which we shall use below is the identity

$$\Delta_d = 2\Delta_\partial = 2\Delta_{\bar\partial} = \Delta_{d^c} \tag{5.1}$$

between the Laplacians of the various operators $d, \partial, \bar\partial, d^c$. It is perhaps worthwhile to give a few alternate forms of the Kähler condition, and from these explain how the identity (5.1) may be proved.

We shall write an Hermitian metric in the two forms

$$ds^2 = \sum_{i,j} h_{i,j}\, dz_i\, d\bar z_j = \sum_i \omega_i \bar\omega_i$$

where $(z_1, \dots, z_n)$ are local holomorphic coordinates, $h = (h_{i,j})$ is a positive Hermitian matrix, and where $\omega_1, \dots, \omega_n$ are differential forms of type $(1,0)$ constituting a C^∞ unitary co-frame for the metric. The associated $(1,1)$ form is

$$A = \frac{\sqrt{-1}}{2}\sum_{i,j} h_{i,j}\, dz_i \wedge d\bar z_j,$$

and our basic definition is, as above:

(i) *The metric is Kähler if* $dA = 0$.

To give the second definition; we recall the basic theorem in local Hermitian geometry that there exist unique differential forms $\omega_{i,j}, \tau_i$ satisfying

$$d\omega_i = \sum_j \omega_{i,j} \wedge \omega_j + \tau_i$$

$$\omega_{i,j} + \bar\omega_{j,i} = 0, \qquad \tau_i \text{ has type } (2,0) \tag{5.2}$$

(c.f. Chern [3]). The vector $\tau = (\tau_1, \dots, \tau_n)$ is the torsion tensor, and $\omega = (\omega_{i,j})$ gives the connection matrix for the unique connection ∇_H in the $(1,0)$ tangent bundle $T'(M)$ satisfying

$$d(\xi, \eta) = (\nabla_H \xi, \eta) + (\xi, \nabla_H \eta),$$

$$\nabla''_H \xi = \bar{\partial} \xi$$

where ξ, η are C^∞ sections of $T'(M)$.

(*Explanations.* We are viewing connections as differential operators from sections to sections tensor 1-forms; the connection matrix means relative to the basis of $T'(M)$ dual to $\omega_1, \dots, \omega_n$; and finally ∇''_H is the $(0,1)$ component of ∇_H.)

Our second definition is:

(ii) *The metric is Kähler if the torsion $\tau = 0$.*

For our third definition, we consider the standard connection ∇_R in the real tangent bundle $T(M)$ associated to the Riemannian metric underlying the given Hermitian metric.

(iii) *The metric is Kähler if $\nabla_R J = 0$; equivalently, parallel translation should preserve the almost complex structure.*

Finally, we shall say that the metric is Kähler in case it osculates to 2nd order to the flat metric on $\mathbb{C}^n$. More precisely,

(iv) *The metric is Kähler if, for each point $x \in M$, there is a holomorphic coordinate system $(z_1, \dots, z_n)$ centered at x and such that*

$$h_{i,j}(z, \bar{z}) = \delta_{i,j} + [2], \tag{5.3}$$

where $[2]$ means terms of order two or more.

Proof of Equivalence of Definitions. (i) $\Rightarrow$ (ii) Since A is real, $dA = 0 \iff \partial A = 0$. Let the $(0,1)$-component of $\omega_{i,j}$ be $\alpha_{i,j}$. Then by the basic structure equations, (5.2), $\omega_{i,j} = \alpha_{i,j} - \bar{\alpha}_{j,i}$,

$$\begin{aligned}
\partial A &= \partial \left(\frac{\sqrt{-1}}{2} \sum_i \omega_i \wedge \bar{\omega}_i \right) \\
&= \frac{\sqrt{-1}}{2} \sum_i \left(\sum_j (-\bar{\alpha}_{j,i} \wedge \omega_j \wedge \bar{\omega}_i) + \tau_i \wedge \bar{\omega}_i - \omega_i \wedge \overline{(\alpha_{i,j} \wedge \omega_j)} \right) \\
&= \frac{\sqrt{-1}}{2} \sum_i \tau_i \wedge \bar{\omega}_i - \sum_{i,j} (\bar{\alpha}_{j,i} \wedge \omega_j \wedge \bar{\omega}_i + \omega_i \wedge \bar{\alpha}_{i,j} \wedge \bar{\omega}_j) \\
&= \frac{\sqrt{-1}}{2} \sum_i \tau_i \wedge \bar{\omega}_i \\
&= 0 \iff \tau = 0.
\end{aligned}$$

(ii) $\Rightarrow$ (iii) Write $z_i = x_i + \sqrt{-1} x_{n+i}$, $\omega_i = \varphi_i + \sqrt{-1} \varphi_{n+1}$ where x_α, φ_α are real (using the index range $1 \le i, j \le n$ and $1 \le \alpha, \beta \le 2n$). Then

$$ds^2 = \sum_{\alpha,\beta} g_{\alpha,\beta} \, dx_\alpha \, dx_\beta = \sum_\alpha \varphi_\alpha \varphi_\alpha$$

is the underlying Riemann metric, and by the basic theorem in local Riemannian geometry there exists a unique matrix $\varphi = (\varphi_{\alpha,\beta})$ of 1-forms satisfying

$$d\varphi_\alpha = \sum_\beta \varphi_{\alpha,\beta} \wedge \varphi_\beta$$

$$\varphi_{\alpha,\beta} + \varphi_{\beta,\alpha} = 0. \tag{5.4}$$

Here φ is the connection matrix for the Riemannian connection ∇_R applied to the dual frame to $\varphi_1, \dots, \varphi_{2n}$. Comparing the structure Eqs. (5.2) and (5.4), we see that if $\tau = 0$, then upon writing $\omega = \omega' + \sqrt{-1}\omega''$ we get

$$\varphi = \begin{pmatrix} \omega' & \omega'' \\ -\omega'' & \omega' \end{pmatrix}.$$

This says exactly that, on the complexification $T_{\mathbb{C}}(M) \cong T'(M) \oplus \overline{T'(M)}$, $\nabla_R = \nabla_H \oplus \overline{\nabla_H}$, which implies that ∇_R commutes with J. (This proof may be given verbally by stating that *the Riemannian and Hermitian connections coincide in the Kähler case*.)

(iii) $\Rightarrow$ (iv) First, we observe that (iv) is equivalent to saying that, for each $x \in M$ there is a C^∞ coordinate system $z = (z_1, \dots, z_n)$ centered at x which is holomorphic to 2nd order. This means that z maps a neighborhood U of x differentially to $\mathbb{C}^n$, $dz_1(x), \dots, dz_n(x)$ span $T'_x(M)$, and all $\bar\partial(dz_i)(x) = 0$, such that (5.3) holds. Assuming (iii), we let $(x_1, \dots, x_{2n})$ be geodesic coordinates centered at x. We may arrange that at x, $J(dx_i) = dx_{n+i}$, and then set $z_i = x_i + \sqrt{-1}x_{n+i}$. Since $(x_1, \dots, x_{2n})$ are geodesic coordinates, $g_{\alpha,\beta} = \delta_{\alpha,\beta} + [2]$. Thus the connection matrix for ∇_R is zero at x, and $\nabla_R J = 0$ implies $(dJ)(x) = 0$. It follows that $\bar\partial(dz_i)(x) = 0$, and (iv) follows immediately.

(iv) $\Rightarrow$ (i) This is the main point. Given an Hermitian metric satisfying (iv), we see that

Any identity involving the metric and only its first derivatives is valid on $M \iff$ *it is valid for the flat metric on* $\mathbb{C}^n$. (5.5)

Applying this to the identity $dA = 0$, which is obviously true on $\mathbb{C}^n$, we conclude the equivalence of (i)-(iv).

We shall discuss how the principle (5.5) may be used to prove (5.1), as well as some other Kähler identities. We first consider the commutation relations

$$[\bar\partial, \partial^*] = 0 = [\partial, \bar\partial^*]. \tag{5.6}$$

On the face of it these involve second derivatives of the metric, but actually only first derivatives appear for the reason that mixed partials are equal (think of the Poisson bracket of vector fields). Thus, (5.6) is valid on a Kähler manifold since it is obviously true in $\mathbb{C}^n$ (the metric has constant coefficients). Expanding out the definition of Δ_d now gives

$$\begin{aligned} \Delta_d &= (\partial + \bar\partial)(\partial^* + \bar\partial^*) + (\partial^* + \bar\partial^*)(\partial + \bar\partial) \\ &= (\partial\partial^* + \partial^*\partial) + (\bar\partial\bar\partial^* + \bar\partial^*\bar\partial) \qquad \text{(by (5.6))}; \end{aligned}$$

or

$$\Delta_d = \Delta_\partial + \Delta_{\bar\partial}. \tag{5.7}$$

From (5.7) we see that Δ_d preserves (p,q) type, since this is obviously true for Δ_∂ and $\Delta_{\bar\partial}$. Thus

$$\Delta_d = J^{-1}\Delta_d J = \Delta_{d^c}. \tag{5.8}$$

Writing $\partial = \frac{1}{2}(d - \sqrt{-1}d^c)$, $\partial^* = \frac{1}{2}(d^* - \sqrt{-1}d^c)$, we see that

$$\begin{aligned}\partial\partial^* + \partial^*\partial &= \frac{1}{4}(dd^* + d^*d + d^cd^{c*} + d^{c*}d^c)\\ &= \frac{1}{2}\Delta_d \qquad \text{(by (5.8))}\end{aligned}$$

Combining this with (5.7) gives (5.1).

In closing, it is worth noting that the most interesting properties of Kähler metrics, such as the fact that $L(\psi) = A\wedge\psi$ takes harmonic forms to harmonic forms, are generally proved from the identity

$$[\Lambda, d] = d^{c*} \tag{5.9}$$

or equivalently

$$[\Lambda, \bar\partial] = -\sqrt{-1}\partial^* \tag{5.10}$$

where $\Lambda = *L*$ is the adjoint of L. Now (5.10) obviously involves only 1st derivatives of the metric and so by (5.5) it will suffice to prove it on $\mathbb{C}^n$. This is not too hard since $dz_1, \dots, dz_n$ gives a flat unitary co-frame; one uses the formulae

$$\bar\partial(f dz^I \wedge d\bar z^J) = \sum_{j\notin J}(-1)^p \frac{\partial f}{\partial \bar z_j} dz^I \wedge d\bar z_j \wedge d\bar z^J,$$

$$\Lambda(f dz^I \wedge d\bar z^J) = \sum_{k\in I\cap J} \pm f dz^{I-\{k\}} \wedge d\bar z^{J-\{k\}},$$

$$\partial^*(f dz^I \wedge d\bar z^J) = \sum_{i\in I} \pm \frac{\partial f}{\partial \bar z_i} dz^{I-\{i\}} \wedge d\bar z^J$$

where $I = (i_1, \dots, i_p)$, $dz^I = dz_{i_1}\wedge\cdots\wedge dz_{i_p}$, etc. and just computes. Using (5.10),

$$\bar\partial\partial^* + \partial^*\bar\partial = \sqrt{-1}(\bar\partial\Lambda\bar\partial - \bar\partial^2\Lambda + \Lambda\bar\partial^2 - \bar\partial\Lambda\bar\partial) = 0,$$

which is (5.6). Similarly, $dA = 0 \Rightarrow d^cA = 0$ and $d^{c*} = [\Lambda, d] = 0$ since ΛA is a constant. Thus A is harmonic for d^c, and hence for d. Further discussion of the basic Kähler identities is given in Weil's book [15], p. 179-194.

The main fact we shall use about compact Kähler manifolds is the following lemma:

(5.11) Lemma (dd^c **Lemma).** *If α is a differential form such that $d\alpha = 0$ and $d^c\alpha = 0$, and such that $\alpha = d\gamma$, then $\alpha = dd^c\beta$ for some β.*

Proof. If D is one of the operators $d, \partial, \bar\partial, d^c$ with Laplacian Δ_D, the Hodge decomposition for the Laplacian gives

$$\psi = \mathcal{H}_D(\psi) + \Delta_D G_D(\psi) \tag{5.12}$$

where $\mathcal{H}_D(\psi)$, the harmonic part of ψ, is the projection of ψ on $\mathcal{H}_D = \ker\Delta_D$, and where the Green's operator G_D is Δ_D^{-1} on $\mathcal{H}_D^\perp$. Suppose that $\alpha = d\gamma$. Then by (5.12)

$$\alpha = dd^*G_d\alpha. \tag{5.13}$$

Since $d^c\alpha = 0$, $\alpha = \mathcal{H}_{d^c}(\alpha) + d^c d^{c*} G_{d^c}\alpha = d^c d^{c*} G_{d^c}\alpha$ because $\mathcal{H}_{d^c}(\alpha) = \mathcal{H}_d(\alpha) = 0$ by (5.1). Plugging this in (5.13) gives

$$\begin{aligned}\alpha &= dd^* G_d (d^c d^{c*} G_{d^c}\alpha)\\ &= dd^c (d^* G_d d^{c*} G_{d^c}\alpha) \qquad \text{(using (5.6))}\\ &= dd^c\beta\end{aligned}$$

as required.

(5.14) Remark. This lemma holds for real as well as complex forms, since the validity of a linear statement such as

$$\mathrm{Ker}(d) \cap \mathrm{Ker}(d^c) \cap \mathrm{Im}(d) = \mathrm{Im}(dd^c)$$

is unchanged by a field extension.

For complex forms, being d and d^c-closed is the same as being ∂ and $\bar\partial$-closed. Being dd^c-exact is the same as being $\partial\bar\partial$-exact, since $dd^c = 2i\partial\bar\partial$.

There are statements similar in nature to the dd^c-lemma which are equivalent to it. The following is a partial list of them, set in the context of a general double complex.

(5.15) Lemma. *Let $(K^{*,*}, d', d'')$ be a bounded double complex of vector spaces (or objects of any abelian category), and let (K^*, d) be the associated simple complex, $(d = d' + d'')$. For each integer n, the following conditions are equivalent:*

$$(a)_n \text{ - in } K^n, \quad \mathrm{Ker}(d') \cap \mathrm{Ker}(d'') \cap \mathrm{Im}(d) = \mathrm{Im}(d'd''),$$

$$(b)_n \text{ - in } K^n, \quad \mathrm{Ker}(d'') \cap \mathrm{Im}(d') = \mathrm{Im}(d'd'') \text{ and}$$

$$\mathrm{Ker}(d') \cap \mathrm{Im}(d'') = \mathrm{Im}(d'd''),$$

$$(c)_n \text{ - in } K^n, \quad \mathrm{Ker}(d') \cap \mathrm{Ker}(d'') \cap (\mathrm{Im}(d') + \mathrm{Im}(d'')) = \mathrm{Im}(d'd''),$$

$$(a^*)_{n-1} \text{ - in } K^{n-1}, \quad \mathrm{Im}(d') + \mathrm{Im}(d'') + \mathrm{Ker}(d) = \mathrm{Ker}(d'd''),$$

$$(b^*)_{n-1} \text{ - in } K^{n-1}, \quad \mathrm{Im}(d'') + \mathrm{Ker}(d') = \mathrm{Ker}(d'd'') \text{ and}$$

$$\mathrm{Im}(d') + \mathrm{Ker}(d'') = \mathrm{Ker}(d'd''), \text{ and}$$

$$(c^*)_{n-1} \text{ - in } K^{n-1}, \quad \mathrm{Im}(d') + \mathrm{Im}(d'') + (\mathrm{Ker}(d') \cap \mathrm{Ker}(d'')) = \mathrm{Ker}(d'd'').$$

Proof. In (a)(b)(c) the inclusions $\supset$ are obvious. In (a)*(b)*(c) *the inclusions $\subset$ are. The statements (a), (b) and (c) are dual to (a)*, (b)*, and (c)* respectively, e.g., $(a)_n$ for K amounts to $(a^*)_{-n}$ for K^0 in the dual category. This duality, the symmetry in the roles of d' and d'', and the fact that $(c) \Rightarrow (a)$ is obvious, leaves us to prove the implications $(a)_n \Rightarrow (b)_n$ (first part), $(b)_n \Rightarrow (c)_n$, and $(b)_n$ (second part) $\Rightarrow (b^*)_{n-1}$ (first part).

In the proofs, we will express ourselves as if in a category of modules. For $p + q = n$, we denote by $x^{p,q}$ the component in $K^{p,q}$ of $x \in K^n$.

$(a)_n \Rightarrow (b)_n$ (first part): If $d''x = 0$ and $x = d'y$, then the same holds for $x^{p,q}$ and $y^{p-1,q}$, and $(a)_n$ applies to $dy^{p-1,q}$: $dy^{p-1,q} = d'd''z$, hence $x^{p,q} = d'd''z^{p-1,q-1}$.

$(b)_n \Rightarrow (c)_n$: If $d'x = d''x = 0$ and $x = d'y + d''z$, then $(b)_n$ applies to $d'y$ and $d''z$: they are $d'd''$-exact, and x is too.

$(b)_n$ (second part) $\Rightarrow (b^*)_{n-1}$ (first part): If $d'd''x = 0$, then $(b)_n$ applies to $d''x$: $d''x = d'd''y$ and $x = (x + d'y) - d'y$, with $d''(x + d'y) = 0$.

(5.16) Remark. If maps $C_i \xrightarrow{f_i} A$, $A \xrightarrow{g_i} B_j$ are given, with $f_i g_j = 0$, we put

$$H((g_j);(f_i)) = \bigcap \mathrm{Ker}(g_j) \,/\, \sum \mathrm{Im}(f_i).$$

With this notation, if the equivalent conditions of 5.15 hold for every n, the natural maps in the following commutative diagram are all isomorphisms:

$$\begin{array}{ccccc} & & H(d', d''; d'd'') & & \\ & \swarrow & \downarrow & \searrow & \\ H(d'; d') & & H(d; d) & & H(d''; d'') \\ & \searrow & \downarrow & \swarrow & \\ & & H(d'd''; d', d'') & & \end{array}$$

(5.17) Proposition. *Let (K^{**}, d', d'') be a bounded double complex of vector spaces, and (K^*, d) be the associated simple complex. The following conditions are equivalent.*

(i) *The equivalent conditions of (5.15) hold for every n.*

(ii) *K^{**} is a sum of double complexes of the following two types:*

(α) complexes which have only a single component, with $d' = d'' = 0$,

(β) complexes which are a square of isomorphisms

$$\begin{array}{ccc} K^{p-1,q} & \xrightarrow{d'} & K^{p,q} \\ \uparrow{\scriptstyle d''} \cong & & \cong \uparrow{\scriptstyle d''} \\ K^{p-1,q} & \xrightarrow{d'} & K^{p,q-1}. \end{array}$$

(iii) *The spectral sequence defined by the filtration associated to either degree (denoted ${}'F$ or ${}''F$) degenerates at E_1 ($E_1 = E_\infty$). The two induced filtrations, on H^n are n-opposite: i.e.,*

$${}'F^p \oplus {}''F^q \xrightarrow{\sim} H^n \qquad \text{for } p + q - 1 = n.$$

Proof. (1) $\Rightarrow$ (ii). Let $S^{p,q}$ be a supplement of $\mathrm{Ker}(d'd'')$ in $K^{p,q}$, and $A^{p,q}$ be a supplement of $\mathrm{Im}(d'd'')$ in $\mathrm{Ker}(d') \cap \mathrm{Ker}(d'') \subset K^{p,q}$. By 5.16, $A^{p,q}$ is also a supplement of $\mathrm{Im}(d') + \mathrm{Im}(d'') \subset K^{pq}$: we have

$$(1) \quad K^{p,q} = S^{p,q} \oplus A^{p,q} \oplus (\mathrm{Im}(d') + \mathrm{Im}(d'')),$$

$$(2) \quad d'A^{p,q} = d''A^{p,q} = 0,$$

$$(3) \quad S^{p,q} \cap \mathrm{Ker}(d'd'') = 0.$$

Further, $\mathrm{Im}(d') = d'(S^{p-1,q} + A^{p,q} + \mathrm{Im}(d') + \mathrm{Im}(d'')) = d'S^{p-1,q} + d'd''S^{p-1,q-1}$ and similarly for d'', hence

$$K^{p,q} = S^{p,q} \oplus A^{p,q} \oplus (d'S^{p-1,q} + d''S^{p,q-1} + d'd''S^{p-1,q-1}).$$

Formula (3) shows that the sum is direct, and this provides us with the required decomposition.

(ii) $\Rightarrow$ (iii) It is enough to prove (iii) for the elementary summands (α) and (β). For these it is obvious.

(iii) $\Rightarrow$ (i) Let us prove that if x, of type (p,q), is closed and d' exact, it is $d'd''$-exact. We have $x = d'y$, with y of type $(p-1,q)$, and x is cohomologous to $x - dy = -d''y$ of type $(p-1, q+1)$: the cohomology class of x is both in ${}'F^{p-1}$ and ${}''F^q$, hence zero.

The differential of a chain complex is strict rel. to a filtration F if and only if the corresponding spectral sequence degenerates at E_1. In our case then, $x = da = db$, with $a \in {}'F^p$ and $b \in {}''F^q$. We have $d(a-b) = 0$. The class of $a-b$ is the sum of a class in ${}'F^p$, and a class in ${}''F^q$: $a - b = a_1 - b_1 + dc$, with $a_1 \in {}'F^p$, $b_1 \in {}''F^q$, $da_1 = db_1 = 0$. In particular, $d''a_1^{p,q-1} = 0$ and $x = d''a^{p,q-1} = d''(a_1^{p,q-1} + d'c^{p-1,q-1} + d''c^{p,q-2}) = -d'd''c^{p-1,q-1}$.

(5.18) *Remark.* Proposition (5.17) holds in any semi-simple abelian category, and the equivalence (i) $\Rightarrow$ (iii) holds in any abelian category.

Before applying these results to the de Rham complex let us recall two well-known definitions in the theory of complex manifolds.

(5.19) Let V be a finite dimensional real vector space and F^* a decreasing filtration on $V \otimes \mathbb{C}$,

$$V \otimes \mathbb{C} = F^0 \supset F^1 \supset \cdots \supset F^n \supset F^{n+1} = 0.$$

F^* induces a *Hodge structure of weight* n if $V \otimes_{\mathbb{R}} \mathbb{C}$ is a direct sum $\bigoplus_{p+q=n} V^{p,q}$, with $V^{p,q} = F^p(V \otimes \mathbb{C}) \cap \bar{F}^q(V \otimes \mathbb{C}) \longrightarrow V \otimes \mathbb{C}$. Here, as always, $\bar{F}^*$ means the complex conjugate filtration to F^*.

(5.20) Let M be a complex manifold. Define $F^p(\mathcal{E}_M^* \otimes \mathbb{C})$ to be $\bigoplus_{i \geq p} \mathcal{E}_M^{i,j}$. This is a decreasing filtration, and $d(F^p) \subset F^p$. Thus F^* induces a spectral sequence. It is the *Fröhlicher (or Hodge to de Rham) spectral sequence*; we have

$$E_1^{p,q} = \frac{\operatorname{Ker} \bar{\partial}\big|_{\mathcal{E}_M^{p,q}}}{\bar{\partial}(\mathcal{E}_M^{p,q-1})}.$$

We call the latter group $H_{\bar{\partial}}^{p,q}(M)$.

On the sheaf level we have a resolution

$$0 \longrightarrow \Omega^p \xrightarrow{\bar{\partial}} \mathcal{E}^{p,0} \xrightarrow{\bar{\partial}} \mathcal{E}^{p,1} \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \mathcal{E}^{p,n} \longrightarrow 0.$$

Here $\mathcal{E}^{p,i}$ means the sheaf of local sections of $\Lambda^p T^{1,0}(M) \wedge \Lambda^i T^{0,1}(M)$, and Ω^p means the sheaf of local holomorphic p-forms. The usual sheaf theory arguments allow us then to identify $H_{\bar{\partial}}^{p,q}(M)$ with $H^q(\Omega^p)$.

(5.21) (Special case of 5.17). The following conditions on the forms of a complex manifold are equivalent.

(1) The equivalent conditions (5.15) for all n hold for the de Rham complex, $\mathcal{E}_M^{*,*}$, ((a) is the dd^c-lemma, and (b) is called the $\partial\bar{\partial}$-lemma).

(2) a) The Fröhlicher (or Hodge to de Rham) spectral sequence degenerates at E_1, and

b) the filtration induced by $F^*(\mathcal{E}_M^* \otimes \mathbb{C})$ on $H^n(M;\mathbb{Q}) \otimes \mathbb{C} = H^n(M;\mathbb{C})$ gives a Hodge structure of weight n, for every $n \geq 0$.

The main application of this is the following.

(5.22) **Theorem.** *Suppose $f : M^n \longrightarrow N^n$ is a holomorphic, birational, mapping between compact, complex manifolds. If the dd^c-lemma holds for M^n, then it also holds for N^n.*

For a proof of this see [21] Sections 4 and 5.

Idea of proof. The main idea is to use Serre duality to prove that $f^* : H^*(N;\Omega^*) \longrightarrow H^*(M;\Omega^*)$ is a split injection. Thus the Fröhlicher spectral sequence for M contains that of N as a direct summand. From this one deduces that if (5.20) (2) holds for M, then it holds for N.

(5.23) **Corollary.** *The dd^c-lemma and hence (5.21) (2) hold for any compact, complex manifold which can be blown up to a Kähler manifold. In particular, they hold for any Moišezon space, [8].*

Section 6. The Main Theorem and Two Proofs

Main Theorem. *(i) Let M be a compact complex manifold for which the dd^c-lemma holds (e.g. M Kähler, or M a Moišezon space). Then the real homotopy type of M is a formal consequence of the cohomology ring $H^*(M;\mathbb{R})$.*

(ii) If the dd^c-lemma also holds for N, and $f : M \longrightarrow N$ is a holomorphic mapping, then the induced map on real homotopy types is a formal consequence of the induced map on real cohomology.

Corollary 1. *If M is simply connected, then the real homotopy groups $\pi_*(M) \otimes_{\mathbb{Z}} \mathbb{R}$, viewed as a graded Lie algebra using Whitehead products, depends only on the cohomology ring $H^*(M;\mathbb{R})$. Moreover, all Massey products of any order are zero over $\mathbb{Q}$.*

Corollary 2. *The real form of the canonical tower of nilpotent quotients of the fundamental group of M is determined by $H^1(M;\mathbb{R})$ and the cup product mapping $H^1 \otimes H^1 \longrightarrow H^2$.*

First Proof (the d^c-Diagram Method). Let $\mathcal{E}^*_M$ be the real de Rham complex, $\mathcal{E}^c_M$ the sub-complex of d^c-closed forms, and $H_{d^c}(M)$ the quotient complex $\mathcal{E}^c_M/d^c\mathcal{E}_M$. We have the diagram

$$\{\mathcal{E}^*_M, d\} \underset{i}{\longleftarrow} \{\mathcal{E}^c_M, d\} \underset{\rho}{\longrightarrow} \{H_{d^c}(M), d\}.$$

Assuming the dd^c-lemma we prove the following.

Claim 1) i^* and ρ^* are isomorphisms on cohomology.

2) The differential induced by d on $H_{d^c}(M)$ is 0.

Proof. a) i^* is *onto.* Let α be a cohomology class in $H^*(\mathcal{E}_M, d)$ and x a closed form in the class of α. We must vary x by an exact form to make it d^c-closed. Consider $d^c x$. It satisfies the hypothesis of the dd^c lemma, thus $d^c x = dd^c\omega$. Let $y = x + d\omega$. Then $d^c y = d^c x + d^c d\omega = 0$.

b) i^* is *one-to-one.* If y is a closed form in $\mathcal{E}^c_M$ which is exact in $\mathcal{E}_M$, then $d^c y = 0 = dy$ and $y = dz$. Thus $y = dd^c\omega$, or $y = d(d^c\omega)$, and $d^c(d^c\omega) = 0$.

c) *The induced differential on H_{d^c} is 0.* If $d^c y = 0$, then by application of the dd^c lemma to dy, we have $dy = dd^c\omega$. Hence $dy \in \operatorname{Im} d^c$. This proves that, in H_{d^c}, $[dy]$ is 0.

d) ρ^* is *onto.* Let $[y] \in H_{d^c}(M)$. Then the representative y is d^c-closed. dy satisfies the hypothesis of the dd^c lemma hence $dy = dd^c\omega$. Let $x = y + d^c\omega$. Thus $dx = 0$ and $[x] = [y] \in H_{d^c}(M)$.

e) *ρ^* is one-to-one.* Suppose $y \in \mathcal{E}^c_M$, $dy = 0$, and $[y] \in H_{d^c}(M)$ is 0. Then $y = d^c\omega$ and hence by the dd^c lemma $y = dd^c z$. Thus y is exact in $\mathcal{E}^c_M$.

This proves the claim and consequently part (1) of the theorem. Part (2) follows from the functorial nature of the diagram. That is to say that if $f : M \to N$ is a holomorphic map between complex manifolds for which the dd^c-lemma holds, then we have a commutative diagram

$$\begin{array}{ccccc} \mathcal{E}^*_N & \overset{i}{\leftarrow} & \mathcal{E}^c_N & \overset{\rho}{\to} & H^c_d(N) \\ {\scriptstyle f^*}\downarrow & & {\scriptstyle f^*}\downarrow & & {\scriptstyle f^*}\downarrow \\ \mathcal{E}^*_M & \overset{i}{\leftarrow} & \mathcal{E}^c_M & \overset{\rho}{\to} & H^c_d(M). \end{array}$$

Thus, the homotopy equivalence of algebras $\mathcal{E}^*_N \simeq H^*(N)$ and $\mathcal{E}^*_M \simeq H^*(M)$ transforms the map f on forms to its induced map on cohomology up to homotopy.

Q.E.D.

Second Proof (the Principle of Two Types). By way of an introduction to the principle of two types, and also to explain its name, let us consider the following situation on a complex manifold for which the dd^c-lemma holds: We have classes $\alpha, \beta \in H^1(M)$ such that $\alpha\beta = 0$ in $H^2(M)$, and we wish to prove that the Massey triple product $\langle\alpha, \alpha, \beta\rangle$ is zero. Using the Hodge decomposition on cohomology, we may suppose that α, β have pure type. If they are both of type $(1,0)$ $\alpha \wedge \beta$ is an exact holomorphic 2-form. By the dd^c-lemma $\alpha \wedge \beta \equiv 0$, and clearly $\langle\alpha, \alpha, \beta\rangle = 0$. By symmetry the only remaining case is $\alpha = \alpha_{1,0}$ and $\beta = \beta_{0,1}$. By the $\partial\bar\partial$-lemma, we may write in two ways $\alpha \wedge \beta = d\gamma_{1,0}$ $\alpha \wedge \beta = d\gamma_{0,1}$. $\gamma_{1,0} - \gamma_{0,1}$ is a closed 1-form. By adding an appropriate closed $(1,0)$ form to $\gamma_{1,0}$ and closed $(0,1)$ form to $\gamma_{0,1}$, we can assume $\gamma_{1,0} - \gamma_{0,1}$ is exact, see (5.23). The Massey product is represented by either $\alpha \wedge \gamma_{1,0}$ or $\alpha \wedge \gamma_{0,1}$, and these classes are homologous. Since $H^2(M)$ has a Hodge direct sum decomposition, (5.23), $\langle\alpha, \alpha, \beta\rangle = 0$. This argument works for any Massey triple product, but the higher Massey products require a systematic induction proof to which we now turn. The proof we give works only for the complex homotopy type. We give only the proof of (1), leaving the similar argument for (2) to the reader.

Suppose $\mathcal{E}^*$ is simply connected. We build a minimal model inductively by dimension

$$\mathcal{M}^2 \subset \mathcal{M}^3 \subset \cdots$$

with the properties:

1) $\mathcal{M}^n \simeq P[V_2 \oplus V_4 \cdots] \otimes \Lambda[V_3 \oplus \cdots]$;
2) each $V_k = C_k \oplus N_k$;
3) $C_k = \bigoplus_{i+j=k} C^{i,j}_k$ and $N_k = \bigoplus_{i+j\geq k+1} N^{i,j}_k$;
4) extending this (i,j) decomposition to all of $\mathcal{M}^n$ multiplicatively gives $\mathcal{M}^n = \bigoplus_{i+j\geq n} \mathcal{M}^{i,j}$ and d is of type $(0,0)$ i.e., $d(\mathcal{M}^{i,j}) \subset \mathcal{M}^{i,j}$;
5) there are homotopic maps $\rho, \rho' : \mathcal{M}^n \to \mathcal{E}^*$ such that $\rho'(\mathcal{M}^{i,j}) \subset \bar F^j(\mathcal{E}^*)$ and $\rho(\mathcal{M}^{i,j}) \subset F^{\mathrm{Min}(i,n-j+1)}(\mathcal{E}^*)$.

From these we build

$$\mathcal{M}^{n+1} = \mathcal{M}^n \otimes_d \Lambda(V_{n+1})$$

with all these properties. We can take $V_{n+1} = (\operatorname{Ker}\rho^*)_{n+2} \oplus (\operatorname{Coker}\rho^*)_{n+1}$. Let the first space be N_{n+1} and the second C_{n+1}.
By 4)

$$H^k(\mathcal{M}^n) = \bigoplus_{i+j\geq k} H^{i,j}(\mathcal{M}^n)$$

and by 5) $\rho^* = \rho'^*$ maps

$$H^{i,j}(\mathcal{M}^n) \cap H^k(\mathcal{M}^n) \text{ into } \begin{cases} 0 & \text{if } i+j>k, \\ H^{i,j}(M) & \text{if } i+j=k. \end{cases}$$

Applying this for $k = n+1$, we have

$$C_{n+1} = (\operatorname{Coker}\rho^*)_{n+1} = \bigoplus_{i+j=n+1} C^{i,j}_{n+1}.$$

Applying this for $k = n+2$, we have

$$N_{n+1} = (\operatorname{Ker}\rho^*)_{n+2} = \bigoplus_{i+j\geq n+2} N^{i,j}_{n+1}.$$

We define $d|C = 0$ and $d : N^{i,j}_{n+1} \to$ closed forms in $\mathcal{M}^{i,j}$ in dimension $n+2$ which generate $N^{i,j}_{n+1}$. Define $\rho|C = \rho'|C$ so that it sends $C^{i,j}$ into closed forms in $\mathcal{E}^{i,j}$ representing a basis for $H^{i,j}(M)$. For $x \in N^{i,j}_{n+1}$, $\rho(dx)$ is an exact form in $F^{\operatorname{Min}(i,n+2-j+1)}$. Thus we can choose $\rho(x) \in F^{\operatorname{Min}(i,n-j+3)} \subset F^{\operatorname{Min}(i,n-j+2)}$ so that $d\rho(x) = \rho(dx)$. Likewise we can choose $\rho'(x) \in \bar{F}^j(\mathcal{E}^*)$ so that $d\rho'(x) = \rho(dx)$. With these choices there is a cohomological obstruction to extending the homotopy over N_{n+1}. It assigns to $x \in N^{i,j}_{n+1}$ a cohomology class $\sigma(x) \in H^{n+1}(M)$. Since

$$H^{n+1}(M) = \bar{F}^j(H^{n+1}(M)) + F^{\operatorname{Min}(i,n+2-j)}(H^{n+1}(M))$$

we can change $\rho(x)$ by a closed class in $F^{n+2-j}(\mathcal{E}^*)$ and $\rho'(x)$ by a closed class in $\bar{F}^j(\mathcal{E}^*)$ to kill $\sigma(x)$. The homotopy from ρ to ρ' then extends over N_{n+1}. This completes the inductive proof of the existence of such $\mathcal{M}$, ρ, and ρ' in the one-connected case. The non simply connected case is similar.
Given such $\mathcal{M}$ and $\rho, \rho' : \mathcal{M} \to \mathcal{E}^*$ then $\rho^* = \rho'^*$ on cohomology. For any $\alpha \in I(\bigoplus N_i)$ with degree $\alpha = r$ we break α up into its (i,j) components

$$\alpha = \sum_{i+j\geq r+1} \alpha_{i,j}.$$

If α is closed, then each $\alpha_{i,j}$ is also closed. $\rho(\alpha_{i,j}) \in F^{r-j+1}(\mathcal{E}^*)$ and $\rho'(\alpha_{i,j}) \in \bar{F}^j(\mathcal{E}^*)$. Thus

$$\rho^*(\alpha_{i,j}) = \rho'^*(\alpha_{i,j}) \in F^{r-j+1}(H^r(M)) \cap \bar{F}^j(H^r(M)).$$

(5.23) implies that $\rho^*(\alpha_{i,j}) = \rho'^*(\alpha_{i,j}) = 0$. This verifies that $\mathcal{M}$ is formal.

Section 7. An Application

The ability to replace the forms on a Kähler manifold functorially by the cohomology allows us to show that in certain situations spectral sequences

collapse. The example we give here is a spectral sequence which the first author [3] showed to degenerate by a different technique (still using Hodge theory on Kähler manifolds, however). Let $X = D_1 \cup \cdots \cup D_j$ be a "union of divisors with normal crossings". That is each D_i is a compact Kähler manifold and $D_{i_1} \cap \cdots \cap D_{i_t}$ is a subcomplex manifold of each D_i which is the transverse intersection of $D_{i_1}, \ldots, D_{i_t}$.

We define a differential algebra of forms on X by

$$\mathcal{E}^\ell(X) = \{(\alpha_1, \ldots, \alpha_j) \mid \alpha_i \in \mathcal{E}^\ell(D_i) \text{ and } \alpha_i|_{D_i \cap D_j} = \alpha_j|_{D_j \cap D_i}\}.$$

The differential and multiplication in $\mathcal{E}^*(X)$ are the obvious coordinate-wise ones. One checks easily (either via sheaf theory or more directly by the Meyer-Vietoris sequence) that $\mathcal{E}^*(X)$ calculates the cohomology ring of X.

If $I = (i_1, \ldots, i_t)$ is an unordered set of t distinct integers $1 \leq i_r \leq j$ then define $D_I = \bigcap_{r=1,\ldots,t} D_{i_r}$. Define $\widetilde{D}^{(t)}$ to be the disjoint union of all the D_I for the length of I, $|I|$, equal to t,

$$\widetilde{D}^{(t)} = \coprod_{I, |I|=t} D_I.$$

Given a D_I there are $t!$ ways to order the intersecting divisors $D_{i_1} \cap \cdots \cap D_{i_t}$. We identify $\mathcal{E}^*(D_{i_1} \cap \cdots \cap D_{i_t})$ with $\mathcal{E}^*(D_{i_{\sigma(1)}} \cap \cdots \cap D_{i_{\sigma(t)}})$ by $(-1)^{\text{sign}(\sigma)}$ for σ any permutation. Call the resulting quotient $\mathcal{E}^*(D_I)$. (Of course, $\mathcal{E}^*(D_I) \cong \mathcal{E}^*(D_{i_1} \cap \cdots \cap D_{i_t})$ for any ordering.) Define $\mathcal{E}^*(\widetilde{D}^{(t)}) = \bigoplus_{I, |I|=t} \mathcal{E}^*(D_I)$, and define $\delta : \mathcal{E}^*(\widetilde{D}^{(t)}) \to \mathcal{E}^*(\widetilde{D}^{(t+1)})$ by the following. If $\beta \in \mathcal{E}^\ell(D_{i_1} \cap \cdots \cap D_{i_t})$, then

$$\delta(\beta) = (-1)^\ell \sum_{k \notin (i_1, \ldots, i_t)} \beta|_{(D_{i_1} \cap \cdots \cap D_{i_t} \cap D_k)}.$$

Clearly $d\delta + \delta d = 0$ and $\delta^2 = 0$.

Thus we can form the double complex associated to

$$E^* = \{\mathcal{E}^*(\widetilde{D}^{(1)}) \xrightarrow{\delta} \mathcal{E}^*(\widetilde{D}^{(2)}) \longrightarrow \cdots \xrightarrow{\delta} \mathcal{E}^*(\widetilde{D}^{(j)}) \longrightarrow 0\}.$$

Since locally

$$0 \longrightarrow \mathcal{E}^*(X) \longrightarrow \mathcal{E}^*(\widetilde{D}^{(1)}) \longrightarrow \cdots \longrightarrow \mathcal{E}^*(\widetilde{D}^{(j)}) \longrightarrow 0$$

is exact, there is a natural isomorphism $H^*(\mathcal{E}^*(X)) \xrightarrow{\cong} H^*(E^*)$.

The filtration of E^* by "the number of divisors" gives a spectral sequence abutting to $H^*(X)$ with $E_1^{p,q} = \bigoplus_{I, |I|=q} H^p(D_I)$ and with $d_1 : E_1^{p,q} \to E_1^{p,q+1}$ given by the restriction map (up to sign).

(7.1) Theorem. (Deligne [3]). *This spectral sequence collapses at E_2, i.e. $E_2 = E_\infty$. Thus one can calculate the cohomology of X from the cohomologies of the "various pieces" and the restriction maps.*

Proof. Form the analogous double compexes based on

1) $\operatorname{Ker} d^c$ on each intersection, and
2) H^* of each intersection.

Call these $E^*(\operatorname{Ker} d^c)$ and $E^*(H^*)$.

We have a natural diagram of double complexes

$$E^*(H^*) \longleftarrow E^*(\operatorname{Ker} d^c) \longrightarrow E^*.$$

If we form the three spectral sequences by filtering by the number of divisors, then we get a diagram of spectral sequences. At $E_1^{*,*}$ all three are isomorphic

by the diagram for Kähler manifolds. Hence all three are isomorphic for $E_k^{*,*}$, $1 \leq k \leq \infty$. In $E^*(H^*)$, the differential $d = 0$, hence the total differential in the double complex is δ. Clearly, then $E^*(H^*)$ collapses at E_2. Consequently, so does E^*.

Q.E.D.

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P. Deligne
Institut des Hautes Etudes Scientifiques
F-91440 Bures-sur-Yvette
France

Ph. Griffiths
Department of Mathematics
Harvard University
Cambridge, Mass 02138
USA

J. Morgan
Institut des Hautes Etudes Scientifiques
F-91440 Bures-sur-Yvette
France
and Department of Mathematics
Columbia University
New York, N.Y. 10027
USA

D. Sullivan
Institut des Hautes Etudes Scientifiques
F-91440 Bures-sur-Yvette
France

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The Support of the Character of a Supercuspidal Representation

Pierre Deligne

Note (*) by Mr. Pierre Deligne, communicated by Mr. Jean-Pierre Serre.

The character of a supercuspidal representation of a semisimple group over a local field is supported on the union of the compact open subgroups.

1 Review of monogenic groups

Let V be a finite-dimensional vector space over a field k, and let $g \in \mathrm{GL}(V)$. If k is perfect, the Zariski closure $\langle g \rangle$ of the group generated by g is the product of a multiplicative group $\langle g \rangle_m$ and a unipotent group $\langle g \rangle_{\mathrm{un}}$. If $g = su = us$ is the Jordan decomposition of g into its semisimple and unipotent parts, then $\langle g \rangle_m = \langle s \rangle$ and $\langle g \rangle_{\mathrm{un}} = \langle u \rangle$. Without any hypothesis on k, define the subgroup $\langle g \rangle_m$ of $\langle g \rangle$ to be the Zariski closure of the union of the subgroups of $\langle g \rangle$ that are kernels of multiplication by n, for n relatively prime to the characteristic exponent of k. This definition is compatible with extension of scalars and is equivalent to the preceding one when k is perfect. Over an algebraic closure $\bar{k}$ of k, the character group of $\langle g \rangle_m$ is the subgroup X of $\bar{k}^*$ generated by the eigenvalues of g, and

$$s \in \langle g \rangle_m(\bar{k}) = \mathrm{Hom}(X, \bar{k}^*)$$

corresponds to the identity inclusion of X.

Suppose that k is complete with respect to a valuation and has uniformizer π. Let v be the valuation of $\bar{k}$, with values in $\mathbf{Q}$, normalized by $v(\pi) = 1$, and let $n > 0$ be an integer such that $nv(X) \subset \mathbf{Z}$. The homomorphism $nv(x) : X \to \mathbf{Z}$ transposes to $m_g : \mathbf{G}_m \to \langle g \rangle_m$.

For a linear algebraic group G over k and $g \in G(k)$, the preceding constructions may be repeated after choosing an embedding $G \subset \mathrm{GL}(V)$. The homomorphism $m_g : \mathbf{G}_m \to \langle g \rangle_m \subset G$ does not depend on the chosen embedding (for fixed n). For every linear representation W of G, it defines an action of $\mathbf{G}_m$ on W, that is, a grading of W (SGA 3 I 4.7.3). The degree-d part is the sum of the generalized eigenspaces of g corresponding to eigenvalues $\alpha \in \bar{k}$ such that $nv(\alpha) = d$.

Suppose that G is reductive. Every homomorphism m from $\mathbf{G}_m$ to G then defines two opposite parabolic subgroups of G, the parabolics contracted and dilated by m. We write P_g^+ for the parabolic contracted by m_g (also called contracted by g). It has the following equivalent characterizations:

(a) An element $x \in G(\bar{k})$ lies in P_g^+ (respectively, in its unipotent radical U_g^+) if and only if the morphism

$$\lambda \longmapsto m_g(\lambda) x m_g(\lambda)^{-1} : \mathbf{G}_m \longrightarrow G$$

extends to $\mathbf{G}_a$ [respectively, and $\lim_{\lambda \to 0} m_g(\lambda) x m_g(\lambda)^{-1} = e$] — equivalently, if and only if $\lim_{n \to \infty} g^n x g^{-n}$ exists (respectively, equals e).

(b) $\mathrm{Lie}\, P_g^+$ (respectively, $\mathrm{Lie}\, U_g^+$) is the sum, in $\mathrm{Lie}\, G$, of the generalized eigenspaces of $\mathrm{ad}\, g$ corresponding to eigenvalues of valuation ≥ 0 (respectively, > 0).

Set $P_g^- = P_{g^{-1}}^+$ [the parabolic dilated by g and contracted by $m_{g^{-1}} = (m_g)^{-1}$], and denote its unipotent radical by U_g^-. The parabolics P_g^+ and P_g^- are opposite, and we set $M_g = P_g^+ \cap P_g^-$.

2 Statement and proof of the theorem

Assume that k is locally compact and that G is reductive. Let τ denote the projection of G onto its adjoint group G^{ad}. An admissible $(\mathscr{H}, \rho)$ representation of $G(k)$ is *supercuspidal* if its matrix coefficients are supported in inverse images of compact subsets of $G^{\mathrm{ad}}(k)$.

If K is a compact subgroup of $G(k)$, let dk denote the normalized Haar measure on K, and let $[K]$ denote its image in the algebra of compactly supported distributions on $G(k)$. The endomorphism

$$\rho(K) \coloneqq \rho([K])$$

is a projector from $\mathscr{H}$ onto the subspace $\mathscr{H}^K$ of K-invariants. Equivalently, supercuspidality of ρ means that, for every compact open subgroup K,

$$\rho(K)\rho(g)\rho(K) = 0$$

whenever $\tau(g) \in G^{\mathrm{ad}}$ lies outside a suitable compact subset (depending on K). Let G_c^{ad} denote the union of the compact subgroups (or, equivalently, the compact open subgroups) of $G^{\mathrm{ad}}(k)$.

THEOREM — *If $(\mathscr{H}, \rho)$ is an admissible supercuspidal representation of $G(k)$, its character[(1)] is supported in $\tau^{-1}(G_c^{\mathrm{ad}})$.*

Choose a system of local coordinates near e, that is, a lattice $L \subset \mathrm{Lie}(G)$ and an analytic isomorphism α from L onto a neighborhood of e, such that

(1) $\alpha(0) = e$, and $d\alpha$ at 0 is the identity of $\mathrm{Lie}(G)$.

Let $g \in G(k)$. Choose L so that

(2) $L = (L \cap \mathrm{Lie}\, U_g^+) \oplus (L \cap \mathrm{Lie}\, M_g) \oplus (L \cap \mathrm{Lie}\, U_g^-)$. Denote this decomposition by $L^+ \oplus L^0 \oplus L^-$.

(3) $\mathrm{ad}\, g(L^+) \subset L^+$, $\mathrm{ad}\, g(L^0) = L^0$, and $\mathrm{ad}\, g^{-1}(L^-) \subset L^-$.

Since P_g^+ and P_g^- are stable under $\mathrm{ad}\, g$, condition (3) may be rewritten as

(3') $\mathrm{ad}\, g(L \cap \mathrm{Lie}\, P_g^+) \subset L$ and $\mathrm{ad}\, g^{-1}(L \cap \mathrm{Lie}\, P_g^-) \subset L$.

Let a subscript i denote intersection with $G_i = \alpha(\pi^i L)$. Under hypotheses (2) and (3), there exists i_0 such that

(4) for $i \geq i_0$, G_i is a subgroup of $G(k)$, $G_i = U_{gi}^+ \cdot M_{gi} \cdot U_{gi}^-$, and

(5) $\mathrm{Int}\, g(U_{gi}^+) \subset U_{gi}^+$, $\mathrm{Int}\, g(M_{gi}) = M_{gi}$, and $\mathrm{Int}\, g^{-1}(U_{gi}^-) \subset U_{gi}^-$.

This follows from the decomposition

$$U_g^+ \times M_g \times U_g^- \hookrightarrow G : (u^+, m, u^-) \longmapsto u^+ m u^-$$

of a neighborhood of e.

Moreover, for i_0 sufficiently large, there is a neighborhood V of g such that (2)–(5) remain valid when g is replaced by $g' \in V$. For (2) and (3'), one uses the continuous dependence of P_g^+ and $\mathrm{ad}\, g$ on g; for (4) and (5), one uses the analytic dependence of m_g, and hence of $P_g^{\pm}$, on g. Thus, in the decomposition $x = u^+ m u^-$, the components u^+, m, and u^- are analytic functions of the pair (x, g).

LEMMA — *If conditions (1)–(5) hold, $i \geq i_0$, and $\tau(g) \notin G_c^{\mathrm{ad}}$, then $\rho(G_i)\rho(g)\rho(G_i)$ is nilpotent.*

For the decomposition $U_g^+ \times M_g \times U_g^- \to G$ of the big cell of G, Haar measures satisfy

$$dg = \lambda(m)\, du^+\, dm\, du^-,$$

where λ is a character of M_g. On every compact subgroup of M_g, $\lambda = 1$; hence

$$[G_i] = [U_{gi}^+] \star [M_{gi}] \star [U_{gi}^-].$$

We prove by induction on N that

$$([G_i]g[G_i])^N = [G_i]g^N[G_i]$$

in the algebra of compactly supported distributions on G (we write g for δ_g). Indeed,

$$\begin{aligned}([G_i]g^N[G_i])([G_i]g[G_i]) &= [G_i]g^N[G_i]g[G_i] \\ &= [G_i]g^N[U_{gi}^+][M_{gi}][U_{gi}^-]g[G_i] \\ &= [G_i][g^N U_{gi}^+ g^{-N}][g^N M_{gi} g^{-N}]g^N g[g^{-1}U_{gi}^- g][G_i] \\ &= [G_i]g^{N+1}[G_i].\end{aligned}$$

If $\tau(g) \notin G_c^{\mathrm{ad}}$, the set of $\tau(g^N)$, $N \geq 1$, is not relatively compact. Since ρ is supercuspidal, there is therefore an N such that

$$(\rho(G_i)\rho(g)\rho(G_i))^N = \rho(G_i)\rho(g^N)\rho(G_i) = 0.$$

Proof of the theorem — For $g \in G(k)$ such that $\tau(g) \notin G_c^{\mathrm{ad}}$, choose L, α, i_0, and V as above, with $gG_{i_0} \subset V$. Let Θ denote the character of ρ. For every $g' \in gG_i$ and every $i \geq i_0$, we then have

$$\Theta(g'[G_i]) = \Theta([G_i]g'[G_i]) = \mathrm{Tr}(\rho(G_i)\rho(g')\rho(G_i)) = 0$$

by the lemma. Since the subgroups G_i form a fundamental system of neighborhoods of 0, Θ vanishes on gG_i, and its support does not contain g.

Remark 1 — The same theorem, with the same proof, holds for an analytic group that is a finite étale covering of an open subgroup of G. The hypothesis becomes "the image in G^{ad} of the support of the matrix coefficients is relatively compact"; the conclusion becomes "the image in G^{ad} of the support of Θ lies in G_c^{ad}".

Remark 2 — Let p be the residual characteristic of k. The proof makes essential use of the Haar measures $[K]$ only for K small, and in particular for a p-group. The theorem and its proof therefore hold for representations with coefficients in any field of characteristic different from p.

Source notes

(*) Session of 24 May 1976.

(1) Once a Haar measure dg on G has been chosen, the character Θ of ρ is the distribution on G whose value on a test function φ is

$$\mathrm{Tr}\left(\int \varphi(g)\rho(g)\,dg\right).$$

More intrinsically, it is the generalized function [the linear form on the space of measures $\varphi(g)\,dg$, where $\varphi(g)$ is locally constant with compact support and dg is a Haar measure] given by this same formula.

Institut des Hautes Études scientifiques,
Le Bois-Marie,
91440 Bures-sur-Yvette.

Local Constants in the Functional Equation of the Artin L-Function of an Orthogonal Representation

Pierre Deligne (Bures-sur-Yvette)

To Jean-Pierre Serre, in token of admiration.

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1. Statement of the theorem

(1.1) Let K be a global field (an A-field in Weil's sense), let K' be a finite Galois extension of K, and let V be a complex representation of the Galois group $\mathrm{Gal}(K'/K)$. We follow the conventions of [1] in defining the corresponding Artin L-function: the factors at infinity are included, and the local nonarchimedean factors are defined in terms of geometric Frobenius elements (the inverses of Frobenius substitutions). We further normalize the reciprocity isomorphism of local class field theory so that it sends geometric Frobenius elements to uniformizers (cf. [1] 3.6). These sign questions are in fact immaterial: our essential calculations will be made modulo 2. The function $L(V,s)$ satisfies a functional equation

$$L(V,s) = \varepsilon(V,s)L(V^*,1-s), \tag{1.1.1}$$

where $\varepsilon(V,s)$ is the product of a constant and an exponential factor, and V^* is the dual of V.

Let f be the norm of the Artin conductor of V, and let D be the absolute value of the discriminant of K (respectively q^{2g-2} for a function field); define

$$W(V) = \varepsilon\Big(V, \tfrac{1}{2}\Big). \tag{1.1.2}$$

Then

$$\varepsilon(V,s) = W(V)\big(f \cdot D^{\dim(V)}\big)^{\frac{1}{2}-s}, \tag{1.1.3}$$

$$W(V)\cdot W(V^*) = 1. \tag{1.1.4}$$

If V is isomorphic to V^*, i.e. if the character of V is real-valued, (1.1.4) shows that $W(V) = \pm 1$. Fröhlich and Queyrut [2] showed that if V is in fact orthogonal, i.e. realizable over $\mathbf{R}$, then $W(V) = +1$.

(1.2) Let K be a local field (= a nondiscrete locally compact field), let K' be a finite Galois extension of K contained in a separable closure $\bar{K}$ of K, let V be a complex representation of the Galois group $G = \mathrm{Gal}(K'/K)$, let $\psi : K \to \mathbf{C}^*$ be a nontrivial additive character, and let dx be a Haar measure on K. In fact, K' plays no role: all that is used is the representation of $\mathrm{Gal}(\bar{K}/K)$ obtained from V by inflation; it is again denoted by V. I refer to [1] §§ 4, 5 for the definition of $\varepsilon(V, \psi, dx, s)$. In [1], more generally, V is taken to be a representation of the Weil group $W(\bar{K}/K)$ (of which $\mathrm{Gal}(\bar{K}/K)$ is a completion in the nonarchimedean case and a quotient in the archimedean case); one defines $\varepsilon(V, \psi, dx)$, and sets $\varepsilon(V, \psi, dx, s) = \varepsilon(V \otimes \omega_s, \psi, dx)$, where ω_s is the quasicharacter $\|x\|^s$ of $K^* \simeq W(\bar{K}/K)^{\mathrm{ab}}$. As a function of s, $\varepsilon(V, \psi, dx, s)$ is the product of a constant and an exponential factor.

Starting from a global situation as in 1.1, let K_v run through the completions of K, let V_v be the restriction of V to a decomposition group at v, let the ψ_v be the local components of a character of A_K/K, and suppose that

$$\int_{A_K/K} \bigotimes_v dx_v = 1.$$

Then

$$\varepsilon(V, s) = \prod_v \varepsilon(V_v, \psi_v, dx_v, s). \tag{1.2.1}$$

Both locally and globally, ε is multiplicative in V: $\varepsilon(V \oplus V') = \varepsilon(V) \cdot \varepsilon(V')$. This makes it possible to define ε for virtual representations (elements of the Grothendieck group $R(G, \mathbf{C})$ of the category of representations of G).

Return to the local case. If V is a virtual representation of dimension 0, $\varepsilon(V, \psi, dx, s)$ is independent of dx, which is omitted from the notation ([1] 5.3). If V has dimension 0 and determinant $\chi : K^* \simeq W(\bar{K}/K)^{\mathrm{ab}} \to \mathbf{C}^*$, then ([1] 5.4)

$$\varepsilon(V, \psi(ax), s) = \chi(a)\varepsilon(V, \psi, s). \tag{1.2.2}$$

In particular, if $\chi = 1$, $\varepsilon(V, \psi, s)$ is independent of ψ; ψ is then omitted from the notation.

The Artin conductor f of V is by definition 1 when K is archimedean. When K is nonarchimedean, it is the number denoted $q^{a(V)}$ in [1] (q = the number of elements of the residue field). For V of dimension 0 and determinant 1, set

$$W(V) = \varepsilon\left(V, \tfrac{1}{2}\right). \tag{1.2.3}$$

Then

$$\varepsilon(V, s) = W(V) f^{\frac{1}{2} - s}, \tag{1.2.4}$$

$$W(V) \cdot W(V^*) = 1. \tag{1.2.5}$$

If V is isomorphic to its dual, i.e. if its character is real, (1.2.5) shows that $W(V) = \pm 1$.

The group $R(G, \mathbf{R})$ of real virtual representations of G, the Grothendieck group of the category of real representations of G, is a subgroup of the group of virtual representations of G having real character. The elements of

this subgroup are said to be realizable over $\mathbf{R}$. We propose to calculate $W(V_v)$ when V_v has dimension 0, determinant 1, and is realizable over $\mathbf{R}$.

(1.3) Let G be a finite group. A real representation $\rho : G \to GL(V)$ has a Stiefel-Whitney class

$$w^*(V) = 1 + w^1(V) + \cdots \in H^*(G, \mathbf{Z}/2) = \prod_i H^i(G, \mathbf{Z}/2).$$

It may be defined as follows: the representation V defines a vector bundle (indeed, a local system) $\mathscr{V}$ on the classifying space BG of the group G, and $w^*(V)$ is the Stiefel-Whitney class of $\mathscr{V}$ in $H^*(BG, \mathbf{Z}/2) = H^*(G, \mathbf{Z}/2)$. The same construction applies to real virtual representations: w^* is the composite homomorphism

$$R(G, \mathbf{R}) \longrightarrow KO(BG) \xrightarrow{w^*} H^*(BG, \mathbf{Z}/2)^\times = H^*(G, \mathbf{Z}/2)^\times.$$

Via the isomorphism $H^1(G, \mathbf{Z}/2) = \mathrm{Hom}(G, \{\pm 1\})$, $w^1(V)$ identifies with the character $g \mapsto \det \rho(g)$. The group $H^2(G, \mathbf{Z}/2)$ classifies central extensions of G by the two-element group. If $w^1(V) = 0$, then $w^2(V)$ is the class of the extension obtained by pulling back under ρ the double cover $\mathrm{Spin}(V, Q)$ of $SO(V, Q)$, where Q is any positive-definite G-invariant quadratic form on V.

(1.4) Retain the notation of 1.2, and let $\mathrm{cl}(w^2(V))$ be the image of the class $w^2(V) \in H^2(G, \mathbf{Z}/2)$ under the composite map

$$H^2(G, \mathbf{Z}/2) \xrightarrow{\text{infl}} H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2) \xrightarrow{(1)} H^2(\mathrm{Gal}(\bar{K}/K), \bar{K}^*) \xrightarrow{\text{inv}} \mathbf{Q}/\mathbf{Z}, \tag{1.4.1}$$

where (1) is induced by $n \mapsto (-1)^n : \mathbf{Z}/2 \to \bar{K}^*$, and where inv is supplied by local class field theory. If K is complex or has characteristic 2, then $\mathrm{cl}\, w^2(V) = 0$. In the other cases, (1.4.1) identifies $H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2)$ with $\{0, \frac{1}{2}\} \subset \mathbf{Q}/\mathbf{Z}$.

Our principal result is the following:

(1.5) Theorem. If V is a real virtual representation of dimension 0 and determinant 1, then

$$W(V) = \exp\bigl(2\pi i\ \mathrm{cl}(w^2(V))\bigr).$$

Thus, in characteristic 2, $W(V) = 1$. In characteristic $\neq 2$, $W(V) = 1$ or -1 according as $w^2(V)$ is or is not trivial in $H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2)$.

(1.6) From this theorem one can deduce the theorem of Queyrut and Fröhlich:

(a) For V a real virtual representation of dimension 0 and determinant 1 over a global field K of characteristic $\neq 2$, the $\mathrm{cl}(w^2(V_v))$ are the local invariants of $w^2(V) \in H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2)$, and the reciprocity law ensures that their sum in $\mathbf{Q}/\mathbf{Z}$ is zero; hence the product of the $W(V_v)$ is one. In equal characteristic 2, all the $W(V_v)$ are 1.

(b) If V has dimension one and is defined by a character of order two (respectively, is trivial), the corresponding L-function is a quotient of two zeta functions (respectively, a zeta function), and one uses the fact that the global constant of a zeta function is $+1$.

(c) The conclusion follows by observing that every real virtual representation is a sum of representations of types (a) and (b).

(1.7) We shall also prove the variants of these results obtained by replacing Galois groups by Weil groups (§ 5).

2. Lemmas on real representations of finite groups

(2.1) Proposition. Let G be a finite group, H a subgroup, V a real virtual representation of H, and $\mathrm{Tr} : H^2(H, \mathbf{Z}/2) \to H^2(G, \mathbf{Z}/2)$ the trace morphism (corestriction). If the dimension of V is divisible by 4 and its determinant is 1 ($w^1(V) = 0$), then

$$w^1\left(\mathrm{Ind}_H^G(V)\right) = 0 \quad \text{and} \quad w^2\left(\mathrm{Ind}_H^G(V)\right) = \mathrm{Tr}\, w^2(V).$$

Let us translate this statement into topological terms. Let B_G be a classifying space for G, let E_G be the universal cover of B_G (the universal principal homogeneous G-space), and set $B_H = E_G/H$. This is a classifying space for H. Let f denote the projection $f : B_H \to B_G$. The diagrams

$$\begin{array}{ccc} R(H,\mathbf{R}) & \longrightarrow & KO(B_H) \\ \downarrow \mathrm{Ind} & & \downarrow f_* \\ R(G,\mathbf{R}) & \longrightarrow & KO(B_G) \end{array} \qquad \text{and} \qquad \begin{array}{ccc} H^*(B_H,\mathbf{Z}/2) & = & H^*(H,\mathbf{Z}/2) \\ \downarrow \mathrm{tr} & & \downarrow f_* \\ H^*(B_G,\mathbf{Z}/2) & = & H^*(G,\mathbf{Z}/2) \end{array}$$

commute (the definition of the maps f_* is recalled below), so that 2.1 follows from the following proposition.

(2.2) Proposition. Let $f : X \to Y$ be a finite covering of topological spaces. If $v \in KO(X)$ has dimension divisible by 4 and zero first Stiefel-Whitney class, then $w^1(f_*v) = 0$ and

$$w^2(f_*v) = f_*(w^2(v)).$$

Recall the definition of the maps f_*.

(a) In K-theory: f_* is induced by the operation that associates to a vector bundle $\mathscr{V}$ on X the vector bundle $f_*\mathscr{V}$ on Y whose fiber at $y \in Y$ is the direct sum

$$\bigoplus_{f(x)=y} \mathscr{V}_x.$$

(b) In cohomology. f_* is the composite of the isomorphism $H^*(X, \mathbf{Z}/2) = H^*(Y, f_*\mathbf{Z}/2)$ and the morphism induced functorially by the trace morphism:

$$\mathrm{Tr}_f : f_*\mathbf{Z}/2_X \longrightarrow \mathbf{Z}/2_Y : \text{“sum of the values over the sheets.”}$$

We reduce to the case in which f has a constant number k of sheets and v is defined by an orthogonal vector bundle $\mathscr{V}$ of constant dimension n. Let P' be the corresponding $O(n)$-torsor and let $[P']$ be its class in $H^1(X, O(n))$. The descriptions in 1.3 of w^1 and w^2 have the following analogue.

(a) $w^1 \in H^1(X, \mathbf{Z}/2)$ is the image of $[P']$ under

$$\det : H^1(X, O(n)) \longrightarrow H^1(X, \{\pm 1\}) = H^1(X, \mathbf{Z}/2).$$

If $w^1 = 0$, P' lifts to an $SO(n)$-torsor P, of class $[P]$.

(b) The exact sequence

$$0 \longrightarrow \mathbf{Z}/2 \longrightarrow \mathrm{Spin}(n) \longrightarrow SO(n) \longrightarrow 0$$

induces $\partial : H^1(X, SO(n)) \to H^2(X, \mathbf{Z}/2)$, and, if $w^1 = 0$, $w^2 = \partial[P]$.

The construction $\mathscr{V} \mapsto f_*\mathscr{V}$ associates to an $O(n)$-torsor on X an $O(nk)$-torsor on Y. It factors as follows: the local isomorphisms from the constant orthogonal vector bundle $(\mathbf{R}^n)^k$ to $f_*\mathscr{V}$ which, at each point y, carry each of the k factors of $(\mathbf{R}^n)^k$ to one of the factors in the decomposition

$$(f_*\mathscr{V})_y = \bigoplus_{f(x)=y} \mathscr{V}_x,$$

identify with the local sections of a torsor under the semidirect product $O(n)^k \rtimes \mathfrak{S}_k$ of $O(n)^k$ by the symmetric group on k letters. The required $O(nk)$-torsor is obtained from this torsor and the morphism $O(n)^k \rtimes \mathfrak{S}_k \to O(nk)$.

If $w^1(\mathscr{V}) = 0$, the structure group of $f_*\mathscr{V}$ can be reduced to $SO(n)^k \rtimes \mathfrak{S}_k$. If n is even, this group is contained in $SO(nk)$, and $w^1(f_*\mathscr{V}) = 0$.

Suppose that the structure group of $\mathscr{V}$ has been reduced to $SO(n)$, giving an $SO(n)$-torsor P and an $SO(n)^k \rtimes \mathfrak{S}_k$-torsor Q on Y, of class $[Q]$. Consider the exact sequence

$$0 \longrightarrow (\mathbf{Z}/2)^k \longrightarrow \mathrm{Spin}(n)^k \rtimes \mathfrak{S}_k \longrightarrow SO(n)^k \rtimes \mathfrak{S}_k \longrightarrow 0.$$

The quotient $SO(n)^k \rtimes \mathfrak{S}_k$ acts on $(\mathbf{Z}/2)^k$ through $\mathfrak{S}_k$, and the local system $((\mathbf{Z}/2)^k)^Q$ on Y identifies with $f_*\mathbf{Z}/2$. Via the isomorphism $H^*(X, \mathbf{Z}/2) \simeq H^*(Y, f_*\mathbf{Z}/2)$, the class $w^2(\mathscr{V}) = \partial[P]$ identifies with $\partial[Q] \in H^2(Y, ((\mathbf{Z}/2)^k)^Q)$, as is shown by a cocycle computation.

The proposition now follows from the existence, when n is divisible by 4, of a middle arrow making the diagram

$$\begin{array}{ccccccccc} 0 & \to & (\mathbf{Z}/2)^k & \to & \mathrm{Spin}(n)^k \rtimes \mathfrak{S}_k & \to & SO(n)^k \rtimes \mathfrak{S}_k & \to & 0 \\ & & \downarrow \Sigma & & \downarrow & & \downarrow & & \\ 0 & \to & \mathbf{Z}/2 & \to & \mathrm{Spin}(nk) & \to & SO(nk) & \to & 0 \end{array} \tag{2.2.1}$$

commute, where Σ is the sum-of-coordinates homomorphism.

To construct (2.2.1), it is enough to construct a lift

$$\begin{array}{ccc} \mathfrak{S}_k & \longrightarrow & SO(nk) \\ & \searrow\!\!\dashrightarrow & \uparrow \\ & & \mathrm{Spin}(nk). \end{array}$$

The obstruction to constructing it is defined for even n, and lies in $H^2(\mathfrak{S}_k, \mathbf{Z}/2)$. It is additive in n, because the diagram

$$\begin{array}{ccccccccc} 0 & \to & \mathbf{Z}/2 \times \mathbf{Z}/2 & \to & \mathrm{Spin}(n'k) \times \mathrm{Spin}(n''k) & \to & SO(n'k) \times SO(n''k) & \to & 0 \\ & & \downarrow a+b & & \downarrow & & \downarrow & & \\ 0 & \to & \mathbf{Z}/2 & \to & \mathrm{Spin}((n'+n'')k) & \to & SO((n'+n'')k) & \to & 0 \end{array}$$

commutes. It is therefore zero when n is divisible by 4.

(2.3) Lemma. The real virtual representations of dimension 0 and determinant 1 of a finite group G form an ideal of $R(G, \mathbf{R})$. This ideal is stable under inflation, restriction, and induction.

That they form an ideal follows from the fact that $\dim : R(G, \mathbf{R}) \to \mathbf{Z}$ is a homomorphism, that $\det(V + W) = \det(V) \cdot \det(W)$, and that

$$\det(V \otimes W) = \det(V)^{\dim(W)} \cdot \det(W)^{\dim(V)}.$$

Stability under inflation and restriction is immediate, while stability under induction follows from 2.2 (the first assertion).

An analogous lemma holds for complex representations. It is proved in the same way ([1] 1.2).

(2.4) Let D_n be the dihedral group, the semidirect product of $\mathbf{Z}/2$ by $\mathbf{Z}/n$, and let $\chi : \mathbf{Z}/n \to \mathbf{C}^*$ be a character. The representation $\mathrm{Ind}(\chi)$ is realizable over $\mathbf{R}$: it is realized on the real vector space $\mathbf{C}$, on which $\mathbf{Z}/n$ acts by χ and $\mathbf{Z}/2$ acts by complex conjugation. Its determinant is independent of χ. The virtual representation

$$r(\chi) = \mathrm{Ind}_{\mathbf{Z}/n}^{D_n}([\chi] - [1])$$

is therefore realizable over $\mathbf{R}$, has dimension 0, and has trivial determinant.

(2.5) Proposition. Let G be a finite group. Every real virtual representation of G of dimension 0 and trivial determinant is a sum

(a) of the realification of a complex virtual representation of dimension 0, and

(b) of a $\mathbf{Z}$-linear combination of representations induced from representations $r(\chi)$ of dihedral quotients of subgroups of G.

Recall that the "realification" of a complex representation V of a group G is the representation of G on the same space V, regarded as a real vector space by restriction of scalars from $\mathbf{C}$ to $\mathbf{R}$.

(2.6) Lemma. Let G be a finite group having a nilpotent subgroup G' of index 1 or 2. Every real virtual representation of G is a sum

(a) of the realification of a complex virtual representation of G,

(b) of a $\mathbf{Z}$-linear combination of representations induced from representations $r(\chi)$,

(c) of a $\mathbf{Z}$-linear combination of one-dimensional real representations.

Let us deduce the proposition from the lemma.

Brauer's induction theorem (as generalized by Witt, J. Crelle, 190 (1952); see, for example, Curtis-Reiner, Representation theory of finite groups and associative algebras, Intersc. Publ., N.Y. 1962) asserts that the representation 1 of G is a $\mathbf{Z}$-linear combination $\sum_i n_i \mathrm{Ind}_{H_i}^G(\rho_i)$ of representations induced from real representations of subgroups H_i of G, each H_i being a semidirect product of a p-group P by a cyclic group C, with P acting on C only by ± 1. Lemma 2.3 and the identity

$$V = V \otimes 1 = V \otimes \sum_i n_i \mathrm{Ind}_{H_i}^G(\rho_i) = \sum_i n_i \mathrm{Ind}_{H_i}^G(V \otimes \rho_i)$$

bring us back to the case where G itself is such a semidirect product, hence of the type considered in 2.6. For a virtual V of dimension 0 and determinant 1, consider the decomposition 2.6

$$V = A + B + C = (A - \dim A \cdot 1) + B + (C + \dim A \cdot 1).$$

The representation $(A - \dim A \cdot 1)$ is of type (a) in the proposition, and B is of type (b). Since V, $(A - \dim A \cdot 1)$, and B have dimension 0 and determinant 1, $(C + \dim A \cdot 1)$ has the same properties; this reduces us to proving the proposition for $(C + \dim A \cdot 1)$, hence for a $\mathbf{Z}$-linear combination of one-dimensional real representations. Thus let V be of the form

$$V = \sum_{i=1}^{k} [\varepsilon_i] - \sum_{i=1}^{l} [\eta_i]$$

with ε_i and η_i characters of order dividing 2, $k = l$ (dimension 0), and $\prod \varepsilon_i = \prod \eta_i$ (determinant 1).

We have

$$\begin{aligned}\sum_{i=1}^{k} [\varepsilon_i] &= k \cdot 1 + \sum_{i=1}^{k} ([\varepsilon_i] - 1) \\ &= k \cdot 1 - \sum_{i=1}^{k} ([\varepsilon_i] - 1) \otimes \left(\left[\prod_{j<i} \varepsilon_j \right] - 1 \right) + \sum_{i=1}^{k} ([\varepsilon_i] - 1) \otimes \left[\prod_{j<i} \varepsilon_j \right] \\ &= k \cdot 1 + \left(\left[\prod \varepsilon_i \right] - 1 \right) - \sum_{i=1}^{k} ([\varepsilon_i] - 1) \left(\left[\prod_{j<i} \varepsilon_j \right] - 1 \right).\end{aligned}$$

It follows that V has an expression of the form

$$V = \sum \pm \left([\alpha_i \beta_i] - [\alpha_i] - [\beta_i] + 1 \right),$$

where the α_i and β_i have order dividing 2. The terms in this sum are zero if $\alpha_i = 1$ or $\beta_i = 1$, are of type (a) if $\alpha_i = \beta_i$, and are of type (b) for a dihedral group $D_4 = (\mathbf{Z}/2)^2$ otherwise.

(2.7) Lemma. Lemma 2.6 holds for a representation of dimension 2.

Such a representation is in fact either

(a) the realification of a one-dimensional complex representation,

(b) dihedral, hence the sum of a virtual representation $r(\chi)$ and $[\varepsilon] + 1$, for a character ε of order two.

(2.8) Lemma. Lemma 2.6 holds for a representation of G induced from a one-dimensional real representation of a subgroup of G.

We proceed by induction on the index of the subgroup H.

Let $H' = G' \cap H$. If H has index ≤ 2, apply Lemma 2.7. Otherwise, there is a subgroup J of G', belonging to the ascending central series, such that $K = H'J/H'$ is nontrivial and abelian. Let l be a prime dividing the order of K, and let K_l be the kernel of multiplication by l. The group H/H' acts on K and K_l.

Since H/H' has order ≤ 2, all its irreducible representations (over $\mathbf{Q}$, or mod l) are one-dimensional, and the $\mathbf{F}_l$-vector space K_l contains a line stable under H/H'. Let I' be its inverse image in $H'J$. The group H normalizes I', so that $I = I'H$ is a subgroup. We have $I' = G' \cap I$, and $H' \subset I' \subset G'$, with I'/H' cyclic of order l.

Let $\varepsilon : H \to \{\pm 1\}$ be the character of the representation of H under consideration. The transitivity formula

$$\mathrm{Ind}_H^G(\varepsilon) = \mathrm{Ind}_I^G \mathrm{Ind}_H^I(\varepsilon)$$

and the induction hypothesis applied to I reduce us to showing that the representation $\mathrm{Ind}_H^I(\varepsilon)$ of I satisfies Lemma 2.6. If $l = 2$, this representation has dimension two, and Lemma 2.7 applies. If $l \neq 2$, the nilpotence of G' ensures that I' normalizes $\varepsilon|_{H'}$: otherwise, G' would have a subquotient that is a noncentral extension of a group of order l by a group of type $(2, \dots, 2)$, and such a group is not nilpotent.

Let us calculate the restriction of $\mathrm{Ind}_H^I(\varepsilon)$ to I'. It is $\mathrm{Ind}_{H'}^{I'}(\varepsilon)$, obtained by inflation from $\mathrm{Ind}_{H'/\ker(\varepsilon)}^{I'/\ker(\varepsilon)}(\varepsilon)$. The group $I'/\ker(\varepsilon)$ is the product of a cyclic group of order l by $H'/\ker(\varepsilon)$, of order ≤ 2. Thus, as a complex representation, the representation of I' is a sum of distinct complex characters, and I/I' acts on the set of these characters either as the identity or by $\chi \mapsto \chi^{-1}$ (according to its action on I'/H'). Grouping χ and χ^{-1} together, we find that the representation $\mathrm{Ind}_H^I(\varepsilon)$ of I is a sum of real representations of dimension ≤ 2, and Lemma 2.7 applies.

(2.9) Let us prove Lemma 2.6. We proceed by induction on the order of G.

We may, and do, assume that

(1) V is an irreducible representation (over $\mathbf{R}$);

(2) V is even absolutely irreducible (otherwise V is of type (a));

(3) V is faithful (otherwise the induction hypothesis is applied to a quotient of G);

(4) V is not induced from a real representation of a proper subgroup of G (otherwise the induction hypothesis is applied to that subgroup, together with Lemma 2.8);

(5) G has no abelian subgroup of index ≤ 2 (otherwise, by (2), $\dim(V) \leq 2$, and Lemma 2.7 applies).

Let B be an abelian normal subgroup of G, contained in G', and maximal with these properties. If the complex character χ of B occurs in the restriction of V to B, then V is known to be induced from a real representation of the subgroup H of G that stabilizes $\{\chi, \bar{\chi}\}$. By (1) and (4), $H = G$, and only χ and $\bar{\chi}$ occur in $V|_B$. Let G'' be the stabilizer of χ. We have $B \subset G' \cap G''$, and B is central in G''.

Case 1. $B \neq G' \cap G''$. By (3), B is central in the nilpotent group $G' \cap G''$. The group $G/(G' \cap G'')$ is of type $(\mathbf{Z}/2)^r$ (with $r = 0, 1$, or 2); all its irreducible representations (over $\mathbf{Q}$, or mod l) are one-dimensional. The argument at the beginning of the proof of Lemma 2.8 therefore applies again and shows that there is a subgroup B' of $G' \cap G''$, normal in G, with B'/B cyclic. This group B' is abelian, contradicting the maximality of B.

Case 2. $B = G' \cap G''$. Then B has index ≤ 2 in G'', which is therefore abelian. This contradicts (5).

3. Proof of 1.5

(3.1) Return to the notation of 1.2 and 1.4. For V of dimension 0 and determinant 1, the number $W(V) = \mathrm{sgn}(\varepsilon(V))$ is invariant under induction and multiplicative in V. By 2.1 and local class field theory (namely, the formula $\mathrm{inv}\,\mathrm{Tr} = \mathrm{inv}$), $\mathrm{cl}\, w^2(V)$ is likewise invariant under induction and additive in V. Proposition 2.5 therefore reduces the proof of the theorem to the following two cases.

Case 1. V is the realification of a complex virtual representation W of dimension zero.

Let $\chi : K^* \to \mathbf{C}^*$ be the determinant of W. For every ψ, $\varepsilon(V) = \varepsilon(W,\psi)\cdot\varepsilon(\bar{W},\psi)$; and

$$\varepsilon(\bar{W},\psi) = \chi(-1)\cdot\varepsilon(\bar{W},\psi(-x)) = \chi(-1)\overline{\varepsilon(W,\psi)} \qquad (1.2.2),$$

whence $W(V) = \chi(-1)$.

In particular, $W(V) = 1$ when K has characteristic two, as it should. Now suppose that K has characteristic $\neq 2$. The class $w^2(V)$ is the reduction modulo 2 of the first Chern class $c^1(W)$. The class $c^1(W) \in H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z})$ is, up to sign, the image of $\chi \in H^1(\mathrm{Gal}(\bar{K}/K), \mathbf{C}^*)$ under the connecting homomorphism ∂ in the long exact sequence arising from the exponential exact sequence

$$0 \longrightarrow \mathbf{Z} \xrightarrow{2\pi i} \mathbf{C} \xrightarrow{\exp} \mathbf{C}^* \longrightarrow 0.$$

The class $w^2(V)$ is similarly obtained from χ and the sequence

$$0 \longrightarrow \mathbf{Z}/2 \longrightarrow \mathbf{C}^* \xrightarrow{z^2} \mathbf{C}^* \longrightarrow 0.$$

It is zero if and only if χ has a square root, i.e., since -1 is the unique element of order 2 in $\widehat{K^*} = \mathrm{Gal}(\bar{K}/K)^{\mathrm{ab}}$, if and only if $\chi(-1) = 1$. This proves the theorem.

Case 2. V is obtained by inflation from a representation $r(\chi)$ of a dihedral quotient of $\mathrm{Gal}(\bar{K}/K)$.

For archimedean K, necessarily $V = 0$. We may therefore assume, and do assume, that K is nonarchimedean.

Let G be a finite group that is an extension of $\mathbf{Z}/2$ by $\mathbf{Z}/n$. If $u \in G$ does not belong to the subgroup $\mathbf{Z}/n$, its image under the transfer $G^{\mathrm{ab}} \to \mathbf{Z}/n$ is u^2. Thus G is dihedral if and only if the transfer is trivial.

Recall also that, for an extension L of K, the transfer $\mathrm{Gal}(\bar{K}/K)^{\mathrm{ab}} \to \mathrm{Gal}(\bar{K}/L)^{\mathrm{ab}}$ corresponds to the inclusion $K^* \to L^*$. The situation in Case 2 can therefore be described as follows: begin with a separable quadratic extension L/K and a character $\chi : L^* \to \mathbf{C}^*$ trivial on K^*. The representation $\mathrm{Ind}\,\chi$ of $\mathrm{Gal}(\bar{K}/K)$ is then realizable over $\mathbf{R}$, is dihedral, and $V = \mathrm{Ind}([\chi] - 1)$.

(3.2) Theorem (Fröhlich and Queyrut [2]). For V as above, and for a nonzero element Δ of L such that $\mathrm{Tr}_{L/K}(\Delta) = 0$ (Δ is well defined up to multiplication by an element of K^*), one has

$$W(V) = \chi(\Delta).$$

Let ψ be a nontrivial additive character of K, let $\psi_L = \psi \circ \mathrm{Tr}_{L/K}$, and let dx be a Haar measure on L. Then

$$\begin{aligned}\varepsilon(V,s) &= \varepsilon(\mathrm{Ind}([\chi]-1),\psi,s) = \varepsilon([\chi]-1,\psi_L,s)\\ &= \varepsilon(\chi,\psi_L,dx,s)\cdot\varepsilon(1,\psi_L,dx,s)^{-1}.\end{aligned}$$

Define the Fourier transform on the Schwartz-Bruhat functions on L by

$$\widehat{f}(y) = \int f(x)\psi_L(xy)\,dx.$$

The ε-factors are defined by Tate's functional equation

$$\frac{\int \widehat{f}(x)\|x\|^{1-s}\chi^{-1}(x)\,d^*x}{L(\chi^{-1},1-s)} = \varepsilon(\chi,\psi_L,dx,s)\frac{\int f(x)\|x\|^{s}\chi(x)\,d^*x}{L(\chi,s)} \tag{3.2.1}$$

(the same Haar measure d^*x on L^* is used on both sides). The two sides of this formula are generally defined by analytic continuation in s. However, if $f(0) = \widehat{f}(0) = 0$, the integrals reduce to finite sums. Let $x \mapsto \bar{x}$ be the nontrivial K-automorphism of L. We have $\chi^{-1}(x) = \chi(\bar{x})$, so that $L(\chi,s) = L(\chi^{-1},s)$. Setting $s = \frac{1}{2}$ in (3.2.1), we find that, when $f(0) = \widehat{f}(0) = 0$,

$$\int \widehat{f}(x)\|x\|^{1/2}\chi^{-1}(x)\,d^*x = \varepsilon\Big(\chi,\psi_L,dx,\tfrac{1}{2}\Big)\int f(x)\|x\|^{1/2}\chi(x)\,d^*x.$$

We have

$$\int f(x)\|x\|^{1/2}\chi(x)\,d^*x = \int_{L^*/K^*}\|x\|^{1/2}\chi(x)(d^*x/d^*a)\int_{K^*} f(ax)\,da,$$

where da is a Haar measure on K, and

$$d^*a = \frac{da}{\|a\|_K}$$

(note that $\|a\|_L = \|a\|_K^2$). Similarly,

$$\begin{aligned}\int \widehat{f}(x)\|x\|^{1/2}\chi^{-1}(x)\,d^*x &= \int_{L^*/K^*}\|x\|^{1/2}\chi^{-1}(x)(d^*x/d^*a)\int_K \widehat{f}(ax)\,da\\ &= \int_{L^*/K^*}\|x\|^{1/2}\chi(x)(d^*x/d^*a)\int_K \widehat{f}(a\bar{x})\,da.\end{aligned}$$

For a constant c depending on ψ_L, dx, da, and Δ, one has

$$\int_K \widehat{f}(a\bar{x})\,da = c\int_K f(ax\Delta)\,da,$$

so, putting $c_1 = c\cdot\|\Delta\|^{-1/2}$,

$$\begin{aligned}\int \widehat{f}(x)\|x\|^{1/2}\chi^{-1}(x)\,d^*x &= \int_{L^*/K^*}\|x\|^{1/2}\chi(x)(d^*x/d^*a)c\cdot\int_K f(ax\Delta)\,da\\ &= c\cdot\|\Delta\|^{-1/2}\chi(\Delta)\int_{L^*/K^*}\|x\|^{1/2}\chi(x)(d^*x/d^*a)\int_K f(ax)\,da\\ &= c_1\chi(\Delta)\int f(x)\|x\|^{1/2}\chi(x)\,d^*x.\end{aligned}$$

For suitable choices of f, the integrals can be made nonzero. Hence

$$\varepsilon\Big(\chi,\psi_L,dx,\tfrac{1}{2}\Big) = c_1\chi(\Delta)$$

and

$$W(V) = \varepsilon\Big(V,\tfrac{1}{2}\Big) = \chi(\Delta).$$

(3.3) Return to Case 2 in the proof of the theorem. If K has characteristic 2, then $\Delta \in K^*$, and Theorem 3.2 shows that $W(V) = 1$, as it should. Now suppose that K has characteristic $\neq 2$.

The group $O(2)$ is the semidirect product of $\{e, s\}$ (with s a reflection) by $SO(2)$. Let $\widetilde{O}(2)$ be the semidirect product of $\{e, s\}$ by the double cover of $SO(2)$. For a representation $\rho : G \to O(2)$ of the finite group G, the obstruction to lifting ρ to $\widetilde{O}(2)$ (the class of the central extension of G by $\mathbf{Z}/2$ obtained by pulling back $\widetilde{O}(2)$ under ρ) is of the form $\alpha w_1^2 + w_2$, with $\alpha = 0$ or 1. In fact, $\alpha = 0$, as is seen by testing on $G = \{e, s\}$.

Put $w^1 = w^1(\mathrm{Ind}\,\chi) = w^1(\mathrm{Ind}\,1)$ and $w^2 = w^2(\mathrm{Ind}\,\chi)$. We have

$$\begin{aligned} w^*\,\mathrm{Ind}([\chi] - 1) &= (w^*\,\mathrm{Ind}(\chi))(w^*\,\mathrm{Ind}\,1)^{-1} \\ &= (1 + w^1 + w^2 + \cdots)(1 - w^1 + (w^1)^2 - \cdots) = 1 + w^2 + \cdots . \end{aligned}$$

The class $w^2\,\mathrm{Ind}([\chi] - 1) \in H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2)$ is therefore zero if and only if the representation

$$\mathrm{Ind}\,\chi : \mathrm{Gal}(\bar{K}/K) \longrightarrow O(2)$$

lifts to $\widetilde{O}(2)$. This means that $\chi : L^* \to \mathbf{C}^*$ has a square root trivial on K^*, or equivalently that $\chi(\Delta) = 1$, where Δ is the unique element of order 2 in L^*/K^*. This completes the proof of the theorem.

4. Weil Groups and the Hasse Reciprocity Law

(4.1) Let K be a local field and $\bar{K}$ a separable closure of K. For the definition of the (absolute) Weil group $W(\bar{K}/K)$, see [1] 2.2. If K is nonarchimedean, it is a subgroup of $\mathrm{Gal}(\bar{K}/K)$, an extension of $\mathbf{Z}$ by inertia. Its cohomology with values in a discrete module M is defined by

$$H^i(W(\bar{K}/K), M) = \varinjlim_{U} H^i(W(\bar{K}/K)/U, M^U)$$

(U an open subgroup of the inertia group, normal in $W(\bar{K}/K)$). If K is archimedean, $W(\bar{K}/K)$ is a Lie group, an extension of $\mathrm{Gal}(\bar{K}/K)$ by $\bar{K}^*$; its cohomology is that of the classifying space $BW(\bar{K}/K)$.

Let n be an integer prime to the characteristic exponent of K, and let μ_n be the group of n-th roots of unity in $\bar{K}$. It is a $W(\bar{K}/K)$-module.

(4.2) Construction. One constructs an isomorphism

$$\mathrm{inv} : H^2(W(\bar{K}/K), \mu_n) \xrightarrow{\sim} (\mathbf{Q}/\mathbf{Z})_n.$$

(4.2.1) Lemma. If K is nonarchimedean and M is a discrete torsion $\mathrm{Gal}(\bar{K}/K)$-module, then

$$H^i(\mathrm{Gal}(\bar{K}/K), M) \xrightarrow{\sim} H^i(W(\bar{K}/K), M).$$

Let I be the inertia group. The map in 4.2.1 is the abutment of a morphism of spectral sequences

$$E_2^{pq} = H^p(\widehat{\mathbf{Z}}, H^q(I, M)) \Rightarrow H^{p+q}(\mathrm{Gal}(\bar{K}/K), M),$$

$$E_2^{pq} = H^p(\mathbf{Z}, H^q(I, M)) \Rightarrow H^{p+q}(W(\bar{K}/K), M),$$

and one uses the fact that $\mathbf{Z}$ and $\widehat{\mathbf{Z}}$ have the same cohomology with values in a discrete torsion module (on which $\widehat{\mathbf{Z}}$ acts continuously).

For nonarchimedean K, therefore,

$$H^2(\mathrm{Gal}(\bar{K}/K), \mu_n) \xrightarrow{\sim} H^2(W(\bar{K}/K), \mu_n).$$

One then uses the Kummer exact sequence

$$0 \longrightarrow \mu_n \longrightarrow \bar{K}^* \xrightarrow{n} \bar{K}^* \longrightarrow 0$$

and Theorem 90 to identify this H^2 with the kernel of multiplication by n in

$$H^2(\mathrm{Gal}(\bar{K}/K), \bar{K}^*) = \mathrm{Br}(K) = \mathbf{Q}/\mathbf{Z}.$$

Given the state of the literature, this defines 4.2 up to sign. The sign will be discussed in 4.7. In the following section, in any case, we shall use only the case $n = 2$.

For archimedean K, let $\mathbf{Z}(1) = 2\pi i\mathbf{Z} \subset \bar{K}$. We shall prove that $H^3(W(\bar{K}/K), \mathbf{Z}(1)) = 0$ and construct an isomorphism

$$\mathrm{inv} : H^2(W(\bar{K}/K), \mathbf{Z}(1)) \xrightarrow{\sim} \mathbf{Z}. \tag{4.2.2}$$

The long exact cohomology sequence associated with the short exact sequence

$$0 \longrightarrow \mathbf{Z}(1) \xrightarrow{n} \mathbf{Z}(1) \xrightarrow{\exp(x/n)} \mu_n \longrightarrow 0$$

will give the required isomorphism

$$\mathrm{inv} : H^2(W(\bar{K}/K), \mu_n) \xrightarrow{\sim} \mathbf{Z}/n \xrightarrow[x \mapsto x/n]{\sim} (\mathbf{Q}/\mathbf{Z})_n.$$

If K is complex, then $\bar{K} = K$, and $H^2(W(\bar{K}/K), \mathbf{Z}(1))$ classifies extensions of K^* by $\mathbf{Z}(1)$, i.e. homomorphisms from $\pi_1(K^*)$ to $\mathbf{Z}(1)$. This is the group $\mathbf{Z}$, generated by the class of the extension

$$0 \longrightarrow \mathbf{Z}(1) \longrightarrow \bar{K} \xrightarrow{\exp} \bar{K}^* \longrightarrow 0. \tag{4.2.3}$$

The cohomology of $W(\bar{K}/K)$ is that of infinite-dimensional complex projective space: the successive H^i, with values in $\mathbf{Z}(1)$, are

$$\mathbf{Z}(1), 0, \mathbf{Z}, 0, \mathbf{Z}(1), \ldots .$$

For real K, use the spectral sequence

$$E_2^{pq} = H^p(\mathbf{Z}/2, H^q(\bar{K}^*, \mathbf{Z}(1))) \Rightarrow H^{p+q}(W(\bar{K}/K), \mathbf{Z}(1)).$$

$H^q(\bar{K}^*, \mathbf{Z}(1))$	H^0H^q	H^1H^q	H^2H^q	H^3H^q	H^4H^q	⋯
0	0	0	0	0	0	⋯
$\mathbf{Z}$	$\mathbf{Z}$	0	$\mathbf{Z}/2$	0	$\mathbf{Z}/2$	⋯
0	0	0	0	0	0	⋯
$\mathbf{Z}(1)$	0	$\mathbf{Z}/2$	0	$\mathbf{Z}/2$	0	⋯

We have $E_2 = E_3$, since $E_2^{pq} = 0$ for odd q.

(4.2.4) Lemma. The homomorphism

$$H^2(W(\bar{K}/K), \mathbf{Z}(1)) \longrightarrow H^2(\bar{K}^*, \mathbf{Z}(1))$$

is not surjective.

It is enough to see that extension 4.2.3 is not the restriction to $\bar{K}^*$ of an extension E of $W(\bar{K}/K)$ by $\mathbf{Z}(1)$. The group $W(\bar{K}/K)$ is generated by $\bar{K}^*$ and F, with relations $F^2 = -1$ and $FzF^{-1} = \bar{z}$. If E existed, one could lift F to $\widetilde{F}$ in E. We would have $\widetilde{F}z\widetilde{F}^{-1} = \bar{z}$ $(z \in \bar{K})$ and $\widetilde{F}^2 \equiv \pi i \pmod{\mathbf{Z}(1)}$. It would follow that $\widetilde{F}^2$ did not commute with $\widetilde{F}$, which is absurd.

By the lemma, $d_3 : E_3^{0,2} \to E_3^{3,0}$ is nonzero, and the first H^i are

$$H^0 = 0.$$

$$H^1 = \mathbf{Z}/2 \quad \text{(equal to } H^1(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}(1))\text{)}.$$

$H^2 = \mathbf{Z}$, and $H^2(W(\bar{K}/K), \mathbf{Z}(1)) \hookrightarrow H^2(\bar{K}^*, \mathbf{Z}(1))$ as a subgroup of index two, the inclusion being identified with $n \mapsto 2n$.
$H^3 = 0$.

(4.3) Remark. One can show that the cohomology is periodic with period 4. We shall not use this fact.

(4.4) Remark. Let K' be a finite extension of K in $\bar{K}$, and let $\alpha \in H^2(W(\bar{K}/K'), \mu_n)$. Then $\operatorname{inv} \operatorname{Tr} \alpha = \operatorname{inv} \alpha$. This follows from the analogous statement for $\operatorname{Br}(K)$ when K is nonarchimedean, and from the same formula for $\alpha \in H^2(W(\bar{K}/K'), \mathbf{Z}(1))$ when K is archimedean. The latter follows from the formulas $\operatorname{Tr} \operatorname{res} \beta = 2\beta$ and $\operatorname{inv} \operatorname{res} \beta = 2 \operatorname{inv} \beta$ for $\beta \in H^2(W(\bar{K}/K), \mathbf{Z}(1))$, with K real and K' complex.

(4.5) Example. Let $\chi : K^* \to \mathbf{C}^*$ be a quasicharacter. The class $\partial\chi$ of the extension of K^* by $\mathbf{Z}$ obtained by pulling back along χ the extension

$$0 \longrightarrow \mathbf{Z} \xrightarrow{2\pi i} \mathbf{C} \xrightarrow{\exp} \mathbf{C}^* \longrightarrow 0 \tag{4.5.1}$$

is an element of $H^2(K^*, \mathbf{Z})$. If α is a root of unity in K whose order divides n, its image under $\mathbf{Z} \to \mu_n : m \mapsto \alpha^m$ gives

$$(\chi, \alpha) \in H^2(W(\bar{K}/K), \mu_n).$$

(4.6) Proposition. We have

$$\exp\bigl(2\pi i \operatorname{inv}(\chi, \alpha)\bigr) = \chi(\alpha).$$

The archimedean case is verified by a direct calculation. When K is nonarchimedean, neither side changes if χ is multiplied by an unramified character. We may therefore suppose that χ has finite order. It then extends to

$$\chi : \operatorname{Gal}(\bar{K}/K)^{\mathrm{ab}} \longrightarrow \mathbf{C}^*,$$

which defines $[\chi] \in H^1(\operatorname{Gal}(\bar{K}/K), \mathbf{C}^*)$. Its image $\partial\chi \in H^2(\operatorname{Gal}(\bar{K}/K), \mathbf{Z})$, under the connecting homomorphism ∂ associated with (4.5.1), is the class of the extension of $\operatorname{Gal}(\bar{K}/K)$ by $\mathbf{Z}$ obtained by pulling back (4.5.1) along χ. Its image under $\mathbf{Z} \to \mu_n$, $m \mapsto \alpha^m$, is the cup product $\partial\chi \cup \alpha \in H^2(\operatorname{Gal}(\bar{K}/K), \mu_n)$. Its restriction to $W(\bar{K}/K)$ is (χ, α), and $\operatorname{inv}(\chi, \alpha)$ is the invariant of the cup product $\partial\chi \cup \alpha \in H^2(\operatorname{Gal}(\bar{K}/K), \bar{K}^*)$. In this form, the formula

$$\exp\bigl(2\pi i \operatorname{inv}(\partial\chi \cup \alpha)\bigr) = \chi(\alpha)$$

holds for every $\alpha \in K^*$: this is the definition of the reciprocity isomorphism.

(4.7) Signs. Since we have already specified our normalization (= choice of sign) of the reciprocity isomorphism, the validity of 4.6 fixes our normalization of inv for nonarchimedean K.

(4.8) Let K be a global field. For a finite Galois extension L of K, Weil defined the Weil group $W_{L/K}$, an extension of $\operatorname{Gal}(L/K)$ by the idèle class group $C(L)$ of L. It is an inverse limit of Lie groups; its cohomology is by definition the direct limit of the cohomologies of their classifying spaces. Let $\bar{K}$ be a separable closure of K, and let M be a discrete $\operatorname{Gal}(\bar{K}/K)$-module. We set

$$H^*(W(\bar{K}/K), M) = \varinjlim_{L \subset \bar{K}} H^*(W_{L/K}, M^{\operatorname{Gal}(\bar{K}/L)}).$$

Let n be prime to the characteristic exponent of K, and let $\alpha \in H^2(W(\bar{K}/K), \mu_n)$. The restriction of α to $H^2(W(\bar{K}_v/K_v), \mu_n)$ is zero for almost all places v of K (because it factors through an $H^2(\mathbf{Z}, \mu_n) = 0$).

The maps inv of 4.2 therefore give a map

$$\mathrm{inv} : H^2(W(\bar{K}/K), \mu_n) \longrightarrow \bigoplus_v (\mathbf{Q}/\mathbf{Z})_n.$$

(4.9) Theorem. The sequence

$$0 \longrightarrow H^2(W(\bar{K}/K), \mu_n) \xrightarrow{\mathrm{inv}} \bigoplus_v (\mathbf{Q}/\mathbf{Z})_n \xrightarrow{\Sigma} (\mathbf{Q}/\mathbf{Z})_n \longrightarrow 0$$

is exact.

Let $\chi : C(K) \to \mathbf{C}^*$ be a quasicharacter. The class $\partial\chi$ of the extension of $C(K)$ by $\mathbf{Z}$ obtained by pulling back along χ the extension

$$0 \longrightarrow \mathbf{Z} \xrightarrow{2\pi i} \mathbf{C} \xrightarrow{\exp} \mathbf{C}^* \longrightarrow 0$$

is an element of $H^2(C(K), \mathbf{Z})$. If α is a root of unity in K whose order divides n, its image under $\mathbf{Z} \to \mu_n : m \mapsto \alpha^m$ gives

$$(\chi, \alpha) \in H^2(C(K), \mu_n).$$

(4.9.1) Lemma. We have

$$\sum_v \mathrm{inv}_v(\chi, \alpha) = 0.$$

By 4.6, this is a reformulation of $\prod_v \chi_v(\alpha_v) = 1$.

(4.9.2) Lemma. There exists $c \in H^2(W(\bar{K}/K), \mu_n)$ having prescribed local invariants at the infinite places (or, indeed, at any finite set of places) and satisfying

$$\sum_v \mathrm{inv}_v(c) = 0.$$

If $\mu_n \subset K$, one may take c of the form (χ, α). Otherwise, take a finite extension K'/K such that K' contains μ_n, and take c of the form $\mathrm{Tr}_{K'/K}(\chi, \alpha)$; the reciprocity law follows from 4.4 and 4.9.1.

(4.9.3) Lemma. Let $c \in H^2(W(\bar{K}/K), \mu_n)$. If $\mathrm{inv}_v(c) = 0$ for complex v, and has order dividing two for real v, then c comes from $\bar{c} \in H^2(\mathrm{Gal}(\bar{K}/K), \mu_n)$.

When K has nonzero characteristic, one passes from a Weil group to a Galois group by replacing a quotient $\mathbf{Z}$ by $\widehat{\mathbf{Z}}$, and it is enough to argue as in 4.2.1.

Now suppose that K is a number field. There exists a finite totally imaginary Galois extension L of K, containing μ_n if desired, such that c is defined by an extension of $W_{L/K}$ by μ_n. There is even a modulus $\mathfrak{m}$ such that c comes from an extension E', by μ_n, of the quotient ${}_{\mathfrak{m}}W_{L/K}$ of $W_{L/K}$, which is an extension of $\mathrm{Gal}(L/K)$ by $C_{\mathfrak{m}}(L)$. The identity component of $C_{\mathfrak{m}}(L)$ is the quotient $L^*_{\mathbf{R}}/U_{\mathfrak{m}}$ of $(L \otimes \mathbf{R})^*$ by the group of units congruent to 1 modulo $\mathfrak{m}$. The hypothesis ensures that the extension induced by E' of $L^*_{\mathbf{R}}$ by μ_n is trivial. The extension of $L^*_{\mathbf{R}}/U_{\mathfrak{m}}$ by μ_n therefore comes from $\phi : U_{\mathfrak{m}} \to \mu_n$. By a theorem of Chevalley (deux théorèmes d'arithmétique, J. Math. Soc. Jap. 3 (1951), pp. 36–44), $\mathrm{Ker}(\phi)$ is a congruence subgroup: for a suitable modulus $\mathfrak{m}' \geq \mathfrak{m}$, c is defined by an extension E of ${}_{\mathfrak{m}'}W_{L/K}$ by μ_n, trivial on the identity component of ${}_{\mathfrak{m}'}W_{L/K}$. The extension E therefore comes from the Galois group $\pi_0({}_{\mathfrak{m}'}W_{L/K})$; this proves 4.9.3.

Set

$$(\mathbf{Q}/\mathbf{Z})_{n,v} = \begin{cases} \text{the subgroup of } \mathbf{Q}/\mathbf{Z} \text{ of order } n, & v \text{ nonarchimedean,} \\ \text{the subgroup of order } (n,2), & v \text{ real,} \\ \text{the subgroup of order } 1, & v \text{ complex,} \end{cases}$$

and

$$A = \left(\bigoplus_v (\mathbf{Q}/\mathbf{Z})_n\right) \Big/ \left(\bigoplus_v (\mathbf{Q}/\mathbf{Z})_{n,v}\right).$$

The proof of 4.9 is completed by considering the following diagram, in which exactness of the last row is the Hasse reciprocity law:

$$\begin{array}{ccccccccc} & & & & 0 & & & & \\ & & & & \uparrow & & & & \\ & & A & = & A & & & & \\ & & \uparrow & & \uparrow & & & & \\ & H^2(W(\bar{K}/K),\mu_n) & \xrightarrow{\text{inv}} & \bigoplus_v (\mathbf{Q}/\mathbf{Z})_n & \xrightarrow{\Sigma} & (\mathbf{Q}/\mathbf{Z})_n & \longrightarrow & 0 \\ & \uparrow & & \uparrow & & \| & & \\ 0 \longrightarrow & H^2(\mathrm{Gal}(\bar{K}/K),\mu_n) & \xrightarrow{\text{inv}} & \bigoplus_v (\mathbf{Q}/\mathbf{Z})_{n,v} & \xrightarrow{\Sigma} & (\mathbf{Q}/\mathbf{Z})_n & \longrightarrow & 0 \\ & & & \uparrow & & & & \\ & & & 0 & & & & \end{array}$$

5. Orthogonal representations of Weil groups

(5.1) Rather than considering real representations of Galois groups, it is more general and more natural to consider complex orthogonal representations of Weil groups.

Let K be a local field, let $\bar{K}$ be a separable closure of K, let V be a complex vector space equipped with a nondegenerate quadratic form Q, and let $\rho : W(\bar{K}/K) \to O(V,Q)$ be a continuous orthogonal representation of $W(\bar{K}/K)$. The Stiefel–Whitney classes $w^i \in H^i(W(\bar{K}/K), \mathbf{Z}/2)$ are still defined. Since the space of nondegenerate quadratic forms on V invariant under $W(\bar{K}/K)$ is connected, they depend only on the linear representation underlying ρ. The total Stiefel–Whitney class has the following additivity properties:

(5.1.1) If V is the orthogonal direct sum of subrepresentations V' and V'', then

$$w^*(V) = w^*(V') \cdot w^*(V'').$$

(5.1.2) If W is an invariant totally isotropic subspace of V, then

$$w^*(V) = w^*(W \oplus V/W^\perp) \cdot w^*(W^\perp/W).$$

The quadratic form Q on V induces a duality between W and $V/W^\perp$—hence a quadratic form on $W \oplus V/W^\perp$—whereas $Q|W^\perp$ has kernel W—hence a quadratic form on $W^\perp/W$. This gives meaning to 5.1.2. To prove the formula, it is enough to prove the analogous statement for a complex orthogonal vector bundle $\mathscr{V}$ on a paracompact topological space and for $\mathscr{W} \subset \mathscr{V}$

totally isotropic. Here $\mathscr{V}$ is isomorphic to $(\mathscr{W} \oplus \mathscr{V}/\mathscr{W}^{\perp}) \oplus (\mathscr{W}^{\perp}/\mathscr{W})$, and one applies the analogue of 5.1.1.

Let $RO(W(\bar{K}/K), \mathbf{C})$ be the Grothendieck group of semisimple complex orthogonal representations of $W(\bar{K}/K)$, with respect to orthogonal direct sum. It is a subgroup of $R(W(\bar{K}/K), \mathbf{C})$, the Grothendieck group of the category of semisimple complex representations of $W(\bar{K}/K)$. Property (5.1.1) makes it possible to define the total Stiefel–Whitney class of an element of $RO(W(\bar{K}/K), \mathbf{C})$ (a virtual orthogonal representation).

If V is an orthogonal representation of $W(\bar{K}/K)$, its semisimplification V' is again orthogonal, and (5.1.2) gives $w^*(V) = w^*(V')$. This justifies the restriction above to semisimple representations.

For $x \in H^2(W(\bar{K}/K), \mathbf{Z}/2)$, the class $\mathrm{cl}(x)$ is 0 when K has characteristic 2, and otherwise is the invariant of its image in $H^2(W(\bar{K}/K), \mu_2)$ (with $\mathbf{Z}/2$ and μ_2 canonically isomorphic). When K is not complex, one has

$$H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2) \xrightarrow{\sim} H^2(W(\bar{K}/K), \mathbf{Z}/2),$$

and this definition is compatible with 1.4. For K of characteristic different from 2, archimedean or not, cl is an isomorphism

$$\mathrm{cl} : H^2(W(\bar{K}/K), \mathbf{Z}/2) \xrightarrow{\sim} (\mathbf{Q}/\mathbf{Z})_2.$$

For a complex virtual representation V of $W(\bar{K}/K)$, of dimension 0 and determinant 1 and isomorphic to its dual, define, as in 1.2, a number $W(V) = \pm 1$. Again one has:

(5.2) Proposition. Let V be a virtual orthogonal representation of dimension 0 and determinant 1. Then

$$W(V) = \exp\big(2\pi i\ \mathrm{cl}(w^2(V))\big).$$

For V of the form $W \oplus W^*$ (where W is a virtual representation of dimension 0 and determinant χ), one proceeds as in 3.1, Case 1, using the formulas

$$w^2(V) = c^1(W) \pmod 2$$

and

$$\varepsilon(W, \psi) \cdot \varepsilon(W^*(1), \psi(-x)) = 1.$$

This suffices to prove 5.2 when K is complex. When K is real, it remains to treat the dihedral case of a representation $r(\chi)$, where χ is a character of $\bar{K}^*$ trivial on K^*. Theorem 3.2 remains valid, with the same proof [this time, for f and $\hat{f}$, smooth and rapidly decreasing and having a zero of high multiplicity at 0, the integrals converge in a wide strip], and one repeats the proof of 3.3.

When K is nonarchimedean, one verifies using ([1] 4.10) that every semisimple orthogonal representation V of $W(\bar{K}/K)$ is the sum of a representation of the form $W \oplus W^*$ and a representation factoring through a finite quotient of $\mathrm{Gal}(\bar{K}/K)$. For virtual V of dimension 0, one obtains a similar decomposition with W of dimension 0, and the proposition therefore follows from Theorem 1.5.

(5.3) Corollary. Let K be a global field, let L be a finite extension of K, and let V be an orthogonal representation of the Weil group $W_{L/K}$. The constant $W(V)$ in the functional equation of the corresponding L-function, defined as in (1.1.2), is $+1$.

The proof proceeds as in 1.6, now using 5.2 and the reciprocity law 4.9 (for $n = 2$).

(5.4) ℓ-adic variant. Let K be a nonarchimedean local field, and let V be a complex virtual representation of $W(\bar{K}/K)$, of dimension 0, with trivial determinant, and

isomorphic to its dual. The definition of $\varepsilon(V) = \varepsilon(V, 0)$ is purely algebraic, i.e. independent of the topology of $\mathbf{C}$. If the conductor of V is not a square, the same is not true of $W(V) = \varepsilon(V) \cdot f^{-1/2}$. However, if V is orthogonal, Serre proved [3] that the conductor exponent is even; in this case $W(V)$ therefore has a purely algebraic meaning. The same holds for $w^2(V)$. By transport of structure, it follows that the formula (to which a meaning has just been given)

$$W(V) = (-1)^{2\,\mathrm{cl}(w^2(V))} \tag{5.4.1}$$

holds for a virtual orthogonal representation V of $W(\bar{K}/K)$, of dimension 0 and trivial determinant, over any algebraically closed field k of characteristic 0. The hypothesis "algebraically closed" is unnecessary, as is seen by replacing k with its algebraic closure.

(5.5) Let (V, ρ) be a continuous ℓ-adic representation of $W(\bar{K}/K)$, and fix ψ and dx. Following [1] 8.4.2, one associates with it a "representation" (ρ', N) of $'W(\bar{K}/K)$, consisting of

(a) a representation ρ' of $W(\bar{K}/K)$ on V, continuous for the discrete topology on V;

(b) a nilpotent endomorphism N of V, such that

$$\rho'(w) N \rho'(w)^{-1} = q^{v(w)} N.$$

If ρ preserves a quadratic form Q, then so does ρ', and $N \in \mathrm{Lie}(O(Q))$.

The constant $\varepsilon(V, \psi, dx, t)$ (where t is morally q^{-s}), defined by [1] 8.12, is the product of $\varepsilon(\rho', \psi, dx, t)$ and

$$\det\bigl(-Ft,\ V^{\rho'(I)} / \operatorname{Ker}(N)^{\rho'(I)}\bigr).$$

This monomial depends only on $V^{\rho'(I)}$, on which F and N act (with $FNF^{-1} = q^{-1}N$). The lemma below shows that, if V is orthogonal, it has even degree and takes the value 1 at $t = q^{-1/2}$.

For a virtual ℓ-adic orthogonal representation V of degree 0 and determinant one, the conductor of V therefore still has even exponent, and $W(V)$ may be defined as before. Moreover, $W(V) = W(V^{\mathrm{ss}})$, where V^{ss} is the semisimplification of V. This result may be used to calculate the right-hand side of (1.2.1) for an ℓ-adic orthogonal representation V of the Weil group of a global field; the method of [2] § 9 calculates the left-hand side, and one finds that (1.2.1) remains valid in this case.

(5.6) Lemma. Let k be a field, let $q \in k^*$, let V be a vector space over k, let Q be a nondegenerate quadratic form on V, and let $F \in O(V, Q)$ and $N \in \mathrm{Lie}(O(Q))$. Suppose that $FNF^{-1} = q^{-1}N$. Then $\dim(V / \operatorname{Ker} N)$ is an even integer $2m$, and

$$\det(-Ft, V / \operatorname{Ker} N) = q^m t^{2m}.$$

Let B be the bilinear form associated with Q. By hypothesis, the form $\langle x, y\rangle = B(Nx, y)$ is alternating, and $\langle Fx, Fy\rangle = q\langle x, y\rangle$. The kernel of $\langle x, y\rangle$ is $\operatorname{Ker} N$, and F induces a symplectic similitude of multiplier q on $V / \operatorname{Ker} N$. The lemma follows.

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Pierre Deligne
I.H.E.S.
35, Route de Chartres
F-91440 Bures-sur-Yvette/France

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Representations of reductive groups over finite fields

By P. DELIGNE and G. LUSZTIG

Introduction

Let us consider a connected, reductive algebraic group G, defined over a finite field $\mathbb{F}_q$, with Frobenius map F. We shall be concerned with the representation theory of the finite group G^F, over fields of characteristic 0.

In 1968, Macdonald conjectured, on the basis of the character tables known at the time (GL_n, Sp_4), that there should be a well defined correspondence which, to any F-stable maximal torus T of G and a character θ of T^F in general position, associates an irreducible representation of G^F; moreover, if T modulo the centre of G is anisotropic over $\mathbb{F}_q$, the corresponding representation of G^F should be cuspidal (see Seminar on algebraic groups and related finite groups, by A. Borel et al., Lecture Notes in Mathematics, 131, pp. 117 and 101). In this paper we prove Macdonald's conjecture. More precisely, for T as above and θ an arbitrary character of T^F we construct virtual representations R_T^θ which have all the required properties.

These are defined in Chapter 1 as the alternating sum of the cohomology with compact support of the variety of Borel subgroups of G which are in a fixed relative position with their F-transform, with coefficients in certain G^F-equivariant locally constant l-adic sheaves of rank one. This generalizes a construction made by Drinfeld for SL_2 (see Ch. 2).

In Chapter 4 we prove a character formula for R_T^θ, based on the fixed point formula of Chapter 3. This character formula is in terms of certain "Green functions" on the unipotent elements; in Chapters 6, 7, 8 we prove that these Green functions satisfy all the axioms predicted by Springer, Kilmoyer and Macdonald ([9], [12]).

By 6.8, $\pm R_T^\theta$ is irreducible if θ is in general position and the vanishing theorem (9.9) gives an explicit model for it provided that q is not too small (if G is a classical group or G_2 any q will do; in the general case $q \geq 30$ is sufficient).

In Chapter 10 we study the irreducible components of the Gelfand-Graev representation of G^F, assuming that the centre of G is connected. The proof uses the results of Chapter 5 together with the disjointness theorem (6.2).

Finally, in Chapter 11 we discuss the case of the Suzuki and Ree groups.

It would be very desirable to find formulas for the Green functions more explicit than (4.1.2). Such formulas are known for GL_n (Green [4]), Sp_4 (Srinivasan [13]) and G_2 (Chang, Ree [2]). Very recently, Kazhdan has proved, using results of Springer, that the Green functions can be expressed as exponential sums on the Lie algebra (see [12]) provided that the characteristic is good and q is not too small.

Some of the results in this paper were announced in [7], [8].

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Conventions and standard notations

0.1. In this paper, k is always an algebraically closed field and p is its characteristic exponent. The following assumptions and definitions will be in force in Chapters 4 to 8, in Chapter 10, and in parts of Chapters 1 and 9.

(0.1.1) p is a prime, and k is an algebraic closure of the prime field $\mathbb{F}_p$ of characteristic p. For q a power of p, $\mathbb{F}_q$ is the subfield with q elements of k. G is a reductive algebraic group over k, obtained by extension of scalars from G_0 over $\mathbb{F}_q$. We denote by F the corresponding Frobenius endomorphism $F : G \to G$ (as well as the Frobenius endomorphism of any k-scheme defined over $\mathbb{F}_q$).

0.2. l is a prime different from p, and $\overline{\mathbb{Q}}_l$ is an algebraic closure of the l-adic field. When there is no ambiguity on l, we will write simply $H_c^i(X)$ (resp. $H^i(X)$) for $H_c^i(X, \overline{\mathbb{Q}}_l)$ (resp. $H^i(X, \overline{\mathbb{Q}}_l)$) ($l$-adic cohomology of X, a scheme over k). The groups $\mathbb{Z}/l^n(1)$ are the groups $\mu_{l^n}(k)$ of l^n-roots of unity in k; they form a projective system, with transition maps $x \mapsto x^{l^{n-m}}$; the projective limit is $\mathbb{Z}_l(1)$. One defines $\mathbb{Z}_l(n) = \mathbb{Z}_l(1)^{\otimes n}$, and $\mathbb{Z}_l(-n)$ the dual of $\mathbb{Z}_l(n)$. The symbol (n) is for a Tate twist: tensoring with $\mathbb{Z}_l(n)$. We will

make an accidental use of the similarly defined group $\widehat{\mathbb{Z}}_{p'}(1) = \varprojlim_{(n,p)=1} \mu_n(k)$.

0.3. For H a finite group, $\mathscr{R}(H)$ is the Grothendieck group of the finite dimensional representations of H over $\overline{\mathbb{Q}}_l$. If f and f' are class functions on H, with values in a cyclotomic field (often $\subset \overline{\mathbb{Q}}_l$), the inner product $\langle f, f' \rangle_H$ (or simply $\langle f, f' \rangle$) is

$$\frac{1}{|H|} \sum_{x \in H} f(x)\, \overline{f'(x)},$$

where the bar is the automorphism inducing $\zeta \mapsto \zeta^{-1}$ on the roots of unity. The same notation $\langle\ ,\ \rangle$ will be used for the inner product of elements of $\mathscr{R}(H) \otimes \mathbb{Q}$, identified with their characters.

0.4. The length function on a Coxeter group W (with canonical generators $s_1, \ldots, s_n$) will be denoted by $l(\ )$. A reduced expression for $w \in W$ is a decomposition $w = s_{i_1} \cdots s_{i_k}$ with $l(w) = k$.

0.5. Miscellaneous:

pr_i: ith projection (as in $\mathrm{pr}_1 : X \times X \to X$);

X^T: the fixed point subscheme of the endomorphism T of the scheme X (for instance: $G^F = G_0(\mathbb{F}_q)$);

p' (in index): away from p (as in $\widehat{\mathbb{Z}}_{p'}$, completion away from p) or the subgroup of elements of order prime to p (as in $(\mathbb{Q}/\mathbb{Z})_{p'}$);

ad g: the inner automorphism $x \mapsto gxg^{-1}$, or maps deduced from it;

1, or e: the identity element of a group;

$\mathrm{Tr}(f, V^*)$, for f an endomorphism of a graded vector space, is

$$\sum (-1)^i \mathrm{Tr}(f, V^i);$$

reductive: reductive groups are meant to be connected and smooth.

The Jordan decomposition of an element $g \in G$ (G as in (0.1.1)) is $g = su$ where s is a semisimple element and u is a unipotent element commuting with s.

0.6. We will often identify a scheme over k with the set of its k-rational points. This should cause no confusion.

1. Some basic definitions

1.1. Suppose that in some category we are given a family $(X_i)_{i \in I}$ of objects and a compatible system of isomorphisms $\varphi_{ji} : X_i \xrightarrow{\sim} X_j$. This is as good as giving a single object X, the "common value" or "projective limit" of the family. This projective limit is provided with isomorphisms $\sigma_i : X \xrightarrow{\sim} X_i$ such that $\varphi_{ji}\sigma_i = \sigma_j$. We will use such a construction to define the maximal torus $\mathbf{T}$ and the Weyl group $\mathbf{W}$ of a connected reductive algebraic group G over k.

As index set I, we take the set of pairs (T, B) consisting of a maximal torus T and a Borel subgroup B containing T. For $i \in I$, $i = (T, B)$, we take

$T_i = T, \quad W_i = N(T)/T.$

The isomorphism φ_{ji} is the isomorphism induced by ad g where g is any element of G conjugating i into j; these elements g form a single right T_i-coset, so that φ_{ji} is independent of the choice of g.

One similarly defines the root system of $\mathbf{T}$, its set of simple roots, the action of $\mathbf{W}$ on $\mathbf{T}$ and the fundamental reflections in $\mathbf{W}$.

Let $F : G \to G$ be an isogeny. For any $i \in I$, $i = (T, B)$, $F(i) = (F(T), F(B))$ is again in I; F induces an isogeny $F : T_i \to T_{F(i)}$ and an isomorphism $F : W_i \to W_{F(i)}$. The corresponding endomorphism (resp. automorphism) $\sigma_{F(i)}^{-1} F \sigma_i$ of the torus $\mathbf{T}$ (resp. of the Weyl groups $\mathbf{W}$) is independent of the choice of i; we say that it is induced by F.

1.2. Let X (or X_G) be the set of all Borel subgroups of G. The group G acts on X by conjugation, and X is a smooth projective homogeneous space of G. For each Borel subgroup B of G, B is the stabilizer of the corresponding point of X, hence there is a natural isomorphism $G/B \to X : g \mapsto gBg^{-1}$.

The set of orbits of G in $X \times X$ can be identified with the Weyl group $\mathbf{W}$ of G as follows: for any $i = (T, B)$ as in (1.1), use the composite bijection

$$\mathbf{W} \underset{\sigma_i}{\tilde{\to}} N(T)/T \tilde{\to} B\backslash G/B \underset{(e,g)}{\tilde{\longrightarrow}} G\backslash(G/B \times G/B) \tilde{\to} G\backslash(X \times X);$$

this is independent of the choice of (T, B). We denote by $O(w)$ the orbit corresponding to $w \in \mathbf{W}$; this is the orbit of $(B, \dot{w}B\dot{w}^{-1})$, where $\dot{w} \in N(T)$ represents w. We shall say that two Borel subgroups B', B'' of G are in relative position w, $w \in \mathbf{W}$, if and only if $(B', B'') \in O(w)$. In diagrams, we will picture this by

$$B' \overset{w}{\to} B''.$$

The basic properties of Bruhat decomposition can be expressed as follows:

(a) If $w = w_1 w_2$, with $l(w) = l(w_1) + l(w_2)$ then

(a_1) $(B', B'') \in O(w_1)$ and $(B'', B''') \in O(w_2) \Rightarrow (B', B''') \in O(w)$;

(a_2) if $(B', B''') \in O(w)$, there is one and only one B'' such that $(B', B'') \in O(w_1)$ and $(B'', B''') \in O(w_2)$.

On the scheme level:

$$O(w_1) \times_X O(w_2) \tilde{\to} O(w).$$

(b) Let s be an elementary reflection and let B be a Borel subgroup.

(b_1) $P = B \cup BsB$ is a (minimal parabolic) subgroup, i.e., if $(B', B'') \in O(s)$ and $(B'', B''') \in O(s)$, then either $(B', B''') \in O(s)$ or $B' = B'''$.

(b_2) The quotient $\overline{L}$ of $L = P/U_P$ (U_P = unipotent radical of P) by its centre is isomorphic to $PGL(2)$. The space $X_{\overline{L}}$ is hence a projective line. The inverse image map $X_{\overline{L}} \to X$ has as image the set of Borel subgroups B' in relative position e or s with B.

(b_3) Via either projection, $\overline{O(s)} = O(s) \cup O(e) \subset X \times X$ is hence a fibre

space over X, with fibre P_1. It is provided with the section $O(e)$; this reduces its structural group from the projective to the affine group. The complement $O(s)$ of that section is a fibre space with fibre the affine line.

1.3. The assumptions (0.1.1) are in force in the rest of this chapter. The scheme X is hence defined over $\mathbb{F}_q$ and provided with a Frobenius map $F : X \to X$.

DEFINITION 1.4. For w in the Weyl group $\mathbf{W}$ of G, $X(w) \subset X$ is the locally closed subscheme of X consisting of all Borel subgroups B of G such that B and $F(B)$ are in relative position w.

One can also regard $X(w)$ as the intersection, in $X \times X$, of $O(w)$ with the graph of Frobenius. It is easily checked that this intersection is transverse. The orbit $O(w)$ being smooth of dimension $\dim(X)+l(w)$, it follows that $X(w)$ is smooth and purely of dimension $l(w)$. The subscheme $X(w)$ of X is G^F-stable. Hence, for each prime number $l \neq p$, G^F acts on the l-adic cohomology with compact supports of $X(w)$.

DEFINITION 1.5. $R^1(w)$ is the virtual representation

$$\sum (-1)^i H_c^i(X(w), \overline{\mathbb{Q}}_l)$$

of G^F (an element of the Grothendieck group of representations of G^F over $\overline{\mathbb{Q}}_l$).

For $w = e$, $X(w)$ is of dimension 0; it is the set of rational Borel subgroups, and $R^1(w)$ is induced by the unit representation of B^F, where B is an F-stable Borel subgroup.

It will follow from (3.3) that the character of $R^1(w)$ has integral values, independent of l. This will justify omitting l from the notation; $R^1(w)$ could be also regarded as a complex virtual representation of G^F.

An element of the Weyl group $\mathbf{W}$ is said to be F-conjugate to $w \in \mathbf{W}$, if it is of the form $w_1 w F(w_1)^{-1}$, for some $w_1 \in \mathbf{W}$. We denote by $W_F^{\sim}$ the set of F-conjugacy classes in $\mathbf{W}$. We shall recall in (1.14), that the G^F-conjugacy classes of F-stable (i.e., $\mathbb{F}_q$-rational) maximal tori in G are parametrized by $W_F^{\sim}$.

THEOREM 1.6. $R^1(w)$ depends only on the F-conjugacy class of w.

Let w and w' be F-conjugate.

Case 1. $w = w_1 w_2$, $w' = w_2 F(w_1)$ and $l(w) = l(w_1) + l(w_2) = l(w_2) + l(F(w_1)) = l(w')$ (0.4). For $B \in X(w)$, there is a unique Borel subgroup σB such that $(B, \sigma B) \in O(w_1)$ and $(\sigma B, FB) \in O(w_2)$; we have the diagram of relative positions:

$$\begin{array}{ccc} B & \xrightarrow{w_1 w_2} & F(B) \\ {\scriptstyle w_1}\downarrow & \nearrow_{w_2} & \downarrow{\scriptstyle F(w_1)} \\ \sigma B & \xrightarrow{w_2 F(w_1)} & F(\sigma B) \end{array}$$

(the bottom one because $l(w_2F(w_1)) = l(w_2) + l(F(w_1)))$, and $\sigma B \in X(w')$. By the same argument applied to $w_2F(w_1)$ and $F(w_1)F(w_2) = F(w)$, we get a map $\tau : X(w') \to X(Fw)$. The diagram

$$\begin{array}{ccc} X(w) & \overset{\sigma}{\to} & X(w') \\ {\scriptstyle F}\downarrow & \swarrow_{\tau} & \downarrow{\scriptstyle F} \\ X(Fw) & \xrightarrow{\sigma^{(q)}} & X(Fw') \end{array}$$

is commutative. The vertical maps induce equivalences of étale sites, hence so do τ and σ [SGA 1, IX, 4.10]. The resulting isomorphism

$$H_c^*(X(w')) \xrightarrow{\sim} H_c^*(X(w))$$

is G^F-equivariant, whence (1.6) in this case.

Case 2. For some fundamental reflection s, we have $w' = swF(s)$ and $l(w') = l(w) + 2$. For $B \in X(w')$, we can find two Borel subgroups γB, δB such that $(B, \gamma B) \in O(s)$, $(\gamma B, \delta B) \in O(w)$, $(\delta B, FB) \in O(F(s))$; moreover, γB and δB are uniquely determined by these requirements. We define a partition $X(w') = X_1 \cup X_2$ by

$$X_1 = \{B \in X(w') \mid \delta B = F(\gamma B)\}, \qquad X_2 = \{B \in X(w') \mid \delta B \neq F(\gamma B)\}.$$

Note that X_1 is a closed subscheme of $X(w')$, while X_2 is an open one. We have $\gamma : X_1 \to X(w)$ and, for $B' \in X(w)$, $\gamma^{-1}(B')$ is the set of all Borel subgroups B such that $(B, B') \in O(s)$; hence $\gamma^{-1}(B')$ is an affine line over k, and X_1 is (via γ) an affine line bundle over $X(w)$. It follows that γ induces an isomorphism

$$H_c^i(X_1) \xrightarrow{\sim} H_c^{i-2}(X(w))(-1), \qquad i \geq 0. \tag{1.6.1}$$

If $B \in X_2$, we have

$$(\delta B, FB) \in O(F(s)), \quad (FB, F(\gamma B)) \in O(F(s)), \quad \delta B \neq F(\gamma B),$$

hence $(\delta B, F(\gamma B)) \in O(F(s))$. Since

$$(F(\gamma B), F(\delta B)) \in O(F(w)), \qquad l(F(sw)) = l(F(s)) + l(F(w)),$$

it follows that $(\delta B, F(\delta B)) \in O(F(sw))$. Thus we have $\delta : X_2 \to X(F(sw))$. Let

$$X_2' = \{(B, \widetilde{B}) \in X_2 \times X(sw) \mid F(\widetilde{B}) = \delta B\}$$

and let $\delta' : X_2' \to X(sw)$ be defined by $\delta'(B, \widetilde{B}) = \widetilde{B}$; we define also $\varphi : X_2' \to X_2$ by $\varphi(B, \widetilde{B}) = B$. We have a cartesian

diagram

$$\begin{array}{ccc} X_2' & \xrightarrow{\delta'} & X(sw) \\ {\scriptstyle\varphi}\downarrow & & \downarrow{\scriptstyle F} \\ X_2 & \xrightarrow{\delta} & X(F(sw)). \end{array} \tag{1.6.2}$$

For any $\widetilde{B} \in X(sw)$ we denote by $s\widetilde{B}$ the unique Borel subgroup such that $(\widetilde{B}, s\widetilde{B}) \in O(s)$, $(s\widetilde{B}, F\widetilde{B}) \in O(w)$. It is easy to see that $\delta'^{-1}(\widetilde{B})$ can be identified with the set of Borel subgroups B such that $(\widetilde{B}, B) \in O(s)$, $B \neq s\widetilde{B}$; it follows that $\delta'^{-1}(\widetilde{B})$ is an affine line with a point removed. Thus X_2' is (via δ') a line bundle over $X(sw)$ with the zero-section removed. Since the vertical arrows in (1.6.2) induce equivalences of étale sites, we have a canonical exact sequence (see [5]):

$$\cdots \xrightarrow{\partial} H_c^{i-1}(X(sw)) \longrightarrow H_c^i(X_2) \longrightarrow H_c^{i-2}(X(sw))(-1) \xrightarrow{\partial} H_c^i(X(sw)) \longrightarrow \cdots. \tag{1.6.3}$$

It can be proved that the maps ∂ in (1.6.3) are zero; this fact will not be used here.

Note that (1.6.1) and (1.6.3) are G^F-equivariant and that G^F acts trivially on $\overline{\mathbb{Q}}_l(-1)$. It follows that for any $g \in G^F$:

$$\operatorname{tr}(g^*, H_c^*(X_2)) = 0,$$

$$\operatorname{tr}(g^*, H_c^*(X_1)) = \operatorname{tr}(g^*, H_c^*(X(w))).$$

If one uses the exact sequence

$$\cdots \longrightarrow H_c^{i-1}(X_1) \longrightarrow H_c^i(X_2) \longrightarrow H_c^i(X(w')) \longrightarrow H_c^i(X_1) \longrightarrow \cdots,$$

it follows that

$$\begin{aligned} \operatorname{tr}(g^*, H_c^*(X(w'))) &= \operatorname{tr}(g^*, H_c^*(X_1)) + \operatorname{tr}(g^*, H_c^*(X_2)) \\ &= \operatorname{tr}(g^*, H_c^*(X(w))) \end{aligned}$$

and 1.6 is proved in this case.

The general case. It suffices to treat the case where $w' = swF(s)$ for a fundamental reflection s. By permuting, if necessary, w and w' ($w = sw'F(s)$), we may even assume that $l(w') \geq l(w)$. If $l(w') > l(w)$ we are in case 2. If $l(w') = l(w)$, the following lemma shows that either we are in case 1 (with w and w' possibly interchanged) or that $w = w'$.

LEMMA 1.6.4. Let s, t be two fundamental reflections in W and let $w \in W$ be such that $l(w) = l(swt)$. Then either $w = swt$, or

$$l(sw) = l(w) - 1, \qquad \text{or} \qquad l(wt) = l(w) - 1.$$

Let $w = s_1 s_2 \cdots s_k$ be a reduced expression for w (0.4). Assume that $l(wt) = l(w) + 1$; then $wt = s_1 s_2 \cdots s_k t$ is also a reduced expression. We have $l(swt) = l(wt) - 1$. It follows (cf. Bourbaki, [1, Ch. IV, §1, Lemme 3]) that either there exists j, $1 \leq j \leq k$, with

$$ss_1 \cdots s_{j-1} = s_1 \cdots s_{j-1} s_j,$$

or we have

$$ss_1 \cdots s_k = s_1 \cdots s_k t.$$

In the first case, we have $w = ss_1 \cdots s_{j-1} s_{j+1} \cdots s_k$ and $l(sw) = l(w) - 1$; in the second case we have $w = swt$ and the lemma is proved.

1.7. Let us choose a maximal torus T^* in G and a Borel subgroup $B^* \subset G$ containing T^*, with unipotent radical U^*. The quotient $E = G/U^*$ is a T^*-torsor (= right principal homogeneous space of T^*) over $X = G/B^*$. For $x \in X$, the fibre $E(x)$ of the projection $E \to X$ is

$$E(x) = \{g \in G \mid ge^* = x\}/U^*,$$

where e^* is the point of X corresponding to B^*.

Let $\dot{w} \in N(T^*)$ define the element w in the Weyl group $\mathbf{W}$ via the isomorphism $\sigma(T^*, B^*) : \mathbf{W} \xrightarrow{\sim} N(T^*)/T^*$. If $x, y \in X$ are in relative position w, the g's in G such that $ge^* = x$ and $g\dot{w}e^* = y$ form a torsor $A(x, y)$ under

$$B^* \cap \dot{w} B^* \dot{w}^{-1} = T^* \cdot (U^* \cap \dot{w} U^* \dot{w}^{-1}).$$

For $g \in A(x, y)$, the class of $g\dot{w}$ in $E(y)$ depends only on the class of g in $E(x)$. This defines a map $E(x) \to E(y)$, which we denote as right multiplication by $\dot{w}$. We have the formulas

$$(ut)\dot{w} = (u\dot{w}) \operatorname{ad} \dot{w}^{-1}(t), \tag{1.7.1}$$

$$u(\dot{w}t) = (u\dot{w})t. \tag{1.7.2}$$

We will express (1.7.1) by saying that $\cdot\dot{w}$ is a w-map of T^*-torsors. It is induced by a w-map of T^*-torsors over $O(w)$

$$\cdot\dot{w} : \mathrm{pr}_1^* E \longrightarrow \mathrm{pr}_2^* E.$$

Assume that $w = w_1 w_2$, $\dot{w} = \dot{w}_1 \dot{w}_2$ and that $l(w) = l(w_1) + l(w_2)$; then for $x, y, z \in X$ with $(x, y) \in O(w_1)$ and $(y, z) \in O(w_2)$, we have $(x, z) \in O(w)$ and

$$u\dot{w} = (u\dot{w}_1)\dot{w}_2. \tag{1.7.3}$$

1.8. We now assume that T^* and B^* are F-stable. The identification $\sigma(T^*, B^*)$ of T^* and $N(T^*)/T^*$ with the torus $\mathbf{T}$ and the Weyl group $\mathbf{W}$ is then compatible with F. For w in the Weyl group, we denote by $\mathbf{T}(w)$ the torus $\mathbf{T}$, provided with the rational structure for which the Frobenius is $\operatorname{ad}(\dot{w})F$. We have

$$\mathbf{T}(w)^F \simeq \{t \in T^* \mid \operatorname{ad}(\dot{w})F(t) = t\}.$$

For $x \in X$, the Frobenius map induces a map $F : E(x) \to E(F(x))$, with $F(ut) = F(u)F(t)$. For $x \in X(w)$, we put

$$E(x,\dot{w}) = \{u \in E(x) \mid F(u) = u\dot{w}\}.$$

This is a $\mathbf{T}(w)^F$-torsor. The $E(x,\dot{w})$ are the fibres of a map

$$\pi : \widetilde{X}(\dot{w}) \longrightarrow X(w),$$

with $\widetilde{X}(\dot{w}) \subset E \mid X(w)$ a $\mathbf{T}(w)^F$-torsor over $X(w)$. The action of G on E restricts to an action of G^F on $\widetilde{X}(\dot{w})$.

Up to isomorphism, the G^F-equivariant $\mathbf{T}(w)^F$-torsor $\widetilde{X}(\dot{w})$ over $X(w)$ is independent of the lifting $\dot{w}$ of w in $N(T^*)$: for $\dot{w}' = \dot{w}t$, there exists t_1 with $\operatorname{ad}\dot{w}^{-1}(t_1)F(t_1)^{-1} = t$ and the map $u \mapsto ut_1$ induces an isomorphism

$$\widetilde{X}(\dot{w}) \xrightarrow{\sim} \widetilde{X}(\dot{w}').$$

The groups G^F and $\mathbf{T}(w)^F$ act on $H_c^*(\widetilde{X}(\dot{w}),\overline{\mathbb{Q}}_l)$ by transport of structure. For any $\theta \in \operatorname{Hom}(\mathbf{T}(w)^F,\overline{\mathbb{Q}}_l^*)$ we denote by $H_c^*(\widetilde{X}(\dot{w}),\overline{\mathbb{Q}}_l)_\theta$ the subspace of $H_c^*(\widetilde{X}(\dot{w}),\overline{\mathbb{Q}}_l)$ on which $\mathbf{T}(w)^F$ acts by θ.

DEFINITION 1.9. $R^\theta(w)$ is the virtual representation

$$\sum(-1)^i H_c^i(\widetilde{X}(\dot{w}),\overline{\mathbb{Q}}_l)_\theta$$

of G^F (an element of the Grothendieck group of representations of G^F over $\overline{\mathbb{Q}}_l$).

The character θ can be used to transform the $\mathbf{T}(w)^F$-torsor $\widetilde{X}(\dot{w})$ into a local system of $\overline{\mathbb{Q}}_l$-vector spaces of rank one $\mathscr{F}_\theta$ over $X(w)$, provided with

$$\theta : \widetilde{X}(\dot{w}) \longrightarrow \mathscr{F}_\theta, \qquad \theta(xt) = \theta(x)\theta(t).$$

The morphism $\pi : \widetilde{X}(\dot{w}) \to X(w)$ is finite and

$$\pi_*\overline{\mathbb{Q}}_l = \bigoplus_\theta \mathscr{F}_\theta.$$

The sheaf $\mathscr{F}_\theta$ is the subsheaf of $\pi_*\overline{\mathbb{Q}}_l$ on which $\mathbf{T}^F$ acts by θ, hence

$$H_c^*(\widetilde{X}(\dot{w}),\overline{\mathbb{Q}}_l)_\theta = H_c^*(X(w),\mathscr{F}_\theta).$$

In particular, for $\theta = 1$,

$$R^1(w) = \sum(-1)^i H_c^i(X(w),\overline{\mathbb{Q}}_l)$$

so that Definition 1.9 is compatible with (1.5).

Example 1.10. For $w = \dot{w} = e$, $\pi : \widetilde{X}(\dot{w}) \to X(w)$ becomes the projection

$$\pi : G^F/U^{*F} \longrightarrow G^F/B^{*F},$$

and $R^\theta(w)$ is the representation of G^F on the space of functions on G^F satisfying

$$f(gtu) = \theta(t)^{-1}f(g),$$

G^F acting by $(g * f)(x) = f(g^{-1}x)$ (induced representation).

1.11. The Borel subgroup $\operatorname{ad} gB^*$ is in $X(w)$ if and only if $\operatorname{ad} gB^*$ and $\operatorname{ad} FgB^*$ are in relative position w, i.e., if and only if $g^{-1}Fg \in B^*\dot{w}B^*$ (where $\dot{w} \in N(T^*)$ represents w):

$$X(w) = \{g \in G \mid g^{-1}Fg \in B^*\dot{w}B^*\}/B^*. \tag{1.11.1}$$

If a Borel subgroup B is in $X(w)$, one can find $g \in G$ such that $\operatorname{ad} gB^*$ is

B and $\operatorname{ad} g \operatorname{ad} \dot{w} B^* = FB$:

$$X(w) = \{g \in G \mid g^{-1}Fg \in \dot{w}B^*\}/\big(B^* \cap \operatorname{ad} \dot{w} B^*\big), \tag{1.11.2}$$

where

$$B^* \cap \operatorname{ad} \dot{w} B^* = T^* \cdot (U^* \cap \operatorname{ad} \dot{w} U^*).$$

Changing g to gt, we can normalize g so that we have also $g^{-1}Fg \in \dot{w}U^*$:

$$X(w) = \{g \in G \mid g^{-1}Fg \in \dot{w}U^*\}/\big(\mathbf{T}(w)^F \cdot (U^* \cap \operatorname{ad} \dot{w} U^*)\big). \tag{1.11.3}$$

A point in $\widetilde{X}(\dot{w})$ is defined by a Borel subgroup B, plus $g \in G$ such that $\operatorname{ad} gB^* = B$, $\operatorname{ad} g \operatorname{ad} \dot{w} B^* = FB$ and $g\dot{w} = Fg \bmod U^*$:

$$\widetilde{X}(\dot{w}) = \{g \in G \mid g^{-1}Fg \in \dot{w}U^*\}/\big(U^* \cap \operatorname{ad} \dot{w} U^*\big). \tag{1.11.4}$$

COROLLARY 1.12. The following assertions are equivalent:

(i) $X(w)$ is affine;

(ii) $\widetilde{X}(\dot{w})$ is affine;

(iii) Let ρ be the action of $U^* \cap \operatorname{ad} \dot{w} U^*$ on U^* defined by

$$\rho(u)v = \operatorname{ad} \dot{w}^{-1}(u) v F(u^{-1});$$

then $U^*/\rho(U^* \cap \operatorname{ad} \dot{w} U^*)$ is affine.

Put $S = \{g \in G \mid g^{-1}Fg \in \dot{w}U^*\}$. The map $f : S \to U^* : g \mapsto \dot{w}^{-1}g^{-1}Fg$ induces an isomorphism $G^F\backslash S \to U^*$ and is such that, for $u \in U^* \cap \operatorname{ad} \dot{w} U^*$,

$$f(gu) = \rho(u)^{-1} f(g).$$

Hence,

$$G^F\backslash \widetilde{X}(\dot{w}) \simeq U^*/(U^* \cap \operatorname{ad} \dot{w} U^*).$$

As $\widetilde{X}(\dot{w})/\mathbf{T}(w)^F = X(w)$, it only remains to use the fact that a space and a quotient of it by a finite group are simultaneously affine or not.

We will have to use another description of $\pi : \widetilde{X}(\dot{w}) \to X(w)$. First, an easy lemma:

LEMMA 1.13. Let J be the set of pairs (T, B), T an F-stable maximal torus and B a Borel subgroup containing T. The map h which to (T, B) associates the relative position of B and FB induces a bijection

$$G^F\backslash J \xrightarrow{\sim} \mathbf{W}.$$

The proof will be given in (1.15).

If we use $(T, B) \in J$ to identify $\mathbf{W}$ with $N(T)/T$, we have

$$h(T, \operatorname{ad} \dot{w} B) = w^{-1} h(T, B) F(w), \tag{1.13.1}$$

where $\dot{w} \in N(T)$ represents w, hence

COROLLARY 1.14. The map h induces a bijection

$$\{G^F\text{-conjugacy classes of } F\text{-stable maximal tori}\} \xrightarrow{\sim} \mathbf{W}_F^{\sim}.$$

Here is another description of h: for $(T, B) \in J$, if $\sigma : \mathbf{T} \to T$ and $\sigma : \mathbf{W} \to W$

($W = N(T)/T$) are defined by (T, B), then

$$\sigma F \sigma^{-1} = \operatorname{ad} h(T,B) \circ F : \mathbf{T} \longrightarrow \mathbf{T}.$$

To give $h(T,B)$ is the same as to give $\operatorname{ad} h(T,B) \circ F \in \mathbf{W} \circ F \subset \operatorname{End}(\mathbf{T})$, and to give the F-conjugacy class of $h(T,B)$ is the same as to give $\operatorname{ad} h(T,B) \circ F$ up to $\mathbf{W}$-conjugacy.

1.15. The space of maximal tori of G can be identified with the homogeneous space $G/N(T^*)$, and the space of maximal tori marked by a containing Borel subgroup can be identified with G/T^*. The group T^* being the connected component of $N(T^*)$, (1.13) is a special case of the general result described below.

Let G_0 be a connected algebraic group over $\mathbb{F}_q$ and let $x : \widetilde{E}_0 \to E_0$ be a morphism of G_0-homogeneous spaces. We denote by $G, \widetilde{E}, E$ the corresponding objects over k, and we assume that the stabilizer $S(\tilde{e})$ of $\tilde{e} \in \widetilde{E}$ is the connected component of the stabilizer $S(\pi(\tilde{e}))$ of $\pi(\tilde{e}) \in E$. Since any G_0-homogeneous space has a rational point, the existence of $\widetilde{E}_0$ imposes no condition on E_0.

The groups $S(e)/S(e)^0$ form a local system on E, which becomes constant on $\widetilde{E}$; we denote by W its constant value on $\widetilde{E}$. For $\tilde{e} \in \widetilde{E}$, we have an isomorphism

$$\alpha(\tilde{e}) : W \xrightarrow{\sim} S(\pi(\tilde{e}))/S(\pi(e)),$$

and, for $\widetilde{f} = g\tilde{e}$, we have $\alpha(\widetilde{f}) = \operatorname{ad} g\, \alpha(\tilde{e})$. We let the group W act on $\widetilde{E}$ on the right, by

$$\tilde{e}\, w = \alpha(\tilde{e})(w)\tilde{e};$$

in this way, $\widetilde{E}$ becomes a W-torsor (= principal homogeneous space) over E.

The group W is acted on by F, with $F(\tilde{e}\, w) = F(\tilde{e})F(w)$. The set $W_F^{\sim}$ of F-conjugacy classes in W is the set of orbits of the action of W on itself by

$$w \longmapsto w_1 w F(w_1)^{-1}.$$

PROPOSITION 1.16. For $e \in E^F$ and $\tilde{e} \in E$ above it, define $h(e, \tilde{e}) \in W$ by

$$F(\tilde{e}) = \tilde{e} \cdot h(e, \tilde{e}).$$

(i) The map h induces a bijection from the set of G^F-orbits in

$$\{(e, \tilde{e}) \mid e \in E^F,\ \pi(\tilde{e}) = e\}$$

to W.

(ii) We have

$$h(e, \tilde{e}w) = w^{-1} h(e, \tilde{e}) F(w).$$

Hence the map h induces a bijection from $G^F \backslash E^F$ to the set of F-conjugacy classes in W.

We will only prove (i). Let Y be the set of $\tilde{e} \in E$ such that $\pi(\tilde{e}) \in E^F$. If $\tilde{e}_0 \in E^F$, the map $g \mapsto g\tilde{e}_0$ identifies X with

$$\{g \in G \mid g^{-1} F g \in S(\pi(\tilde{e}_0))\}/S(\tilde{e}_0).$$

The Lang isogeny $g \mapsto g^{-1} F g$ is an isomorphism $G^F \backslash G \to G$, hence the map $g e_0 \mapsto g^{-1} F g$ induces a bijection

$$G^F \backslash X \xrightarrow{\sim} S(\pi(\tilde{e}_0))/\big(S(\tilde{e}_0) \text{ acting by } x \mapsto s^{-1} x F(s)\big).$$

The orbits of this action of $S(\tilde{e}_0)$ are just the usual cosets, and the resulting

bijection $G^F\backslash X \xrightarrow{\sim} W$ is the h above.

DEFINITION 1.17. Let T be an F-stable maximal torus and let B be a Borel subgroup containing T, with unipotent radical U; let w be the relative position of B and FB.

(i) $X_{T\subset B}$ is $X(w)$. The map $g \mapsto \operatorname{ad} gB$ induces isomorphisms

$$\begin{aligned} X_{T\subset B} &= \{g \in G \mid g^{-1}Fg \in B \cdot F(B)\}/B \\ &= \{g \in G \mid g^{-1}Fg \in FB\}/(B \cap FB) \\ &= \{g \in G \mid g^{-1}Fg \in FU\}/\big(T^F \cdot (U \cap FU)\big). \end{aligned}$$

(ii) $\widetilde{X}_{T\subset B}$ is

$$\{g \in G \mid g^{-1}Fg \in FU\}/(U \cap FU).$$

We have a projection map $\pi : \widetilde{X}_{T\subset B} \to X_{T\subset B}$, for which $\widetilde{X}_{T\subset B}$ is a G^F-equivariant T^F-torsor over $X_{T\subset B}$; G^F acts by left multiplication and T^F by right multiplication (it normalizes $U \cap FU$).

1.18. Let $\dot{w} \in N(T^*)$ be a representative of w. If $x' \in G$ is such that $\operatorname{ad} x'(T^*, B^*) = (T, B)$ then B and $FB = \operatorname{ad} Fx'B^*$ are in relative position w, and FB contains T, hence $FB = \operatorname{ad} x' \operatorname{ad} \dot{w}B^*$ and

$$x'^{-1}F(x') \in \dot{w}B^* \cap N(T^*) = \dot{w}T^*.$$

Replacing x' by $x = x't$ ($t \in T^*$) one can achieve $x^{-1}F(x) = \dot{w}$. The x such that $\operatorname{ad} x(T^*, B^*) = (T, B)$ and $x^{-1}F(x) = \dot{w}$ form a T^{*F}-torsor. For such an x, $\operatorname{ad} x$ induces an isomorphism $\mathbf{T}(w) \to T$ (hence $\mathbf{T}(w)^F \to T^F$); this isomorphism is independent of x.

PROPOSITION 1.19. Let T, B, U, w be as in (1.17) and $x, \dot{w}$ as in (1.18). The map $g \mapsto gx^{-1}$ induces an isomorphism from the G^F-equivariant $\mathbf{T}(w)^F$-torsor $\widetilde{X}(\dot{w})$ over $X(w)$ (or rather its model (1.11)) to the G^F-equivariant T^F-torsor $\widetilde{X}_{T\subset B}$ over $X_{T\subset B}$ (or rather its model (1.17)).

This is a straightforward computation.

1.20. The cohomology of $\widetilde{X}_{T\subset B}$ is acted on by G^F and T^F. For any character $\theta : T^F \to \overline{\mathbb{Q}}_l^*$ we put as in (1.8):

$$R^\theta_{T\subset B} = \sum_i (-1)^i H^i_c(\widetilde{X}_{T\subset B}, \overline{\mathbb{Q}}_l)_\theta$$

(an element in the Grothendieck group of representations of G^F over $\overline{\mathbb{Q}}_l$).

By (1.19), for x as in (1.18), we have

$$R^\theta_{T\subset B} = R^{\theta\circ\operatorname{ad} x}(w).$$

We shall see in Chapter 4 that $R^\theta_{T\subset B}$ is independent of B. For $g \in G^F$ such that $\operatorname{ad} g$ carries T, B, θ to T', B', θ', we have clearly

$$R^\theta_{T\subset B} = R^{\theta'}_{T'\subset B'},$$

hence $R^\theta_{T\subset B}$ will eventually depend only on the G^F-conjugacy class of T and on the orbit of θ under $(N(T)/T)^F$.

The end of this chapter will be used in the proof of 7.10 only.

1.21. Isogenies. Let T be an F-stable maximal torus of G, and let Z be the centre of G. We denote $G \times^Z T$ the quotient of $G \times T$ by the subgroup

$$\{(z, z^{-1}) \mid z \in Z\}.$$

Let B be a Borel subgroup containing T. The action $x \mapsto gxt$ of $G^F \times T^F$ on $\widetilde{X}_{T\subset B}$ is induced by an action of $(G \times^Z T)^F$, given by the same formula.

COROLLARY 1.22. On $H^*(\widetilde{X}_{T\subset B}, \overline{\mathbb{Q}}_l)_\theta$, Z^F acts by the character $\theta \mid Z^F$. In particular, on any irreducible representation occurring in $R^\theta_{T\subset B}$, Z^F acts by $\theta \mid Z^F$.

Let $\pi : \widetilde{G} \to G$ be the simply connected covering of the derived group of G, $\widetilde{T} = \pi^{-1}(T)$, $\widetilde{B} = \pi^{-1}(B)$ and let $\widetilde{Z}$ be the centre of $\widetilde{G}$.

PROPOSITION 1.23. One has

$$T^F/\pi(\widetilde{T}^F) \xrightarrow{\sim} G^F/\pi(\widetilde{G}^F).$$

Injectivity is clear; we have to check the surjectivity of the map

$$(T \times \widetilde{G})^F \longrightarrow G^F$$

induced by

$$\varphi : T \times \widetilde{G} \longrightarrow G : \quad t, \tilde{g} \longmapsto t\pi(\tilde{g}).$$

Via φ, $T \times \widetilde{G}$ is a $\widetilde{T}$-torsor over G (with

$$(t, \tilde{g}) * \tilde{t} = (t\tilde{t}, \tilde{t}^{-1}\tilde{g})),$$

and one applies Lang's theorem to the connected group $\widetilde{T}$.

1.24. Let $h : A \to B$ be a homomorphism of finite groups and let X be a space on which A acts. The induced space $\mathrm{Ind}_A^B(X)$ (unique up to unique isomorphism) is any B-space I, provided with an A-equivariant map $\varphi : X \to I$, such that for any B-space Y,

$$\mathrm{Hom}_B(I, Y) \xrightarrow[\varphi]{\sim} \mathrm{Hom}_A(X, Y).$$

One has

$$\mathrm{Ind}_A^B(X) = \coprod_{b\in B/A} b\varphi(X), \qquad \varphi(X) \simeq \ker(h)\backslash X.$$

PROPOSITION 1.25. The $(G \times^Z T)^F$-space $\widetilde{X}_{T\subset B}$ is induced by the $(\widetilde{G} \times^{\widetilde{Z}} \widetilde{T})^F$-space $\widetilde{X}_{\widetilde{T}\subset\widetilde{B}}$.

LEMMA 1.26. In the diagram

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & \ker & \longrightarrow & \widetilde{T}^F & \longrightarrow & T^F & \longrightarrow & \mathrm{coker} & \longrightarrow & 0 \\
 & & {\scriptstyle(1)}\downarrow & & {\scriptstyle(t,1)}\downarrow & & {\scriptstyle(t,1)}\downarrow & & {\scriptstyle(2)}\downarrow & & \\
0 & \longrightarrow & \ker & \longrightarrow & (\widetilde{T} \times^{\widetilde{Z}} \widetilde{G})^F & \longrightarrow & (T \times^Z G)^F & \longrightarrow & \mathrm{coker} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & & & & & \\
 & & (\widetilde{G}/\widetilde{Z})^F & = & (G/Z)^F & & & & & &
\end{array}$$

the maps (1) and (2) are isomorphisms.

Indeed, $\widetilde{T} \times^{\widetilde{Z}} \widetilde{G}$ (resp. $T \times^Z G$) is a $\widetilde{T}$ (resp. T)-torsor over $\widetilde{G}/\widetilde{Z} = G/Z$, hence, since T and $\widetilde{T}$ are connected, $(\widetilde{T} \times^{\widetilde{Z}} \widetilde{G})^F$ is a $\widetilde{T}^F$-torsor over $(\widetilde{G}/\widetilde{Z})^F$, and $(T \times^Z G)^F$ is the induced T^F-torsor.

Proof of 1.25. By 1.26, we are reduced to prove that $\widetilde{X}_{T\subset B}$, as a T^F-space, is induced by the $\widetilde{T}^F$-space $\widetilde{X}_{\widetilde{T}\subset\widetilde{B}}$. The spaces $\widetilde{X}_{T\subset B}$ and $\widetilde{X}_{\widetilde{T}\subset\widetilde{B}}$ are

indeed respectively T^F and $\widetilde{T}^F$-torsor over $X_{T\subset B} = X_{\widetilde{T}\subset\widetilde{B}}$, and $X_{\widetilde{T}\subset\widetilde{B}}$ is hence the T^F-torsor induced by the $\widetilde{T}^F$-torsor $X_{T\subset B}$.

COROLLARY 1.27. Let θ be a character of $G^F/\pi(\widetilde{G}^F)$. We denote again by θ its restriction to T^F. One has

$$R^{\theta\eta}_{T\subset B} = \theta \otimes R^{\eta}_{T\subset B}.$$

It follows from 1.25 that

$$H^*(\widetilde{X}_{T\subset B}, \overline{\mathbb{Q}}_l) = \mathrm{Ind}^{(G\times^Z T)^F}_{(\widetilde{G}\times^{\widetilde{Z}}\widetilde{T})^F}(\widetilde{X}_{\widetilde{T}\subset\widetilde{B}}, \overline{\mathbb{Q}}_l).$$

The character $\theta(gt)$ of $(G\times^{\widetilde{Z}} T)^F$ is trivial on the image of $\widetilde{G}\times^{\widetilde{Z}}\widetilde{T}$. The induced representation we consider is hence isomorphic to its tensor product with $\theta(gt)$, and 1.27 is a formal consequence of this.

2. Examples in the classical groups

2.1. Let V be an n-dimenssional vector space over k and put $G = \mathrm{GL}(V)$. If $b = (b_1, \dots, b_n)$ is a basis of V, we may take for T^* the group of diagonal matrices and for B^* the group of upper triangular matrices. The Weyl group lifts into the subgroup of $N(T^*)$ consisting of the $\dot{w}$'s inducing a permutation of basis vectors. In this case, $\mathbf{T}, \mathbf{W}$ (1.1), X (1.2), E, $\cdot\dot{w}$ (1.7), have the following alternative description.

(a) $\mathbf{T} = \mathbf{G}_m^n$, $\mathbf{W} = \mathfrak{S}_n$, the fundamental reflections are the transpositions $(i, i+1)$ and the action of $\mathbf{W}$ on $\mathbf{T}$ is by permutation.

(b) X is the space of complete flags $D_1 \subset \cdots \subset D_{n-1}$ in V: a flag D is an increasing filtration of V with $\dim D_i = i$ for $1 \le i \le n-1$.

(c) E is the space of complete flags marked by non-zero vectors

$$e_i \in D_i/D_{i-1} = \mathrm{Gr}_i^D(V) \qquad (1 \le i \le n),$$

where we use the convention $D_0 = 0$, $D_n = V$; $\mathbf{T}$ acts on E by

$$(D, (e_i))(\lambda_i) = (D, (\lambda_i e_i)).$$

This is a G-equivariant $\mathbf{T}$-torsor over X.

(d) If D' and D'' are two flags, their relative position is labelled by the permutation w such that

$$\mathrm{Gr}^{D'}_{w(i)}\,\mathrm{Gr}_i^{D''}(V) \neq 0.$$

The isomorphisms

$$\mathrm{Gr}^{D'}_{w(i)}(V) \simeq \mathrm{Gr}_i^{D''}\,\mathrm{Gr}^{D'}_{w(i)}(V) \simeq \mathrm{Gr}^{D'}_{w(i)}\,\mathrm{Gr}_i^{D''}(V) \simeq \mathrm{Gr}_i^{D''}(V)$$

induce a w-isomorphism between the $\mathbf{T}$-torsor $E(D')$ of markings of D' and $E(D'')$:

$$e \longmapsto e\cdot\dot{w}.$$

When w is the n-cycle $(1, \dots, n)$, D' and D'' are in relative position w if and only if

$$D''_i + D'_i = D'_{i+1} \quad (1 \le i < n-1), \qquad D''_{n-1} + D'_1 = V.$$

2.2. We now take k and $\mathbb{F}_q$ as in (0.1.1) and assume that V is provided with an $\mathbb{F}_q$-structure. Frobenius maps are then defined. For $w = (1, \dots, n)$, the condition for a flag D to be in relative position w with its image FD by Frobenius is that D be the flag

$$D_1 \subset D_1 + FD_1 \subset D_1 + FD_1 + F^2D_1 \subset \cdots \qquad \text{and that} \qquad V = \bigoplus_0^{n-1} F^i D_1.$$

If we denote by $\mathbf{P}(V)$ the set of homogeneous lines in V, the map $D \mapsto D_1$ is an isomorphism from $X(w)$ to the set of all $x \in \mathbf{P}(V)$ which do not lie on any $\mathbb{F}_q$-rational hyperplane. A marking e of F is such that $F(e) = e \cdot \dot{w}$ if and only if

$$\begin{aligned} e_2 &\equiv F(e_1) \pmod{e_1}, \\ e_3 &\equiv F^2(e_1) \pmod{e_1, F(e_1)}, \quad \ldots, \\ e_n &\equiv F^{n-1}(e_1) \pmod{e_1, F(e_1), \ldots, F^{n-2}(e_1)} \end{aligned}$$

and

$$e_1 \equiv F^n(e_1) \pmod{F(e_1), \ldots, F^{n-1}(e_1)};$$

e is defined by $e_1 \in D_1$ subject to the condition that

$$e_1 \wedge F(e_1) \wedge \cdots \wedge F^{n-1}(e_1) = F^n(e_1) \wedge F(e_1) \wedge \cdots \wedge F^{n-1}(e_1),$$

i.e.:

$$F(e_1 \wedge \cdots \wedge F^{n-1}(e_1)) = (-1)^{n-1}(e_1 \wedge \cdots \wedge F^{n-1}(e_1)). \tag{2.2.1}$$

If (x_i) are the coordinates of e_1 with respect to some rational basis, the condition (2.2.1) can be rewritten

$$(-1)^{n-1}\left(\det(x_i^{q^{j-1}})_{1\le i,j\le n}\right)^{q-1} = 1. \tag{2.2.2}$$

The form on the left is invariant under $\mathrm{GL}(n, \mathbb{F}_q)$. Up to a scalar factor, it is the product of all non-zero $\mathbb{F}_q$-rational linear forms. The map $(D, e) \mapsto e_1$ induces an isomorphism of $\widetilde{X}(\dot{w})$ with the affine hypersurface (2.2.2). This hypersurface is stable under $x \mapsto \lambda x$ for $\lambda \in \mathbb{F}_{q^n}^*$, and this is the action of $\mathbf{T}(w)^F$.

Our work has been inspired by results of Drinfeld, who proved that the discrete series representations of $\mathrm{SL}(2, \mathbb{F}_q)$ occur in the cohomology of the affine curve

$$xy^q - x^q y = 1$$

(the form

$$xy^q - x^q y = \begin{vmatrix} x & x^q \\ y & y^q \end{vmatrix}$$

is $\mathrm{SL}_2(\mathbb{F}_q)$-invariant).

Let $\varphi(q, r, n)$ be the number of $\mathbb{F}_{q^r}$-rational points of $X(w)$, $r \ge 1$. We have the following

PROPOSITION 2.3.

$$\varphi(q, r, n) = \prod_{1\le i\le n-1} (q^r - q^i).$$

Proof. We define a partition

$$\mathbf{P}(V) = X_0 \cup X_1 \cup \cdots \cup X_{n-1}$$

as follows: X_i is the set of points $x \in \mathbf{P}(V)$ such that $x, F(x), F^2(x), \ldots$ span a linear subspace of dimension i of $\mathbf{P}(V)$. This partition is invariant under Frobenius, and clearly X_i has precisely

$$\varphi(q, r, i+1)\frac{(1-q^{n-i})\cdots(1-q^n)}{(1-q)\cdots(1-q^{i+1})}$$

$\mathbb{F}_q$-rational points. It follows that

$$\frac{q^{nr}-1}{q-1} = \sum_{0\le i\le n-1} \varphi(q,r,i+1)\frac{(1-q^{n-i})\cdots(1-q^n)}{(1-q)\cdots(1-q^{i+1})}.$$

This shows that, for fixed n, $\varphi(q,r,n)$ is a polynomial of degree $(n-1)$ in $Q=q^r$, with coefficient polynomials in q, and leading term Q^{n-1}. Since $\varphi(q,r,n)=0$ for $1\le r\le n-1$, this polynomial must be divisible by $(Q-q)$, $(Q-q^2)$, ..., $(Q-q^{n-1})$. It follows that

$$\varphi(q,r,n) = (Q-q)(Q-q^2)\cdots(Q-q^{n-1})$$

and the proposition is proved.

2.4. Let V be a vector space as in (2.2), with a fixed non-degenerate symplectic form $\langle\ ,\ \rangle$ defined over $\mathbb{F}_q$. We must have $n=2m$. The symplectic group $\mathrm{Sp}(V)$ is then a group as in (0.1.1). Let Y be the set of all complete isotropic flags $D_1\subset D_2\subset\cdots\subset D_m$ in V ($\dim D_i=i$) such that

$$D_1\ne FD_1\subset D_2,\quad D_2\ne FD_2\subset D_3,\quad \dots,\quad D_{m-1}\ne FD_{m-1}\subset D_m,\quad D_m\ne FD_m.$$

Then Y can be identified with $X(w)$, where w is a Coxeter element in the Weyl group $\mathbf{W}$ of $\mathrm{Sp}(V)$; if we identify $\mathbf{W}$ with the group of all permutations σ of $-m,\dots,-2,-1,1,2,\dots,m$ such that $\sigma(-i)=-\sigma(i)$ for all i, then w is the permutation

$$\begin{aligned} -i &\longmapsto -i+1 \quad (2\le i\le m), & -1&\longmapsto m,\\ i &\longmapsto i-1 \quad (2\le i\le m), & 1&\longmapsto -m. \end{aligned}$$

On the other hand, Y (hence also $X(w)$) can be identified with the set of $x\in\mathbf{P}(V)$ such that

$$\langle x,F(x)\rangle = \langle x,F^2(x)\rangle = \cdots = \langle x,F^{m-1}(x)\rangle = 0, \qquad \langle x,F^m(x)\rangle\ne 0.$$

For example, if $\dim V=4$, this is just the set of all $x\in\mathbf{P}(V)$ such that $\langle x,F(x)\rangle=0$, $\langle x,F^2(x)\rangle\ne 0$. Note that the equation $\langle x,F(x)\rangle=0$ defines a non-singular surface S in $\mathbf{P}(V)$ and that $\langle x,F^2(x)\rangle\ne 0$ means that we remove from S a union of rational curves, one for each isotropic plane in V, defined over $\mathbb{F}_q$. One can prove that in this case ($n=4$), the number of $\mathbb{F}_{q^r}$-rational points of $X(w)$ is given by:

$$q^{2r}-\frac{q}{2}(1+q)^2q^r+\frac{q}{2}(1-q)^2(-q)^r+q^4 = \begin{cases}(q^r-q^2)^2, & r \text{ even},\\ (q^r-q)(q^r-q^3), & r \text{ odd}.\end{cases}$$

3. A fixed point formula

3.1. Let X be a scheme, separated and of finite type over k and let $\sigma: X\to X$ be an automorphism of finite order of X. We decompose σ as $\sigma=s\cdot u$ where s and u are powers of σ respectively of order prime to p and

a power of p. The main result of this chapter is the following:

THEOREM 3.2. With the above notations (see also 0.5),

$$\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \mathrm{Tr}(u^*, H_c^*(X^s, \overline{\mathbb{Q}}_l)).$$

The first step is to prove that the left hand side is an integer independent of l. One might conjecture that for any endomorphism f of X and each i, $\mathrm{Tr}(f^*, H_c^i(X, \overline{\mathbb{Q}}_l))$ is integral and independent of l, but such a more general and precise result is known only for X proper and smooth (Katz and Messing, Inv. Math. 23 (1974), 73-77).

PROPOSITION 3.3. With the notations of 3.1, $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is an integer independent of l ($l \neq p$).

A standard specialization argument allows us to assume that $p > 1$ and that k is an algebraic closure of the prime field $\mathbb{F}_p$. This is anyway the only case we will need in the rest of the paper. The scheme X and σ can then be defined over some finite extension $\mathbb{F}_q \subset k$ of $\mathbb{F}_p$; we denote by $F : X \to X$ the corresponding Frobenius endomorphism.

Let us first assume that X is quasi-projective. Then, for $n \geq 1$, the composite $F^n \circ \sigma$ is the Frobenius map relative to some new way of lowering the field of definition of X from k to $\mathbb{F}_{q^n}$ and the Lefschetz fixed point formula for Frobenius ([5] and [11]) shows that $\mathrm{Tr}((F^n\sigma)^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is the number of fixed points of $F^n\sigma$ ($n \geq 1$). The automorphisms F^* and σ^* of the cohomology commute; as a function of n, $\mathrm{Tr}((F^n\sigma)^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is hence of the form $\sum_\lambda \alpha_\lambda \lambda^n$ where λ runs through the multiplicative group $\overline{\mathbb{Q}}_l^*$ of $\overline{\mathbb{Q}}_l$ and where α_λ is zero for all λ except for a finite number of them.

The functions $\mathbb{N}^+ \to \overline{\mathbb{Q}}_l^* : n \mapsto \lambda^n$ (for $\lambda \in \overline{\mathbb{Q}}_l^*$) are linearly independent. This can be checked either by using Vandermonde determinants or by appealing to Dedekind's theorem on the linear independence of characters (Bourbaki, Algèbre V, § 7, 5). For $n \geq 1$ the numbers $\sum_\lambda \alpha_\lambda \lambda^n = |X^{F^n\sigma}|$ are rational; hence for any automorphism τ of $\overline{\mathbb{Q}}_l$ one has

$$\sum_\lambda \tau(\alpha_\lambda)\tau(\lambda)^n = \sum_\lambda \tau(\alpha_{\tau^{-1}(\lambda)})\lambda^n = \sum_\lambda \alpha_\lambda \lambda^n \qquad (n \geq 1),$$

and, by the linear independence of the functions λ^n,

$$\alpha_{\tau(\lambda)} = \tau(\alpha_\lambda).$$

In particular, $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \sum_\lambda \alpha_\lambda$ is invariant by any automorphism of $\overline{\mathbb{Q}}_l$, hence rational. Similarly, as $\mathrm{Tr}((F^n\sigma)^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = |X^{F^n\sigma}|$ is independent of l, for any isomorphism $\tau : \overline{\mathbb{Q}}_l \to \overline{\mathbb{Q}}_{l'}$ one has

$$\tau\,\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_{l'})),$$

hence the rational number $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is independent of l.

Since the automorphism σ has finite order, $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is a sum of roots of unity, hence an algebraic integer. Being rational it is an ordinary integer.

If we do not assume X to be quasi-projective, we can either repeat the previous argument by working with algebraic spaces or reduce to the quasi-projective case: if (X_i) is a finite partition of X into locally closed quasi-projective subschemes stable under σ, then (cf. [5])

$$\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \sum_i \mathrm{Tr}(\sigma^*, H_c^*(X_i, \overline{\mathbb{Q}}_l)). \tag{3.3.1}$$

3.4. Proof of 3.2. Let (X_i) be a partition of X into locally closed subschemes stable under σ and such that, on each X_i, σ defines a free action of a cyclic group. By applying (3.3.1) to the decompositions $X = \bigcup X_i$ and $X^s = \bigcup X_i^s$, one reduces to the case where σ generates a free action of a cyclic group H. In this case, either

(a) σ is of order a power of p, hence $s = \mathrm{Id}$, $u = \sigma$ and (3.2) is trivial, or

(b) the order of σ is divisible by a prime number $l' \neq p$ and we must prove that $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = 0$. As

$$\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_{l'})) \tag{3.3}$$

we may as well assume that $l = l'$. Let us now grant the

PROPOSITION 3.5. Let H be a finite group acting freely on X. Then

$$\chi(h) = \mathrm{Tr}(h^*, H_c^*(X, \overline{\mathbb{Q}}_l))$$

is the character of a virtual projective $\mathbb{Z}_l[H]$-module.

If H is abelian, and if H' is its largest subgroup of order prime to l, a representation of H over $\mathbb{Z}_l$ gives rise to a projective $\mathbb{Z}_l[H]$-module if and only if it is induced from a representation of H'. Its character vanishes then on $H - H'$ and we get the vanishing (b) by applying (3.5) to the cyclic group generated by σ.

3.6. Let A be a torsion ring with unit and let $\mathscr{F}$ be a sheaf of left A-modules on a scheme Y, with Y separated and of finite type over k. The A-modules $H_c^i(Y, \mathscr{F})$ are then the cohomology modules of a finer object $R\Gamma_c(Y, \mathscr{F})$ in the derived category $D^b(A)$ of the category of A-modules. The following is the key to the proof of 3.5.

PROPOSITION 3.7. Assume A to be right and left noetherian. If $\mathscr{F}$ is a constructible sheaf of projective A-modules, then $R\Gamma_c(Y, \mathscr{F})$ can be represented by a finite complex of projective A-modules of finite type.

We repeat the proof in [10, XVII (5.2.10)].

(a) The $H_c^i(Y, \mathscr{F})$ are A-modules of finite type, and vanish for $i > 2\dim(Y)$ ([10, XVII (5.2.8.1) and (5.3.6)]).

(b) For any right A-module of finite type N, one has

$$R\Gamma_c(Y, N \otimes_A \mathscr{F}) \simeq N \otimes_A^L R\Gamma_c(Y, \mathscr{F})$$

([10, XVII (5.2.9)]); the proof rests on replacing N by a free resolution of N, to reduce to the trivial case where N is free of finite type; we are allowed to use such left infinite resolutions because $R\Gamma_c$ is of finite cohomological dimension. The assumption that N is of finite type is in fact unnecessary.

(c) By (a) we can represent $R\Gamma_c(Y, \mathscr{F})$ by a complex of A-modules $K^\bullet$, with $K^i = 0$ for $i \notin [0, 2\dim Y]$ and K^i free of finite type for $i > 0$. For any right A-module of finite type N and any $i > 0$,

$$\mathrm{Tor}_i^A(N, K^0) = H^{-i}(N \otimes^L R\Gamma_c(Y, \mathscr{F})) =_{(b)} H^{-i}R\Gamma_c(Y, N \otimes \mathscr{F}) = 0.$$

The A-module K^0 is hence flat; it is of finite type because $H^0(K^\bullet)$ is, hence it is projective.

3.8. Proof of 3.5. We will assume that X is quasi-projective (the general case can be handled as in 3.3). Put $Y = X/H$ and denote by π the projection $\pi : X \to Y$. The group H acts on $\pi_*\mathbb{Z}/l^n$ and this action turns $\pi_*\mathbb{Z}/l^n$ into a locally constant sheaf of free $\mathbb{Z}/l^n[H]$-modules of rank one. By 3.7, $R\Gamma_c(Y, \pi_*\mathbb{Z}/l^n)$ can be represented by a complex $K_n^\bullet$ of projective $\mathbb{Z}/l^n[H]$-modules of finite type. One has

$$R\Gamma_c(Y, \pi_*\mathbb{Z}/l^n) \simeq R\Gamma_c(Y, \pi_*\mathbb{Z}/l^{n+1}) \otimes^L_{\mathbb{Z}/l^{n+1}} \mathbb{Z}/l^n$$

(3.7 (b)), and one checks easily ([11, XV, 3.3, Lemme 1]) that once $K_n^\bullet$ is chosen, one can choose $K_{n+1}^\bullet$ such that $K_n^\bullet$ is the reduction mod l^n of $K_{n+1}^\bullet$. Taking a projective limit, we get a complex $K_\infty^\bullet$ of projective $\mathbb{Z}_l[H]$-modules and an H-equivariant isomorphism

$$H_c^*(X, \mathbb{Z}_l) = \varprojlim H_c^*(X, \mathbb{Z}/l^n) = \varprojlim H_c^*(Y, \pi_*\mathbb{Z}/l^n) = H^*(K_\infty^\bullet)$$

(the middle equality because π is finite). We then have

$$\chi(h) = \sum_i (-1)^i \mathrm{Tr}(h, K_\infty^i).$$

3.9. Let X/k be as in (3.1), T a finite abelian group of order prime to p, G a finite group and ρ an action of $T \times G$ on X. We assume that the action of T on X is free. We let $T \times G$ act on $H_c^*(X)$ by transport of structure. For any character θ of T with values in $\overline{\mathbb{Q}}_l^*$, we denote by H_θ^* the subspace of $H_c^*(X)$ on which T acts by θ; on H_θ^*, t^* is multiplication by $\theta(t)^{-1}$.

Let us assume for simplicity that X is quasi-projective. We denote by π the projection $\pi : X \to Y = X/T$. The group G acts on Y. For each $g \in G$, we denote by $I(g)$ the set of connected components of the fixed point set Y^g

and by Y_i^g the component corresponding to $i \in I(g)$.

For $y \in Y$, $\pi^{-1}(y)$ is a principal homogeneous set for the action of the abelian group T. If $y \in Y^g$, g acts on $\pi^{-1}(y)$ and commutes with T, it hence acts like some $t(g,y) \in T$:

$$gx = t(g,x)x, \qquad x \in \pi^{-1}(y);$$

$t(g,y)$ is constant for y in each connected component Y_i^g of Y^g; we denote it by $t(g,i)$.

The Theorem 3.2 will be used in the following form.

COROLLARY 3.10. With the above notations, let $g = su$ be the decomposition of $g \in G$ as the product of commuting elements respectively of order prime to p and a power of p. Then,

$$\mathrm{Tr}(g^*, H_\theta^*) = \sum_{i \in I(s)} \theta(t(s,i)^{-1}) \,\mathrm{Tr}(u^*, H_c^*(Y_i^s)).$$

For $t \in T$, the decomposition 3.1 of $\sigma = t \cdot g$ is $tg = (st) \cdot u$. We have

$$X^{st} = \coprod_{i \in I(s)} \pi^{-1}(Y_i^s)^{st} = \coprod_{\substack{i \in I(s) \\ t = t(s,i)^{-1}}} \pi^{-1}(Y_i^s),$$

hence

$$\begin{aligned}
\mathrm{Tr}(g^*, H_\theta^*) &= \frac{1}{|T|} \sum_t \theta(t)^{-1} \,\mathrm{Tr}(g^* t^{*-1}, H_c^*(X)) \\
&= \frac{1}{|T|} \sum_t \theta(t) \,\mathrm{Tr}(g^* t^*, H_c^*(X)) \\
&= \frac{1}{|T|} \sum_t \theta(t) \sum_{\substack{i \in I(s) \\ t = t(s,i)^{-1}}} \mathrm{Tr}(u^*, H_c^*(\pi^{-1}(Y_i^s))) \\
&= \frac{1}{|T|} \sum_{i \in I(s)} \theta(t(s,i)^{-1}) \,\mathrm{Tr}(u^*, H_c^*(\pi^{-1}(Y_i^s))).
\end{aligned}$$

The decomposition (3.1) of the automorphism tu of $\pi^{-1}(Y_i^s)$ is $t \cdot u$. For $t \neq e$, t has no fixed point, hence

$$\mathrm{Tr}((tu)^*, H_c^*(\pi^{-1}(Y_i^s))) = 0 \qquad (t \neq e).$$

The endomorphism

$$\frac{1}{|T|} \sum_t t^* \quad \text{of} \quad H_c^*(\pi^{-1}(Y_i^s))$$

is a retraction onto the subspace $\pi^* H_c^*(Y_i^s)$, hence

$$\frac{1}{|T|} \,\mathrm{Tr}(u^*, H_c^*(\pi^{-1}(Y_i^s))) = \mathrm{Tr}(u^*, H_c^*(Y_i^s))$$

and, substituting into the expression for $\mathrm{Tr}(g^*, H_\theta^*)$, we get (3.10).

3.11. The end of this chapter will not be used in this paper. Let G be

a finite group acting on X/k as in (3.1). For simplicity we assume X to be quasi-projective. Assume further that G acts freely on X, and put $Y = X/G$. The covering X of Y is said to be tame if Y can be imbedded as a Zariski open dense subset of a proper scheme $\overline{Y}$ in such a way that the p-Sylow-subgroups of G act freely on the normalisation $\overline{X}$ of $\overline{Y}$ in X:

$$\begin{array}{ccc} X & \overset{j}{\hookrightarrow} & \overline{X} \\ \downarrow & & \downarrow \\ Y & \hookrightarrow & \overline{Y}. \end{array}$$

PROPOSITION 3.12. With the above notations, if X/Y is tame, then the virtual representation $\sum_i (-1)^i H_c^i(X)$ of G is a multiple of the regular representation.

It suffices to prove that $\mathrm{Tr}(g^*, H_c^*(X)) = 0$ for $g \neq e$. If g is not of order a power of p, this follows from (3.2). If g is of order a power of p, g acts without fixed points on $\overline{X}$ and

$$\mathrm{Tr}(g^*, H_c^*(X)) = \mathrm{Tr}(g^*, H^*(\overline{X}, j_!\overline{\mathbb{Q}}_l)) = 0$$

by the Lefschetz fixed point theorem applied to $\overline{X}$ and the sheaf $j_!\overline{\mathbb{Q}}_l$.

3.13. Historical remark. The method we have followed, using (3.5), was first used by Zarelua (On finite groups of transformations, Proc. Int. Symp. on Topology and its Applications, 1968, pp. 334-339) and independently by J. L. Verdier to prove that if a finite group G acts freely on a topological space X of finite cohomological dimension, and if the $H^i(X, \mathbb{Z})$ are of finite type, then $\sum_i (-1)^i H^i(X, \mathbb{Z}) \otimes \mathbb{Q}$ is a multiple of the regular representation of G.

4. The character formula

The assumptions (0.1.1) are in force in this chapter and the next four.

DEFINITION 4.1. Let T be an F-stable maximal torus of G. The Green function $Q_{T,G}(u)$ is the restriction to the unipotent elements of the character of the virtual representation $R^1_{T\subset B}$, where B is any Borel subgroup containing T.

This Green function does not depend on B (by (1.6) and (1.17 (i))); it only depends on the G^F-conjugacy classes of T and u. The natural map $x \mapsto \overline{x}$ from G to its adjoint group induces a bijection on the unipotent sets, and

$$Q_{T,G}(u) = Q_{\overline{T},G^{\mathrm{ad}}}(\overline{u}). \tag{4.1.1}$$

The Green function is integer-valued (3.3) and is a restriction of a character

of G^F, hence $Q_{T,G}(u) = Q_{T,G}(u^n)$ if $(n,p) = 1$.

Let α be the smallest integer ≥ 1 such that F^α is the identity on $\mathbf{W}$. From the proof of (3.3) we see that $\sum_{n\geq 1} X_{T\subset B}^{F^{\alpha n}u} t^n$ is a rational function of t and that

$$Q_{T,G}(u) = -\left\{\sum_{n\geq 1} X_{T\subset B}^{F^{\alpha n}u} t^n\right\}_{t=\infty}. \tag{4.1.2}$$

In this chapter, we will express the character of $R_{T\subset B}^\theta$ in terms of θ and of Green functions.

THEOREM 4.2. Let $x = su$ be the Jordan decomposition of $x \in G^F$. Then

$$\mathrm{Tr}(x, R_{T\subset B}^\theta) = \frac{1}{|Z^0(s)^F|} \sum_{\substack{g\in G^F \\ \mathrm{ad}\, gT\subset Z^0(s)}} Q_{\mathrm{ad}\, gT, Z^0(s)}(u)\, \mathrm{ad}\, g(\theta)(s).$$

Let ε_x be the function on G^F whose value is 1 at $x \in G^F$ and 0 elsewhere. One can rewrite 4.2 as giving the following formula for the character of $R_{T\subset B}^\theta$:

$$\mathrm{Tr}(\ , R_{T\subset B}^\theta) = \sum_{t\in T^F} \frac{1}{|Z^0(t)^F|}\theta(t) \sum_{\substack{u\in Z^0(t)^F \\ g\in G^F}} Q_{T,Z^0(t)}(u)\varepsilon_{\mathrm{ad}\, g(tu)} \tag{4.2.1}$$

where we put $Q_{T,Z^0(t)}(u) = 0$ whenever u is not unipotent.

COROLLARY 4.3. $R_{T\subset B}^\theta$ is independent of the choice of B ($B \supset T$).

From now on we shall write R_T^θ for $R_{T\subset B}^\theta$. We will deduce 4.2 from 3.7 and the following geometrical facts.

PROPOSITION 4.4. (i) If a Borel subgroup B_1 of G contains s, then $B_1 \cap Z^0(s)$ is a Borel subgroup of $Z^0(s)$.

(ii) Pick $T \subset B$ in G. Any Borel subgroup B_1 such that $s \in B_1$ is of the form $\mathrm{ad}\, gB$ with $\mathrm{ad}\, gT \subset Z^0(s)$ (i.e., $g^{-1}sg \in T$). The left coset

$$\tau_{T,B}(B_1) \overset{\text{def.}}{=} Z^0(s)\cdot g$$

depends only on T, B, and B_1.

(iii) The map $B_1' \mapsto B_1' \cap Z^0(s)$ is an isomorphism from the space of Borel subgroups containing s, such that $\tau_{T,B}(B_1') = \tau_{T,B}(B_1)$ and the space of Borel subgroups of $Z^0(s)$.

Proof. (i) is well known. Put $B_1 = \mathrm{ad}\, g_1 B$. We have $g_1^{-1}sg_1 \in B$ hence there exists u in the unipotent radical of B such that $u^{-1}g_1^{-1}sg_1u \in T$. We take $g = g_1u$. Put $T' = \mathrm{ad}\, gT$. If $g' = hg$ is also such that $\mathrm{ad}\, g'B = B_1$ (resp. $\mathrm{ad}\, g'T \subset Z^0(s)$) then $h \in B_1$ (resp. $h \in Z^0(s)N(T')$, by the conjugacy of maximal tori in $Z^0(s)$). If $W = N(T')/T'$ is the Weyl group of G and $W^0 = (N(T')\cap Z^0(s))/T'$ that of $Z^0(s)$, the Bruhat decomposition for $Z^0(s)$ reads

$$Z^0(s) = (B_1 \cap Z^0(s))W^0(B_1 \cap Z^0(s)),$$

and

$$(B_1 Z^0(s)) \cap N(T') = B_1 W^0 (B_1 \cap Z^0(s)) \cap N(T') \subset B_1 W^0 B_1 \cap N(T') = W^0(T') \subset Z^0(s).$$

Hence

$$B_1 \cap (Z^0(s) \cdot N(T')) \subset Z^0(s)$$

and

$$B_1 \cap (Z^0(s) \cdot N(T')) = B_1 \cap Z^0(s).$$

The element g such that $\mathrm{ad}\, gB = B_1$ and $\mathrm{ad}\, gT \subset Z^0(s)$ is unique modulo $B_1 \cap Z^0(s)$, hence (ii) and (iii).

A $Z^0(s)$-left coset $\tau = Z^0(s)g$, with $\mathrm{ad}\, gT \subset Z^0(s)$, defines an isomorphism $\overline{\tau}$ from the maximal torus of G to that of $Z^0(s)$; $\mathrm{ad}\, g$ maps T (contained in the Borel subgroup B) to $\mathrm{ad}\, gT \subset Z^0(s)$ (contained in the Borel subgroup $\mathrm{ad}\, gB \cap Z^0(s)$). This construction will allow us to compare the relative position of Borel subgroups in G and $Z^0(s)$.

PROPOSITION 4.5. Let B_1' and B_1'' be Borel subgroups containing s,

$$\tau' = \tau_{T,B}(B_1'), \qquad \tau'' = \tau_{T,B}(B_1'');$$

let w be the relative position of B_1' and B_1'' (in G) and w_1 that of $B_1' \cap Z^0(s)$ and $B_1'' \cap Z^0(s)$ (in $Z^0(s)$). Then

$$w_1 = \overline{\tau}' w \overline{\tau}''^{-1}.$$

Replacing (T, B) by a conjugate, we may assume that $B = B_1'$ and that $T \subset B_1' \cap Z^0(s) \cap B_1''$. We will identify the maximal torus of G and that of $Z^0(s)$ with T, using $T \subset B$ and $T \subset (B \cap Z^0(s))$. We then have $\overline{\tau}' = \mathrm{Id}$. Replacing B_1'' by an $N(T) \cap Z^0(s)$-conjugate, we may further assume that $w_1 = 1$, i.e., $B_1' \cap Z^0(s) = B_1'' \cap Z^0(s)$. We must prove that $w = \overline{\tau}''$, which is clear.

LEMMA 4.6. If B and B', containing T, are in the same relative position as B_1 and B_1' containing s, then $\tau_{T,B}(B_1) = \tau_{T,B'}(B_1')$.

Replacing (B, T, B') by a conjugate, we may again assume that $B = B_1$ and that $T \subset B_1 \cap Z^0(s) \cap B_1'$. In this case $B' = B_1'$.

To prove (4.2) we first compute the space $X^s_{T \subset B}$ of Borel subgroups B_1 containing s and such that B_1 and FB_1 are in the same relative position as B and FB.

PROPOSITION 4.7. Let $X^s_{T \subset B}(g)$ be the subspace of $X^s_{T \subset B}$ consisting of those B_1 with $\tau_{T,B}(B_1) = Z^0(s)g$. Then

$$X^s_{T \subset B} = \coprod_{\substack{g \in Z^0(s)^F \backslash G^F \\ \mathrm{ad}\, gT \subset Z^0(s)}} X^s_{T \subset B}(g) \tag{4.7.1}$$

and, for $g \in G^F$, the map $B_1 \mapsto B_1 \cap Z^0(s)$ induces an isomorphism

$$X^s_{T\subset B}(g) \xrightarrow{\sim} X_{\mathrm{ad}\, gT\subset \mathrm{ad}\, gB\cap Z^0(s)}. \tag{4.7.2}$$

By 4.6, we have $F\tau_{T,B}(B_1) = \tau_{T,FB}(FB_1) = \tau_{T,B}(B_1)$. By Lang's Theorem, the $Z^0(s)$-principal homogeneous space $\tau_{T,B}(B_1)$, being F-stable, has a rational point, hence (4.7.1). By (4.4 (iii)) and (4.5), the map (4.7.2) is an isomorphism from $X^s_{T\subset B}(g)$ to some $X(w)$ (relative to $Z^0(s)$); to check which w appears, we observe that $\mathrm{ad}\, gB \in X^s_{T\subset B}(g)$.

Proof of 4.2. The theorem is now an immediate application of (4.7) and (3.10) applied to the action (1.17) of $G^F \times T^F$ on $\widetilde{X}_{T\subset B}$.

5. Characters of tori

We will assume chosen isomorphisms

$$k^* \xrightarrow{\sim} (\mathbb{Q}/\mathbb{Z})_{p'}, \tag{5.0.1}$$

and

$$(\text{roots of unity of order prime to } p \text{ in } \overline{\mathbb{Q}}_l^*) \xrightarrow{\sim} (\mathbb{Q}/\mathbb{Z})_{p'}. \tag{5.0.2}$$

5.1. Let T be a torus over k. Besides the character group $X(T) = \mathrm{Hom}(T, \mathbb{G}_m)$, we will consider its dual $Y(T) = \mathrm{Hom}(\mathbb{G}_m, T)$. The duality is given by

$$\langle \alpha, h\rangle = \alpha \circ h \in \mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_m) = \mathbb{Z}$$

($\alpha \in X(T)$ and $h \in Y(T)$). The maps $(h, x) \mapsto h(x)$ and $t \mapsto (\alpha \mapsto \alpha(t))$ induce isomorphisms

$$Y(T) \otimes k^* = T(k) = \mathrm{Hom}(X(T), k^*). \tag{5.1.1}$$

If T is a maximal torus in G, the roots of T in G are elements of $X(T)$, while the coroots belong to $Y(T)$. If α is a root, the corresponding coroot H_α is characterised as follows: there is a homomorphism $u : \mathrm{SL}(2) \to G$, which maps the subgroup $\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$ onto the root subgroup U_α, and whose restriction to the group of diagonal matrices (identified with $\mathbb{G}_m$ by $x \mapsto \begin{pmatrix} x & 0 \\ 0 & x^{-1} \end{pmatrix}$) is H_α.

5.2. Using the chosen isomorphism (5.0.1), we can rewrite (5.1.1) as

$$T(k) = Y(T) \otimes (\mathbb{Q}/\mathbb{Z})_{p'} \simeq (Y(T) \otimes \mathbb{Q}/Y(T))_{p'}. \tag{5.2.1}$$

If T is obtained by extension of scalars from a torus $T_0/\mathbb{F}_q$, the Frobenius map F induces a map $F : Y(T) \to Y(T)$ (Y is a covariant functor) and T^F is the subgroup $\mathrm{Ker}(F-1)$ of $T(k) = Y(T) \otimes (\mathbb{Q}/\mathbb{Z})_{p'}$; since F is divisible by p, it is also the kernel of $F-1$ in $Y(T) \otimes \mathbb{Q}/\mathbb{Z}$:

$$0 \longrightarrow T^F \longrightarrow Y(T) \otimes \mathbb{Q}/\mathbb{Z} \xrightarrow{F-1} Y(T) \otimes \mathbb{Q}/\mathbb{Z} \longrightarrow 0. \tag{5.2.2}$$

By applying the Snake Lemma to

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Y(T) & \longrightarrow & Y(T)\otimes\mathbb{Q} & \longrightarrow & Y(T)\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0 \\ & & {\scriptstyle F-1}\downarrow & & {\scriptstyle F-1}\downarrow & & {\scriptstyle F-1}\downarrow & & \\ 0 & \longrightarrow & Y(T) & \longrightarrow & Y(T)\otimes\mathbb{Q} & \longrightarrow & Y(T)\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0, \end{array}$$

we get another exact sequence

$$0 \longrightarrow Y(T) \xrightarrow{F-1} Y(T) \longrightarrow T^F \longrightarrow 0. \tag{5.2.3}$$

The maps in (5.2.2), (5.2.3) depend on the choice (5.0.1). An intrinsic expression for these sequences would be as follows: T^F is the image of the middle map in the exact sequence

$$0 \longrightarrow Y(T)_{\widehat{p'}}(1) \xrightarrow{F-1} Y(T)_{\widehat{p'}}(1) \xrightarrow{(F-1)^{-1}} Y(T)_{\widehat{p'}}(1) \otimes \mathbb{Q}/\mathbb{Z} \xrightarrow{F-1} Y(T)_{\widehat{p'}}(1) \otimes \mathbb{Q}/\mathbb{Z} \longrightarrow 0.$$

Let us map the sequences (5.2.2), (5.2.3) into $\mathbb{Q}/\mathbb{Z}$. The isomorphism (5.0.2) provides an isomorphism

$$(T^F)^{\sim} \overset{\text{def}}{=} \mathrm{Hom}(T^F, \overline{\mathbb{Q}}_l^*) = \mathrm{Hom}(T^F, \mathbb{Q}/\mathbb{Z}),$$

and we get exact sequences

$$^*0 \longrightarrow X(T) \xrightarrow{F-1} X(T) \longrightarrow (T^F)^{\sim} \longrightarrow 0, \tag{5.2.2}$$

$$^*0 \longrightarrow (T^F)^{\sim} \longrightarrow X(T) \otimes \mathbb{Q}/\mathbb{Z} \xrightarrow{F-1} X(T) \otimes \mathbb{Q}/\mathbb{Z} \longrightarrow 0. \tag{5.2.3}$$

The dual torus T^* of T is defined by the rule $X(T^*) = Y(T)$, (hence $Y(T^*) = X(T)$); its $\mathbb{F}_q$-structure is defined by (F on $Y(T^*)$) = (${}^tF$ on $X(T)$). Via some isomorphism

$$(T^F)^{\sim} = T^{*F}, \tag{5.2.4}$$

the sequences (5.2.3), (5.2.2) for T^* are identical with $(5.2.2)^*$, $(5.2.3)^*$.

5.3. If we go from $\mathbb{F}_q$ to $\mathbb{F}_{q^n}$, F is replaced by F^n. Composition with the norm map

$$N = \frac{F^n - 1}{F - 1} = \sum_0^{n-1} F^i \,:\, T^{F^n} \longrightarrow T^F$$

is an injection ${}^tN : (T^F)^{\sim} \to (T^{F^n})^{\sim}$. We have commutative diagrams

$$\begin{array}{ccccccccccccc}
0 & \longrightarrow & Y & \xrightarrow{F^n-1} & Y & \longrightarrow & T^{F^n} & \longrightarrow & Y\otimes\mathbb{Q}/\mathbb{Z} & \xrightarrow{F^n-1} & Y\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0 \\
 & & {\scriptstyle N}\downarrow & & \| & & {\scriptstyle N}\downarrow & & {\scriptstyle N=\sum_0^{n-1}F^i}\downarrow & & \| & & \\
0 & \longrightarrow & Y & \longrightarrow & Y & \longrightarrow & T^F & \longrightarrow & Y\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & Y\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0,
\end{array}$$

$$\begin{array}{ccccccccccccc}
0 & \longrightarrow & X & \longrightarrow & X & \longrightarrow & (T^F)^\sim & \longrightarrow & X\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & X\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0 \\
 & & \| & & {\scriptstyle {}^tN=\sum_0^{n-1}{}^tF^i}\downarrow & & {\scriptstyle {}^tN}\downarrow & & \| & & {\scriptstyle {}^tN}\downarrow & & \\
0 & \longrightarrow & X & \longrightarrow & X & \longrightarrow & (T^{F^n})^\sim & \longrightarrow & X\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & X\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0.
\end{array}$$

where X and Y stand for $X(T)$ and $Y(T)$. In particular, for θ a character of T^F, θ and $\theta\circ N$ have the same image in $X(T)\otimes\mathbb{Q}/\mathbb{Z}$, and (5.2.4) gives rise to a commutative diagram

$$\begin{array}{ccc}
(T^F)^\sim & \simeq & (T^*)^F \\
{\scriptstyle {}^tN}\downarrow & & {\scriptstyle N}\downarrow \\
(T^{F^n})^\sim & \simeq & (T^*)^{F^n}.
\end{array}$$

PROPOSITION 5.4. Let T and T' be two F-stable maximal tori of G, and let θ,θ' be characters of $T^F,(T')^F$. We identify them with characters of $Y(T)$ and $Y(T')$, by (5.2.3). The following conditions are equivalent:

(i) For some $x\in G$, with $\mathrm{ad}\,x(T)=T'$, the map induced by $\mathrm{ad}\,x : Y(T)\to Y(T')$ carries θ to θ';

(ii) For some n, the pairs $(T,\theta\circ N)$, $(T',\theta'\circ N)$, where N is the norm from T^{F^n} to T^F (resp. $(T')^{F^n}$ to $(T')^F$) are G^{F^n}-conjugate.

By 5.3, the condition (i) is invariant under the replacement of F by F^n and of θ,θ' by $\theta\circ N,\theta'\circ N$. It hence suffices to check (5.4) in the trivial case where T and T' are split.

DEFINITION 5.5. The pairs (T,θ), (T',θ') are said to be geometrically conjugate when the equivalent conditions of (5.4) hold.

5.6. Let (T,θ) be as above. The choice of a Borel subgroup B containing T defines an isomorphism of T with the torus $\mathbf{T}$; the corresponding isomorphism $Y(T)\to Y(\mathbf{T})$ carries θ to a character of $Y(\mathbf{T})$ with values in the roots of unity of order prime to p in $\overline{\mathbb{Q}}_l$. The isomorphism (5.0.2) identifies it with an element of $X(\mathbf{T})\otimes(\mathbb{Q}/\mathbb{Z})_{p'}$. Its class $[\theta]$ in $(X(\mathbf{T})\otimes(\mathbb{Q}/\mathbb{Z})_{p'})/\mathbf{W}$ is independent of the choice of B. It is invariant under F. Let us put

$$\mathcal{S}\overset{\text{def}}{=}[(X(\mathbf{T})\otimes(\mathbb{Q}/\mathbb{Z})_{p'})/\mathbf{W}]^F\xrightarrow{\sim}[(X(\mathbf{T})\otimes\mathbb{Q}/\mathbb{Z})/\mathbf{W}]^F.$$

PROPOSITION 5.7. (i) The map $\theta\mapsto[\theta]$ induces a bijection from the set of geometric conjugacy classes of pairs (T,θ) to $\mathcal{S}$.

(ii) (Compare Steinberg [15, p. 93]). The number of geometric conjugacy classes of pairs (T,θ) is $|Z^{0F}|q^l$ where Z^0 is the identity component of the centre of G and l is the semisimple rank of G.

(i) The injectivity is clear on 5.4(i). Let us prove surjectivity. If the class mod $\mathbf{W}$ of $x \in X(\mathbf{T}) \otimes (\mathbb{Q}/\mathbb{Z})_{p'}$ is invariant by F, one indeed has ${}^t(wF)x = x$ for some $w \in \mathbf{W}$, and x corresponds to a character of $\mathbf{T}(w)^F$, hence of T^F, for some F-stable maximal torus T in G (1.14).

(ii) As in (5.2), we may now identify geometric conjugacy classes of pairs (T,θ) with $\mathbb{F}_q$-rational points of the quotient $\mathbf{T}^*/\mathbf{W}$. Our task is to compute

$$|(\mathbf{T}^*/\mathbf{W})^F| = \sum_i (-1)^i \operatorname{Tr}(F^*, H_c^i(\mathbf{T}^*/\mathbf{W})) = \sum_i (-1)^i \operatorname{Tr}(F^*, H_c^i(\mathbf{T}^*)^{\mathbf{W}}).$$

Let G' be the derived group of G; the torus $\mathbf{T}^*$ is isogenous to (hence has the same cohomology as) the product of Z^{0*} with $\mathbf{T}'^*$ ($\mathbf{T}'$ being the torus of G'). Since any torus has as many rational points as its dual, we get by Künneth's Theorem

$$|(\mathbf{T}^*/\mathbf{W})^F| = |Z^{0F}| \sum_i (-1)^i \operatorname{Tr}(F^*, H_c^i(\mathbf{T}'^*)^{\mathbf{W}}).$$

We have

$$H^*(\mathbf{T}'^*, \overline{\mathbb{Q}}_l) = \Lambda^* H^1(\mathbf{T}'^*, \overline{\mathbb{Q}}_l) = \Lambda^*(Y(\mathbf{T}') \otimes \overline{\mathbb{Q}}_l(-1));$$

hence by Poincaré duality,

$$H_c^{2l-i}(\mathbf{T}'^*, \overline{\mathbb{Q}}_l) = \operatorname{Hom}(\Lambda^i Y(\mathbf{T}'), \overline{\mathbb{Q}}_l(i-l)),$$

and

$$\operatorname{Tr}(F^*, H_c^{2l-i}(\mathbf{T}'^*)^{\mathbf{W}}) = q^{l-i} \operatorname{Tr}(F, \Lambda^i Y(\mathbf{T}')^{\mathbf{W}}).$$

The next lemma shows that this trace is zero for $i \neq 0$, hence the q^l factor.

LEMMA 5.8. Let $W \subset \operatorname{Aut}(V)$ be a finite group generated by reflections of a real vector space V of dimension d. We assume that $V^W = 0$. Then $(\Lambda^i V)^W = 0$ for $i \neq 0$.

For a proof, see Bourbaki [1, Ch. V, Ex. 3 of § 2].

5.9. Let T be an F-stable maximal torus in G and let α be a root of T. For θ a character of T^F, we will say that θ is orthogonal to H_α: $\langle H_\alpha, \theta\rangle = 0$ if θ, viewed as a character of $Y(T)$ (5.2.3), is identically 1 on H_α. The character θ of $Y(T)$ is then invariant by the corresponding reflection s_α.

5.10. Let $\pi : \widetilde{G} \to G$ be the simply connected covering of the derived group of G, and let $\widetilde{T}$ be the inverse image of T in $\widetilde{G}$. The group $\widetilde{T}$ is

connected: it is a maximal torus in $\widetilde{G}$. It acts on $T \times \widetilde{G}$ by

$$\tilde{t} * (t, g) = (t\pi(\tilde{t})^{-1}, \tilde{t}g),$$

and the map $(t, g) \mapsto tg$ induces an isomorphism $\widetilde{T}\backslash(T \times \widetilde{G}) \xrightarrow{\sim} G$, hence (Lang's Theorem) $\widetilde{T}^F\backslash(T^F \times \widetilde{G}^F) \xrightarrow{\sim} G^F$. We have

$$T^F/\pi(\widetilde{T}^F) \xrightarrow{\sim} G^F/\pi^F(\widetilde{G}). \tag{5.10.1}$$

PROPOSITION 5.11. (i) A character of T^F is the restriction to T^F of a character of $G^F/\pi(\widetilde{G}^F)$ if and only if it is orthogonal to all coroots.

(ii) Let θ be a character of $G^F/\pi(\widetilde{G}^F)$, T and T' two F-stable maximal tori and let $x \in G$ be such that $\operatorname{ad} x(T) = T'$. Then $\operatorname{ad} x : Y(T) \to Y(T')$ carries the restriction of θ to T^F (viewed as a character of $Y(T)$) onto the restriction of θ to $(T')^F$: the restrictions of θ to the F-stable maximal tori are all geometrically conjugate.

The assertion (i) follows from (5.10.1), and the fact that $Y(\widetilde{T}) \subset Y(T)$ is spanned by the coroots H_α. To prove (ii), we will use the criterion 5.4 (ii). Let n be such that $x \in G^{F^n}$. We will prove that if $y \in T^{F^n}$, $N(y)$ and $N(\operatorname{ad} x(y))$ have the same image in $G^F/\pi(\widetilde{G}^F)$, hence that $\operatorname{ad} x$ transforms $(\theta \mid T^F) \circ N$ into $(\theta \mid (T')^F) \circ N$.

The map

$$t \longmapsto N(t)^{-1}N(\operatorname{ad} x(t)) : T \longrightarrow G$$

lifts uniquely into a map $\varphi : T \to \widetilde{G}$ mapping e to e. Indeed,

(a) it factors through T/Z;

(b) the map $t \mapsto N(t)^{-1}N(\operatorname{ad} x(t)) : \widetilde{T} \to \widetilde{G}$ factors through $\widetilde{T}/\widetilde{Z} = T/Z$ and the desired lifting is the composite $T \to T/Z = \widetilde{T}/\widetilde{Z} \to \widetilde{G}$.

Similarly, the map

$$(a, t) \longmapsto a^{-1}N(t)^{-1}N(\operatorname{ad} x(t)) \operatorname{ad} x(a) : T \times T \longrightarrow G$$

lifts into $\psi : T \times T \to \widetilde{G}$, with $\psi(e, t) = \varphi(t)$. The identity

$$\begin{aligned} F(N(t)^{-1}N(\operatorname{ad} x(t))) &= N(Ft)^{-1}N(F \operatorname{ad} x(t)) \\ &= (t^{-1}F^n t)^{-1}N(t)^{-1}N(\operatorname{ad} x(t)) \operatorname{ad} x(t^{-1})F^n \operatorname{ad} x(t) \\ &= (t^{-1}F^n t)^{-1}N(t)^{-1}N(\operatorname{ad} x(t)) \operatorname{ad} x(t^{-1}F^n t) \end{aligned}$$

lifts into

$$F\varphi(t) = \psi(t^{-1}F^n t, \varphi(t)).$$

Putting $t = y$, we have $y^{-1}F^n y = e$, hence $F\varphi(y) = \varphi(y)$ and $\varphi(y) \in \widetilde{G}^F$. We have $N(y)^{-1}N(\operatorname{ad} x(y)) \in \pi(\widetilde{G}^F)$, as required.

Remark 5.12. The argument above can be used to show the existence of a norm map

$$N : G^{F^n}/\pi(\widetilde{G}^{F^n}) \longrightarrow G^F/\pi(\widetilde{G}^F)$$

which for each F-stable maximal torus T gives rise to a commutative diagram

$$\begin{array}{ccc} T^{F^n}/\pi(\widetilde{T}^{F^n}) & \xrightarrow{\sim} & G^{F^n}/\pi(\widetilde{G}^{F^n}) \\ {\scriptstyle N}\downarrow & & {\scriptstyle N}\downarrow \\ T^{F}/\pi(\widetilde{T}^{F}) & \xrightarrow{\sim} & G^{F}/\pi(\widetilde{G}^{F}). \end{array}$$

THEOREM 5.13. Let (T,θ) be as above. We use (5.2.3) to identify θ with a character of $Y(T)$. If the centre of G is connected, then the stabilizer of θ in the Weyl group $W = N(T)/T$ is generated by the reflections s_α, for H_α a coroot orthogonal to θ.

The key point in the proof is the same as in Steinberg's Theorem that if the derived group G' is simply connected, then the centralizer of any semisimple element is connected.

Let us identify θ with the corresponding element in $X(T)\otimes(\mathbb{Q}/\mathbb{Z})_{p'}$ (invariant under F). Let $\theta_1 \in X(T)\otimes\mathbb{Q}$ be a representative for it. This θ_1 has no p in the denominator. We have the dictionary:

(a) $\langle H_\alpha,\theta\rangle = 0 \iff \langle H_\alpha,\theta_1\rangle \in \mathbb{Z}$;

(b) θ is fixed by $w \in W \iff$ for some $x \in X(T)$, $w\theta_1 + x = \theta_1$.

Let X^{ad} be the subgroup of X generated by the roots. It is the character group of the image T^{ad} of T in the adjoint group. The character group of the centre $Z = \mathrm{Ker}(T \to T^{\mathrm{ad}})$ of G is X/X^{ad}; the character group of Z/Z^0_{red} is the torsion subgroup of X/X^{ad}, hence

(c) Z is connected and smooth (resp. connected) if and only if X/X^{ad} is torsion free (resp. has no torsion prime to p).

For each root α, $s_\alpha(\theta_1) = \theta_1 - \langle H_\alpha,\theta_1\rangle\alpha$. The number $\langle H_\alpha,\theta_1\rangle$ has no p in the denominator; hence $\langle H_\alpha,\theta_1\rangle\alpha \in X(T)$ if and only if $\langle H_\alpha,\theta_1\rangle \in \mathbb{Z}$; $s_\alpha(\theta) = \theta$ if and only if $\langle H_\alpha,\theta\rangle = 0$. It remains to check that the stabilizer of θ is generated by the reflections it contains.

Fix N such that $(1/p^N)X^{\mathrm{ad}} \supset X \cap (X^{\mathrm{ad}}\otimes\mathbb{Q})$ and such that $p^N \equiv 1$ (mod the order of θ). For any $w \in W$, $w\theta_1 - \theta_1$ is in $X^{\mathrm{ad}}\otimes\mathbb{Q}$, hence if $w\theta_1 - \theta_1$ is in X, $w(p^N\theta_1) - (p^N\theta_1) \in X^{\mathrm{ad}}$. Replacing θ_1 by $p^N\theta_1$, which is also a representative of θ, we may assume that θ is fixed by $w \in W$ if and only if $w\theta_1 + x = \theta_1$ for some $x \in X^{\mathrm{ad}}$. The stabilizer of θ in W is the image in W of the stabilizer of θ_1 in the affine Weyl group. It remains to apply Bourbaki [1, Ch. VI, Ex. 1 of § 2].

Remark 5.14. The proof shows that, if we use a suitable Borel subgroup $B \supset T$ to identify W with $\mathbf{W}$, the stabilizer of θ becomes the subgroup of $\mathbf{W}$ generated by some of the reflections corresponding either to a simple root or to the negative of the highest coroot.

DEFINITION 5.15. (i) The character θ of T^F is nonsingular if it is not

orthogonal to any coroot.

(ii) θ is in general position if it is not kept fixed by any non-trivial element of $(N(T)/T)^F$.

PROPOSITION 5.16. If the centre of G is connected, a character is non-singular if and only if it is in general position.

The stabilizer of θ in $N(T)/T$ is a group generated by reflections, and is stable by Frobenius. We have to prove that the subgroup fixed by Frobenius is non-trivial. This follows from the next lemma.

LEMMA 5.17. Let $V \neq \{0\}$ be a euclidian vector space, and let W be a finite group generated by orthogonal reflections. We assume that $V^W = \{0\}$. If a belongs to the normalizer A of W in $O(V)$, the centralizer W^a of a in W is non-trivial.

(a) Reduction to the irreducible case. Let $V = \bigoplus V_i$ be the decomposition of V as a direct sum of irreducible root systems; we have $W = \prod W_i$. The automorphism a permutes the V_i; by taking a direct factor, we may assume that it permutes them cyclically: $V = \bigoplus_{i \in \mathbb{Z}/n} V_i$ and $aV_i = V_{i+1}$. An element $w = (w_i) \in W = \prod W_i$ is in W^a if and only if $w_i = \mathrm{ad}\, a^i(w_0)$ and w_0 is fixed by a^n; we are reduced to proving that $W_0^{a^n}$ is non-trivial.

(b) In the irreducible case, one always has either $A = W \cup -W$ or $-1 \in W$; both cases are clear.

COROLLARY 5.18. For any G, if θ is in general position, then θ is non-singular.

Let us embed G in a group G_1 with connected centre and the same derived group. For instance, one can take $G_1 = G \times T/\{(z, z^{-1}) \mid z \in Z(G)\}$. The torus T is then contained in an F-stable maximal torus T_1 of G_1, and θ is the restriction to T^F of some character θ_1 of T_1^F. If θ is in general position, θ_1 is so a fortiori, hence non-singular (5.16). A character θ_1 of T_1^F is non-singular if and only if its restriction to T^F is, hence (5.18).

When all roots of G have the same length, geometric conjugacy can be given the following convenient description.

DEFINITION 5.19. (Assuming all roots to be of the same length) Let T be an F-stable maximal torus and let θ be a character of T^F. The connected centralizer $S(T, \theta)$ of (T, θ) in G is the F-stable reductive subgroup of G with maximal torus T and with root subgroups relative to T the U_α for which $\langle H_\alpha, \theta \rangle = 0$.

If $\langle H_\alpha, \theta \rangle = \langle H_\beta, \theta \rangle = 0$ and $\alpha + \beta$ is a root, then $H_{\alpha+\beta} = H_\alpha + H_\beta$,

hence $\langle H_{\alpha+\beta}, \theta\rangle = 0$ and the definition makes sense. Let $\pi : \widetilde{S(T,\theta)} \to S(T,\theta)$ be the simply connected covering of the derived subgroup of $S(T,\theta)$. By 5.11 (i), θ extends to a character θ_S of $S(T,\theta)^F/\pi(\widetilde{S(T,\theta)}^F)$.

PROPOSITION 5.20. If the centre of G is connected and all roots are of the same length, then a pair (T',θ') is geometrically conjugate to (T,θ) if and only if $(S(T,\theta),\theta_S)$ and $(S(T',\theta'),\theta'_{S'})$ are G^F-conjugate.

The "if" part follows from 5.11. Conversely, let (T',θ') be geometrically conjugate to (T,θ): for some $x \in G$, $\operatorname{ad} x(T) = T'$ and $\operatorname{ad} x : Y(T) \to Y(T')$ maps θ to θ' (via (5.2.3)). If $F' : T' \to T'$ is the Frobenius map of T', $\operatorname{ad} x^{-1}(F') = \operatorname{ad} wF$ for some w in the Weyl group of T. We have ${}^tF\theta = \theta$, and ${}^t(\operatorname{ad} wF)\theta = \theta$, hence θ is fixed by w, which by part (i) belongs to the Weyl group of T in $S(T,\theta)$. Let $T'' = \operatorname{ad} y(T)$, $y \in S(T,\theta)$, be an F-stable torus in $S(T,\theta)$, with Frobenius F'', such that $(\operatorname{ad} y)^{-1}(F'') = (\operatorname{ad} x)^{-1}(F')$. Replacing (T,θ) by $(T'', \operatorname{ad} y(\theta))$ (by applying 5.11) and replacing x by xy^{-1}, we are reduced to the case where $\operatorname{ad} xF = F' \operatorname{ad} x$. In this case, (T,θ) and (T',θ') are G^F-conjugate; the identity $\operatorname{ad} x(Ft) = F' \operatorname{ad} x(t)$ for $t \in T$ amounts to $x^{-1}Fx \in T$; and replacing x by xt^{-1} with $t \in T$, $t^{-1}Ft = x^{-1}Fx$, makes x rational.

When roots are not all of the same length, one has to work in the dual group G^* of G.

DEFINITION 5.21. A group G^* dual to G is a reductive group G^* defined over $\mathbb{F}_q$, whose maximal torus $\mathbf{T}^*$ is provided with an isomorphism with the dual of the maximal torus $\mathbf{T}$ of G, this isomorphism carrying simple roots to simple coroots.

Here are some properties of this duality. Let G and G^* be dual.

(5.21.1) G and G^* have the same Weyl group $\mathbf{W}$. The action of $\mathbf{W}$ on $Y(\mathbf{T}^*)$ is the contragredient of its action on $Y(\mathbf{T})$.

(5.21.2) However, Frobenius does not act in the same way on $\mathbf{W}$ in G and G^*: it acts in inverse ways (the origin of this is that F on $Y(\mathbf{T}^*)$ is the transpose of F on $Y(\mathbf{T})$).

(5.21.3) The conjugacy classes of pairs (T,B) (T an F-stable maximal torus, B a Borel subgroup containing it) in G and G^* correspond. With the notations of (1.13), (1.14), to the class of (T,B) we associate the class of (T^*,B^*) with

$$^t(\operatorname{ad} h(T,B)^{-1} \circ F) = \operatorname{ad} h(T^*,B^*)^{-1} \circ {}^tF;$$

the tori T and T^* are in duality.

(5.21.4) This induces a bijection between the rational conjugacy classes of maximal tori in G and G^*. If T and T' are in corresponding classes, we have a natural class (mod $(N(T)/T)^F$) of isomorphisms between T^* and T'.

(5.21.5) Let T be an F-stable torus on G, and let θ be a character of T^F. Fix T', a corresponding torus in G^*. The character θ defines an element of T^{*F}, then a $(N(T')/T')^F$-conjugacy class of elements θ' of T'. In that way, we get a bijection between G^F-conjugacy classes of pairs (T,θ) as above, and G^{*F}-conjugacy classes of pairs (T',θ'), T' an F-stable maximal torus of G^* and θ' an element of T'^F. This correspondence is compatible with extension of scalars from $\mathbb{F}_q$ to $\mathbb{F}_{q^n}$ (cf. 5.3.).

(5.21.6) By forgetting T', we see that each G^F-conjugacy class of pairs (T,θ) defines an element $\theta' \in G^{*F}$, well-defined up to G^{*F}-conjugacy.

PROPOSITION 5.22. Two pairs (T_1,θ_1) and (T_2,θ_2) are geometrically conjugate if and only if θ_1' and θ_2' are geometrically conjugate.

An extension of scalars (cf. 5.4, 5.5) reduces us to the case where T_1 and T_2 are split. Conjugating, we may assume further that $T_1 = T_2$. In this case, geometric conjugacy is W-conjugacy, and the proposition is clear.

PROPOSITION 5.23. Let G and G^* be dual. Then, the centre of G is connected and smooth (resp. connected) if and only if the derived group of G^* is simply connected (resp. if its simply connected covering is unseparable).

Put $Y = Y(T^*)$, T^* the maximal torus of G^*; let Y_1 be the subgroup generated by the coroots; and put $Y_2 = Y \cap Y_1 \otimes \mathbb{Q}$. Then, Y_2 is the Y-group of the maximal torus of the derived group, and Y_1 that of the maximal torus of its universal covering. The group of this covering is the dual of the Pontrjagin dual of Y_2/Y_1, and (5.23) follows by comparison with point (c) in the proof of (5.13).

COROLLARY 5.24. If the centre of G is connected, two pairs (T_1,θ_1), (T_2,θ_2) as in (5.22) are geometrically conjugate if and only if θ_1' and θ_2' are G^{*F}-conjugate.

Indeed, by a theorem of Steinberg, geometric conjugacy amounts in this case (5.23) to conjugacy, for semi-simple elements of G^* (their centralizers are connected).

DEFINITION 5.25. Let (T',θ') correspond to (T,θ) as in (5.21.5). The pair (T,θ) is maximally split if the $\mathbb{F}_q$-rank of T is equal to that of the centralizer of θ', i.e., if T' is maximally split in that centralizer.

Any geometric conjugacy class of pairs (T,θ) contains maximally split pairs.

PROPOSITION 5.26. If the centre of G is connected, two geometrically conjugate maximally split pairs (T_1,θ_1), (T_2,θ_2) are G^F-conjugate.

Via (5.21.5) and (5.24), this amounts to the known fact that two maximally split maximal tori in the centralizer $Z(\theta')$ of a semi-simple element θ' of G^{*F} are conjugate by an element of $Z(\theta)^F$.

PROPOSITION 5.27. If (T,θ) is maximally split, and the image of T in the adjoint group is anisotropic, then θ is non-singular.

Fix (T',θ') as in (5.21.5). By assumption, $Z^0(\theta')$ contains no non-central F-stable split torus, hence $Z^0(\theta') = T'$. The roots of T' in $Z^0(\theta')$ correspond to the coroots of T orthogonal to θ, hence the proposition.

COROLLARY 5.28. If (T,θ) is maximally split, T is containd in an F-stable Levi subgroup L of an F-stable parabolic subgroup P, and, in L, (T,θ) is non-singular.

One chooses P and L so that the image of T in the adjoint group of L is anisotropic. Being maximally split in G, (T,θ) is a fortiori maximally split in L, and one applies (5.27).

The following result will be used in the next chapter.

PROPOSITION 5.29. Let T' be a subtorus of a torus T, with T defined over $\mathbb{F}_q$ (no assumption on T'). Let θ be a character of T^F. It is trivial on $T' \cap T^F$ if and only if, when viewed as a character of $Y(T)$, it is trivial on

$$((F-1)Y(T') \otimes \mathbb{Q} \cap Y(T)).$$

The condition is that $\theta(x) = 0$ for

$$(F-1)^{-1}(x) \in (Y(T') \otimes \mathbb{Q} + Y(T)),$$

i.e., for

$$x \in ((F-1)Y(T') \otimes \mathbb{Q} \cap Y(T)) + ((F-1)Y(T)).$$

The character θ being trivial on $(F-1)Y(T)$, this means $\theta(x)$ is trivial for $x \in ((F-1)Y(T') \otimes \mathbb{Q} \cap Y(T))$.

6. Intertwining numbers

6.1. Let T, T' be two F-stable maximal tori in G and let θ, θ' be characters of T^F and T'^F. We put $N_G(T,T') = \{g \in G \mid Tg = gT'\}$ and

$$W_G(T,T') = T\backslash N_G(T,T') = N_G(T,T')/T'.$$

We will drop the index G if there is no ambiguity. F acts on $W(T,T')$, and

$$W(T,T')^F = T^F\backslash N(T,T')^F = N(T,T')^F/T'^F.$$

THEOREM 6.2. If θ^{-1} is not geometrically conjugate to θ', then

$$\left[H_c^*(\widetilde{X}_{T\subset B})_\theta \otimes H_c^*(\widetilde{X}_{T'\subset B'})_{\theta'}\right]^{G^F} = 0,$$

i.e., if an irreducible representation of G^F occurs in $H_c^*(\widetilde{X}_{T\subset B})_\theta$, its dual does not occur in $H_c^*(\widetilde{X}_{T'\subset B'})_{\theta'}$.

The virtual representation $R_T^{\theta^{-1}}$ is the dual of R_T^θ (as can be seen on the character formula 4.2), hence we have as a corollary:

COROLLARY 6.3. If θ and θ' are not geometrically conjugate, no irreducible representation of G^F can occur in both virtual representations R_T^θ and $R_{T'}^{\theta'}$.

The proof of 6.2 will make use of the following homotopy argument:

PROPOSITION 6.4. Let H be a connected algebraic group acting on a scheme Y, separated and of finite type over k. For any $h \in H$, the action of h on $H_c^*(Y, \mathbb{Z}/n)$ is trivial.

Let π be the projection of $H \times Y$ onto H and let $f : H \times Y \to H \times Y$ be defined by $f(h, y) = (h, hy)$:

$$\begin{array}{ccc} & H\times Y & \xrightarrow{f} & H\times Y \\ & \searrow_{\pi} & & \swarrow_{\pi} \\ [-1ex] & & H & \end{array}$$

By the change-of-basis theorem in cohomology with compact support applied to

$$\begin{array}{ccc} H\times Y & \longrightarrow & Y \\ \downarrow & & \downarrow \\ H & \longrightarrow & \operatorname{Spec} k, \end{array}$$

$R^i\pi_!\mathbb{Z}/n$ is the constant sheaf $H_c^i(Y, \mathbb{Z}/n)$ on H. The automorphism f acts on it and, at $h \in H$, it acts the way h acts on $H_c^i(Y, \mathbb{Z}/n)$. At the identity element of H, the action of f is trivial. An endomorphism of a constant sheaf over a connected base is constant, hence f acts trivially everywhere, and the proposition is proved.

COROLLARY 6.5. The conclusion of (6.4) holds in l-adic cohomology.

The proof is a passage to limit.

6.6. Proof of 6.2. On $\widetilde{X}_{T\subset B}\times\widetilde{X}_{T'\subset B'}$ we have commuting actions of G^F, T^F, and T'^F (G^F acts diagonally). By Künneth's Theorem, we have to prove that

$$\left[H_c^*(\widetilde{X}_{T\subset B}\times\widetilde{X}_{T'\subset B'})_{\theta,\theta'}\right]^{G^F} = 0,$$

where the symbol $M_{\theta,\theta'}$, applied to any $T^F \times T'^F$-module M, means the subspace

of M, where T^F and T'^F act by θ and θ'.

$$\begin{cases}\text{Let } U \text{ (resp. } U') \text{ be the unipotent radical of } B \text{ (resp. } B').\\ \text{Let } S_{T\subset B} = \{g \in G \mid g^{-1}Fg \in FU\}, \quad S_{T'\subset B'} = \{g' \in G \mid g'^{-1}Fg' \in FU'\}.\end{cases}$$

The unipotent group $(U \cap FU) \times (U' \cap FU')$ acts freely on $S_{T\subset B} \times S_{T'\subset B'}$ and the orbit space is $\widetilde{X}_{T\subset B} \times \widetilde{X}_{T'\subset B'}$ (see (1.17)). It follows that

$$H_c^i(\widetilde{X}_{T\subset B} \times \widetilde{X}_{T'\subset B'})(-d) \simeq H_c^{i+2d}(S_{T\subset B} \times S_{T'\subset B'}),$$

where d is the dimension of $(U \cap FU) \times (U' \cap FU')$. This isomorphism is compatible with the action of $G^F \times T^F \times T'^F$; moreover this group acts trivially on $\overline{\mathbb{Q}}_l(-d)$. Hence it is sufficient to prove that

$$H_c^*(S_{T\subset B} \times S_{T'\subset B'})_{\theta,\theta'}^{G^F} = H_c^*((S_{T\subset B} \times S_{T'\subset B'})/G^F)_{\theta,\theta'} = 0.$$

The map

$$(g, g') \longmapsto (x, x', y), \qquad x = g^{-1}Fg, \quad x' = g'^{-1}Fg', \quad y = g^{-1}g',$$

defines an isomorphism of $(S_{T\subset B} \times S_{T'\subset B'})/G^F$ with

$$\bar{S} = \{(x, x', y) \in FU \times FU' \times G \mid xF(y) = yx'\}.$$

Under this isomorphism, $T^F \times T'^F$ acts on $\bar{S}$ by the formula

$$(x, x', y) \longmapsto (t^{-1}xt, t'^{-1}x't', t^{-1}yt'), \qquad (t, t') \in T^F \times T'^F. \tag{6.6.1}$$

Let U'^- be opposed to U' with respect to T'. Any $g \in G$ can be written uniquely in the form $g = u_g n_g u'_g$, with with $u_g \in U \cap n_g U'^- n_g^{-1}$, $n_g \in N(T, T')$ (see (6.1)), $u'_g \in U'$; this follows easily from Bruhat's Lemma. For any $w \in W(T, T')$, let G_w be the set of all $g \in G$ such that n_g represents w. Then $(G_w)_{w\in W(T,T')}$ is a finite partition of G into locally closed subschemes. It has the property (6.6.2) below, expressing the fact that the closure of a Bruhat cell is a union of Bruhat cells.

(6.6.2.) For a suitable ordering of $W(T, T')$, the unions

$$\Sigma_w = \bigcup_{w'<w} G_{w'}$$

are closed.

Let $\bar{S}_w = \{(x, x', y) \in \bar{S} \mid y \in G_w\}$. Then $(\bar{S}_w)_{w\in W(T,T')}$ is a finite partition of $\bar{S}$ into locally closed subschemes, stable under $T^F \times T'^F$. It inherits a property analogous to (6.6.2), that is, the unions $\bigcup_{w'<w} \bar{S}_{w'}$ are closed for any w.

The spectral sequence associated to the filtration of $\bar{S}$ by these unions shows that, in order to prove that $H_c^*(\bar{S})_{\theta,\theta'} = 0$, it is sufficient to prove that $H_c^*(\bar{S}_w)_{\theta,\theta'} = 0$ for any $w \in W(T, T')$.

Let

$$H_w = \{(t, t') \in T \times T' \mid t'F(t')^{-1} = F(\dot{w})^{-1}tF(t)^{-1}F(\dot{w})\},$$

where $\dot{w} \in N(T, T')$ represents w. Then H_w is a closed subgroup of $T \times T'$, containing $T^F \times T'^F$. We define an action of H_w on $\bar{S}_w$ as follows. Let $(t, t') \in H_w$; for $(x, x', y) \in \bar{S}_w$ define

$$f_{t,t'}(x,x',y) = (\tilde{x},\tilde{x}',\tilde{y}),$$

where

$$\tilde{x} = t^{-1}xF(u_y)tF(t^{-1}u_y^{-1}t),$$
$$\tilde{x}' = t'^{-1}x'F(u'_y)^{-1}t'F(t'^{-1}u'_yt'),$$
$$\tilde{y} = t^{-1}yt'.$$

It is easy to check that $(\tilde{x},\tilde{x}',\tilde{y}) \in \bar{S}_w$, hence $f_{t,t'} : \bar{S}_w \longrightarrow \bar{S}_w$. It is also easy to check that

$$f_{t_1,t'_1} \circ f_{t_2,t'_2} = f_{t_1t_2,t'_1t'_2}$$

for $(t_i,t'_i) \in H_w$, $i = 1,2$, hence $f_{t,t'}$ defines an action of H_w on $\bar{S}_w$. It is clear that this action extends the action (6.6.1) of $T^F \times T'^F$ on $\bar{S}_w$. The theorem now follows from 6.5 and the

LEMMA 6.7. If the character $\theta\theta'$ of $T^F \times T'^F$ is trivial on $H_w^0 \cap (T^F \times T'^F)$, then $\theta'^{-1} = \theta \circ \mathrm{ad}(F(w))$ (as characters of $Y(T')$).

The subgroup H_w of $T \times T'$ is the kernel of the composite map

$$T \times T' \xrightarrow{x^{-1}Fx} T \times T' \xrightarrow{t^{-1}\,\mathrm{ad}\,F(w)(t')} T.$$

Applying the functor Y, we get that $Y(H_w^0)$ is the kernel K of

$$Y(T) \times Y(T') \xrightarrow{Fx-x} Y(T) \times Y(T') \xrightarrow{\mathrm{ad}\,F(w)(x')-x} Y(T).$$

By (5.29), the triviality of $\theta\theta'$ on $H_w^0 \cap (T^F \times T'^F)$ means that, when we view $\theta\theta'$ as a character of $Y(T) \times Y(T')$, it is trivial on

$$(F-1)(K \otimes \mathbb{Q}) \cap (Y(T) \times Y(T')).$$

Since the map $F-1$ is injective, this intersection is

$$\mathrm{Ker}(\mathrm{ad}\,F(w)(x') - x : Y(T) \times Y(T') \longrightarrow Y(T))$$

and the assumption becomes the triviality of $\theta\theta'$ on $\mathrm{Ker}(\mathrm{ad}\,F(w)(x') - x)$, i.e., the identity

$$\theta \circ \mathrm{ad}\,F(w)(\theta') = 1.$$

We will now conjugate (6.2) with the character formula (4.2) to get quantitative results.

THEOREM 6.8. Let $\theta \in (T^F)^\vee$, $\theta' \in (T'^F)^\vee$. Then

$$\langle R_T^\theta, R_{T'}^{\theta'} \rangle_{G^F} = \#\{w \in W(T,T')^F \mid \mathrm{ad}\,w(\theta') = \theta\}.$$

THEOREM 6.9.

$$\frac{1}{|G^F|} \sum_{\substack{u \in G^F \\ \text{unipotent}}} Q_{T,G}(u)Q_{T',G}(u) = \frac{|N(T,T')^F|}{|T^F|\,|T'^F|}. \tag{6.9.1}$$

We will prove (6.8) and (6.9) simultaneously, by induction on the dimension of G.

LEMMA 6.10. If (6.9) holds, for G replaced by $Z^0(s)$, where $s \in G^F$ is any semisimple element not contained in the centre Z of G, then

$$\langle R_T^\theta, R_{T'}^{\theta'} \rangle_{G^F} = \#\{w \in W(T,T')^F \mid \mathrm{ad}\,w(\theta') = \theta\} + \alpha \tag{6.10.1}$$

where

$$\alpha = \sum_{s\in T^F\cap Z}\theta(s)\theta'(s^{-1})\left(\frac{1}{|G^F|}\sum_{\substack{u\in G^F\\ \text{unipotent}}} Q_{T,G}(u)Q_{T',G}(u) - \frac{|N(T,T')^F|}{|T^F|\,|T'^F|}\right).$$

In particular, (6.9) implies (6.8).

According to (4.2) we have:

$$\begin{aligned}\langle R_T^\theta, R_{T'}^{\theta'}\rangle &= \frac{1}{|G^F|}\sum_{g_1\in G^F}\operatorname{Tr}(g_1, R_T^\theta)\operatorname{Tr}(g_1^{-1}, R_{T'}^{\theta'})\\ &= \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\frac{1}{|Z^0(s)^F|^2}\sum_{\substack{g,g'\in G^F\\ g^{-1}sg\in T^F\\ g'^{-1}sg'\in T'^F}}\theta(g^{-1}sg)\theta'(g'^{-1}sg')^{-1}\\ &\qquad\times\sum_{\substack{u\in Z^0(s)^F\\ \text{unipotent}}} Q_{gTg^{-1},Z^0(s)}(u)Q_{g'T'g'^{-1},Z^0(s)}(u).\end{aligned}$$

With our assumption, this can be written as

$$\alpha + \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\frac{1}{|Z^0(s)^F|}\sum_{\substack{g,g'\in G^F\\ g^{-1}sg\in T^F\\ g'^{-1}sg'\in T'^F}}\theta(g^{-1}sg)\theta'(g'^{-1}sg')^{-1}\times\frac{|N_{Z^0(s)}(gTg^{-1}, g'T'g'^{-1})^F|}{|T^F|\,|T'^F|}.$$

The formula $(g,g',n_1)\longmapsto(g,n,n_1)$, $n=g'^{-1}n_1g$, establishes a one-to-one correspondence between the sets

$$\{(g,g',n_1)\in G^F\times G^F\times G^F \mid g^{-1}sg\in T^F,\ g'^{-1}sg'\in T'^F,\ n_1\in N_{Z^0(s)}(gTg^{-1}, g'T'g'^{-1})^F\},$$

$$\{(g,n,n_1)\in G^F\times N_G(T,T')^F\times Z^0(s)^F \mid g^{-1}sg\in T^F\}.$$

It follows that

$$\begin{aligned}\langle R_T^\theta, R_{T'}^{\theta'}\rangle &= \alpha + \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\frac{1}{|Z^0(s)^F|}\sum_{\substack{g\in G^F\\ n\in N_G(T,T')^F\\ g^{-1}sg\in T^F}}\theta(g^{-1}sg)\theta'(ng^{-1}sgn^{-1})^{-1}\frac{|Z^0(s)^F|}{|T^F|\,|T'^F|}\\ &= \alpha + \frac{1}{|G^F|}\sum_{\substack{t\in T^F\\ n\in N_G(T,T')^F}}\theta(t)\theta'(ntn^{-1})^{-1}\frac{|G^F|}{|T^F|\,|T'^F|}\\ &= \alpha + \frac{1}{|T'^F|}\sum_{n\in N_G(T,T')^F}\frac{1}{|T^F|}\sum_{t\in T^F}\theta(t)\theta'(ntn^{-1})^{-1}\\ &= \alpha + \#\{w\in W(T,T')^F \mid \operatorname{ad} w(\theta')=\theta\},\end{aligned}$$

and (6.10.1) is proved.

Proof of 6.9. Let $\bar G = G/Z$; the centre of $\bar G$ consists of the identity element. Both sides of (6.9.1) remain unchanged when G is replaced by $\bar G$; for

the left hand side this follows from (4.1.1), while for the right hand side this is easy to check. Hence in order to prove (6.9), we may assume that the centre of G has a single element; moreover we may assume by induction that the conclusion of (6.9) is true for G replaced by $Z^0(s)$, where $s \in G^F$ is an arbitrary non-central semisimple element, and for any two F-stable maximal tori in $Z^0(s)$. (To start the induction we may take G to be the group with only one element in which case (6.9) is clear.) It follows from (6.3) that for any non-trivial character θ of T^F one has

$$\langle R_T^\theta, R_{T'}^1 \rangle = 0.$$

Substituting this information into (6.10), we see that if T^F has any non-trivial character, then $\alpha = 0$ which proves (6.9). A similar argument applies when $|T'^F| \neq 1$. It remains to check the special case where $|T^F| = |T'^F| = 1$. In this case, we must have $q = 2$, and T, T' must be $\mathbb{F}_q$-split tori. In particular, T and T' are conjugate under G^F, hence

$$\#\{w \in W(T,T')^F \mid \operatorname{ad} w(\theta') = \theta\} = |W(T)^F| = |W(T)|.$$

Moreover R_T^1 and $R_{T'}^1$ are both equal to the representation of G^F induced by the unit representation of B^F, where B is an F-stable Borel subgroup of G. It follows that

$$\langle R_T^1, R_{T'}^1 \rangle = |B^F \backslash G^F / B^F| = |W(T)|. \tag{1.10}$$

Substituting in (6.10) we get (6.9) in this case. This completes the proof of (6.9) (and of (6.8) by (6.10)).

7. Computations on semisimple elements

The following result describes the Euler characteristic of the scheme $X_{T\subset B}$. Let $\sigma(G)$ (resp. $\sigma(T)$) be the $\mathbb{F}_q$-rank of G (resp. T).

THEOREM 7.1.

$$\chi(X_{T\subset B}) = Q_{T,G}(e) = (-1)^{\sigma(G)-\sigma(T)} \frac{|G^F|}{\mathrm{St}_G(e)|T^F|}.$$

(Here St_G is the Steinberg representation of G^F, identified with its character.)

We may assume that the centre of G is reduced to a single element and that the statement is true when G is replaced by $Z^0(s)$, where $s \in G^F$ is any non-central semisimple element.

We first assume that T^F has a non-trivial character θ. The Steinberg representation occurs in

$$R_{T^*}^1 = \mathrm{Ind}_{B^{*F}}^{G^F}(1)$$

($T^* \subset B^*$ as in 1.8), hence, by (6.3), it does not occur in R_T^θ: $\langle R_T^\theta, \mathrm{St}_G \rangle = 0$. It is known that

$$\mathrm{St}_G(g) = \begin{cases} (-1)^{\sigma(G)-\sigma(Z^0(g))}\,\mathrm{St}_{Z^0(g)}(e), & g \in G^F \text{ semisimple}, \\ 0, & g \in G^F \text{ non-semisimple}. \end{cases}$$

If we use 4.2, it follows:

$$\sum_{s\in G^F}(-1)^{\sigma(G)-\sigma(Z^0(s))}\frac{\mathrm{St}_{Z^0(s)}(e)}{|Z^0(s)^F|}\sum_{\substack{g\in G^F\\ g^{-1}sg\in T}}Q_{gTg^{-1},Z^0(s)}(e)\theta(g^{-1}sg)=0$$

(sum over the semisimple elements s of G^F).

By our assumption, we may substitute

$$Q_{gTg^{-1},Z^0(s)}(e)=(-1)^{\sigma(Z^0(s))-\sigma(T)}\frac{|Z^0(s)^F|}{\mathrm{St}_{Z^0(s)}(e)|T^F|},$$

for all $s\neq e$:

$$\frac{(-1)^{\sigma(G)-\sigma(T)}}{|T^F|}\sum_{\substack{s\in G^F\\ s\neq e}}\sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}}\theta(g^{-1}sg)+\mathrm{St}_G(e)Q_{T,G}(e)=0,$$

$$(-1)^{\sigma(G)-\sigma(T)}\frac{|G^F|}{|T^F|}\sum_{\substack{t\in T^F\\ t\neq e}}\theta(t)+\mathrm{St}_G(e)Q_{T,G}(e)=0.$$

Since the character θ in non-trivial, $\sum_{t\in T^F}\theta(t)=0$ and the desired formula for $Q_{T,G}(e)$ follows.

In the case where $|T^F|=1$, T must be an $\mathbb{F}_q$-split torus and $q=2$. In this case, $Q_{T,G}(e)=|G^F|/|B^{*F}|$, where B^* is any F-stable Borel subgroup (1.8). This agrees with the statement of the theorem and ends the proof.

COROLLARY 7.2. For any semisimple element $s\in G^F$,

$$\mathrm{Tr}(s,R_T^\theta)=(-1)^{\sigma(Z^0(s))-\sigma(T)}\frac{1}{\mathrm{St}_{Z^0(s)}(e)|T^F|}\sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}}\theta(g^{-1}sg).$$

This follows from (7.1) and 4.2. An equivalent statement is:

PROPOSITION 7.3.

$$(-1)^{\sigma(G)-\sigma(T)}R_T^\theta\otimes\mathrm{St}_G=\mathrm{Ind}_{T^F}^{G^F}(\theta).$$

PROPOSITION 7.4. If θ is a non-singular character of T^F (5.19) then $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ can be represented by an actual G^F-module. If θ is in general position (5.15), $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ is irreducible.

By embedding G in a group G_1 with connected centre and the same derived group, as in (5.18), we reduce to the case where θ is in general position. It remains to use (6.8) and (7.1).

PROPOSITION 7.5. Let $s\in G^F$ be a semisimple element. The character of

$$\frac{1}{\mathrm{St}_G(s)}\sum_{\substack{T\\ s\in T}}\sum_{\theta\in(T^F)^\vee}\theta(s)^{-1}(-1)^{\sigma(G)-\sigma(T)}R_T^\theta \tag{7.5.1}$$

equals $|Z(s)^F|$ on elements conjugate to s in G^F and zero on all other elements of G^F.

Let μ be the character of (7.5.1) and let μ' be the class function on G^F defined by

$$\mu'(g) = \begin{cases} |Z(s)^F|, & g \text{ conjugate to } s \text{ in } G^F, \\ 0, & \text{otherwise.} \end{cases}$$

In order to prove that $\mu = \mu'$ it is sufficient to prove that

$$\langle \mu, \mu \rangle = \langle \mu, \mu' \rangle = \langle \mu', \mu' \rangle \qquad \text{(see (0.3)).}$$

We have $\langle \mu', \mu' \rangle = |Z(s)^F|$; from (6.8) we see that

$$\begin{aligned} \langle \mu, \mu \rangle &= \frac{1}{\mathrm{St}_G(s)^2} \sum_{\substack{T \ni s \\ T' \ni s}} \sum_{\substack{\theta \in (T^F)^\vee \\ \theta' \in (T'^F)^\vee}} \theta(s^{-1})\theta'(s) \#\{n \in N(T,T')^F \mid \mathrm{ad}\, n(\theta') = \theta\} |T^F|^{-1} \\ &= \frac{1}{\mathrm{St}_G(s)^2} \sum_{\substack{T \ni s \\ \theta \in (T^F)^\vee \\ n \in G^F \\ nsn^{-1} \in T}} \theta(s^{-1})\theta(nsn^{-1}) |T^F|^{-1} \\ &= \frac{1}{\mathrm{St}_G(s)^2} \#\{T \ni s,\ n \in G^F \mid sn = ns\} \\ &= \frac{|Z(s)^F|}{\mathrm{St}_G(s)^2} \#\{F\text{-stable maximal tori in } Z^0(s)\} = |Z(s)^F| \end{aligned}$$

(by [15, Cor. 14.16]).

By (7.2), we have

$$\begin{aligned} \langle \mu, \mu' \rangle = \mu(s) &= \frac{1}{\mathrm{St}_G(s)} \sum_{\substack{T \\ s \in T}} \sum_{\theta \in (T^F)^\vee} \theta(s^{-1}) \frac{(-1)^{\sigma(G)-\sigma(Z^0(s))}}{\mathrm{St}_{Z^0(s)}(e) |T^F|} \sum_{\substack{g \in G^F \\ g^{-1}sg \in T^F}} \theta(g^{-1}sg) \\ &= \frac{1}{\mathrm{St}_G(s)^2} \sum_{\substack{T \\ s \in T}} \sum_{\substack{g \in G^F \\ g^{-1}sg \in T^F}} |T^F|^{-1} \theta(s^{-1})\theta(g^{-1}sg) \\ &= \frac{1}{\mathrm{St}_G(s)^2} \#\{T \ni s,\ g \in G^F \mid sg = gs\} \\ &= \frac{|Z(s)^F|}{\mathrm{St}_G(s)^2} \#\{F\text{-stable maximal tori in } Z^0(s)\} = |Z(s)^F|, \end{aligned}$$

and the proposition is proved.

COROLLARY 7.6. For any $\rho \in \mathscr{R}(G^F)$ and any semisimple element $s \in G^F$,

$$\mathrm{Tr}(s, \rho) = \frac{1}{\mathrm{St}_G(s)} \sum_{\substack{T \\ s \in T}} \sum_{\theta \in (T^F)^\vee} \theta(s)(-1)^{\sigma(G)-\sigma(T)} \langle \rho, R_T^\theta \rangle. \tag{7.6.1}$$

In particular, if $s \in G^F$ and is regular semisimple, contained in a unique torus T,

$$\mathrm{Tr}(s, \rho) = \sum_{\theta \in (T^F)^\vee} \theta(s) \langle \rho, R_T^\theta \rangle, \tag{7.6.2}$$

$$\dim \rho = \frac{1}{\mathrm{St}_G(e)} \sum_{T} \sum_{\theta \in (T^F)^\vee} (-1)^{\sigma(G)-\sigma(T)} \langle \rho, R_T^\theta \rangle. \tag{7.6.3}$$

COROLLARY 7.7. For any irreducible representation ρ of G^F there exists an F-stable maximal torus T and a character θ of T^F such that $\langle \rho, R_T^\theta \rangle \neq 0$.

This follows from (7.6.3).

DEFINITION 7.8. An irreducible representation ρ of G^F is said to be unipotent if $\langle \rho, R_T^1 \rangle \neq 0$ for some F-stable maximal torus T.

By (6.3) and (7.7), ρ is unipotent if and only if $\langle \rho, R_T^\theta \rangle = 0$ for any F-stable maximal torus T and any $\theta \in (T^F)^\vee$, $\theta \neq 1$. For example, any irreducible component of $\mathrm{Ind}_{B^{*F}}^{G^F}(1)$ (B^{*F} an F-stable Borel subgroup) is a unipotent representation. For unipotent representations, (7.6) becomes:

PROPOSITION 7.9. Let ρ be a unipotent representation of G^F and let $s \in G^F$ be a semisimple element. Then

$$\mathrm{Tr}(s, \rho) = \frac{1}{\mathrm{St}_G(s)} \sum_{\substack{T \\ s \in T}} (-1)^{\sigma(G)-\sigma(T)} \langle \rho, R_T^1 \rangle.$$

In particular, if $s \in G^F$ is regular semisimple contained in a unique maximal torus T, $\mathrm{Tr}(s, \rho) = \langle \rho, R_T^1 \rangle$ and

$$\dim \rho = \frac{1}{\mathrm{St}_G(e)} \sum_{T} (-1)^{\sigma(G)-\sigma(T)} \langle \rho, R_T^1 \rangle.$$

It follows that if B is an F-stable Borel subgroup of G and T is an F-stable maximal torus in B, we have $\mathrm{Tr}(s, \rho) = \langle \rho, \mathrm{Ind}_{B^F}^{G^F}(1) \rangle$; in the case where $\langle \rho, \mathrm{Ind}_{B^F}^{G^F}(1) \rangle \neq 0$, this is a result of Curtis, Kilmoyer and Seitz (see C. W. Curtis, On the Values of Certain Irreducible Characters of Finite Chevalley Groups, Ist. Naz. di Alta Mat. Symp. Mat. XIII, 1974, 343–355).

PROPOSITION 7.10. Let G^{ad} be the adjoint group of G. Then, the restriction to G^F of a unipotent representation of $(G^{\mathrm{ad}})^F$ is irreducible; non-isomorphic unipotent representations have non-isomorphic restrictions; and every unipotent representation of G is such a restriction.

It suffices to check that, if σ, τ are unipotent representations of $(G^{\mathrm{ad}})^F$, then $\langle \sigma, \tau \rangle_{G^{\mathrm{ad}}F} = \langle \sigma, \tau \rangle_{G^F}$. Let G_1 be the image of G^F in $(G^{\mathrm{ad}})^F$. One has

$$\begin{aligned} \langle \sigma, \tau \rangle_{G^F} = \langle \sigma, \tau \rangle_{G_1} &= \left\langle \mathrm{Ind}_{G_1}^{G^{\mathrm{ad}}F}(\mathrm{Res}\, \sigma), \tau \right\rangle_{G^{\mathrm{ad}}F} \\ &= \sum_{\theta \in (G^{\mathrm{ad}}F/G_1^F)^\vee} \langle \sigma \otimes \theta, \tau \rangle_{G^{\mathrm{ad}}F}. \end{aligned}$$

By the exclusion theorem 6.3 and (1.23), (1.27), $\langle \sigma \otimes \theta, \tau \rangle_{G^{\mathrm{ad}}F} = 0$ for $\theta \neq 1$, hence the proposition.

PROPOSITION 7.11. Let ρ be a virtual representation of G^F such that $\mathrm{Tr}(su, \rho) = \mathrm{Tr}(s, \rho)$ for any $s \in G^F$ semisimple, $u \in G^F$ unipotent, $su = us$. Let T be an F-stable maximal torus and let θ be a character of T. Then

$$\langle \rho, R_T^\theta \rangle_{G^F} = (-1)^{\sigma(G)-\sigma(T)} \langle \rho \otimes \mathrm{St}_G, R_T^\theta \rangle_{G^F} = \langle \rho, \theta \rangle_{T^F}.$$

The Steinberg character is integral valued, hence by (7.3),

$$(-1)^{\sigma(G)-\sigma(T)}\langle \rho \otimes \mathrm{St}_G, R_T^\theta\rangle_{G^F} = \langle \rho, (-1)^{\sigma(G)-\sigma(T)}\,\mathrm{St}_G \otimes R_T^\theta\rangle_{G^F} = \langle \rho, \mathrm{Ind}_{T^F}^{G^F}(\theta)\rangle_{G^F} = \langle \rho, \theta\rangle_{T^F}.$$

We now prove the identity

$$\langle \rho, R_T^\theta\rangle_{G^F} = \langle \rho, \theta\rangle_{T^F}. \tag{7.11.1}$$

The special case of this identity

$$\langle 1, R_T^1\rangle_{G^F} = 1 \tag{7.11.2}$$

follows from the fact that the Euler characteristic of $X_{T\subset B}/G^F$ (which is the same as the Euler characteristic of

$$\{g \in G \mid g^{-1}Fg \in FU\}/G^F \times T^F = FU/T^F$$

by (1.17)) equals 1. If we use (4.2), (7.11.2) can be written in the form

$$\frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}} \frac{1}{|Z^0(s)^F|} \sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}} \sum_{\substack{u\in Z^0(s)^F\\ \text{unipotent}}} Q_{gTg^{-1},Z^0(s)}(u) = 1. \tag{7.11.3}$$

This implies, by induction on the dimension of G, that

$$\frac{1}{|G^F|}\sum_{\substack{u\in G^F\\ \text{unipotent}}} Q_{T,G}(u) = \frac{1}{|T^F|}. \tag{7.11.4}$$

Indeed, by substituting (7.11.4) in (7.11.3) we find a true identity:

$$\frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}} \frac{1}{|T^F|}\#\{g \in G^F \mid g^{-1}sg \in T^F\} = 1.$$

We apply (4.2) again and use (7.11.4) to compute:

$$\begin{aligned}
\langle \rho, R_T^\theta\rangle_{G^F} &= \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}} \frac{1}{|Z^0(s)^F|} \sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}} \sum_{\substack{u\in Z^0(s)^F\\ \text{unipotent}}} Q_{gTg^{-1},Z^0(s)}(u)\theta(g^{-1}sg)^{-1}\,\mathrm{Tr}(s,\rho)\\
&= \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}} \sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}} \frac{1}{|T^F|}\theta(g^{-1}sg)^{-1}\,\mathrm{Tr}(s,\rho)\\
&= \frac{1}{|G^F|}\sum_{t\in T^F} \theta(t)^{-1}\,\mathrm{Tr}(t,\rho) = \langle \rho, \theta\rangle_{T^F}.
\end{aligned}$$

PROPOSITION 7.12. Let ρ be as in (7.11). Then

$$\rho = \sum_{(T)} \frac{1}{|W(T)^F|} \sum_{\theta\in(T^F)^\vee} \langle \rho, \theta\rangle_{T^F} R_T^\theta, \tag{7.12.1}$$

$$\rho \otimes \mathrm{St}_G = \sum_{(T)} \frac{(-1)^{\sigma(G)-\sigma(T)}}{|W(T)^F|} \sum_{\theta\in(T^F)^\vee} \langle \rho, \theta\rangle_{T^F} R_T^\theta, \tag{7.12.2}$$

where $\sum_{(T)}$ means sum over all G^F-conjugacy classes of F-stable maximal tori T.

The character of $\rho \otimes \mathrm{St}_G$ is a linear combination of the form $\sum_{(T,\theta)} c_{T,\theta} R_T^\theta$ (sum over all G^F-conjugacy classes of pairs (T,θ)). The coefficients $c_{T,\theta}$ are

determined by (7.11) and (7.12.2) follows.

Now let ρ' (resp. ρ'') be the right hand side of (7.12.1) (resp. (7.12.2)). We have clearly

$$\langle \rho', \rho' \rangle = \langle \rho'', \rho'' \rangle \tag{7.12.3}$$

and by (7.11),

$$\langle \rho, \rho' \rangle = \langle \rho \otimes \mathrm{St}_G, \rho'' \rangle. \tag{7.12.4}$$

We note also that

$$\langle \rho, \rho \rangle = \langle \rho \otimes \mathrm{St}_G, \rho \otimes \mathrm{St}_G \rangle. \tag{7.12.5}$$

This follows from the fact that the number of unipotents in $Z^0(s)^F$ equals $\mathrm{St}_G(s)^2$ for any semisimple element $s \in G^F$ (see [15, Thm. 15.1]). By (7.12.2) we have

$$\langle \rho \otimes \mathrm{St}_G - \rho'', \rho \otimes \mathrm{St}_G - \rho'' \rangle = 0.$$

This, together with (7.12.3), (7.12.4) and (7.12.5) implies:

$$\langle \rho - \rho', \rho - \rho' \rangle = 0,$$

hence

$$\rho = \rho'.$$

Remark 7.13. It is known that

$$\langle \rho, \rho \rangle_{G^F} = \sum_{(T)} \frac{1}{|W(T)^F|} \langle \rho, \rho \rangle_{T^F}$$

(N. Kawanaka, A theorem on finite Chevalley groups, Osaka J. Math. 10 (1973), 1–13); this could be also deduced from (7.12.1) and (6.8).

COROLLARY 7.14.

$$1 = \sum_{(T)} \frac{1}{|W(T)^F|} R_T^1. \tag{7.14.1}$$

$$\mathrm{St}_G = \sum_{(T)} \frac{(-1)^{\sigma(G)-\sigma(T)}}{|W(T)^F|} R_T^1. \tag{7.14.2}$$

In particular,

$$\langle \mathrm{St}_G, R_T^1 \rangle = (-1)^{\sigma(G)-\sigma(T)}.$$

Here is an alternative proof of (7.14.1). In the language of (1.4), (7.14.1) asserts that

$$\sum_{w \in W} R^1(w) = |W| \, 1.$$

Since $(X(w))_{w \in W}$ is a partition of the flag manifold into locally closed subschemes, we have

$$\sum_{w\in W} R^1(w) = \sum_i (-1)^i H_c^i(X_G)$$

as virtual representations of G^F. The action of G^F on X_G is the restriction of the action of the connected group G; by (6.5), G^F acts trivially on $H_c^*(X_G)$ and it remains to use the fact that the Euler characteristic of X_G equals $|W|$.

COROLLARY 7.15.

$$\mathrm{St}_G = \sum_{(T)} \frac{(-1)^{\sigma(G)-\sigma(T)}}{|W(T)^F|} \mathrm{Ind}_{T^F}^{G^F}(1), \tag{7.15.1}$$

$$\mathrm{St}_G \otimes \mathrm{St}_G = \sum_{(T)} \frac{1}{|W(T)^F|} \mathrm{Ind}_{T^F}^{G^F}(1). \tag{7.15.2}$$

This follows from (7.1) by tensoring with St_G, and using (7.3). The formula (7.15.1) is due to B. Srinivasan (On the Steinberg character of a finite simple group of Lie type, J. Australian Math. Soc. 12 (1971), 1–14).

8. Induced and cuspidal representations

8.1. Let P be an F-stable parabolic subgroup of G and let $T \subset P$ be an F-stable maximal torus. We denote by U_P the unipotent radical of P. The quotient group P/U_P is a connected reductive algebraic group acted on by F. Let $\pi : P \to P/U_P$ be the canonical projection. π induces an isomorphism $T \xrightarrow{\sim} \pi(T)$ hence also an isomorphism $T^F \xrightarrow{\sim} \pi(T)^F$. Let θ be a character of T^F and let $\bar\theta$ be the corresponding character of $\pi(T)^F$. We denote by $R_{T,P}^{\theta}$ the image of the virtual representation $R_{\pi(T)}^{\bar\theta}$ of $(P/U_P)^F$ under the canonical embedding $\mathscr{R}((P/U_P)^F) \subset \mathscr{R}(P^F)$. With these notations, we have the following generalisation of 1.10:

PROPOSITION 8.2.

$$R_T^{\theta} = \mathrm{Ind}_{P^F}^{G^F}(R_{T,P}^{\theta}).$$

We choose a Borel subgroup B in G such that $T \subset B \subset P$. Let $\mathscr{P}$ be the set of all parabolic subgroups P' in G such that P' and P are conjugate under G^F; this is a finite set. We have

$$\widetilde{X}_{T\subset B} = \coprod_{P'\in\mathscr{P}} \widetilde{X}(P')$$

where

$$\widetilde{X}(P') = \{g \in G \mid g^{-1}Fg \in FU,\ gPg^{-1} = P'\}/U \cap FU,$$

with U the unipotent radical of B.

(Note that $g^{-1}F(g) \in FU$ implies $F(gPg^{-1}) = gPg^{-1}$.) Let $P' \in \mathscr{P}$ and let $g_1 \in G^F$ be such that $g_1 P g_1^{-1} = P'$. If $g \in G$, $g^{-1}F(g) \in FU$, $gPg^{-1} = P'$, we have

$$g_1^{-1}g \in P, \qquad (g_1^{-1}g)^{-1}F(g_1^{-1}g) \in FU;$$

by sending g to $\pi(g_1^{-1}g)$ we get an isomorphism

$$\widetilde{X}(P') \xrightarrow{\sim} \widetilde{X}_{\pi(T)\subset\pi(B)}.$$

It follows easily that the G^F-module $H_c^i(\widetilde{X}_{T\subset B})$ is canonically isomorphic to the G^F-module induced by $H_c^i(\widetilde{X}_{\pi(T)\subset\pi(B)})$ regarded as a P^F-module; moreover this isomorphism is compatible with the action of T^F. The proposition follows.

THEOREM 8.3. Let T be an F-stable maximal torus in G such that T is not contained in any F-stable, proper parabolic subgroup of G. Let θ be a non-singular character of T^F (cf. 5.19). Then $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ can be represented by a cuspidal G^F-module.

Applying (7.5) for s equal to the identity element of G^F, we see that the regular representation of G^F is equal to:

$$\mathrm{Ind}_e^{G^F}(1) = \frac{1}{\mathrm{St}_G(e)} \sum_T \sum_{\theta\in(T^F)^\vee} (-1)^{\sigma(G)-\sigma(T)} R_T^\theta.$$

We apply this formula with G replaced by P/U_P (P as in (8.1)); we then regard the regular representation of $(P/U_P)^F$ as a representation of P^F on which U_P^F acts trivially (i.e., as $\mathrm{Ind}_{U_P^F}^{P^F}(1)$) and we induce it to G^F:

$$\begin{aligned}\mathrm{Ind}_{U_P^F}^{G^F}(1) &= \mathrm{Ind}_{P^F}^{G^F}\big(\mathrm{Ind}_{U_P^F}^{P^F}(1)\big)\\ &= \frac{1}{\mathrm{St}_{P/U_P}(e)} \sum_{T'\subset P/U_P} \sum_{\theta'\in(T'^F)^\vee} (-1)^{\sigma(P/U_P)-\sigma(T')} \mathrm{Ind}_{P^F}^{G^F}(R_{T'}^{\theta'}),\end{aligned}$$

where $R_{T'}^{\theta'}$ is regarded as an element in $\mathscr{R}(P^F)$ in the natural way. We now use (8.2) and the fact that any F-stable maximal torus T' in P/U_P can be lifted to an F-stable maximal torus T in P in precisely $|U_P^F|$ different ways:

$$\mathrm{Ind}_{U_P^F}^{G^F}(1) = \frac{1}{\mathrm{St}_G(e)} \sum_{T\subset P} \sum_{\theta\in(T^F)^\vee} (-1)^{\sigma(G)-\sigma(T)} R_T^\theta. \tag{8.3.1}$$

If $P \neq G$ and T is as in the statement of the theorem, we see from (8.3.1) and (6.8) that

$$\Big\langle (-1)^{\sigma(G)-\sigma(T)}R_T^\theta, \mathrm{Ind}_{U_P^F}^{G^F}(1)\Big\rangle = 0 \qquad \text{for any } \theta \in (T^F)^\vee. \tag{8.3.2}$$

If θ is non-singular, $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ can be represented by an actual G^F-module (7.4). This G^F-module is cuspidal by (8.3.2).

9. A Vanishing Theorem

9.1. Let G be a reductive algebraic group over k. Let $w = s_1\cdots s_n$ be a minimal expression for an element of the Weyl group W of G (0.4). The following desingularization of the closure $\overline{O(w)}$ of $O(w) \subset X \times X$ (1.2) has been considered by H. C. Hansen [6] and M. Demazure [3].

DEFINITION 9.2. $\overline{O}(s_1,\dots,s_n)$ is the space of sequences $(B_0,\dots,B_n)$ of

Borel subgroups, with B_{i-1} and B_i in relative position e or s_i.

The maps

$$\overline{O}(s_1,\dots,s_n) \longrightarrow \overline{O}(s_1,\dots,s_{n-1}) \longrightarrow \cdots \longrightarrow \overline{O}(\,) = X$$

express $\overline{O} = \overline{O}(s_1,\dots,s_n)$ as an iterated fibre space over X, with fibre $\mathbb{P}^1$ (1.2). The subspace $D_i \subset \overline{O}$ where $B_{i-1} = B_i$ is the inverse image of a section of $\overline{O}(s_1,\dots,s_i) \to \overline{O}(s_1,\dots,s_{i-1})$; it is a smooth divisor, and $D = \bigcup D_i$ is a divisor with normal crossings. The map $(B_0,\dots,B_n) \mapsto (B_0,B_n)$ induces an isomorphism $\overline{O} - D \xrightarrow{\sim} O(w)$, and hence maps $\overline{O}$ to $\overline{O(w)}$: $\overline{O}$ is a resolution of singularities for $\overline{O(w)}$.

9.3. Let T^*, B^* and U^* be as in (1.7). For any character $\lambda : T^* \to \mathbb{G}_m$, the T^*-torsor E over X defined in (1.7) gives rise to a line bundle E_λ, provided with $\lambda : E \to E_\lambda$ such that $\lambda(et) = \lambda(e)\lambda(t)$. For $\dot{w} \in N(T^*)$ with image w in $W = N(T^*)/T^*$, the w-map of T^*-torsors over $O(w) \subset X \times X$ constructed in (1.7) induces an isomorphism

$$\Psi(\dot{w}) : \mathrm{pr}_1^* E_\lambda \xrightarrow{\sim} \mathrm{pr}_2^* E_{\lambda \circ \mathrm{ad}\,\dot{w}} = \mathrm{pr}_2^* E_{w^{-1}(\lambda)}$$

which makes the following diagram commute:

$$\begin{array}{ccc} \mathrm{pr}_1^* E & \xrightarrow{\cdot\dot{w}} & \mathrm{pr}_2^* E \\ \lambda \downarrow & & \downarrow w^{-1}(\lambda) \\ \mathrm{pr}_1^* E_\lambda & \xrightarrow{\Psi(\dot{w})} & \mathrm{pr}_2^* E_{w^{-1}(\lambda)}. \end{array}$$

We will investigate its behaviour at infinity.

9.4. Let $X(T^*)$ be the character group of T^* and let $C \subset X(T^*) \otimes \mathbb{R}$ be the fundamental chamber (corresponding to B^*). If C_1 and C_2 are two chambers, we write $D(C_1,C_2)$ for the intersection of the (closed) radicial half spaces containing both C_1 and C_2, and $D^0(C_1,C_2)$ for its interior.

PROPOSITION 9.5. Let $\overline{O(w)}'$ be the normalization of the closure $\overline{O(w)}$ of $O(w)$ in $X \times X$. The map $\Psi(\dot{w}) : \mathrm{pr}_1^* E_\lambda \to \mathrm{pr}_2^* E_{w^{-1}(\lambda)}$ extends over $\overline{O(w)}'$ if and only if $\lambda \in D(C,-wC)$. It vanishes outside of $O(w)$ if and only if $\lambda \in D^0(C,-wC)$.

Let us consider $\Psi(\dot{w})$ as a rational section of $\mathrm{pr}_1^* E_{\lambda^{-1}} \otimes \mathrm{pr}_2^* E_{w^{-1}(\lambda)}$; it suffices to prove that, with the notations of (9.2), for each i, the order v_i of $\Psi(\dot{w})$ along the divisor $D_i \subset \overline{O}(s_1,\dots,s_n)$ is ≥ 0 for $\lambda \in D(C,-wC)$, > 0 for $\lambda \in D^0(C,-wC)$. The convex region $D(C,-wC)$ is the intersection of the radicial half spaces containing C but not wC. Put $w_i = s_1 \cdots s_i$. The gallery $(C, w_1C, w_2C, \dots, wC)$ connects C to wC; its walls are all the radicial hyperplanes that separate C and wC. The wall between $w_{i-1}C$ and w_iC is defined

by the root $w_{i-1}(\alpha_i)$, where α_i is the simple root corresponding to s_i; C and $w_{i-1}C$ are on the same side of it. The convex region $D(C, -wC)$ is hence defined by the inequalities

$$\langle \mu, H_{w_{i-1}(\alpha_i)} \rangle \geq 0.$$

(Recall that H_α denotes the coroot corresponding to a root α, cf. (5.1).) We will prove that

$$v_i = \langle \lambda, H_{w_{i-1}(\alpha_i)} \rangle. \tag{9.5.1}$$

Let us lift the decomposition $w = (s_1 \cdots s_{i-1}) s_i (s_{i+1} \cdots s_n)$ into a decomposition $\dot{w} = \dot{w}_{i-1} \dot{s}_i \dot{w}_i'$ in $N(T^*)$. We have

$$\Psi(\dot{w}) = \Psi(\dot{w}_i') \Psi(\dot{s}_i) \Psi(\dot{w}_{i-1}).$$

When we approach a general point of D_i, the isomorphisms $\Psi(\dot{w}_i')$ and $\Psi(\dot{w}_{i-1})$ extend, and we are left to prove that the order along D_i of

$$\Psi(\dot{s}_i) \,:\, E_{w_{i-1}^{-1}(\lambda)} \longrightarrow E_{w_i^{-1}(\lambda)}$$

is

$$\langle \lambda, H_{w_{i-1}(\alpha_i)} \rangle = \langle w_{i-1}^{-1}(\lambda), H_{\alpha_i} \rangle;$$

we are reduced to the case where w is a fundamental reflection: all computations occur within a minimal parabolic subgroup P, or P/U_P, or the derived group of P/U_P, or its universal covering SL(2). Let us make a direct check for $G = \mathrm{SL}(2)$, $w \neq e$ and λ the fundamental weight.

(a) We take SL(2) in its obvious representation k^2, $T^* =$ diagonal matrices, $B^* =$ upper triangular matrices. The weight λ is

$$\begin{pmatrix} a & * \\ 0 & a^{-1} \end{pmatrix} \longmapsto a, \qquad \text{and} \qquad \langle \lambda, H_\alpha \rangle = 1.$$

(b) $X = \mathbb{P}^1$, the space of homogeneous lines of k^2 and the fibre of E_λ at x is the corresponding line.

(c) Up to a scalar, $\Psi(\dot{w})$ associates to $u \in (E_\lambda)_x$ the linear form $u \wedge v$ on $(E_\lambda)_y$, for $x \neq y$. It vanishes simply for $y \to x$.

9.6. The assumptions (0.1.1) are in force from now on. Under the assumptions of (1.8), we have

$$F^* E_\lambda = E_{F^*\lambda} \qquad (\text{where } F^*\lambda = \lambda \circ F).$$

On the inverse image $\overline{X(w)}'$ of the graph of the Frobenius map $F : X \to X$ in $\overline{O(w)}'$, the line bundle $\mathrm{pr}_1^* E_{\lambda^{-1}} \otimes \mathrm{pr}_2^* E_{w^{-1}(\lambda)}$ is hence isomorphic to

$$\mathrm{pr}_1^* (E_{F^* w^{-1}\lambda - \lambda}).$$

It will be ample if $E_{F^* w^{-1}\lambda - \lambda}$ is ample on X, i.e., if $F^* w^{-1}\lambda - \lambda \in -C^0$. On the

other hand, if $\lambda \in D^0(C, -wC)$, the section $\Psi(\dot{w})$ of it has as zero set the complement of $X(w)$ in the projective variety $\overline{X(w)}'$. If both conditions can be simultaneously fulfilled, then $X(w)$ is affine. In terms of $\mu = -w^{-1}(\lambda)$, the conditions read

$$\mu \in D^0(C, -w^{-1}C), \tag{9.6.1}$$

$$F^*\mu - w\mu \in C^0. \tag{9.6.2}$$

THEOREM 9.7. If there is $\mu \in X(T^*) \otimes \mathbb{R}$ satisfying (9.6.1) and (9.6.2) then $X(w)$ is affine. As a consequence, $X(w)$ is affine as soon as q is larger than the Coxeter number h of G.

It remains to prove that if $q \geq h$, then some μ satisfies (9.6.1) and (9.6.2). The first condition will be fulfilled if $\mu \in C^0$, that is, if $\langle \mu, H_\alpha \rangle > 0$ for each simple root α. We take μ so that $\langle \mu, H_\alpha \rangle = 1$ for each simple root α. We then have $\langle F\mu, H_\alpha \rangle = q$, and $\langle w\mu, H_\alpha \rangle = \langle \mu, w^{-1}H_\alpha \rangle$. If $H = \sum n_\alpha H_\alpha$ is the highest coroot, $\sum n_\alpha = h - 1$ and

$$\langle F\mu - w\mu, H_\alpha \rangle \geq q - \langle \mu, H \rangle = q - h + 1 > 0,$$

hence (9.6.2).

THEOREM 9.8. If the character $\theta : T(w)^F \to \overline{\mathbb{Q}}_l^*$ is non-singular, then

$$H_c^*(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta \xrightarrow{\sim} H^*(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta.$$

COROLLARY 9.9. If $X(w)$ is affine and θ is non-singular, then

$$H_c^i(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta = 0 \qquad \text{for } i \neq l(w).$$

Let us deduce (9.9) from (9.8). With the notations of (1.9), (9.8) means that

$$H_c^*(X(w), \mathscr{F}_\theta) \xrightarrow{\sim} H^*(X(w), \mathscr{F}_\theta). \tag{9.9.1}$$

If $X(w)$ is affine, then $H^i(X(w), \mathscr{F}_\theta) = 0$ for $i > \dim X(w) = l(w)$ ([10, XIV, 3.2]). The sheaf $\mathscr{F}_\theta$ is locally constant, its dual is $\mathscr{F}_{\theta^{-1}}$ and $X(w)$ is smooth and purely of dimension $l(w)$. By Poincaré duality, $H_c^i(X(w), \mathscr{F}_\theta)$ is dual (up to a twist) to $H^{2l(w)-i}(X(w), \mathscr{F}_{\theta^{-1}})$, hence vanishes for $2l(w) - i > l(w)$, i.e., for $i < l(w)$. This proves the corollary.

9.10. We first construct a nice compactification of $X(w)$. Let $w = s_1 \cdots s_n$ be a minimal expression for w. We define $\overline{X}(s_1, \ldots, s_n)$ to be the space of sequences $(B_0, \ldots, B_n)$ of Borel subgroups of G, with $B_n = FB_0$ and with B_{i-1} and B_i in relative position s_i or e. It is the inverse image of the graph of Frobenius by the map

$$\alpha : (B_0, \ldots, B_n) \longmapsto (B_0, B_n) : \overline{O}(s_1, \ldots, s_n) \longrightarrow X \times X.$$

In other words, if $\Gamma \xrightarrow{i} X \times X$ is the inclusion in $X \times X$ of the graph of Frobenius, it is the fibre product $\overline{O} \times_{X\times X} \Gamma$:

$$\begin{array}{ccc} \overline{X}(s_1, \dots, s_n) & \longrightarrow & \Gamma \\ \downarrow & & \downarrow i \\ \overline{O}(s_1, \dots, s_n) & \longrightarrow & X \times X. \end{array}$$

LEMMA 9.11. Γ is transverse to $\overline{O}(s, \dots, s)$, as well as to any intersection of the divisors D_i. The fibre product $\overline{X}(s, \dots, s_n)$ is hence a smooth compactification of $X(w)$, with a divisor with normal crossings (sum of the traces $\overline{D}_i$ of the D_i) at infinity.

We will show that Γ is transverse to any smooth G-equivariant scheme $\pi : Y \to X \times X$ over $X \times X$ (with G acting diagonally on $X \times X$); that is, if $\pi(y) \in \Gamma$, the sum of the tangent space to Γ at $\pi(y)$ and of the image of $d\pi$ is the whole tangent space of $X \times X$ at $\pi(y)$. Indeed

(a) the tangent space of Γ at $\pi(y)$ is $T_x \times \{0\}$;

(b) the image of $d\pi$ contains the image of Lie(G) by the derivative at $e \in G$ of $g \mapsto g\pi y$ (by the equivariance of Y). Since the space X is homogeneous with reduced stabilizers this image projects onto T_x by the second projection.

9.12. We now investigate how the covering $\widetilde{X}(\dot{w})$ (with structural group $T(w)^F$) ramifies along the divisor $\overline{D}_i$. The structural group being of order prime to p, the ramification is tame. On each connected component of $\overline{D}_i$, it gives rise to a homomorphism $\tau_i : \widehat{\mathbb{Z}}(1) \to T(w)^F$.

Let $x(t) \in \overline{X}(s_1, \dots, s_n)$ be a one parameter family with $x(0)$ a point of $\overline{D}_i$ (not on any $\overline{D}_j$, $j \neq i$), and with the tangent vector $\dot{x}(0)$ transverse to $\overline{D}_i$. In technical terms: for $S =$ spectrum of the Henselization of $k[t]$ at (t), x is a morphism $x : S \to \overline{X}(s_1, \dots, s_n)$ such that the closed point $t = 0$ is mapped to a point of $\overline{D}_i$ not on any $\overline{D}_j$, $j \neq i$, and that the inverse image of $\overline{D}_i$ is the reduced scheme $t = 0$. Put $x(t) = (B_0(t), \dots, B_n(t))$ and let E_i be the pull back by $t \mapsto B_i(t)$ of the T-torsor E on X (1.7). If we factor $\dot{w}$ as in 9.5, the isomorphism $\Psi(\dot{w}) : E_0 \to E_n$ (over the generic point of S) factors as

$$\Psi(\dot{w}) = \Psi(\dot{w}_i')\Psi(\dot{s}_i)\Psi(\dot{w}_{i-1})$$

and $\Psi(\dot{w}_{i-1})$ and $\Psi(\dot{w}_i')$ extend over S; $\Psi(\dot{s}_i)$ doesn't; rather, the composite $x \mapsto \Psi(\dot{s}_i)(xH_{-\alpha}(t))$ does (9.5), hence $\Psi(\dot{w})$ is of the form

$$\Psi(\dot{w})(x) = \Psi_0\big(xH_{w_{i-1}(\alpha_i)}(t)\big)$$

where $\Psi_0 : E_0 \to E_n$ extends over S. The pull back under x of the $T(w)^F$-torsor $\widetilde{X}(\dot{w})$, to the generic point of S, is given by the equation

$$F(u) = \Psi_0\big(uH_{w_{i-1}(\alpha_i)}(t)\big). \tag{9.12.1}$$

Let u_0 be a solution, over S, of the equation

$$F(u_0) = \Psi_0(u_0)$$

(this exists, because S is strictly Henselian and F and Ψ_0 extend over S); if we put $u = u_0 y$, (9.12.1) becomes

$$y^{-1} \operatorname{ad} \dot{w} F(y) = H_{w_{i-1}(\alpha_i)}(t). \tag{9.12.2}$$

In other words:

LEMMA 9.13. The $T(w)^F$-torsor $\widetilde{X}(\dot{w})$ ramifies along $\overline{D}_i$ in the same way as the pull back under $H_{w_{i-1}(\alpha_i)} : \mathbb{G}_m \to T$ of the Lang covering of $T(w)$ ramifies at 0.

The tame fundamental group of $\mathbb{G}_m$ is $\widehat{\mathbb{Z}}_{p'}(1)$, that of T is $Y^\wedge_{p'}(1)$ (for Y the dual of the character group of T) and $H_{w_{i-1}(\alpha_i)}$ induces $x \mapsto x H_{w_{i-1}(\alpha_i)}$:

$$\widehat{\mathbb{Z}}_{p'}(1) \longrightarrow Y^\wedge_{p'}(1).$$

The Lang covering of $T(w)^F$ gives rise to the map $Y^\wedge_{p'}(1) \to T(w)^F$ described in (5.2). Hence the pull back (9.13) corresponds to the composite

$$\widehat{\mathbb{Z}}_{p'}(1) \xrightarrow{H_{w_{i-1}(\alpha_i)}} Y^\wedge_{p'}(1) \longrightarrow T(w)^F.$$

For θ a character of $T(w)^F$, $\mathscr{F}_\theta$ will ramify along $\overline{D}_i$ if and only if θ is not trivial on $H_{w_{i-1}(\alpha_i)}\widehat{\mathbb{Z}}_{p'}(1)$. In that case, all higher direct images $R^j j_* \mathscr{F}_\theta$ of $\mathscr{F}_\theta$ ($i \geq 0$) under $j : X(w) \to \overline{X}(s_1, \dots, s_n)$ will vanish on $\overline{D}_i$. In particular, we have

LEMMA 9.14. If θ is non-singular, $j_* \mathscr{F}_\theta = j_! \mathscr{F}_\theta$ (extension by zero) and $R^j j_* \mathscr{F}_\theta = 0$ for $i > 0$.

Proof of 9.8 (in the guise 9.9.1). By (9.14), the Leray spectral sequence for j reads

$$H^*(\overline{X}(s_1, \dots, s_n), j_! \mathscr{F}_\theta) \xrightarrow{\sim} H^*(X(w), \mathscr{F}_\theta).$$

The space $\overline{X}(s_1, \dots, s_n)$ being a compactification of $X(w)$, the left hand side is by definition $H^*_c(X(w), \mathscr{F}_\theta)$.

Remark 9.15.1. By using arguments parallel to those of (1.6), one can show that the conclusion of (9.9): "for θ non-singular, $H^i_c(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta = 0$ for $i \neq l(w)$" holds true as soon as there is w' in the F-conjugacy class of w such that $X(w')$ is affine. The criterion (9.7) can be used to check that this is always the case for the classical groups and for G_2 except the case ($G_2, q = 2, w =$ Coxeter) for which a different method applies.

Remark 9.15.2. Let (T', θ') be maximally split (5.17); under an isomorphism $T(w) \xrightarrow{\sim} T'$ (inducing $T(w)^F \xrightarrow{\sim} T'^F$), as in (1.18), θ' becomes a character θ of $T(w)^F$. Proposition 5.24 and the proof of the Induction Theorem

8.2 show that the conclusion of (9.8) still holds for θ. The same applies to (9.9).

THEOREM 9.16. If $u \in G^F$ is a regular unipotent element, then, for any F-stable maximal torus T, we have $Q_{T,G}(u) = 1$.

We must prove that for any $w \in \mathbf{W}$, $\mathrm{Tr}(u, H_c^*(X(w))) = 1$. We know that $\mathrm{Tr}(u, H_c^*(X(w)))$ is an integer, and that

$$\sum_{w \in \mathbf{W}} \mathrm{Tr}(u, H_c^*(X(w))) = |\mathbf{W}|$$

(see (7.14) and its proof). Hence it suffices to prove the

LEMMA 9.17. Under the assumptions of (9.16), $\mathrm{Tr}(u, H_c^*(X(w))) > 0$, for any $w \in \mathbf{W}$.

Take $w = s_1 \cdots s_n$ as in (9.10) and let $\overline{X}(s_1, \dots, s_n)$ be the corresponding compactification of $X(w)$. For $P \subset [1, n]$ we write $\overline{D}_P$ for the intersection, in $\overline{X}(s_1, \dots, s_n)$, of the divisors $\overline{D}_i$ ($i \in P$) (9.11). For $P = \emptyset$, $\overline{D}_P$ is $\overline{X}(s_1, \dots, s_n)$. The additivity property of cohomology with compact support shows that

$$\mathrm{Tr}(u, H_c^*(X(w))) = \sum_{P \subset [1,n]} (-1)^{|P|} \mathrm{Tr}(u, H^*(\overline{D}_P)). \tag{9.17.1}$$

Let b be the unique fixed point of u in X. Then $a = (b, \dots, b)$ is the unique fixed point of u in $\overline{X}(s_1, \dots, s_n)$; it is contained in each $\overline{D}_P$. The variety $\overline{D}_P$ being smooth and compact, $\mathrm{Tr}(u, H^*(\overline{D}_P))$ is just the multiplicity of the unique fixed point a of u acting on $\overline{D}_P$; to compute it, we may replace $\overline{D}_P$ by its completion $\widehat{D}_P$ at a. Let (x_i) be a formal coordinate system for $\overline{X}(s_1, \dots, s_n)$ at a, such that $x_i = 0$ is an equation for $\widehat{D}_i$. In terms of these coordinates, u maps the point with coordinates $x_1, \dots, x_n$ into the point with coordinates $P_1(x_1, \dots, x_n), \dots, P_n(x_1, \dots, x_n)$, for some formal power series P_i. Since the divisor $\widehat{D}_i$ is preserved by u, P_i is divisible by x_i. The jacobian matrix of u at $(0, \dots, 0)$ is hence diagonal; u being of order a power of p, it is the identity, and we have

$$P_i = x_i(1 + Q_i).$$

with $Q_i(0, \dots, 0) = 0$. The fixed point scheme of u is given by the equations $x_i Q_i = 0$.

Let E_i be the divisor $Q_i = 0$. It is non-empty. The formula (9.17.1) can be rewritten

$$\begin{aligned}
\mathrm{Tr}(u, H_c^*(X(w))) &= \sum_{P \subset [1,n]} (-1)^{|P|} \prod_{i \in P} \widehat{D}_i \prod_{j \notin P} (\widehat{D}_j + E_j) \\
&= \sum_{P \subset Q \subset [1,n]} (-1)^{|P|} \prod_{i \in Q} \widehat{D}_i \prod_{j \notin Q} E_j \\
&= \sum_{Q \subset [1,n]} \left(\sum_{P \subset Q} (-1)^{|P|} \right) \prod_{i \in Q} \widehat{D}_i \prod_{j \notin Q} E_j \\
&= \prod_j E_j > 0,
\end{aligned}$$

(the product of n divisors stands for the multiplicity of their intersection).

PROPOSITION 9.18. Let $s \in G^F$ be a semisimple element. Let ν be the class function on G^F defined by:

$$\nu(g) = \begin{cases} |Z^0 Z^0(s)^F|\, q^{l(Z^0(s))} \dfrac{|Z(s)^F|}{|Z^0(s)^F|}, & \text{if } g \in G^F \text{ is regular with semisimple} \\ & \text{part conjugate in } G^F \text{ to } s, \\ 0, & \text{otherwise.} \end{cases}$$

Then ν is the character of

$$\frac{1}{|Z^0(s)^F|} \sum_{\substack{T \\ s \in T}} |T^F| \sum_{\theta \in (T^F)^\sim} \theta(s)^{-1} R_T^\theta.$$

(Here $Z^0 Z^0(s)$ is the connected centre of the connected centralizer of s, and $l(Z^0(s))$ is the semisimple rank of $Z^0(s)$.) The proof is similar to that of (7.5); it uses (9.16) and (4.2) instead of (7.2).

For any $\rho \in \mathscr{R}(G^F)$ we have

$$\langle \rho, \nu \rangle = \frac{|Z^0 Z^0(s)^F|}{|Z^0(s)^F|} q^{l(Z^0(s))} \sum_{u \in A(s)} \rho(su) \tag{9.18.1}$$

where $A(s)$ is the set of regular unipotent elements in $Z^0(s)^F$.

COROLLARY 9.19. For any $\rho \in \mathscr{R}(G^F)$, the average value of the character of ρ on the regular elements in G^F with semisimple part s, equals

$$\frac{1}{|A(s)|} \sum_{u \in A(s)} \mathrm{Tr}(su, \rho) = \frac{1}{|Z^0(s)^F|} \sum_{\substack{T \\ s \in T}} |T^F| \sum_{\theta \in (T^F)^\sim} \theta(s) \langle \rho, R_T^\theta \rangle.$$

This follows from (9.18), (9.18.1) and the following lemma.

LEMMA 9.20. The number of regular unipotent elements in G^F equals

$$|G^F| / |Z^{0F}|\, q^l.$$

(Here Z^0 is the connected centre of G and l is the semisimple rank of G.)

To count regular unipotents in G^F, we observe that any such element is contained in a unique Borel subgroup and then count the regular unipotents in G^F contained in a fixed F-stable Borel subgroup.

10. A decomposition of the Gelfand-Graev representation

10.1. Let $(G^F)^\sim$ be the set of isomorphism classes of irreducible representations of G^F (over $\overline{\mathbb{Q}}_l$). For any $\rho \in (G^F)^\sim$, there exists an F-stable maximal torus T and $\theta \in (T^F)^\sim$ such that $\langle \rho, R_T^\theta \rangle \neq 0$ (7.7); moreover the geometric conjugacy class $[\theta]$ of (T, θ) (see 5.5), is uniquely determined by ρ (6.3). We thus get a well-defined surjective map

$$(G^F)^\sim \longrightarrow \mathfrak{S} \tag{10.1.1}$$

where $\mathfrak{S}$ is the set of all geometric conjugacy classes of pairs (T,θ).

10.2. In the rest of this chapter we shall assume that the centre Z of G is connected. Let T^*, B^* be as in (1.8). Let U^* be the unipotent radical of B^* and let $U^*_\bullet$ be the subgroup of U^* generated by the root subgroups corresponding to non-simple roots. The quotient $U^*/U^*_\bullet$ is commutative and is a direct product over the simple roots α: $\prod_\alpha U^*_\alpha$, with U^*_α one-dimensional. Let I be the set of orbits of F on the simple roots. For any $i \in I$, let $U^*_i = \prod_{\alpha\in i} U^*_\alpha$; then $U^*/U^*_\bullet = \prod_{i\in I} U^*_i$. This decomposition is F-stable, hence we have also $U^{*F}/U^{*F}_\bullet = \prod_{i\in I} U^{*F}_i$.

We consider the Gelfand-Graev representation $\Gamma_G = \mathrm{Ind}_{U^{*F}}^{G^F}(\chi)$ where χ is any character of U^{*F} which is trivial on $U^{*F}_\bullet$ and defines a non-trivial character of U^{*F}_i for all $i \in I$. All such χ are conjugate under T^{*F}, since Z is connected, hence the G^F-module Γ_G is well-defined up to isomorphism.

Let Δ_G be the class function on G^F which equals $|Z^F|q^l$ on any regular unipotent element in G^F and vanishes on all other elements. (l is the semisimple rank of G.) The following result shows that this is the character of a virtual representation of G^F (which will be also denoted by Δ_G):

PROPOSITION 10.3. For any subset $J \subset I$, let $P(J) \supset B^*$ be the parabolic subgroup generated by B^* and by the root subgroups corresponding to minus the simple roots in F-orbits in J. Let $L(J)$ be $P(J)$ modulo its unipotent radical. Then

$$\Delta_G = \sum_{J\subset I} (-1)^{|J|} \mathrm{Ind}_{P(J)^F}^{G^F}(\Gamma_{L(J)}). \tag{10.3.1}$$

(Note that the centre of $L(J)$ is connected, since that of G is, hence $\Gamma_{L(J)}$ is well-defined.)

Remark 10.4. The following identity is a formal consequence of (10.3.1):

$$\Gamma_G = \sum_{J\subset I} (-1)^{|J|} \mathrm{Ind}_{P(J)^F}^{G^F}(\Delta_{L(J)}).$$

10.5. Proof of 10.3. For any subset $J \subset I$, let $\mathscr{C}(J)$ be the set of characters χ of U^{*F} with $\chi|U^{*F}_\bullet = 1$ and such that χ induces a non-trivial character of U^*_i ($i \in I$) if and only if $i \in J$. It is easy to see that, for any $\chi \in \mathscr{C}(J)$, we have

$$\mathrm{Ind}_{P(J)^F}^{G^F}(\Gamma_{L(J)}) = \mathrm{Ind}_{U^{*F}}^{G^F}(\chi).$$

Let $u_0 \in U^{*F}$ be a fixed regular unipotent element; in other words, the image of u_0 in U^{*F}_i is non-zero for any $i \in I$. It follows that

$$\sum_{\chi\in\mathscr{C}(J)} \chi(u_0) = (-1)^{|J|}.$$

Hence (10.3.1) is equivalent to:

$$\Delta_G = \sum_{\chi\in\mathscr{C}} \chi(u_0)\,\mathrm{Ind}_{U^{*F}}^{G^F}(\chi) \tag{10.5.1}$$

where $\mathscr{C} = \bigcup_{J\subset I} \mathscr{C}(J)$. The character of the right hand side of (10.5.1) vanishes at non-unipotent elements. Its value at $u \in U^{*F}$ is:

$$\begin{aligned} &|U^{*F}|^{-1} \sum_{\substack{g\in G^F \\ g^{-1}ug\in U^{*F}}} \sum_{\chi\in\mathscr{C}} \chi(g^{-1}ugu_0) \\ &= q^l |U^{*F}|^{-1} \#\{g \in G^F \mid g^{-1}ug \in U^{*F},\ g^{-1}ugu_0 \in U_\bullet^{*F}\}. \end{aligned} \tag{10.5.2}$$

This is zero unless $u \in U^{*F}$ is regular unipotent; we now assume that $u \in U^{*F}$ is regular unipotent. For any $g \in G^F$ with $g^{-1}ug \in U^{*F}$, we have $u \in B^* \cap gB^*g^{-1}$; but B^* is the only Borel subgroup containing u, hence $g \in B^{*F}$. If $g = tu'$, $t \in T^{*F}$, $u' \in U^{*F}$, the relation $g^{-1}ugu_0 \in U_\bullet^{*F}$ determines t uniquely modulo Z^F and leaves u' arbitrary. Thus the expression (10.5.2) becomes

$$q^l |U^{*F}|^{-1} |Z^F| |U^{*F}| = |Z^F| q^l,$$

and (10.3) is proved.

We shall now prove the following

LEMMA 10.6. Let M be a virtual representation of G^F such that

$$\langle M, M\rangle = |Z^F| q^l \qquad \text{and} \qquad \langle M, R_T^\theta\rangle = \varepsilon_T^\theta = \pm 1$$

for any F-stable maximal torus $T \subset G$ and any $\theta \in T^{\vee F}$. Then

$$M = \sum_{x\in\mathfrak{S}} \sigma_x M_x$$

where $\sigma_x = \pm 1$ and M_x are distinct irreducible representations of G^F such that

$$M_x = \sigma_x \sum_{\substack{(T,\theta) \bmod G^F \\ [\theta]=x}} \frac{\varepsilon_T^\theta}{\langle R_T^\theta, R_T^\theta\rangle} R_T^\theta.$$

Let $\{M\}$ be the set of all $\rho \in (G^F)^\sim$ such that $\langle \rho, M\rangle \neq 0$; $\{M\}$ has at most $|Z^F| q^l$ elements. By our assumption and by 5.7(i), the map $\{M\} \to \mathfrak{S}$ (the restriction of (10.1.1)) is surjective. Since $|\mathfrak{S}| = |Z^F| q^l$ (5.7), it is bijective and $\{M\}$ has exactly $|Z^F| q^l$ elements. Let $M_x \in \{M\}$ be the element corresponding to $x \in \mathfrak{S}$ under this bijection; we must have $M = \sum_{x\in\mathfrak{S}} \sigma_x M_x$, $\sigma_x = \pm 1$, since $\langle M, M\rangle = |\{M\}|$. If (T,θ) is such that $[\theta] = x$, we have

$$\langle M, R_T^\theta\rangle = \sigma_x \langle M_x, R_T^\theta\rangle = \varepsilon_T^\theta;$$

from the orthogonality of the R_T^θ's (6.8) we have

$$M_x = \sigma_x \sum_{\substack{(T,\theta) \bmod G^F \\ [\theta]=x}} \frac{\varepsilon_T^\theta}{\langle R_T^\theta, R_T^\theta\rangle} R_T^\theta + M_x'$$

where M_x' is orthogonal to all the R_T^θ's. It remains to prove that $M_x' = 0$. Since M_x is irreducible, this would follow if we prove that $\langle M_x - M_x', M_x - M_x'\rangle = 1$; we certainly have $\langle M_x - M_x', M_x - M_x'\rangle \leq 1$, hence it is sufficient to prove

that

$$\sum_{x\in\mathfrak{S}}\langle M_x - M'_x, M_x - M'_x\rangle = |Z^F|q^l$$

or, in other words, that

$$\sum_{(T,\theta)\bmod G^F}\frac{1}{\langle R_T^\theta, R_T^\theta\rangle} = |Z^F|q^l.$$

By (6.8), this is equivalent to the identity

$$\sum_{T\bmod G^F}\frac{1}{|W(T)^F|}|T^F| = |Z^F|q^l,$$

which is the same as

$$\frac{1}{|\mathbf{W}|}\sum_{w\in\mathbf{W}}\det(q - w\tau) = q^l$$

where w is regarded as an automorphism of the lattice $X_0 = X(T/Z)$ and τ is defined by $F^* = q\cdot\tau^{-1}$. The last identity (compare Steinberg [15, p. 91]) can be written as

$$\sum_{0\le i\le l}(-1)^i q^{l-i}\frac{1}{|\mathbf{W}|}\sum_{w\in\mathbf{W}}\mathrm{Tr}(w\tau, \Lambda^i(X_0)) = q^l$$

and follows from the fact that $\Lambda^i(X_0)^{\mathbf{W}} = 0$ for $i > 0$ (5.8).

THEOREM 10.7. We recall that Z is connected. Let $x\in\mathfrak{S}$, and x' be a representative of the corresponding semisimple conjugacy class in the dual group (5.24). We define δ_x to be the $\mathbf{F}_q$-rank of the centralizer of x'.

(i) The formulae

$$\rho_x = \sum_{\substack{(T,\theta)\bmod G^F\\ [\theta]=x}}\frac{(-1)^{\sigma(G)-\sigma(T)}}{\langle R_T^\theta, R_T^\theta\rangle}R_T^\theta \tag{10.7.1}$$

and

$$\rho'_x = (-1)^{\sigma(G)-\delta_x}\sum_{\substack{(T,\theta)\bmod G^F\\ [\theta]=x}}\frac{1}{\langle R_T^\theta, R_T^\theta\rangle}R_T^\theta \tag{10.7.2}$$

define irreducible representations of G^F. The $|Z^F|q^l$ elements $\rho_x\in(G^F)^\sim$, $x\in\mathfrak{S}$ (resp. $\rho'_x\in(G^F)^\sim$, $x\in\mathfrak{S}$) are distinct. The maps $x\mapsto\rho_x$, $x\mapsto\rho'_x$ are two cross-sections of the map (10.1.1).

(ii) One has

$$\Gamma_G = \sum_{x\in\mathfrak{S}}\rho_x = \sum_{(T,\theta)\bmod G^F}\frac{(-1)^{\sigma(G)-\sigma(T)}}{\langle R_T^\theta, R_T^\theta\rangle}R_T^\theta, \tag{10.7.3}$$

$$\Delta_G = \sum_{x\in\mathfrak{S}}(-1)^{\sigma(G)-\delta_x}\rho'_x = \sum_{(T,\theta)\bmod G^F}\frac{1}{\langle R_T^\theta, R_T^\theta\rangle}R_T^\theta. \tag{10.7.4}$$

(iii) Assume that all the roots have the same length. Fix (T,θ) in the geometric conjugacy class x, let S be the connected centralizer of (T,θ) (5.19) and let θ_S be the corresponding character of S^F. We have

$$\rho_x \otimes \mathrm{St}_G = \mathrm{Ind}_{S^F}^{G^F}(\theta_S \otimes \mathrm{St}_S \otimes \mathrm{St}_S) \tag{10.7.5}$$

$$\rho'_x \otimes \mathrm{St}_G = \mathrm{Ind}_{S^F}^{G^F}(\theta_S \otimes \mathrm{St}_S). \tag{10.7.6}$$

(iv) Let S^* be the centralizer of x'. We still have

$$\dim \rho_x = \frac{|G^F|/\mathrm{St}_G(e)}{|S^{*F}|/\mathrm{St}_{S^*}(e)} \cdot \mathrm{St}_{S^*}(e), \tag{10.7.7}$$

$$\dim \rho'_x = \frac{|G^F|/\mathrm{St}_G(e)}{|S^{*F}|/\mathrm{St}_{S^*}(e)}. \tag{10.7.8}$$

If we tensor the right hand side of (10.7.1) by St_G we find (by 7.3):

$$\sum_{\substack{(T,\theta) \bmod G^F \\ [\theta]=x}} \frac{1}{\langle R_T^\theta, R_T^\theta\rangle} \mathrm{Ind}_{T^F}^{G^F}(\theta). \tag{10.7.9}$$

Let us first assume that all roots have the same length. By 5.14(ii), the G^F-conjugacy classes of pairs (T,θ), $T \subset G$, with $[\theta] = x$ are in one-to-one correspondence with S^F-conjugacy classes of pairs $(T',\theta_S|_{T'})$, with T' an F-stable maximal torus in S. Moreover, by 5.14(i) and (6.8), we have $\langle R_T^\theta, R_T^\theta\rangle = |W_S(T')^F|$. Thus the expression (10.7.9) equals

$$\begin{aligned}
&\sum_{\substack{T'\subset S \\ \bmod S^F}} \frac{1}{|W_S(T')^F|} \mathrm{Ind}_{T'^F}^{G^F}(\theta_S|_{T'}) \\
&= \sum_{\substack{T'\subset S \\ \bmod S^F}} \frac{1}{|W_S(T')^F|} \mathrm{Ind}_{S^F}^{G^F}\left(\mathrm{Ind}_{T'^F}^{S^F}(1)\otimes\theta_S\right) \\
&= \mathrm{Ind}_{S^F}^{G^F}\left(\sum_{\substack{T'\subset S \\ \bmod S^F}} \frac{1}{|W_S(T')^F|} \mathrm{Ind}_{T'^F}^{S^F}(1)\otimes\theta_S\right) \\
&= \mathrm{Ind}_{S^F}^{G^F}(\theta_S \otimes \mathrm{St}_S \otimes \mathrm{St}_S), \qquad \text{by 7.14.}
\end{aligned}$$

Similarly, by tensoring the right hand side of (10.7.2) by St_G, we find

$$\mathrm{Ind}_{S^F}^{G^F}\left(\sum_{\substack{T'\subset S \\ \bmod S^F}} \frac{(-1)^{\sigma(S)-\sigma(T')}}{|W_S(T')^F|} \mathrm{Ind}_{T'^F}^{S^F}(1)\otimes\theta_S\right) = \mathrm{Ind}_{S^F}^{G^F}(\theta_S \otimes \mathrm{St}_S), \qquad \text{by 7.14.}$$

In general, the G^F-conjugacy classes of pairs (T,θ), with $[\theta] = x$, are in one-to-one correspondence with the G^{*F}-conjugacy classes of pairs (T',θ'), with θ' conjugate to x' (5.24), i.e., with the S^{*F}-conjugacy classes of pairs (T',x'), $(T' \subset S^*)$. The dimension of (10.7.9) is that of

$$\sum_{T'\subset S^*} \frac{1}{|W_{S^*}(T')^F|} \mathrm{Ind}_{T'^F}^{G^{*F}}(1)$$

(because $|G^{*F}| = |G^F|$), while that of (10.7.2) $\otimes\, \mathrm{St}_G$ is

$$\sum_{T'\subset S^*} \frac{(-1)^{\sigma(S)-\sigma(T')}}{|W_{S^*}(T')^F|} \mathrm{Ind}_{T'^F}^{G^F}(1)$$

and the same proof as above gives (10.7.7), (10.7.8).

We now show that (10.6) can be applied with $M = \Gamma_G$ or Δ_G. We have

$$\langle \Delta_G, \Delta_G \rangle = \frac{1}{|G^F|}(|Z^F|q^l)^2 \#\{\text{regular unipotents in } G^F\}$$
$$= \frac{1}{|G^F|}(|Z^F|q^l)^2 \frac{|G^F|}{|Z^F|q^l} = |Z^F|q^l \qquad \text{(by 9.20).}$$

The analogous identity $\langle \Gamma_G, \Gamma_G \rangle = |Z^F|q^l$ follows from R. Steinberg ([14]). Now, for any (T, θ), $\langle \Delta_G, R_T^\theta \rangle$ equals the average value of the character of R_T^θ on the regular unipotents in G^F hence equals 1, by (9.16).

We now prove that $\langle \Gamma_G, R_T^\theta \rangle = (-1)^{\sigma(G)-\sigma(T)}$ for any (T, θ). Using a result of Rodier (cf. T. A. Springer, Caractères de groupes de Chevalley finis, Sém. Bourbaki 429, Fév. 1973) and the induction formula 8.2, we are reduced to the case where T is not contained in any proper F-stable parabolic subgroup of G. Using (10.3.1) and the already known fact that $\langle \Delta_G, R_T^\theta \rangle = 1$, we see that it is enough to prove

$$\left\langle \mathrm{Ind}_{P(J)^F}^{G^F}(\Gamma_{L(J)}), R_T^\theta \right\rangle = 0, \qquad J \neq I. \tag{10.7.8}$$

(Note that in this case $(-1)^{\sigma(G)-\sigma(T)} = (-1)^{|I|}$.) Assuming that the theorem is already proved for groups of dimension strictly smaller than that of G and, in particular, for $L(J)$, we see from (10.7.5) and (8.2) that $\mathrm{Ind}_{P(J)^F}^{G^F}(\Gamma_{L(J)})$ is a linear combination of terms of the form $R_{T'}^{\theta'}$, with $T' \subset P(J)$, $FT' = T'$, so that (10.7.8) follows from the orthogonality formula (6.8).

Now (10.7.1), (10.7.2), (10.7.5), (10.7.6) follow directly from (10.6) applied to $M = \Gamma_G$ or Δ_G, except for an indeterminacy of sign. To remove this indeterminacy of sign, we only have to check that the right hand sides of (10.7.1) and (10.7.2) have strictly positive dimension; but they are both of the form $|G^F|/|S^{*F}|$ up to a power of q, by the first part of the proof. The theorem is proved.

COROLLARY 10.8. For any $\rho \in (G^F)^\sim$, the average value of the character of ρ on the regular unipotents in G^F equals 0, 1, or -1. This value is ± 1 if and only if $\rho = \rho_x'$ for some $x \in \mathfrak{S}$.

Indeed, this average value is just $\langle \rho, \Delta_G \rangle$.

Remark 10.9. If we assume that p is a good prime for G, then all regular unipotents in G^F fall in a single G^F-conjugacy class so that the character of ρ at any regular unipotent in G^F equals 0, 1, or -1. This was first proved in the article by J. A. Green, G. I. Lehrer, and G. Lusztig (On the degrees of certain group characters, to appear in The Quarterly Journal of Mathematics).

PROPOSITION 10.10. Let $x \in \mathfrak{S}$. Let (T, θ) be a maximally split pair in x

(see (5.17); (T,θ) is uniquely defined up to G^F-conjugacy by (5.18)). Then the virtual representation $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ of G^F can be represented by an actual G^F-module, in which ρ_x and ρ'_x occur with multiplicity 1.

By (5.23), there exists an F-stable parabolic subgroup $P \subset G$ such that T is contained in some F-stable Levi subgroup L of P and (T,θ) is non-singular in L. L has connected centre, hence, by 5.20, (T,θ) is non-degenerate in L. Let $R_{T,L}^\theta$ be the virtual representation corresponding to (T,θ) with respect to L (see 1.20). By 7.4,

$$(-1)^{\sigma(L)-\sigma(T)}R_{T,L}^\theta = (-1)^{\sigma(G)-\sigma(T)}R_{T,L}^\theta$$

is irreducible. By 8.2,

$$(-1)^{\sigma(G)-\sigma(T)}R_T^\theta = \mathrm{Ind}_{P^F}^{G^F}\big((-1)^{\sigma(G)-\sigma(T)}R_{T,L}^\theta\big),$$

hence it can be realized as an actual G^F-module.

It remains to observe that for any (T,θ) in x (not necessarily maximally split) we have (by (10.7.1), (10.7.2), and (6.8)):

$$\langle \rho_x, R_T^\theta\rangle = (-1)^{\sigma(G)-\sigma(T)},$$

$$\langle \rho'_x, R_T^\theta\rangle = (-1)^{\sigma(G)-\delta_x}.$$

11. Suzuki and Ree groups

Our methods apply also to the Suzuki and Ree groups ${}^2B_2(q)$, ${}^2G_2(q)$, ${}^2F_4(q)$ (with q an odd power of $\sqrt{2}$, resp. $\sqrt{3}$, $\sqrt{2}$).

These groups are the fixed point set of an exceptional isogeny $F' : G \to G$, with G respectively of type B_2, G_2 and F_4, and F'^2 being the Frobenius endomorphism relative to a rational structure over the field with q^2 elements (R. Steinberg [15]). For the sake of induction, one should more generally consider groups of the form $G^{F'}$, with G reductive and F'^2 the Frobenius endomorphism relative to a rational structure over $\mathbf{F}_{q^2}$ (q an odd power of $\sqrt{2}$ or $\sqrt{3}$). Besides the Suzuki and Ree groups, this includes for instance groups of the form $G = G_1 \times G_1$, with $F'(x,y) = (Fy,x)$; for such a group, $G^{F'} = G_1^F$.

All definitions and results in Chapter 1 can be given with F replaced by F' throughout; in particular, for any F'-stable maximal torus T in G and any Borel subgroup B containing T, the subscheme $X_{T\subset B}$ of X_G and the $G^{F'}$-equivariant $T^{F'}$-torsor $\widetilde{X}_{T\subset B}$ over $X_{T\subset B}$ are well-defined (1.17). Thus for any $\theta \in (T^{F'})^\sim$, the virtual representation $R_{T\subset B}^\theta$ of $G^{F'}$ is well-defined (1.20). For $G = G_1\times G_1$, $F'(x,y) = (Fy,x)$, this coincides with the similar representation of $G_1^F = G^{F'}$.

All the results in Chapters 4, 5, 9 (except (9.18), (9.19)), the disjointness

Theorem 6.2 (with its Corollary 6.3), the results (7.13) and (8.2) continue to be true. On the other hand, the proofs of the orthogonality relations (6.8), (6.9), and of the dimension formula (7.1) require the assumption $|T^{F'}| > 1$. If $q > 2$ this is automatically satisfied: $|T^{F'}|$ is of the form $\prod(q - \rho_i)$, where $|\rho_i| = 1$, hence $|T^{F'}| > \prod(q-1)$. Thus, if we exclude the case where q equals $\sqrt{2}$ or $\sqrt{3}$, all the results in Chapters 6, 7, 8, 10 hold, as well as (9.18), (9.19). They can all be deduced from the orthogonality relations (6.8), (6.9). Even in the case $q = \sqrt{2}$ or $\sqrt{3}$, we can prove them for ${}^2B_2$ and ${}^2G_2$, but we cannot handle ${}^2F_4(\sqrt{2})$.

ADDED IN PROOF (2nd Dec. 1975). We can now handle ${}^2F_4(\sqrt{2})$, too; this will be considered by one of us, in a future article.

INSTITUT DES HAUTES ÉTUDES SCIENTIFIQUES, FRANCE

MATHEMATICS INSTITUTE, UNIVERSITY OF WARWICK, ENGLAND

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CIRCLE DIFFEOMORPHISMS
[after M. R. HERMAN]
by Pierre DELIGNE

Let f be an orientation-preserving diffeomorphism of the circle $\mathbf{R}/\mathbf{Z}$.
If the rotation number $\alpha \in \mathbf{R}/\mathbf{Z}$ of f is irrational (i.e. if f has no periodic point) and the derivative Df of f has bounded variation (i.e. if the second derivative $D^2 f$—taken in the sense of distributions—is a measure), Denjoy proved that f is topologically conjugate to the rotation

$$R_\alpha : x \longmapsto x + \alpha :$$

there exists an orientation-preserving homeomorphism $h : \mathbf{R}/\mathbf{Z} \to \mathbf{R}/\mathbf{Z}$ such that $hfh^{-1} = R_\alpha$. This homeomorphism is unique up to composition on the left by a rotation, because the centralizer of an irrational rotation consists only of rotations.
It amounts almost to the same thing to prove a regularity property for h, or the same property uniformly for all iterates f^n of f. To pass from h to the f^n, one writes

$$f^n = (h^{-1} R_\alpha h)^n = h^{-1} R_{n\alpha} h.$$

Conversely, it is useful to change notation and pass to the universal cover $\mathbf{R}$ of the circle $\mathbf{R}/\mathbf{Z}$: f becomes an increasing diffeomorphism of $\mathbf{R}$, and h an increasing homeomorphism of $\mathbf{R}$, defined up to addition of a constant; both are of the form $x + \varphi(x)$, with φ periodic of period 1, and

$$hfh^{-1} = R_\alpha : x \longmapsto x + \alpha,$$

with $\alpha \in \mathbf{R}$ (the rotation number). Set

$$h_n = \frac{1}{n} \sum_{i=0}^{n-1} (f^i - i\alpha).$$

We shall see that h, suitably normalized, is the limit of the h_n (3.1). Suppose now that the f^n are uniformly $C^{r+\beta}$, i.e. that f is C^r and that the r-th derivatives $D^r f^n$ satisfy, uniformly in n, a Hölder condition of exponent β ($0 < \beta \leq 1$):

$$|D^r f^n(x) - D^r f^n(y)| \leq A|x - y|^\beta.$$

The $f^i - R_{i\alpha}$ satisfy the same condition and are, moreover, bounded (by one). The same is true of the $h_n - \mathrm{Id}$, which are their barycenters. Thus the $h_n - \mathrm{Id}$ form a relatively compact set of functions on $\mathbf{R}/\mathbf{Z}$ in the C^r topology, and one can extract a convergent subsequence from the sequence h_n. The conjugacy h is therefore C^r, and passage to the limit shows that $D^r h$ satisfies the same Hölder condition as the $D^r f^n$.

Herman proved a theorem slightly stronger than the following.

THEOREM.— If the continued-fraction expansion of α has bounded coefficients and f is C^∞, then h is C^∞.

First (§§ 1 to 6), one proves that, as q ranges over the denominators of the convergents p_n/q_n of the continued-fraction expansion of α, $|Df^q|$ tends rapidly to 0 (at rate $1/q^\varepsilon$, with $\varepsilon > 0$). Such a result is plausible: the rotation $R_{q\alpha}$ is close to the identity, and $f^q = h^{-1} R_{q\alpha} h$.

It follows that h is $C^{1+\varepsilon'}$, with $\varepsilon' > 0$; then, from the fact that f is $C^{1+\varepsilon'}$-conjugate to R_α, one deduces that the conjugates $f_n = h_n f h_n^{-1}$ of f tend to R_α in the $C^{2+\varepsilon''}$ topology ($\varepsilon'' > 0$) (§§ 7 and 8).

An implicit-function theorem of Arnold-Moser ([1], [3]), improved by Herman following Rüssmann [4], finally shows that if f is sufficiently close to R_α in the $C^{2+\varepsilon}$ topology ($\varepsilon > 0$), then f is C^∞-conjugate to a rotation.

Terminology and notation

Write $\|\psi\|_\infty$ and $\|\psi\|_1$ for the L^∞ and L^1 norms of a function ψ on $\mathbf{R}/\mathbf{Z}$: $\sup |\psi(x)|$ and $\int_{\mathbf{R}/\mathbf{Z}} |\psi(x)|\, dx$. For $0 \leq \beta \leq 1$, write $|\psi|_\beta$ for its Hölder norm of exponent β:

$$|\psi|_\beta = \sup_{x \neq y} \frac{|\psi(x) - \psi(y)|}{|x - y|^\beta}.$$

Write $\mathrm{Var}(\psi)$ for the total variation of ψ. If ψ is C^1, this is $\|D\psi\|_1$.
R_s = the "rotation" $x \longmapsto x + s$.
$f : \mathbf{R} \to \mathbf{R}$ is an increasing diffeomorphism of the form $x + \varphi(x)$, with φ periodic of period 1. Its iterates are denoted by f^n ($n \in \mathbf{Z}$). By hypothesis, f and f^{-1} are C^1; in addition, Df (and hence $\log Df$) is assumed to have bounded variation.

$$V = \mathrm{Var}(\log Df).$$

α = the rotation number of f. We have $\|f^n - R_{n\alpha}\|_\infty < 1$. We assume that α is irrational.
μ = the probability measure on $\mathbf{R}/\mathbf{Z}$ invariant under f. If h conjugates f to R_α, it carries μ to dx: $\mu = dh$ (the distributional derivative, or Stieltjes integral).
h = the increasing homeomorphism of $\mathbf{R}/\mathbf{Z}$ conjugating f to R_α, normalized by the condition

$$\int (h(x) - x)\, \mu = 0.$$

$$h_n = \frac{1}{n} \sum_{i=0}^{n-1} (f^i - i\alpha).$$

This is a barycenter of increasing diffeomorphisms, hence an increasing diffeomorphism. We have $h = \lim h_n$ (3.1).

$$f_n = h_n f h_n^{-1}.$$

A rational approximation to $s \in \mathbf{R}$ is an irreducible fraction p/q such that $|s - p/q| < 1/q^2$, i.e. $|qs - p| < 1/q$. Every irrational number has infinitely many rational approximations (Dirichlet).
q denotes the denominator of a rational approximation to α (and p its numerator).
§ 1. First bound for $|Df^q|$; the Denjoy-Koksma inequality
THEOREM 1.1 (Denjoy-Koksma).— Let $s \in \mathbf{R}$, let m be the denominator of a rational approximation to s, and let ψ be a function of bounded variation on $\mathbf{R}/\mathbf{Z}$. Then

$$\left| \sum_{i=0}^{m-1} \psi(x + is) - m \int_{\mathbf{R}/\mathbf{Z}} \psi(t)\, dt \right| \leq \mathrm{Var}(\psi).$$

After the change of variable $t \longmapsto \pm(t-x)$, it suffices to assume that $x = 0$ and $n/m \leq s \leq n/m + 1/m^2$ (with $(n,m) = 1$). For $0 \leq i \leq m-1$, we have $in/m \leq is < in/m + 1/m$, and the points in/m run through $\frac{1}{m}\mathbf{Z}/\mathbf{Z}$. Hence (the sums run from $i = 0$ to $i = m-1$)

$$\sum_{i=0}^{m-1} \psi(is) - m \int_{\mathbf{R}/\mathbf{Z}} \psi(x)\, dx = \sum_{i=0}^{m-1} \left(\psi(is) - m \int_{in/m}^{in/m+1/m} \psi(x)\, dx \right),$$

and

$$\left| \psi(is) - \int_{in/m}^{in/m+1/m} \psi(x)\, (m\, dx) \right|$$

is bounded by the variation of ψ between in/m and $in/m + 1/m$, proving the theorem.
Recall that the variation $\int_{\mathbf{R}/\mathbf{Z}} |d\psi|$ is invariant under change of variables. Take $s = \alpha$, and make the change of variable that conjugates the rotation R_α to f; it carries dx to the measure μ, and one obtains:
COROLLARY 1.2.— For a function ψ of bounded variation on $\mathbf{R}/\mathbf{Z}$,

$$\left| \sum_{i=0}^{q-1} \psi \circ f^i - q \int_{\mathbf{R}/\mathbf{Z}} \psi(x)\, \mu \right| \leq \mathrm{Var}(\psi).$$

1.3. We have

$$Df^n = Df \circ f^{n-1} \cdot Df^{n-1} = Df \circ f^{n-1} \cdot Df \circ f^{n-2} \cdots Df,$$

whence

$$\log Df^n = \sum_{i=0}^{n-1} \log Df \circ f^i. \tag{1.3.1}$$

Applying the bound 1.2 with $\psi = \log Df$ gives

$$\left| \log Df^q - q \int_{\mathbf{R}/\mathbf{Z}} \log Df \cdot \mu \right| \leq V.$$

If the integral were nonzero, $\log Df^q$ would tend uniformly to $\pm\infty$ as $q \to \infty$, and Df^q would tend uniformly to 0 or ∞. Since

$$\int_{\mathbf{R}/\mathbf{Z}} Df^q(x)\, dx = 1,$$

this is impossible, and therefore:
COROLLARY 1.4

$$\int_{\mathbf{R}/\mathbf{Z}} \log Df \cdot \mu = 0.$$

COROLLARY 1.5

$$e^{-V} \leq Df^q \leq e^V \qquad (\text{when } |\alpha - p/q| < 1/q^2).$$

Remark 1.6.— The proof method of Theorem 1.1 proves 1.2 directly for every increasing homeomorphism f (and for μ an invariant measure). If $V = \int |d \log Df| < \infty$, one obtains 1.5. In his note [2], Denjoy starts from 1.5 to prove that f is topologically conjugate to a rotation.

1.7. Let s be irrational, let q_i ($i \geq 0$) be an increasing sequence of rational approximations to s, and let a_i be the integer part of q_i/q_{i-1} ($i \geq 1$). Assume that $q_0 = 1$. Every integer $N < q_n$ can be written

$$N = \sum_{i<n} b_i q_i \qquad \text{with } b_i \leq a_{i+1} :$$

if $n = 0$, this is clear; if $n > 0$, write $N = b_{n-1}q_{n-1} + N'$ with $N' < q_{n-1}$, and conclude by induction. The sum $\sum_{i<N} \psi(x + is)$ then decomposes into $(\sum b_i)$ partial sums, each having one of the q_i as its length, and (1.1) gives

$$\left| \sum_{i=0}^{N-1} \psi(x + is) - N \int_{\mathbf{R}/\mathbf{Z}} \psi(t)\, dt \right| \leq \sum_i b_i \operatorname{Var}(\psi) \leq \sum_{q_i \leq N} a_{i+1} \operatorname{Var}(\psi), \tag{1.7.1}$$

and, for $s = \alpha$, (1.2) and (1.3) give

$$\left\| \sum_{i=0}^{N-1} \psi \circ f^i - N \int \psi\, \mu \right\|_\infty \leq \sum_{q_i \leq N} a_{i+1} \operatorname{Var}(\psi), \tag{1.7.2}$$

$$\| \log Df^N \|_\infty \leq \sum_{q_i \leq N} a_{i+1} V. \tag{1.7.3}$$

§ 2. Bound for $|D^2 f^q|$

2.1. Differentiate formula (1.3.1) twice:

$$\log Df^n = \sum_{i<n} \log Df \circ f^i :$$

$$D \log Df^n = \sum_{i<n} D \log Df \circ f^i \cdot Df^i, \tag{2.1.1}$$

that is,

$$D^2 f^n = Df^n \cdot \sum_{i<n} D \log Df \circ f^i \cdot Df^i. \tag{2.1.2}$$

$$\begin{aligned} D^2 \log Df^n &= \sum_{i<n} \left(D^2 \log Df \circ f^i \cdot (Df^i)^2 + D \log Df \circ f^i \cdot D^2 f^i \right) \\ &= \sum_{i<n} D^2 \log Df \circ f^i \cdot (Df^i)^2 + \sum_{j<i<n} D \log Df \circ f^i \cdot Df^i \cdot D \log Df \circ f^j \cdot Df^j. \end{aligned} \tag{2.1.3}$$

In the last term one recognizes (Oh! miracle) the cross terms in the expansion of the square of the right-hand side of (2.1.1). Completing the square gives

$$D^2 \log Df^n = \frac{1}{2}(D \log Df^n)^2 - \frac{1}{2} \sum_{i<n} (D \log Df \circ f^i \cdot Df^i)^2 + \sum_{i<n} D^2 \log Df \circ f^i \cdot (Df^i)^2. \tag{2.1.4}$$

If in (2.1.4) one bounds the $|Df^i|$ by $\sup_{i<n} \|Df^i\|_\infty$, one obtains

$$\|D^2 \log Df^n\|_\infty \leq \frac{1}{2} \|D \log Df^n\|_\infty^2 + \frac{1}{2} An \sup_{i<n} \|Df^i\|_\infty^2, \tag{2.1.5}$$

where

$$A = \|D \log Df\|_\infty^2 + 2\|D^2 \log Df\|_\infty. \tag{2.1.6}$$

2.2. By Hadamard, if ψ is a C^2 function on $\mathbf{R}/\mathbf{Z}$, then (cf. 7.1.4)

$$\|D\psi\|_\infty \leq \left(2\|\psi\|_\infty \|D^2\psi\|_\infty\right)^{1/2}. \tag{2.2.1}$$

Assume that f is C^3, and apply this interpolation inequality to $\psi = \log Df^q$, taking account of (1.5) and (2.1.5):

$$\|D \log Df^q\|_\infty \leq \left(V \|D \log Df^q\|_\infty^2 + qAV \sup_{i<q} \|Df^i\|_\infty^2 \right)^{1/2}. \tag{2.2.2}$$

If $V < 1$, this gives an estimate for $\|D \log Df^q\|_\infty$ (which occurs on both sides of the formula): squaring, one finds

$$\|D \log Df^q\|_\infty^2 \leq \frac{AV}{1-V} \cdot q \cdot \sup_{i<q} \|Df^i\|_\infty^2.$$

Writing $D \log Df^q = D^2 f^q / Df^q$, and applying 1.5 once again, one finally obtains:
THEOREM 2.3.— Assume that f is C^3 and that $V < 1$. Let

$$B = \left(\frac{AV}{1-V} \right)^{1/2} \cdot e^V,$$

where A is defined by (2.1.6). If q is the denominator of a rational approxi-

mation to α, then

$$\|D^2 f^q\|_\infty \leq B\, q^{1/2} \sup_{i<q} \|Df^i\|_\infty.$$

The essential gain in this estimate is that a factor $\sqrt{q}$, rather than q, has been obtained.

§ 3. Var$(\log Df_n)$ tends to 0

In order to apply 2.3, it is necessary to reduce to the case $V < 1$. Later, in order to control the $\|Df^i\|_\infty$, $(i < q)$, we shall even need to reduce to the case in which V is small. The main result 3.3 of this section will make this possible.

PROPOSITION 3.1.— The h_n converge uniformly to h.

Put $\psi(x) = h(x) - x$. By definition, h is normalized so that

$$\int_{\mathbf{R}/\mathbf{Z}} \psi(x)\, \mu = 0.$$

The formula $R_\alpha h = hf$ may be written $h = h \circ f - \alpha = f - \alpha + \psi \circ f$. Likewise, since h conjugates f^i to $R_{i\alpha}$, we have

$$h = f^i - i\alpha + \psi \circ f^i.$$

Take the mean of these identities for $0 \leq i < n$:

$$h = \frac{1}{n} \sum_{i<n} (f^i - i\alpha) + \frac{1}{n} \sum_{i<n} \psi \circ f^i = h_n + \frac{1}{n} \sum_{i<n} \psi \circ f^i.$$

It remains to show that $\frac{1}{n} \sum_{i<n} \psi \circ f^i$ converges uniformly to $\int \psi\, \mu = 0$. This can be deduced from 1.7 (ψ has variation ≤ 2), or, by a change of variable (using h), this convergence can be reduced to a theorem of H. Weyl:

Reminder 3.2 (H. Weyl).— Let φ be a continuous function on $\mathbf{R}/\mathbf{Z}$, and let α be irrational. The sums

$$S_n(\varphi) = \frac{1}{n} \sum_{i<n} \varphi(x + i\alpha)$$

converge uniformly to

$$\int_{\mathbf{R}/\mathbf{Z}} \varphi(x)\, dx.$$

This is seen by approximating φ by trigonometric polynomials P. One checks directly that, for a trigonometric polynomial, $S_n(P)$ tends to $\int_{\mathbf{R}/\mathbf{Z}} P(x)\, dx$. Since $|S_n(\varphi)| \leq \|\varphi\|_\infty$, it follows that

$$\limsup \left\| S_n(\varphi) - \int_{\mathbf{R}/\mathbf{Z}} \varphi(x)\,dx \right\|_\infty = \limsup \left\| S_n(\varphi - P) - \int_{\mathbf{R}/\mathbf{Z}} (\varphi(x) - P(x))\,dx \right\|_\infty \leq 2\|\varphi - P\|_\infty,$$

which is arbitrarily small.

THEOREM 3.3.— If f is C^2, the variation

$$V_n = \|D \log Df_n\|_1$$

tends to 0 as $n \to \infty$.

Let us compute Df_n. We have

$$h_n \circ f = \frac{1}{n}\sum_{i=0}^{n-1}(f^{i+1} - i\alpha) = \frac{1}{n}\sum_{i=1}^{n}(f^i - i\alpha) + \alpha = h_n + \alpha + \frac{1}{n}(f^n - x - n\alpha),$$

whence

$$f_n = R_\alpha + \frac{1}{n}(f^n - x - n\alpha) \circ h_n^{-1}$$

and

$$Df_n = 1 + \left[\frac{1}{n}(Df^n - 1)\cdot(Dh_n)^{-1}\right] \circ h_n^{-1}.$$

Thus

$$Df_n - 1 = \frac{Df^n - 1}{\sum_{i<n} Df^i} \circ h_n^{-1}.$$

Furthermore, since $Df^n = Df \circ f^{n-1} \cdot Df^{n-1}$ and

$$(Df^{n-1})^{-1} \circ f^{1-n} = D(f^{1-n}),$$

we have

$$\frac{Df^n}{\sum_{i<n} Df^i} \circ f^{1-n} = \frac{Df}{\sum_{i<n} Df^i \circ f^{1-n} \cdot D(f^{1-n})} = \frac{Df}{\sum_{i<n} Df^{-i}},$$

that is,

$$Df_n - 1 = \left(\frac{Df}{\sum_{i<n} Df^{-i}} \circ f^{n-1} - \frac{1}{\sum_{i<n} Df^i}\right) \circ h_n^{-1}. \tag{3.3.1}$$

Lemma 3.4.— The sum $\sum_{i<n} Df^i$ tends uniformly to ∞.

Indeed, all the Df^i are positive, and infinitely many of them are $\geq e^{-V}$ (1.5). Applying this lemma to f and f^{-1}, we deduce from 3.4 that:

Lemma 3.5.— The Df_n converge uniformly to 1; it is therefore equivalent to prove that the V_n tend to 0, or that the variation of Df_n tends to 0.

Since variation is invariant under change of variable, we have

$$\begin{aligned}\mathrm{Var}(Df_n) = \mathrm{Var}(Df_n - 1) &\leq \mathrm{Var}\left(\frac{Df}{\sum_{i<n} Df^{-i}}\right) + \mathrm{Var}\left(\frac{1}{\sum_{i<n} Df^{i}}\right)\\ &= \left\| D\frac{Df}{\sum_{i<n} Df^{-i}}\right\|_1 + \left\| D\frac{1}{\sum_{i<n} Df^{i}}\right\|_1\\ &\leq \left\|\frac{D^2 f}{\sum_{i<n} Df^{-i}}\right\|_1 + \|Df\|_\infty \left\| D\frac{1}{\sum_{i<n} Df^{-i}}\right\|_1 + \left\| D\frac{1}{\sum_{i<n} Df^{i}}\right\|_1.\end{aligned}$$

By 3.4, the function in the first term tends uniformly to 0. Theorem 3.1 therefore follows from the following result, applied to f and f^{-1}.

THEOREM 3.6.— If f is C^2,

$$D\left(\frac{1}{\sum_{i<n} Df^{i}}\right)$$

is uniformly bounded and tends almost everywhere to 0. Consequently, this function tends to 0 in the L^1 norm. Let G be the automorphism

$$\varphi \longmapsto \varphi \circ f \cdot Df = f^*(\varphi\, dx)/dx$$

of $L^1(\mathbf{R}/\mathbf{Z})$. It preserves the L^1 norm and maps positive functions to positive functions. The Chacon-Ornstein ergodic theorem can therefore be applied to it (see, for example, Garsia, Topics in almost everywhere convergence): for every φ in L^1, the quantities

$$\psi_n(\varphi) = \frac{\sum_{i=0}^{n-1} G^i(\varphi)}{\sum_{i=0}^{n-1} G^i(1)} = \frac{\sum_{i=0}^{n-1} \varphi \circ f^i \cdot Df^i}{\sum_{i=0}^{n-1} Df^i}$$

converge almost everywhere to a limit $\psi(\varphi)$.

We shall apply this theorem to $\varphi = D\log Df$. Put

$$\psi_n = \frac{\sum_{i<n} D\log Df \circ f^i \cdot Df^i}{\sum_{i<n} Df^i},$$

and denote by ψ the almost-sure limit of the ψ_n. The $|\psi_n|$, and hence $|\psi|$, are bounded above by $\|D\log Df\|_\infty$.

Lemma 3.7.—

$$D\left(\frac{1}{\sum_{i<n} Df^i}\right)$$

is uniformly bounded and tends almost everywhere (and hence also in the L^1 norm) to $-\frac{1}{2}\psi$.
With the summation indices running from 0 to $n-1$, we have

$$-D\left(\frac{1}{\sum Df^i}\right) = \frac{\sum D^2 f^i}{(\sum Df^i)^2} = \frac{\sum_{i>j} Df^i \cdot D\log Df\circ f^j \cdot Df^j}{(\sum Df^i)^2}.$$

The quantities

$$\lambda_{ij} = \frac{Df^i Df^j}{\left(\sum_{i<n} Df^i\right)^2} \qquad (j<i<n)$$

are positive and have sum ≤ 1; hence one already sees that

$$-D\left(\frac{1}{\sum Df^i}\right) = \sum \lambda_{ij} D\log Df\circ f^j$$

is bounded in absolute value by $\|D\log Df\|_\infty$.
With the summation indices still running from 0 to $n-1$, we have

$$\begin{aligned}\sum_{i>j} Df^i \cdot D\log Df\circ f^j \cdot Df^j &= \sum_i Df^i\cdot\psi_i\cdot\sum_{j<i} Df^j \\ &= \psi\cdot\sum_{j<i} Df^i Df^j + \sum_i(\psi-\psi_i)Df^i\sum_{j<i}Df^j \\ &= \frac{1}{2}\psi\cdot\left(\sum Df^i\right)^2 - \frac{1}{2}\psi\sum(Df^i)^2 + \sum_i(\psi-\psi_i)Df^i\sum_{j<i}Df^j,\end{aligned}$$

and therefore

$$D\left(\frac{1}{\sum Df^i}\right) + \frac{1}{2}\psi = \frac{1}{2}\psi\cdot\frac{\sum(Df^i)^2}{(\sum Df^i)^2} + \frac{\sum_i(\psi-\psi_i)Df^i\sum_{j<i}Df^j}{(\sum Df^i)^2}.$$

Bound $\sum(Df^i)^2 = \sum Df^i\cdot Df^i$ by $\sup Df^i\cdot\sum Df^i$, and $\sum_{j<i}Df^j$ by $\sum Df^j$. This gives

$$\left|D\left(\frac{1}{\sum Df^i}\right)+\frac{1}{2}\psi\right| \leq \frac{1}{2}|\psi|\frac{\sup Df^i}{\sum Df^i} + \frac{\sum|\psi-\psi_i|Df^i}{\sum Df^i} = A_n + B_n.$$

For every j between 0 and $n-1$,

$$\frac{Df^j}{\sum Df^i} = \left(\sum Df^i \cdot (Df^j)^{-1}\right)^{-1} = \left(\sum Df^{i-j} \circ f^j\right)^{-1} \leq \inf\left(\frac{1}{\sum_{i<n-j} Df^i} \circ f^j, \frac{1}{\sum_{i\leq j} Df^{-i}} \circ f^j\right).$$

By 3.4, A_n therefore tends uniformly to 0. As for B_n, the following lemma, whose verification is left to the reader, shows that B_n tends to 0 at every point at which ψ_n tends to ψ.

Lemma 3.8.— Let a_n be a sequence tending to 0, and let (b_n) be the terms of a divergent positive series. Then

$$\frac{\sum_{i<n} a_i b_i}{\sum_{i<n} b_i}$$

tends to 0.

Finally, let us deduce 3.6 from 3.7. Since $1/\sum_{i<n} Df^i$ tends to 0,

$$D\left(1/\sum_{i<n} Df^i\right)$$

tends to 0 in the sense of distributions. By 3.7, this sequence tends in L^1 to $-\frac{1}{2}\psi$. Hence $\psi = 0$, and 3.7 applies.

Supplement 3.9.— Herman can deduce from 3.7 that an increasing C^2 diffeomorphism of $\mathbf{R}/\mathbf{Z}$ with irrational rotation number is dx-ergodic: every Borel subset (indeed, every dx-measurable subset) of $\mathbf{R}/\mathbf{Z}$ that is invariant under f has Lebesgue measure 0 or 1.

Remark 3.10.— The Chacon-Ornstein theorem is not effective: it gives no estimate for the rate at which, for each ε, the ψ_n converge uniformly to ψ outside a set of measure ε. Consequently, I do not know how to deduce an effective bound for the V_n from the proof above.

§ 4. Approximation of a number by rationals

Except in 4.5, the general conventions concerning α, p, q are suspended in this section. We begin with some reminders about continued fractions, for which Hardy and Wright may be consulted.

4.1. Let $[a_0, \ldots, a_n, \ldots] = a_0 + 1/(a_1 + 1/\cdots$ ($a_i \in \mathbf{Z}$, $a_i > 0$ for

$i > 0$) be the continued-fraction expansion of an irrational number α. Let $p_n/q_n = [a_0, \ldots, a_n]$ be the nth convergent. We have $p_n/q_n < \alpha$ for even n, and $p_n/q_n > \alpha$ for odd n. If, for $i = -1, -2$, (p_i, q_i) is defined by

$$\begin{pmatrix} p_{-2} & p_{-1} \\ q_{-2} & q_{-1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

then for every $n \geq 0$,

$$\begin{pmatrix} p_{n-1} & p_n \\ q_{n-1} & q_n \end{pmatrix} = \begin{pmatrix} p_{n-2} & p_{n-1} \\ q_{n-2} & q_{n-1} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & a_n \end{pmatrix}, \tag{4.1.1}$$

that is,

$$\begin{cases} p_n = a_n p_{n-1} + p_{n-2}, \\ q_n = a_n q_{n-1} + q_{n-2}. \end{cases} \tag{4.1.2}$$

It follows from (4.1.2) that

$$a_{n+1} q_n \leq q_{n+1} \leq (a_{n+1} + 1) q_n, \tag{4.1.3}$$

$$2q_n \leq (a_{n+1} a_{n+2} + 1) q_n \leq q_{n+2}, \tag{4.1.4}$$

$$q_n \geq \sqrt{2}^{\,n} \quad \text{for } n \geq 2 \quad (\text{since } q_0 = 1,\ q_3 \geq 3 > 2^{3/2}). \tag{4.1.5}$$

By induction, (4.1.1) gives $p_{n+1} q_n - p_n q_{n+1} = \pm 1$, i.e.

$$\left| \frac{p_n}{q_n} - \frac{p_{n+1}}{q_{n+1}} \right| = \frac{1}{q_n q_{n+1}}. \tag{4.1.6}$$

Since α lies between p_n/q_n and p_{n+1}/q_{n+1}, it follows from (4.1.6) that the p_n/q_n are rational approximations to α, from below for even n and from above for odd n.

4.2. The following relations hold between convergents and rational approximations:

(4.2.1) Every rational approximation to s is either a convergent or is of the form $[a_0, \ldots, a_{n-1}, a_n \pm 1]$. If the rational approximations are arranged in order of increasing denominators, then among any three successive ones at least one is a convergent.

(4.2.2) Of two successive convergents, at least one satisfies

$$\left|\alpha - \frac{p}{q}\right| < \frac{1}{2q^2}.$$

A rational approximation such that $|s - p/q| < 1/(2q^2)$ is a convergent.

4.3. Set

$$\|x\| = \inf_{n \in \mathbf{Z}} |x - n|.$$

The denominators q_n are the numbers q such that $\|q\alpha\| < \|q'\alpha\|$ for every $q' < q$. On $\mathbf{R}/\mathbf{Z}$, three successive points $q_n\alpha$ are arranged as follows (the drawing is for $a_{n+2} = 2$):

$$q_{n+1}\alpha \quad | \quad 0 \quad | \quad q_{n+2}\alpha \quad \underbrace{\qquad\qquad}_{a_{n+2}(q_{n+1}\alpha)} \quad q_n\alpha. \tag{4.3.1}$$

If the case $n = 0$, $a_1 = q_1 = 1$, is excluded, then $\|q_n\alpha\|$ is the distance from 0 to $q_n\alpha$, as it appears in this figure, and

$$a_{n+2} \text{ is the integer part of } \frac{\|q_n\alpha\|}{\|q_{n+1}\alpha\|} \quad (\text{for } n \geq 1), \tag{4.3.2}$$

$$\|q_{n+2}\alpha\| \leq \frac{1}{a_{n+2}+1}\|q_n\alpha\| \leq \frac{1}{2}\|q_n\alpha\|. \tag{4.3.3}$$

Set $p'_n/q'_n = [a_0, \ldots, a_n + 1]$. The arrangement is

$$\frac{p_{n+1}}{q_{n+1}} \quad | \quad \alpha \quad | \quad \frac{p'_{n+1}}{q'_{n+1}} \quad | \quad \frac{p_n}{q_n}. \tag{4.3.4}$$

Hence, by (4.1.6) and (4.1.3),

$$\frac{1}{q_n q'_{n+1}} \leq \left|\alpha - \frac{p_n}{q_n}\right| \leq \frac{1}{q_n q_{n+1}}, \tag{4.3.5}$$

that is,

$$\frac{1}{(a_{n+1}+2)q_n} \leq \|q_n\alpha\| \leq \frac{1}{a_{n+1}q_n} \quad (\text{except for } n = 0,\ q_0 = q_1 = 1). \tag{4.3.6}$$

Comparing (4.1.3) and (4.3.6), and taking (4.2.2) into account, one sees that the following requirements are equivalent:

(a) $\|q\alpha\|$ is never too small;

(b) a_n is never too large;

(c) the denominators of the rational approximations to α do not grow too quickly.

DEFINITION 4.4.— An irrational number α has bounded density if its density

$$d = \sup_{n>0} \frac{1}{n} \sum_{i=1}^{n} a_i$$

is finite.
Apply 1.7 with $s = \alpha$ of density $d < \infty$, taking the q_i to be the denominators of the successive convergents. For $n \geq 2^{N/2}$, one has $N \leq q_n$ by (4.1.5), and

$$\sum_{q_i \leq N} a_{i+1} \leq nd.$$

Formula 1.7.3 gives the following result.
PROPOSITION 4.5.— There are universal constants C, D such that, when α has bounded density d,

$$\|D^N f\|_\infty \leq C^{dV} \cdot N^{DdV}.$$

4.6. Let $T : [0,1[\to [0,1[$ send x to the fractional part of x^{-1}. The probability measure

$$\nu = \frac{1}{\log 2} \frac{dx}{1+x}$$

on $[0,1[$ satisfies $T_*\nu = \nu$. Khintchine proved that T is ν-ergodic. The n-th coefficient in the continued-fraction expansion of an irrational $x \in [0,1]$ is

$$a_n(x) = [(T^{n-1}(x))^{-1}] \qquad (n \geq 1).$$

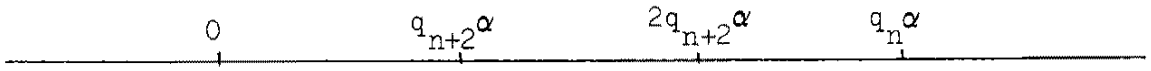
For almost every x (with respect to ν—or equivalently Lebesgue measure), an integer k therefore occurs in the sequence $a_n(x)$, $n \geq 1$, with frequency

$$\lim_{N\to\infty} \frac{1}{N} \big(\#\{n \leq N \mid a_n(x) = k\}\big) = \int_{[x^{-1}]=k} \nu \sim \frac{1}{k^2}.$$

Since $\sum_k (1/k^2)\, k$ diverges, the set of numbers of bounded density has Lebesgue measure zero.
Proposition 4.5 is the only step in Herman's proof at which one must impose on α a hypothesis that is not satisfied by almost every number.

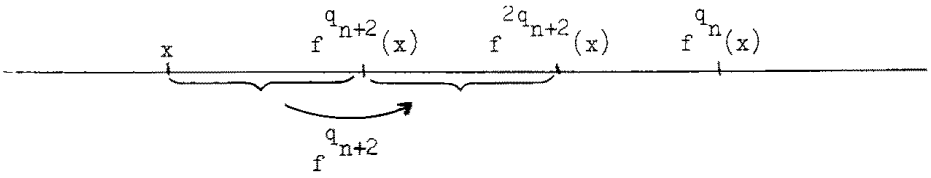
§ 5. Estimate of $|f^{q_n} - x - p_n|$

Write p_n/q_n for the n-th convergent of α. By 4.3.3, the following arrangement holds on the circle $\mathbf{R}/\mathbf{Z}$:



Authority-derived positional diagram P15-A01.

By conjugacy, the same arrangement holds for the iterates of f:



Authority-derived positional diagram P15-A02.

On $\mathbf{R}$, the same arrangement holds if f^{q_i} is replaced by $f^{q_i} - p_i$. The axis must be oriented from left to right for odd n, and from right to left for even n.

Since $Df^{q_i} \geq e^{-V}$, it follows that

$$(1 + e^{-V})\,|f^{q_{n+2}}(x) - p_{n+2} - x| \leq |f^{q_n}(x) - p_n - x|\,,$$

and hence:

PROPOSITION 5.1.— The quantities $\|f^{q_n} - x - p_n\|_\infty$ decrease at least as fast as a geometric progression.

A slight refinement of this argument proves the following theorem.

THEOREM 5.2.— Suppose that α satisfies

$$(*) \qquad \lim_{A\to\infty} \limsup_{N\to\infty} \frac{\sum\limits_{\substack{a_i>A \\ 1\leq i\leq N}} \log(a_i+1)}{\sum\limits_{1\leq i\leq N} \log(a_i+1)} = 0,$$

or, equivalently,

$$(*)' \qquad \text{For every } \varepsilon \text{ there is } A \text{ such that } \prod_{\substack{a_i>A \\ 1\leq i\leq N}} (a_i+1) \leq O(q_N^{\varepsilon}).$$

There is then a function $V \longmapsto \varepsilon(V)$, tending to 0 with V, such that

$$|f^{q_n}(x) - p_n - x| \le O\Big(q_n^{-1+\varepsilon(V)}\Big)\,.$$

Sketch of proof.
a) Let p/q and p'/q' be two rational approximations to α from above, and put $a = q\alpha - p$, $b = q'\alpha - p'$, and $c = a/b$. If $0 \le rb \le a$, where r is an integer ≥ 1, then on $\mathbf{R}/\mathbf{Z}$ one has the following arrangement (drawn for $r = 3$):

Authority-derived positional diagram P16-A01.

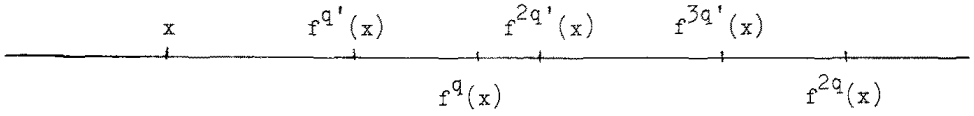
Since $Df^{q'} \ge e^{-V}$,

$$\left(\sum_{i=0}^{r-1} e^{-iV}\right)\left|f^{q'}(x) - p' - x\right| \le |f^q(x) - p - x|\,.$$

b) More generally, if $rb \le sa$, i.e. if $r/s \le c$, then

$$\left(\sum_{i=0}^{r-1} e^{-iV}\right)\left|f^{q'}(x) - p' - x\right| \le \left(\sum_{i=0}^{s-1} e^{iV}\right)|f^q(x) - p - x|\,. \tag{5.2.1}$$

For $r = 3$, $s = 2$, the drawing is:



Authority-derived positional diagram P16-A02.

c) Given ε and A, there is B such that

$$\forall x\ (2 \le x \le A) \quad \exists r, s \quad (r \le B,\ s \le B,\ x^{1-\varepsilon} < r/s < x).$$

There is then V_0 such that, if $V < V_0$, for all $r, s < B$ with $r \ge 2s$,

$$\frac{\displaystyle\sum_{i=0}^{s-1} e^{iV}}{\displaystyle\sum_{i=0}^{r-1} e^{-iV}} \le \left(\frac{s}{r}\right)^{1-\varepsilon}.$$

d) If $V < V_0$ and $2 \le c \le A$, (5.2.1) gives

$$\left|f^{q'}(x) - p' - x\right| \le c^{-1+\varepsilon'}\,|f^q(x) - p - x|\,, \tag{5.2.2}$$

where $(1 - \varepsilon') = (1 - \varepsilon)^2$.

e) Take the sequence of rational approximations p_{2n+1}/q_{2n+1}, and set

$$b_n = |q_{2n+1}\alpha - p_{2n+1}| = \|\|\|q_{2n+1}\alpha\|\|\|.$$

Then

$$2 \leq \frac{b_n}{b_{n+1}} \leq (a_{2n+2}+1)(a_{2n+3}+1).$$

If $V < V_0$, it follows that

$$\begin{cases} |f^{q_{2n+3}}(x) - p_{2n+3} - x| \leq \left(\dfrac{b_{n+1}}{b_n}\right)^{1-\varepsilon'} |f^{q_{2n+1}}(x) - p_{2n+1} - x| & (1) \\ \qquad\qquad \text{if } (a_{2n+2}+1)(a_{2n+3}+1) \leq A, & \\ |f^{q_{2n+3}}(x) - p_{2n+3} - x| \leq |f^{q_{2n+1}}(x) - p_{2n+1} - x| & (2) \\ \qquad\qquad \text{in every case.} & \end{cases}$$

Chaining these inequalities gives

$$|f^{q_{2n+1}}(x) - p_{2n+1} - x| \leq O\left(\left[\prod (a_{2i}+1)(a_{2i+1}+1)\right]^{1-\varepsilon'} \cdot b_n^{1-\varepsilon'}\right),$$

where the product is over $i \leq n$ such that $(a_{2i}+1)(a_{2i+1}+1) \geq A$.
It is bounded by the product $B_n = \prod (a_j+1)^2$, taken over $j \leq 2n+1$ such that $a_j + 1 \geq A^{1/2}$; also, b_n is bounded by $1/q_{2n+1}$. Thus

$$|f^{q_{2n+1}}(x) - p_{2n+1} - x| \leq O\left(B_n q_{2n+1}^{-1+\varepsilon'}\right).$$

The hypothesis on α allows B_n to be absorbed into ε', which proves the result.
Remark 5.3.— Condition (*) holds for almost every number and for numbers of bounded density.
Remark 5.4.— Unless the a_i are bounded, this method unfortunately does not provide a lower bound for $|f^{q_n}(x) - p_n - x|$.

§ 6. Estimate of $|Df^{q_n} - 1|$

Let $\varphi_n = f^{q_n} - p_n - x$. We shall bound $|Df^{q_n} - 1| = |D\varphi_n|$ by Hadamard's interpolation formula (2.2.1). For α of bounded density and $f \in C^3$,

$$\|\varphi_n\|_\infty \leq O\left(q_n^{-1+\varepsilon(V)}\right) \quad \text{with } \varepsilon(V) \to 0 \text{ as } V \to 0 \qquad (5.2),$$

$$\|D^2\varphi_n\|_\infty \leq O\left(q_n^{1/2} \cdot \sup_{i<q_n} \|Df^i\|_\infty\right) \quad \text{for } V < 1 \qquad (2.3),$$

$$\sup_{i<q_n} \|Df^i\|_\infty \leq O\left(q_n^{\varepsilon'(V)}\right) \quad \text{with } \varepsilon'(V) \to 0 \text{ as } V \to 0 \qquad (4.5).$$

Consequently,

$$\|D\varphi_n\|_\infty \leq O\left(q_n^{-1/4+\varepsilon''(V)}\right), \qquad \varepsilon''(V) = \tfrac{1}{2}(\varepsilon(V) + \varepsilon'(V)).$$

For every $\varepsilon > 0$, there is m such that

$$\varepsilon''(\text{Var} \log Df_m) < \varepsilon \qquad (3.3),$$

and hence

$$\|Df_m^{q_n} - 1\|_\infty \leq O\left(q_n^{-1/4+\varepsilon}\right).$$

By conjugacy ($f^{q_n} = h_m f_m^{q_n} h_m^{-1}$, $f_m^{q_n}$ is close to $\mathbf{R}/\mathbf{Z}$ by 5.2, and h_m is C^3), we finally obtain:

THEOREM 6.1.— If f is C^3 and α has bounded density, then for every $\varepsilon > 0$,

$$\|Df^{q_n} - 1\|_\infty \leq O\left(q_n^{-1/4+\varepsilon}\right).$$

§ 7. h is $C^{1+\beta}$ for $\beta < 1/3$

7.1. We shall use the following properties of the Hölder seminorm $|\varphi|_\beta$.

(7.1.1) If φ has period 1 and $g : \mathbf{R}/\mathbf{Z} \to \mathbf{R}/\mathbf{Z}$, then

$$|\varphi \circ g|_\beta \leq |\varphi|_\beta \|Dg\|_\infty^\beta.$$

(7.1.2) $|\varphi|_\beta$ is an increasing log-convex function of β.

Indeed, $\log |\varphi|_\beta$ is the supremum of the increasing linear functions

$$\log \frac{|\varphi(x)-\varphi(y)|}{|x-y|^\beta}.$$

$$|\varphi|_0 \le 2\|\varphi\|_\infty \qquad \text{and} \qquad |\varphi|_1 = \|D\varphi\|_\infty. \tag{7.1.3}$$

$$\|D\varphi\|_\infty \le \left(\frac{1+\beta}{2\beta}\right)^{\beta/(\beta+1)} \cdot \left(|\varphi|_0^\beta |D\varphi|_\beta\right)^{1/(1+\beta)}. \tag{7.1.4}$$

Suppose that the maximum of $|D\varphi|$ is attained at x_0, and, to fix ideas, that $D\varphi(x_0) \ge 0$. Then

$$D\varphi(x) \ge \|D\varphi\|_\infty - |D\varphi|_\beta (x-x_0)^\beta$$

and

$$\varphi(x) - \varphi(x_0) \ge \|D\varphi\|_\infty (x-x_0) - |D\varphi|_\beta \frac{(x-x_0)^{1+\beta}}{1+\beta}, \qquad x \ge x_0.$$

Evaluate this lower bound at x such that $\|D\varphi\|_\infty - |D\varphi|_\beta (x-x_0)^\beta = 0$. Combining it with the symmetric result for $x < x_0$, one obtains

$$\frac{1}{2}|\varphi|_0 \ge \|D\varphi\|_\infty X - |D\varphi|_\beta \frac{X^{1+\beta}}{1+\beta} = X\left(\|D\varphi\|_\infty - \frac{|D\varphi|_\beta X^\beta}{1+\beta}\right)$$

for X such that $\|D\varphi\|_\infty = |D\varphi|_\beta X^\beta$. Formula (7.1.4), of which Hadamard's formula (2.2.1) is the special case $\beta = 1$, follows.

(7.1.5) $\|\varphi\|_\beta = \|\varphi\|_\infty + |\varphi|_\beta$ is a Banach-algebra norm on the algebra of C^β functions. The formula $\|\varphi_1\varphi_2\|_\beta \le \|\varphi_1\|_\beta \|\varphi_2\|_\beta$ follows from the printed identity

$$\varphi_1(x)\varphi_2(x) - \varphi_1(y)\varphi_2(y) = \varphi_1(x)(\varphi_2(x)-\varphi_2(y)) + (\varphi_1(x)-\varphi_2(x))\varphi_2(y).$$

7.2. Assume that α has bounded density and that $f \in C^3$. Let $\varepsilon < 1/4$. By 6.1,

$$|\log Df^{q_n}| \le A \cdot q_n^{-\varepsilon}. \tag{7.2.1}$$

Let N be an integer and write $N = \sum b_i q_i$ as in 1.7. Breaking the sum

$$\log Df^N = \sum_{i<N} \log Df \circ f^i$$

into partial sums as in 1.7 gives

$$\log Df^N = \sum_i \left(\sum \text{ of } b_i \text{ terms of the form } \log Df^{q_i} \circ f^j\right). \tag{7.2.2}$$

Applying (7.2.1):

$$\|\log Df^N\|_\infty \leq A\sum_{i=0}^{\infty} a_{i+1}q_i^{-\varepsilon}. \tag{7.2.3}$$

This series converges if α has bounded density (and for almost every number), hence:
PROPOSITION 7.3.— $\|\log Df^n\|_\infty$ is uniformly bounded.
7.4. Suppose $V<1$, and interpolate by 7.1.2 between the estimate in 6.1 for $\|Df^{q_n}-1\|_\infty$ and the estimate in 2.3 for $\|D(Df^{q_n}-1)\|_\infty$. For $\beta<1/3$, this gives

$$|Df^{q_n}|_\beta \leq O(q_n^{-\varepsilon}) \qquad \text{with } \varepsilon>0. \tag{7.4.1}$$

Take $|\ |_\beta$ in (7.2.2). The compositions by f^j are harmless by (7.1.1) and 7.3, so

$$|Df^n|_\beta \leq O\Big(\sum a_{i+1}q_i^{-\varepsilon}\Big) = O(1)\,:$$

the f^n are uniformly $C^{1+\beta}$, as are the h_n, and therefore h is $C^{1+\beta}$, as explained in the introduction. Finally, 3.3 removes the hypothesis $V<1$.
PROPOSITION 7.5.— For every $\beta<1/3$, h is $C^{1+\beta}$.
§ 8. $C^{2+\beta}$ convergence of the f_n
THEOREM 8.1.— Let $0<\beta'<\beta\leq 1$. Suppose that f is $C^{2+\beta}$ and h is $C^{1+\beta}$. Then the f_n converge to R_α in the $C^{2+\beta'}$ topology.
The convergence is already known to be C^1 (3.5). Here this is also immediate from (3.3.1), since the Df^i are bounded below:

$$Df^i > A > 0.$$

The interpolation formulas (7.1.2) and (7.1.4) therefore reduce the problem to showing that the f_n remain bounded in the $C^{2+\beta}$ topology, i.e. that the $|D^2 f_n|_\beta$ remain bounded.
The f^n are uniformly $C^{1+\beta}$. Thus

$$\left|\frac{1}{n}\sum_{i<n} Df^i\right|_\beta$$

is bounded. Since the Df^i are bounded below, and

$$|\varphi^{-1}|_\beta \leq \frac{|\varphi|_\beta}{(\inf|\varphi|)^2},$$

we also have boundedness of

$$\left| \frac{n}{\sum_{i<n} Df^i} \right|_\beta .$$

By the same method as in 3.6, using the fact that the h_n and f^n are uniformly C^1 (which justifies the changes of variables), it remains to bound uniformly in the norm $\| \ \|_\beta$ of (7.1.5) the functions

$$-D\left(\frac{1}{\sum_{i<n} Df^i}\right) = \sum_{j<i<n} \lambda_{ij}\, D \log Df \circ f^j$$

where

$$\lambda_{ij} = \frac{Df^i Df^j}{\left(\sum_{i<n} Df^i\right)^2} = \frac{1}{n^2}\left[Df^i \cdot Df^j \cdot \left(\frac{n}{\sum_{i<n} Df^i}\right)^2\right] \qquad \text{(proof of } (3.7)).$$

The expression in brackets and the functions $D \log Df \circ f^j$ are uniformly bounded in the norm $\| \ \|_\beta$, so this is immediate.

§ 9. Application of an implicit-function theorem

Herman's final result is the following.

THEOREM 9.1.— If α has bounded density and f is C^r, $r \geq 3$ ($r = n + \beta$, with integer n and $0 \leq \beta < 1$), then h is $C^{r-1-\varepsilon}$ for every $\varepsilon > 0$.

He obtains it by combining 7.5 and 8.1 with an implicit-function theorem:

THEOREM 9.2.— Let α be irrational and let $1 \leq \theta < \theta'$. Suppose there is C such that

$$|q\alpha - p| \geq C \cdot q^{-\theta}$$

for all integers p, q. There is then a neighborhood V of R_α in the $C^{2\theta'}$ topology such that, for f in this neighborhood and of class C^r (respectively C^∞, C^ω), the equation

$$f = R_\lambda \circ g \circ R_\alpha \circ g^{-1}$$

with $\lambda \in \mathbf{R}$ and g an increasing homeomorphism has a $C^{r-\theta'}$ solution (respectively a C^∞, C^ω solution).

To obtain 9.1, take $1 < \theta < \theta' = 1 + \varepsilon$, take α to be the rotation number of f, reduce by 7.5 and 8.1 to the case where $f \in V$, and observe that the rotation number of

$$R_\lambda \circ g \circ R_\alpha \circ g^{-1}$$

can equal α only when $\lambda = 0$.
As already stated, the proof of 9.2 is inspired by [1], [3], and [4].

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GROUP THEORY. — Non-residually finite central extensions of arithmetic groups. Note (*) by Mr. Pierre Deligne, Member of the Academy.

[[English translation of the French abstract]]

A variant of known methods for treating the congruence subgroup problem gives results such as the following: for n $\geq$ 2, the group $\widetilde{Sp}(2n, \mathbb{Z})$, the inverse image of Sp(2n, $\mathbb{Z}$) in the universal covering of Sp(2n, $\mathbb{R}$) (a central extension of Sp(2n, $\mathbb{Z}$) by π_1 Sp(2n, $\mathbb{R}$) = $\mathbb{Z}$), is not Hausdorff for the topology defined by finite-index subgroups. More precisely, every finite-index subgroup of $\widetilde{Sp}(2n, \mathbb{Z})$ contains $2\pi_1$ Sp(2n, $\mathbb{R}$).

[[English abstract printed in the 1978 source, retained as source matter]]

A variant of known methods to handle the congruence subgroup problem yields results like the following. Let $\widetilde{Sp}(2n, \mathbb{Z})$ be the inverse image of Sp(2n, $\mathbb{Z}$) in the universal covering of Sp(2n, $\mathbb{R}$); it is a central extension of Sp(2n, $\mathbb{Z}$) by π_1 Sp(2n, $\mathbb{R}$) $\simeq \mathbb{Z}$. If n $\geq$ 2, the group $\widetilde{Sp}(2n, \mathbb{Z})$ is not residually finite : any subgroup of finite index contains $2\pi_1$ Sp(2n, $\mathbb{R}$).

1. S-ARITHMETIC GROUPS. — Let G be an almost simple, simply connected group isotropic over a global field F. To simplify the statements, we shall also suppose that G(F) is perfect (i.e. equal to its commutator subgroup); this is the case except, perhaps, when G is a trialitarian form of D_4. Let S also be a finite set of places of F containing all the infinite places, and suppose that

(i) $\sum_{v\in S} \operatorname{rank}(G/F_v) \geq 2$.

For each place v of F, put $G_v = G(F_v)$. For each finite set T of places, likewise put $G_T = \prod_{v\in T} G_v$ and $G_{T'} = \prod_{v\notin T} G_v$ (restricted product): for $\mathbb{A}$, the ring of adèles of F, one has $G(\mathbb{A}) = G_T \times G_{T'}$.

A subgroup of G(F) is called an S-congruence subgroup if it is the inverse image of a compact open subgroup of $G_{S'}$. It is called S-arithmetic if it is commensurable with an S-congruence subgroup, i.e. if it contains, with finite index, a subgroup of finite index in an S-congruence group. The S-congruence subgroups (respectively, S-arithmetic subgroups) form a neighborhood basis of the identity for a topology (of a topological group) $\mathcal{T}$(c) [respectively, $\mathcal{T}$(a)] on G(F). For a fixed S-congruence subgroup Γ, one still obtains a neighborhood basis by considering only the S-congruence subgroups normal in Γ (respectively, only the normal finite-index subgroups of Γ). This ensures the existence of the completion $\widehat{G(F)}$(c) [respectively, $\widehat{G(F)}$(a)] of G(F) for $\mathcal{T}$(c) [respectively, $\mathcal{T}$(a)]. The topology $\mathcal{T}$(c) is the topology induced from that of $G_{S'}$. Since G(F) is dense in $G_{S'}$ (the strong approximation theorem), one has $\widehat{G(F)}(c) = G_{S'}$. The completion $\widehat{G(F)}$(a) of G(F) is an extension of $\widehat{G(F)}(c) = G_{S'}$ by the profinite group C(S, G) = lim proj Γ/Δ (the projective limit over pairs $\Delta \subset \Gamma$, with Δ normal of finite index in Γ and Γ an S-congruence group).

For G of rank $\geq$ 2, Raghunathan (1) proved that:

(ii) $\widehat{G(F)}$(a) is a central extension of $\widehat{G(F)}(c) = G_{S'}$,

and he completes the verification that the same conclusion holds under hypothesis (i) alone.

Let $\widetilde{G}_S$ be a topological central extension of G_S by a discrete group A, and let $\widetilde{G(F)}$ be the inverse image of G(F) in $\widetilde{G}_S$. A subgroup of $\widetilde{G(F)}$ will be called an S-congruence subgroup if it is the inverse image of an S-congruence subgroup of G(F), and S-arithmetic if it is commensurable with an S-congruence subgroup. We again denote the corresponding topologies by $\mathcal{T}$(c) and $\mathcal{T}$(a), and the separated completions by $\widetilde{\widehat{G(F)}}$(c), $\widetilde{\widehat{G(F)}}$(a), for these

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topologies. One again has $\widehat{\widetilde{G(F)}}(\mathrm{c}) = G_{S'}$, while $\widehat{\widetilde{G(F)}}(\mathrm{a})$ is an extension of $G_{S'}$ by a profinite group $\widetilde{C}(\mathrm{S}, \mathrm{G})$, itself an extension of C(S, G) by

$\widehat{A} = \lim \mathrm{proj}\ \mathrm{A}/(\mathrm{A} \cap \Gamma)$ (Γ S-arithmetic).

PROPOSITION 1. — If condition (ii) holds, $\widehat{\widetilde{G(F)}}(\mathrm{a})$ is a central extension of $\widehat{\widetilde{G(F)}}(\mathrm{c}) = G_{S'}$.

Recall that if a group $\widetilde{H}$ is a central extension of a group H, the commutator map $\widetilde{H} \times \widetilde{H} \to \widetilde{H}$ factors through a map, again denoted $(\ ,\ ) : \mathrm{H} \times \mathrm{H} \to \widetilde{H}$. In particular, since $\widehat{\widetilde{G(F)}}(\mathrm{a})$ is a central extension of $\widehat{G(F)}(\mathrm{a})$ by $\widehat{A}$, the commutator map $\widetilde{C}(\mathrm{S}, \mathrm{G}) \times \widehat{\widetilde{G(F)}}(\mathrm{a}) \to \widehat{\widetilde{G(F)}}(\mathrm{a})$ factors through $\mathrm{C}(\mathrm{S}, \mathrm{G}) \times \widehat{G(F)}(\mathrm{a})$. By (ii), it factors through a bimultiplicative map

$\mathrm{C}(\mathrm{S}, \mathrm{G}) \times \widehat{G(F)}(\mathrm{a}) \to \widehat{A}$;

since G(F) is perfect and dense in $\widehat{G(F)}(\mathrm{a})$, this map is trivial.

More concretely, the proposition means that for every S-arithmetic subgroup Δ one has:

(1 a) For every conjugate Δ^g of Δ ($\mathrm{g} \in \widehat{G(F)}$), there exists an S-congruence group Γ such that $\Gamma \cap \Delta = \Gamma \cap \Delta^g$.

(1 b) $\widehat{G(F)}$ acts trivially on the finite group $C_\Delta = \lim \mathrm{proj}\ \Gamma/(\Gamma \cap \Delta)$.

These conditions are also equivalent to:

(1 c) For every $\mathrm{g} \in \mathrm{G}(\mathrm{F})$, there exists an S-congruence group Γ such that $(\mathrm{g}, \Gamma) \subset \Delta$.

One has $\widetilde{C}(\mathrm{S}, \mathrm{G}) = \lim \mathrm{proj}\ C_\Delta$.

2. CENTRAL EXTENSIONS OF ADELIC GROUPS. — From now on we shall suppose that:

(iii) the group $\widetilde{G}_S$ is perfect.

The most interesting case would be that in which $\widetilde{G}_S$ is the universal central extension of G_S by a discrete group (the "universal covering" of G_S), but we do not wish to assume the existence of such a universal extension.

A topological central extension E of $\mathrm{G}(\mathbb{A})$ by a discrete group X is determined by the extensions E_S and $E_{S'}$ of G_S and $G_{S'}$ by X, obtained as the inverse images of G_S and $G_{S'}$ in E: because G_S is perfect, one has

$\mathrm{E} \leftarrow\sim (E_S \times E_{S'})/\{(\mathrm{x}, -\mathrm{x}) \mid \mathrm{x} \in \mathrm{X}\}$.

If the extension E_S is dominated by $\widetilde{G}_S$, i.e. is obtained from $\widetilde{G}_S$ by pushout along $\mathrm{f} : \mathrm{A} \to \mathrm{X}$, then f is unique, and the data of E amount to the data of the extension $E_{S'}$ together with f (an equivalence of categories); one has

$\mathrm{E} \leftarrow\sim \widetilde{G}_S \times E_{S'}/\{(\mathrm{a}, -\mathrm{f}(\mathrm{a})) \mid \mathrm{a} \in \mathrm{A}\}$.

Let $\mathcal{L}$ be the category of topological central extensions of $\mathrm{G}(\mathbb{A})$ by a finite group, split over G(F), and with E_S dominated by $\widetilde{G}_S$:

$1 \to \mathrm{X} \to \mathrm{E} \to \mathrm{G}(\mathbb{A}) \to 1$
$\nwarrow \mathrm{s} \uparrow$
$\mathrm{G}(\mathrm{F})$

It would amount to the same thing to require "splittable over G(F)": since G(F) is perfect, the splitting, if it exists, is unique. The category $\mathcal{L}$ is filtered on the left. For (E, s) in $\mathcal{L}$, the closure in E of (image of $\widetilde{G}_S$) $\cdot$ sG(F) maps onto $\mathrm{G}(\mathbb{A})$: its image is closed, because X is finite, and it contains $G_S \cdot \mathrm{G}(\mathrm{F})$, which is dense (strong approximation). The subcategory L of $\mathcal{L}$ consisting of the pairs (E, s) for which (image of $\widetilde{G}_S$) $\cdot$ sG(F) is dense in E is therefore cofinal; since there is at most one arrow between any two objects of L, it reduces to a filtered ordered set.

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PROPOSITION 2. — Under hypotheses (ii) and (iii), we construct a natural isomorphism between $\widetilde{C}$(S, G) and the projective limit over $\mathcal{L}$, or L, lim proj X. Under this isomorphism, the natural map from A to $\widetilde{C}$(S, G) is the limit of the maps $-$f : A $\to$ X, where f is such that E_S is obtained from $\widetilde{G}_S$ by f.

Let E be a topological central extension of G($\mathbb{A}$) by a discrete group X. Giving a splitting s of E over G(F) amounts to giving an isomorphism between the extension of G(F) by X induced from E_S and the opposite of the extension induced from $E_{S'}$. When E_S is dominated by $\widetilde{G}_S$ and f is as above, this amounts to giving a commutative diagram

$1 \to \mathrm{X} \to E_{S'} \to G_{S'} \to 1$
$\uparrow -\mathrm{f} \uparrow \mathrm{t} \uparrow$
$1 \to \mathrm{A} \to \widetilde{G(F)} \to \mathrm{G(F)} \to 1.$

Hence there is an equivalence between the category $\mathcal{L}$ and the category $\mathcal{L}'$ of commutative diagrams

(1) $1 \to \mathrm{X} \to E_{S'} \text{—q}\to G_{S'} \to 1$
$\uparrow \mathrm{t} \uparrow$
$\widetilde{G(F)} \to \mathrm{G(F)}$

(where $E_{S'}$ is a topological central extension of $G_{S'}$ by a finite group): E is reconstructed as $\widetilde{G}_S \times E_{S'}/\{(\mathrm{a}, \mathrm{ta}) \mid \mathrm{a} \in \mathrm{A}\}$, and the splitting over G(F) is given by s(g) = $(\tilde{g}, \mathrm{t}\tilde{g})$ modulo the elements (a, ta), for $\tilde{g} \in \widetilde{G(F)} \subset \widetilde{G}_S$ mapping to g in G(F). The subcategory L of $\mathcal{L}$ corresponds to the subcategory L′ of diagrams (1) in which t has dense image.

Let $(E_{S'}, \mathrm{t})$ be in L′, and let $\mathcal{T}$ be the topology on $\widetilde{G(F)}$ induced by that of $E_{S'}$. If K is a compact open subgroup of $E_{S'}$ disjoint from X, the intersections of $\Delta = t^{-1}\mathrm{K}$ with the S-congruence subgroups form a neighborhood basis for $\mathcal{T}$. For Γ an S-congruence subgroup defined by a compact open subgroup $\mathrm{K}_1 \subset \mathrm{qK}$, one furthermore has

(2) $\mathrm{X} \to\sim \mathrm{XK/K} \leftarrow\sim q^{-1}\mathrm{K}_1/(\mathrm{K} \cap q^{-1}\mathrm{K}_1) \leftarrow\sim \Gamma/(\Delta \cap \Gamma) \leftarrow\sim C_\Delta$.

Since X is finite, Δ is S-arithmetic. Conversely, Proposition 1 ensures that a neighborhood basis of S-arithmetic groups is obtained in this way: the filtered ordered set L′ identifies with the set of germs (for the S-congruence topology) of S-arithmetic groups, and the isomorphism of Proposition 2 is defined as the projective limit of the isomorphisms (2).

3. QUASI-SPLIT GROUPS. — Let μ be the group of roots of unity in F and; for every non-complex place v of F, let μ_v be the group of roots of unity in F_v. For complex v, put $\mu_v = \{1\}$. Deodhar proved that, for G quasi-split, there exists a universal topological central extension $\widetilde{G}_v$ of G_v by a discrete group (the "universal covering"), and that for v non-real its kernel $\pi_1(G_v)$ is canonically a quotient of μ_v. I completed this result by proving that in fact $\pi_1(G_v) = \mu_v$. For real v, one has $\pi_1(G_v) = \mathbb{Z}$ or $\mathbb{Z}/2$. In every case, μ_v occurs as a quotient of $\pi_1(G_v)$.

For almost every v, the extension $\widetilde{G}_v$ splits over the natural maximal compact subgroup of G_v. This makes it possible to define the restricted product of the $\widetilde{G}_v$, an extension of G($\mathbb{A}$) by the direct sum of the $\pi_1(G_v)$. Push it out by

$\mathrm{R} : \oplus_v \pi_1(G_v) \to \oplus_v \mu_v \text{—}\sigma \to \mu : \sigma((x_v)) = \prod_v x_v^{[\mu_v:\mu]}$.

This gives a central extension of G($\mathbb{A}$) by μ, and it follows from Deodhar (2) and C. Moore (3) that this is the universal topological central extension of G($\mathbb{A}$) by a discrete group, split over G(F):

(3) $1 \to \mu \to \widetilde{G(\mathbb{A})} \to \mathrm{G}(\mathbb{A}) \to 1$
$\nwarrow \uparrow$
G(F)

Take $\widetilde{G}_S$ to be the universal covering of G_S, and let R_S be the restriction of R to $\pi_1 G_S = \prod_{v \in S} \pi_1(G_v)$. Proposition 2 identifies the completion $\widehat{\widetilde{G(F)}}$(a) of $\widetilde{G(F)}$ with the inverse image $\widetilde{G(\mathbb{A})}_{S'}$ of $G_{S'} = \widehat{\widetilde{G(F)}}$(c) in (3), and gives an exact commutative diagram

(4) $1 \to \mu \to \widetilde{G(\mathbb{A})}_{S'} \to G_{S'} \to 1$
$\uparrow -R_S \uparrow \mathrm{i} \uparrow$
$\prod_{v \in S} \pi_1(G_v) \to \widehat{\widetilde{G(F)}}(\mathrm{a}) \to \widehat{G(F)}(\mathrm{a}) \to 1$

In particular:

THEOREM. — Let G/F be quasi-split, and let $\widetilde{G(F)}$ be the inverse image of G(F) in the universal covering of G_S. If $\sum_{v \in S} \mathrm{rank}(G_v) \geq 2$, the intersection of the S-arithmetic subgroups of $\widetilde{G(F)}$ is the kernel of $R_S : \prod_{v \in S} \pi_1(G_v) \to \mu$.

For Sp(2n, $\mathbb{Q}$) and S = ∞, this is the result announced in the introduction. For SL(n, K), with K real quadratic and S = ∞, the resulting statement generalizes that of J. Millson (4).

PROBLEM. — Does every discrete finite-covolume subgroup of the universal covering of Sp(2n, $\mathbb{R}$) (n $\geq$ 2) contain $2\pi_1$ Sp(2n, $\mathbb{R}$)? (5).

4. THE SYSTEM OF FACTORS OF THE θ FUNCTIONS. — When condition (i) is not satisfied, Proposition 1 is no longer available; it nevertheless remains possible to study S-arithmetic subgroups satisfying a condition of type (1 a).

Let S therefore be a nonempty finite set of places containing all the infinite places, and let T $\supset$ S be such that (G, T) satisfies condition (ii). The interesting case is that in which S consists of a single place at which G has rank 1. Let $\widetilde{G}_S$ again be a topological central extension of G_S by a discrete group, and let $\widetilde{G(F)}$ be the inverse image of G(F) in $\widetilde{G}_S$. For an S-arithmetic subgroup Δ of $\widetilde{G(F)}$, consider the condition:

($\star$) there exists a T-arithmetic subgroup U of $\widetilde{G(F)}$ such that, for every conjugate Δ^g of Δ (g $\in$ U), there exists an S-congruence group Γ for which $\Gamma \cap \Delta = \Gamma \cap \Delta^g$ (U-invariance of the germ of Δ).

PROPOSITION 3. — The completion $\widehat{\widetilde{G(F)}}$(a*) of $\widetilde{G(F)}$ for the topology of S-arithmetic subgroups satisfying ($\star$) is a central extension of $\widehat{\widetilde{G(F)}}$(c) = $G_{S'}$.

More concretely, we must verify that if (U, Δ) satisfies ($\star$), then Δ satisfies (1 c). To prove this, we may first shrink U and Δ. In particular, we may suppose that $\Delta \subset$ U (replacing Δ by $\Delta \cap$ U). The group U then acts by conjugation on the

finite group C_Δ^U = lim proj $(\Gamma \cap U)/(\Gamma \cap \Delta)$ (Γ S-congruence); replacing U by the kernel of this action and Δ by some $\Delta \cap \Gamma$, we may further suppose that U acts trivially on C_Δ^U.

Since U is T-arithmetic, its completion U(c) $\subset G_{S'}$ has finite index in a subgroup $G_{T-S} \times L$, with $L \subset G_{T'}$ compact open; since G_{T-S} has no open subgroup of finite index, U(c) itself has the form $G_{T-S} \times L$. Condition ($\star$) ensures that the subgroups $\Delta \cap \Gamma$ (Γ S-congruence) form a neighborhood basis of the identity for a group topology on U. The corresponding completion U(Δ) is an extension of U(c) by C_Δ^U—and hence a central extension of $G_{T-S} \times L$. Replace L by an open subgroup L_1 over which this extension is trivial, and replace U and Δ by the inverse images of L_1. Since G_{T-S} is perfect, a central extension of $G_{T-S} \times L$ is determined by its restrictions to the two factors; this reduces us to the case in which the central extension U(Δ) of $G_{T-S} \times L$ is the inverse image of a central extension $\widetilde{G}_{T-S}$ of G_{T-S} by C_Δ^U. By construction, U lifts to $\widetilde{G}_{T-S}$, and Δ is the inverse image of a compact open subgroup K of $\widetilde{G}_{T-S} \times L$:

$$\begin{array}{ccc} (5)\ K & \hookrightarrow & \widetilde{G}_{T-S} \times L \\ \uparrow & & \uparrow \\ \Delta & \longrightarrow & U \end{array}$$

Let $\widetilde{G(F)}_T$ be the central extension of G(F) obtained as the inverse image of G(F) in $\widetilde{G}_T = \widetilde{G}_S \times \widetilde{G}_{T-S}$, and let $\widetilde{U}$ be the lifting (5) of U in $\widetilde{G(F)}_T$. For g $\in$ G(F), Proposition 1 gives (g, Γ_T) $\subset \widetilde{U}$ for a T-congruence group Γ_T defined by a sufficiently small $L' \subset L$. If the S-congruence subgroup Γ defined by $K' \subset G_{T-S} \times L'$, whose inverse image in $\widetilde{G}_{T-S} \times L'$ is K″, is small enough that (g, K′) $\subset$ K, then (g, Γ) $\subset \Delta$, which verifies (1 c).

The kernel $\widetilde{C}^*$(S, G) = Ker($\widetilde{\widetilde{G(F)}}$(a*) $\to$ $\widetilde{G(F)}$(c)) is still described by Proposition 2 (with the same proof), and the calculation can be carried through in the quasi-split case (cf. §3). Here is an application, with G = SL(2, $\mathbb{Q}$), S = $\{\infty\}$, and T = $\{\infty, p\}$. Let $\widetilde{SL(2,\mathbb{R})}$ be the double covering of SL(2, $\mathbb{R}$). The factor systems of the θ functions, restricted to arbitrarily small congruence subgroups of SL(2, $\mathbb{Z}$), may be interpreted as defining a germ τ (for the congruence-subgroup topology) of liftings of SL(2, $\mathbb{Z}$) into $\widetilde{SL(2,\mathbb{R})}$. This germ is invariant under GL(2, $\mathbb{Q}$). If a modular form of half-integral weight k/2 is defined to be a function on the Poincaré half-plane which, under the action of a congruence subgroup of SL(2, $\mathbb{Z}$), transforms like the θ function of a quadratic form in k variables, this amounts to saying that if

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{Q})$$

has positive determinant and f is modular of weight k/2, then $f(-z^{-1})$ and $(cz+d)^{k/2}$ f[(az + b)/(cz + d)] are likewise modular. Conversely:

PROPOSITION 4. — For every prime p, the germ τ is the unique germ of liftings of SL(2, $\mathbb{Z}$) into $\widetilde{SL(2,\mathbb{R})}$ that is invariant under a finite-index subgroup of SL(2, $\mathbb{Z}$[1/p]).

The arguments leading to the theorem of §3 show that every arithmetic subgroup of SL(2, $\mathbb{Z}$) whose germ is invariant under a finite-index subgroup of SL(2, $\mathbb{Z}$[1/p]) is a congruence subgroup.

Let $\Gamma \subset$ SL(2, $\mathbb{Z}$) be a congruence subgroup, and let $\tau_1 = \tau_2$ be two liftings of Γ into $\widetilde{SL(2,\mathbb{R})}$.

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Let Δ be the subgroup of Γ on which $\tau_1 = \tau_2$. If the germs of τ_1 and τ_2 are distinct and invariant under a {p}-arithmetic group, then Δ satisfies $(\star)$ and is not a congruence subgroup.

(*) Session of July 3, 1978.

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(5) I learn from Mostow that such a subgroup Γ contains, in any case, a finite-index subgroup of π_1 Sp(2n, $\mathbb{R}$): one verifies that the projection of Γ into Sp(2n, $\mathbb{R}$) is discrete, since otherwise the Lie algebra of its closure would be nontrivial and central.

Institut des Hautes Études scientifiques, 91440 Bures-sur-Yvette

PIERRE DELIGNE · D030

Values of L-functions and periods of integrals

P. Deligne

Standalone English edition

34 authority pages (printed pages 313–346). The French main paper is translated into English; the embedded Koblitz–Ogus appendix is replayed in its source English. The text is organized in exact authority-page units; repeated proceedings copy matter is excluded and documented in the separate apparatus.

Edition note

This maintenance edition is a source-checked, math-typeset reconstruction keyed one-to-one to the controlling 34-page authority scan. The comparator is retained for comparison only. The Koblitz–Ogus appendix on authority pages 31–34 is source-English in both language editions. Page-unit markers preserve the authority and printed-page topology. The certified image fallback for diagram (1.3.1) is embedded at its exact page unit, with both the lossless crop and presentation derivative retained in the source package.

In this article I state a conjecture (1.8, 2.8) relating the values of certain L-functions at certain integral points to periods of integrals.

The L-functions considered are those of motives—a word to which no precise meaning will be attached. These include, in particular, Artin L-functions, L-functions attached to algebraic Hecke characters (= Grössencharakter of type A_0), and those attached to primitive holomorphic modular forms on the Poincaré upper half-plane (= new forms; all the functions L, L_k, attached to the symmetric powers Sym^k of the corresponding l-adic representation are considered).

This article owes its existence to D. Zagier: to his insistence that a conjecture be sought, and to the experimental confirmation he supplied, as soon as it was formulated, for the functions L_3 and L_4 attached to $\Delta = \Sigma\, \tau(n)q^n$ (see [18]). It was this confirmation that gave me the confidence needed to verify that the conjecture was compatible with Shimura's results [13] on values of L-functions of algebraic Hecke characters.

0. Motives. The reader is invited not to consult this section except as needs arise. We recall part of the formalism of motives, due to Grothendieck. For proofs, I refer to [8].

0.1. Grothendieck's definition of motives over a field k has the following form. (a) Let $\mathscr{V}(k)$ be the category of smooth projective varieties over k. One constructs an additive category $\mathscr{M}'(k)$, for which the groups Hom(M, N) are vector spaces over $\mathbb{Q}$, equipped with

α. A tensor product $\otimes$, associative, commutative, and distributive with respect to addition of objects ("associative" and "commutative" are not properties of the functor $\otimes$, but data subject to certain compatibilities—cf. Saavedra [19]);

β. A contravariant functor H^* from $\mathscr{V}(k)$ to $\mathscr{M}'(k)$, bijective on objects and transforming disjoint sums into sums and products into tensor products (the datum of an isomorphism of functors $H^*(X \times Y) = H^*(X) \otimes H^*(Y)$, compatible with associativity and commutativity).

The essential point is to define $\mathrm{Hom}(H^*(X), H^*(Y))$. For the definition, used—

by Grothendieck, of this group as a group of classes of correspondences between X and Y, see 0.6.

(b) Recall (SGA 4 IV 7.5) that an additive category is Karoubian if every projector (= idempotent endomorphism) is defined by a direct-sum decomposition, and that every additive category has a Karoubian envelope, obtained by formally adjoining the images of projectors. The category $\mathscr{M}_{\text{eff}}(k)$ of effective motives over k is the Karoubian envelope of $\mathscr{M}'(k)$.

(c) The Tate motive $\mathbb{Z}(-1)$ is defined as a suitable direct factor $H^2(\mathbb{P}^1)$ of $H^*(\mathbb{P}^1)$. One checks that the symmetry (deduced from the commutativity of $\otimes$), $\mathbb{Z}(-1) \otimes \mathbb{Z}(-1) \to \mathbb{Z}(-1) \otimes \mathbb{Z}(-1)$, is the identity, and that the functor $M \to M \otimes \mathbb{Z}(-1)$ is fully faithful.

(d) The category $\mathscr{M}(k)$ of motives over k is obtained from $\mathscr{M}_{\text{eff}}(k)$ by making the functor $M \to M \otimes \mathbb{Z}(-1)$ invertible.

Let (n) denote the (−n)th iterate of the auto-equivalence $M \to M \otimes \mathbb{Z}(-1)$. The category $\mathscr{M}(k)$ contains $\mathscr{M}_{\text{eff}}(k)$ as a full subcategory, and every object of $\mathscr{M}(k)$ is of the form M(n), with M in $\mathscr{M}_{\text{eff}}(k)$ and n an integer.

By definition, if F is an additive functor from $\mathscr{M}'(k)$ to a Karoubian category $\mathscr{A}$, it extends to $\mathscr{M}_{\text{eff}}(k)$. If $\mathscr{A}$ is endowed with an auto-equivalence $A \to A(-1)$, and the extension of F with an isomorphism of functors $F(M(-1)) = F(M)(-1)$, it extends to $\mathscr{M}(k)$.

EXAMPLE 0.1.1. If k′ is an extension of k, and F is the functor $H^*(X) \to H^*(X \otimes_k k')$ from $\mathscr{M}'(k)$ to $\mathscr{M}'(k')$, one obtains the scalar-extension functor from $\mathscr{M}(k)$ to $\mathscr{M}(k')$. If k′ is a finite extension of k, one has Grothendieck's restriction-of-scalars functor

$\Pi_{k'/k}: \mathscr{V}(k') \to \mathscr{V}(k)$: $(X \to \text{Spec}(k')) \mapsto (X \to \text{Spec}(k') \to \text{Spec}(k))$.

It extends by F: $H^*(X) \mapsto H^*(\Pi_{k'/k}X)$, yielding a restriction-of-scalars functor $R_{k'/k}: \mathscr{M}(k') \to \mathscr{M}(k)$.

EXAMPLE 0.1.2. Let $\mathscr{H}$ be a "cohomology theory" with values in a Karoubian category $\mathscr{A}$, functorial for morphisms in $\mathscr{M}'(k)$. The functor $\mathscr{H}$ extends to $\mathscr{M}_{\text{eff}}(k)$. If $\mathscr{A}$ has a tensor product, $\mathscr{H}$ satisfies a Künneth formula, and tensoring with $\mathscr{H}(\mathbb{Z}(-1))$ is an auto-equivalence of $\mathscr{A}$, then it extends to $\mathscr{M}(k)$. This extension is the functor "realization of a motive in the theory $\mathscr{H}$."

We shall write (n) for the auto-equivalence of $\mathscr{A}$ obtained as the (−n)th iterate of tensoring with $\mathscr{H}(\mathbb{Z}(-1))$. For the determination of $\mathscr{H}(\mathbb{Z}(-1))$ in various theories $\mathscr{H}$, see 3.1.

0.2. We shall use the following realizations: 0.2.1. Betti realization H_B. Corresponding to $k = \mathbb{C}$, $\mathscr{A}$ = vector spaces over $\mathbb{Q}$, $\mathscr{H}$ = rational cohomology: $X \mapsto H^*(X(\mathbb{C}), \mathbb{Q})$; 0.2.2. de Rham realization H_{DR}. Corresponding to k of characteristic 0, $\mathscr{A}$ = vector spaces over k, $\mathscr{H}$ = de Rham cohomology: $X \mapsto H^*(X, \Omega_X{}^*)$; 0.2.3. l-adic realization H_l. Corresponding to algebraically closed k of characteristic $\neq l$, $\mathscr{A}$ = vector spaces over $\mathbb{Q}_l$, $\mathscr{H}$ = l-adic cohomology: $X \mapsto H^*(X, \mathbb{Q}_l)$.

And their variants: 0.2.4. Hodge realization. Here k is an algebraic closure of $\mathbb{R}$, and $\mathscr{A}$ is the category of vector spaces over $\mathbb{Q}$ whose scalar extension $V \otimes k$ is endowed with a bigrading

$V \otimes k = \oplus_{p,q} V^{p,q}$, with $V^{q,p}$ the complex conjugate of $V^{p,q}$. For the cohomology theory, one takes the functor $X \mapsto H^*(X(k), \mathbb{Q})$, endowed with the bigrading of its scalar extension $H^*(X(k), k)$ supplied by Hodge theory. The Betti realization underlies the Hodge realization.

0.2.5. For arbitrary k and a complex embedding σ of k, write $H_\sigma(M)$ for the Betti realization of the motive over $\mathbb{C}$ obtained from M by scalar extension through σ: $k \to \mathbb{C}$. Let c denote complex conjugation. Transport of structure gives an isomorphism F_∞: $H_\sigma(M) \cong H_{c\sigma}(M)$, and $F_\infty \otimes c$ sends ${H_\sigma}^{p,q}$ to ${H_{c\sigma}}^{q,p}$. If σ is real, F_∞ is an involution of $H_\sigma(M)$, whose complexification interchanges ${H_\sigma}^{p,q}(M)$ and ${H_\sigma}^{q,p}(M)$.

The σ-Hodge realization is $H_\sigma(M)$, endowed with its Hodge bigrading and, when σ is real, the involution F_∞. For $k = \mathbb{Q}$, or $k = \mathbb{R}$ with σ the identity embedding, the index σ is replaced by B. For $k = \mathbb{R}$ and $M = H^*(X)$, the involution F_∞ on ${H_B}^*(M) = H^*(X(\mathbb{C}), \mathbb{Q})$ is induced by complex conjugation F_∞: $X(\mathbb{C}) \to X(\mathbb{C})$.

0.2.6. de Rham realization. de Rham cohomology carries a natural filtration, the Hodge filtration, arising as the abutment of the hypercohomology spectral sequence

${E_1}^{p,q} = H^q(X, \Omega^p) \Rightarrow {H_{DR}}^{p+q}(X)$.

This gives a filtration F on $H_{DR}(M)$.

0.2.7. l-adic realization. If X is a variety over a field k, with algebraic closure $\bar{k}$, the Galois group $Gal(\bar{k}/k)$ acts by transport of structure on $H^*(X_{\bar{k}}, \mathbb{Q}_l)$. If M is a motive over k and $H_l(M)$ is defined as the l-adic realization of the motive over $\bar{k}$ obtained from it by scalar extension, this gives an action of $Gal(\bar{k}/k)$ on $H_l(M)$.

0.2.8. Finite adelic realization. For algebraically closed k of characteristic 0, the l-adic cohomologies can be assembled into a finite adelic cohomology

$H^*(X, \mathbb{A}^f) = (\prod_l H^*(X, \mathbb{Z}_l)) \otimes_{\mathbb{Z}} \mathbb{Q}$.

The variants 0.2.7 and 0.2.8 may be combined.

0.2.9. In Examples 0.2.1, 0.2.2, 0.2.3 and their variants, the category $\mathscr{A}$ may be replaced by the category $\mathscr{A}^*$ of graded objects of $\mathscr{A}$, using the natural grading of H^*. In order that the Künneth formula yield an isomorphism of functors $\mathscr{H}(M \otimes N) \cong \mathscr{H}(M) \otimes \mathscr{H}(N)$, compatible with associativity and commutativity, the commutativity datum in $\mathscr{A}^*$ must be the one given by the Koszul rule.

0.3. For $k = \mathbb{C}$, the comparison theorem between classical and étale cohomology gives an isomorphism $H^*(X(\mathbb{C}), \mathbb{Q}) \otimes \mathbb{Q}_l \cong H^*(X, \mathbb{Q}_l)$. If X is defined over $\mathbb{R}$, this isomorphism carries F_∞ (0.2.5) to the action (0.2.7) of complex conjugation. For a motive one likewise has $H_B(M) \otimes \mathbb{Q}_l \cong H_l(M)$.

0.4. For $k = \mathbb{C}$, the holomorphic de Rham complex on X^{an} is a resolution of the constant sheaf $\mathbb{C}$. By GAGA one therefore has an isomorphism

$H^*(X(\mathbb{C}), \mathbb{Q}) \otimes \mathbb{C} = H^*(X(\mathbb{C}), \mathbb{C}) \cong \mathbb{H}^*(X^{an}, \Omega^{*an}) \cong H^*(X, \Omega^*)$.

For a motive one obtains an isomorphism, compatible with the Hodge filtrations in 0.2.4—$F^p = \oplus_{p' \geq p} V^{p',q'}$—and 0.2.6: $H_B(M) \otimes \mathbb{C} \cong H_{DR}(M)$.

More generally, let M be a motive over a field k and σ a complex embedding of k. Applying the preceding construction to the motive over $\mathbb{C}$ obtained from M by scalar extension gives an isomorphism

(0.4.1) I: $H_\sigma(M) \otimes \mathbb{C} \cong H_{DR}(M) \otimes_{k,\sigma} \mathbb{C}$.

In the particular case $k = \mathbb{Q}$, this gives two natural $\mathbb{Q}$-structures on the realization of M in complex cohomology: one, $H_B(M)$, linked to the description of cohomology in terms of cycles; the other, $H_{DR}(M)$, linked to its description in terms of algebraic differential forms.

0.5. The behavior of these realizations and compatibilities under extension of the scalar field is clear. For restriction of scalars, the results are as follows. Let k' be a finite extension of k, let M' be a motive over k', and let $M = R_{k'/k}(M')$.

(0.5.1) $H_\sigma(M) = \oplus_\tau H_\tau(M')$,

where the sum extends over the set $J(\sigma)$ of complex embeddings of k' extending σ. This isomorphism is compatible with the Hodge bigradings and with F_∞.

(0.5.2) $H_{DR}(M) = H_{DR}(M')$ (restriction of scalars from k' to k).

This isomorphism is compatible with the Hodge filtration.

(0.5.3) $H_l(M) = \mathrm{Ind}\, H_l(M')$ (the representation induced from $\mathrm{Gal}(\bar{k}/k')$ to $\mathrm{Gal}(\bar{k}/k)$).

(0.5.4) Via the isomorphism $k' \otimes_{k,\sigma} \mathbb{C} = \mathbb{C}^{J(\sigma)}$, and the resulting isomorphism $H_{DR}(M') \otimes_{k,\sigma} \mathbb{C} = \oplus_{\tau \in J(\sigma)} H_{DR}(M') \otimes_{k',\tau} \mathbb{C}$, (0.4.1) for M and σ is the sum of the isomorphisms (0.4.1) for M' and τ ($\tau \in J(\sigma)$):

```
H_σ(M) ⊗ ℂ                --(0.4.1), ≅-->  H_DR(M) ⊗_{k,σ} ℂ
      ‖ (0.5.1)                                ‖ (0.5.2)
⊕_τ H_τ(M′) ⊗ ℂ            --⊕(0.4.1), ≅--> ⊕_τ H_DR(M′) ⊗_{k′,τ} ℂ.
```

0.6. For X smooth and projective over k, let $Z^d(X)$ be the $\mathbb{Q}$-vector space with basis the irreducible closed subschemes of codimension d in X, and let ${Z_R}^d(X)$ be its quotient by an equivalence relation R. In characteristic 0, one of Grothendieck's definitions of motives is obtained by setting, for connected X and Y,

$\mathrm{Hom}(H^*(Y), H^*(X)) = {Z_R}^{dim(Y)}(X \times Y)$,

with R equal to homological equivalence. In characteristic p there are two difficulties to note: it is not known whether homological equivalence in l-adic cohomology ($l \neq p$) is independent of l, and defining the class of a cycle raises problems in crystalline cohomology.

0.7. Let X be a smooth projective complex variety. A Hodge cycle of codimension d on X will mean an element of $H^{2d}(X(\mathbb{C}), \mathbb{Q})$ of type (d, d), or equivalently an element of $H^{2d}(X(\mathbb{C}), \mathbb{Q})(d)$ of type (0, 0).

Let k be an algebraically closed field and X a smooth projective variety over k. Set

$H^{2d}(X, k \times \mathbb{A}^f)(d) = H_{DR}{}^{2d}(X)(d) \times H^{2d}(X, \mathbb{A}^f)(d)$.

For $k = \mathbb{C}$, the image in

$H^{2d}(X, k \times \mathbb{A}^f)(d) = H^{2d}(X(\mathbb{C}), \mathbb{Q})(d) \otimes (k \times \mathbb{A}^f)$

of a Hodge cycle is again called a Hodge cycle. For algebraically closed k admitting complex embeddings, an absolute Hodge cycle of codimension d on X is an element of $H^{2d}(X, k \times \mathbb{A}^f)(d)$ whose image in $H^{2d}(X \otimes_k \mathbb{C}, \mathbb{C} \times \mathbb{A}^f)(d)$ is a Hodge cycle for every complex embedding σ of k. One verifies the following.

PROPOSITION 0.8. (i) The $\mathbb{Q}$-vector space $Z_{ha}{}^d(X)$ of absolute Hodge cycles is invariant under extension of scalars from k to an algebraically closed field k′ that still admits a complex embedding.

(ii) Suppose k is algebraically closed of characteristic 0 and X is defined over an algebraically closed subfield k_0 of k admitting a complex embedding, with $X = X_0 \otimes_{k_0} k$. Set

$Z_{ha}{}^d(X) = Z_{ha}{}^d(X_0) \subset H^{2d}(X_0, k_0 \times \mathbb{A}^f)(d) \subset H^{2d}(X, k \times \mathbb{A}^f)(d)$.

By (i), this definition is independent of the choices of X_0 and k_0.

(iii) If X is defined over a subfield k_0 of k, with $X = X_0 \otimes_{k_0} k$, then $\mathrm{Aut}(k/k_0)$, acting on $H^{2d}(X, k \times \mathbb{A}^f)(d)$, stabilizes $Z_{ha}{}^d(X)$. Its action on $Z_{ha}{}^d(X)$ factors through a finite group corresponding to a finite extension k'_0 of k_0. Set $Z_{ha}{}^d(X_0) = Z_{ha}{}^d(X)^{Aut(k/k_0)}$.

0.9. A useful notion of motive is obtained by setting, for connected X and Y,

$\mathrm{Hom}(H^*(Y), H^*(X)) = Z_{ha}{}^{dim(Y)}(X \times Y)$.

The Künneth components of the diagonal of $X \times X$ are absolute Hodge cycles. This makes it possible to decompose $H^*(X)$ into a sum of motives $H^i(X)$, to grade the category of motives—with $H^i(X)$ of weight i—and to modify the commutativity constraint for $\otimes$ as in [19, VI, 4.2.1.4]. With these conventions, the tensor category of motives over k is Tannakian and isomorphic to the category of representations of a reductive proalgebraic group.

For $k = \mathbb{C}$, the following conjecture, weaker than the Hodge conjecture, is equivalent to saying that the Hodge-realization functor is an equivalence between the category of motives in 0.9 and the category of Hodge structures that occur as direct factors of the cohomology of algebraic varieties, or are obtained from such factors by Tate twist.

Hope 0.10. Every Hodge cycle is absolute.

If X is an abelian variety with complex multiplication by an imaginary quadratic field K, with Lie(X) free over $k \otimes K$, the methods of B. Gross [7] prove that certain nontrivial Hodge cycles are absolute Hodge cycles.

Added in proof: starting from this special case, I have been able to verify 0.10 for abelian varieties. I have also verified that the tensor category of motives (0.9) generated by $H^n(X)$, for X an abelian variety, contains the motives $H^*(X)$ for X a K3 surface or a Fermat hypersurface.

0.11. The principal defect of definition 0.9 of motives is that it does not lend itself to reduction modulo p. It is not known whether a motive in the sense of 0.9 over a number field F gives a compatible system of l-adic representations of $\mathrm{Gal}(\bar{F}/F)$.

0.12. We shall use the word "motive" freely, without concerning ourselves with placing the motives under consideration within Grothendieck's framework. What matters for us is to have realizations $\mathscr{H}(M)$, for the theories $\mathscr{H}$ considered in 0.2, and to have for these groups the same formalism as for the $\mathscr{H}^*(X)$.

1. Statement of the conjecture (rational case).

1.1. Let M be a motive over $\mathbb{Q}$. We shall assume that the l-adic realizations $H_l(M)$ of M form a strictly compatible system of l-adic representations, in the sense of Serre [11, I.11]. Namely, there is a finite set S of prime numbers such that each $H_l(M)$ is unramified outside $S \cup \{l\}$, and, if $F_p \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ denotes a geometric Frobenius element at p—the inverse of a Frobenius substitution φ_p—then the polynomial $\det(1 - F_p\, t, H_l(M)) \in \mathbb{Q}_l[t]$, for $p \notin S \cup \{l\}$, has rational coefficients and is independent of l. Let $Z_p(M, t)$ denote its inverse, and set $L_p(M, s) = Z_p(M, p^{-s})$.

The Dirichlet series with rational coefficients given by the Euler product

$L_S(M, s) = \prod_{p \notin S} L_p(M, s)$

converges for Re(s) sufficiently large. For arbitrary s, $L_S(M, s)$ is defined by analytic continuation, which is expected to exist.

Our aim is to state a conjecture giving the value of $L_S(M, s)$ at certain integral points, up to a rational factor. Since p^{-s} is rational for integral s, the choice of S is immaterial, except that enlarging S can introduce unwanted zeros.

1.2. To write the conjectural functional equation of L-functions properly, the Euler product $L_S(M, s)$ must be completed by local factors at $p \in S$ and at infinity. Defining the local factors at $p \in S$ requires an additional hypothesis, which we assume:

(1.2.1) Let p be a prime number, $D_p \subset \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ a decomposition group at p, $I_p \subset D_p$ its inertia subgroup, and $F_p \in D_p$ a geometric Frobenius. The polynomial $\det(1 - F_p\, t, H_l(M)^{I_p}) \in \mathbb{Q}_l[t]$, for $l \neq p$, has rational coefficients and is independent of l.

Set $Z_p(M, t) = \det(1 - F_p\, t, H_l(M)^{I_p})^{-1} \in \mathbb{Q}(t)$, and $L_p(M, s) = Z_p(M, p^{-s})$. Define

(1.2.2) $L(M, s) = \prod_p L_p(M, s)$.

The factor at infinity $L_\infty(M, s)$, essentially a product of Γ-functions, depends on the Hodge realization of M—in fact only on the isomorphism class of the complex vector space $H_B(M) \otimes \mathbb{C}$, endowed with its Hodge decomposition and the involution F_∞. Its definition is recalled in 5.2.

Setting $\Lambda(M, s) = L_\infty(M, s)L(M, s)$, the conjectural functional equation of L-functions is

(1.2.3) $\Lambda(M, s) = \varepsilon(M, s)\Lambda(\check{M}, 1 - s)$,

where $\check{M}$ is the dual of M—its realizations are dual to those of M—and $\varepsilon(M, s)$, as a function of s, is the product of a constant and an exponential. Its definition is recalled in 5.2. It depends on an additional hypothesis.

DEFINITION 1.3. An integer n is critical for M if neither $L_\infty(M, s)$ nor $L_\infty(\check{M}, 1 - s)$ has a pole at $s = n$.

Our aim is to conjecture the value of L(M, n) for critical n, up to multiplication—

by a rational number. We have L(M(n), s) = L(M, n + s) (3.1.2), and likewise for L_∞. This allows us to consider only the numbers L(M) $=_{\text{def}}$ L(M, 0). We shall say that M is critical if 0 is critical for M. One verifies that M is critical if and only if the Hodge numbers $h^{pq} =_{\text{def}} \dim H^{pq}(M)$ of M, for $p \neq q$, can be nonzero only for (p, q) in the shaded part of the diagram below, and F_∞ acts on $H^{p,p}$ as the identity if $p < 0$ and as -1 if $p \geq 0$.

(1.3.1)

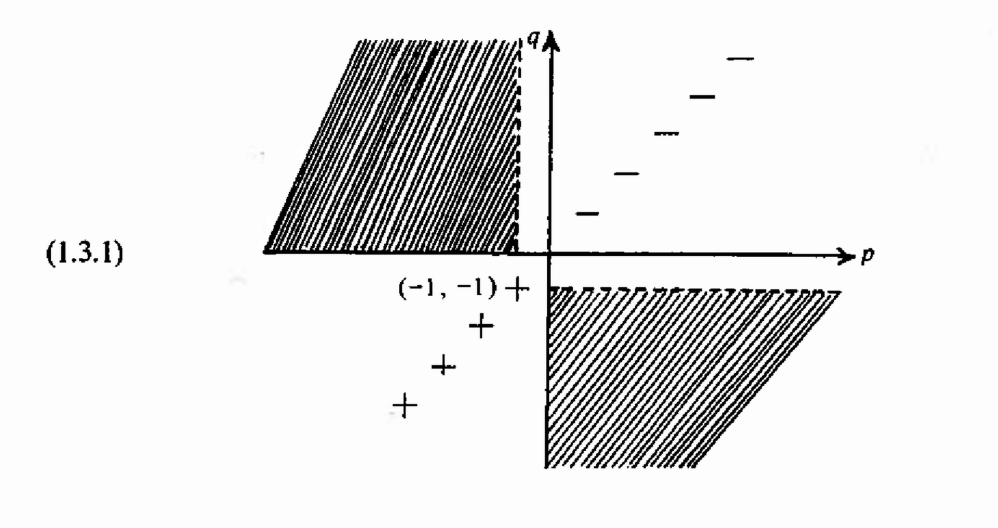

Diagram (1.3.1), certified presentation derivative; the lossless authority crop is retained with the sources.

Suppose that M is homogeneous of weight w, so $h^{pq} = 0$ for $p + q \neq w$, and set $\mathfrak{R}(M) = -w/2$. The Weil conjecture implies that, for S sufficiently large, the Dirichlet series $L_S(M, s)$ converges absolutely when $\mathfrak{R}(M) + \mathrm{Re}(s) > 1$. When $\mathfrak{R}(M) + \mathrm{Re}(s) = 1$, one is on the boundary of the half-plane of convergence, and one conjectures (a) that $L_S(M, s)$ does not vanish; (b) that $L_S(M, s)$ is holomorphic, except when M has even weight $-2n$ and contains $\mathbb{Z}(n)$ as a factor, in which case one expects a pole at $s = 1 - n$, a noncritical value.

The analogy with function fields and the known cases suggest that the local factors $L_p(M, s)$, for arbitrary p including infinity, have poles only when $\mathfrak{R}(M) + \mathrm{Re}(s) \leq 0$. If this is so, (a), (b), and the conjectural functional equation (1.2.3) imply that L(M) is neither zero nor infinite for critical M with $\mathfrak{R}(M) \neq 1/2$. When $\mathfrak{R}(M) = 1/2$, L(M) sometimes vanishes, and Conjecture 1.8 is then vacuous.

PROPOSITION 1.4. Let M be a motive over $\mathbb{R}$. Under the isomorphism (0.4.1)

$H_B(M) \otimes \mathbb{C} \xrightarrow{\cong} H_{DR}(M) \otimes_{\mathbb{R}} \mathbb{C}$,

$H_{DR}(M)$ identifies with the subspace of $H_B(M) \otimes \mathbb{C}$ fixed by $c \mapsto F_\infty \bar{c}$.

Take $M = H^*(X)$. Complex conjugation on $H_{DR}(M)_{\mathbb{C}} = H^*(X_{\mathbb{C}}, \Omega^*)$ follows functorially from the antilinear automorphism F_∞ of the scheme $X_{\mathbb{C}}$. Tracing back through the composite arrow (0.4) defining I, it is identified with the involution induced by the automorphism $(F_\infty, F_\infty{}^*)$ of $(X(\mathbb{C}), \Omega^{*an})$, and then with the composite of $F_\infty{}^*$: $H^*(X(\mathbb{C}), \mathbb{C}) \to H^*(X(\mathbb{C}), \mathbb{C})$ and complex conjugation on the coefficients. This proves 1.4.

1.5. For a motive M over $\mathbb{R}$, let $H_B{}^+(M)$, respectively $H_B{}^-(M)$, denote the subspace of $H_B(M)$ fixed by F_∞, respectively the subspace on which $F_\infty = -1$. Set $d(M) = \dim H_B(M)$ and $d^\pm(M) = \dim H_B{}^\pm(M)$. If M is a motive over k and σ is a complex embedding of k factoring through $\mathbb{R}$, write $H_\sigma{}^\pm(M)$, $d_\sigma{}^\pm(M)$, and $d_\sigma(M)$ for these objects attached to the motive over $\mathbb{R}$ obtained from M through σ. For $k = \mathbb{Q}$, the reference to σ is omitted.

The following corollary follows immediately from 1.4 and from the fact that both F_∞ and complex conjugation interchange H^{pq} and H^{qp}.

COROLLARY 1.6. Let M be a motive over $\mathbb{R}$. For the real structure $H_{DR}(M)$ on $H_B(M) \otimes \mathbb{C}$, the subspaces H^{pq} are defined over $\mathbb{R}$; the subspace $H_B{}^+(M) \otimes \mathbb{R}$ is real, and $H_B{}^-(M) \otimes \mathbb{R}$ is purely imaginary.

1.7. In the remainder of this section we consider only motives over $\mathbb{Q}$ and, unless explicitly stated otherwise, assume them homogeneous. If their weight w is even, we also assume that F_∞ acts on $H^{pp}(M)$, where $w = 2p$, by a scalar, either $+1$ or -1. This hypothesis holds for critical M. Since F_∞ interchanges H^{pq} and H^{qp}, the dimensions $d^+(M)$ and $d^-(M)$ are, in some order, $\sum_{p>q} h^{pq}$ and $\sum_{p\geq q} h^{pq}$. In particular, these dimensions equal those of the subspaces F^+ and F^- occurring in the Hodge filtration. Setting $H_{\overline{DR}}{}^\pm(M) = H_{DR}(M)/F^\mp$, one also has $\dim H_{\overline{DR}}{}^\pm(M) = d^\pm(M)$.

It follows that the composite maps

(1.7.1) $I^\pm\colon H_B{}^\pm(M)_{\mathbb{C}} \to H_B(M)_{\mathbb{C}} \xrightarrow{-I,\cong} H_{DR}(M)_{\mathbb{C}} \to H_{\overline{DR}}{}^\pm(M)_{\mathbb{C}}$

are isomorphisms. Set

(1.7.2) $c^\pm(M) = \det(I^\pm)$, (1.7.3) $\delta(M) = \det(I)$,

where the determinant is computed in rational bases of $H_B{}^\pm$ and $H_{\overline{DR}}{}^\pm$, respectively H_B and H_{DR}. The definition of $\delta(M)$ does not require the hypotheses imposed on M.

By 1.6, I^+ is real—that is, it induces $I^+\colon H_B{}^+(M)_{\mathbb{R}} \to H_{\overline{DR}}{}^+(M)_{\mathbb{R}}$—whereas I^- is purely imaginary. Thus $c^+(M)$, $i^{d^-(M)}c^-(M)$, and $i^{d^-(M)}\delta(M)$ are nonzero real numbers. Up to multiplication by a rational number, they depend only on M.

The periods of M are classically the numbers $\langle \omega, c\rangle$ with $\omega \in H_{DR}(M)$ and $c \in H_B(M)^\vee$. For example, if X is an algebraic variety over $\mathbb{Q}$, ω is an n-form on X defined over $\mathbb{Q}$, and c is an n-cycle on $X(\mathbb{C})$, then $\langle \omega, c\rangle = \int_c \omega$ is a period of $H^n(X)$. Let us express $c^\pm(M)$ in terms of periods. The dual of $H_{\overline{DR}}{}^\pm(M)$ is the subspace $F^\pm$ of $H_{DR}(M^\vee)$, where $M^\vee$ is the dual motive. If the chosen basis of $H_{\overline{DR}}{}^\pm(M)$ has dual basis (ω_i) in $F^\pm(H_{DR}(M^\vee))$, and (c_i) is the chosen basis of $H_B{}^\pm(M)$, then the matrix of $I^\pm$ is $(\langle \omega_i, c_j\rangle)$, and $c^\pm(M) = \det((\langle \omega_i, c_j\rangle))$.

Conjecture 1.8. If M is critical, $L(M)$ is a rational multiple of $c^+(M)$.

2. Statement of the conjecture (general case).

2.1. Until now we have considered only L-functions given by Dirichlet series with rational coefficients. To do better, we must consider motives with coefficients in number fields.

There are two equivalent ways to construct the category of motives over k,

with coefficients in a number field E, from the category of motives over k. The construction applies to any additive Karoubian category in which Hom(X, Y) is a $\mathbb{Q}$-vector space.

A. A motive over k with coefficients in E is a motive M over k endowed with an E-module structure $E \to \mathrm{End}(M)$.

B. The category $\mathscr{M}_{k,E}$ of motives over k with coefficients in E is the Karoubian envelope of the category whose objects are motives over k—when regarded as an object of $\mathscr{M}_{k,E}$, a motive M is denoted M_E—and whose morphisms are

$\mathrm{Hom}(X_E, Y_E) = \mathrm{Hom}(X, Y) \otimes E$.

Passage from B to A. For a motive X and a finite-dimensional $\mathbb{Q}$-vector space V, let $X \otimes V$ denote the motive isomorphic to a sum of $\dim(V)$ copies of X and characterized by $\mathrm{Hom}(Y, X \otimes V) = \mathrm{Hom}(Y, X) \otimes V$, or equivalently by $\mathrm{Hom}(X \otimes V, Y) = \mathrm{Hom}(V, \mathrm{Hom}(X, Y))$. Associate to M_E the motive $M \otimes E$ with its natural E-module structure.

Passage from A to B. If M has an E-module structure, M_E carries two E-module structures: the one induced by the structure on M and the one carried by every object of $\mathscr{M}_{k,E}$. The object of $\mathscr{M}_{k,E}$ corresponding to M is the largest direct factor of M_E on which these structures agree. More explicitly, $E \otimes E$ is a product of fields, including a copy of E in which $x \otimes 1$ and $1 \otimes x$ both project to x. The corresponding idempotent e acts on M_E, which is an $E \otimes E$-module, and its image is the object of $\mathscr{M}_{k,E}$ corresponding to M.

Motives with coefficients in E are most often given in form A. Form B has the advantage of remaining meaningful when E is not of finite rank over $\mathbb{Q}$. It is useful for understanding the tensor formalism: tensor products and duals of motives with coefficients in E are defined by functoriality and the formulas

$X_E \otimes_E Y_E = (X \otimes Y)_E$ and $(X_E)^\vee = (X^\vee)_E$.

In language A, $X \otimes_E Y$ is the largest direct factor of $X \otimes Y$ on which the two E-module structures agree, and $X^\vee$ is the ordinary dual of X endowed with the transposed E-module structure. Applying these observations to vector spaces over $\mathbb{Q}$ rather than to motives recovers the isomorphism between the E-dual of an E-vector space and its $\mathbb{Q}$-dual, given by

$\omega \mapsto$ the form $\mathrm{Tr}_{E/\mathbb{Q}}(\langle \omega, v \rangle)$.

In 0.1.1 we defined restriction and extension functors for the base field k. They carry motives with coefficients in E to motives with coefficients in E. There are also restriction and extension functors for the coefficient field; let F be a finite extension of E.

Extension of coefficients. In language A this is $X \mapsto X \otimes_E F$; in language B it is $X_E \mapsto X_F$.

Restriction of coefficients. In language A one restricts the F-module structure to E.

Care must be taken not to confuse the roles of k and E. A standard example is H_1 of an abelian variety over k with complex multiplication by an order in E. In terms of the abelian variety, taken up to isogeny, the preceding functors become extension of the base field, Weil restriction of scalars, the construction $\otimes_E F$, which multiplies the dimension by [F:E], and restriction to E of the F-module structure.

2.2. Let M be a motive over $\mathbb{Q}$ with coefficients in a number field E. For each prime number l, the l-adic realization $H_l(M)$ is a module over the l-adic completion E_l of E. This completion is the product of the completions E_λ over the prime ideals λ above l, giving a decomposition of $H_l(M)$ into a product of E_λ-modules $H_\lambda(M)$.

One conjectures for the $H_\lambda(M)$ a compatibility analogous to 1.2.1. If it holds, then for each complex embedding σ of E one may define a Dirichlet series with coefficients in σE, convergent for Re(s) sufficiently large:

$L(\sigma, M, s) = \prod_p L_p(\sigma, M, s)$, where

(2.2.1) $L_p(\sigma, M, s) = \sigma Z_p(M, p^{-s})$, with $Z_p \in E(t) \subset E_\lambda(t)$, given by $Z_p(M, t) = \det(1 - F_p\, t, H_\lambda(M)^{I_p})^{-1}$ for $\lambda \nmid p$.

As σ varies, these Dirichlet series are obtained from one another by conjugating their coefficients. We regard the system $L(\sigma, M, s)$ as a function $L^*(M, s)$ with values in the $\mathbb{C}$-algebra $E \otimes \mathbb{C}$, identified with $\mathbb{C}^{Hom(E,\mathbb{C})}$ by

(2.2.2) $E \otimes \mathbb{C} \xrightarrow{\sim} \mathbb{C}^{Hom(E,\mathbb{C})}$: $e \otimes z \mapsto (z, \sigma(e))_\sigma$.

This function can also be defined directly by an Euler product. As in 1.1, it is expected to admit analytic continuation in s.

The Euler product $L(\sigma, M, s)$ must be completed by an infinite factor $L_\infty(\sigma, M, s)$ depending on the Hodge realization of M. Setting $\Lambda(\sigma, M, s) = L_\infty(\sigma, M, s)L(\sigma, M, s)$, the conjectural functional equation is

(2.2.3) $\Lambda(\sigma, M, s) = \varepsilon(\sigma, M, s)\Lambda(\sigma, \check{M}, 1 - s)$,

where $\varepsilon(\sigma, M, s)$, as a function of s, is the product of a constant and an exponential. The definitions of L_∞ and ε are recalled in 5.2. As above, the systems Λ and ε, for varying σ, are regarded as functions Λ^* and ε^* with values in $E \otimes \mathbb{C}$.

It follows from 2.5 below that $L_\infty(\sigma, M, s)$ is independent of σ, and that the infinite factor for the motive $R_{E/\mathbb{Q}}M$ obtained by restriction of the coefficient field is the $[E:\mathbb{Q}]$-th power of $L_\infty(\sigma, M, s)$. This justifies the following.

PROPOSITION-DEFINITION 2.3. Let M be a motive over $\mathbb{Q}$ with coefficients in E. An integer n is critical for M if the following equivalent conditions hold: (i) n is critical for $R_{E/\mathbb{Q}}M$; (ii) neither $L_\infty(\sigma, M, s)$ nor $L_\infty(\sigma, \check{M}(1), s)$ has a pole at $s = n$. M is called critical if 0 is critical for M.

Our aim is to conjecture the value $L^*(M) =_{def} L^*(M, 0)$, for critical M, up to multiplication by an element of E. Equivalently, the values $L(\sigma, M) =_{def} L(\sigma, M, 0)$ are to be conjectured simultaneously, up to multiplication by a system of numbers $\sigma(e)$, with $e \in E$.

2.4. The Betti realization $H_B(M)$ is an E-vector space. Its dimension is the rank of M over E. The involution F_∞ is E-linear, so its + and − parts are E-vector subspaces. Their dimensions are denoted $d^+(M)$ and $d^-(M)$.

The complexification of $H_B(M)$ is a free $E \otimes \mathbb{C}$-module. Identifying $E \otimes \mathbb{C}$ with $\mathbb{C}^{Hom(E,\mathbb{C})}$ by (2.2.2), one obtains a decomposition

$H_B(M) \otimes \mathbb{C} = \oplus_\sigma H_B(\sigma, M)$,

where

$H_B(\sigma, M) = (H_B(M) \otimes \mathbb{C}) \otimes_{E\otimes\mathbb{C},\sigma} \mathbb{C}$,

or equivalently

$H_B(\sigma, M) = H_B(M) \otimes_{E,\sigma} \mathbb{C}$.

Since the Hodge summands $H_B{}^{pq}(M)$ are stable under E, each $H_B(\sigma, M)$ inherits a Hodge decomposition $H_B(\sigma, M) = \oplus H_B{}^{pq}(\sigma, M)$. The involution F_∞ interchanges H^{pq} and H^{qp}; in particular it stabilizes $H_B{}^{pp}$, which it splits into + and − parts. Let $h^{pq}(\sigma, M)$ be the dimension of $H_B{}^{pq}(\sigma, M)$, and $h^{pp\pm}(\sigma, M)$ that of $H_B{}^{pp\pm}(\sigma, M)$. The following proposition allows σ to be omitted from the notation.

PROPOSITION 2.5. The numbers $h^{pq}(\sigma, M)$ and $h^{pp\pm}(\sigma, M)$ are independent of σ.

We may and do assume that M is homogeneous. Then $H^{pq}(M)$ is the complexification of the E-vector space $Gr_F{}^p(H_{DR}(M))$; it is a free $E \otimes \mathbb{C}$-module, proving the first assertion. For the second, $h^{pp\pm}(\sigma, M)$ is the excess of $d^\pm(M)$ over $\sum_{p>q} h^{pq}(\sigma, M)$.

2.6. Let M be a motive over $\mathbb{Q}$ with coefficients in E. For the remainder of this section, unless explicitly stated otherwise, assume that $R_{E/\mathbb{Q}}M$ satisfies the hypotheses of 1.7. The spaces $F^\pm$ and $H_{\overline{DR}}{}^\pm$ are now E-vector spaces. The isomorphisms (0.4.1) and (1.7.1),

```
(2.6.1)    I: H_B(M) ⊗ ℂ → H_DR(M) ⊗ ℂ,
(2.6.2)    I^±: H_B^±(M) ⊗ ℂ → H_{D̄R}^±(M) ⊗ ℂ,
```

are isomorphisms of $E \otimes \mathbb{C}$-modules between complexifications of E-vector spaces. Set

$c^\pm(M) = \det(I^\pm) \in (E \otimes \mathbb{C})^*$, $\delta(M) = \det(I) \in (E \otimes \mathbb{C})^*$,

where the determinant is computed in E-rational bases of $H_B{}^\pm(M)$ and $H_{\overline{DR}}{}^\pm(M)$, respectively $H_B(M)$ and $H_{DR}(M)$. The definition of $\delta(M)$ does not require the hypotheses imposed on M. Up to multiplication by an element of E^*, these numbers depend only on M. Again by 1.6, $c^+(M)$, $i^{d^-(M)}c^-(M)$, and $i^{d^-(M)}\delta(M)$ lie in $(E \otimes \mathbb{R})^*$.

Conjecture 2.7. Set $\mathfrak{R}(M) = -w/2$. If M is critical, (i) $L(\sigma, M, s)$ never has a pole at $s = 0$ and can vanish there only when $\mathfrak{R}(M) = 1/2$; (ii) the order of vanishing of $L(\sigma, M, s)$ at $s = 0$ is independent of σ.

For (i), see the discussion at the end of 1.3. B. Gross suggested that (ii) should be reasonable.

Conjecture 2.8. If M is critical and $L(\sigma, M) \neq 0$, then $L^*(M)$ is the product of $c^+(M)$ and an element of E^*.

REMARK 2.9. A motive M over a number field k with coefficients in E also defines an L-function $L^*(M, s)$. These functions are covered by the conjecture because $L^*(M, s) = L^*(R_{k/\mathbb{Q}}M, s)$, where $R_{k/\mathbb{Q}}$ denotes restriction of scalars from k to $\mathbb{Q}$; apply (0.5.3).

The functions $L(\sigma, M, s)$ are Euler products indexed by the finite places of k. They must be completed by the infinite factors $L_v(\sigma, M, s)$, indexed by the infinite places and defined in 5.2. In general they depend on σ. Only $L_\infty(\sigma, R_{k/\mathbb{Q}}M, s)$, the product over the infinite places v of the $L_v(\sigma, M, s)$, is independent of σ.

REMARK 2.10. Let F be an extension of E, let ι: $E \to F$ be the structural morphism, and let ι_C: $E \otimes \mathbb{C} \hookrightarrow F \otimes \mathbb{C}$ be its complexification. Then $L^*(M \otimes_E F, s) = \iota_C\, L^*(M, s)$ and $c^\pm(M \otimes_E F) = \iota_C\, c^\pm(M)$. Thus the conjecture is compatible with extension of the coefficient field. If F/E is Galois with group G, Hilbert's Theorem 90, $H^1(G, F^*) = 0$, gives $((F \otimes \mathbb{C})^*/F^*)^G = (E \otimes \mathbb{C})^*/E^*$, so the conjecture is invariant under extension of coefficients.

REMARK 2.11. If E is an extension of F, then $L^*(R_{E/F}M, s) = N_{E/F}L^*(M, s)$. Since the periods $c^\pm$ satisfy the same identity, the conjecture is compatible with restriction of coefficients.

REMARK 2.12. Let D be a division algebra of rank d^2 over E. A motive M over $\mathbb{Q}$ with a D-module structure and rank n over D defines a Dirichlet series with coefficients in E whose Euler factors almost all have degree nd. For a finite place λ of E, $H_\lambda(M)$ is a free module over $D_\lambda = D \otimes_E E_\lambda$, and definition (2.2.1) is used with

$$Z_p(M, t) = \det \operatorname{red}(1 - F_p\, t, H_\lambda(M)^{I_p})^{-1}.$$

If D_λ is a matrix algebra over E_λ and e is an indecomposable idempotent, the reduced determinant of an endomorphism A of a D_λ-module H is the determinant, over E_λ, of the restriction of A to eH; the general case is obtained by descent.

The freedom afforded by 2.10 to extend the coefficient field also brings these L-functions under Conjecture 2.8. Choose an extension ι: $E \hookrightarrow F$ that splits D and an indecomposable idempotent e of $D \otimes_E F$. Then $\iota_C\, L^*(M, s) = L^*(e(M \otimes_E F), s)$.

One may also define $c^+(M)$ directly in this setting, as explained in 2.13 below.

The remainder of this section is not used later.

2.13. Let D be a simple algebra over a field E, or even an Azumaya algebra over a ring. For the tensor formalism it is convenient to regard D-modules as "fictitious E-vector spaces":

(a) To give an E-vector space V is equivalent to giving, for every étale extension F/E, an F-vector space V_F and compatible isomorphisms $V_G = V_F \otimes_F G$ for extensions G/F. One takes $V_F = V \otimes_E F$. By descent it is enough to give V_F for sufficiently large F.

(b) Let W be a D-module. For every étale extension F/E and every F-isomorphism $D \otimes F \cong \operatorname{End}_F(L)$, with L free, set $W_{F,L} = \operatorname{Hom}_{D\otimes F}(L, W \otimes F)$, where all unlabelled tensor products are over E. There is an isomorphism of $D \otimes$ F-modules

$$W \otimes_E F = L \otimes_F W_{F,L}.$$

If F is large enough to split D, L is unique up to a nonunique isomorphism; the nonuniqueness is due to scalar homotheties, which act trivially on End(L). This is why the data $W_{F,L}$ are not of type (a); they form a "fictitious vector space over E."

Let (W^{α}) be a family of D-modules, and let T be a tensor operation. If the scalar homotheties of L act trivially on $T(W_{F,L}{}^{\alpha})$, the F-vector space $T(W_{F,L}{}^{\alpha})$ is independent of the choice of L, and one obtains a system of type (a), hence an E-vector space $T(W^{\alpha})$.

EXAMPLE. If W′ and W″ are two D-modules of rank $n\cdot[D{:}E]^{1/2}$ over E, one may take $T = \mathrm{Hom}(\wedge^n W'_{F,L}, \wedge^n W''_{F,L})$; one obtains a one-dimensional E-vector space $\delta(W', W'')$, and every homomorphism $f\colon W' \to W''$ has a reduced determinant $\det \mathrm{red}(f) \in \delta(W', W'')$.

To define $c^+(M)$, apply this construction to the D-modules $H_B{}^+(M)$ and $H_{\overline{DR}}{}^+(M)$. Set $\delta = \delta(H_B{}^+(M), H_{\overline{DR}}{}^+(M))$. The reduced determinant of the isomorphism of $D \otimes \mathbb{C}$-modules $I^+\colon H_B{}^+(M)_{\mathbb{C}} \to\simeq H_{\overline{DR}}{}^+(M)_{\mathbb{C}}$ lies in the free rank-one $E \otimes \mathbb{C}$-module $\delta \otimes \mathbb{C}$. Set $\det \mathrm{red}(I^+) = c^+(M)\cdot e$ for a basis e of δ.

3. Example: the ζ-function. 3.1. To understand the various realizations of the Tate motive $\mathbb{Z}(1)$, the simplest course is to write $\mathbb{Z}(1) = H_1(\mathbb{G}_m)$. Since the multiplicative group is not a projective variety, this does not fit Grothendieck's framework, which instead requires $\mathbb{Z}(1)$ to be defined as the dual of the direct factor $H^2(\mathbb{P}^1)$ of $H^*(\mathbb{P}^1)$.

The realization of $\mathbb{Z}(1)$ in $\mathbb{Z}_l$-cohomology is the Tate module $T_l(\mathbb{G}_m)$ of $\mathbb{G}_m$:

$\mathbb{Z}_l(1) = \mathrm{proj\ lim}\ \mu_{l^n}$.

In the Hodge realization, $H_B(\mathbb{Z}(1)) = H_1(\mathbb{C}^*)$ is isomorphic to $\mathbb{Z}$—or rather to $\mathbb{Q}$ in rational homology. This group is purely of type $(-1, -1)$, and $F_\infty = -1$.

In the de Rham realization, $H_{\mathrm{DR}}(\mathbb{Z}(1))$ is the dual of $H_{\mathrm{DR}}{}^1(\mathbb{G}_m)$, isomorphic to $\mathbb{Q}$ and generated by the class of dz/z. The unique period of $H^1(\mathbb{G}_m)$ is

(3.1.1) $\oint dz/z = 2\pi i$.

On $\mathbb{Z}_l(1)$, arithmetic Frobenius φ_p ($p \neq l$) acts by multiplication by p. Geometric Frobenius therefore acts by multiplication by p^{-1}; this justifies the identity cited in 1.3:

(3.1.2) $L(M(n), s) = L(M, n + s)$.

Since F_∞ acts as $(-1)^n$ on $H_B(\mathbb{Z}(n))$, $H_B{}^{\varepsilon}(\mathbb{Z}(n))$ vanishes for $\varepsilon = -(-1)^n$, and $c^{\varepsilon}(\mathbb{Z}(n)) = 1$. For $\varepsilon = (-1)^n$, (3.1.1) gives $c^{\varepsilon}(\mathbb{Z}(n)) = \delta(\mathbb{Z}(n)) = (2\pi i)^n$:

(3.1.3) $c^{\varepsilon}(\mathbb{Z}(n)) = (2\pi i)^n$ for $\varepsilon = (-1)^n$, $c^{\varepsilon}(\mathbb{Z}(n)) = 1$, $\delta(\mathbb{Z}(n)) = (2\pi i)^n$ for $\varepsilon = -(-1)^n$.

3.2. The function $\zeta(s)$ is the L-function attached to the unit motive $\mathbb{Z}(0) = H^*(\mathrm{Point})$. The critical integers for $\mathbb{Z}(0)$ are the even integers > 0 and the odd integers ≤ 0. Because $\zeta(s)$ has a pole at $s = 1$, 0 is not a trivial zero, and it would be reasonable to define the critical integers so as to include 0. Equation (3.1.3) and the known values

of $\zeta(n) = L(\mathbb{Z}(n))$ for critical n verify 1.8: $\zeta(n)$ is rational for odd $n \leq 0$, and is a rational multiple of $(2\pi i)^n$ for even $n \geq 0$.

4. Compatibility with the Birch and Swinnerton-Dyer conjecture. 4.1. Let A be an abelian variety over $\mathbb{Q}$, of dimension d. The Birch and Swinnerton-Dyer conjecture [15] asserts in particular: (a) $L(H^1(A), 1)$ is nonzero if and only if $A(\mathbb{Q})$ is finite. (b) Let ω be a generator of $H^0(A, \Omega^d)$. Then $L(H^1(A), 1)$ is the product of $\int_{A(\mathbb{R})} |\omega|$ and a rational number.

The motive $H^1(A)(1)$ is isomorphic to the dual $H_1(A)$ of $H^1(A)$; this expresses the existence of a polarization, an autoduality of $H^1(A)$ with values in $\mathbb{Z}(-1)$. By 1.7, $c^+(H^1(A)(1))$ is therefore computed as follows: if $\omega_1,\cdots,\omega_d$ is a basis of $H^0(A, \Omega^1) = F^+H_{DR}{}^1(A)$, and $c_1,\cdots,c_d$ is a basis of $H_1(A(\mathbb{C}), \mathbb{Q})^+$, then

(4.1.1) $c^+(H^1(A)(1)) = \det\langle \omega_i, e_j \rangle$.

Again denoting by e_i a representative cycle, one has $\langle \omega_i, e_j \rangle = \int_{e_j} \omega_i$. Take as (e_i) a $\mathbb{Z}$-basis of $H_1(A(\mathbb{R})^\circ, \mathbb{Z}) \subset H_1(A(\mathbb{C}), \mathbb{Z})$. The Pontryagin product of the e_i is represented by the d-cycle $A(\mathbb{R})^\circ$ in $A(\mathbb{C})$, with a suitable orientation; if ω is the exterior product of the ω_i, the determinant (4.1.1) is $\int_{A(\mathbb{R})^\circ} \omega$. Thus

$\int_{A(\mathbb{R})} |\omega| = [A(\mathbb{R}): A(\mathbb{R})^\circ] \, |\int_{A(\mathbb{R})^\circ} \omega|$,

and 1.8 for $H^1(A)(1)$ is therefore equivalent to 4.1(b) above.

4.2. The Birch and Swinnerton-Dyer conjecture gives the exact value of $L(H^1(A), 1)$; the description of the rational factor in 4.1(b) in terms of the motive $H^1(A)(1)$ has the following form: (a) The datum $M = H^1(A)(1)$ is equivalent to that of A up to isogeny. One must begin by choosing A. This amounts to choosing an integral lattice in $H_B(M)$ whose l-adic completions are stable under $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$. (b) One then chooses ω, for example as the exterior product of the elements of a basis of an integral lattice in $H_{DR}(M)$. This ω determines the period $c^+(M)$ and, for each prime p, a rational factor $c_p(M)$. Almost all $c_p(M)$ equal 1, and the product formula ensures that $c^+(M)\cdot\prod_p c_p(M)$ is independent of ω. (c) Another rational number h(M) is defined in terms of cohomological invariants of A; the desired rational number is $h(M)\cdot\prod_p c_p(M)^{-1}$.

The invariance of the conjecture under isogeny is, moreover, a nontrivial theorem.

4.3. To generalize 4.1(a) to an arbitrary motive M of weight -1, one would need an analogue of $A(\mathbb{Q})$. The group $A(\mathbb{Q})$ may be interpreted as the group of extensions of $\mathbb{Z}(0)$ by $H_1(A)$ in the category of 1-motives [6, §10] over $\mathbb{Q}$. This suggests considering the group of extensions of $\mathbb{Z}(0)$ by M in a category of mixed motives parallel to mixed Hodge structures, but not even a conjectural definition of such a category is available.

I shall only observe that, in all the usual cohomology theories, a cycle Y of dimension d, homologous to zero, on a proper algebraic variety

and smooth X determines a torsor under $H^{2d-1}(X)(d)$: there is an exact cohomology sequence

$0 \to H^{2d-1}(X)(d) \to H^{2d-1}(X - Y)(d) \xrightarrow{-\partial} {H_Y}^{2d}(X)(d) \to H^{2d}(X)(d)$.

The cycle Y defines a cohomology class $cl(Y) \in {H_Y}^{2d}(X)(d)$ whose image in $H^{2d}(X)(d)$ is zero, and one takes $\partial^{-1}cl(Y)$. This construction is the analogue of assigning a point of the Jacobian to a degree-zero divisor on a curve.

5. Compatibility with the functional equation. Let E be a number field. In $(E \otimes \mathbb{C})^*$, write $\sim$ for the equivalence relation defined by the subgroup E^*.

PROPOSITION 5.1. Let M be a motive over $\mathbb{Q}$ with coefficients in E. Assume the hypotheses of 1.7. Then

$c^+(M) \sim (2\pi i)^{-d^-(M)} \cdot \delta(M) \cdot c^+(\check{M}(1))$.

For a free module L of rank n over a commutative ring A, where A is E or $E \otimes \mathbb{C}$, set $\det L = \wedge^n L$. By localization this definition extends to L projective of finite type; that generalization is not needed for the proof of 5.1. There are canonical isomorphisms

(5.1.1) $\det(L^\vee) = \det(L)^{-1}$,

and, for any direct factor P of L,

(5.1.2) $\det(L) = \det(P) \cdot \det(L/P)$.

Here $\cdot$ denotes a tensor product and $^{-1}$ denotes a dual. Sign issues arise; they are best resolved by regarding det(L) as an invertible graded module placed in degree rank L. Since the final results are taken modulo $\sim$, we shall not be concerned with them.

For X and Y of the same rank, set $\delta(X, Y) = \mathrm{Hom}(\det X, \det Y) = \det(X)^{-1} \cdot \det(Y)$. The determinant of $f: X \to Y$ lies in $\delta(X, Y)$. Equations (5.1.1) and (5.1.2) give isomorphisms

(5.1.3) $\delta(X, Y) = \delta(Y^\vee, X^\vee)$,

and, for $F \subset X$ and $G \subset Y$,

(5.1.4) $\delta(X, Y) = \delta(F, Y/G) \cdot \delta(G, X/F)^{-1}$.

LEMMA 5.1.5. Under the isomorphism (5.1.3), $\det(u) = \det({}^t u)$.

LEMMA 5.1.6. Let F and G be direct factors of X and Y. Suppose that the isomorphism $f: X \to Y$ induces isomorphisms $f_F: F \to Y/G$ and ${f_G}^{-1}: G \to X/F$. Under the isomorphism (5.1.4), $\det(f) = \det(f_F)\,\det({f_G}^{-1})^{-1}$.

The verification of these lemmas is left to the reader. For 5.1.6, note only that, with $G^1 = f^{-1}(G)$ and $F^1 = f(F)$, one has $X = F \oplus G^1$ and $Y = G \oplus F^1$, and f exchanges F with F^1 and G with G^1.

Apply Lemma 5.1.6 to the complexifications of ${H_B}^+(M) \subset H_B(M)$ and $F^- \subset H_{DR}(M)$. Under the isomorphism (5.1.4), the determinant of $I: H_B(M) \otimes \mathbb{C} \to H_{DR}(M) \otimes \mathbb{C}$ is the product of the determinant of $I^+: {H_B}^+(M) \otimes \mathbb{C} \to$

$(H_{DR}(M)/F^-) \otimes \mathbb{C}$ and the inverse of the determinant of the morphism induced by I^{-1}:

$J^-: F^- \otimes \mathbb{C} \to (H_B(M)/H_B{}^+(M)) \otimes \mathbb{C}$.

The morphism J^- is the transpose of the morphism I^- for the dual motive $\check{M}$. Applying 5.1.5 and choosing E-bases of the spaces $\delta(X, Y)$ involved, one finally obtains

(5.1.7) $\delta(M) \sim c^+(M) \cdot c^-(\check{M})^{-1}$.

Proposition 5.1 follows by applying to $\check{M}$ the following consequence of 3.1:

(5.1.8) $c^\pm(M) = (2\pi i)^{-d^\pm(M)\cdot n}\, c^{\pm(-1)^n}(M(n))$.

For later use, also note the analogous formula

(5.1.9) $\delta(M) = (2\pi i)^{-d(M)\cdot n}\, \delta(M(n))$.

5.2. We recall the exact form of the conjectural functional equation for motivic L-functions ([12], [4]). We work in the general case of a motive over a number field k with coefficients in a field E equipped with a complex embedding σ (cf. 2.9). First, the general form: (a) For each place v of k, define a local factor $L_v(\sigma, M, s)$. For finite v, the definition $L_v(\sigma, M, s) = \sigma Z_v(M, Nv^{-s})$, with $Z_v(M, t) \in E(t)$, depends on a compatibility hypothesis analogous to (1.2.1). One has $Z_v(M, t) = \det(1 - F_{vt}, H_\lambda(M)^{I_v})^{-1}$. For an infinite place v induced by a complex embedding τ, $L_v(\sigma, M, s)$ is obtained by decomposing $H_\tau(M) \otimes_{E,\sigma} \mathbb{C}$ as a direct sum of minimal subspaces stable under the projectors defining the Hodge decomposition and, for real v, under F_∞; one associates to each subspace the "Γ-factor" in Table (5.3) and takes their product. For complex v, the minimal subspaces are one-dimensional, of type (p, q). For real v, they are either two-dimensional of type {(p, q), (q, p)}, $p \neq q$, or one-dimensional of type (p, p), with $F_\infty = \pm 1$. Write $\Lambda(\sigma, M, s)$ for the product of the $L_v(\sigma, M, s)$. (b) Let Ψ be a nontrivial character of the idele-class group $\mathbb{A} \otimes k/k$, with components Ψ_v. For every place v of k, also choose a Haar measure dx_v on k_v. Assume that for almost every v, dx_v gives mass 1 to the integers of k_v, and that $\otimes\, dx_v$ is the Tamagawa measure, assigning mass 1 to the idele-class group.

Define local constants $\varepsilon_v(\sigma, M, s, \Psi_v, dx_v)$, almost all equal to 1 and each, as a function of s, a constant times an exponential. Let $\varepsilon(\sigma, M, s)$ be their product; it is independent of Ψ and the dx_v. (c) The conjectural functional equation is

$\Lambda(\sigma, M, s) = \varepsilon(\sigma, M, s)\Lambda(\sigma, \check{M}, 1 - s)$.

Defining the ε_v requires an additional compatibility hypothesis among the $H_l(M)$. It assigns to v, σ, and M an isomorphism class of complex representations of the Weil group $W(\bar{k}_v/k_v)$ for infinite v, and of the enlarged Weil group $W'(\bar{k}_v/k_v)$ for finite v; one takes the ε-factor of [4, 8.12.4], with $t = p^{-s}$.

For infinite v, this amounts to decomposing $H_\tau(M) \otimes_{E,\sigma} \mathbb{C}$ as in (a), and associating

a factor ε'_v to each subspace of the decomposition, then taking their product. The table of ε'_v for a particular choice of Ψ_v and dx_v is given in 5.3.

For finite v, first restrict the representations $\mathrm{H}_l(\mathrm{M})$ to a decomposition group $\mathrm{Gal}(\bar{k}_v/\mathrm{k}_v) \subset \mathrm{Gal}(\bar{k}/\mathrm{k})$, and then to the Weil group $\mathrm{W}(\bar{k}_v/\mathrm{k}_v)$. Applying [4, 8.3, 8.4] gives from $\mathrm{H}_l(\mathrm{M})$ an isomorphism class ρ_l of representations of $\mathrm{W}'(\bar{k}_v/\mathrm{k}_v)$ over E_l. It is permissible, and useful, to replace it by its F-semisimplification [4, 8.6]. These representations, as l varies, are required to be compatible: after extending scalars from E_l to $\mathbb{C}$ by $\tilde{\sigma}$: $\mathrm{E}_\lambda \to \mathbb{C}$ extending σ, the isomorphism class obtained must be independent of l and σ. This is the required isomorphism class.

For the reader, we unpack the internal references to [4]. A representation of W′ consists of a representation ρ of the Weil group in GL(V), together with a nilpotent endomorphism N of V. In terms of the local constant of ρ in [4, 4.1], that of (ρ, N) is

$\varepsilon((\rho, \mathrm{N}), \mathrm{s}, \Psi, \mathrm{dx}) = \varepsilon(\rho \otimes \omega_s, \Psi\, \mathrm{dx}) \cdot \det(-\mathrm{F}\, \mathrm{Nv}^{-s}, \mathrm{V}^{\rho(I)}/\mathrm{Ker}(\mathrm{N})^{\rho(I)})$.

REMARK 5.2.1. The function of s

$\varepsilon((\rho, \mathrm{N}), \mathrm{s}, \Psi, \mathrm{dx})\mathrm{L}((\rho, \mathrm{N})^\vee, 1 - \mathrm{s})\mathrm{L}((\rho, \mathrm{N}), \mathrm{s})^{-1}$

is the same for (ρ, N) and $(\rho, 0)$. Thus the conjectural functional equation for L-functions can be stated assuming compatibility only among the semisimplifications of the restrictions of $\mathrm{H}_l(\mathrm{M})$ to a decomposition group.

5.3. The following table gives the local factors and constants attached to the various types of minimal subspaces in the Hodge realization. For the constants, Ψ_v and dx_v are chosen as follows: real v: $\Psi_v(\mathrm{x}) = \exp(2\pi\mathrm{ix})$, measure dx; complex v: $\Psi_v(\mathrm{z}) = \exp(2\pi\mathrm{i}\, \mathrm{Tr}_{\mathbb{C}/\mathbb{R}}(\mathrm{z}))$, measure $|\mathrm{dz} \wedge \mathrm{d}\,\bar{z}|$, so that for $\mathrm{z} = \mathrm{x} + \mathrm{iy}$ these are $\exp(4\pi\mathrm{ix})$ and 2dxdy.

Use the notation $\Gamma_{\mathbb{R}}(\mathrm{s}) = \pi^{-s/2}\, \Gamma(\mathrm{s}/2)$, $\Gamma_{\mathbb{C}}(\mathrm{s}) = 2 \cdot (2\pi)^{-s}\, \Gamma(\mathrm{s})$.

place type Γ-factor constant (for Ψ, dx above) complex (p, q) or (q, p), $\mathrm{p} \le \mathrm{q}$ $\Gamma_{\mathbb{C}}(\mathrm{s} - \mathrm{p})\, \mathrm{i}^{q-p}$ real {(p, q), (q, p)}, $\mathrm{p} < \mathrm{q}$ $\Gamma_{\mathbb{C}}(\mathrm{s} - \mathrm{p})\, \mathrm{i}^{q-p+1}$ (p, p), $\mathrm{F}_\infty = (-1)^{p+\varepsilon}$, $\varepsilon = 0$ or 1 $\Gamma_{\mathbb{R}}(\mathrm{s}+\varepsilon-\mathrm{p})\, \mathrm{i}^{\varepsilon}$

The case of interest here is $\mathrm{k} = \mathbb{Q}$. One may then take $\Psi_\infty(\mathrm{x}) = \exp(2\pi\mathrm{ix})$, $\Psi_p(\mathrm{x}) = \exp(-2\pi\mathrm{ix})$, using the identification of $\mathbb{Q}_p/\mathbb{Z}_p$ with the p-primary part of $\mathbb{Q}/\mathbb{Z}$; dx_∞ is Lebesgue measure dx, and dx_p is Haar measure on $\mathbb{Q}_p$ giving $\mathbb{Z}_p$ mass 1.

PROPOSITION 5.4. If M is critical of weight w, then, modulo a rational number independent of σ,

$\mathrm{L}_\infty(\sigma, \check{M}(1))\mathrm{L}_\infty(\sigma, \mathrm{M})^{-1} \sim (2\pi)^{-d^-(M)} \cdot (2\pi)^{-wd(M)/2}$.

By 2.5, the $\mathrm{L}_\infty(\sigma,)$ are independent of σ. It is therefore enough to verify 5.4 for a fixed σ. The formula is compatible with the substitution $\mathrm{M} \mapsto \check{M}(1)$: $\check{M}(1)$ has

weight $-2-w$, its d^- is $d^+(M)$, and abbreviating $d(M)$ and $d^\pm(M)$ as d and $d^\pm$, one has $d = d^+ + d^-$ and

$(-d^- - wd/2) + (-d^+ - ((-2-w)d)/2) = 0$.

It is therefore enough to verify 5.4 for $w \geq -1$. For an integer s, modulo $\mathbb{Q}^*$,

(5.4.1) $\Gamma_{\mathbb{R}}(s) \sim (2\pi)^{-s/2}$ for even $s > 0$, $\Gamma_{\mathbb{R}}(s) \sim (2\pi)^{(1-s)/2}$ for odd s, $\Gamma_{\mathbb{C}}(s) \sim (2\pi)^{-s}$ for $s > 0$.

For $w \geq -1$, the power of 2π contributed by each subspace of $H_B(M) \otimes \mathbb{C}$, as in 5.2(a), is therefore

$L_\infty(M)\ L_\infty(\check{M}(1))\ L_\infty(\check{M}(1))L_\infty(M)^{-1}$ {(pq), (qp)}, $p \leq q$ p $-1-p$ $-1-w$ (pp), p even ≥ 0, $F_\infty = -1$ $p/2$ $-1-p/2$ $-1-w/2$ (pp), p odd ≥ 0, $F_\infty = -1$ $(1+p)/2$ $(-1-p)/2$ $-1-w/2$

The proposition follows immediately. Set $\det M = \wedge^{d(M)}M$, the exterior power over E.

PROPOSITION 5.5. $\varepsilon^*(M) \sim \varepsilon^*(\det M)$.

For the dx_v chosen as suggested in 5.4, we prove more precisely the equivalences

(5.5.1) ${\varepsilon_v}^*(M, \Psi_v, dx_v) \sim {\varepsilon_v}^*(\det M, \Psi_v, dx_v)$.

Set $\eta_v(\sigma, M, \Psi_v) = \varepsilon_v(\sigma, M, \Psi_v, dx_v)\ {\varepsilon_v}^{-1}(\sigma, \det M, \Psi_v, dx_v)$. This is equivalent to

(5.5.2) $\tau\eta_v(\sigma, M, \Psi_v) = \eta_v(\tau\sigma, M, \tau\Psi_v)$

for every automorphism τ of $\mathbb{C}$.

It follows from [4, 5.4] that, if $a \in {\mathbb{Q}_v}^*$ has absolute value 1 and $(\Psi_v \cdot a)(x) = \Psi_v(ax)$, then

(5.5.3) $\eta_v(\sigma, M, \Psi_v) = \eta_v(\sigma, M, \Psi_v \cdot a)$ (for $\|a\|_v = 1$).

For an automorphism τ of $\mathbb{C}$ and finite v, $\tau\Psi_v$ is of the form $\Psi_v \cdot a$ with $\|a\|_v = 1$; likewise, $\bar{\Psi}_\infty = \Psi_\infty \cdot (-1)$.

For finite v, the definition of ε_v is purely algebraic, so $\tau\eta_v(\sigma, M, \Psi_v) = \eta_v(\tau\sigma, M, \tau\Psi_v)$, and (5.5.2) follows from (5.5.3).

For $v = \infty$, one also has $\bar{\eta}_\infty(\sigma, M, \bar{\Psi}_\infty) = \eta_\infty(\bar{\sigma}, M, \bar{\Psi}_\infty) = \eta_\infty(\bar{\sigma}, M, \Psi_\infty)$. If Ψ_∞ is chosen as in 5.3, η_∞ is a power of i independent of σ; hence $\eta_\infty(\sigma, M, \Psi_\infty) = \pm 1$, independently of σ, which proves (5.5.2).

THEOREM 5.6. Subject to Conjecture 6.6 on the nature of rank-one motives, Conjecture 2.8 is compatible with the conjectural functional equation of L-functions: one has

$L_\infty{}^*(M)c^+(M) \sim \varepsilon^*(M)L_\infty{}^*(\check{M}(1))c^+(\check{M}(1))$.

By 5.1, 5.3, and 5.5, this formula is equivalent to

$(2\pi i)^{-d^-(M)} \cdot \delta(M) \sim (2\pi)^{-d^-(M)} \cdot (2\pi)^{-wd(M)/2} \cdot \varepsilon^*(\det M)$.

Set $D = \det M$ and $\varepsilon = d^-(D)$. Then $\delta(M) = \delta(D)$, $d^-(M) \equiv \varepsilon \pmod 2$, and $wd(M)$ is the weight $w(D)$ of D, so that the formula is again equivalent to

(5.6.1) $\varepsilon^*(D) \sim (2\pi)^{w(D)/2} \cdot i^\varepsilon \cdot \delta(D)$.

In 6.5 we shall prove that (5.6.1) holds for a class of rank-one motives which conjecturally (6.6) includes them all.

6. Example: Artin L-functions. DEFINITION 6.1. The category of Artin motives is the Karoubian envelope of the dual of the category whose objects are zero-dimensional varieties over $\mathbb{Q}$ and whose morphisms are correspondences defined over $\mathbb{Q}$.

By definition, every zero-dimensional variety X defines an Artin motive H(X); H is a fully faithful contravariant functor,

H: (zero-dimensional varieties, correspondences) $\longrightarrow$ (Artin motives),

and every Artin motive is a direct factor of some H(X).

6.2. Let us make this definition explicit. Let $\bar{\mathbb{Q}}$ be an algebraic closure of $\mathbb{Q}$, and let G be the Galois group $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$. A zero-dimensional variety is the spectrum of a finite product A of number fields, and Galois theory, in the form given to it by Grothendieck, says that the functor

$X = \mathrm{spec}(A) \mapsto X(\bar{\mathbb{Q}}) = \mathrm{Hom}(A, \bar{\mathbb{Q}})$:

(category of zero-dimensional varieties over $\mathbb{Q}$ and morphisms of schemes) $\longrightarrow$ (category of finite sets with a continuous G-action)

is an equivalence of categories. The inverse functor sends I to the spectrum of the ring of G-invariant functions from I to $\bar{\mathbb{Q}}$.

A correspondence from a zero-dimensional variety X to another Y is a formal rational linear combination of connected components of $X \times Y$. The map

$\sum a_i Z_i \mapsto \sum a_i(\text{characteristic function of } Z_i(\bar{\mathbb{Q}}) \subset (X \times Y)(\bar{\mathbb{Q}}))$

identifies correspondences with rational-valued G-invariant functions on $(X \times Y)(\bar{\mathbb{Q}}) = X(\bar{\mathbb{Q}}) \times Y(\bar{\mathbb{Q}})$, and composition of correspondences with matrix multiplication.

Let H be the contravariant functor sending X to the vector space $\mathbb{Q}^{X(\bar{\mathbb{Q}})}$ with its natural G-action; a correspondence $F: X \to Y$ is sent to the morphism $F^*: \mathbb{Q}^{Y(\bar{\mathbb{Q}})} \to \mathbb{Q}^{X(\bar{\mathbb{Q}})}$ whose matrix is ${}^tF$. This functor is fully faithful and identifies the category of Artin motives with that of rational representations of G.

6.3. In this model, if $\bar{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ inside $\mathbb{C}$, the "Betti realization" functor H_B is the "underlying vector space" functor. The Hodge structure is purely of type (0, 0), and the involution F_∞ is the action of complex conjugation $F_\infty \in G$. Indeed, there is an isomorphism, functorial for correspondences,

$H^*(X(\mathbb{C}), \mathbb{Q}) = \mathbb{Q}^{X(\mathbb{C})} = \mathbb{Q}^{X(\bar{\mathbb{Q}})} = H(X)$.

The "l-adic realization" functor H_l is $H_l(V) = V \otimes \mathbb{Q}_l$. Indeed, there is an isomorphism, functorial for correspondences,

$H^*(X(\bar{\mathbb{Q}}), \mathbb{Q}_l) = {\mathbb{Q}_l}^{X(\bar{\mathbb{Q}})} = \mathbb{Q}^{X(\bar{\mathbb{Q}})} \otimes \mathbb{Q}_l$.

Let us similarly compute the de Rham realization. For $X = \mathrm{Spec}(A)$, ${H_{DR}}^*(X) = A$. Since $A = \mathrm{Hom}_G(X(\bar{\mathbb{Q}}), \bar{\mathbb{Q}}) = (\mathbb{Q}^{X(\bar{\mathbb{Q}})} \otimes \bar{\mathbb{Q}})^G$, we obtain

(6.3.1) $H_{DR}(V) = (V \otimes \bar{\mathbb{Q}})^G$.

Formula (6.3.1) realizes $H_{DR}(V)$ as a subspace of $V \otimes \bar{\mathbb{Q}}$. This subspace is a $\mathbb{Q}$-structure: $(V \otimes \bar{\mathbb{Q}})^G \otimes \bar{\mathbb{Q}} \to\simeq V \otimes \bar{\mathbb{Q}}$. After extending scalars to $\bar{\mathbb{Q}}$, $H_B(V)$ and $H_{DR}(V)$ are therefore canonically isomorphic. Extending scalars further to $\mathbb{C}$ gives the isomorphism (0.4.1); this may be checked for $V = H(X)$.

6.4. Let E be a finite extension of $\mathbb{Q}$. The category of Artin motives with coefficients in E is obtained from the category of Artin motives as in 2.1. We identify it with the category of finite-dimensional E-vector spaces endowed with a G-action.

Rank-one Artin motives with coefficients in E correspond to characters ε: $G \to E^*$. We now compute their periods. Let ε: $G \to E^*$ and let f be its conductor; ε factors through a character, again denoted ε, of the quotient $(\mathbb{Z}/f\mathbb{Z})^* = \mathrm{Gal}(\mathbb{Q}(\exp(2\pi i/f))/\mathbb{Q})$ of G. Let $[\varepsilon]$ denote the one-dimensional E-vector space E_ε on which G acts through ε. The Gauss sum

$g = \sum \varepsilon(u) \otimes \exp(2\pi i u/f) \in [\varepsilon] \otimes \bar{\mathbb{Q}}$

is nonzero and G-invariant; it is an E-basis of $H_{DR}([\varepsilon])$. The determinant of I: $H_B([\varepsilon]) \otimes \mathbb{C} \to\simeq H_{DR}([\varepsilon]) \otimes \mathbb{C}$, computed in the bases 1 and g, is g^{-1}. For every complex embedding σ of E one knows that $\sigma g \cdot \bar{\sigma} g = f$, a rational number independent of σ. Hence

(6.4.1) $\delta([\varepsilon]) \sim \sum \varepsilon^{-1}(u) \otimes \exp(-2\pi i u/f) \in (E \otimes \mathbb{C})^*$.

PROPOSITION 6.5. Let D be the motive $[\varepsilon](n)$. It is a rank-one motive over $\mathbb{Q}$ with coefficients in E and weight $-2n$. Write $\varepsilon(-1) = (-1)^\eta$, with $\eta = 0$ or 1. Then

$\varepsilon^*(D) \sim (2\pi)^{-n}\, i^{\eta - n}\, \delta(D)$.

The constant in the functional equation of the Dirichlet L-function

$L(\sigma, [\varepsilon], s) = \sum \sigma\varepsilon^{-1}(n) \cdot n^{-s}$,

where σ is a complex embedding of E, is known to be

$\varepsilon(\sigma, [\varepsilon], s) = i^\eta \cdot f^s \cdot \sum \sigma\varepsilon(u)^{-1} \exp(-2\pi i u/f)$.

On the other hand, (5.1.9) gives $\delta(D) = (2\pi i)^n\, \delta([\varepsilon])$. Applying (6.4.1), and observing that for integral $s = n$ the number f^n is rational and independent of σ, proves the proposition.

This proposition verifies (5.6.1) for the motives $[\varepsilon](n)$. To complete the proof of 5.6, it remains only to state:

Conjecture 6.6. Every rank-one motive over $\mathbb{Q}$ with coefficients in E is of the form $[\varepsilon](n)$, where ε is a character of $G = \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ with values among the roots of unity in E, and n is an integer.

PROPOSITION 6.7. Conjecture 2.8 is true for Artin L-functions.

For Artin motives, the functional equation of the L-functions is available. Moreover, the determinant of an Artin motive, after a Tate twist, is of the type predicted in 6.6. The arguments of Sections 5 and 6 therefore show the compatibility of 2.8 with the functional equation, and it is enough to prove 2.8 for motives V(n), where V is an Artin motive and $n \leq 0$ is an integer. If V(n) is critical, F_∞ acts on V by multiplication by $-(-1)^n$ and $c^+ = 1$ (cf. 3.1); one must prove that for every embedding $\sigma\colon E \to \mathbb{C}$ and every automorphism τ of $\mathbb{C}$,

$\tau L(\sigma, V(n)) = L(\tau\sigma, V(n))$.

This follows from Siegel's results [14]. See [2, 1.2].

7. L-functions attached to modular forms. 7.1. Put $q = e^{2\pi i z}$, and let $f = \sum a_{nq}{}^n$ be a primitive holomorphic cuspidal modular form (new form) of weight $k \geq 2$, conductor N, and character ε. The Dirichlet series $\sum a_{nn}{}^{-s}$ has an Euler-product expansion whose local factor at $p \nmid N$ is

$(1 - a_{pp}{}^{-s} + \varepsilon(p)p^{k-1}p^{-2s})^{-1}$.

Let E be the subfield of $\mathbb{C}$ generated by the a_n. The form f should give rise to a rank-two motive M(f) with coefficients in E, of Hodge type $\{(k-1, 0), (0, k-1)\}$, determinant $[\varepsilon^{-1}](1-k)$ in the notation of 6.4, and L-function given by the Dirichlet series $\sum a_{nn}{}^{-s}$.

I have not tried to define M(f) as a motive in Grothendieck's sense. One difficulty is that M(f) appears naturally as a direct factor in the cohomology of a noncompact variety, or as a direct factor in the cohomology of a complete modular curve with coefficients in the direct image of a locally constant sheaf—more properly, a local system of motives—on the finite part of that modular curve. This lies outside Grothendieck's formalism, but it permits the realizations of M(f) to be defined.

7.2. First suppose that $k = 2$ and ε is trivial. Let X be the Poincaré upper half-plane, let N be the conductor of f, and let $\Gamma_0(N)$ be the subgroup of $SL(2, \mathbb{Z})$ consisting of matrices whose reduction modulo N has the form $((*, *), (0, *))$. Put

$\omega_f = \sum a_{nq}{}^n \cdot dq/q = \sum a_{nq}{}^n \cdot 2\pi i\, dz$.

The form ω_f is a holomorphic differential on the completed curve $\bar{M}(\Gamma_0(N))$ associated with $M(\Gamma_0(N)) = X/\Gamma_0(N)$. It is an eigenvector for the Hecke correspondences:

(7.2.1) $T_n{}^*\omega_f = a_n\omega_f$ (for n prime to N),

and is characterized up to a factor by (7.2.1). This follows from strong multiplicity one and the theory of primitive forms. The reader may instead supplement (7.2.1) by the analogous condition for n not prime to N, and do the same below; the assertion then becomes elementary, since the completed condition (7.2.1) determines the Taylor expansion of ω_f at the cusp $i\infty$ up to a factor.

Abbreviate $M(\Gamma_0(N))$ to M and $\bar{M}(\Gamma_0(N))$ to $\bar{M}$. These curves have a natural $\mathbb{Q}$-structure for which the Hecke correspondences are defined over $\mathbb{Q}$. The form ω_f is defined over E, and its conjugate by an automorphism σ of $\mathbb{C}$ is $\omega_{\sigma f}$, obtained by applying σ to the coefficients.

In the language of 2.1(B), define M(f) to be the submotive of $H^1(\bar{M})_E$ that is the kernel of the endomorphisms $T_n{}^* - a_n$—if one wishes to use only

kernels of projectors, replace $T_n^* - a_n$ by $P(T_n - a_n\Delta)^*$ for a suitable polynomial P. The motive M(f) has the properties stated in 7.1.

In each cohomology theory $\mathscr{H}$ under consideration, the map

(7.2.2) $\mathscr{H}_c^1(M) \longrightarrow \mathscr{H}^1(\bar{M})$

is surjective, and the systems of eigenvalues of the T_n^* on its kernel do not occur in $\mathscr{H}^1(\bar{M})$. The $\mathscr{H}$-realization of M(f) is therefore still the common kernel, inside $\mathscr{H}_c^1(M) \otimes E$, of the $T_n^* - a_n$.

The cohomology group $H_c^1(M, \mathbb{Q})$ carries a mixed Hodge structure of type $\{(0, 0), (0, 1), (1, 0)\}$, with $H^1(\bar{M}, \mathbb{Q})$ as its quotient of type $\{(0, 1), (1, 0)\}$. In particular, (7.2.2) induces an isomorphism on the F^1-subspaces of the Hodge filtration; every holomorphic differential ω on $\bar{M}$ therefore defines a compactly supported cohomology class on M. This can be seen directly. If $\mathscr{J} \subset \mathscr{O}$ is the ideal of the cusps, de Rham cohomology $H_c^*(M)$ is the hypercohomology on $\bar{M}$ of the complex $\mathscr{J} \to \Omega^1$; analytically this complex resolves the constant sheaf $\mathbb{C}$ on M extended by zero to $\bar{M}$, and H^1 receives $H^0(\bar{M}, \Omega^1)$.

For simplicity suppose $E = \mathbb{Q}$, and compute $c^+(M(f)(1))$. The motive M(f)(1) is dual to M(f), so c^+ is a period of M(f) by 1.7: one integrates ω_f against a rational homology class of $\bar{M}$ fixed by F_∞. Lifting M(f) to $\mathscr{H}_c^1(M)$ shows that one may instead integrate ω_f against a homology class without compact support on M fixed by F_∞. All nonzero integrals of this type are commensurable.

To compute F_∞, it is useful to write M as the quotient of $X^\pm = \mathbb{C} - \mathbb{R}$ by the subgroup of $GL(2, \mathbb{Z})$ whose reduction modulo N has the form $((*, *), (0, *))$. Complex conjugation is induced by $z \mapsto \bar{z}$, and the image of $i\mathbb{R}^+$ in $M(\mathbb{C})$ is a noncompact cycle fixed by F_∞. The formula

$L(M(f), 1) = -\int_0^{i\infty} \omega_f$

therefore verifies 1.8 for M(f)(1): the right-hand side is either zero or equivalent to $c^+(M(f)(1))$.

REMARK 7.3. One also has

$c^\pm(M(f)(1)) \sim \int_{a/b}^{i\infty} \omega_f \pm \int_{-a/b}^{i\infty} \omega_f$.

7.4. For arbitrary ε and E, $\Gamma_0(N)$ must be replaced by $\Gamma_1(N)$, the subgroup of $SL(2, \mathbb{Z})$ consisting of matrices whose reduction modulo N has the form $((1, *), (0, 1))$. As above, for computing F_∞ it is more convenient to work with $X^\pm$ and the subgroup of $GL(2, \mathbb{Z})$ whose reduction modulo N has the form $((1, *), (0, *))$. A dualization also appears, hidden in 7.2 by the symmetry of the correspondence T_n. For a suitable definition of T_n one has:

(a) M(f) is the common kernel of the $T_n^* - a_n$ in $H^1(\bar{M})$, where $\bar{M} = X/\Gamma_1(N)$.

(b) One has ${}^tT_n^*\omega_f = a_n\omega_f$ and $T_n^*\omega_f = \bar{a}_n\omega_f$, where $\bar{a}_n = \varepsilon(n)^{-1}a_n$. Thus ω_f belongs to the de Rham realization of $M(f) = M(f)^\vee(-1)$.

It follows that $c^+(M(f)(1)) \in (E \otimes \mathbb{C})^*/E^* \cong \mathbb{C}^{Hom(E,\mathbb{C})}/E^*$ is given by the system of periods

$c^+(M(f)(1)) \sim (\int_0^{i\infty} \omega_{\sigma f})_\sigma$,

provided this system is nonzero. If one of these integrals vanishes, they all vanish. This happens exactly when, in the part of noncompact homology

spanned by the cycle $i\mathbb{R}^+$ and its transforms under the T_n, the system of eigenvalues a_n for the T_n does not occur. This proves 2.8, and partly 2.7, for M(f)(1).

REMARK 7.5. The systems of T_n-eigenvalues in $\mathrm{Ker}(\mathscr{H}_c{}^1(M) \to \mathscr{H}^1(\bar{M}))$ are associated with Eisenstein series, whereas those occurring in $\mathscr{H}^1(\bar{M})$ are associated with cusp forms. The two sets are therefore disjoint. It follows that the mixed Hodge structure on $H_c{}^1(M, \mathbb{Q})$ is a direct sum of Hodge structures: the extension of $H^1(\bar{M}, \mathbb{Q})$ by $\mathrm{Ker}(H_c{}^1(M, \mathbb{Q}) \to H^1(\bar{M}, \mathbb{Q}))$ splits. Generalized to any congruence subgroup of $SL(2, \mathbb{Z})$, this is equivalent to Manin's theorem [9] that the difference of two cusps always has finite order in the Jacobian (cf. [6, 10.3.4, 10.3.8, and 10.1.3]). The proof just given does not differ in substance from Manin's.

7.6. For arbitrary weight k, the cohomology of $\bar{M}$ must be replaced by cohomology with coefficients in a suitable sheaf. To describe the construction, replace M and $\bar{M}$ by M_n and $\bar{M}_n$ for the congruence subgroup $\Gamma(n)$, where n is a multiple of N and $n \geq 3$. One then descends to M and $\bar{M}$ by taking invariants under a suitable finite group. Let $g: \mathscr{E} \to M_n$ be the universal elliptic curve and j the inclusion $M_n \to \bar{M}_n$. The relevant cohomology is

$H^1(\bar{M}_n, j_*\mathrm{Sym}^{k-2}(R^1g_*\mathbb{Q}))$.

The realizations of M(f) are direct factors of this group, computed in the corresponding cohomology theory. As before they may be lifted to

$H_c{}^1(M_n, \mathrm{Sym}^{k-2}(R^1g_*\mathbb{Q}))$.

To compute $c^{\pm}$ of the dual of M(f), one integrates the class in $H_c{}^1$ defined by f against noncompact homology classes on M_n with coefficients in the local system dual to $\mathrm{Sym}^{k-2}(R^1g_*\mathbb{Q})$. Taking the cycle that is the image of $i\mathbb{R}^+$, together with various sections of the dual local system forming a basis, gives the Eichler integrals

$\int_0{}^{i\infty} f(q) \cdot dq/q \cdot (2\pi iz)^l$ $(0 \leq l \leq k-2)$,

even periods for even l and odd periods for odd l. Critical values L(M(f), n) can be expressed in terms of these periods.

PROPOSITION 7.7. Let M be a rank-two motive with coefficients in E and Hodge type {(a, b), (b, a)}, where $a \neq b$. Then $d^{\pm}\mathrm{Sym}^n(M)$ and $c^{\pm}\mathrm{Sym}^{nM}$ are given by the following formulas:

(1) if $n = 2l + 1$, then $d^{\pm} = l + 1$ and

$c^{\pm}\mathrm{Sym}^{nM} = c^{\pm}(M)^{(l+1)(l+2)/2}\, c^{\mp}(M)^{l(l+1)/2}\, \delta(M)^{l(l+1)/2}$;

(2) if $n = 2l$, then $d^+ = l + 1$, $d^- = l$, and

$c^+\mathrm{Sym}^{nM} = (c^+(M)c^-(M))^{l(l+1)/2}\, \delta(M)^{l(l+1)/2}$, $c^-\mathrm{Sym}^{nM} = (c^+(M)c^-(M))^{l(l+1)/2}\, \delta(M)^{l(l-1)/2}$.

This is a question of elementary linear algebra, which we treat by "dimensional analysis."

(a) $H_B(M)$ is a rank-two E-vector space with an involution F_∞ having matrix ((0, 1), (1, 0)) in a suitable basis e^+, e^-. The quantities $d^{\pm}\mathrm{Sym}^{nM}$ are the dimensions of the plus and minus parts of the nth symmetric power, with respective bases

$\{(e^+)^n, (e^+)^{n-2}(e^-)^2, \ldots\}$ and $\{(e^+)^{n-1}e^-, (e^+)^{n-3}(e^-)^3, \ldots\}$.

The stated dimensions follow.

(b) Let ω, η be a basis of the dual of $H_{DR}(M)$, with ω annihilating $F^+H_{DR}(M)$. The subspace $F^{\pm}$ of $H_{DR}(Sym^{nM})^\vee \sim Sym^{nH}{}_{DR}(M)^\vee$ has basis

$\omega^n, \omega^{n-1}\eta, \ldots, \omega^{n-d^{\pm}+1}\eta^{d^{\pm}-1}$,

and

(7.7.1)$^+$ $c^+Sym^{nM} = \langle \omega^n \wedge (\omega^{n-1}\eta) \wedge \ldots, (e^+)^n \wedge ((e^+)^{n-2}(e^-)^2) \wedge \ldots\rangle$,

(7.7.1)$^-$ $c^-Sym^{nM} = \langle \omega^n \wedge (\omega^{n-1}\eta) \wedge \ldots, ((e^+)^{n-1}e^-) \wedge ((e^+)^{n-3}(e^-)^3) \wedge \ldots\rangle$.

(c) Put $V = H_B(M) \otimes \mathbb{C} \sim H_{DR}(M) \otimes \mathbb{C}$; this is an $E \otimes \mathbb{C}$-module. The formulas above show that $c^{\pm}Sym^{nM}$ depends only on V as an $E \otimes \mathbb{C}$-module, on its basis e^+, e^-, on $\omega \in V^*$, and on the image $\bar\eta$ of η in $V^*/\langle\omega\rangle$. The $d^{\pm}$-vector on the left of the pairing in (7.7.1) does not change when η is replaced by $\eta + \lambda\omega$. Moreover, $\langle \omega, e^+\rangle$ and $\langle \omega, e^-\rangle$ are invertible and $\bar\eta$ is a basis of $V^*/\langle\omega\rangle$. Up to isomorphism, the system $(V, e^+, e^-, \omega, \bar\eta)$ is therefore described by

$c^+ = \langle \omega, e^+\rangle$, $c^- = \langle \omega, e^-\rangle$, $\delta = \langle \omega \wedge \eta, e^+ \wedge e^-\rangle$

in $(E \otimes \mathbb{C})^*$. Replacing $e^+, e^-, \omega, \bar\eta$ by $\lambda e^+, \mu e^-, \alpha\omega, \nu\bar\eta/\lambda\mu$ replaces c^+, c^-, and δ by $\lambda\alpha c^+$, $\mu\alpha c^-$, and $\nu\delta$. When $c^+ = c^- = \delta = 1$, the right-hand side of (7.7.1)$^{\pm}$ lies in $\mathbb{Q}^*$. In general, $c^{\pm}Sym^{nM}$ is therefore a rational multiple of the product of $c^+(M)$, $c^-(M)$, and $\delta(M)/c^+(M)c^-(M)$, raised to the respective degrees with which e^+, e^-, and $\bar\eta$ occur in (7.7.1). Half the calculation may be avoided by observing that replacing F_∞ by $-F_\infty$ exchanges e^+ and e^-, hence c^+ and c^-, preserves δ, and exchanges (7.7.1)$^+$ and (7.7.1)$^-$ when n is odd while preserving them when n is even.

For $n = 2l + 1$, $d^+ = d^- = l + 1$:

deg $\bar\eta$ in (7.7.1)$^{\pm} = 0 + 1 + \cdots + l = l(l + 1)/2$, deg e^+ in (7.7.1)$^+ = (2l + 1) + (2l - 1) + \cdots + 1 = (l + 1)^2$ = deg e^- in (7.7.1)$^-$, deg e^+ in (7.7.1)$^- = 2l + (2l - 2) + \cdots + 0 = l(l + 1)$ = deg e^- in (7.7.1)$^+$.

For $n = 2l$, $d^+ = l + 1$ and $d^- = l$:

deg $\bar\eta$ in (7.7.1)$^+ = 0 + 1 + \cdots + l = l(l + 1)/2$, deg $\bar\eta$ in (7.7.1)$^- = 0 + 1 + \cdots + (l - 1) = l(l - 1)/2$, deg $e^{\pm}$ in (7.7.1)$^+$ $= 2l + (2l - 2) + \cdots + 0 = l(l + 1)$, deg $e^{\pm}$ in (7.7.1)$^- = (2l - 1) + (2l - 3) + \cdots + 1 = l^2$.

7.8. This proposition gives a conjecture for the values at critical integers of $L(Sym^{nM}(f), s)$. The formulas are as follows:

(7.8.1) $L(\sigma, M(f), s) = \sum \sigma a_n\, n^{-s} = \prod_p L_p(\sigma, M(f), s)$,

where for almost every p,

$L_p(M(f), s) = (1 - a_{pp}{}^{-s} + \varepsilon(p)p^{k-1-2s})^{-1} = ((1 - \alpha'_{pp}{}^{-s})(1 - \alpha''_{pp}{}^{-s}))^{-1}$.

Here ε is a Dirichlet character. Since $\wedge^2 M(f) = [\varepsilon^{-1}](1 - k)$, one obtains

(7.8.2) $\delta(M(f)) \sim (2\pi i)^{1-k} \cdot \sum_{u \bmod F} \varepsilon(u) \otimes \exp(-2\pi i u/F)$,

if ε has conductor F, and

(7.8.3) $L^*(M(f), m) \sim (2\pi i)^m\, c^{\pm}(M(f))$, $\pm = (-1)^m$, for $1 \le m \le k - 1$.

(7.8.4) $L(\sigma, \mathrm{Sym}^{nM}(f), s) = \prod_p L_p(\sigma, \mathrm{Sym}^{nM}(f), s)$,

where for almost every p,

$L_p(\mathrm{Sym}^{nM}(f), s)^{-1} = \prod_{i=0}^{n} (1 - \alpha'_p{}^{i}\, \alpha''_p{}^{n-i}\, p^{-s})$.

Conjecturally, for critical m,

$L^*(\mathrm{Sym}^{nM}(f), m) \sim (2\pi i)^{m\, d^{\pm} Sym^n M(f)} \cdot c^{\pm}\mathrm{Sym}^{nM}(f), \pm = (-1)^m$,

where $c^{\pm}$ and $d^{\pm}$ are given in terms of $c^{\pm}(M)$ and $\delta(M)$, characterized by (7.8.2) and (7.8.3), through the formulas of 7.7.

For numerical evidence supporting this conjecture, see [18].

8. Algebraic Hecke characters. In [3, paragraph 5], whose notation we shall use, the reader will find the essential definitions concerning algebraic Hecke characters (= Grössencharaktere of type A_0).

Conjecture 8.1. Let k and E be two finite extensions of $\mathbb{Q}$, and let χ be an algebraic Hecke character of k with values in E.

(i) There exists a rank-one motive $M(\chi)$ over k, with coefficients in E, such that, for every place λ of E, the λ-adic representation $H_\lambda M(\chi)$ is the one defined by χ: geometric Frobenius at a prime $\mathfrak{P}$ prime to the conductor of χ and to the residue characteristic l of λ acts by multiplication by $\chi(\mathfrak{P})$.

(ii) This motive is characterized up to isomorphism by this property.

(iii) Every rank-one motive is of the form $M(\chi)$.

(iv) Decompose $k \otimes E$ as a product of fields, $k \otimes E = \prod K_i$, and write the algebraic part χ_{alg}: $k^* \to E^*$ of χ in the form

$\chi_{alg}(x) = \prod N_{K_i/E}(x)^{n_i}$.

The decomposition of $k \otimes E$ induces a decomposition of $H_{DR}(M(\chi))$ into

$H_{DR}(M(\chi))_i := H_{DR}(M(\chi)) \otimes_{k\otimes E} K_i$;

with this notation, the Hodge filtration is the filtration by

$\oplus_{n_i \geq p} H_{DR}(M(\chi))_i$.

The uniqueness in 8.1(ii) imposes the following formalism on the $M(\chi)$:

(8.1.1) $M(\chi'\chi'') \sim M(\chi') \otimes M(\chi'')$.

(8.1.2) If ι: $E \to E'$ is a finite extension of E, then $M(\iota\chi)$ is obtained from $M(\chi)$ by extension of coefficients from E to E′.

(8.1.3) If k′ is a finite extension of k, then $M(\chi \circ N_{k'/k})$ is obtained from $M(\chi)$ by extension of scalars from k to k′.

REMARK 8.2. Introduce the notation c = complex conjugation, $\bar{\mathbb{Q}}$ = the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, $S = \mathrm{Hom}(k, \bar{\mathbb{Q}}) = \mathrm{Hom}(k, \mathbb{C})$, and $J = \mathrm{Hom}(E, \bar{\mathbb{Q}}) = \mathrm{Hom}(E, \mathbb{C})$. The decomposition of $k \otimes E$ into the fields K_i corresponds to the partition of $S \times J$ into $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-orbits.

Every algebraic homomorphism η: $k^* \to E^*$ can be written in the form $\prod N_{K_i/E}(x)^{n_i}$. If the $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-orbit of $(\sigma, \tau) \in S \times J$ corresponds to K_i, that is, if $\sigma \otimes \tau$: $k \otimes E \to \mathbb{C}$ factors through K_i, write $n(\eta; \sigma, \tau)$, or simply $n(\sigma, \tau)$, for the integer n_i. The function $n(\sigma, \tau)$ is constant on $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$-orbits. If η is the algebraic part of a Hecke character χ, then in addition

(8.2.1) The integer $w = n(\eta; \sigma, \tau) + n(\eta; c\sigma, \tau)$ is independent of σ and τ.

This is the weight of χ, and of $M(\chi)$. We shall sometimes write $n(\chi; \sigma, \tau)$ for $n(\chi_{alg}; \sigma, \tau)$.

Conversely, a homomorphism η satisfying (8.2.1) is nearly of the form χ_{alg} for a suitable Hecke character χ: (a) some power of it is of this form; (b) there is a finite extension $\iota\colon E \to E'$ of E such that $\iota\eta$ is of this form; (c) there is a finite extension k' of k such that $\eta \circ N_{k'/k}$ is of this form.

Rule 8.1(iv) determines the Hodge bigrading of $H_\sigma M(\chi)$ from its E-module structure: the direct factor $H_\sigma M(\chi) \otimes_{E,\tau} \mathbb{C}$ of $H_\sigma M(\chi) \otimes \mathbb{C}$ has Hodge type (p, q), where $p = n(\chi; \sigma, \tau)$ and $q = w - p = n(\chi; \sigma, c\tau)$.

EXAMPLE 8.3. Let A be an abelian variety over k with complex multiplication by E. Suppose $H_1(A)$ has rank one over E (CM type). It follows from the theory of Shimura and Taniyama that $H_1(A)$ satisfies condition 8.1(i) for an algebraic Hecke character χ of k with values in E, and that the algebraic part χ_{alg} of χ is read from the $k \otimes E$-module Lie(A):

$\chi_{alg}(x) = \det_E(x \otimes 1, \mathrm{Lie}(A))^{-1}$.

EXAMPLE 8.4. Let k be the field of n-th roots of unity, let V be the Fermat hypersurface with projective equation $\sum_{i=0}^{m} X_i^n = 0$, and let M be the motive "primitive cohomology in the middle dimension of V^m." Let G be the quotient of μ_n^{m+1} by its diagonal subgroup. This group acts on V by $(\alpha_i) * (X_i) = (\alpha_i X_i)$, and on M by transport of structure. Decomposing M by means of the product decomposition of the group algebra $\mathbb{Q}[G]$, one obtains motives satisfying 8.1(i) for suitable algebraic Hecke characters, namely those introduced by Weil in his study of Jacobi sums.

EXAMPLE 8.5. The motive $\mathbb{Z}(-1)$ satisfies 8.1(i) for χ equal to the norm. For χ of finite order, a suitable Artin motive satisfies 8.1(i).

8.6. For every $\sigma \in S$, $H_\sigma(M(\chi))$ has rank one over E. Choose a basis e_σ of each $H_\sigma(M(\chi))$. Besides its E-module structure, the sum of the $H_\sigma(M(\chi)) \otimes \mathbb{C}$ has a natural $k \otimes \mathbb{C} = \mathbb{C}^S$-module structure; altogether it is a free rank-one $k \otimes E \otimes \mathbb{C}$-module, for which $e = \sum e_\sigma$ is a basis.

8.7. The de Rham realization $H_{DR}(M(\chi))$ is a free rank-one $k \otimes E$-module. Choose a basis ω. The sum of the maps $I_\sigma\colon H_\sigma(M(\chi)) \otimes \mathbb{C} \xrightarrow{\sim} H_{DR}(M(\chi)) \otimes_{k,\sigma} \mathbb{C}$ is the isomorphism (0.4) of $k \otimes E \otimes \mathbb{C}$-modules

$I\colon \oplus_\sigma H_\sigma(M(\chi)) \otimes \mathbb{C} \xrightarrow{\sim} H_{DR}(M(\chi)) \otimes_{\mathbb{Q}} \mathbb{C}$.

Put $p'(\chi) = \omega/I(e) \in (k \otimes E \otimes \mathbb{C})^*$. This period depends on χ, ω, and e. Modulo $(k \otimes E)^*$ and E^{*S}, it depends only on χ. Write $p'(\chi; \sigma, \tau)$, or simply $p'(\sigma, \tau)$, for the component indexed by (σ, τ) of its image under the isomorphism $k \otimes E \otimes \mathbb{C} \xrightarrow{\sim} \mathbb{C}^{S \times J}$.

8.8. Let A be as in 8.3 and calculate the periods $p'(\sigma, \tau)$, modulo $\bar{\mathbb{Q}}^*$, for the motive $H^1(A)$. First extend scalars from k to $\bar{\mathbb{Q}}$ using σ. If $n(\sigma, \tau) = 1$, there is a holomorphic 1-form ω defined over $\bar{\mathbb{Q}}$ such that $u^*\omega = \tau(u)\omega$ for $u \in E$, and, for $Z \in H_1(A(\mathbb{C}))$, one has $p'(\sigma, \tau) \sim \int_Z \omega$. The general case follows from

$p'(\sigma, \tau) \cdot p'(\sigma, c\tau) \sim 2\pi i$.

8.9. Conjecture 8.1(ii) asserts in particular that two motives satisfying 8.1(i) have the same period p. For the motives of 8.4, the periods are expressed

in terms of values of the Γ function and, when working modulo $\bar{\mathbb{Q}}^*$, 8.1 suggests B. Gross's conjecture [7] relating certain periods to products of Γ-values.

Comparison of 8.4 and 8.5 leads to Conjectures 8.11 and 8.13 below. The result announced after 0.10 proves them.

8.10. Let N be an integer, $k = \mathbb{Q}(\exp(2\pi i/N))$, $\mathfrak{P}$ a prime ideal of k prime to N, $k_{\mathfrak{P}}$ its residue field, and $q = N\mathfrak{P} = |k_{\mathfrak{P}}|$. Let t denote the inverse of reduction modulo $\mathfrak{P}$, from the N-th roots of unity in $k_{\mathfrak{P}}$ to those in k.

For $a \in N^{-1}\mathbb{Z}/\mathbb{Z}$, $a \neq 0$, consider the Gauss sum

$g(\mathfrak{P}, a, \Psi) = -\sum_{x \in k_{\mathfrak{P}}^*} t(x^{-a(q-1)})\, \Psi(x)$.

The sum is over $k_{\mathfrak{P}}^*$, and $\Psi\colon k_{\mathfrak{P}} \to \mathbb{C}^*$ is a nontrivial additive character.

Let $\mathbf{a} = \sum n(a)\delta_a$ in the free abelian group with basis $N^{-1}\mathbb{Z}/\mathbb{Z} - \{0\}$. If $\sum n(a)a = 0$, the product of the $g(\mathfrak{P}, a, \Psi)^{n(a)}$ is independent of Ψ, and set

$g(\mathfrak{P}, \mathbf{a}) = \prod_a g(\mathfrak{P}, a, \Psi)^{n(a)}$.

Weil [16] proved that, as a function of $\mathfrak{P}$, $g(\mathfrak{P}, \mathbf{a})$ is an algebraic Hecke character $\chi_{\mathbf{a}}$ of k with values in k. Let $\langle a \rangle$ be the representative between 0 and 1 of a in $\mathbb{Z}/N$. If $\mathbf{a} = \sum n(a)\delta_a$ satisfies

(*) For every $u \in (\mathbb{Z}/N)^*$, $\sum n(a)\langle ua \rangle = 0$,

Weil's determination of the algebraic part of $\chi_{\mathbf{a}}$ implies that $\chi_{\mathbf{a}}$ has finite order; write $\chi_{\mathbf{a}}$ also for the character of $\mathrm{Gal}(\bar{\mathbb{Q}}/k)$ whose value at geometric Frobenius at $\mathfrak{P}$ is $\chi_{\mathbf{a}}(\mathfrak{P})$.

Put $\Gamma(\mathbf{a}) = \prod \Gamma(\langle a \rangle)^{n(a)}$. A first part of the conjecture, the algebraicity of $\Gamma(\mathbf{a})$ when $\mathbf{a}$ satisfies (*), was proved by Koblitz and Ogus; see the appendix. More precisely, one would like:

Conjecture 8.11. If $\mathbf{a}$ satisfies (*), then $\sigma\Gamma(\mathbf{a}) = \chi_{\mathbf{a}}(\sigma) \cdot \Gamma(\mathbf{a})$ for every $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/k)$.

If $\mathbf{a}$ is invariant under a subgroup H of $(\mathbb{Z}/N)^*$, 8.11 can be sharpened by replacing $\mathrm{Gal}(\bar{\mathbb{Q}}/k)$ with $\mathrm{Gal}(\bar{\mathbb{Q}}/k^H)$, and by using SGA $4^{1/2}$ 6.5 to define a character of this group with values in the roots of unity of k^H. This is done below.

8.12. Let H be a subgroup of $(\mathbb{Z}/N)^*$, $\mathfrak{P}$ a prime ideal of k^H prime to N, κ its residue field, and $\Psi\colon \kappa \to \mathbb{C}^*$ a nontrivial additive character. For $a \in N^{-1}\mathbb{Z}/\mathbb{Z} - \{0\}$, let $\mathfrak{P}_{a,i}$ be the prime ideals of $k^H(\exp(2\pi i a))$ above $\mathfrak{P}$. If κ' is the residue field at $\mathfrak{P}_{a,i}$ and $|\kappa'| = q'$, write $g(\mathfrak{P}_{a,i}, a, \Psi)$ for the Gauss sum

$-\sum_{x \in \kappa'^*} t(x^{-a(q'-1)})\, \Psi(\mathrm{Tr}_{\kappa'/\kappa} x)$.

The product $g(\mathfrak{P}, a, \Psi) = \prod_i g(\mathfrak{P}_{a,i}, a, \Psi)$ depends only on the H-orbit of a.

Let $\mathbf{a} = \sum n(a)\delta_a$ be H-invariant and satisfy $\sum n(a)a = 0$. For every H-orbit O in $N^{-1}\mathbb{Z}/\mathbb{Z} - \{0\}$, write n(O) for the constant value of n(a) on O, and put

$g(\mathfrak{P}, \mathbf{a}) = \prod_{a \bmod H} g(\mathfrak{P}, a, \Psi)^{n(a)}$.

As a function of $\mathfrak{P}$, $g(\mathfrak{P}, \mathbf{a})$ is an algebraic Hecke character $\chi_{\mathbf{a}}$ of k^H with values in k^H. To prove this, by additivity one reduces to $n(a) \geq 0$. Apply [3, 6.5] with $F = k^H$, $\bar{F} = \bar{\mathbb{Q}}$, k equal to the present k, and I a disjoint union of H-orbits, with n(O) copies of O. For $i \in I$ whose image in $N^{-1}\mathbb{Z}/\mathbb{Z}$ is a/N, take λ_i to be the composite

$\widehat{\mathbb{Z}}(1)_F \to (\mathbb{Z}/N)(1)_F = \mu_N(k) \to^{x \mapsto x^a} k^*$.

The resulting Hecke character is the product of $\chi_{\mathbf{a}}$ with the "signature" character of the

permutation representation of H on I″. A special case of this result already appears in Weil [17]. The character $\chi_{\mathbf{a}}$ of 8.10 is the composite of the $\chi_{\mathbf{a}}$ above with the norm N_{k/k^H}.

Conjecture 8.13. Let $\mathfrak{P}$ be a prime ideal of k^H prime to N, and let $F_{\mathfrak{P}}$ be a geometric Frobenius at $\mathfrak{P}$. If the H-invariant **a** satisfies (*), then

$F_{\mathfrak{P}}\Gamma(\mathbf{a}) = g(\mathfrak{P}, \mathbf{a}) \cdot \Gamma(\mathbf{a})$.

In particular, if the group of roots of unity of k^H has order N′, then $\Gamma(\mathbf{a})^{N'} \in k^H$.

One can deduce from 8.13 the following, apparently more general, variant. Replace condition (*) by

(*′) $\sum n(a)\langle ua\rangle = k$, an integer independent of $u \in (\mathbb{Z}/N)^*$.

The Hecke character $N\mathfrak{P}^{-k} \cdot g(\mathfrak{P}, \mathbf{a})$ then has finite order. Identify it with a character χ of $\mathrm{Gal}(\bar{\mathbb{Q}}/k^H)$; one expects $\sigma((2\pi i)^{-k}\,\Gamma(\mathbf{a})) = \chi(\sigma) \cdot ((2\pi i)^{-k}\,\Gamma(\mathbf{a}))$.

8.14. If k is a totally imaginary quadratic extension of a totally real field—that is, in the terminology used here, a field of CM type—Shimura [13] determined the critical values of the L-functions of algebraic Hecke characters of k up to multiplication by an algebraic number. He expresses them in terms of periods of CM abelian varieties with complex multiplication by k. In the remainder of the section, we show that his theorem is compatible with Conjecture 2.8, which expresses them in terms of periods of rank-one motives over k.

8.15. Let χ and $M(\chi)$ be as in 8.1. Exclude the case 8.5 and suppose $R_{k/\mathbb{Q}}M(\chi)$ satisfies 1.7. Then k is totally imaginary, and the Hodge numbers $h^{p,p}$ vanish: with the notation of 8.1 and 8.2, no n_i equals w/2. Our first task is to compute $c^+R_{k/\mathbb{Q}}M(\chi)$ in terms of the periods $p'(\chi; \sigma, \tau)$. Recall that it is the determinant of the isomorphism of $E \otimes \mathbb{C}$-modules

$I^+\colon H_B{}^+R_{k/\mathbb{Q}}M(\chi) \otimes \mathbb{C} \xrightarrow{\sim} H_{DR}{}^+R_{k/\mathbb{Q}}M(\chi) \otimes \mathbb{C}$,

calculated in bases defined over E.

Choose the e_σ of 8.6 so that $F_\infty e_\sigma = e_{c\sigma}$. The vectors $e_\sigma \pm e_{c\sigma}$ then form a basis of $H_B{}^{\pm}R_{k/\mathbb{Q}}M(\chi) \subset H_{BRk/\mathbb{Q}}M_\chi = \oplus_{\sigma\in S}H_\sigma M(\chi)$. Note that $d^+ = d^- = \tfrac{1}{2}[k{:}\mathbb{Q}]$. Let $\bar{S}$ be the quotient of S by $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$. To calculate $c^+ = \det(I^+)$, use the basis $(e_\sigma + e_{c\sigma})$ of $H_B{}^+$, indexed by $\bar{S}$.

By 8.1(iv), the Hodge filtration of $H_{DRk/\mathbb{Q}}M(\chi) = H_{DRM}(\chi)$ is read from its $k \otimes E$-module structure. Let $(k \otimes E)^+$ be the direct factor of $k \otimes E$ that is the product of the K_i with $n_i < w/2$. The quotient $H_{DR}{}^+R_{k/\mathbb{Q}}M(\chi)$ is the corresponding direct factor:

$H_{DR}{}^+R_{k/\mathbb{Q}}M(\chi) = H_{DRM}(\chi) \otimes_{k\otimes E} (k \otimes E)^+$.

Let ω be as in 8.7, and use the $k \otimes E \otimes \mathbb{C} = \mathbb{C}^{S\times J}$-module structure of $H_{DRM}(\chi)$ to decompose ω as $\omega = \sum\omega_{\sigma,\tau}$. By definition $I(e_\sigma) = \sum_\tau p'(\sigma,\tau)^{-1}\omega_{\sigma,\tau}$, hence

$I^+(e_\sigma + e_{c\sigma}) = \sum_{n(\sigma,\tau)<w/2} p'(\sigma,\tau)^{-1}\omega_{\sigma,\tau} + \sum_{n(c\sigma,\tau)<w/2} p'(c\sigma,\tau)^{-1}\omega_{c\sigma,\tau}$.

This is the sum, indexed by $\tau \in J$, of terms $p'(\sigma, \tau)^{-1}\omega_{\sigma,\tau}$ when $n(\sigma, \tau) < w/2$, and $p'(c\sigma, \tau)^{-1}\omega_{c\sigma,\tau}$ when $n(\sigma, \tau) > w/2$. For $\bar{\sigma} \in \bar{S}$ represented by σ, put

$\omega_{\bar{\sigma}} = \sum_{n(\sigma,\tau)<w/2} \omega_{\sigma,\tau} + \sum_{n(c\sigma,\tau)<w/2} \omega_{c\sigma,\tau}$.

The vectors $I^+(e_\sigma + e_{c\sigma})$ are multiples of the $\omega_{\bar{\sigma}}$ by elements of $E \otimes \mathbb{C}$; thus the $\omega_{\bar{\sigma}}$ form a basis of ${H_{DR}}^+$. In the bases $e_\sigma + e_{c\sigma}$ and $\omega_{\bar{\sigma}}$, the matrix of I^+ is diagonal, and its determinant $\det'(I^+) \in (E \otimes \mathbb{C})^* = \mathbb{C}^{*J}$ has coordinates

$\det'(I^+)_\tau = \prod_{n(\sigma,\tau)<w/2} p'(\sigma, \tau)^{-1}$.

Consider the maps $E^{\bar{S}} \to E^S$ given by $1_{\bar{\sigma}} \mapsto 1_\sigma + 1_{c\sigma}$, for $\bar{\sigma}$ the image of σ; the isomorphism $E^S \otimes \mathbb{C} \xrightarrow{\sim} k \otimes E \otimes \mathbb{C}$ induced by $k \otimes \mathbb{C} \cong \mathbb{C}^S$; and the projection of $k \otimes E$ onto $(k \otimes E)^+$. Their composite is an isomorphism of $E \otimes \mathbb{C}$-modules

$E^{\bar{S}} \otimes \mathbb{C} \xrightarrow{\sim} (k \otimes E)^+ \otimes \mathbb{C}$.

Let $D(\chi)$ be its determinant, calculated in bases defined over E on both sides. Identifying ${H_{DR}}^+$ with $k \otimes E$ by means of the basis ω, this is the determinant of the identity map on ${H_{DR}}^+ \otimes \mathbb{C}$, calculated in the basis $\omega_{\bar{\sigma}}$ at the source and in a basis defined over E at the target. Let $D_\tau(\chi)$ denote its components in $\mathbb{C}^J$. Since $c^+ = \det'(I^+) \cdot D(\chi)$, one obtains:

PROPOSITION 8.16. One has

$c^+ R_{k/\mathbb{Q}} M(\chi) = (\prod_{n(\sigma,\tau)<w/2} p'(\sigma, \tau)^{-1} \cdot D_\tau(\chi))_{\tau \in J}$.

REMARK 8.17. Suppose k is of CM type, a quadratic extension of the totally real field k_0. Then $\bar{S}$ identifies with the set of complex embeddings of k_0, and the diagram

commutes. The composite $k_0 \otimes E \to (k \otimes E)^+$ is therefore an isomorphism, and $D(\chi)$ is also the determinant of $E^{\bar{S}} \otimes \mathbb{C} \xrightarrow{\sim} k_0 \otimes E \otimes \mathbb{C}$ obtained by extending scalars from $\mathbb{Q}$ to E in the isomorphism $\mathbb{C}^{\bar{S}} \xrightarrow{\sim} k_0 \otimes \mathbb{C}$. Thus $D(\chi)$, well defined modulo E^*, has a representative in $(\mathbb{Q} \otimes \mathbb{C})^* = \mathbb{C}^* \subset (E \otimes \mathbb{C})^*$, namely the determinant of the inverse of the matrix $(\sigma\alpha)$, where $\sigma \in \bar{S}$ and α runs through a basis of k_0 over $\mathbb{Q}$. The isomorphism $\mathbb{C}^{\bar{S}} \xrightarrow{\sim} k_0 \otimes \mathbb{C}$ transforms the quadratic form $\sum {x_\sigma}^2$ into $\mathrm{Tr}(xy)$. Hence $(\det(\sigma\alpha))^2$ identifies with the discriminant of k_0:

$D(\chi) \sim$ a square root of the discriminant of k_0.

8.18. Let $p''(\chi; \sigma, \tau)$ denote the image of $p'(\chi; \sigma, \tau)$ in $\mathbb{C}^*/\bar{\mathbb{Q}}^*$. It depends only on χ, σ, and τ. If an algebraic homomorphism η: $k^* \to E^*$ satisfies (8.2.1), some power of it is the algebraic part of a Hecke character: $\eta^N = \chi_{\mathrm{alg}}$. Moreover, if ${\chi'}_{\mathrm{alg}} = {\chi''}_{\mathrm{alg}}$, then χ' and χ'' differ only by a finite-order character, and ${\chi'}^M = {\chi''}^M$ for a suitable M. It follows from (8.1.1) that $p''(\chi^M; \sigma, \tau) = p''(\chi; \sigma, \tau)^M$, which permits the following unambiguous definition:

$p(\eta; \sigma, \tau) = p(\chi; \sigma, \tau)^{1/N}$, when $\eta^N = \chi_{\text{alg}}$.

These periods satisfy the following formalism:

(8.18.1) $p(\eta'\eta''; \sigma, \tau) = p(\eta'; \sigma, \tau)p(\eta''; \sigma, \tau)$.

(8.18.2) $p(\eta; \sigma, \tau)$ is unchanged when E is replaced by an extension E′ and τ by one of its extensions to E′.

(8.18.3) $p(\eta; \sigma, \tau)$ is unchanged when k is replaced by an extension k′, σ by one of its extensions to k′, and χ by $\chi \circ N_{k'/k}$.

(8.18.4) If α is an automorphism of k and β an automorphism of E, then

$p(\eta; \sigma, \tau) = p(\beta\eta\alpha^{-1}; \sigma\alpha^{-1}, \tau\beta^{-1})$.

(8.18.5) The complex conjugate of $p(\eta; \sigma, \tau)$ is $p(\eta; \bar{\sigma}, \bar{\tau})$.

(8.18.6) For $k = F = \mathbb{Q}$, $p(\mathrm{Id}; \mathrm{Id}, \mathrm{Id}) = 2\pi i$.

Formulas (8.15.1)–(8.15.3) follow from (8.1.1)–(8.1.3); (8.18.4) and (8.18.5) follow by transport of structure; and (8.18.6) follows from (8.5).

8.19. Let η: $k^* \to E^*$ be a homomorphism satisfying (8.2.1). Suppose also that $n(\eta; \sigma, \tau)$ is never $w/2$, which makes it possible to define $(k \otimes E)^+$ as in 8.15. Define η^*: $E^* \to k^*$ by

$\eta^*(y) = \det_k(1 \otimes y, (k \otimes E)^+)$.

This homomorphism again satisfies (8.2.1), and

$n(\eta^*; \sigma, \tau) = 1$ if $n(\eta; \sigma, \tau) < w/2$, 0 if $n(\eta; \sigma, \tau) > w/2$.

If η^* is the algebraic part of a Hecke character χ^*, then $M(\chi^*)$ is the H^1 of an abelian variety over E with complex multiplication by k, whose Lie algebra, as a $k \otimes E$-module, is isomorphic to $(k \otimes E)^+$.

PROPOSITION 8.20. Under the hypotheses and notation of 8.19, regard E as a subfield of $\mathbb{C}$ and write 1 for the identity embedding $E \hookrightarrow \mathbb{C}$. Then

$\prod_{n(\sigma,1)<w/2} p(\eta; \sigma, 1) = \prod_\sigma p(\eta^*; 1, \sigma)^{n(\eta;\sigma,1)}$.

If ι: $E \to E' \subset \mathbb{C}$ is a finite extension of E, the left-hand side does not change when η is replaced by $\iota\eta$, by (8.18.2). One has $(\iota\eta)^* = \eta^* \circ N_{E'/E}$; by (8.18.3), the right-hand side is likewise unchanged.

If ι: $k \to k'$ is a finite extension of degree d, the left-hand side is raised to the d-th power when k is replaced by k′ and η by $\eta \circ N_{k'/k}$: a complex embedding of k is induced by d embeddings of k′, and (8.18.3) applies. The same holds for the right-hand side, by (8.18.2) and the equality $(\eta \circ N_{k'/k})^* = \iota\eta^*$.

These compatibilities reduce us to the case in which E is Galois and k is isomorphic to E. For each isomorphism ω from k to E, put $n(\omega) = n(\eta; 1 \circ \omega, 1)$. Writing the group of homomorphisms $k^* \to E^*$ additively, $\eta = \sum n(\omega)\omega$. Since $p(\eta; 1 \circ \omega, 1) = p(\eta \circ \omega^{-1}; 1, 1)$, by (8.15.4), one obtains

(8.20.1) $\prod_{n(\sigma,1)<w/2} p(\eta; \sigma, 1) = \prod_{n(\omega)<w/2} p(\eta \circ \omega^{-1}; 1, 1) = p(\sum_{n(\omega)<w/2} \eta \circ \omega^{-1}; 1, 1)$.

On the other hand,

$\sum_{n(\omega)<w/2} \eta \circ \omega^{-1} = \sum_{n(\omega_1)<w/2;\,\omega_2} n(\omega_2)\, \omega_2 \circ {\omega_1}^{-1} = \sum_{\omega_2} n(\omega_2) \sum_{n(\omega_1)<w/2} \omega_2 \circ {\omega_1}^{-1} = \sum_{\omega_2} n(\omega_2)\, \omega_2 \circ \eta^*$.

This permits (8.20.1) to be continued as

$= \prod_\omega p(\omega \circ \eta^*; 1, 1)^{n(\omega)} = \prod_\omega p(\eta^*; 1, 1 \circ \omega)^{n(\omega)} = \prod_\sigma p(\eta^*; 1, \sigma)^{n(\eta;\sigma,1)}$,

(a new application of 8.15.4), and proves 8.20.

8.21. Combining 8.16 and 8.20, one obtains, for the component of index 1 of

$c^+ R_{k/\mathbb{Q}} M(\chi) \in (E \otimes \mathbb{C})^*/E^* = \mathbb{C}^{*J}/E^*$,

the following expression, modulo $\bar{Q}^*$:

${c_1}^+ R_{k/\mathbb{Q}} M(\chi) \sim \prod_\sigma p({\chi^*}_{alg}; 1, \sigma)^{-n(\chi;\sigma,1)}$.

If χ (i.e., $M(\chi)$) is critical, Conjecture 2.8 therefore asserts that

$L(1 \circ \chi, 0) \sim \prod_\sigma p({\chi^*}_{alg}; 1, \sigma)^{-n(\chi;\sigma,1)} \pmod{\bar{Q}^*}$.

For E sufficiently large, ${\chi^*}_{alg}$ is the algebraic part of a Hecke character χ^*, and the periods are interpreted as periods of abelian integrals (8.19), (8.3). The resulting statement is the one Shimura proved when k is of CM type, or is abelian over a field of CM type (with a restriction on the weight).

REMARK 8.22. If $\eta : k^* \to E^*$ satisfies (8.2.1), there are subfields k' of k and E' of E, each either of CM type or equal to $\mathbb{Q}$, an inclusion $\iota : E' \to E$, and a factorization $\eta = \iota\eta' N_{k/k'}$. One then has $p(\eta; \sigma, \tau) = p(\eta'; \sigma \mid k', \tau \mid E')$.

If now k and E are of CM type (or $\mathbb{Q}$), and c again denotes their complex conjugation, then $c\sigma = \sigma c$, $c\tau = \tau c$, and $\eta c = c\eta$; hence

$p(\eta; \sigma, \tau\bar{)} = p(\eta; c\sigma, c\tau) = p(\eta; \sigma c, \tau c) = p(c\eta c^{-1}; \sigma, \tau) = p(\eta; \sigma, \tau)$.

Thus the periods, a priori in $\mathbb{C}^*/\bar{Q}^*$, are real, i.e., lie in $\mathbb{R}^*/(\bar{Q}^* \cap \mathbb{R}^*)$.

REMARK 8.23. Let G be the Galois group of the union of the CM-type extensions of $\mathbb{Q}$ in $\bar{Q} \subset \mathbb{C}$, and let $c \in G$ be complex conjugation. It is a central element of G. If φ is a locally constant integer-valued function on G, put $\varphi^*(x) = \varphi(xc)$. Suppose that $\varphi + \varphi^*$ is constant. Let G_1 be a finite quotient of G through which φ factors as a function φ_1 on G_1, and let k be the corresponding field. The hypothesis means that the endomorphism $\sum \varphi_1(\sigma)\sigma$ of k^* satisfies (8.2.1). The period $p(\sum \varphi_1(\sigma)\sigma; 1, 1)$ is independent of the choice of G_1; put $P(\varphi) = p(\sum \varphi_1(\sigma)\sigma; 1, 1)$. The functional P is a homomorphism into $\mathbb{C}^*/\bar{Q}^*$ from the group of locally constant integer-valued functions on G satisfying $\varphi + \varphi^* =$ constant.

Appendix by N. Koblitz and A. Ogus. Algebraicity of some products of values of the Γ function. Let $A_N = N^{-1}\mathbb{Z}/\mathbb{Z} - \{0\}$, and let $U_N = (\mathbb{Z}/N\mathbb{Z})^*$ operate on A_N in the obvious way. If $f : A_N \to \mathbb{C}$, define $\langle f\rangle : U_N \to \mathbb{C}$ by

$\langle f\rangle(u) = \sum_{a \in A_N} \langle a\rangle f(ua)$,

where $\langle a\rangle$ denotes the representative of a between 0 and 1.

We first compute $H_Q = \{f : A_N \to \mathbb{Q} : \langle f\rangle \text{ is constant}\}$.

EXAMPLE. If n is a divisor of N and $a \in A_N$, consider the set

$S_{n,a} = [\{a + k/n : k = 0, 1, \ldots, n - 1\} \cup \{-na\}] \cap A_N$.

It is easy to see that its characteristic function $\varepsilon_{n,a}$ is in H_Q. Note that $S_{1,a} = \{a, -a\}$.

PROPOSITION. H_Q is generated by $\{\varepsilon_{n,a} : n = 1$ or n is prime and $na \neq 0\}$.

PROOF. The orbits of A_N under the action of U_N correspond to the divisors d of N with $1 < d \leq N$: each orbit can be written uniquely in the form $U_N(1/d)$. The stabilizer subgroup I_d of 1/d is $\{u \in U_N : u \equiv 1 \bmod d\}$ and the orbit of 1/d is canonically isomorphic to U_d. Thus, a function f on A_N is determined by the collection of functions $f_d : U_d \to \mathbb{Q}$ defined by $f_d(v) = f(v/d)$.

To prove the proposition, we first complexify, so as to be able to work with characters of U_N.

LEMMA. If $f : A_N \to \mathbb{C}$ and if χ is a character of U_N, then the inner product of $\langle f\rangle$ and χ is given by:

$(\langle f\rangle \mid \chi) = -\sum_d L(0, \chi_d)|I_d| \cdot (f_d \mid \chi_d)$,

where the sum is taken over those divisors d of N such that χ is pulled back from a character χ_d of U_d.

PROOF. We have

$(\langle f\rangle \mid \chi) = \sum_{u\in U_N} \langle f\rangle(u)\bar{\chi}(u) = \sum_{a\in A_N} \sum_{u\in U_N} \langle a\rangle f(ua)\bar{\chi}(u) = \sum_{d|N} \sum_{v\in U_d} \sum_{u\in U_N} \langle v/d\rangle f_d(u_{dv})\bar{\chi}(u)$,

where $u_d \in U_d$ is the image of u.

Now if we choose a set $U'_d \subseteq U_N$ of coset representatives for $U_N \to U_d$, we can write:

$\sum_{u\in U_N} f_d(u_{dv})\bar{\chi}(u) = \sum_{u'\in U'_d} \sum_{u\in I_d} f_d(u'_{dv})\bar{\chi}(u')\bar{\chi}(u) = \sum_{u'\in U'_d} f_d(u'_{dv})\bar{\chi}(u') \sum_{u\in I_d} \bar{\chi}(u)$.

Of course, this sum is zero unless χ is trivial on I_d, i.e., unless χ is pulled back from a character χ_d of U_d, in which case it is

$\sum_{u'\in U'_d} f_d(u'_{dv})\bar{\chi}_d(u')|I_d| = \sum_{w\in U_d} f_d(wv)\bar{\chi}_d(w)\chi_d(v)|I_d| = \chi_d(v)|I_d|(f_d \mid \chi_d)$.

If we substitute this into the above expression for $(\langle f\rangle \mid \chi)$ and note that $\langle v/d\rangle = |v|/d$, where $|v|$ is the smallest positive representative of $v \in U_d$, we find:

$(\langle f\rangle \mid \chi) = \sum_d \sum_{v\in U_d} |v|\chi_d(v)|I_d|d^{-1}(f_d \mid \chi_d) = -\sum_d L(0, \chi_d)|I_d|(f_d \mid \chi_d)$,

as claimed. □

To prove the proposition, we let d_f be the largest divisor of N such that $f_{d_f} \neq 0$; it suffices to prove that any f in H_Q can be written as $g + f'$, with g a linear combination of the ε's and $d_{f'} < d_f$. Let $d = d_f$; we shall show below that f_d is a linear combination of functions h_1 which factor through $U_d \to U_d/\{\pm 1\}$, and functions h_p which factor through $U_d \to U_{d/p}$ for some prime divisor p of d, $p \neq d$. Any function of the first type is invariant under ± 1, and hence the corresponding function on the orbit $U_N(1/d)$ is a linear combination of $\varepsilon_{1,a}$'s. A function of the second type is invariant under the kernel K of $U_d \to U_{d/p}$, which has order p if p divides d/p and order $p - 1$ otherwise. In the first case, $K = \{1 + kd/p : k = 0, \ldots, p - 1\}$, and in

the second, it is this same set with one element deleted, namely, the value of 1 + kd/p which is divisible by p. Thus, the set $S_{p,w/d}$ = (Kw)/d, with the addition of one or two elements in the orbit U_N(p/d). Hence it is clear that the corresponding function on the orbit U_N(1/d) can be written as a linear combination of $\varepsilon_{p,a}$'s and functions supported on the orbit of p/d, and we have obtained our desired decomposition f = g + f′.

It remains for us to prove the claim about f_d. If d is twice an odd number the map $U_d \to U_{d/2}$ is an isomorphism, and the claim is trivial. In the other cases, choose a primitive odd character χ_d of U_d, and let χ be the pull-back of χ_d to U_N. Since χ is nontrivial and ⟨f⟩ is constant, (⟨f⟩ | χ) = 0. Let us look at the expression for this inner product provided by the lemma; the only nonzero terms come from those f_e such that $f_e \neq 0$ and $I_e \subseteq \mathrm{Ker}(\chi)$. But then Ker($\chi$) contains I_e and I_d, hence also $I_{\mathrm{dI}e} = I_m$, where m = g.c.d.(d, e). Since χ is primitive, m = d and d divides e, and since f_e = 0 for e > d,

$0 = (\langle f\rangle \mid \chi) = -L(0, \chi_d)|I_d|(f_d \mid \chi_d)$.

Since χ_d is odd and primitive, we conclude that $(f_d \mid \chi_d) = 0$, i.e., f_d is orthogonal to every odd primitive character of U_d. It follows that f_d can be written as a linear combination of characters which are even (factor through $U_d/\{\pm 1\}$) or imprimitive (factor through some $U_{d/p}$). When d = p, we should also remark that the constant functions on U_p are also invariant under ±1, and hence are already covered by the first case. □

THEOREM. Suppose that f : $A_N \to \mathbb{Q}$ is such that ⟨f⟩ is constant: $\sum_a \langle a\rangle f(ua) = k$ for all $u \in U_N$. Then

$\Gamma(f) := \pi^{-k} \prod_a \Gamma(\langle a\rangle)^{f(a)}$

is an algebraic number.

PROOF. The map ⟨ ⟩ : $H_Q \to \mathbb{Q}$ sending any f to $\sum_a \langle a\rangle f(a)$ is evidently linear, and hence Γ(f + g) = Γ(f)Γ(g), for f and g in H_Q. By the proposition, any f in H_Q is a $\mathbb{Q}$-linear combination of the ε's; hence nf is a sum of ε's for some n. Since $\Gamma(f)^n = \Gamma(nf)$, we are reduced to checking the theorem for the ε's described in the proposition. Noting that $\langle \varepsilon_{1,a}\rangle = 1$ and that $\langle \varepsilon_{p,a}\rangle$ = (p + 1)/2 if pa ≠ 0, one has only to appeal to the identities:

$\Gamma(x)\Gamma(1 - x) = \pi(\sin \pi x)^{-1}$,

$\sum_{k=1}^{p-1} \Gamma(x + k/p) = p^{1/2-x}(2\pi)^{(p-1)/2}\, \Gamma(px)$. □

REMARK. Kubert has recently obtained a much more precise result expressed in a different terminology from which it follows that, if H_Q is $\mathbb{Z}$-valued, then 2f is a $\mathbb{Z}$-linear combination of the ε's (to appear).

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INSTITUT DES HAUTES ÉTUDES SCIENTIFIQUES, BURES SUR YVETTE, FRANCE

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ABSOLUTE HODGE CYCLES AND PERIODS OF INTEGRALS ON ABELIAN VARIETIES

by

Pierre DELIGNE

written up by J. L. BRYLINSKI

We show here how a suitable theory of "motives" makes it possible to recover certain results of Shimura on the periods of abelian varieties of CM type [5]. It seemed instructive to begin with an informal description of the properties of motives and of their realizations; this is done in Section 1. Section 2 interprets certain results of Shimura by means of motives. Section 3 defines absolute Hodge cycles and the theory of motives subordinate to them. Section 4 gives an upper bound for the transcendence degree of the field generated by the periods of an arbitrary abelian variety.

§ 1. THE LANGUAGE OF MOTIVES

Let k be a field of characteristic 0. There are several definitions of "motives over k", which are not known to be equivalent, each having its own

virtues. In Section 3 we shall specify which one is useful to us.

The motives over k form an additive category equipped with a tensor product. Every smooth projective variety X over k defines a motive $h(X)$ over k, the motivic cohomology of X, and every motive over k is obtained from a motive of this form (by cutting it into pieces and by Tate twisting). Finally, the various cohomology theories for algebraic varieties give rise to "realization" functors for motives.

Every abelian variety A over k defines a motive $T(A)$ over k, its motivic H_1, and the functor $A \mapsto T(A)$ is fully faithful. The advantage of embedding the category of abelian varieties in that of motives is that one can take tensor products. But the objects obtained are not geometric: there are no "functions" on a motive.

The rank of a motive is the dimension of any one of its realizations, which we now describe.

If $k = \mathbf{C}$, the Betti realization of a motive M is a free abelian group of finite rank $H_B(M)$, endowed with a Hodge structure, that is, with a decomposition

$$H_B(M) \otimes_{\mathbf{Z}} \mathbf{C} = \bigoplus_{p+q=n} H^{p,q} \qquad (\text{with } \overline{H^{p,q}} = H^{q,p}).$$

Let us illustrate this with the motive $T(A)$, which is the H_1 of an abelian variety A. We define $H_B(T(A)) = H_1(A, \mathbf{Z})$, the homology of $A = A(\mathbf{C})$ viewed as a topological space.

Let $\mathrm{Lie}(A)$ be the Lie algebra of the analytic group A, and let $\int : H_1(A, \mathbf{Z}) \to \mathrm{Lie}(A)$ be the homomorphism "integration over a 1-cycle". There is an exact sequence of complex analytic groups

$$0 \longrightarrow H_1(A, \mathbf{Z}) \xrightarrow{\int} \mathrm{Lie}(A) \xrightarrow{\exp} A \longrightarrow 0.$$

The Hodge structure on $H_1(A, \mathbf{C})$ is such that $H_1(A, \mathbf{C}) = H^{0,-1} \oplus H^{-1,0}$. One obtains $H^{0,-1}$ as the kernel of the map $H_1(A, \mathbf{C}) \to \mathrm{Lie}(A)$, the $\mathbf{C}$-linear extension of

$\int : H_1(A, \mathbf{Z}) \to \mathrm{Lie}(A)$.

We now work over a field of characteristic 0 and choose an algebraic closure $\bar{k}$ of k. For every prime number ℓ, the ℓ-adic realization of a motive M is a free $\mathbf{Z}_\ell$-module of finite type $H_\ell(M)$ (also endowed with an action of the Galois group of $\bar{k}$ over k). When $k = \mathbf{C}$, there is an isomorphism

$$H_\ell(M) \simeq H_B(M) \otimes \mathbf{Z}_\ell.$$

If A is an abelian variety over k and $\bar{k}$ is an algebraic closure of k, the ℓ-adic realization of $T(A)$ is denoted $T_\ell(A)$; it is the $\mathbf{Z}_\ell$-module

$$T_\ell(A) = \varprojlim_n A_{\ell^n}(\bar{k}) \quad \big(\text{where } A_{\ell^n}(\bar{k}) \text{ is the group of points of order } \ell^n \text{ in } A(\bar{k})\big).$$

If $k = \mathbf{C}$, the isomorphism $T_\ell(A) \simeq H_B(T(A)) \otimes \mathbf{Z}_\ell$ is easy to describe. For every n, the projection of $\gamma \otimes x \in H_B(T(A)) \otimes \mathbf{Z}_\ell$ to $A_{\ell^n}(\mathbf{C})$ is computed as follows: if $m \in H_B(T(A))$ is such that $x - m \in \ell^n(H_B(T(A)) \otimes \mathbf{Z}_\ell)$, the exponential of $m/\ell^n \in H_1(A, \mathbf{Q})$ is the required point of order ℓ^n.

The de Rham realization of M is a finite-dimensional vector space $H_{DR}(M)$ over k, equipped with a decreasing filtration $F^\bullet$ (the Hodge filtration). When $k = \mathbf{C}$, there is an isomorphism

$$H_B(M) \otimes \mathbf{C} \simeq H_{DR}(M).$$

Under this isomorphism, the Hodge filtration is given by $F^p = \bigoplus_{p' > p} H^{p',q}$. In the case of the motive $T(A)$, recall [7] the existence of a "universal extension" E of A by a vector group, fitting into an exact sequence of algebraic groups

$$0 \longrightarrow H^1(A, \mathcal{O}_A)^* \longrightarrow E \longrightarrow A \longrightarrow 0.$$

We have $H_{DR}(T(A)) = \mathrm{Lie}(E)$, and the Hodge filtration is

$$0 = F^1 \subset F^0 = H^1(A, \mathcal{O}_A)^* \subset F^{-1} = \mathrm{Lie}(E).$$

If $k = \mathbf{C}$, then $H_1(E, \mathbf{Z}) \xrightarrow{\sim} H_1(A, \mathbf{Z})$, giving a morphism

$$H_1(A, \mathbf{C}) \xleftarrow{\sim} H_1(E, \mathbf{C}) \xrightarrow{\int} \mathrm{Lie}(E) = H_{DR}(T(A)).$$

This is the desired isomorphism.

Examples: **1)** Here $k = \mathbf{Q}$. We shall define the Tate motive $\mathbf{Z}(1)$; it is the H_1 of the algebraic group $\mathbf{G}_m$. To obtain its various realizations, substitute $\mathbf{G}_m$ for A in the definition of $T(A)$. This gives

- $H_B(\mathbf{Z}(1)) = 2\pi i \cdot \mathbf{Z}$, the kernel of $\exp : \mathbf{C} \longrightarrow \mathbf{C}^\times$;
  $H_B(\mathbf{Z}(1)) \otimes \mathbf{C} = H^{-1,-1}$;
- $H_\ell(\mathbf{Z}(1)) = \mathbf{Z}_\ell(1) = T_\ell(\mathbf{G}_m) = \varprojlim_n \mu_{\ell^n}(\bar{\mathbf{Q}})$;
- $H_{DR}(\mathbf{Z}(1)) = \mathbf{Q}$.

The isomorphism $H_B(\mathbf{Z}(1)) \otimes \mathbf{C} \xrightarrow{\sim} H_{DR}(\mathbf{Z}(1)) \otimes \mathbf{C}$ sends $(2\pi i) \otimes 1$ to $2\pi i$. Recall the isomorphisms $H_1(\mathbf{G}_m) \simeq H_1(\mathbf{P}^1 - \{0, \infty\}) \simeq H_2(\mathbf{P}^1)$. Moreover, for every smooth, proper, connected algebraic curve over $\mathbf{Q}$, $H_2(X)$ is isomorphic to $H_2(\mathbf{P}^1)$ (and hence to $\mathbf{Z}(1)$). For every integer d, define a motive $\mathbf{Z}(d)$ so that $\mathbf{Z}(d) = \mathbf{Z}(d-1) \otimes \mathbf{Z}(1)$. We have $\mathbf{Z}(m+n) \simeq \mathbf{Z}(m) \otimes \mathbf{Z}(n)$ for $m, n \in \mathbf{Z}$.

2) Let E be an elliptic curve over $\mathbf{Q}$. The motive $T(E)$ has rank 2. Hence the motive $\bigwedge^2 T(E)$ has rank 1, and

$$\bigwedge^2 T(E) = \bigwedge^2 H_1(E) = H_2(E) = \mathbf{Z}(1).$$

Let us specify the isomorphism $\bigwedge^2 T(E) \simeq \mathbf{Z}(1)$ in the various realizations.

On $H_1(E(\mathbf{C}), \mathbf{R}) \simeq \mathrm{Lie}(E)$, the complex structure defines an orientation. If (e_1, e_2) is an oriented $\mathbf{Z}$-basis of $H_B(T(E)) = H_1(E(\mathbf{C}), \mathbf{Z})$, then $e_1 \wedge e_2$ is a basis of $H_B(\bigwedge^2 T(E))$; it is associated with $2\pi i \in H_B(\mathbf{Z}(1))$.

The isomorphism $\bigwedge^2(T_\ell(E)) \simeq \mathbf{Z}_\ell(1)$ is given by the alternating Weil pairing [6]. Finally,

$$\bigwedge^2 H_{DR}(T(E)) = \mathrm{Hom}(H^2_{DR}(E), \mathbf{Q}).$$

Let ω and η be the usual differentials of the first and second kind. $H^2_{DR}(E)$ is generated by the cup product $\omega \wedge \eta$ of their cohomology classes. Under the isomorphism

$$H_B(\bigwedge^2 T(E)) \otimes \mathbf{C} \longrightarrow H_{DR}(\bigwedge^2 T(E)) \otimes \mathbf{C},$$

$e_1 \wedge e_2$ maps to the homomorphism sending $\omega \wedge \eta \in H^2_{DR}(E)$ to

$$\begin{vmatrix} \omega_1 & \omega_2 \\ \eta_1 & \eta_2 \end{vmatrix} \in \mathbf{C}.$$

Compatibility of the Betti and de Rham realizations of $\bigwedge^2(T(E)) \simeq \mathbf{Z}(1)$ is equivalent to the Legendre relation

$$\omega_1 \cdot \eta_2 - \omega_2 \cdot \eta_1 = 2\pi i.$$

3) Let A be an abelian variety over $\mathbf{Q}$. The cohomology class of a hyperplane section of A in a projective embedding gives a morphism of motives $\bigwedge^2 T(A) \to \mathbf{Z}(1)$. In Betti cohomology this corresponds to an alternating form $2\pi i \cdot E$ on $H_B(T(A))$ with values in $2\pi i \cdot \mathbf{Z}$. In de Rham cohomology one likewise has an alternating form E_{DR} on $H_{DR}(T(A))$ with values in $\mathbf{Q}$. It is classical that F^0 is a totally isotropic subspace. It follows that one can find differentials of the first kind $\omega_1, \ldots, \omega_g$ on A and differentials of the second kind $\eta_1, \ldots, \eta_g$ on A such that

$$E_{DR}(x,y) = \sum_{i=1}^{g} \big((\omega_i, x)(\eta_i, y) - (\omega_i, y) \cdot (\eta_i, x) \big).$$

Compatibility between E and E_{DR} can be expressed as follows. Let $(e_1, \ldots, e_{2g})$ be a basis of $H_B(T(A))$, let Ω be the $n \times 2n$ matrix with $\Omega_{ij} = \int_{e_j} \omega_i$, and let N be the $n \times 2n$ matrix with $N_{ij} = \int_{e_j} \eta_i$. Then

$${}^t N \cdot \Omega - {}^t \Omega \cdot N = 2i\pi \cdot E.$$

Finally, the ℓ-adic realization gives an alternating form on $T_\ell(A)$ with values in $\mathbf{Z}_\ell(1)$: it is the alternating Weil form associated with the divisor of the hyperplane section [6].

§ 2. MOTIVES ASSOCIATED WITH ABELIAN VARIETIES OF CM TYPE

Let k be an algebraically closed subfield of $\mathbf{C}$ – for example $\bar{\mathbf{Q}}$ – and let $\mathcal{A}$ be the subcategory of the category of motives over k generated by the $T(A)$ of abelian varieties (under tensor products, passage to duals, and direct factors). The functor "Betti realization of the motive obtained from M/k by extension of scalars to $\mathbf{C}$" sends $\mathcal{A}$ to the category of Hodge structures. It follows formally from Theorem 1 of Section 3 that this functor is fully faithful.

Let E be a CM field, let S be the set of these complex embeddings, and let $- : S \to S$ denote complex conjugation. To each function $\varphi : S \to \mathbf{Z}$ satisfying

$$(*) \qquad \varphi(\sigma) + \varphi(\bar{\sigma}) \quad \text{is independent of } \sigma \in S,$$

associate a Hodge structure H_φ equipped with a module structure over the ring of integers $\mathcal{O}_E$ of E:

$$H_\varphi = \mathcal{O}_E, \quad \text{with the evident module structure, and in the decomposition}$$
$$H_\varphi \otimes \mathbf{C} = E \otimes \mathbf{C} \simeq \mathbf{C}^S, \quad \text{the } \mathbf{C}\text{-factor indexed by } \sigma \in S \text{ has Hodge type}$$
$$(\varphi(\sigma), \varphi(\bar{\sigma})).$$

One verifies that the H_φ lie in the image of $\mathcal{A}$, and indeed in the image of the subcategory generated by abelian varieties of CM type. This allows one to define the motive M_φ over $\bar{\mathbf{Q}}$ as the motive whose Betti realization is H_φ.

The de Rham realization $H_{DR}(M_\varphi)$ is a vector space over $\bar{\mathbf{Q}}$ equipped with an action of E. For each $\sigma \in S$, let $H_{DR}(M_\varphi)_\sigma$ be the $\bar{\mathbf{Q}}$-subspace on which $x \in E$ acts as $\sigma^{-1}(x) \in \bar{\mathbf{Q}}$. Each of these vector spaces is one-dimensional, generated by an element ω_σ, and $H_{DR}(M_\varphi)$ is their direct sum. Under the isomorphism

$$H_B(M_\varphi) \otimes \mathbf{C} \xrightarrow{\sim} H_{DR}(M_\varphi) \otimes \mathbf{C},$$

an element e of $E = H_B(M_\varphi) \otimes \mathbf{Q}$ corresponds to a family (e_σ) with $e_\sigma \in H_{DR}(M_\varphi)_\sigma$. It follows immediately that the complex number $p(\sigma, \varphi)$ defined by $p(\sigma, \varphi) e_\sigma = \omega_\sigma$ is independent, up to multiplication by $\bar{\mathbf{Q}}^\times$, both of e and of the choice of ω_σ. This justifies the notation.

The relation between the $p(\sigma, \varphi)$ and Shimura's invariants $p_E(\sigma, \varphi)$ (see [5]) is

$$p_E(\sigma, \varphi) = \frac{p(\sigma, \varphi)}{(2\pi i)^{\varphi(\sigma)}} \quad (\text{modulo } \bar{\mathbf{Q}}^\times),$$

which is merely a translation of the definitions.

The identities among Shimura's periods correspond to algebraic properties of the motives M_φ. Before summarizing them in a table, introduce some notation for an extension $F|E$ of CM fields. Let S_F (respectively S_E) be the set of complex embeddings of F (respectively E). For $\sigma \in S_F$, write $\sigma|E \in S_E$ for the restriction of the embedding σ to E.

For $\varphi : S_E \to \mathbf{Z}$ satisfying $(*)$, let $\mathrm{Inf}(\varphi)$ be the function from S_F to $\mathbf{Z}$ defined by $\mathrm{Inf}(\varphi)(\sigma) = \varphi(\sigma|E)$. This function satisfies $(*)$. For $\varphi : S_F \to \mathbf{Z}$ satisfying $(*)$, let $\mathrm{Res}(\varphi)$ be the function from S_E to $\mathbf{Z}$ defined by

$$\mathrm{Res}(\varphi)(\sigma) = \sum_{\tau|E=\sigma} \varphi(\tau);$$

it satisfies $(*)$.

Identities among periods	**Relations among motives**
1. $p(\sigma, \varphi' + \varphi'') = p(\sigma, \varphi') \cdot p(\sigma, \varphi'')$	1. $M_{\varphi'+\varphi''} = M_{\varphi'} \otimes_E M_{\varphi''}$
2. $F\|E$ an extension of CM fields, $\sigma \in S_F$, $\varphi : S_E \to \mathbf{Z}$ satisfying $(*)$: $p(\sigma, \mathrm{Inf}(\varphi)) = p(\sigma\|E, \varphi)$	2. $M_{\mathrm{Inf}(\varphi)} = M_\varphi \otimes_E F$
3. $\sigma \in S_E$, $\varphi : S_F \to \mathbf{Z}$ satisfying $(*)$: $p(\sigma, \mathrm{Res}\,\varphi) = \prod_{\tau\|E=\sigma} p(\tau, \varphi)$	3. M_φ is an F-motive of rank 1 and hence an E-motive of rank $d = [F : E]$ $M_{\mathrm{Res}(\varphi)} = \bigwedge_E^d M_\varphi$
4. $p(\sigma, \underline{1}) = 2\pi i$	4. $M_{\underline{1}} = \mathbf{Z}(-1)$ (the dual of the Tate motive)

These identities among periods recover Theorems 1 through 4 of Shimura's article [5]. Note that in [5] Shimura relates the periods $p_K(\sigma, \varphi)$ to values, at a special point, of Hilbert modular forms rational over $\bar{\mathbf{Q}}$.

§ 3. ABSOLUTE HODGE CYCLES AND MOTIVES

Let X be a smooth projective variety over $\mathbf{C}$. A Hodge cycle of codimension d on X is an element of $H^{2d}(X, \mathbf{Q})$ of type (d, d), or equivalently an element of

$$H^{2d}(X, \mathbf{Q})(d) \stackrel{\mathrm{def}}{=} H^{2d}(X, \mathbf{Q}) \otimes \mathbf{Z}(d)$$

of type $(0, 0)$. Observe that for $x \in H^{2d}(X, \mathbf{Q})(d)$, saying that x is of type $(0, 0)$ is equivalent to saying that x lies in F^0. The cohomology class of an algebraic subvariety of X is a Hodge cycle. Hodge conjectured that every Hodge cycle is a linear combination

of cohomology classes of subvarieties. Since this conjecture appears inaccessible, we introduce a notion that "arithmetizes" that of a Hodge cycle.

The cycles defined here will have all the visible attributes of algebraic cycles, namely ℓ-adic and de Rham cohomology realizations. An absolute Hodge cycle of codimension d on a smooth projective variety X over a field k of characteristic 0 is a collection of cohomology classes in the groups

$$H^{2d}(X, \mathbf{Q}_\ell)(d) \quad \text{(respectively } H^{2d}_{DR}(X)(d)\text{)}$$

satisfying the following two conditions:

- the element of $H^{2d}_{DR}(X)(d)$ belongs to F^0;
- for every embedding $k \hookrightarrow \mathbf{C}$, the different cohomology classes correspond (through the isomorphisms of Section 1) to a single rational class in the Betti cohomology of $X \otimes_k \mathbf{C}$.

There are inclusions (algebraic cycles) $\subset$ (absolute Hodge cycles) $\subset$ (Hodge cycles). We use this definition to specify the category of motives used in Sections 1 and 2. First let $\mathcal{V}(k)$ be the category of smooth projective varieties over k. One constructs an additive category $\mathcal{M}(k)$, in which the groups $\mathrm{Hom}(M, N)$ are vector spaces over $\mathbf{Q}$, together with the following data:

a) a tensor product $\otimes$, associative, commutative, and distributive with respect to addition of objects;

b) a contravariant functor H^* from $\mathcal{V}(k)$ to $\mathcal{M}(k)$, bijective on objects, compatible with sums, and carrying products to tensor products.

In $\mathcal{M}(k)$, for connected X and Y, one has

$$\mathrm{Hom}(H^*(X), H^*(Y)) = Z^{\dim(Y)}_{h.a}(X \times Y),$$

where $Z^d_{h.a}(Z)$ denotes the vector space over $\mathbf{Q}$ of absolute Hodge cycles of codimension d on Z. The category of motives $\mathcal{M}(k)$ is then constructed exactly as in [2], with algebraic cycles replaced by absolute Hodge cycles. By this new definition, the Künneth components of the diagonal of $X \times X$ are absolute Hodge, and in $\mathcal{M}(k)$ there is a decomposition

$$H^*(X) = \bigoplus_i H^i(X).$$

The following result was the subject of lectures at the I.H.E.S. in 1979.

Theorem 1: Every Hodge cycle on an abelian variety is absolute Hodge.

We note the following.

Corollary (Borovoï): Let A be an abelian variety over a field k of characteristic 0, and let ξ be a Hodge cycle on A. The corresponding ℓ-adic classes ξ_ℓ of ξ are fixed by an open subgroup of $\mathrm{Gal}(\bar{k}|k)$.

§ 4. PERIODS OF INTEGRALS ON ABELIAN VARIETIES

Let A be a complex abelian variety, and let $H_1(A, \mathbf{C}) = H^{0,-1} \oplus H^{-1,0}$ be the Hodge decomposition of $H_B(A)$ (cf. Section 1). Regard $GL(H_1(A))$ as an algebraic group over $\mathbf{Q}$. There is a morphism of complex algebraic groups

$$\mathbf{G}_m \xrightarrow{\mu} GL(H_1(A, \mathbf{C}))$$

such that $\mu(z)$ acts by the homothety of ratio z on $H^{-1,0}$ and by the identity on $H^{0,-1}$. The Hodge group G of A is the Zariski closure of the subgroup $\mu(\mathbf{G}_m)$ in $GL(H_1(A))$. In other words, G is an algebraic subgroup over $\mathbf{Q}$ of $GL(H_1(A))$. A regular function on a Zariski-open subset of G that is rational over $\mathbf{Q}$ vanishes on G if and only if it vanishes on $\mu(\mathbf{G}_m)$. One checks easily that G contains the homotheties of $H_1(A)$. Here is a second, very useful description of G: consider all rational tensors of type $(-N/2, -N/2)$ in the tensor products $\bigotimes^N H_1(A)$. Then G is the subgroup of $GL(H_1(A))$ formed by those linear transformations g for which there is a scalar μ such that $g \cdot t = \mu^m t$ for every tensor t of type (m, m). Finally, one shows that G is the group fixing all absolute Hodge cycles of type $(0, 0)$ in the mixed tensor spaces on $H_1(A)$ (mixed means covariant and contravariant).

Now suppose that A is defined over a subfield k of $\mathbf{C}$. Put $g = \dim(A)$.

Theorem 2: Let $\omega_1, \ldots, \omega_g$ be differentials of the first kind on A, and let $\eta_1, \ldots, \eta_g$ be differentials of the second kind on A, such that

$(\omega_1, \ldots, \omega_g, \eta_1, \ldots, \eta_g)$ is a basis of $H^1_{DR}(A)$. Let K be the subfield of $\mathbf{C}$ generated by k and the periods $\int_\gamma \omega_i$, $\int_\gamma \eta_i$ $(1 \leq i \leq g,\ \gamma \in H_1(A, \mathbf{Z}))$; the transcendence degree of K over k is bounded above by the dimension of G.

Proof: K is the field of rationality of the isomorphism

$$H_{1,B}(A) \otimes_{\mathbf{Q}} \mathbf{C} \xrightarrow{\sim} H_{1,DR}(A) \otimes_k \mathbf{C}$$

given by "integration" over a cycle. This isomorphism preserves the cohomology classes of absolute Hodge cycles. Let P be the algebraic variety of isomorphisms having this property. From the definition of G and Theorem 1, it follows that P is a principal homogeneous space under G, defined over k, and therefore $\dim(P) = \dim(G)$. We conclude by the following lemma.

Lemma: Let z be a point of $\mathbb{A}^N(\mathbf{C})$; the transcendence degree of $k(z)$ over k is equal to the dimension of the k-Zariski closure of $\{z\}$;

This is a tautology. Q.E.D.

Remark: Example 3 of Section 1 shows that $2\pi i \in K$. One can moreover show that the transcendence degree of K over $k(2\pi i)$ is bounded above by $\dim(G) - 1$.

Corollary: With the notation of Theorem 2, suppose in addition that A is simple of CM type. The transcendence degree of K over k is bounded above by $g + 1$.

Proof: It is enough to show that $\dim(G) < g + 1$. Now $\mu(\mathbf{G}_m)$ commutes with the action of the field E on $H_1(A, \mathbf{Q})$. It follows that G also commutes with this action and that G is an algebraic torus. Moreover, G preserves, up to a factor, the alternating form associated with a hyperplane section (see Example 3 of Section 1). This proves the assertion.

Observe that in this last case one has the following lower bound:

$$\dim(G) > 2 + \log_2(g),$$

due to Ribet [3].

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P. Deligne
Institut des Hautes Etudes Scientifiques
91440 Bures-sur-Yvette

J.-L. Brylinski
Ecole polytechnique, Centre de Mathématiques
91128 Palaiseau cedex

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CUBIC GAUSS SUMS AND COVERS OF SL(2), FOLLOWING S. J. PATTERSON

by P. Deligne

0. Introduction.

A Gauss sum is the scalar product of an additive character and a multiplicative character of a finite field. If F_q is a finite field with q elements, χ a character of F_q^*, and ψ a character of the additive group of F_q, set

$$g(\chi,\psi) = \sum_{x \in F_q^*} \chi(x)\,\psi(x).$$

We assume in principle that χ and ψ are nontrivial; the sum $g(\chi,\psi)$ then has absolute value $q^{1/2}$. It is called an n-th order sum if χ has order n. For $a \in F_q^*$, one has

(0.1) $g(\chi,\psi(ax)) = \chi(a)^{-1}\, g(\chi,\psi)$.

In particular, if χ has order n, $g(\chi,\psi)^n$ is independent of ψ.

To each prime number $p \neq 2$ one can attach the quadratic Gauss sum

$$g_p = \sum_{x \in F_p^*} (x/p)\exp(2\pi i\, x/p),$$

where (x/p) is the Legendre symbol : +1 if x is a square modulo p and -1 otherwise. Gauss calculated this sum : it is $\sqrt{p}$ or $i\sqrt{p}$ according as $p \equiv 1$ or -1 modulo 4.

If, similarly, one wishes to attach an n-th order Gauss sum to a prime ideal $\mathfrak{p}$ of the ring of integers O of a number field F, one must define a multiplicative character, and an additive character of the residue field $O/\mathfrak{p}$.

Suppose that F contains a primitive n-th root of unity, and put $\mu := \mu_n(F)$. If $\mathfrak{p}$ is prime to n, there is an isomorphism $\mu \simeq \mu_n(O/\mathfrak{p})$.

Let t denote the inverse isomorphism, and put $q = N\mathfrak{p}$. We write $(x/\mathfrak{p})_n$, or simply $(x/\mathfrak{p})$, for the character $t(x^{(q-1)/n})$ of $(O/\mathfrak{p})^*$ with values in μ. More generally, if $\mathfrak{a}$ is an ideal of O prime to x, with prime-ideal factorization $\mathfrak{a} = \prod \mathfrak{p}_i^{a_i}$, we write $(x/\mathfrak{a})_n$, or simply $(x/\mathfrak{a})$, for the product $\prod (x/\mathfrak{p}_i)^{a_i}$ (for x prime to $\mathfrak{a}$). For $\mathfrak{a} = (a)$, we also write (x/a). Its relation with the Hilbert symbol (section 2) is the formula

$$(x/a) = \prod_{v|a} (x,a)_v.$$

The reciprocity law for the Hilbert symbol yields a reciprocity law for these symbols.

(*) text revised in July 1979.

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The symbol $(x/\mathfrak{a})$ is a character of $(O/\mathfrak{a})^*$ (and already of $(O/\prod \mathfrak{p}_i)^*$) with values in μ. To obtain from it a complex-valued character, it suffices to compose it with an isomorphism ε from μ to the group of n-th roots of unity in C.

We shall denote by e the character $\exp(2\pi ix)$ of R/Z, the induced character on Q/Z, those deduced from it on the Q_p/Z_p through the isomorphism $Q/Z \simeq \oplus\, Q_p/Z_p$, and those deduced from it by traces : for example, the character $\exp(2\pi i\, Tr_{F/Q}\, x)$ of F.

There are two methods for constructing additive characters of residue fields.

First method : For an ideal $\mathfrak{p}$ of O, with residue field $O/\mathfrak{p}$ of characteristic p, define the additive character ψ as the composite of the trace $O/\mathfrak{p} \to Z/pZ$ and the character $e(x/p)$ of Z/pZ.

Second method : We spell this out only in the following special case : n = 3 and F is the field $Q(\sqrt{\ } - 3)$ of cube roots of unity. Throughout the Introduction, unless expressly stated otherwise, we shall consider only this case. Every ideal $\mathfrak{a}$ prime to 3 has a unique generator a congruent to 1 mod 3, and as additive character of $O/\mathfrak{a}$ we take $e(a^{-1}x)$.

For prime $\mathfrak{p}$, the cubic Gauss sums obtained by using the additive characters from the two methods coincide : for $\mathfrak{p} = (\pi)$, with $\pi \equiv 1$ (3), an easy calculation (cf. (0.1)) shows that one is the product of the other by $(-1/\pi)$ if $\mathfrak{p}$ has degree 2, and by $(\overline{\pi}/\pi)$ if $\mathfrak{p}$ has degree 1. These cube roots of unity are trivial : $(-1/\pi)$ because $(-1)^2 = 1$, and $(\overline{\pi}/\pi)$ because it is real ; indeed, $\overline{(\overline{\pi}/\pi)} = (\pi/\overline{\pi})$ (by transport of structure), while $(\overline{\pi}/\pi) = (\pi/\overline{\pi})$ (by cubic reciprocity).

The second method makes it possible to attach a sum, still called a "Gauss sum," to an ideal $\mathfrak{a}$ that need not be prime. For $\mathfrak{a} = (a)$, with $a \equiv 1$ (3), put

$$(0.2)\quad g(\mathfrak{a}) = \sum_{x \in (O/\mathfrak{a})^*} (x/\mathfrak{a}) e(a^{-1}x).$$

One has :

(0.3) If $\mathfrak{a}$ is squarefree, $g(\mathfrak{a}) = (N\mathfrak{a})^{1/2}$. Otherwise, $g(\mathfrak{a}) = 0$. (0.4) Twisted multiplicativity : If $\mathfrak{a} = \mathfrak{a}' \cdot \mathfrak{a}''$, with $\mathfrak{a}'$ and $\mathfrak{a}''$ relatively prime, and with generators a' and a" congruent to 1 mod 3, then

$$g(\mathfrak{a}) = (a'/\mathfrak{a}'')(a''/\mathfrak{a}')\, g(\mathfrak{a}')\, g(\mathfrak{a}'').$$

Cubic reciprocity gives $(a'/\mathfrak{a}'') = (a''/\mathfrak{a}')$. Thus the factor $(a'/\mathfrak{a}'')(a''/\mathfrak{a}')$ may be replaced by $(a'/\mathfrak{a}'')^2$.

For arbitrary F and n, Weil [22] showed that the n-th power of the Gauss sum $g(\mathfrak{p})$, as a function of $\mathfrak{p}$, is an algebraic Hecke character. Notice that this n-th power does not depend on the choice of additive character. If this Hecke character is regarded as known, then, in order to calculate the sum $g(\mathfrak{p})$ itself, one must determine which n-th root of $g(\mathfrak{p})^n$

This is what Gauss does in the quadratic case ; and, when n is even, n = 2m, his result reduces the calculation of $g(\mathfrak{p})^m$ to that of an algebraic Hecke character. There are also p-adic results (Stickelberger, recently sharpened by Gross-Koblitz [4]) and, for n = 3,4 and F = $Q(\sqrt[n]{1})$, an expression in terms of a product of division values of elliptic functions (Matthews [13]).

In [19], Patterson proves that the argument $g(\mathfrak{a})/|g(\mathfrak{a})|$ of $g(\mathfrak{a})$, for $\mathfrak{a}$ prime to 3 and squarefree, is equidistributed on the unit circle. He uses Weyl's criterion, and must therefore prove that, for every nonzero integer T,

$$(0.5)\quad \Big|\sum_{N\mathfrak{a}\le X} (g(\mathfrak{a})/|g(\mathfrak{a})|)^T\Big| \le o(X).$$

The Wiener-Ikehara Tauberian theorem allows such an estimate to be deduced from the holomorphy, for Re(s) ≥ 1, of the Dirichlet series

(0.6) $\sum (g(\mathfrak{a})/|g(\mathfrak{a})|)^T \cdot N(\mathfrak{a})^{-s}$.

The study of these series is the heart of the problem.

In principle, the same arguments apply to the product $g(\mathfrak{a})\chi(\mathfrak{a})$, where χ is a Hecke character. The conclusion becomes the following. Consider only squarefree ideals $\mathfrak{a} = (a)$, with $a \equiv 1$ (3), for which the argument of a lies in an interval I of the unit circle. The argument of $g(\mathfrak{a})^3 = -a^2\overline{a}$ then lies in the opposite interval (-1)I, and that of $g(\mathfrak{a})$ lies in $\{z \mid z^3 \in (-1)I\}$, that is, in the union of three disjoint intervals I_i (i = 1,2,3). The argument of $g(\mathfrak{a})$ occurs one third of the time in each of them : for $N\mathfrak{a} < X$, the number of occurrences in any one of them is $1/3(the$ number of $\mathfrak{a})$ + o(X). The same conclusion holds after imposing an additional congruence condition on a. This result may be interpreted as saying that there is no "simple" rule, even a statistical one, for determining the argument of the Gauss sum.

In [5], Heath-Brown and Patterson prove analogous results for the cubic Gauss sum $g(\mathfrak{p})$ ($\mathfrak{p}$ prime) ; this contradicts a suggestion of Kummer that Hasse imprudently raised to the rank of a conjecture. The Dirichlet series (0.6), with T = 1, again plays an essential role. After shifting s, it becomes

(0.7) $\sum g(\mathfrak{a})\, N(\mathfrak{a})^{-s}$;

this is the series we shall discuss.

The Dirichlet series (0.7) is not multiplicative. We now give an adelic expression for the twisted multiplicativity (0.4). For notation, see 0.0 and section 2. Let A^f be the ring of finite adeles of F, and let (,) be the cubic Hilbert symbol on $(A^f)^*$. Define a central extension $\widetilde{(A^f)^*}$ of $(A^f)^*$ by μ, equipped with a section a → [a] = (a,1), by setting $(a,\zeta')(b,\zeta'') = (ab,(a,b)\zeta'\zeta'')$. Then

(0.8) [a] [b] = [ab] · (a,b).

This is the extension defined by the 2-cocycle (a,b). Reciprocity for Hilbert symbols ensures that $[\alpha][\beta] = [\alpha\beta]$ for $\alpha, \beta \in F^*$: [ ] splits

the extension over F^*. For each finite place ν of F, define in the same way an extension $\widetilde{F_\nu{}^*}$ of $F_\nu{}^*$ by μ; if $\nu \nmid 3$, then [a][b] = [ab] for a,b $\in O_\nu{}^*$; if $\nu|3$, one must assume a,b $\equiv \pm 1$ (3). The restricted product of the $\widetilde{F_\nu{}^*}$, relative to $[O_\nu{}^*]$, is an extension of $(A^f)^*$ by $\mu^{(\Sigma)}$ ($\sum$ is the set of finite places). Pushing out by the "product" map $\mu^{(\Sigma)} \to \mu$ gives the extension $\widetilde{(A^f)^*}$.

For a character ε of μ, a function f on $\widetilde{F_\nu{}^*}$ (respectively $\widetilde{(A^f)^*}$) will be called an ε-function if $f(\zeta x) = \varepsilon(\zeta) f(x)$ for $\zeta \in \mu$. An ε-function is uniquely determined by its restriction to $[F_\nu{}^*]$ (respectively $[(A^f)^*]$), and it will often be identified with that restriction : its support will be described as a subset of $F_\nu{}^*$ (respectively $(A^f)^*$)....

For each ν, let f_ν be an ε-function on $\widetilde{F_\nu{}^*}$, right invariant under an open subgroup of $\widetilde{F_\nu{}^*}$. Suppose that, for almost every ν, f_ν is right invariant under $[O_\nu{}^*]$ and equals 1 on $[O_\nu{}^*]$. Under these hypotheses, the product of the f_ν – *a function on the restricted product of the* $\widetilde{F_\nu{}^*}$ – *comes* from an ε-function f on $\widetilde{(A^f)^*}$, right invariant under an open subgroup. An ε-function obtained in this way will be called multiplicative, and we write f = $\prod f_\nu$.

(0.9) Example : Take ε to be the identity extension of μ. For $\nu \nmid 3$, let f_ν be the ε-function on $\widetilde{F_\nu{}^*}$ with support {x | $\nu(x) = 0$ or 1}, equal to 1 on $[O_\nu{}^*]$, and given on uniformizers by

$$f_\nu([a]) = \sum_{x \in (O_\nu/(a))^*} (\mathrm{x,a})\, e(a^{-1}x).$$

If a is a uniformizer and $u \in O_\nu{}^*$, then $f_\nu([au]) = (u,a) f_\nu([a])$; hence f_ν is right invariant under $[O_\nu{}^*]$. For $\nu|3$, put $f_\nu([a]) = 1$ if $a \equiv 1$ (3), and $f_\nu([a]) = 0$ otherwise. Let $f = \prod f_\nu$. The Dirichlet series (0.7) is

$$\sum_{\lambda \in F^*} f([\lambda])\, N\lambda^{-s}.$$

The analytic properties of the Dirichlet series (0.7) are deduced from relations with certain "automorphic" functions on GL(2, C). Identify the homogeneous space PGL(2, C)/U(2) with upper half-space $H = \{(z,v) \mid z \in C, v \in R, v > 0\}$ by the Iwasawa decomposition

$$(\mathrm{z,v}) \mapsto \begin{pmatrix}1 & \mathrm{z}\\ 0 & 1\end{pmatrix}\begin{pmatrix}\mathrm{v} & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = (\mathrm{v\ z}).$$

In this model, the upper unipotent group acts by $(z,v) \to (z+t, v)$, and

$$\begin{pmatrix}\mathrm{a} & 0\\ 0 & 1\end{pmatrix}(\mathrm{z,v}) = (\mathrm{az}, |a|\mathrm{v}),\ a \in C^*.$$

Let $\Gamma(3)$ be the subgroup of GL(2, O) consisting of matrices congruent to 1 modulo 3. Kubota discovered that one obtains a character of this group by setting, for

γ = (a b),
(c d)

$\chi(\gamma) = (c/a)_3$ if c $\neq$ 0, and $\chi(\gamma) = 1$ otherwise.

Put $\Gamma_2 = \Gamma(3) \cdot \mathrm{GL}(2, Z)$. Patterson verifies that χ extends to a character of Γ_2 that is trivial on $\mathrm{GL}(2, Z)$; this extension is again denoted χ. The kernel of χ is visibly not a congruence subgroup. Nevertheless,

(0.10) For every $\delta \in \mathrm{GL}(2, F)$, the characters $\chi(\gamma)$ and $\chi(\delta^{-1}\gamma\delta)$ coincide on a congruence subgroup of $\Gamma_2 \cap \delta\Gamma_2\delta^{-1}$.

Again denote by ε the identity embedding of μ, and introduce the multiplicative ε^2-function $\tau = \prod \tau_v$ characterized by the following properties (cf. [18] I, Th. 8.1, p. 152). (a) For $v \nmid 3$, τ_v is supported in O_v and is right invariant under $[O_v{}^*]$. (b) For $v|3$, τ_v is supported in $\{x \mid v(x) \geq \{\} - 3\}$ and is right invariant under $[\{x \mid x \equiv \pm 1\ (3)\}]$. (c) For d in O_v, and x lying respectively over O_v when $v \nmid 3$ and over $\{x \mid v(x) \geq \{\} - 3\}$ when $v|3$, one has $\tau_v(x[d^3]) = \tau(x)\|d\|_v$. (d) If $v \nmid 3$, τ_v equals 1 on $[O_v{}^*]$, equals $\|a\|_v f_v(a)^-$ on uniformizers (f_v as in 0.10), and equals 0 on elements of valuation 2. (e) If $v|3$ and $x = \pm\omega^j\lambda^k$, where $\omega = \exp(2\pi i/3)$ and $\lambda = \surd -3$, then

$\tau_v(x) = 1$ if x = $\pm\lambda^{-3}$;
$\tau_v(x) = 1/3 \exp(\text{-j } 2\pi i/9)$ if x = $\pm\omega^j\lambda^{-1}$, $j = -1, 0,$ or 1;
$\tau_v(x) = 0$ if (j,k) is not congruent modulo 3 to
(0,0), (0,2), (1,2), or (2,2).

For $w = (z, v) \in H$, put

$$K(w) = v\,K_{1/3}(4\pi v)e(z),$$

where $K_{1/3}$ is the "usual Bessel function of the second kind" ([18] I, pp. 129, 134).

One of the principal results of [18] is that the series

(0.11) $\theta(w) = \sigma v^{2/3} + \sum \tau([\mu])\,\mathrm{K}(\,(\mu\ 0)\,\mathrm{w})$,
$\mu \in F^*\ (0\ 1)$

$$\sigma = 2^{-1}3^{-3/2},$$

satisfies $\theta(\gamma w) = \chi(\gamma)\theta(w)$ for $\gamma \in \Gamma_2$ (and in particular $\theta(0, v) = \theta(0, v^{-1})$), and is the residue (in s) of an Eisenstein series E_s – *defined* by analytic continuation in a parameter s—at a value of s where a simple pole occurs. Up to a factor, the Eisenstein series is obtained from $\varphi_s(w) = v^s$ by summing $\overline{\chi(\gamma)}\varphi_s(\gamma w)$ over $\gamma \in \Gamma_2/\Gamma_2 \cap B^+$ (notation : see 0.0.1). The pole is at $s = 4/3$.

Up to elementary factors, Γ and ζ, the Dirichlet series (0.7) is obtained from $\theta(w)$ by Mellin transformation. This gives its analytic behavior and a functional equation.

Patterson calls θ a "cubic θ-series," by analogy with the classical case of holomorphic modular forms of weight $1/2$, where the series $\theta(z) = \sum q^{n^2}$ $(q = e^{2\pi iz})$ has an analogous description as a residue of Eisenstein series.

Some of these results can be predicted by the representation theory of covers of the groups $\mathrm{GL}(2, F_v)$. Here is the method. As above, put $\varphi_s(w) = v^s$ and, for a congruence subgroup Γ, consider the Eisenstein series

$$E_{s,\Gamma}(w) = \sum_{\gamma \in \Gamma/\Gamma \cap B^+} \varphi_s(\gamma w).$$

It converges for Re(s) large and admits analytic continuation as a meromorphic function of s. The study of the "constant terms" shows that, for $Re(s) \geq 4/3$, it is meromorphic in s, with at worst a simple pole at $s = 4/3$. Let E_Γ be the residue at $s = 4/3$, and let $\mathscr{E}$ be the space of linear combinations of the functions $E(\gamma w)$, for congruence subgroups Γ and $\gamma \in \mathrm{GL}(2, F)$.

The group $\mathrm{GL}(2, F)$ acts on $\mathscr{E}$ by $(\gamma * f)(w) = f(\gamma^{-1}w)$. For every $f \in \mathscr{E}$ there is a congruence subgroup $\Gamma \subset \Gamma(3)$ such that $\gamma * f = \chi(\gamma)f$ for $\gamma \in \Gamma$. This follows from (0.10). In particular, $\gamma * f = f$ for $\gamma \in \Gamma \cap \mathrm{Ker}(\chi)$, which allows the action of $\mathrm{GL}(2, F)$ on $\mathscr{E}$ to be extended continuously to an action of the completion $\mathrm{GL}(2, F)^\wedge$ of $\mathrm{GL}(2, F)$ for the topology defined by the subgroups $\Gamma \cap \mathrm{Ker}(\chi)$, with $\Gamma \subset \Gamma(3)$ a congruence subgroup. The study of the "constant terms," together with a "local" study, shows that the representation $\mathscr{E}$ is irreducible and determines its isomorphism class.

For $f \in \mathscr{E}$ satisfying $f(z, v) = f(z + \lambda, v)$ for λ in a lattice $L \subset O$, define the function $W^*(f)$ on H by

$W^*(f)(z, v) = 1/vol(C/L) \int_{C/L} f(z + z_1, v)e(-z_1)dz_1$.

This function is independent of the choice of L. Since φ_s, and hence $E_{s,\Gamma}$, are eigenfunctions of the Laplacian, $W^*(f)$ is K(w) times a constant W(f). Fourier inversion on C/L, together with a study of the constant terms, gives an expansion

$$\mathrm{f} = \sigma(f)v^{2/3} + \sum_{\mu \in F^* \backslash (0\ 1)/(0\ 1)} \mathrm{W}(\ (\mu\ 0) * \mathrm{f}\ )\, \mathrm{K}(\ (\mu\ 0)\ \mathrm{w}),$$

for a suitable constant $\sigma(f)$.

The functional W is a linear form on $\mathscr{E}$. It satisfies

$$\mathrm{W}(\ (1\ v) * \mathrm{f}\ ) = e(v)W(f).$$
$$(0\ 1)$$

for $\nu \in$ F. This, essentially (up to a nuisance due to the place dividing 3), determines it up to a scalar - hence a determination of the nonconstant terms in the Fourier expansions of the f in $\mathscr{C}$.

These methods give the following result. Let T be the set of linear combinations of multiplicative ε^2-functions $\tau' = \prod \tau'_\nu$ on $(\widetilde{A^f})^*$, where a) for almost every ν, one has $\tau'_\nu = \tau_\nu$; b) for every ν, τ'_ν has compact support, or is a right translate of τ_ν. For each $\tau' \in$ T there is then a unique constant $\sigma(\tau')$ such that the function

$$\theta(\tau'; w) = \sigma(\tau')\ v^{2/3} + \sum_{\mu \in F^*\backslash(0\ 1)} \tau'([\mu])\ \mathrm{K}(\ (\mu\ 0)\ \mathrm{w})$$

satisfies $\theta(\tau'; \gamma w) = \chi(\gamma)\ \theta(\tau'; w)$, for γ in a suitable congruence subgroup. The space $\mathscr{C}$ is the set of the $\theta(\tau'; w)$.

The group $\mathrm{GL}(2, F)^\wedge$ is not a restricted product of local groups. It may, however, be compared with such a product, thereby reducing the proof of the preceding results to the study of local groups and of the "principal series" of their representations. The main part of the exposé is devoted to this study.

The method followed by Patterson [18] is different. He begins by proving functional equations for Dirichlet series such as (0.7), and from them deduces the automorphic character of θ by a variant ([18] 7.1) of Weil's method [23]. He deduces the required functional equations from the functional equation (for s $\to$ 2 - s) of Eisenstein series ([11],[12]), by examining what it implies for the Fourier coefficients : each of them, as a function of s, is a Dirichlet series reminiscent of (0.7). He hopes that this method will also provide the required analytic properties of Dirichlet series analogous to (0.7), in order to generalize [5] to cases where n > 3. The method presented here, by contrast, depends crucially on n being odd and on $(n-1)/2 = 1$.

In the text we shall dwell on the local theory and only sketch how it applies to residues of Eisenstein series. We shall discuss neither the double role, indicated above, of series (0.7), for me one of the great mysteries of the theory, nor the determination of the constant term, nor the ingenious method by which Patterson [18] identifies θ, proved automorphic, with the residue of the Eisenstein series.

0.0 **Notations :**

0.0.1 We shall denote the following subgroups of GL(2) as follows :
Z : the center (scalar matrices).
T : diagonal matrices. We put diag (a,b) = (a 0).
(0 b)
B,U : Borel subgroup ; its unipotent radical. In particular :

B^+ : (* *) ; B^- : (* 0) ; U^+ : (1 *) ; U^- : (1 0).
(0 *) (* *) (0 1) (* 1)

0.0.2 From section 2 onward, F denotes either a local field (= locally compact and non-discrete) or a global field (= a finite extension of Q or of a field $F_p(t)$). We choose an integer n, not divisible by the characteristic of F, such that F contains a primitive n-th root of unity. We write $\mu_n(F)$, or simply μ, for the group of n-th roots of unity in F.

For global F and a place ν of F, the inclusion of F in its completion F_ν induces an isomorphism from $\mu_n(F)$ to $\mu_n(F_\nu)$, by means of which we identify these two groups.

0.0.3 For a global field F, we write A for its ring of adeles, F_∞ for the product of the F_ν over the real or complex places ν, and A^f for the ring of finite adeles :

$$\mathrm{A} = F_\infty \times A^f .$$

If F is a number field, O denotes its ring of integers. If F is a nonarchimedean local field, O denotes its valuation ring. Variant : notation F_ν, O_ν.

0.0.4 For a local (respectively global) field F, from section 3 onward we shall have to consider a central extension of G(F) (respectively G(A)) by μ. A ~ denotes the inverse image of a subgroup in this extension. For example : $T(F)^\sim$, $T(A)^\sim$, $G(A^f)^\sim$, ...

0.0.5 From section 4 onward, fix a faithful character $\varepsilon : \mu \hookrightarrow C^*$. A representation of a central extension of a group by μ will be called an ε-representation if each $\zeta \in \mu$ acts by multiplication by $\varepsilon(\zeta)$. A function on the extension will be called an ε-function if $f(\zeta x) = \varepsilon(\zeta) f(x)$ for $\zeta \in \mu$.

0.0.6 A representation π of a totally disconnected locally compact group is called algebraic if the stabilizer of every vector is an open subgroup, and admissible if, in addition, the subspace fixed by an open subgroup is finite-dimensional. For the extension of this terminology to other groups, such as $\mathrm{GL}(2, A)^\sim$, I refer to [6].

0.0.7 From section 5 onward, it will sometimes be useful to write the multiplication of characters additively, and to write s for the character $\|x\|^s$; for example, to write $(\alpha_2 - \alpha_1 - 1)(\lambda)$ for $\alpha_2(\lambda)\ \alpha_1(\lambda)^{-1}\ \|\lambda\|^{-1}$. The real part of a character σ is the real number s such that $\sigma(x)\|x\|^{-s}$ is unitary.

1. Central extensions of SL(2)

Let F be a field. Recall the

Definition 1.1 : (i) A symbol is a map s from $F^* \times F^*$ to an abelian group (written multiplicatively), which is bimultiplicative and satisfies s(a,b) = 1 when a + b = 1.

(ii) $K_2(F)$ denotes the quotient of $F^* \otimes F^*$ by the subgroup generated by the a ⊗ b, for a,b ∈ F^* with a + b = 1, and {x,y} denotes the image in $K_2(F)$ of x ⊗ y ; it is the universal symbol.

Example 1.2 Let n be an integer ≥ 1, prime to the characteristic exponent of F, and suppose that F contains the n-th roots of unity. Let $\overline{F}$ be a separable closure of F. The long exact $\mathrm{Gal}(\overline{F}/F)$-cohomology sequence deduced from the Kummer exact sequence $1 \to \mu_n \to \overline{F}^* \to \overline{F}^* \to 1$ gives isomorphisms $F^*/F^{*n} \simeq H^1(\mu_n)$ and $H^2(\mu_n) \simeq \mathrm{Br}(F)_n$, and the cup product

$$H^1(\mu_n) \otimes H^1(\mu_n) \to H^2(\mu_n \otimes \mu_n) = H^2(\mu_n) \otimes \mu_n$$

gives a bimultiplicative map from $F^* \times F^*$ to $\mathrm{Br}(F)_n \otimes \mu_n$. It is a symbol ([21] XIV 2).

1.3 Let G be an algebraic group over F, isomorphic to SL(2). There is then a canonical central extension $G(F)^\wedge$ of G(F) by $K_2(F)$. For every symbol s : $F^* \times F^* \to$ A, it yields a central extension $G(F)^s$ of G(F) by A.

I did not take G = SL(2) from the outset so that I could say that the adjoint group $\mathrm{PGL}(2,F)$ acts on $\mathrm{SL}(2,F)^s$ by transport of structure. This action lifts the action by "inner" automorphisms of $\mathrm{PGL}(2,F)$ on $\mathrm{SL}(2,F)$ (rather than "inner," one should say : induced by an inner automorphism of $\mathrm{GL}(2,F)$). It is trivial on A.

Remark 1.4 (Not used below). Exclude the cases in which F has 2 or 3 elements. Then $\mathrm{SL}(2,F)$ is perfect, and according to Moore [17] and Matsumoto [14], the universal central extension of $\mathrm{SL}(2,F)$ is a central extension by the symplectic analogue ${}_{-1}L_2(F)$ of $K_2(F)$. The group $\mathrm{PGL}(2,F)$ acts on ${}_{-1}L_2(F)$ (through its quotient by $\mathrm{PSL}(2,F)$), and $K_2(F)$ is the group of coinvariants (the largest quotient inheriting a trivial action). According to Karoubi [24] III p. 78, if F is not of characteristic 2, the group ${}_{-1}L_2' := \mathrm{Ker}({}_{-1}L_2(F) \to K_2(F))$ fits into an exact sequence

$F^* \dashrightarrow (2\text{-torsion of } K_2(F)) \to {}_{-1}L_2' \to W \xrightarrow{r} K_2(F)/2K_2(F) \to 0,$

where W is the group of virtual quadratic forms of dimension 0 and discriminant 1. I imagine that r coincides with the Stiefel-Whitney class defined by Milnor [15] ; according to Milnor, $\mathrm{Ker}(r)$ would then be the cube of the augmentation ideal (dim = 0) in the ring of virtual quadratic forms.

1.5 The group $GL(2,F)$ is the semidirect product of the subgroup of diagonal matrices $\mathrm{diag}(a,1)$ by the normal subgroup $SL(2,F)$. Likewise take the semi-direct product of F^* by $SL(2,F)^s$, with a $\in F^*$ acting as the diagonal matrix $\mathrm{diag}(a,1)$ in $PGL(2,F)$. This is a central extension $GL(2,F)^s$ of $GL(2,F)$ by A. For s the universal symbol, it will be denoted $GL(2,F)^\wedge$.

After the preliminaries in 1.6, we shall give in 1.7-1.9 some properties of these central extensions. They suffice to characterize them.

1.6 Preliminaries : (a) If $H^\sim$ is a central extension of a group H, and a,b $\in$ H, the commutator (ã,b̃) of lifts ã and b̃ of a and b depends only on a and b. We denote it simply by (a,b). Likewise, the inner automorphism $int_{\tilde{a}}$ of $H^\sim$ depends only on a. We denote it simply by int_a.

(b) If U is a subgroup of H, normalized by B $\subset$ H and generated by the commutators (b,u) (b $\in$ B, u $\in$ U), there is at most one lift u $\to$ ũ of U into $H^\sim$, normalized by B. If the homomorphism u $\to$ ũ commutes with int_b (b $\in$ B), the lift of a commutator (b,u) is necessarily, as in (a) above,

$$(b,u)^\sim = int_b(u)^\sim \, \tilde{u}^{-1} = int_b(\tilde{u}) \cdot \tilde{u}^{-1} = (\tilde{b},\tilde{u}).$$

1.7 Excluding the cases in which F has 2 or 3 elements, 1.6(b) applies to a Borel subgroup B of $SL(2,F)$ and its unipotent radical U, i.e. to the stabilizer B of a (homogeneous) line D $\subset F^2$ and to its pointwise stabilizer U. It is easy to show that, for every central extension of $SL(2,F)$, U has a lift normalized by B (unique by 1.3). We write it u $\to$ ũ. One has ẽ = e, and if u $\neq$ e fixes a line, that line is uniquely determined. Thus the map u $\to$ ũ is well defined on the union of the conjugates of U (the unipotent elements). It is invariant under conjugation. In the case of the extension 1.3, it commutes with the action of $PGL(2,F)$.

1.8 The group $GL(2,F)^s$ was defined as a semidirect product $F^* \times SL(2,F)^s$. We denote by d_1 the corresponding inclusion of F^*, and by d_2 its conjugate by

$$w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \text{ (cf. 1.9).}$$

For a,b $\in F^*$, $d_1(a)$ and $d_2(b)$ are respectively lifts of $\mathrm{diag}(a,1)$ and $\mathrm{diag}(1,b)$ in $GL(2,F)$. These matrices commute. Hence the commutator of $d_1(a)$ and $d_2(b)$ lies in A. One has

$$(d_1(a), d_2(b)) = s(a,b),$$

and this formula uniquely determines the extension of $SL(2,F)$ under consideration.

Here is a more complete formula sheet.

1.9 **Formula sheet : Put**

$$e^+(a) := \text{the lift of 1.7 of } \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \qquad e^-(a) := \text{the lift of 1.7 of } \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}$$

$$\text{w(a)} := e^{+}(a)e^{-}(-1/a)e^{+}(a), \text{ a lift of } \begin{pmatrix} 0 & a \\ (a^{-1}) & 0 \end{pmatrix}$$

$d_2(a) :=$ w(1) $d_1(a)$ $w(1)^{-1}$

h(a) $:= d_1(a)$ $d_2(a^{-1})$

For every a $\in$ F^*, the images in SL(2, F) of $e^{+} = e^{+}(a)$, $e^{-} = e^{-}(-1/a)$, and w = w(a) belong to the Tits trijection. In SL(2, F)^ one has

(1.9.1) $e^{\pm} = int_w(e^{\mp})$

(1.9.2) w = e^{+} e^{-} e^{+} = e^{-} e^{+} e^{-}

(expressing $int_w(w)$ = w). Each of the elements w, e^{+}, e^{-} determines the other two. It follows that w(a) = $d_1(a)w(1)d_1(a)^{-1}$, and

(1.9.3) w(a) $w(1)^{-1} = d_1(a)$ $d_2(a^{-1}) =$: h(a).

In terms of h, formula

(1.9.4) $(d_1(a), d_2(b))$ = s(a,b)

may be rewritten as

(1.9.5) h(ab) = s(a,b) h(a) h(b).

Example 1.10 Take F = R, A = $\pm$1, and s(a,b) = 1 if a or b > 0, and -1 otherwise. The extension $\mathrm{SL}(2,R)^s$ is then the nontrivial double cover $\mathrm{SL}(2,R)^{\sim}$ of $\mathrm{SL}(2,R)$. It has the following description :

(a) Let $(R^2 - 0)^{\sim}$ be a nontrivial double cover of R^2 - {0}. Since $\pi_1(R^2 - \{0\}) = Z$, it is unique up to isomorphism.

(b) The group $\mathrm{SL}(2,R)^{\sim}$ is the group of elements of $\mathrm{SL}(2,R)$, equipped with a lift to $(R^2 - 0)^{\sim}$ of their action on R^2.

The same description holds for the cover $\mathrm{GL}(2,R)^s$ of $\mathrm{GL}(2,R)$.

1.11 An analogous description applies over an arbitrary field and, without being indispensable, is convenient for computation. It requires part of the general theory of the Zariski sheaves $\underline{a}K_q$ on a scheme S associated with the presheaves U $\to$ $K_q(\Gamma(U,O))$. Specifically, we shall use special cases of the following isomorphism, deduced from Quillen's resolution of $\underline{a}K_q$ (the Gersten conjecture) : for T smooth of pure codimension d in a scheme S smooth over F, one has

$$(1.11.1)\ \underline{H}^i_T(S, \underline{a}K_q) = \begin{cases} \underline{a}K_{q-d} & \text{for i = d} \\ 0 & \text{for i} \neq \text{d (and if d > q).} \end{cases}$$

Special case : (a) for the affine plane A^2 = Spec F[X,Y],

(1.11.2) $H^0(A^2 - \{0\}, \underline{a}K_2) = K_2(F)$

(1.11.3) $H^1(A^2 - \{0\}, \underline{a}K_2) = Z\ (= K_0(F))$.

(b) For T of codimension 1 in S, the case i = 1 of (1.11.1) gives a residue morphism Res : $H^0(S-T, \underline{a}K_2) \to H^0(T, \underline{a}K_1) = H^0(T, O_T^*)$. Suppose T is connected, and let v denote the corresponding valuation ; then

$$\text{Res}\{f,g\} = (-1)^{v(f)v(g)} f^{v(g)}\ g^{-v(f)} \text{ restricted to T.}$$

The residue morphism vanishes on the image of $H^0(S, \underline{a}K_2)$. If P is a torsor under $\underline{a}K_2$ on S and e is a section of P on S - T, this also makes it possible to define Res e $\in H^0(T, O_T^*)$, which vanishes if and only if e extends across T.

Now let P be a torsor under $\underline{a}K_2$ on A^2 - {0}. Since the isomorphism (1.11.3) is canonical, P is isomorphic to its transform by any element of $\mathrm{GL}(2,F)$, and, by (1.11.2), the group of elements of $\mathrm{GL}(2,F)$, equipped with a lift to the torsor P of their action on A^2 - {0}, is a central extension of $\mathrm{GL}(2,F)$ by $K_2(F)$. We shall verify that, when the class of P is a generator of $H^1(A^2 - 0, \underline{a}K_2) = Z$, this is the extension $\mathrm{GL}(2,F)^\wedge$.

1.12 With respect to the cover of A^2 - {0} by the open sets Y $\neq$ 0 and X $\neq$ 0, the generator of H^1 is represented by the Čech cocycle {X,Y}. Take P to be the torsor equipped with a section e_X on X $\neq$ 0 and a section e_Y on Y $\neq$ 0, with

$$e_X - e_Y = \{X,Y\}.$$

Note that $e_Y - e_X = \{Y,X\}$, so that this description of P is symmetric in the coordinates X and Y. One has

$$\text{Res } e_X = \text{Res}\{X,Y\} = Y^{-1} \text{ (on X = 0),}$$

and symmetrically. To identify the central extension of $\mathrm{GL}(2,F)$ defined by P with the extension $\mathrm{GL}(2,F)^\wedge$, set

$$d_1(a) = \text{the lift to P of diag}(a,1) \text{ that is the identity}$$

at the fixed point (0,1) ; hence, since w sends (0,1) to (1,0),

$$d_2(a) = \text{the analogous lift for diag}(1,a) \text{ and (1,0).}$$

To verify (1.9.4), note that $d_1(a)$ transforms e_Y into $e_Y + \{a,Y\}$ (whose residue is the transform aX^{-1} of the residue X^{-1} of e_Y), whereas $d_2(a)$ transforms e_Y into itself.

1.13 Let $A^{2\vee}$ be the affine scheme Spec F[a,b], corresponding to the dual of the natural representation of $\mathrm{GL}(2,F)$. To $\ell \in A^{2\vee} - \{0\}$, associate the group of trivializations of P on $\ell = 1$. It is a principal homogeneous space under $K_2(F)$, and, by sheafifying this construction, one obtains a torsor $P^\vee$ under $\underline{a}K_2$ on $A^{2\vee} - \{0\}$. By construction, P and $P^\vee$ become isomorphic on the locus $\ell(x) = 1$

of $A^{2\vee} \times A^2$, and $\mathrm{GL}(2,F)$ acts on P and $P^\vee$ while respecting this isomorphism.

Here is a more complete formula sheet.

1.14 Formula sheet :

P : a torsor on A^2 - {0}, where A^2 is the plane with coordinates X,Y. $P^\vee$: a torsor on $A^{2\vee}$ - {0}, where $A^{2\vee}$ is the dual plane, with coordinates a,b. e_X, e_Y : sections of P on X $\neq$ 0, Y $\neq$ 0. e_a, e_b : sections of $P^\vee$ on a $\neq$ 0, b $\neq$ 0.

The defining formulas are (a) e_X - e_Y = {X,Y} ; (b) P and $P^\vee$ become isomorphic on the locus aX + bY = 1 in $A^2 \times A^{2\vee}$. Under this isomorphism, e_a at (a,b) coincides with e_X at $(1/a, 0)$, and e_b at (a,b) coincides with e_Y at $(0, 1/b)$.

One checks that the resulting formula sheet is symmetric in P and $P^\vee$ and in the coordinates : it is preserved by the substitutions (X,Y ;a,b) $\mapsto$ (a,b ;X,Y) and (Y,X ;b,a). One has

(1.14.1) e_X - e_Y = {X,Y} and e_a - e_b = {a,b} ;

(1.14.2) e_a - e_Y = -{a,Y} on the locus aX + bY = 1, and likewise

$$e_b \text{ - } e_X = \text{-}\{b,X\} ;$$

(1.14.3) $d_1(x)_*$ and $e^-(y)_*$ fix e_X and e_a,
and likewise

$$d_2(x)_* \text{ and } e^+(y)_* \text{ fix } e_Y \text{ and } e_b.$$

2. Review : the Hilbert symbol

The notation of 0.0.2 is in force.

2.1 Let F be a local field. If F is not complex, then $\mathrm{Br}(F)_n = Z/nZ$, and the symbol of 1.2 is the Hilbert symbol

$$\{\ ,\ \} : F^* \times F^* \to\ \mu.$$

For complex F, the Hilbert symbol is defined to be trivial. As n varies, these symbols, denoted $\{\ ,\ \}_n$, satisfy $a, b_n = a, b^m_{nm}$.

Recall 2.2 : The pairing induced by the Hilbert symbol,

$$\{\ ,\ \} : F^*/F^{*n} \times F^*/F^{*n} \to\ \mu,$$

is nondegenerate : it identifies each factor F^*/F^{*n} with the Pontryagin dual $\mathrm{Hom}(F^*/F^{*n},\ \mu_n(F))$ of the other.

In the definition of $\{\ ,\ \}_n$ we adopt Serre's signs [21], i.e. those for which (2.4.1) below is true.

Example 2.3 : F = R, n = 2 : this is the symbol of 1.10.

Example 2.4 : The tame case is that in which F is non-Archimedean and n is prime to its residual characteristic. Place ourselves in this case, and let O be the valuation ring of F, v its valuation, k = O/m the residue field, q = $|k|$, and r the reduction map modulo m. One has n | q-1 ; r induces an isomorphism $\mu_n(F) \simeq\ \mu_n(k)$, and

(2.4.1) $ra, b_n = r((-1)^{v(a)v(b)}\ a^{v(b)}\ b^{-v(a)})^{(q-1)/n}$.

2.5 Let F be a global field. Denote by $\{\ ,\ \}_v\{\}$ the Hilbert symbol for the completion F_v of F at a place v. It takes values in $\mu_n(F_v)$, identified with μ by (0.0.2). For two idèles x and y of F, one has $x_v, y_{v\,v} = 1$ at almost every place v (cf. 2.4). This makes it possible to define {x,y} as the product of the $x_v, y_{v\,v}$ over all places.

Recall 2.6 : The induced pairing

$$\{\ ,\ \} : A^*/A^{*n} \times A^*/A^{*n} \to\ \mu$$

is nondegenerate. It identifies the locally compact group A^*/A^{*n} with its own Pontryagin dual $\mathrm{Hom}(A^*/A^{*n},\ \mu)$.

This follows from 2.2 and from the fact that in the tame case 2.4, hence for almost every v, the image of ${O_v}^*$ in ${F_v}^*/{F_v}^{*n}$ is maximal totally isotropic for $\{\ ,\ \}_v\{\}$.

The reciprocity law for Hilbert symbols says that if the idèles x and y are principal (x,y $\in\ F^*$), then {x,y} = 1. More precisely :

Recall 2.7 : (i) $F^{*n} = F^* \cap A^{*n}$. (ii) The image F^*/F^{*n} of F^* in A^*/A^{*n} is maximal totally isotropic. It is a discrete subgroup with compact quotient, and $\{\ ,\ \}$ identifies the Pontryagin dual

of the discrete group F^*/F^{*n} with the quotient $A^*/A^{*n}\cdot F^*$.

(ii) The Image F^*/F^{*n} of F^* in A^*/A^* is maximal totally isotropic. It is a discrete subgroup with compact quotient, and { , } identifies the Pontryagin dual of the discrete group F^*/F^{*n} with the quotient $A^*/A^{*n}\cdot F^*$.

Example 2.8 Take for F the field $Q(\sqrt{}-3)$ of cube roots of unity, and let n = 3. Let K be the compact open subgroup of $A^*/A^{*3} = A^{f*}/A^{f*3}$ that is the image of the product of the ${O_\nu}^* \subset {F_\nu}^*$ for $\nu \nmid 3$, and of $\{x \in {F_\nu}^* \mid x \equiv 1(3)\}$ for $\nu \mid 3$. The group A^*/A^{*3} is the product of the discrete group F^*/F^{*3} and the compact group K; each of these two subgroups is maximal isotropic, and { , } identifies either one with the Pontryagin dual of the other.

3. The metaplectic group

3.1 Let F be a local field. A ~ will denote the central extensions of $\mathrm{SL}(2,F)$ and $\mathrm{GL}(2,F)$ by μ deduced from the Hilbert symbol.

3.2 In the tame case (2.4), the central extension $\mathrm{SL}(2,F)^{\sim}$ of $\mathrm{SL}(2,F)$ is trivial over the maximal compact subgroups. The trivialization is moreover unique : if the residue field does not have 2 or 3 elements, $\mathrm{SL}(2,O)$ is perfect and, in all cases, its largest abelian quotient has order prime to n. Over the matrices (1 a) and (1 0) (a ∈ O), it coincides with the trivialization of 1.7. (0 1) (a 1)

Likewise, the central extension $\mathrm{GL}(2,F)^{\sim}$ of $\mathrm{GL}(2,F)$ is trivialized over $\mathrm{GL}(2,O)$, by the unique trivialization extending $\mathrm{diag}(a,1) \mapsto d_1(a)$ (a ∈ O^*).

3.3 Now take F global. At almost every place v one is in the tame case, and 3.2 makes it possible to define the restricted product $\mathrm{SL}(2,A)^{\sim}$ of the $\mathrm{SL}(2,F_v)^{\sim}$ relative to the compact subgroups lifting $\mathrm{SL}(2,O_v)$. This is an extension of $\mathrm{SL}(2,A)$ by the direct sum of the $\mu_n(F_v)$. Pushing out by the product map $\prod : \oplus\, \mu_n(F_v) \to \mu_n(F) =: \mu$, one obtains a central extension $\mathrm{SL}(2,A)^{\sim}$ of $\mathrm{SL}(2,A)$ by μ, and the reciprocity law for Hilbert symbols ensures that this extension splits over $\mathrm{SL}(2,F)$. Since $\mathrm{SL}(2,F)$ is perfect, the splitting is unique. On the unipotent radical of a Borel subgroup it is induced by the splitting obtained from the local splittings (1.7) :

$$
\begin{array}{ccccccccc}
 & & 1 & \to & \mu & \to & \mathrm{SL}(2,F_v)^{\sim} & \to & \mathrm{SL}(2,F_v) \to 1 \\
 & & & & \| & & \downarrow & & \downarrow \\
(3.3.1) & & 1 & \to & \mu & \to & \mathrm{SL}(2,A)^{\sim} & \to & \mathrm{SL}(2,A) \to 1 \\
 & & & & & \nwarrow & [\,] & \uparrow & \\
 & & & & & & \mathrm{SL}(2,F) & &
\end{array}
$$

3.4 Take n = 3, F = $Q(\sqrt[3]{1}) \subset$ C, and ε = Id. One has $\mathrm{SL}(2,A)^{\sim} = \mathrm{SL}(2,C) \times \mathrm{SL}(2,A^f)^{\sim}$, whence, with the notation of the Introduction,

$$\mathrm{SL}(2,A)^{\sim} / \mathrm{SU}(2) \times \{1\} \simeq \mathrm{H} \times \mathrm{SL}(2,A^f)^{\sim} ;$$

since $\mathrm{SL}(2,C)/SU(2) \simeq \mathrm{PGL}(2,C)/U(2)$. Restriction to H × {1} therefore associates with every function f on $\mathrm{SL}(2,A)^{\sim}$ that is right-invariant under SU(2) a function on H.

Scholium 3.5 The restriction map above gives a bijective correspondence between (a) the space of ε-functions on $\mathrm{SL}(2,A)^{\sim}$, left-invariant under $[\mathrm{SL}(2,F)]$ and right-invariant under SU(2) × (a suitable open subgroup K of $\mathrm{SL}(2,A^f)^{\sim}$),

(b) the space of functions on H which, for γ in a suitable congruence subgroup Γ of $\mathrm{SL}(2,O)$, satisfy

$$f(\gamma w) = \chi(\gamma)\ \mathrm{f(w)}.$$

The point is that $\mathrm{SL}(2, A^f)^{\sim}$ is identified with the completion of $\mathrm{SL}(2, F)$ for the topology whose subgroups are the kernels of χ inside congruence subgroups of $\mathrm{SL}(2, O)$. See [14].

The group $\mathrm{SL}(2, A^f)^{\sim}$ acts by right translations on the space in (a). From the classical point of view (b), this corresponds to, and makes more precise, the existence of Hecke operators. To express and use (0.11), it is moreover useful to pass to $\mathrm{GL}(2)$.

3.6 For a global field F, again arbitrary, define $\mathrm{GL}(2, A)^{\sim}$ in either of two ways : (a) by pushing out the restricted product of the $\mathrm{GL}(2, F_v)^{\sim}$ by $\sum : \oplus\ \mu_n(F_v) \to\ \mu$; (b) as the semidirect product of A^* by $\mathrm{SL}(2, A)^{\sim}$.

There is again a diagram (3.3.1), with SL replaced by GL ; the splitting [ ] coincides with d_1 on $\mathrm{diag}(a, 1)$.

3.7. Suppose again that F is a local field. From the Hilbert symbol one deduces a morphism of Zariski sheaves on $A^2 - 0$, from $\underline{a}K_2$ to the sheaf of locally constant functions on $F^2 - 0$, for the usual topology, with values in μ. This makes it possible to deduce from P (1.11) a torsor P_n under μ on $F^2 - 0$, equipped with its usual topology. For each homogeneous line D there is a set of distinguished trivializations of P_n on F^2-D : those that are images of a trivialization of P on A^2-D. If d = 0 is an equation of D, any two differ by a function $\zeta \cdot a, d(x)$ with values in μ.

The description 1.11 of $\mathrm{GL}(2, F)^{\wedge}$ gives here : $\mathrm{GL}(2, F)^{\sim}$ is the group of elements of $\mathrm{GL}(2, F)$ equipped with a lift to P_n of their action on F^2, this lift respecting the system of distinguished trivializations above.

3.8 In particular, $Z(F)^{\sim}$ acts on P_n. The action will be written as scalar multiplication : $z \cdot p$. For a,b $\in\ F^{*n}$, put $diag^{\sim}(a, b) = d_1(a)d_2(b)$; $diag^{\sim}$ is a homomorphism $F^{*n} \times F^{*n} \to T(F)^{\sim}$. For a $\in\ F^{*n}$ and p $\in\ P_n$, put $a \cdot p = diag^{\sim}(a, a)p$. This formula lifts to P_n the homothety of ratio a on F^2.

3.9 As in 1.13, there is a torsor $P_n^{\vee}$ on the dual and an isomorphism between P_n and $P_n^{\vee}$ on the locus $\ell(x) = 1$, all $\mathrm{GL}(2, F)^{\sim}$-equivariantly. Formula sheet 1.14 remains valid.

For a $\in\ F^{*n}$, write $a \cdot p$ for the action of $\mathrm{diag}(a^{-1}, a^{-1})$ on $P_n^{\vee}$; it lifts to $P_n^{\vee}$ the homothety v $\mapsto$ av on $F^{2\vee}$.

3.10 **In the rest of the exposé, the torsors P_n and $P_n^{\vee}$ will simply be denoted P and $P^{\vee}$.**

3.11 For global F, and at almost every place v, there is a trivialization of P_v on $(A^2-0)(O_v) = (O_v{}^* \times O_v) \cup (O_v \times O_v{}^*)$: for tame v, e_X on $O_v{}^* \times O_v$ and e_Y on $O_v \times O_v{}^*$ coincide on $O_v{}^* \times O_v{}^*$. This makes it possible to adelize 3.7 and obtain a torsor P under μ on $(A^2-0)(A)$. The action of $\mathrm{GL}(2,A)$ on A^2 lifts to an action of $\mathrm{GL}(2,A)^{\sim}$ on this torsor. The torsor is trivialized on $(A^2-0)(F) = F^2-0$ by e_X and e_Y, and $[\mathrm{GL}(2,F)]$ respects this trivialization.

4. The group of diagonal matrices.

4.1 Heisenberg groups : Let $H^{\sim}$ be a locally compact group that is an extension of a locally compact, totally disconnected, commutative group H by its center Z. Fix a character ω of Z. Assume that ω is unitary on commutators, and that the bicharacter $\omega((x,y))$ of H identifies H with its own Pontryagin dual. This implies that $Z/\operatorname{Ker}(\omega)$ is the center of $H^{\sim}/\operatorname{Ker}(\omega)$.

For every closed subgroup L of H, write $L^{\sim}$ for its inverse image in $H^{\sim}$. We shall say that L is totally isotropic if $\omega((x,y))$ is trivial on L, i.e. if $L^{\sim}/\operatorname{Ker}(\omega)$ is commutative.

We are interested in representations of $H^{\sim}$ with central character ω. Such a representation will be called admissible if, for every compact open totally isotropic subgroup K $\subset$ H, its restriction to $K^{\sim}$ is a sum of one-dimensional representations, each occurring with finite multiplicity.

A function f on $H^{\sim}$ will be called ω-locally constant if, for every x in $H^{\sim}$, there exist a K as above and a character χ of $K^{\sim}$ extending ω such that f(kx) = $\chi(k)f(x)$ for k $\in$ $K^{\sim}$.

Finally, fix a maximal totally isotropic subgroup L of H and a character ω' of $L^{\sim}$ extending ω. Under the preceding hypotheses it is known that :

(4.1.1) The group $H^{\sim}$ has an irreducible admissible representation V with central character ω. It is unique up to isomorphism.

(4.1.2) There is a nonzero linear form a on V such that $a(\ell v) = \omega'(\ell)a(v)$ for $\ell \in L^{\sim}$. It is unique up to a scalar.

(4.1.3) The map sending v to the function a(gv) on $H^{\sim}$ is an isomorphism of V with the space of ω-locally constant functions on $H^{\sim}$ satisfying $f(\ell g) = \omega'(\ell)f(g)$ for $\ell \in L^{\sim}$, and having compact support modulo $L^{\sim}$.

In particular, if H is finite, V has dimension equal to the index of a maximal totally isotropic subgroup of H, i.e. the square root of the order of H.

(4.2) Let F be a local field, and write simply $T^{\sim}$ for $T(F)^{\sim}$. By 1.8.1 and 2.2, the center of $T^{\sim}$ is the subgroup $\operatorname{diag}(F^{*n} \times F^{*n})^{\sim}$. The quotient of $T^{\sim}$ by its center is finite and commutative, and the commutator map has values in μ. The theory of Heisenberg groups gives the following.

Proposition 4.3 (i) The isomorphism class of an irreducible admissible ε-representation of $T^{\sim}$ is determined uniquely by its central character. (ii) Let ω be an ε-character of the center of $T^{\sim}$, let T' be a subgroup of T such that $T'^{\sim}$ is a maximal commutative subgroup of $T^{\sim}$, and let ω' extend ω to $T'^{\sim}$.

The induced representation $Ind_{T'^{\sim}}^{T^{\sim}}(\omega')$ is irreducible, with central character ω.

4.4 The center of $T^{\sim}$ is $(F^{*n} \times F^{*n})^{\sim}$. Its index in $T^{\sim}$ is $[F^* : F^{*n}]^2$, and the irreducible admissible ε-representations of $T^{\sim}$ therefore have dimension $[F^* : F^{*n}]$. The extension $(F^{*n} \times F^{*n})^{\sim}$ of $F^{*n} \times F^{*n}$ by μ is split by $diag^{\sim}$ (3.8). This identifies ε-characters ω of the center of $T^{\sim}$ with pairs (α_1, α_2) of characters of F^{*n}, by

(4.4.1) $\omega(diag^{\sim}(a,b)) = \alpha_1(a)\alpha_2(b)$.

We shall use this identification. By 4.3, the irreducible admissible ε-representations of $T^{\sim}$ are parametrized by these pairs (α_1, α_2). It is sometimes more convenient to parametrize them by pairs (χ_1, χ_2) of characters of F^* that are trivial on $\mu_n(F) \subset F^*$, by setting

(4.4.2) $\chi_i(x) = \alpha_i(x^n)$ (i = 1,2).

To realize the ε-representations of $T^{\sim}$, two choices of T' as in 4.3(ii) are convenient. In each case there is a natural splitting of the extension $T'^{\sim}$ of T' by μ, so giving an extension ω' of ω to $T'^{\sim}$ amounts to extending (α_1, α_2) from $F^{*n} \times F^{*n}$ to T'.

a) Subgroup $F^* \times F^{*n}$; splitting (a,b) $\mapsto d_1(a)d_2(b)$. b) In the tame case : subgroup $O^*F^{*n} \times O^*F^{*n}$; splitting (a,b) $\mapsto d_1(a)d_2(b)$.

In case (b), when χ_1 and χ_2 are unramified, there is a unique extension ω' of ω that is trivial on $d_1(O^*)d_2(O^*)$.

4.5 Now suppose F is global. The center of $T(A)^{\sim}$ is the inverse image of T(A) = diag$(A^{*n} \times A^{*n})$, and the inverse image of $T(F)T(A)^n$ is a maximal commutative subgroup. By 4.1 one has :

Theorem 4.6. Let α_1 and α_2 be two characters of A^{*n}/F^{*n}, and let V_{α_1,α_2} be the space of ε-functions on $[T(F)]\backslash T(A)^{\sim}$, right-invariant under an open subgroup of $T(A^f)^{\sim}$, and satisfying

$$f(g \cdot diag^{\sim}(a_1, a_2)) = f(g)\alpha_1(a_1)\alpha_2(a_2)$$
$$\text{if } a_1, a_2 \in A^{*n}.$$

Then V_{α_1,α_2} is an irreducible admissible ε-representation of $T(A)^{\sim}$.

These representations are the automorphic representations of T(A). It is sometimes more convenient to index them by pairs (χ_1, χ_2) of Hecke characters that are trivial on the local n-th roots of unity; put $\chi_i(a) = \alpha_i(a^n)$.

5. The principal series

5.0. Let F be a local field. Write G for $GL(2, F)$, and simplify the notation by omitting F : thus $T^{\sim}$ will stand for $T(F)^{\sim}$, and so on.

For F = C one has $G^{\sim}$ = G × μ, and the theory of representations of $G^{\sim}$ reduces to that of G. For F = R one has n = 1 or 2. We shall not dwell on these familiar cases : throughout the rest of this section, except in 5.12, F is assumed non-Archimedean.

5.1. Some arguments from the representation theory of reductive groups over F generalize unchanged to central extensions such as $SL(2, F)^{\sim}$ or $GL(2, F)^{\sim}$; others do not. Those that generalize are based on the Bruhat decomposition ; the structure of the unipotent radical U of a parabolic P (root subgroups, ...) ; the Levi decomposition P = MU and the $contracting/dilating$ character of the action of M on U ; and the relation between the group and its Lie algebra. Examples include general results on Jacquet modules and supercuspidal representations. Difficulties arise from the commutativity of tori and the structure of Hecke algebras.

5.2. Write simply B and U for B^+ and U^+. The group $B^{\sim}$ is the semidirect product $T^{\sim} \cdot U$ of $T^{\sim}$ by the lift (1.7) of U. If ρ is a representation of $T^{\sim}$, use the same symbol ρ for the representation of $B^{\sim}$ that is trivial on U and is given by $\rho(tu) = \rho(t)$.

The modulus of $B^{\sim}$ is thereby obtained from the quasi-character δ of $T^{\sim}$, defined by $\delta(t) = \|ab^{-1}\|$ when t lies above $\mathrm{diag}(a, b) \in$ T. If $\rho : T^{\sim} \to GL(V)$ is an algebraic representation of $T^{\sim}$, the induced representation $Ind_{B^{\sim}}^{G^{\sim}}(V, \rho)$, or simply Ind(V), is the representation by right translation on $B^{\sim}\backslash G^{\sim}$ on the space of locally constant functions $f : G^{\sim} \to$ V satisfying

(5.2.1) f(utg) = $\delta(t)^{1/2}\rho(t)f(g)$.

5.3. Interpret this definition through the dictionary between :

(a) locally constant sheaves of complex vector spaces on P^1, equipped with an (”algebraic”) action of $G^{\sim}$;

(b) algebraic representations (V, ρ) of $B^{\sim}$;

given by the equivalence of categories $\mathscr{F} \mapsto$ the fiber of $\mathscr{F}$ at the point (1,0) of P^1, whose stabilizer is $B^{\sim}$.

If to a section f of $\mathscr{F}$ one associates the function f(g) equal to the germ at (1,0) of $g^{-1}f$, then $\Gamma(\mathscr{F})$ is identified with the space of locally constant functions $f : G^{\sim} \to$ V satisfying f(bg) = $\rho(b)f(g)$. For V = C and the representation given by the character ut $\mapsto$ $\delta(t)$, one obtains the sheaf of locally constant measures, i.e. locally of the form $\|\omega\|$ for a differential form ω. Finally,

the space of functions 2.2.1 is interpreted as the space of locally constant densities of weight $1/2$ with values in the equivariant sheaf defined by V.

5.4. Let us spell out, for the group $G^{\sim}$, what the general theory of Jacquet modules gives. For a very clear account, see [1] (it discusses $\mathrm{GL}(n)$, but the arguments are general).

(A) On the category of algebraic representations of U, the coinvariant functor W $\mapsto$ W_U is exact.

This exactness makes it easy, up to an extension, to calculate $Ind(V,\rho)_U$. Write proj for the projection to coinvariants. As in 2.3, write $Ind(V,\rho) = H^0(P^1, \mathscr{F})$ for a suitable equivariant sheaf. The exact sequence

$$0 \to H_c^0(P^1 - (1,0), \mathscr{F}) \to H^0(P^1, \mathscr{F}) \to \mathscr{F}_{(1,0)} \to 0$$

is $B^{\sim}$-equivariant. Passing to coinvariants gives an analogous exact sequence. In the quotient, $\mathscr{F}_{(1,0)} \simeq \mathscr{F}_{(1,0),U}$, identified with V by proj f $\mapsto$ f(e). In the subobject, $H_c^0(P^1-(1,0),\mathscr{F})_U$ is again identified with V by integration over U (of which $P^1 - (1,0)$ is a principal homogeneous space). The functoriality will be made explicit below.

(B) The induction functor from algebraic representations of $T^{\sim}$ to those of $G^{\sim}$ has as left adjoint the functor r sending W to the coinvariant quotient W_U, with $T^{\sim}$ acting by

$$t \cdot proj(w) = \delta(t)^{-1/2} proj(tw).$$

After choosing w $\in N(T^{\sim}) - T^{\sim}$ and a Haar measure on U, calculation (A) gives an exact sequence of representations of $T^{\sim}$:

(5.4.1) $0 \to (V,\rho \circ int_w) \to$ r $Ind(V,\rho) \to (V,\rho) \to 0$,

where the projection to (V,ρ) is the adjunction map proj(f) $\mapsto$ f(e), and, when f(e) = 0, the image of proj(f) in $(V,\rho \circ int_w)$ is

(5.4.2) proj(f) $\mapsto \int$ f(wu)du (for f(e) = 0).

The condition f(e) = 0 ensures that the integrand has compact support.

(C) Let (V,ρ) be an irreducible admissible representation of $T^{\sim}$. Then every nonzero subquotient W of $Ind(V,\rho)$ satisfies rW $\neq 0$.

The idea of the proof (cf. [2], 2.9) is that if rW = 0, then W is projective, hence also a subrepresentation of $Ind(V,\rho)$, contradicting :

(D) For (V, ρ) as above, and any nonzero subrepresentation W $\subset$ $Ind(V, \rho)$, the map rW $\to$ (V, ρ) is onto. Otherwise the f $\in$ W would all be identically zero !

5.5. Let (V, ρ) be an irreducible admissible ε-representation of $T^{\sim}$, with central character (α_1, α_2) (4.3, 4.4).

If $\alpha_1 \neq \alpha_2$, the representations ρ and $\rho \circ int_w$ have distinct central characters, and the sequence 5.4.1 splits uniquely. The splitting

(5.5.1) r $Ind(V, \rho) \to (V, \rho \circ int_w)$

regularizes the integral 5.4.2 for f not necessarily satisfying f(e) = 0. For $\alpha_1(a) \neq \alpha_2(a)$, it is given by

(5.5.2) f $\mapsto$ $(1 - \alpha_1(a)^{-1}\alpha_2(a))^{-1} \int f_a(wu)du$,

with

$$f_a(g) = \text{f(g) - } \alpha_1(a)^{-1}\|a\|^{-1/2} f(gd_1(a))$$

(which vanishes at g = e).

By adjunction, (5.5.1) gives a morphism

(5.5.3) $I : Ind(V, \rho) \to Ind(V, \rho \circ int_w)$,

given by the intertwining integral f $\mapsto$ $\int f(wug)du$, regularized by (5.5.2). Note that 5.5.2 gives an unconditional definition (even when $\alpha_1 = \alpha_2$) of "$(1 - \alpha_1(a)^{-1}\alpha_2(a))I$". One checks :

Lemma 5.6. If $\alpha_1 = \alpha_2$ and a has nonzero valuation, then

(i) r $Ind(V, \rho)$ is an indecomposable representation of $T^{\sim}$;

(ii) the expression "$(1 - \alpha_1(a)^{-1}\alpha_2(a))I$" above is a nonzero scalar multiple of the identity.

5.7. Let (V, ρ) and (α_1, α_2) be as in 5.5. By 5.4(A),(C), the functor r embeds the lattice of subrepresentations of $Ind(V, \rho)$ into that of r $Ind(V, \rho)$; and by (D), if W is a proper subrepresentation, then rW defines a splitting of the exact sequence 5.4.1. Hence :

a) if $\alpha_1 = \alpha_2$, 5.6 gives the irreducibility of $Ind(V, \rho)$;

b) if $\alpha_1 \neq \alpha_2$, $Ind(V, \rho)$ has at most one proper subrepresentation, and is therefore indecomposable ;

c) if W is a proper subrepresentation of $Ind(V, \rho)$, then rW is the kernel of 5.5.1, and W lies in the kernel of the intertwining operator I. The $T^{\sim}$-morphism rI annihilates rW $\simeq (V, \rho)$, and its image is a splitting of (5.4.1) for $(V, \rho \circ int_w)$. Thus the image of I is a proper subrepresentation of $Ind(V, \rho \circ int_w)$.

If α is a character of F^*, write $\alpha \circ det$ for the character of $G^{\sim}$ given by $G^{\sim} \to \mathrm{GL}(2, F) \to^{det} F^*$. One has $Ind(V, \rho) \otimes (\alpha \circ det) = Ind(V, \rho \cdot \alpha \circ det)$, and the central character of $\rho \cdot \alpha \circ det$ is $(\alpha_1 \alpha, \alpha_2 \alpha)$. Such a twist does not change whether $Ind(V, \rho)$ is reducible or irreducible. If $\sigma = \alpha_1 \alpha_2^{-1}$ is unitary, such a twist can make (V, ρ), and hence $Ind(V, \rho)$, unitary. Altogether :

Proposition 5.8 (i) The representation $Ind(V, \rho)$ is indecomposable. If $\sigma = \alpha_1 \alpha_2^{-1}$ is unitary, it is irreducible.

(ii) If $\sigma \neq 1$, one of the following holds :

(a) $Ind(V, \rho)$ is irreducible, and I is an isomorphism ;

(b) $Ind(V, \rho)$ and $Ind(V, \rho \circ int_w)$ are reducible, and I induces an isomorphism from the unique nontrivial quotient of $Ind(V, \rho)$ to the unique nontrivial subrepresentation of $Ind(V, \rho \circ int_w)$.

5.9. The Jacquet module of a representation controls the asymptotic behavior of its matrix coefficients. It follows easily that, when $Ind(V, \rho)$ is reducible, exactly one of the two representations of which it is an extension has square-integrable coefficients (i.e. belongs to the discrete series). It will be called special. The irreducible representations $Ind(V, \rho)$, together with the nonspecial constituents of the reducible $Ind(V, \rho)$, form the principal series.

5.10. For Re $\sigma > 1$ (notation 0.0.7), the intertwining integral converges. Standard procedures make it possible to treat σ as a parameter ; then 5.5.2 gives the analytic continuation of this integral. An elegant way to proceed was pointed out to me by Bernstein. It consists in considering representations with coefficients not in C, but in an algebra such as $C[T, T^{-1}]$. Its advantage over ordinary analytic continuation is that it shows the intertwining integral I to depend rationally on σ, not merely meromorphically. The regularization formula 5.5.2 shows that, on putting $L(\sigma) = 1$ when $\sigma(x^n)$ is ramified, and $L(\sigma) = (1 - \sigma(\pi^n))^{-1}$ when σ is unramified and π is a uniformizer, the operator $I/L(\sigma)$ even depends polynomially on σ.

Iterating the intertwining integral gives an endomorphism of $Ind(V, \rho)$. When $Ind(V, \rho)$ is irreducible, this is multiplication by a nonzero scalar.

When $Ind(V, \rho)$ is reducible, the iterated operator is again a scalar (by analytic continuation) ; it is therefore 0.

5.10. C. Moen determined when $Ind(V, \rho)$ is reducible. I prefer to state his result in terms of the characters χ_i of F^*, defined by $\chi_i(x) = \alpha_i(x^n)$.

Theorem 5.11 (C. Moen). The representation $Ind(V, \rho)$ is irreducible except when

$$\chi_1 \chi_2^{-1} = \|x\|^{\pm 1}.$$

When $\chi_1 \chi_2^{-1} = \|x\|$, it is the subrepresentation of $Ind(V, \rho)$ that belongs to the discrete series. When $\chi_1 \chi_2^{-1} = \|x\|^{-1}$, it is the quotient representation.

5.12. This theorem determines the complementary series. If $Ind(V, \rho)$ is unitary, it is isomorphic to the contragredient of its complex conjugate, and either χ_1 and χ_2 are unitary (the principal series in the strict sense), or $\chi_2 = \overline{\chi_1}^{-1}$. A twist (5.7) reduces the latter case to $\chi_1 = \|x\|^s$ and $\chi_2 = \|x\|^{-s}$, with s $\in$ R. The corresponding induced representation π_s is real, with contragredient π_{-s}, and the intertwining operator $I/L(\sigma)$ supplies an invariant bilinear form on π_s. For s = 0 it is positive definite. By analytic continuation it remains so as long as $I/L(\sigma)$ is invertible, i.e. for $|s| < 1$. At s = 1 the two constituents of π_s are unitary (take s = 1 for one and s = -1 for the other). Beyond this, π_s cannot be unitary ; otherwise it would remain so as s $\to \infty$, which is impossible, since the coefficients are unbounded for large s.

5.13. Y. Flicker [3] reduced the classification of ε-representations of $G^\sim$ to the classification of representations of G. He establishes a bijection, characterized in terms of characters, between isomorphism classes of irreducible admissible ε-representations of $G^\sim$ and isomorphism classes of irreducible admissible representations π of G satisfying :

a) if π is in the principal series, $\pi = \pi(\chi_1, \chi_2)$, and χ_1 and χ_2 are trivial on $\mu_n(F) \subset F^*$;

b) if π is in the discrete series, its central character ω_π is trivial on $\mu_n(F) \subset F^*$.

If $\pi^\sim$ corresponds to π, the central character of $\pi^\sim$ (a character of F^{*n}) is $\omega_\pi(x^n)$. If $\pi^\sim$ is in the principal series defined by the representation of $T^\sim$ with central character (α_1, α_2), and $\chi_i = \alpha_i(x^n)$, then π is $\pi(\chi_1, \chi_2)$. In particular, if $(\chi_1, \chi_2) = (\|x\|^{1/2}, \|x\|^{-1/2})$, the irreducible quotient of $Ind(V, \rho)$ corresponds to the trivial representation of G.

5.14. Let ψ be a nontrivial character of U, and repeat the calculation in 5.7 of $(Ind(V,\rho))_U$ in order to calculate $(Ind(V,\rho)\otimes\overline{\psi})_U$. With the notation of 5.7, $\mathscr{F}_{(1,0)}$ contributes nothing, since the action of U there is trivial. Integration of the term $H_c^0(P^1-(1,0),\mathscr{F})$ again gives a copy of V :

Proposition 5.15. There exists a unique linear map

$$\varphi : Ind(V,\rho) \to \text{ V}$$

given by $\varphi(f)=\int f(wu)\psi(u)du$ for f(e) = 0 and satisfying $\varphi(uf)=\psi(u)\varphi(f)$. It identifies V with $(Ind(V,\rho)\otimes\overline{\psi})_U$.

If $\psi(u_1)\neq 1$, one also has

(5.15.1) $\varphi(f)=(1-\psi(u_1))^{-1}\int(f(wu)-f(wuu_1))\overline{\psi}(u)du$.

As in 5.10, this formula may be regarded as an analytic continuation in σ of the integral $\int f(wu)du$, which converges for Re $\sigma > 1$.

5.16 For a representation W of $G^{\sim}$, one has on $(\text{W}\otimes\overline{\psi})_U$, if not an action of $T^{\sim}$, at least an action of the centralizer $Z^{\sim}$ of ψ in $T^{\sim}$; define τ_ψ W to be the representation of $Z^{\sim}$ on $(\text{W}\otimes\overline{\psi})_U$ given by $z\cdot proj(f)=proj(z\cdot f)$. Since the character δ is trivial on Z, the morphism $\varphi : \tau_\psi\, Ind(V,\rho)\to$ V of 5.15 is a morphism of representations of $Z^{\sim}$.

5.17 As in 4.2, one checks that the irreducible ε-representations of $Z^{\sim}$ are classified by their central character. If n is odd (respectively, if n = 2m), the center of $Z^{\sim}$ is the inverse image of the subgroup F^{*n} (respectively F^{*m}) of the center F^* of G. As a representation of $Z^{\sim}$, V

(a) for odd n, is a sum of $[F^* : F^{*n}]^{1/2}$ irreducible representations, all isomorphic, with central character induced by ω_ρ ;

(b) for even n = 2m, has as its irreducible subrepresentations the irreducible representations of $Z^{\sim}$ whose central character extends ω_ρ (there are $[F^{*m} : F^{*n}]$ nonisomorphic ones). Each occurs with multiplicity $[F^* : F^{*m}]^{1/2}$.

5.18. To calculate the functor r_ψ on the subquotients of the principal series, it suffices, by 5.8, to calculate the morphism induced by the intertwining integral—

continued

(5.18.1) r_ψ I : $r_\psi\ Ind(V, \rho) \to\ r_\psi\ Ind(V, \rho \circ int_w)$

for ρ with central character ω_ρ defined by (χ_1, χ_2), where $\chi_1 \chi_2^{-1} = \|x\|$. By 2.14, this $Z^\sim$-morphism is identified with an endomorphism of V.

Theorem 5.19. Let ρ be a representation of $T^\sim$ with central character $(\alpha_1, \alpha_2) = (\alpha \cdot \|x\|^{1/n}, \alpha)$. Putting $\chi_i = \alpha_i(x^n)$, one has $(\chi_1, \chi_2) = (\chi \cdot \|x\|, \chi)$. Let W be the irreducible quotient of $Ind(\rho)$. In the tame case, for n odd, the representation $(W \otimes \overline{\psi})_U$ of $Z^\sim$ is a sum of $(n-1)/2$ mutually isomorphic representations.

When n = 3, $(n-1)/2 = 1$; this uniqueness result explains the “multiplicative” appearance of the statements in the Introduction.

6. Models, and proofs.

6.1 Let α_1 and α_2 be two characters of F^{*n}. We shall describe an irreducible ε-representation ρ of $T^{\sim}$ with central character (α_1, α_2), and describe the induced representation $Ind(\rho)$ as a space of functions on the cover P of $F^2 - 0$. It is the space of locally constant ε^{-1}-functions $(f(\zeta p) = \varepsilon(\zeta)^{-1} f(p))$ having a suitable homogeneity with respect to the "homotheties" v $\mapsto$ λv for $\lambda \in F^{*n}$ (3.8).

Choose an extension, still denoted α_2, of α_2 to F^*, and describe ρ by 4.4(a). The representation $Ind(\rho)$ is then obtained by induction in stages : from $d_1(F^{*n})d_2(F^*)\mu$ to $T^{\sim}$, and then from $B^{\sim}$ to $G^{\sim}$, starting with (α_1, α_2). It is the space of ε-functions f on $G^{\sim}$ such that

$$\text{f(u } d_1(a)d_2(b)g) = \alpha_1(a)\alpha_2(b)\|a/b\|^{1/2} f(g)$$

for u $\in$ U, a $\in$ F^{*n}, and b $\in$ F^*. If the character $\alpha_2(b)\|b\|^{-1/2}$ is trivial, this is identified with a space of functions on $d_2(F^*) \cdot U \backslash \mathrm{GL}(2, F)^{\sim}$. This quotient is precisely the cover P of $F^2 - 0$: associate to g $\in$ $\mathrm{GL}(2, F)^{\sim}$ the point $g^{-1}((1, 0), e_X)$. Reducing to this case by twisting by a determinant character,

det : $\mathrm{GL}(2, F)^{\sim} \to \mathrm{GL}(2, F) \to F^*$,

one obtains the following model.

a) The representation space consists of locally constant ε^{-1}-functions on P with homogeneity $(\alpha_2 - \alpha_1 - 1)$:

$$f(\lambda v) = (\alpha_2 - \alpha_1 - 1)(\lambda) f(v) \text{ for } \lambda \in F^{*n}, \text{ in the notation of 0.0.7.}$$

b) The group acts by

$$\text{(g*f)(v)} = f(g^{-1}v) \cdot (\alpha_2 - 1/2)(det\ \text{g}).$$

6.2 Another model of the same representation is obtained from the dual plane. Choose this time an extension of α_1 to F^*, and take

a) as space, the locally constant ε^{-1}-functions f on $P^{\vee}$ satisfying

$$f(\lambda v) = (\alpha_2 - \alpha_1 - 1)(\lambda) f(v) \text{ for } \lambda \in F^{*n} \text{ (0.7,3.9)};$$

b) as action,

$$(g * \varphi)(v) = \varphi(g^{-1}v)(\alpha_1 + 1/2)(det\ \text{g}).$$

6.3 In these models, the intertwining operator from the induced representation with parameters (α_1, α_2) to that with parameters (α_2, α_1) is a Radon transform. To f on P, of homogeneity $(\alpha_2 - \alpha_1 - 1)$, it associates Rf on $P^\vee$, of homogeneity $(\alpha_1 - \alpha_2 - 1)$. The operator R is as follows. A point $\ell \in P^\vee$ determines

a) a point $\overline{\ell}$ in $A^{2\vee} - 0$, and a line $\overline{\ell}(\overline{v}) = 1$ in A^2;

b) a trivialization of P over this line. We write $< \ell, v > = 1$ for the relation that $\overline{\ell}(\overline{v}) = 1$ and v lies in the section of P over the line $\overline{\ell}(\overline{v}) = 1$ determined by ℓ.

Set

$$(Rf)(\ell) = \int_{<\ell, v>=1} \mathrm{f(v)}\ (dxdy/d\ell).$$

This formula presupposes the choice of a Haar measure on F. With a pardonable abuse of notation, it may also be written

$$(Rf)(\ell) = \iint \mathrm{f(v)}\ \delta(<\ell, v> - 1)\ \mathrm{dxdy}.$$

The same formula defines the Radon transform from functions on $P^\vee$ to functions on P; this is again a model of the intertwining operator.

The regularization (5.5.2) of the integral is the usual "finite part," except when $\alpha_1 = \alpha_2$, the case corresponding to the pole of the intertwining operator.

6.4 Let us determine the map Φ of 5.15. Write f(x,y;p) for a function on P, where $\mathrm{p} \in$ P lies above (x,y). For an ε^{-1}-function f on P, of homogeneity $(\alpha_2 - \alpha_1 - 1)$, as in 6.2, put

$$\mathrm{L(f;y)} = \int f(x, y; e_y)\ \psi(x/y)\ \mathrm{dx}.$$

For large x, and for λ in a suitable open subgroup of O^{*n}, one has $f(x, y; e_y) = f(\lambda x, y; e_y)$. This permits the regularization of the integral as in 5.15.1. One has

(6.4.1) $L(e^+(u)f; y) = \psi(u)L(f; y)$, and

(6.4.2) $L(f; \lambda y) = (\alpha_2 - \alpha_1)(\lambda)L(f; y)$ for $\lambda \in F^{*n}$,

(6.4.3) $L(d_1(z)d_2(z)f; y) = \|z\|\varepsilon(z, y)^{-1}L(f; z^{-1}y)$, for $\mathrm{z} \in F^*$.

The space of functionals satisfying 6.4.1 has as a basis the L(f;y), with y running over F^*/F^{*n}. These are the coordinates of Φ (5.15) in a suitable basis.

6.5 **In the dual-plane model, the coordinates of Φ are likewise**

$$\mathrm{L(f;a)} = \int f(a, b; e_a)\ \overline{\psi}(b/a)\ \mathrm{db}.$$

One has

(6.5.1) $L(e^{+}(u)f;a) = \psi(u)L(f;a)$,

(6.5.2) $L(f;\lambda a) = (\alpha_2 - \alpha_1)(\lambda)L(f;a)$ for $\lambda \in F^{*n}$,

(6.5.3) $L(d_1(z)d_2(z)f;a) = \|z\|\varepsilon(z,z^{-1}a)^{-1}L(f;z^{-1}a)$.

The action in 5.18 of the intertwining operator is given by

(6.5.4) $\mathrm{L(Rf;a)} = \int \varepsilon(a,y)\,\overline{\psi}(1/ay)L(f;y)d^{*}y$,

$$= \sum_{y \in F^{*}/F^{*n}} L(f;y)\varepsilon(a,y)$$

$$\cdot \int_{F^{*n}} \overline{\psi}(\lambda/ay)(\alpha_1 - \alpha_2)(\lambda)d^{*}\lambda,$$

where the integral is defined by analytic continuation in $(\alpha_1 - \alpha_2)$. There is a pole at $\alpha_1 = \alpha_2$, as required.

6.6 **By** 5.18 **this reduces the proof of** 5.19 **to the following problem. Consider**

V' : the space of locally constant functions L(y) on F^{*} satisfying (6.4.2), on which Z^{-} acts by the right-hand side of 6.4.3;

V" : the space of locally constant functions L(a) on F^{*} satisfying (6.5.2), on which Z^{-} acts by the right-hand side of 6.5.3;

R : V' → V", given by 6.5.4 :

$$\mathrm{(RL)(a)} = \sum_{y \in F^{*}/F^{*n}} L(y)\varepsilon(a,y)$$

$$\cdot \int_{F^{*}} \overline{\psi}(\lambda/ay)(\alpha_1 - \alpha_2)(\lambda)d^{*}\lambda\,;$$

and seek the representation Im R of Z^{-} when R is not an isomorphism, i.e. when the character $(\alpha_1 - \alpha_2)$ is $\|\lambda\|^{\pm 1/n}$.

6.7 Under the hypotheses of 5.19, it is enough to restrict R to the invariants of $d_1(a)d_2(a)|a \in O^{*}$. Using as bases the functions L(y) or L(a) supported on a coset of $O^{*} \cdot F^{*n}$, R is represented by a matrix that decomposes along the diagonal into one 1×1 block and $(n-1)/2$ blocks of size 2×2 (in a suitable ordering). The calculation is then easy.

7. Eisenstein Series.

7.1 The "Eisenstein-series" operator E is an intertwining operator from

$$Ind_{B^+(A)^-}^{G(A)^-}(V_{\alpha_1,\alpha_2})$$

into the space of functions on $G(F)\backslash G(A)^-$. The space of the induced representation is identified with the space of ε-functions on $U(A)T(F)\backslash G(A)^-$, right-invariant under an open compact subgroup of $G(A^f)^-$ and K_∞-finite, and satisfying

$$f(diag^-(a_1,a_2)g) = \alpha_1(a_1)\alpha_2(a_2)\|a_1a_2^{-1}\|^{1/2}\ \mathrm{f(g)}$$
$$(a_1,a_2 \in A^{*n}).$$

The operator is

$$\sum_{\gamma \in B(F)\backslash G(F)} f(\gamma g),$$

when this converges, and in general is obtained by analytic continuation in α_1, α_2. Put $\chi(x) = \alpha_1(x^n)\alpha_2(x^n)^{-1}$. Classical arguments ([11]) produce a pole at $\chi = \|x\|$. If $\bar{\mathrm{E}}$ is the residue of E, $\bar{\mathrm{E}}$ intertwines the irreducible quotient of $Ind_{B^+(A)^-}^{G(A)^-}(V_{\alpha_1,\alpha_2})$, for $\chi = \|x\|$, into the space of functions on $G(F)\backslash G(A)^-$.

7.2 This determines the isomorphism class of the representation of $G(A)^-$ that is the image of $\bar{\mathrm{E}}$. Let ψ be an additive character of A/F. The "Fourier coefficient" operator

$\mathrm{f} \mapsto \mathrm{W(f;g)} = \int_{U(F)\backslash U(A)} \mathrm{f(ug)}\ \overline{\psi}(u)\ \mathrm{du}$

commutes with right translations and satisfies $\mathrm{W(f;ug)} = \psi(u)W(f;g)$. One also has $\mathrm{W(f;zg)} = \mathrm{W(f;g)}$ for $\mathrm{z} \in \mathrm{Z(F)}$.

7.3 For n = 3, these properties and the uniqueness results of 5.19 should make it possible to identify it - and to obtain the results announced in the Introduction. Another method consists in changing tack and systematically replacing $\mathrm{GL}(2,F)$, $\mathrm{GL}(2,F_v)$, or $\mathrm{GL}(2,A)$ by the subgroup of elements whose determinant is a cube. For this new group, the inverse image of the center is commutative, which simplifies the uniqueness results : the spaces $V_{U,\psi}$, previously irreducible representations of the inverse image of the center, become one-dimensional. Put

$f^0(g) = \int_{U(F)\backslash U(A)} \mathrm{f(ug)\ du}$ (constant term).

One also has

$$\mathrm{f(g)} = f^0(g) + \sum_{\lambda \in F^*} \mathrm{W(f;\ diag}(\lambda^2, \lambda)g).$$

If f, as a vector in the representation of $G(A)^{-}$ in the image of $\bar{\mathrm{E}}$, is decomposable, the function on $A^{f *\sim}$, $W(f; d_1(a^2) d_2(a))$, is multiplicative in the sense of the Introduction, from which the results follow.

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THE FUNDAMENTAL GROUP OF THE COMPLEMENT OF A PLANE CURVE HAVING ONLY ORDINARY DOUBLE POINTS IS ABELIAN

[after W. FULTON]

by Pierre DELIGNE

Introduction

THEOREM 1.- Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a plane curve whose only singularities are double points with distinct tangents. Then the fundamental group of $\mathbb{P}^2(\mathbb{C})$ - C is abelian.

This theorem was stated by Zariski in 1929 ([8]). See also [9], Ch. VIII. His proof used Severi's theorem, according to which the space of irreducible plane curves of a given degree having a given number of double points is irreducible (Vorlesungen über algebraische Geometrie, 1921, Anhang F). It is still unknown whether Severi's assertion is true or false.

The π_1 considered in the theorem is the genuine fundamental group, the topologists' one. If $\mathbb{C}$ is replaced by an arbitrary algebraically closed field k of characteristic 0, the same statement holds for the algebraic fundamental group, the one classifying étale coverings: the Lefschetz principle reduces us to assuming $k = \mathbb{C}$, in which case the algebraic fundamental group is the profinite completion of the genuine fundamental group. In arbitrary characteristic one has the variant:

THEOREM 2 (Fulton [3]).- Let k be an algebraically closed field and, over k, let $C \subset \mathbb{P}^2$ be a plane curve whose only singularities are double points with distinct tangents. Then the tame fundamental group of $\mathbb{P}^2$ - C is abelian.

In other words, coverings (in the sense of 0.2) of $\mathbb{P}^2$, ramified only along C and tamely ramified, are abelian. Recall that tameness of the ramification is tested after base change from $\mathbb{P}^2$ to its localizations at the generic points of the irreducible components of C. These localizations are traits (spectra of discrete valuation rings).

Suppose k has characteristic $p > 0$. If C is as in Theorem 2, there always exists in relative projective space $\mathbb{P}^2$ over Spec(W(k)) a relative normal-crossings divisor $\widetilde{C}$ whose special fiber is C (see, for example, Mumford, Appendix to Ch. VIII of [9]). This makes it possible to deduce Theorem 2 from Theorem 1 by a specialization argument.

Once one knows that $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C) is abelian, homological arguments immediately give its structure: if C is the union of irreducible curves of degrees d_1, ..., d_r, then $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C) is the quotient of $\mathbb{Z}^r$ by the subgroup generated by the element (d_1, ..., d_r).

In his proof of Theorem 2, Fulton uses, on the one hand, methods introduced by Abhyankar [1] to prove the special case of Theorem 2 in which the irreducible components of C are assumed smooth - methods very clearly presented here by Serre ([7]) - and, on the other hand, Theorem 3 below. An analysis of the proof then enabled the expositor to give a variant yielding Theorem 1.

Abhyankar and Fulton both use instances of the principle that, in algebraic geometry, what is very mobile is connected. Let C $\subset \mathbb{P}^2$ be as in Theorem 2, let D be an irreducible component of C, and let f : X $\to \mathbb{P}^2$ be a covering ramified only along C and tamely ramified. Since D $\subset \mathbb{P}^2$ is very mobile, f^{-1}(D) is connected. More precisely, the divisor D $\subset \mathbb{P}^2$ is ample, hence the divisor f^{-1}(D) $\subset$ X is also ample because f is finite, and one applies Bertini. By a local study of f, Abhyankar shows that if D is smooth then (f^{-1}(D))red is smooth as well, and concludes that f^{-1}(D) is irreducible.

In the general case, let g : $\widetilde{D} \to$ D be the normalization of D. The same local study shows that the reduced fiber product (X $\times_{\mathbb{P}^2} \widetilde{D}$)red is smooth. It is therefore the normalization of f^{-1}(D). If it can be shown to be connected, it follows that f^{-1}(D) is irreducible, and one can repeat the end of Abhyankar's proof as presented in [7].

Recall the definition of the fiber product: if h : X $\times \widetilde{D} \to \mathbb{P}^2 \times \mathbb{P}^2$ is the product of the maps f : X $\to \mathbb{P}^2$ and g : $\widetilde{D} \to$ D $\subset \mathbb{P}^2$, then the fiber product X $\times_{\mathbb{P}^2} \widetilde{D}$ is the inverse image $h^{-1}(\Delta)$ of the diagonal. Since $\mathbb{P}^2$ has many automorphisms, the diagonal $\Delta \subset \mathbb{P}^2 \times \mathbb{P}^2$ is very mobile, and $h^{-1}(\Delta)$ is connected. This completes the proof. More precisely, one applies to h the special case r = 2, m = 2 of the following theorem:

THEOREM 3 (W. Fulton and J. Hansen [4]).- Let P = $(\mathbb{P}^m)^r$ be the product of r copies of projective space $\mathbb{P}^m$, let δ be the diagonal map from $\mathbb{P}^m$ to P, let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be a complete irreducible variety, let h : Z $\to$ P, let n = dim h(Z), and let c = (r - 1)m be the codimension of Δ in P. If n - c $\geq$ 1, then $h^{-1}(\Delta)$ is connected.

This theorem has many other applications, for which I refer to [4] and to the article of T. Gaffney and R. Lazarsfeld [5].

At the end of this introduction, we explain by an example the method used to transpose Fulton's proof of Theorem 2 into a proof

of Theorem 1. The example is this: how to transpose Abhyankar's arguments to prove Theorem 1 in the special case where C is smooth (hence irreducible).

Let U be a tubular neighborhood of C in $\mathbb{P}^2(\mathbb{C})$, let U* = U - C, and let u ∈ U*. For every covering f : X → $\mathbb{P}^2(\mathbb{C})$ ramified only along C, one has

(f^{-1}(C) connected) ⇔ (f^{-1}(U) connected) ⇔ (f^{-1}(U*) connected).

The key result - that for every covering X the curve f^{-1}(C) is connected - is therefore equivalent to the following: for every finite quotient H of $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u), the composite map

π_1(U*,u) —j*→ $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u) → H

is surjective. A Morse-theoretic argument proves more: the map j* : π_1(U*,u) → $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u) is surjective.

After choosing a Riemannian metric on $\mathbb{P}^2(\mathbb{C})$, for $\varepsilon > 0$ sufficiently small one may take U to be the image under the exponential map exp of the subbundle of balls of radius ε in the normal bundle of C in $\mathbb{P}^2(\mathbb{C})$. For ε sufficiently small, exp identifies U* with an oriented punctured-disk bundle over C, giving an exact sequence

$\mathbb{Z} \to \pi_1$(U*,u) → π_1(C,$\bar{u}$) → 1

($\bar{u}$ is the image of u). The image of $\mathbb{Z}$ is central; the image of 1 ∈ $\mathbb{Z}$ is represented by a small loop γ going once around C in the positive direction.

Since j* is surjective, γ also defines a central element of $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u). The quotient of $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u) by the normal subgroup generated by γ (in this case, the cyclic subgroup generated by γ) is $\pi_1(\mathbb{P}^2(\mathbb{C})$,u) = {e}, and the special case of the theorem under consideration follows.

0. Terminology

0.1. Connected means connected and nonempty. The same convention applies to irreducible.

0.2. A covering of a scheme S, assumed normal and connected, is a morphism f : T → S whose source T is normal and connected and whose restriction to a dense open subset of T is étale.

0.3. Throughout the rest of this exposition we work over $\mathbb{C}$. We shall write simply $\mathbb{P}^n$ for $\mathbb{P}^n(\mathbb{C})$. All schemes considered are assumed separated.

1. Theorems of Lefschetz and Bertini type

The special case of 1.1 below in which Z $\subset \mathbb{P}^n$ is the complement of a hypersurface is proved by Hamm and Lê [6], using Morse theory. Our aim in this section is to deduce from it the variant 1.6 of Theorem 3.

Assertion 1.1.- Let Z $\subset \mathbb{P}^n$ be a nonempty Zariski-open subset. If H is a general hyperplane, the natural map π_i(Z $\cap$ H) $\to \pi_i$(Z) is bijective for i < n - 1 and surjective for i = n - 1.

Assertion 1.1 is stated in [2], but Cheniot's proof covers only the case in which the complement of Z has codimension $\geq$ 2. Under this hypothesis, Z and Z $\cap$ H are simply connected, and it is harmless to work, as he does, with the H_i rather than the π_i. The case i = 1 can also be deduced from [6] or [2 bis].

Recall that the adjective "general," qualifying an element H of an irreducible algebraic variety S (here the dual projective space $\mathbb{P}^n\vee$), is a quantifier. It means that there exists a nonempty (hence dense) Zariski-open subset S' of S such that for every H in S', ...

1.2.- Let Y $\subset \mathbb{P}^n \times \mathbb{P}^n\vee$ be the space of pairs (z,H) such that z $\in$ Z and z $\in$ H. The isotopy theorem, applied to the second projection pr_2 : Y $\to \mathbb{P}^n\vee$, gives a nonempty Zariski-open subset U of $\mathbb{P}^n\vee$ over which Y is a topologically locally trivial fiber space. It is enough to take U to be the set of hyperplanes transverse to the strata of an algebraic Whithney stratification of the complement of Z in $\mathbb{P}^n$. Locally over U, after choosing a local trivialization of this fiber space, the inclusion Z $\cap$ H = $\mathrm{pr}_2^{-1}(\{\mathrm{H}\})$ into Z appears as a continuously H-dependent map from a fixed space into Z. Thus the conclusion of 1.1 can hold for one H in U only if it holds for all of them, and 1.1 is equivalent to saying that it holds for every H in U.

Suppose that Z depends on parameters: take an irreducible scheme S and an open subset Z of S $\times \mathbb{P}^n$; set $\mathrm{pr}_2\ \mathrm{pr}_1^{-1}(\{\mathrm{s}\})$ = Z(s), and assume every Z(s) is nonempty. Let Y $\subset$ S $\times \mathbb{P}^n \times \mathbb{P}^n\vee$ be the space of triples (s,z,H) such that z $\in$ Z(s) and z $\in$ H. The isotopy theorem, applied to the projection of Y onto S $\times \mathbb{P}^n\vee$ (respectively, of Z onto S), gives a nonempty Zariski-open subset U of S $\times \mathbb{P}^n\vee$ (respectively V of S) over which Y (respectively Z) is a topologically locally trivial fiber space.

As above, if s $\in$ V and (s,H) $\in$ U, the conclusion of 1.1 holds for Z(s) and H. This parameterized variant of 1.1 will be used implicitly in the proof of 1.4.

I conjecture more generally:

Conjecture 1.3.- Let Z be smooth and connected of dimension n, let f : Z $\to \mathbb{P}^N$ be a morphism with finite fibers, let c be an integer, and let L be a linear subspace of $\mathbb{P}^N$ of

codimension c, with n* = n - c. Choose a Riemannian metric on $\mathbb{P}^N$, and let V(ε) be the set of points p of $\mathbb{P}^N$ such that d(p,L) < ε. Then, for ε sufficiently small, the natural map

$\pi_i(f^{-1}(V(\varepsilon))) \to \pi_i(Z)$

is bijective for i < n* and surjective for i = n*.

For general L, the inclusion of f^{-1}(L) into f^{-1}(V(ε)) is, for ε sufficiently small, a homotopy equivalence: 1.3 is indeed a generalization of 1.1.

If f is no longer assumed quasi-finite, and if the set of points of $\mathbb{P}^N$ where the fiber of f has dimension k has dimension φ(k), then n* should be redefined as

n* = 2 dim Z - sup(2k + φ(k) + inf(φ(k), c - 1)) - 1.

Conjecture 1.3 should moreover follow from a more precise statement saying that Z is obtained from f^{-1}(V(ε)) by attaching handles that kill spheres of dimension $\geq$ n*.

We shall be content with weaker results than 1.3. We deduce them from the special case of 1.1 in which Z is the complement of a hypersurface.

Lemma 1.4.- Let Z be normal and connected, let f : Z $\to$ $\mathbb{P}^N$ be a morphism, let n be the dimension of f(Z), and let c be an integer. Assume n - c $\geq$ 1. If L is a general linear subspace of $\mathbb{P}^N$ of codimension c, then f^{-1}(L) is connected and, for z $\in$ f^{-1}(L), the natural map from $\pi_1(f^{-1}(L),z)$ to π_1(Z,z) is surjective.

One may replace Z by a nonempty Zariski-open subset Z': for general L, f^{-1}(L) is still normal, and Z' (respectively Z' $\cap$ f^{-1}(L)) is obtained from Z (respectively f^{-1}(L)) by removing an algebraic subset with empty interior. By normality, this does not change π_0, can only enlarge π_1, and one considers the commutative diagram

$\pi_1(Z' \cap f^{-1}(L),z) \twoheadrightarrow \pi_1(f^{-1}(L),z)$
$\downarrow \downarrow$
$\pi_1(Z',z) \twoheadrightarrow \pi_1(Z,z)$.

In particular, we may and do assume that Z is smooth. A reader interested only in the proof of Theorem 1 could in fact have assumed this from the outset; throughout the remainder of the section, that reader may replace “normal” by “smooth.”

After shrinking Z, we may and do assume that f(Z) is locally closed and that Z is a topologically locally trivial fiber space over f(Z).

If c = 1 and f is dominant, then f(Z) is a Zariski-open subset of $\mathbb{P}^N$ - after shrinking Z, we may assume it is the complement of a hypersurface - and it remains only to apply 1.1 to f(Z) and compare the long exact homotopy sequences for Z $\to$ f(Z) and f^{-1}(L) $\to$ L $\cap$ f(Z).

Continue to assume c = 1. If f is not dominant, let A be a

linear subspace of dimension N - n - 1, let B be the projective space of linear subspaces of dimension N - n containing A, and let q : $\mathbb{P}^N$ - A $\to$ B send x to the subspace containing it. For A disjoint from $\overline{f(Z)}$, the restriction of q to $\overline{f(Z)}$ is a finite morphism, hence dominant because dim $\overline{f(Z)}$ = dim B, and qf : Z $\to$ B is dominant. Applying the lemma to qf with c = 1 gives the conclusion of the lemma for a general hyperplane L among those containing A. Since A is only required to lie in a dense open subset of a suitable Grassmannian, the conclusion of the lemma also holds for general L (cf. 1.2).

The proof is completed by induction on c: a general linear subspace of codimension c - 1 in a general hyperplane is a general linear subspace of codimension c (cf. 1.2).

Lemma 1.5.- Let Z, f, n, and c be as in 1.4. Every linear subspace L of $\mathbb{P}^N$ of codimension c has a fundamental system of open neighborhoods V such that f^{-1}(V) is connected and, for z $\in$ f^{-1}(V), the natural map from $\pi_1(f^{-1}(V),z)$ to π_1(Z,z) is surjective.

Let Gr be the Grassmannian of linear subspaces of $\mathbb{P}^N$ of codimension c, let Y $\subset$ Z $\times$ Gr be the set of pairs (z,L) such that f(z) $\in$ L, and let G be a nonempty Zariski-open subset of Gr such that over G, Y is a topologically locally trivial fiber space. The conclusion of 1.4 then holds for L $\in$ G (cf. 1.2).

For every open subset U of Gr, write U* for the open subset of $\mathbb{P}^N$ that is the union of the L $\in$ U. We shall prove that if U is connected, the conclusion of 1.5 holds for V = U*. The lemma will follow.

Let us prove that if U is connected, f^{-1}(U*) is connected as well. The intersection U' = U $\cap$ G is connected, since it is obtained from U by removing a triangulable subset of real codimension $\geq$ 2. By 1.4, the union f^{-1}(U'*) of the f^{-1}(L) for L $\in$ U' is therefore connected. We shall show that it is dense in f^{-1}(U*).

The image f(Z) of Z is not contained in the complement F of G*: since n - c $\geq$ 0, the L $\in$ G meet f(Z). If z $\in$ Z, the image of any neighborhood of z is not contained in F; otherwise the image of Z would be contained in F by analytic continuation. If z $\in$ f^{-1}(U*), there are therefore points z' arbitrarily near z such that f(z') $\in$ G*. In the set of L $\in$ Gr containing z', the trace of G is dense, since it is nonempty. If z $\in$ f^{-1}(L), with L $\in$ U, this makes it possible to find z' as above lying in some f^{-1}(L'), with L' in G arbitrarily close to L. In particular one may take L' $\in$ U', in which case z' $\in$ f^{-1}(U'*). This proves the density of f^{-1}(U'*) in f^{-1}(U*) and the connectedness of f^{-1}(U*).

Finally let L $\in$ U'. We have f^{-1}(L) $\subset$ f^{-1}(U*) $\subset$ Z. By 1.4, $\pi_1(f^{-1}(L))$ maps onto π_1(Z). A fortiori the same is true for $\pi_1(f^{-1}(U^*))$.

We shall deduce from this a variant of Theorem 3.

THEOREM 1.6.- Let P = $(\mathbb{P}^m)^r$ be the product of r copies of projective space $\mathbb{P}^m$, let δ be the diagonal map from $\mathbb{P}^m$ to P, let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be normal and connected, let h : Z $\to$ P, let n = dim h(Z), and let c = (r - 1)m be the codimension of Δ in P. If n - c $\geq$ 1, then Δ has a fundamental system of open neighborhoods V such that h^{-1}(V) is connected. If h^{-1}(V) is connected and z $\in$ h^{-1}(V), the natural map from $\pi_1(h^{-1}(V),z)$ to π_1(V,z) is surjective.

As in 1.4, one may shrink Z, and in particular assume that Z is smooth. We do so. Let $(H_\alpha)_0 \leq \alpha \leq$m be a family of hyperplanes in $\mathbb{P}^m$ such that

a) the intersection of the H_α is empty: in suitable projective coordinates (X_α), they are given by $X_\alpha = 0$;

b) h(Z) is not contained in any pr_$i^{-1}(H_\alpha)$. After shrinking Z, we may and do assume that $\overline{h(Z)}$ is disjoint from the pr_$i^{-1}(H_\alpha)$;

c) $\overline{h(Z)} \cap \Delta$ is not contained in the union of the $\delta(H_\alpha)$: since Δ belongs to a connected family of cycles filling all of P, one has $\overline{h(Z)} \cap \Delta \neq \emptyset$, and it is possible to choose H_α satisfying c).

Let $A_\alpha = \mathbb{P}^m$ - H_α, let $B_\alpha = A_\alpha^r \subset$ P, and let $\Delta_\alpha = \delta(A_\alpha)$ be the diagonal of B_α. The spaces A_α and B_α are affine spaces and Δ_α is an affine subspace of B_α. The projective completion of A_α is $\mathbb{P}^m$. Denote by $\bar{B}$_α the projective completion of B_α, and by Δ_$\alpha \subset \bar{B}$_α the completion of Δ_α.

Fix α, and write simply H, A, B, Δ_A, $\bar{B}$, Δ_A for H_α, A_α, B_α, Δ_α, $\bar{B}$_α, and Δ_α.

Lemma 1.7.- For every neighborhood U of $\Delta \subset$ P, there exists a neighborhood V of Δ_A in $\bar{B}$ such that V $\cap$ B $\subset$ U $\cap$ B.

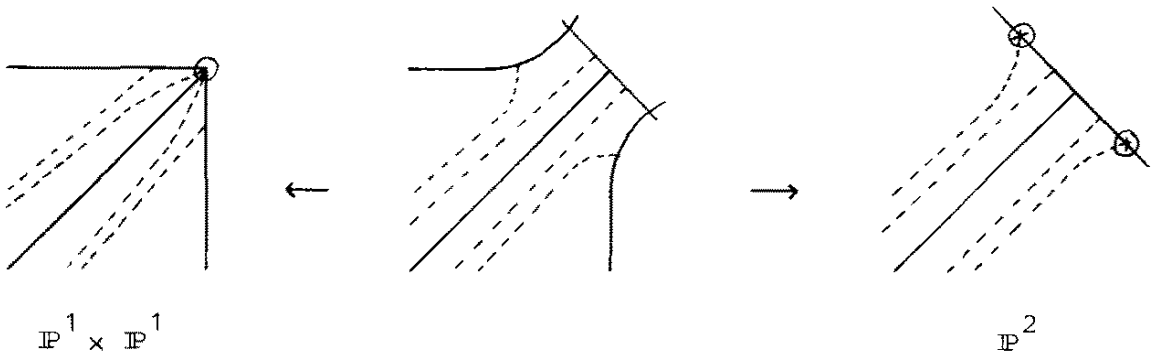
Choose on A the structure of a Hermitian vector space compatible with its affine structure. A fundamental system of neighborhoods U of Δ (respectively V of Δ_A) then has traces on B of the following form:

U $\cap$ B = $\{(x_i)$ | If $|x_i| \leq$ R for some i, then $\forall$j $\forall$k $|x_j - x_k| < \varepsilon$; if the $|x_i|$ are $\geq$ R, then for all j,k the angle between the complex lines generated by x_j and x_k is $< \varepsilon\}$

V $\cap$ B = $\{(x_i)$ | If $|x_i| \leq$ R for some i, then $\forall$j $\forall$k $|x_j - x_k| < \varepsilon$; if the $|x_i|$ are $\geq$ R, then $\forall$j $\forall$k $|x_j - x_k| \leq \varepsilon$ inf$\{|x_i|\}\}$

The lemma follows.

When r = m = 1, the two compactifications $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$ of $\mathbb{A}^2$ are topped by a third as follows: the drawing represents the locus at infinity and the diagonal; the circled points are those that must be blown up to obtain the third compactification; U and V are indicated by the dotted lines:



A similar construction can be made for arbitrary r and m.

Return to the proof of 1.6. Apply 1.5 to h : Z $\to \bar{B}_\alpha$ and to Δ_α. It follows that Δ_α has an arbitrarily small open neighborhood V_α such that $h^{-1}(V_\alpha) = h^{-1}(V_\alpha \cap B_\alpha)$ is connected and its π_1 maps onto that of Z. By 1.7, whatever the neighborhood U of Δ in P, V_α can be chosen so that $V_\alpha \cap B_\alpha \subset$ U.

The union V of the open subsets $V_\alpha \cap B_\alpha$ of P is an arbitrarily small neighborhood of Δ. Hypothesis b) on the H_α ensures that the intersection of the $V_\alpha \cap B_\alpha$ meets h(Z). Hence h^{-1}(V), a union of connected open subsets $h^{-1}(V_\alpha)$ with nonempty common intersection, is connected. Since the π_1 of one of the $h^{-1}(V_\alpha)$ maps onto that of Z, a fortiori the same is true for h^{-1}(V), and this completes the proof.

2. Proof of Theorem 1

2.1.- Let C $\subset \mathbb{P}^2$ be a reduced plane curve, let D be an irreducible component of C, let C' = $\overline{(C - D)}$ be the union of the other components, let Sing(C) be the set of singular points of C, and let D* = D - Sing(C). Choose a base point O $\in \mathbb{P}^2$ - C, and let γ be a loop in $\mathbb{P}^2$ - C of the following type: γ goes from O along a path ch to a point p' near a point p of D*, goes once around D* in the positive direction, and returns to O along ch$\hat{}${-1}. Since D* is connected, all loops of this type are conjugate. The quotient of $\pi_1(\mathbb{P}^2$ - C,O) by the normal subgroup they generate is $\pi_1(\mathbb{P}^2$ - C').

PROPOSITION 2.2.- Suppose that the singular points of C lying on D are double points with distinct tangents. Then γ is central in $\pi_1(\mathbb{P}^2$ - C,O).

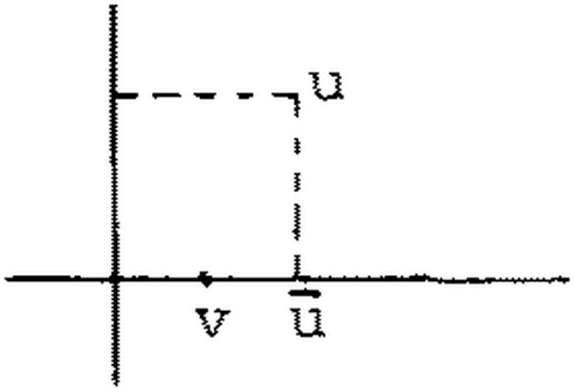
Let j be the inclusion of $\mathbb{P}^2$ - C into $\mathbb{P}^2$, let g : $\widetilde{D} \to$ D be the normalization of D, and apply 1.6 to

h = j $\times$ g : ($\mathbb{P}^2$ - C) $\times \widetilde{D} \to \mathbb{P}^2 \times \mathbb{P}^2$.

To describe conveniently the inverse images of certain neighborhoods of the diagonal, choose on $\mathbb{P}^2$ a Riemannian structure such that, near each

point P ∈ D ∩ Sing(C), $\mathbb{P}^2$ is Euclidean and the two branches of C through P are orthogonal real 2-planes.

If (u,v) ∈ ($\mathbb{P}^2$ - C) × $\widetilde{D}$ lies in the inverse image of a sufficiently small neighborhood of the diagonal, that is, if u and v are sufficiently close, then u is near D and one can drop a short perpendicular from u to D. When u is near Sing(D), there are even two such perpendiculars, one for each branch. Choose the one whose foot $\bar{u}$ is close to v on $\widetilde{D}$:



The point $\bar{u}$ lies in D*.

Conversely, let $\varepsilon > 0$ be sufficiently small, and let q : U* → D* be the punctured-disk subbundle of the normal bundle of D* in $\mathbb{P}^2$ consisting of nonzero normal vectors of length $< \varepsilon$. Let E be the set of pairs (x,v) with x ∈ U*, v ∈ $\widetilde{D}$, and d(q(x),v) $< \varepsilon$ (distance measured on $\widetilde{D}$). The map pr_1 : E → U* is a homotopy equivalence and has the space of pairs (x,q(x)) as a section. The map exp : (x,v) ↦ (exp(x),v) from E to ($\mathbb{P}^2$ - C) × $\widetilde{D}$ identifies E with the inverse image of a neighborhood of Δ, and as $\varepsilon \to 0$ one obtains inverse images of a fundamental system of neighborhoods. Applying 1.6, one finds that for every O ∈ U*, exp and q induce a surjection

(exp,q)* : π_1(U*,O) ↠ π_1($\mathbb{P}^2$ - C,exp(O)) × π_1($\widetilde{D}$,q(O)).

There is an exact homotopy sequence

$\mathbb{Z} \to \pi_1(U^*,O) \to \pi_1(D^*,q(O)) \to 1$.

The image of $\mathbb{Z}$ is central; the image of 1 ∈ $\mathbb{Z}$ is represented by a small loop going once around D* in the positive direction. Since (exp,q)*, and hence exp* : π_1(U*,O) → π_1($\mathbb{P}^2$ - C,exp(O)), is surjective, this loop also defines a central element of π_1($\mathbb{P}^2$ - C,exp(O)). It is γ.

2.3 Proof of Theorem 1: For every irreducible component D of C, let γ(D) be the corresponding central element of π_1($\mathbb{P}^2$ - C,O). The quotient of π_1($\mathbb{P}^2$ - C,O) by the - central - subgroup generated by the γ(D) is π_1($\mathbb{P}^2$,O) = {e}. The theorem follows.

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I have just received (1/10/80) a very fine account surveying the methods studied here, their variants, and their many applications: W. FULTON and R. LAZARSFELD - Connectivity and its applications in algebraic geometry.

Pierre DELIGNE
Institut des Hautes Etudes
Scientifiques
35 route de Chartres
F-91440 BURES-SUR-YVETTE

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Exposé V

ORDINARY CRYSTALS AND CANONICAL COORDINATES

by P. DELIGNE

with the collaboration of L. ILLUSIE (*)

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Throughout this exposé, k denotes an algebraically closed field of characteristic $p > 0$, and W the ring of Witt vectors over k.

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(*) Associated Research Team of the C.N.R.S. no. 653

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0. INTRODUCTION

Let X_0/k be an ordinary elliptic curve. By Serre-Tate theory [14, V], the formal moduli space M of deformations of X_0 over local Artinian W-algebras with residue field k is isomorphic to the formal multiplicative group $\widehat{G}_m/W$. More precisely, the p-divisible group

$$G_0 = \cup \operatorname{Ker}(p^n : X_0 \to X_0)$$

is isomorphic to $(\mathbb{Q}_p/\mathbb{Z}_p) \times \mu_{p^\infty}$, and the choice of an isomorphism α between the étale part of G_0 and $\mathbb{Q}_p/\mathbb{Z}_p$ canonically identifies, for every local Artinian W-algebra R with residue field k, the set of liftings of X_0 over R with the group $\operatorname{Ext}^1((\mathbb{Q}_p/\mathbb{Z}_p)_R, (\mu_{p^\infty})_R)$, which in turn is canonically isomorphic to the group of units of R congruent to 1 modulo the maximal ideal. Thus one obtains an isomorphism, depending only on α, between M and $(\widehat{G}_m)_W$; in particular, the universal deformation X of X_0 over M defines a "canonical coordinate" q on M such that $M \simeq \operatorname{Spf} W[[q-1]]$. The parameter q, which describes the p-divisible group G associated with X as an extension of $\mathbb{Q}_p/\mathbb{Z}_p$ by μ_{p^∞}, also admits another interpretation in terms of the module $H^1_{\mathrm{DR}}(X/M)$, equipped with its F-crystal structure and its Hodge filtration $H^0(X, \Omega^1_{X/M}) \subset H^1_{\mathrm{DR}}(X/M)$. The datum of an isomorphism α as above gives, by duality, an isomorphism α' between the formal group $\widehat{X}/M$ associated with X and the formal group $(\widehat{G}_m)_M$; hence it gives a differential form $b \in H^0(X, \Omega^1_{X/M})$ corresponding under α' to the invariant form on $(\widehat{G}_m)_M$. Let a $\in H^1_{\mathrm{DR}}(X/M)$ satisfy $\nabla a = 0$, and let (a, b) be a symplectic basis of $H^1_{\mathrm{DR}}(X/M)$, i.e. $\langle a, b\rangle = 1$ and $\langle a, a\rangle = 0$. If

$$\nabla : H^1_{\mathrm{DR}}(X/M) \to \Omega^1_{M/W} \otimes H^1_{\mathrm{DR}}(X/M)$$

denotes the Gauss-Manin connection, then necessarily

$$(0.1) \qquad \nabla b = \eta \otimes a,$$

with $\eta \in \Omega^1_{M/W}$. One can show (see the Appendix and the following Exposé, by N. Katz) that

$$(0.2) \qquad \eta = d \log q,$$

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where q is the parameter defined above, and hence coincides with the parameter independently defined by Dwork ([8], [6]) using (0.2). The basis (a, b) above also has remarkable properties with respect to the Frobenius operation F on $H^1_{\mathrm{DR}}(X/M)$: if one chooses, as the endomorphism of M lifting Frobenius on $M \otimes k$, the endomorphism given by $q \mapsto q^p$, then F is expressed by

$$(0.3) \qquad F\,a = a, \qquad F\,b = p\,b.$$

The preceding constructions generalize to ordinary abelian varieties [8]; see also [9], [11] for arithmetic applications in the case of elliptic curves.

The purpose of this exposé is to show that an analogous theory can be developed for ordinary K3 surfaces, at least when $p \neq 2$. Let X_0/k be an ordinary K3 surface, i.e. one for which Frobenius on $H^2(X_0, \mathcal{O})$ is nonzero, and let S be the formal moduli space of deformations of X_0 over local Artinian W-algebras with residue field k (IV 1.2). If $p \neq 2$, we prove that S is canonically endowed with a formal group structure isomorphic to $(\widehat{G}_m)^{20}_W$. In particular, the universal deformation X of X_0 over S defines "canonical coordinates" q_i $(1 \leq i \leq 20)$ such that

$$S \simeq \mathrm{Spf}\, W[[q_1 - 1, \ldots, q_{20} - 1]],$$

and these form a basis of the characters of S. We further show that, if L_0 is a nontrivial invertible sheaf on X_0, the hypersurface $\Sigma(L_0)$ of S over which L_0 extends to $X \times_S \Sigma(L_0)$ (IV 1.5) is the kernel of a character of S, defined in an almost tautological way by the crystalline Chern class of L_0.

In the absence of a Serre-Tate theory for liftings of K3 surfaces, the formal group structure on S is defined in terms of the F-crystal structure on $H^2_{\mathrm{DR}}(X/S)$. In fact, the case of abelian varieties and that of K3 surfaces can be included in a single structure theorem for a certain class of ordinary F-crystals. The study of these crystals is the subject of

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No. 1. The geometric applications are given in No. 2. As regards the theorem concerning the hypersurfaces $\Sigma(L_0)$, the essential ingredient is the use of the crystalline Chern class to control the obstruction to extension.

1. CANONICAL PARAMETERS OF ORDINARY HODGE F-CRYSTALS

1.1. Review of definitions.

1.1.1. For the definitions and general properties of F-crystals, see Katz [8], [10]. Throughout No. 1 we fix the formal power-series rings

$$A_0 = k[[t_1, \dots, t_n]], \qquad A = W[[t_1, \dots, t_n]],$$

where $(t_1, \dots, t_n)$ is a finite sequence of indeterminates (for $n = 0$, we take $A_0 = k$ and $A = W$). A crystal over A_0 (understood to be in finite free modules) is, by definition, a finite free A-module H equipped with a connection

$$\nabla : H \to \Omega^1_{A/W} \otimes H$$

that is integrable and p-adically topologically nilpotent. Integrability of ∇ means that, on putting $D_i = d/\mathrm{d}t_i$, one has

$$\nabla(D_i)\nabla(D_j) = \nabla(D_j)\nabla(D_i)$$

for all $i \neq j$. This makes the PD-differential operators on A, i.e. the polynomials in the D_i with coefficients in A, act on H by the formula

$$\nabla(\textstyle\sum_m a_m D^m) = \sum_m a_m (\nabla(D))^m,$$

with the usual condensed notation $D^m = D_1^{m_1} \cdots D_n^{m_n}$, etc. The topological nilpotence of ∇ means that, for every i, $\nabla(D_i)^m$ tends to zero in the p-adic topology of $\mathrm{End}_W(H)$ as m tends to $+\infty$. By Berthelot [1, II 4], the

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datum of a connection ∇ as above is equivalent to the following datum: for every pair of W-homomorphisms (f, g) from A to a local, Noetherian, complete W-algebra B, such that f and g are congruent modulo an ideal I of B endowed with divided powers compatible with the standard divided powers of p, an isomorphism

$$\chi(f, g) : f^*H \simeq g^*H, \qquad (*)$$

subject to the transitivity conditions made explicit in [8, 1.2], namely $\chi(g, h)\chi(f, g) = \chi(f, h)$, $\chi(\mathrm{fk}, \mathrm{gk}) = k^*\chi(f, g)$, and $\chi(\mathrm{id}, \mathrm{id}) = \mathrm{id}$. The following formula describes concretely the part of this dictionary that we need.

LEMMA 1.1.2. With the notation above, let

$$f^* : H \to f^*H$$

denote the adjunction map, where $f^*H = f_*f^*H$ by abuse of notation. The composite A-linear homomorphism

$$\chi(f, g)f^* : H \xrightarrow{f^*} f^*H \xrightarrow{\chi(f, g)} g^*H$$

is given by

$$(1.1.2.1) \qquad \chi(f, g)f^*(x) = \sum_m (f(t) - g(t))^{[m]}\, g^*(\nabla(D)^m x),$$

where the sum ranges over all multi-indices $m = (m_1, \dots, m_n) \in N^n$, with the condensed notation

$$(f(t) - g(t))^{[m]} = (f(t_1) - g(t_1))^{[m_1]} \cdots (f(t_n) - g(t_n))^{[m_n]}$$

(the brackets denote a divided power in I), and

$$\nabla(D)^m = \nabla(D_1)^{m_1} \cdots \nabla(D_n)^{m_n}.$$

(N.B. The series on the right-hand side of (1.1.2.1) converges by the topological nilpotence of ∇.)

This statement is essentially contained in [1, II 4.3]. It suffices to verify it after reduction modulo p^r for every r. We shall again write

———————

(*) We also write f (respectively g) : Spec(B) → Spec(A) for the corresponding morphism of schemes.

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by abuse of notation $(f, g) : A \rightrightarrows B$ for the $W_r -$ homomorphisms obtained from the given homomorphisms by reduction modulo p^r. Let D be the divided-power envelope of the augmentation ideal J of $A \widehat{\otimes}_{W_r} A$ $(= W_r[[t_1, \dots, t_n, u_1, \dots, u_n]])$, and write d_0 (respectively d_1) : $A \to D$ for the homomorphism given by a $\mapsto$ a $\otimes$ 1 (respectively 1 $\otimes$ a), defining the left (respectively right) A-module structure on D. Since J, generated by the $t_i - u_i$, is regular, it follows from [1, I 4.5.1] that, for the left structure, D identifies with the divided-power algebra
$A\langle \xi_1, \dots, \xi_n \rangle$,
where $\xi_i = d_1 t_i - d_0 t_i$ is the image of $u_i - t_i$ in D. By the universal property of divided-power envelopes, and because B is complete, there is a PD-morphism $s : D \to B$ such that $s\, d_0 = g$ and $s\, d_1 = f$. On the other hand, by [1, II 4.3.8], the connection ∇ defines a PD-stratification

$$\varepsilon : D \otimes_A H\,(= d_1^* M) \simeq H \otimes_A D\,(= d_0^* M)$$

such that the $d_1 -$ linear composite

$$\theta : H \xrightarrow{d_1^*} d_1^* H \xrightarrow{\varepsilon} d_0^* H$$

is given by

$$(*) \qquad \theta(x) = \sum_m \nabla(D^m) x \otimes \xi^{[m]},$$

with the evident condensed notation. By definition, $\chi(f, g)$ is the isomorphism obtained from ε by pullback along s:

$$\chi(f, g) = s^* \varepsilon : f^* H \simeq g^* H.$$

If f (respectively g) is regarded as defining a right (respectively left) A-module structure on B, there is a commutative diagram

$$\begin{array}{ccccc} & & H & \xrightarrow{d_1^*} & D \otimes_A H \xrightarrow{\varepsilon} H \otimes_A D \\ & & & & \| \qquad\qquad \downarrow s \otimes 1 \quad \downarrow 1 \otimes s \\ H - f^* & \longrightarrow & B \otimes_A H & - \chi(f, g) \longrightarrow & H \otimes_A B, \end{array}$$

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which shows, in view of (*), that the lower horizontal composite is given by

$$\chi(f, g) f^*(x) = \sum_m \nabla(D)^m x \otimes s(\xi)^{[m]}.$$

But

$$s(\xi_i) = s(d_1 t_i - d_0 t_i) = f(t_i) - g(t_i),$$

and therefore

$$\chi(f, g) f^*(x) = \sum_m \nabla(D)^m x \otimes (f(t) - g(t))^{[m]}$$
$$= \sum_m (f(t) - g(t))^{[m]} g^*(\nabla(D)^m x).$$

1.1.3. An F-crystal over A_0 is, by definition, a crystal (H, ∇) over A_0 equipped, for every lifting $\varphi : A \to A$ of Frobenius on A_0 compatible with the Frobenius endomorphism σ of W, with a horizontal homomorphism

$$F(\varphi) : \varphi^* H \to H$$

($\varphi^* H$ being equipped with the connection $\varphi^* \nabla$), such that $F(\varphi) \otimes \mathbb{Q}_p$ is an isomorphism and such that, for every pair of liftings (φ, ψ), there is a commutative triangle
(1.1.3.1)

$$\begin{array}{ccc} \varphi^* H & \xrightarrow{F(\varphi)} & H \\ & \searrow^{\chi(\varphi,\psi)} & \nearrow_{F(\psi)} \\ & \psi^* H, & \end{array}$$

where $\chi(\varphi, \psi)$ is the canonical isomorphism defined by ∇, using the fact that φ is congruent to ψ modulo the divided-power ideal pA.
The horizontality of $F(\varphi)$ is expressed by the commutativity of the square
(1.1.3.2)

$$\begin{array}{ccc} \varphi^* H - \varphi^* \nabla & \longrightarrow & \Omega^1_{A/W} \otimes \varphi^* H \\ \downarrow{\scriptstyle F(\varphi)} & & \downarrow{\scriptstyle 1 \otimes F(\varphi)} \\ H - \nabla & \longrightarrow & \Omega^1_{A/W} \otimes H. \end{array}$$

Since, by definition, $\varphi^* \nabla$ makes the square

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$$\begin{array}{ccc} & \overset{\nabla}{} & \\ H & \longrightarrow & \Omega^1_{A/W} \otimes H \\ & & \downarrow \\ & \varphi^* & \\ & \downarrow & \\ & \downarrow & \\ \varphi^* H & \longrightarrow & \Omega^1_{A/W} \otimes \varphi^* H \end{array} \qquad \begin{array}{c} \downarrow \\ \varphi^* \otimes \varphi^* \\ \downarrow \\ \varphi^* \nabla \end{array} \qquad \downarrow$$

commute, one obtains the formula

$$(1.1.3.3) \qquad \nabla F(\varphi)\varphi^* x = \sum_i \varphi^* \omega_i \otimes F(\varphi)\varphi^* h_i,$$

for $x \in H$ such that $\nabla x = \sum_i \omega_i \otimes h_i$.
On the other hand, the variation law for $F(\varphi)$ in (1.1.3.1) is expressed by the formula

$$(1.1.3.4) \qquad F(\psi)\psi^* x = F(\varphi)\varphi^* x + \sum_{|n|>0} (\psi(t) - \varphi(t))^{[n]} F(\varphi)\varphi^*(\nabla(D)^n x),$$

with the notation of 1.1.2. Indeed, from the transitivity conditions satisfied by χ,

$$\begin{aligned} F(\psi)\psi^* x &= F(\psi)\chi(\varphi,\psi)\chi(\psi,\varphi)\psi^* x \\ &= F(\varphi)\chi(\psi,\varphi)\psi^* x \qquad (1.1.3.1), \end{aligned}$$

whence 1.1.3.4, by 1.1.2.1 applied with $f = \psi$ and $g = \varphi$.

1.1.4. Let $s_0 : A_0 \to k'$ be a point of $\mathrm{Spec}(A_0)$ with values in an algebraically closed extension of k. If φ lifts Frobenius as above, it is known [8, 1.1] that there is a unique lifting of s_0 to $s : A \to W(k')$ such that $s\varphi = \sigma s$ (where σ denotes Frobenius on $W(k')$); this is called the Teichmüller lifting of s_0 relative to φ. Let H be an F-crystal over A_0. Then $F(\varphi)$ gives a $\sigma-$linear endomorphism of s^*H, and hence an F-crystal s^*H over k'. If φ' is another lifting of Frobenius, and s' is the corresponding Teichmüller lifting of s_0, the F-crystals s^*H and s'^*H are canonically isomorphic by (1.1.3.1) (cf. [8, 1.4]). We shall write s_0^*H for s^*H, and say that s_0^*H is the F-crystal induced by H at the point s_0.

1.1.5. Finally, recall that to every F-crystal over k one associates a Newton polygon and a Hodge polygon; for their definitions and properties we refer the reader to [12], [8], and [10].

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1.2. Unit F-crystals.

1.2.1. An F-crystal (H, ∇) over A_0 is called a unit F-crystal if, for a lifting φ to A of Frobenius on A_0, $F(\varphi)$ is an isomorphism. By (1.1.3.1), this is equivalent to requiring $F(\varphi)$ to be an isomorphism for every lifting φ.

Every finite free $\mathbb{Z}_p -$ module M gives a unit F-crystal over A_0, namely $(H = M \otimes A, \nabla = 1 \otimes d, F(\varphi) = 1 \otimes \varphi)$. In fact, every unit F-crystal over A_0 is of this form:

PROPOSITION 1.2.2 (cf. [8, §3]). Let H be a unit F-crystal over A_0. There exists a basis (e_i) $(1 \leq i \leq r)$ of H over A such that $\nabla e_i = 0$ and $F(\varphi)\varphi^* e_i = e_i$ for every lifting φ of Frobenius on A_0.

In other words, the basis (e_i) defines an isomorphism from the trivial F-crystal $\mathbb{Z}_p^r \otimes A$ defined above onto H.

Before proving 1.2.2, let us first make two remarks.

a) If $x \in H$ is such that, for some lifting φ, $F(\varphi)\varphi^* x = x$, then $\nabla x = 0$. Indeed, since φ lifts Frobenius,

$$\varphi^*(\Omega^1_{A/W}) \subset p\Omega^1_{A/W};$$

hence p divides $\nabla x = \nabla F(\varphi)\varphi^* x$ by (1.1.3.3). The same argument shows inductively that ∇x is divisible by p^n for every n, and therefore $\nabla x = 0$.

b) If $x \in H$ satisfies $F(\varphi)\varphi^* x = x$ for one lifting φ, then $F(\psi)\psi^* x = x$ for every lifting ψ. In view of a), this follows from the variation law for $F(\varphi)$, (1.1.3.4).

It is therefore enough to find a basis (e_i) of H over A such that $F(\varphi)\varphi^* e_i = e_i$ for one chosen lifting φ, for example the one given by $t_j \mapsto t_j^p$ $(1 \leq j \leq n)$. Let $s : A \to W$ be the augmentation given by $s(t_i) = 0$. Since $s\varphi = \sigma s$, $F(\varphi)\varphi^*$ induces on s^*H a $\sigma -$ linear automorphism Φ. Since k is algebraically closed, there exists a basis of s^*H fixed by Φ (start with a basis fixed by Φ modulo p, which exists by Lang's lemma, and modify it successively so as

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to make it fixed by Φ modulo p^n for every n). In other words, there exists a basis (f_i) $(1 \leq i \leq r)$ of H such that, on writing $F(\varphi)\varphi^* = F$, one has

Ff = (I+M)f,

with $M \in M_r(A)$ and $M = 0$ mod(t) $(= (t_1, \ldots, t_n)A)$. It is enough to show that f may be replaced by e = (I+N)f, where $N \in M_r(A)$ and $N = 0$ mod(t), so that Fe = e. Written out, this gives the equation

$$(*) \qquad N = M + \varphi(N) + \varphi(N)M.$$

The map $N \mapsto M + \varphi(N) + \varphi(N)M$ is contractive for the (t)-adic topology, since $\varphi((t)) \subset (t)^p$. Thus (*) has a unique solution, completing the proof of 1.2.2.

1.2.3. We shall say that an F-crystal (H, ∇) over A_0 is topologically nilpotent if, for a lifting φ of Frobenius on A_0, $F(\varphi)\varphi^*$ is p-adically topologically nilpotent. This condition is independent of the choice of φ, as follows from (1.1.3.4) (or, more precisely, from the analogue of (1.1.3.4) for iterates of $F(\varphi)\varphi^*$ and $F(\psi)\psi^*$). It is not true that every F-crystal over A_0 is an extension of a topologically nilpotent F-crystal by a unit F-crystal. More precisely, one has the following criterion, a special case of a theorem of Katz [10, 2.4.2]:

PROPOSITION 1.2.4. Let H be an F-crystal over A_0. Let F_0 denote the endomorphism of $H_0 = H \otimes_A A_0$ induced by $F(\varphi)\varphi^*$, where φ lifts Frobenius F_{A_0} on A_0 (F_0 is F_{A_0} − linear and independent of φ). The following conditions are equivalent:

(i) The stable ranks of (H_0, F_0) at the closed point and at an algebraically closed geometric point lying over the generic point of Spec(A_0) are equal.

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(ii) There exists an extension of F-crystals over A_0,

$$0 \to U \to H \to E \to 0,$$

where U is a unit F-crystal and E is a topologically nilpotent F-crystal.

Recall what is meant by "stable rank." If L is a finite-dimensional vector space over a perfect field of characteristic p, equipped with a p-linear endomorphism Φ, then L has a canonical decomposition

$$L = L^{\mathrm{ss}} \oplus L^{\mathrm{nilp}},$$

where $L^{\mathrm{ss}} = \cap \operatorname{Im} \Phi^n$ and $L^{\mathrm{nilp}} = \cup \operatorname{Ker} \Phi^n$; the dimension of L^{ss} is, by definition, the stable rank of (L, Φ).

Although this is a special case of [10, 2.4.2], let us briefly give a proof of 1.2.4 (*), since the present situation is distinctly simpler than that of (loc. cit.). It is clear that (ii) implies (i). Conversely, put $\overline{H}_0 = H_0 \otimes_{A_0} k = H \otimes_A k$. Lift to H_0 a basis of $\overline{H}_0$ adapted to the decomposition of $\overline{H}_0$ under $F_0 \otimes k$,

$$\overline{H}_0 = \overline{H}_0^{\mathrm{ss}} \oplus \overline{H}_0^{\mathrm{nilp}}.$$

Then F_0 is represented by a matrix

(a c)
$(b\ d)$,

with $b = 0$ mod(t), and a invertible of rank $r = \dim \overline{H}_0^{\mathrm{ss}}$. In a new basis of H_0 given by the change-of-basis matrix

$$P = \begin{pmatrix} 1_r & 0 \\ x & 1 \end{pmatrix},$$

the matrix of F_0 is

$$\begin{pmatrix} a' & c' \\ b' & d' \end{pmatrix} = P^{-1} \begin{pmatrix} a & c \\ b & d \end{pmatrix} P^{(p)},$$

and $x = 0$ mod(t) can be chosen so that $b' = 0$. Indeed, x must satisfy $x = \mathrm{f(x)}$, where

$$f(x) = b\,a^{-1} - \mathrm{xcx}^{(p)} a^{-1} + d\,x^{(p)} a^{-1}.$$

This equation has a unique solution because f is contractive for the (t)-adic topology. In this new basis, the matrix of F_0 therefore has the form

$$\begin{pmatrix} a' & c' \\ 0 & d' \end{pmatrix},$$

with a' invertible of rank r. Hypothesis (i) then implies that the endomorphism d' is nilpotent, because this is true after localizing d' in an algebraically closed extension of the fraction field of A_0. Choose a lifting φ of F_{A_0}. In a basis of H lifting the basis of H_0

———————

(*) appearing in unpublished notes of Katz predating (loc. cit.).

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chosen above, the matrix of $F = F(\varphi)\varphi^*$ is written (with changed notation)

(a c)
(b d),

where $b = 0 \bmod p$ and d is topologically nilpotent. Applying the fixed-point method once again (now with a map contractive for the p-adic topology), one sees that a new basis $(e_1, \dots, e_r, e_{r+1}, \dots, e_{r+s})$ of H can be found, given by a change-of-basis matrix

$$\begin{pmatrix} 1_r & 0 \\ x & 1_s \end{pmatrix},$$

with $x = 0 \bmod p$, such that in this new basis the matrix of F is

$$\begin{pmatrix} a' & c' \\ 0 & d' \end{pmatrix},$$

where a' is invertible (of rank r) and d' is topologically nilpotent. The sub-F-module

$$U = \oplus_{1 \le i \le r} \mathrm{A}e_i$$

is equal to $\cap_{n \ge 0} \operatorname{Im} F^n$ and hence, by (1.1.3.2), is horizontal, i.e.

$$\nabla U \subset \Omega^1_{A/W} \otimes U.$$

By (1.1.3.1), U is therefore a unit sub-F-crystal of H, and by construction the F-crystal H/U is topologically nilpotent. This completes the proof.

REMARK 1.2.5. Under the hypotheses of 1.2.4, the F-crystal U is characterized, in terms of H, by $U = \cap \operatorname{Im} F^n$ (where $F = F(\varphi)\varphi^*$). We shall call U the unit sub-F-crystal of H.

1.3. Ordinary F-crystals.

1.3.1. Let H be an F-crystal over A_0, and set $H_0 = H \otimes_A A_0$. Let φ be a lifting to A of Frobenius F_{A_0}, and put

$$H_0^{(p)} = F^*_{A_0} H_0 = \varphi^* H \otimes_A A_0.$$

For $i \in \mathbb{N}$, define

$$(1.3.1.1) \qquad M^i H_0^{(p)} = \{x \in H_0^{(p)} \mid \text{there exists } y \in \varphi^* H \text{ lifting } x \text{ such that } F(\varphi)\, y \in p^i H\},$$

$$(1.3.1.2) \qquad \mathrm{Fil}^i H_0 = H_0 \cap M^i H_0^{(p)}$$

(where H_0 is regarded as a submodule of $F_{A_{0*}} H_0^{(p)}$ through the adjunction map),

$$(1.3.1.3) \qquad \mathrm{Fil}_i H_0 = \{x \in H_0 \mid \text{there exists } y \in H \text{ lifting } x \text{ such that } p^i y \in \operatorname{Im} F(\varphi)\}.$$

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By (1.3.1.1), the submodules $\mathrm{Fil}^i H_0$ (respectively $\mathrm{Fil}_i H_0$) do not depend on the choice of φ; they form a decreasing (respectively increasing) filtration of H_0, called the Hodge filtration (respectively the conjugate filtration), cf. [15, 1.9], where this filtration is denoted F^i_{Hodge} (respectively F^{-i}_{con}). One can also show [15, 2.2] that

$$(1.3.1.4) \qquad \mathrm{Fil}^i H_0 = \{x \in H_0 \mid \text{there exists } y \in H \text{ lifting } x \text{ such that } F(\varphi)\varphi^* y \in p^i H\}.$$

The terminology comes from the Mazur-Ogus theorem [3, 8.26], which asserts that if X_0/A_0 is a proper smooth scheme (or formal scheme) such that

(i) the crystalline cohomology $H^*(X_0/A)$ is a free A-module,

(ii) the spectral sequence

$$E_1^{ij} = H^j(X_0, \Omega^i_{X_0/A_0}) \Rightarrow H^*_{\mathrm{DR}}(X_0/A_0)$$

degenerates at E_1 and E_1 is free over A_0 and compatible with base change, then, on putting $H = H^n(X_0/A)$ (and hence $H_0 = H^n_{\mathrm{DR}}(X_0/A_0)$), with n fixed, the filtration $\mathrm{Fil}^i H_0$ (respectively $\mathrm{Fil}_i H_0$) coincides with the canonical filtration on the abutment of the preceding spectral sequence (respectively of the conjugate spectral sequence

$$E_2^{ij} = H^i(X_0, H^j(\Omega^\bullet_{X_0/A_0})) \Rightarrow H^*_{\mathrm{DR}}(X_0/A_0)).$$

Recall [15, 1.6] that the filtrations 1.3.1.2 and 1.3.1.3 are finite, separated, and exhaustive, and that for a given n the conditions $\mathrm{Fil}^{n+1} H_0 = 0$ and $\mathrm{Fil}_n H_0 = H_0$ are equivalent; one then says that H has level $\leq n$.

We shall write

$$(1.3.1.5) \qquad \mathrm{gr}^i H_0 \qquad (\text{respectively } \mathrm{gr}_i H_0)$$

for the graded module associated with the filtration $\mathrm{Fil}^i H_0$ (respectively $\mathrm{Fil}_i H_0$). If $\mathrm{gr}_i H_0$ is free over A_0, then so is $\mathrm{gr}^i H_0$; formation of $\mathrm{gr}_i H_0$ and $\mathrm{gr}^i H$ commutes with base change; and for every i, $\mathrm{gr}_i H_0$ and $\mathrm{gr}^i H_0$ are canonically isomorphic. In particular they have the same rank h_i, called the i-th Hodge number of H [15, 1.12, 2.3].

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Recall also [15, 1.6, 2.6] that the conjugate filtration is horizontal for the connection ∇_0 induced on H_0 (and has associated graded object of zero p-curvature), whereas the Hodge filtration satisfies Griffiths transversality:

$$(1.3.1.6) \qquad \nabla_0 \operatorname{Fil}^i H_0 \subset \Omega^1_{A_0/k} \otimes \operatorname{Fil}^{i-1} H_0.$$

The notion of an ordinary F-crystal was isolated by Mazur [12] and Ogus, to whom the following characterization is due [15, 3.1.3]:

PROPOSITION 1.3.2. Let H be an F-crystal over A_0 such that $\operatorname{gr}\bullet H_0$ is free over A_0. The following conditions are equivalent:

(i) the Newton and Hodge polygons of the $F-$crystal $e_0^* H$ induced by H at the closed point $e_0 : A_0 \to k$ (1.1.4) coincide;

(i') for every point s_0 of $\operatorname{Spec}(A_0)$ with values in an algebraically closed extension k' of k, the Newton and Hodge polygons of the F-crystal $s_0^*(H)$ (1.1.4) coincide;

(ii) the Hodge and conjugate filtrations of H_0 are opposite, i.e. for every i,

$$H_0 = \operatorname{Fil}_i H_0 \oplus \operatorname{Fil}^{i+1} H_0;$$

(iii) there exists a unique filtration of H by sub-F-crystals

$$(1.3.2.1) \qquad 0 \subset U_0 \subset U_1 \subset \ldots \subset U_i \subset U_{i+1} \subset \ldots$$

such that U_i lifts $\operatorname{Fil}_i H_0$ and U_i/U_{i-1} is of the form $V_i(-i)$, where V_i is a unit F-crystal and $(-i)$ denotes the Tate twist that replaces F by $p^i F$.

DEFINITION 1.3.3. An F-crystal H over A_0 is called ordinary if $\operatorname{gr}\bullet H_0$ is free and H satisfies the equivalent conditions of 1.3.2.

Let us prove 1.3.2. The implication $(i') \Rightarrow (i)$ is immediate, and it is clear that (iii) implies (i'). We prove $(i) \Rightarrow$ (ii). Suppose H has level $\leq n$ and that the conclusion is known for F-crystals of level $\leq n-1$. Let F denote the p-linear endomorphism of H_0 induced

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by $F(\varphi)$ for a lifting φ (F does not depend on this choice). By definition (1.3.1),

$$(1.3.3.1) \qquad \mathrm{Fil}_0 H_0 = \mathrm{Im}\ F : H_0^{(p)} \to H_0, \qquad \mathrm{Fil}^1 H_0 = \mathrm{Ker}\ F : H_0 \to H_0.$$

Hypothesis (i) first implies that the stable rank of $e_0^* H_0$ is $h_0 = \mathrm{rank}\ \mathrm{Fil}_0 H_0$ (the slope-zero parts of the Newton and Hodge polygons of $e_0^* H$ coincide). Let s_0 be a point of $\mathrm{Spec}(A_0)$ with values in an algebraically closed extension of the fraction field of A_0. The proof of 1.2.4 shows that

$$\mathrm{rank.st}(s_0^* H_0) \geq \mathrm{rank.st}(e_0^* H_0)$$

(rank.st = stable rank), while (1.3.3.1) also gives

$$\mathrm{rank.st}(s_0^* H_0) \leq h_0.$$

Hence

$$\mathrm{rank.st}(e_0^* H_0) = \mathrm{rank.st}(s_0^* H_0) = h_0.$$

It follows, on the one hand, that

$$\mathrm{Fil}_0(s_0^* H_0) = (s_0^* H_0)^s s, \qquad \mathrm{Fil}^1(s_0^* H_0) = (s_0^* H_0)^n \mathrm{ilp},$$

so $s_0^* \mathrm{Fil}_0 H_0 \cap s_0^* \mathrm{Fil}^1 H_0 = 0$ and therefore

$$(1.3.3.2) \qquad H_0 = \mathrm{Fil}_0 H_0 \oplus \mathrm{Fil}^1 H_0.$$

On the other hand, by 1.2.4 there is an exact sequence of F-crystals over A_0,

$$(1.3.3.3) \qquad 0 \to U_0 \to H \to E \to 0,$$

where U_0 is the unit sub-F-crystal of H and E is a topologically nilpotent F-crystal. Since by definition $\mathrm{rank}(U_0) = \mathrm{rank.st}(e_0^* H_0) = h_0$, U_0 lifts $\mathrm{Fil}_0 H_0$; hence the projection $H \to E$ induces an isomorphism $\mathrm{Fil}^1 H_0 \simeq E_0$. This means that $F(\varphi) : \varphi^* E \to E$ is divisible by p, so $E = E'(-1)$, where E' is an F-crystal of level $\leq n-1$. Clearly $e_0^* E$, and hence also $e_0^* E'$, has the same Newton and Hodge polygons; thus, by the induction hypothesis applied to E', one

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concludes that the Hodge and conjugate filtrations of $E_0 = E \otimes A_0$ are opposite. Thus (ii) will be proved once we verify that the projection $H \to E$ induces isomorphisms, for $i \geq 0$,

$$\mathrm{Fil}_i H_0 / \mathrm{Fil}_0 H_0 \simeq \mathrm{Fil}_i E_0, \qquad \mathrm{Fil}^{i+1} H_0 \simeq \mathrm{Fil}^{i+1} E_0.$$

These maps are plainly injective, and their surjectivity follows easily from the fact that U_0 is a unit F-crystal. We now prove (ii) $\Rightarrow$ (iii). Again suppose that H has level $\leq n$ and that the conclusion is known for F-crystals of level $\leq n-1$. Uniqueness in (1.3.2.1) is clear by induction, since U_0 must be the unit sub-F-crystal of H. For existence, note that (1.3.3.2) holds by hypothesis. By (1.3.3.1),

$$e_0^* \mathrm{Fil}_0 H_0 = (e_0^* H_0)^s s, \qquad s_0^* \mathrm{Fil}_0 H_0 = (s_0^* H_0)^s s$$

(the decomposition into semisimple and nilpotent parts is unique), so the stable ranks of $e_0^* H_0$ and $s_0^* H_0$ are equal. By 1.2.4 we again obtain the exact sequence (1.3.3.3). As seen above, the Hodge and conjugate filtrations of E_0 are obtained from those of H_0 by passage to the quotient, so they are opposite. Since $E = E'(-1)$, with E' of level $\leq n-1$, induction gives a filtration of E by sub-F-crystals

$$0 = W_0 \subset W_1 \subset \ldots \subset W_i \subset W_{i+1} \subset \ldots$$

such that W_i lifts $\mathrm{Fil}_i E_0$ and W_i / W_{i-1} is of the form $T_i(-i)$, with T_i a unit F-crystal. For $i \geq 1$, let U_i be the inverse image of W_i in H. The U_i for $i \geq 1$, together with U_0, form a filtration (1.3.2.1) satisfying the stated conditions in (iii). This completes the proof of 1.3.2.

REMARK 1.3.4. If $A_0 = k$, the filtration (1.3.2.1) admits a unique splitting

$$(1.3.4.1) \qquad U_i = \oplus_{j \leq i} V_j(-j),$$

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where V_j is a unit F-crystal. This is a special case of Katz's theorem [10, 1.6.1], but it is easy to verify directly: uniqueness is clear, since for $j < i$ one has $\mathrm{Hom}(V_i(-i), V_j(-j)) = 0$; existence of the splitting follows readily by induction on the level, using a fixed-point argument analogous to those used in the proof of 1.2.4.

1.3.5. Let H be an F-crystal over A_0. A Hodge filtration on H means a finite decreasing filtration by free A-submodules

$$\mathrm{Fil}^0 H = H \supset \mathrm{Fil}^1 H \supset \ldots \supset \mathrm{Fil}^i H \supset \mathrm{Fil}^{i+1} H \supset \ldots$$

satisfying the following two conditions:

(i) for every i, $\mathrm{Fil}^i H$ lifts $\mathrm{Fil}^i H_0$;

(ii) ("transversality") for every i,

$$(1.3.5.1) \qquad \nabla \mathrm{Fil}^i H \subset \Omega^1_{A/W} \otimes \mathrm{Fil}^{i-1} H.$$

A Hodge F-crystal over A_0 is an F-crystal over A_0 equipped with a Hodge filtration.

A standard example, in the geometric situation considered in 1.3.1, is given by a lifting of X_0/A_0 to a proper smooth formal scheme X/A such that the Hodge spectral sequence of X/A,

$$E_1^{ij} = H^j(X, \Omega^i_{X/A}) \Rightarrow H^*_{\mathrm{DR}}(X/A) \,(= H^*(X_0/A)),$$

degenerates at E_1, with E_1 free over A. The Hodge filtration on $H^n_{\mathrm{DR}}(X/A)$, as the abutment of this spectral sequence, satisfies conditions (i) and (ii) above.

PROPOSITION 1.3.6. Let $(H, \mathrm{Fil}^\bullet H)$ be a Hodge F-crystal over A_0. If H is ordinary (1.3.3), the filtrations U_i (1.3.2.1) and $\mathrm{Fil}^\bullet H$ are opposite; that is, for every i, $H = U_i \oplus \mathrm{Fil}^{i+1} H$. Hence there is a decomposition

$$(1.3.6.1) \qquad H = \oplus_i H^i, \qquad H^i = U_i \cap \mathrm{Fil}^i H,$$

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with H^i of rank h_i (1.3.1.5).

This is an immediate consequence of 1.3.2 and 1.3.5(i) (condition (ii) is not used).

In the preceding example, suppose the F-crystal $H = H^n_{\mathrm{DR}}(X/A)$ is ordinary. For every $r \geq 1$, the conjugate spectral sequence

$$(*)_r \qquad E_2^{ij} = H^i(X \otimes W_r,\ H^j(\Omega^{\bullet}_{X\otimes W_r/A\otimes W_r})) \Rightarrow H^*_{\mathrm{DR}}(X \otimes W_r/A \otimes W_r)$$

defines a filtration $(U_{i,r})$ on $H^n_{\mathrm{DR}}(X \otimes W_r/A \otimes W_r)$. One may hope that the projective limit of the spectral sequences $(*)_r$ is a spectral sequence whose abutment filtration on H^n is the limit of the $(U_{i,r})$ and coincides with the filtration (U_i).

1.4. Ordinary Hodge F-crystals of level ≤ 1.

1.4.1. Let H be an ordinary Hodge F-crystal of level ≤ 1 over A_0. By 1.3.2, there is an extension of F-crystals

$$(1.4.1.1) \qquad 0 \to U \to H \to V(-1) \to 0,$$

where U and V are unit F-crystals and U lifts $\mathrm{Fil}_0 H_0$.

Let r (respectively s) be the rank of U (respectively V). The module H also has the free submodule $\mathrm{Fil}^1 H$, which lifts $\mathrm{Fil}^1 H_0$ and therefore satisfies, by (1.3.1.4),

$$(1.4.1.2) \qquad F(\varphi)\varphi^* \mathrm{Fil}^1 H \subset \mathrm{pH}$$

for every lifting φ of Frobenius. Moreover, by 1.3.6, H decomposes as an A-module direct sum

$$(1.4.1.3) \qquad H = U \oplus \mathrm{Fil}^1 H.$$

In particular, $\mathrm{Fil}^1 H$ has rank s and maps isomorphically onto V.

THEOREM 1.4.2. With the notation of 1.4.1: (i) There exist a basis $a = (a_i)_{1 \leq i \leq r}$ of the A-module U and a basis $b = (b_i)_{1 \leq i \leq s}$ of the A-module $\mathrm{Fil}^1 H$ satisfying the conditions

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$$(1.4.2.1) \qquad \nabla a_i = 0, \qquad 1 \le i \le r,$$

$$(1.4.2.2) \qquad \nabla b_i = \sum_{1 \le j \le r} \eta_{ij} \otimes a_j, \qquad \eta_{ij} \in \Omega^1_{A/W}, \qquad 1 \le i \le s,$$

$$(1.4.2.3) \qquad F(\varphi)\varphi^* a_i = a_i, \qquad 1 \le i \le r,$$

$$(1.4.2.4) \qquad F(\varphi)\varphi^* b_i = p\, b_i + p \sum_{1 \le j \le r} u_{ij}(\varphi) a_j, \qquad u_{ij}(\varphi) \in A, \qquad 1 \le i \le s,$$

for every lifting φ of Frobenius. Moreover, the forms η_{ij} are closed and satisfy, for every lifting φ of Frobenius,

$$(1.4.2.5) \qquad \varphi^* \eta_{ij} = p \eta_{ij} + p\, \mathrm{d}u_{ij}(\varphi).$$

(ii) Once the bases a and b have been chosen as in (i), there is a unique family of formal power series $\tau_{ij} \in \mathrm{K}[[t]]$ (where K is the fraction field of W) such that, for all i and j,

$$(1.4.2.6) \qquad \eta_{ij} = d\tau_{ij},$$

$$(1.4.2.7) \qquad \varphi^* \tau_{ij} - p\tau_{ij} - \mathrm{p}u_{ij}(\varphi) = 0$$

for every lifting φ of Frobenius. One has

$$(1.4.2.8) \qquad \tau_{ij}(0) \in \mathrm{pW}.$$

Moreover, if $p \neq 2$, the series

$$(1.4.2.9) \qquad q_{ij} = \exp(\tau_{ij})$$

are defined, belong to A, and satisfy $q_{ij}(0) = 1 \bmod \mathrm{pW}$.

We prove (i). By the structure theorem for unit F-crystals (1.2.2), there is a basis a $= (a_i)_{1 \le i \le r}$ of U satisfying (1.4.2.1) and (1.4.2.3), and a basis $b = (b_i)_{1 \le i \le s}$ of $\mathrm{Fil}^1 H$ such that, if b'_i denotes the image of b_i in $V(-1)$, then $\nabla b'_i = 0$ and $F(\varphi)\varphi^* b'_i = p\, b'_i$ for every i and every lifting φ of Frobenius. Hence ∇b_i and $F(\varphi)\varphi^* b_i$ have the forms (1.4.2.2) and (1.4.2.4). Applying ∇ to (1.4.2.2), integrability gives $d\eta_{ij} = 0$ for all i, j. Applying ∇ to (1.4.2.4) and using horizontality of F (1.1.3.3) gives the relations (1.4.2.5).

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We prove (ii). Since the forms η_{ij} are closed, the Poincaré lemma gives series $\tau_{ij} \in K[[t]]$, determined up to an additive constant, such that (1.4.2.6) holds. Relation (1.4.2.5) then implies

$$\varphi^*\tau_{ij} - p\tau_{ij} - \mathrm{pu}_{ij}(\varphi) \in K,$$

and hence

$$(1.4.2.10) \qquad \varphi^*\tau_{ij} - p\tau_{ij} - \mathrm{pu}_{ij}(\varphi) = (\varphi^*\tau_{ij})(0) - p\tau_{ij}(0) - \mathrm{pu}_{ij}(\varphi)(0).$$

Write $x \mapsto x^\sigma$ for the Frobenius automorphism of K. First fix the lifting φ of Frobenius given by $\varphi(t_i) = t_i^p$, and normalize τ_{ij} by the condition

$$(1.4.2.11) \qquad (\varphi^*\tau_{ij})(0) - p\tau_{ij}(0) - \mathrm{pu}_{ij}(\varphi)(0) = 0,$$

which may also be written, since $(\varphi^*\tau_{ij})(0) = \tau_{ij}(0)^\sigma$,

$$(1.4.2.12) \qquad \tau_{ij}(0) = \sum_{n\geq 1} V^n u_{ij}(\varphi)(0)$$

(where $\mathrm{Vx} = \mathrm{pF}^{-1}x = \mathrm{px}^{\sigma^{-1}}$). In particular, (1.4.2.8) holds. We show that if ψ is any lifting of Frobenius, then (1.4.2.7), with φ replaced by ψ, is satisfied. In view of (1.4.2.10), again with φ replaced by ψ, it is equivalent to verify

$$(1.4.2.13) \qquad (\psi^*\tau_{ij})(0) - p\tau_{ij}(0) - \mathrm{pu}_{ij}(\psi)(0) = 0.$$

If $\psi(0) = 0$ (that is, ψ is compatible with the augmentation $e : W[[t]] \to W$), then $(\psi^*\tau_{ij})(0) = \tau_{ij}(0)^\sigma$, and (1.4.2.13) is equivalent to (1.4.2.12) with φ replaced by ψ. It therefore remains only to check that $u_{ij}(\varphi)(0) = u_{ij}(\psi)(0)$. The augmentation e is the Teichmüller lifting (1.1.4) of $k[[t]] \to k$ simultaneously for φ and ψ; hence $F(\varphi)\varphi^*$ and $F(\psi)\psi^*$ induce the same σ-linear endomorphism F of e^*H. Applying e^* to (1.4.2.4), written once for φ and once for ψ, gives

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$$F(e^*b_i) = p(e^*b_i) + p\sum_{1\le j\le r} u_{ij}(\varphi)(0)(e^*a_j)$$
$$= p(e^*b_i) + p\sum_{1\le j\le r} u_{ij}(\psi)(0)(e^*a_j),$$

so $u_{ij}(\varphi)(0) = u_{ij}(\psi)(0)$, proving (1.4.2.13) in this case.
Now let ψ be an arbitrary lifting of Frobenius. We may write

$$\psi(t_i) = \psi_0(t_i) + \mathrm{pw}_i, \qquad 1 \le i \le n,$$

with $w = (w_1, \dots, w_n) \in W^n$, where $\psi_0 : A \to A$ is a lifting of Frobenius such that $\psi_0(0) = 0$. Applying (1.1.3.4), with $\varphi = \psi_0$ and $x = b_i$, and using (1.4.2.4), gives

$$(*) \qquad \sum_{1\le j\le r} \mathrm{pu}_{ij}(\psi)a_j = \sum_{1\le j\le r} \mathrm{pu}_{ij}(\psi_0)a_j$$
$$+ \sum_{|n|>0} p^{[n]}(w^\sigma)^n F(\psi_0)\psi_0^*(\nabla(D)^n b_i).$$

Let us compute $\nabla(D)^n b_i$. If $D_k = \partial/\partial t_k$, then by (1.4.2.2) and (1.4.2.6),

$$\nabla(D_k)b_i = \langle D_k, \sum_j d\tau_{ij} \otimes a_j\rangle = \sum_j (D_k\tau_{ij})a_j.$$

Since $\nabla a_j = 0$, it follows that

$$\nabla(D^n)b_i = \sum_j (D^n\tau_{ij})a_j$$

for every multi-index n with $|n| > 0$. Substitution in (*) yields
$\mathrm{pu}_{ij}(\psi) = \mathrm{pu}_{ij}(\psi_0) + \sum_{|n|>0} p^{[n]}(w^\sigma)^n\ \psi_0^*(D^n\tau_{ij})$,
and the identity (1.4.2.13) to be proved becomes

$$(**) \qquad (\psi^*\tau_{ij})(0) - p\tau_{ij}(0) - \mathrm{pu}_{ij}(\psi_0)(0)$$
$$- \sum_{|n|>0} p^{[n]}(w^n(D^n\tau_{ij})(0))^\sigma = 0.$$

Write $\psi = \psi_0 \circ a$, where a is the W-algebra endomorphism of A given by $a(t_i) = t_i + \mathrm{pw}_i$. Then

$$(\psi^*\tau_{ij})(0) = (\psi_0^*(a^*\tau_{ij}))(0) = (a^*\tau_{ij})(0)^\sigma = \tau_{ij}(\mathrm{pw})^\sigma.$$

Moreover ψ_0 satisfies (1.4.2.13), as shown above, because $\psi_0(0) = 0$ and $(\psi_0^*\tau_{ij})(0) = \tau_{ij}(0)^\sigma$. Thus

$$p\tau_{ij}(0) = \tau_{ij}(0)^\sigma - \mathrm{pu}_{ij}(\psi_0)(0).$$

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We may therefore rewrite (**) as

$$\tau_{ij}(pw)^{\sigma} = \tau_{ij}(0)^{\sigma} + \sum_{|n|>0} p^{[n]}(w^n(D^n\tau_{ij})(0))^{\sigma},$$

i.e.

$$\tau_{ij}(pw) = \tau_{ij}(0) + \sum_{|n|>0} p^{[n]} w^n (D^n\tau_{ij})(0).$$

This is precisely the Taylor expansion of $\tau_{ij}(pw)$. It proves (**), and therefore (1.4.2.13), in every case. To prove the last assertion of 1.4.2 we shall use the following result of Dwork:

LEMMA 1.4.3 ([5], Lemma 1) (*). Let φ denote the σ − linear automorphism of K[[t]] (where $t = (t_1, \dots, t_n)$ and K is the fraction field of W) given by $t_i \mapsto t_i^p$. Let $f \in$ K[[t]] with f(0) = 1. The following conditions are equivalent:

$$(i)\ f \in W[[t]];$$

(ii) $(\varphi^* f)/f^p = 1 + pu$, with $u \in$ W[[t]] and u(0) = 0.

The implication $(i) \Rightarrow$ (ii) is immediate. Recall the proof of (ii) $\Rightarrow$ (i). Write $f = \sum_i a_i t^i$ $(a_0 = 1)$, and suppose it is known that $a_i \in W$ for |i| < r, where $|i| = i_1 + \dots + i_n$. Denote by ( )<j the part of total degree < j. Then

$((\sum_{|i|\geq 0} a_i t^i)^p) < r$
$= ((\sum_{|i|\leq r-1} a_i t^i)^p) < r + p\sum_{|i|=r} a_i t^i$
$= (\sum_{|i|\leq r-1} a_i^p t^{pi}) < r + p\sum_{|i|=r} a_i t^i \bmod pW[[t]]$.

Next, writing $f^p/(\varphi^* f) = 1 + p\sum_{|i|\geq 1} d_i t^i$ with $d_i \in W$, one obtains

$((\sum_{|i|\geq 0} a_i^{\sigma} t^{pi})(1 + p\sum_{|i|\geq 1} d_i t^i)) < r$
$= ((\sum_{|i|\leq r-1} a_i^{\sigma} t^{pi})(1 + p\sum_{|i|\geq 1} d_i t^i)) < r$
$= (\sum_{|i|\leq r-1} a_i^{\sigma} t^{pi}) < r \bmod pW[[t]]$
$= (\sum_{|i|\leq r-1} a_i^p t^{pi}) < r \bmod pW[[t]]$.

———————

(*) See also Hazewinkel [7] for a generalization.

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Comparing the two expressions obtained for $(f^p) < r$ gives

$$p \sum_{|i|=r} a_i t^i \in \mathrm{pW}[[t]],$$

so $a_i \in W$ for every i with |i| = r. This proves (ii) $\Rightarrow$ (i).

Before returning to the proof of 1.4.2, we record two immediate consequences of 1.4.3.

COROLLARY 1.4.4. Let $g \in$ K[[t]] with g(0) = 0. The following conditions are equivalent:

$$(i)\ \exp(g) \in W[[t]];$$

(ii) $\varphi^* g - \mathrm{pg} \in \mathrm{pW}[[t]]$.

Indeed, put $f = \exp(\mathrm{g})$. Since for $n \geq 1$ the quantities $p^n/n!$ (and a fortiori p^n/n) belong to pW, condition (ii) is equivalent to $\varphi^* f / f^p = 1 + \mathrm{pu}$, with $u \in$ W[[t]] and u(0) = 0.

COROLLARY 1.4.5. Suppose $p \neq 2$. Let $g \in$ K[[t]] satisfy g(0) $\in$ pW and $\varphi^* g - \mathrm{pg} \in \mathrm{pW}[[t]]$. Then the series $f = \exp(\mathrm{g})$ is defined, belongs to W[[t]], and satisfies f(0) = 1 mod p.

Indeed, let $h = g - \mathrm{g}(0)$. Since $p \neq 2$ and g(0) $\in$ pW, exp(g(0)) is defined, and hence so is exp(g) = exp(g(0))exp(h). Write g(0) = a, with a $\in W$. Then

$$\varphi^* h - \mathrm{ph} = \varphi^* g - \mathrm{pg} - (a^\sigma - \mathrm{pa}) \in \mathrm{pW}[[t]]$$

because $a^\sigma - \mathrm{pa} \in$ pW and $\varphi^* g - \mathrm{pg} \in$ pW[[t]]. Applying 1.4.4 to h gives exp(h) $\in$ W[[t]], and since exp(a) = 1 mod pW, it follows that $f = \exp(\mathrm{g}) \in$ W[[t]] and f(0) = 1 mod pW.

End of the proof of 1.4.2. It remains only to apply 1.4.5 to τ_{ij}: its hypotheses hold by (1.4.2.7) and (1.4.2.8).

1.4.6. If H is a Hodge F-crystal as in 1.3.5, the connection ∇ induces, by (1.3.5.1), an A-linear homomorphism

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$$(1.4.6.1) \qquad \operatorname{gr}\nabla : \operatorname{gr}^i H \to \Omega^1_{A/W} \otimes \operatorname{gr}^{i-1} H$$

(where $\operatorname{gr}^i = \operatorname{Fil}^i / \operatorname{Fil}^{i+1}$). In particular, in the situation of 1.4.1, ∇ induces an A-linear homomorphism

$$(1.4.6.2) \qquad \operatorname{gr}\nabla : \operatorname{Fil}^1 H \to \Omega^1_{A/W} \otimes U.$$

Since $U \simeq H / \operatorname{Fil}^1 H$ by (1.4.1.3), this gives an A-linear homomorphism

$$(1.4.6.3) \qquad \operatorname{gr}\nabla : T_{A/W} \to \operatorname{Hom}_A(\operatorname{Fil}^1 H, U),$$

where $T_{A/W} = (\Omega^1_{A/W})\vee$.

COROLLARY 1.4.7. Under the hypotheses of 1.4.1, suppose that (1.4.6.3) is an isomorphism and that p is different from 2. Then the elements $q_{ij} - 1$ ($1 \le i \le s$, $1 \le j \le r$) defined in (1.4.2.9), together with p, form a regular system of parameters for $A = \mathrm{W}[[t]]$; that is, the W-homomorphism

$$W[[x_{ij}]]_{(1\le i\le s, 1\le j\le r)} \to A$$

sending x_{ij} to $q_{ij} - 1$ is an isomorphism. If φ denotes the lifting of Frobenius defined by $\varphi(q_{ij}) = q_{ij}^p$, then

$$(1.4.7.1) \qquad F(\varphi)\varphi^* b_i = p\, b_i$$

for every i, with the notation of 1.4.2.

Let $m = (p, t_1, \dots, t_n)$ be the maximal ideal of $A = \mathrm{W}[[t]]$. Since $q_{ij}(0) = 1$ mod pW, the q_{ij} are units of A congruent to 1 mod m. To prove the first assertion it is enough to show that the forms $\eta_{ij} = d \log q_{ij}$ form an A-basis of $\Omega^1_{A/W}$.

Every homomorphism $f : \operatorname{Fil}^1 H \to U$ is determined by a matrix (f_{ij}) such that $f(b_i) = \sum_j f_{ij} a_j$. Let D_{ij} be the basis of $T_{A/W}$ characterized by

$$(\operatorname{gr}\nabla)(D_{ij})_k \ell = \{0 \text{ if } (k,\ell) \ne (i,j),$$
$$1 \text{ if } (k,\ell) = (i,j).\}$$

For every $D \in T_{A/W}$, (1.4.2.2) gives

$(\operatorname{gr}\nabla)(D) b_i = \sum_j \eta_{ij}(D) a_j$,

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thus
(gr ∇)$(D_{ij})_k \ell = \eta_k \ell(D_{ij})$.

The η_{ij} therefore form a basis of $\Omega^1_{A/W}$, namely the basis dual to the basis D_{ij}. On the other hand, the relation $\varphi^* q_{ij} = q_{ij}^p$ implies $\varphi^* \log q_{ij} = p \log q_{ij}$; but since $\log q_{ij} = \tau_{ij}$, (1.4.2.7) implies $u_{ij}(\varphi) = 0$, and hence (1.4.7.1) follows from (1.4.2.4).

1.4.8. Let, as in 1.3.6, $(H, \mathrm{Fil}\bullet H)$ be a Hodge F-crystal over A_0 (= k[[t]]) such that H is ordinary. Suppose that H has level $\leq N$, i.e. (1.3.1) $\mathrm{Fil}^i H_0 = 0$ for $i \geq$ n+1, so that the decomposition (1.3.6.1) is written

$$(1.4.8.1) \qquad H = \oplus_{0 \leq i \leq N} H^i, \qquad H^i = U_i \cap \mathrm{Fil}^i H, \qquad \mathrm{rg}(H^i) = h_i.$$

Suppose moreover that $p \neq 2$. Applying 1.4.2 to the ordinary Hodge F-crystals (of level ≤ 1) $(U_{i+2}/U_i, (\mathrm{Fil}\bullet H \cap U_{i+2})/(\mathrm{Fil}\bullet H \cap U_i))$,
one obtains:

COROLLARY 1.4.9. Under the hypotheses of 1.4.8, there exist a basis $(e_1^i, \ldots, e_{h_i}^i)_{0 \leq i \leq N}$ of H and elements $q_{i;\alpha,\beta}$ of A ($i \leq N-1$, $\alpha \leq h_{i+1}$, $\beta \leq h_i$) such that

$$(1.4.9.1) \qquad \{q_{i;\alpha,\beta}(0) \in 1 + \mathrm{pW},$$
$$\nabla e_i^0 = 0 \qquad (1 \leq i \leq h_0),$$
$$\nabla e_i^m = \sum\nolimits_{1 \leq j \leq h_{m-1}} (d \log q_{m-1;i,j}) e_j^{m-1} \qquad (1 \leq m \leq N,\ 0 \leq i \leq h_m).\}$$

One may use 1.4.9 to bound the growth of the horizontal sections of H. Recall [8, 3.1] that these "converge in the open unit polydisc," and, more precisely, that the module $(H \otimes_A K\{\{t_1, \ldots, t_n\}\}, \nabla)$ possesses a basis of horizontal sections; this is a property valid for every F-crystal, independently of any ordinarity hypothesis. That said, it follows readily from 1.4.9 that, for every i, the horizontal sections of $U_i \otimes_A$ K{{t}} belong to $U_i \otimes_A L_i$, where

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$$0 = L_{-1} \subset L_0 \subset \ldots \subset L_i \subset \ldots$$

is the sequence of A-submodules of K{{t}} defined recursively by

$$(f \in L_i) \Leftrightarrow (f \text{ belongs to the } A-\text{submodule of } K\{\{t\}\} \text{ generated by the series } g \text{ such that } dg = \sum a_\alpha\, d\log u_\alpha, \text{ with } a_\alpha \in L_{i-1} \text{ and } u_\alpha \in A,\ u_\alpha(0) \in 1 + pW).$$

In the situation of 1.4.8, however, one cannot expect a statement analogous to 1.4.7 (the hypothesis that (1.4.6.3) is an isomorphism has no generalization in level ≥ 2).

2. GEOMETRIC APPLICATIONS

We retain the notation A_0, A of 1.1.1.

2.1. Canonical coordinates.

A) Abelian varieties (cf. [8, §8]).

2.1.1. Let X/A be a formal abelian scheme of relative dimension g. The A-module

$$H = H^1_{DR}(X/A),$$

endowed with the Gauss-Manin connection ∇, is an F-crystal over A_0 of rank 2g. It is known (see, for example, [13, Addendum]) that the Hodge spectral sequence

$$E_1^{ij} = H^j(X, \Omega^i_{X/A}) \Rightarrow H^*_{DR}(X/A)$$

degenerates at E_1, and that the term E_1^{ij} is free over A and is compatible with every base change. It follows from (1.3.5) that H, endowed with the Hodge filtration

$$H = \mathrm{Fil}^0 H \supset \mathrm{Fil}^1 H\ (= H^0(X, \Omega^1_{X/A})) \supset 0,$$

is a Hodge F-crystal over A_0. Note also that, if $X_0 = X \otimes A_0$, the Hodge spectral sequence and the conjugate spectral sequence of X_0/A_0 degenerate (at E_1 and E_2, respectively), and that their initial terms are

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free and compatible with every base change. In particular, the associated graded object for the conjugate filtration, $\mathrm{gr}_\bullet(H \otimes A_0)$, is free. On the other hand, if $e_0 : A_0 \to k$ is the augmentation, the F-crystal $e_0^* H$ induced over k (1.1.4) is precisely $H^1(X \otimes k/W)$, because, if $e : A \to W$ is the augmentation, then $e^* H = H^1_{\mathrm{DR}}(X \otimes W/W) = H^1(X \otimes k/W)$ by the definition of crystalline cohomology.

2.1.2. Suppose that the abelian variety $X \otimes k$ is ordinary, which means, equivalently, that $(X \otimes \mathrm{k})(\mathrm{k}) \simeq (\mathbb{Z}/p)^g$, that Frobenius on $H^1(X \otimes k, \mathcal{O})$ is bijective, or that $H^1(X \otimes k/W)$ has the same Newton and Hodge polygons. From what has just been recalled, the F-crystal H is then ordinary in the sense of 1.3.3; and since it is a Hodge F-crystal of level ≤ 1, 1.4.2 applies to it. In the decomposition $H = U \oplus \mathrm{Fil}^1 H$, both U and $\mathrm{Fil}^1 H$ are free of rank g. There therefore exist a basis $(a_i)_{1 \leq i \leq g}$ of U, a basis $(b_i)_{1 \leq i \leq g}$ of $\mathrm{Fil}^1 H$, and series $\tau_{ij} \in \mathrm{K}[[\mathrm{t}]]$ $(1 \leq j \leq g, 1 \leq j \leq g)$ satisfying conditions (1.4.2.1) through (1.4.2.8). Moreover, if $p \neq 2$, one has the series $q_{ij} = \exp(\tau_{ij}) \in A$, with $q_{ij}(0) = 1$ mod pW.

2.1.3. Suppose that X/A is the versal formal deformation of $X \otimes k$, so that $A = W[[t_1, \ldots, t_{g^2}]]$. Then (1.4.6.3) is an isomorphism, identified with the Kodaira-Spencer isomorphism

$$T_{A/W} \simeq H^1(X, T_{X/A}) \simeq \mathrm{Hom}(H^0(X, \Omega^1_{X/A}), H^1(X, \mathcal{O}))$$

(cf. (IV 2.3)). Thus, if $p \neq 2$, the elements $q_{ij} - 1 \in A$ form, by 1.4.7, "canonical" coordinates on A (i.e. $A \simeq W[[q_{ij} - 1]]$); and if φ denotes the lifting of Frobenius given by $\varphi(q_{ij}) = q_{ij}^p$, then, with the notation of 2.1.2,

$$F(\varphi)\varphi^* a_i = a_i, \qquad F(\varphi)\varphi^* b_i = p\, b_i \qquad (1 \leq i \leq g).$$

Let $e : A \to W$ be the Teichmüller lifting of e_0 corresponding to φ, i.e. such that $e(q_{ij}) = 1$. It is not difficult to show - we

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shall establish an analogous result below for K3 surfaces - that e is independent of the choice of (a, b, q), and that there is a canonical isomorphism (defined using (a, b, q), but independent of it) between S and G_W, where G is the formal torus over $\mathbb{Z}_p$ with character group $\mathrm{Hom}_{\mathbb{Z}_p}(P_0, P_1)$, P_0 (respectively P_1) denoting the $\mathbb{Z}_p-$submodule (free of rank g) of $H^1(X \otimes k/W)$ consisting of the x such that Fx = x (respectively Fx = px). This isomorphism sends e to the identity element of G_W. One can show (see the Appendix and the following Exposé, by N. Katz) that this formal group structure on S coincides with the one defined, in the manner of Serre-Tate, by the versal formal lifting of the p-divisible group associated with $X \otimes k$. In particular, the abelian variety e^*X/W is the canonical lifting of $(X \otimes$ k)/k [14, V §3]. Recall that this lifting corresponds to the trivial lifting (i.e. as a product) of the p-divisible group associated with $X \otimes k$, or, equivalently, to the lifting of $\mathrm{Fil}^1_{\mathrm{Hdg}}\ H^1_{\mathrm{DR}}(X \otimes k/k)$ to the direct factor $P_1 \otimes W$ of slope 1 in $H^1(X \otimes k/W)$ (cf. 1.3.4).

2.1.4. Variant. Let X/A be a proper smooth curve with geometrically connected fibers of genus $g \geq 1$. Then $H = H^1_{\mathrm{DR}}(X/A)$, endowed with the Gauss-Manin connection ∇ and the Hodge filtration $\mathrm{Fil}^1\ H = H^0(X, \Omega^1_{X/A})$, is a Hodge F-crystal of level 1. If $X \otimes k$ is assumed ordinary, 1.4.2 applies and gives the results of Katz [8, §8]: if $(a_i)_{1 \leq i \leq g}$ is a basis of horizontal sections fixed by F in the unit subcrystal U, then one may take as the basis $(b_i)_{1 \leq i \leq g}$ of $\mathrm{Fil}^1\ H$ the basis dual to (a_i) under Poincaré duality ((a, b) is then a symplectic basis of H), because compatibility of the duality with F and ∇ immediately shows that the conditions of (1.2.4 (i)) are satisfied.

B) K3 surfaces.

2.1.5. Let X/A be a proper smooth formal scheme such that $X_k = X \otimes k$ is a K3 surface. Then, by (IV 2.2, 2.3), the

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A-module

$$H = H^2_{\mathrm{DR}}(X/A),$$

endowed with the Gauss-Manin connection ∇ and the Hodge filtration

$$H = \mathrm{Fil}^0 H \supset \mathrm{Fil}^1 H \supset \mathrm{Fil}^2 H \supset 0,$$

is a Hodge F-crystal over A_0. The submodules $\mathrm{Fil}^i H$ are free of ranks 22, 21, and 1 for $i = 0,1,2$, and the associated graded object $\mathrm{gr}^\bullet H$ is free. Moreover, the cup product

$$\langle , \rangle : H \otimes H \to H^4_{\mathrm{DR}}(X/A) \simeq A$$

is a perfect duality, horizontal for ∇, i.e. satisfying

$$\langle \nabla x, y\rangle + \langle x, \nabla y\rangle = \nabla\langle x, y\rangle,$$

and compatible with F, i.e. such that, for every lifting of Frobenius φ,

$$\langle F(\varphi)\varphi^* x, F(\varphi)\varphi^* y\rangle = F(\varphi)\varphi^*\langle x, y\rangle.$$

Observe that $F(\varphi)\varphi^*|_{H^4_{\mathrm{DR}}(X/A)} = p^2\alpha$, where α is an automorphism; that is, $H^4_{\mathrm{DR}}(X/A)(2)$ is a unit F-crystal (of rank 1), which we identify with the trivial F-crystal A by means of a horizontal basis fixed by F. Recall also that

$$\mathrm{Fil}^1 H = (\mathrm{Fil}^2 H)^\perp .$$

Finally, recall that, if $e_0 : A_0 \to k$ is the augmentation, the F-crystal $e_0^* H$ (1.1.4) is precisely the crystalline cohomology $H^2(X_k/W)$.

2.1.6. Suppose now that X_k is ordinary, which means, by definition, that the Newton polygon of $H^2(X_k/W)$ coincides with its Hodge polygon, i.e. has slopes (0,1,2) with multiplicities (1,20,1). Equivalently, Frobenius on $H^2(X_k, \mathcal{O})$ is nonzero. Since $e_0^* H = H^2(X_k/W)$ and $\mathrm{gr}^\bullet H$ is free, the F-crystal H is then ordinary in the sense of 1.3.3, and consequently admits a filtration by sub-F-crystals

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$$0 \subset U_0 \subset U_1 \subset U_2 = H$$

such that U_i/U_{i-1} is of the form $V_i(-i)$, where V_i is a unit F-crystal of rank 1, 20, and 1 for $i = 0,1,2$. Moreover, since U_i lifts $\mathrm{Fil}_i H_0$, compatibility of F with the duality implies

$$U_1 = U_0 \perp .$$

Finally, by (1.3.6), the filtrations (U_i) and $(\mathrm{Fil}^i H)$ are opposite; that is,

$$H = H_0 \oplus H_1 \oplus H_2,$$

with

$$H_0 = U_0, \qquad H_1 = U_1 \cap \mathrm{Fil}^1 H, \qquad H_2 = \mathrm{Fil}^2 H, \qquad H_1 \simeq U_1/U_0.$$

THEOREM 2.1.7. Assume $p \neq 2$. Let $X/A = W[[t_1, \dots, t_{20}]]$ be the universal formal deformation of an ordinary K3 surface X_k/k (IV 1.3). Then, with the notation of 2.1.5 and 2.1.6, there exist a basis $(a, b_1, \dots, b_{20}, c)$ of H adapted to the decomposition $H = H_0 \oplus H_1 \oplus H_2$ and satisfying $\langle a, c \rangle = 1$, and elements $q_i \in A$ $(1 \leq i \leq 20)$ having the following properties:

$$(i)\ q_i(0) = 1 \bmod \mathrm{pW}, \text{ and the } q_i \text{ define an isomorphism}$$

$$A \simeq W[[q_1 - 1, \dots, q_{20} - 1]]$$

(i.e. the forms $d \log q_i$ form a basis of $\Omega^1_{A/W}$).

$$\text{(ii)} \begin{cases} \nabla a = 0, \\ \nabla b_i = (d \log q_i) a \qquad (1 \leq i \leq 20), \\ \nabla c = -\sum (d \log q_i) b_i \vee , \end{cases}$$

where $(b_i \vee)$ denotes the basis of H_1 dual to (b_i) for the restriction to H_1 of the cup-product form.
(iii) If $\varphi : A \to A$ denotes the lifting of Frobenius such that $\varphi(q_i) = q_i^p$, then

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{ $F(\varphi)\varphi^* a = \mathrm{a}$,
$F(\varphi)\varphi^* b_i = p\, b_i$ $(1 \le i \le 20)$,
$F(\varphi)\varphi^* c = p^2 c$. }

By 2.1.6, the F-crystal U_1, endowed with the filtration $\mathrm{Fil}^i H \cap U_1$, is an ordinary Hodge F-crystal of level 1. Moreover, since X/A is the universal formal deformation of X_k, the map

$\mathrm{gr}^1 \nabla : T_{A/W} \to \mathrm{Hom}_A(H_1, H_0) = \mathrm{Hom}_A(\mathrm{gr}^1 H, \mathrm{gr}^0 H)$

is an isomorphism by (IV 2.4). Applying 1.4.2 and 1.4.7 to U_1, there therefore exist a basis $(a, b_1, \dots, b_{20})$ of U_1 adapted to the decomposition $U_1 = H_0 \oplus H_1$ and elements $q_i \in A$ satisfying conditions (i) through (iii) of 2.1.7. It remains to show that, if the basis c of H_2 is chosen so that $\langle a, c\rangle = 1$, the last formulas in (ii) and (iii) are satisfied. First, by the transversality condition (1.3.5.1), one has

$$\nabla c = \alpha \otimes c + \sum_{1 \le i \le 20} \beta_i \otimes b_i \vee .$$

The relation

$$0 = \nabla\langle a, c\rangle = \langle \nabla a, c\rangle + \langle a, \nabla c\rangle = \langle a, \nabla c\rangle$$

implies $\alpha = 0$. Then, from

$$0 = \nabla\langle b_i, c\rangle = \langle \nabla b_i, c\rangle + \langle b_i, \nabla c\rangle$$

and the second formula in (ii), one obtains $\beta_i = -\, d \log q_i$, which gives the last formula in (ii). On the other hand, from $F(\varphi)\varphi^* a = \mathrm{a}$ and the relation

$$\langle F(\varphi)\varphi^* a, F(\varphi)\varphi^* c\rangle = p^2\langle a, c\rangle = p^2,$$

one obtains $F(\varphi)\varphi^* c = p^2 c$, completing the proof of 2.1.7.

2.1.8. Let X_k/k be an ordinary K3 surface. Set

$$(2.1.8.1) \qquad P_0 = H^2(X_k/W)_{F=1}, \qquad P_1 = H^2(X_k/W)_{F=p}, \qquad P_2 = H^2(X_k/W)_{F=p^2},$$

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where the notation $M_{F=a}$ denotes $\{x \in M \mid Fx = ax\}$. By 1.3.4, P_i is a free $\mathbb{Z}_p$ − module of rank 1, 20, and 1 for $i = 0,1,2$, and the F-crystal $H^2(X_k/W)$ has a unique decomposition

$$(2.1.8.2) \qquad H^2(X_k/W) = ((W,\sigma)\otimes P_0) \oplus ((W,p\sigma)\otimes P_1) \oplus ((W,p^2\sigma)\otimes P_2).$$

Suppose moreover that $p \neq 2$. With the notation of 2.1.7, let

$$(2.1.8.3) \qquad e_q : A \to W$$

denote the Teichmüller lifting of the augmentation $e_0 : A_0 \to k$ corresponding to the lifting φ considered in (iii), and hence given by $e_q(q_i) = 1$. Then X induces a formal scheme $e_q^* X/W$, and

$$(2.1.8.4) \qquad e_q^* H = H^2_{\mathrm{DR}}(e_q^* X/W) = H^2(X_k/W).$$

Moreover, by (2.1.7 (iii)), $e_q^* a$ is a basis of P_0, $(e_q^* b_i)_{1\le i\le 20}$ is a basis of P_1, $e_q^* c$ is a basis of P_2, and $\langle e_q^* a, e_q^* c\rangle = 1$. In particular, the Hodge filtration on $H^2_{\mathrm{DR}}(e_q^* X/W)$ is given by

$$(2.1.8.5) \qquad \mathrm{Fil}^1 H^2_{\mathrm{DR}}(e_q^* X/W) = (W\otimes P_1)\oplus(W\otimes P_2), \qquad \mathrm{Fil}^2 H^2_{\mathrm{DR}}(e_q^* X/W) = W\otimes P_2.$$

We shall deduce from this:

PROPOSITION 2.1.9. Under the hypotheses of 2.1.7, the lifting e_q (2.1.8.3) is independent of the family $q = (q_i)$ defined in 2.1.7.

We shall therefore write e in place of e_q. By analogy with the theory of the canonical lifting of ordinary abelian varieties [14, V §3], the formal scheme $e^* X/W$, which we shall denote simply by X_W, will be called the canonical lifting of X_k.

2.1.10. To prove 2.1.9, we shall use a “linear” description of formal liftings over W of a K3 surface over k. Let Y/W be a proper flat formal scheme such that $Y_k = Y \otimes k$ is a K3 surface. Then $H^2_{\mathrm{DR}}(Y/W)$ is canonically identified with $H = H^2(Y_k/W)$, and $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y/W)$ is a line (by definition, a direct summand of rank 1) in H, isotropic (i.e. contained in its orthogonal), lifting $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_k/k)$.

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THEOREM 2.1.11. Suppose $p \neq 2$. Let Y_0 be a K3 surface over k. Set $H = H^2(Y_0/W)$. The map assigning to a proper flat formal scheme Y/W lifting Y_0 the line $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y/W)$ is a bijection from the set of liftings of Y_0 over W (i.e. the W-valued points of the formal moduli scheme of Y_0) to the set of liftings of $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k)$ as an isotropic line in H.

First observe that the two types of liftings under consideration (liftings of Y_0 and liftings of $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k)$) are rigid, i.e. have no infinitesimal automorphisms. Let Y_n be a lifting of Y_0 over W_{n+1}. The set of liftings of Y_n over W_{n+2} is a torsor under $H^1(Y_0, T_{Y_0/k})$. On the other hand, let L_n be an isotropic line in $H \otimes W_{n+1}$ lifting $L_0 = \mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k)$. Then the set of liftings of L_n as an isotropic line in $H \otimes W_{n+2}$ is a torsor under $\mathrm{Hom}(L_0, L_0{}^{\perp}/L_0)$: without the isotropy condition one would obtain a torsor under $\mathrm{Hom}(L_0, H_0/L_0)$, where $H_0 = H \otimes k = H^2_{\mathrm{DR}}(Y_0/k)$; since $p \neq 2$, imposing isotropy on the lifting gives the stated result (if $H \otimes W_{n+2} = L_{n+1} \oplus M_{n+1}$, with L_{n+1} isotropic and lifting L_n, an isotropic line lifting L_n is generated by a vector $x + p^{n+1} y$, where x is a basis of L_n, and one must have $2\langle x, y\rangle = 0 \bmod p$). Now $L_0{}^{\perp} = \mathrm{Fil}^1 H^2(Y_0/k)$, hence

$$\mathrm{Hom}(L_0, L_0{}^{\perp}/L_0) = \mathrm{Hom}(\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k), \mathrm{gr}^1 H^2_{\mathrm{DR}}(Y_0/k)),$$

and, by (IV 2.4), there is a canonical isomorphism (given by the cup product)

$$H^1(Y_0, T_{Y_0/k}) \simeq \mathrm{Hom}(\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k), \mathrm{gr}^1 H^2_{\mathrm{DR}}(Y_0/k)).$$

The conclusion of 2.1.11 follows readily.

REMARK 2.1.12. Note the analogy between 2.1.11 and the theory of liftings of p-divisible groups and abelian varieties (lifting the Hodge filtration in the “crystal” extension

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universal) [14].

Proof of 2.1.9. In view of 2.1.11, it is enough to observe that $e_q^* X$, and hence e_q, is completely characterized by the second equality in (2.1.8.5).

We shall now see that 2.1.7 gives a canonical formal-group structure over W on the formal moduli space $S = \mathrm{spf}(A)$, with origin the canonical lifting X_W (2.1.9). For this we shall need the following lemma:

LEMMA 2.1.13. Let $(a', b'_1, \dots, b'_{20}, c')$ be a basis of H and let $(q'_i)_{1 \le i \le 20}$ be a family of elements of A satisfying the same conditions (i), (ii), and (iii) of 2.1.7 as $(a, (b_i), c)$ and (q_i). Then there exist $\alpha \in \mathbb{Z}_p^*$ and $\beta = (\beta_{ij}) \in \mathrm{GL}(20, \mathbb{Z}_p)$ such that

$$
\begin{aligned}
(2.1.13.1) \qquad & \{ a' = \alpha a, \qquad c' = c/\alpha, \\
& q'_i = \prod_{1 \le j \le 20} q_j^{(\beta_{ji}/\alpha)}, \\
& b'_i = \sum_{1 \le j \le 20} \beta_{ji} b_j . \}
\end{aligned}
$$

First, since $(a', (b'_i), c')$ is a basis of H adapted to the decomposition $H = H_0 \oplus H_1 \oplus H_2$ and $\langle a', c' \rangle = 1$, there exist $\alpha \in A^*$ and $\beta = (\beta_{ij}) \in \mathrm{GL}(20,A)$ such that

$$
a' = \alpha a, \qquad c' = c/\alpha, \qquad b'_i = \sum_{1 \le j \le 20} \beta_{ji} b_j .
$$

By 2.1.9, write $e = e_q = e_{q'} : A \to W$ for the lifting (2.1.8.3). Since $\nabla a = \nabla a' = 0$, one has $d\alpha = 0$, so α is "constant," with value $e^*\alpha \in W$. Let φ' be the lifting of Frobenius such that $\varphi'(q'_i) = {q'_i}^p$. One has

$$
a' = F(\varphi')\varphi'^* a' = F(\varphi)\varphi^* a = \alpha^\sigma F(\varphi)\varphi^* a = \alpha^\sigma a = \alpha a,
$$

hence $\alpha^\sigma = \alpha$, i.e. $\alpha \in \mathbb{Z}_p$. On the other hand,

$$
\begin{aligned}
\nabla b'_i &= \sum\nolimits_j (d\beta_{ji}) \otimes b_j + \sum\nolimits_j \beta_{ji} (d \log q_j) \otimes a \\
&= (d \log q'_i) \otimes a' = (d \log q'_i) \otimes \alpha a,
\end{aligned}
$$

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hence

$$(*) \qquad d\beta_{ji} = 0 \qquad \text{for every } i, j,$$

and

$$(**) \quad \alpha\, d \log q'_i = \sum_j \beta_{ji}\, d \log q_j \qquad (1 \le i \le 20).$$

Formula (*) means that β_{ji} is constant, with value $e^*\beta_{ji} \in W$. But
$p\, b'_i = F(\varphi')\varphi'^* b'_i = \sum_j \beta_{ji}^{\sigma}\ F(\varphi')\varphi'^* b_j$.
Applying e^* and taking account of

$$e^* F(\varphi')\varphi'^* b_j = e^* F(\varphi)\varphi^* b_j = e^* p\, b_j,$$

one obtains $\beta_{ji}^{\sigma} = \beta_{ji}$ for every i, j, and hence $\beta \in \mathrm{GL}(20, \mathbb{Z}_p)$. From (**) one deduces that there is a $C \in W^*$ such that

$$q'_i = C \prod_j q_j^{(\beta_{ji}/\alpha)}.$$

Since $e^* q'_i = e^* q_i = 1$, one has $C = 1$, which completes the proof.
THEOREM 2.1.14. Suppose $p \neq 2$. Let X_k be an ordinary K3 surface over k, and let $S = \mathrm{Spf}(A)$ be the corresponding formal moduli space over W. Let G be the formal torus over $\mathbb{Z}_p$ whose character group is $\mathrm{Hom}_{\mathbb{Z}_p}(P_0, P_1)$, with the notation of 2.1.8. There is a canonical isomorphism between S and G_W under which the canonical lifting $X_W \in S$ (2.1.9) corresponds to the identity element $1 \in G_W$.
Let a = $(a, b, (q_i))$ be as in 2.1.7. The family (q_i) gives an isomorphism $u_a : S \simeq (\widehat{G}_m)^{20}_W = \mathrm{Spf}(W[[T_i - 1]])$, $T_i \mapsto q_i$, while the basis (a, b) gives an isomorphism $\nu_a : \mathbb{Z}_p^{20} \simeq \mathrm{Hom}(P_0, P_1)$, such that $\nu_a(x_1, \ldots, x_{20})(a) = \sum x_i b_i$. If a' = $(a', b', (q'_i))$ is another choice, there exist $\alpha \in \mathbb{Z}_p^*$ and $\beta = (\beta_{ij}) \in \mathrm{GL}(20, \mathbb{Z}_p)$ satisfying (2.1.13.1). Write

$$f : (\widehat{G}_m)^{20}_{\mathbb{Z}_p} \simeq (\widehat{G}_m)^{20}_{\mathbb{Z}_p}$$

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for the isomorphism given by $T_i \mapsto \prod_j T_j^{(\beta_{ji}/\alpha)}$. The isomorphism

$$\mathrm{Hom}(f, \widehat{G}_m) : \mathbb{Z}_p^{20} \simeq \mathbb{Z}_p^{20}$$

induced by f on character groups is multiplication by the matrix β/α. The formulas (2.1.13.1) give commutative squares

$$\begin{array}{ccccc} u_a & & \nu_a & & \\ S & \simeq \to & (\widehat{G}_m)^{20} & \mathbb{Z}_p^{20} & \simeq \to \mathrm{Hom}(P_0, P_1) \\ \| & & \downarrow f_W & \uparrow \beta/\alpha & \| \\ S & \simeq \to & (\widehat{G}_m)^{20} & \mathbb{Z}_p^{20} & \simeq \to \mathrm{Hom}(P_0, P_1). \\ u_{a'} & & \nu_{a'} & & \end{array}$$

In other words, the isomorphism $S \simeq G_W$ given by (u_a, ν_a) is independent of a; it is the asserted canonical isomorphism. By definition X_W corresponds to the augmentation $q_i \mapsto 1$, and 2.1.14 is proved.

DEFINITION 2.1.15. A system a = (a, b, c, q) satisfying the conditions of 2.1.7 will be called a system of canonical coordinates (or parameters) on S.

Observe that the choice of a corresponds to the choice of a basis of P_0 (and is therefore defined up to a factor in $\mathbb{Z}_p^*$) and determines c; on the other hand, once a has been chosen, the datum of b corresponds to the choice of a basis of P_1 (and is therefore defined up to a factor in $\mathrm{GL}(20, \mathbb{Z}_p)$) and determines the isomorphism $(q_i) : S \simeq (\widehat{G}_m)_W^{20}$.

Problems 2.1.16. a) Can one give an "intrinsic" definition of the formal-group structure on S constructed in 2.1.14, for example by exhibiting it as the representative of a suitable functor? (*)

b) Extend the theory of canonical coordinates to the case $p = 2$. This presents several difficulties. First, of course, one must find an adequate substitute for the definition of the q_{ij} in 1.4.2. On the other hand, in the K3 case there are additional difficulties connected with the fact that it is not known, for $p = 2$, whether the quadra-

(*) (added in January 1981), this question has just been answered affirmatively by Nygaard.

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tic form $\langle x, x\rangle$ on $H^2_{\mathrm{DR}}(X_k/k)$ is identically zero (recall that, if X is a K3 surface over the complex numbers, the form $\langle x, x\rangle$ on $H^2(X, \mathbb{Z})$ is even [17]). It can nevertheless be shown [4] that, for Y_0/k ordinary, or more generally if the kernel of the form $\langle x, x\rangle$ on $H^2_{\mathrm{DR}}(Y_0/k)$ is not equal to $\mathrm{Fil}^1 H^2_{\mathrm{DR}}(Y_0/k)$, the map considered in 2.1.11, assigning to a formal lifting Y the isotropic line $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y/W) \subset H^2(Y_0/W)$, is surjective with finite fibers: for Y_0 ordinary, the line $W \otimes P_2$ of (2.1.8.5) then defines a finite family of "canonical liftings" of Y_0.

2.2. Liftings of invertible sheaves.

2.2.1. Let $X/S = \mathrm{Spf}(A)$ be the universal formal deformation of a K3 surface X_k/k, and let L_0 be a nontrivial invertible sheaf on X_k. It is known (IV 1.6) that there is a largest closed formal subscheme $T = \Sigma(L_0) \subset S$ such that L_0 extends to an invertible sheaf L on X_T (*), and that T is defined by an equation $f = 0$ with p not dividing f. Assume $p \neq 2$ and X_k ordinary. We shall show that f can then be made explicit in terms of the crystalline Chern class

$$(2.2.1.1) \qquad c_1(L_0) \in H^2(X_k/k).$$

Recall (IV 2.9) that, with the notation of (2.1.8.1),

$$(2.2.1.2) \qquad c_1(L_0) \in P_1.$$

In particular, if (a, b, c, q) is a system of canonical coordinates on S (2.1.15), and $e : A \to W$ is the augmentation given by $q_i \mapsto 1$, one may write

$$(2.2.1.3) \qquad c_1(L_0) = \sum_{1 \le i \le 20} x_i (e^* b_i),$$

with $x_i \in \mathbb{Z}_p$. We then have the following result, whose proof will occupy the remainder of the exposé:

(*) $X_{S'}$ denotes the formal scheme obtained from X by the base change $S' \to S$.

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THEOREM 2.2.2. With the hypotheses and notation of 2.2.1, T is defined by the equation

$$(2.2.2.1) \qquad \prod_{1\leq i\leq 20} q_i^{x_i} = 1.$$

In other words, if, by means of the basis e^*a of P_0, $c_1(L_0)$ is identified with a character x of the formal torus S (2.1.14), T is the formal subgroup defined by

$$(2.2.2.2) \qquad T = \mathrm{Ker}(x).$$

In particular, if p does not divide $c_1(L_0)$ (which also means [16, 1.4] that p does not divide the class of L_0 in $\mathrm{NS}(X_k)$), T is a smooth formal subtorus over W of relative dimension 19.

2.2.3. Let us first indicate how one may heuristically convince oneself of the validity of 2.2.2. Over the complex numbers, T would be the locus where the horizontal section of H^2_{DR} through $c_1(L_0)$ remains of type (1,1), i.e. lies in $\mathrm{Fil}^1 H^2_{\mathrm{DR}}$. Let $\underline{x} \in H^2_{\mathrm{DR}}(X/S)\otimes K[[q_i-1]]$ be the horizontal section such that $e^*\underline{x} = c_1(L_0)$. An immediate calculation from (2.1.7 (ii)) gives

$$(2.2.3.1) \qquad \underline{x} = (-\sum_{1\leq i\leq 20} x_i \log q_i)a + \sum_{1\leq i\leq 20} x_i b_i.$$

Thus, heuristically, $-\sum x_i \log q_i = 0$, "i.e." $\prod q_i^{x_i} = 1$, is the desired equation. Naturally this argument is insufficient, but it can be adapted by using the crystalline Chern class to control, step by step, the obstruction to extending L_0.

For this purpose we shall need some reminders concerning Chern classes and obstructions, supplementing (IV 2.8, 2.9).

2.2.4. Let $i : Y_0 \hookrightarrow Y$ be a closed immersion of schemes (or formal schemes) defined by an ideal I, and let E_0 be an invertible sheaf on Y_0, of class $c\ell(E_0) \in H^1(Y_0, \mathcal{O}^*_{Y_0})$. The exact sequence of abelian sheaves on Y

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$$(2.2.4.1) \qquad 0 \to (1+I)^* \to \mathcal{O}_Y^* \to \mathcal{O}_{Y_0}^* \to 0$$

shows that the obstruction $\omega^*(E_0, i)$ to extending E_0 to an invertible sheaf on Y is the image of $c\ell(E_0)$ under the connecting morphism in the exact cohomology sequence deduced from (2.2.4.1):

$$(2.2.4.2) \qquad \omega^*(E_0, i) = d\, c\ell(E_0) \in H^2(Y, (1+I)^*).$$

If $Y \to Z$ is a morphism, one may consider the "multiplicative" de Rham complex

$$\Omega^{\cdot *}_{Y/Z} = (\mathcal{O}_Y^* \text{—dlog} \to \Omega^1_{Y/Z} \text{—}d \to \Omega^2_{Y/Z} \text{—}d \to \cdots),$$

and (2.2.4.1) refines to an exact sequence of complexes

$$(2.2.4.3) \qquad 0 \to (1+I)\Omega^{\cdot *}_{Y/Z} \to \Omega^{\cdot *}_{Y/Z} \to \mathcal{O}_{Y_0}^* \to 0,$$

where

$$(1+I)\Omega^{\cdot *}_{Y/Z} \quad := \quad ((1+I)^* \text{—dlog} \to \Omega^1_{Y/Z} \text{—}d \to \Omega^2_{Y/Z} \text{—}d \to \cdots).$$

The obstruction $\omega^*(E_0, i)$ is the image, under the natural map, of the class

$$(2.2.4.4) \qquad c^*(E_0, i, Y/Z) \in H^2(Y, (1+I)\Omega^{\cdot *}_{Y/Z})$$

obtained from $c\ell(E_0)$ as the connecting image in the exact cohomology sequence associated with (2.2.4.3).

2.2.5. Assume $I^2 = 0$. Then

$$(1+I)^* = 1 + I \quad \simeq \quad I, \qquad \log(1+x) = x,$$

and similarly $(1+I)\Omega^{\cdot *}_{Y/Z}$ is identified with

$$I\Omega^{\cdot}_{Y/Z} \quad := \quad (I \text{—}d \to \Omega^1_{Y/Z} \text{—}d \to \Omega^2_{Y/Z} \text{—}d \to \cdots).$$

Write

$$(2.2.5.1) \qquad \omega(E_0, i) \in H^2(Y, I), \qquad c(E_0, i, Y/Z) \in H^2(Y, I\Omega^{\cdot}_{Y/Z})$$

for the images of $\omega^*(E_0, i)$ and $c^*(E_0, i, Y/Z)$ under these identifications. The obstruction $\omega(E_0, i)$ is again the image of $c(E_0, i, Y/Z)$ under the natural map.

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Assume further that $Z = (Z, K, \delta)$ is a PD-scheme on which p is nilpotent [1], or that Z is a P-adic base (with $p \in P$) in the sense of [3, 7.17]: the case actually of interest to us is that in which Z is a p-adically complete formal W-scheme, endowed with the standard divided powers on $p\mathcal{O}_Z$. If Y is smooth over Z, and if the trivial divided powers γ on I (i.e. $\gamma^i(x) = 0$ for i>1) are compatible with those of Z, then

$$H^2(Y, I\Omega^{\cdot}_{Y/Z}) = H^2(Y_0/Z, J_{Y_0/Z})$$

(where $J_{Y_0/Z}$, as usual, denotes the crystalline ideal that is the kernel of $\mathcal{O}_{Y_0/Z} \to \mathcal{O}_{Y_0}$), and the class $c(E_0, i, Y/Z)$ of (2.2.5.1) is precisely the crystalline Chern class of L_0 relative to Y_0/Z, defined in [2]:

$$(2.2.5.2) \qquad c(E_0, i, Y/Z) = c_1(E_0)_{Y_0/Z} \in H^2(Y_0/Z, J_{Y_0/Z}).$$

We shall apply these remarks when there are Cartesian squares

$$\begin{array}{ccccc} & & i & & \\ Y_0 & \longrightarrow & Y & \longrightarrow & X \\ & & \downarrow f_0 & & \downarrow f \quad \downarrow \\ & & \downarrow & & \downarrow \quad \downarrow \\ Z_0 & \xrightarrow{j} & Z & \longrightarrow & S, \end{array}$$

(2.2.5.3)

where X/S is the universal formal deformation of X_k, Z is an affine p-adically complete formal W-scheme, and j is a closed immersion defined by a square-zero ideal J, so that $I = f^*(J)$. Since $R^1 f_{0*}(\mathcal{O}_{Y_0}) = 0$ and hence $H^1(Y_0, \mathcal{O}) = 0$, the canonical maps

$$H^2(Y, I) \to H^2(Y, \mathcal{O}), \qquad H^2(Y, I\Omega^{\cdot}_{Y/Z}) \to H^2_{\mathrm{DR}}(Y/Z)$$

are injective; we shall therefore regard $\omega(E_0, i)$ (respectively $c(E_0, i, Y/Z)$) as an element of $H^2(Y, \mathcal{O})$ (respectively $H^2_{\mathrm{DR}}(Y/Z)$). Moreover, if $p \neq 2$, or if $p\mathcal{O}_Z = 0$, the trivial divided powers on I are compatible with the standard divided powers on $p\mathcal{O}_Z$. From what has

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just been seen, the obstruction $\omega(E_0, i)$ is therefore given by the following recipe:

LEMMA 2.2.6. In the situation of (2.2.5.3), if $p \neq 2$, or if $p\mathcal{O}_Z = 0$, the obstruction $\omega(E_0, i)$ to extending E_0 to an invertible sheaf on Y is the image, under the canonical map $H^2_{\mathrm{DR}}(Y/Z) \to H^2(Y, \mathcal{O}_Y)$, of the crystalline Chern class of E_0 relative to Z, $c_1(E_0)_{Y_0/Z} \in H^2_{\mathrm{DR}}(Y/Z)$.

In other words, under the hypotheses of 2.2.6, E_0 extends to an invertible sheaf on Y if and only if

$$c_1(E_0)_{Y_0/Z} \in \mathrm{Fil}^1 H^2_{\mathrm{DR}}(Y/Z).$$

2.2.7. The usefulness of the recipe in 2.2.6 is that it gives a computational principle for the Chern class $c_1(E_0)_{Y_0/Z}$, which we now recall.

Under the hypotheses of 2.2.6, one may consider the Chern class

$$c_1(E_0)_{f_0/W} \in H^2(Y_0/W), \tag{2.2.7.1}$$

and its image

$$c_1(E_0)_{f_0/W} \in H^0(Z_0/W, R^2 f_{0*}(\mathcal{O}_{Y_0/W})) \tag{2.2.7.2}$$

under the canonical map

$$H^2(Y_0/W) \to H^0(Z_0/W, R^2 f_{0*}(\mathcal{O}_{Y_0/W})).$$

On the other hand, for every PD-thickening Z_1 of Z_0 one may consider the Chern class

$$c_1(E_0)_{Y_0/Z_1} \in H^2(Y_0/Z_1). \tag{2.2.7.3}$$

The relation between (2.2.7.2) and (2.2.7.3) is as follows: the crystal $R^2 f_{0*}(\mathcal{O}_{Y_0/W})$ has a "value" $R^2 f_{0*}(\mathcal{O}_{Y_0/W})(Z_1)$ on Z_1, and, for affine Z_1, the section

$$c_1(E_0)_{f_0/W}(Z_1) \in H^0(Z_1, R^2 f_{0*}(\mathcal{O}_{Y_0/W})(Z_1)) = H^2(Y_0/Z_1)$$

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defined by (2.2.7.2) is (2.2.7.3). Suppose now that the map $Z_0 \to S$ in (2.2.5.3) factors as

$$(2.2.7.4) \qquad Z_0 \xrightarrow{j'} Z' \to S,$$

where Z' is a smooth, p-adically complete formal W-scheme and j' is a closed immersion. Let $\widehat{Z'} = D_{Z_0}(Z')$ be the p-adically completed PD-envelope of Z_0 in Z'. By Berthelot [1] (or [3, §7]), the crystal $R^2 f_{0*}(\mathcal{O}_{Y_0/W})$ is described by an $\mathcal{O}_{\widehat{Z'}}$ – module with integrable connection relative to W. In the present case, since there are Cartesian squares

$$\begin{array}{ccccc} Y_0 & \longrightarrow & \widehat{Y'} & \longrightarrow & X \\ & & \downarrow & & \downarrow \quad \downarrow \\ & & \downarrow & & \downarrow \quad \downarrow \\ Z_0 & \longrightarrow & \widehat{Z'} & \longrightarrow & S, \end{array}$$

(2.2.7.5)

this module is none other than $H^2_{\mathrm{DR}}(X/S) \otimes \mathcal{O}_{\widehat{Z'}}$, equipped with the Gauss-Manin connection. The class (2.2.7.2) is interpreted as a horizontal section of this module. When, in the situation of (2.2.5.3), Z is a closed subscheme of Z', the universal property of PD-envelopes maps Z into $\widehat{Z'}$, and the section of $H^2_{\mathrm{DR}}(X/S) \otimes \mathcal{O}_Z = H^2_{\mathrm{DR}}(Y/Z)$ induced by (2.2.7.2) is precisely $c_1(E_0)_{Y_0/Z}$. We shall see below how to use the horizontality of the section (2.2.7.2) of $H^2_{\mathrm{DR}}(X/S) \otimes \mathcal{O}_{\widehat{Z'}}$ to calculate it when E_0 extends the given sheaf L_0 on X_k, using suitably chosen embeddings j' as in (2.2.7.4).

2.2.8. For this purpose we shall need one last, somewhat technical, ingredient concerning PD-envelopes. Write

$$(2.2.8.1) \qquad W\langle\langle t\rangle\rangle = W\langle\langle t_1, \dots, t_n\rangle\rangle$$

for the p-adically completed PD-envelope of the ideal $(t_1, \dots, t_n)$ of W[[t]] = $W[[t_1, \dots, t_n]]$: it is the subring of K[[t]] consisting of series $\sum a_i t^i/i!$ with $a_i \in W$ tending to 0. Now consider

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the ideal $t^r = (t_1^{r_1}, \dots, t_n^{r_n})$ of W[[t]], and the p-adically completed PD-envelope of t^r in W[[t]]:

$$(2.2.8.2) \qquad D_{(t^r)}(W[[t]]).$$

Using the compatibility of the formation of PD-envelopes with flat extensions [1, I 2.7.4] for the finite flat morphism W[[t]] $\to$ W[[t]], $t_i \mapsto t_i^{r_i}$, one obtains the following description of (2.2.8.2): its underlying additive group consists of formal series, with coefficients in W tending to 0, in the $t^{(m)}$, where

$$t^{(m)} = t^s (t^r)^{[q]}, \qquad \text{if } m = r\,q + s, \qquad 0 \le s < r,$$

and multiplication is the evident multiplication formally given by $(t^r)^{[q]} = t^{r\,q}/q!$.
The map

$$(2.2.8.3) \qquad D_{(t^r)}(W[[t]]) \to W\langle\langle t \rangle\rangle$$

defined by the inclusion $(t^r) \subset (t)$ sends the basis element $t^{(m)}$ of multi-degree m to $(m!/q!)t^{[m]}$; in particular, it is injective.
2.2.9. Proof of 2.2.2. Since L_0 is nontrivial, it is known (IV 3.4) that $c_1(L_0) \neq 0$. Let p^m be the greatest power of p dividing $x = c_1(L_0)$, so that $x = p^m y$ with $p \nmid y$. Thus y is part of a basis of the character group of S, and the coordinates (a, b, q) may be chosen so that $y = (1, 0, \dots, 0)$, i.e. $x = p^m(e^* b_1)$. Put

$$T' = \mathrm{Spf}(W[[q_i - 1]]/(q_1^{p^m} - 1)).$$

We must prove that $T = T'$. We proceed in four steps.
a) Extension of L_0 over $X_{\mathrm{Spec}(k[q_1-1]/(q_1^{p^m}-1))}$. This can be done in two ways. The quicker is to observe that, by [16, 1.4], since p^m divides $c_1(L_0)$, p^m also divides the class of L_0 in $\mathrm{NS}(X_k)$. If F denotes Frobenius on X_k, L_0 is therefore the inverse image under F^m of an invertible sheaf L_0', and

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consequently L_0 extends to every infinitesimal k-neighborhood of X_k of order $\leq p^m - 1$. One may also proceed directly, extending L_0 step by step over X_{Y_n}, where $Y_n = \mathrm{Spec}(k[t_1]/t_1^n)$ and $q_1 - 1 = t_1$. Assume $m > 0$ and that L_0 has been extended to an invertible sheaf L_{Y_n} on X_{Y_n} for $n < p^m$; we show that L_{Y_n} extends to an invertible sheaf on $X_{Y_{n+1}}$. Let $c(L_{Y_n})_{Y_{n+1}} \in H^2_{\mathrm{DR}}(X_{Y_{n+1}}/Y_{n+1})$ be the crystalline Chern class of L_{Y_n}. By 2.2.6, it is enough to show that the image of this class in $H^2(X_{Y_{n+1}}, \mathcal{O})$ is zero. Now Y_n is the formal subscheme of $\mathrm{Spf}(W[[t_1]])$ defined by the ideal (p, t_1^n), and, by 2.2.7, if D_n denotes the p-adically completed PD-envelope of this ideal in $W[[t_1]]$, i.e. $D_n = D_{(t_1^n)}(W[[t_1]])$, then $c(L_{Y_n})_{Y_{n+1}}$ is induced by the horizontal section $c(L_{Y_n})_{D_n} \in H^2_{\mathrm{DR}}(X/S) \otimes D_n$ of the type in (2.2.7.2). Since, by 2.2.8, the map $D_n \to D_1 = W\langle\langle t_1 \rangle\rangle$ is injective, $c(L_{Y_n})_{D_n}$ is determined by its image in $H^2_{\mathrm{DR}}(X/S) \otimes D_1$. That image is precisely the horizontal section $c(L_0)_{D_1}$ defined by L_0: this follows from the functoriality of the class (2.2.7.1), in view of the commutative diagram

$$\begin{array}{ccc} Y_n & \longrightarrow & \mathrm{Spf}(W[[t_1]]) \\ & & \downarrow \quad \nearrow \quad \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(W) \end{array}$$

and the fact that L_{Y_n} extends L_0. Since $c(L_0)$ is the horizontal class extending $c_1(L_0) = p^m e^* b_1$, an immediate calculation using 2.1.7 gives

$$c(L_0)_{D_1} = -(p^m \log q_1) a_{D_1} + p^m (b_1)_{D_1}$$

(where $(a_{S'}, b_{S'})$ denotes the inverse image of (a, b) on S' for $S' \to S$). Hence, by the injectivity of $D_n \to D_1$,

$$c(L_{Y_n})_{D_n} = -(\log q_1^{p^m}) a_{D_n} + p^m (b_1)_{D_n}$$

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(observe that $q_1^{p^m} - 1$ belongs to the ideal generated by p and $t_1^n = (q_1 - 1)^n$, so that the logarithm is meaningful by virtue of the divided powers on (p, t_1^n)). The map $D_n \to \mathcal{O}_{Y_{n+1}}$ defined by the universal property of PD-envelopes sends $x^{[i]}$ (for $x \in (p, t_1^n)$) to x mod (p, t_1^{n+1}) for $i = 1$ and to 0 for $i \geq 2$. In particular, it sends $\log q_1^{p^m}$ to $(q_1^{p^m} - 1)$ mod $(p, t_1^{n+1}) = 0$, since $n < p^m$. In other words, the image $c(L_{Y_n})_{Y_{n+1}} \in H^2_{\mathrm{DR}}(X/S) \otimes \mathcal{O}_{Y_{n+1}}$ of $c(L_{Y_n})_{D_n}$ lies in Fil^1; therefore, by what was said above, L_{Y_n} extends to an invertible sheaf on $X_{Y_{n+1}}$.

b) Extension of L_0 over X_Z, where $Z =$ $\mathrm{Spf}(W[[q_1 - 1]]/(q_1^{p^m} - 1))$. Let L_Y denote the extension, obtained in a), of L_0 over X_Y. Put

$$Z_n = \mathrm{Spec}(W_n[[q_1 - 1]]/(q_1^{p^m} - 1)).$$

Assume that L_Y has been extended to an invertible sheaf L_{Z_n} on X_{Z_n}; we show that L_{Z_n} extends to an invertible sheaf on $X_{Z_{n+1}}$. By 2.2.6 (which applies because $p \neq 2$), it is enough to verify that the Chern class $c(L_{Z_n})_{Z_{n+1}} \in H^2_{\mathrm{DR}}(X_{Z_{n+1}}/Z_{n+1})$ lies in Fil^1. It is induced by the class $c(L_{Z_n})_D \in H^2_{\mathrm{DR}}(X_D/D)$, where D is the p-adically completed PD-envelope of Y_n in $W[[q_1 - 1]]$. If $D_I(\,)$ denotes the p-adically completed PD-envelope of the ideal I, then

$D \coloneqq D_{(p^n, q_1^{p^m} - 1)}(W[[q_1 - 1]])$
$= D_{(p, q_1^{p^m} - 1)}(W[[q_1 - 1]])$
$= D_{(p, (q_1 - 1)^{p^m})}(W[[q_1 - 1]])$
$= D_{(t_1^{p^m})}(W[[t_1]])$

where $t_1 = q_1 - 1$. By 2.2.8, the map $D \to D_{(t_1)}(W[[t_1]]) = W\langle\langle t_1 \rangle\rangle$ defined by the inclusion $(p^n, q_1^{p^m} - 1) \subset (p, q_1 - 1)$ is therefore injective, and the same argument as in a) shows that

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$$c(L_{Z_n})_D = -(\log q_1^{p^m})a_D + p^m(b_1)_D.$$

Consequently, $c(L_{Z_n})_{Z_{n+1}}$, the image of $c(L_{Z_n})_D$ under the map $D \to \mathcal{O}_{Z_{n+1}}$ defined by the universal property of PD-envelopes (which sends $x^{[i]}$ (for $x \in (p^n, q_1^{p^m} - 1)$) to the image of x in $\mathcal{O}_{Z_{n+1}}$ for $i = 1$ and to 0 for $i > 1$), is

$$\begin{aligned} c(L_{Z_n})_{Z_{n+1}} &= -(q_1^{p^m} - 1)a_{Z_{n+1}} + p^m(b_1)_{Z_{n+1}} \\ &= p^m(b_1)_{Z_{n+1}} \end{aligned}$$

(since $q_1^{p^m} - 1 = 0$ on Z_{n+1}). Thus $c(L_{Z_n})_{Z_{n+1}}$ belongs to $\mathrm{Fil}^1 H^2_{\mathrm{DR}}(X_{Z_{n+1}}/Z_{n+1})$, and consequently L_{Z_n} extends to an invertible sheaf on $X_{Z_{n+1}}$. Hence L_Y extends to an invertible sheaf L_Z on X_Z.

c) Extension of L_0 over $X_{T'}$. Put $t_i = q_i - 1$. Let $n = (n_2, \dots, n_{20})$ be a sequence of integers ≥ 1, and put

$$T'_n = \mathrm{Spf}(W[[t]]/(q_1^{p^m} - 1, t_2^{n_2}, \dots, t_{20}^{n_{20}}))$$

(so $T'_{(1,\dots,1)} = Z$). Assume L_Z has been extended to an invertible sheaf $L_{T'_n}$ on $X_{T'_n}$. By an argument analogous to that used in *b*), if $2 \leq i \leq 20$ and $n + 1_i = (n_2, \dots, n_{i-1}, n_i + 1, n_{i+1}, \dots, n_{20})$, then $L_{T'_n}$ extends to an invertible sheaf on $X_{T'_{(n+1_i)}}$. Induction therefore shows that L_Z extends to an invertible sheaf $L_{T'}$ on $X_{T'}$.

d) End of the proof. To prove $T' = T$, it remains to check that if $T'' \supset T'$ is a closed formal subscheme of S, defined by an ideal I, such that $L_{T'}$ extends to an invertible sheaf on $X_{T''}$, then $T'' = T'$. Equivalently, if $T'' \neq T'$ and $I^2 = 0$, the obstruction to extending $L_{T'}$ to an invertible sheaf on $X_{T''}$ must be nonzero, i.e. the Chern class $c(L_{T'})_{T''} \in H^2_{\mathrm{DR}}(X_{T''}/T'')$ must not belong to Fil^1. But the same

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calculation as before gives

$$c(L_{T'})_{T''} = -(q_1^{p^m} - 1)a_{T''} + p^m(b_1)_{T''}.$$

Since $T'' \neq T'$, the image of $q_1^{p^m} - 1$ in T'' is nonzero; hence $c(L_{T'})_{T''} \notin \mathrm{Fil}^1$, which completes the proof.

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P. DELIGNE
Institut des Hautes Etudes
Scientifiques
35, Route de Chartres
91440 BURES/YVETTE (France)

L. ILLUSIE
Université de Paris-Sud
Centre d'Orsay
Mathématique, bât. 425
91405 ORSAY (France)

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APPENDIX TO EXPOSE *V*

Nicholas $M.$ Katz

A1. UNIQUENESS OF GROUP STRUCTURES

Let k be a perfect field, $W = W(k)$ its ring of Witt vectors, and $\sigma : W \cong \to W$ the absolute Frobenius automorphism of W. Let M be a finite-dimensional formal Lie variety over W, i.e. $M = \mathrm{Spf}(A)$ with A non-canonically isomorphic to $W[[T_1, \dots, T_n]]$, $n = \dim M$. Suppose we are given a W-morphism of formal Lie varieties

$$\Phi : M \to M^{(\sigma)}$$

whose reduction modulo p is the absolute Frobenius endomorphism

$$\mathrm{Frob} : M \otimes_W k \to (M \otimes_W k)^{(\sigma)}.$$

UNIQUENESS LEMMA A1.1.(1) Given (M, Φ) as above, there exists at most one structure of commutative formal Lie group over W on the formal Lie variety M for which the given map $\Phi : M \to M^{(\sigma)}$ is a group homomorphism. (2) If this structure exists, it makes M into a toroidal formal group, and the given $\Phi : M \to M^{(\sigma)}$ is the unique group homomorphism lifting Frobenius. (3) If (M_1, Φ_1) and (M_2, Φ_2) both admit group structures as in (2) above, then a morphism

$$f : M_1 \to M_2$$

of formal Lie varieties over W is a group homomorphism if and only if the diagram

$$\begin{array}{ccc} & f & \\ M_1 & \longrightarrow & M_2 \\ \downarrow \Phi_1 & & \downarrow \Phi_2 \\ \downarrow & & \downarrow \\ M_1^{(\sigma)} & -f^{(\sigma)} \to & M_2^{(\sigma)} \end{array}$$

commutes.

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PROOF. We begin by proving (2). Thus let G be a finite-dimensional commutative formal Lie group over W, together with a homomorphism

$$\Phi : G \to G^{(\sigma)}$$

which lifts Frobenius. We must show that G is toroidal, and that Φ is unique. By the rigidity of toroidal groups, it suffices to show that $G \otimes k$ is toroidal, and for this it suffices to show that Ker(Frob) is toroidal. For this, we first observe that $\Phi : G \to G^{(\sigma)}$ is finite (because it is finite modulo p, being a lifting of Frobenius) and flat (because it is a finite morphism between regular local rings of the same dimension). Therefore Ker(Φ) is a finite flat commutative group-scheme over W whose reduction mod p is Ker(Frob). According to Fontaine, if we denote by N the contravariant Dieudonné module of Ker(Frob), then the lifting Ker(Φ) is described by a W-submodule $L \subset N$ which satisfies

$$a) \quad L/\mathrm{pL} \cong N/\mathrm{FN}$$
$$b) \quad V|L \text{ is injective.}$$

But N is killed by F, and is of finite length over W. Therefore a) implies that $L = N$, and b) then shows that V is injective, and hence bijective on N. Therefore Ker(Frob) is toroidal, as required.
We next prove (3). By extending scalars $W(k) \to W(\overline{k})$, we may suppose k algebraically closed. Then M_1 and M_2 become isomorphic to products of $\widehat{G}_m$, and our commutative diagram - in the category of formal Lie varieties - becomes

$$\begin{array}{ccc} & f & \\ (\widehat{G}_m)^{n_1} & \longrightarrow & (\widehat{G}_m)^{n_2} \\ \downarrow p & & \downarrow p \\ \downarrow & & \downarrow \\ (\widehat{G}_m)^{n_1} & -f^{(\sigma)} \longrightarrow & (\widehat{G}_m)^{n_2}. \end{array}$$

If f is a group homomorphism, then $f = f^{(\sigma)}$, and the diagram commutes. To prove the converse, we argue as follows.
In terms of "multiplicative" coordinates T on $(\widehat{G}_m)^{\mathrm{hi}}$, $i = 1,2$, our

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hypothesis on f is:

$$f^{(\sigma)}(T^p) = (f(T))^p, \quad f(1) \equiv 1 \bmod p.$$

Iterating, we find

$$f^{(\sigma^n)}(T^{\mathrm{pn}}) = (f(T))^{\mathrm{pn}},$$

so in particular

$$f^{(\sigma^n)}(1) = (f(1))^{\mathrm{pn}} \to 1 \text{ as } n \to \infty.$$

Therefore f(1) = 1. Now take logarithms, i.e. let

$$F : (\widehat{G}_a)^{n_1} \to (\widehat{G}_a)^{n_2}$$

be the unique pointed morphism (over the fraction field of W) for which

$$F(\log T) = \log(f(T));$$

then we have

$$F^{(\sigma)}(\mathrm{pX}) = \mathrm{pF}(X).$$

Therefore F is linear, and has coefficients in Q_p. As these coefficients are intrinsically the matrix entries of the tangent map of f at the origin, they lie in W as well, hence F is a linear map with coefficients in $Z_p = W \cap Q_p$. Therefore f(T) = exp(F(log T)) is a homomorphism.

Finally, we obtain (1) as the special case $(M_1, \Phi_1) = (M_2, \Phi_2)$, $f = \mathrm{id}$, of (3). Q.E.D.

COROLLARY A1.2. Suppose that k is algebraically closed, and that (M, Φ) admits a group structure as above. Then

(1) The character group $X(M) = \mathrm{Hom}_{W-\mathrm{gp}}(M, \widehat{G}_m)$ is a free $Z_p -$ module of rank $n = \dim M$, and the natural map (of functors in $Z_p -$ modules)

$$M \to \mathrm{Hom}_{Z_p}(X(M), \widehat{G}_m)$$

is an isomorphism.

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(2) Let q be a function on M (i.e. $q \in A$ where $M = \mathrm{Spf}(A)$) with $q = 1$ modulo the maximal ideal of A. Then $q \in X(M)$ if and only if q transforms under Φ by

$$\Phi^*(q^{(\sigma)}) = q^p.$$

(3) Let ω be a (continuous) one-form on A, i.e. $\omega \in (\Omega^1_{A/W})^{\mathrm{contin}}$. Then $\omega = \mathrm{dq}/q$ with $q \in X(M) = \mathrm{Hom}(M, \widehat{G}_m)$ if and only if

$$\Phi^*(\omega^{(\sigma)}) = p\omega.$$

If ω satisfies this, the associated $q \in X(M)$ is unique; it is given by the formula

$$q(X) = \exp(\textstyle\int_O^X \omega), \quad O = \text{the origin in } M.$$

(4) Let $\tau \in A \,\widehat{\otimes}_W$ W[1/p] be a function on $M \otimes_W$ W[1/p]. Then $\tau = \log(\mathrm{q})$ for some $q \in X(M)$ if and only if τ satisfies

$$\tau(O) = 0, \quad \Phi^*(\tau^{(\sigma)}) = p\tau.$$

If τ satisfies this, then the associated q is unique; it is given by the formula

$$q = \exp(\tau).$$

(5) Let $q_1, \ldots, q_n$ be $n = \dim M$ elements of $X(M)$, $\omega_1, \ldots, \omega_n$ the corresponding differentials $\omega_i = \mathrm{dq}_i/q_i$ and $\tau_1, \ldots, \tau_n$ the corresponding "functions with denominators" $\tau_i = \log q_i$. Then the following conditions are equivalent:

a) $q_1, \ldots, q_n$ form a Z_p base of $X(M)$

b) the natural map

$$M \to \mathrm{Spf}(W[[q_1 - 1, \ldots, q_n - 1]])$$

is an isomorphism

c) $\omega_1, \ldots, \omega_n$ form an $A-$base of $(\Omega^1_{A/W})^{\mathrm{contin}}$

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$$d)\ \omega_1, \ldots, \omega_n \text{ form a } Z_p - \text{base of}$$

$$\{\omega \in \Omega^1 \mid \Phi^*(\omega^{(\sigma)}) = p\omega\}$$

$$e)\ \tau_1, \ldots, \tau_n \text{ form a } Z_p - \text{base of}$$

$$\{\tau \in A \hat{\otimes} W[1/p] \mid \tau(O) = 0,\ \Phi^*(\tau^{(\sigma)}) = p\tau\}.$$

PROOF. Because M is toroidal, and k is algebraically closed, M is (non-canonically) isomorphic to $(\widehat{G}_m)^n$. This makes (1) obvious. Assertion (2) is a particular case of part (3) of the uniqueness lemma, namely $M_1 = M$, $M_2 = \widehat{G}_m$. Assertion (3) becomes obvious if we choose a Z_p – basis $q_1, \ldots, q_n$ of $X(M)$, i.e. if we choose an isomorphism

$$M \cong (\widehat{G}_m)^n,$$

and write ω as an A-linear combination of the differentials dq_i/q_i:

$$\omega = \Sigma f_i \, \mathrm{dq}_i/q_i.$$

The condition

$$\Phi^*(\omega^{(\sigma)}) = p\omega$$

means precisely that the coefficient functions f_i each satisfy

$$\Phi^*(f_i^{(\sigma)}) = f_i,$$

which in turn implies that each coefficient function f_i is simply a constant in Z_p. Therefore

$$\omega = \mathrm{dq}/q \quad \text{for} \quad q = \prod (q_i)^{\mathrm{fi}};$$

because q(O) = 1, we obtain by integration the formula

$$\exp(\textstyle\int_O^X \omega) = q(X).$$

For assertion (4), first note that the Dieudonné-Dwork integrality criterion (cf. 1.4.4) guarantees that the series q, defined as

$$q \quad \mathrm{dfn} = \quad \exp(\tau)$$

is integral (i.e. lies in A) and lies q(O) = 1. From the equality

$$\Phi^*(\tau^{(\sigma)}) = p\tau$$

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(and not simply their congruence modulo pA) we see that

$$\Phi^*(q^{(\sigma)}) = q^p,$$

and we conclude by (2) that q lies in $X(M)$.
In assertion (5), the equivalence of a), b) and c) is physically obvious for $(\widehat{G_m})^r$, and the equivalence of a) with d) and e) is obvious from parts 3) and 4) above. Q.E.D.

A2. SHARPENINGS OF 1.4

COROLLARY A2.1 OF 1.4.2. With the hypotheses and notations of 1.4.2, suppose that there exists a lifting Φ_{can} of Frobenius on $\mathrm{Spf}(A)$ such that

$$F(\Phi_{\text{can}})(\Phi^*_{\text{can}}(\mathrm{Fil}^1)^{(\sigma)}) \subset \mathrm{Fil}^1 .$$

Let O denote the W-valued point of $\mathrm{Spf}(A)$ which is the $\Phi_{\text{can}}-$ Teichmuller representative of the augmentation $A \to k$. Then in formulaire 1.4.2.1-7 we have the further precisions

$$\begin{cases} \nabla a_i = 0 \\ \nabla b_i = \sum_j \eta_{ij} \otimes a_j \\ F(\Phi_{\text{can}})(\Phi^*_{\text{can}}(a_i^{(\sigma)})) = a_i \\ F(\Phi_{\text{can}})(\Phi^*_{\text{can}}(b_i^{(\sigma)})) = p\, b_i \\ \Phi^*_{\text{can}}(\eta_{ij}^{(\sigma)}) = p\eta_{ij}, \quad d\eta_{ij} = 0 \end{cases}$$

$$\begin{cases} d\tau_{ij} = \eta_{ij} \\ \Phi^*_{\text{can}}(\tau_{i,j}^{(\sigma)}) = p\tau_{ij} \\ \tau_{ij}(O) = 0 \end{cases}$$

$$\begin{cases} q_{ij} = \exp(\tau_{ij}) \text{ is defined, lies in } A, \text{ and} \\ q_{ij}(O) = 1 \\ \Phi^*_{\text{can}}(q_{ij}^{(\sigma)}) = q_{ij}^p. \end{cases}$$

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FURTHER COROLLARY A2.2. (Analogue of (4.7)(1) Given Φ_{can} as above, suppose in addition that (1.4.6.3) : $T_{A/W} \to \text{Hom}_A(\text{Fil}^1, U)$ is an isomorphism. Then $(\text{Spf}(A), \Phi_{\text{can}})$ admits a group structure, and the morphism

$$\text{Spf}(A) \to \prod_{i,j} \widehat{G}_m$$

defined by the $q_{i,j}$ is an isomorphism of groups.

(2) If $p \neq 2$, then Φ_{can} is the unique lifting of Frobenius such that

$$F(\Phi_{\text{can}})(\Phi_{\text{can}}^*(\text{Fil}^1)^{(\sigma)}) \subset \text{Fil}^1 .$$

PROOF. The formulaire is simply obtained from the one given in 1.4.2.1-7 by noting that the $u_{i,j}(\Phi_{\text{can}})$ all vanish. The first assertion of the further corollary has already been proven when $p \neq 2$, but the condition $p \neq 2$ was used only to assure that $\exp(\tau_{i,j}(O))$ make sense. As our hypotheses on Φ_{can} give $\tau_{i,j}(O) = 0$, this problem will not arise, and the proof given goes through tel quel. The unicity of Φ_{can} in case $p \neq 2$ resulte from the observation that the series $q_{\text{ij}} = \exp(\tau_{\text{ij}})$ are then definible without reference to a particular choice of Φ, and furnish an isomorphism

$$M \cong \text{Spf}(W[[q_{\text{ij}} - 1]]);$$

our Φ_{can} is the lifting of Frobenius given by

$$\Phi_{\text{can}}(q_{\text{ij}}^{(\sigma)}) = (q_{\text{ij}})^p . \quad Q.E.D.$$

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A3. FORMAL MODULI OF ORDINARY ABELIAN VARIETIES ; THE SERRE-TATE & DWORK GROUP STRUCTURES

Let k be an algebraically closed field of characteristic $p > 0$, and X_o/k an ordinary abelian variety over k. Let M be its formal deformation space, and X/M the corresponding formal abelian scheme. One knows that M is a $g^2 -$ dimensional formal Lie variety over $W = W(k)$. Because X_o/k is ordinary, the formal group $\widehat{X}$ of X/M is non-canonically isomorphic to $(\widehat{G}_m)^g$ over M. The "canonical subgroup" $H_{\text{can}} \subset X$ is defined to be the kernel of $[p]$ in $\widehat{X}$; it is the unique finite flat subgroup-scheme of X/M which mod p becomes the kernel of the relative Frobenius endomorphism of $X \otimes_W k/M \otimes_W k$. We denote by

$$F_{\text{can}} : X \to X/H_{\text{can}}$$

the projection onto the quotient by H_{can}. The quotient X/H_{can} over M is a deformation of $X_o^{(p)}/k$, so its "classifying map" is a morphism of formal Lie varieties

$$\Phi_{\text{can}} : M \to M^{(\sigma)},$$

"defined by" an isomorphism of formal abelian schemes over M

$$\Phi_{\text{can}}^*(X^{(\sigma)}) \cong X/H_{\text{can}}.$$

Thus Φ_{can} is a lifting of the absolute Frobenius endomorphism of $M \otimes k$, and

$$F_{\text{can}} : X \to X/H_{\text{can}} \cong \Phi_{\text{can}}^*(X^{(\sigma)}) \quad \swarrow \text{ (to } M\text{)}$$

is a lifting of the relative (to $M \otimes k$) Frobenius endomorphism of $X \otimes k/M \otimes k$. Therefore this morphism induces on H^1_{DR} an M-linear map

$$F^*_{\text{can}} : \Phi^*_{\text{can}}\, \sigma^*\, H^1_{\text{DR}}(X/M) \to H^1_{\text{DR}}(X/M)$$

which is none other than the crystalline map $F(\Phi_{\text{can}})$. Because F_{can} is

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a physical morphism, the induced map F^*_{can} respects the Hodge filtration of H^1_{DR}'s. Therefore we have

$$F(\Phi_{\text{can}})(\Phi^*_{\text{can}}(\text{Fil}^1)^{(\sigma)}) \subset \text{Fil}^1 .$$

THEOREM A3.1. The structure of group imposed on M by the Serre-Tate description of M as

$$\text{Ext}(X_o(p^\infty)^{\text{et}},\ X_o(p^\infty)^{\text{conn}})$$

coincides with the structure of group on M for which the $q_{\text{ij}} = \exp(\tau_{\text{ij}})$ define an isomorphism of groups

$$M \cong \prod_{i,j} \widehat{G}_m .$$

PROOF. The morphism $\Phi_{\text{can}} : M \to M^{(\sigma)}$ is also a group homomorphism for the Serre-Tate group structure on M. Q.E.D.

A4. FORMAL MODULI OF ORDINARY K3 SURFACES ; SHARPENING OF 2.1.7, 2.1.14

COROLLARY A4.1 (of 2.1.7, 2.1.14). With the hypotheses and notations of 2.1.7 and 2.1.14, there exists a unique morphism

$$\Phi_{\text{can}} : \text{Spf}(A) \to \text{Spf}(A)^{(\sigma)}$$

lifting Frobenius for which the induced crystalline map $F(\Phi_{\text{can}})$ on $H^2_{\text{DR}}(X/A)$

$$F(\Phi_{\text{can}}) : \Phi^*_{\text{can}}\,\sigma^*\,H^2_{\text{DR}}(X/A) \to H^2_{\text{DR}}(X/A)$$

preserves the Hodge filtration, i.e. satisfies

$$\begin{cases} F(\Phi_{\text{can}})\Phi^*_{\text{can}}((\text{Fil}^2)^{(\sigma)}) \subset \text{Fil}^2 \\ F(\Phi_{\text{can}})\Phi^*_{\text{can}}((\text{Fil}^2)^{(\sigma)}) \subset \text{Fil}^1 . \end{cases}$$

The group structure on $\text{Spf}(A)$ defined by $q_1, \ldots, q_{20}$ is the unique one for which Φ_{can} is a group homomorphism.

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PROOF. The proof of 2.1.7 shows that any Φ_{can} whose associated $F(\Phi_{\text{can}})$ preserves the Hodge filtration satisfies

$$\Phi_{\text{can}}^{*}(q_i^{(\sigma)}) = (q_i)^p \quad \text{for} \quad i = 1, \ldots, 20.$$

Therefore Φ_{can} is a group homomorphism; as it is completely specified by its effect on the q_i, it is unique. Part (iii) of 2.1.7 shows that such a Φ_{can}, preserving the Hodge filtration, does in fact exist. Q.E.D.

EXPOSÉ IV

LIFTING K3 SURFACES TO CHARACTERISTIC ZERO

by P. DELIGNE

(written up by Luc ILLUSIE)*

Rudakov and Shafarevitch [14] (see also Nygaard [12]) established that K3 surfaces admit no nonzero vector fields.

The purpose of this exposé is to prove, from this, that every polarized K3 surface in characteristic $p > 0$ lifts to characteristic zero (1.8). The proof rests on the fact that the Rudakov–Shafarevitch result entails, for K3 surfaces, the existence of a good deformation theory, making it possible in particular to "control" the locus in the versal formal space along which a given nontrivial invertible sheaf deforms. The only ingredients used are, on the one hand, the usual relations in de Rham cohomology between the Gauss–Manin connection and the Kodaira–Spencer operation, and, on the other hand, the existence of a crystalline Chern class for invertible sheaves.

We denote by k an algebraically closed field of characteristic $p > 0$, and write $W = W(k)$ for the ring of Witt vectors over k.

*Research Team associated with the C.N.R.S., no. 653.

1. STATEMENT OF THE LIFTING THEOREM

In this section, X_0 denotes a K3 surface over k.

PROPOSITION 1.1.

a) The Hodge spectral sequence

$$E_1^{ij} = H^j(X_0, \Omega^i_{X_0/k}) \Longrightarrow H^*_{\mathrm{DR}}(X_0/k)$$

degenerates at E_1, and the matrix of Hodge numbers $h^{ij} = \dim_k H^j(X_0, \Omega^i_{X_0/k})$ is

$$\begin{matrix} 1 & 0 & 1 \\ 0 & 20 & 0. \\ 1 & 0 & 1 \end{matrix}$$

b) Let $T = T_{X_0/k} = (\Omega^1_{X_0/k})^\vee$ be the tangent bundle of X_0. Then $H^i(X_0, T) = 0$ for $i = 0$ or 2, and $\dim_k H^1(X_0, T) = 20$.

c) The crystalline cohomology W-modules $H^i(X_0/W)$ are free, of ranks $1, 0, 22, 0, 1$ for $i = 0, 1, 2, 3, 4$.

By [14], $H^0(X_0, T) = 0$. By the definition of a K3 surface, $\Omega^2_{X_0/k}$ is trivial; hence

$$T_{X_0/k} \simeq \Omega^1_{X_0/k}, \tag{1.1.1}$$

and consequently $H^0(X_0, \Omega^1_{X_0/k}) = 0$, so that $H^2(X_0, \Omega^1_{X_0/k}) = 0$ by Serre duality. Moreover, $H^1(X_0, \mathcal{O}) = 0$ and $H^2(X_0, \mathcal{O}) \simeq k$ by the definition of a K3 surface. Thus the table of the Hodge numbers h^{ij}, for $(i, j) \neq (1, 1)$, is the one displayed in 1.1(a). It follows that the Hodge spectral sequence degenerates at E_1. Recall also (SGA 5, VII 4.11) that

$$\sum (-1)^{i+j} h^{ij} = c_2,$$

and that $c_2 = 24$ by Riemann–Roch. Hence $h^{11} = 20$, which gives (b), in view of (1.1.1). It follows from (a) that the spaces $H^i_{\mathrm{DR}}(X_0/k)$ have dimensions

$1, 0, 22, 0, 1$ for $i = 0, 1, 2, 3, 4$. Assertion (c) follows from the "universal-coefficients" exact sequence [2, VII 1.1.11]:

$$0 \longrightarrow H^i(X_0/W) \otimes k \longrightarrow H^i_{\mathrm{DR}}(X_0/k) \longrightarrow \mathrm{Tor}_1^W(H^{i+1}(X_0/W), k) \longrightarrow 0.$$

COROLLARY 1.2. The versal formal space S of deformations of X_0 over local Artinian W-algebras with residue field k [15] is universal and formally smooth of dimension 20; that is, it is W-isomorphic to $\mathrm{Spf}(W[[t_1, \ldots, t_{20}]])$.

This is an immediate consequence of 1.1(b).

In what follows in this section, we write

$$\underline{X}/S \tag{1.3}$$

for the universal deformation of X_0 over S.

1.4. Let L_0 be an invertible sheaf on X_0. Write $\mathrm{Def}(X_0, L_0)$ for the functor on the category $\mathcal{A}$ of local Artinian W-algebras with residue field k that assigns to each $A \in \mathcal{A}$ the set of isomorphism classes of pairs (X, L) consisting of a flat (hence smooth) deformation X of X_0 over A and an extension of L_0 to an invertible sheaf L on X. Write $\mathrm{Def}(X_0)$ for the functor assigning to each $A \in \mathrm{Ob}\,\mathcal{A}$ the set of isomorphism classes of flat deformations of X_0 over A; by 1.2 this functor is (pro-)represented by S. There is a morphism "forget the extension of L_0"

$$\mathrm{Def}(X_0, L_0) \longrightarrow \mathrm{Def}(X_0). \tag{1.4.1}$$

PROPOSITION 1.5. The functor $\mathrm{Def}(X_0, L_0)$ is pro-representable, and the morphism (1.4.1) is a closed immersion defined by one equation.

The first assertion means that there is a largest closed formal subscheme

$$\Sigma(L_0) \subset S \tag{1.5.0}$$

such that L_0 extends to an invertible sheaf over $\underline{X} \times_S \Sigma(L_0)$, and such that this extension is unique.

It is readily verified that the functor $\mathrm{Def}(X_0, L_0)$ satisfies Schlessinger's conditions for the existence of a hull. Thus there is a formal scheme $S' = \mathrm{Spf}(R')$, where R' is a complete Noetherian local W-algebra with residue field k, and a deformation $(\underline{X}', \underline{L}')$ of (X_0, L_0) over S' such that, for every $A \in \mathrm{Ob}\,\mathcal{A}$, the associated map

$$\mathrm{Hom}(R', A) \longrightarrow \mathrm{Def}(X_0, L_0)(A) \tag{1.5.1}$$

is surjective, and is bijective for $A = k[\varepsilon]$, $\varepsilon^2 = 0$. Let R be the ring of S. Since S pro-represents $\mathrm{Def}(X_0)$, $\underline{X}'$ defines a homomorphism $u : R \to R'$ such that the composite

$$\mathrm{Hom}(R', A) \xrightarrow{1.5.1} \mathrm{Def}(X_0, L_0)(A) \xrightarrow{1.4.1} \mathrm{Def}(X_0)(A) = \mathrm{Hom}(R, A) \tag{1.5.2}$$

is $\mathrm{Hom}(u, A)$. To establish the first assertion of 1.5, it is therefore enough to prove that u is surjective, since then (1.5.2) is injective and hence (1.5.1) is bijective. By a familiar lemma [15, 1.1], this is equivalent to proving that, if $\mathfrak{m}$ (respectively $\mathfrak{m}'$) is the maximal ideal of R (respectively R'), the map $\mathfrak{m}/(pR + \mathfrak{m}^2) \to \mathfrak{m}'/(pR' + \mathfrak{m}'^2)$ induced by u is surjective; equivalently, that the tangent map at the origin of (1.4.1),

$$\mathrm{Hom}(u, k[\varepsilon]) : \mathrm{Def}(X_0, L_0)(k[\varepsilon]) \longrightarrow \mathrm{Def}(X_0)(k[\varepsilon]), \tag{1.5.3}$$

is injective. Let T' be the sheaf on X_0 of automorphisms of the trivial deformation of (X_0, L_0) over $k[\varepsilon]$. There is an exact sequence

$$0 \longrightarrow \mathcal{O}_{X_0} \longrightarrow T' \longrightarrow T \longrightarrow 0, \tag{1.5.4}$$

where $T' \to T$ is defined by forgetting L_0, and $T = T_{X_0}$ is viewed as the sheaf of automorphisms of the trivial deformation of X_0 over $k[\varepsilon]$. It is standard that (1.5.3) identifies canonically with the map

$H^1(X_0, T') \to H^1(X_0, T)$ induced by $T' \to T$. Since $H^1(X_0, \mathcal{O}) = 0$, the long exact cohomology sequence of (1.5.4) shows that (1.5.3) is injective. It remains to prove that the closed immersion $S' \hookrightarrow S$ is defined by a single equation, i.e. that $I = \ker(u)$ is principal. Put $S'' = \mathrm{Spf}(R/\mathfrak{m}I)$; this is a square-zero thickening of S' in S, with ideal $I/\mathfrak{m}I$. The obstruction to extending the invertible sheaf $\underline{L}'$ above $\underline{X} \times_S S''$ is an element

$$a \in H^2(X_0, I/\mathfrak{m}I) = H^2(X_0, \mathcal{O}) \otimes I/\mathfrak{m}I,$$

which, after choosing a basis of $H^2(X_0, \mathcal{O})$, we regard as an element of $I/\mathfrak{m}I$. Let $\Sigma = \mathrm{Spf}(R/(\mathfrak{m}I + (f)))$, where $f \in I$ lifts a. Then $S' \subset \Sigma \subset S'' \subset S$, and by construction (and functoriality of the obstruction) $\underline{L}'$ extends to $\underline{X} \times_S \Sigma$. The universal property of S' therefore implies $S' = \Sigma$, i.e. $\mathfrak{m}I + (f) = I$. Nakayama's lemma now shows that f generates I, completing the proof.

We may now state the main result of the exposé.

THEOREM 1.6. Let L_0 be a nontrivial invertible sheaf on X_0. With the notation of 1.5, the formal scheme $\Sigma(L_0)$ pro-representing $\mathrm{Def}(X_0, L_0)$ is flat over W, of relative dimension 19.

In other words, if f is an equation for $\Sigma(L_0)$ in S (1.5), then p does not divide f. Equivalently, $\Sigma(L_0)$ does not contain the reduction S_0 of S modulo p, i.e. L_0 cannot extend over $\underline{X} \times_S S_0$.

The proof of 1.6 will be given in Section 2. We end this section with a few consequences of 1.6.

COROLLARY 1.7. Let L_0 be a nontrivial invertible sheaf on X_0. There exist a trait T finite over W, a deformation of X_0 to a formal scheme $\underline{X}$ flat over T, and an extension of L_0 to an invertible sheaf $\underline{L}$ on $\underline{X}$.

We must prove that there is a W-morphism $T \to \Sigma(L_0)$, where T is a trait finite over W. Since p is not a zero divisor in the ring R' of $\Sigma(L_0)$, there exist (EGA 0_{IV} 16.4.1) elements $x_1, \ldots, x_n$ of the maximal ideal of R' which, together with p, form a system of parameters for R' (thus $n = 19$). The quotient $B = R'/(x_1, \ldots, x_n)$ is quasi-finite over W, hence finite over W. There is therefore a local W-homomorphism from B to a complete discrete valuation ring C finite over W, and the composite $R' \to B \to C$ gives the required morphism.

Applying Grothendieck's algebraization theorem (EGA III 5.4.5), one deduces from 1.7 the announced lifting theorem.

COROLLARY 1.8. Let L_0 be an ample invertible sheaf on X_0. There exist a trait T finite over W, a deformation of X_0 to a proper smooth scheme X over T, and an extension of L_0 to an ample invertible sheaf L on X.

REMARK 1.9. It is not known whether every K3 surface over k lifts to a proper smooth scheme over W. Ogus [13, §2] shows that: (a) every K3 surface over k lifts over W, except possibly in the "superspecial", non-elliptic case, which in fact does not exist if one assumes Artin's conjecture [1]; (b) if $p > 2$, every K3 surface over k lifts over $W[\sqrt{p}]$. Thus, at present, only the special case of 1.8 in which $p = 2$ and X_0 is superspecial is not covered by these results. We also point out that the cited paper of Ogus contains useful further information on the structure of $\Sigma(L_0)$; see also the following exposé for the case in which X_0 is ordinary.

COROLLARY 1.10. If k is the algebraic closure of a finite field over which X_0 is defined, the corresponding Frobenius acts semisimply on $H^2(X_0, \mathbf{Q}_\ell)$, where $\ell \neq p$.

This follows from [5]. With the notation of 1.8, the generic fibre of X is a K3 surface (the property of a surface of being K3 is stable under generization, since it is expressed by "K algebraically equivalent to zero and $\chi(\mathcal{O}) = 2$"). Hence X_0 satisfies the hypothesis of [5, 1.2]; the conclusion follows from [5, 6.6] and from the fact that Frobenius acts semisimply on the ℓ-adic H^1 of an abelian variety over a finite field ([16], [11, p. 203]). Notice that 1.10 is in fact independent of the assertion $H^0(X_0, T_{X_0}) = 0$: if X_0 admitted a nonzero vector field, then X_0 would be unirational by [14], and the conclusion of 1.10 would still hold (trace argument).

2. DE RHAM COHOMOLOGY OF $\underline{X}/S$ AND PROOF OF 1.6

We retain the notation of Section 1: X_0 is a K3 surface over k, and $\underline{X}/S$ denotes its universal W-deformation (1.3). Readers familiar with de Rham cohomology may skip 2.1–2.10, which merely recall standard facts concerning the Gauss–Manin connection, the Hodge filtration, the Frobenius operation, and the Chern class of an invertible sheaf.

2.1. Let $\Omega^\bullet_{\underline{X}/S}$ be the de Rham complex of the relative formal scheme $\underline{X}/S$ (by definition, for each i, $\Omega^i_{\underline{X}/S}$ is the inverse limit of the usual modules of differentials $\Omega^i_{\underline{X}\times_S S'/S'}$, as S' runs through the infinitesimal neighbourhoods of $\mathrm{Spec}(k)$ in S). Let $f : \underline{X} \to S$ be the projection. By definition, the de Rham cohomology of $\underline{X}/S$ consists of the $\mathcal{O}_S$-modules

$$H^i_{\mathrm{DR}}(\underline{X}/S) \overset{\mathrm{def}}{=} R^i f_*(\Omega^\bullet_{\underline{X}/S}), \tag{2.1.1}$$

whereas the Hodge cohomology of $\underline{X}/S$ consists of the $\mathcal{O}_S$-modules

$$H^i(\underline{X}, \Omega^j_{\underline{X}/S}) \overset{\mathrm{def}}{=} R^i f_*(\Omega^j_{\underline{X}/S}). \tag{2.1.2}$$

Since the $\mathcal{O}_{\underline{X}}$-modules $\Omega^i_{\underline{X}/S}$ are locally free of finite type, the $\mathcal{O}_S$-modules (2.1.1) and (2.1.2) are coherent by Grothendieck's finiteness theorem (EGA III 3.4.2). There is also the usual spectral sequence (the "Hodge spectral sequence")

$$E_1^{ij} = H^j(\underline{X}, \Omega^i_{\underline{X}/S}) \Longrightarrow H^*_{\mathrm{DR}}(\underline{X}/S). \tag{2.1.3}$$

PROPOSITION 2.2.

a) The spectral sequence (2.1.3) degenerates at E_1; the $\mathcal{O}_S$-modules $H^j(\underline{X}, \Omega^i_{\underline{X}/S})$ are free of finite type, and the canonical maps

$$H^j(\underline{X}, \Omega^i_{\underline{X}/S}) \otimes k \longrightarrow H^j(X_0, \Omega^i_{X_0/k})$$

are isomorphisms.

b) The $\mathcal{O}_S$-modules $H^i_{\mathrm{DR}}(\underline{X}/S)$ are free of finite type, and the canonical maps

$$H^i_{\mathrm{DR}}(\underline{X}/S) \otimes k \longrightarrow H^i_{\mathrm{DR}}(X_0/k)$$

are isomorphisms.

c) The cup product

$$\langle\ ,\ \rangle : H^2_{\mathrm{DR}}(\underline{X}/S) \otimes H^2_{\mathrm{DR}}(\underline{X}/S) \longrightarrow H^4_{\mathrm{DR}}(\underline{X}/S)$$

is a perfect duality.

Because the table of Hodge numbers of X_0/k is "interlaced with zeros" (1.1(a)), the usual criterion (EGA III 7.5.4), applied to the cohomological functors $M \mapsto H^\bullet(\underline{X}, \Omega^i_{\underline{X}/S} \otimes f^*M)$, immediately gives the second assertion of (a), and the first follows. Assertion (b) follows from (a), and implies (c) by Poincaré duality for $H^*_{\mathrm{DR}}(X_0/k)$.

2.3. Let $\Omega^\bullet_{S/W}$ be the de Rham complex of the "formal" differentials of S/W: $\Omega^i_{S/W} = \bigwedge^i \Omega^1_{S/W}$, where $\Omega^1_{S/W}$, the inverse limit of the completed modules of differentials $\Omega^1_{S_n/W_n}$ (with S_n/W_n the reduction of S/W modulo p^{n+1}), is free over $\mathcal{O}_S$ with basis $dt_1, \ldots, dt_{20}$ if $S \simeq W[[t_1, \ldots, t_{20}]]$. The $H^i_{\mathrm{DR}}(\underline{X}/S)$ carry a canonical integrable connection, the Gauss–Manin connection,

$$\nabla : H^i_{\mathrm{DR}}(\underline{X}/S) \longrightarrow \Omega^1_{S/W} \otimes H^i_{\mathrm{DR}}(\underline{X}/S). \tag{2.3.1}$$

The simplest definition of (2.3.1) paraphrases the Katz–Oda construction [ ], starting from the canonical extension

of $\mathcal{O}_{\underline{X}}$-modules

$$0 \longrightarrow f^*\Omega^1_{S/W} \longrightarrow \Omega^1_{\underline{X}/W} \longrightarrow \Omega^1_{\underline{X}/S} \longrightarrow 0, \tag{2.3.0}$$

where $\Omega^1_{\underline{X}/W}$ is the module of completed differentials. One may also use the fact that $H^i_{\mathrm{DR}}(\underline{X}/S)$ is the value at S of a crystal in $\mathcal{O}$-modules on the crystalline site of S_0/W:

$$H^i_{\mathrm{DR}}(\underline{X}/S) = R^i(f_0)_{\mathrm{cris}*}(\mathcal{O}_{\underline{X}_0/W})(S), \tag{2.3.2}$$

where

$$(f_0 : \underline{X}_0 \to S_0) = f \times_S S_0, \qquad S_0 = S \otimes_W k \simeq \mathrm{Spf}(k[[t_1, \dots, t_{20}]]), \tag{2.3.2.1}$$

and $(f_0)_{\mathrm{cris}} : (\underline{X}_0/W)_{\mathrm{cris}} \to (S_0/W)_{\mathrm{cris}}$ is the corresponding morphism of crystalline sites. This is a "formal variant" of Berthelot's result [2, V 3.6], readily deduced from it by applying it to the morphisms induced by f_0 over the infinitesimal neighbourhoods of $\mathrm{Spec}(k)$ in S_0.

Either definition shows that the cup product of 2.2(c) is horizontal:

$$\langle \nabla x, y\rangle + \langle x, \nabla y\rangle = \nabla\langle x, y\rangle \tag{2.3.3}$$

for all $x, y \in H^2_{\mathrm{DR}}(\underline{X}/S)$.

Let $F^i_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}/S)$ be the Hodge filtration, the abutment of (2.1.3). If $\Omega^{\geq i}_{\underline{X}/S} = (0 \to \Omega^i_{\underline{X}/S} \to \Omega^{i+1}_{\underline{X}/S} \to \cdots)$ denotes the de Rham complex truncated by zero in degrees $< i$, then the inclusion $\Omega^{\geq i}_{\underline{X}/S} \to \Omega^\bullet_{\underline{X}/S}$ gives, by 2.2(a), an isomorphism

$$H^2(\underline{X}, \Omega^{\geq i}_{\underline{X}/S}) \simeq F^i_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}/S). \tag{2.3.4}$$

We have

$$H^2_{\mathrm{DR}}(\underline{X}/S) = F^0_{\mathrm{Hdg}} \supset F^1_{\mathrm{Hdg}} \supset F^2_{\mathrm{Hdg}} \supset F^3_{\mathrm{Hdg}} = 0, \tag{2.3.5}$$

the F^i_{Hdg} are free $\mathcal{O}_S$-modules, and degeneration of (2.1.3) gives isomorphisms

$$H^{2-i}(\underline{X}, \Omega^i_{\underline{X}/S}) \simeq \mathrm{gr}^i_F H^2_{\mathrm{DR}}(\underline{X}/S), \qquad (F = F_{\mathrm{Hdg}}). \tag{2.3.6}$$

In particular, F^1_{Hdg} (respectively F^2_{Hdg}) is free of rank 21 (respectively 1). From (2.3.4) one sees that the orthogonal of F^2_{Hdg} for the cup product of 2.2(c) contains F^1_{Hdg}, and therefore equals it:

$$F^1_{\mathrm{Hdg}} = (F^2_{\mathrm{Hdg}})^{\perp} \qquad (\text{and hence } F^2_{\mathrm{Hdg}} = (F^1_{\mathrm{Hdg}})^{\perp}). \tag{2.3.7}$$

The Katz–Oda description of the Gauss–Manin connection shows that

$$\nabla F^i_{\mathrm{Hdg}} H^2_{\mathrm{DR}}(\underline{X}/S) \subset \Omega^1_{S/W} \otimes F^{i-1}_{\mathrm{Hdg}} H^2_{\mathrm{DR}}(\underline{X}/S) \tag{2.3.8}$$

("Griffiths transversality"; see, for example, [8, 1.4]). Passing to quotients, ∇ therefore induces an $\mathcal{O}_S$-linear map

$$\mathrm{gr}^i \nabla : \mathrm{gr}^i_F H^2_{\mathrm{DR}}(\underline{X}/S) \longrightarrow \Omega^1_{S/W} \otimes \mathrm{gr}^{i-1}_F H^2_{\mathrm{DR}}(\underline{X}/S). \tag{2.3.9}$$

Correspondingly there is an $\mathcal{O}_S$-linear map

$$\nabla_i : T_{S/W} \longrightarrow \mathrm{Hom}(\mathrm{gr}^i_F H^2_{\mathrm{DR}}(\underline{X}/S), \mathrm{gr}^{i-1}_F H^2_{\mathrm{DR}}(\underline{X}/S)), \tag{2.3.10}$$

where $T_{S/W} = (\Omega^1_{S/W})^{\vee}$. By (2.3.6), the target of (2.3.10) identifies canonically, for the fixed i, with

$$\mathrm{Hom}\big(H^{2-i}(\underline{X}, \Omega^i_{\underline{X}/S}), H^{3-i}(\underline{X}, \Omega^{i-1}_{\underline{X}/S})\big).$$

Let $T_{\underline{X}/S} = (\Omega^1_{\underline{X}/S})^{\vee}$. If

$$\mathrm{Kod}(\underline{X}/S) : T_{S/W} \longrightarrow H^1(\underline{X}, T_{\underline{X}/S}) \tag{2.3.11}$$

denotes the Kodaira–Spencer map associated with $\underline{X}/S/W$, defined by the extension (2.3.0) (cf. [8, 1.3]), then, under the preceding identification, (2.3.10) belongs to the commutative triangle

$$\begin{array}{ccc} T_{S/W} & \xrightarrow{\mathrm{Kod}(\underline{X}/S)} & H^1(\underline{X}, T_{\underline{X}/S}) \\ & \searrow^{\nabla_i} & \downarrow \\ & & \mathrm{Hom}(H^{2-i}(\underline{X}, \Omega^i_{\underline{X}/S}), H^{3-i}(\underline{X}, \Omega^{i-1}_{\underline{X}/S})), \end{array} \tag{2.3.12}$$

where the vertical arrow is defined by cup product, via the interior product $T_{\underline{X}/S} \otimes \Omega^i_{\underline{X}/S} \to \Omega^{i-1}_{\underline{X}/S}$. This compatibility is verified exactly as

in [8, 1.4.1.7].

By (2.3.7), the cup product of 2.2(c), after passage to gr_F, induces a perfect duality

$$\mathrm{gr}_F^i H^2_{\mathrm{DR}}(\underline{X}/S) \otimes \mathrm{gr}_F^{2-i} H^2_{\mathrm{DR}}(\underline{X}/S) \longrightarrow H^4_{\mathrm{DR}}(\underline{X}/S) \simeq H^2(\underline{X}, \Omega^2_{\underline{X}/S}), \tag{2.3.13}$$

corresponding under (2.3.6) to the cup product

$$H^{2-i}(\underline{X}, \Omega^i_{\underline{X}/S}) \otimes H^i(\underline{X}, \Omega^{2-i}_{\underline{X}/S}) \longrightarrow H^2(\underline{X}, \Omega^2_{\underline{X}/S}).$$

It follows from (2.3.3) that, for every $D \in T_{S/W}$,

$$\nabla_1(D) = -\nabla_2(D)^\vee, \tag{2.3.14}$$

i.e. $\langle \nabla_2(D)x, y\rangle + \langle x, \nabla_1(D)y\rangle = 0$ for $x \in \mathrm{gr}_F^2$, $y \in \mathrm{gr}_F^1$.

The following statement expresses the mobility of the Hodge filtration under Gauss–Manin.

PROPOSITION 2.4. The maps (2.3.10) and (2.3.11) are isomorphisms.

First, the sheaf $\Omega^2_{\underline{X}/S}$, the extension to $\underline{X}$ of the trivial sheaf $\Omega^2_{X_0/k}$, is necessarily trivial (1.5); hence

$$T_{\underline{X}/S} \simeq \Omega^1_{\underline{X}/S}. \tag{2.4.1}$$

Consequently, by 2.2(a), $H^1(\underline{X}, T_{\underline{X}/S})$ is free of rank 20, and the canonical map $H^1(\underline{X}, T_{\underline{X}/S}) \otimes k \to H^1(X_0, T_{X_0/k})$ is an isomorphism. A standard argument shows that

$$\mathrm{Kod}(\underline{X}/S) \otimes_W k : T_{S/W} \otimes_W k \longrightarrow H^1(X_0, T_{X_0/k}) \tag{2.4.2}$$

is precisely the tangent map at the origin of the canonical isomorphism $S \simeq \mathrm{Def}(X_0)$ (1.4), so $\mathrm{Kod}(\underline{X}/S)$ is an isomorphism. By (2.3.12), it remains to prove that the vertical arrow in (2.3.12) is an isomorphism; by (2.3.14), it is enough to do so for $i = 1$. By 2.2(a) and the observation at the beginning of the proof, it suffices to show that the cup product

$$H^1(X_0, T_{X_0/k}) \otimes H^1(X_0, \Omega^1_{X_0/k}) \longrightarrow H^2(X_0, \mathcal{O}) \tag{2.4.3}$$

is nondegenerate. But the choice of a basis of $H^0(X_0, \Omega^2_{X_0/k})$ identifies $T_{X_0/k}$ with $\Omega^1_{X_0/k}$ and identifies (2.4.3) with Serre duality

$$H^1(X_0, \Omega^1_{X_0/k}) \otimes H^1(X_0, \Omega^1_{X_0/k}) \longrightarrow H^2(X_0, \Omega^2_{X_0/k}),$$

which proves the assertion.

COROLLARY 2.4.4. The map $\mathrm{gr}^1 \nabla$ (respectively $\mathrm{gr}^2 \nabla$) in (2.3.9) is an isomorphism (respectively is injective with free cokernel).

Consequently the same assertion holds for $\mathrm{gr}^1(\nabla|S_0)$ (respectively $\mathrm{gr}^2(\nabla|S_0)$), where $\nabla|S_0$ is the Gauss–Manin connection on $H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$. In view of the duality (2.3.13), 2.4.4 follows immediately from the fact that (2.3.10) is an isomorphism.

2.5. With the notation of (2.3.2.1), let $F_{\underline{X}_0/S_0} : \underline{X}_0 \to \underline{X}_0^{(p)}$ be the relative Frobenius of $\underline{X}_0/S_0$, defined by the usual Cartesian diagram

$$\begin{array}{ccccc} \underline{X}_0 & \longleftarrow & \underline{X}_0^{(p)} & \xleftarrow{F_{\underline{X}_0/S_0}} & \underline{X}_0 \\ \downarrow & & \downarrow & \swarrow & \\ S_0 & \longleftarrow & S_0 & & \end{array} \tag{2.5.1}$$

where the horizontal composites are the absolute Frobenius maps. The crystalline interpretation (2.3.2) of $H^i_{\mathrm{DR}}(\underline{X}/S)$ shows, by functoriality of crystalline cohomology, that $F_{\underline{X}_0/S_0}$ induces, for every lift $\varphi : S \to S$ of the absolute Frobenius of S_0 compatible with the canonical Frobenius σ of W, an $\mathcal{O}_S$-linear horizontal homomorphism

$$F(\varphi) : \varphi^* H^i_{\mathrm{DR}}(\underline{X}/S) \longrightarrow H^i_{\mathrm{DR}}(\underline{X}/S). \tag{2.5.2}$$

This homomorphism is an isogeny (i.e. $F(\varphi) \otimes_{\mathbf{Z}_p} \mathbf{Q}_p$ is an isomorphism), and its dependence on φ is controlled by ∇ (cf. [7] and the following exposé). In what follows we fix a lift φ, for example the one given by $\varphi(t_i) = t_i^p$ $(1 \leq i \leq 20)$, with $S = \operatorname{Spf} W[[t_1, \ldots, t_{20}]]$, so that $\varphi(\sum a_\alpha t^\alpha) = \sum a_\alpha^\sigma t^{p\alpha}$. We write $F(\varphi) = \widetilde{F}$ and denote by

$$F : H^i_{\mathrm{DR}}(\underline{X}/S) \longrightarrow H^i_{\mathrm{DR}}(\underline{X}/S) \tag{2.5.3}$$

the σ-linear homomorphism obtained by composing $\widetilde{F}$ with the adjunction map $H^i_{\mathrm{DR}}(\underline{X}/S) \to \varphi^* H^i_{\mathrm{DR}}(\underline{X}/S)$, $x \mapsto 1 \otimes x$. By construction, F is compatible with the cup product of 2.2(c):

$$\langle Fx, Fy \rangle = F\langle x, y \rangle \tag{2.5.4}$$

for all $x, y \in H^2_{\mathrm{DR}}(\underline{X}/S)$.

The following statement is a special case of a theorem of Mazur–Ogus ([10], [4, 8.26])*.

PROPOSITION 2.6. With the notation of 2.3 and 2.5,

$$F^1_{\mathrm{Hdg}} H^2_{\mathrm{DR}}(\underline{X}/S) \subset \{x \in H^2_{\mathrm{DR}}(\underline{X}/S) \mid Fx \in pH^2_{\mathrm{DR}}(\underline{X}/S)\},$$

and the two sides have the same image in $H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$.

In other words, if $F : H^2_{\mathrm{DR}}(\underline{X}_0/S_0) \to H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$ denotes the p-linear Frobenius endomorphism (the reduction modulo p of (2.5.3)), and if $F^i_{\mathrm{Hdg}} H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$ is the Hodge filtration (the reduction modulo p of that on $H^2_{\mathrm{DR}}(\underline{X}/S)$), then

$$F^1_{\mathrm{Hdg}} H^2_{\mathrm{DR}}(\underline{X}_0/S_0) = \ker\big(F : H^2_{\mathrm{DR}}(\underline{X}_0/S_0) \to H^2_{\mathrm{DR}}(\underline{X}_0/S_0)\big). \tag{2.6.1}$$

We recall the proof. By definition, F is the composite of the canonical p-linear injection $H^2_{\mathrm{DR}}(\underline{X}_0/S_0) \to H^2_{\mathrm{DR}}(\underline{X}_0^{(p)}/S_0) = F^*_{S_0} H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$ defined by the Cartesian square (2.5.1), and the $\mathcal{O}_{S_0}$-linear map $\widetilde{F} : H^2_{\mathrm{DR}}(\underline{X}_0^{(p)}/S_0) \to H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$ defined by the relative Frobenius $F_{\underline{X}_0/S_0}$. It is immediate that

*This is a formal variant of [4, 8.26], from which it follows easily.

$$F^1_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}_0/S_0) = H^2_{\mathrm{DR}}(\underline{X}_0/S_0) \cap F^1_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}_0^{(p)}/S_0),$$

so it is enough to prove

$$F^1_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}_0^{(p)}/S_0) = \ker\big(\widetilde{F} : H^2_{\mathrm{DR}}(\underline{X}_0^{(p)}/S_0) \to H^2_{\mathrm{DR}}(\underline{X}_0/S_0)\big). \tag{2.6.2}$$

By 2.2, the Hodge spectral sequence of $\underline{X}_0^{(p)}/S_0$ degenerates at E_1, and therefore there is an exact sequence

$$0 \longrightarrow F^1_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}_0^{(p)}/S_0) \longrightarrow H^2_{\mathrm{DR}}(\underline{X}_0^{(p)}/S_0) \xrightarrow{\pi} H^2(\underline{X}_0^{(p)}, \mathcal{O}) \longrightarrow 0, \tag{2.6.3}$$

where π is the canonical projection. Since $F_{\underline{X}_0/S_0}$ vanishes on $\Omega^i_{\underline{X}_0^{(p)}/S_0}$ for $i \geq 1$, there is a commutative square

$$\begin{array}{ccc} H^2_{\mathrm{DR}}(\underline{X}_0^{(p)}/S_0) & \xrightarrow{\quad\pi\quad} & H^2(\underline{X}_0^{(p)}, \mathcal{O}) \\ \Big\downarrow{\scriptstyle \widetilde{F}} & & \Big\downarrow{\scriptstyle \widetilde{F}} \\ H^2_{\mathrm{DR}}(\underline{X}_0/S_0) & \xleftarrow{\quad(*)\quad} & H^2(\underline{X}_0, \mathcal{H}^0(\Omega^\bullet_{\underline{X}_0/S_0})) \end{array}$$

where $(*)$ is induced by the inclusion $\mathcal{H}^0(\Omega^\bullet_{\underline{X}_0/S_0}) \hookrightarrow \Omega^\bullet_{\underline{X}_0/S_0}$. Degeneration at E_1 of the Hodge spectral sequence of $\underline{X}_0/S_0$ entails, by the standard argument using the Cartier operator, degeneration at E_2 of the conjugate spectral sequence

$$E_2^{ij} = H^i(\underline{X}_0, \mathcal{H}^j(\Omega^\bullet_{\underline{X}_0/S_0})) \Longrightarrow H^*_{\mathrm{DR}}(\underline{X}_0/S_0).$$

In particular, $(*)$ is injective. The right vertical arrow is an isomorphism (a special case of the Cartier isomorphism), so $\widetilde{F}$ and π have the same kernel. Together with (2.6.3), this proves the assertion.

REMARKS 2.7.

a) By [4, 8.26], one also has

$$F^2_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}_0/S_0) = \mathrm{Im}\big(\{x \in H^2_{\mathrm{DR}}(\underline{X}/S) \mid Fx \in p^2 H^2_{\mathrm{DR}}(\underline{X}/S)\} \longrightarrow H^2_{\mathrm{DR}}(\underline{X}_0/S_0)\big), \tag{2.7.1}$$

but we shall not need this. b) The F-crystal $H^4_{\mathrm{DR}}(\underline{X}/S)$ is isomorphic to the "Tate crystal" $\mathcal{O}(-2)$, namely $\mathcal{O}_S$ equipped with $\nabla = d$ and $F = p^2\varphi$. In the absence, in the literature, of a trace morphism $H^4_{\mathrm{DR}}(\underline{X}/S) \to \mathcal{O}_S$, one may nevertheless

see this as follows. First, $H^4_{\mathrm{DR}}(\underline{X}/S)$ is the image under cup product of $F^1H^2_{\mathrm{DR}}(\underline{X}/S)\otimes F^1H^2_{\mathrm{DR}}(\underline{X}/S)$. By (2.5.4) and 2.6, it follows that $F(H^4_{\mathrm{DR}}(\underline{X}/S))\subset p^2H^4_{\mathrm{DR}}(\underline{X}/S)$. Since $H^4_{\mathrm{DR}}(\underline{X}/S)\otimes W = H^4(X_0/W)$ is isomorphic to $W(-2)$, the crystal $H^4_{\mathrm{DR}}(\underline{X}/S)$ has the form $U(-2)$, where U is a unit-root F-crystal; every unit-root F-crystal over S is trivial [7], proving the assertion.

2.8. The last ingredient needed in the proof of 1.6 is the crystalline Chern class of an invertible sheaf (cf. [3]). Let $\mathrm{Fil}^1\,\Omega^\bullet_{\underline{X}/S}$ be the subcomplex of $\Omega^\bullet_{\underline{X}/S}$ which is the kernel of the canonical projection $\Omega^\bullet_{\underline{X}/S}\to\mathcal{O}_{\underline{X}_0}$, i.e.

$$\mathrm{Fil}^1\,\Omega^\bullet_{\underline{X}/S} = (p\mathcal{O}_{\underline{X}} \xrightarrow{d} \Omega^1_{\underline{X}/S} \xrightarrow{d} \Omega^2_{\underline{X}/S}). \tag{2.8.1}$$

The defining exact sequence for Fil^1,

$$0 \longrightarrow \mathrm{Fil}^1\,\Omega^\bullet_{\underline{X}/S} \longrightarrow \Omega^\bullet_{\underline{X}/S} \longrightarrow \mathcal{O}_{\underline{X}_0} \longrightarrow 0, \tag{2.8.2}$$

gives an exact sequence

$$0 \longrightarrow 1+\mathrm{Fil}^1\,\Omega^\bullet_{\underline{X}/S} \longrightarrow (\Omega^\bullet_{\underline{X}/S})^* \longrightarrow \mathcal{O}^*_{\underline{X}_0} \longrightarrow 0, \tag{2.8.3}$$

where

$$(\Omega^\bullet_{\underline{X}/S})^* = (\mathcal{O}^*_{\underline{X}} \xrightarrow{d\log} \Omega^1_{\underline{X}/S} \xrightarrow{d} \Omega^2_{\underline{X}/S})$$

is the "multiplicative" de Rham complex. By definition, the Chern-class homomorphism

$$c_1 : \mathrm{Pic}(\underline{X}_0) = H^1(\underline{X}_0, \mathcal{O}^*_{\underline{X}_0}) \longrightarrow H^2(\underline{X}, \mathrm{Fil}^1\,\Omega^\bullet_{\underline{X}/S}) \tag{2.8.4}$$

is the composite of the connecting morphism of (2.8.3) and the map induced by the morphism of complexes

$$\log : 1+\mathrm{Fil}^1\,\Omega^\bullet_{\underline{X}/S} \longrightarrow \mathrm{Fil}^1\,\Omega^\bullet_{\underline{X}/S}, \tag{2.8.5}$$

which is the identity in degrees ≥ 1 and sends $1+x$ to $\sum(-1)^i x^{i+1}/(i+1)$ in degree zero. The legitimacy of (2.8.5) is immediate.

Since $H^1(\underline{X}_0, \mathcal{O}) = 0$ (2.2(a)), the canonical map

$$H^2(\underline{X}, \mathrm{Fil}^1\, \Omega^\bullet_{\underline{X}/S}) \longrightarrow H^2_{\mathrm{DR}}(\underline{X}/S) \tag{2.8.6}$$

is injective. We shall therefore sometimes regard c_1 in (2.8.4) as taking values in $H^2_{\mathrm{DR}}(\underline{X}/S)$.

PROPOSITION 2.9. Let $\underline{L}_0 \in \mathrm{Pic}(\underline{X}_0)$ and put $x = c_1(\underline{L}_0)$. Then

a) $Fx = px$;
b) $\nabla x = 0$.

For these formulas it is convenient to interpret c_1 crystallinely. The construction of [3, 2.1] gives a homomorphism

$$c_{1/W} : \mathrm{Pic}(\underline{X}_0) \longrightarrow H^2_{\mathrm{cris}}(\underline{X}_0, J_{\underline{X}_0/W}), \tag{2.9.1}$$

where $J_{\underline{X}_0/W}$ is the crystalline sheaf defined by the canonical exact sequence

$$0 \longrightarrow J_{\underline{X}_0/W} \longrightarrow \mathcal{O}_{\underline{X}_0/W} \longrightarrow \mathcal{O}_{\underline{X}_0} \longrightarrow 0. \tag{2.9.2}$$

By (2.3.2),

$$H^2(\underline{X}, \mathrm{Fil}^1\, \Omega^\bullet_{\underline{X}/S}) = R^2(f_0)_{\mathrm{cris}*}(J_{\underline{X}_0/W})(S),$$

and hence

$$H^0_{\mathrm{cris}}(S_0, R^2(f_0)_{\mathrm{cris}*} J_{\underline{X}_0/W}) = \{x \in H^2(\underline{X}, \mathrm{Fil}^1\, \Omega^\bullet_{\underline{X}/S}) \mid \nabla x = 0\}. \tag{2.9.3}$$

It is readily checked that (2.8.4) is the composite of (2.9.1) and the canonical homomorphism

$$H^2_{\mathrm{cris}}(\underline{X}_0, J_{\underline{X}_0/W}) \longrightarrow H^0_{\mathrm{cris}}(S_0, R^2(f_0)_{\mathrm{cris}*} J_{\underline{X}_0/W}), \tag{2.9.4}$$

under the identification (2.9.3). This proves 2.9(b). The preceding description also shows functoriality in $\underline{X}_0/S_0$ of (2.8.4), which implies 2.9(a), because, for the absolute Frobenius $F_{\underline{X}_0}$ of $\underline{X}_0$,

$$Fx = c_1(F^*_{\underline{X}_0} \underline{L}_0) = c_1(\underline{L}_0^{\otimes p}) = px.$$

REMARK 2.9.5. As regards (b), a reader reluctant to use crystalline considerations may verify it "by hand" from the explicit Katz–Oda description [9] of the Gauss–Manin connection. Let D be a derivation of S/W. Choose an affine open covering $\mathcal{U}$ of $\underline{X}_0$, a cocycle (g_{ij}) with values in $\mathcal{O}^*_{\underline{X}_0}$ representing $\underline{L}_0$, and lifts D_i of D to derivations of $\underline{X}/W$ on U_i. Lifting g_{ij} to a section $\widetilde{g}_{ij}$ of $\mathcal{O}^*_{\underline{X}}$ on $U_i \cap U_j$ defines a 2-cocycle h_{ijk} with values in $1 + p\mathcal{O}_{\underline{X}}$; the class $x = c_1(\underline{L}_0)$ is represented by

$$(\log h_{ijk}) + (d \log g_{ij}) \in Z^2(\mathcal{U}, \mathrm{Fil}^1\, \Omega^\bullet_{\underline{X}/S}).$$

On the other hand, by [9], $\nabla(D)x$ is the class of the 2-cocycle $(i < j < k)$

$$y = (D_i \log h_{ijk}) + (dD_i \log g_{ij}) - ((D_i \log - D_j \log) g_{jk}).$$

One checks that y is the coboundary of the 1-cochain $(D_i \log \widetilde{g}_{ij}) \in C^1(\mathcal{U}, \Omega^1_{\underline{X}/S})$, and therefore $\nabla(D)x = 0$.

2.10. Functoriality of the crystalline Chern class implies that, for every W-valued point e of S, there is a commutative square

$$\begin{array}{ccc} \mathrm{Pic}(\underline{X}_0) & \xrightarrow{c_1} & H^2_{\mathrm{DR}}(\underline{X}/S) = H^2_{\mathrm{cris}}(\underline{X}_0/S) \\ \Big\downarrow{\scriptstyle \underline{L} \mapsto \underline{L}|_{X_0}} & & \Big\downarrow{\scriptstyle e^*} \\ \mathrm{Pic}(X_0) & \xrightarrow[c_1]{} & H^2_{\mathrm{DR}}(X/W) = H^2_{\mathrm{cris}}(X_0/W), \end{array} \tag{2.10.1}$$

where X/W is obtained from $\underline{X}/S$ by base change along e. Since X_0 is a K3 surface, $\mathrm{Pic}(X_0)$ coincides with the Néron–Severi group $\mathrm{NS}(X_0)$ and is torsion free; hence the lower horizontal arrow in (2.10.1) is injective (3.4).

2.11. Proof of 1.6. Suppose that L_0 extends to an invertible sheaf $\underline{L}_0$ on $\underline{X}_0$, and let $x = c_1(\underline{L}_0) \in H^2_{\mathrm{DR}}(\underline{X}/S)$. Commutativity of (2.10.1) gives $e^*x = c_1(L_0)$. Since L_0 is nontrivial by hypothesis, $c_1(L_0) \neq 0$ by the preceding remark, and hence $x \neq 0$. By 2.9, $Fx = px$ and $\nabla x = 0$. Let p^n be the highest power of p dividing x, and put $y = p^{-n}x$; the image y_0 of y in $H^2_{\mathrm{DR}}(\underline{X}_0/S_0) = H^2_{\mathrm{DR}}(\underline{X}/S) \bmod p$ is nonzero. We still have $Fy = py$ and $\nabla y = 0$. By (2.6.1), the first relation implies $y_0 \in F^1_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$. Since $y_0 \neq 0$ and $\nabla y_0 = 0$, one cannot have $y_0 \in F^2_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$, for this would contradict the injectivity of

$$\mathrm{gr}^2\,\nabla : F^2_{\mathrm{Hdg}}H^2_{\mathrm{DR}}(\underline{X}_0/S_0) \longrightarrow \Omega^1_{S_0/k} \otimes \mathrm{gr}^1_F\, H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$$

from 2.4.4. Thus the image of y_0 in $\mathrm{gr}^1_F\, H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$ is nonzero; but $\nabla y_0 = 0$, contradicting the fact that

$$\mathrm{gr}^1\,\nabla : \mathrm{gr}^1_F\, H^2_{\mathrm{DR}}(\underline{X}_0/S_0) \longrightarrow \Omega^1_{S_0/k} \otimes \mathrm{gr}^0_F\, H^2_{\mathrm{DR}}(\underline{X}_0/S_0)$$

is an isomorphism. This contradiction proves 1.6.

3. APPENDIX; CRYSTALLINE CHERN CLASSES AND INTERSECTIONS*

The purpose of this appendix is to establish that, on a proper smooth surface over k, the intersection number of two divisors is the trace of the cup product of their crystalline Chern classes (this compatibility apparently does not occur in the literature).

3.1. Let X be a proper smooth scheme over k. Recall [2, VI 3.3.6] that to every closed subscheme Y of X, smooth over k and of codimension d in X, one can associate a cohomology class

$$c\ell(Y) \in H^{2d}(X/W). \tag{3.1.1}$$

If Y' and Y'' are smooth closed subschemes of codimensions d' and d'', meeting transversely, then [2, VI 4.3.15]

*By P. Deligne and L. Illusie.

$$c\ell(Y')\,c\ell(Y'') = c\ell(Y' \cap Y'') \tag{3.1.2}$$

in $H^{2(d'+d'')}(X/W)$. If X' is proper and smooth over k, $f : X' \to X$ is a k-morphism, and Y is a smooth closed subscheme of codimension d such that $Y \times_X X'$ is smooth of codimension d in X', then [2, VI 4.3.13]

$$c\ell(Y \times_X X') = f^* c\ell(Y). \tag{3.1.3}$$

Finally, if a is a rational point of X/k, then [2, VII 3.1.7]

$$\mathrm{Tr}_{X/W}(c\ell(a)) = 1. \tag{3.1.4}$$

In particular, if X is connected, $c\ell(a)$ is independent of a, since $\mathrm{Tr}_{X/W} : H^{2n}(X/W) \to W$ is an isomorphism, where $n = \dim X$ [2, VII 2.1.1].

3.2. To every vector bundle L on X one can associate Chern classes [3, 2.4]

$$c_i(L) \in H^{2i}(X/W), \tag{3.2.1}$$

which vanish for $i > \mathrm{rk}(L)$. These classes depend functorially on L: if $f : X' \to X$ is a k-morphism, then (loc. cit.)

$$c_i(f^*L) = f^* c_i(L). \tag{3.2.2}$$

If D is a smooth effective divisor on X, there are therefore both the cohomology class $c\ell(D) \in H^2(X/W)$ and the Chern class $c_1(\mathcal{O}_X(D)) \in H^2(X/W)$.

PROPOSITION 3.2.3. Let D be a smooth effective divisor on X. If $\mathcal{O}_X(D)$ is very ample, then

$$c\ell(D) = c_1(\mathcal{O}_X(D)). \tag{3.2.3.1}$$

REMARK 3.2.4. Berthelot (unpublished) verified that (3.2.3.1) holds without the hypothesis that $\mathcal{O}_X(D)$ be very ample. The proof is rather complicated. For our purposes, 3.2.3 is sufficient.

Let us prove 3.2.3. By 3.1.3 and 3.2.2, we reduce at once to the case in which X is projective space $\mathbf{P}_k^N$ and D is a hyperplane. Put $\mathbf{P}_W^N = X'$, and let D' be a hyperplane of X' lifting D. Then $H^2(X/W) = H^2_{\mathrm{DR}}(X'/W)$. By [2, VI 3.3.5], $c\ell(D)$ is the de Rham cohomology class of D', which is computed from the extension of $\Omega^\bullet_{D'/W}[-1]$ by $\Omega^\bullet_{X'/W}$ supplied by the de Rham complex of X'/W with logarithmic poles along D'. If $(t_0, \ldots, t_N)$ are homogeneous coordinates on X', it follows readily that $c\ell(D)$ is represented, on the standard covering, by the 1-cocycle

$$\left(\frac{dt_j}{t_j} - \frac{dt_i}{t_i}\right)_{i<j}.$$

On the other hand, by [3, 2.3], the Chern class $c_1(\mathcal{O}_{X'}(1)) \in H^2_{\mathrm{DR}}(X'/W)$ is obtained from the cocycle t_j/t_i by the homomorphism $d\log : \mathcal{O}^*_{X'} \to \Omega^\bullet_{X'/W}[1]$. Thus $c_1(\mathcal{O}_X(D)) = c\ell(D)$.

THEOREM 3.3. Let X be a proper smooth surface over k. If D_1, D_2 are divisors on X, then

$$(D_1 \cdot D_2) = \mathrm{Tr}_{X/W}\big(c_1(\mathcal{O}_X(D_1))\, c_1(\mathcal{O}_X(D_2))\big),$$

where $(D_1 \cdot D_2)$ denotes their intersection number.

If D_1, D_2 are effective, smooth, very ample, and transverse, the conclusion follows from (3.1.2), (3.1.4), and 3.2.3. To reduce to this case, choose a very ample divisor H on X. For n sufficiently large, $D_1 + nH$ and $D_2 + nH$ are very ample. By Bertini, there are effective smooth very ample divisors D_1', D_1'', D_2', D_2'' such that

$$\mathcal{O}_X(D_1) = \mathcal{O}_X(D_1' - D_1''), \qquad \mathcal{O}_X(D_2) = \mathcal{O}_X(D_2' - D_2''),$$

and one may further suppose that the four pairs

$$(D_1', D_2'), \quad (D_1', D_2''), \quad (D_1'', D_2'), \quad (D_1'', D_2'')$$

are transverse. Additivity of c_1 and bilinearity of the intersection pairing reduce the theorem to the special case already considered, completing the proof.

3.4. Again let X be a proper smooth scheme over k. It is not known whether, for two smooth closed subschemes of X of the same codimension, algebraic equivalence implies equality of their crystalline cohomology classes (this is known only for zero-dimensional subschemes; cf. 3.1.4!). On the other hand, if $\mathrm{NS}(X) = \mathrm{Pic}(X)/\mathrm{Pic}^0(X)$ is the Néron–Severi group, the homomorphism

$$c_1 : \mathrm{Pic}(X) \longrightarrow H^2(X/W)$$

vanishes on $\mathrm{Pic}^0(X)$, because $\mathrm{Pic}^0(X)$ is p-divisible and $H^2(X/W)$ is of finite type over W. It therefore induces

$$c_1 : \mathrm{NS}(X) \longrightarrow H^2(X/W). \tag{3.4.1}$$

Suppose now that X is a surface. By the Hodge index theorem, the intersection form on $\mathrm{NS}(X) \otimes \mathbf{Q}$ is nondegenerate. In view of 3.3, it follows that

$$c_1 \otimes \mathbf{Q} : \mathrm{NS}(X) \otimes \mathbf{Q} \longrightarrow H^2(X/W) \otimes \mathbf{Q} \tag{3.4.2}$$

is injective.

REMARK 3.5. One can show [6, II 5.10] that (3.4.1) defines an injection $\mathrm{NS}(X) \otimes \mathbf{Z}_p \hookrightarrow H^2(X/W)$. It is probably not true in general that (3.4.1) induces an injection $\mathrm{NS}(X)/p\,\mathrm{NS}(X) \to H^2(X/W)/pH^2(X/W)$, i.e. that the de Rham Chern-class map sends $\mathrm{NS}(X)/p\,\mathrm{NS}(X)$ injectively into $H^2_{\mathrm{DR}}(X/k)$. This does hold if $H^*(X/W)$ is torsion free and the Hodge-to-de Rham spectral sequence of X/k degenerates at E_1; see [13, 1.4] and [6, II 5.18]. These conditions are satisfied, for example, by a K3 surface.

P. DELIGNE
Institut des Hautes Etudes
Scientifiques
35, Route de Chartres
91440 BURES/YVETTE (France)

L. ILLUSIE
Université de Paris-Sud
Centre d'Orsay
Mathématique, bât. 425
91405 ORSAY (France)

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Duality for Representations of a Reductive Group over a Finite Field

Pierre Deligne
I.H.E.S., Bures-sur-Yvette, France
George Lusztig
Department of Mathematics, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139

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1

The purpose of this paper is to construct a duality operation for representations of a reductive group over a finite field. Its effect, very roughly speaking, is to interchange irreducible representations of small degree with ones of large degree (for example, the unit and Steinberg representation.) At level of characters, this operation has been also considered by Alvis [1], Curtis [2], and Kawanaka [4].

We shall now fix some notation. $\mathbf{G}$ will denote a connected reductive group defined over a finite field $\mathbb{F}_q$, (W, S) its Weyl group, and $\bar{S}$ the set of orbits of the Frobenius map on S. The subsets of $\bar{S}$ parametrize the classes of parabolic subgroups of $\mathbf{G}$ which are defined over $\mathbb{F}_q$; let $\mathcal{P}_I$ be the class corresponding to $I \subset \bar{S}$. (Thus $\mathcal{P}_\varnothing$ is the class of Borel subgroups and $\mathcal{P}_{\bar{S}} = \{\mathbf{G}\}$.) We denote by G the group of $\mathbb{F}_q$-rational points of $\mathbf{G}$ and by $\mathcal{P}_I$ the set of $\mathbb{F}_q$-defined parabolic subgroups in that class. If $\mathbf{P} \in \mathcal{P}_I$, we denote by P its group of $\mathbb{F}_q$-rational points and by U_P the group of $\mathbb{F}_q$-rational points of its unipotent radical. Bold letters denote algebraic subgroups; the corresponding roman letters denote their groups of rational points. Let K be an algebraically closed field of characteristic zero. All G-modules will be over K.

Let E be a G-module. For each $I \subset \bar{S}$ we define

$$E_{(I)} = \bigoplus_{\mathbf{P} \in \mathcal{P}_I} E^{U_P},$$

where E^{U_P} denotes the set of U_P-invariant vectors of E. We regard $E_{(I)}$ as a

G-module in a natural way. If $I \subset I'$, there is a canonical linear map $\varphi^I_{I'}\colon E_{(I)} \to E_{(I')}$. It is defined as follows: if $e \in E^{U_P}$ ($\mathbf{P} \in \mathcal{P}_I$), e can be also regarded as an element of $E^{U_{P'}}$, where $\mathbf{P'} \in \mathcal{P}_{I'}$ is uniquely defined by the condition $\mathbf{P} \subset \mathbf{P'}$. (Note that $U_P \supset U_{P'}$, hence $E^{U_P} \subset E^{U_{P'}}$.) The map $\varphi^I_{I'}$ takes $e \in E^{U_P}$ to e regarded as an element of $E^{U_{P'}}$. It is clear that if $I \subset I' \subset I''$, then

$$\varphi^{I'}_{I''}\varphi^I_{I'} = \varphi^I_{I''}. \tag{1.1}$$

Now let $\tilde{E}_{(I)} = E_{(I)} \otimes \Lambda^{|I|}(K^I)$. (We regard $\Lambda^{|I|}(K^I)$ as a G-module with trivial action.) If $I \subset I'$ and $|I'| = |I| + 1$, we have a natural map $\varepsilon^I_{I'}\colon \Lambda^{|I|}(K^I) \xrightarrow{\sim} \Lambda^{|I'|}(K^{I'})$ given by $\omega \mapsto \omega \wedge f$ where f is the unique element in $I' - I$. We define $\widetilde{\varphi}^I_{I'}\colon \tilde{E}_{(I)} \to \tilde{E}_{(I')}$ by $\widetilde{\varphi}^I_{I'} = \varphi^I_{I'} \otimes \varepsilon^I_{I'}$.

We now consider the sequence of maps of G-modules:

$$0 \to \tilde{E}_{(\varnothing)} \xrightarrow{d_0} \bigoplus_{|I|=1} \tilde{E}_{(I)} \xrightarrow{d_1} \bigoplus_{|I|=2} \tilde{E}_{(I)} \xrightarrow{d_2} \cdots \to \tilde{E}_{(\bar{S})} \to 0 \tag{1.2}$$

where the maps d have components $\widetilde{\varphi}^I_{I'}$ ($I \subset I'$, $|I'| = |I| + 1$). Using (1.1), we see that (1.2) is a complex.

2

We have the following:

THEOREM. Assume that E is irreducible. Let i_0 be the smallest integer ≥ 0 such that $\bigoplus_{|I|=i_0} \tilde{E}_{(I)} \neq 0$. Then the sequence

$$\bigoplus_{|I|=i_0} \tilde{E}_{(I)} \xrightarrow{d_{i_0}} \bigoplus_{|I|=i_0+1} \tilde{E}_{(I)} \xrightarrow{d_{i_0+1}} \cdots \to \tilde{E}_{(\bar{S})} \to 0 \tag{2.1}$$

is exact.

In particular, the homology of the complex (1.2) is concentrated in a single degree.

3

We shall prove that the conclusion of Theorem 2 holds in the case where instead of assuming E irreducible we shall assume that:

$$E \text{ admits a direct sum decomposition } E = \bigoplus_{\mathbf{P} \in \mathcal{P}_{I_0}} E_{\mathbf{P}} \text{ (for some } I_0 \subset \bar{S}) \tag{3.1}$$

such that

(a) $gE_{\mathbf{P}} = E_{g\mathbf{P}g^{-1}}$ $(\forall g \in G, \mathbf{P} \in \mathcal{P}_{I_0})$;
(b) for each $\mathbf{P} \in \mathcal{P}_{I_0}$, the P-module $E_{\mathbf{P}}$ factors through P/U_P and is irreducible, cuspidal as a P/U_P-module.
This would imply the theorem since any irreducible G-module is a direct summand of a G-module E as in (3.1).

In this and the following section, we assume that E is as in (3.1).

Given $J \subset \bar{S}$, we define $\mathcal{Z}_J$ to be the set of G-orbits σ on $\mathcal{P}_J \times \mathcal{P}_{I_0}$ which have the following property. There exist $\mathbf{P} \in \mathcal{P}_{I_0}$ and $\mathbf{Q} \in \mathcal{P}_J$, with $(\mathbf{Q}, \mathbf{P}) \in \sigma$, such that $\mathbf{Q}$ contains a Levi subgroup of $\mathbf{P}$.

If $\mathbf{Q} \in \mathcal{P}_J$, we have

$$E^{U_Q} = \Big(\bigoplus_{\mathbf{P}\in\mathcal{P}_{I_0}} E_{\mathbf{P}}\Big)^{U_Q} = \Big(\bigoplus_{\substack{\mathbf{P}\in\mathcal{P}_{I_0}\\(\mathbf{Q}\cap\mathbf{P})U_P=P}} E_{\mathbf{P}}\Big)^{U_Q} \oplus \Big(\bigoplus_{\substack{\mathbf{P}\in\mathcal{P}_{I_0}\\(\mathbf{Q}\cap\mathbf{P})U_P\subsetneq P}} E_{\mathbf{P}}\Big)^{U_Q}.$$

The second summand is zero (since $E_{\mathbf{P}}$ is cuspidal for P/U_P, so that $E_{\mathbf{P}}^{U_Q\cap P} = 0$). Thus

$$E_{(J)} = \bigoplus_{\mathbf{Q}\in\mathcal{P}_J} E^{U_Q} = \bigoplus_{\mathbf{Q}\in\mathcal{P}_J} \Big(\bigoplus_{\substack{\mathbf{P}\in\mathcal{P}_{I_0}\\(\mathbf{Q}\cap\mathbf{P})U_P=P}} E_{\mathbf{P}}\Big)^{U_Q} = \bigoplus_{\sigma\in\mathcal{Z}_J} E^{\sigma} \tag{3.2}$$

where

$$E^{\sigma} = \bigoplus_{\mathbf{Q}\in\mathcal{P}_J} \Big(\bigoplus_{\substack{\mathbf{P}\in\mathcal{P}_{I_0}\\(\mathbf{Q},\mathbf{P})\in\sigma\\(\mathbf{Q}\cap\mathbf{P})U_P=P}} E_{\mathbf{P}}\Big)^{U_Q} \qquad (\sigma \in \mathcal{Z}_J).$$

Now assume that $J \subset J'$ and $\sigma \in \mathcal{Z}_J$; we define $\delta^J_{J'}(\sigma) = \sigma' = \{(\mathbf{Q}', \mathbf{P}) \in \mathcal{P}_{J'} \times \mathcal{P}_{I_0} \mid \exists \mathbf{Q} \in \mathcal{P}_J,\ \mathbf{Q} \subset \mathbf{Q}',\ (\mathbf{Q}, \mathbf{P}) \in \sigma\}$. Then $\sigma' \in \mathcal{Z}_{J'}$ and we say that σ' is the face of type J' of σ. In this case, we define a linear map $\Psi^{\sigma}_{\sigma'}\colon E^{\sigma} \to E^{\sigma'}$ as follows. If $e = (e_{\mathbf{Q},\mathbf{P}}) \in E^{\sigma}$, $e_{\mathbf{Q},\mathbf{P}} \in E_{\mathbf{P}}$ for every $(\mathbf{Q}, \mathbf{P}) \in \sigma$, we set $\Psi^{\sigma}_{\sigma'}(e) = (e'_{\mathbf{Q}',\mathbf{P}}) \in E^{\sigma'}$ where, for any $(\mathbf{Q}', \mathbf{P}) \in \sigma'$, we have

$$e'_{\mathbf{Q}',\mathbf{P}} = \sum_{\substack{\mathbf{Q}\in\mathcal{P}_J\\\mathbf{Q}\subset\mathbf{Q}'\\(\mathbf{Q},\mathbf{P})\in\sigma}} e_{\mathbf{Q},\mathbf{P}} \in E_{\mathbf{P}}.$$

Let us admit the following

LEMMA 3.3. For any σ, σ' as above, the map $\Psi^{\sigma}_{\sigma'}\colon E^{\sigma} \to E^{\sigma'}$ is an isomorphism.

Note also that given $J \subset J' \subset J''$ and $\sigma \in \mathcal{Z}_J$, $\sigma' \in \mathcal{Z}_{J'}$ as above we can

define $\sigma'' \in \mathcal{Z}_{J''}$ to be the face of type J'' of σ (or of σ') and we have

$$\Psi^{\sigma}_{\sigma''} = \Psi^{\sigma'}_{\sigma''}\Psi^{\sigma}_{\sigma'}.$$

Using (3.2), the complex (1.2) can be rewritten as

$$0 \to \cdots \to 0 \to \bigoplus_{|J|=|I_0|} \bigoplus_{\sigma\in\mathcal{Z}_J} E^\sigma \otimes \Lambda^{|J|}(K^J) \xrightarrow{d} \bigoplus_{|J|=|I_0|+1} \bigoplus_{\sigma\in\mathcal{Z}_J} E^\sigma \otimes \Lambda^{|J|}(K^J) \xrightarrow{d} \cdots \tag{3.4}$$

where the maps d have components $\Psi^{\sigma}_{\sigma'} \otimes \varepsilon^J_{J'}$ ($J \subset J'$, $|J'| = |J| + 1$, $\sigma \in \mathcal{Z}_J$, $\sigma' \in \mathcal{Z}_{J'}$ the face of type J' of σ). Let ρ be the unique element in $\mathcal{Z}_{\bar{S}}$. We have $E^\rho = E$. Using Lemma 3.3, we may identify E^σ with E for any $\sigma \in \mathcal{Z}_J$, via the isomorphism Ψ^{σ}_{ρ}; the complex (3.4) becomes the tensor product of E with the complex

$$0 \to \cdots \to 0 \to \bigoplus_{|J|=|I_0|} K^{\mathcal{Z}_J} \otimes \Lambda^{|J|}(K^J) \to \bigoplus_{|J|=|I_0|+1} K^{\mathcal{Z}_J} \otimes \Lambda^{|J|}(K^J) \to \cdots \tag{3.5}$$

whose differential has components $\delta^J_{J'} \otimes \varepsilon^J_{J'}$ ($J \subset J'$, $|J'| = |J| + 1$). This complex has the following simple interpretation. Let T be an $\mathbb{F}_q$-split torus of maximal possible dimension in the adjoint group $\mathbf{G}_{\mathrm{ad}}$. Let Y be the lattice of one-parameter subgroups of T. The root hyperplanes in the real vector space $Y \otimes \mathbb{R}$ give rise to a partition of $Y \otimes \mathbb{R}$ into simplicial cones. These are in a natural 1–1 correspondence with the parabolic subgroups in $\mathbf{G}$ which are defined over $\mathbb{F}_q$ and whose image in $\mathbf{G}_{\mathrm{ad}}$ contain T. Let us fix $\mathbf{P}_0 \in \mathcal{P}_{I_0}$. It corresponds to a cone in $Y \otimes \mathbb{R}$, which spans a linear subspace $L \subset Y \otimes \mathbb{R}$ of codimension $|I_0|$. The cones contained in L are in 1–1 correspondence with those parabolic subgroups $\mathbf{Q} \subset \mathbf{G}$ (defined over $\mathbb{F}_q$) which contain a Levi subgroup of $\mathbf{P}_0$ and whose image in $\mathbf{G}_{\mathrm{ad}}$ contain T, hence they are in 1–1 correspondence with $\bigcup_J \mathcal{Z}_J$. Thus $\bigcup_{J\neq\bar{S}} \mathcal{Z}_J$ can be regarded as the set of simplices in a triangulation of a unit sphere with centre 0 in L; and (3.5) is the (reduced) chain complex of this triangulation. It follows that its homology is concentrated in a single degree (corresponding to $|J| = |I_0|$) where it is isomorphic to K.

4

It remains to prove Lemma 3.3. We will give the proof in six steps. Parts of the argument are similar in spirit to arguments in [3].

Step 1. Given $\sigma \in \mathcal{Z}_J$ and $(\mathbf{Q}, \mathbf{P}) \in \sigma$, we consider the parabolic subgroup $\mathbf{R} = (\mathbf{Q} \cap \mathbf{P})U_Q \subset \mathbf{Q}$. Then $\mathbf{R} \in \mathcal{P}_{\widetilde{J}}$ for some $\widetilde{J} \subset J$ and σ is the face of

type J of some $\widetilde{\sigma} \in \mathcal{Z}_{\widetilde{J}}$. Also $\mathbf{R}$, $\mathbf{P}$ have a common Levi subgroup and hence are associated. We first show that

$$\widetilde{\Psi}_\sigma^{\widetilde{\sigma}} \colon \bigoplus_{\mathbf{R} \in \mathcal{P}_{\widetilde{J}}} \Big(\bigoplus_{\substack{\mathbf{P} \in \mathcal{P}_{I_0} \\ (\mathbf{R},\mathbf{P}) \in \widetilde{\sigma}}} E_{\mathbf{P}} \Big)^{U_R} \longrightarrow \bigoplus_{\mathbf{Q} \in \mathcal{P}_J} \Big(\bigoplus_{\substack{\mathbf{P} \in \mathcal{P}_{I_0} \\ (\mathbf{Q},\mathbf{P}) \in \sigma}} E_{\mathbf{P}} \Big)^{U_Q}$$

is an isomorphism.

Now the pairs $(\mathbf{R}, \mathbf{P})$ such that $(\mathbf{R}, \mathbf{P}) \in \widetilde{\sigma}$ are in 1–1 correspondence with the pairs $(\mathbf{Q}, \mathbf{P})$ such that $(\mathbf{Q}, \mathbf{P}) \in \sigma$. Hence $\widetilde{\Psi}_\sigma^{\widetilde{\sigma}}$ is injective. If we fix $\mathbf{Q} \in \mathcal{P}_J$, there is a natural 1–1 correspondence between the set $\{\mathbf{R} \in \mathcal{P}_{\widetilde{J}} \mid \mathbf{R} \subset \mathbf{Q}\}$ and the set of orbits of U_Q on the set $\{\mathbf{P} \in \mathcal{P}_{I_0} \mid (\mathbf{Q}, \mathbf{P}) \in \sigma\}$. Hence the space E^σ is a direct sum over all pairs $\mathbf{R} \subset \mathbf{Q}$ ($\mathbf{R} \in \mathcal{P}_{\widetilde{J}}$, $\mathbf{Q} \in \mathcal{P}_J$) of pieces of form $\big(\bigoplus_{\mathbf{P} \in B} E_{\mathbf{P}}\big)^{U_Q}$, where $\mathbf{P}$ runs through a fixed U_Q-orbit. But this piece has the same dimension as $E_{\mathbf{P}}$. (Note that for $\mathbf{P}$ in that U_Q-orbit we have $P \cap U_Q \subset U_P$ and U_P acts trivially on $E_{\mathbf{P}}$.) Thus $\dim E^\sigma = \dim E_{\mathbf{P}} \cdot \#(\mathcal{P}_{\widetilde{J}}) = \dim E_{\mathbf{P}} \cdot \#(\mathcal{P}_{I_0}) = \dim E$. Similarly, $\dim E^{\widetilde{\sigma}} = \dim E$. It follows that $\varphi_\sigma^{\widetilde{\sigma}}$ is an isomorphism.

Step 2. Assume that $|I_0| = |\bar{S}| - 1$. Let $\mathcal{P}_J$ be the class of parabolic subgroups opposed to $\mathcal{P}_{I_0}$, and let $\sigma = \{(\mathbf{Q}, \mathbf{P}) \in \mathcal{P}_J \times \mathcal{P}_{I_0} \mid \mathbf{Q} \text{ opposed to } \mathbf{P}\}$. We will show that $\varphi_\rho^\sigma \colon E^\sigma \to E^\rho = E$ is an isomorphism. It is certainly non-zero. Thus, if E is irreducible, φ_ρ^σ is indeed an isomorphism. If E is reducible, it has exactly two composition factors, and J is necessarily equal to I_0. For each $\mathbf{P} \in \mathcal{P}_{I_0}$, there exists an isomorphism

$$\eta_{\mathbf{P}} \colon E_{\mathbf{P}} \xrightarrow{\sim} \Big(\bigoplus_{\substack{\mathbf{P}' \\ (\mathbf{P}',\mathbf{P}) \in \sigma}} E_{\mathbf{P}'} \Big)^{U_P}$$

of P/U_P-modules unique up to a scalar. We may assume that $\eta_{g\mathbf{P}g^{-1}} = g\eta_{\mathbf{P}}g^{-1}$ ($\forall \mathbf{P} \in \mathcal{P}_{I_0}$, $\forall g \in G$). These maps together define an isomorphism of G-modules $\eta = \bigoplus_{\mathbf{P} \in \mathcal{P}_{I_0}} \eta_{\mathbf{P}}$:

$$E = \bigoplus_{\mathbf{P} \in \mathcal{P}_{I_0}} E_{\mathbf{P}} \xrightarrow{\sim} \bigoplus_{\mathbf{P} \in \mathcal{P}_{I_0}} \Big(\bigoplus_{\substack{\mathbf{P}' \\ (\mathbf{P}',\mathbf{P}) \in \sigma}} E_{\mathbf{P}'} \Big)^{U_P}.$$

The composition $\varphi_\rho^\sigma \circ \eta$ is a semisimple G-endomorphism of E. It clearly has trace zero (its diagonal blocks are zero). Let λ', λ'' be its eigenvalues on the two irreducible pieces E', E'' of E. Then $\lambda' \dim E' + \lambda'' \dim E'' = 0$. Since $\dim E' \neq 0$, $\dim E'' \neq 0$, λ' and λ'' are either both zero or both non-zero. They cannot be both zero since $\varphi_\rho^\sigma \circ \eta \neq 0$. Hence they are both non-zero. It follows that $\varphi_\rho^\sigma \circ \eta$ and hence φ_ρ^σ are isomorphisms.

Step 3. Assume that $|I_0| \leq |\bar{S}| - 1$. Let $J \subset \bar{S}$ be such that $|J| = |I_0| + 1$,

and let $\tau \in \mathcal{Z}_J$. There are precisely two elements $\sigma' \in \mathcal{Z}_{J'}$, $\sigma'' \in \mathcal{Z}_{J''}$ such that J', $J'' \subset J$, $|J'| = |J''| = |I_0|$, and such that τ is the J-face of σ' and the J-face of σ''. We shall prove that $\varphi_\tau^{\sigma'}$ and $\varphi_\tau^{\sigma''}$ are isomorphisms. One of these maps, say $\varphi_\tau^{\sigma'}$, is of the type considered in Step 1 ($\sigma = \tau$); hence it is known to be an isomorphism. We now turn to $\varphi_\tau^{\sigma''}$. Define

$$E' = \bigoplus_{\mathbf{R}' \in \mathcal{P}_{J'}} E'_{\mathbf{R}'}$$

where, by definition,

$$E'_{\mathbf{R}'} = \Big(\bigoplus_{\substack{\mathbf{P} \in \mathcal{P}_{I_0} \\ (\mathbf{R}', \mathbf{P}) \in \sigma'}} E_{\mathbf{P}} \Big)^{U_{R'}}.$$

Then E' is a G-module in a natural way. It satisfies a property similar to property (3.1) for E. Let α be the set of pairs $(\mathbf{R}'', \mathbf{R}') \in \mathcal{P}_{J''} \times \mathcal{P}_{J'}$ such that $\mathbf{R}''$, $\mathbf{R}'$ are both contained in the same $\mathbf{Q} \in \mathcal{P}_J$ and are opposed in $\mathbf{Q}$. We define E'^{α} and $\Psi^\alpha \colon E'^{\alpha} \to E'$ in the same way as E^σ and $\varphi_\rho^\sigma \colon E^\sigma \to E$. We have a natural commutative diagram

$$\begin{array}{ccc} E'^{\alpha} & \xrightarrow{\Psi^\alpha} & E' \ (= E^{\sigma'}) \\ \big\downarrow & & \big\downarrow \varphi_\tau^{\sigma'} \\ E^{\sigma''} & \xrightarrow{\varphi_\tau^{\sigma''}} & E^\tau \end{array}$$

(Note that, given $(\mathbf{R}'', \mathbf{P}) \in \sigma''$ there is a unique $\mathbf{R}' \in \mathcal{P}_{J'}$ such that $(\mathbf{R}'', \mathbf{R}') \in \alpha$, $(\mathbf{R}', \mathbf{P}) \in \sigma'$; conversely, if $(\mathbf{R}'', \mathbf{R}') \in \alpha$, $(\mathbf{R}', \mathbf{P}) \in \sigma'$ then $(\mathbf{R}'', \mathbf{P}) \in \sigma''$. This gives rise to the isomorphism $E'^{\alpha} \xrightarrow{\sim} E^{\sigma''}$.) Now the right-hand vertical map in the diagram is known to be an isomorphism; moreover, by Step 2, the top map is an isomorphism. It follows that the bottom map $\varphi_\tau^{\sigma''}$ is an isomorphism.

Step 4. Let $\sigma \in \mathcal{Z}_J$, $|J| = |I_0|$. (Thus σ corresponds to an open cone in L.) We show that $\varphi_\rho^\sigma \colon E^\sigma \to E$ is an isomorphism. We can find a sequence $\sigma = \sigma_1, \sigma_2, \ldots, \sigma_n = \rho$ of elements corresponding to open cones in L such that (when regarded as cones in L) two consecutive ones have a common face of codimension 1; moreover, we take n as small as possible. Let τ be the element corresponding to the common face of codimension 1 of $\sigma = \sigma_1$ and σ_2. By induction on n, we may assume that $\varphi_\rho^{\sigma_2}$ is an isomorphism. From Step 3, we know that φ_τ^σ and $\varphi_\tau^{\sigma_2}$ are isomorphisms. From $\varphi_\rho^{\sigma_2} = \varphi_\rho^\tau \varphi_\tau^{\sigma_2}$, it follows that φ_ρ^τ is an isomorphism. From $\varphi_\rho^\sigma = \varphi_\rho^\tau \varphi_\tau^\sigma$ it follows that φ_ρ^σ is an isomorphism.

Step 5. Let $J \subset \bar{S}$ and $\sigma \in \mathcal{Z}_J$. We want to show that φ_ρ^σ is an isomorphism. We can find $\widetilde{J} \subset J$ and $\widetilde{\sigma} \in \mathcal{Z}_{\widetilde{J}}$ such that $|\widetilde{J}| = |I_0|$, such that σ is the face of type J of $\widetilde{\sigma}$, and such that $\varphi_\sigma^{\widetilde{\sigma}}$ is of the type considered in Step 1 (hence an isomorphism).

We have $\varphi_\rho^\sigma \varphi_\sigma^{\tilde{\sigma}} = \varphi_\rho^{\tilde{\sigma}}$. By Step 4, $\varphi_\rho^{\tilde{\sigma}}$ is an isomorphism. It follows that φ_ρ^σ is an isomorphism.

Step 6. Let $J \subset J'$ and $\sigma \in \mathcal{Z}_J$; let $\sigma' \in \mathcal{Z}_{J'}$ be the face of type J' of σ. We want to show that $\varphi_{\sigma'}^\sigma$ is an isomorphism. We have $\varphi_\rho^{\sigma'} \varphi_{\sigma'}^\sigma = \varphi_\rho^\sigma$. By Step 5, φ_ρ^σ and $\varphi_\rho^{\sigma'}$ are isomorphisms. It follows that $\varphi_{\sigma'}^\sigma$ is an isomorphism.

This completes the proof of Lemma 3.3 and hence that of the theorem.

5

We have

COROLLARY (see Alvis [1]). In the setting of the theorem, let $E^\#$ denote the kernel of d_{i_0} in (2.1). Then
(a) $E^\#$ is an irreducible G-module,
(b) $(E^\#)^\#$ is isomorphic to E as a G-module,
(c) $(-1)^{i_0} E^\# = \sum_{I \subset \bar{S}} (-1)^{|I|} E_{(I)}$ in the Grothendieck group of virtual G-modules.

Statement (c) is obvious.

Let E' be a G-module of the type considered in (3.1) which has a direct sum decomposition $E' = E_1 \oplus \cdots \oplus E_n$ with E_i irreducible G-modules and $E_1 \cong E$. The proof of Theorem 2 shows that $(E')^\#$ can be defined in the same way as $E^\#$, and that it has the following properties:

$$(E')^\# \cong E_1^\# \oplus \cdots \oplus E_n^\#, \qquad E'^\# \cong E'. \tag{5.1}$$

For each $1 \le i, j \le n$, we have from (c)

$$\langle E_i, E_j^\# \rangle = \langle E_i^\#, E_j \rangle = \sum_{I \subset \bar{S}} (-1)^{|I|+i_0} \langle E_i^{U_{P_I}}, E_j^{U_{P_I}} \rangle_{P_I} \tag{5.2}$$

(where $\mathbf{P}_I \in \mathcal{P}_I$). It follows that

$$\langle E_i^\#, E' \rangle = \sum_j \langle E_i^\#, E_j \rangle = \sum_j \langle E_i, E_j^\# \rangle = \langle E_i, (E')^\# \rangle = \langle E_i, E' \rangle$$

and, in particular, $E_i^\# \neq 0$. If $E_i^\#$ is not irreducible for some i, then $E_1^\# \oplus \cdots \oplus E_n^\#$ would have more irreducible components than $E_1 \oplus \cdots \oplus E_n$, contradicting (5.1). Thus, $E_i^\#$ is irreducible for each i. Hence, there exists a permutation π of $\{1, 2, \ldots, n\}$ such that $E_i^\# = E_{\pi(i)}$ for all i. From (5.2), we have

$$\langle E_{\pi^2(i)}, E_i \rangle = \langle (E_i^\#)^\#, E_i \rangle = \langle E_i^\#, E_i^\# \rangle = \langle E_{\pi(i)}, E_{\pi(i)} \rangle = 1.$$

It follows that $E_{\pi^2(i)}$ is isomorphic to E_i. The corollary is proved.

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DIVISION ALGEBRAS AND THE HAUSDORFF-BANACH-TARSKI PARADOX

by Pierre DELIGNE and Dennis SULLIVAN

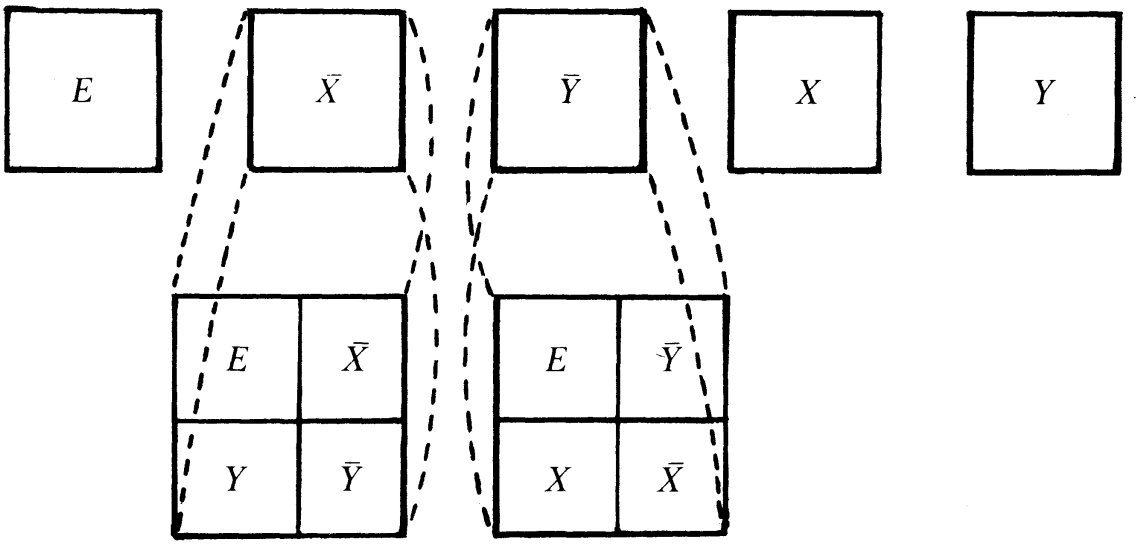
In this note we observe that a question raised by Dekker (1956) about rotations inspired by the Hausdorff-Banach-Tarski paradox can be answered using algebraic number theory. For motivation, we recall a form of the paradox.

Partition the free group in two generators F into the five sets E, X, $\bar{X}$, Y, $\bar{Y}$ consisting respectively of the identity and of the elements which, when written in reduced form, begin with x, x^{-1}, y or y^{-1}. If F acts freely on a sphere S, by rigid rotations, and if, using the axiom of choice, we choose a transversal T, i.e. a set with exactly one point in each orbit, the five subsets T, XT, $\bar{X}T$, YT, $\bar{Y}T$ form a partition of S. For economy of notation, we will write again E, X, $\bar{X}$, Y, $\bar{Y}$ for T, XT, $\bar{X}T$, YT, $\bar{Y}T$. The rotation by x moves $\bar{X}$ onto

$$S - X = E \cup \bar{X} \cup Y \cup \bar{Y}.$$

Similarly, y moves $\bar{Y}$ onto

$$S - Y = E \cup X \cup \bar{X} \cup \bar{Y}.$$



Thus we can reassemble from these 11 (actually 5) pieces 2 congruent spheres plus one congruent copy of the set E. This is a form of the Hausdorff-Banach-Tarski paradox which comes quickly from a free action of a free group

on two generators (see Appendix A for the more precise form). Thus we have the question of Dekker (communicated by Jan Mycielski): do such actions really exist? The sphere must be odd-dimensional for topological reasons: a fixed point free map $f : S^d \to S^d$ must have vanishing Lefschetz number

$$L(f) = \sum (-1)^i \operatorname{Trace}\big(f; H_i(S^d)\big) = 1 + (-1)^d \deg(f).$$

For a free action of a free group, the square of a non-trivial element will act by a map of degree +1 and this forces d odd. The dimension must be > 1. These are the only conditions.

THEOREM. *For $n \geq 2$, there is a free non-abelian group of rigid rotations acting freely on the odd dimensional sphere S^{2n-1}.*

Remark. The corresponding orthogonal matrices can be chosen to have algebraic entries, and the group of matrices corresponds to a subgroup of the non-zero elements in a division algebra over a number field.

Remark. The theorem was proved by Dekker for n even [D].

Remark. Let $u : O(p) \times O(q) \to O(p+q)$ be the natural embedding:

$$u(A, B) = \text{matrix} \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

If $A \in O(p)$ and $B \in O(q)$ don't have any nonzero fixed vector, neither has $u(A, B)$. If $\sigma_1 : F \to O(p)$ and $\sigma_2 : F \to O(q)$ define free actions on S^{p-1} and S^{q-1}, $u(\sigma_1, \sigma_2)$ hence defines a free action of F on S^{p+q-1}. Using this remark, one could reduce the theorem to the two particular cases $n = 2$ and $n = 3$.

Proof. Let $k \subset \mathbf{R}$ be a real algebraic number field, and $k' \subset \mathbf{C}$ be a quadratic extension of k. We assume that $k' \not\subset \mathbf{R}$, i.e. that $k' \otimes_k \mathbf{R} = \mathbf{C}$. Let D be a division algebra of dimension n^2 over its center k', equipped with an anti-involution $*$ inducing on k' the complex conjugation. The $\mathbf{R}$-algebra $D \otimes_k \mathbf{R}$ is a simple algebra over its center $k' \otimes_k \mathbf{R} = \mathbf{C}$, hence isomorphic to $M(n, \mathbf{C})$. We assume that, for a suitable isomorphism between $D \otimes_k \mathbf{R}$ and $M(n, \mathbf{C})$, $*$ becomes transpose conjugate.

In term of an isomorphism as above, the elements u of D satisfying $uu^* = 1$ become unitary matrices. They operate on the unit sphere in $\mathbf{C}^n$. Furthermore, if $u \neq 1$, $u - 1$ is invertible in D so that the corresponding matrix does not have 1 as an eigenvalue. It hence acts without fixed point on the sphere.

The group $\Gamma := \{u \in D \mid uu^* = 1\}$ is the group of k-rational points of a k-form of the real algebraic group $U(n)$. For $n > 1$, the perfect subgroup $SU(n)$

of $U(n)$ is not trivial, and $U(n)$ is not solvable. The group Γ is dense in $U(n)$: skew adjoints elements of D are dense in the skew adjoint matrices in $M(n, \mathbf{C})$, and the Cayley transform $t \mapsto \dfrac{t-1}{t+1}$ is an homeomorphism from the space of skew-adjoint matrices in $M(n, \mathbf{C})$ to an open dense subset of $U(n)$, carrying skew adjoint elements of D into Γ. From this density, it results that, if $n > 1$, the linear group Γ is not solvable. By [Tits], it contains a non abelian free subgroup.

It remains to construct pairs $(D, *)$. A division algebra D with center k' admits an anti-involution $*$ inducing on k' the non trivial element $\mathrm{Gal}(k'/k)$, if and only if its class $\mathrm{cl}(D)$ in the Brauer group $\mathrm{Br}(k')$ of k' has a trivial image by the norm map $N_{k'/k} : \mathrm{Br}(k') \to \mathrm{Br}(k)$—see Appendix B. Class field theory provides an explicit computation of $\mathrm{Br}(k)$, and of $N_{k'/k}$, and tells which elements of $\mathrm{Br}(k')$ come from division algebras. From the explicit description it provides, existence of such D follows. A direct construction is given in Appendix C. When we choose an isomorphism of $D \otimes_k \mathbf{R}$ with $M(n, \mathbf{C})$, the involution $*$ becomes adjunction with respect to some hermitian form ϕ on $\mathbf{C}^n$, not necessarily positive definite: $\phi(ax, y) = \phi(x, a^* y)$. If h is self adjoint in D, $\mathrm{int}(h^{-1}) \circ *$ is adjunction, with respect to the form $\phi_h(x, y) = \phi(hx, y)$. For suitable h, ϕ_h is positive definite and $(D, \mathrm{int}(h^{-1}) \circ *)$ is of the type sought.

Appendix A

Consider $\phi : S' \cup S'' \to S - E$ as in the introduction, with S' and S'' two copies of the sphere S, and $\psi : S \to S'$ the obvious bijection. Consider as in the Schröder-Bernstein theorem the set S_e of points p in S with an even number of ancestors, namely for which there exists an integer $n \geq 0$ with $p \in \mathrm{Im}(\phi \circ \psi)^n$ and $p \notin \mathrm{Im}(\psi \circ (\phi \circ \psi)^n)$. Consider also the set S_o of those p in S for which there exists $n \geq 0$ with $p \in \mathrm{Im}(\psi \circ (\phi \circ \psi)^n)$ and $p \notin \mathrm{Im}(\phi \circ \psi)^{n+1}$, and finally the set S_∞ of those p such that $p \in \mathrm{Im}(\phi \circ \psi)^n$ for any $n \geq 0$. Consider similarly

$$S' \cup S'' = (S' \cup S'')_e \cup (S' \cup S'')_o \cup (S' \cup S'')_\infty.$$

Then ψ induces a bijection from $S_e \cup S_\infty$ onto $(S' \cup S'')_o \cup (S' \cup S'')_\infty$ and ϕ^{-1} from S_o onto $(S' \cup S'')_e$. Combining these two we have a bijection $\chi : S \to S' \cup S''$ and a partition of S into finitely many pieces, the restriction of χ to each of these being a rotation.

Appendix B

Let K be a separable quadratic extension of a field k. We denote $x \mapsto \bar{x}$ the non trivial element $\mathrm{Gal}(K/k)$. Let D be a simple algebra with dimension n^2 over its center K. We will check the criterion of the text, for the existence of an involution of the second kind on D, i.e. of an anti-involution $*$ of D, inducing $x \mapsto \bar{x}$ on K. The criterion is that $N_{K/k}\,\mathrm{cl}(D) = 0$ in $\mathrm{Br}(k)$.

Let us localize, for the étale topology, over $\mathrm{Spec}(k)$. This means making large enough étale extensions of scalars $\otimes_k k'$, and keeping track of the functoriality in k'. The field K becomes the separable quadratic extension $K' = K \otimes_k k'$ of k'. The algebra D becomes $D' = D \otimes_k k'$, and is of the form $D' = \mathrm{End}_{K'}(V')$, for V' a free module K'. The module V' is not determined uniquely by D', only up to homotheties (the corresponding projective space is uniquely determined).

For any K-module M, let $\bar{M}$ be the module deduced from M by the extension of scalars $\bar{} : K \to K$, i.e. the module, unique up to unique isomorphism, provided with an anti-linear isomorphism $x \mapsto \bar{x} : M \xrightarrow{\sim} \bar{M}$. Similarly for K'-modules. If $D' = \mathrm{End}(V')$, then $\bar{D}' = \mathrm{End}(\bar{V}')$, and

$$(D \otimes_K \bar{D})' = \mathrm{End}(V' \otimes \bar{V}').$$

Let W' be the fixed subspace of the anti-linear automorphism of $V' \otimes \bar{V}'$ defined by $v \otimes \bar{w} \mapsto w \otimes \bar{v}$. It is the space of Hermitian forms on the dual of V'. One has $W' \otimes_{k'} K' = V' \otimes \bar{V}'$. If $D_1 \subset D \otimes_K \bar{D}$ is the fixed subspace of the anti-linear automorphism of $D \otimes_K \bar{D}$ defined by $x \otimes \bar{y} \mapsto y \otimes \bar{x}$, then D'_1 is the k'-form of the K'-algebra $(D \otimes_K \bar{D})' = \mathrm{End}(V' \otimes \bar{V}')$ deduced from the k'-form W' of the K'-module $V' \otimes \bar{V}'$: $D'_1 = \mathrm{End}_{k'}(W')$.

Involutions of the second kind on D' correspond one to one to non degenerate Hermitian forms on V', taken up to a factor (in k'^*). Those, in turn, by the "dual form" construction, correspond to "non degenerate" elements of W'. Again, one has to take them up to a factor. The projective space $\mathbf{P}(W')$ over k' is determined up to unique isomorphism by D'. It is hence (this is the point of localisation) defined over k: $\mathbf{P}(W') = P \otimes_k k'$, functorially in k'. The k-points of P (rather, the non degenerate points) parametrize the involutions of the second kind on D.

The functorial isomorphism $D'_1 = \mathrm{End}_{k'}(W')$ shows that P is the form of projective space (Severi-Brauer variety) attached to D_1. It has a rational point, and is then the ordinary projective space, if and only if D_1 is a matrix algebra.

This shows that D has involutions of the second kind if and only if the class of D_1 in $\mathrm{Br}(k)$ is trivial. This class is the required norm $N_{K/k}(\mathrm{cl}(D))$. In the localization spirit, this can be deduced from the fact that the homothety by $\lambda \in K'^*$ of V' induces on W' the homothety by $N_{K'/k'}(\lambda) \in k'^*$.

Appendix C

For $n \geq 3$, examples can be obtained as follows: take $k' = \mathbf{Q}[\zeta]$, with $\zeta = \exp(2\pi i/n)$, and $k = k' \cap \mathbf{R}$. Fix $a, b \in k^*$ and let D be the k'-algebra generated by X, Y, subject to

$$X^n = a, \qquad Y^n = b$$

$$XY = \zeta YX.$$

It admits the anti-involution $*$, inducing complex conjugation on k', defined by $\zeta^* = \zeta^{-1}$, $X^* = X$, $Y^* = Y$. The algebra D is of the type we require, provided it is a division algebra. This happens already with $a, b \in \mathbf{Z}$: take for a a prime congruent to 1 mod n, and for b an integer whose residue mod a has in the cyclic group of order n $(\mathbf{Z}/(a))^*/(\mathbf{Z}/(a))^{*n}$ an image of exact order n. For instance $n = 3$, $a = 7$, $b = 2$. For $n = 2$, one proceeds similarly with $k' = \mathbf{Q}[i]$, $\zeta = -1$, a congruent to 1 mod 4 and b not a square mod a. For instance, $a = 5$, and $b = 2$. In each case, the assumption on a ensures that k' embed in the a-adic completion $\mathbf{Q}_a$ of $\mathbf{Q}$, and the fact that D is a division algebra can be seen locally at a: $D \otimes_{k'} \mathbf{Q}_a$ is a division algebra with center $\mathbf{Q}_a$.

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Pierre Deligne
Dennis Sullivan

Institut des Hautes Etudes Scientifiques
35, route de Chartres
91440 Bures-sur-Yvette
France